Grundlehren der mathematischen Wissenschaften 364
A Series of Comprehensive Studies in Mathematics

Bertram Huppert

Finite Groups I

Springer

Grundlehren der mathematischen Wissenschaften

A Series of Comprehensive Studies in Mathematics

Volume 364

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As far as the development of mathematics permits, the direction of GL remains true to the original spirit of Courant. Many of the oldest volumes are popular to this day and some have not been superseded. One should perhaps never advertise a contemporary book as a classic but many recent volumes and many forthcoming volumes will surely earn this attribute through their use by generations of mathematicians.

Bertram Huppert

Finite Groups I

Translated by Christopher A. Schroeder

Springer

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ISSN 0072-7830 ISSN 2196-9701 (electronic)
Grundlehren der mathematischen Wissenschaften
ISBN 978-3-031-87528-1 ISBN 978-3-031-87529-8 (eBook)
https://doi.org/10.1007/978-3-031-87529-8

Mathematics Subject Classification (2020): 20-01, 20-02

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Foreword

In honor of Bertram Huppert (*1927–2023*).

Since its publication in 1967, Bertram Huppert's *Endliche Gruppen I* has remained a vital mathematical resource because it is clear, precise and complete. With this in mind, I have aspired to an especially faithful translation. The theorem numbers remain the same, so the German original and this translation can be referenced interchangeably. The only significant change I have made is to update the following notation:

Object	German original	English translation
Group	$\mathfrak{G}$	G
Order of group	g	n or $\lvert G\rvert$
Index of subgroup	$\lvert G : H\rvert$	$[G : H]$
Identity subgroup	$\mathfrak{E}$	1
Group element	G	g
Group identity	E	1
Order of group element	$O(G)$	$\lvert g\rvert$
Ring	$\mathfrak{R}$	R

Open problems that have been resolved by more recent work are updated in the footnotes. Corrections from the errata are incorporated and are also noted in the text. The remaining errors are my own.

I would like to thank Thomas Keller for supervising the translation, Rémi Lodh for his editorial assistance and Gunter Malle for his heroic proofreading of the entire manuscript. I also thank Tim Burness, Bettina Eick, Bob Guralnick, Evgeny Khukhro, Klaus Lux, Gunter Malle, Alexander Moretó, Chris Parker and Gunnar Traustason for their help updating the text. Above all, I am grateful for Professor Huppert's permission to undertake the translation. With his passing, I deeply appreciate this connection to his life and work, and I would be thrilled if this translation enables more readers to spend time in his company.

New York, 10.22.2024 *Christopher A. Schroeder*

Preface



(Jan van Eyck, on several paintings after 1433)

When I began the preliminary work for this book in 1958, it still seemed possible to give a somewhat complete presentation of the structure of finite groups in a single volume. The tempestuous development of the theory since then, of which the bibliography gives an impression, has made this objective impossible. In addition to the foundational material, the present volume contains the theory of nilpotent, p-nilpotent and solvable groups, as well as ordinary representation theory. As recent developments have not focused on these areas at the time of writing, a somewhat complete overview of the state of these topics can be given here. (The theory of formations and Fitting classes, which arose in the few years before publication of this book, could only be discussed in part.) The second and third volumes are planned to contain the theory of subnormal subgroups, a more refined theory of p-length, multiply-transitive permutation groups and several newer applications of character theory. Due to the wealth of results in these areas, it is not possible for us to aspire to a complete treatment.

Several topics were excluded from this volume:

1. A unified treatment of the now well-known series of finite simple groups that follow from the methods of Chevalley would have required considerable prior knowledge of Lie algebras. I restricted myself in Chapter II to the projective and symplectic groups. The simple groups of Mathieu and Suzuki will be discussed in Volume 3.

2. The theory of p-groups of exponent p and the connection between nilpotent groups and Lie rings were not touched upon. This theory and its methods have few connections to current questions regarding the structure of finite groups, and their applications lie largely in the theory of infinite groups.

3. The cohomology theory of finite groups has thus far found its most important application in class field theory and not in the structure theory of finite groups. Therefore, I have developed only as much cohomology theory (dispensing with the functorial description of cohomology groups) that is required to handle the topics covered in this book (extension theory, automorphisms of p-groups, Schur multipliers) in §16 of Chapter I.

Although this book requires no previous exposure to groups, it is intended for readers with a basic knowledge of algebra. Chapter I develops the fundamental principles and elementary tools of group theory in full, although concisely and without motivation. Tools from ring theory, such as the theory of principal ideal domains (Chapter I, §13) and semisimple algebras (Chapter V, §2–§4), are developed in full. In contrast, I occasionally used simple observations about $\mathfrak{p}$-adic fields and algebraic number fields without proof.

I did not attempt to provide exercises consistently throughout the book. In Chapter I, numerous observations and proof variants are offered in the exercises, whereas later chapters are nearly devoid of problems. The results of exercises are rarely referenced in the main text.

Numerous people supported me in the writing of this book. R. Baer, N. Itô, O. H. Kegel and J. Rose read different versions of several chapters closely. N. Blackburn and H. Lüneberg read the complete manuscript. I was able to thoroughly discuss almost every chapter in the manuscript or in the galley proof with W. Gaschütz, F. Gross and J. Neubüser. This led to numerous improvements, including correcting errors and improving clarity. I thank N. Blackburn, R. Carter, E. Dade, W. Gaschütz, P. Hall, P. Roquette, D. Taunt, J. G. Thompson and H. Wielandt for unpublished results, proofs or examples. A. Brandis, K. Doerk, W. Gaschütz, K. D. Graf, F. Gross, B. Klaiber, J. Neubüser and A. Schlette helped with the lengthy corrections. I deeply thank all of the above for their assistance. In addition, I have to thank the following institutions for the opportunity to work on this book and for fruitful discussions with colleagues: the British Council for a stay in Manchester 1958/59; the University of Illinois in Urbana for a guest professor position 1963/64; the California Institute of Technology in Pasadena for a guest professor position in Fall 1964. Finally, I thank the publisher for generously accepting the many changes that I requested during proofreading and for the excellent quality of the production of this book.

My very special thanks go to my teacher Helmut Wielandt. His lectures, which introduced me to the theory of finite groups, were often the starting point for writing sections and sometimes entire chapters of this book. They have frequently influenced the content and style of my presentation.

Mainz, 9.15.1967 *Bertram Huppert*

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Chapter I
Foundations

We begin Chapter I by developing in a concise but complete fashion the basic concepts of group theory (§1–§4). These sections contain numerous elementary theorems and lemmas that will be used repeatedly later on; this applies in particular to the product formula in Lemma 2.12 a), the Dedekind identity in Lemma 2.12 c), Lemma 2.13 and Theorems 4.5, 4.8 and 4.9. The representations of a group as a permutation group (§6) yield the fundamental Sylow theorems (§7) as an important consequence. Several elementary criteria for solvability will be deduced from Sylow's theorems in §8.

The remaining sections of Chapter I are dedicated in large part to various constructions in finite group theory: direct products (§9 and §12), extension theory (§14) and wreath products (§15). We describe two constructions that are derived from the direct product, namely the direct product with identified central subgroups (also known as the central product) in Theorem 9.10 and the direct product with identified factor groups (also known as the subdirect product) in Theorem 9.11. We return multiple times to the important question of when a given group can be built from simpler groups by one of these constructions. In this way, we obtain in §13 the fundamental theorem of finitely generated abelian groups as a special case of the theory of modules over principal ideal domains. The Zassenhaus theorem (§18), which provides sufficient conditions for an extension to be a semidirect product, is of fundamental importance for the entire theory of finite groups. In §16, we treat the theory of group cohomology only to the extent needed for a further study of p-group automorphisms (Chapter III, §19) and Schur multipliers (Chapter V, §23).

In several places, it was ideal or even necessary to remove our restriction to finite groups. The theory of direct products, composition series and the direct decomposition of groups with operators and of modules (§9–§13) is developed so that many of the theorems about modules required in representation theory (Chapter V) can be deduced as special cases. Finally, it was unavoidable to discuss free groups in order to develop the description of groups by generators and relations in §19.

We recommend that the inexperienced reader restrict themselves to §1–§9 and §18 on a first reading; a proof of Theorem 17.5 without the use of cohomology theory is sketched in Exercise 70.

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B. Huppert, *Finite Groups I*, Grundlehren der mathematischen Wissenschaften 364,
https://doi.org/10.1007/978-3-031-87529-8_1

1 The group axioms

Group Axioms 1.1. *A nonempty set G is called a group if the following conditions are satisfied:*

a) *To every ordered pair (a, b) of elements $a, b \in G$ there is assigned a unique element $c \in G$. We write $c = ab$ and call c the product of a and b.*
b) *The associative property $(ab)c = a(bc)$ holds for all $a, b, c \in G$.*
c) *There exists an element $e \in G$ with the property that $ea = a$ for all $a \in G$. We call e an identity element of G. (We will show that there is only one such element, and we will then denote it by* 1.)
d) *For every element $a \in G$, there exists an element $b \in G$ such that $ba = e$.*

We first derive a few simple corollaries from the axioms.

Corollary 1.2. *The formation of products in a group is independent of the parentheses; more exactly:*

For arbitrary elements $a_1, a_2, \ldots$ of G we define the subsets $\mathscr{P}_k(a_1, \ldots, a_k)$ of G recursively by

$$\mathscr{P}_1(a_1) = \{a_1\}, \quad \mathscr{P}_2(a_1, a_2) = \{a_1 a_2\}$$

and

$$\mathscr{P}_k(a_1, \ldots, a_k) = \{xy \mid x \in \mathscr{P}_m(a_1, \ldots, a_m), y \in \mathscr{P}_n(a_{m+1}, \ldots, a_{m+n}), k = m + n\}.$$

(The elements of $\mathscr{P}_k(a_1, \ldots, a_k)$ are therefore the products $a_1 \cdots a_k$ with all possible meaningful parentheses.) We claim that the subset $\mathscr{P}_k(a_1, \ldots, a_k)$ contains exactly one element for every k. We denote this element by $a_1 \cdots a_k$.

Proof. We prove by induction on k that every $\mathscr{P}_k(a_1, \ldots, a_k)$ contains exactly one element. For $k = 1, 2$ this is clear from the definition. Let $k \geq 3$, and let $g = xy \in \mathscr{P}_k(a_1, \ldots, a_k)$ with $x \in \mathscr{P}_m(a_1, \ldots, a_m)$ and $y \in \mathscr{P}_n(a_{m+1}, \ldots, a_k)$. By our induction hypothesis, $x = a_1 z$ with $z = a_2 \cdots a_m$; if $m = 1$, $z = e$ is omitted. Using Group Axiom 1.1 b), we have

$$g = xy = (a_1 z)y = a_1(zy).$$

It then follows by induction that $zy \in \mathscr{P}_{k-1}(a_2, \ldots, a_k) = \{a_2 \cdots a_k\}$, and we are done. □

Corollary 1.3.

a) *If e is an identity element of the group G, then $ae = a$ for all $a \in G$.*
b) *If $ba = e$, then $ab = e$.*
c) *If $ax = ay$ or $xa = ya$, then $x = y$.*

Proof. a) By the Group Axiom 1.1 d), there exist elements $x, y \in G$ such that $xa = e$ and $yx = e$. It follows that

$$ye = y(xa) = (yx)a = ea = a.$$

Since $ee = e$, we obtain

$$a = ye = y(ee) = (ye)e = ae,$$

as claimed.

b) By the Group Axiom 1.1 d), there exists some $x \in G$ such that $xb = e$. Using Corollary 1.2, we have

$$\begin{aligned} e = xb = xeb = x(ba)b \\ = (xb)(ab) = e(ab) = ab. \end{aligned}$$

c) Let $ba = e$. Then we also have $ab = e$ and

$$x = ex = (ba)x = b(ax) = b(ay) = (ba)y = ey = y,$$

and similarly

$$x = xe = x(ab) = (xa)b = (ya)b = y(ab) = ye = y. \qquad \square$$

Corollary 1.4. *Let G be a group. For each pair of elements $a, b \in G$, there exists a unique element $x \in G$ such that $ax = b$. Likewise, there exists a unique element $y \in G$ such that $ya = b$.*

Proof. Let c be an element such that $ca = e$, which exists by the Group Axiom 1.1 d). By Corollary 1.3 b), we also have $ac = e$. Then $x = cb$ is an element with the desired property, for we have

$$ax = a(cb) = (ac)b = eb = b.$$

If x_1 and x_2 are elements satisfying $ax_1 = ax_2 = b$, then it follows immediately from Corollary 1.3 c) that $x_1 = x_2$. $\square$

It follows directly from Corollary 1.4 that G possesses a unique identity element, which we denote by 1 in the rest of the text. Furthermore, we have the following:

Corollary 1.5.

a) *For each $a \in G$, there is a unique element $x \in G$ such that $xa = 1$. Then $ax = 1$, as well. We call x the inverse of a and write $x = a^{-1}$.*

b) *We have $(ab)^{-1} = b^{-1}a^{-1}$ and $(a^{-1})^{-1} = a$.*

Proof. Part a) is a consequence of Corollary 1.4 and Corollary 1.3 b). Then part b) follows by Corollary 1.2 and the uniqueness of inverse elements since

$$(b^{-1}a^{-1})(ab) = (b^{-1}(a^{-1}a))b = b^{-1}b = 1, \quad aa^{-1} = 1. \qquad \square$$

Definition 1.6. We define the powers of the group element $g \in G$ recursively by $g^0 = 1$ and $g^{i+1} = g^i g$; for $i < 0$ we define $g^i = (g^{-i})^{-1}$. It is easy to see that $g^{i+j} = g^i g^j$ for all integers i and j.

Definition 1.7. We denote the cardinality of a set M by $|M|$. If G is a group, we refer to $|G|$ as the order of G. We say that G is a finite group if $|G|$ is finite.

Oftentimes, we are not interested in the nature of the specific elements of a group, but rather only in the algebraic relationships between elements. In order to allow for this, we introduce the following equivalence relation:

Definition 1.8. Two groups G_1 and G_2 are said to be isomorphic if there is a bijective map π from G_1 to G_2 such that

$$(gh)^\pi = g^\pi h^\pi$$

for all $g, h \in G_1$. In this case, we write $G_1 \cong G_2$. Isomorphism is clearly an equivalence relation. We understand group properties to be the properties satisfied by G and every group isomorphic to G.

We note the following:

Corollary 1.9. *If π is an isomorphism from G_1 to G_2 and 1_i is the identity element of G_i, then $1_1^\pi = 1_2$. Furthermore, we have $(g^{-1})^\pi = (g^\pi)^{-1}$ for all $g \in G_1$.*

Proof. The claim $1_1^\pi = 1_2$ follows from the fact that, for all $g \in G_1$,

$$1_1^\pi g^\pi = (1_1 g)^\pi = g^\pi$$

together with the uniqueness guaranteed by Corollary 1.4. It follows from

$$(g^{-1})^\pi g^\pi = (g^{-1} g)^\pi = 1_1^\pi = 1_2$$

that $(g^{-1})^\pi = (g^\pi)^{-1}$. □

Definition 1.10. A group G is called abelian (or commutative) if for all $a, b \in G$ we have $ab = ba$. In this case, every product $a_1 \cdots a_n$ with $a_i \in G$ is independent of the ordering of the a_i.

2 Subgroups

Definition 2.1. A subset U of the group G is called a subgroup of G whenever U is a group with respect to the product defined on G. This means:

If $u_1, u_2 \in U$, then $u_1 u_2 \in U$; if $u \in U$, then $u^{-1} \in U$; the identity element 1 of G lies in U. (Of course, 1 is then the identity element of U.)

We write $U \leq G$ whenever U is a subgroup of G; if U is a proper subgroup of G, then we write $U < G$. Every group contains the subgroup $\{1\}$ consisting solely of the identity element. In an abuse of notation, we will denote this group by 1. We occasionally refer to the groups 1 and G as the trivial subgroups of G.

The following is often useful:

Lemma 2.2. *If U is a finite subset of the group G and $u_1u_2 \in U$ for all $u_1, u_2 \in U$, then U is a subgroup of G.*

Proof. Let u_0 be an arbitrary element of U. The products uu_0 with $u \in U$ are distinct by Corollary 1.3 c), so they yield exactly $|U|$ elements of U. Hence, $U = \{uu_0 \mid u \in U\}$. In particular, there is a $u_1 \in U$ such that $u_1u_0 = u_0 = 1u_0$. Then $u_1 = 1$ by Corollary 1.3 c). Furthermore, there exists some $u_2 \in U$ such that $u_2u_0 = 1$, and therefore $u_2 = (u_0)^{-1}$ lies in U as well. □

Theorem 2.3. *If U_i is a subgroup of G for every i in an index set I, then the intersection $\bigcap_{i \in I} U_i$ is a subgroup of G.*

Proof. If a and b lie in all subgroups U_i, then a^{-1} and ab lie in all U_i, and so also in $\bigcap_{i \in I} U_i$. Clearly, 1 also lies in the intersection. □

Definition 2.4. Let M be a subset of the group G. We denote by $\langle M \rangle$ the intersection of all subgroups of G that contain M and refer to it as the subgroup of G generated by M. It is clear that $\langle M \rangle$ is the smallest subgroup of G that contains the set M. If we define $M^{-1} = \{m^{-1} \mid m \in M\}$, then we have

$$\langle M \rangle = \{1, x_1 \cdots x_s \mid x_i \in M \cup M^{-1},\ s = 1, 2, \dots\}.$$

If $M = \{a, b, \dots\}$, we also write $\langle M \rangle = \langle a, b, \dots \rangle$. If $G = \langle M \rangle$, then we say that M is a generating set for G. A group generated by one element is called cyclic. Defining $a^0 = 1$, the group $\langle a \rangle$ then consists exactly of the elements a^i with $i = 0, \pm 1, \pm 2, \dots$, which are not necessarily all distinct.

Theorem 2.5. *Let U be a subgroup of G. The sets*

$$gU = \{gu \mid u \in U\} \quad (\textit{with } g \in G)$$

are called the left cosets of U in G. For all $g, h \in G$, we have either $gU \cap hU = \emptyset$ or $gU = hU$. The group G is the disjoint union of the distinct left cosets gU. If a set R contains exactly one element of every left coset of U, then we say R is a set of left coset representatives (or a left transversal) for U in G. We refer to the partition

$$G = \bigcup_{r \in R} rU$$

as the decomposition of G into left cosets of U. All cosets of U have the same cardinality $|U|$. We refer to the cardinality $|R|$ of R as the index of U in G and denote it by $[G : U]$. (Analogous statements hold for the right cosets Ug.)

Proof. We define an equivalence relation on the elements of G by $x \sim y$ if $x^{-1}y \in U$. This is actually an equivalence relation:

The relation is reflexive since $x^{-1}x = 1 \in U$ implies $x \sim x$. If $x \sim y$, then $x^{-1}y \in U$. It follows that $(x^{-1}y)^{-1} = y^{-1}x \in U$ and so $y \sim x$, which means the relation is symmetric. Finally, if $x \sim y$ and $y \sim z$, then $x^{-1}y \in U$ and $y^{-1}z \in U$, and we deduce that $x^{-1}z = (x^{-1}y)(y^{-1}z) \in U$. So $x \sim z$ and the relation is transitive.

It is well known that we can now write G as the disjoint union of the equivalence classes of our relation. The class containing $g \in G$ is clearly the left coset gU. The other claims are now clear. □

Theorem 2.6.

a) *Let U be a subgroup of G. If $G = \bigcup_{r \in R} Ur$ is the decomposition of G into right cosets of U, then $G = \bigcup_{r \in R} r^{-1}U$ is the decomposition of G into left cosets of U.*
(*In particular, we need not distinguish between a left and right index of U in G.*)

b) *If $G = \bigcup_{r \in R} Ur$ is the decomposition of G into right cosets of the subgroup U and $U = \bigcup_{s \in S} Vs$ is the decomposition of U into right cosets of V, then $G = \bigcup_{s \in S, r \in R} Vsr$ is the decomposition of G into rights cosets of V. In particular, we have $[G : V] = [G : U][U : V]$.*

Proof. a) The map α defined by $g^\alpha = g^{-1}$ is a bijective map from G to itself, and $(Ur)^\alpha = r^{-1}U$.

b) If $Vs_1r_1 = Vs_2r_2$, then by multiplying on the left by U we have $Ur_1 = Ur_2$. So $r_1 = r_2$, and then $s_1 = s_2$. It is trivial that $G = \bigcup_{s \in S, r \in R} Vsr$. □

The following theorem of Lagrange is fundamental to the theory of finite groups. It brings an arithmetical element into the theory.

Main Theorem 2.7 (Lagrange). *Let G be a finite group and U a subgroup of G. Then $|G| = |U|[G : U]$. In particular, the order and index of a subgroup of a finite group are divisors of the group order.*

Proof. This is the special case $V = 1$ of Theorem 2.6 b). □

Definition 2.8. Let g be an element of the group G. The smallest natural number n such that $g^n = 1$ is called the order $|g|$ of g; if there is no such natural number, we write $|g| = \infty$. The least common multiple of the orders $|g|$ of the elements $g \in G$, if it exists, is called the exponent $\exp G$ of G.

Theorem 2.9. *Let G be a group. Suppose $g \in G$ has finite order $|g| = n < \infty$.*

a) *If $g^m = 1$, then $n \mid m$.*

b) *We have $\langle g \rangle = \{1, g, g^2, \ldots, g^{n-1}\}$ and $|\langle g \rangle| = |g| = n$. In particular, if G is a finite group, then $|g|$ is a divisor of $|G|$.*

c) *We have $|g^k| = n/(n, k)$ for every integer k.*

Proof. a) Let $m = nr + s$ with $0 \le s < n$. Then

$$1 = g^m = g^{nr+s} = (g^n)^r g^s = g^s.$$

The minimality of $n = |g|$ then forces $s = 0$, so that $n \mid m$.

b) Let $i+j = nr+s$ with $0 \le s < n$. Then $g^{i+j} = g^s$. The finite set $\{g^s \mid 0 \le s < n\}$ is then a group by Lemma 2.2, so it is exactly $\langle g \rangle$. Assume that $g^i = g^j$ with $0 \le i \le j < n$. Then $g^{j-i} = 1$. It follows from a) that $n \mid j - i$, so $i = j$. Hence, the g^i with $0 \le i \le n - 1$ are pairwise distinct. Since $g^{nk+r} = g^r$, we have

$$\langle g \rangle = \{1, g, g^2, \ldots, g^{n-1}\}.$$

It follows from Theorem 2.7 that $|g| = |\langle g \rangle|$ is a divisor of $|G|$.

c) Since $n \,\Big|\, k\frac{n}{(n,k)}$, we have that

$$g^{k\frac{n}{(n,k)}} = 1.$$

Then a) implies that $|g^k|$ divides $\frac{n}{(n,k)}$. Conversely, it follows from $g^{kt} = 1$ and a) that $n \mid kt$, so $\frac{n}{(n,k)} \,\Big|\, t$. Therefore, $|g^k| = \frac{n}{(n,k)}$. □

Theorem 2.10. *Every group of prime order is cyclic.*

Proof. Let $|G| = p$ be a prime number, and let $g \in G$ be distinct from the identity. We set $U = \langle g \rangle$. Then $|U|$ is a divisor of $|G| = p$ by Theorem 2.7. Since $1 < |U|$ it follows that $|U| = |G|$, so $U = G$ and G is cyclic. □

Theorem 2.10 is the first in a long series of theorems that deduce statements about G from the prime factorization of $|G|$.

Definition 2.11. Let A and B be subsets of the group G. We define

$$AB = \{ab \mid a \in A,\ b \in B\}.$$

We emphasize that AB is not necessarily a subgroup of G, even if A and B are subgroups of G! (See §5, Exercise 18.)

The following is often useful:

Lemma 2.12. *Let A and B be subgroups of G.*

a) *Let $A = \bigcup_{r \in R} r(A \cap B)$ be the decomposition of A into cosets of $A \cap B$. Then $AB = \bigcup_{r \in R} rB$ is a disjoint union. In particular, if A and B are finite, then*

$$|AB| = \frac{|A||B|}{|A \cap B|}.$$

b) *AB is a subgroup of G if and only if $AB = BA$.*

c) *(Dedekind identity) If $A \le C \le G$ and $C \subseteq AB$, then $C = AB \cap C = A(B \cap C)$.*

Proof. a) Clearly, $AB = \bigcup_{r \in R} rB$. We show that the union is disjoint. Assume that $rB \cap r'B \neq \emptyset$. Then there exist elements $b, b' \in B$ with $rb = r'b'$. Since $R \subseteq A$, it follows that

$$(r')^{-1}r = b'b^{-1} \in A \cap B,$$

so $r = r'$.

b) First assume that AB is a subgroup of G. Then

$$\begin{aligned} AB &= \{ab \mid a \in A,\ b \in B\} = \{(ab)^{-1} \mid a \in A,\ b \in B\} \\ &= \{b^{-1}a^{-1} \mid b \in B, a \in A\} = BA. \end{aligned}$$

Conversely, if $AB = BA$ then every product ba with $a \in A$ and $b \in B$ can be written as $ba = a'b'$ with $a' \in A$ and $b' \in B$.

For elements $a_i \in A$ and $b_i \in B$ $(i = 1, 2)$, it follows that

$$(a_1b_1)(a_2b_2) = (a_1a_2')(b_1'b_2) \in AB$$

and

$$(ab)^{-1} = b^{-1}a^{-1} = (a^{-1})'(b^{-1})' \in AB.$$

So AB is a subgroup of G.

c) Let $c \in C$. Since $C \subseteq AB$, $c = ab$ for some $a \in A$, $b \in B$. Since $A \leq C$, $b = a^{-1}c \in B \cap C$, so $C \subseteq A(B \cap C)$. The reverse inclusion $A(B \cap C) \subseteq C$ is trivial as the group C contains A and $B \cap C$, so C contains their product. □

Lemma 2.13. *Let A and B be subgroups of G with finite index. Then:*

a) $[G : A \cap B] \leq [G : A][G : B]$.

b) *If $[G : A]$ and $[G : B]$ are coprime, then $[G : A \cap B] = [G : A][G : B]$. If in addition G is finite, then $G = AB$.*

Proof. a) Let d be the number of left cosets of B in G that lie in AB. By Lemma 2.12 a), we have $[A : A \cap B] = d \leq [G : B]$. Then by Theorem 2.6 b),

$$[G : A \cap B] = [G : A][A : A \cap B] \leq [G : A][G : B].$$

b) Since $[G : A]$ divides $[G : A \cap B]$, $[G : B]$ divides $[G : A \cap B]$ and $([G : A], [G : B]) = 1$, we have that $[G : A][G : B]$ is a divisor of $[G : A \cap B]$. It therefore follows from a) that $[G : A \cap B] = [G : A][G : B]$. If G is finite, then Lemma 2.12 a) implies that

$$|AB| = \frac{|A||B|}{|A \cap B|} = |G|,$$

and therefore $G = AB$. □

Lemma 2.14. *For all $a, g \in G$, we define $g^a = a^{-1}ga$. We call the map $g \mapsto g^a$ conjugation by a. The following identities hold:*
(1) $g^{ab} = (g^a)^b$
(2) $(gh)^a = g^a h^a$.
(3) $(g^a)^{-1} = (g^{-1})^a$.
We define g^{a+b} to be $g^{a+b} = g^a g^b$. However, it is important to note that $g^{a+b} \neq g^{b+a}$ in general since the elements g^a and g^b do not necessarily commute. In any case, it is always true that:
(4) $g^{ab+ac} = (g^a)^{b+c}$ *and*
(5) $g^{ba+ca} = (g^{b+c})^a$.

Proof. Identity (1) follows from

$$g^{ab} = (ab)^{-1}g(ab) = b^{-1}(a^{-1}ga)b = (g^a)^b$$

and (2) from

$$(gh)^a = a^{-1}gha = (a^{-1}ga)(a^{-1}ha) = g^a h^a.$$

The equality

$$(g^a)^{-1} = (a^{-1}ga)^{-1} = a^{-1}g^{-1}a = (g^{-1})^a$$

gives us (3). The equalities

$$g^{ab+ac} = g^{ab}g^{ac} = (g^a)^b(g^a)^c = (g^a)^{b+c}$$

and

$$g^{ba+ca} = g^{ba}g^{ca} = (g^b g^c)^a = (g^{b+c})^a$$

give (4) and (5). □

Definition 2.15. If M is a nonempty subset of G and $g \in G$, then we define

$$M^g = \{m^g \mid m \in M\}.$$

We refer to the sets M^g as the conjugates of M in G.

If U is a subgroup of G, then U^g is a subgroup of G isomorphic to U: The group properties of U^g follow from Lemma 2.14 (2) and (3). The map π from U onto U^g defined by $u^\pi = u^g$, which is inverted by the map α defined by $(u^g)^\alpha = (u^g)^{g^{-1}} = u$, is an isomorphism from U to U^g by Lemma 2.14 (2).

A special case of Definition 2.15 is particularly important:

Theorem 2.16. *Let $M = \{m\}$ be a set containing a single element. Then we call the set consisting of the elements that are G-conjugate to m, namely $\{m^g \mid g \in G\}$, the conjugacy class of m in G. The conjugacy classes are the equivalence classes of the following equivalence relation:*

We say $x \sim y$ if there exists some $g \in G$ such that $x^g = y$. If G is finite, then the partition of G into its conjugacy classes gives rise to a disjoint union

$$G = \bigcup_{i=1}^{h} K_i$$

and the equation

$$|G| = \sum_{i=1}^{h} |K_i| \quad (\textit{the so-called class equation}),$$

where the K_i are the distinct conjugacy classes of G. We refer to the number h of conjugacy classes as the class number of G.

Proof. We only need to show that the given relation is an equivalence relation. It follows from $x^1 = x$ that $x \sim x$. If $x \sim y$, then $x^g = y$ for some $g \in G$, and so $y^{g^{-1}} = x$ implies $y \sim x$; so the relation is symmetric. Finally, if $x \sim y$ and $y \sim z$, then

$$x^a = y \quad \text{and} \quad y^b = z$$

for suitable $a, b \in G$. Then by Lemma 2.14 (1), we have

$$x^{ab} = (x^a)^b = y^b = z,$$

so $x \sim z$. □

Definition 2.17.

a) Let M be a subset of G. Then $\mathbf{N}_G(M) = \{g \in G \mid M^g = M\}$ is called the normalizer of M in G. Clearly, $\mathbf{N}_G(M)$ is a subgroup of G. If U is a subgroup of G, then $U \leq \mathbf{N}_G(U)$. Note that $\mathbf{N}_G(M^x) = \mathbf{N}_G(M)^x$ for all $x \in G$.

b) The set $\mathbf{C}_G(M) = \{g \in G \mid gm = mg \text{ for all } m \in M\}$ is called the centralizer of M in G. Clearly, $\mathbf{C}_G(M)$ is a subgroup of G and $\mathbf{C}_G(M) \leq \mathbf{N}_G(M)$. If U is an abelian subgroup of G, then $U \leq \mathbf{C}_G(U)$.

c) We define $\mathbf{C}_G(G) = \mathbf{Z}(G)$ and call this the center of G. The center $\mathbf{Z}(G)$ of G consists of the elements of G that commute with all the elements of G.

Theorem 2.18. *The index $[G : \mathbf{N}_G(M)]$ is the number of distinct conjugates of the set M in G.*

Proof. If $x = ny$ with $n \in \mathbf{N}_G(M)$, then

$$M^x = M^{ny} = M^y.$$

Conversely, if $M^x = M^y$, then $M^{xy^{-1}} = M$. So $xy^{-1} \in \mathbf{N}_G(M)$ and then $\mathbf{N}_G(M)x = \mathbf{N}_G(M)y$. We conclude that a right coset of $\mathbf{N}_G(M)$ gives rise to exactly one conjugate of M, and distinct cosets give rise to distinct conjugates. □

Theorem 2.19. *Let U and V be subgroups of G. The double cosets of G with respect to U and V are the sets*

$$UgV = \{ugv \mid u \in U,\ v \in V\} \quad (\textit{with } g \in G).$$

If $UgV \cap UhV \neq \emptyset$, then $UgV = UhV$. So the double cosets give rise to a partition of G. If G is finite, then

$$G = \bigcup_{i=1}^{n} Ug_iV \quad (\textit{disjoint})$$

and

$$|G| = \sum_{i=1}^{n} \frac{|U||V|}{|U^{g_i} \cap V|}.$$

Proof. Assume that $u_1gv_1 = u_2hv_2$ with $u_i \in U$ and $v_i \in V$ $(i = 1, 2)$. Then it follows that

$$g = u_1^{-1}u_2hv_2v_1^{-1}$$

and therefore

$$UgV = U(u_1^{-1}u_2hv_2v_1^{-1})V = UhV.$$

So the given union is certainly disjoint. Since $|UgV| = |g^{-1}UgV|$ and $|U^g| = |U|$, the claim regarding the orders follows from Lemma 2.12 a) because

$$|UgV| = |U^gV| = \frac{|U^g||V|}{|U^g \cap V|} = \frac{|U||V|}{|U^g \cap V|}. \qquad \square$$

Theorem 2.20. *Let $G = \langle g \rangle$ be a cyclic group of order n. For every divisor d of n, there exists exactly one subgroup of G of order d; namely, $\langle g^k \rangle$ with $k = \frac{n}{d}$.*

Proof. By Theorem 2.9 c), we have $|\langle g^k \rangle| = \frac{n}{k} = d$. So $U = \langle g^k \rangle$ is indeed a subgroup of G of order d.

Let $V < G$ and $|V| = d$. Let t be the greatest common divisor of all a_i with $0 \leq a_i < n$ and $g^{a_i} \in V$. It is well known that there exist integers b_i $(i = 1, \ldots, m)$ such that $t = \sum_{i=1}^{m} a_ib_i$. It follows that

$$g^t = \prod_{i=1}^{m} g^{a_ib_i} \in V.$$

If $g^a \in V$, then $t \mid a$ and therefore $g^a \in \langle g^t \rangle$. We conclude that $V = \langle g^t \rangle$. It follows from Theorem 2.9 c) that $d = |V| = \frac{n}{(n,t)}$. Let $t = (n, t)s$. Then $t = \frac{ns}{d} = ks$ and $V = \langle g^t \rangle \leq \langle g^k \rangle = U$. So $U = V$, and U is the unique subgroup of G of order d. $\square$

Exercises

1) If $g^2 = 1$ for all group elements $g \in G$, then G is abelian.
2) (Itô, Szép [286]) Let $H < G$, and let $H_1, \ldots, H_n$ be the distinct conjugates of H in G, written in some arbitrary but fixed order. Show that

$$\langle H_1, \ldots, H_n \rangle = \{h_1 h_2 \cdots h_n \mid h_i \in H_i\}.$$

(Hint: To every decomposition $x = h_{i_1} \cdots h_{i_s}$ with $h_{i_j} \in H_{i_j}$, assign the s-tuple $m(x) = (i_1, \ldots, i_s)$. Order the $m(x)$ according to their length s and those of equal length lexicographically. Suppose there is an x that cannot be written in the desired form $h_1 \cdots h_n$ and consider a decomposition of x with $m(x)$ minimal.)
3) Let A and B be subgroups of G. Then $A \cup B$ is a subgroup of G if and only if $A \leq B$ or $B \leq A$.
4) If U is a proper subgroup of the finite group G, then the union $\bigcup_{g \in G} U^g$ of all conjugates of U in G is always a proper subset of G.
5) Let U and V be subgroups of G, and suppose $Ug = Vh$ for elements $g, h \in G$. Then $U = V$.
6) Let $a, b \in G$ with $(|a|, |b|) = 1$ and $ab = ba$. Then $|ab| = |a||b|$, and the group $\langle a, b \rangle$ is cyclic and generated by ab.

3 Normal subgroups, factor groups and homomorphisms

Definition 3.1. A subgroup N of G is called a normal subgroup of G (sometimes also called an invariant subgroup) if $N^g \leq N$ for all $g \in G$. In this case, we write $N \trianglelefteq G$, or $N \lhd G$ if N is distinct from G. Every group G contains the trivial normal subgroups G and 1. If the only normal subgroups of G are 1 and G, then G is said to be simple. Clearly, every group of prime order is simple.

Theorem 3.2. *Let N be a subgroup of G. The following are equivalent:*
a) $N \trianglelefteq G$.
b) $N^g = N$ *for all* $g \in G$.
c) $gN = Ng$ *for all* $g \in G$.
d) *Every left coset of N is a right coset of N.*

Proof. a) $\Rightarrow$ b): We have for all $g \in G$ that $N^g \leq N$ and $N^{g^{-1}} \leq N$. Consequently,

$$N = (N^{g^{-1}})^g \leq N^g.$$

So $N^g = N$.

b) $\Rightarrow$ c): We have

$$gN = g(N^g) = g(g^{-1}Ng) = Ng.$$

c) $\Rightarrow$ d): This is trivial.
d) $\Rightarrow$ a): For all $g \in G$, there exists by assumption some $g^* \in G$ such that

$$gN = Ng^*.$$

So we have $1 \in gNg^{-1} = Ng^*g^{-1}$. It follows that

$$N \leq Ng^*g^{-1} = gNg^{-1},$$

and thus $N \trianglelefteq G$. □

Remark 3.3. If N is a subgroup of G with $[G : N] = 2$, then $N \lhd G$.

Proof. For every $g \in G - N$, we have

$$G = N \cup gN = N \cup Ng.$$

Therefore, $gN = G - N = Ng$ and then $N \lhd G$ by Theorem 3.2. □

Theorem 3.4. *A subgroup N of G is normal in G if and only if the assignment*

$$(gN)(hN) = ghN$$

on the set of left cosets of N in G defines a multiplication. With this multiplication, the cosets form a group.

Proof. a) Assume that the above definition of the product of two cosets of N is well defined; that is, it does not depend on g and h, but rather only on the cosets gN and hN. For $g' = gn$ with $n \in N$, we must then have

$$ghN = g'hN = gnhN.$$

This forces $h^{-1}nh \in N$ for all $n \in N$ and $h \in G$. We therefore have $N^h \leq N$ for all $h \in G$. So N is a normal subgroup of G.

b) Let N be a normal subgroup of G and $n_i \in N$ $(i = 1, 2)$. Since

$$gn_1hn_2 = ghn_1^h n_2 \in ghN,$$

the multiplication defined on cosets of N by $(gN)(hN) = ghN$ is well defined. It is certainly associative since the multiplication of group elements is associative. Then we have $(gN)N = gN$ and $(g^{-1}N)(gN) = N$. Therefore, N is the identity element of the group constructed from cosets and $g^{-1}N$ is the inverse of gN. □

(If $N \trianglelefteq G$, then one can also understand the equality $(gN)(hN) = ghN$ in the sense of Definition 2.11 as

$$(gN)(hN) = \{xy \mid x \in gN,\ y \in hN\}$$

since $gn_1hn_2 = ghn_1^h n_2 \in ghN$ for $n_i \in N$.)

Definition 3.5. If N is a normal subgroup of G, then the group constructed from the cosets of N with respect to the multiplication

$$(gN)(hN) = ghN$$

is called the factor group (or quotient group) of G by N, written G/N. Clearly, $|G/N| = [G : N]$.

Definition 3.6.

a) A map π from the group G to the group H is called a homomorphism from G to H if for all $g, g' \in G$ we have

$$(gg')^\pi = g^\pi (g')^\pi.$$

We define

$$G^\pi = \{g^\pi \mid g \in G\}$$

and call G^π the image of G under π.

b) A homomorphism π from G to H is called an epimorphism if π is a map from G onto H; that is, if $H = G^\pi$.

c) A homomorphism π from G to H is called a monomorphism if π is a bijection from G to G^π.

d) A homomorphism from G to G is called an endomorphism of G.

Remark 3.7. Let π be a homomorphism from G to H.

a) If 1 is the identity element of G, then 1^π is the identity element of H.

b) For all $g \in G$, $(g^{-1})^\pi = (g^\pi)^{-1}$.

c) $G^\pi = \{g^\pi \mid g \in G\}$ is a subgroup of H.

d) If M is a generating set for G, then the homomorphism π is uniquely determined by the values of m^π for all $m \in M$.

Proof. The claims a) and b) follow immediately from

$$g^\pi = (1g)^\pi = 1^\pi g^\pi$$

and

$$1^\pi = (g^{-1}g)^\pi = (g^{-1})^\pi g^\pi.$$

Since $g^\pi g'^\pi = (gg')^\pi$ and $(g^\pi)^{-1} = (g^{-1})^\pi$, it follows that $G^\pi \leq H$. If $g = m_1^{a_1} \cdots m_s^{a_s}$ with $m_i \in M$ and $a_i = \pm 1$, then $g^\pi = (m_1^\pi)^{a_1} \cdots (m_s^\pi)^{a_s}$. □

In what follows, we will denote the identity element of any group by 1.

Homomorphism Theorem 3.8.

a) *If K is a normal subgroup of G, then the map α defined by $g^\alpha = gK$ is an epimorphism from G onto G/K. We call α the natural homomorphism from G onto G/K.*

b) *Let π be a homomorphism from G to H. Let*

$$K = \{g \in G \mid g^\pi = 1\}.$$

We call K the kernel $\ker \pi$ *of π. Then K is a normal subgroup of G, and there is a monomorphism φ from G/K to H with $\alpha\varphi = \pi$. In particular, we have*

$$G/K \cong G^\pi.$$

Proof. a) Since

$$(gg')^\alpha = (gg')K = (gK)(g'K) = g^\alpha g'^\alpha,$$

the map α is an epimorphism from G onto G/K.

b) If $k \in K$, then it follows from $k^\pi = 1$ that

$$(g^{-1}kg)^\pi = (g^\pi)^{-1}k^\pi g^\pi = 1,$$

so k^g is also in K. Hence, $K^g \subseteq K$. If $k, k' \in K$, then

$$(kk')^\pi = k^\pi k'^\pi = 1$$

and

$$(k^{-1})^\pi = (k^\pi)^{-1} = 1.$$

So $kk' \in K$ and $k^{-1} \in K$, which means K is a subgroup of G. Hence, K is a normal subgroup of G.

Consider the map φ from G/K to H defined by $(gK)^\varphi = g^\pi$. If $gK = g'K$, then $g = g'k$ for some $k \in K$ and therefore

$$g^\pi = g'^\pi k^\pi = g'^\pi.$$

So φ is well defined. Since

$$((gK)(g'K))^\varphi = (gg'K)^\varphi = (gg')^\pi = g^\pi g'^\pi = (gK)^\varphi (g'K)^\varphi,$$

we see that φ is a homomorphism from G/K to H. If $(gK)^\varphi = (g'K)^\varphi$, then $g^\pi = g'^\pi$ and thus $gg'^{-1} \in K$. So $gK = g'K$. Hence, φ is a monomorphism from G/K to H.

For all $g \in G$, we have

$$g^\pi = (gK)^\varphi = g^{\alpha\varphi}.$$

Since φ is both a monomorphism and an epimorphism from $G/K = G^\alpha$ to G^π, we have

$$G^\pi \cong G^\alpha = G/K. \qquad \square$$

Theorem 3.9. *If M_i is a normal subgroup of G for all $i \in I$, then the intersection $\bigcap_{i\in I} M_i$ and the subgroup $\langle M_i \mid i \in I\rangle$ generated by the M_i are normal subgroups of G.*

Proof. We have

$$\left(\bigcap_{i\in I} M_i\right)^g = \bigcap_{i\in I} M_i^g = \bigcap_{i\in I} M_i$$

and

$$\langle M_i \mid i \in I\rangle^g = \langle M_i^g \mid i \in I\rangle = \langle M_i \mid i \in I\rangle. \qquad \square$$

Theorem 3.10. *Let N be a normal subgroup of G.*

a) *The map α defined by $U^\alpha = U/N$ is a bijective correspondence between the subgroups U of G with $N \le U \le G$ and the subgroups of $\overline{G} = G/N$. This map respects intersection and generating sets.*

b) *For $N \le U \le G$, we have $[G : U] = [G/N : U/N]$.*

c) *If $N \le M \trianglelefteq G$, then we have $M/N \trianglelefteq G/N$ and*

$$G/N \Big/ M/N \cong G/M.$$

Proof. a) One sees immediately that it follows from $N \trianglelefteq G$ and $N \le U \le G$ that $N \trianglelefteq U$. Let α be the map from the collection of subgroups U with $N \le U \le G$ to subgroups of G/N defined by

$$U^\alpha = U/N = \{uN \mid u \in U\}.$$

We define a map ϵ in the opposite direction in the following way:

If $\overline{U} \le \overline{G} = G/N$, then define

$$\overline{U}^\epsilon = \{g \mid gN \in \overline{U}\}.$$

Clearly, $N \le \overline{U}^\epsilon \le G$. We have

$$\overline{U}^{\epsilon\alpha} = \{gN \mid g \in \overline{U}^\epsilon\} = \{gN \mid gN \in \overline{U}\} = \overline{U}.$$

Furthermore, we have

$$U^{\alpha\epsilon} = (U/N)^\epsilon = \{g \mid gN = uN \in U/N \text{ for some } u \in U\}.$$

If $gN = uN$, then $g = un$ for some $n \in N \le U$, and so $g \in U$. This shows that $U^{\alpha\epsilon} = U$.

The map α is therefore one-to-one since it is inverted by ϵ. Furthermore, α is a map onto all subgroups of $\overline{G}$ since $\overline{U}^{\epsilon\alpha} = \overline{U}$. Finally, for all $N \le U$ and $N \le V$ we have

$$U^\alpha \cap V^\alpha = \{xN \mid x \in U \cap V\} = (U \cap V)/N = (U \cap V)^\alpha$$

and

$$\begin{aligned}\langle U, V\rangle^\alpha = \langle U, V\rangle/N &= \left\{\Big(\prod_i x_i\Big)N \;\Big|\; x_i \in U \cup V\right\} \\ &= \left\{\prod_i (x_i N) \;\Big|\; x_i N \in U/N \cup V/N\right\} \\ &= \langle U/N, V/N\rangle = \langle U^\alpha, V^\alpha\rangle.\end{aligned}$$

b) It follows from the coset decomposition $G = \bigcup_{i\in I} g_i U$ that

$$G/N = \bigcup_{i\in I} (g_i N)(U/N).$$

We show that this is a disjoint union:

Suppose that

$$xN \in (g_i N)(U/N) \cap (g_j N)(U/N).$$

Then we have

$$xN = g_i u_1 N = g_i N u_1 = g_j u_2 N$$

for some $u_k \in U$, therefore

$$g_j^{-1} g_i \in u_2 N u_1^{-1} \subseteq U,$$

and then $i = j$. It follows that

$$[G/N : U/N] = |I| = [G : U].$$

c) Finally, let $N \leq M \trianglelefteq G$. We define the map φ from G/N onto G/M by $(gN)^\varphi = gM$. Then φ is well defined: If $g_1 N = g_2 N$, then $g_2^{-1} g_1 \in N \leq M$ and so $g_1 M = g_2 M$. Since

$$(g_1 N g_2 N)^\varphi = (g_1 g_2 N)^\varphi = g_1 g_2 M = g_1 M g_2 M = (g_1 N)^\varphi (g_2 N)^\varphi,$$

the map φ is a homomorphism from G/N onto G/M. The kernel of φ consists of the cosets gN with $g \in M$, so M/N is the kernel of φ. By the Homomorphism Theorem 3.8, it follows that $M/N \trianglelefteq G/N$ and

$$G/M \cong G/N \Big/ M/N. \qquad \square$$

Lemma 3.11. *If N is a normal subgroup of G and U is a subgroup of G, then the subgroup $\langle N, U\rangle$ is equal to the product UN defined in Definition 2.11. In particular, $NU = UN$.*

Proof. It is enough to show that UN is a subgroup of G. We proceed as follows: For $u_i \in U$ and $n_i \in N$ $(i = 1, 2)$, we have

$$u_1 n_1 u_2 n_2 = u_1 u_2 n_1^{u_2} n_2,$$

and this lies again in UN since $N \trianglelefteq G$ implies $n_1^{u_2} \in N$. Furthermore, we have that

$$(u_1 n_1)^{-1} = n_1^{-1} u_1^{-1} = u_1^{-1}(u_1 n_1^{-1} u_1^{-1})$$

lies in UN. □

Theorem 3.12. *If N is a normal subgroup of G and U is a subgroup of G, then $N \cap U \trianglelefteq U$ and*

$$UN/N \cong U/(U \cap N).$$

Proof. If $u \in U \cap N$ and $u' \in U$, it follows that $u^{u'} \in U \cap N$. Thus, $U \cap N$ is a normal subgroup of U.

Let α be the map from UN/N to $U/(U \cap N)$ defined by $(uN)^\alpha = u(U \cap N)$. Then α is well defined:

If $u_1 N = u_2 N$, then $u_2^{-1} u_1 \in U \cap N$ and so $u_1(U \cap N) = u_2(U \cap N)$.

Clearly, α is an epimorphism from UN/N to $U/(U \cap N)$. The kernel of α consists of the cosets uN with $u(U \cap N) = U \cap N$, and this is equivalent to $uN = N$. □

Theorem 3.13. *If M and N are normal subgroups of G with $M \cap N = 1$, then $mn = nm$ for all $m \in M$ and $n \in N$. In this case, we say that M and N commute.*

Proof. The element $n^{-1}m^{-1}nm = n^{-1}n^m$ lies in N. Since

$$n^{-1}m^{-1}nm = (m^{-1})^n m$$

also lies in M, it follows that $n^{-1}m^{-1}nm = 1$, which was our claim. □

Definition 3.14. If M is a subset of G, we define

$$M^G = \langle m^g \mid m \in M,\ g \in G\rangle.$$

Note that M^G is clearly the smallest normal subgroup of G containing M.

Exercises

7) Let π be a homomorphism from G to H. For all $g \in G$, $|g^\pi|$ divides $|g|$.
8) Let $N \lhd G$ and $g \in G$ with $(|g|, |G/N|) = 1$. Then $g \in N$.

4 Automorphisms

Definition 4.1. An isomorphism from G to itself is called an automorphism of G. The automorphisms of G form a group $\mathbf{Aut}(G)$ with respect to the following product:

For $\alpha, \beta \in \mathbf{Aut}(G)$, define the map $\alpha\beta$ by $g^{\alpha\beta} = (g^\alpha)^\beta$ for all $g \in G$.

The group $\mathbf{Aut}(G)$ is called the automorphism group of G. If G is finite, then clearly so is $\mathbf{Aut}(G)$.

Theorem 4.2. *Every map α of the form $x^\alpha = x^g$ for some $g \in G$ is an automorphism of G, a so-called inner automorphism. The inner automorphisms form a normal subgroup $\mathbf{Inn}(G)$ of $\mathbf{Aut}(G)$. The group $\mathbf{Inn}(G)$ is called the inner automorphism group of G. It is isomorphic to the quotient $G/\mathbf{Z}(G)$ of G by its center $\mathbf{Z}(G)$.*

Proof. That every α of the given form is an automorphism of G follows immediately from

$$(xy)^\alpha = (xy)^g = x^g y^g = x^\alpha y^\alpha$$

and the bijectivity of α.

Let α and β be inner automorphisms of G with $x^\alpha = x^g$ and $x^\beta = x^h$. Then

$$x^{\alpha\beta} = (x^\alpha)^\beta = (x^g)^h = x^{gh}.$$

Thus, $\alpha\beta$ is the inner automorphism of G induced by gh. Furthermore, one sees easily that α^{-1} is the inner automorphism induced by g^{-1}.

Let φ be an arbitrary automorphism of G and α the inner automorphism defined by $x^\alpha = x^g$. Then we have

$$x^{\varphi^{-1}\alpha\varphi} = (g^{-1}x^{\varphi^{-1}}g)^\varphi = (g^\varphi)^{-1}xg^\varphi.$$

Therefore, $\varphi^{-1}\alpha\varphi$ is the inner automorphism induced by g^φ and the inner automorphisms form a normal subgroup $\mathbf{Inn}(G)$ of $\mathbf{Aut}(G)$.

The mapping of $g \in G$ to the inner automorphism α defined by $x^\alpha = x^g$ is clearly an epimorphism from G to $\mathbf{Inn}(G)$, and the kernel is the center $\mathbf{Z}(G)$ of G. The claim that $\mathbf{Inn}(G) \cong G/\mathbf{Z}(G)$ then follows from the Homomorphism Theorem 3.8. □

Theorem 4.3. *Let M and N be normal subgroups of G with $M \leq N$. Let A be a subgroup of the automorphism group $\mathbf{Aut}(G)$ of G. Suppose that for every $\alpha \in A$, we have $M^\alpha = M$ and $N^\alpha = N$. Then the map $\overline{\alpha}$ defined by*

$$(nM)^{\overline{\alpha}} = n^\alpha M \quad (\text{for } n \in N)$$

is an automorphism of N/M. The map $\alpha \mapsto \overline{\alpha}$ is a homomorphism that takes A to the automorphism group $\mathbf{Aut}(N/M)$ of N/M. The kernel of this homomorphism consists of the automorphisms $\alpha \in A$ of the form $n^\alpha = n\,m(n)$, where $m(n) \in M$ for all $n \in N$. The $m(n)$ satisfy the functional equation

$$m(n_1n_2) = m(n_1)^{n_2}\, m(n_2).$$

(*We will have much more to say about this functional equation in §17.*)

In the special case $A = \mathbf{Inn}(G)$, *we denote the kernel of the composite homomorphism from* G (*through* $\mathbf{Inn}(G)$) *to* $\mathbf{Aut}(N/M)$ *by*

$$\mathbf{C}_G(N/M) = \{g \in G \mid n^g M = nM \text{ for all } n \in N\}.$$

Then $G/\mathbf{C}_G(N/M)$ *is clearly isomorphic to a subgroup of* $\mathbf{Aut}(N/M)$. (*If* $M = 1$, *then* $\mathbf{C}_G(N/1)$ *is the centralizer* $\mathbf{C}_G(N)$ *defined in Definition 2.17 b).*)

Proof. First we must show that the map $\overline{\alpha}$ is well defined. Let $n' = nm \in nM$ for some $m \in M$. Since $m^\alpha \in M^\alpha = M$, we obtain

$$(n'M)^{\overline{\alpha}} = n'^{\alpha}M = n^\alpha m^\alpha M = n^\alpha M = (nM)^{\overline{\alpha}}.$$

For all $n_1, n_2 \in N$, we have

$$\begin{aligned}(n_1Mn_2M)^{\overline{\alpha}} &= (n_1n_2M)^{\overline{\alpha}} = (n_1n_2)^\alpha M = n_1^\alpha n_2^\alpha M\\ &= n_1^\alpha M n_2^\alpha M = (n_1M)^{\overline{\alpha}}(n_2M)^{\overline{\alpha}}.\end{aligned}$$

Clearly, $\overline{\alpha}$ is bijective. Therefore, $\overline{\alpha}$ is an automorphism of N/M. Furthermore, we have for all $\alpha, \beta \in A$ that

$$(nM)^{\overline{\alpha\beta}} = n^{\alpha\beta}M = (n^\alpha M)^{\overline{\beta}} = (nM)^{\overline{\alpha}\overline{\beta}}.$$

Thus, the map $\alpha \mapsto \overline{\alpha}$ is a homomorphism from A to $\mathbf{Aut}(N/M)$. It is clear that the kernel of this homomorphism consists of those automorphisms $\alpha \in A$ of the form $n^\alpha = n\, m(n)$ for some $m(n) \in M$. We also have

$$\begin{aligned}n_1n_2\, m(n_1n_2) = (n_1n_2)^\alpha &= n_1^\alpha n_2^\alpha = n_1 m(n_1)\; n_2 m(n_2)\\ &= n_1n_2\, m(n_1)^{n_2} m(n_2),\end{aligned}$$

and so $m(n_1n_2) = m(n_1)^{n_2}\, m(n_2)$. □

Theorem 4.4. *Let* N *be a normal subgroup of* G, *and let* α *be an automorphism of* G *such that* $n^\alpha = n$ *for all* $n \in N$ *and* $gN = g^\alpha N$ *for all* $g \in G$. *Then* $g^\alpha = g\, t(g)$, *where* $t(g)$ *lies in* $\mathbf{Z}(N)$, *depends only on the coset* gN *and satisfies the functional equation*

$$t(gh) = t(g)^h t(h)$$

from Theorem 4.3. All automorphisms α *of the given form commute. If* m *is the exponent of* $\mathbf{Z}(N)$, *then* $\alpha^m = 1$. *In particular, if* $(|\alpha|, |\mathbf{Z}(N)|) = 1$, *then* $\alpha = 1$.

Proof. Let $t(g)$ be defined via the equation $g^\alpha = gt(g)$. Since $g^\alpha N = gN$, the element $t(g)$ lies in N. For all $g \in G$ and all $n \in N$, we have

$$gnt(gn) = (gn)^\alpha = g^\alpha n^\alpha = gt(g)n.$$

On the other hand,

$$(gn)^\alpha = (gng^{-1}g)^\alpha = (gng^{-1})^\alpha g^\alpha = gng^{-1}gt(g) = gnt(g).$$

It follows that

$$nt(gn) = t(g)n = nt(g),$$

so that $t(gn) = t(g)$ for all $n \in N$ and $t(g) \in \mathbf{Z}(N)$.

From this we obtain

$$g^{\alpha^m} = gt(g)^m = g,$$

so that $\alpha^m = 1$. It is clear that the α commute since the elements $t(g) \in \mathbf{Z}(N)$ commute. □

The following theorem is often useful.

Theorem 4.5. *Let U be a subgroup of G, let $\mathbf{C}_G(U)$ be the centralizer of U and $\mathbf{N}_G(U)$ the normalizer of U in G. Then the factor group $\mathbf{N}_G(U)/\mathbf{C}_G(U)$ is isomorphic to a subgroup of the automorphism group $\mathbf{Aut}(U)$ of U.*

Proof. We identify every element $n \in \mathbf{N}_G(U)$ with an automorphism α of U by defining $u^\alpha = u^n$. This gives rise to a homomorphism from $\mathbf{N}_G(U)$ to $\mathbf{Aut}(U)$. The kernel of this homomorphism is clearly the centralizer $\mathbf{C}_G(U)$ of U in G. □

Theorem 4.6. *Let G be a cyclic group of order n. Then every automorphism of G has the form $g^\alpha = g^k$, where $(k, n) = 1$. The automorphism group $\mathbf{Aut}(G)$ of G is isomorphic to the multiplicative group consisting of the residue classes modulo n of those integers coprime to n. In particular, $|\mathbf{Aut}(G)| = \varphi(n)$, where φ is the Euler φ-function from number theory.*

Proof. Let $G = \langle g \rangle$, let α be an automorphism of G and let $g^\alpha = g^k$. Then by Theorem 2.9 c),

$$n = |g| = |g^\alpha| = |g^k| = \frac{n}{(n, k)}.$$

So k is coprime to n.

Conversely, every k coprime to n gives rise to an automorphism α of G by defining $(g^i)^\alpha = g^{ik}$. One sees immediately that the composition of automorphisms corresponds to multiplication of the k's and that all k which are congruent mod n give rise to the same automorphism of G. □

Definition 4.7. A subgroup U of G is called a characteristic subgroup of G if $U^\alpha = U$ for every automorphism α of G. (Every characteristic subgroup of G is clearly a normal subgroup of G.) If the only characteristic subgroups of G are 1 and G, we say that G is characteristically simple.

The following lemma is trivial but useful.

Lemma 4.8.
a) *Let N be a normal subgroup of G, and let M be a characteristic subgroup of N. Then M is a normal subgroup of G.*
b) *If $A \leq B \leq G$ with A characteristic in B and B characteristic in G, then A is characteristic in G.*

Proof. a) For every $g \in G$, the map α defined by $n^\alpha = n^g$ for all $n \in N$ is an automorphism of N. Since M is characteristic in N, it follows that $M = M^\alpha = M^g$, and therefore $M \trianglelefteq G$.

b) Let α be an automorphism of G. Since B is a characteristic subgroup of G, we have $B^\alpha = B$. Since the restriction of α to B is an automorphism of B and A is a characteristic subgroup of B, it follows that $A^\alpha = A$. So A is characteristic in G. □

We note that if $M \lhd N$ and $N \lhd G$, it does not always follow that $M \lhd G$. We plan to investigate further the problems that arise from this fact in future volumes. If A and B are characteristic in G and $A < B < G$, then A is not necessarily characteristic in B.

Theorem 4.9. *Let $N \lhd G$, and assume that $|N|$ and $|G/N|$ are finite and coprime.*
a) *If $M \leq G$ and $|M|$ divides $|N|$, then $M \leq N$.*
b) *The only subgroup of G of order $|N|$ is N. In particular, N is a characteristic subgroup of G.*

Proof. a) By Lemma 3.11, we have $NM \leq G$. So $|NM/N|$ is a divisor of $|G/N|$. By Theorem 3.12, we have

$$|NM/N| = |M/(M \cap N)|$$

and so $|NM/N|$ is also a divisor of $|N|$. It follows from our assumption that $|NM/N| = 1$, so $M \leq N$.

b) follows immediately from a). □

Example 4.10.
a) Let p be an odd prime number, and let G be the group of all linear functions $x \mapsto ax + b$ $(a, b, x \in \mathrm{GF}(p), a \neq 0)$ over the finite field $\mathrm{GF}(p)$. Then every automorphism of G is an inner automorphism.
b) (G. E. Wall) Let G be the group of all linear functions $x \mapsto ax + b$ over the ring of integers modulo 8 with $a \equiv 1 \bmod 2$. (The order of G is 2^5.) Then the function α that maps $g \in G$ with $xg = ax + b$ to g^α with

$$xg^\alpha = ax + b + \frac{1}{2}(a^2 - 1)$$

is an automorphism of G that maps every element of G onto a conjugate, but α is not an inner automorphism of G.

Proof. a) The subgroup P of translations $x \mapsto x + a$ is a normal subgroup of G of order p. Since $|G| = p(p-1)$, P is characteristic in G by Theorem 4.9.

Let r be a primitive $(p-1)$-th root of unity in GF(p). Then G is clearly generated by the functions f_1 and f_2 defined by

$$xf_1 = x + 1 \quad \text{and} \quad xf_2 = rx.$$

Every automorphism α of G is then uniquely determined by f_1^α and f_2^α.

Let $xf_1^\alpha = x + a$. Then we can write a in the form $a = r^s$. We have $f_2^{-s} f_1 f_2^s = f_1^\alpha$. If ϵ is the inner automorphism of G induced by conjugation by f_2^s, then the automorphism $\varphi = \alpha\epsilon^{-1}$ fixes f_1. Let

$$xf_2^\varphi = bx + c.$$

It follows from $f_2^{-1} f_1 f_2 = f_1^r$ that, upon application of φ, we have

$$f_1^r = (f_2^\varphi)^{-1} f_1 f_2^\varphi$$

and therefore

$$x + r = xf_1^r = x(f_2^\varphi)^{-1} f_1 f_2^\varphi = x + b.$$

So we obtain $b = r$.

Let $g \in G$ be chosen such that $xg = x + t$ with $(1 - r)t = c$. We claim that this element g induces the automorphism φ via conjugation. Note that $f_1^g = f_1$ and $f_2^g = g^{-1} f_2 g = f_2^\varphi$. Therefore, $\varphi = \alpha\epsilon^{-1}$ is an inner automorphism of G, and so $\alpha = \varphi\epsilon$ is as well.

b) Let $g : x \mapsto ax + b$ and $h : x \mapsto cx + d$ be arbitrary elements of G. Then we have

$$xg^\alpha h^\alpha = acx + d + bc + \frac{1}{2}(c^2 - 1) + \frac{c}{2}(a^2 - 1).$$

Furthermore,

$$xgh = acx + d + bc$$

and therefore

$$x(gh)^\alpha = acx + d + bc + \frac{1}{2}(a^2c^2 - 1).$$

Since

$$c^2 + c(a^2 - 1) - a^2c^2 = (c - c^2)(a^2 - 1) \equiv 0 \mod 16,$$

it follows that $(gh)^\alpha = g^\alpha h^\alpha$. So α is an automorphism of G.

Defining $g : x \mapsto ax+b$ and $t : x \mapsto x+\frac{1}{2}(a+1)$, we have $g^\alpha = tgt^{-1}$. Therefore, α maps every conjugacy class of G onto itself.

If α were induced by an inner automorphism $g \mapsto g^h$ with $h : x \mapsto sx+t$, then for all a, b with $a \equiv 1 \bmod 2$ we would have

$$b + \frac{1}{2}(a^2-1) \equiv -at + sb + t \mod 8.$$

Setting $a = 1$ forces $b \equiv sb \bmod 8$ for all b. But then

$$\frac{1}{2}(a^2-1) \equiv -(a-1)t \mod 8$$

for all a with $a \equiv 1 \bmod 2$. However, this doesn't hold for any t. Therefore, α is not an inner automorphism of G. □

Exercises

9) Let $\mathbf{Z}(G) = 1$. Show that the centralizer of $\mathbf{Inn}(G)$ in $\mathbf{Aut}(G)$ is 1.

10) Let G be a nonabelian simple group. Then every automorphism of $\mathbf{Aut}(G)$ is an inner automorphism of $\mathbf{Aut}(G)$.

11) Let U be a subgroup of G and α an automorphism of G such that $Ug^\alpha = Ug$ for all $g \in G$. Then α has the form $g^\alpha = u(g)g$ for some $u(g) \in \bigcap_{x \in G} U^x$.

12) Let α be an automorphism of the finite group G that maps every element $g \in G$ onto an element in the conjugacy class of g. Then every prime divisor of the order of α is also a divisor of $|G|$.

(Hint: One may assume without loss of generality that $|\alpha| = p \nmid |G|$. Let

$$F = \{g \in G \mid g^\alpha = g\}.$$

Show that if C is a conjugacy class of G with $C \cap F = \emptyset$, then $p \mid |C|$, which yields the contradiction $p \mid |G|$. Hence, $C \cap F \neq \emptyset$ for every conjugacy class C of G. From this one obtains a contradiction using Exercise 4 of §2.)

13) a) G is abelian if and only if there exists an automorphism α of G such that $g^\alpha = g^{-1}$ for all $g \in G$.

b) Let G be an abelian group with $2 \nmid |G|$, and let α be an automorphism of G such that $\alpha^2 = 1$. Then every $g \in G$ can be written uniquely as $g = g_1 g_{-1}$, where $g_1^\alpha = g_1$ and $g_{-1}^\alpha = g_{-1}^{-1}$.

c) The statement in b) does not hold in general for groups of even order.

14) Let α be an automorphism of G such that $\alpha^2 = 1$, and suppose $2 \nmid |G|$. Define

$$G_1 = \{g \in G \mid g^\alpha = g\} \quad \text{and} \quad G_{-1} = \{g \in G \mid g^\alpha = g^{-1}\}.$$

Then $G = G_1 G_{-1}$ and $G_1 \cap G_{-1} = 1$.

(Hint: Determine x by $x^2 = g^{-\alpha} g$, then show $x^\alpha = x^{-1}$ and $(gx^{-1})^\alpha = gx^{-1}$.)

5 Permutation groups

Definition 5.1. Let Ω be a finite set with $|\Omega| = n$. In order to differentiate notationally between the elements of Ω and group elements, we call the elements of Ω points and often write them as $1, 2, \ldots, n$. The set of all bijective maps from Ω to itself is a group with the product gh of two maps g and h defined by

$$i^{gh} = (i^g)^h \quad (i \in \Omega).$$

We call this group the symmetric group S_n of degree n (on Ω). Clearly, $|S_n| = n!$. We call the elements of S_n permutations and often specify them in the form

$$g = \begin{pmatrix} i \\ i^g \end{pmatrix}.$$

If Δ is a subset of Ω and $g \in S_n$, then we write

$$\Delta^g = \{i^g \mid i \in \Delta\}.$$

Theorem 5.2. *A permutation of the form*

$$x = \begin{pmatrix} a_1 & a_2 & \cdots & a_{k-1} & a_k & a_{k+1} & \cdots & a_n \\ a_2 & a_3 & \cdots & a_k & a_1 & a_{k+1} & \cdots & a_n \end{pmatrix}$$

(or more succinctly: $x = (a_1, a_2, \ldots, a_k)$) is called a k-cycle or a cycle of length k. We also call 2-cycles transpositions. Every permutation g can be written uniquely (up to ordering) as a product of cycles that have no points in common. The points $i = i^g$ fixed by g appear as cycles of length 1 and will often not be written. If the cycles of g have lengths $z_1, z_2, \ldots, z_m$, then the order $|g|$ of g is the least common multiple of the z_i. We call the m-tuple $(z_1, z_2, \ldots, z_m)$ the cycle type of the permutation g.

Proof. Beginning with any point a_1, we define $a_i = a_1^{g^{i-1}}$. Let k be the integer such that $a_1, \ldots, a_k$ are pairwise distinct and $a_k^g = a_i$ for some $i \leq k$. Assume for contradiction that $1 < i$. Then it follows that

$$a_k^g = a_i = a_{i-1}^g,$$

and therefore $a_k = a_{i-1}$, contradicting the choice of k. We conclude that $a_k^g = a_1$, and so the points $a_1, \ldots, a_k$ are cyclically permuted by g. Now it is clear how to construct the unique cycle decomposition of g in finitely many steps.

Let $g = g_1 \cdots g_m$, where the cycle g_j has length z_j and the cycles have no points in common. Let n be the order of g. Since the g_j commute, we obtain $1 = g^n = g_1^n \cdots g_m^n$. It follows that $g_j^n = 1$ for every j since the g_j have no points in common. Therefore, every z_j is a divisor of n. Conversely, since $g_j^{z_j} = 1$ the least common multiple of the z_j is a multiple of the order n of g. □

Lemma 5.3. *If g has the cycle decomposition $g = (a_1, \ldots, a_k) \cdots (s_1, \ldots, s_m)$ and*

$$t = \begin{pmatrix} a_i \\ b_i \end{pmatrix} \cdots \begin{pmatrix} s_j \\ t_j \end{pmatrix},$$

then $g^t = t^{-1}gt$ has the cycle decomposition $g^t = (b_1, \ldots, b_k) \cdots (t_1, \ldots, t_m)$. In particular, it follows that the conjugacy classes of S_n consist exactly of those permutations of the same cycle type.

Proof. The claim follows by direct calculation. □

Definition 5.4. Let g be a permutation of $\{1, \ldots, n\}$. Then we have

$$\prod_{i<j} (i^g - j^g) = \epsilon \prod_{i<j} (i - j) \quad (i, j = 1, \ldots, n)$$

for some $\epsilon = \pm 1$. We define $\epsilon = \operatorname{sgn}(g)$. The permutations with $\operatorname{sgn}(g) = 1$ are called even and those with $\operatorname{sgn}(g) = -1$ are called odd.

Theorem 5.5. *For $n > 1$, the permutations $g \in S_n$ with $\operatorname{sgn}(g) = 1$ constitute a normal subgroup A_n of S_n of index 2. We call A_n the alternating group of degree n.*

Proof. Clearly, $\operatorname{sgn}(g) = (-1)^{a(g)}$, where $a(g)$ is the number of pairs (i, j) with $i < j$ and $i^g > j^g$. It follows that, for all $g, h \in S_n$,

$$\begin{aligned} \operatorname{sgn}(g) \operatorname{sgn}(gh) \prod_{i<j} (i - j) &= \operatorname{sgn}(g) \prod_{i<j} (i^{gh} - j^{gh}) = \prod_{i^g<j^g} (i^{gh} - j^{gh}) \\ &= \operatorname{sgn}(h) \prod_{i^g<j^g} (i^g - j^g), \end{aligned}$$

and so $\operatorname{sgn}(gh) = \operatorname{sgn}(g) \operatorname{sgn}(h)$. Therefore, the map φ defined by $g^\varphi = \operatorname{sgn}(g)$ is a homomorphism from S_n to the cyclic group $\{\pm 1\}$ of order 2. There always exist permutations g with $\operatorname{sgn}(g) = -1$, for example $g = (1, 2)$. Hence, we obtain an epimorphism from S_n onto the cyclic group of order 2. We call the kernel of this epimorphism A_n, and the Homomorphism Theorem 3.8 yields the claims $A_n \lhd S_n$ and $[S_n : A_n] = 2$. □

Definition 5.6. A subgroup of S_n is called a permutation group of degree n.

Definition 5.7. Let G_1 and G_2 be two permutation groups that permute the sets of points Ω_1 and Ω_2, respectively. We say that G_1 and G_2 are permutation isomorphic if there is a bijective map ϵ from Ω_1 to Ω_2 and a group isomorphism φ from G_1 to G_2 such that for all $i \in \Omega_1$ and all $g \in G_1$ we have

$$(i^g)^\epsilon = (i^\epsilon)^{g^\varphi}.$$

If $\Omega_1 = \Omega_2$, then ϵ is a permutation of Ω_1 and the conditions imply that G_1 and G_2 are conjugate in the symmetric group on Ω_1.

Definition 5.8. Let G be a permutation group on Ω. We say that the points $i, j \in \Omega$ are equivalent and write $i \sim j$ if there exists a permutation $g \in G$ such that $i^g = j$. This yields an equivalence relation on Ω:

If $i \sim j$, then $i^g = j$ for some $g \in G$. Then $j^{g^{-1}} = i$ and so $j \sim i$. If $i \sim j$ and $j \sim k$, then there exist $g, h \in G$ such that $i^g = j$ and $j^h = k$. Then we have $i^{gh} = k$ and therefore $i \sim k$. Finally, $i^1 = i$, so $i \sim i$.

This equivalence relation yields a disjoint decomposition of Ω into equivalence classes T_i. We call the T_i the orbits of G on Ω. Clearly, $|\Omega| = \sum_i |T_i|$. We say that the permutation group G is transitive on Ω if Ω is a single G-orbit; that is, for every pair $i, j \in \Omega$ there exists some $g \in G$ such that $i^g = j$.

Definition 5.9. If G is a permutation group on Ω and Δ is a subset of Ω, then we set

$$G_\Delta = \{g \in G \mid i^g = i \text{ for all } i \in \Delta\}$$

and

$$\Delta^G = \{i^g \mid g \in G,\, i \in \Delta\}.$$

Clearly, G_Δ is a subgroup of G.

In the special case that $\Delta = \{a\}$, we also write $G_\Delta = G_a$ and call G_a the stabilizer of a in G.

If $\Delta^g = \Delta$ for all $g \in G$, then we define g^Δ to be the restriction of g to Δ. Then the map $g \mapsto g^\Delta$ is an epimorphism from G onto the permutation group

$$G^\Delta = \{g^\Delta \mid g \in G\}$$

on Δ. In particular, if Δ is an orbit of G, then we call G^Δ the constituent of G on Δ.

Theorem 5.10.

a) *If a is a point in the orbit T of the permutation group G, then G_a is a subgroup of G of index $[G : G_a] = |T|$.*

b) *If $a \in \Omega$ and g is an element of the permutation group G on Ω, then $G_a^g = G_{a^g}$.*

Proof. a) Let $G = \bigcup_i G_a\, g_i$ be the decomposition of G into cosets of G_a. It follows from $g = hg_i \in G_a g_i$ with $h \in G_a$ that

$$a^g = (a^h)^{g_i} = a^{g_i}.$$

So, every permutation in a fixed coset of G_a yields the same image of a. Conversely, if $a^g = a^h$ then $a^{gh^{-1}} = a$, so $gh^{-1} \in G_a$ and then $G_a g = G_a h$. Hence, different cosets of G_a yield different images of a. Altogether, this gives

$$|T| = |\{a^g \mid g \in G\}| = [G : G_a].$$

b) follows immediately from a direct calculation. □

We mention the following important special case:

Theorem 5.11. *Let G be a transitive permutation group on Ω with $|\Omega| = n$. Then for every $a \in \Omega$ we have the equality $|G| = n|G_a|$, and all subgroups G_a are conjugate in G. Therefore, the degree of a transitive permutation group is always a divisor of the group order.*

Definition 5.12. The transitive permutation group G on Ω is called regular if $G_a = 1$ for all points $a \in \Omega$; that is, if every nonidentity permutation in G leaves no points fixed. (Since all G_a are conjugate in G by Theorem 5.11, the regularity of G is already guaranteed if $G_a = 1$ for a single $a \in \Omega$.)

Theorem 5.13. *If G is an abelian transitive permutation group, then G is regular.*

Proof. Let a and b be points in the set Ω. Since G is transitive, there exists some $g \in G$ such that $a^g = b$. It follows from Theorem 5.10 b) that $G_a^g = G_b$. However, $G_a^g = G_a$ since G is abelian. Therefore, G_a fixes all the points in Ω, so $G_a = 1$ and G is regular. □

Exercises

15) Let g be a permutation of cycle type $(z_1, \ldots, z_k)$ in the symmetric group S_n. Show that

$$\operatorname{sgn}(g) = (-1)^t, \quad \text{where} \quad t = \sum_{i=1}^{k} (z_i - 1).$$

Calculate the size of the conjugacy class of g in S_n.

16) Let g be as in Exercise 15, and let $g \in A_n$ ($n > 2$). Show that the conjugacy class of g in S_n splits in A_n into two conjugacy classes of equal size if and only if all z_i are odd and pairwise distinct.

17) Every permutation of the form $t = (i, j)$ is called a transposition. Show that:
   a) The transpositions $t_i = (i, i+1)$ with $1 \le i \le n-1$ generate the entire symmetric group S_n on the set $\{1, \ldots, n\}$.
   b) Let $g = s_1 \cdots s_m$ for some transpositions $s_i \in S_n$. Then $g \in A_n$ if and only if m is even.

18) Let $A = \langle (1,2) \rangle$ and $B = \langle (1,3) \rangle$. Show that AB is not a subgroup of S_3.

19) Show that the alternating group A_5 is simple. (Hint: If $N \trianglelefteq A_5$, then N is a union of conjugacy classes of A_5.)

20) The alternating group A_4 does not have a subgroup of order 6. (Thus, the naive converse of Lagrange's Theorem 2.7 does not hold; more on this in Chapter VI, §1.)

6 Permutation representations

Definition 6.1. Every homomorphism from a group G to the symmetric group S_n on a set Ω with $|\Omega| = n$ is called a permutation representation of G of degree n on Ω. A permutation representation is called faithful if it is a monomorphism.

We describe an important algorithm for constructing permutation representations:

Main Theorem 6.2. *Let H be a subgroup of G of index n, and let $G = \bigcup_{i=1}^{n} Hg_i$ be the decomposition of G into cosets of H. We obtain an epimorphism $\mathbf{P}$ from G to a transitive subgroup of S_n on the set $\{Hg_i \mid i = 1, \ldots, n\}$ if we assign to every $g \in G$ the permutation*

$$\mathbf{P}(g) = \begin{pmatrix} Hg_i \\ Hg_ig \end{pmatrix}.$$

(*Here we write the map $\mathbf{P}$ on the left; we will never compose such maps.*)

The kernel of the epimorphism $\mathbf{P}$ is the intersection of all conjugates H^g of H in G. We call this epimorphism $\mathbf{P}$ the permutation representation of G on the cosets of H.

Proof. For all $g, g' \in G$, we have

$$(Hg_i)(gg') = ((Hg_i)g)g'.$$

Hence, $\mathbf{P}$ is actually a homomorphism from G to S_n. It follows immediately from

$$(Hg_i)(g_i^{-1}g_j) = Hg_j$$

that the image of G under $\mathbf{P}$ is a transitive permutation group.

Suppose $\mathbf{P}(g) = 1$, so that $Hg_ig = Hg_i$ for all $i = 1, \ldots, n$. This is equivalent to

$$g \in \bigcap_{i=1}^{n} g_i^{-1}Hg_i = \bigcap_{x \in G} H^x.$$

Therefore, the kernel of $\mathbf{P}$ has the specified form. □

Next, we identify an important special case.

Theorem 6.3 (Cayley). *The map $\mathbf{R}$ taking $g \in G$ to the permutation*

$$\mathbf{R}(g) = \begin{pmatrix} x \\ xg \end{pmatrix}$$

on the set G yields an isomorphism from the finite group G of order n to a regular subgroup of the symmetric group S_n (on the set G). We call $\mathbf{R}$ the regular representation of G.

Proof. This is the special case $H = 1$ of Theorem 6.2. The image is regular since if $xg = x$ for some $x \in G$, then $g = 1$. □

Remark 6.4. Let G be a transitive permutation group of degree n on the set $\Omega = \{1, \dots, n\}$. Let $H = G_1$ be the stabilizer of 1. Choose representatives g_i of the right cosets of H so that $1^{g_i} = i$. Then the construction of Theorem 6.2 yields (up to permutation isomorphism) the original permutation group.

Proof. We have $\mathbf{P}(g) = \binom{Hg_i}{Hg_{i'}}$, where i' is determined by $(Hg_i)g = Hg_{i'}$. The elements of the coset Hg_ig of H map 1 to

$$1^{g_i g} = i^g,$$

and those in $Hg_{i'}$ map 1 to i'. It follows that $i' = i^g$. Defining ϵ to be $i^\epsilon = Hg_i$ yields the desired permutation isomorphism. □

Theorem 6.5. *Let G be embedded in the symmetric group S_n of degree $n = |G|$ by the regular representation $\mathbf{R}$ with $\mathbf{R}(g) = \binom{x}{xg}$. Then the centralizer $\mathbf{C}_{S_n}(G)$ of G in S_n consists of those permutations of the form $\binom{x}{gx}$. Therefore, $\mathbf{C}_{S_n}(G)$ is likewise a regular permutation group. In particular, if G is abelian then $\mathbf{C}_{S_n}(G) = G$.*

Proof. Let $c = \binom{x}{x^\epsilon}$ be an element in $\mathbf{C}_{S_n}(G)$. Then for all $g \in G$ we have

$$\mathbf{R}(g)c = c\mathbf{R}(g),$$

which means

$$(xg)^\epsilon = x^\epsilon g$$

for all $x, g \in G$. Setting $x = 1$, it follows that

$$g^\epsilon = 1^\epsilon g,$$

and c has the desired form $c = \binom{x}{ax}$, where $a = 1^\epsilon$. The converse statement, that all permutations of the form $\binom{x}{ax}$ lie in $\mathbf{C}_{S_n}(G)$, is clear. □

Theorem 6.2 is a powerful tool for proving the existence of nontrivial normal subgroups of a finite group. We first mention the following trivial but useful theorem:

Theorem 6.6. *Let H be a subgroup of G of index n. Then G contains a normal subgroup N such that n divides $|G/N|$ and $|G/N|$ divides $n!$.*

Proof. We construct the permutation representation of G on the cosets of H according to Theorem 6.2. Let the kernel be N. Then G/N is isomorphic to a transitive permutation group of degree n. Now on the one hand, $|G/N|$ is a divisor of $|S_n| = n!$ by Theorem 2.7, and on the other hand n is a divisor of $|G/N|$ by the transitivity of G/N and Theorem 5.11. □

Theorem 6.7. *Suppose $|G| = 2n$ with n odd. Then G contains a normal subgroup of index 2.*

Proof. First we show that G has elements of order 2. Consider the decomposition $G = A \cup B$ with $A = \{g \in G \mid g = g^{-1}\}$ and $B = \{g \in G \mid g \neq g^{-1}\}$. Since g^{-1} lies in B if g does, $|B|$ is even. Then $|A| = 2n - |B|$ is also even. Since $1 \in A$, we have $|A| > 1$, and so there exists some g with $g = g^{-1} \neq 1$.

Let $g \in G$ be an element of order 2. Let $\mathbf{R}$ be the regular representation of G. Consider the cycle decomposition of the permutation $\mathbf{R}(g)$. Since g has order 2, this decomposition consists only of cycles of lengths 1 and 2 by Theorem 5.2. Since $g \neq 1$, the permutation $\mathbf{R}(g)$ has no 1-cycles. Therefore, $\mathbf{R}(g)$ is the product of n cycles of length 2. So $\operatorname{sgn}(\mathbf{R}(g)) = (-1)^n = -1$. The mapping of h to $\operatorname{sgn}(\mathbf{R}(h))$ for all $h \in G$ is an epimorphism of G onto the cyclic group of order 2, and the kernel of this epimorphism is a normal subgroup of index 2 in G. □

Even permutation representations that are not transitive are occasionally useful:

Theorem 6.8. *Let G be finite and $\Omega = G$. Then the map $\mathbf{T}$ defined by*

$$\mathbf{T}(g) = \begin{pmatrix} x \\ x^g \end{pmatrix}$$

is a homomorphism from G to the symmetric group S_n of degree $n = |G|$. The orbits of the permutation group

$$\mathbf{T}(G) = \{\mathbf{T}(g) \mid g \in G\}$$

are the conjugacy classes K_i of G introduced in Theorem 2.16. They have sizes $|K_i| = [G : \mathbf{C}_G(k_i)]$ for any $k_i \in K_i$. Furthermore, we have the class equation

$$|G| = \sum_{i=1}^{h} |K_i|.$$

Proof. It is clear that the map $\mathbf{T}$ is a homomorphism from G to S_n and that the orbits of $\mathbf{T}(G)$ are exactly the conjugacy classes of G. The statement about the sizes of the K_i is in Theorem 2.18. (Note that if $M = \{m\}$, then $\mathbf{N}_G(M) = \mathbf{C}_G(M)$!) □

There is an important consequence of the class equation that we would like to familiarize ourselves with here:

Theorem 6.9. *If the order of G is a power of a prime p and N is a nonidentity normal subgroup of G, then $N \cap \mathbf{Z}(G) > 1$. In particular, every nonidentity group G of prime power order has center $\mathbf{Z}(G) > 1$.*

Proof. Since $N \trianglelefteq G$, the subgroup N is a union of conjugacy classes of G, say $N = \bigcup_{i=1}^{s} K_i$. Let $K_1 = \{1\}$. Since $|K_i| = [G : \mathbf{C}_G(k_i)]$ for any $k_i \in K_i$, the sizes of the conjugacy classes $|K_i|$ are powers p^{n_i} of the prime p. It follows that

$$|N| = p^n = 1 + \sum_{i=2}^{s} p^{n_i}.$$

If $p^{n_i} > 1$ for all i with $2 \le i \le s$, then we obtain the contradiction $0 \equiv 1 \bmod p$. Therefore, there is some K_i with $1 < i \le s$ such that $|K_i| = 1$. If $K_i = \{k\}$, it follows that $1 \ne k \in N \cap \mathbf{Z}(G)$. □

We will undertake a systematic analysis of Theorem 6.9 in Chapter III. Here we only mention the following:

Theorem 6.10. *If $|G| = p^2$ and p is a prime, then G is abelian.*

Proof. By Theorem 6.9, we have $|\mathbf{Z}(G)| > 1$. Let $1 \ne a \in \mathbf{Z}(G)$. If $G = \langle a \rangle$, then we are done. If $G \ne \langle a \rangle$, then there exists an element $b \in G - \langle a \rangle$. It follows that

$$p = |\langle a \rangle| < |\langle a, b \rangle| \text{ divides } |G| = p^2,$$

and therefore $g = \langle a, b \rangle$. Since b commutes with the central element a, the group G is abelian. □

Exercises

21) Let $|G| = 2^n m$ with m odd. Suppose G has an element of order 2^n. Then there exists some $N \lhd G$ such that $|G/N| = 2^n$. (Hint: Use the proof method of Theorem 6.7.)
22) Show that every finite division ring is commutative. (Hint, following Witt [479]: Let K be a finite division ring with $|\mathbf{Z}(\mathsf{K})| = q = p^m$ and $|\mathsf{K}| = q^n$. Since the centralizers of elements in $\mathsf{K}^\times$ are subdivision rings containing $\mathbf{Z}(\mathsf{K})$, the class equation yields

$$|\mathsf{K}^\times| = q^n - 1 = q - 1 + \sum_i \frac{q^n - 1}{q^{k_i} - 1},$$

where $k_i \mid n$. From this, conclude that $n = 1$.)
23) The permutation representations $\mathbf{P}_1$ and $\mathbf{P}_2$ of G on the sets Ω_1 and Ω_2 are said to be permutation isomorphic if there is a bijective map β from Ω_1 onto Ω_2 such that

$$\beta \mathbf{P}_2(g) = \mathbf{P}_1(g)\beta$$

for all $g \in G$. Prove the following:
   a) Let $H < G$. Then the permutation representations $\mathbf{P}_1$ and $\mathbf{P}_2$ defined by

$$\mathbf{P}_1(g) = \begin{pmatrix} Hx \\ Hxg \end{pmatrix} \quad \text{and} \quad \mathbf{P}_2(g) = \begin{pmatrix} xH \\ g^{-1}xH \end{pmatrix}$$

   are permutation isomorphic.
   b) The permutation representations $\mathbf{P}_1$ and $\mathbf{P}_2$ defined by

$$\mathbf{P}_1(g) = \begin{pmatrix} H^x \\ H^{xg} \end{pmatrix} \quad \text{(on the set } \{H^x \mid x \in G\}\text{)}$$

and

$$\mathbf{P}_2(g) = \begin{pmatrix} \mathbf{N}_G(H)x \\ \mathbf{N}_G(H)xg \end{pmatrix} \quad \text{(on the set } \{\mathbf{N}_G(H)x \mid x \in G\}\text{)}$$

are permutation isomorphic.

c) The representations of G as permutation groups on the cosets of the subgroups U and V are permutation isomorphic if and only if U and V are conjugate in G.

24) Let $\mathbf{P}$ be the permutation representation

$$\mathbf{P}(g) = \begin{pmatrix} Hx \\ Hxg \end{pmatrix}$$

of G on the cosets of H. We denote by $\pi(g)$ the number of points fixed by $\mathbf{P}(g)$. Let $C = \{g^x \mid x \in G\}$ be the conjugacy class of g in G. Then

$$\pi(g) = \frac{[G : H]|C \cap H|}{|C|}.$$

(Hint: Count the pairs (Hx, y) with $Hxy = Hx$ and $y \in C$ in two different ways.)

25) The alternating group A_5 has subgroups of indices 6 and 10, but no subgroups of indices 3 and 4.

7 The Sylow theorems

In this section, all groups are assumed to be finite.

Definition 7.1. Let G be a finite group of order $|G| = p^a n$, where p is a prime number not dividing n. Then every subgroup $P \leq G$ of order $|P| = p^a$ is called a Sylow p-subgroup of G.

Theorem 7.2. *Let p be a prime number. Let $|G| = p^a n$, and let $N(p^a)$ be the number of subgroups of G of order p^a. Then $N(p^a) \equiv 1$ mod p; in particular, $N(p^a) \geq 1$.* (*We do not require here that n and p are coprime.*)

Proof. (Wielandt). Let

$$\Omega = \left\{ M \subseteq G \,\middle|\, |M| = p^a \right\}$$

be the set of all subsets $M \subseteq G$ of size $|M| = p^a$. Then clearly $|\Omega| = \binom{|G|}{p^a}$.

We give G the structure of a permutation group on Ω by defining $M^{\mathbf{T}(g)} = Mg$. By Definition 5.8, Ω is partitioned into G-orbits T_i. We have $|\Omega| = \sum_i |T_i|$ and

$|T_i| = [G : U_i]$ by Theorem 5.10, where

$$U_i = \{g \in G \mid M_i g = M_i\}$$

for a set M_i in T_i.

Since $M_i U_i = M_i$, we have $M_i = \bigcup_{j=1}^{k_i} g_{ij} U_i$ for some g_{ij}. Therefore,

$$|M_i| = p^a = k_i |U_i|, \quad \text{so} \quad |U_i| = p^{b_i} \leq p^a.$$

If $|U_i| < p^a$, then $|T_i| = [G : U_i] \equiv 0 \bmod pn$. Note that $|U_i| = p^a$ if and only if $|T_i| = n$ since $|T_i| = [G : U_i]$. Therefore, we obtain

$$|\Omega| = \binom{|G|}{p^a} \equiv \sum_{|T_i|=n} |T_i| \mod pn.$$

If $|T_i| = n$, then $|U_i| = p^a$ and $M_i = m_i U_i$ for some $m_i \in M_i$. Then

$$M_i m_i^{-1} = m_i U_i m_i^{-1} = V_i$$

is a subgroup of order p^a in T_i, and the cosets $V_i g$ yield exactly the elements in T_i. Conversely, every subgroup U of G of order $|U| = p^a$ gives rise to an orbit $T = \{Ug \mid g \in G\}$ of size n. Distinct subgroups U_i with $|U_i| = p^a$ give rise to distinct orbits T_i; for if $U_i = U_j g$, then $1 = u_j g$ for some $u_j \in U_j$, so $g = u_j^{-1} \in U_j$ and then $U_i = U_j$. We therefore obtain

$$\binom{np^a}{p^a} \equiv \sum_{|T_i|=n} |T_i| = nN(p^a) \mod pn.$$

Now we choose G to be a cyclic group of order np^a, following a suggestion of J. McLaughlin. We have by Theorem 2.20 that $N(p^a) = 1$. Thus, we have

$$\binom{np^a}{p^a} \equiv n \mod pn.$$

Therefore, for arbitrary G we have that $N(p^a) \equiv 1 \bmod p$, as claimed. □

The first fundamental theorem of Sylow now follows:

Theorem 7.3 (First Sylow theorem). *Every finite group contains Sylow p-subgroups for every prime p. The number N_p of Sylow p-subgroups is congruent to* $1 \bmod p$.

Theorem 7.4 (Cauchy). *If the prime p is a divisor of the order of G, then there exist elements of order p in G.*

Proof. According to Theorem 7.2, there exists a subgroup of order p, which is generated by an element of order p. □

The next theorem gives an overview of all the Sylow p-subgroups.

Theorem 7.5 (Second Sylow theorem). *Let G be a finite group.*
a) *If P is a Sylow p-subgroup of G and U is a subgroup of G of p-power order, then there exists an element $g \in G$ such that $U \leq P^g$.*
b) *All Sylow p-subgroups of G are conjugate in G. Therefore, $[G : \mathbf{N}_G(P)]$ is the number of Sylow p-subgroups of G.*

Proof. a) Consider the decomposition of G into the double cosets of P and U. By Theorem 2.19, we have

$$|G| = \sum_i \frac{|P||U|}{|P^{g_i} \cap U|}$$

for some $g_i \in G$. Assume for contradiction that $P^{g_i} \cap U < U$ for all i. Then $[U : P^{g_i} \cap U] \neq 1$ is a power of p, and it follows that

$$[G : P] = \sum_i [U : P^{g_i} \cap U] \equiv 0 \mod p,$$

contradicting that $[G : P]$ is coprime to p. Therefore, there exists some i such that $U = P^{g_i} \cap U \leq P^{g_i}$.

b) follows immediately from a) if we take U to be a Sylow p-subgroup of G. □

Theorem 7.6. *Let P be a Sylow p-subgroup of G, and let $\mathbf{N}_G(P) \leq U \leq G$. Then $\mathbf{N}_G(U) = U$.*

Proof. Let $x \in \mathbf{N}_G(U)$, so $U^x = U$. Then

$$P^x \leq \mathbf{N}_G(P)^x \leq U^x = U.$$

Hence, P and P^x are Sylow p-subgroups of U. By Theorem 7.5, P and P^x are conjugate in U. So there exists an element $u \in U$ such that $P^x = P^u$. It follows that $P^{xu^{-1}} = P$, then $xu^{-1} \in \mathbf{N}_G(P) \leq U$ and so $x \in U$. Therefore, $\mathbf{N}_G(U) = U$. □

Theorems 7.3 through 7.6 are of fundamental importance to the theory of finite groups. We will use these theorems repeatedly, and a sizeable portion of our work will be concerned with extending them. We note a few simple but useful consequences:

Lemma 7.7. *Let N be a normal subgroup of G, and let P be a Sylow p-subgroup of G. Then:*
a) *$N \cap P$ is a Sylow p-subgroup of N.*
b) *PN/N is a Sylow p-subgroup of G/N.*
c) *$\mathbf{N}_G(P)N/N = \mathbf{N}_{G/N}(PN/N)$.*
(*Note that the claim in a) is no longer correct if N is simply a subgroup of G!*)

Proof. a) Since $P \leq PN \leq G$, the index $[PN : P] = [N : N \cap P]$ is coprime to p, being a divisor of $[G : P]$. In addition, $|N \cap P|$ is a power of p. Therefore, $N \cap P$ is a Sylow p-subgroup of N.

b) We have that $|PN/N| = |P/(N \cap P)|$ is a power of p, and

$$[G/N : PN/N] = [G : PN]$$

is coprime to p, being a divisor of $[G : P]$. Therefore, PN/N is a Sylow p-subgroup of G/N.

c) For $x \in \mathbf{N}_G(P)$, we have

$$(x^{-1}N)(PN/N)(xN) = P^xN/N = PN/N.$$

Therefore, we have

$$\mathbf{N}_G(P)N/N \leq \mathbf{N}_{G/N}(PN/N).$$

Conversely, if $xN \in \mathbf{N}_{G/N}(PN/N)$, then $P^xN = PN$. Therefore, P^x is a Sylow p-subgroup of the subgroup PN of G. By Theorem 7.5, there exists an element gn in PN with $g \in P$ and $n \in N$ such that

$$P^{gn} = P^n = P^x.$$

It follows that $xn^{-1} \in \mathbf{N}_G(P)$, so $x \in \mathbf{N}_G(P)N$ and then $xN \in \mathbf{N}_G(P)N/N$. □

Theorem 7.8 (The Frattini argument). *Let N be a normal subgroup of G, and let P be a Sylow p-subgroup of N. Then $G = N\mathbf{N}_G(P)$.*

Proof. Let g be any element of G. Then P^g lies in $N^g = N$, so P^g is a Sylow p-subgroup of N. By Theorem 7.5, there exists an $n \in N$ such that $P^g = P^n$. It follows that $gn^{-1} \in \mathbf{N}_G(P)$, and then $g \in \mathbf{N}_G(P)N = N\mathbf{N}_G(P)$. Since this holds for all $g \in G$, we have $G = \mathbf{N}_G(P)N$. □

In some cases, the statement of Theorem 7.3 regarding the number of Sylow p-subgroups can be sharpened:

Theorem 7.9. *Let $P_1, \ldots, P_N$ be all the Sylow p-subgroups of G. If for every $i \neq j$ the inequality*

$$[P_i : P_i \cap P_j] \geq p^d$$

holds, then $N \equiv 1 \bmod p^d$.

Proof. Consider the decomposition of G into double cosets of a Sylow p-subgroup P and its normalizer $\mathbf{N}_G(P)$ (see Theorem 2.19). We have

$$G = \bigcup_{i=1}^{s} Px_i\mathbf{N}_G(P) \quad (\text{with } x_1 = 1)$$

and

$$|G| = \sum_{i=1}^{s} \frac{|P||\mathbf{N}_G(P)|}{|P^{x_i} \cap \mathbf{N}_G(P)|}.$$

Since P is the unique Sylow p-subgroup of $\mathbf{N}_G(P)$ by Theorem 7.5 b), it follows from Theorem 7.5 a) that

$$P^{x_i} \cap \mathbf{N}_G(P) = P^{x_i} \cap P.$$

Therefore, we have

$$[G : \mathbf{N}_G(P)] = N = \sum_{i=1}^{s} [P : P^{x_i} \cap P].$$

For $i > 1$ we have $x_i \notin \mathbf{N}_G(P)$, so $P^{x_i} \neq P$ and by assumption

$$[P : P^{x_i} \cap P] \geq p^d.$$

Hence, $N \equiv 1 \bmod p^d$. □

Exercises

26) Prove the following using a decomposition into double cosets, but without the use of the Sylow theorems:
Let $H < G$, and let P be a Sylow p-subgroup of G. Then there is a $g \in G$ such that $H \cap P^g$ is a Sylow p-subgroup of H.

27) Let V be the vector space of dimension n over $\mathrm{GF}(p)$, and let $\mathrm{GL}_n(p)$ be the group of all bijective linear maps from V to itself. Show that:
a) $|\mathrm{GL}_n(p)| = (p^n - 1)(p^n - p) \cdots (p^n - p^{n-1})$.
b) Let $\{v_1, \ldots, v_n\}$ be a basis of V over $\mathrm{GF}(p)$. Then the linear maps t defined by

$$v_1 t = v_1$$
$$v_i t - v_i \in \langle v_1, \ldots, v_{i-1} \rangle \quad (2 \leq i \leq n)$$

constitute a Sylow p-subgroup of $\mathrm{GL}_n(p)$.

28) Give another proof of the existence of Sylow p-subgroups using a combination of Exercises 26 and 27.

29) Prove the following without using theorems from §7: Assume G is abelian, and let p be a prime divisor of $|G| = n$. Then there exists a $g \in G$ such that $|g| = p$.
(Hint: Let $G = \{g_1, \ldots, g_n\}$ and $|g_i| = a_i$. Show that every element of G appears equally often in the set

$$\{g_1^{x_1} \cdots g_n^{x_n} \mid 0 \leq x_i < a_i\}.)$$

30) Give another proof of the existence of Sylow p-subgroups in the following way: Let G have smallest order among the counterexamples that do not have Sylow p-subgroups. Show that:
   a) If $U < G$, then $[G : U] \equiv 0 \bmod p$.
   b) G has no nonidentity normal subgroups of p-power order.
   c) Using parts a) and b) and the class equation from Theorem 6.8, conclude that G is abelian and use Exercise 29.

31) (Brodkey [73]) Assume that the Sylow p-subgroups of G are abelian. Show that if the intersection of all Sylow p-subgroups of G is 1, then there exist two Sylow p-subgroups P_1 and P_2 of G such that $P_1 \cap P_2 = 1$.
   (Hint: Let G be a counterexample with the smallest number N_p of Sylow p-subgroups. Suppose $P_2 \cap \cdots \cap P_k = A > 1$, but $P_1 \cap A = 1$. Consider $\mathbf{C}_G(A)$.)
   For a significantly deeper statement regarding the intersection of Sylow p-subgroups, see Itô [279].

8 Solvable groups

Definition 8.1. If a and b are elements of a group G, then we define

$$[a, b] = a^{-1}b^{-1}ab$$

and call $[a, b]$ the commutator of a with b. Furthermore, we define

$$G' = \langle [a, b] \mid a, b \in G \rangle$$

and call G' the commutator subgroup of G. One sees immediately that G' is a characteristic subgroup of G since automorphisms send commutators to commutators. (Bear in mind that G' is generated by the set of commutators of G; the set of commutators itself is not generally a subgroup of G!) We define the higher commutator subgroups $G^{(i)}$ recursively by $G^{(0)} = G$, $G^{(1)} = G'$ and $G^{(i+1)} = (G^{(i)})'$. It is clear that

$$G \geq G' \geq G^{(2)} \geq G^{(3)} \geq \cdots,$$

and all the $G^{(i)}$ are characteristic subgroups of G. We often write G'' instead of $G^{(2)}$. If A and B are subgroups of G, we define

$$[A, B] = \langle [a, b] \mid a \in A, b \in B \rangle.$$

If $B \leq \mathbf{N}_G(A)$, then $[a, b] = a^{-1}a^b \in A$ for all $a \in A$ and $b \in B$, so then $[A, B] \leq A$.

Theorem 8.2. *Let U be a subgroup of G. Then U is a normal subgroup of G with abelian factor group G/U if and only if $G' \leq U$.*

Proof. If G/U is abelian, then for all $a, b \in G$ we have

$$abU = (aU)(bU) = (bU)(aU) = baU.$$

Therefore, $[a, b] = a^{-1}b^{-1}ab \in U$ and so $G' \leq U$.

Conversely, suppose $G' \leq U \leq G$. For all $u \in U$ and $g \in G$ we have

$$u^g = g^{-1}ug = u[u, g] \in UG' = U.$$

Thus, $U \trianglelefteq G$. Since $[a, b] \in G' \leq U$, it follows that

$$(aU)(bU) = (bU)(aU),$$

and so G/U is abelian. □

Definition 8.3. A group G is called solvable if $G^{(k)} = 1$ for some natural number k.

Lemma 8.4. *Let $N \trianglelefteq G$. For all $i = 1, 2, \ldots$ we have*

$$(G/N)^{(i)} = G^{(i)}N/N.$$

Proof. We have

$$(G/N)' = \langle [aN, bN] \mid a, b \in G \rangle = \langle [a, b]N \mid a, b \in G \rangle = G'N/N.$$

Now suppose for induction that $(G/N)^{(i)} = G^{(i)}N/N$ for some $i \geq 1$. Then

$$(G/N)^{(i+1)} = (G^{(i)}N/N)' = (G^{(i)}N)'N/N \geq G^{(i+1)}N/N.$$

On the other hand,

$$\begin{aligned} G^{(i)}N/N \Big/ G^{(i+1)}N/N &\cong G^{(i)}(G^{(i+1)}N)/G^{(i+1)}N \\ &\cong G^{(i)}/(G^{(i)} \cap G^{(i+1)}N), \end{aligned}$$

and this is abelian since

$$(G^{(i)})' = G^{(i+1)} \leq G^{(i)} \cap G^{(i+1)}N.$$

By Theorem 8.2, it follows that

$$(G/N)^{(i+1)} = (G^{(i)}N/N)' \leq G^{(i+1)}N/N,$$

which proves the claim. □

Theorem 8.5. *If G is solvable, then every subgroup of G and every factor group of G is also solvable.*

Proof. If $U \leq G$, then $U' \leq G'$ and in general by induction on i we have $U^{(i)} \leq G^{(i)}$ as well. If G is solvable, then $G^{(k)} = 1$ for some k. Then $U^{(k)} \leq G^{(k)} = 1$, so U is solvable.

If G is solvable with $G^{(k)} = 1$, then it follows by Lemma 8.4 that

$$(G/N)^{(k)} = G^{(k)}N/N = N/N = 1.$$

Thus, G/N is solvable. □

Theorem 8.6.

a) *If N is a normal subgroup of G such that N and G/N are solvable, then G is also solvable.*

b) *Suppose $G^{(k)} = G^{(k+j)}$ for all $j \geq 0$. (Certainly there exists such a k for a finite group G.) Let $N \trianglelefteq G$. Then G/N is solvable if and only if $G^{(k)} \leq N$.*

c) *If G/M and G/N are solvable, then $G/(M \cap N)$ is also solvable.*

d) *If M and N are solvable normal subgroups of G, then the product $MN = \langle M, N\rangle$ is also a solvable normal subgroup of G. The product of all solvable normal subgroups of a finite group G is therefore a solvable normal subgroup of G.*

Proof. a) If N and G/N are solvable, then for some natural numbers k and m we have

$$N^{(k)} = 1 \quad \text{and} \quad (G/N)^{(m)} = 1.$$

It follows from

$$(G/N)^{(m)} = G^{(m)}N/N = 1$$

that $G^{(m)} \leq N$, and so $G^{(m+k)} = (G^{(m)})^{(k)} \leq N^{(k)} = 1$. So G is solvable.

b) If $G^{(k)} \leq N$, then by Lemma 8.4

$$(G/N)^{(k)} = G^{(k)}N/N = 1,$$

and G/N is solvable.

Conversely, if G/N is solvable, say

$$(G/N)^{(m)} = G^{(m)}N/N = 1,$$

then $G^{(m)} \leq N$ and

$$G^{(k)} = G^{(k+m)} = (G^{(m)})^{(k)} \leq N.$$

c) If G/M and G/N are solvable, then $G^{(k)} \leq M \cap N$ by b), and therefore $G/(M \cap N)$ is also solvable.

d) We have $\langle M, N\rangle = MN$ and $MN/N \cong M/(M \cap N)$. Then MN/N is solvable, as it is an epimorphic image of M, and N is solvable. So by a), MN is solvable. □

Definition 8.7. Let $G \neq 1$ be solvable. The smallest natural number k such that $G^{(k)} = 1$ is called the derived length of G. We say the trivial group 1 has derived length 0. The groups of derived length 1 are clearly the nonidentity abelian groups. We call the groups of derived length 2 metabelian.

Theorem 8.8. *Let $|G| = p^n$, where p is a prime number, and let $U < G$. Then $U < \mathbf{N}_G(U)$.*

Proof. Let $G = \bigcup_{i=1}^{s} Ux_iU$ be the decomposition of G into double cosets of the pair of subgroups U, U. Let $x_1 = 1$. Then by Theorem 2.19, we have

$$|G| = p^n = \sum_{i=1}^{s} \frac{|U||U|}{|U^{x_i} \cap U|}.$$

Let $|U| = p^m$. It follows that

$$p^{n-m} = 1 + \sum_{i=2}^{s} [U : U^{x_i} \cap U].$$

Since $[U : U^{x_i} \cap U]$ is a power of a prime p and $m < n$, there must be an x_j with $j \geq 2$ such that $|U^{x_j} \cap U| = |U|$. But this means $U^{x_j} = U$, and so $x_j \in \mathbf{N}_G(U)$. Since $U = Ux_1U \neq Ux_jU$, we have $x_j \notin U$. Therefore, $U < \mathbf{N}_G(U)$. □

We will give another proof of Theorem 8.8 in Theorem III.2.3.

Theorem 8.9.

a) *If the order of G is a power of a prime p, then every maximal subgroup of G has index p and is normal in G. (A subgroup U of G is called maximal if $U < G$ and for all $V \leq G$ such that $U < V \leq G$, we have $V = G$.)*

b) *Every group of prime power order is solvable.*

Proof. a) Let U be a maximal subgroup of G. By Theorem 8.8, we have $U < \mathbf{N}_G(U)$, so that $\mathbf{N}_G(U) = G$ and $U \lhd G$. By Theorem 3.10, G/U has no nontrivial subgroups, so G/U has prime order by Theorem 7.4. Therefore, $|G/U| = p$.

b) Let U be a maximal subgroup of G. By a), $U \lhd G$ and $|G/U| = p$. By induction on the order of the group, we can assume that U is solvable. Since groups of prime order are cyclic by Theorem 2.10, G/U is cyclic and so G/U is also solvable. Then G itself is solvable by Theorem 8.6 a). □

Theorem 8.10. *Let p and q be prime numbers with $p \geq q$, and let $|G| = pq$. Then G is solvable, and G has a normal Sylow p-subgroup. If $q = p$, then G is abelian. If $p \neq q$ and $q \nmid p - 1$, then G is cyclic.*

Proof. If $|G| = p^2$, then we know by Theorem 6.10 that G is abelian. Now suppose that $p > q$, and let P be a Sylow p-subgroup of G. Then $|P| = p$ and $P \leq \mathbf{N}_G(P) \leq G$, so $[G : \mathbf{N}_G(P)]$ is a divisor of $[G : P] = q$. In addition, $[G : \mathbf{N}_G(P)] \equiv 1 \bmod p$ by Theorems 7.3 and 7.5 b). Since $p + 1 > q$ it follows that $[G : \mathbf{N}_G(P)] = 1$, so $P \lhd G$. Now, G/P and P are groups of prime order, so they are cyclic and therefore solvable. By Theorem 8.6 a), G is also solvable. By Theorem 4.5, $G/\mathbf{C}_G(P)$ is isomorphic to a subgroup of the automorphism group $\mathbf{Aut}(P)$ of P. Since $P \leq \mathbf{C}_G(P)$, we have on the one hand that $|G/\mathbf{C}_G(P)|$ divides q. On the other hand, we have that

$$|G/\mathbf{C}_G(P)| \text{ divides } |\mathbf{Aut}(P)| = p-1$$

by Theorems 4.5 and 4.6. If $p-1$ and q are coprime, then $G = \mathbf{C}_G(P)$ and $P \leq \mathbf{Z}(G)$. Let $P = \langle g \rangle$, and let $x \in G$ be an element of order q; such an x exists by Theorem 7.4. Since $P \leq \mathbf{Z}(G)$, we have $gx = xg$. One sees easily that gx has order pq (see Exercise 6). We obtain $G = \langle gx \rangle$, and so G is cyclic. □

The following is somewhat deeper, but can be proved with the same methods.

Theorem 8.11. *If p and q are prime numbers and $|G| = p^a q$, then G is solvable.*

Proof. We may assume that $p \neq q$; otherwise we apply Theorem 8.9. If P is a Sylow p-subgroup of G, then $[G : \mathbf{N}_G(P)]$ divides q. We may assume that $[G : \mathbf{N}_G(P)] = q$: Otherwise, $P \lhd G$, so P and G/P are solvable. This implies G is solvable by Theorem 8.6 a) and we are done.

Let $D = P_1 \cap P_2$ be an intersection of maximal order of distinct Sylow p-subgroups P_1 and P_2 of G. We consider two cases:

Case 1: $D = 1$. Then every two of the q distinct Sylow p-subgroups of G have only the identity element in common. Thus, G has exactly

$$1 + q(p^a - 1) = |G| - (q-1)$$

elements of p-power order. Then there are only $q-1$ elements of G remaining. They necessarily have order q and constitute, with 1, the single Sylow q-subgroup Q of G. Therefore, $Q \lhd G$. Since $|G/Q| = p^a$, G is solvable by Theorems 8.9 b) and 8.6 a).

Case 2: $1 < D$. By Theorem 8.8, we have

$$D < \mathbf{N}_{P_i}(D) = V_i \leq P_i \quad (i = 1, 2).$$

Therefore, $D \lhd \langle V_1, V_2 \rangle = T$. Suppose for contradiction that $|T| = p^t$. Then there is a Sylow p-subgroup P_3 of G such that $T \leq P_3$ by Theorem 7.5 a). It follows that

$$P_1 \cap P_3 \geq P_1 \cap T \geq V_1 > D,$$

and so $P_1 = P_3$ by the maximality of D. In the same way, we conclude that $P_2 = P_3$, which implies the contradiction $P_1 = P_2$.

Therefore, $|T| = p^t q$ for some t. Let Q be a Sylow q-subgroup of T. By Lemma 2.12 a), we have

$$|QP_1| = qp^a = |G|,$$

and therefore $G = QP_1$.

We set $K = D^G = \langle D^g \mid g \in G \rangle$. Clearly, $D \leq K \trianglelefteq G$. Let $g = xy$ with $x \in Q \leq T \leq \mathbf{N}_G(D)$ and $y \in P_1$. Then

$$D^g = (D^x)^y = D^y \leq P_1,$$

and therefore $K \leq P_1 < G$. Thus, $1 < K \lhd G$. By induction, K and G/K are solvable, so G is solvable as well by Theorem 8.6 a). □

We note the following trivial consequence:

Remark 8.12. The symmetric groups S_n are solvable for $n \leq 4$.

We will see that S_n is not solvable for $n \geq 5$ as a consequence of Theorem 8.14. The following is another consequence of the Sylow theorems.

Theorem 8.13. *If $|G| = pqr$ for three primes $p > q > r$, then G is solvable.*

Proof. We designate the number of Sylow p-subgroups of G as N_p. We may assume that $N_p > 1$, $N_q > 1$ and $N_r > 1$; otherwise, G would have a normal subgroup of prime order, and the factor group would be solvable by Theorem 8.10.

Since $N_p \equiv 1 \bmod p$ and $N_p = [G : \mathbf{N}_G(P)]$ divides qr (for every Sylow p-subgroup P of G), it follows from $p > q > r$ that $N_p = qr$. Since $N_q \equiv 1 \bmod q$ and N_q divides pr, it follows from $q > r$ that $N_q = p$ or $N_q = pr$. Hence, $N_q \geq p$ in every case. Finally, $N_r \geq q$. Since the cyclic Sylow subgroups of G have prime order, they intersect only at the identity. Now we count elements: We have

$$\begin{aligned} N_p(p-1) &= qr(p-1) && \text{elements of order } p, \\ N_q(q-1) &\geq p(q-1) && \text{elements of order } q \text{ and} \\ N_r(r-1) &\geq q(r-1) && \text{elements of order } r. \end{aligned}$$

It follows that

$$|G| = pqr \geq 1 + qr(p-1) + p(q-1) + q(r-1) = pqr + 1 + pq - p - q,$$

which yields the contradiction $p - 1 \geq q(p-1)$. □

Later, we will greatly generalize Theorems 8.11 and 8.13 using some powerful tools. Namely, we will prove:

Theorem V.7.3 (Burnside): Every group of order $p^a q^b$ (p and q primes) is solvable.

Theorem IV.2.9 (O. Hölder, see Exercise 40): Every group of square-free order is solvable.

We should also mention at this point that W. Feit and J. Thompson proved the following old conjecture of Burnside: Every group of odd order is solvable.

It is not difficult to show, using the solvability criteria of Theorems 8.11 and 8.13 (see Exercise 33), that every group G of order $|G| < 60$ is solvable. Now we show that there exists exactly one simple group of order 60 (so it is necessarily not solvable):

Theorem 8.14. *If G is a simple group of order 60, then $G \cong A_5$.*

Proof. a) First we show that A_5 is simple.

If t is an element of order 3 in A_5, then t is a 3-cycle and leaves exactly two points i, j fixed. For every i, j there exist two (mutually inverse) such 3-cycles. Therefore, A_5 contains exactly 20 elements of order 3 and 10 Sylow 3-subgroups. In addition, A_5 contains exactly 24 5-cycles, which are distributed in 6 Sylow 5-subgroups.

Let $1 < N \trianglelefteq A_5$.

Assume first that 3 divides $|N|$, and let T be a Sylow 3-subgroup of N. Then N contains all 10 Sylow 3-subgroups of A_5, and it follows that

$$|N| = [N : \mathbf{N}_N(T)]|\mathbf{N}_N(T)| = 10|\mathbf{N}_N(T)| \equiv 0 \bmod 30.$$

So 30 divides $|N|$.

If 5 divides $|N|$, then N contains all 6 Sylow 5-subgroups of A_5, and again we find that 30 divides $|N|$.

Therefore, we find in both cases that 30 divides $|N|$. Then N contains all 20 elements of order 3 and all 24 elements of order 5 of A_5. This leads to

$$|N| \geq 1 + 20 + 24 = 45.$$

Since $|N|$ divides 60 we must have $N = A_5$.

It remains to consider the case that $|N|$ divides 4. The orbits of N have lengths 1, 2 or 4 by Theorem 5.10 a). Therefore, one of the 5 points, say i, is fixed by N. Since $N \trianglelefteq A_5$, i^g is fixed by $N^g = N$ for all $g \in A_5$. Since A_5 is clearly transitive, N must fix all points and it follows that $N = 1$, a contradiction.

b) Let G be a simple group of order 60. In order to prove that $G \cong A_5$, it suffices to prove that G contains a subgroup U of index 5: Then by constructing the permutation representation of G (of degree 5) on the cosets of U according to Theorem 6.2, we obtain an isomorphism from the simple group G onto a subgroup V of index 2 in S_5. Since $[S_5 : V] = 2$, $V \lhd S_5$ by Theorem 3.3. Suppose for contradiction that $V \neq A_5$. Then $S_5 = VA_5$ and therefore

$$[S_5 : V] = [A_5 : V \cap A_5] = 2.$$

Using Theorem 3.3 again, we conclude that $V \cap A_5 \lhd A_5$. But this contradicts the simplicity of A_5, which we proved in a).

So assume for contradiction that G has no subgroup of index 5. Let p divide $|G|$, and let N_p be the number of Sylow p-subgroups of G. We have $N_p \equiv 1 \bmod p$ by Theorem 7.3. By assumption, $N_p = [G : \mathbf{N}_G(P)] \neq 5$ for every Sylow p-subgroup P of G. If $N_p \leq 4$, then the permutation representation of G acting on the set $\{P_i \mid i = 1, \ldots, N_p\}$ of Sylow p-subgroups yields a homomorphism

$$\mathbf{T}(g) = \begin{pmatrix} P_i \\ P_i^g \end{pmatrix}$$

from G to a symmetric group S_k with $k = N_p \leq 4$. Since G is simple, we obtain the contradiction that $|G| = 60$ divides 4!. We conclude that $N_p > 5$ for all p. Since $N_p \equiv 1 \bmod p$, we obtain $N_5 = 6$, $N_3 = 10$ and $N_2 = 15$.

Assume for contradiction that P_1 and P_2 are two distinct Sylow 2-subgroups of G such that $1 < D = P_1 \cap P_2$. Since P_1 and P_2 are abelian by Theorem 6.10, $D \lhd \langle P_1, P_2 \rangle = T$. By the simplicity of G, we have $T < G$. Since T contains more than one Sylow 2-subgroup, T contains at least three Sylow 2-subgroups by

Theorem 7.3, and it follows that $12 \leq |T|$. This forces the contradiction $[G : T] \leq 5$. Therefore, $P_1 \cap P_2 = 1$ for $P_1 \neq P_2$.

Now we count elements: G contains

$$\begin{aligned} N_5(5-1) &= 24 && \text{elements of order 5,} \\ N_3(3-1) &= 20 && \text{elements of order 3 and} \\ N_2(4-1) &= 45 && \text{elements of 2-power order.} \end{aligned}$$

However,

$$24 + 20 + 45 > 60,$$

a contradiction. □

Since $A_5 \leq S_n$ for $n \geq 5$, it follows from Theorem 8.14 that the symmetric groups S_n are not solvable for $n \geq 5$.

Exercises

32) If $|G| = p^2q$, where p and q are primes, then G has a normal Sylow subgroup.
33) If $|G| = p^2q^2$, where p and q are primes, then G is solvable.
34) If $|G| \leq 200$ and $|G| \neq 60, 120, 168$ or 180, then G is solvable.
35) There are no simple groups of orders 120 or 180.
36) Let $|G| = n$. Suppose that for every divisor d of n, G has at most one subgroup of order d. Then G is cyclic.
37) Let K be a field, and let G be a finite subgroup of the multiplicative group $\mathsf{K}^\times$ of K. Then G is cyclic.
38) (Theorem of Frobenius). Let $C = \{x^g \mid g \in G\}$ be a conjugacy class of G with $|C| = c$. Let $|G| = n$. Then the number of $g \in G$ with $g^m \in C$ is divisible by (mc, n).

(Hint: For every subset K of G and natural number k, define

$$\mathscr{A}(K, k) = \{g \in G \mid g^k \in K\}$$

and $|\mathscr{A}(K, k)| = a(K, k)$. Prove the claim by induction on $n+m$ in the following way:

a) For $c > 1$, show that $\mathscr{A}(x, m) \subseteq \mathbf{C}_G(x) < G$ and then

$$\left(m, \frac{n}{c}\right) \mid a(X, m) = \frac{a(C, m)}{c}.$$

b) Let $m = m_1m_2$ with $(m_1, m_2) = 1$ and $m_i \neq 1$ $(i = 1, 2)$. Let $D = \mathscr{A}(C, m_1)$. Then $\mathscr{A}(C, m) = \mathscr{A}(D, m_2)$. Use the induction hypothesis!

c) Let $m = p^a$ $(a > 0)$ with $p \mid |x|$. For $g \in \mathscr{A}(x, m)$, we have $|g| = p^a|x|$ and $|\mathscr{A}(x, m) \cap \langle g \rangle| = m$.

d) Let $(m, |x|) = 1$. Let $Z = \{y \in \mathbf{Z}(G) \mid (m, |y|) = 1\}$. Show that $a(y_1, m) = a(y_2, m)$ for $y_i \in Z$. Conclude from the class equation

$$n = \sum_{K_i \not\subseteq Z} a(K_i, m) + |Z| a(y, m)$$

that (m, n) divides $|Z|a(y, m)$.)

39) Let G be a group in which every Sylow subgroup is cyclic. Let $|G| = \prod_{i=1}^{r} p_i^{a_i}$ with $p_1 > p_2 > \cdots > p_r$. Let $m = p_1^{a_1} \cdots p_j^{a_j} p_{j+1}^{b}$ with $1 \leq j < r$ and $b \leq a_{j+1}$. Then there are exactly m elements $g \in G$ such that $g^m = 1$.

(Hint: Let $p = p_{j+1}$ for $b < a_{j+1}$ and $p = p_{j+2}$ for $b = a_{j+1}$. Suppose it has already been shown that $x^{mp} = 1$ has exactly mp solutions. From this assumption it follows, with the help of Exercise 38, that

$$|\{g \in G \mid g^m = 1\}| = km$$

with $1 \leq k < p$. The $mp - mk$ elements with $g^m \neq 1 = g^{mp}$ are exactly the generators of the cyclic subgroups Z of G with $|Z| \nmid m$ and $|Z| \mid mp$. Then conclude that $m(p - k) \equiv 0 \bmod p - 1$.)

40) Using Exercise 39, show that if all Sylow subgroups of G are cyclic, then G is solvable.

(In particular, every group of square-free order is solvable, generalizing Theorem 8.13. We will give a different proof of this fact in Theorem IV.2.9.)

41) Let $G = S_4$. Show that:
   a) We have $G' = A_4$, and G'' is a normal subgroup K of S_4 of order 4. (The group K is called the Klein 4-group.)
   b) A_4 and K are the only nontrivial normal subgroups of S_4.
   c) Describe the Sylow subgroups of S_4 and their intersections.

9 Direct products

Definition 9.1. Let $\{G_i\}_{i \in I}$ be a collection of subgroups of the (not necessarily finite) group G indexed by the set I. We say that G is the direct product of the G_i when the following conditions are satisfied:

(1) $G_i \trianglelefteq G$ for all $i \in I$.

(2) Every element $g \in G$ can be written in exactly one way (up to ordering of the factors) as $g = \prod_{i \in I} g_i$ with $g_i \in G_i$ and $g_i \neq 1$ for only finitely many i. In particular, $G_i \cap G_j = 1$ for $i \neq j$. Thus, $g_i g_j = g_j g_i$ for all $g_i \in G_i$ and $g_j \in G_j$ with $i \neq j$ by Theorem 3.13.

If these conditions hold, we write $G = \bigtimes_{i \in I} G_i$.

If the groups G_i are abelian and are written additively, most of the time we write $G = \bigoplus_{i \in I} G_i$ and say that G is the direct sum of the G_i.

Theorem 9.2. *Let the G_i be arbitrary groups for all $i \in I$. Let G be the set of all maps f from I to the set $\bigcup_{i \in I} G_i$ with $f(i) \in G_i$ for all i and $f(i) \neq 1$ for at most finitely many i. We define a multiplication on G by*

$$(fg)(i) = f(i)g(i).$$

Then:

a) *G is the direct product of the subgroups*

$$\overline{G}_i = \{f \mid f(j) = 1 \text{ for } j \neq i\},$$

and $\overline{G}_i \cong G_i$.

b) *If the group H is the direct product of subgroups H_i $(i \in I)$ and α_i is an isomorphism (homomorphism) from G_i onto H_i, then the map α defined by*

$$f^{\alpha}(i) = f(i)^{\alpha_i}$$

is an isomorphism (homomorphism) from G onto H.

(*That is, the direct product of the G_i is uniquely determined up to isomorphism by the isomorphism types of the G_i.*)

Proof. a) It is clear that the set G is a group under the given multiplication; the identity element of G is the map f with $f(i) = 1$ for all $i \in I$. Furthermore, $\overline{G}_i$ is a subgroup of G, and the map ϵ_i defined by

$$f^{\epsilon_i} = f(i)$$

is an isomorphism from $\overline{G}_i$ to G_i.

Let $f \in \overline{G}_i$ and $g \in G$. Then

$$(g^{-1}fg)(j) = g(j)^{-1}f(j)g(j) = 1 \quad \text{for} \quad i \neq j.$$

So $\overline{G}_i$ is a normal subgroup of G.

Let $f \in G$ with $f(i) = g_i \in G_i$ for i in a finite subset J of I, and suppose $f(i) = 1$ for $i \in I - J$. For $j \in J$, we define $f_j \in \overline{G}_j$ by

$$f_j(i) = \begin{cases} 1 & \text{for } i \neq j \\ g_j & \text{for } i = j. \end{cases}$$

Then clearly $f = \prod_{j \in J} f_j$.

If $f = \prod_j f'_j$ with $f'_j \in \overline{G}_j$, then it follows that

$$g_i = f(i) = \prod_j f'_j(i) = f'_i(i)$$

for $i \in J$ and

$$1 = f(i) = f'_i(i)$$

for $i \in I - J$. Therefore, the decomposition $f = \prod_i f_i$ into factors $f_i \in \overline{G}_i$ is uniquely determined. Since the conditions (1) and (2) of Definition 9.1 are satisfied, G is the direct product of the $\overline{G}_i$.

b) We have

$$(ff')^\alpha(i) = (ff')(i)^{\alpha_i} = f(i)^{\alpha_i} f'(i)^{\alpha_i} = (f^\alpha f'^\alpha)(i).$$

So α is an epimorphism from G to H.

Suppose that every α_i is an isomorphism from G_i to H_i. If $f^\alpha = 1$, then it follows for all $i \in I$ that

$$1 = f^\alpha(i) = f(i)^{\alpha_i}, \quad \text{and so} \quad f(i) = 1.$$

So α is an isomorphism from G to H. □

In the rest of this section, we mostly consider direct products of finitely many (although not necessarily finite) factors G_i $(i = 1, \dots, n)$ and write

$$G_1 \times \cdots \times G_n.$$

We often write the elements of $G_1 \times \cdots \times G_n$ as n-tuples

$$(g_1, \dots, g_n) \quad \text{with} \quad g_i = f(i).$$

The multiplication rule is then

$$(g_1, \dots, g_n)(g'_1, \dots, g'_n) = (g_1 g'_1, \dots, g_n g'_n).$$

Theorem 9.3. *Suppose the G_i $(i = 1, \dots, n)$ are subgroups of G that satisfy the following conditions:*
(1) $G_i \trianglelefteq G$.
(2) *Letting $G_1 \cdots G_j = \{g_1 \cdots g_j \mid g_i \in G_i\}$, we have*

$$G_1 \cdots G_{i-1} \cap G_i = 1 \text{ for } i = 2, \dots, n.$$

(3) $G = G_1 \cdots G_n$.
Then $G = G_1 \times \cdots \times G_n$.

Proof. Since the G_i are normal subgroups of G, the subgroup

$$G_1 \cdots G_j = \langle G_1, \dots, G_j \rangle$$

is also a normal subgroup of G. Since $G_1 \cdots G_{i-1} \cap G_i = 1$, the groups G_i and $G_1 \cdots G_{i-1}$ commute by Theorem 3.13. Thus, G_i and G_j commute for $i \neq j$.

Suppose

$$g_1 \cdots g_n = h_1 \cdots h_n$$

with g_i and h_i in G_i. Then it follows that

$$(g_1 h_1^{-1}) \cdots (g_n h_n^{-1}) = 1.$$

We set $f_i = g_i h_i^{-1}$. Assume for induction that $1 = f_{j+1} = \cdots = f_n$. Then we have

$$f_1 \cdots f_{j-1} = f_j^{-1} \in G_1 \cdots G_{j-1} \cap G_j = 1.$$

Then $f_j = 1$, and it follows that $g_i = h_i$ for all i. We have shown that the conditions of Definition 9.1 are satisfied, and we conclude that

$$G = G_1 \times \cdots \times G_n. \qquad \square$$

Theorem 9.4. *Assume that the G_i $(i = 1, \ldots, n)$ are normal subgroups of G with pairwise coprime orders m_i, and suppose $|G| = \prod_{i=1}^{n} m_i$. Then:*

a) $G = G_1 \times \cdots \times G_n$.
b) $\mathbf{Aut}(G) \cong \mathbf{Aut}(G_1) \times \cdots \times \mathbf{Aut}(G_n)$.

Proof. a) We will show that the conditions (1)–(3) of Theorem 9.3 are satisfied. Condition (1) follows directly from our assumptions.

Assume for induction that $|G_1 \cdots G_{i-1}| = m_1 \cdots m_{i-1}$. Since $m_1 \cdots m_{i-1}$ and m_i are coprime, we have

$$G_1 \cdots G_{i-1} \cap G_i = 1,$$

and therefore it follows by Lemma 2.12 a) that

$$|G_1 \cdots G_i| = m_1 \cdots m_i.$$

Condition (2) of Theorem 9.3 then follows immediately. Finally, we have

$$|G_1 \cdots G_n| = m_1 \cdots m_n = |G|,$$

so that $G = G_1 \cdots G_n$.

The claim now follows from Theorem 9.3.

b) Let α be an automorphism of G. By Theorem 4.9, the subgroups G_i are characteristic subgroups of G, so $G_i^\alpha = G_i$ for $i = 1, \ldots, n$.

Let α_i be the restriction of α to G_i. The map μ defined by

$$\alpha^\mu = (\alpha_1, \ldots, \alpha_n)$$

is clearly a monomorphism from $\mathbf{Aut}(G)$ to $\mathbf{Aut}(G_1) \times \cdots \times \mathbf{Aut}(G_n)$.

On the other hand, every element $(\alpha_1, \ldots, \alpha_n) \in \mathbf{Aut}(G_1) \times \cdots \times \mathbf{Aut}(G_n)$ induces an automorphism α of $G_1 \times \cdots \times G_n$ defined by

$$(g_1, \ldots, g_n)^\alpha = (g_1^{\alpha_1}, \ldots, g_n^{\alpha_n}).$$

Therefore, $\mathbf{Aut}(G) \cong \mathbf{Aut}(G_1) \times \cdots \times \mathbf{Aut}(G_n)$. □

Theorem 9.5. *If G is a finite group in which all Sylow subgroups are normal, then G is the direct product of its Sylow subgroups. We call such groups nilpotent.*

Proof. With these assumptions, the Sylow subgroups satisfy the conditions of Theorem 9.4. □

The following observation is often useful:

Lemma 9.6. *Let M and N be normal subgroups of G. Then $G/(M \cap N)$ is isomorphic to a subgroup of $G/M \times G/N$.*

Proof. The map ϵ from G to $G/M \times G/N$ defined by

$$g^\epsilon = (gM, gN)$$

is clearly a homomorphism from G to $G/M \times G/N$ with kernel $M \cap N$. □

Using Lemma 9.6, we immediately obtain the following striking result:

Theorem 9.7. *Let $\mathscr{E}$ be a group property that satisfies the following conditions:*
(1) *If G has property $\mathscr{E}$, then so do all subgroups of G.*
(2) *If G_1 and G_2 have property $\mathscr{E}$, then so does the direct product $G_1 \times G_2$.*
Then we have the following: If M and N are normal subgroups of G such that G/M and G/N have property $\mathscr{E}$, then $G/(M \cap N)$ also has property $\mathscr{E}$.

Some examples of group properties satisfying conditions (1) and (2) are commutativity, finiteness, solvability, nilpotency, $|G|$ only divisible by primes contained in some fixed set of primes.

As a useful consequence, we note:

Theorem 9.8. *If G is finite, then the intersection of all normal subgroups of G of p-power index is again a normal subgroup of p-power index.*

A trivial calculation yields:

Lemma 9.9. *We have $\mathbf{Z}(\bigtimes_{i \in I} G_i) = \bigtimes_{i \in I} \mathbf{Z}(G_i)$ and $\left(\bigtimes_{i \in I} G_i\right)' = \bigtimes_{i \in I} G_i'$. It follows immediately that if $G_1, \ldots, G_n$ are solvable, then so is $G_1 \times \cdots \times G_n$.*

The next two theorems describe some useful constructions that are derived from the direct product.

Theorem 9.10. *Let G_i $(i = 1, 2)$ be two groups with subgroups $U_i \leq \mathbf{Z}(G_i)$. Suppose μ is an isomorphism from U_1 to U_2. Then*

$$D = \{(u_1, (u_1^\mu)^{-1}) \mid u_1 \in U_1\}$$

is a normal subgroup of $G_1 \times G_2$. We define

$$G_1 \curlyvee G_2 = (G_1 \times G_2)/D$$

and claim:

a) *There exist monomorphisms α_i from G_i to $G_1 \curlyvee G_2$ $(i = 1, 2)$, the $G_i^{\alpha_i}$ are commuting normal subgroups of $G_1 \curlyvee G_2$, and*

$$G_1 \curlyvee G_2 = G_1^{\alpha_1} G_2^{\alpha_2}.$$

b) $G_1^{\alpha_1} \cap G_2^{\alpha_2} = U_1^{\alpha_1} = U_2^{\alpha_2}$, *and* $u_1^{\alpha_1} = u_1^{\mu\alpha_2}$ *for all* $u_1 \in U_1$.

We call $G_1 \curlyvee G_2$ the central product of G_1 and G_2 obtained by identifying the central subgroups U_1 and U_2 via the isomorphism μ.

Proof. First we show that D is a subgroup of $G_1 \times G_2$. We have

$$(u_1, (u_1^\mu)^{-1})(u_1', (u_1'^\mu)^{-1}) = (u_1 u_1', ((u_1 u_1')^\mu)^{-1}) \in D$$

and

$$(u_1, (u_1^\mu)^{-1})^{-1} = (u_1^{-1}, ((u_1^{-1})^\mu)^{-1}) \in D.$$

It follows from Lemma 9.9 that

$$D \leq \mathbf{Z}(G_1) \times \mathbf{Z}(G_2) = \mathbf{Z}(G_1 \times G_2),$$

and so $D \trianglelefteq G_1 \times G_2$.

For $g_1 \in G_1$, we define $\overline{g_1} = (g_1, 1)$ and $\overline{G_1} = \{(g_1, 1) \mid g_1 \in G_1\}$, and we define $\overline{g_2}$ and $\overline{G_2}$ in the corresponding way. Clearly, $\overline{G_i} \cong G_i$ for $i = 1, 2$.

We define the map α_i from G_i to $G_1 \curlyvee G_2$ by

$$g_i^{\alpha_i} = \overline{g_i} D \quad (g_i \in G_i).$$

Then we have

$$G_i^{\alpha_i} = \overline{G_i} D/D \cong \overline{G_i}/(\overline{G_i} \cap D).$$

If $(u_1, (u_1^\mu)^{-1}) \in \overline{G_1} \cap D$, it follows that $u_1^\mu = 1$, and then $u_1 = 1$ as well. We conclude that α_i is an isomorphism from G_i to $G_i^{\alpha_i}$.

The rest of the claims in a) are trivial.

Let

$$\overline{g_1} D = \overline{g_2} D \in G_1^{\alpha_1} \cap G_2^{\alpha_2} \quad \text{with } g_i \in G_i.$$

Then $\overline{g_1}\,\overline{g_2}^{-1} \in D$, and so $\overline{g_1}\,\overline{g_2}^{-1} = (u_1, (u_1^\mu)^{-1})$ for some $u_1 \in U_1$. It follows that

$$g_1 = u_1 \in U_1 \quad \text{and} \quad g_2 = u_1^\mu \in U_1^\mu = U_2.$$

Then we have

$$\overline{g_1}D = \overline{g_2}D \in U_1^{\alpha_1} \cap U_2^{\alpha_2}$$

and therefore

$$G_1^{\alpha_1} \cap G_2^{\alpha_2} \leq U_1^{\alpha_1} \cap U_2^{\alpha_2}.$$

Since

$$u_1^{\alpha_1} = \overline{u_1}D = \overline{u_1}(\overline{u_1}(\overline{u_1^\mu})^{-1})^{-1}D = \overline{u_1^\mu}D = u_1^{\mu\alpha_2},$$

we certainly have

$$U_1^{\alpha_1} = U_2^{\alpha_2} \leq G_1^{\alpha_1} \cap G_2^{\alpha_2}.$$

From this it follows that

$$G_1^{\alpha_1} \cap G_2^{\alpha_2} = U_1^{\alpha_1} = U_2^{\alpha_2}. \qquad \square$$

Next, we discuss a construction that is dual to Theorem 9.10:

Theorem 9.11. *Let G_i $(i = 1, 2)$ be groups, and let the μ_i be epimorphisms from G_i to H. Let N_i be the kernel of μ_i. Then*

$$G_1 \curlywedge G_2 = \{(g_1, g_2) \mid g_i \in G_i,\ g_1^{\mu_1} = g_2^{\mu_2}\}$$

is a subgroup of $G_1 \times G_2$. There are epimorphisms α_i from $G_1 \curlywedge G_2$ to G_i with

$$\ker \alpha_1 = K_1 = \{(1, n_2) \mid n_2 \in N_2\} \cong N_2,$$
$$\ker \alpha_2 = K_2 = \{(n_1, 1) \mid n_1 \in N_1\} \cong N_1.$$

Furthermore, there is an epimorphism μ from $G_1 \curlywedge G_2$ to H with

$$\ker \mu = K_1 \times K_2.$$

We call $G_1 \curlywedge G_2$ the subdirect product of G_1 and G_2 obtained by identifying along the factor group H.

Proof. It is clear that $G_1 \curlywedge G_2$ is a subgroup of $G_1 \times G_2$. Let π_i be the epimorphism from $G_1 \times G_2$ to G_i defined by

$$(g_1, g_2)^{\pi_i} = g_i.$$

The restriction of π_i to $G_1 \curlywedge G_2$ is clearly an epimorphism α_i from $G_1 \curlywedge G_2$ to G_i with

$$\ker \alpha_1 = \{(1, g_2) \mid g_2^{\mu_2} = 1^{\mu_1} = 1, \text{ so } g_2 \in N_2\}.$$

We define an epimorphism μ from $G_1 \curlywedge G_2$ to H by $(g_1, g_2)^\mu = g_1^{\mu_1} = g_2^{\mu_2}$. The kernel of μ is clearly $K_1 \times K_2$. □

We are now in a position to determine the structure of characteristically simple finite groups completely in terms of simple groups:

Theorem 9.12.

a) *If G is a finite, characteristically simple group, then G is a direct product of isomorphic simple groups.*

b) *If $G = G_1 \times \cdots \times G_r$ is the direct product of nonabelian simple groups G_i, then the only normal subgroups of G are the partial products $\bigtimes_{j \in J} G_j$ with $J \subseteq \{1, \ldots, r\}$.*
In particular, the G_i are the only minimal normal subgroups of G (that is, the minimal elements of the set $\{N \mid 1 < N \trianglelefteq G\}$), and every normal subgroup of G is a direct factor of G.

c) *The direct product of finitely many isomorphic simple groups is always characteristically simple.*

Proof. a) Let N be a minimal normal subgroup of G. We consider the images N^α of N under the automorphisms α of G. Let M be a largest possible subgroup of G that is a direct product of some of the N^α. Suppose for contradiction there exists some α with $N^\alpha \not\leq M$. Since N^α is also a minimal normal subgroup of G and $M \trianglelefteq G$, we have $M \cap N^\alpha = 1$. But then by Theorem 9.3, we have that

$$\langle M, N^\alpha \rangle = M \times N^\alpha$$

is a direct product of subgroups N^μ with $\mu \in \mathbf{Aut}(G)$ that properly contains M. Since M was chosen to be maximal, this is a contradiction. Therefore, $N^\alpha \leq M$ for all $\alpha \in \mathbf{Aut}(G)$. The group

$$M = \langle N^\alpha \mid \alpha \in \mathbf{Aut}(G) \rangle$$

is a nonidentity characteristic subgroup of G. Since G is characteristically simple, it follows that $G = M = N_1 \times \cdots \times N_r$ for some $N_i = N^{\alpha_i}$. Since every normal subgroup of N_i is a normal subgroup of G and the N_i are minimal normal subgroups of G, the N_i must be simple.

b) We show that if N is a normal subgroup of $G = G_1 \times \cdots \times G_r$ and there is an element $n = g_1 \cdots g_r \in N$ with $g_i \in G_i$ and $g_j \neq 1$, then $G_j \leq N$. From this the claim b) clearly follows.

Since $\mathbf{Z}(G_j) = 1$, we have $g_j \notin \mathbf{Z}(G_j)$. So there exists some $x \in G_j$ such that

$$[x, n] = [x, g_j] = c \neq 1.$$

Since $N \trianglelefteq G$, we have $c \in N \cap G_j$. The simplicity of G_j implies that

$$1 \neq N \cap G_j = G_j \trianglelefteq N,$$

which is what we wanted to show.

c) First let $G = G_1 \times \cdots \times G_r$ for some isomorphic, nonabelian simple groups G_i. Every nontrivial normal subgroup of G contains at least one of the G_i by b). If π is a permutation of $\{1, \ldots, r\}$, then the map α defined by

$$(g_1, \ldots, g_r)^\alpha = (g_{1\pi}, \ldots, g_{r\pi})$$

is an automorphism of $G = G_1 \times \cdots \times G_r$ with $G_i^\alpha = G_{i\pi}$. Thus, 1 and G are the only characteristic subgroups of G.

Now let $G = G_1 \times \cdots \times G_r$ for some cyclic subgroups $G_i = \langle g_i \rangle$ of prime order p. The map ϵ defined by

$$(g_1^{a_1}, \ldots, g_r^{a_r})^\epsilon = (a_1, \ldots, a_r)$$

is clearly an isomorphism from G to the additive group of the vector space V of dimension r over the prime field $\mathrm{GF}(p)$. The automorphisms of V are exactly the bijective, additive (so linear over $\mathrm{GF}(p)$) maps of V onto itself. If x and y are two nonzero vectors in V, then it is well known that there is always a bijective linear map from V to V that takes x to y. The automorphism group of V is therefore transitive on the nonzero vectors, and in particular V is characteristically simple. □

Theorem 9.13. *Let G be a nonidentity solvable group, and let N be a minimal normal subgroup of G. Then N is an abelian group of prime exponent (a so-called elementary abelian group). In particular, every simple solvable group has prime order.*

Proof. Since N is a minimal normal subgroup of G, the subgroup N is characteristically simple by Theorem 4.8 a). So we have by Theorem 9.12 that

$$N = K_1 \times \cdots \times K_r,$$

where the K_i are isomorphic simple groups. The K_i are solvable by Theorem 8.5. So $K_i' < K_i$, and then $K_i' = 1$ by the simplicity of the K_i. Now it follows from Theorem 7.4 that $|K_i|$ is prime. □

Theorem 9.14 (Rose [360]). *Let G be an arbitrary group. Then*

$$D = \{(g, g) \mid g \in G\}$$

is a subgroup of

$$G \times G = \{(g_1, g_2) \mid g_i \in G\}.$$

If U is a subgroup of G, we define

$$U_1 = \{(u, 1) \mid u \in U\} \quad \textit{and} \quad U_2 = \{(1, u) \mid u \in U\}.$$

Then:

a) *The map μ defined by $K^\mu = K_1 D = K_2 D$ is a bijective map from the set of normal subgroups of G onto the set of subgroups V of $G \times G$ with $D \leq V \leq G \times G$. We have*

$$(K \cap L)^\mu = K^\mu \cap L^\mu \quad \textit{and} \quad (KL)^\mu = K^\mu L^\mu.$$

b) *If G is simple, then D is a maximal subgroup of $G \times G$.*

c) *The normalizer of D in $G \times G$ is*

$$\mathbf{Z}(G)^\mu = \{(g_1, g_2) \mid g_i \in G,\ g_1 g_2^{-1} \in \mathbf{Z}(G)\}.$$

Proof. a) For every subgroup V of $G \times G$ containing D, we define the subgroup V^τ of G by

$$V^\tau = \{x \mid (x, 1) \in V\}.$$

If $(v_1, v_2) \in V$, then since $D \leq V$ we have

$$(v_1, v_2)(v_2, v_2)^{-1} = (v_1 v_2^{-1}, 1) \in V.$$

So $v_1 v_2^{-1} \in V^\tau$ and

$$(v_1, v_2) = (v_1 v_2^{-1}, 1)(v_2, v_2) \in (V^\tau)_1 D = V^{\tau\mu}.$$

Trivially, $V^{\tau\mu} \leq V$, so that $V^{\tau\mu} = V$.

For $v \in V^\tau$, we have

$$(v, 1) \in V.$$

Thus for $g \in G$,

$$(g, g)^{-1}(v, 1)(g, g) = (v^g, 1) \in V,$$

so that $v^g \in V^\tau$. Hence, V^τ is a normal subgroup of G.

Since $V^{\tau\mu} = V$, μ is a surjection from the set of normal subgroups of G onto the set of subgroups of $G \times G$ containing D. If $K \trianglelefteq G$, then

$$K^\mu = K_1 D = \{(g_1, g_2) \mid g_1 g_2^{-1} \in K\}.$$

It follows from $K^\mu = L^\mu$ that $K = L$. So μ is a bijection.

Furthermore, we have

$$(KL)^\mu = (KL)_1 D = K_1 L_1 D = K_1 D L_1 D = K^\mu L^\mu.$$

If V and W are subgroups of $G \times G$ containing D, then so is

$$V^{\tau\mu} \cap W^{\tau\mu} = V \cap W = (V \cap W)^{\tau\mu} = (V^\tau \cap W^\tau)^\mu.$$

Thus, μ is a lattice isomorphism.

b) Clearly, b) is a trivial consequence of a).

c) We have $\mathbf{N}_{G\times G}(D) = K_1 D$, where $K_1 = \mathbf{N}_{G\times G}(D) \cap G_1$. As a result, $[D, K_1] \leq D \cap G_1 = 1$. If $(x, 1) \in K_1$, then for all $g \in G$ we have

$$(x, 1)^{-1}(g, g)(x, 1) = (g^x, g) = (g, g).$$

Therefore, $K = \mathbf{Z}(G)$ and $\mathbf{N}_{G\times G}(D) = \mathbf{Z}(G)^\mu$. □

Exercises

42) Let $G_i = \langle g_i \rangle$ $(i = 1, 2)$ be cyclic groups of order p^{a_i} with $a_1 \geq a_2 > n$.

a) Let

$$U_i = \langle g_i^{p^{a_i - n}} \rangle$$

be the subgroup of order p^n in G_i. Let μ be the isomorphism from U_1 to U_2 defined by

$$(g_1^{p^{a_1 - n}})^\mu = g_2^{p^{a_2 - n}}.$$

Show that the direct product of G_1 and G_2 with subgroups U_i identified via the isomorphism μ is isomorphic to the direct product of cyclic groups of orders p^{a_1} and $p^{a_2 - n}$.

b) Let $Z = \langle z \rangle$ be the cyclic group of order p^n, and define epimorphisms μ_i from G_i to Z by $g_i^{\mu_i} = z$. Then $G_1 \leftthreetimes G_2$ is isomorphic to the direct product of cyclic groups of orders p^{a_1} and $p^{a_2 - n}$.

43) Let $G_i = \langle g_i \rangle$ $(i = 1, \ldots, 4)$ be cyclic of orders p^{a_i} with $a_1 \geq a_3 > a_2 \geq a_4$. We define $H_1 = G_1 \times G_2$ and $H_2 = G_3 \times G_4$. Let

$$U_1 = \langle g_1^{p^{a_1 - 1}} \rangle \times \langle g_2^{p^{a_2 - 1}} \rangle \quad \text{and} \quad U_2 = \langle g_3^{p^{a_3 - 1}} \rangle \times \langle g_4^{p^{a_4 - 1}} \rangle.$$

Furthermore, let μ and ϵ be isomorphisms from U_1 to U_2 defined by

$$(g_1^{p^{a_1 - 1}})^\mu = g_3^{p^{a_3 - 1}}, \quad (g_2^{p^{a_2 - 1}})^\mu = g_4^{p^{a_4 - 1}}$$

and

$$(g_1^{p^{a_1 - 1}})^\epsilon = g_4^{p^{a_4 - 1}}, \quad (g_2^{p^{a_2 - 1}})^\epsilon = g_3^{p^{a_3 - 1}}.$$

Show that the direct product of H_1 and H_2 with subgroups U_i identified via μ, resp. ϵ, has the following structure: It is a direct product of cyclic groups of orders p^{a_1}, p^{a_2}, $p^{a_3 - 1}$, $p^{a_4 - 1}$ via μ and orders p^{a_1}, $p^{a_2 - 1}$, p^{a_3}, $p^{a_4 - 1}$ via ϵ.

44) a) Let G_i $(i = 1, 2, 3)$ be groups with pairwise isomorphic central subgroups U_i. Interpret and prove

$$(G_1 \curlyvee G_2) \curlyvee G_3 = G_1 \curlyvee (G_2 \curlyvee G_3).$$

b) Let G_i $(i = 1, 2, 3)$ be groups with epimorphisms onto H. Interpret and prove

$$(G_1 \curlywedge G_2) \curlywedge G_3 = G_1 \curlywedge (G_2 \curlywedge G_3).$$

45) The group G is said to be semisimple if it is the direct product of nonabelian simple groups. Show that if G/M and G/N are semisimple, then $G/(M \cap N)$ is also semisimple. (Note that Theorem 9.7 cannot be used here!)

10 Groups with operators and modules

In §10–§13, all groups are allowed to be infinite unless explicitly stated otherwise.

Definition 10.1.

a) Let M be a group. We denote by $\mathrm{Hom}(M, M)$ the set of all endomorphisms of M (see Definition 3.6 d)).

b) If every element μ of a set Ω is an endomorphism $m \mapsto m^\mu$ of M, then we call the pair (M, Ω) an Ω-group or a group with operators. A subgroup $U \leq M$ is called Ω-admissible (or an Ω-subgroup) if $U^\mu \leq U$ for all $\mu \in \Omega$. Clearly, the intersection of admissible subgroups and the group generated by admissible subgroups are always admissible. If N is an Ω-admissible normal subgroup of M, then we give the factor group M/N the structure of an Ω-group upon defining $(mN)^\mu = m^\mu N$. This is well defined since whenever $m_1 N = m_2 N$, we have that $m_1^{-1} m_2 \in N$ and so $(m_1^{-1})^\mu m_2^\mu \in N$ as N is an Ω-subgroup.

c) A map α from an Ω-group M_1 to another Ω-group M_2 is called an Ω-homomorphism if α is a homomorphism from M_1 to M_2 and for all $\mu \in \Omega$ and $m \in M_1$ we have

$$m^{\mu\alpha} = m^{\alpha\mu}.$$

If α is one-to-one, then we call α an Ω-monomorphism. We define an Ω-isomorphism analogously.

d) If M is an Ω-group, then $\mathrm{Hom}_\Omega(M, M)$ is defined to be the collection of all Ω-homomorphisms from M to M. If $\alpha, \delta \in \mathrm{Hom}_\Omega(M, M)$, then $\alpha\delta \in \mathrm{Hom}_\Omega(M, M)$ as well since for all $\mu \in \Omega$ we have

$$(m^\mu)^{\alpha\delta} = (m^{\mu\alpha})^\delta = (m^\alpha)^{\mu\delta} = (m^\alpha)^{\delta\mu} = (m^{\alpha\delta})^\mu.$$

e) We say α is a normal endomorphism of M if α commutes with all inner automorphisms of M; that is, for all $m, x \in M$ we have

$$(x^\alpha)^{-1} m^\alpha x^\alpha = (m^x)^\alpha = (m^\alpha)^x = x^{-1} m^\alpha x.$$

f) If the M_i are Ω-groups for all $i \in I$, then the direct product

$$\bigtimes_{i\in I} M_i = \{f \mid f : I \to \bigcup_{i\in I} M_i,\ f(i) \in M_i,\ f(i) \neq 1 \text{ for finitely many } i\}$$

becomes an Ω-group upon defining

$$f^\mu(i) = f(i)^\mu$$

for all $i \in I$ and all $\mu \in \Omega$.

g) We always denote by 0 the epimorphism μ from M onto the identity subgroup 1 defined by $m^\mu = 1$ for all $m \in M$.

By directly inspecting the proofs, one sees that Theorems 3.8–3.12 still hold when the words "group" and "isomorphism" are replaced by "Ω-group" and "Ω-isomorphism".

Lemma 10.2. *If M is an Ω-group and $\alpha \in \mathrm{Hom}_\Omega(M, M)$, then*

$$M^\alpha = \{m^\alpha \mid m \in M\} = \mathrm{im}\,\alpha \quad \textit{and} \quad K_\alpha = \{m \mid m^\alpha = 1\} = \ker \alpha$$

are Ω-subgroups of M.

Proof. We have

$$m_1^\alpha m_2^\alpha = (m_1 m_2)^\alpha \in M^\alpha \quad \text{and} \quad (m^\alpha)^{-1} = (m^{-1})^\alpha \in M^\alpha.$$

So M^α is a subgroup of M. In addition, for all $\mu \in \Omega$,

$$(m^\alpha)^\mu = (m^\mu)^\alpha \in M^\alpha,$$

and therefore M^α is an Ω-subgroup of M.

If $m_1^\alpha = m_2^\alpha = 1$, then $(m_1 m_2)^\alpha = 1 = (m_1^{-1})^\alpha$ and $(m_1^\mu)^\alpha = (m_1^\alpha)^\mu = 1$. Thus, K_α is also an Ω-subgroup of M. □

Lemma 10.3. *If α is a normal endomorphism of M, then*

$$m^\alpha m^{-1} \in \mathbf{C}_M(M^\alpha)$$

for all $m \in M$.

Proof. For all $m, x \in M$, we have

$$m^{-1} x^\alpha m = (x^\alpha)^m = (x^m)^\alpha = (m^\alpha)^{-1} x^\alpha m^\alpha.$$

So $m^\alpha m^{-1}$ commutes with all x^α. □

Definition 10.4. We say that the Ω-group M satisfies the maximal (resp. minimal) condition for Ω-subgroups if every nonempty set of Ω-subgroups of M contains a maximal (resp. minimal) element. We say $M \neq 1$ is irreducible if M and 1 are the only Ω-admissible normal subgroups of M. We say $M \neq 1$ is a directly indecomposable Ω-group whenever it cannot be written as a direct product of two nontrivial Ω-subgroups; that is, if $M = M_1 \times M_2$ for some Ω-subgroups M_1 and M_2, then $M_1 = 1$ or $M_2 = 1$.

The following theorems of Schur and Fitting are fundamental.

Lemma 10.5 (Schur's lemma). *Let M be an irreducible Ω-group, and let α be a normal Ω-endomorphism of M. If $\alpha \neq 0$, then α is an automorphism of M and $\alpha^{-1} \in \mathrm{Hom}_\Omega(M, M)$.*

Proof. By Lemma 10.2, M^α is an Ω-subgroup of M. Since α is a normal endomorphism of M, we have for all $m, x \in M$ that

$$(m^\alpha)^x = (m^x)^\alpha \in M^\alpha.$$

So M^α is a normal subgroup of M. Since $\alpha \neq 0$, we have $M^\alpha \neq 1$. Then the irreducibility of M forces $M^\alpha = M$, and so α is a map from M onto M. By Lemma 10.2, $\ker\alpha$ is an Ω-admissible normal subgroup of M, and since $\alpha \neq 0$ we certainly have $\ker\alpha \neq M$. So $\ker\alpha = 1$. Hence, α is a bijective map from M to M, so α^{-1} exists. To finish, we show that α^{-1} lies in $\mathrm{Hom}_\Omega(M, M)$:

Since $x^{\alpha\mu} = x^{\mu\alpha}$ for all $x \in M$ and $\mu \in \Omega$, it follows that

$$x^{\alpha\mu\alpha^{-1}} = x^\mu.$$

Setting $x^\alpha = y$, we have

$$y^{\mu\alpha^{-1}} = y^{\alpha^{-1}\mu}$$

for all $y \in M^\alpha = M$. □

Theorem 10.6 (Fitting's theorem). *Let M be an Ω-group satisfying the maximal and minimal conditions for Ω-subgroups. Then the following hold:*

a) *If $\alpha \in \mathrm{Hom}_\Omega(M, M)$ is a normal Ω-endomorphism, then for all sufficiently large natural numbers k we have*

$$M = M^{\alpha^k} \times \ker\alpha^k.$$

b) *If $\ker\alpha = 1$ or $M^\alpha = M$, then α is an automorphism of M. (That is, if M satisfies the maximal and minimal conditions, then the normal endomorphisms behave in many respects like linear maps on finite-dimensional vector spaces.)*

Proof. a) The set of Ω-subgroups M^{α^i} $(i = 0, 1, \dots)$ has a minimal element, say M^{α^n}, by the minimal condition. So for all $k \geq 0$ we have

$$M^{\alpha^{k+n}} = (M^{\alpha^k})^{\alpha^n} \leq M^{\alpha^n}$$

and therefore

$$M^{\alpha^n} = M^{\alpha^{n+1}} = \cdots .$$

Likewise, the set $\{\ker \alpha^i \mid i = 1, 2, \dots\}$ has a maximal element, say $\ker \alpha^m$. Since

$$\ker \alpha^m \leq \ker \alpha^{m+1} \leq \cdots$$

we have

$$\ker \alpha^m = \ker \alpha^{m+1} = \cdots .$$

Let $k \geq \max\{m, n\}$. We claim that $M = M^{\alpha^k} \times \ker \alpha^k$ for this k: If $g \in M^{\alpha^k} \cap \ker \alpha^k$, then $g^{\alpha^k} = 1$ and $g = x^{\alpha^k}$ for some $x \in M$. Then

$$1 = g^{\alpha^k} = x^{\alpha^{2k}}, \text{ so } x \in \ker \alpha^{2k} = \ker \alpha^k .$$

It then follows that $g = x^{\alpha^k} = 1$. We conclude that $M^{\alpha^k} \cap \ker \alpha^k = 1$.

Since $g^{\alpha^k} \in M^{\alpha^k} = M^{\alpha^{2k}}$, every g^{α^k} can be written in the form $g^{\alpha^k} = h^{\alpha^{2k}}$ for some $h \in M$. Then we have

$$g = (gh^{-\alpha^k})h^{\alpha^k},$$

where $h^{\alpha^k} \in M^{\alpha^k}$ and $gh^{-\alpha^k} \in \ker \alpha^k$ since $(gh^{-\alpha^k})^{\alpha^k} = 1$. This proves that

$$M = M^{\alpha^k} \ker \alpha^k .$$

Since α is a normal endomorphism of M, the subgroups M^{α^k} and $\ker \alpha^k$ are normal subgroups of M. It follows by Theorem 9.3 that $M = M^{\alpha^k} \times \ker \alpha^k$.

b) The map α is an automorphism of M if and only if $M^\alpha = M$ and $\ker \alpha = 1$. We must show that each of these conditions implies the other.

If $\ker \alpha = 1$, then $\ker \alpha^k = 1$ for all k, and therefore $M = M^{\alpha^k}$ for suitable k by a). Since

$$M \geq M^\alpha \geq M^{\alpha^k},$$

we have $M = M^\alpha$.

If $M = M^\alpha$, then $M = M^{\alpha^k}$ for all k. Using a), we conclude that $\ker \alpha^k = 1$ for sufficiently large k. Since $\ker \alpha \leq \ker \alpha^k$, we must have $\ker \alpha = 1$. □

Lemma 10.7 (Fitting's lemma). *Let M be a directly indecomposable Ω-group satisfying the maximal and minimal conditions for Ω-subgroups. If α is a normal Ω-endomorphism in $\mathrm{Hom}_\Omega(M, M)$, then either $\alpha^k = 0$ for sufficiently large k or α is an automorphism of M.*

Proof. By Theorem 10.6, we have for suitable k that

$$M = M^{\alpha^k} \times \ker \alpha^k.$$

Since M is directly indecomposable, it follows that either $\ker \alpha^k = M$, so $\alpha^k = 0$, or $M = M^{\alpha^k}$. Since

$$M \geq M^{\alpha} \geq M^{\alpha^k},$$

it follows in the second case that $M = M^{\alpha}$, and then α is an automorphism of M by Theorem 10.6. □

Definition 10.8. Two elements $\alpha, \delta \in \mathrm{Hom}(M, M)$ are called additive if the map ϵ from M to M defined by

$$m^{\epsilon} = m^{\alpha} m^{\delta}$$

for all $m \in M$ lies in $\mathrm{Hom}(M, M)$. We then write $\epsilon = \alpha + \delta$.

The elements α_i $(i = 1, \ldots, n)$ in $\mathrm{Hom}(M, M)$ are said to be additive if α_i and α_j are additive whenever $i \neq j$. It then follows immediately from Lemma 10.9 that $\alpha = \sum_{i=1}^{n} \alpha_i$ defined by

$$m^{\alpha} = \prod_{i=1}^{n} m^{\alpha_i}$$

is an endomorphism of M.

Lemma 10.9. *The elements $\alpha, \delta \in \mathrm{Hom}(M, M)$ are additive if and only if M^{α} and M^{δ} commute. For additive α and δ, we have $\alpha + \delta = \delta + \alpha$. If $\alpha, \delta \in \mathrm{Hom}_{\Omega}(M, M)$ are additive, then $\alpha + \delta \in \mathrm{Hom}_{\Omega}(M, M)$. The distributive laws $(\alpha + \beta)\gamma = \alpha\gamma + \beta\gamma$ and $\gamma(\alpha + \beta) = \gamma\alpha + \gamma\beta$ always hold.*

Proof. If α and δ are additive, then we have for all $m, n \in M$ that

$$\begin{aligned}(mn)^{\alpha+\delta} &= m^{\alpha+\delta} n^{\alpha+\delta} = m^{\alpha} m^{\delta} n^{\alpha} n^{\delta} \\ &= (mn)^{\alpha} (mn)^{\delta} = m^{\alpha} n^{\alpha} m^{\delta} n^{\delta}.\end{aligned}$$

It follows that $m^{\delta} n^{\alpha} = n^{\alpha} m^{\delta}$. The reverse implication follows similarly. Since

$$(m^{\mu})^{\alpha+\delta} = m^{\mu\alpha} m^{\mu\delta} = m^{\alpha\mu} m^{\delta\mu} = (m^{\alpha+\delta})^{\mu}$$

for all $\mu \in \Omega$, it follows that $\alpha + \delta \in \mathrm{Hom}_{\Omega}(M, M)$ for all $\alpha, \delta \in \mathrm{Hom}_{\Omega}(M, M)$ additive. A trivial calculation proves the distributive laws. □

We will require the following result in our study of direct product decompositions in §12.

Lemma 10.10. *Let M be a directly indecomposable Ω-group satisfying the maximal and minimal conditions for Ω-subgroups. Let ϵ and μ be additive normal Ω-endomorphisms in $\mathrm{Hom}_\Omega(M, M)$, and assume $\alpha = \epsilon + \mu$ is an automorphism of M. Then either ϵ or μ is an automorphism of M.*

Proof. We define $\epsilon' = \epsilon\alpha^{-1}$ and $\mu' = \mu\alpha^{-1}$. Since

$$[M^{\epsilon'}, M^{\mu'}] = [M^{\epsilon}, M^{\mu}]^{\alpha^{-1}} = 1,$$

we have that $M^{\epsilon'}$ and $M^{\mu'}$ commute. Hence, ϵ' and μ' are additive by Lemma 10.9. Now we have $\epsilon' + \mu' = 1$, and therefore

$$\epsilon'\mu' = (1 - \mu')\mu' = \mu'(1 - \mu') = \mu'\epsilon',$$

where $1 - \mu'$ is defined by

$$m^{1-\mu'} = m(m^{-1})^{\mu'} = m(m^{\mu'})^{-1}.$$

Assume for contradiction that neither ϵ' nor μ' is an automorphism of M. By Theorem 10.7, we have $\epsilon'^k = \mu'^k = 0$ for sufficiently large k. It follows that

$$1 = (\epsilon' + \mu')^{2k} = \sum_{j=0}^{2k} \binom{2k}{j} \epsilon'^{2k-j}\mu'^{j} = 0,$$

which contradicts that $M \neq 1$. So either ϵ' or μ' is an automorphism, which means either $\epsilon = \epsilon'\alpha$ or $\mu = \mu'\alpha$ is an automorphism. □

Definition 10.11. Let $\mathfrak{m}$ be an abelian group written additively. Let R be a ring with unity 1, and let $\mathfrak{m}$ be an R-group in the sense of Definition 10.1. That is, we have:
(1) $(m_1 + m_2)r = m_1 r + m_2 r$ for all $r \in R$ and all $m_i \in \mathfrak{m}$.
In addition, we demand that
(2) $m1 = m$ for all $m \in \mathfrak{m}$.
(3) $(mr)r' = m(rr')$ for all $m \in \mathfrak{m}$ and all $r, r' \in R$.
(4) $m(r + r') = mr + mr'$ for all $m \in \mathfrak{m}$ and all $r, r' \in R$.

If conditions (1)–(4) are satisfied, then $\mathfrak{m}$ is called a right R-module. We define left R-modules in the analogous way. Given a collection $\mathfrak{m}_i$ $(i \in I)$ of right R-modules, $\bigoplus_{i\in I} \mathfrak{m}_i$ becomes a right R-module upon defining $(fr)(i) = f(i)r$ for $f \in \bigoplus_{i\in I} \mathfrak{m}_i$.

Since $\mathfrak{m}$ is abelian, all endomorphisms of $\mathfrak{m}$ are normal and additive by Theorem 10.9. Also note that

$$m(\alpha + \epsilon)\mu = (m\alpha + m\epsilon)\mu = m\alpha\mu + m\epsilon\mu = m(\alpha\mu + \epsilon\mu)$$

and

$$m\alpha(\epsilon + \mu) = m\alpha\epsilon + m\alpha\mu = m(\alpha\epsilon + \alpha\mu)$$

for all $\alpha, \epsilon, \mu \in \mathrm{Hom}_R(\mathfrak{m}, \mathfrak{m})$. Therefore, $\mathrm{Hom}_R(\mathfrak{m}, \mathfrak{m})$ is a ring.

Example 10.12. Let R be a ring. The additive group of R admits the structure of a right R-module, which we denote by R_r, via the map $x \mapsto xr$ $(r \in R)$.

Schur's Lemma 10.5 specializes to the following in the case of modules.

Lemma 10.13 (Schur's lemma for modules). *If* $\mathfrak{m}$ *is an irreducible right* R*-module, then* $\mathrm{Hom}_R(\mathfrak{m}, \mathfrak{m})$ *is a division ring.*

Definition 10.14. Let R be a ring, and let $\mathfrak{m}$ be a right R-module. A set $\{m_i \mid i \in I\}$ is called a generating set for $\mathfrak{m}$ if:
(1) $m_i \in \mathfrak{m}$ for all $i \in I$.
(2) Every $m \in \mathfrak{m}$ can be written as $m = \sum_{i \in I} m_i r_i$ for some $r_i \in R$, where only finitely many of the r_i are nonzero.

Definition 10.15. Let R be a ring. A set $\mathfrak{f}$ is called a free right R-module in the free generators f_i $(i \in I)$ if the following conditions hold:
(1) $\mathfrak{f}$ is a right R-module and $\{f_i \mid i \in I\}$ is a generating set for $\mathfrak{f}$ as a right R-module.
(2) If $\mathfrak{m}$ is generated as a right R-module by elements m_i with $i \in I$, then there exists an R-epimorphism μ from $\mathfrak{f}$ to $\mathfrak{m}$ with $f_i\mu = m_i$.

The question of the existence of free right R-modules is answered by the following theorem.

Theorem 10.16.
a) *The direct sum of* $|I|$ *right* R*-modules isomorphic to* R_r *is a free right* R*-module in the free generators* f_i*, where*

$$f_i(i) = 1 \quad \textit{and} \quad f_i(j) = 0 \quad \textit{for} \quad i \neq j \ (i, j \in I).$$

b) *All free right* R*-modules in* $|I|$ *free generators are isomorphic.*

Proof. a) Let $\mathfrak{f}$ be the direct sum of $|I|$ right R-modules isomorphic to R_r constructed according to Theorem 9.2 and Definition 10.11. For every $g \in \mathfrak{f}$, we have

$$g = \sum_{i \in I} f_i g(i) \quad (g(i) \in R).$$

Hence, $\mathfrak{f}$ is generated by the f_i as a right R-module. If

$$g = \sum_{i \in I} f_i r_i \quad \text{with} \quad r_i \in R,$$

it follows that

$$g(j) = \sum_{i \in I} f_i(j) r_i = r_j.$$

Therefore, the expression $g = \sum_{i \in I} f_i r_i$ with $r_i \in R$ is unique.

Let $\mathfrak{m}$ be a right R-module with generating set $\{m_i \mid i \in I\}$. Then the map μ defined by

$$\left(\sum_{i\in I} f_i r_i\right)\mu = \sum_{i\in I} m_i r_i$$

is an R-epimorphism from $\mathfrak{f}$ to $\mathfrak{m}$ of the desired form.

b) Assume $\mathfrak{f}$ and $\mathfrak{f}'$ are free right R-modules in the free generators f_i and f_i' with $i \in I$. By Definition 10.15, there exist R-epimorphisms μ and ϵ defined by

$$f_i\mu = f_i' \quad \text{and} \quad f_i'\epsilon = f_i$$

for all $i \in I$. It follows that

$$f_i\mu\epsilon = f_i \quad \text{and} \quad f_i'\epsilon\mu = f_i'$$

for all $i \in I$. Since the f_i, resp. f_i', generate $\mathfrak{f}$, resp. $\mathfrak{f}'$, it follows that $\mu\epsilon = 1$ on $\mathfrak{f}$ and $\epsilon\mu = 1$ on $\mathfrak{f}'$. Therefore, $\mathfrak{f}$ and $\mathfrak{f}'$ are isomorphic right R-modules. □

Exercises

46) Let M and N be Ω-groups with $N \lhd M$.
   a) If the maximal or minimal condition holds in M, then it holds in M/N and N as well.
   b) If the maximal or minimal condition holds in N and M/N, then it holds in M as well.

47) Let $\mathfrak{m}$ be a finitely-generated right R-module. If the maximal or minimal condition holds in R for right ideals, then it holds in $\mathfrak{m}$ for submodules.

48) The right R-module $\mathfrak{m}$ satisfies the maximal condition for submodules if and only if every submodule of $\mathfrak{m}$ is finitely generated.

49) If α is a normal endomorphism of G, then the map $-1 + \alpha$ defined by

$$g^{-1+\alpha} = g^{-1}g^{\alpha}$$

for all $g \in G$ is an endomorphism of G.

11 The Jordan–Hölder theorem

Definition 11.1. Let M be an Ω-group. A finite chain of Ω-subgroups $M_i \leq M$ with

$$M = M_0 \trianglerighteq M_1 \trianglerighteq \cdots \trianglerighteq M_r = 1$$

is called an Ω-series of M. (Note that the M_i are not necessarily normal in M!) An Ω-series of M

$$M = M_0 \rhd M_1 \rhd \cdots \rhd M_r = 1$$

is called an Ω-composition series of M if the factor groups M_i/M_{i+1} are irreducible Ω-groups ($i = 0, 1, \ldots, r-1$).

Lemma 11.2. *If an Ω-group $M \neq 1$ satisfies the maximal and minimal conditions for Ω-subgroups, then M has at least one Ω-composition series.*

Proof. By the maximal condition, there exists in every Ω-subgroup $N \neq 1$ of M an Ω-subgroup N_1 with $N_1 \lhd N$ and N/N_1 irreducible; indeed, we may take N_1 to be a maximal element of the nonempty set

$$\{K \mid K \lhd N,\ K \text{ an } \Omega\text{-subgroup of } M\}.$$

Let $\mathscr{S}$ be the set of all Ω-subgroups N of M for which there exists a finite chain

$$M = M_0 \rhd M_1 \rhd \cdots \rhd M_s = N$$

of Ω-subgroups M_i such that M_i/M_{i+1} is irreducible. The set $\mathscr{S}$ contains M itself, so it is nonempty. By the minimal condition, there exists a minimal element S in $\mathscr{S}$. If $S \neq 1$, then there exists an Ω-subgroup S_1 with $S_1 \lhd S$ and S/S_1 irreducible. But then $S_1 \in \mathscr{S}$, contradicting the minimality of S in $\mathscr{S}$. Therefore, $S = 1$ and M has an Ω-composition series. □

Lemma 11.3. *Let $U_1 \trianglelefteq U \leq M$ and $V_1 \trianglelefteq V \leq M$. Then*

$$U_1(U \cap V)/U_1(U \cap V_1) \cong V_1(U \cap V)/V_1(U_1 \cap V).$$

If all the groups are Ω-admissible, then this isomorphism is in fact an Ω-isomorphism.

Proof. Since $U_1 \trianglelefteq U$, we have $U_1 \cap V \trianglelefteq U \cap V$. By Theorem 3.12,

$$U_1(U \cap V)/U_1 \cong (U \cap V)/(U \cap V \cap U_1) = (U \cap V)/(U_1 \cap V).$$

In the same way, $U \cap V_1 \trianglelefteq U \cap V$ and therefore

$$(U_1 \cap V)(U \cap V_1) \trianglelefteq U \cap V.$$

Under the isomorphism from $U_1(U \cap V)/U_1$ to $(U \cap V)/(U_1 \cap V)$, the factor $U_1(U \cap V_1)/U_1$ is mapped onto $(U_1 \cap V)(U \cap V_1)/(U_1 \cap V)$. Therefore, we have by Theorem 3.10 c) that

$$\begin{aligned}(U\cap V)\Big/(U_1\cap V)(U\cap V_1) &\cong (U\cap V)/(U_1\cap V)\Big/(U_1\cap V)(U\cap V_1)/(U_1\cap V)\\ &\cong U_1(U\cap V)/U_1\Big/U_1(U\cap V_1)/U_1\\ &\cong U_1(U\cap V)\Big/U_1(U\cap V_1).\end{aligned}$$

In the same way, we obtain

$$(U\cap V)\Big/(U_1\cap V)(U\cap V_1)\cong V_1(U\cap V)\Big/V_1(U_1\cap V),$$

and we are done. □

Theorem 11.4 (Refinement theorem of Schreier and Zassenhaus). *Let*

$$M=M_0\trianglerighteq M_1\trianglerighteq\cdots\trianglerighteq M_r=1$$

and

$$M=N_0\trianglerighteq N_1\trianglerighteq\cdots\trianglerighteq N_s=1$$

be two Ω*-series of* M*. Then the subgroups*

$$M_{i,j}=M_i(M_{i-1}\cap N_j)\quad\text{and}\quad N_{i,j}=N_j(N_{j-1}\cap M_i)$$

give rise to Ω*-series of* M *with*

$$M_{i-1}=M_{i,0}\trianglerighteq M_{i,1}\trianglerighteq\cdots\trianglerighteq M_{i,s}=M_i$$

and

$$N_{j-1}=N_{0,j}\trianglerighteq N_{1,j}\trianglerighteq\cdots\trianglerighteq N_{r,j}=N_j.$$

Furthermore, we have

$$M_{i,j-1}/M_{i,j}\cong N_{i-1,j}/N_{i,j}.$$

Proof. It is clear that the $M_{i,j}$ and the $N_{i,j}$ are Ω-admissible subgroups of M. It follows from $N_{j+1}\trianglelefteq N_j$ that

$$N_{j+1}\cap M_{i-1}\trianglelefteq N_j\cap M_{i-1}$$

and then

$$M_{i,j}=M_i(M_{i-1}\cap N_j)\trianglerighteq M_i(M_{i-1}\cap N_{j+1})=M_{i,j+1}.$$

So the given groups do form Ω-series of M. By Lemma 11.3, we have

$$M_{i,j-1}/M_{i,j} = M_i(M_{i-1} \cap N_{j-1})/M_i(M_{i-1} \cap N_j)$$
$$\cong N_j(M_{i-1} \cap N_{j-1})/N_j(M_i \cap N_{j-1}) = N_{i-1,j}/N_{i,j}. \qquad \square$$

Theorem 11.5 (Jordan–Hölder theorem). *Let M be an Ω-group satisfying the maximal and minimal conditions for Ω-subgroups. Let*

$$M = M_0 \rhd M_1 \rhd \cdots \rhd M_r = 1$$

and

$$M = N_0 \rhd N_1 \rhd \cdots \rhd N_s = 1$$

be two Ω-composition series of M. Then $r = s$, and there is some labeling such that the factor groups M_i/M_{i+1} and N_i/N_{i+1} are isomorphic Ω-groups. We call the irreducible factor groups M_i/M_{i+1} the Ω-composition factors of M.

Proof. We perform the construction of Theorem 11.4. However, a composition series cannot be further refined, so we obtain

$$M_{i-1}/M_i = M_{i,j-1}/M_{i,j} \cong N_{i-1,j}/N_{i,j} = N_{j-1}/N_j.$$

Then since

$$M_{i-1} = M_{i,j-1} = M_i(M_{i-1} \cap N_{j-1})$$

and

$$M_i = M_{i,j} = M_i(M_{i-1} \cap N_j),$$

the labels i and $j = j(i)$ can be put in one-to-one correspondence by specifying that

$$M_{i-1} \cap N_j \leq M_i \quad \text{but} \quad M_{i-1} \cap N_{j-1} \not\leq M_i$$

and

$$N_{j-1} \cap M_i \leq N_j \quad \text{but} \quad N_{j-1} \cap M_{i-1} \not\leq N_j. \qquad \square$$

Next, we identify some important special cases:

Definition 11.6.

a) If Ω is the empty set, then the associated Ω-composition series of M is simply called a composition series of M.
b) If Ω is the set of all inner automorphisms of M, then an Ω-composition series of M is called a chief series of M. We call the Ω-composition factors chief factors.
c) If Ω is the set of all automorphisms of M, then an Ω-composition series of M is called a characteristic series of M.

The following theorem is a special case of Theorem 11.5.

Theorem 11.7. *Let M be a finite group. Then all composition series (chief series, characteristic series) of M have the same length and have the same composition factors up to reordering.*

For modules, the conclusion is as follows.

Theorem 11.8. *Let $\mathfrak{m}$ be a right R-module satisfying the maximal and minimal conditions for submodules. Then all R-composition series*

$$\mathfrak{m} = \mathfrak{m}_0 > \mathfrak{m}_1 > \cdots > \mathfrak{m}_r = 0$$

have the same length, and the irreducible factor modules $\mathfrak{m}_i/\mathfrak{m}_{i+1}$ are uniquely determined by $\mathfrak{m}$, up to ordering. We call the right R-modules $\mathfrak{m}_i/\mathfrak{m}_{i+1}$ the irreducible constituents of $\mathfrak{m}$.

Now we are able to give a new characterization of finite solvable groups.

Theorem 11.9. *Let G be a finite group. Then G is solvable if and only if all composition factors of G are cyclic (of prime order).*

Proof. a) Let

$$G = G_0 \rhd G_1 \rhd \cdots \rhd G_r = 1$$

be a composition series of G with cyclic factor groups G_i/G_{i+1}. By induction, we may assume that the i-th commutator subgroup $G^{(i)}$ of G lies in G_i. Since G_i/G_{i+1} is abelian, it follows by Theorem 8.2 that

$$G^{(i+1)} \leq G_i' \leq G_{i+1}.$$

We obtain finally that $G^{(r)} \leq G_r = 1$, and therefore G is solvable.

b) Let G be solvable, and let

$$G = G_0 \rhd G_1 \rhd \cdots \rhd G_r = 1$$

be a composition series of G. The factor groups G_i/G_{i+1} are therefore simple groups. Since G is solvable, the G_i/G_{i+1} are solvable by Theorem 8.5. We conclude by Theorem 9.13 that G_i/G_{i+1} has prime order. □

12 Direct decompositions

In this section, we always assume that Ω contains all inner automorphisms of M. All Ω-subgroups of M are then normal in M, and all Ω-endomorphisms are normal.

Lemma 12.1. *Suppose $M = M_1 \times \cdots \times M_r$ is a direct decomposition of the Ω-group M into Ω-subgroups M_i. We define the projections π_i associated to this decomposition by*

$$(m_1 \cdots m_r)^{\pi_i} = m_i \quad \text{for } m_j \in M_j.$$

Then the π_i are additive and normal endomorphisms in $\mathrm{Hom}_\Omega(M, M)$. *We have*

$$1 = \pi_1 + \cdots + \pi_r,$$

$\pi_i \pi_j = 0$ *for* $i \neq j$ *and* $\pi_i^2 = \pi_i$.

Proof. If $m = m_1 \cdots m_r$ and $n = n_1 \cdots n_r$ with $m_i, n_i \in M_i$, then

$$mn = m_1 n_1 \cdots m_r n_r,$$

and therefore

$$(mn)^{\pi_i} = m_i n_i = m^{\pi_i} n^{\pi_i}.$$

Furthermore, for $\mu \in \Omega$, $M_i^\mu \leq M_i$ implies that

$$(m^\mu)^{\pi_i} = (m_1^\mu \cdots m_r^\mu)^{\pi_i} = m_i^\mu = m^{\pi_i \mu}.$$

So the π_i are elements of $\mathrm{Hom}_\Omega(M, M)$. Since $M^{\pi_i} = M_i$ and the elements of M_i and M_j commute for $i \neq j$, it follows from Lemma 10.9 that the π_i are additive. The equalities

$$1 = \pi_1 + \cdots + \pi_r, \quad \pi_i \pi_j = 0 \quad \text{for } i \neq j \quad \text{and} \quad \pi_i^2 = \pi_i$$

are clear. □

Theorem 12.2. *Let $M \neq 1$ be an Ω-group satisfying the maximal and minimal conditions for Ω-subgroups. Then M admits a direct decomposition*

$$M = M_1 \times \cdots \times M_r$$

into finitely many directly indecomposable Ω-groups M_i.

Proof. a) First we show that every nonidentity Ω-subgroup N of M can be directly decomposed into $N = T \times T^*$, where $T > 1$ is a directly indecomposable Ω-group and T^* is an Ω-group. Let

$$\mathscr{S} = \big\{A \mid 1 < A,\ N = A \times A^*,\ A \text{ and } A^* \text{ are } \Omega\text{-subgroups of } M\big\}.$$

The set $\mathscr{S}$ contains N, so is nonempty. By the minimal condition, $\mathscr{S}$ has minimal elements, and these are clearly directly indecomposable direct factors of N.

b) Now let

$$\mathscr{T} = \big\{B \mid B = M_1 \times \cdots \times M_t,\ M = B \times B^*;\ M_i, B \text{ and } B^* \ \Omega\text{-groups}, \\ M_i \text{ directly indecomposable}\big\}.$$

By a), the set $\mathscr{T}$ is nonempty. By the maximal condition, there exists a maximal element $T \in \mathscr{T}$. If $T < M$, then we have $M = T \times T^*$ with $T^* > 1$. By a), there is a direct decomposition $T^* = A \times A^*$, where $A > 1$ is a directly indecomposable Ω-group and A^* is an Ω-group. It follows that

$$T < T \times A \in \mathscr{T},$$

which contradicts the choice of T. Therefore

$$T = M = M_1 \times \cdots \times M_r,$$

and we have the desired decomposition of M. □

Theorem 12.3 (Krull–Remak–Schmidt theorem). *Let $M \neq 1$ be an Ω-group satisfying the maximal and minimal conditions for Ω-subgroups. Let*

$$M = M_1 \times \cdots \times M_r = N_1 \times \cdots \times N_s$$

be two direct decompositions of M into directly indecomposable Ω-subgroups M_i and N_j. Then $r = s$. For some labeling of the N_i, we have

$$M = N_1 \times \cdots \times N_{i-1} \times M_i \times \cdots \times M_r$$

for all $i = 1, \ldots, r$. Furthermore, there exists a normal Ω-automorphism α of M such that $M_i^\alpha = N_i$ for $i = 1, \ldots, r$.

Proof. Assume for induction that we have found a normal Ω-automorphism α_i of M such that

$$M_j^{\alpha_i} = N_j \text{ for } j \leq i-1 \quad \text{and} \quad M_j^{\alpha_i} = M_j \text{ for } j \geq i.$$

(If $i = 1$, this is trivially true.)

Then we have

$$M = M^{\alpha_i} = M_1^{\alpha_i} \times \cdots \times M_r^{\alpha_i} = N_1 \times \cdots \times N_{i-1} \times M_i \times \cdots \times M_r.$$

Let $\pi_1, \ldots, \pi_r$ be the projections associated to the direct decomposition

$$M = N_1 \times \cdots \times N_{i-1} \times M_i \times \cdots \times M_r$$

constructed in Lemma 12.1, and let $\mu_1, \ldots, \mu_s$ be the projections associated to

$$M = N_1 \times \cdots \times N_s.$$

Then we have

$$1 = \sum_{j=1}^{s} \mu_j \quad \text{and} \quad \pi_i = \left(\sum_{j=1}^{s} \mu_j \right) \pi_i = \sum_{j=1}^{s} \mu_j \pi_i.$$

Since

$$M^{\mu_j} = N_j,$$

we have $\mu_j = \mu_j\pi_j$ for $j \leq i-1$, and then $\mu_j\pi_i = \mu_j\pi_j\pi_i = 0$. Hence, there remains only

$$\pi_i = \sum_{j=i}^{s} \mu_j\pi_i.$$

The endomorphisms $\mu_j\pi_i$ are additive, and so

$$[m^{\mu_j\pi_i}, n^{\mu_k\pi_i}] = [m^{\mu_j}, n^{\mu_k}]^{\pi_i} = 1$$

for $j \neq k$ and all $m, n \in M$.

If μ is an endomorphism of M that maps M_i into itself, then we denote the restriction of μ to M_i by $\overline{\mu}$. On the directly indecomposable group M_i, we have

$$\overline{\pi_i} = 1 = \sum_{j=i}^{s} \overline{\mu_j\pi_i}.$$

Repeated application of Lemma 10.10 shows that at least one of the $\overline{\mu_j\pi_i}$ $(j = i, \ldots, s)$ is an automorphism of M_i. We relabel if necessary so that $\overline{\mu_i\pi_i}$ is an automorphism of M_i. Now,

$$\mu_i : M_i \to M_i^{\mu_i}$$

is an isomorphism from M_i to $M_i^{\mu_i} = N_i$, and

$$\pi_i : M_i^{\mu_i} \to M_i^{\mu_i\pi_i} = M_i$$

is a monomorphism from $M_i^{\mu_i}$ onto M_i.

Let $K_i = \{m \mid m^{\pi_i} = 1\}$ be the kernel of π_i. Then we have

$$M_i^{\mu_i} \cap K_i = 1.$$

We claim that $N_i = M_i^{\mu_i} \times (K_i \cap N_i)$.

Let $n \in N_i$. Then $n^{\pi_i} \in M_i$. Since $M_i^{\mu_i\pi_i} = M_i$, we have

$$n^{\pi_i} = k^{\mu_i\pi_i}$$

for some $k \in M_i$. We write

$$n = n(k^{\mu_i})^{-1}k^{\mu_i}.$$

Then $k^{\mu_i} \in M_i^{\mu_i}$ and $(n(k^{\mu_i})^{-1})^{\pi_i} = n^{\pi_i}k^{-\mu_i\pi_i} = 1$, and so $nk^{-\mu_i} \in K_i \cap N_i$.

Noting that $M_i^{\mu_i} \cap (K_i \cap N_i) = 1$, we obtain the desired direct decomposition

$$N_i = M_i^{\mu_i} \times (K_i \cap N_i).$$

Since N_i is directly indecomposable and $M_i^{\mu_i} > 1$, it follows that $N_i = M_i^{\mu_i}$. That is, μ_i is an isomorphism from M_i onto N_i.

The π_j ($j \neq i$) are additive with $\pi_i\mu_i$:

For $j < i$, we have

$$M^{\pi_i\mu_i} = M_i^{\mu_i} = N_i \quad \text{and} \quad M^{\pi_j} = N_j.$$

It follows from this and Lemma 10.9 that $\pi_i\mu_i$ and π_j for $j < i$ are additive.

For $j > i$, we have $M^{\pi_j} = M_j$, and this commutes with

$$M^{\pi_i\mu_i} = M_i^{\mu_i},$$

since for $m_i \in M_i$ and $m_j \in M_j$, we have

$$m_i^{\mu_i} = (m_i^{m_j})^{\mu_i} = (m_i^{\mu_i})^{m_j}.$$

Therefore, the map

$$\delta_i = \pi_1 + \cdots + \pi_{i-1} + \pi_i\mu_i + \pi_{i+1} + \cdots + \pi_r$$

is an Ω-endomorphism of M. Hence,

$$\begin{aligned} N_j^{\delta_i} &= N_j^{\pi_j} = N_j \quad \text{for } j < i, \\ M_i^{\delta_i} &= M_i^{\pi_i\mu_i} = M_i^{\mu_i} = N_i \quad \text{and} \\ M_j^{\delta_i} &= M_j^{\pi_j} = M_j \quad \text{for } j > i. \end{aligned}$$

In this way, we obtain

$$M^{\delta_i} = N_1 \times \cdots \times N_i \times M_{i+1} \times \cdots \times M_r.$$

We conclude that this product is direct since

$$N_1 \times \cdots \times N_{i-1} \times M_{i+1} \times \cdots \times M_r = K_i = \ker \pi_i$$

and

$$N_i \cap K_i = M_i^{\delta_i} \cap K_i = M_i^{\mu_i} \cap K_i = 1.$$

Now we show that δ_i is an automorphism of M. By Theorem 10.6 b), it is enough to show that the kernel of δ_i is 1. So suppose that

$$(n_1 \cdots n_{i-1} m_i \cdots m_r)^{\delta_i} = n_1 \cdots n_{i-1} m_i^{\mu_i} m_{i+1} \cdots m_r = 1.$$

From the given direct decomposition of M^{δ_i}, it follows that

$$n_1 = \cdots = n_{i-1} = m_i^{\mu_i} = m_{i+1} = \cdots = m_r = 1.$$

Since μ_i is an isomorphism from M_i to N_i, we have $m_i = 1$ as well.

Now we set $\alpha_{i+1} = \alpha_i \delta_i$. The map $\alpha = \alpha_{r+1}$ is then a normal Ω-endomorphism of M that satisfies all the conditions of the theorem. □

Now we state Theorem 12.3 for modules.

Theorem 12.4. *Let $\mathfrak{m} \neq 0$ be a right R-module satisfying the maximal and minimal conditions for submodules. If*

$$\mathfrak{m} = \mathfrak{m}_1 \oplus \cdots \oplus \mathfrak{m}_r = \mathfrak{n}_1 \oplus \cdots \oplus \mathfrak{n}_s$$

are two direct decompositions of $\mathfrak{m}$ into directly indecomposable modules $\mathfrak{m}_i$, $\mathfrak{n}_j$, then $r = s$ and there is a labeling such that $\mathfrak{m}_i \cong \mathfrak{n}_i$.

One obtains sharper statements for noncommutative groups:

Theorem 12.5. *Let G be a group satisfying the maximal and minimal conditions for normal subgroups.*

a) *If*

$$G = G_1 \times \cdots \times G_r = H_1 \times \cdots \times H_s$$

are two direct decompositions of G into nonidentity directly indecomposable groups G_i, H_j, then $r = s$ and there is a labeling such that $G_i \cong H_i$. Indeed, there exists a normal automorphism α of G such that $G_i^\alpha = H_i$.

b) *The direct decomposition*

$$G = G_1 \times \cdots \times G_r$$

into directly indecomposable factors G_i is unique (up to permuting the G_i) if and only if there are no nontrivial homomorphisms from G_i to the center $\mathbf{Z}(G_j)$ of G_j for $i \neq j$.

Proof. a) is a special case of Theorem 12.3 in which Ω is the set of all inner automorphisms of G.

b) Assume

$$G = G_1 \times \cdots \times G_r = H_1 \times \cdots \times H_r$$

are two direct decompositions of G, and let α be a normal automorphism of G such that $G_i^\alpha = H_i$. By Lemma 10.3, we have for all $g_1 \in G_1$ that

$$g_1^\alpha g_1^{-1} = s_1 \cdots s_r \in \mathbf{C}_G(G^\alpha) = \mathbf{Z}(G)$$

for some $s_i \in G_i$. Therefore, the s_i lie in $\mathbf{Z}(G_i)$ by Lemma 9.9. The map $\tau_i = (\alpha - 1)\pi_i$ with

$$g_1^{\tau_i} = s_i \quad (i \geq 2)$$

is clearly a homomorphism from G_1 to $\mathbf{Z}(G_i)$. If $G_1 \neq H_1$, then there exists some $i > 1$ with $\tau_i \neq 0$. Then τ_i is the desired nontrivial homomorphism from G_1 to $\mathbf{Z}(G_i)$.

Conversely, assume that τ is a nontrivial homomorphism from G_1 to $\mathbf{Z}(G_2)$. We define

$$G_1^* = \{g_1 g_1^\tau \mid g_1 \in G_1\}.$$

Now, G_1^* is a normal subgroup of G distinct from G_1, and every $g \in G$ can be uniquely decomposed into

$$g = g_1 \cdots g_r = (g_1 g_1^\tau)(g_2 g_1^{-\tau}) g_3 \cdots g_r \quad (g_i \in G_i).$$

Therefore, $G = G_1^* \times G_2 \times \cdots \times G_r$, and the direct decomposition of G is not unique.□

In the special case of finite groups, the previous theorem takes the following form:

Theorem 12.6. *Let G be a finite group. The direct decomposition*

$$G = G_1 \times \cdots \times G_r$$

into directly indecomposable factors G_i is unique (up to permuting the G_i) if and only if $|G_i/G_i'|$ is coprime to $|\mathbf{Z}(G_j)|$ for all $i \neq j$.

Proof. The condition is sufficient:

If α is a homomorphism from G_i to $\mathbf{Z}(G_j)$, then G_i' lies in the kernel of α. Therefore, $|G_i^\alpha|$ divides both $|G_i/G_i'|$ and $|\mathbf{Z}(G_j)|$. So $G_i^\alpha = 1$. The direct decomposition is then unique by Theorem 12.5.

Conversely, if p is a common prime divisor of $|G_i/G_i'|$ and $|\mathbf{Z}(G_j)|$, then there exists a normal subgroup N of G_i and a subgroup U of $\mathbf{Z}(G_j)$ such that

$$|G_i/N| = |U| = p.$$

Then $G_i/N \cong U$, and there is an epimorphism from G_i to the subgroup U of $\mathbf{Z}(G_j)$. Therefore, the direct decomposition of G is not unique in this case by Theorem 12.5.□

13 Modules over principal ideal domains and abelian groups

In this section, the ring R is always assumed to be commutative. A commutative ring R with no zero divisors is called a principal ideal domain if every ideal of R has the form $Ra = \{ra \mid r \in R\}$ for some $a \in R$. The ring of integers is well known to be a principal ideal domain, as well as the ring of polynomials in one variable over a field. Another example of a principal ideal domain is the ring of $\mathfrak{p}$-adic integers. This ring will play a role in Chapter V.

First we recall some concepts from the theory of commutative rings:

Definition 13.1. Let R be a commutative ring with 1 that has no zero divisors. (Such a ring R is called an integral domain.)

a) An ideal $\mathfrak{p}$ is called a prime ideal if whenever $a, b \in R$ and $ab \in \mathfrak{p}$, either $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$.
b) An element $e \in R$ is called a unit of R if $Re = R$; that is, there exists an element $c \in R$ such that $ce = 1$.
c) Let a and b be elements of R. We say that a is a divisor of b, or $a \mid b$, if there exists an element $c \in R$ such that $ac = b$.
d) Suppose p is not a unit of R. If $p = ab$ always implies that a or b is a unit, then we say that p is a prime element of R.
e) Elements $a_i \in R$ $(i = 1, \ldots, n)$ are said to be coprime if whenever $c \in R$ and $c \mid a_i$ $(i = 1, \ldots, n)$, we have that c is a unit of R.
f) An ideal $\mathfrak{m}$ is said to be a maximal ideal of R if $\mathfrak{m} \subset R$ and there is no ideal $\mathfrak{a}$ such that $\mathfrak{m} \subset \mathfrak{a} \subset R$.

We collect the fundamental properties of principal ideal domains in the following theorem.

Theorem 13.2. *Let R be a principal ideal domain that is not a field. Then:*

a) (*Maximal condition*) *Every nonempty set of ideals of R contains a maximal element. In particular, every proper ideal of R lies in a maximal ideal of R.*
b) *Every maximal ideal of R is a prime ideal and has the form Rp for some prime element $p \in R$.*
c) *Let $\{\mathfrak{p}_i = Rp_i \mid i \in I\}$ be the set of all maximal ideals of R. If $0 \neq a \in R$, then there is a decomposition*

$$a = e \prod_{i \in I} p_i^{a_i},$$

where e is a unit of R and $a_i \neq 0$ for only finitely many of the nonnegative integers a_i.
d) (*Uniqueness of prime factorization*) *If*

$$e \prod_{i \in I} p_i^{a_i} = e' \prod_{i \in I} p_i^{b_i}$$

where $e, e' \in R$ are units and $a_i, b_i \in \mathbb{Z}$, then $a_i = b_i$ for all $i \in I$.
e) *If $r_i \in R$ $(i = 1, \ldots, n)$ are coprime, then there exist elements $s_i \in R$ such that*

$$\sum_{i=1}^{n} r_i s_i = 1.$$

f) *Every prime ideal of R distinct from 0 and R is a maximal ideal.*

Proof. a) Assume for contradiction that the set $\mathscr{S}$ of ideals of R contains no maximal element. Then for every $\mathfrak{a} \in \mathscr{S}$ there exists some $\mathfrak{a}' \in \mathscr{S}$ such that $\mathfrak{a} \subset \mathfrak{a}'$. (Note

that the construction of the map $\mathfrak{a} \mapsto \mathfrak{a}'$ requires the axiom of choice!) Now we construct a countable increasing chain of ideals

$$\mathfrak{c}_1 \subset \mathfrak{c}_2 \subset \cdots$$

by defining $\mathfrak{c}_1 = 0$ and $\mathfrak{c}_{i+1} = (\mathfrak{c}_i)'$. One easily sees that $\bigcup_{i=1}^{\infty} \mathfrak{c}_i = \mathfrak{d}$ is an ideal of R. Let $\mathfrak{d} = Rd$. Since $d \in \bigcup_i \mathfrak{c}_i$, there is some j such that $d \in \mathfrak{c}_j$. Then we obtain the contradiction

$$\mathfrak{c}_{j+1} \subseteq \mathfrak{d} = Rd \subseteq \mathfrak{c}_j.$$

Thus, the set $\mathscr{S}$ has at least one maximal element.

Let $\mathfrak{a}$ be a proper ideal of R. Then the set

$$\mathscr{S} = \{\mathfrak{b} \mid \mathfrak{b} \text{ an ideal of } R \text{ and } \mathfrak{a} \subseteq \mathfrak{b} \subset R\}$$

contains $\mathfrak{a}$, so is nonempty. Every maximal element of $\mathscr{S}$ is a maximal ideal of R containing $\mathfrak{a}$.

b) Let $\mathfrak{m}$ be a maximal ideal of R. Assume for contradiction that $\mathfrak{m}$ is not a prime ideal. Then there exist $a, b \in R$ such that $ab \in \mathfrak{m}$ but $a \notin \mathfrak{m}$ and $b \notin \mathfrak{m}$. By the maximality of $\mathfrak{m}$, it follows that

$$\mathfrak{m} \subset \mathfrak{m} + Ra = \mathfrak{m} + Rb = R.$$

So we have

$$1 = m + ra = m' + r'b$$

for some $m, m' \in \mathfrak{m}$ and $r, r' \in R$. From this follows the contradiction

$$1 = mm' + mr'b + ram' + rr'ab \in \mathfrak{m}.$$

(We have not yet used the assumption that R is a principal ideal domain in b).)

Therefore, if $\mathfrak{m}$ is a maximal ideal of R, then $\mathfrak{m} = Rp$ is a prime ideal. Since R is not a field, $\mathfrak{m} \neq 0$ and so $p \neq 0$. Suppose $p = ab$ with $a, b \in R$. Since $\mathfrak{m}$ is a prime ideal of R, it follows without loss of generality that $a \in \mathfrak{m} = Rp$. So there exists some $c \in R$ such that

$$a = cp = (cb)a.$$

Since R has no zero divisors, we have $cb = 1$. Hence, b is a unit and p is a prime element.

c) Let $0 \neq a \in R$ and

$$\mathscr{S} = \{Rb \mid a = b \prod_{i \in I} p_i^{a_i}, \text{ where } a_i \neq 0 \text{ for only finitely many } i\}.$$

Clearly, $Ra \in \mathscr{S}$. Let $\mathfrak{c} = Rc$ be a maximal element of $\mathscr{S}$, which exists by a). Assume for contradiction that $\mathfrak{c} \neq R$. Then $\mathfrak{c}$ lies in a maximal ideal $\mathfrak{p}_j$ ($j \in I$). So $c = p_j d$ for some $d \in R$, and it follows from $a = dp_j \prod_{i \in I} p_i^{b_i}$ that $Rc \subseteq Rd \in \mathscr{S}$. The maximality of Rc in $\mathscr{S}$ forces $Rc = Rd$. So there exists an $r \in R$ such that $d = cr = drp_j$; that is, with $1 = rp_j$. This yields the contradiction $Rp_j = R$.

We conclude that $\mathfrak{c} = Rc = R$, and so c is a unit of R. So we obtain the decomposition

$$a = c \prod_{i \in I} p_i^{a_i}$$

of the desired form.

d) Assume for contradiction that we have

$$e \prod_{i \in I} p_i^{a_i} = e' \prod_{i \in I} p_i^{b_i}$$

with $a_1 > b_1$. Then

$$ep_1^{a_1 - b_1} \prod_{1 \neq i \in I} p_i^{a_i} = e' \prod_{1 \neq i \in I} p_i^{b_i}.$$

Thus,

$$e' \prod_{i \neq 1} p_i^{b_i} \in Rp_1.$$

Since Rp_1 is a prime ideal, it follows that either $e' \in Rp_1$ or $p_j \in Rp_1$ for some j with $1 \neq j \in I$. Both are contradictions since

$$Re' = R \neq Rp_1 \neq Rp_j.$$

e) The ideal

$$\mathfrak{d} = \left\{ \sum_{i=1}^{n} r_i t_i \mid t_i \in R \right\}$$

is a principal ideal, so $\mathfrak{d} = Rd$ for some $d \in R$. Since $r_i \in \mathfrak{d}$, we have $d \mid r_i$ and then d is a unit since the r_i are coprime. It follows that $1 \in \mathfrak{d} = Rd$ and that

$$1 = \sum_{i=1}^{n} r_i s_i$$

for some $s_i \in R$.

f) Let $\mathfrak{p} = Rp$ be a prime ideal of R with $0 \neq \mathfrak{p} \neq R$. By c), there is a decomposition

$$p = e \prod_{i \in I} p_i^{a_i}.$$

Since p is not a unit of R, there is some j with $a_j > 0$. Since $\mathfrak{p}$ is a prime ideal, $p_j \in \mathfrak{p}$ for some j. It follows from $\mathfrak{p}_j = Rp_j \subseteq \mathfrak{p}$ and the maximality of $\mathfrak{p}_j$ (see c)) that $\mathfrak{p} = \mathfrak{p}_j$. □

We now turn our attention to R-modules:

Lemma 13.3. *If μ is an epimorphism from the right R-module $\mathfrak{m}$ to the free right R-module $\mathfrak{f}$, then there is a submodule $\mathfrak{n}$ of $\mathfrak{m}$ such that $\mathfrak{m} = \ker \mu \oplus \mathfrak{n}$ and $\mathfrak{n} \cong \mathfrak{f}$.*

Proof. Let f_i ($i \in I$) be free generators of $\mathfrak{f}$, and let the m_i be some preimages of the f_i in $\mathfrak{m}$, so $f_i = m_i\mu$. By the definition of a free R-module in Definition 10.15, the map ϵ defined by $f_i\epsilon = m_i$ is an R-homomorphism from $\mathfrak{f}$ to $\mathfrak{m}$. We claim that

$$\mathfrak{m} = \ker \mu \oplus \mathfrak{f}\epsilon.$$

Clearly, $\epsilon\mu = 1$ on $\mathfrak{f}$, and it follows that $\ker \mu \cap \mathfrak{f}\epsilon = 0$. Let $m \in \mathfrak{m}$. Then

$$(m - m\mu\epsilon)\mu = m\mu - (m\mu)(\epsilon\mu) = m\mu - m\mu = 0,$$

so that $m - m\mu\epsilon \in \ker \mu$. We obtain

$$\mathfrak{m} = \ker \mu + \mathfrak{f}\epsilon,$$

and so $\mathfrak{m} = \ker \mu \oplus \mathfrak{f}\epsilon$. □

Theorem 13.4.

a) *Let R be a principal ideal domain, and let $\mathfrak{f}$ be a free right R-module in n free generators. Then every submodule of $\mathfrak{f}$ is a free right R-module in at most n free generators.*

b) *Let R be an integral domain, and assume that every finitely generated ideal of R is a principal ideal. Let $\mathfrak{f}$ be a free right R-module in n free generators. Then every finitely generated submodule of $\mathfrak{f}$ is a free right R-module in at most n free generators. (Part b) also holds even if one doesn't assume $\mathfrak{f}$ is finitely generated; the proof requires a well-ordering theorem, of course, but is otherwise identical to the following proof.)*

Proof. We prove parts a) and b) simultaneously. Let $f_1, \ldots, f_n$ be free generators of $\mathfrak{f}$, and let $\mathfrak{m}$ be a submodule of $\mathfrak{f}$; in case b) we assume $\mathfrak{m}$ is finitely generated. We let

$$\mathfrak{f}_i = \left\{ \sum_{j \le i} f_j r_j \mid r_j \in R \right\}.$$

Certainly $\mathfrak{m} = \mathfrak{m} \cap \mathfrak{f}_n$ is finitely generated in case b). Assume for induction that $\mathfrak{m} \cap \mathfrak{f}_i$ is finitely generated in case b).

Let $a \in \mathfrak{m} \cap \mathfrak{f}_i$. Then we have a decomposition $a = b + f_i r$ with $b \in \mathfrak{f}_{i-1}$ for some uniquely determined $r \in R$. The map μ defined by $a\mu = r$ is an R-epimorphism from $\mathfrak{m} \cap \mathfrak{f}_i$ onto an ideal $\mathfrak{a}_i$ of R, and we have $\ker \mu = \mathfrak{m} \cap \mathfrak{f}_{i-1}$. Since $\mathfrak{a}_i$ is the epimorphic image of the finitely generated module $\mathfrak{m} \cap \mathfrak{f}_i$, the ideal $\mathfrak{a}_i$ is finitely generated if we are in case b). Therefore, $\mathfrak{a}_i$ is in either case a principal ideal. As R has no zero divisors, we have either $\mathfrak{a}_i = 0$ or $\mathfrak{a}_i = Ra_i$ is a free R-module with generator a_i. By Lemma 13.3, we have

$$\mathfrak{m} \cap \mathfrak{f}_i = (\mathfrak{m} \cap \mathfrak{f}_{i-1}) \oplus \mathfrak{b}_i,$$

where $\mathfrak{b}_i$ is either the zero module or a free right R-module in one free generator. From this decomposition, it follows in case b) that $\mathfrak{m} \cap \mathfrak{f}_{i-1}$ is finitely generated since it is a homomorphic image of $\mathfrak{m} \cap \mathfrak{f}_i$.

From $\mathfrak{m} \cap \mathfrak{f}_i = (\mathfrak{m} \cap \mathfrak{f}_{i-1}) \oplus \mathfrak{b}_i$ and $\mathfrak{m} \cap \mathfrak{f}_0 = 0$,[1] it follows immediately by induction on i that $\mathfrak{m} \cap \mathfrak{f}_i = \bigoplus_{j=1}^{i} \mathfrak{b}_j$. Therefore, $\mathfrak{m} = \mathfrak{m} \cap \mathfrak{f}_n = \bigoplus_{j=1}^{n} \mathfrak{b}_j$. □

Definition 13.5. Let $\mathfrak{m}$ be a right R-module. We say $\mathfrak{m}$ is a torsion module if for every $m \in \mathfrak{m}$ there exists some $0 \neq r \in R$ such that $mr = 0$. We say $\mathfrak{m}$ is torsion free if $0 \neq m \in \mathfrak{m}$ and $mr = 0$ implies $r = 0$.

Theorem 13.6. *Let R be an integral domain, and let $\mathfrak{m}$ be a right R-module. Then the set*

$$\mathfrak{t} = \{m \in \mathfrak{m} \mid mr = 0 \text{ for some } 0 \neq r \in R\}$$

is a submodule of $\mathfrak{m}$, the so-called torsion submodule of $\mathfrak{m}$. The factor module $\mathfrak{m}/\mathfrak{t}$ is torsion free.

Proof. If $m_1 r_1 = 0 = m_2 r_2$ ($0 \neq r_i \in R$, $m_i \in \mathfrak{m}$), then commutativity and the absence of zero divisors in R implies

$$(m_1 + m_2) r_1 r_2 = 0 \quad \text{with} \quad 0 \neq r_1 r_2 \in R.$$

Furthermore, if $mr = 0$, then for all $r' \in R$ we have

$$(mr')r = (mr)r' = 0.$$

So $\mathfrak{t}$ is a submodule of $\mathfrak{m}$.

If $(m + \mathfrak{t})r = mr + \mathfrak{t} = \mathfrak{t}$ with $r \neq 0$ (note the definition of $\mathfrak{m}/\mathfrak{t}$ as an R-module!), then $mr \in \mathfrak{t}$, so $(mr)r' = m(rr') = 0$ for some $r' \neq 0$ in R. But this shows that $m \in \mathfrak{t}$. □

Theorem 13.7. *Let R be a principal ideal domain, and let $\mathfrak{m}$ be a finitely generated, torsion-free right R-module. Then $\mathfrak{m}$ is a free right R-module in finitely many free generators.*

[1] Correction 1 from errata.

Proof. Let $\mathfrak{m} = \left\{ \sum_{i=1}^{k} m_i r_i \mid r_i \in R \right\}$ with $m_i \neq 0$ for $i = 1, \ldots, k$. Let $\mathfrak{b}/m_k R$ be the torsion submodule of $\mathfrak{m}/m_k R$. Then $\mathfrak{m}/\mathfrak{b}$ is torsion free: If $mr \in \mathfrak{b}$ with $r \neq 0$, then for some $s \neq 0$ in R we have

$$m(rs) = (mr)s \in m_k R \quad \text{and} \quad rs \neq 0.$$

So $m \in \mathfrak{b}$.

We prove Theorem 13.7 by induction on the number of generators k of $\mathfrak{m}$. If $k = 1$, then the map $s \mapsto m_1 s$ is an R-isomorphism from the regular right R-module R_r to $\mathfrak{m}$ since $\mathfrak{m}$ is torsion free. So $\mathfrak{m}$ is free.

We have already noted that $\mathfrak{m}/\mathfrak{b}$ is torsion free and generated by the $k-1$ elements $m_1 + \mathfrak{b}, \ldots, m_{k-1} + \mathfrak{b}$. Therefore, $\mathfrak{m}/\mathfrak{b}$ is free in finitely many generators by the induction hypothesis. By Lemma 13.3, $\mathfrak{m} = \mathfrak{b} \oplus \mathfrak{f}$ for some free module $\mathfrak{f} \cong \mathfrak{m}/\mathfrak{b}$.

Now we show that $\mathfrak{b}$ is also free:

If $b \in \mathfrak{b}$ with $br = m_k s$ and $br' = m_k s'$, then

$$0 = m_k(r's - rs').$$

Since $\mathfrak{m}$ is torsion free, this yields $r's - rs' = 0$. Hence, the map δ from $\mathfrak{b}$ to the field of fractions Q of the integral domain R defined by $b\delta = \frac{s}{r}$ is well defined. It is clear that δ is an R-isomorphism from $\mathfrak{b}$ to the R-module Q.

Since $\mathfrak{b}$ is a homomorphic image of $\mathfrak{m}$, the R-module $\mathfrak{b}$ is finitely generated, say $\mathfrak{b} = \left\{ \sum_{i=1}^{n} b_i t_i \mid t_i \in R \right\}$. Let

$$b_i \delta = \frac{s_i}{r_i}$$

and let $r = r_1 \cdots r_n$. Then the map ϵ defined by $b\epsilon = (b\delta)r$ is an R-isomorphism from $\mathfrak{b}$ to the right R-module R_r. As R is a principal ideal domain, $\mathfrak{b}\epsilon$ is a free module in one generator. □

Now we can partially describe the structure of finitely generated modules over a principal ideal domain:

Theorem 13.8. *Let R be a principal ideal domain, and let $\mathfrak{m}$ be a finitely generated right R-module. Let $\mathfrak{t}$ be the torsion submodule of $\mathfrak{m}$. Then $\mathfrak{m} = \mathfrak{t} \oplus \mathfrak{s}$, where $\mathfrak{s}$ is a free right R-module and the number of free generators of $\mathfrak{s}$ is uniquely determined by $\mathfrak{m}$.*

Proof. By Theorem 13.6, $\mathfrak{m}/\mathfrak{t}$ is torsion free. Since $\mathfrak{m}/\mathfrak{t}$ is an epimorphic image of $\mathfrak{m}$, it is finitely generated. By Theorem 13.7, $\mathfrak{m}/\mathfrak{t}$ is therefore free. By Lemma 13.3, it follows that $\mathfrak{m} = \mathfrak{t} \oplus \mathfrak{s}$ for some free, finitely generated module $\mathfrak{s}$.

Let

$$\mathfrak{m} = \mathfrak{t} \oplus \mathfrak{s} = \mathfrak{t} \oplus \mathfrak{s}'$$

be two direct decompositions of $\mathfrak{m}$ with free modules

$$\mathfrak{s} = \bigoplus_{i=1}^{n} s_i R \quad \text{and} \quad \mathfrak{s}' = \bigoplus_{j=1}^{m} s'_j R.$$

Then we have $\mathfrak{s} \cong \mathfrak{m}/\mathfrak{t} \cong \mathfrak{s}'$. It suffices to prove, then, that the number of generators in a free generating set of a free module $\mathfrak{f} = \bigoplus_{i=1}^{n} f_i R$ is uniquely determined by $\mathfrak{f}$.

By Theorem 13.2 a), there exists a maximal ideal $\mathfrak{p}$ in R. Then $R/\mathfrak{p} = \mathsf{K}$ is a field (here we are crucially using the commutativity of R; over noncommutative rings it can even occur that all free modules in finitely many generators are isomorphic!). We have

$$\mathfrak{f}/\mathfrak{f}\mathfrak{p} \cong \bigoplus_{i=1}^{n} f_i R/f_i\mathfrak{p}.$$

Clearly, $\mathfrak{f}/\mathfrak{f}\mathfrak{p}$ is in fact a K-module, which is to say a vector space over $\mathsf{K} = R/\mathfrak{p}$. Since $f_i R/f_i\mathfrak{p} \cong R/\mathfrak{p}$ is a one-dimensional vector space over K, the dimension n of $\mathfrak{f}/\mathfrak{f}\mathfrak{p}$ over K is well known to be uniquely determined. □

(To finish the proof of Theorem 13.8, one can also use Theorem 11.5 and the observation that $\mathfrak{f}/\mathfrak{f}\mathfrak{p}$ has a composition series of length n as an R-module.)

It remains to clarify the structure of finitely generated torsion modules. In preparation, we prove the following:

Theorem 13.9 (Primary decomposition of torsion modules). *Let R be a principal ideal domain that is not a field, and let $\mathfrak{m}$ be a torsion R-module. Then*

$$\mathfrak{m} = \bigoplus \mathfrak{m}_p,$$

where

$$\mathfrak{m}_p = \{m \in \mathfrak{m} \mid mp^i = 0 \text{ for some } i \text{ depending on } m\};$$

in this expression, Rp runs through the set of all maximal ideals of the ring R. If $\mathfrak{m}$ is finitely generated, then so is $\mathfrak{m}_p$, and there exists some $i = i_p$ such that $mp^{i_p} = 0$ for all $m \in \mathfrak{m}_p$.

Proof. Let $0 \neq m \in \mathfrak{m}$. We define the annihilator $\mathbf{Ann}(m)$ of m to be the following subset of R:

$$\mathbf{Ann}(m) = \{r \in R \mid mr = 0\}.$$

Clearly, $\mathbf{Ann}(m)$ is an ideal of R that is distinct from 0 and R. So $\mathbf{Ann}(m) = Ra$ for some $0 \neq a \in R$. Since $0 = ma \neq m$, the element a is not a unit of R. Let $a = e \prod_{i=1}^{s} p_i^{k_i}$ be the prime factorization of a, which exists by Theorem 13.2 c). We

let $a_i = \prod_{j \neq i} p_j^{k_j}$ $(i = 1, \ldots, s)$. Then 1 is the greatest common divisor of the a_i. By Theorem 13.2 e), there exist elements $b_i \in R$ such that

$$1 = \sum_{i=1}^{s} a_i b_i.$$

We define $m_i = m a_i b_i$, so that $m = \sum_{i=1}^{s} m_i$. Therefore, $m_i p_i^{k_i} = m a b_i = 0$. So $m_i \in \mathfrak{m}_{p_i}$ and $\mathfrak{m} = \sum_p \mathfrak{m}_p$.

Suppose $0 = \sum_j m_j$ with $m_j \in \mathfrak{m}_{p_j}$ and $m_j p_j^{k_j} = 0$. Again we define

$$a_i = \prod_{j \neq i} p_j^{k_j}.$$

Then $m_j a_i = 0$ for $i \neq j$. Since a_i and $p_i^{k_i}$ are coprime, there exist elements $c_i, d_i \in R$ such that $1 = a_i c_i + p_i^{k_i} d_i$ by Theorem 13.2 e). Now it follows that

$$0 = \left(\sum_{j=1}^{s} m_j \right) a_i c_i = m_i a_i c_i = m_i - m_i p_i^{k_i} d_i = m_i.$$

We have shown that $\mathfrak{m} = \bigoplus_p \mathfrak{m}_p$.

Assume that $\mathfrak{m}$ is finitely generated. Then the $\mathfrak{m}_p$ are also finitely generated since they are homomorphic images of $\mathfrak{m}$. If $\{m_1, \ldots, m_n\}$ is a generating set of $\mathfrak{m}_p$, then there is a power p^i of p such that $m_j p^i = 0$ for every $j = 1, \ldots, n$. Then $\mathfrak{m}_p p^i = 0$ as well. □

Theorem 13.10. *Let R be a principal ideal domain, let $p \in R$ be a prime element and let $\mathfrak{m}$ be a finitely generated right R-module such that $\mathfrak{m} p^s = 0$. Then there exist $y_i \in \mathfrak{m}$ such that*

$$\mathfrak{m} \cong \bigoplus_{i=1}^{k} y_i R \quad \text{and} \quad y_i R \cong R/p^{a_i} R.$$

The a_i are uniquely determined (up to their ordering) by $\mathfrak{m}$.

Proof. a) First we show there exists such a decomposition. Since the prime ideal Rp is maximal by Theorem 13.2 f), $R/Rp = \mathsf{K}$ is a field. Since $\mathfrak{m}$ is finitely generated, the finitely generated K-vector space $\mathfrak{m}/\mathfrak{m}p$ has finite dimension $\dim_{\mathsf{K}}(\mathfrak{m}/\mathfrak{m}p)$. We define an invariant d for all finitely generated right R-modules $\mathfrak{v}$ by

$$d(\mathfrak{v}) = \dim_{\mathsf{K}}(\mathfrak{v}/\mathfrak{v}p).$$

Assume $\mathfrak{m}p^s = 0 \neq \mathfrak{m}p^{s-1}$. Then there exists some $x \in \mathfrak{m}$ such that $xp^s = 0 \neq xp^{s-1}$. First we show that $d(\mathfrak{m}/xR) < d(\mathfrak{m})$.

Certainly $x \notin \mathfrak{m}p$, since otherwise $xp^{s-1} \in \mathfrak{m}p^s = 0$. So we have

$$\mathfrak{m}p + xR \supset \mathfrak{m}p$$

and therefore

$$\begin{aligned} d(\mathfrak{m}/xR) &= \dim_{\mathsf{K}}(\mathfrak{m}/xR\,/\,(\mathfrak{m}p + xR)/xR) = \dim_{\mathsf{K}}(\mathfrak{m}/(\mathfrak{m}p + xR)) \\ &< \dim_{\mathsf{K}}(\mathfrak{m}/\mathfrak{m}p) = d(\mathfrak{m}). \end{aligned}$$

If $m \in \mathfrak{m}$, then $\mathbf{Ann}(m) = Rp^t$ for some t: Suppose $y = ap^k \in \mathbf{Ann}(m)$ with $p \nmid a$. Since $(a, p^s) = 1$, there exist elements $b, c \in R$ such that $1 = ab + p^s c$ by Theorem 13.2 e). It follows that

$$0 = (map^k)b = m(1 - p^s c)p^k = mp^k - mp^{s+k}c = mp^k,$$

and so $p^k \in \mathbf{Ann}(m)$.

Now we prove the claimed decomposition of $\mathfrak{m}$ by induction on $d(\mathfrak{m})$:

For the base case $d(\mathfrak{m}) = 0$, we have $\mathfrak{m} = \mathfrak{m}p = \mathfrak{m}p^s = 0$ and the claim is trivially true. Now suppose $x \notin \mathfrak{m}p$. Since

$$d(\mathfrak{m}/xR) < d(\mathfrak{m}),$$

we may assume by the induction hypothesis that

$$\mathfrak{m}/xR \cong \bigoplus_{i=1}^{k-1} \overline{y}_i R \quad \text{with} \quad \overline{y}_i R \cong R/p^{a_i} R.$$

Since $(\mathfrak{m}/xR)p^s = 0$, we have $a_i \leq s$. Let the y_i be any preimages in $\mathfrak{m}$ of the $\overline{y}_i$. Then we have

$$y_i p^{a_i} = x c_i p^{b_i}$$

for some b_i with $b_i \leq s$ and some $c_i \in R - pR$. Since $\mathfrak{m}p^s = 0$, it follows that

$$0 = y_i p^{a_i} p^{s-a_i} = x c_i p^{s-a_i+b_i}.$$

It follows from $\mathfrak{m}p^s = 0$ and $xp^{s-1} \neq 0$ that $\mathbf{Ann}(x) = p^s R$. So necessarily $a_i \leq b_i$. For the elements

$$y_i' = y_i - x c_i p^{b_i - a_i},$$

we have $y_i' p^{a_i} = 0$.

It is certainly true that $\mathfrak{m} = xR + \sum_{i=1}^{k-1} y_i' R$. Suppose that

$$0 = xr + \sum_{i=1}^{k-1} y_i' r_i .$$

Then it follows that

$$\sum_{i=1}^{k-1} y_i' r_i \equiv 0 \mod xR,$$

and therefore $r_i \in p^{a_i} R$. By what we have already proved, this implies

$$y_i' r_i = 0 = xr .$$

Therefore,

$$\mathfrak{m} = xR \oplus \bigoplus_{i=1}^{k-1} y_i' R,$$

where $xR \cong R/p^s R$ and $y_i' R \cong R/p^{a_i} R$, as claimed.

b) It remains to show that the a_i are uniquely determined by the structure of $\mathfrak{m}$. Suppose then that

$$\mathfrak{m} = \bigoplus_{i=1}^{k} x_i R \quad \text{with} \quad x_i R \cong R/p^{a_i} R$$

and $a_1 \geq a_2 \geq \cdots \geq a_k$. Let n_j be the number of i such that $a_i \geq j$. So $n_1 = k$. Now we have

$$\mathfrak{m} p^j = \bigoplus_{a_i \geq j+1} x_i p^j R$$

and it follows that $\dim_{\mathsf{K}}(\mathfrak{m} p^j / \mathfrak{m} p^{j+1}) = n_{j+1}$.

Thus, the n_j are uniquely determined by $\mathfrak{m}$. Then the number k_j of i with $a_i = j$ is uniquely determined since $k_j = n_j - n_{j+1}$. □

We also note the following:

Theorem 13.11. *Let R be a principal ideal domain. Then every submodule of a finitely generated right R-module is finitely generated.*

Proof. The claim holds for free modules by Theorem 13.4 a). This also proves the general case since every finitely generated module is the epimorphic image of a finitely generated free module. □

The considerations of this section specialize to statements about abelian groups by taking the ring R to be the ring $\mathbb{Z}$ of integers. In particular, we inherit the notions of free abelian group, torsion-free group and torsion group.

An important consequence of Theorems 13.8–13.10 is the fundamental theorem of finitely generated abelian groups. We write our groups multiplicatively in the statement of the theorem.

Theorem 13.12 (Fundamental theorem of finitely generated abelian groups). *Let A be a finitely generated abelian group. Then*

$$A = \langle a_1 \rangle \times \cdots \times \langle a_n \rangle,$$

where $|a_i| = p_i^{n_i}$ for $1 \leq i \leq k$ and $|a_i| = \infty$ for $k+1 \leq i \leq k+r = n$. The number r and the prime powers $p_i^{n_i}$ are uniquely determined by A. We call r the rank $r(A)$ of A and the $p_i^{n_i}$ the invariants of A. The group A is clearly determined up to isomorphism by its rank and its invariants. We call

$$(p_1^{n_1}, \ldots, p_k^{n_k}, \underbrace{\infty, \ldots, \infty}_{r})$$

the type of A. We call the set $\{a_1, \ldots, a_n\}$ a basis of the abelian group A.

Remark 13.13. A finite abelian group is said to be elementary abelian if it has prime exponent p. The type of an elementary abelian group A has the form $(p, \ldots, p)$, and there is a direct decomposition

$$A = \langle a_1 \rangle \times \cdots \times \langle a_n \rangle$$

with $|a_i| = p$. The map ϵ defined by

$$(a_1^{k_1} \cdots a_n^{k_n})^{\epsilon} = (k_1, \ldots, k_n)$$

is an isomorphism from A to the additive group of the vector space $\mathsf{V} = \mathsf{V}(n, p)$ of dimension n over the field $\mathrm{GF}(p)$. Since linearity and additivity have the same meaning over a finite field of prime order, the automorphism group of an elementary abelian group of order p^n is isomorphic to the group $\mathrm{GL}_n(p)$ of all linear maps from the vector space $\mathsf{V} = \mathsf{V}(n, p)$ onto itself. We will make much use of this interpretation of automorphisms of elementary abelian groups as linear maps.

The following theorem will occasionally be useful in Chapter V when we discuss projective representations.

Theorem 13.14. *Let A be a finitely generated abelian group.*

a) *If F is a free abelian subgroup of A with $|A/F| < \infty$, then $r(A) = r(F)$.*
b) *If $B \leq A$, then $r(B) + r(A/B) = r(A)$. In particular, $r(B) = r(A)$ if and only if A/B is finite.*
c) *If A/B is finite and B can be generated by $r(A)$ elements, then B is a free abelian group in $r(A)$ free generators.*
d) *If μ is an epimorphism from A onto A, then μ is an isomorphism.*
e) *If A and B are finitely generated abelian groups, B is an epimorphic image of A and A is an epimorphic image of B, then $A \cong B$.*

Proof. a) Let $A = T \times S$ with $|T| < \infty$ and S a free abelian group in $r(A)$ free generators. Since $T \cap F = 1$, it follows that

$$\langle T, F \rangle = T \times F \leq A.$$

By the Dedekind identity, Lemma 2.12 c), we have

$$T \times F = T \times \overline{S},$$

where $\overline{S} = (T \times F) \cap S \leq S$. The subgroup $\overline{S}$ of S is free by Theorem 13.4, and

$$r(F) = r(T \times F) = r(\overline{S}) \leq r(S) = r(A).$$

So $r(F) \leq r(A)$.

In addition, we have

$$|S/\overline{S}| = |(T \times S)/(T \times \overline{S})| = |A/(T \times F)| \leq |A/F| < \infty.$$

If $|S/\overline{S}| = m$, then the map ϵ defined by $s^\epsilon = s^m$ is a monomorphism from S to $\overline{S}$. Then we obtain from Theorem 13.4 the reverse inequality

$$r(A) = r(S) = r(S^\epsilon) \leq r(\overline{S}) = r(F).$$

b) By Theorem 13.11, B is finitely generated. By Theorem 13.12, there exist free abelian subgroups F_1 of B and S/B of A/B such that $|B/F_1| < \infty$ and $|A/S| < \infty$. By Lemma 13.3, $S = B \times F_2$ with F_2 free. Since $F_1 \cap F_2 \leq B \cap F_2 = 1$, we have that $F = F_1 \times F_2$ is a free subgroup of A.

Now we will show that $|A/F|$ is finite. It then follows from a) that

$$\begin{aligned} r(A) = r(F) &= r(F_1) + r(F_2) = r(B) + r(S/B) \\ &= r(B) + r(A/B). \end{aligned}$$

Since A and A/F are finitely generated, it suffices to prove that A/F is a torsion group. Let $|B/F_1| = m$ and $|A/S| = n$. For $a \in A$, we have

$$a^n = bf \in S = B \times F_2$$

and

$$a^{nm} = b^m f^m \in F_1 \times F_2.$$

c) Since A/B is finite, part b) implies $r(A) = r(B) = r$. Let $B = F \times T$ be the decomposition of B into its torsion group T and a free abelian group F. If p is a prime divisor of $|T|$, then we define

$$B^p = \{b^p \mid b \in B\}.$$

Then $B/B^p \cong F/F^p \times T/T^p$ is an elementary abelian group of exponent p and order p^k with $k \geq r+1$. In particular, B/B^p cannot be generated by r elements. But since B can be generated by r elements by assumption, $T = 1$ and B is a free abelian group.

d) (Blackburn) Let μ be an epimorphism from A onto A. Then $D = \bigcup_{n=1}^{\infty} \ker \mu^n$ is a subgroup of A. By Theorem 13.11, D is finitely generated, say $D = \langle d_1, \ldots, d_m \rangle$. Let $d_i \in \ker \mu^{n_i}$ $(i = 1, \ldots, m)$ and $n = \max_i n_i$. Then

$$\ker \mu^{n+1} \leq D \leq \ker \mu^n, \quad \text{so} \quad \ker \mu^{n+1} = \ker \mu^n.$$

Let $b \in \ker \mu$. Since μ^n is again an epimorphism from A to A, $b = c^{\mu^n}$ for some $c \in A$. Then $c^{\mu^{n+1}} = b^{\mu} = 1$. So $c \in \ker \mu^{n+1} = \ker \mu^n$, and therefore $b = c^{\mu^n} = 1$. Since $\ker \mu = 1$, we conclude that μ is an isomorphism from A to A.

e) If μ_1 is an epimorphism from A to B and μ_2 is an epimorphism from B to A, then $\mu_1\mu_2$ is an epimorphism from A to A. Then $\mu_1\mu_2$ is an isomorphism by d), so μ_1 is itself an isomorphism. □

In addition to the finitely generated abelian groups, we will later require another class of abelian groups. Again we will consider the general case of modules over principal ideal domains.

Definition 13.15. A right R-module $\mathfrak{m}$ is called injective when it has the following property: If $\mathfrak{b}$ is a submodule of the right R-module $\mathfrak{a}$ and α is an R-homomorphism from $\mathfrak{b}$ to $\mathfrak{m}$, then there exists an extension μ of α to an R-homomorphism from $\mathfrak{a}$ to $\mathfrak{m}$.

Definition 13.16. A right R-module $\mathfrak{m}$ is called divisible if for every $m \in \mathfrak{m}$ and every nonzero $r \in R$, there exists some $x \in \mathfrak{m}$ such that $m = xr$.

The following theorem shows that these two notions coincide for principal ideal domains.

Theorem 13.17. *Let R be a principal ideal domain. A right R-module $\mathfrak{m}$ is injective if and only if it is divisible.*

Proof. a) Let $\mathfrak{m}$ be divisible, let π be a homomorphism from $\mathfrak{b}$ to $\mathfrak{m}$ and let $\mathfrak{b}$ be a submodule of $\mathfrak{a}$. We consider the pairs $(\mathfrak{r}, \mu)$ such that $\mathfrak{b} \subseteq \mathfrak{r} \subseteq \mathfrak{a}$, where $\mathfrak{r}$ is a submodule of $\mathfrak{a}$ and μ is an R-homomorphism from $\mathfrak{r}$ to $\mathfrak{m}$ that agrees with π on $\mathfrak{b}$. We impose a partial order on these pairs by saying $(\mathfrak{r}_1, \mu_1) \leq (\mathfrak{r}_2, \mu_2)$ whenever $\mathfrak{r}_1 \subseteq \mathfrak{r}_2$ and $\mu_2|_{\mathfrak{r}_1} = \mu_1$. It is clear that the set of pairs equipped with this partial order is inductive. By Zorn's Lemma, there is a maximal pair $(\mathfrak{r}, \mu)$. Assume for contradiction that $\mathfrak{r} \subset \mathfrak{a}$ and let $a \in \mathfrak{a} - \mathfrak{r}$. We distinguish two cases:

Case 1: Assume $aR \cap \mathfrak{r} = 0$. In this case, we define an R-homomorphism α from $aR + \mathfrak{r} = aR \oplus \mathfrak{r}$ to $\mathfrak{m}$ by

$$(ar + x)\alpha = x\mu$$

for all $x \in \mathfrak{r}$. Then we have

$$(\mathfrak{r}, \mu) < (aR + \mathfrak{r}, \alpha),$$

which contradicts the maximality of $(\mathfrak{r}, \mu)$.

Case 2: Assume $aR \cap \mathfrak{r} \neq 0$. Then the set

$$\mathfrak{i} = \{r \mid ar \in \mathfrak{r}\}$$

is a nonzero ideal of R, so it is a principal ideal $\mathfrak{i} = iR$ for some $i \in R$. Since $\mathfrak{m}$ is divisible, there is some $y \in \mathfrak{m}$ such that $yi = (ai)\mu$. For $z = x + ar \in \mathfrak{r} + aR$, we now define α by

$$z\alpha = x\mu + yr.$$

The map α is well defined: It follows from $x = ar \in \mathfrak{r} \cap aR$ that $r = is \in iR$ for some $s \in R$, and then

$$x\mu = (ais)\mu = (ai)\mu s = yis = yr = (ar)\alpha.$$

Clearly, α is an R-homomorphism from $aR + \mathfrak{r}$ to $\mathfrak{m}$. From this, we obtain the contradiction

$$(\mathfrak{r}, \mu) < (aR + \mathfrak{r}, \alpha).$$

So $\mathfrak{r} = \mathfrak{a}$, and $\mathfrak{m}$ is injective.

b) The proof that every injective module is divisible is much simpler and uses only the fact that R has no zero divisors.

Assume $\mathfrak{m}$ is an injective right R-module, and let $\mathfrak{a} = R_r$ be the free right R-module in one generator. Let $\mathfrak{b} = bR$ be the submodule of $\mathfrak{a}$ generated by $b \neq 0$. Let $m \in \mathfrak{m}$. Then the map π defined by $(bx)\pi = mx$ for all $x \in R$ is an R-homomorphism from $\mathfrak{b}$ to $\mathfrak{m}$. (The map π is well defined since R has no zero divisors.) If μ is an extension of π to $\mathfrak{a}$, then

$$m = b\pi = b\mu = (1\mu)b,$$

which is what we wanted to show. □

Some examples of divisible abelian groups (that is, divisible $\mathbb{Z}$-modules) are the multiplicative groups of algebraically closed fields and the additive groups of fields of characteristic 0.

The divisible abelian groups can be classified. They are Cartesian products of (not necessarily finitely many) additive groups of the rational numbers and so-called Prüfer p-groups $\mathbb{Z}_{p^\infty}$, which for a prime p consist of the roots of unity of p-power order equipped with the multiplication of the complex numbers. We will not require this fact in what follows (for more information, see Scott [379], pg. 99).

Next, we determine the automorphism groups of cyclic groups. In preparation, we prove the following:

Lemma 13.18. *Let p be a prime number and r any natural number.*
a) *For $p > 2$, $k \geq 0$ and $i \geq 2$, we have*

$$\binom{p^k r}{i} p^i \equiv 0 \mod p^{k+2}.$$

b) *For $i \geq 1$ and $k \geq 0$, we have*

$$\binom{2^k r}{i} 2^i \equiv 0 \mod 2^{k+1}.$$

Proof. We calculate in the ring $\mathbb{Z}[x]$ of integer polynomials.
a) We show for $p > 2$ by induction on k that

$$(1 + px)^{rp^k} \equiv 1 + rp^{k+1}x \mod p^{k+2}.$$

Then expanding the left-hand side using the binomial theorem and comparing coefficients yields the claim. For $k = 0$, the given congruence holds by the binomial theorem. Now assume for induction that

$$(1 + px)^{rp^k} = 1 + rp^{k+1}x + p^{k+2}f$$

for some $f \in \mathbb{Z}[x]$. Since the binomial coefficients $\binom{p}{i}$ are divisible by p for $0 < i < p$, it follows from $p(k+1) \geq k+3$ and $j(k+1)+1 \geq k+3$ for $j \geq 2$ that

$$(1 + px)^{rp^{k+1}} \equiv 1 + p^{k+2}(rx + pf) \equiv 1 + rp^{k+2}x \mod p^{k+3}.$$

b) For $p = 2$, we show in the same way that

$$(1 + 2x)^{r2^k} \equiv 1 \mod 2^{k+1}. \qquad \square$$

Theorem 13.19. *Let $G(n)$ be the multiplicative group consisting of the residue classes modulo n of the integers coprime to n. (By Theorem 4.6, $G(n)$ is isomorphic to the automorphism group of the cyclic group of order n.) Then:*
a) *If $n = n_1 \cdots n_r$ with the n_i pairwise coprime, then*

$$G(n) \cong G(n_1) \times \cdots \times G(n_r).$$

b) *For $p > 2$, $G(p^m)$ is cyclic of order $p^{m-1}(p-1)$.*
c) *For $m \geq 3$, $G(2^m)$ has type $(2^{m-2}, 2)$. The residue classes -1 and $5 \bmod 2^m$ form a basis of $G(2^m)$. Furthermore, $|G(4)| = 2$.*

Proof. a) Let $Z = \langle z \rangle$ be the cyclic group of order n. We define $z_i = z^{k_i}$ with $k_i = \frac{n}{n_i}$. Defining $Z_i = \langle z_i \rangle$, we have $|Z_i| = n_i$ and $|Z| = n = n_1 \cdots n_r$. By Theorem 9.4, we have

$$Z = Z_1 \times \cdots \times Z_r \quad \text{and} \quad \mathbf{Aut}(Z) \cong \mathbf{Aut}(Z_1) \times \cdots \times \mathbf{Aut}(Z_r).$$

b) We first show that $G(p)$ is cyclic of order $p-1$ (see also Exercise 37). Suppose that

$$G(p) = S_1 \times \cdots \times S_k$$

is a direct decomposition of $G(p)$ into cyclic factors S_i with orders z_i. If the z_i are pairwise coprime, then $G(p)$ is cyclic (see Exercise 6). If not, then there exists a prime number q such that $q \mid z_i$ and $q \mid z_j$ for some pair i, j with $i \neq j$. Then $G(p)$ contains at least q^2 solutions to $x^q = 1$. Since the residue classes mod p constitute a field, this is not possible, however, since every polynomial of degree q has at most q zeros over a field. Therefore, $G(p)$ is cyclic.

The map δ defined by

$$(x + p^m\mathbb{Z})^\delta = x + p\mathbb{Z}$$

is an epimorphism from $G(p^m)$ onto $G(p)$. The kernel of δ consists of the residue classes $x + p^m\mathbb{Z}$ such that $x \equiv 1 \mod p$, and it has order $\varphi(p^m)/\varphi(p) = p^{m-1}$, where φ is the Euler φ-function. We have

$$(1+p)^{p^{m-2}} = 1 + p^{m-1} + \cdots + \binom{p^{m-2}}{i} p^i + \cdots .$$

By Lemma 13.18 a), $\binom{p^{m-2}}{i} p^i \equiv 0 \mod p^m$ for $p > 2$ and $i \geq 2$. Therefore,

$$(1+p)^{p^{m-2}} \equiv 1 + p^{m-1} \not\equiv 1 \mod p^m.$$

Therefore, the residue class of $1 + p$ lying in $\ker\delta$ has order p^{m-1}. Hence, $\ker\delta$ is generated by the class of $1 + p$, so is cyclic. Since $|G(p^m)/\ker\delta| = |G(p)| = p - 1$, $\ker\delta$ is the Sylow p-subgroup of $G(p^m)$. Let

$$G(p^m) = G_1 \times \cdots \times G_r \times \ker\delta$$

be the direct decomposition of $G(p^m)$ into its Sylow subgroups according to Theorem 13.9. We have

$$G(p^m)/\ker\delta \cong G_1 \times \cdots \times G_r \cong G(p),$$

and $G(p)$ is cyclic. So $G(p^m)$ is the direct product of cyclic groups of orders $p - 1$ and p^{m-1}. So $G(p^m)$ is itself cyclic (see Exercise 6).

c) For $2^m > 4$, the map δ defined by

$$(x + 2^m\mathbb{Z})^\delta = x + 4\mathbb{Z}$$

is an epimorphism from $G(2^m)$ onto $G(4)$ with kernel of order $\varphi(2^m)/2 = 2^{m-2}$. Note that

$$(1+2^2)^{2^{m-3}} = 1 + 2^{m-1} + \cdots + \binom{2^{m-3}}{i} 2^{2i} + \cdots .$$

By Lemma 13.18 b), we have

$$\binom{2^{m-3}}{i} 2^i \equiv 0 \mod 2^{m-2}, \quad \text{so} \quad \binom{2^{m-3}}{i} 2^{2i} \equiv 0 \mod 2^m \quad \text{for} \quad i \geq 2.$$

Therefore, there remains only

$$(1+2^2)^{2^{m-3}} \equiv 1 + 2^{m-1} \not\equiv 1 \mod 2^m.$$

We conclude that $\ker \delta$ is generated by the residue class $\overline{5} = 5 + 2^m\mathbb{Z}$. Let $-\overline{1} = -1 + 2^m\mathbb{Z}$. Since $\langle -\overline{1} \rangle \cap \ker \delta = 1$, it follows that

$$|\langle -\overline{1} \rangle \times \langle \overline{5} \rangle| = 2^{m-1} = |G(2^m)|. \qquad \square$$

Exercises

50) Let R be a principal ideal domain, and let $\mathfrak{a} \neq 0$ be an ideal in R. Then $R/\mathfrak{a}$ satisfies the minimal condition for ideals.

51) Let A be a finite abelian p-group. We define

$$\Omega_i(A) = \{a \in A \mid a^{p^i} = 1\}$$

and

$$\mho_i(A) = \{a^{p^i} \mid a \in A\}.$$

Then

$$A/\Omega_i(A) \cong \mho_i(A) \quad \text{and} \quad A/\mho_i(A) \cong \Omega_i(A).$$

If $|\Omega_i(A)/\Omega_{i-1}(A)| = p^{a_i}$, then $a_1 \geq a_2 \geq \cdots$.

52) Let A be a finite abelian group. There exists a direct decomposition

$$A = Z_1 \times \cdots \times Z_n$$

with cyclic $Z_i \neq 1$ and $|Z_{i+1}|$ dividing $|Z_i|$ for all i. The $|Z_i|$ are uniquely determined, but in general the Z_i are not.

53) Let A be an abelian group of type (p^3, p). Show that A contains an element of order p^2 that sits in none of the cyclic direct factors of A.

54) Let A be an abelian p-group of order p^n. Show that p^{n-1} divides $|\mathbf{Aut}(A)|$.

14 Extension theory

In this section, we consider the construction of a group from a normal subgroup and its factor group. We begin with the following theorem.

Theorem 14.1. *Let N be a normal subgroup of G, let $G/N = H$ and let $G = \bigcup_{h\in H} g(h)\,N$ with $g(h)\,N = h$ be the decomposition of G into left cosets of N.* (*Here, we index the cosets of N in G using the elements of the factor group G/N.*) *Since*

$$(g(h_1)N)(g(h_2)N) = h_1h_2 = g(h_1h_2)N,$$

we have

$$g(h_1)\,g(h_2) = g(h_1h_2)\,n(h_1,h_2)$$

for uniquely determined $n(h_1,h_2) \in N$. These elements satisfy the following functional equations:

(1) $n(h_1,h_2h_3)\,n(h_2,h_3) = n(h_1h_2,h_3)\,n(h_1,h_2)^{g(h_3)}$ *for all* $h_i \in H$.

(2) $n^{g(h_1)\,g(h_2)} = n^{g(h_1h_2)\,n(h_1,h_2)}$ *for all* $n \in N$ *and all* $h_i \in H$.

(3) *If one chooses* $g(1) = 1$, *then*

$$n(1,h) = n(h,1) = 1 \quad \textit{for all} \quad h \in H.$$

The function $n : H\times H \to N$ is called the factor system for the set of representatives $\{g(h) \mid h \in H\}$.

Proof. We have

$$\begin{aligned}\big(g(h_1)\,g(h_2)\big)\,g(h_3) &= g(h_1h_2)\,n(h_1,h_2)\,g(h_3)\\ &= g(h_1h_2)\,g(h_3)\,n(h_1,h_2)^{g(h_3)}\\ &= g(h_1h_2h_3)\,n(h_1h_2,h_3)\,n(h_1,h_2)^{g(h_3)},\end{aligned}$$

which by associativity is equal to

$$\begin{aligned}g(h_1)\,\big(g(h_2)\,g(h_3)\big) &= g(h_1)\,g(h_2h_3)\,n(h_2,h_3)\\ &= g(h_1h_2h_3)\,n(h_1,h_2h_3)\,n(h_2,h_3).\end{aligned}$$

This proves (1). The claims in (2) and (3) are trivial. □

Now we turn these considerations around and build from N and H a group G with $G/N^* \cong H$ and $N^* \cong N$.

Theorem 14.2 (O. Schreier). *Let H and N be two* (*not necessarily finite*) *groups, and suppose we have a map*

$$(h_1, h_2) \mapsto n(h_1, h_2)$$

from $H \times H$ to N, a so-called factor system. Suppose to every $h \in H$ we have associated an automorphism $\alpha(h)$ of N. Finally, assume for all $n \in N$ and all $h_i, h \in H$ that

(1) $n(h_1, h_2h_3)\, n(h_2, h_3) = n(h_1h_2, h_3)\, n(h_1, h_2)^{\alpha(h_3)}$.

(2) $n^{\alpha(h_1)\,\alpha(h_2)} = (n^{\alpha(h_1h_2)})^{n(h_1,h_2)}$.

(3) $n(h, 1) = n(1, h) = 1$.

If we define a multiplication on the product set

$$G = \{(h, n) \mid h \in H, n \in N\}$$

by

$$(h_1, n_1)\,(h_2, n_2) = (h_1h_2,\ n(h_1, h_2)\, n_1^{\alpha(h_2)} n_2),$$

then G is a group. The set

$$N^* = \{(1, n) \mid n \in N\}$$

is then a normal subgroup of G isomorphic to N, and $G/N^* \cong H$. We call G the extension of N by H with factor system $n(_, _)$ and automorphisms $\alpha(h)$.

Proof. The multiplication defined on G is associative since

$$\begin{aligned}\big((h_1, n_1)\,(h_2, n_2)\big)\,(h_3, n_3) &= (h_1h_2,\ n(h_1, h_2)\, n_1^{\alpha(h_2)} n_2)\,(h_3, n_3)\\ &= (h_1h_2h_3,\ n(h_1h_2, h_3)\,(n(h_1, h_2)\, n_1^{\alpha(h_2)} n_2)^{\alpha(h_3)} n_3)\end{aligned}$$

and

$$\begin{aligned}(h_1, n_1)\,\big((h_2, n_2)\,(h_3, n_3)\big) &= (h_1, n_1)\,(h_2h_3,\ n(h_2, h_3)\, n_2^{\alpha(h_3)} n_3)\\ &= (h_1h_2h_3,\ n(h_1, h_2h_3)\, n_1^{\alpha(h_2h_3)}\, n(h_2, h_3)\, n_2^{\alpha(h_3)}\, n_3).\end{aligned}$$

These expressions are equal since

$$\begin{aligned}&n(h_1h_2, h_3)\, n(h_1, h_2)^{\alpha(h_3)}\, n_1^{\alpha(h_2)\alpha(h_3)}\\ =&n(h_1, h_2h_3)\, n(h_2, h_3)\, (n_1^{\alpha(h_2h_3)})^{n(h_2,h_3)}\\ =&n(h_1, h_2h_3)\, n_1^{\alpha(h_2h_3)}\, n(h_2, h_3).\end{aligned}$$

Furthermore, we have

$$(1, 1)(h, n) = (1h,\ n(1, h)\, 1^{\alpha(h)}\, n) = (h, n),$$

and upon defining

$$x = (n(h^{-1}, h)^{-1} n^{-1})^{\alpha(h)^{-1}}$$

we have

$$(h^{-1}, x)(h, n) = (1,\ n(h^{-1}, h)\, x^{\alpha(h)}\, n) = (1, 1).$$

So G is a group.

It follows from $n(1, 1) = 1$ and

$$n^{\alpha(1)^2} = (n^{\alpha(1)})^{n(1,1)} = n^{\alpha(1)},$$

that $\alpha(1)^2 = \alpha(1)$, so $\alpha(1) = 1$.

Therefore,

$$(1, n_1)(1, n_2) = (1,\ n(1, 1)\, n_1^{\alpha(1)}\, n_2) = (1, n_1 n_2),$$

and so $N^* = \{(1, n) \mid n \in N\}$ is a subgroup of G isomorphic to N. The map ϵ defined by $(h, n)^\epsilon = h$ is clearly an epimorphism from G onto H with kernel N^*. Therefore, N^* is a normal subgroup of G with $G/N^* \cong H$. □

Lemma 14.3. *Let N be a normal subgroup of G with $G/N = H$. Let $G = \bigcup_{h \in H} g(h)\, N$ be the decomposition of G into left cosets of N, and let*

$$g(h_1)\, g(h_2) = g(h_1 h_2)\, n(h_1, h_2).$$

Suppose we consider another set $\{\widetilde{g}(h) \mid h \in H\}$ of coset representatives of N in G, where $\widetilde{g}(h) = g(h)\, a(h)$ for some $a(h) \in N$. Then we have

$$\widetilde{g}(h_1)\, \widetilde{g}(h_2) = \widetilde{g}(h_1 h_2)\, \widetilde{n}(h_1, h_2),$$

where

$$\widetilde{n}(h_1, h_2) = a(h_1 h_2)^{-1}\, n(h_1, h_2)\, a(h_1)^{g(h_2)}\, a(h_2).$$

Proof. We calculate

$$\begin{aligned}
\widetilde{g}(h_1)\, \widetilde{g}(h_2) &= g(h_1)\, a(h_1)\, g(h_2)\, a(h_2) \\
&= g(h_1)\, g(h_2)\, a(h_1)^{g(h_2)}\, a(h_2) \\
&= g(h_1 h_2)\, n(h_1, h_2)\, a(h_1)^{g(h_2)}\, a(h_2) \\
&= g(h_1 h_2)\, a(h_1 h_2)\, \widetilde{n}(h_1, h_2) = \widetilde{g}(h_1 h_2)\, \widetilde{n}(h_1, h_2).
\end{aligned}$$

□

Next, we consider some special cases in which the previous formulas become simpler.

Example 14.4. Let G be an extension of N by H with the so-called trivial factor system $n(_, _)$ defined by $n(h_1, h_2) = 1$ for all $h_1, h_2 \in H$. Then equations (1) and (3) from Theorem 14.2 are trivially satisfied, while equation (2) becomes

$$n^{\alpha(h_1)\,\alpha(h_2)} = n^{\alpha(h_1h_2)}$$

and we have $\alpha(1) = 1$. With this, the map $h \mapsto \alpha(h)$ becomes a homomorphism from H to the automorphism group $\mathbf{Aut}(N)$ of N. The elements $(h, 1)$ with $h \in H$ constitute a subgroup H^* of G isomorphic to H since

$$(h_1, 1)(h_2, 1) = (h_1h_2, 1).$$

Hence, $G = H^*N^*$ and $H^* \cap N^* = 1$.

In this case, we say G is a semidirect product (or split extension) of N with H via the automorphisms $\alpha(h)$.

If $\alpha(h) = 1$ for all $h \in H$, then we obtain the direct product $N \times H$.

Conversely, if N is a normal subgroup of G and G contains a subgroup H such that $G = HN$ and $H \cap N = 1$, then the elements of G can be uniquely written in the form hn with $h \in H$ and $n \in N$. The multiplication satisfies the equation

$$(h_1n_1)\,(h_2n_2) = h_1h_2n_1^{h_2}n_2.$$

If we define the automorphisms $\alpha(h)$ of N by $n^{\alpha(h)} = n^h$ for all $h \in H$, then the map $(h, n) \mapsto hn$ is an isomorphism from the semidirect product of N with H via the automorphisms $\alpha(h)$ to the group G.

We single out this important special case in the following definition.

Definition 14.5. Let N be a normal subgroup of G. A subgroup H of G such that $G = HN$ and $H \cap N = 1$ is called a complement of N in G. The elements of H are clearly a transversal for the cosets of N in G.

Lemma 14.6. *Let N be a normal subgroup of G with $G/N = H$ and $G = \bigcup_{h\in H} g(h)\,N$. Let $n : H \times H \to N$ be the factor system corresponding to the transversal $\{g(h) \mid h \in H\}$. Then there exists a complement of N in G if and only if there exist elements $a(h) \in N$ such that*

$$a(h_1h_2)^{-1}\,n(h_1, h_2)\,a(h_1)^{g(h_2)}\,a(h_2) = 1 \quad \textit{for all } h_1, h_2 \in H.$$

Proof. A complement of N is a transversal with the trivial factor system. By Lemma 14.3, corresponding to the transversal $\{\widetilde{g}(h) \mid h \in H\}$ with $\widetilde{g}(h) = g(h)a(h)$ there is the factor system $\widetilde{n}(_, _)$ with

$$\widetilde{n}(h_1, h_2) = a(h_1h_2)^{-1}\,n(h_1, h_2)\,a(h_1)^{g(h_2)}\,a(h_2).$$

Therefore, the existence of a transversal $\{\widetilde{g}(h) \mid h \in H\}$ with $\widetilde{n}(h_1, h_2) = 1$ for all $h_1, h_2 \in H$ is equivalent to the solvability of the given equations. □

Example 14.7. Assume that the normal subgroup N is abelian. Then equation (2) from Theorem 14.2 simplifies to

$$n^{\alpha(h_1)\alpha(h_2)} = n^{\alpha(h_1h_2)}.$$

Then $h \mapsto \alpha(h)$ is a homomorphism from H to the automorphism group $\mathbf{Aut}(N)$ of N, and N is an H-group in the sense of Definition 10.1. The condition in Lemma 14.6 for the existence of a complement of N can then be written as

$$n(h_1, h_2)^{-1} = a(h_1, h_2)^{-1}\, a(h_1)^{\alpha(h_2)}\, a(h_2).$$

Example 14.8. Suppose the group G has a normal subgroup N with cyclic factor group $G/N = \langle gN \rangle = \langle \overline{g} \rangle$ of order m. Then $g^m = a \in N$ is an element of N. If we choose the elements $1, g, g^2, \ldots, g^{m-1}$ as a transversal for N, then we obtain the factor system

$$n(\overline{g}^i, \overline{g}^j) = \begin{cases} 1 & i + j < m \\ a & i + j \geq m \end{cases} \qquad (0 \leq i, j < m).$$

Furthermore, $a^g = (g^m)^g = a$.

Conversely, if $a \in N$ is given along with automorphisms $\alpha(\overline{g}^i)$ of N such that

$$\begin{aligned} a^{\alpha(\overline{g}^i)} &= a, \\ \alpha(\overline{g}^i)\alpha(\overline{g}^j) &= \alpha(\overline{g}^{i+j}) \quad \text{for} \quad i + j < m \end{aligned}$$

and

$$n^{\alpha(\overline{g}^i)\alpha(\overline{g}^j)} = (n^{\alpha(\overline{g}^k)})^a \quad \text{for} \quad i + j = m + k \geq m,$$

then if we define

$$n(\overline{g}^i, \overline{g}^j) = \begin{cases} 1 & i + j < m \\ a & i + j \geq m, \end{cases}$$

we obtain a factor system. Since $\alpha(\overline{g}^i) = \alpha(\overline{g})^i$ for $i < m$, the extension is completely determined by the automorphism $\alpha(\overline{g})$ of N and the element $a \in N$, where $n^{\alpha(\overline{g})^m} = n^a$ and $a = a^{\alpha(\overline{g})}$.

Theorem 14.9.

a) *For each prime $p > 2$, there exists up to isomorphism exactly one nonabelian group of order p^{m+1} $(m \geq 2)$ containing a cyclic normal subgroup $N = \langle n \rangle$ of order p^m, namely the semidirect product of N with the cyclic group $\langle s \rangle$ such that*

$$n^s = n^{1+p^{m-1}} \quad \textit{and} \quad s^p = 1.$$

b) *For $m \geq 3$, there exist exactly four nonisomorphic nonabelian groups of order 2^{m+1} with a cyclic normal subgroup $N = \langle n \rangle$ of order 2^m, namely:*
(1) *The dihedral group $D = \langle n, s \rangle$ with relations*

$$n^s = n^{-1} \quad \textit{and} \quad n^{2^m} = s^2 = 1.$$

(2) *The generalized quaternion group* $Q = \langle n, s\rangle$ *with relations*

$$n^s = n^{-1}, \quad n^{2^m} = 1, \quad s^2 = n^{2^{m-1}}.$$

(3) $G = \langle n, s\rangle$ *with relations*

$$n^s = n^{1+2^{m-1}} \quad \textit{and} \quad n^{2^m} = s^2 = 1.$$

(4) *The quasidihedral* (*also called semidihedral*) *group* $G = \langle n, s\rangle$ *with relations*

$$n^s = n^{-1+2^{m-1}} \quad \textit{and} \quad n^{2^m} = s^2 = 1.$$

The groups of types (1), (3) *and* (4) *are semidirect products, while that of type* (2) *is not. For* $m = 2$, *only types* (1) *and* (2) *exist. The generalized quaternion group of order* 8 *is called the quaternion group.*

Proof. a) Let $p > 2$. Let $N = \langle n\rangle$ be the cyclic normal subgroup of G with index p, and let $s \in G$ such that $G/N = \langle sN\rangle$. Then $n^s = n^k$ for some integer k. Since $s^p \in N$, we have

$$n^{s^p} = n^{k^p} = n,$$

so that $k^p \equiv 1 \bmod p^m$. It follows that $k \equiv 1 \bmod p^{m-1}$, so $k \equiv 1 + xp^{m-1} \bmod p^m$. Since G is nonabelian, certainly $x \not\equiv 0 \bmod p$. If we define y such that $xy \equiv 1 \bmod p$, then we have

$$k^y \equiv 1 + xyp^{m-1} \equiv 1 + p^{m-1} \mod p^m.$$

Then $G/N = \langle s^y N\rangle$, as well. Replacing s with s^y, we obtain

$$n^s = n^{1+p^{m-1}}.$$

Now we try to replace s with some $n^z s$ such that $(n^z s)^p = 1$. This requires that

$$\begin{aligned}(n^z s)^p &= n^z (n^z)^{s^{-1}} (n^z)^{s^{-2}} \cdots (n^z)^{s^{-(p-1)}} s^p \\ &= n^z n^{z(1-p^{m-1})} n^{z(1-2p^{m-1})} \cdots n^{z(1-(p-1)p^{m-1})} s^p = 1.\end{aligned}$$

Since $p > 2$, we have

$$\begin{aligned}1 + (1 - p^{m-1}) + (1 - 2p^{m-1}) + \cdots + (1 - (p-1)p^{m-1}) \\ \equiv p - \frac{p(p-1)}{2} p^{m-1} \equiv p \mod p^m.\end{aligned}$$

We define $a = s^p$, so it remains to solve $n^{zp} a = 1$. We have

$$a^s = a = a^{1+p^{m-1}}, \quad \text{so} \quad a^{p^{m-1}} = 1.$$

Hence, $a \in N$ is a p-th power, and $n^{zp} = a^{-1}$ has a solution. So we obtain for $p > 2$ only the semidirect product corresponding to the automorphism α defined by

$$n^\alpha = n^{1+p^{m-1}}.$$

b) Let $p = 2$. By Theorem 13.19, the only elements of order 2 in the automorphism group of the cyclic group N for $2^m > 4$ are the automorphisms α, ϵ, δ defined by

$$n^\alpha = n^{-1}, \quad n^\epsilon = n^{1+2^{m-1}}, \quad n^\delta = n^{-1+2^{m-1}} \quad (\delta = \alpha\epsilon).$$

For $2^m = 4$, we have $\alpha = \epsilon$ and $\delta = 1$.

Let $G = N \cup sN$ and $s^2 = a \in N$. Then again $a^s = a$. If $a^s = a^\alpha$, it follows that

$$a = a^s = a^{-1}, \quad \text{so} \quad a^2 = 1.$$

There remain only the possibilities that $a = 1$ and $a = n^{2^{m-1}}$. The first case yields the dihedral group D with

$$n^s = n^{-1}, \quad n^{2^m} = s^2 = 1,$$

and the second case yields the generalized quaternion group Q with

$$n^s = n^{-1}, \quad n^{2^m} = 1, \quad s^2 = n^{2^{m-1}}.$$

Next, we show that for the other two automorphisms ϵ and δ we obtain a semidirect product.

Using ϵ, we obtain

$$(n^x s)^2 = n^x (n^x)^{s^{-1}} s^2 = n^{x(2+2^{m-1})} a.$$

Furthermore,

$$a^\epsilon = a^{1+2^{m-1}} = a, \quad \text{so} \quad a^{2^{m-1}} = 1.$$

Therefore, a is a square in N, and

$$a = n^{-x(2+2^{m-1})}$$

has a solution since $m - 1 \geq 2$. So $\langle n^x s \rangle$ is a complement of N in G.

Using δ, we obtain

$$(n^x s)^2 = n^{x2^{m-1}} a$$

and

$$a^\delta = a^{-1+2^{m-1}} = a, \quad \text{so} \quad a^{-2+2^{m-1}} = 1.$$

Since $m \geq 3$, we have $a^2 = 1$. Therefore, a is a 2^{m-1}-th power in N, and

$$a = n^{-x2^{m-1}}$$

has a solution.

The four groups we have constructed are pairwise nonisomorphic for $m > 2$:

(1) Let $D = \langle n, s \rangle$ be the dihedral group of order 2^{m+1}. Then D has exactly $2^m + 1$ elements of order 2, namely

$$n^{2^{m-1}} \quad \text{and all} \quad n^i s \quad (0 \leq i \leq 2^m - 1).$$

(2) The generalized quaternion group $Q = \langle n, s \rangle$ has exactly one element of order 2, namely $n^{2^{m-1}}$.

(3) Let $G = \langle n, s \rangle$ with

$$n^s = n^{1+2^{m-1}} \quad \text{and} \quad n^{2^m} = s^2 = 1.$$

Then G has exactly 3 elements of order 2, namely $n^{2^{m-1}}$, s and $n^{2^{m-1}} s$.

(4) Let $G = \langle n, s \rangle$ be the quasidihedral group. Then G has exactly $2^{m-1} + 1$ elements of order 2, namely $n^{2^{m-1}}$ and all $n^i s$ with $2 \mid i$. □

Theorem 14.10. *There exist only the following nonabelian groups of order p^3, up to isomorphism:*

a) *For $p = 2$, the dihedral group of order 8 and the quaternion group.*

b) *For $p > 2$, the group $G = \langle n, s \rangle$ with relations*

$$n^s = n^{1+p}, \quad n^{p^2} = s^p = 1$$

from Theorem 14.9 a), as well as the group $G = \langle a, b, c \rangle$ of exponent p with relations

$$a^p = b^p = c^p = 1, \quad b^a = bc, \quad c^a = c^b = c.$$

(Since $c = b^{-1}a^{-1}ba$, we also have $G = \langle a, b \rangle$.)

Proof. a) Let $|G| = 2^3$. If G has exponent 2, then we have for all $x, y \in G$ that

$$xy = (xy)^{-1} = y^{-1}x^{-1} = yx,$$

and G is abelian. Therefore, there exists an $x \in G$ with $|x| = 4$. By Theorem 3.3, $\langle x \rangle \lhd G$ and we may apply Theorem 14.9 b).

b) Let G be nonabelian with $|G| = p^3$ and $p > 2$. By Theorem 6.9, we have $p \leq |\mathbf{Z}(G)| \leq p^2$. If $G/\mathbf{Z}(G)$ were cyclic, then for some $x \in G$ we would have $G = \langle \mathbf{Z}(G), x \rangle$, and then G would be abelian. Therefore, $G/\mathbf{Z}(G)$ is not cyclic. It follows that $|G/\mathbf{Z}(G)| = p^2$, and $G/\mathbf{Z}(G)$ is abelian by Theorem 6.10. There are two cases:

Case 1: Assume G has exponent p. Let $\mathbf{Z}(G) = \langle c \rangle$ and $b \in G - \mathbf{Z}(G)$. Then $N = \langle b, c \rangle$ is an elementary abelian subgroup of G of index p. By Theorem 8.9, in fact $N \lhd G$. Let $a \in G - N$. Since $G/\mathbf{Z}(G)$ is abelian but G is not, $b^a = bc^i$ with $i \not\equiv 0 \bmod p$. If we replace c with c^i, we preserve this equality. Therefore, G is the semidirect product of N with the cyclic group $\langle a \rangle$ via the automorphism α of N defined by

$$b^\alpha = bc, \quad c^\alpha = c.$$

Case 2: Assume G has exponent $p^2 = |a|$. Then $\langle a \rangle$ is a normal subgroup of index p in G by Theorem 8.9. Then by Theorem 14.9 a), there exists one such group up to isomorphism. □

Exercises

55) Let Q be the quaternion group and D the dihedral group of order 8. Show that

$$Q \curlyvee Q \cong D \curlyvee D \ncong Q \curlyvee D.$$

(All products are constructed with identified centers.)

56) If G is a group from Theorem 14.9 b), then $|\mathbf{Aut}(G)|$ is a power of 2, except when G is the quaternion group of order 8, in which case $|\mathbf{Aut}(G)| = 24$.

57) Let $p > 2$, $n > 2$ and $G = \langle a, b \rangle$ with relations

$$a^{p^n} = 1, \quad b^{p^n} = a^{p^{n-1}}, \quad a^b = a^{1+p}.$$

Then G has no cyclic normal subgroup with a cyclic complement.

58) There is no group G such that $G/\mathbf{Z}(G)$ is a generalized quaternion group.

59) (K. H. Hofmann, Mostert [250]) Let N be an abelian normal subgroup of G and $G = \bigcup_{x \in G/N} g(x)N$ with $x = g(x)N$. We define the map $F : G \times G/N \to N$ by

$$F(y, x) = y\, g(x)\, g(\overline{y}x)^{-1},$$

where $\overline{y} = yN$. Then:

a) $F(gh, x) = F(h, x)^{g^{-1}} F(g, x)$ and $F(n, x) = n$ for all $n \in N$.

b) If $q = |G/N|$ is coprime to $|N|$, then we define m by the equation $qm \equiv 1 \bmod |N|$ and define

$$s(g) = \left(\prod_{x \in G/N} F(g, x) \right)^m.$$

Then $s(gh) = s(h)^{g^{-1}} s(g)$ and $s(n) = n$ for all $n \in N$.

c) If the assumptions of b) hold, then the map t defined by $t(g) = s(g)^{-1}\, g$ is an endomorphism of G with $t(n) = 1$ for all $n \in N$. Then

$$V = \{t(g) \mid g \in G\}$$

is a complement of N in G. (This is a proof of Theorem 17.5.)

15 Wreath products

In this section, we describe a special case of the extension theory developed in the last section for which the construction of examples (in particular of solvable groups) is especially useful.

Definition 15.1. Let G be a group, and let H be a (not necessarily transitive) permutation group on the finite set Ω. The wreath product $G \wr H$ of G with H is the set

$$\{(f, h) \mid h \in H,\ f : \Omega \to G\}$$

with the multiplication

$$(f_1, h_1)(f_2, h_2) = (f', h_1 h_2),$$

where $f'(i) = f_1(i) f_2(i^{h_1})$ for $i \in \Omega$.

It is an easy calculation to show that this multiplication is associative, that $(f, 1)$ with $f(i) = 1$ for all $i \in \Omega$ is the identity element of $G \wr H$, and that the inverse of (f, h) is $(\widetilde{f}, h^{-1})$ with $\widetilde{f}(i) = f(i^{h^{-1}})^{-1}$. Therefore, $G \wr H$ is a group.

The structure of $G \wr H$ is clarified by the following theorem.

Theorem 15.2. *Let H be a permutation group on Ω with $|\Omega| = n$. The wreath product $G \wr H$ contains a normal subgroup*

$$D = D_1 \times \cdots \times D_n,$$

where

$$D_i = \{(f, 1) \mid f(j) = 1 \text{ for } j \neq i\} \cong G.$$

The subgroup

$$H^* = \{(e, h) \mid h \in H,\ e(i) = 1 \text{ for all } i \in \Omega\}$$

is a complement of D in $G \wr H$ that is isomorphic to H. Thus, $|G \wr H| = |G|^n |H|$. Conjugation by $h^* = (e, h) \in H^*$ permutes the direct factors D_i of D according to $D_i^{h^*} = D_{i^h}$.

Proof. Let $\Omega = \{1, \dots, n\}$. We define

$$D = \{(f, 1) \mid f : \Omega \to G\}$$

and

$$D_i = \{(f, 1) \mid f(j) = 1 \text{ for } j \neq i\}.$$

Clearly, $D_i \cong G$ and $D = D_1 \times \cdots \times D_n$.

The map ϵ defined by $(f, h)^\epsilon = h$ is an epimorphism from $G \wr H$ onto H with kernel D. Since

$$(f, h) = (f, 1)(e, h) \quad \text{with} \quad e(i) = 1 \quad \text{for all} \quad i \in \Omega,$$

the group

$$H^* = \{(e, h) \mid e(i) = 1 \text{ for all } i \in \Omega\}$$

is a complement of D in $G \wr H$.

For $h^* = (e, h) \in H^*$ and $d = (f, 1) \in D_i$, we have

$$(h^*)^{-1}\, d\, h^* = (f', 1),$$

where $f'(j) = f(j^{h^{-1}})$. Thus, $D_i^{h^*} = D_{i^h}$. □

We will often write $D = G \wr 1$, where 1 is to be understood as the trivial permutation group on the set Ω.

Theorem 15.3. *Let H be a permutation group on Ω and G a group. Assume we have a permutation representation of G on a set Γ,*

$$g \mapsto \begin{pmatrix} i \\ i^g \end{pmatrix} \quad (g \in G,\ i \in \Gamma).$$

Then the action of $G \wr H$ on the product set $\Gamma \times \Omega$ defined by

$$(i, j)(f, h) = (i^{f(j)}, j^h)$$

for all $(i, j) \in \Gamma \times \Omega$ yields a permutation representation of $G \wr H$ on $\Gamma \times \Omega$. If G acts faithfully on Γ, then $G \wr H$ acts faithfully on $\Gamma \times \Omega$. If G acts transitively on Γ and H acts transitively on Ω, then $G \wr H$ acts transitively on $\Gamma \times \Omega$.

Proof. We have

$$\begin{aligned}\big((i, j)\,(f, h)\big)\,(f', h') &= (i^{f(j)}, j^h)\,(f', h') = (i^{f(j)f'(j^h)}, j^{hh'}) \\ &= (i, j)\,\big((f, h)\,(f', h')\big).\end{aligned}$$

Since the map (f, h) on $\Gamma \times \Omega$ has an inverse, the given action induces a permutation representation of $G \wr H$ on $\Gamma \times \Omega$.

Assume that G acts faithfully on Γ. Suppose that

$$(i,j)\,(f,h) = (i^{f(j)}, j^h) = (i,j)$$

for all $(i,j) \in \Gamma \times \Omega$. Then $h = 1$ and $f(j) = 1$ for all $j \in \Omega$. So $G \wr H$ acts faithfully on $\Gamma \times \Omega$.

Assume that G acts transitively on Γ and H acts transitively on Ω. Let (i,j) and (i',j') in $\Gamma \times \Omega$ be given. Then there is some $h \in H$ with $j^h = j'$ and some $g \in G$ with $i^g = i'$. Let f be a map from Ω to G with $f(j) = g$. Then

$$(i,j)\,(f,h) = (i^g, j^h) = (i',j').$$

So $G \wr H$ acts transitively on $\Gamma \times \Omega$. □

If we started with the permutation groups G on Γ and H on Ω, we could have defined the wreath product $G \wr H$ as a permutation group on $\Gamma \times \Omega$ according to Theorem 15.3. If K is a permutation group on the set Δ, then $(G \wr H) \wr K$ is a permutation group on $(\Gamma \times \Omega) \times \Delta$. We claim that:

Theorem 15.4. *If G, H and K are permutation groups on the sets Γ, Ω and Δ, respectively, then the permutation groups $(G \wr H) \wr K$ and $G \wr (H \wr K)$ coincide if we identify $(\Gamma \times \Omega) \times \Delta$ and $\Gamma \times (\Omega \times \Delta)$ in the natural way:*

$$\big((\gamma,\omega),\delta\big) = \big(\gamma,(\omega,\delta)\big)$$

for all $\gamma \in \Gamma$, $\omega \in \Omega$ and $\delta \in \Delta$.

Proof. Let $x = (f,k) \in (G \wr H) \wr K$ with

$$f(\delta) = (f_\delta, h_\delta) \in G \wr H \quad \text{for} \quad \delta \in \Delta.$$

Then

$$((\gamma,\omega),\delta)\,x = ((\gamma,\omega)^{f(\delta)}, \delta^k) = ((\gamma^{f_\delta(\omega)}, \omega^{h_\delta}), \delta^k).$$

On the other hand, let

$$y = (g,(h,k)) \in G \wr (H \wr K)$$

with $h(\delta) = h_\delta$ and $g((\omega,\delta)) = f_\delta(\omega)$, then

$$(\gamma,(\omega,\delta))\,y = (\gamma^{g((\omega,\delta))}, (\omega,\delta)^{(h,k)}) = (\gamma^{f_\delta(\omega)}, (\omega^{h_\delta}, \delta^k)).$$

This proves the claim. □

Lemma 15.5. *If G_p is a Sylow p-subgroup of the finite group G and H_p is a Sylow p-subgroup of the finite group H, then $G_p \wr H_p$ is a Sylow p-subgroup of $G \wr H$. In the construction of $G_p \wr H_p$, one should interpret H_p as a subgroup of the permutation group H on Ω.*

Proof. Letting $|\Omega| = n$, we have $|G_p \wr H_p| = |G_p|^n |H_p|$. This is the highest power of p dividing $|G \wr H|$ since

$$|G \wr H| = |G|^n |H|. \qquad \square$$

The following special case is useful in numerous constructions.

Definition 15.6. Let G be an arbitrary group, and let H be a finite group. We define the regular wreath product $G \wr_r H$ to be the wreath product of G with the regular representation of H (see Theorem 6.3). Therefore, we have

$$|G \wr_r H| = |G|^{|H|} |H|.$$

This wreath product is the set

$$G \wr_r H = \{(f, h) \mid h \in H,\ f : H \to G\}$$

with the multiplication

$$(f_1, h_1)(f_2, h_2) = (f', h_1 h_2), \quad \text{where} \quad f'(h) = f_1(h)\, f_2(h h_1).$$

It is important to note that for the regular wreath product there is no associativity when combining the constructions as in Theorem 15.4. Namely, we have

$$|(G \wr_r H) \wr_r K| = |G \wr_r H|^{|K|} |K| = |G|^{|H||K|} |H|^{|K|} |K|,$$

while

$$|G \wr_r (H \wr_r K)| = |G|^{|H \wr_r K|} |H \wr_r K| = |G|^{|H|^{|K|} |K|} |H|^{|K|} |K|.$$

Next, we prove some lemmas about the wreath product that will be useful in later constructions.

Lemma 15.7. *Let H be a permutation group on the set Ω.*

a) *If α is an epimorphism from G onto G^α, then the map $\overline{\alpha}$ defined by*

$$(f, h)^{\overline{\alpha}} = (g, h), \quad \textit{where} \quad g(i) = f(i)^\alpha$$

for all $i \in \Omega$ is an epimorphism from $G \wr H$ onto $G^\alpha \wr H$.

b) *If μ is an automorphism of H, then the map $\overline{\mu}$ defined by*

$$(f, h)^{\overline{\mu}} = (g, h^\mu), \quad \textit{where} \quad g(y) = f(y^{\mu^{-1}}) \quad \textit{for all} \quad y \in H$$

is an automorphism of $G \wr_r H$.
(Here, we must restrict ourselves to the regular wreath product.)

c) *Let α be an automorphism of G, and let μ be an automorphism of H. If $\overline{\alpha}$ and $\overline{\mu}$ are constructed as in a) and b), then $\overline{\alpha}\,\overline{\mu} = \overline{\mu}\,\overline{\alpha}$.*

Proof. a) is trivial.

b) It is clear that μ is bijective. We have

$$(f_1, h_1)^{\overline{\mu}}\,(f_2, h_2)^{\overline{\mu}} = (g_1, h_1^{\mu})\,(g_2, h_2^{\mu}) = (g', (h_1h_2)^{\mu}),$$

where

$$g'(y) = g_1(y)\,g_2(yh_1^{\mu}) = f_1(y^{\mu^{-1}})\,f_2(y^{\mu^{-1}}h_1)$$

and

$$((f_1, h_1)\,(f_2, h_2))^{\overline{\mu}} = (k, h_1h_2)^{\overline{\mu}} = (m, (h_1h_2)^{\mu}),$$

where

$$m(y) = k(y^{\mu^{-1}}) = f_1(y^{\mu^{-1}})\,f_2(y^{\mu^{-1}}h_1).$$

So $\overline{\mu}$ is an automorphism of $G \wr_r H$.

c) We have

$$(f, h)^{\overline{\alpha}\,\overline{\mu}} = (g, h^{\mu}) = (f, h)^{\overline{\mu}\,\overline{\alpha}}, \quad \text{where} \quad g(y) = f(y^{\mu^{-1}})^{\alpha}. \qquad \square$$

Lemma 15.8.

a) *Let $G_1 \leq G$ and $H_1 \leq H$. There exists a monomorphism from $G_1 \wr H_1$ to $G \wr H$. Here, we interpret H and H_1 as permutation groups on the same set.*

b) *Let $G_1 \leq G$ and $H_1 \leq H$. There exists a monomorphism from $G_1 \wr_r H_1$ to $G \wr_r H$.*

Proof. a) The trivial embedding gives the desired homomorphism.

b) This is not a special case of a)! For in the construction of $G \wr_r H$, we interpret H as a regular permutation group on H, while we interpret H_1 in $G \wr_r H_1$ as a regular permutation group on H_1.

Let $H = \bigcup_i r_i H_1$ be the decomposition of H into cosets of H_1. If f is a map from H_1 to G_1, then we define the map $\overline{f}$ from H to G (in fact, to G_1) by

$$\overline{f}(r_ih_1) = f(h_1) \quad \text{for} \quad h_1 \in H_1.$$

We define

$$(f, h_1)^{\delta} = (\overline{f}, h_1).$$

Then we have

$$(f, h_1)^{\delta}(f', h_1')^{\delta} = (g, h_1h_1'),$$

where

$$g(r_ih) = \overline{f}(r_ih)\,\overline{f}'(r_ihh_1) = f(h)\,f'(hh_1),$$

and

$$((f, h_1)\,(f', h_1'))^{\delta} = (g', h_1 h_1'),$$

where

$$g'(r_i h) = f(h)\, f'(h h_1).$$

Hence, δ is a homomorphism from $G_1 \wr_r H_1$ to $G \wr_r H$. The kernel is clearly $\ker \delta = 1$.□

The following fact is sometimes useful for answering questions about embeddings and anticipates the considerations of Chapter IV, §1.

Theorem 15.9. *Let $N \trianglelefteq G$, and assume G/N is finite. Then there exists a monomorphism from G to $N \wr_r (G/N)$.*

Proof. Let $G = \bigcup_{\overline{g} \in G/N} N\, r(\overline{g})$ be the decomposition of G into cosets of N. For all $x \in G$, we define the elements $h_{\overline{g}}(x) \in N$ by

$$r(\overline{g})\, x = h_{\overline{g}}(x)\, r(\overline{g}\,\overline{x}).$$

We define the map $\tau : G \to N \wr_r G/N$ by

$$x^{\tau} = (f, \overline{x}), \quad \text{where} \quad \overline{x} = xN \quad \text{and} \quad f(\overline{g}) = h_{\overline{g}}(x).$$

Then

$$x^{\tau} y^{\tau} = (f_1, \overline{x})\,(f_2, \overline{y}) = (f', \overline{x}\,\overline{y}),$$

where

$$f'(\overline{g}) = f_1(\overline{g})\, f_2(\overline{g}\,\overline{x}) = h_{\overline{g}}(x)\, h_{\overline{g}\,\overline{x}}(y).$$

Since

$$\begin{aligned} h_{\overline{g}}(xy)\, r(\overline{g}\,\overline{x}\,\overline{y}) &= r(\overline{g})\, x\, y = h_{\overline{g}}(x)\, r(\overline{g}\,\overline{x}) y \\ &= h_{\overline{g}}(x)\, h_{\overline{g}\,\overline{x}}(y)\, r(\overline{g}\,\overline{x}\,\overline{y}), \end{aligned}$$

we have $f'(\overline{g}) = h_{\overline{g}}(xy)$, and so τ is a homomorphism from G to $N \wr_r (G/N)$. If $x \in \ker \tau$, then $r(\overline{g})\, x = r(\overline{g})$, so $x = 1$. □

Next, we introduce a generalization of the wreath product.

Definition 15.10. Let G and H be groups. Let U be a subgroup of H with finite index, and suppose we are given a homomorphism from U to the automorphism group $\mathbf{Aut}(G)$ of G. We write the action of U on G by $g \mapsto g^u$ for all $u \in U$. Let $H = \bigcup_{r \in R} Ur$ be the decomposition of H into cosets of U; we may choose 1 as a

representative of U. We define the twisted wreath product $G \wr_U H$ of G with H over U to be the set

$$\{(f, h) \mid h \in H,\ f : R \to G\}$$

with the following multiplication:

If $r \in R$, $h_1 \in H$ and $rh_1 = u'r'$ with $u' \in U$ and $r' \in R$, then

$$(f_1, h_1)(f_2, h_2) = (f', h_1h_2),$$

where $f'(r) = f_1(r)f_2(r')^{u'^{-1}}$.

A direct calculation shows that this multiplication is associative. The identity element is $(e, 1)$, where $e(r) = 1$ for all $r \in R$. Furthermore, the inverse of (f, h) is $(\widetilde{f}, h^{-1})$, where

$$\widetilde{f}(r') = (f(r)^{-1})^{u'} \quad \text{for} \quad rh = u'r'.$$

Therefore, $G \wr_U H$ is a group.

If $g^u = g$ for all $g \in G$ and all $u \in U$, then one obtains the wreath product of G with the permutation representation of H on the cosets of U in H. If $U = H$, then one obtains the semidirect product of G with H.

As in Definition 15.1, one can show that:

Theorem 15.11. *The set*

$$G^* = \{(f, 1) \mid f : R \to G\}$$

is a normal subgroup of $G \wr_U H$ isomorphic to $G \times \cdots \times G$ (with $[H : U]$ factors). Furthermore, the set

$$H^* = \{(e, h) \mid h \in H,\ e(r) = 1 \text{ for all } r \in R\}$$

is a complement of G^* in $G \wr_U H$ isomorphic to H.

With this construction, we obtain an embedding theorem similar to that of Theorem 15.9:

Theorem 15.12 (B. H. Neumann). *Let $G = NH$ be a semidirect product of the normal subgroup $N \trianglelefteq G$ with H, and suppose H is isomorphically embedded in the group W via the identity map. Then there exists a monomorphism μ from G to $N \wr_H W$.*

Proof. Let $W = \bigcup_{r \in R} Hr$ be the decomposition of W into cosets of H with $1 \in R$. For $n \in N$ and $h \in H$, we define $(nh)^\mu = (f, h)$, where $f(1) = n$ and $f(r) = 1$ for $r \notin H$. Then we have

$$(n_1h_1)^\mu(n_2h_2)^\mu = (f_1, h_1)(f_2, h_2) = (f', h_1h_2),$$

where

$$f'(r) = f_1(r) f_2(r')^{h'^{-1}} \quad \text{for} \quad rh_1 = h'r' \quad \text{with} \quad h' \in H.$$

For $r = 1$, we have $h_1 = h'$ and therefore

$$f'(1) = f_1(1) f_2(1)^{h_1^{-1}} = n_1 n_2^{h_1^{-1}},$$

while for $r \neq 1$ we have $r' \neq 1$ and so $f'(r) = 1$. Thus,

$$(n_1 h_1)^\mu \, (n_2 h_2)^\mu = (f', h_1 h_2) = (n_1 n_2^{h_1^{-1}} h_1 h_2)^\mu = (n_1 h_1 n_2 h_2)^\mu.$$

So μ is a homomorphism from G to $N \wr_H W$. Clearly, μ is a monomorphism. □

Exercises

60) Show that the regular wreath product $G \wr_r H$ has the following property: Let K be a group, α a homomorphism from G to K and μ a homomorphism from H to K. Assume $K = \langle G^\alpha, H^\mu \rangle$. Suppose that whenever $h, h' \in H$ are distinct, the groups

$$(G^\alpha)^{h^\mu} \quad \text{and} \quad (G^\alpha)^{h'^\mu}$$

commute. Then there exists an epimorphism $\tau : G \wr_r H \to K$ defined by

$$(f, h)^\tau = (f(h_1)^\alpha)^{h_1^\mu} \cdots (f(h_n)^\alpha)^{h_n^\mu} h^\mu,$$

where $|H| = n$ and $H = \{h_1, \ldots, h_n\}$.

(It is actually because of this "universal property" of $G \wr_r H$ that Lemma 15.7 holds.)

61) Let μ be an epimorphism from H onto $\overline{H}$. If G is abelian, then by Exercise 60, there exists an epimorphism $\tilde{\mu}$ from $G \wr_r H$ onto $G \wr_r \overline{H}$ given by

$$(f, h)^{\tilde{\mu}} = (\overline{f}, h^\mu).$$

State the function $\overline{f} : \overline{H} \to G$ explicitly.

16 Cohomology theory

In this entire section, let R be a ring with unity 1. All R-modules are right R-modules on which 1 acts as the identity map. If G is a group, we will assign to every G-module $\mathfrak{m}$ a series of abelian groups $\mathbf{H}^n(G, \mathfrak{m})$ $(n = 1, 2, \ldots)$, the so-

called cohomology groups of $\mathfrak{m}$. We will give group theoretical interpretations of $\mathbf{H}^1(G, \mathfrak{m})$ and $\mathbf{H}^2(G, \mathfrak{m})$ in §17. The higher cohomology groups have not played a role in the structure theory of finite groups at the time of writing.

Definition 16.1.

a) A chain complex of R-modules is a sequence $\mathscr{K} = \{\mathfrak{a}_n, \partial_n\}$ of R-modules $\mathfrak{a}_n$ for $n \geq 0$ and R-homomorphisms $\partial_n \in \mathrm{Hom}_R(\mathfrak{a}_n, \mathfrak{a}_{n-1})$ for $n \geq 1$; the map ∂_0 takes $\mathfrak{a}_0$ onto the 0 module. In addition, we require that $\partial_n \partial_{n-1} = 0$ for all $n \geq 1$. We define

$$\mathbf{H}_n(\mathscr{K}) = \ker \partial_n / \mathrm{im}\, \partial_{n+1} \quad (n \geq 1)$$

and call $\mathbf{H}_n(\mathscr{K})$ the n-th homology group of $\mathscr{K}$.

b) A cochain complex of R-modules is a sequence $\mathscr{K} = \{\mathfrak{a}_n, \delta_n\}$ of R-modules $\mathfrak{a}_n$ and R-homomorphisms $\delta_n \in \mathrm{Hom}_R(\mathfrak{a}_n, \mathfrak{a}_{n+1})$ for $n \geq 0$ such that $\delta_n \delta_{n+1} = 0$ for all $n \geq 0$. We define

$$\mathbf{H}^n(\mathscr{K}) = \ker \delta_n / \mathrm{im}\, \delta_{n-1} \quad (n \geq 1).$$

The group $\mathbf{H}^n(\mathscr{K})$ is called the n-th cohomology group of $\mathscr{K}$.

c) An exact sequence of R-modules is a (finite or infinite) sequence of R-modules $\mathfrak{a}_n$ $(n = 0, 1, \dots)$ and maps $\mu_n \in \mathrm{Hom}_R(\mathfrak{a}_n, \mathfrak{a}_{n-1})$ for $n \geq 1$ such that $\mathrm{im}\, \mu_n = \ker \mu_{n-1}$ for all $n \geq 2$.

d) Let $\mathfrak{m}$ be an R-module. A free resolution $\{\mathfrak{f}_n, \partial_n\}$ of $\mathfrak{m}$ is an exact sequence

$$\cdots \longrightarrow \mathfrak{f}_n \xrightarrow{\partial_n} \mathfrak{f}_{n-1} \xrightarrow{\partial_{n-1}} \cdots \longrightarrow \mathfrak{f}_1 \xrightarrow{\partial_1} \mathfrak{f}_0 \xrightarrow{\partial_0} \mathfrak{m} \longrightarrow 0$$

with free R-modules $\mathfrak{f}_n$.

In what follows, we formulate the theorems in terms of chain complexes and homology groups. Corresponding results hold for cochain complexes and cohomology groups.

Definition 16.2.

a) Let $\mathscr{K} = \{\mathfrak{a}_n, \partial_n\}$ and $\mathscr{K}' = \{\mathfrak{a}'_n, \partial'_n\}$ be chain complexes of R-modules. A homomorphism φ from $\mathscr{K}$ to $\mathscr{K}'$ is defined to be a sequence $\{\varphi_n \mid n = 0, 1, \dots\}$ of homomorphisms $\varphi_n \in \mathrm{Hom}_R(\mathfrak{a}_n, \mathfrak{a}'_n)$ such that $\partial_n \varphi_{n-1} = \varphi_n \partial'_n$ for $n \geq 1$; that is, the following diagram commutes:

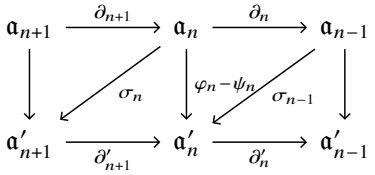
$$\begin{array}{ccc} \mathfrak{a}_n & \xrightarrow{\partial_n} & \mathfrak{a}_{n-1} \\ \downarrow{\scriptstyle \varphi_n} & & \downarrow{\scriptstyle \varphi_{n-1}} \\ \mathfrak{a}'_n & \xrightarrow{\partial'_n} & \mathfrak{a}'_{n-1} \end{array}$$

We denote the set of all homomorphisms from $\mathscr{K}$ to $\mathscr{K}'$ by $\mathrm{Hom}(\mathscr{K}, \mathscr{K}')$.

b) Let $\mathscr{K}$ and $\mathscr{K}'$ be chain complexes of R-modules, and let φ and ψ be homomorphisms from $\mathscr{K}$ to $\mathscr{K}'$. We say that the sequence $\{\sigma_n \mid n \geq 0\}$ of maps $\sigma_n \in \mathrm{Hom}_R(\mathfrak{a}_n, \mathfrak{a}'_{n+1})$ is a homotopy from φ to ψ if for all $n \geq 0$ we have

$$\varphi_n - \psi_n = \partial_n \sigma_{n-1} + \sigma_n \partial'_{n+1},$$

where we define $\sigma_{-1} = 0$ (see the following diagram).



In this case, we write $\varphi \sim \psi$. Clearly, $\sim$ is an equivalence relation.

Lemma 16.3.

a) *Let $\mathscr{K}$ and $\mathscr{K}'$ be chain complexes of R-modules, and let φ be a homomorphism from $\mathscr{K}$ to $\mathscr{K}'$. Then for all $n \geq 1$, the homomorphism φ induces a homomorphism φ_n^* from $\mathbf{H}_n(\mathscr{K})$ to $\mathbf{H}_n(\mathscr{K}')$.*

b) *Let $\varphi \in \mathrm{Hom}(\mathscr{K}, \mathscr{K}')$ and $\psi \in \mathrm{Hom}(\mathscr{K}', \mathscr{K})$ with $\varphi\psi \sim 1_{\mathscr{K}}$ and $\psi\varphi \sim 1_{\mathscr{K}'}$. Then the φ_n^* constructed in a) is an isomorphism from $\mathbf{H}_n(\mathscr{K})$ to $\mathbf{H}_n(\mathscr{K}')$ for all $n \geq 1$.*

Proof. a) For every $x \in \ker \partial_n$, we define

$$(x + \mathrm{im}\, \partial_{n+1})\varphi_n^* = x\varphi_n + \mathrm{im}\, \partial'_{n+1}.$$

If $x = y\partial_{n+1} \in \mathrm{im}\, \partial_{n+1}$, then we have

$$x\varphi_n = y\partial_{n+1}\varphi_n = y\varphi_{n+1}\partial'_{n+1} \in \mathrm{im}\, \partial'_{n+1}.$$

So the map φ_n^* is well defined. Clearly, φ_n^* is then a homomorphism from $\mathbf{H}_n(\mathscr{K})$ to $\mathbf{H}_n(\mathscr{K}')$.

b) Assume $\varphi\psi \sim 1_{\mathscr{K}}$, so

$$\varphi_n\psi_n - 1 = \partial_n\sigma_{n-1} + \sigma_n\partial_{n+1}.$$

Then for $x \in \ker \partial_n$, we have

$$\begin{aligned}(x + \mathrm{im}\, \partial_{n+1})\, \varphi_n^*\psi_n^* &= x\varphi_n\psi_n + \mathrm{im}\, \partial_{n+1} \\ &= x + x\partial_n\sigma_{n-1} + x\sigma_n\partial_{n+1} + \mathrm{im}\, \partial_{n+1} \\ &= x + \mathrm{im}\, \partial_{n+1}.\end{aligned}$$

In the same way, it follows that $\psi_n^*\varphi_n^*$ is the identity map on $\mathbf{H}_n(\mathscr{K}')$. Hence, φ_n^* is an isomorphism from $\mathbf{H}_n(\mathscr{K})$ to $\mathbf{H}_n(\mathscr{K}')$. □

Lemma 16.4. *Let* $\mathfrak{f}$ *be a free* R*-module, and let both* $\mathfrak{a}$ *and* $\mathfrak{b}$ *be* R*-modules. Let* $\varphi \in \mathrm{Hom}_R(\mathfrak{f}, \mathfrak{b})$*, and let* ϵ *be an epimorphism from* $\mathfrak{a}$ *onto* $\mathfrak{b}$*. Then there exists some* $\psi \in \mathrm{Hom}_R(\mathfrak{f}, \mathfrak{a})$ *such that* $\psi\epsilon = \varphi$ *on all of* $\mathfrak{f}$*.*

(*A module* $\mathfrak{f}$ *is said to be projective if such a map* ψ *exists for every* $\mathfrak{a}$, $\mathfrak{b}$, φ *and* ϵ.)

Proof. Let $\{f_i \mid i \in I\}$ be a free generating set of $\mathfrak{f}$. Since ϵ is an epimorphism, there exists some $a_i \in \mathfrak{a}$ such that $a_i\epsilon = f_i\varphi$. Since $\mathfrak{f}$ is free in the free generators f_i, there exists some $\psi \in \mathrm{Hom}_R(\mathfrak{f}, \mathfrak{a})$ such that $f_i\psi = a_i$ by Definition 10.15. Therefore, $f_i\psi\epsilon = a_i\epsilon = f_i\varphi$ for all i, and so $\psi\epsilon = \varphi$. □

Theorem 16.5. *Let* $\mathscr{K} = \{\mathfrak{f}_n, \partial_n\}$ *and* $\mathscr{K}' = \{\mathfrak{f}'_n, \partial'_n\}$ *be free resolutions of the* R*-modules* $\mathfrak{m}$ *and* $\mathfrak{m}'$*, respectively. Let* $\mu \in \mathrm{Hom}_R(\mathfrak{m}, \mathfrak{m}')$*. Then there exists some* $\alpha \in \mathrm{Hom}(\mathscr{K}, \mathscr{K}')$ *such that* $\alpha_0\partial'_0 = \partial_0\mu$*. Any two such* α *are homotopic.*

Proof. a) We define $\mathfrak{f}_{-1} = \mathfrak{m}$, $\mathfrak{f}'_{-1} = \mathfrak{m}'$ and $\alpha_{-1} = \mu$. Since $\mathfrak{f}_0$ is free and ∂'_0 is an epimorphism, there exists some $\alpha_0 \in \mathrm{Hom}_R(\mathfrak{f}_0, \mathfrak{f}'_0)$ such that $\alpha_0\partial'_0 = \partial_0\mu$ by Lemma 16.4.

Let $n \geq 1$, and assume for induction that $\alpha_0, \ldots, \alpha_{n-1}$ have already been determined.

$$\begin{array}{ccccc}
\mathfrak{f}_n & \xrightarrow{\partial_n} & \mathfrak{f}_{n-1} & \xrightarrow{\partial_{n-1}} & \mathfrak{f}_{n-2} \\
\downarrow{\scriptstyle \alpha_n} & & \downarrow{\scriptstyle \alpha_{n-1}} & & \downarrow{\scriptstyle \alpha_{n-2}} \\
\mathfrak{f}'_n & \xrightarrow[\partial'_n]{} & \mathfrak{f}'_{n-1} & \xrightarrow[\partial'_{n-1}]{} & \mathfrak{f}'_{n-2}
\end{array}$$

Since $\partial_n\partial_{n-1} = 0$, we have

$$\partial_n\alpha_{n-1}\partial'_{n-1} = \partial_n\partial_{n-1}\alpha_{n-2} = 0.$$

And therefore

$$\mathrm{im}\,\partial_n\alpha_{n-1} \subseteq \ker\partial'_{n-1} = \mathrm{im}\,\partial'_n.$$

Since $\mathfrak{f}_n$ is a free module, there exists an $\alpha_n \in \mathrm{Hom}_R(\mathfrak{f}_n, \mathfrak{f}'_n)$ such that $\alpha_n\partial'_n = \partial_n\alpha_{n-1}$ by Lemma 16.4.

b) Let $\alpha, \beta \in \mathrm{Hom}(\mathscr{K}, \mathscr{K}')$ such that $\alpha_0\partial'_0 = \beta_0\partial'_0 = \partial_0\mu$. Our goal is to find $\sigma_n \in \mathrm{Hom}_R(\mathfrak{f}_n, \mathfrak{f}'_{n+1})$ such that

$$\alpha_n - \beta_n = \partial_n\sigma_{n-1} + \sigma_n\partial'_{n+1} \quad (n \geq 0).$$

We define $\sigma_{-1} = 0$, as required.

$$\begin{array}{ccccccc}
\mathfrak{f}_1 & \xrightarrow{\partial_1} & \mathfrak{f}_0 & \xrightarrow{\partial_0} & \mathfrak{m} & \longrightarrow & 0 \\
 & \swarrow{\scriptstyle \sigma_0} & \downarrow{\scriptstyle \alpha_0-\beta_0} & & \downarrow{\scriptstyle \mu} & & \\
\mathfrak{f}'_1 & \xrightarrow[\partial'_1]{} & \mathfrak{f}'_0 & \xrightarrow[\partial'_0]{} & \mathfrak{m}' & \longrightarrow & 0
\end{array}$$

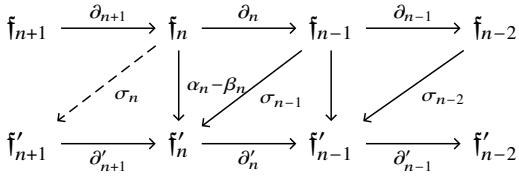
First we determine σ_0:

Let $f \in \mathfrak{f}_0$ and $x = f(\alpha_0 - \beta_0) \in \operatorname{im}(\alpha_0 - \beta_0)$. Then

$$x\partial_0' = f(\alpha_0 - \beta_0)\partial_0' = f(\partial_0\mu - \partial_0\mu) = 0.$$

So $x \in \ker \partial_0' = \operatorname{im} \partial_1'$. Since $\mathfrak{f}_0$ is free, there exists a homomorphism σ_0 in $\operatorname{Hom}_R(\mathfrak{f}_0, \mathfrak{f}_1')$ such that $\alpha_0 - \beta_0 = \sigma_0\partial_1'$ by Lemma 16.4. With this, we have determined σ_0.

For $n \geq 1$, suppose for induction that $\sigma_0, \ldots, \sigma_{n-1}$ have already been found.



Let

$$x = f(\alpha_n - \beta_n - \partial_n\sigma_{n-1}) \quad \text{with} \quad f \in \mathfrak{f}_n.$$

Since

$$(\alpha_n - \beta_n)\partial_n' = \partial_n(\alpha_{n-1} - \beta_{n-1})$$

and

$$\sigma_{n-1}\partial_n' = \alpha_{n-1} - \beta_{n-1} - \partial_{n-1}\sigma_{n-2},$$

we have

$$\begin{aligned} x\partial_n' &= f\partial_n(\alpha_{n-1} - \beta_{n-1}) - f\partial_n(\alpha_{n-1} - \beta_{n-1} - \partial_{n-1}\sigma_{n-2}) \\ &= f\partial_n\partial_{n-1}\sigma_{n-2} = 0. \end{aligned}$$

Hence,

$$x \in \ker \partial_n' = \operatorname{im} \partial_{n+1}'.$$

Since $\mathfrak{f}_n$ is free, there exists some $\sigma_n \in \operatorname{Hom}_R(\mathfrak{f}_n, \mathfrak{f}_{n+1}')$ such that

$$\sigma_n\partial_{n+1}' = \alpha_n - \beta_n - \partial_n\sigma_{n-1}$$

by Lemma 16.4. With this, σ_n has also been determined. □

The most important rings for our purposes are by far the group rings, which we now introduce.

Definition 16.6.

a) Let G be a group and S a commutative ring with unity. Let $S[G]$ be the free S-module in the free generators $\{e(g) \mid g \in G\}$. We define an associative and

two-sided distributive multiplication on $S[G]$ by

$$\left(\sum_{g\in G} a_g e(g)\right)\left(\sum_{g\in G} b_g e(g)\right) = \sum_{g\in G}\left(\sum_{xy=g} a_x b_y\right)e(g),$$

where $a_g, b_g \in S$ and only finitely many of the a_g, b_g are nonzero.

In this way, $S[G]$ becomes a ring with unity $e(1)$. Since $e(gg') = e(g)e(g')$, we may identify $e(g)$ with g. So we will now write elements of $S[G]$ in the form

$$a = \sum_{g\in G} a_g g \quad (a_g \in S),$$

where only finitely many of the a_g are nonzero. We call $S[G]$ the group ring of G over S.

b) If $\mathfrak{m}$ is an abelian G-group (in the sense of Definition 10.1) with $m1 = m$ and $m(gg') = (mg)g'$ for all $m \in \mathfrak{m}$ and all $g, g' \in G$,[2] then $\mathfrak{m}$ becomes a $\mathbb{Z}[G]$-module upon defining

$$m\sum_{g\in G} a_g g = \sum_{g\in G} a_g\,(mg).$$

We regard the ring of integers $\mathbb{Z}$ as a trivial $\mathbb{Z}[G]$-module; that is, $zg = z$ for all $z \in \mathbb{Z}$ and all $g \in G$.

Theorem 16.7. *Let $\mathfrak{a}, \mathfrak{a}', \mathfrak{a}''$ and $\mathfrak{b}$ be R-modules.*

a) *Let $\mu \in \mathrm{Hom}_R(\mathfrak{a}, \mathfrak{a}')$. Then there exists a homomorphism μ^* of abelian groups from $\mathrm{Hom}_R(\mathfrak{a}', \mathfrak{b})$ to $\mathrm{Hom}_R(\mathfrak{a}, \mathfrak{b})$ such that*

$$a(\varphi\mu^*) = (a\mu)\varphi$$

for all $\varphi \in \mathrm{Hom}_R(\mathfrak{a}', \mathfrak{b})$ and all $a \in \mathfrak{a}$.

b) *If $\mu_1 \in \mathrm{Hom}_R(\mathfrak{a}, \mathfrak{a}')$ and $\mu_2 \in \mathrm{Hom}_R(\mathfrak{a}', \mathfrak{a}'')$, then*

$$(\mu_1\mu_2)^* = \mu_2^*\mu_1^*.$$

($\mathrm{Hom}_R(_, \mathfrak{b})$[3] *is a so-called contravariant functor.*)

c) *If $\mu_1, \mu_2 \in \mathrm{Hom}_R(\mathfrak{a}, \mathfrak{a}')$, then*

$$(\mu_1 + \mu_2)^* = \mu_1^* + \mu_2^*.$$

d) *If $\mu \in \mathrm{Hom}_R(\mathfrak{a}, \mathfrak{a})$ such that $a\mu = na$ for some natural number $n \in \mathbb{N}$ and all $a \in \mathfrak{a}$, then $\varphi\mu^* = n\varphi$ for all $\varphi \in \mathrm{Hom}_R(\mathfrak{a}, \mathfrak{b})$.*

Proof. a) The claim is trivial.

b) For all $\varphi \in \mathrm{Hom}_R(\mathfrak{a}'', \mathfrak{b})$ and all $a \in \mathfrak{a}$, we have

[2] Correction 2 from errata.

[3] Correction 3 from errata.

$$a(\varphi(\mu_1\mu_2)^*) = (a\mu_1\mu_2)\varphi = (a\mu_1)(\varphi\mu_2^*) = a((\varphi\mu_2^*)\mu_1^*).$$

So we have $(\mu_1\mu_2)^* = \mu_2^*\mu_1^*$.

c) For $\varphi \in \mathrm{Hom}_R(\mathfrak{a}', \mathfrak{b})$ and $a \in \mathfrak{a}$, we have

$$\begin{aligned} a(\varphi(\mu_1 + \mu_2)^*) &= (a(\mu_1 + \mu_2))\varphi = (a\mu_1 + a\mu_2)\varphi \\ &= (a\mu_1)\varphi + (a\mu_2)\varphi = a(\varphi\mu_1^*) + a(\varphi\mu_2^*). \end{aligned}$$

d) We have

$$a(\varphi\mu^*) = (a\mu)\varphi = (na)\varphi = n(a\varphi) = a(n\varphi). \qquad \square$$

Theorem 16.8. *Let $R = \mathbb{Z}[G]$, and let $\mathscr{K} = \{\mathfrak{f}_n, \partial_n\}$ be a free resolution of $\mathbb{Z}$ as an R-module. Let $\mathfrak{m}$ be an R-module. By Theorem 16.7 we obtain a cochain complex (but not necessarily an exact sequence!)*

$$\cdots \longleftarrow \mathrm{Hom}_R(\mathfrak{f}_n, \mathfrak{m}) \xleftarrow{\partial_n^*} \mathrm{Hom}_R(\mathfrak{f}_{n-1}, \mathfrak{m}) \xleftarrow{\partial_{n-1}^*} \cdots \xleftarrow{\partial_1^*} \mathrm{Hom}_R(\mathfrak{f}_0, \mathfrak{m}).$$

For $n \geq 1$, we define

$$\mathbf{H}^n(G, \mathfrak{m}) = \ker \partial_{n+1}^* / \mathrm{im}\, \partial_n^*$$

and call $\mathbf{H}^n(G, \mathfrak{m})$ the n-th cohomology group of $\mathfrak{m}$. Up to isomorphism, $\mathbf{H}^n(G, \mathfrak{m})$ is independent of the free resolution of $\mathbb{Z}$.

Proof. Since $\partial_n\partial_{n-1} = 0$, it follows by Theorem 16.7 b) that $\partial_{n-1}^*\partial_n^* = 0$. Therefore,

$$\cdots \longleftarrow \mathrm{Hom}_R(\mathfrak{f}_n, \mathfrak{m}) \xleftarrow{\partial_n^*} \mathrm{Hom}_R(\mathfrak{f}_{n-1}, \mathfrak{m}) \xleftarrow{\partial_{n-1}^*} \cdots \xleftarrow{\partial_1^*} \mathrm{Hom}_R(\mathfrak{f}_0, \mathfrak{m})$$

is a cochain complex and possesses cohomology groups in the sense of Definition 16.1 b).

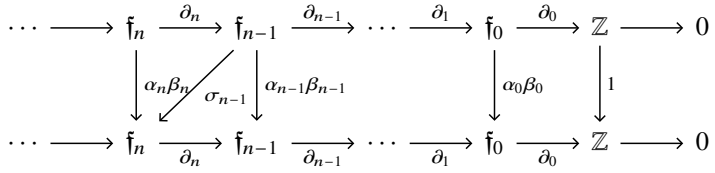
Let $\mathscr{K} = \{\mathfrak{f}_n, \partial_n\}$ and $\mathscr{K}' = \{\mathfrak{f}_n', \partial_n'\}$ be free resolutions of $\mathbb{Z}$. By Theorem 16.5 (with $\mathfrak{m} = \mathfrak{m}' = \mathbb{Z}$ and $\mu = 1$), there exists an $\alpha \in \mathrm{Hom}(\mathscr{K}, \mathscr{K}')$ and a $\beta \in \mathrm{Hom}(\mathscr{K}', \mathscr{K})$ such that $\alpha_0\partial_0' = \partial_0$ and $\beta_0\partial_0 = \partial_0'$. Then $\alpha\beta \in \mathrm{Hom}(\mathscr{K}, \mathscr{K})$ and

$$\alpha_0\beta_0\partial_0 = \alpha_0\partial_0' = \partial_0 = \partial_0 1_{\mathbb{Z}} = (1_{\mathscr{K}})_0\partial_0.$$

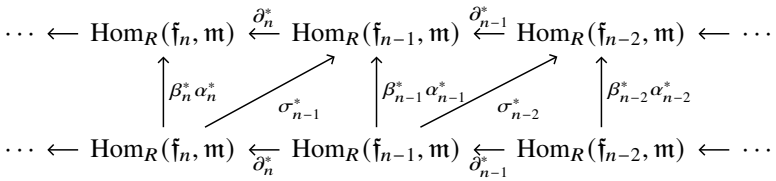
We may conclude that $\alpha\beta \sim 1_{\mathscr{K}}$ by Theorem 16.5 (with $\mathscr{K} = \mathscr{K}'$). In the same way, we obtain $\beta\alpha \sim 1_{\mathscr{K}'}$. So let

$$\alpha_n\beta_n = 1 + \partial_n\sigma_{n-1} + \sigma_n\partial_{n+1}.$$

From the diagram



and $\partial_n\alpha_{n-1}\beta_{n-1} = \alpha_n\partial'_n\beta_{n-1} = \alpha_n\beta_n\partial_n$, we obtain using Theorem 16.7 the diagram



Then by Theorem 16.7 b) and c), we obtain the relations

$$\beta^*_{n-1}\alpha^*_{n-1}\partial^*_n = \partial^*_n\beta^*_n\alpha^*_n$$

and

$$\beta^*_n\alpha^*_n = 1 + \sigma^*_{n-1}\partial^*_n + \partial^*_{n+1}\sigma^*_n,$$

so that $\beta^*\alpha^* \sim 1$. It follows in the same way that $\alpha^*\beta^* \sim 1$. Therefore, we find for $n \geq 1$ that

$$\ker \partial^*_{n+1}/\operatorname{im} \partial^*_n \cong \ker \partial'^*_{n+1}/\operatorname{im} \partial'^*_n$$

by Lemma 16.3 b) (for cohomology groups), as claimed. □

In calculating cohomology groups, we will make use of the fact that we may choose any free resolution of $\mathbb{Z}$. We first give a particularly simple resolution in the case of cyclic groups:

Theorem 16.9. *Let $G = \langle g \rangle$ be a cyclic group of order n. We define the following elements in the group ring $R = \mathbb{Z}[G]$ of G over the integers:*

$$N = 1 + g + g^2 + \cdots + g^{n-1} \quad \text{and} \quad D = g - 1.$$

The maps $\varphi, \psi \in \operatorname{Hom}_R(R, R)$ are defined by $r\varphi = rN$ and $r\psi = rD$ for $r \in R$. Furthermore, let $\epsilon : R \to \mathbb{Z}$ be the ring homomorphism defined by

$$\left(\sum_{i=0}^{n-1} a_i g^i\right)\epsilon = \sum_{i=0}^{n-1} a_i \quad (a_i \in \mathbb{Z}).$$

Then the following sequence is a free resolution of $\mathbb{Z}$:

$$\cdots \xrightarrow{\varphi} R \xrightarrow{\psi} R \xrightarrow{\varphi} R \xrightarrow{\psi} R \xrightarrow{\varphi} R \xrightarrow{\psi} R \xrightarrow{\epsilon} \mathbb{Z} \longrightarrow 0.$$

Proof. For $r = \sum_{i=0}^{n-1} a_i g^i$, we have

$$rD = \sum_{i=0}^{n-1} (a_{i-1} - a_i) g^i,$$

where we define $a_{-1} = a_{n-1}$. Furthermore,

$$rN = \left(\sum_{i=0}^{n-1} a_i \right) N.$$

Therefore, we have

$$\ker \psi = \left\{ \sum_{i=0}^{n-1} a_i g^i \;\middle|\; a_0 = \cdots = a_{n-1} \right\} = \{aN \mid a \in \mathbb{Z}\} = \operatorname{im} \varphi$$

and

$$\begin{aligned} \ker \epsilon &= \left\{ \sum_{i=0}^{n-1} a_i g^i \;\middle|\; \sum_{i=0}^{n-1} a_i = 0 \right\} = \ker \varphi \\ &= \{(a_0 + (a_0 + a_1)g + \cdots + (a_0 + \cdots + a_{n-1})g^{n-1})D \mid a_i \in \mathbb{Z}\} \\ &= \operatorname{im} \psi. \end{aligned}$$

Since R is a free R-module in one generator, the given exact sequence is a free resolution of $\mathbb{Z}$. □

Theorem 16.10. *Let $G = \langle g \rangle$ be a cyclic group of order n, and let $\mathfrak{m}$ be a G-module. We define $N = \sum_{i=0}^{n-1} g^i$. Then for $n \geq 1$, we have*

$$\mathbf{H}^{2n-1}(G, \mathfrak{m}) \cong \{m \in \mathfrak{m} \mid mN = 0\} / \{m(1-g) \mid m \in \mathfrak{m}\},$$

and

$$\mathbf{H}^{2n}(G, \mathfrak{m}) \cong \{m \in \mathfrak{m} \mid mg = m\} / \{mN \mid m \in \mathfrak{m}\}.$$

Proof. We use the free resolution of $\mathbb{Z}$ given in Theorem 16.9. Then we obtain according to Theorem 16.7 the cochain complex

$$\cdots \xleftarrow{\psi^*} \operatorname{Hom}_R(R, \mathfrak{m}) \xleftarrow{\varphi^*} \operatorname{Hom}_R(R, \mathfrak{m}) \xleftarrow{\psi^*} \operatorname{Hom}_R(R, \mathfrak{m}) \xleftarrow{\varphi^*} \cdots$$
$$\cdots \xleftarrow{\psi^*} \operatorname{Hom}_R(R, \mathfrak{m}).$$

Since $\{1\}$ is a free generating set for the free R-module R, the map τ from $\operatorname{Hom}_R(R, \mathfrak{m})$ to $\mathfrak{m}$ defined by $\alpha\tau = 1\alpha$ for $\alpha \in \operatorname{Hom}_R(R, \mathfrak{m})$ is clearly an iso-

morphism from the abelian group $\mathrm{Hom}_R(R, \mathfrak{m})$ to the additive group of $\mathfrak{m}$. Then for $\alpha \in \mathrm{Hom}_R(R, \mathfrak{m})$, we have by Theorem 16.7 a) that

$$(\alpha\varphi^*)\tau = 1(\alpha\varphi^*) = (1\varphi)\alpha = (1N)\alpha = (1\alpha)N = (\alpha\tau)N,$$

and analogously $(\alpha\psi^*)\tau = (\alpha\tau)(g-1)$. For $n \geq 1$, it follows that

$$\mathbf{H}^{2n-1}(G, \mathfrak{m}) \cong \ker\varphi^*/\mathrm{im}\,\psi^* \cong \{m \in \mathfrak{m} \mid mN = 0\}/\{m(g-1) \mid m \in \mathfrak{m}\}$$

and

$$\mathbf{H}^{2n}(G, \mathfrak{m}) \cong \ker\psi^*/\mathrm{im}\,\varphi^* \cong \{m \in \mathfrak{m} \mid mg = m\}/\{mN \mid m \in \mathfrak{m}\}. \qquad \square$$

The description of the cohomology groups of a cyclic group given in Theorem 16.10 plays an important role in class field theory, see Serre's *Local Fields*. We will use this result in §19 of Chapter III, where we study the automorphism groups of p-groups.

The group theoretical applications of cohomology theory are mostly concerned with the so-called standard complex, which we now describe:

Theorem 16.11. *Let G be an arbitrary group. We set $R = \mathbb{Z}[G] = \mathfrak{s}_0$. For $n \geq 1$, let $\mathfrak{s}_n$ be the free $\mathbb{Z}[G]$-module generated by the ordered n-tuples $[g_1, \ldots, g_n]$ with $g_i \in G$ $(i = 1, \ldots, n)$. We define the maps $\partial_n \in \mathrm{Hom}_R(\mathfrak{s}_n, \mathfrak{s}_{n-1})$ and $\partial_0 \in \mathrm{Hom}_R(\mathfrak{s}_0, \mathbb{Z})$ as follows:*

$$g\partial_0 = 1 \quad \text{for all } g \in G;$$
$$[g]\partial_1 = 1 - g;$$

for $n > 1$, let

$$[g_1, \ldots, g_n]\partial_n = [g_2, \ldots, g_n] + \sum_{i=1}^{n-1} (-1)^i [g_1, \ldots, g_{i-1}, g_i g_{i+1}, g_{i+2}, \ldots, g_n] + (-1)^n [g_1, \ldots, g_{n-1}]g_n.$$

Then $\{\mathfrak{s}_n, \partial_n\}$ is a free resolution of the $\mathbb{Z}[G]$-module $\mathbb{Z}$. We call $\{\mathfrak{s}_n, \partial_n\}$ the (inhomogeneous) standard complex of G.

Proof. Since the $[g_1, \ldots, g_n]$ are a free generating set of $\mathfrak{s}_n$ as an R-module, there exists exactly one $\partial_n \in \mathrm{Hom}_R(\mathfrak{s}_n, \mathfrak{s}_{n-1})$ such that $[g_1, \ldots, g_n]\partial_n$ is as given above. We must prove that $\mathrm{im}\,\partial_{n+1} = \ker\partial_n$ for $n \geq 0$. Clearly,

$$\mathrm{im}\,\partial_1 = \ker\partial_0 = \left\{ \sum_{g \in G} a_g g \;\middle|\; \sum_{g \in G} a_g = 0 \right\}.$$

For $n \geq 1$, the elements $[g_1, \ldots, g_n]g$ are a free generating set for $\mathfrak{s}_n$ as a $\mathbb{Z}$-module. Hence, there is a map $\rho_n \in \mathrm{Hom}_{\mathbb{Z}}(\mathfrak{s}_n, \mathfrak{s}_{n+1})$ with

$$([g_1, \ldots, g_n]g)\rho_n = [g^{-1}g_n^{-1} \cdots g_1^{-1}, g_1, \ldots, g_n]g.$$

We further define $\rho_{-1} \in \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathfrak{s}_0)$ by $1\rho_{-1} = 1$ and $\rho_0 \in \mathrm{Hom}_{\mathbb{Z}}(\mathfrak{s}_0, \mathfrak{s}_1)$ by $g\rho_0 = [g^{-1}]g$. We now show that

$$\rho_n\partial_{n+1} + \partial_n\rho_{n-1} = 1 \qquad (*)$$

for all $n \geq 0$. For $n \leq 1$, this follows by a direct calculation, so assume that $n \geq 2$. Since ∂_n and ∂_{n+1} are both $\mathbb{Z}[G]$-homomorphisms, we have

$$\begin{aligned}
&([g_1, \ldots, g_n]\, g)\,(\rho_n\partial_{n+1} + \partial_n\rho_{n-1}) \\
&= ([g^{-1}g_n^{-1} \cdots g_1^{-1},\, g_1,\, \ldots,\, g_n]\, g)\,\partial_{n+1} + \\
&\quad + \Big\{ [g_2,\, \ldots,\, g_n]\, g + \sum_{i=1}^{n-1} (-1)^i [g_1,\, \ldots,\, g_i g_{i+1},\, \ldots,\, g_n]\, g + \\
&\quad + (-1)^n [g_1,\, \ldots,\, g_{n-1}]\, g_n g \Big\} \rho_{n-1} \\
&= [g_1,\, \ldots,\, g_n]\, g - [g^{-1}g_n^{-1} \cdots g_2^{-1},\, g_2,\, \ldots,\, g_n]\, g + \\
&\quad + \sum_{i=1}^{n} (-1)^{i+1} [g^{-1}g_n^{-1} \cdots g_1^{-1},\, g_1,\, \ldots,\, g_i g_{i+1},\, \ldots,\, g_n]\, g + \\
&\quad + (-1)^{n+1} [g^{-1}g_n^{-1} \cdots g_1^{-1},\, g_1,\, \ldots,\, g_{n-1}]\, g_n g + \\
&\quad + [g^{-1}g_n^{-1} \cdots g_2^{-1},\, g_2,\, \ldots,\, g_n]\, g + \\
&\quad + \sum_{i=1}^{n} (-1)^i [g^{-1}g_n^{-1} \cdots g_1^{-1},\, g_1,\, \ldots,\, g_i g_{i+1},\, \ldots,\, g_n]\, g + \\
&\quad + (-1)^n [g^{-1}g_n^{-1} \cdots g_1^{-1},\, g_1,\, \ldots,\, g_{n-1}]\, g_n g \\
&= [g_1, \ldots, g_n]\, g.
\end{aligned}$$

For $n \geq 1$, we have $\ker \partial_n = \mathrm{im}\, \partial_{n+1}$:
For $x \in \ker \partial_n$, it follows from $(*)$ that

$$x = x\,(\rho_n\partial_{n+1} + \partial_n\rho_{n-1}) = x\rho_n\partial_{n+1} \in \mathrm{im}\, \partial_{n+1}.$$

So $\ker \partial_n \subseteq \mathrm{im}\, \partial_{n+1}$. To finish, we need to show that $\mathrm{im}\, \partial_{n+1} \subseteq \ker \partial_n$; that is, $\partial_{n+1}\partial_n = 0$. For $n = 0$, this has already been shown. So assume $n \geq 1$. Assume for induction that $\partial_n\partial_{n-1} = 0$. Then we obtain with the help of $(*)$ that

$$\begin{aligned}
\rho_n\partial_{n+1}\partial_n &= (1 - \partial_n\rho_{n-1})\partial_n = \partial_n - \partial_n(1 - \partial_{n-1}\rho_{n-2}) \\
&= \partial_n\partial_{n-1}\rho_{n-2} = 0.
\end{aligned}$$

Let $[g_1, \ldots, g_{n+1}] \in \mathfrak{s}_{n+1}$. If we set $g^{-1} = g_1 \cdots g_{n+1}$, then

$$[g_1, \ldots, g_{n+1}] = ([g_2, \ldots, g_{n+1}]\, g)\, \rho_n\, g^{-1}.$$

It follows that

$$[g_1, \ldots, g_{n+1}]\, \partial_{n+1}\partial_n = ([g_2, \ldots, g_{n+1}]\, g)\, \rho_n\, \partial_{n+1}\, \partial_n g^{-1} = 0.$$

Since the $[g_1, \ldots, g_{n+1}]$ generate the module $\mathfrak{s}_{n+1}$, we have $\partial_{n+1}\partial_n = 0$ on $\mathfrak{s}_{n+1}$. □

Definition 16.12. Let $\mathfrak{m}$ be a G-module. We define the abelian groups $\mathbf{C}^n(G, \mathfrak{m})$, written additively, for $n \geq 0$ as follows:

(1) $\mathbf{C}^0(G, \mathfrak{m}) = \mathfrak{m}$.
(2) For $n \geq 1$, let $\mathbf{C}^n(G, \mathfrak{m})$ be the set of all maps from the n-fold direct product $G \times \cdots \times G$ to $\mathfrak{m}$. The addition in $\mathbf{C}^n(G, \mathfrak{m})$ is pointwise addition of functions.

We define $\delta_n \in \mathrm{Hom}_{\mathbb{Z}}(\mathbf{C}^n(G, \mathfrak{m}), \mathbf{C}^{n+1}(G, \mathfrak{m}))$ by

$$(f\delta_0)(g) = f - fg \quad \text{for} \quad f \in \mathbf{C}^0(G, \mathfrak{m}) = \mathfrak{m}$$

and

$$\begin{aligned}(f\delta_n)(g_1, \ldots, g_{n+1}) = {} & f(g_2, \ldots, g_{n+1}) + \\ & + \sum_{i=1}^{n} (-1)^i\, f(g_1, \ldots, g_{i-1}, g_i g_{i+1}, g_{i+2}, \ldots, g_{n+1}) + \\ & + (-1)^{n+1}\, f(g_1, \ldots, g_n)\, g_{n+1}\end{aligned}$$

for $f \in \mathbf{C}^n(G, \mathfrak{m})$ with $n \geq 1$. We define $\mathbf{Z}^n(G, \mathfrak{m}) = \ker \delta_n$ for $n \geq 0$, $\mathbf{B}^n(G, \mathfrak{m}) = \operatorname{im} \delta_{n-1}$ for $n \geq 1$ and $\mathbf{B}^0(G, \mathfrak{m}) = 0$. We call the elements of $\mathbf{C}^n(G, \mathfrak{m})$ n-cochains, the elements of $\mathbf{Z}^n(G, \mathfrak{m})$ n-cocycles and the elements of $\mathbf{B}^n(G, \mathfrak{m})$ n-coboundaries.

Theorem 16.13. *Let $\mathfrak{m}$ be a G-module. For all $n \geq 1$, we have*

$$\mathbf{H}^n(G, \mathfrak{m}) \cong \mathbf{Z}^n(G, \mathfrak{m})/\mathbf{B}^n(G, \mathfrak{m}).$$

Proof. Let $R = \mathbb{Z}[G]$. We use the free resolution $\{\mathfrak{s}_n, \partial_n\}$ of $\mathbb{Z}$ from Theorem 16.11. We obtain, according to Theorem 16.7, the cochain complex

$$\cdots \longleftarrow \mathrm{Hom}_R(\mathfrak{s}_n, \mathfrak{m}) \xleftarrow{\partial_n^*} \mathrm{Hom}_R(\mathfrak{s}_{n-1}, \mathfrak{m}) \xleftarrow{\partial_{n-1}^*} \cdots \xleftarrow{\partial_1^*} \mathrm{Hom}_R(\mathfrak{s}_0, \mathfrak{m}).$$

We define the homomorphisms τ_n of abelian groups from $\mathrm{Hom}_R(\mathfrak{s}_n, \mathfrak{m})$ to $\mathbf{C}^n(G, \mathfrak{m})$ by

$$\varphi\tau_0 = 1\varphi \quad \text{for} \quad \varphi \in \mathrm{Hom}_R(\mathfrak{s}_0, \mathfrak{m})$$

and

$$(\varphi\tau_n)(g_1,\dots,g_n) = [g_1,\dots,g_n]\varphi$$

for $\varphi \in \mathrm{Hom}_R(\mathfrak{s}_n, \mathfrak{m})$ with $n \geq 1$. As $\mathfrak{s}_n$ (resp. $\mathfrak{s}_0$) is a free R-module in the free generators $[g_1,\dots,g_n]$ (resp. 1), the elements $[g_1,\dots,g_n]\varphi$ (resp. 1φ) in $\mathfrak{m}$ can be chosen arbitrarily. Hence, τ_n is a $\mathbb{Z}$-isomorphism from $\mathrm{Hom}_R(\mathfrak{s}_n, \mathfrak{m})$ onto $\mathbf{C}^n(G, \mathfrak{m})$. For $n \geq 1$ and $\varphi \in \mathrm{Hom}_R(\mathfrak{s}_n, \mathfrak{m})$, we have

$$\begin{aligned}
&(\varphi\partial^*_{n+1}\tau_{n+1})\,(g_1,\dots,g_{n+1}) = [g_1,\dots,g_{n+1}]\,(\varphi\partial^*_{n+1})\\
&= ([g_1,\dots,g_{n+1}]\,\partial_{n+1})\varphi\\
&= [g_2,\dots,g_{n+1}]\,\varphi +\\
&\quad + \sum_{i=1}^{n}(-1)^i\,[g_1,\dots,g_{i-1},g_ig_{i+1},g_{i+2},\dots,g_{n+1}]\,\varphi +\\
&\quad + (-1)^{n+1}\,([g_1,\dots,g_n]\,g_{n+1})\,\varphi\\
&= (\varphi\tau_n)\,(g_2,\dots,g_{n+1}) + \sum_{i=1}^{n}(-1)^i\,(\varphi\tau_n)\,(g_1,\dots,g_ig_{i+1},\dots,g_{n+1}) +\\
&\quad + (-1)^{n+1}\,(\varphi\tau_n)\,(g_1,\dots,g_n)\,g_{n+1}\\
&= (\varphi\tau_n\delta_n)\,(g_1,\dots,g_{n+1}).
\end{aligned}$$

Hence, we find that

$$(\ker\partial^*_{n+1})\tau_n = \ker\delta_n \quad \text{and} \quad (\mathrm{im}\,\partial^*_n)\tau_n = \mathrm{im}\,\delta_{n-1},$$

and therefore

$$\mathbf{H}^n(G,\mathfrak{m}) = \ker\partial^*_{n+1}/\mathrm{im}\,\partial^*_n \cong \ker\delta_n/\mathrm{im}\,\delta_{n-1}. \qquad \square$$

In nearly every group theoretical application of cohomology, the description of cohomology groups given in Theorem 16.13 is used. However, for the proof of the main theorem in this section, another description is more useful:

Theorem 16.14. *Let $\mathfrak{m}$ be a G-module. We define the abelian groups $\widetilde{\mathbf{C}}^n(G,\mathfrak{m})$, written additively, for $n \geq 0$ as the set of all maps f from the $(n+1)$-fold direct product $G\times\cdots\times G$ to $\mathfrak{m}$ such that*

$$f(g_0g,\dots,g_ng) = f(g_0,\dots,g_n)\,g$$

for all $g_0,\dots,g_n,g \in G$. Furthermore, we define the maps

$$\widetilde{\delta}_n \in \mathrm{Hom}_{\mathbb{Z}}(\widetilde{\mathbf{C}}^n(G,\mathfrak{m}), \widetilde{\mathbf{C}}^{n+1}(G,\mathfrak{m})) \quad \textit{by}$$

$$(f\widetilde{\delta}_n)\,(g_0,\dots,g_{n+1}) = \sum_{i=0}^{n+1}(-1)^i\,f(g_0,\dots,\hat{g}_i,\dots,g_{n+1});$$

where $\widehat{\ }$ *means that the corresponding argument is removed from the expression. For* $n \geq 1$*, we define*

$$\widetilde{\mathbf{Z}}^n(G, \mathfrak{m}) = \ker \widetilde{\delta}_n \quad \textit{and} \quad \widetilde{\mathbf{B}}^n(G, \mathfrak{m}) = \operatorname{im} \widetilde{\delta}_{n-1}.$$

Then for all $n \geq 1$*, we have*

$$\mathbf{H}^n(G, \mathfrak{m}) \cong \widetilde{\mathbf{Z}}^n(G, \mathfrak{m}) / \widetilde{\mathbf{B}}^n(G, \mathfrak{m}).$$

Proof. We define the maps

$$\mu_n \in \operatorname{Hom}_{\mathbb{Z}}(\mathbf{C}^n(G, \mathfrak{m}), \widetilde{\mathbf{C}}^n(G, \mathfrak{m})) \quad \text{and} \quad \tau_n \in \operatorname{Hom}_{\mathbb{Z}}(\widetilde{\mathbf{C}}^n(G, \mathfrak{m}), \mathbf{C}^n(G, \mathfrak{m}))$$

by

$$\begin{aligned} (f\mu_0)(g_0) &= f g_0 \quad \text{for } f \in \mathbf{C}^0(G, \mathfrak{m}) = \mathfrak{m}; \\ \widetilde{f}\tau_0 &= \widetilde{f}(1) \quad \text{for } \widetilde{f} \in \widetilde{\mathbf{C}}^0(G, \mathfrak{m}); \end{aligned}$$

while for $n \geq 1$ and $f \in \mathbf{C}^n(G, \mathfrak{m})$, we define

$$(f\mu_n)\,(g_0, \ldots, g_n) = f(g_0 g_1^{-1}, g_1 g_2^{-1}, \ldots, g_{n-1} g_n^{-1})\, g_n;$$

and for $n \geq 1$ with $\widetilde{f} \in \widetilde{\mathbf{C}}^n(G, \mathfrak{m})$, we define

$$(\widetilde{f}\tau_n)\,(g_1, \ldots, g_n) = \widetilde{f}(g_1 \cdots g_n, g_2 \cdots g_n, \ldots, g_{n-1} g_n, g_n, 1).$$

A trivial calculation shows that $\mu_n \tau_n = 1$ and $\tau_n \mu_n = 1$. Therefore, μ_n is an isomorphism of abelian groups from $\mathbf{C}^n(G, \mathfrak{m})$ to $\widetilde{\mathbf{C}}^n(G, \mathfrak{m})$. For $f \in \mathbf{C}^n(G, \mathfrak{m})$, we have

$$\begin{aligned} (f\mu_n\widetilde{\delta}_n)\,(g_0, \ldots, g_{n+1}) &= \sum_{i=0}^{n+1} (-1)^i\, (f\mu_n)\,(g_0, \ldots, \hat{g}_i, \ldots, g_{n+1}) \\ &= \sum_{i=0}^{n} (-1)^i\, f(g_0 g_1^{-1}, \ldots, g_{i-2} g_{i-1}^{-1}, g_{i-1} g_{i+1}^{-1}, g_{i+1} g_{i+2}^{-1}, \ldots, g_n g_{n+1}^{-1})\, g_{n+1} + \\ &\quad + (-1)^{n+1}\, f(g_0 g_1^{-1}, \ldots, g_{n-1} g_n^{-1})\, g_n \end{aligned}$$

and

$$\begin{aligned} (f\delta_n\mu_{n+1})\,(g_0, \ldots, g_{n+1}) &= (f\delta_n)\,(g_0 g_1^{-1}, g_1 g_2^{-1}, \ldots, g_n g_{n+1}^{-1})\, g_{n+1} \\ &= f(g_1 g_2^{-1}, \ldots, g_n g_{n+1}^{-1})\, g_{n+1} + \\ &\quad + \sum_{i=1}^{n} (-1)^i\, f(g_0 g_1^{-1}, \ldots, g_{i-1} g_{i+1}^{-1}, \ldots, g_n g_{n+1}^{-1})\, g_{n+1} + \\ &\quad + (-1)^{n+1} f(g_0 g_1^{-1}, \ldots, g_{n-1} g_n^{-1})\, g_n. \end{aligned}$$

So $\mu_n\widetilde{\delta}_n = \delta_n\mu_{n+1}$. It follows that $\ker\widetilde{\delta}_n = (\ker\delta_n)\mu_n$ and $\operatorname{im}\widetilde{\delta}_n = (\operatorname{im}\delta_n)\mu_{n+1}$. Therefore,

$$\mathbf{H}^n(G,\mathfrak{m}) \cong \mathbf{Z}^n(G,\mathfrak{m})/\mathbf{B}^n(G,\mathfrak{m}) \cong \widetilde{\mathbf{Z}}^n(G,\mathfrak{m})/\widetilde{\mathbf{B}}^n(G,\mathfrak{m})$$

for $n \geq 1$ by Theorem 16.13. □

Definition 16.15. Let $U \leq G$, and let $\mathfrak{m}$ be a G-module.

a) We define the map $\widetilde{\rho}_n : \widetilde{\mathbf{C}}^n(G,\mathfrak{m}) \to \widetilde{\mathbf{C}}^n(U,\mathfrak{m})$ by

$$(f\widetilde{\rho}_n)(u_0,\ldots,u_n) = f(u_0,\ldots,u_n)$$

for $f \in \widetilde{\mathbf{C}}^n(G,\mathfrak{m})$ and $u_0,\ldots,u_n \in U$. Clearly, $\widetilde{\rho}_n\widetilde{\delta}_n = \widetilde{\delta}_n\widetilde{\rho}_{n+1}$. Therefore, the map r'_n defined by

$$(f + \widetilde{\mathbf{B}}^n(G,\mathfrak{m}))r'_n = f\widetilde{\rho}_n + \widetilde{\mathbf{B}}^n(U,\mathfrak{m})$$

for $f \in \widetilde{\mathbf{Z}}^n(G,\mathfrak{m})$ induces a homomorphism $r_n : \mathbf{H}^n(G,\mathfrak{m}) \to \mathbf{H}^n(U,\mathfrak{m})$. We call r_n the restriction map from $\mathbf{H}^n(G,\mathfrak{m})$ to $\mathbf{H}^n(U,\mathfrak{m})$.

We also define the maps $\rho_n : \mathbf{C}^n(G,\mathfrak{m}) \to \mathbf{C}^n(U,\mathfrak{m})$ using the maps μ_n from the proof of Theorem 16.14 by $\rho_n\mu_n = \mu_n\widetilde{\rho}_n$, so ρ_n is clearly the restriction of the functions in $\mathbf{C}^n(G,\mathfrak{m})$ to $U \times \cdots \times U$.

b) Assume $[G : U] = k$ is finite, and let $G = \bigcup_{j=1}^{k} r_jU$ be the decomposition of G into cosets of U. For $f \in \widetilde{\mathbf{C}}^n(U,\mathfrak{m})$, we define the function $f\tilde{\iota}_n$ on $G \times \cdots \times G$ by

$$(f\tilde{\iota}_n)(g_0,\ldots,g_n) = \sum_{j=1}^{k} f(\overline{g_0r_j}^{-1}g_0r_j,\ \ldots,\ \overline{g_nr_j}^{-1}g_nr_j)\,r_j^{-1},$$

where $\overline{g_ir_j} \in \{r_1,\ldots,r_k\}$ is uniquely determined by $g_ir_jU = \overline{g_ir_j}U$.

In the next lemma, we collect the properties of $\tilde{\iota}_n$ that we will need.

Lemma 16.16. *Let $\tilde{\iota}_n$ be defined as in Definition 16.15 b).*

a) *For $f \in \widetilde{\mathbf{C}}^n(U,\mathfrak{m})$, the map $f\tilde{\iota}_n$ is independent of the choice of transversal $\{r_1,\ldots,r_k\}$.*

b) *For $f \in \widetilde{\mathbf{C}}^n(U,\mathfrak{m})$, we have $f\tilde{\iota}_n \in \widetilde{\mathbf{C}}^n(G,\mathfrak{m})$.*

c) *We have $\tilde{\iota}_n\widetilde{\delta}_n = \widetilde{\delta}_n\tilde{\iota}_{n+1}$.*

Proof. a) Suppose $\{r'_1,\ldots,r'_k\}$ is another transversal of U in G, where $r'_j = r_ju_j$ for some $u_j \in U$. For $f \in \widetilde{\mathbf{C}}^n(U,\mathfrak{m})$, we have

$$\sum_{j=1}^{k} f(\overline{g_0 r'_j}^{-1} g_0 r'_j, \ldots, \overline{g_n r'_j}^{-1} g_n r'_j)\, r'^{-1}_j$$
$$= \sum_{j=1}^{k} f(\overline{g_0 r_j}^{-1} g_0 r_j u_j, \ldots, \overline{g_n r_j}^{-1} g_n r_j u_j)\, u_j^{-1} r_j^{-1}$$
$$= \sum_{j=1}^{k} f(\overline{g_0 r_j}^{-1} g_0 r_j, \ldots, \overline{g_n r_j}^{-1} g_n r_j)\, r_j^{-1}.$$

b) Let $g \in G$, and let $g r_j = r_{j'} u_j$ for some $u_j \in U$ $(j = 1, \ldots, k)$. For $f \in \widetilde{\mathbf{C}}^n(U, \mathfrak{m})$, it follows upon using a) that

$$\begin{aligned}(f\tilde{\iota}_n)(g_0 g, \ldots, g_n g) &= \sum_{j=1}^{k} f(\overline{g_0 g r_j}^{-1} g_0 g r_j, \ldots, \overline{g_n g r_j}^{-1} g_n g r_j)\, r_j^{-1} \\ &= \sum_{j=1}^{k} f(\overline{g_0 r_{j'}}^{-1} g_0 r_{j'} u_j, \ldots, \overline{g_n r_{j'}}^{-1} g_n r_{j'} u_j)\, r_j^{-1} \\ &= \sum_{j=1}^{k} f(\overline{g_0 r_{j'}}^{-1} g_0 r_{j'}, \ldots, \overline{g_n r_{j'}}^{-1} g_n r_{j'})\, r_{j'}^{-1} g \\ &= (f\tilde{\iota}_n)(g_0, \ldots, g_n)\, g.\end{aligned}$$

c) For $f \in \widetilde{\mathbf{C}}^n(U, \mathfrak{m})$, we have

$$\begin{aligned}(f\tilde{\iota}_n\widetilde{\delta}_n)(g_0, \ldots, g_{n+1}) &= \sum_{i=0}^{n+1} (-1)^i (f\tilde{\iota}_n)(g_0, \ldots, \hat{g}_i, \ldots, g_{n+1}) \\ &= \sum_{i=0}^{n+1} \sum_{j=1}^{k} (-1)^i f(\overline{g_0 r_j}^{-1} g_0 r_j, \ldots, \widehat{\overline{g_i r_j}^{-1} g_i r_j}, \ldots, \overline{g_{n+1} r_j}^{-1} g_{n+1} r_j)\, r_j^{-1} \\ &= \sum_{j=1}^{k} (f\widetilde{\delta}_n)(\overline{g_0 r_j}^{-1} g_0 r_j, \ldots, \overline{g_{n+1} r_j}^{-1} g_{n+1} r_j)\, r_j^{-1} \\ &= (f\widetilde{\delta}_n\tilde{\iota}_{n+1})(g_0, \ldots, g_{n+1}).\end{aligned}$$

□

Definition 16.17. Let $U \leq G$. We define the homomorphism

$$j'_n : \widetilde{\mathbf{Z}}^n(U, \mathfrak{m})/\widetilde{\mathbf{B}}^n(U, \mathfrak{m}) \to \widetilde{\mathbf{Z}}^n(G, \mathfrak{m})/\widetilde{\mathbf{B}}^n(G, \mathfrak{m})$$

by

$$(f + \widetilde{\mathbf{B}}^n(U, \mathfrak{m}))\, j'_n = f\tilde{\iota}_n + \widetilde{\mathbf{B}}^n(G, \mathfrak{m}),$$

which is well defined by Lemma 16.16 c). We call the homomorphism j_n from $\mathbf{H}^n(U,\mathfrak{m})$ to $\mathbf{H}^n(G,\mathfrak{m})$ induced by j'_n the corestriction map.

We define the map $\iota_n : \mathbf{C}^n(U,\mathfrak{m}) \to \mathbf{C}^n(G,\mathfrak{m})$, using the map μ_n from the proof of Theorem 16.14, by $\mu_n\tilde{\iota}_n = \iota_n\mu_n$. We obtain the formula

$$\begin{aligned}&(f\iota_n)(g_1,\dots,g_n)\\ &= \sum_{j=1}^{k} f(\overline{g_1\cdots g_n r_j}^{-1}\, g_1\, \overline{g_2\cdots g_n r_j},\ \overline{g_2\cdots g_n r_j}^{-1}\, g_2\, \overline{g_3\cdots g_n r_j},\ \dots\\ &\qquad \dots,\ \overline{g_n r_j}^{-1}\, g_n\, r_j)\, r_j^{-1}.\end{aligned}$$

This formula is certainly more cumbersome than the formula for $\tilde{\iota}_n$ in Definition 16.15 b).

Main Theorem 16.18. *Let $[G:U] = k$ be finite, and let $\mathfrak{m}$ be a G-module. For all $n \geq 1$ and all $h \in \mathbf{H}^n(G,\mathfrak{m})$, we have $hr_n j_n = kh$.*

Proof. We show that for all $f \in \widetilde{\mathbf{Z}}^n(G,\mathfrak{m})$, we have $f\widetilde{\rho}_n\tilde{\iota}_n - kf \in \widetilde{\mathbf{B}}^n(G,\mathfrak{m})$: Since $f \in \widetilde{\mathbf{Z}}^n(G,\mathfrak{m})$, we have in particular that

$$0 = (f\widetilde{\delta}_n)(g_0,\ \dots,\ g_i,\ \overline{g_i r_j}^{-1} g_i,\ \dots,\ \overline{g_n r_j}^{-1}\, g_n)$$

for all $g_0,\dots,g_n \in G$, all $j = 1,\dots,k$ and all $i = 0,1,\dots,n$. Therefore,

$$\sum_{m=0}^{n+1} a_{ijm} = 0,$$

where

$$a_{ijm} = (-1)^{i+m} f(g_0,\ \dots,\ g_i,\ \overline{g_i r_j}^{-1} g_i,\ \dots,\ \overline{g_n r_j}^{-1} g_n);$$

in this expression for a_{ijm}, the m-th argument is to be omitted, and the numbering of the arguments begins with 0. It follows that

$$(1)\quad \sum_{j=1}^{k}\sum_{i=0}^{n}\sum_{m=0}^{n+1} a_{ijm} = 0.$$

Now we collect the terms in (1) in a suitable way:

$$\begin{aligned}(2)\quad \sum_{j=1}^{k} a_{0j0} &= \sum_{j=1}^{k} f(\overline{g_0 r_j}^{-1} g_0,\ \dots,\ \overline{g_n r_j}^{-1} g_n)\\ &= \sum_{j=1}^{k} f(\overline{g_0 r_j}^{-1}\, g_0\, r_j,\ \dots,\ \overline{g_n r_j}^{-1}\, g_n\, r_j)\, r_j^{-1}\\ &= (f\widetilde{\rho}_n\tilde{\iota}_n)(g_0,\ \dots,\ g_n).\end{aligned}$$

(3) $\sum_{j=1}^{k} a_{nj,n+1} = (-1)^{2n+1} \sum_{j=1}^{k} f(g_0, \ldots, g_n) = -k\, f(g_0, \ldots, g_n).$

(4) $a_{mjm} = f(g_0, \ldots, g_{m-1}, \overline{g_m r_j}^{-1} g_m, \ldots, \overline{g_n r_j}^{-1} g_n) = -a_{m-1,jm}.$

(5) For $1 \le i \le n$, we define the functions h_i on $G \times \overset{n}{\cdots} \times G$ by

$$\begin{aligned} &h_i(x_0, \ldots, x_{n-1}) \\ &= (-1)^i \sum_{j=1}^{k} f(x_0, \ldots, x_{i-1}, \overline{x_{i-1} r_j}^{-1} x_{i-1}, \ldots, \overline{x_{n-1} r_j}^{-1} x_{n-1}). \end{aligned}$$

Then for all $g \in G$, we have

$$\begin{aligned} &h_i(x_0 g, \ldots, x_{n-1}\, g) \\ &= (-1)^i \sum_{j=1}^{k} f(x_0 g, \ldots, x_{i-1} g, \overline{x_{i-1} g r_j}^{-1} x_{i-1}\, g, \ldots, \overline{x_{n-1} g r_j}^{-1} x_{n-1}\, g). \end{aligned}$$

If we determine $r_{j'}$ by $r_{j'} = \overline{g r_j}$, it follows that

$$\begin{aligned} &h_i(x_0 g, \ldots, x_{n-1} g) \\ &= (-1)^i \sum_{j=1}^{k} f(x_0, \ldots, x_{i-1}, \overline{x_{i-1} r_{j'}}^{-1} x_{i-1}, \ldots, \overline{x_{n-1} r_{j'}}^{-1} x_{n-1})\, g \\ &= h_i(x_0, \ldots, x_{n-1})\, g. \end{aligned}$$

Therefore, h_i lies in $\widetilde{\mathbf{C}}^{n-1}(G, \mathfrak{m})$. We have

$$\begin{aligned} &(h_i \widetilde{\delta}_{n-1})(g_0, \ldots, g_n) \\ &= (-1)^i \sum_{j=1}^{k} \Big\{ f(g_1, \ldots, g_i, \overline{g_i r_j}^{-1} g_i, \ldots, \overline{g_n r_j}^{-1} g_n) - \\ &\quad - f(g_0, g_2, \ldots, g_i, \overline{g_i r_j}^{-1} g_i, \ldots, \overline{g_n r_j}^{-1} g_n) \pm \cdots + \\ &\quad + (-1)^{i-1} f(g_0, \ldots, g_{i-2}, g_i, \overline{g_i r_j}^{-1} g_i, \ldots, \overline{g_n r_j}^{-1} g_n) + \\ &\quad + (-1)^i f(g_0, \ldots, g_{i-1}, \overline{g_{i-1} r_j}^{-1} g_{i-1}, \overline{g_{i+1} r_j}^{-1} g_{i+1}, \ldots, \overline{g_n r_j}^{-1} g_n) \pm \\ &\quad \pm \cdots + (-1)^n f(g_0, \ldots, g_{i-1}, \overline{g_{i-1} r_j}^{-1} g_{i-1}, \ldots, \overline{g_{n-1} r_j}^{-1} g_{n-1}) \Big\} \\ &= (-1)^i \sum_{j=1}^{k} \big\{ (-1)^i a_{ij0} - (-1)^{i+1} a_{ij1} + \cdots + (-1)^{3i-2} a_{ij,i-1} \\ &\quad + (-1)^i a_{i-1,j,i+1} + \cdots + (-1)^{2n+i} a_{i-1,j,n+1} \big\} \\ &= \sum_{j=1}^{k} \big\{ a_{ij0} + a_{ij1} + \cdots + a_{ij,i-1} + a_{i-1,j,i+1} + \cdots + a_{i-1,j,n+1} \big\}. \end{aligned}$$

Altogether, we then obtain

$$(f\widetilde{\rho}_n\tilde{\iota}_n)(g_0, \ldots, g_n) - k\, f(g_0, \ldots, g_n) = -\sum_{i=1}^{n} (h_i\, \widetilde{\delta}_{n-1})(g_0, \ldots, g_n).$$

Therefore, $f\widetilde{\rho}_n\tilde{\iota}_n - kf \in \widetilde{\mathbf{B}}^n(G, \mathfrak{m})$. □

If we carry over the maps $\widetilde{\rho}_n$ and $\tilde{\iota}_n$ to $\mathbf{C}^n(G, \mathfrak{m})$ with the help of μ_n, then we obtain

$$f\rho_n\iota_n - kf \in \mathbf{B}^n(G, \mathfrak{m})$$

for all $f \in \mathbf{Z}^n(G, \mathfrak{m})$.

Theorem 16.18 has many important consequences, some of which we mention now.

Theorem 16.19. *Let G be a finite group, and let $\mathfrak{m}$ be a G-module.*
a) *The exponent of $\mathbf{H}^n(G, \mathfrak{m})$ is a divisor of $|G|$.*
b) *If $\mathfrak{m}$ is finitely generated, then $\mathbf{H}^n(G, \mathfrak{m})$ is finite.*

Proof. a) By Theorem 16.10, $\mathbf{H}^n(1, \mathfrak{m}) = 0$ for all $n \geq 1$. (It is also not difficult to see this directly.) We set $U = 1$ in Theorem 16.18. For all $h \in \mathbf{H}^n(G, \mathfrak{m})$, it then follows that $hr_n \in \mathbf{H}^n(1, \mathfrak{m}) = 0$, and therefore

$$|G|h = hr_n j_n = 0.$$

b) Let $\{m_1, \ldots, m_k\}$ be a generating set for $\mathfrak{m}$ as a G-module. Then $\widetilde{\mathbf{C}}^n(G, \mathfrak{m})$ is a finitely generated abelian group. Since

$$f(g_0 g, \ldots, g_n g) = f(g_0, \ldots, g_n)\, g$$

for $f \in \widetilde{\mathbf{C}}^n(G, \mathfrak{m})$, the group $\widetilde{\mathbf{C}}^n(G, \mathfrak{m})$ is generated by the functions that take one of the values m_i $(i = 1, \ldots, k)$ on exactly one of the $|G|^{n+1}$ elements of $G \times \cdots \times G$ and are zero otherwise. By Theorem 13.11, the subgroup $\widetilde{\mathbf{Z}}^n(G, \mathfrak{m})$ of $\widetilde{\mathbf{C}}^n(G, \mathfrak{m})$ is finitely generated. Therefore,

$$\mathbf{H}^n(G, \mathfrak{m}) \cong \widetilde{\mathbf{Z}}^n(G, \mathfrak{m})/\widetilde{\mathbf{B}}^n(G, \mathfrak{m})$$

is finitely generated, as well. Since $\mathbf{H}^n(G, \mathfrak{m})$ is a torsion group by a), $\mathbf{H}^n(G, \mathfrak{m})$ is finite. □

Theorem 16.20. *Let U be a subgroup of G with finite index k. Let $\mathfrak{m}$ be a G-module such that $\mathbf{H}^n(U, \mathfrak{m}) = 0$ for some $n \geq 1$. Suppose there exists a $\tau \in \mathrm{Hom}_{\mathbb{Z}}(\mathfrak{m}, \mathfrak{m})$ such that $(km)\tau = m$ for all $m \in \mathfrak{m}$. Then $\mathbf{H}^n(G, \mathfrak{m}) = 0$.*

Proof. Let $h \in \mathbf{H}^n(G, \mathfrak{m})$. Then $hr_n \in \mathbf{H}^n(U, \mathfrak{m}) = 0$, so that by Theorem 16.18 we have

$$kh = hr_n j_n = 0.$$

We will show that $h = 0$. The endomorphism τ of $\mathfrak{m}$ induces an endomorphism τ_n^* of $\mathbf{H}^n(G, \mathfrak{m})$ by Theorem 16.5 and Lemma 16.3. The endomorphism φ of $\mathfrak{m}$ defined by $m\varphi = km$ for all $m \in \mathfrak{m}$ induces on $\mathbf{H}^n(G, \mathfrak{m})$ the "multiplication by k" map. It follows from $\varphi\tau = 1$ that $(kh)\tau_n^* = h$ for all $h \in \mathbf{H}^n(G, \mathfrak{m})$. So $\mathbf{H}^n(G, \mathfrak{m}) = 0$. □

Theorem 16.21. *Let G be a finite group, and let P be a Sylow p-subgroup of G. Let $\mathfrak{m}$ be a finitely generated G-module. Then the Sylow p-subgroup of the finite abelian group $\mathbf{H}^n(G, \mathfrak{m})$ is isomorphic to a subgroup of $\mathbf{H}^n(P, \mathfrak{m})$.*

Proof. Let r_n be the restriction of $\mathbf{H}^n(G, \mathfrak{m})$ to $\mathbf{H}^n(P, \mathfrak{m})$, and let $h \in \ker r_n$. By Theorem 16.18, we have

$$[G : P]h = hr_n j_n = 0.$$

Therefore, the intersection of the Sylow p-subgroup of $\mathbf{H}^n(G, \mathfrak{m})$ and $\ker r_n$ is 0. Hence, r_n is a monomorphism from the Sylow p-subgroup of $\mathbf{H}^n(G, \mathfrak{m})$ to the p-group $\mathbf{H}^n(P, \mathfrak{m})$. □

Theorem 16.21 will be put to good use in Chapter V when we calculate Schur multipliers.

17 Theorems of Gaschütz and Maschke

In the following applications of group cohomology, the G-module $\mathfrak{m}$ will most often be an abelian normal subgroup A of G on which G and G/A act as $a^g = a^{Ag}$. In this case, the formulas for the coboundary operator δ on $\mathbf{C}^n(G, A)$ from Definition 16.12 take the following form for $n \leq 2$ (which are the interesting values of n for the applications):

$$n = 0 : (f\delta)(g) = f\, f^{-g}$$
$$n = 1 : (f\delta)(g_1, g_2) = f(g_2)\, f(g_1 g_2)^{-1}\, f(g_1)^{g_2}$$
$$n = 2 : (f\delta)(g_1, g_2, g_3) = f(g_2, g_3)\, f(g_1 g_2, g_3)^{-1}\, f(g_1, g_2 g_3)\, f(g_1, g_2)^{-g_3}.$$

Theorem 17.1.

a) *Let $N \trianglelefteq G$, and let α be an automorphism of G with $n^\alpha = n$ for all $n \in N$ and $g^\alpha N = gN$ for all $g \in G$. Then $g^\alpha = g\, f(gN)$ for some 1-cocycle $f \in \mathbf{Z}^1(G/N, \mathbf{Z}(N))$; here, the center $\mathbf{Z}(N)$ of N is regarded as a right G/N-module with the clearly well-defined action $x^{Ng} = x^g$ for $x \in \mathbf{Z}(N)$ and $g \in G$.*

b) *Let $f \in \mathbf{Z}^1(G/N, \mathbf{Z}(N))$. Then the map α defined by $g^\alpha = g\, f(gN)$ is an automorphism of G that leaves N and G/N fixed elementwise.*

c) *Suppose the automorphism α of G and the 1-cocycle $f \in \mathbf{Z}^1(G/N, \mathbf{Z}(N))$ correspond as in a). Then $f \in \mathbf{B}^1(G/N, \mathbf{Z}(N))$ if and only if there is some $h \in N$ such that $g^\alpha = g^h$ for all $g \in G$.*

Proof. a) By Theorem 4.4, we know that $g^\alpha = g\,t(g)$ for some $t(g) \in \mathbf{Z}(N)$, where $t(g)$ depends only on the coset Ng. Furthermore, we have

$$1 = t(g_2)\,t(g_1g_2)^{-1}\,t(g_1)^{g_2}.$$

We define f by $f(gN) = t(g)$, so then $f \in \mathbf{Z}^1(G/N, \mathbf{Z}(N))$.

b) We have

$$1 = (f\delta)(g_1N, g_2N) = f(g_2N)\,f(g_1g_2N)^{-1}\,f(g_1N)^{g_2}.$$

Therefore,

$$\begin{aligned}(g_1g_2)^\alpha &= g_1g_2\,f(g_1g_2N) = g_1g_2\,f(g_1N)^{g_2}\,f(g_2N)\\ &= g_1\,f(g_1N)\,g_2\,f(g_2N) = g_1^\alpha g_2^\alpha.\end{aligned}$$

Hence, α is an endomorphism of G with $g^\alpha N = gN$ for all $g \in G$. For $g_1 = g_2 = 1$, we obtain

$$1 = (f\delta)(N, N) = f(N).$$

Thus, $n^\alpha = n$ for all $n \in N$. It is now easy to see that α is an automorphism of G.

c) First assume that $f = h\delta \in \mathbf{B}^1(G/N, \mathbf{Z}(N))$ with $h \in \mathbf{Z}(N)$. Then

$$f(gN) = (h\delta)(gN) = hh^{-g} = h^{-g}h$$

and

$$g^\alpha = g\,f(gN) = g\,h^{-g}\,h = h^{-1}\,g\,h = g^h$$

with $h \in \mathbf{Z}(N)$.

Conversely, assume that

$$g^\alpha = g\,f(gN) = g^h$$

for some $h \in N$. Since $x^\alpha = x$ for $x \in N$, we have $h \in \mathbf{Z}(N)$. It follows that

$$f(gN) = g^{-1}h^{-1}gh = hh^{-g} = (h\delta)(gN). \qquad \square$$

We will use Theorem 17.1 in Chapter III, §19 when we consider automorphisms of p-groups.

Theorem 17.2. *Let A be an abelian normal subgroup of G. We give A the structure of a right G/A-module by defining $a^{Ag} = a^g$ for all $a \in A$. We set $H = G/A$.*

a) *For every transversal $\{r(h) \mid h \in H\}$ for the cosets of A in G, there corresponds a 2-cocycle $c \in \mathbf{Z}^2(H, A)$ such that*

$$r(h_1)\,r(h_2) = r(h_1h_2)\,c(h_1, h_2).$$

The element $c\,\mathbf{B}^2(H,A)$ in $\mathbf{H}^2(H,A)$ is uniquely determined by the extension G and does not depend on the choice of transversal $\{r(h) \mid h \in H\}$.

b) *G is a semidirect product of A with H if and only if $c \in \mathbf{B}^2(H,A)$.*

c) *If A is an H-group and $\mathbf{H}^2(H,A) = 1$, then every extension of A by H is a semidirect product.*

Proof. a) Let $G = \bigcup_{h\in H} r(h)A$ with $h = r(h)A$ be the decomposition of G into cosets of A, and let

$$r(h_1)\,r(h_2) = r(h_1h_2)\,c(h_1,h_2)$$

with $c(h_1,h_2) \in A$. If we regard A as an H-module as described in the statement of the theorem, then Theorem 14.1 yields the equations

$$\begin{aligned} 1 &= c(h_2,h_3)\,c(h_1h_2,h_3)^{-1}\,c(h_1,h_2h_3)\,c(h_1,h_2)^{-h_3} \\ &= (c\delta)(h_1,h_2,h_3) \end{aligned}$$

since A is abelian. Therefore, $c \in \mathbf{Z}^2(H,A)$.

If $\{r(h)\,a(h) \mid h \in H\}$ with $a(h) \in A$ is another transversal for the cosets of A in G, then by Lemma 14.3 we obtain a factor system c' such that

$$c'(h_1,h_2) = c(h_1,h_2)\,a(h_1)^{h_2}\,a(h_1h_2)^{-1}\,a(h_2).$$

Hence, $c' = c(a\delta)$.

b) G is a semidirect product of A with H if and only if there exists a transversal $\{r'(h) \mid h \in H\}$ corresponding to a factor system c' such that $c'(h_1,h_2) = 1$ for all $h_i \in H$. By a), this is equivalent to $c \in \mathbf{B}^2(H,A)$.

c) follows directly from b). □

Theorem 17.3. *Let $G = AK$ be a semidirect product of the abelian normal subgroup A with K. We give A, written multiplicatively, the structure of a right K-module by defining $a^k = k^{-1}ak$ for all $a \in A$ and $k \in K$.*

a) *Every conjugacy class of complements of A in G corresponds to exactly one element in $\mathbf{H}^1(K,A)$. In particular, $|\mathbf{H}^1(K,A)|$ is the number of G-conjugacy classes of complements of A in G.*

b) *The complements of A in G are all G-conjugate if and only if $\mathbf{H}^1(K,A) = 1$.*

Proof. a) Let $V = \{k\,a(k) \mid k \in K\}$ with $a(k) \in A$ be a complement of A in G. Then

$$k_1\,a(k_1)\,k_2\,a(k_2) = k_1\,k_2\,a(k_1k_2)$$

for all $k_1, k_2 \in K$. It follows that

$$a(k_1)^{k_2}\,a(k_1k_2)^{-1}\,a(k_2) = (a\delta)(k_1,k_2) = 1,$$

and a is a 1-cocycle in $\mathbf{Z}^1(K,A)$. Conversely, if $a \in \mathbf{Z}^1(K,A)$, then the set $V = \{k\,a(k) \mid k \in K\}$ is a complement of A in G by the same calculation.

Suppose $V_2 = V_1^x$. Since $x \in V_1 A = G$, we may assume that $x = a \in A$. Since $k^a A = kA$, we have

$$k\, a_2(k) = (k\, a_1(k))^a = k^a a_1(k)$$

for all $k \in K$. By the commutativity of A, it follows that

$$a_2(k)\, a_1(k)^{-1} = k^{-1}\, k^a = a\, a^{-k},$$

and so

$$a_2\, a_1^{-1} \in \mathbf{B}^1(K, A).$$

Conversely, it follows from

$$a_2 a_1^{-1} \in \mathbf{B}^1(K, A), \quad \text{that is,} \quad a_2(k)\, a_1(k)^{-1} = a a^{-k},$$

that again

$$k\, a_2(k) = (k\, a_1(k))^a.$$

Therefore, the conjugacy classes of complements of A in G are in one-to-one correspondence with the elements of $\mathbf{H}^1(K, A) = \mathbf{Z}^1(K, A)/\mathbf{B}^1(K, A)$. The claim in b) is a special case of a). □

Main Theorem 17.4 (Gaschütz [154]). *Let A be an abelian normal subgroup of G, let $A \le B \le G$ and assume $[G : B] = k$ is finite. Suppose the map $a \mapsto a^k$ (with $a \in A$) has an inverse τ. (This is always the case when A is finite and $(|A|, k) = 1$; then the congruence $ky \equiv 1 \bmod |A|$ has a solution, and τ defined by $a^\tau = a^y$ gives the desired map.)*

a) *If A has a complement in B, then A has a complement in G.*

b) *If A has a complement in B and all complements of A in B are conjugate in B, then all complements of A in G are conjugate in G.*

(*The assumption that A is abelian cannot be removed, see Example 18.7.*)

Proof. a) Let $H = G/A$, and let $\{r(h) \mid h \in H\}$ be a transversal for the cosets of A in G. Let $c \in \mathbf{Z}^2(H, A)$ be the 2-cocycle constructed from $\{r(h) \mid h \in H\}$ according to Theorem 17.2. Then the restriction $c\rho_2$,[4] constructed with respect to the subgroup $\overline{B} = B/A$ of H, is the 2-cocycle for the extension B of A. Since A has a complement in B, we have $c\rho_2 = b\delta_1$ for some $b \in \mathbf{C}^1(\overline{B}, A)$ by Theorem 17.2. By Theorem 16.18, we have

$$c^k \equiv c\rho_2\iota_2 = b\delta_1\iota_2 = (b\iota_1)\delta_1 \equiv 1 \mod \mathbf{B}^2(H, A).$$

Hence, there exists some $f \in \mathbf{C}^1(H, A)$ such that $c^k = f\delta_1$. By assumption, it follows that

[4] Correction 4 from errata.

$$c = c^{k\tau} = (f\delta_1)^\tau.$$

Since the map sending a to a^k is a G-endomorphism of A, the inverse τ is a G-endomorphism as well. So we have

$$\begin{aligned} c(h_1, h_2) = c(h_1, h_2)^{k\tau} &= ((f\delta)(h_1, h_2))^\tau = (f(h_2)\, f(h_1h_2)^{-1}\, f(h_1)^{h_2})^\tau \\ &= f(h_2)^\tau\, f(h_1h_2)^{-\tau}\, f(h_1)^{\tau h_2} = (f^\tau\delta_1)(h_1, h_2). \end{aligned}$$

Hence, $c \in \mathbf{B}^2(H, A)$. Then A has a complement in G by Theorem 17.2.

b) Since all complements of A in B are conjugate by assumption, we have $\mathbf{H}^1(\overline{B}, A) = 1$ by Theorem 17.3. By Theorem 16.20, it follows that $\mathbf{H}^1(H, A) = 1$, so all complements of A in G are conjugate by Theorem 17.3. □

Next, we highlight a special case that we will greatly generalize in the next section.

Theorem 17.5. *Assume A is an abelian normal subgroup of the finite group G with $(|A|, |G/A|) = 1$. Then A has a complement in G and all complements of A in G are conjugate in G.*

Proof. This is the special case $A = B$ in Theorem 17.4. (A direct proof without the use of §16 is contained in Exercise 70.) □

The following fundamental theorem will be used numerous times.

Theorem 17.6 (Maschke, Schur). *Let G be a finite group of order n, and let $\mathfrak{m}$ be a right G-module. Suppose there is a G-endomorphism τ of $\mathfrak{m}$ such that $(n\mathfrak{m})\tau = m$ for all $m \in \mathfrak{m}$. Let $\mathfrak{m} = \mathfrak{m}_1 \oplus \mathfrak{m}_2^*$ be a direct decomposition of $\mathfrak{m}$ as an abelian group, where $\mathfrak{m}_1$ is a G-submodule of $\mathfrak{m}$. Then there exists a direct decomposition $\mathfrak{m} = \mathfrak{m}_1 \oplus \mathfrak{m}_2$ of $\mathfrak{m}$ as a G-module.*

Proof. We give the product set $L = \mathfrak{m} \times G$ the structure of a group by declaring

$$(m_1, g_1)(m_2, g_2) = (m_1g_2 + m_2,\ g_1g_2).$$

One easily checks that L is actually a group. The subsets

$$B = \{(m, 1) \mid m \in \mathfrak{m}\} \quad \text{and} \quad B_1 = \{(m, 1) \mid m \in \mathfrak{m}_1\}$$

are abelian normal subgroups of L. Since $B = B_1 \times B_2^*$ with $B_2^* = \{(m, 1) \mid m \in \mathfrak{m}_2^*\}$, B_1 has a complement in B. The assumptions of Theorem 17.4 with $[L : B] = n$ are satisfied, so B_1 has a complement V in L. Then

$$B = B_1 \times (B \cap V)$$

and $B \cap V \trianglelefteq L$. We set

$$\mathfrak{m}_2 = \{m \mid (m, 1) \in B \cap V\}.$$

For $m \in \mathfrak{m}_2$ and $g \in G$, we have $(0, g)^{-1}(m, 1)(0, g) = (mg, 1) \in B \cap V$, so $mg \in \mathfrak{m}_2$. Hence, $\mathfrak{m}_2$ is a G-module such that $\mathfrak{m} = \mathfrak{m}_1 \oplus \mathfrak{m}_2$. □

Closely related to Theorem 17.6 is the following theorem, which plays a fundamental role in representation theory (see Theorem V.3.4, pg. 495).

Theorem 17.7. *Let G be a finite group of order n, and let K be a field such that* $\operatorname{char}\mathsf{K} \nmid n$ (*in particular, this assumption is always satisfied if* $\operatorname{char}\mathsf{K} = 0$). *Let V be a $\mathsf{K}[G]$-module, and let V_1 be a $\mathsf{K}[G]$-submodule of V. Assume that $(va)g = v(ag)$ for all $v \in \mathsf{V}$, all $a \in \mathsf{K}$ and all $g \in G$. Then there exists a direct decomposition $\mathsf{V} = \mathsf{V}_1 \oplus \mathsf{V}_2$ of V as a $\mathsf{K}[G]$-module.*

Proof. It is well known that there exists a K-subspace W of V such that $\mathsf{V} = \mathsf{V}_1 \oplus \mathsf{W}$. For $w \in \mathsf{W}$ and $g \in G$, there are uniquely determined $v_1 \in \mathsf{V}_1$ and $w' \in \mathsf{W}$ such that $wg = v_1 + w'$. We define the linear maps $s(g) \in \operatorname{Hom}_{\mathsf{K}}(\mathsf{W}, \mathsf{V}_1)$ and $u_2(g) \in \operatorname{Hom}_{\mathsf{K}}(\mathsf{W}, \mathsf{W})$ by $w\,s(g) = v_1$ and $w\,u_2(g) = w'$. Furthermore, let $u_1(g) \in \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_1, \mathsf{V}_1)$ be defined by $v_1\,u_1(g) = v_1 g$ for $v_1 \in \mathsf{V}_1$. Then for all $w \in \mathsf{W}$, we have

$$\begin{aligned} w\,s(g_1g_2) + w\,u_2(g_1g_2) &= w(g_1g_2) \\ &= (wg_1)g_2 = (w\,s(g_1) + w\,u_2(g_1))g_2 \\ &= w\,s(g_1)\,u_1(g_2) + w\,u_2(g_1)\,s(g_2) + w\,u_2(g_1)\,u_2(g_2). \end{aligned}$$

Upon comparing the components in V_1 and W, we see that

(1) $s(g_1g_2) = s(g_1)\,u_1(g_2) + u_2(g_1)\,s(g_2)$ and
(2) $u_2(g_1g_2) = u_2(g_1)\,u_2(g_2)$.

We may give the K-vector space $\mathfrak{s} = \operatorname{Hom}_{\mathsf{K}}(\mathsf{W}, \mathsf{V}_1)$ the structure of a right $\mathsf{K}[G]$-module by defining $sg = u_2(g)^{-1}\,s\,u_1(g)$ for $s \in \operatorname{Hom}_{\mathsf{K}}(\mathsf{W}, \mathsf{V}_1)$, as is easily checked. We define $t(g) \in \mathfrak{s}$ by $t(g) = u_2(g)^{-1}s(g)$. From equations (1) and (2), it follows that

$$\begin{aligned} &t(g_2) - t(g_1g_2) + t(g_1)\,g_2 \\ &= u_2(g_2)^{-1}\,s(g_2) - u_2(g_1g_2)^{-1}\,s(g_1g_2) + u_2(g_2)^{-1}\,u_2(g_1)^{-1}\,s(g_1)\,u_1(g_2) \\ &= u_2(g_1g_2)^{-1}\left(u_2(g_1)\,s(g_2) - s(g_1g_2) + s(g_1)\,u_1(g_2)\right) = 0. \end{aligned}$$

Therefore, $t \in \mathbf{Z}^1(G, \mathfrak{s})$. Since $\operatorname{char}\mathsf{K} \nmid n$, the assumptions of Theorem 16.20 (with $U = 1$) are satisfied, which implies $\mathbf{H}^1(G, \mathfrak{s}) = 0$. Therefore, $t \in \mathbf{B}^1(G, \mathfrak{s})$ and so

$$t(g) = b - bg = b - u_2(g)^{-1}\,b\,u_1(g)$$

for some $b \in \mathbf{C}^0(G, \mathfrak{s}) = \mathfrak{s}$. Hence,

$$s(g) = u_2(g)\,b - b\,u_1(g)$$

for all $g \in G$. We set $\mathsf{V}_2 = \{w(1+b) \mid w \in \mathsf{W}\}$. Then $\mathsf{V} = \mathsf{V}_1 \oplus \mathsf{V}_2$ is clearly a direct decomposition as K-vector spaces. Furthermore,

$$w\,(1+b)\,g = w\,s(g) + w\,u_2(g) + w\,b\,u_1(g)$$
$$= w\,u_2(g)\,(1+b) \in \mathsf{V}_2.$$

Hence, V_2 is a $\mathsf{K}[G]$-submodule of V. □

Example 17.8. The assumption that $\operatorname{char}\mathsf{K} \nmid n$ cannot be removed in Theorem 17.7: Let $G = \langle g \rangle$ be the cyclic group of order p, and let V be the vector space of dimension 2 over the field $\mathrm{GF}(p)$. Let $\{v_1, v_2\}$ be a basis of V. We give V the structure of a G-module by defining

$$v_1 g^i = v_1, \quad v_2 g^i = v_2 + i v_1 \quad (0 \le i \le p-1).$$

It is easy to see that there is no direct decomposition $\mathsf{V} = \mathsf{V}_1 \oplus \mathsf{V}_2$ with $\mathsf{V}_1 = \langle v_1 \rangle$ and V_2 a G-invariant subspace.

Exercises

62) We regard the vector space $\mathsf{V} = \mathsf{V}(n, p^f)$ of dimension n over the field $\mathrm{GF}(p^f)$ as a right module for the general linear group $\mathrm{GL}_n(p^f)$ in the natural way. Show that if $p^f > 2$, then

$$\mathbf{H}^k(\mathrm{GL}_n(p^f), \mathsf{V}) = 0 \quad \text{for} \quad k = 1, 2.$$

(Hint: Use $|\mathbf{Z}(\mathrm{GL}_n(p^f))| = p^f - 1$, the Frattini argument from Theorem 7.8 (pg. 36), and Theorem 17.5.)

63) Let A be an abelian normal subgroup of G with $|G/A| = \prod_i p_i^{a_i}$. Let P_i/A be a Sylow p_i-subgroup of G/A. Show that A has a complement in G if and only if A has a complement in all P_i.

64) (K. H. Hofmann [249]) Suppose the finite group G is a semidirect product of an abelian normal subgroup N with H. Let A be a group of automorphisms of G such that $N^\alpha = N$ for all $\alpha \in A$, and assume $(|A|, |N|) = 1$. Then there exists a complement K of N in G such that $K^\alpha = K$ for all $\alpha \in A$.

65) Let G be a finite group, and let $G = D \times H$ be a direct decomposition of G. Let A be a group of automorphisms of G such that $D^\alpha = D$ for all $\alpha \in A$ and $(|A|, |\mathbf{Z}(D)|) = 1$. Then there is a decomposition $G = D \times K$ such that $K^\alpha = K$ for all $\alpha \in A$.

66) Let μ be a homomorphism from the right G-module $\mathfrak{m}$ to the right G-module $\mathfrak{m}'$ with $\ker \mu = \mathfrak{n}$. Then there is an exact sequence of homomorphisms

$$0 \xrightarrow{\mu_1} \mathbf{Z}^n(G, \mathfrak{n}) \xrightarrow{\mu_2} \mathbf{Z}^n(G, \mathfrak{m}) \xrightarrow{\mu_3} \mathbf{Z}^n(G, \mathfrak{m}').$$

If all the groups appearing in this sequence are finite, then we have

$$|\mathbf{Z}^n(G, \mathfrak{m})| \le |\mathbf{Z}^n(G, \mathfrak{m}')|\,|\mathbf{Z}^n(G, \mathfrak{n})|.$$

67) (Gaschütz [158]). Let G be a finite group with no nontrivial quotients of p-power order. Let $U < G$, and let p^d be the order of the largest elementary abelian quotient of U. Assume G may be generated by k elements. Then

$$d \leq [G : U](k-1) - k + 1.$$

(Compare this with the bound obtained from Schreier's Theorem 19.11, $d \leq [G : U](k-1) + 1$.)

(Hint: Let $G = \bigcup_{r \in R} Ur$ and $[G : U] = n$. Let $\mathsf{K} = \mathrm{GF}(p)$, and let V be the vector space of dimension n over K with basis $\{v_{Ur} \mid r \in R\}$. We give V the structure of a right G-module by defining $v_{Ur}g = v_{Urg}$. Show:

a) Let $\mathsf{W} = \{a \sum_{r \in R} v_{Ur} \mid a \in \mathsf{K}\}$. Then we have

$$|\mathbf{Z}^1(G, \mathsf{V})| \leq |\mathbf{Z}^1(G, \mathsf{V}/\mathsf{W})|\,|\mathbf{Z}^1(G, \mathsf{W})| \leq p^{k([G:U]-1)}.$$

b) Let $f \in \mathbf{Z}^1(G, \mathsf{V})$ and $f(g) = \sum_{r \in R} a_{Ur}(g)\, v_{Ur}$. Then for all $u \in U$ and $g \in G$, we have

$$a_U(gu) = a_U(g) + a_U(u).$$

In particular, the map h defined by $h(u) = a_U(u)$ is a homomorphism from U to the additive group K^+ of K.

c) For $r \in R$ and $g \in G$, let $gr^{-1} = r_1^{-1}u_1$ with $r_1 \in R$ and $u_1 \in U$. Then $a_{Ur}(g) = a_U(r_1^{-1}) + a_U(u_1) - a_U(r^{-1})$.

d) Let h be a homomorphism from U to K^+. We freely choose the values $h(r^{-1}) \in \mathsf{K}$ for $1 \neq r \in R$. Then the function f defined by $f(g) = \sum_{r \in R} a_{Ur}(g)\, v_{Ur}$ with $a_{Ur}(g) = h(r_1^{-1}) + h(u_1) - h(r^{-1})$ (where $gr^{-1} = r_1^{-1}u_1$ as in c)) is a 1-cocycle in $\mathbf{Z}^1(G, \mathsf{V})$. Therefore,

$$|\mathbf{Z}^1(G, \mathsf{V})| = p^d p^{[G:U]-1}.)$$

68) Let P be an abelian p-group, and let A be a group of automorphisms of P with $p \nmid |A|$. Let p^m be the largest invariant of P. Then there is a decomposition $P = P_1 \times P_2$ in which P_1 and P_2 are invariant under A, P_1 has only invariants p^m and all invariants of P_2 are smaller than p^m.

(Hint: For $m > 1$, use induction on the exponent $\exp P = p^m$ of P.)

69) Let P be an abelian p-group, and let A be a group of automorphisms of P with $p \nmid |A|$. Show that:

a) There is a decomposition $P = P_1 \times \cdots \times P_s$, where all P_i are invariant under A and all the invariants of each P_i are equal.

b) $\Omega_1(P)$ and $P/\mho_1(P)$ are isomorphic as A-groups (see Exercise 51, pg. 91).

70) Give a direct proof of Theorem 17.5 in the following way: Let A be an abelian normal subgroup of G with $G/A = H$ and $(|A|, |H|) = 1$. Let $G = \bigcup_{h \in H} r(h)A$ be the decomposition of G into cosets of A.
   a) Let $r(h_1)\, r(h_2) = r(h_1 h_2)\, c(h_1, h_2)$. Choose z such that $|H|z \equiv 1 \bmod |A|$ and define $c(h) = \prod_{x \in H} c(x, h)^{-z}$. Then $\{r(h)\, c(h) \mid h \in H\}$ is a complement of A in G.
   b) Let $\{r(h) \mid h \in H\}$ and $\{r'(h) \mid h \in H\}$ be complements of A in G. Let $r'(h) = r(h)\, a(h)$ for some $a(h) \in A$. Define $s = \prod_{h \in H} a(h)^z$ and prove that $r'(h) = s^{-1}\, r(h)\, s$.

18 The Zassenhaus theorem

In this section, all groups are assumed to be finite.

Our objective is to prove a fundamental criterion for the existence of a complement of a normal subgroup. Our assumptions are as follows:

Let N be a normal subgroup of G with $G/N = H$; assume the orders of N and H are coprime.

Note that every subgroup V of G with $|V| = |H|$ is a complement of N:

Since $(|V|, |N|) = 1$, we have $V \cap N = 1$, and it follows by Lemma 2.12 a) that

$$|VN| = |V||N| = |G|,$$

and so $G = VN$.

Main Theorem 18.1 (Zassenhaus). *Let N be a normal subgroup of a finite group G. Assume that $|N|$ and $|G/N|$ are coprime. Then there exists a complement of N in G.*

Proof. Assume for contradiction that the theorem is false. Let G be a counterexample of smallest possible order. We show:

(1) N is a direct product of its Sylow subgroups:

If P is a Sylow subgroup of N, then by the Frattini argument of Theorem 7.8, we have $G = N\mathbf{N}_G(P)$. It follows that

$$|G/N| = |\mathbf{N}_G(P)/\mathbf{N}_G(P) \cap N|.$$

Hence, the group $\mathbf{N}_G(P)$ and its normal subgroup $\mathbf{N}_N(P) = \mathbf{N}_G(P) \cap N$ satisfy our assumptions. If $\mathbf{N}_G(P) < G$, then there exists a complement V of $\mathbf{N}_N(P)$ in $\mathbf{N}_G(P)$. We have

$$|V| = |\mathbf{N}_G(P)/\mathbf{N}_G(P) \cap N| = |G/N|.$$

Then V is also a complement of N in G by the remark at the beginning of this section. This contradicts the choice of G as a counterexample. Therefore, $\mathbf{N}_G(P) = G$ for all

Sylow subgroups P of N. In particular, all Sylow subgroups of N are normal in N, and so N is the direct product

$$N = P_1 \times \cdots \times P_r$$

of its Sylow subgroups P_i by Theorem 9.5.

(2) N is abelian:

By Theorem 8.9 and Lemma 9.9, N is solvable, so $N' < N$. Assume for contradiction that $1 < N'$. Since $|G/N'| < |G|$, the theorem holds for G/N' with the normal subgroup N/N'. So there exists a complement of N/N' in G/N', which has the form W/N' for some subgroup W of G. Then

$$|W/N'| = |(G/N')\big/(N/N')| = |G/N|$$

is coprime to $|N'|$ and

$$|W| = |W/N'||N'| = |G/N||N'| < |G|.$$

So there exists a complement of N' in W. Such a complement has order $|G/N|$, so is also a complement of N in G. This contradicts our choice of G as a counterexample. Therefore, $N' = 1$ and N is abelian.

By Theorem 17.5, there exists a complement of N in G. This contradicts our assumption that a counterexample exists, and Theorem 18.1 is proved. □

Main Theorem 18.2 (Zassenhaus). *Suppose N is a normal subgroup of the finite group G with $(|N|, |G/N|) = 1$. If N or G/N is solvable, then all complements of N in G are conjugate.*

Proof. a) First assume that N is solvable, so $N' < N$. If $N' = 1$, then N is abelian and the claim follows from Theorem 17.5. If $1 < N'$, then we prove the claim by induction on $|G|$ as follows:

Let V_1 and V_2 be two complements of N in G. Since

$$|V_iN'/N'| = |V_i/V_i \cap N'| = |V_i| = |G/N| = |(G/N')\big/(N/N')|,$$

the V_iN'/N' are complements of N/N' in G/N'. Therefore, the V_iN'/N' are conjugate in G/N'. That is,

$$V_1N' = (V_2N')^x = V_2^xN'$$

for some $x \in G$. Now, V_1 and V_2^x are complements of N' in V_1N' and are conjugate in V_1N' by induction since

$$V_1N' < V_1N = G.$$

So there exists some $y \in V_1N'$ such that

$$V_1 = (V_2^x)^y = V_2^{xy}.$$

b) Now assume that G/N is solvable. Again we prove the claim by induction on $|G|$.

Let M/N be a minimal normal subgroup of G/N. By the Dedekind identity Lemma 2.12 c), we have

$$M = M \cap (V_i N) = (M \cap V_i)N \quad (i = 1, 2),$$

where $M \cap V_i \trianglelefteq V_i$.

If $M < G$, then the $M \cap V_i$ are complements of N in M, and so are conjugate in M by our induction hypothesis. Therefore, there is some $x \in M$ such that

$$M \cap V_1 = (M \cap V_2)^x = M \cap V_2^x.$$

Replacing V_2 with $V_2^{x^{-1}}$ if necessary, we may assume that

$$M \cap V_1 = M \cap V_2.$$

We define $J = M \cap V_1$, and we have $J \trianglelefteq V_i$ $(i = 1, 2)$. Since

$$\mathbf{N}_G(J) = \mathbf{N}_G(J) \cap NV_i = (\mathbf{N}_G(J) \cap N)\, V_i$$

and

$$(\mathbf{N}_G(J) \cap N)J \cap V_i = (\mathbf{N}_G(J) \cap N \cap V_i)\, J = J,$$

the V_i/J are complements of $(\mathbf{N}_G(J) \cap N)J/J$ in $\mathbf{N}_G(J)/J$, and so are conjugate in $\mathbf{N}_G(J)/J$ by induction. Then the V_i are also conjugate in G.

It remains to consider the case that $M = G$. Then G/N is a solvable simple group, so it must have prime order p. As p is not a divisor of $|N|$, the complements of N in G are exactly the Sylow p-subgroups of G, which are all conjugate in G by Theorem 7.5. □

Theorem 18.2 raises the question of whether one can remove the assumption that N or G/N is solvable. This is in fact the case. By the theorem of Feit–Thompson (Feit, Thompson [132]), every group of odd order is solvable. If $|N|$ and $|G/N|$ are coprime, then N or G/N is therefore solvable, and Theorem 18.2 may be applied. Hence, we have:

Main Theorem 18.3. *Let N be a normal subgroup of G such that $(|N|, |G/N|) = 1$. Then all complements of N in G are conjugate.*

A proof of Theorem 18.3 without using the extremely deep Feit–Thompson theorem (whose proof lies outside the scope of this book) is not known. Next, we describe a possible proof of Theorem 18.3 that does not rely on the Feit–Thompson theorem, but to finish it requires the so-called Schreier conjecture.

Lemma 18.4. *Assume the hypothesis of Theorem 18.3. Let G be a minimal counterexample to Theorem 18.3. Then:*

(1) *G/N and N are nonabelian simple groups.*
(2) $\mathbf{C}_G(N) = 1$.

Proof. Assume V_1 and V_2 are two complements of N in G that are not conjugate in G. We show:

a) N is a minimal normal subgroup of G:

Suppose $1 < M < N$ and $M \lhd G$. Then the V_iM/M are complements of N/M in G/M, so are conjugate in G/M. This implies $V_1M = V_2^xM$ for some $x \in G$. As complements of M in the proper subgroup V_1M of G, the subgroups V_1 and V_2^x are conjugate, contradicting the choice of G as a counterexample. Hence, N is a minimal normal subgroup of G.

b) We have $\mathbf{C}_G(N) = 1$:

Since $N \lhd G$, we have $C = \mathbf{C}_G(N) \trianglelefteq G$. Assume for contradiction that $C \neq 1$. Since N is nonabelian by Theorem 17.5 and is characteristically simple by a), we have $N \cap C = \mathbf{Z}(N) = 1$. Hence, $|C| = |CN/N|$ is a divisor of $|G/N| = |V_i|$. As

$$|CV_i| = \frac{|C||V_i|}{|C \cap V_i|}$$

is coprime to $|N|$, it follows that $C \leq V_i$ $(i = 1, 2)$. Hence, the V_i/C are complements of NC/C in G/C, and so are conjugate since $|G/C| < |G|$. Then V_1 and V_2 are conjugate in G, contradicting the choice of G as a counterexample.

c) G/N is simple:

Let M/N be a minimal normal subgroup of G/N with $M < G$. As in part b) of the proof of Theorem 18.2, we may assume that

$$V_1 \cap M = V_2 \cap M.$$

Defining $J = V_i \cap M$, we again obtain the conjugacy of V_1/J and V_2/J in $\mathbf{N}_G(J)/J$, and we have a contradiction. So $M = G$ and G/N is simple.

d) N is simple:

The subgroup N is characteristically simple as it is a minimal normal subgroup of G. By Theorem 9.12 (pg. 53), $N = N_1 \times \cdots \times N_r$ for some isomorphic simple groups N_i. Since N is nonabelian by Theorem 17.5, the N_i are nonabelian. We conclude from Theorem 9.12 that the N_i are the only minimal normal subgroups of N. The N_i are then transitively permuted by the inner automorphisms of G.

Assume for contradiction that $r > 1$. Let $H = \mathbf{N}_G(N_1)$. Since $[G : H] = r > 1$ (see Theorem 5.11, pg. 28) and $N_1 \lhd N$, we certainly have $N \leq H < G$. Let $q \mid |N_1|$, let Q_1 be a Sylow q-subgroup of N_1 and let $T = \mathbf{N}_G(Q_1)$. For $t \in T$, we have

$$N_1^t \cap N_1 \geq Q_1 > 1,$$

and so $N_1^t = N_1$. It follows that $T \leq H$, and then by the Frattini argument of Theorem 7.8, we have $H = N_1T = NT$. We then have $H/N \cong T/(T \cap N)$. Hence, $|T/(T \cap N)|$ and $|T \cap N|$ are coprime. Let U be a complement of $T \cap N$ in T, which exists by Theorem 18.1. Then we have

$$H = NT = N(T \cap N)U = NU$$

and

$$U \cap N = U \cap (N \cap T) = 1.$$

So U is a complement of N in H.

The $V_i \cap H$ $(i = 1, 2)$ are complements of N in H since

$$H = H \cap NV_i = N(H \cap V_i).$$

Since $H < G$, we may conclude that U, $V_1 \cap H$ and $V_2 \cap H$ are conjugate. We may therefore assume that $U = V_1 \cap H = V_2 \cap H$.

Since $G = NV_1$, the N_i are transitively permuted under conjugation by elements of V_1, so $V_1 = \bigcup_{i=1}^{r} (V_1 \cap H)v_i$, where $N_1^{v_i} = N_i$. Then $Q_1^{v_i} = Q_i$ $(i = 1, \ldots, r)$ is a Sylow q-subgroup of N_i, and so

$$Q = Q_1 \times \cdots \times Q_r$$

is a Sylow q-subgroup of N. For any $v \in V_1$, we have $v_i v = u v_j$ for some j and some $u \in V_1 \cap H = U \leq T = \mathbf{N}_G(Q_1)$. This implies

$$Q_i^v = Q_1^{v_i v} = Q_1^{u v_j} = Q_1^{v_j} = Q_j,$$

and therefore $Q^v = Q$. Hence, $V_1 \leq \mathbf{N}_G(Q)$, and we obtain via the Dedekind identity of Lemma 2.12 c) that $\mathbf{N}_G(Q) = V_1 \mathbf{N}_N(Q)$. In the same way, we conclude that $V_2 \leq \mathbf{N}_G(Q^n)$ for some $n \in N$. Hence, V_1 and $V_2^{n^{-1}}$ are complements of $\mathbf{N}_N(Q)$ in $\mathbf{N}_G(Q)$. If we had $\mathbf{N}_G(Q) < G$, then V_1 and $V_2^{n^{-1}}$ would be conjugate. Since this is not the case, $\mathbf{N}_G(Q) = G$ and so $N = Q$ by the minimality of N. So N is a characteristically simple group of q-power order, so is abelian. But then Theorem 17.5 yields the conjugacy of V_1 and V_2, contradicting our assumption.

Therefore $r = 1$ and N is simple. □

Remark 18.5. Theorem 18.3 would follow from the following conjecture[5] of Schreier: If G is a simple group, then the outer automorphism group $\mathbf{Aut}(G)/\mathbf{Inn}(G)$ is solvable.

Let $G = NV$ be a minimal counterexample to Theorem 18.3. By Lemma 18.4, $G/\mathbf{C}_G(N) = G$ is isomorphic to a subgroup of the automorphism group $\mathbf{Aut}(N)$, and N is mapped onto $\mathbf{Inn}(N)$ under this isomorphism. Therefore, G/N is isomorphic to a subgroup of $\mathbf{Aut}(N)/\mathbf{Inn}(N)$. As N is simple by Lemma 18.4, the Schreier conjecture implies that G/N is solvable. Then the complements of N in G are conjugate by Theorem 18.2.

Later we will have more to say about the Schreier conjecture, which incidentally has been proved for all known simple groups.

[5] The Schreier conjecture is now known to be true by the classification of finite simple groups. *trans.*

Theorems 18.1 and 18.2, resp. 18.3, are among the most important theorems in finite group theory. We will have many occasions to call upon them. In most cases, Theorem 18.2, which has been fully proved here, will be enough for our purposes.

Our first useful application of Theorem 18.3 is the following theorem, which gives some information about how fixed points of automorphisms descend:

Theorem 18.6. *Let A be a group of automorphisms of G, and let H be a subgroup of G with $(|A|, |H|) = 1$. Let $g \in G$. Assume that $H^a = H$ and $Hg^a = Hg$ for all $a \in A$. Then there exists some $x \in G$ such that $Hg = Hx$ and $x^a = x$ for all $a \in A$. (That is, every coset of H fixed by A contains an element fixed by A.)*

Proof. (Glauberman [169]). We define $f(a) = g^a g^{-1}$. Then f is a map from A to H satisfying

$$f(a_1 a_2) = g^{a_1 a_2} g^{-a_2} g^{a_2} g^{-1} = f(a_1)^{a_2} f(a_2).$$

Let $L = GA$ be the semidirect product of G with A. Consider the subgroup HA (recall that $H^a = H$ for all $a \in A$!). Since

$$a_1 f(a_1) a_2 f(a_2) = a_1 a_2 f(a_1)^{a_2} f(a_2) = a_1 a_2 f(a_1 a_2),$$

the set

$$K = \{a f(a) \mid a \in A\}$$

is a complement of H in HA. Since $(|A|, |H|) = 1$, there exists some $h \in H$ such that $A^h = K$ by Theorem 18.3. We have

$$a^h = a[a, h] \in aH \cap K = \{a f(a)\},$$

and therefore

$$(hg)^a = h^a g^a = a^{-1} h a f(a) g = h g$$

for all $a \in A$. Setting $x = hg$ gives the claim. □

Example 18.7. (K. H. Hofmann [249]). In this example, we will show that the assumption that A is abelian in Theorem 17.4 cannot be removed (unlike the special case of Theorem 17.5).

Let V be the two-dimensional vector space over the field $\mathsf{K} = \mathrm{GF}(p)$ with $p > 2$. Let f be a skew-symmetric bilinear form on V that is not identically zero. We define a multiplication on the set

$$P = \{(v_1, v_2, w) \mid v_i \in \mathsf{V},\ w \in \mathsf{K}\}$$

by

$$(v_1, v_2, w)(v_1', v_2', w') = (v_1 + v_1', v_2 + v_2', w + w' + f(v_1, v_1') - f(v_2, v_2')).$$

An easy calculation shows that P is a group of order p^5 with commutator subgroup

$$P' = \{(0, 0, w) \mid w \in \mathsf{K}\}.$$

The map α defined by

$$(v_1, v_2, w)^\alpha = (-v_1, v_2, w)$$

is an automorphism of P of order 2. The group

$$D = \{(v, v, 0) \mid v \in \mathsf{V}\}$$

is clearly a complement of the nonabelian normal subgroup

$$N = \{(v, 0, w) \mid v \in \mathsf{V}, w \in \mathsf{K}\}$$

of P.

Let G be the semidirect product of P with the cyclic group $\langle a \rangle$ of order 2 via the automorphism $x^a = x^\alpha$ for all $x \in P$. Since $N^\alpha = N$, we have $N \lhd G$. Again, D is a complement of N in P, and we have $(|N|, [G : P]) = 1$.

Now assume for contradiction that K is a complement of N in G. Since the Sylow 2-subgroups of G are conjugate, we may pass to a conjugate of K if necessary to ensure that $\langle a \rangle < K$. Then

$$K = \langle a \rangle T \quad \text{with} \quad T = P \cap K.$$

For all $x \in P$, we have $[x, a] \in N$. For $t \in T$, it therefore follows that $[t, a] \in K \cap N = 1$, so that T is centralized by a. For $t_1, t_2 \in T$, we obtain

$$[t_1, t_2] \in T \cap P' \leq T \cap N = 1.$$

So T is abelian.

Since $P = NT$ and $N \cap T = 1$, we have

$$T = \{(g(v), v, h(v)) \mid v \in \mathsf{V}\}$$

for some map $g : \mathsf{V} \to \mathsf{V}$ and some map $h : \mathsf{V} \to \mathsf{K}$. We have

$$(g(v), v, h(v)) = (g(v), v, h(v))^\alpha = (-g(v), v, h(v)).$$

Since $p > 2$, we must have $g(v) = 0$ for all $v \in \mathsf{V}$. The commutativity of T forces

$$f(v, v') = f(v', v) = -f(v, v') = 0$$

for all $v, v' \in \mathsf{V}$, contradicting our assumption on f. So N has no complement in G.

Exercises

71) Let A be a group of automorphisms of G with $(|A|, |G|) = 1$. We define $F = \{g \in G \mid g^a = g \text{ for all } a \in A\}$. Then $\mathbf{N}_G(F) = F\mathbf{C}_G(F)$.
72) Consider the setup of Exercise 71. In this case, there exists for every prime number p at least one Sylow p-subgroup P of G such that $P^a = P$ for all $a \in A$. All such Sylow p-subgroups are conjugate under F.

19 Free groups and defining relations

Definition 19.1. A group F is said to be free of rank $|I|$ in the free generators f_i $(i \in I)$ if it satisfies the following properties:
(1) $F = \langle f_i \mid i \in I \rangle$.
(2) If G is a group with generating set $\{g_i \mid i \in I\}$, then there exists an epimorphism μ from F onto G with $f_i^\mu = g_i$ for all $i \in I$. (Compare this definition with that of free modules in Definition 10.15!)

Theorem 19.2.
a) *For every cardinal number $\aleph$, there exists a free group of rank $\aleph$.*
b) *If F_1 and F_2 are free groups of the same rank, then F_1 and F_2 are isomorphic.*

Proof. a) Let I be a set with $|I| = \aleph$. Consider the symbols s_i, s_i^{-1} with $i \in I$. We say an expression is a word in these symbols if it is of the form

$$s_{i_1}^{n_1} \cdots s_{i_k}^{n_k}$$

with $i_j \in I$ and $n_j = \pm 1$. We denote the empty word by 1. We have

$$s_{i_1}^{n_1} \cdots s_{i_k}^{n_k} = s_{j_1}^{m_1} \cdots s_{j_l}^{m_l}$$

if and only if $k = l$ and $i_r = j_r$, $n_r = m_r$ for $r = 1, \ldots, k$. We define the product of two arbitrary words $x = s_{i_1}^{n_1} \cdots s_{i_k}^{n_k}$ and $y = s_{j_1}^{m_1} \cdots s_{j_l}^{m_l}$ by

$$xy = s_{i_1}^{n_1} \cdots s_{i_k}^{n_k} s_{j_1}^{m_1} \cdots s_{j_l}^{m_l},$$

and $x1 = 1x = x$, $1 \cdot 1 = 1$.

We define an elementary transformation to be the passage from $a_1 a_2$ to $a_1 s_i^n s_i^{-n} a_2$ $(n = \pm 1)$ or from $a_1 s_i^n s_i^{-n} a_2$ to $a_1 a_2$. We say words a and b are equivalent $(a \sim b)$ if there is a chain

$$a = w_1, w_2, \ldots, w_s = b$$

of words w_i such that w_{i+1} is obtained from w_i by an elementary transformation. One sees immediately that this is an equivalence relation and that $a_1 \sim a_2$, $b_1 \sim b_2$

implies $a_1 b_1 \sim a_2 b_2$.[6] We denote the equivalence class of a by $[a]$. We define a multiplication on the set of equivalence classes by $[a][b] = [ab]$. This is well defined by what we have already observed. In addition, we have the following:
(1) It follows from the trivial relation $a(bc) = (ab)c$ for words that

$$[a]([b][c]) = [a][bc] = [a(bc)] = [(ab)c] = [ab][c] = ([a][b])[c].$$

So the multiplication is associative.
(2) $[1][a] = [1a] = [a]$.
(3) If $a = s_{i_1}^{n_1} \cdots s_{i_k}^{n_k}$ and we define $b = s_{i_k}^{-n_k} \cdots s_{i_1}^{-n_1}$, then

$$[b][a] = [ba] = [1].$$

So the equivalence classes form a group F with this multiplication.

We show that F is free in the free generators $f_i = [s_i]$ $(i \in I)$. It is clear that F is generated by the f_i.

Let $G = \langle g_i \mid i \in I \rangle$. We define a map ϵ from the set of words to G by

$$(s_{i_1}^{n_1} \cdots s_{i_k}^{n_k})^\epsilon = g_{i_1}^{n_1} \cdots g_{i_k}^{n_k}$$

and $1^\epsilon = 1$. The image only depends on the equivalence class of the word: If b is obtained from a by an elementary transformation

$$a = a_1 s_j^n s_j^{-n} a_2 \mapsto b = a_1 a_2,$$

then

$$a^\epsilon = a_1^\epsilon g_j^n g_j^{-n} a_2^\epsilon = a_1^\epsilon a_2^\epsilon = b^\epsilon.$$

We define the map μ from F to G by

$$(f_{i_1}^{n_1} \cdots f_{i_k}^{n_k})^\mu = (s_{i_1}^{n_1} \cdots s_{i_k}^{n_k})^\epsilon = g_{i_1}^{n_1} \cdots g_{i_k}^{n_k}.$$

It is clear that μ is an epimorphism from F onto G with $f_i^\mu = g_i$.

b) Let F_i $(i = 1, 2)$ be free in the free generators $f_{i,j}$ $(j \in I_i)$, and assume that $|I_1| = |I_2|$. Let ϵ be a bijection from I_1 onto I_2. Then by Definition 19.1, there is an epimorphism μ_1 from F_1 onto F_2 with

$$f_{1,j}^{\mu_1} = f_{2,j^\epsilon}.$$

In the same way, there is an epimorphism μ_2 from F_2 onto F_1 with

$$f_{2,j^\epsilon}^{\mu_2} = f_{1,j}.$$

Now $\mu_1 \mu_2 = 1$ on F_1 and $\mu_2 \mu_1 = 1$ on F_2, so μ_1 is an isomorphism from F_1 to F_2.□

[6] Correction 5 from errata.

Definition 19.3. Let $G = \langle g_i \mid i \in I\rangle$. By Definition 19.1, there is an epimorphism μ from the free group $F = \langle f_i \mid i \in I\rangle$ in the free generators f_i onto G with $f_i^\mu = g_i$. Let $R = \ker \mu$. Note that R consists exactly of those elements

$$f_{i_1}^{n_1} \cdots f_{i_k}^{n_k}$$

which upon "replacing" with the g_i become

$$g_{i_1}^{n_1} \cdots g_{i_k}^{n_k} = 1.$$

We call the elements of R relations for the generators g_i.

Conversely, if we are given a set S of elements of the free group $F = \langle f_i \mid i \in I\rangle$, then let $R = S^F$ be the normal subgroup of F generated by the elements of S and their conjugates. We say F/R is the group defined by the generators $f_i R$ ($i \in I$) and the relations $s \in S$.

Lemma 19.4. *Let $G = \langle g_i \mid i \in I\rangle$, and let $F = \langle f_i \mid i \in I\rangle$ be the free group in the free generators f_i. Let the subset $S \subseteq F$ be a collection of relations for the g_i. Then G is an epimorphic image of the group defined by the generators $f_i S^F$ and the relations S.*

Proof. Let μ be an epimorphism from F onto G with $f_i^\mu = g_i$, and let K be the kernel of μ. By our assumption, $s^\mu = 1$ for all $s \in S$. So S lies in K as well as the normal subgroup $R = S^F$ of F generated by S. Hence, we have

$$G \cong F/K \cong (F/R)\big/(K/R).$$

Therefore, G is an epimorphic image of the group F/R defined by the relations S.□

Determining the structure of a group that is defined by generators and relations is usually extremely difficult. For example, there does not exist any general criterion or algorithm for determining whether the group is finite, or even if it is the trivial group. We consider here a few simple examples that will, in part, be needed later. A thorough treatment of these questions can be found in the survey by Coxeter, Moser [99].

We have defined relations between elements $g_i \in G$ to be elements $\prod_j f_{i_j}^{n_j}$ in a free group. In what follows, we will be somewhat imprecise and speak of relations $\prod_j g_{i_j}^{n_j} = 1$ in G.

Example 19.5. The generators x_1, x_2 and the relations $x_1^2 = x_2^2 = (x_1x_2)^m = 1$ define a finite group of order $2m$, the so-called dihedral group. It contains a cyclic normal subgroup $Z = \langle z\rangle$ of order m, and $\langle x_1\rangle$ is a complement of Z with

$$x_1^{-1} z x_1 = z^{-1}.$$

(We have already seen the special case $m = 2^n$ in Theorem 14.9.)

Proof. Let G be the group defined by the relations

$$x_1^2 = x_2^2 = (x_1x_2)^m = 1.$$

We set $z = x_1x_2$. Then $G = \langle x_1, z\rangle$. We have

$$x_1^{-1}zx_1 = x_2x_1 = x_2^{-1}x_1^{-1} = (x_1x_2)^{-1} = z^{-1}.$$

Thus, every element of G may be written in the form $x_1^i z^j$ with $0 \le i \le 1$ and $0 \le j \le m-1$. It follows that $|G| \le 2m$.

Let $D = \langle a, b\rangle$ be the semidirect product of the cyclic group $\langle a\rangle$ of order m with the cyclic group $\langle b\rangle$ of order 2 by means of the automorphism

$$a^b = a^{-1}.$$

We define $y_1 = ab$ and $y_2 = b$. Then $D = \langle y_1, y_2\rangle$, and the following relations hold:

$$y_1^2 = (ab)^2 = aa^b = 1,$$
$$y_2^2 = b^2 = 1$$

and

$$(y_1y_2)^m = (ab^2)^m = a^m = 1.$$

By Lemma 19.4, D is an epimorphic image of G. Since

$$|G| \le 2m = |D|,$$

it follows that $G \cong D$. □

Example 19.6. Let G be the group defined by generators x_1, x_2 and the relations $x_1^3 = x_2^3 = (x_1x_2)^3 = 1$. Then G is a semidirect product of a free abelian group $A = \langle a_1\rangle \times \langle a_2\rangle$ in the free generators a_1, a_2 with the cyclic group $\langle x_1\rangle$ of order 3 by means of the automorphism

$$a_1^{x_1} = a_1^{-1}a_2^{-1}, \quad a_2^{x_1} = a_1.$$

Proof. We define $a_1 = x_1x_2^2$, $a_2 = x_2x_1x_2$, $a_3 = x_2^2x_1$. Then we have

$$a_1^{x_1} = x_2^2x_1 = a_3 = a_2^{-1}a_1^{-1},$$
$$a_2^{x_1} = x_1^{-1}x_2x_1x_2x_1 = a_1,$$
$$a_3^{x_1} = x_1^{-1}x_2^2x_1^2 = (a_1a_3)^{-1},$$
$$a_1^{x_2} = x_2^{-1}x_1x_2^3 = a_3,$$
$$a_2^{x_2} = x_1x_2^2 = a_1,$$
$$a_3^{x_2} = x_2x_1x_2 = a_2.$$

Furthermore, we have $a_1a_2 = a_3^{-1}$. Hence, $\langle a_1, a_2\rangle$ is a normal subgroup of G. Since $(x_1x_2)^3 = 1$, we have

$$a_2a_1 = x_2x_1x_2x_1x_2^2 = x_1^{-1}(x_1x_2)^3x_2 = x_1^2x_2 = a_1a_2.$$

So $A = \langle a_1, a_2\rangle$ is abelian.[7] It follows from

$$x_1^{-1}a_1 = x_2^2 = x_2^{-1} \in A\langle x_1\rangle$$

that $G = A\langle x_1\rangle$.

Conversely, suppose H is the semidirect product of the free abelian group $B = \langle b_1\rangle \times \langle b_2\rangle$ with the cyclic group $\langle s\rangle$ of order 3 by means of the automorphism

$$b_1^s = b_1^{-1}b_2^{-1}, \quad b_2^s = b_1.$$

Then $y_1 = b_1s$ and $y_2 = s$ satisfy the following relations:

$$y_1^3 = b_1b_1^{s^2}b_1^ss^3 = b_1b_2b_1^{-1}b_2^{-1} = 1, \quad y_2^3 = 1$$

and

$$(y_1y_2)^3 = (b_1s^2)^3 = b_1b_1^sb_1^{s^2}s^6 = 1.$$

Thus, H is an epimorphic image of G, and there is an epimorphism μ from G onto H defined by

$$x_1^\mu = y_1 = b_1s \quad \text{and} \quad x_2^\mu = y_2 = s.$$

It follows that

$$a_1^\mu = (x_1x_2^2)^\mu = b_1s^3 = b_1 \quad \text{and} \quad a_2^\mu = (x_2x_1x_2)^\mu = sb_1s^2 = b_1^{s^{-1}} = b_2.$$

Hence, the free abelian group $\langle b_1, b_2\rangle$ is an epimorphic image of $\langle a_1, a_2\rangle$. So $A = \langle a_1, a_2\rangle$ is also a free abelian group in two free generators. □

Example 19.7. The generators $x_1, \ldots, x_{n-1}$ and the relations

$$\begin{aligned} x_i^2 &= 1 \quad (1 \le i \le n-1),\\ (x_ix_{i+1})^3 &= 1 \quad (i \le n-2),\\ (x_ix_j)^2 &= 1 \quad (j < i-1) \end{aligned}$$

define the symmetric group S_n for $n > 1$. The transpositions $s_i = (i, i+1)$ (with $1 \le i \le n-1$) are a generating set for S_n and satisfy these relations.

Proof. a) Let G be the group defined by the given relations, and consider the subgroup $H = \langle x_1, \ldots, x_{n-2}\rangle$ of G. Assume for induction that the claim has already

[7] Correction 6 from errata.

been proved for $n-1$, but not for n. Then H is an epimorphic image of S_{n-1} by Lemma 19.4, so in particular $|H| \leq (n-1)!$.

We consider the following cosets of H in G:

$$H,\ Hx_{n-1},\ Hx_{n-1}x_{n-2},\ \ldots,\ Hx_{n-1}x_{n-2}\cdots x_2x_1.$$

These cosets are permuted under right multiplication by any x_j since we have:

$$x_ix_j = (x_ix_j)^{-1} = x_jx_i \quad \text{for } j < i-1,$$

and therefore

$$(Hx_{n-1}\cdots x_i)x_j = (Hx_j)x_{n-1}\cdots x_i = Hx_{n-1}\cdots x_i;$$

since $x_jx_{j-1}x_j = x_{j-1}x_jx_{j-1}$, we have for $j > i$ that

$$\begin{aligned}(Hx_{n-1}\cdots x_i)x_j &= Hx_{n-1}\cdots x_{j+1}x_jx_{j-1}x_jx_{j-2}\cdots x_i\\ &= Hx_{n-1}\cdots x_{j+1}x_{j-1}x_jx_{j-1}x_{j-2}\cdots x_i\\ &= Hx_{j-1}x_{n-1}\cdots x_i = Hx_{n-1}\cdots x_i;\end{aligned}$$

and finally

$$\begin{aligned}(Hx_{n-1}\cdots x_i)x_i &= Hx_{n-1}\cdots x_{i+1},\\ (Hx_{n-1}\cdots x_i)x_{i-1} &= Hx_{n-1}\cdots x_{i-1}.\end{aligned}$$

Hence, the given cosets contain all products of the x_i, and thus cover all of G.[8] This gives $|G| \leq |H|n \leq n!$.

b) We still need to show that the symmetric group S_n can be generated by elements s_i $(i = 1, \ldots, n-1)$ that satisfy the relations.

Let $s_i = (i, i+1)$. Then $s_i^2 = 1$. From $s_is_{i+1} = (i, i+2, i+1)$, it follows that $(s_is_{i+1})^3 = 1$. For $j < i-1$, we have $s_is_j = (i, i+1)(j, j+1)$, so that $(s_is_j)^2 = 1$.

It remains to show that the s_i generate all of S_n. Let $U = \langle s_1, \ldots, s_{n-1}\rangle$. Since

$$(i, j) = s_i^{s_{i+1}\cdots s_{j-1}} \quad (i < j),$$

all transpositions lie in U. It follows from the equation

$$(a_1, a_2, \ldots, a_k) = (a_1, a_2)(a_1, a_3)\cdots(a_1, a_k)$$

that all k-cycles lie in U. As every element of S_n is a product of cycles by Theorem 5.2, we have proved that $U = S_n$. □

Example 19.8. The generators $x_1, \ldots, x_{n-2}$ and the relations

$$x_1^3 = x_i^2 = 1 \quad (2 \leq i \leq n-2),$$

[8] Correction 7 from errata.

$$(x_i x_{i+1})^3 = 1 \quad (1 \le i \le n-3),$$
$$(x_i x_j)^2 = 1 \quad (1 \le i \le n-4;\ i+1 < j)$$

define the alternating group A_n for $n \ge 3$.

Proof. a) Let G be the group defined by the given relations. Let $H = \langle x_1, \dots, x_{n-3} \rangle$. By an induction hypothesis, we may assume that H is an epimorphic image of A_{n-1}, so $|H| \le \frac{(n-1)!}{2}$. We consider the following cosets of H in G:

$$H,\ Hx_{n-2},\ Hx_{n-2}x_{n-3},\ \dots,\ Hx_{n-2}\cdots x_2x_1,\ Hx_{n-2}\cdots x_2x_1^2.$$

From the equations

$$\begin{aligned} x_i x_j &= x_j x_i && \text{for} \quad 1 < j < i-1, \\ x_j x_{j-1} x_j &= x_{j-1} x_j x_{j-1} && \text{for} \quad 2 < j, \\ x_2 x_1 x_2 = x_1^{-1} x_2^{-1} x_1^{-1} &= x_1^2 x_2 x_1^2, \\ x_1 x_j = x_j^{-1} x_1^{-1} &= x_j x_1^2 && \text{for} \quad j > 2, \end{aligned}$$

we obtain

$$\begin{aligned} (Hx_{n-2}\cdots x_i)x_1 &= Hx_{n-2}\cdots x_{i+1}x_1^3 x_i = Hx_{n-2}\cdots x_{i+2}x_1x_{i+1}x_i \\ &= \cdots = Hx_{n-2}\cdots x_i \quad \text{for} \quad i > 2. \end{aligned}$$

Right multiplication by x_1 cyclically permutes the cosets

$$Hx_{n-2}\cdots x_2,\ Hx_{n-2}\cdots x_2x_1,\ Hx_{n-2}\cdots x_2x_1^2.$$

In a similar way, we obtain

$$(Hx_{n-2}\cdots x_i)x_2 = Hx_{n-2}\cdots x_i \quad \text{for} \quad i > 3.$$

Right multiplication by x_2 permutes the cosets $Hx_{n-2}\cdots x_3$ and $Hx_{n-2}\cdots x_3x_2$. Furthermore,

$$\begin{aligned} (Hx_{n-2}\cdots x_3x_2x_1)x_2 &= Hx_{n-2}\cdots x_3x_1^2x_2x_1^2 = Hx_{n-2}\cdots x_2x_1^2, \\ (Hx_{n-2}\cdots x_2x_1^2)x_2 &= Hx_{n-2}\cdots x_3x_1x_2x_1 = Hx_{n-2}\cdots x_2x_1. \end{aligned}$$

For $2 < j < i-1$, we have just as in Example 19.7 that

$$(Hx_{n-2}\cdots x_i)x_j = Hx_{n-2}\cdots x_i.$$

The cosets[9]

$$Hx_{n-2}\cdots x_{j+1} \quad \text{and} \quad Hx_{n-2}\cdots x_j$$

[9] Correction 8 from errata.

are permuted under right multiplication with x_j. In addition,

$$\begin{aligned}(Hx_{n-2}\cdots x_i)x_j &= Hx_{n-2}\cdots x_i \quad \text{for} \quad 1 < i < j,\\ (Hx_{n-2}\cdots x_1)x_j &= Hx_{n-2}\cdots x_j x_1^2 = Hx_{n-2}\cdots x_2 x_1^2,\\ (Hx_{n-2}\cdots x_1^2)x_j &= Hx_{n-2}\cdots x_2 x_j x_1 = Hx_{n-2}\cdots x_2 x_1.\end{aligned}$$

So the n given cosets of H cover G, and it follows that

$$|G| \le n|H| \le \frac{n!}{2}.$$

b) We again need to show that the alternating group A_n can be generated by elements s_i $(1 \le i \le n-2)$ that satisfy the relations

$$\begin{aligned}s_1^3 = s_i^2 &= 1 \quad (2 \le i \le n-2),\\ (s_i s_{i+1})^3 &= 1,\\ (s_i s_j)^2 &= 1 \quad (i+1 < j).\end{aligned}$$

We define

$$s_1 = (1,2,3) \quad \text{and} \quad s_i = (1,2)(i+1,i+2) \quad \text{for} \quad 2 \le i \le n-2.$$

One easily observes that the s_i lie in A_n and satisfy the given relations.

The s_i $(1 \le i \le n-2)$ generate all of A_n:

For $n = 3$, this is trivial since $A_3 = \langle s_1 \rangle$. We may assume for induction that $\langle s_1, \ldots, s_{n-3} \rangle$ is the full alternating group on the set $\{1, \ldots, n-1\}$. Let $g \in A_n$ and $n^g = j$. If $j = n$, then g is a product of the s_i $(1 \le 2 \le n-3)$, and we are done. If $j < n$, then there is a product t of the s_i $(1 \le i \le n-3)$ such that $n^{gt} = j^t = n-1$. It follows that

$$n^{gts_{n-2}} = n, \quad \text{and so} \quad gts_{n-2} \in \langle s_1, \ldots, s_{n-3} \rangle. \qquad \square$$

Example 19.9. The generators x_1, x_2 and the relations

$$x_1^5 = x_2^2 = (x_1 x_2)^3 = 1$$

define the alternating group A_5.

Proof. a) Let $G = \langle x_1, x_2 \rangle$ with the given relations. We define

$$a_1 = x_2 x_1^{-1}, \quad a_2 = x_1^{-1} x_2 x_1^3 x_2, \quad a_3 = a_2 x_1 x_2 x_1^3.$$

Then

(1) $a_1 = x_2 x_1^{-1} = (x_1 x_2)^{-1}$, so $a_1^3 = 1$.

(2) $a_2 = x_1^{-1} x_2 x_1^3 x_2 = x_1^{-1} x_2^{-1} x_1^{-1} \cdot x_1^4 x_2 = x_2 x_1 x_2 \cdot x_1^{-1} x_2 = x_2^{x_1^{-1} x_2}$

and therefore $a_2^2 = 1$.

(3) With $a_2 = x_2^{x_1^{-1}x_2}$, it follows that

$$\begin{aligned} a_3^2 &= (x_2x_1x_2x_1^{-1}x_2x_1x_2x_1^3)^2 \\ &= x_2x_1x_2 \cdot x_1^{-1} \cdot x_2x_1x_2 \cdot x_1^3 \cdot x_2x_1x_2 \cdot x_1^{-1} \cdot x_2x_1x_2 \cdot x_1^3 \\ &= x_1^{-1}x_2x_1^{-1} \cdot x_1^{-1} \cdot x_1^{-1}x_2x_1^{-1} \cdot x_1^3 \cdot x_1^{-1}x_2x_1^{-1} \cdot x_1^{-1} \cdot x_1^{-1}x_2x_1^{-1} \cdot x_1^3 \\ &\qquad\qquad (\text{since } x_2x_1x_2 = x_1^{-1}x_2x_1^{-1}) \\ &= x_1^{-1}x_2x_1^{-3} \cdot x_2x_1x_2 \cdot x_1^{-3}x_2x_1^2 = x_1^{-1}x_2x_1^{-4}x_2x_1^{-4}x_2x_1^2 \\ &= x_1^{-2}(x_1x_2)^3x_1^2 = 1. \end{aligned}$$

(4) $(a_2a_3)^2 = (x_1x_2x_1^3)^2 = x_1x_2x_1^4x_2x_1^3 = x_1 \cdot x_2x_1^{-1}x_2 \cdot x_1^3$
$= x_1 \cdot x_1x_2x_1 \cdot x_1^3 = x_1^2x_2x_1^{-1}$,
so then

$$(a_2a_3)^3 = x_1^2x_2x_1^{-1} \cdot x_1x_2x_1^3 = x_1^5 = 1.$$

(5) $a_1a_2 = x_2x_1^{-1}x_1^{-1}x_2x_1^3x_2 = (x_1x_2)^{x_1^3x_2}$ and therefore $(a_1a_2)^3 = 1$.

(6) $$\begin{aligned} (a_1a_3)^{x_1^{-1}x_2x_1^3} &= x_1^2x_2x_1 \cdot x_2x_1^{-1} \cdot x_1^{-1}x_2x_1^3x_2x_1x_2x_1^3 \cdot x_1^{-1}x_2x_1^3 \\ &= x_1^2 \cdot x_2x_1x_2 \cdot x_1^{-2}x_2x_1^3 \cdot x_2x_1x_2 \cdot x_1^2x_2x_1^3 \\ &= x_1^2 \cdot x_1^{-1}x_2x_1^{-1} \cdot x_1^{-2}x_2x_1^3 \cdot x_1^{-1}x_2x_1^{-1} \cdot x_1^2x_2x_1^3 \\ &= x_1x_2x_1^{-3}x_2x_1^2 \cdot x_2x_1x_2 \cdot x_1^3 = x_1x_2x_1^2x_2x_1x_2x_1^2 \\ &= x_1x_2x_1 \cdot x_1x_2x_1x_2x_1 \cdot x_1 = x_1x_2x_1 \cdot x_2x_1 = x_2, \end{aligned}$$
and thus $(a_1a_3)^2 = 1$.

Furthermore, we have $x_1^3 = a_1a_2^{-1}a_3$ and $x_2 = a_1x_1$. It follows that $G = \langle a_1, a_2, a_3\rangle$. By Example 19.8 and Lemma 19.4, G is an epimorphic image of A_5, so $|G| \leq 60$.

b) Let $s_1 = (1,2,3,4,5)$ and $s_2 = (1,2)(3,4)$. Since

$$s_1s_2 = (1)(3)(2,4,5),$$

the given relations hold for the s_i. It remains to show that s_1 and s_2 generate all of A_5. Let $U = \langle s_1, s_2\rangle$. We have

$$t = s_2^{s_1}s_1 = (1,2,4)(3)(5) \in U$$

and therefore

$$s_2^t = (2,4)(1,3) \in U.$$

So $\langle s_2, s_2^t\rangle$ is a subgroup of order 4 in U. Since U contains elements s_1 and s_1s_2 of orders 5 and 3, respectively, $|U|$ is a multiple of 60. Therefore, $U = A_5$. □

The following will occasionally be useful in applications:

Theorem 19.10. *Let $G = \langle g_1, \dots, g_n \rangle$ be a finitely generated group, and let $U \leq G$ be a subgroup of finite index. Then U can be generated by $2n[G : U]$ elements.*

Proof. Let $G = \bigcup_{j=1}^{m} Uv_j$ with $v_1 = 1$ be the decomposition of G into cosets of U. We define

$$g_{n+1} = g_1^{-1}, \dots, g_{2n} = g_n^{-1}$$

and determine $2nm$ elements $u_{i,j} \in U$ by equations

$$v_i g_j = u_{ij} v_k$$

for some k that depends on i and j. We claim that these u_{ij} generate all of U:

Let $u = g_{i_1} \cdots g_{i_s} \in U$ (with $i_k \in \{1, \dots, 2n\}$). Then

$$\begin{aligned} u &= (v_1 g_{i_1}) g_{i_2} \cdots g_{i_s} \\ &= u_{1,i_1} v_{1'} g_{i_2} \cdots g_{i_s} \quad \text{(for some } 1'\text{)} \\ &= u_{1,i_1} u_{1',i_2} v_{2'} g_{i_3} \cdots g_{i_s} = \cdots \\ &= u_{1,i_1} u_{1',i_2} \cdots u_{(s-1)',i_s} v_{s'}. \end{aligned}$$

Since u and the u_{ij} lie in U, it follows that $v'_s = 1$ and the claim is proved. □

The following theorem can be proved by a considerable refinement of the proof of Theorem 19.10. We state it here without proof, but see Theorem 7.2.8 of M. Hall [204] or Weir [455].

Theorem 19.11 (Schreier's subgroup theorem).

a) *If F is a free group, then every subgroup of F is free. If F is a free group in n free generators and U is a subgroup of F with finite index j, then U is a free group in $1 + j(n-1)$ free generators.*

b) *If G can be generated by n elements and $U \leq G$ is a subgroup of index j, then U can be generated by $1 + j(n-1)$ elements.*

Exercises

73) The symmetric group S_n can be defined by the generators r, s and the relations

$$r^n = s^2 = (rs)^{n-1} = (sr^{-1}sr)^3 = (sr^{-j}sr^j)^2 = 1 \quad (2 \leq j \leq n-2).$$

(Hint: Pass to the new generators r_i $(1 \leq i \leq n-1)$ defined by

$$r_1 = s, \quad r_{i+1} = r^{-i} s r^i \quad \text{and} \quad r = r_1 r_2 \cdots r_{n-1}.)$$

74) The symmetric group S_4 can be defined by the generators r, s and the relations

$$r^4 = s^2 = (rs)^3 = 1.$$

75) The alternating group A_4 can be defined by the generators r, s and the relations

$$r^3 = s^3 = (rs)^2 = 1.$$

76) Let F be a free group in the free generators f_i $(i \in I)$, and let $\mathfrak{m}$ be a right F-module. For a given $m_i \in \mathfrak{m}$, there exists a 1-cocyle $f \in \mathbf{Z}^1(F, \mathfrak{m})$ such that $f(f_i) = m_i$. (Hint: Construct a suitable extension of $\mathfrak{m}$ by F.)

Notes for Chapter I

§4: We have taken Example 4.10 b) from Zassenhaus [496]. One can find Exercise 12 worked out in Speiser [392], pg. 123. The assertion of Exercise 14 has been used quite often.

§6: For a solution to Exercise 22, one may consult Zassenhaus [496], pg. 104 or the original work of Witt [479].

§7: One may find the elegant proof of Theorem 7.2 in Wielandt [472]. Two other proofs of Theorem 7.3 are sketched in Exercises 26–28 and 29–30.

§8: For Exercises 38–40, see M. Hall [204], pg. 136–137 and 147.

§9: The construction of the direct product with identified subgroups in Theorem 9.10 is common; for the notation used there (and in Theorem 9.11), I follow a recommendation of H. Wielandt. For an indication of the usefulness of Theorem 9.11, I am grateful to P. Roquette (see the application in Roquette [359] and in Theorem IV.4.6). Theorem 9.14 is taken from Rose [360], where numerous additional statements about the embedding of the diagonal D in $G \times G$ are proved.

§13: The layout of this section until Theorem 13.10 is strongly influenced by Kaplansky [297]; the formulation of the theory of modules over principal ideal domains presented here has the advantage that it may be extended to modules over Dedekind rings without too much trouble. The proof of Theorem 13.7 follows a recommendation of J. Rotman. In the proof of Theorem 13.14, I have used suggestions of N. Blackburn, which improved the presentation.

§14: Extension theory was developed by O. Schreier in Schreier [369]. Exercise 59 follows the methods of Hofmann, Mostert [250].

§15: Wreath products have been used quite often to construct finite and infinite groups; see, for example, P. Hall [221]. The twisted wreath product in Definition 15.10 stems from B. H. Neumann [336].

§16: Theorem 16.18 is contained in Eckmann [119].

§17: Exercise 63 stems from a lecture by H. Wielandt.

§18: The proof of Lemma 18.4 follows P. Hall [219]. One may compare Theorem 18.6 with Wielandt [475].

§19: Example 19.6 played a role in Feit, Thompson [131].

Chapter II
Permutation groups and linear groups

Chapter II studies groups of automorphisms of sets and vector spaces. In §1–§5, we develop the elementary principles of the theory of permutation groups. We restrict ourselves to a discussion of only the most fundamental concepts. For more information, the reader may consult Wielandt [477]. Theorems 3.2 and 3.7 play an important role in the theory of solvable groups. We will have considerably more to say about permutation groups in Volume 3 with the help of character theory.

The automorphism groups of vector spaces equipped with a scalar product are one of the richest sources of nonabelian simple groups. We treat the two easiest cases, namely the projective and symplectic groups, in §6 and §9. The method we use to prove the simplicity of these groups, due to Iwasawa and Tamagawa, is a combination of elementary geometric considerations and simple observations about primitive permutation groups. The crucial step in the proof is showing that the groups under consideration are primitive. It is worth noting that this method of proof also works for the unitary and orthogonal groups. (For the orthogonal groups, one may consult Tamagawa [422].) However, due to the additional difficulties that arise in considering the unitary groups and, especially, the orthogonal groups, we limit ourselves to a few observations in §10. For a complete treatment, the reader may consult Artin [15] and Dieudonné [112, 115].

After some preparation in §7, we determine in §8 all the subgroups of $\mathrm{PSL}_2(p^f)$, the group of linear fractional transformations over $\mathrm{GF}(p^f)$. This exact knowledge of the subgroup structure of the simple group $\mathrm{PSL}_2(p^f)$ is useful in many cases, especially in the construction of counterexamples.

A unified theory of simple linear groups is only possible from the standpoint of Lie algebras. The Cartan–Killing theory shows that over an algebraically closed field of characteristic 0, there exist only the following nonabelian simple Lie algebras: the four series

A_n (Lie algebra of the projective linear group $\mathrm{PGL}_{n+1}(\mathsf{K})$)
B_n (Lie algebra of the orthogonal group of dimension $2n+1$)
C_n (Lie algebra of the symplectic group of dimension $2n$)
D_n (Lie algebra of the orthogonal group of dimension $2n$)

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B. Huppert, *Finite Groups I*, Grundlehren der mathematischen Wissenschaften 364,
https://doi.org/10.1007/978-3-031-87529-8_2

and five exceptional algebras G_2, F_4, E_6, E_7 and E_8. Chevalley has shown how to construct a group over a given finite field from each of these algebras that is finite and, with few exceptions, simple (Chevalley [95], Carter [91]).

The technique for constructing unitary groups from projective groups shows how one can produce new simple groups of so-called twisted type. We will discuss the most interesting example of a such a construction, the simple groups of Suzuki, in Volume 3. They are in many respects the smallest simple groups next to the groups $\mathrm{PSL}_2(p^f)$.

1 Primitive and multiply-transitive permutation groups

In this section, we collect the definitions and elementary theorems in the theory of permutation groups. In the interest of the greater generality that we will require when treating linear groups in §6, §9 and §10, we do not assume that the set on which the group acts is always finite.

By a permutation group on the (not necessarily finite) set Ω we mean a (not necessarily finite) group of bijective maps from Ω to itself. The concepts of orbit, transitivity and permutation isomorphism should be understood as defined in Chapter I, §5 (pg. 25).

Definition 1.1. A transitive permutation group G on the set Ω is called imprimitive if there exists a proper subset $\Delta \subset \Omega$ containing more than one point such that for all $g \in G$, either $\Delta \cap \Delta^g = \emptyset$ or $\Delta = \Delta^g$. In this case, we say Δ is a block for G. If G is transitive and not imprimitive, then G is called primitive. (In particular, the symmetric group S_1 on the set Ω with $|\Omega| = 1$ is primitive.)

Theorem 1.2. *Assume G is imprimitive on Ω, and let Δ be a block for G. Let*

$$H = \{g \in G \mid \Delta^g = \Delta\},$$

and let $G = \bigcup_{r \in R} Hr$ be the decomposition of G into cosets of H.

a) *We have $\Omega = \bigcup_{r \in R} \Delta^r$ and $\Delta^r \cap \Delta^{r'} = \emptyset$ for $r \neq r'$.*

b) *Assume $|\Omega| < \infty$. Then $|\Omega| = |\Delta||R|$, so $|\Delta|$ is a divisor of $|\Omega|$. If we set $|\Delta| = k$ and $|R| = m$, then G is permutation isomorphic to a subgroup of the wreath product $S_k \wr S_m$ of the symmetric groups of degrees k and m, which is to be understood in the sense of Theorem I.15.3 (pg. 102) as a permutation group of degree $km = |\Omega|$. There exists a homomorphism from G to the symmetric group S_m whose kernel is isomorphic to a subgroup of the m-fold direct product of S_k. In particular, $|G|$ is a divisor of $m!(k!)^m$.*

c) *The subgroup $H = \{g \in G \mid \Delta^g = \Delta\}$ is transitive on Δ.*

Proof. a) Let $i_0 \in \Delta$, and let i be an arbitrary point in Ω. By the transitivity of G, there exists some $g \in G$ such that $i = i_0^g \in \Delta^g$. Letting $g = hr$ for some $h \in H$ and $r \in R$, it follows that

$$i \in \Delta^{hr} = \Delta^r.$$

Therefore, certainly $\Omega = \bigcup_{r \in R} \Delta^r$.

If $\Delta^r \cap \Delta^{r'} \neq \emptyset$, then $\Delta^{rr'^{-1}} \cap \Delta \neq \emptyset$. By our assumption on Δ, it then follows that $\Delta^{rr'^{-1}} = \Delta$. Hence, $rr'^{-1} \in H$ and so $r = r'$.

It is clear that for all $g \in G$, Δ^g is also a block for G.

b) Assume $|\Omega| < \infty$, let $|\Delta| = k$ and $[G : H] = |R| = m$. We identify the points in Δ^r with pairs (i, Hr) for $i \in \Gamma = \{1, \ldots, k\}$. Let $g \in G$. Then $rg = hr'$ for some $h \in H$ and $r' \in R$, and it follows that

$$\Delta^{rg} = \Delta^{r'}.$$

That is, G permutes the sets Δ^r according to the permutation representation of G on the cosets of H. In the notation of ordered pairs, this means that

$$(i, Hr)^g = (i', Hrg);$$

where $\binom{i}{i'}$ is a permutation $\pi(g, Hr)$ on $\Gamma = \{1, \ldots, k\}$ that depends on g and Hr. Therefore, we have

$$(i, Hr)^g = (i^{\pi(g,Hr)}, Hrg).$$

Comparing with the formula in Theorem I.15.3 (pg. 102) shows that G is permutation isomorphic to a subgroup of the wreath product $S_k \wr S_m$, which is to be understood as a permutation group on the product set $\Gamma \times \{Hr \mid r \in R\}$. The remaining claims regarding G follow immediately from Theorem I.15.2 (pg. 101).

c) Let $a, b \in \Delta$. By the transitivity of G, there exists some $g \in G$ such that $a^g = b$. Then $b \in \Delta \cap \Delta^g$, so that $\Delta = \Delta^g$ and therefore $g \in H$. That is, H is transitive on Δ.□

Theorem 1.2 often reduces the study of transitive permutation groups to the case of primitive permutation groups. Next, we note that:

Theorem 1.3. *Every transitive group of prime degree is primitive.*

Proof. Let G be a transitive permutation group of degree p. If Δ is a block for G, then $1 < |\Delta| < p$. By Theorem 1.2, $|\Delta|$ is a divisor of p. This is impossible, so G is primitive. □

Theorem 1.4. *Let G be a transitive permutation group on the set Ω with $|\Omega| > 1$. Then G is primitive if and only if the stabilizer*

$$U = \{g \in G \mid 1^g = 1\}$$

of $1 \in \Omega$ is a maximal subgroup of G.

Proof. a) Assume H is a subgroup of G such that $U < H < G$. Let $G = \bigcup_{r \in R} Ur$ and $H = \bigcup_{s \in S} Us$ with $\{1\} \subset S \subset R$. We define

$$\Delta = \{1^h \mid h \in H\} = \{1^s \mid s \in S\}.$$

Then we have

$$\{1\} \subset \Delta \subset \{1^r \mid r \in R\} = \Omega.$$

We claim that Δ is a block for G:

If $\Delta \cap \Delta^g \neq \emptyset$, then there exist elements $h_1, h_2 \in H$ such that

$$1^{h_1} = 1^{h_2 g}.$$

This gives $h_2 g h_1^{-1} \in U < H$, so $g \in H$ and then $\Delta^g = \Delta$. So if U is not a maximal subgroup of G, then G is imprimitive.

b) Assume G is imprimitive, and let Δ be a block for G with $1 \in \Delta$. We define

$$H = \{g \in G \mid \Delta^g = \Delta\}.$$

Then H is a subgroup of G. Since H is transitive on Δ by Theorem 1.2 c), we have

$$\{1\} \subset \Delta = \{1^h \mid h \in H\} \subset \Omega = \{1^g \mid g \in G\}.$$

It follows that $U < H < G$, and therefore U is not a maximal subgroup of G. □

The following theorem is an important criterion for imprimitivity.

Theorem 1.5. *Let G be a transitive permutation group, and let $N > 1$ be a normal subgroup of G. If N is not transitive, then the orbits of N are blocks for G that all have the same size. In particular, every nonidentity normal subgroup of a primitive permutation group is transitive.*

Proof. Let T be an orbit of N with $|T| > 1$. Such an orbit exists since $N \neq 1$. If $g \in G$ and $n \in N$, then $gng^{-1} = n' \in N$ and therefore $T^{gn} = T^{n'g} = T^g$. That is, T^g is mapped onto itself by N. If i^g and j^g are points in T^g, then there exists some $n \in N$ such that $i^n = j$. It follows that

$$(i^g)^{g^{-1}ng} = j^g.$$

Thus, T^g is an orbit of N. So if $T^g \neq T$, then certainly $T^g \cap T = \emptyset$. Hence, T is a block for G. By the transitivity of G, we have $\Omega = \bigcup_{g \in G} T^g$; therefore, the T^g are all the orbits of N. □

Lemma 1.6 (Rudio's lemma). *Let G be a primitive permutation group on the finite set Ω, and let $\emptyset \subset \Delta \subset \Omega$. For every pair of points $a, b \in \Delta$ with $a \neq b$, there exists some $g \in G$ such that $a \in \Delta^g$ and $b \notin \Delta^g$.*

Proof. We define $\Lambda = \bigcap_{a \in \Delta^g} \Delta^g$ and show that $\Lambda = \{a\}$.

First assume that $a \in \Lambda^h = \bigcap_{a \in \Delta^g} \Delta^{gh}$ for some $h \in G$. This means that for all $g \in G$ such that $a \in \Delta^g$, we have $a \in \Delta^{gh}$ as well. Hence, we have that

$$\Lambda^h = \bigcap_{a\in\Delta^g} \Delta^{gh} \supseteq \bigcap_{a\in\Delta^g} \Delta^g = \Lambda.$$

Since $|\Lambda| = |\Lambda^h|$, this forces $\Lambda = \Lambda^h$. (The finiteness of Ω is necessary here!)

Now suppose $c \in \Lambda \cap \Lambda^h$. By the transitivity of G, there exists some $k \in G$ such that $a^k = c$. It follows that

$$a \in \Lambda \cap \Lambda^{k^{-1}} \cap \Lambda^{hk^{-1}}.$$

By the last paragraph, we conclude that $\Lambda^{k^{-1}} = \Lambda = \Lambda^{hk^{-1}}$ and so $\Lambda = \Lambda^h$. We have shown that Λ is a block for G. But since G is primitive and $\Lambda \subseteq \Delta \subset \Omega$, we must have $|\Lambda| = 1$. This means $\Lambda = \{a\} \not\ni b$. So there exists some $g \in G$ such that $a \in \Delta^g$ and $b \notin \Delta^g$. □

Definition 1.7. The permutation group G on the set Ω is called k-transitive ($k \leq |\Omega|$) if for every pair of ordered subsets $(a_1, \ldots, a_k)$ and $(b_1, \ldots, b_k)$ of Ω, there exists some $g \in G$ such that $a_i^g = b_i$ for all $i = 1, \ldots, k$. If G is k-transitive, then G is clearly also $(k-1)$-transitive. If G is $(|\Omega|-1)$-transitive, then G is the full symmetric group on Ω.

Lemma 1.8. *Assume G is transitive on Ω, and let $k > 1$. The group G is k-transitive on Ω if and only if for some $i \in \Omega$ (equivalently, for all $i \in \Omega$) the subgroup*

$$G_i = \{g \in G \mid i^g = i\}$$

is $(k-1)$-transitive on $\Omega - \{i\}$.

Proof. Assume G is k-transitive on Ω, and let $(a_1, \ldots, a_{k-1})$, $(b_1, \ldots, b_{k-1})$ be ordered subsets of $\Omega - \{i\}$. Then there exists an element $g \in G$ such that $i^g = i$ and $a_j^g = b_j$ for $j = 1, \ldots, k-1$. Therefore, $g \in G_i$ and so G_i is $(k-1)$-transitive on $\Omega - \{i\}$ for all $i \in \Omega$.

Conversely, assume there is some $i \in \Omega$ such that G_i is $(k-1)$-transitive on $\Omega - \{i\}$. Let $(a_1, \ldots, a_k)$ and $(b_1, \ldots, b_k)$ be any pair of ordered subsets of Ω. By the transitivity of G, there exist elements $g, h \in G$ such that $a_1^g = i$ and $i^h = b_1$. Since G_i is $(k-1)$-transitive on $\Omega - \{i\}$, there is some $g' \in G_i$ such that $i^{g'} = i$ and

$$a_j^{gg'} = b_j^{h^{-1}}$$

for $j = 2, \ldots, k$. It follows that $a_j^{gg'h} = b_j$ for $j = 1, \ldots, k$, and so G is k-transitive. □

Theorem 1.9. *If G is 2-transitive, then G is primitive.*

Proof. Assume for contradiction that Δ is a block for G with $a, b \in \Delta$, $a \neq b$ and $c \notin \Delta$. By the 2-transitivity of G, there exists some $g \in G$ such that $a^g = a$ and $b^g = c$. It follows that $a \in \Delta \cap \Delta^g$ so that, in particular, $\Delta \cap \Delta^g \neq \emptyset$. However, since $c \notin \Delta$ but $c \in \Delta^g$, we have $\Delta \neq \Delta^g$. This contradicts our assumption that Δ is a block for G. □

Definition 1.10. Let G be a permutation group on Ω. For every natural number k, let $\Omega^{[k]}$ be the collection of ordered subsets of Ω with cardinality k. By defining

$$(a_1, \ldots, a_k)^g = (a_1^g, \ldots, a_k^g),$$

we give the group G the structure of a permutation group on $\Omega^{[k]}$, which we denote by $G^{[k]}$. Note that G and $G^{[k]}$ are isomorphic as abstract groups. Clearly, $G^{[k]}$ is transitive on $\Omega^{[k]}$ if and only if G is k-transitive on Ω.

We assume from now on that the set Ω is finite, although this is not necessary in every instance.

Theorem 1.11. *If G is k-transitive of degree n, then*

$$n(n-1)(n-2)\cdots(n-k+1)$$

is a divisor of $|G|$.

Proof. As G is k-transitive on Ω, $G^{[k]}$ is transitive on $\Omega^{[k]}$. The claim follows by Theorem I.5.11 (pg. 28) since

$$|\Omega^{[k]}| \text{ divides } |G| \quad \text{and} \quad |\Omega^{[k]}| = n(n-1)\cdots(n-k+1). \qquad \square$$

Theorem 1.12. *Assume G is transitive on $\Omega = \{1, \ldots, n\}$, and let $H = G_1$ be the stabilizer of 1. Let $G = \bigcup_{i=1}^{t} Hx_iH$ be the decomposition of G into double cosets of H with $x_1 \in H$ (see Theorem I.2.19, pg. 11). Then the following hold:*

a) *The sets*

$$T_i = \{1^g \mid g \in Hx_iH\} \qquad (i = 1, \ldots, t)$$

are all the orbits of H.

b) *G is 2-transitive if and only if $t = 2$; that is, if and only if $G = H \cup HxH$ for every $x \in G - H$.*

Proof. a) The set T_i is mapped onto itself by H:

Let $j = 1^g \in T_i$ and $h \in H$. Then $gh \in Hx_iH$, and it follows that $1^{gh} = j^h \in T_i$.

T_i is an orbit of H:

Let $j = 1^g \in T_i$. Then $g \in Hx_iH$, and therefore $HgH = Hx_iH$. Letting $k = 1^{g'} \in T_i$, we have $g' \in Hx_iH = HgH$. So $g' = hgh'$ for some $h, h' \in H$ and $k = 1^{g'} = j^{h'}$.

b) is a trivial consequence of a) and Lemma 1.8. $\square$

Lemma 1.13. *Assume G is k-transitive on Ω.*

a) *Let $\Gamma \subseteq \Omega$ with $|\Gamma| = k$, and let $U \leq G_\Gamma$. Assume that if $U^g \leq G_\Gamma$ with $g \in G$, then there always exists some $h \in G_\Gamma$ such that $U^g = U^h$. Then $\mathbf{N}_G(U)$ is k-transitive on the set of fixed points of U, which contains Γ. (The reader may refer to Definition I.5.9, pg. 27 for the notation used here.)*

b) $\mathbf{N}_G(G_\Gamma)$ *is* k*-transitive on the set of fixed points of* G_Γ.
c) *If* P *is a Sylow* p*-subgroup of* G_Γ, *then* $\mathbf{N}_G(P)$ *is* k*-transitive on the set of fixed points of* P.

Proof. a) Let $N = \mathbf{N}_G(U)$, and let Δ be the set of fixed points of U. Then certainly $\Delta = \Delta^n$ for all $n \in N$.

Let $(a_1, \ldots, a_k)$ and $(b_1, \ldots, b_k)$ be elements in $\Delta^{[k]}$. Since G is k-transitive, there exists some $s \in G$ such that $\{b_1^s, \ldots, b_k^s\} = \Gamma$ and some $g \in G$ such that $a_i^g = b_i$ $(i = 1, \ldots, k)$. The points b_i are fixed by U and U^g. Therefore U^s and U^{gs} lie in G_Γ. By our assumption, there exist elements $h_1, h_2 \in G_\Gamma$ such that

$$U^s = U^{h_1} \quad \text{and} \quad U^{gs} = U^{h_2}.$$

It follows that $U^{sh} = U^{gs}$ for $h = h_1^{-1}h_2 \in G_\Gamma$. We define $x = gsh^{-1}s^{-1}$, so that $x \in \mathbf{N}_G(U)$. Furthermore, since $h \in G_\Gamma$ and $b_i^s \in \Gamma$ we have

$$a_i^x = b_i^{sh^{-1}s^{-1}} = b_i^{ss^{-1}} = b_i.$$

Therefore, $\mathbf{N}_G(U)$ is k-transitive on Δ.

b) This is a special case of a) in which we set $U = G_\Gamma$. Part c) follows by choosing U to be a Sylow p-subgroup of G_Γ and recalling the conjugacy of Sylow p-subgroups from Theorem I.7.5 (pg. 35). □

In the next two definitions, we refine the scale of transitivity:

Definition 1.14. Let G be a permutation group on Ω with $|\Omega| = n$.

a) G is called 1-primitive whenever G is primitive on Ω. We define k-primitivity for $2 \le k \le n$ recursively by saying that G is k-primitive whenever G is transitive on Ω and there exists some $i \in \Omega$ such that G_i is $(k-1)$-primitive on $\Omega - \{i\}$. (Since all G_i are conjugate in the transitive group G by Theorem I.5.11, pg. 28, every G_i is $(k-1)$-primitive on $\Omega - \{i\}$ in this case.)
b) G is said to be $\frac{1}{2}$-transitive on Ω if $G \neq 1$, $|\Omega| > 1$ and all orbits of G have the same size (which is necessarily larger than 1). Furthermore, we say that the symmetric group S_1 on Ω with $|\Omega| = 1$ is $\frac{1}{2}$-transitive. We say G is $(k+\frac{1}{2})$-transitive on Ω for $1 \le k < n$ whenever G is transitive on Ω and G_i is $(k-\frac{1}{2})$-transitive on $\Omega - \{i\}$ for some (equivalently all) $i \in \Omega$.

The following lemma is clear.

Lemma 1.15. *Let* k *be a natural number. Every* k*-transitive group is* $(k-\frac{1}{2})$*-transitive, and every* $(k+\frac{1}{2})$*-transitive group is* k*-transitive. Every* k*-primitive group is* k*-transitive, and every* $(k+1)$*-transitive group is* k*-primitive* (*by Theorem 1.9*). *If* G *and* H *are permutation groups on* Ω *with* $G \le H$, *then if* G *is* k*-transitive, resp.* k*-primitive, then* H *is* k*-transitive, resp.* k*-primitive, as well.* (*However, overgroups of* $(k+\frac{1}{2})$*-transitive groups are not necessarily* $(k+\frac{1}{2})$*-transitive!*)

Just as trivial is the following lemma.

Lemma 1.16. *Let G be a permutation group on Ω with $|\Omega| = n$. The following are equivalent:*

a) *G is transitive and $|G| = n$.*
b) *G is regular in the sense of Definition I.5.12 (pg. 28).*

Definition 1.17. The permutation group G is said to be sharply k-transitive on Ω whenever the group $G^{[k]}$ in Definition 1.10 is regular on $\Omega^{[k]}$. By Lemma 1.16, this is equivalent to the property that for every pair of elements $(a_1, \ldots, a_k)$ and $(b_1, \ldots, b_k)$ in $\Omega^{[k]}$, there exists exactly one $g \in G$ such that $a_i^g = b_i$ $(i = 1, \ldots, k)$. Therefore, the order of a sharply k-transitive group of degree n is always $n(n-1)\cdots(n-k+1)$. (Clearly, G is sharply 1-transitive whenever G is regular.)

We will thoroughly investigate the sharply k-transitive permutation groups for $k \geq 2$ in Volume 3.

Example 1.18.

a) The symmetric group S_n is n-primitive and sharply n-transitive.
b) For $n \geq 3$, the alternating group A_n is sharply $(n-2)$-transitive and $(n-2)$-primitive.
c) Let $\Omega = \mathsf{K} \cup \{\infty\}$ be the projective line over the finite field $\mathsf{K} = \mathrm{GF}(p^f)$, and let $\mathrm{PGL}_2(p^f)$ be the group of all invertible linear fractional transformations from Ω to itself (where the symbol ∞ behaves as expected in all calculations). Then $\mathrm{PGL}_2(p^f)$ is sharply 3-transitive on Ω and has order $(p^f+1)p^f(p^f-1)$.
d) Let $\Omega = \mathsf{K} = \mathrm{GF}(p^f)$, and let $\Gamma(p^f)$ be the group of all affine semilinear transformations of K

$$x \mapsto ax^\alpha + b \qquad (0 \neq a \in \mathsf{K},\ \alpha \text{ an automorphism of } \mathsf{K}).$$

Then $\Gamma(p^f)$ is 2-transitive and solvable. The order of $\Gamma(p^f)$ is $p^f(p^f-1)f$. The group $\Gamma(p^f)$ is 3-transitive if and only if $p^f = 3$ or $p^f = 4$; in these cases, $\Gamma(p^f)$ is the symmetric group S_3 resp. S_4. The group $\Gamma(p^f)$ is 2-primitive if and only if $p^f = 2$ or $p^f - 1$ is a prime number. The group $\Gamma(p^f)$ is $\frac{5}{2}$-transitive if and only if $p^f = 2^q$ with q prime or $p^f = 3$.
e) Let $\mathsf{V} = \mathsf{V}(n, \mathsf{K})$ be the vector space of dimension $n > 1$ over the finite field $\mathsf{K} = \mathrm{GF}(p^f)$. Let $A(n, p^f)$ be the group of all affine linear transformations g of the form

$$x^g = xa + b \quad (x, b \in \mathsf{V}),$$

where a is a nonsingular K-linear transformation of V onto itself. Then $A(n, p^f)$ is always 2-transitive and contains the regular abelian normal subgroup

$$T = \{t \in A(n, p^f) \mid x^t = x + b,\ b \in \mathsf{V}\}.$$

If $A(n, p^f)$ is 2-primitive, then $p^f = 2$ and then $A(n, p^f)$ is 3-transitive. The group $A(n, p^f)$ is $\frac{5}{2}$-transitive if and only if $p^f = 2$.

Proof. a) is trivial.

b) The alternating group A_3 is regular and primitive by Theorem 1.3. By induction on n, the group A_n is sharply $(n-2)$-transitive and $(n-2)$-primitive.

c) The group $\mathrm{PGL}_2(p^f)$ is clearly transitive on Ω. The subgroup

$$\left\{ \begin{pmatrix} x \\ ax+b \end{pmatrix} \;\middle|\; a, b \in \mathsf{K}, a \neq 0 \right\}$$

is the stabilizer of $\{\infty\}$ and acts transitively on K. The subgroup

$$\left\{ \begin{pmatrix} x \\ ax \end{pmatrix} \;\middle|\; a \in \mathsf{K}, a \neq 0 \right\}$$

fixes $\{\infty\}$ and $\{0\}$ and is regular on $\mathsf{K} - \{0\}$. Therefore, $\mathrm{PGL}_2(p^f)$ is sharply 3-transitive.

d) Let $G = \Gamma(p^f)$. It is well known that the automorphism group of the field $\mathsf{K} = \mathrm{GF}(p^f)$ is cyclic of order f, so $|G| = p^f(p^f-1)f$. An easy computation shows that the subgroups

$$T = \left\{ \begin{pmatrix} x \\ x+b \end{pmatrix} \;\middle|\; b \in \mathsf{K} \right\} \quad \text{and} \quad L = \left\{ \begin{pmatrix} x \\ ax+b \end{pmatrix} \;\middle|\; a, b \in \mathsf{K}, a \neq 0 \right\}$$

are normal subgroups of G. Note that T is isomorphic to the additive group K^+ of K, L/T is isomorphic to the multiplicative group $\mathsf{K}^\times$ of K, and G/L is isomorphic to the automorphism group of K. Hence, T is elementary abelian while both L/T and G/L are cyclic. Therefore, G is solvable with $G^{(3)} = 1$; if $f = 1$ then in fact $G'' = 1$.

It is clear that G is 2-transitive since the subgroup L of affine linear maps is already 2-transitive. If G is 3-transitive, then $p^f(p^f-1)(p^f-2)$ would be a divisor of $|G| = p^f(p^f-1)f$ by Theorem 1.11. But this is the case only if $p^f = 3$ or $p^f = 4$. By comparing orders, G must then be the symmetric group S_3, resp. S_4.

We have

$$G_0 = \{g \in G \mid x^g = ax^\alpha,\ x \in \mathsf{K},\ a \in \mathsf{K}^\times,\ \alpha \in \mathbf{Aut}(\mathsf{K})\}.$$

Clearly, G_0 is transitive on $\mathsf{K}^\times = \mathsf{K} - \{0\}$. If $|\mathsf{K}^\times| = p^f - 1$ is a prime number, then G_0 is primitive on $\mathsf{K}^\times$ by Lemma 1.3 and so G is 2-primitive on K. This also covers the case that $p^f = 2$, where G is then the 2-primitive symmetric group S_2 on K.

Let $p^f > 2$, and suppose $p^f - 1$ is not a prime number. Let r be a prime divisor of $p^f - 1$. Since the multiplicative group $\mathsf{K}^\times$ of K is cyclic of order $p^f - 1$, it contains a subgroup R of order r. Then

$$R = \{n \in G \mid x^n = cx,\ x \in \mathsf{K},\ c \in \mathsf{R}\}$$

is clearly a normal subgroup of G_0. Since $|R| = r < |\mathsf{K}^\times|$, R is not transitive on $\mathsf{K}^\times$. By Theorem 1.5, we conclude that G_0 is imprimitive on $\mathsf{K}^\times$.

We have shown that if $p^f > 2$, then G is 2-primitive if and only if $p^f - 1$ is a prime number. It is known that this forces $p^f = 3$ or $p^f = 2^q$ for some prime q.

We have

$$G_{0,1} = \{g \in G \mid x^g = x^\alpha, x \in \mathsf{K}, \alpha \in \mathbf{Aut}(\mathsf{K})\}.$$

For $p^f = 3$, we have $G_{0,1} = 1$ and $G_{0,1}$ is $\frac{1}{2}$-transitive on $\mathsf{K} - \{0,1\} = \{-1\}$. Hence, G is $\frac{5}{2}$-transitive. If $\mathsf{K} = \mathrm{GF}(2^q)$ for a prime number q, it follows that $|G_{0,1}| = q$. All elements of K aside from 0 and 1 are primitive elements for K over GF(2), so they have exactly q images under $G_{0,1}$. Therefore, G is also $\frac{5}{2}$-transitive on K.

Conversely, suppose that $G_{0,1}$ is $\frac{1}{2}$-transitive on $\mathsf{K} - \{0,1\}$. Since all elements of the prime field GF(p) of K are fixed by $G_{0,1}$, we have $p^f = 3$ or $p = 2$. For $p^f = 3$ we are done, so suppose that $p = 2$. We have $G_{0,1} = \langle s \rangle$ with $x^s = x^2$. Let q be a prime divisor of f. Then s^q fixes the elements of GF(2^q). Therefore, the permutation s contains a cycle of length q. Since $G_{0,1} = \langle s \rangle$ is $\frac{1}{2}$-transitive on $\mathsf{K} - \{0,1\}$, all the cycles of s aside from (0) and (1) have the same length, namely q. It follows that $s^q = 1$, and therefore $f = |s| = q$.

So G is $\frac{5}{2}$-transitive if and only if $p^f = 3$ or $p^f = 2^q$ for a prime number q.

e) Let $G = A(n, p^f)$. The group G_0 of linear transformations of V is clearly transitive on $\mathsf{V} - \{0\}$. Since G is certainly transitive on V, G is 2-transitive.

Now assume that G is 2-primitive on V. Then G_0 is primitive on $\mathsf{V} - \{0\}$. Let $0 \neq v \in \mathsf{V}$ and $\Delta = \{av \mid a \in \mathsf{K}^\times\}$. If $\Delta \cap \Delta^g \neq \emptyset$ for some $g \in G_0$, then clearly $\Delta = \Delta^g$. As G_0 is primitive and $n > 1$ implies $\Delta \neq \mathsf{V}$, we must have $|\Delta| = |\mathsf{K}^\times| = 1$, so $p^f = 2$.

Now assume $p^f = 2$, and let (v_1, v_2) and (v_1', v_2') be pairs of distinct elements of $\mathsf{V} - \{0\}$. Since $\mathsf{K} = \mathrm{GF}(2)$, the vectors v_1 and v_2 are linearly independent over K, and likewise for v_1' and v_2'. Therefore, there exists a linear map $g \in G_0$ such that $v_i^g = v_i'$ for $i = 1, 2$. So G is 3-transitive on V.

Assume G is $\frac{5}{2}$-transitive on V. Let $0 \neq v \in \mathsf{V}$. Then $G_{0,v}$ is $\frac{1}{2}$-transitive on $\mathsf{V} - \{0, v\}$ and has no fixed points on $\mathsf{V} - \{0, v\}$ since $n > 1$. Therefore, 0 and v are the only scalar multiples of v. So $p^f = 2$, and then G is indeed 3-transitive on V. □

It is known that every 6-transitive permutation group of degree n contains the alternating group A_n, so is at least $(n-2)$-transitive.[1] We will show in Volume 3 that if we assume the Schreier Conjecture holds (see Remark I.18.5, pg. 139), then every at least 8-transitive permutation group of degree n contains the alternating group A_n (Wielandt [476]). At the time of writing, it is not known how to prove that there exists a universal k such that every k-transitive permutation group is an alternating or symmetric group without using the Schreier Conjecture. The following is all that is known: If G is k-transitive of degree n and $k > \frac{n}{3}$, then $A_n \leq G$. (See Carmichael [85]; the sharpest known results are in Wielandt [458].)

The only at least 4-transitive groups that do not contain the alternating group of the same degree are the four Mathieu groups, namely,

M_{24}: 5-transitive of degree 24, $|M_{24}| = 24 \cdot 23 \cdot 22 \cdot 21 \cdot 20 \cdot 3 \cdot 2^4$,

[1] This has been proved using the classification of finite simple groups. For an introduction to the impact of the classification on permutation group theory, see P. J. Cameron, Finite permutation groups and finite simple groups, Bull. London Math. Soc. **13** (1981), no. 1, 1–22. *trans.*

M_{23}: 4-transitive of degree 23, $|M_{23}| = 23 \cdot 22 \cdot 21 \cdot 20 \cdot 3 \cdot 2^4$,
M_{12}: sharply 5-transitive of degree 12, $|M_{12}| = 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8$,
M_{11}: sharply 4-transitive of degree 11, $|M_{11}| = 11 \cdot 10 \cdot 9 \cdot 8$.
We will give these groups a thorough treatment in Volume 3, together with a 3-transitive group M_{22} of degree 22. These five Mathieu groups are simple and do not belong to one of the infinite families of simple groups.

2 Regular normal subgroups of multiply-transitive permutation groups

In this section, we will first address the question of how strongly transitive a subgroup of the automorphism group $\mathbf{Aut}(G)$ of a finite group G can be when regarded as a permutation group on $G - 1$.

Theorem 2.1. *Let* $\mathbf{Aut}(G)$ *be the automorphism group of* $G \neq 1$ *and* $A \leq \mathbf{Aut}(G)$. *We regard* A *as a permutation group on* $G - 1$. *Then the following hold:*
a) *If* A *is transitive, then* G *is elementary abelian.*
b) *If* A *is primitive or* $\frac{3}{2}$*-transitive, then* $|G| = 2^m$ *or* $|G| = 3$.
c) *If* A *is* 2*-primitive or* $\frac{5}{2}$*-transitive, then* $|G| = 4$ *or* $|G| = 3$.
d) A *is never* 4*-transitive.*

Proof. a) Since A is transitive on $G - 1$, all nonidentity elements of G have the same prime order p. Therefore, G is a p-group by Theorem I.7.4 (pg. 34), and $\mathbf{Z}(G) > 1$ by Theorem I.6.9 (pg. 31). Since $\mathbf{Z}(G)$ is a characteristic subgroup of G, it is fixed by A. So $G = \mathbf{Z}(G)$ and G is elementary abelian.

b) First assume that A is primitive on $G - 1$. Let $1 \neq g \in G$ and $\Delta = \{g, g^{-1}\}$. If $\alpha \in A$ and $\Delta \cap \Delta^\alpha \neq \emptyset$, then $g^\alpha = g^{\pm 1}$ and so $\Delta^\alpha = \Delta$. By the primitivity of A, we must have $|\Delta| = 1$ or $\Delta = G - 1$. The first case leads to $g = g^{-1}$ and so $p = 2$, while the second case leads to $|G| = 3$.

Assume A is $\frac{3}{2}$-transitive on $G - 1$, and assume that $p \neq 2$. Then if $1 \neq g \in G$, we have $g^{-1} \neq g$. The stabilizer A_g of g also fixes g^{-1}. Consequently, A_g has an orbit of size 1 on the set $G - \{1, g\}$. By the $\frac{1}{2}$-transitivity of A_g on $G - \{1, g\}$, it follows that

$$G - \{1, g\} = \{g^{-1}\}, \quad \text{and so} \quad |G| = 3.$$

c) Certainly $|G| \geq 3$. If $|G| \geq 4$, then G is elementary abelian and $|G| = 2^m$ by part b).

Assume A is 2-primitive and $U = \{1, g, h, gh\}$ is a subgroup of order 4 in G. Then A_g is primitive on $G - \{1, g\}$. If we define $\Delta = \{h, hg\}$, it follows for all $\alpha \in A_g$ that if $\Delta \cap \Delta^\alpha \neq \emptyset$, then either $h^\alpha = h$ or $h^\alpha = hg$. In either case, we have $\Delta = \Delta^\alpha$. Since A_g is primitive on $G - \{1, g\}$ and $|\Delta| > 1$, this forces

$$G - \{1, g\} = \Delta, \quad \text{and so} \quad |G| = 4.$$

Assume A is $\frac{5}{2}$-transitive, and let U be chosen as in the last paragraph. Since $A_{g,h}$ fixes hg, we have $G - \{1, g, h\} = \{hg\}$ and so $|G| = 4$.

d) This part follows from c) since the automorphism groups of the elementary abelian groups of orders 3 and 4 are certainly not 4-transitive on $G - 1$. □

Lemma 2.2. *Let G be a permutation group on Ω, and suppose N is a regular normal subgroup of G. Let $a \in \Omega$. Then G_a acting on $\Omega - \{a\}$ is permutation isomorphic to G_a acting on $N - 1$ as $n \mapsto n^g$ $(g \in G_a)$.*

Proof. Let $\epsilon : N - 1 \longrightarrow \Omega - \{a\}$ be the map defined by $n^\epsilon = a^n$. Since N is regular, ϵ is a bijection. For $g \in G_a$, we have

$$(n^g)^\epsilon = a^{g^{-1}ng} = a^{ng} = (n^\epsilon)^g,$$

which shows that the map is a permutation isomorphism. □

We immediately obtain the following theorem from Theorem 2.1 and Lemma 2.2.

Theorem 2.3. *Let G be a permutation group of degree n, and suppose N is a regular normal subgroup of G. Then the following hold:*

a) *If G is 2-transitive, then N is elementary abelian and n is a prime power.*
b) *If G is 2-primitive or $\frac{5}{2}$-transitive, then $n = 2^m$ or $n = 3$.*
c) *If G is 3-primitive or $\frac{7}{2}$-transitive, then $n = 4$ or $n = 3$.*
d) *G is never 5-transitive.*

We have already shown in Remark I.8.12 (pg. 43) that S_n, and therefore A_n, is solvable for $n \leq 4$. Now we prove that:

Theorem 2.4. *The alternating group A_n is simple for $n \geq 5$.*

Proof. We prove the simplicity of A_n for $n \geq 5$ by induction on n. The base case $n = 5$ was proved in Theorem I.8.14 (pg. 43).

Let $n \geq 6$, and let N be a minimal normal subgroup of A_n. We identify A_{n-1} inside A_n with $\{g \in A_n \mid 1^g = 1\}$. Then $N \cap A_{n-1} \trianglelefteq A_{n-1}$. Since A_{n-1} is simple by induction, either $N \cap A_{n-1} = 1$ or $N \cap A_{n-1} = A_{n-1}$. Since A_n is primitive, N must act transitively by Theorem 1.5. If $N \cap A_{n-1} = 1$, then N is regular. By Theorem 2.3 d), it follows that A_n is at most 4-transitive. Since A_n is $(n-2)$-transitive by Example 1.18 b), we must have $n = 6$. But this is impossible by Theorem 2.3 a) since 6 is not a prime power.

So it must be the case that $N \cap A_{n-1} = A_{n-1} \leq N$. By the transitivity of N, we have $|N| \geq n|A_{n-1}| = |A_n|$, so that $N = A_n$ and A_n is simple. □

Next, we consider the case that G is 3-transitive and contains a regular normal subgroup. We have already seen instances of this in Example 1.18 e), namely the groups $A(n, 2)$ of all affine linear maps of the form $v^g = va + b$ on the vector space $V(n, 2)$ of dimension n over GF(2). Only one other example is known[2], which we now describe. We will need the following theorem, which is of independent interest.

[2] It is now known that this is the only other example, see Corollary 8.3 of P. J. Cameron and W. M. Kantor, 2-transitive and antiflag transitive collineation groups of finite projective spaces, J. Algebra **60** (1979), no. 2, 384–422. *trans.*

Theorem 2.5. *Let* $\mathsf{V} = \mathsf{V}(4,2)$ *be the vector space of dimension* 4 *over the field* GF(2)*, and let* $\mathrm{GL}_4(2)$ *be the group of all linear maps from* V *onto itself. Then* $\mathrm{GL}_4(2) \cong A_8$.

Proof. The matrices

$$g_1 = \begin{pmatrix} 1&1&1&1\\ 0&0&0&1\\ 1&1&0&0\\ 0&1&0&1 \end{pmatrix} \qquad g_2 = \begin{pmatrix} 0&1&0&1\\ 0&0&1&0\\ 0&1&0&0\\ 1&0&1&0 \end{pmatrix}$$

$$g_3 = \begin{pmatrix} 0&1&1&1\\ 0&1&0&1\\ 1&1&0&0\\ 0&0&0&1 \end{pmatrix} \qquad g_4 = \begin{pmatrix} 1&0&1&0\\ 0&1&0&0\\ 0&0&1&0\\ 0&1&0&1 \end{pmatrix}$$

$$g_5 = \begin{pmatrix} 0&0&1&0\\ 0&1&0&1\\ 1&0&0&0\\ 0&0&0&1 \end{pmatrix} \qquad g_6 = \begin{pmatrix} 0&1&1&1\\ 0&0&1&0\\ 0&1&0&0\\ 1&1&1&0 \end{pmatrix}$$

lie in $\mathrm{GL}_4(2)$. A trivial but tedious calculation shows that these matrices satisfy the relations defining A_8 in Example I.19.8 (pg. 147):

$$\begin{aligned} g_1^3 = g_i^2 = 1 \quad & (2 \le i \le 6),\\ (g_i g_{i+1})^3 = 1 \quad & (1 \le i \le 5),\\ (g_i g_j)^2 = 1 \quad & (i+1 < j). \end{aligned}$$

Therefore, $G = \langle g_1, \ldots, g_6 \rangle$ is an epimorphic image of A_8. Since A_8 is simple by Theorem 2.4, it follows that $G \cong A_8$. The simple calculation of Lemma 6.2 shows that $|\mathrm{GL}_4(2)| = |A_8|$. Since $G \le \mathrm{GL}_4(2)$, we must have $G = \mathrm{GL}_4(2) \cong A_8$. □

In Volume 3, we will obtain a less computational proof of Theorem 2.5 when we consider the Mathieu groups.

Theorem 2.6. *Every subgroup* G *of* $\mathrm{GL}_4(2)$ *that is isomorphic to the alternating group* A_7 *permutes the nonzero vectors of* $\mathsf{V} = \mathsf{V}(4,2)$ 2*-transitively. Clearly, then*

$$S = \{s \mid v^s = vg + b, g \in G, b \in \mathsf{V}\}$$

is a 3*-transitive permutation group on* V *with regular normal subgroup*

$$T = \{t \in S \mid v^t = v + b, b \in \mathsf{V}\}.$$

Proof. Since $\mathrm{GL}_4(2)$ is isomorphic to A_8, the former definitely has subgroups isomorphic to A_7. Let G be one of them. We regard G as a permutation group of degree 15 on the nonzero vectors of $\mathsf{V} = \mathsf{V}(4,2)$.

a) G is transitive:

Let t be an element of order 7 in G. If t has at least 8 fixed points on $\mathsf{V} - \{0\}$, then these fixed points generate V since every proper subspace of V contains at most 7 nonzero vectors. But then $t = 1$. As the cycle decomposition of t can only contain cycles of lengths 1 and 7, t must have exactly one fixed point on $\mathsf{V} - \{0\}$, and the cycle decomposition of t consists of two 7-cycles.

Let s be an element of order 5 in G. If s has at least 5 fixed points, then there exists a subspace of dimension 3 fixed pointwise by s. So with respect to some basis of V, the element s has the form

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ * & * & * & 1 \end{pmatrix},$$

and we obtain the contradiction $s^2 = 1$. So the cycle decomposition of s consists of three 5-cycles.

Therefore, every orbit of G has size k with $k = 5a$ and $k = 7b$ or $k = 7b + 1$. This implies $k = 15$, and G is transitive.

b) G is 2-transitive:

Let H be the stabilizer of one of the 15 points of $\mathsf{V} - \{0\}$, so $|H| = 168$. The cosets Hy that are contained in the double coset HxH correspond exactly to the orbits of H by Theorem 1.12. Let

$$G = H \cup Hx_1H \cup \cdots \cup Hx_rH$$

so that

$$15 = [G : H] = 1 + n_1 + \cdots + n_r,$$

where

$$n_i = |Hx_iH|/|H| = [H : H \cap x_iHx_i^{-1}].$$

By the cycle decomposition of the elements of order 7 we found in a), we must have $n_1 = n_2 = 7$ or $n_1 = 14$.

Assume for contradiction that $n_1 = n_2 = 7$. Then $|H \cap xHx^{-1}| = 24$ for all $x \in G - H$. Since $G \cong A_7$ is simple, G is generated by its elements of order 2. So there exists some $x \in G$ such that $x \notin H$ and $x^2 = 1$. Then the group $D = H \cap xHx^{-1}$ is normalized by x, so that

$$|\langle D, x\rangle| = |D\langle x\rangle| = 2|D| = 48,$$

which contradicts $48 \nmid |G|$. Thus, $n_1 = 14$. Hence, H has one orbit of size 14, and G is 2-transitive on $\mathsf{V} - \{0\}$. □

3 Primitive permutation groups with abelian normal subgroups

Lemma 3.1. *Let A be an abelian, transitive permutation group of degree n on $\Omega = \{1, \dots, n\}$. Then A is equal to its centralizer in the full symmetric group S_n.*

Proof. Suppose $c \in \mathbf{C}_{S_n}(A)$ with $1^c = 1$. Let $i \in \Omega$ be arbitrary, and choose $a \in A$ such that $1^a = i$. Then

$$i^c = 1^{ac} = 1^{ca} = 1^a = i.$$

Since $i \in \Omega$ was arbitrary, we must have $c = 1$. Hence, we have that $|\mathbf{C}_{S_n}(A)| = [\mathbf{C}_{S_n}(A) : \mathbf{C}_{S_n}(A)_1] \leq n = |A|$, where the last equality follows from Theorem I.5.13 (pg. 28). Since $A \leq \mathbf{C}_{S_n}(A)$, we have proved the claim. □

Theorem 3.2 (Galois). *Let G be a primitive permutation group of degree n, and let N be a minimal normal subgroup of G. If N is solvable, then the following hold:*

a) *N is regular and elementary abelian. The degree n of G is a prime power p^m.*
b) *$G = NG_1$ and $N \cap G_1 = 1$.*
c) *$\mathbf{C}_G(N) = N$.*
d) *N is the unique minimal normal subgroup of G.*
e) *G_1 contains no nonidentity normal subgroup of p-power order.*
f) *If G is solvable, then all complements of N are conjugate in G.*

(*The assumptions of the theorem are satisfied, in particular, when G is a primitive solvable permutation group.*)

Proof. a) Since N is a minimal normal subgroup, it is characteristically simple. So N is elementary abelian of prime power order p^m. Since G is primitive, N must be transitive by Theorem 1.5. By Theorem I.5.13 (pg. 28), N is regular and so $n = |N| = p^m$.

b) Since N is regular, we have $N \cap G_1 = 1$. From this it follows that

$$|NG_1| = |N||G_1| = n|G_1| = |G|,$$

and so $G = NG_1$.

c) follows immediately from Lemma 3.1.

d) Suppose that $M \lhd G$ with $M \cap N = 1$. By Theorem I.3.13 (pg. 18), $M \leq \mathbf{C}_G(N) = N$. So $M = 1$, and we conclude that N is the only minimal normal subgroup of G.

e) Let $P \trianglelefteq G_1$ with $|P| = p^r$. Then $NP \trianglelefteq NG_1 = G$ and $|NP| = p^{m+r}$. Since $N \trianglelefteq NP$, we have

$$1 < N \cap \mathbf{Z}(NP) = D \trianglelefteq G$$

by Theorem I.6.9 (pg. 31). Since N is a minimal normal subgroup of G, it follows that $D = N \leq \mathbf{Z}(NP)$. This implies $P \leq \mathbf{C}_G(N) = N$, and so $P \leq N \cap G_1 = 1$.

f) We may assume that $N < G$. Let K/N be a minimal normal subgroup of G/N. Since K/N is a characteristically simple solvable group, it is elementary abelian. So

the order of K/N is then a power q^k of some prime q. Observing that $G/N \cong G_1$, we have $q \neq p$ by part e).[3]

Let V be a complement of N in G, so $G = NV$ and $N \cap V = 1$. We define $Q = K \cap V$. Since $K \trianglelefteq G$, we have $Q \trianglelefteq V$. Now, $K = K \cap NV$ and $N < K$, so by the Dedekind identity of Lemma I.2.12 c) (pg. 7) we have

$$K = N(K \cap V) = NQ,$$

and so $|K/N| = |Q| = q^k$. The group Q is a Sylow q-subgroup of K since $p \neq q$. All Sylow q-subgroups of K are conjugate in K and therefore in G. We will show that $V = \mathbf{N}_G(Q)$, from which it follows that all complements of N in G are conjugate.

We claim that V is a maximal subgroup of G: Suppose $U \leq G$ is a subgroup with $V < U \leq G$. By the Dedekind identity, U has the form $U = VN^*$, where $1 < N^* = N \cap U$. So N^* is normal in U and in the abelian group N. It follows that $N^* \lhd NU = G$. By the minimality of N, we have $N^* = N$ and so $U = G$.

Now, $V \leq \mathbf{N}_G(Q)$ since $Q \trianglelefteq V$. Certainly $\mathbf{N}_G(Q) < G$, because otherwise $Q \lhd G$ and $Q \cap N \leq V \cap N = 1$, contradicting d). Therefore, $V = \mathbf{N}_G(Q)$ by the maximality of V. □

Theorem 3.2 admits a partial converse:

Theorem 3.3. *Let G be solvable, and suppose N is a minimal normal subgroup of G such that $N = \mathbf{C}_G(N)$. Then there exists a complement H of N in G such that $\bigcap_{g \in G} H^g = 1$. The permutation representation of G on the cosets of H is a faithful representation of G as a primitive permutation group.*

Proof. We may assume that $N < G$. Since N is a minimal normal subgroup of the solvable group G, its order is a prime power $|N| = p^m$ by Theorem I.9.13 (pg. 54). Let M/N be a minimal normal subgroup of G/N. Again by Theorem I.9.13, we have $|M/N| = q^k$ for some prime q.

Assume for contradiction that $p = q$. Then $1 < N \cap \mathbf{Z}(M) \trianglelefteq G$ by Theorem I.6.9 (pg. 31). The minimality of N implies that $N \leq \mathbf{Z}(M)$. But then $M \leq \mathbf{C}_G(N) = N$, a contradiction.

So we must have $p \neq q$. Let Q be a Sylow q-subgroup of M, so $M = NQ$. By the Frattini argument of Theorem I.7.8 (pg. 36), we have $G = M\mathbf{N}_G(Q) = N\mathbf{N}_G(Q)$. The commutativity of N implies $N \cap \mathbf{N}_G(Q) \trianglelefteq G$. The minimality of N then forces either $N \leq \mathbf{N}_G(Q)$ or $N \cap \mathbf{N}_G(Q) = 1$. If $N \leq \mathbf{N}_G(Q)$, then $N \cap Q = 1$ implies $Q \leq \mathbf{C}_G(N) = N$ by Theorem I.3.13 (pg. 18). This is a contradiction. Hence, $N \cap \mathbf{N}_G(Q) = 1$ and so $H = \mathbf{N}_G(Q)$ is a complement of N in G. Defining $K = \bigcap_{g \in G} H^g$, we have $K \lhd G$ and $K \cap N = 1$.[4] By Theorem I.3.13, $K \leq \mathbf{C}_G(N) = N$ and so $K = 1$. So the representation of G as a permutation group on the cosets of H is faithful. Now one shows that H is a maximal subgroup of G using the Dedekind

[3] Correction 9 from errata.

[4] Correction 10 from errata.

identity as in the proof of Theorem 3.2 f). Therefore, G acts primitively on the cosets of H in G by Theorem 1.4. □

Example 3.4. The assertions in Theorems 3.2 and 3.3 do not hold in general without the solvability assumptions.

a) It is not difficult to construct an example of a group G containing a minimal normal subgroup $N \lhd G$ such that $\mathbf{C}_G(N) = N$ and N has no complement in G:

Let p be an odd prime, and let $R = \mathbb{Z}/p^2\mathbb{Z}$ be the ring of integers modulo p^2.[5] Let

$$H = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \;\middle|\; a, b, c, d \in R, ad - bc = 1 \right\}.$$

Then

$$M = \{m \in H \mid m = 1 + ph,\ h \in M_{2\times 2}(R)\}$$

is a minimal normal subgroup of H of order p^3. An easy computation shows that $\mathbf{C}_G(M) = M\langle -1\rangle$. Hence, the assumptions of Theorem 3.2 f) are satisfied in $G = H/\langle -1\rangle$ if we take $N = M\langle -1\rangle/\langle -1\rangle$, except perhaps the solvability of G.

Assume for contradiction that N has a complement in G. Let P be a Sylow p-subgroup of G. One can show that P is generated by elements of order p and, for $p > 3$, that P has exponent p: Since $|P| = p^4 < p^{p+1}$ for $p > 3$, the group P is regular by Theorem III.10.2 b); the elements of P with order p together with 1 constitute a subgroup of P by Theorem III.10.5 a), so they generate all of P. However, this cannot be true, for $h = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ is an element of order p^2 in H, and $h\langle -1\rangle$ is an element of the same order in G. (Incidentally, if $p = 3$ then the group G is solvable and N does have a complement in G.)

b) Consider the semidirect product of the vector space $\mathsf{V}(3, 2)$ with its full automorphism group $\mathrm{GL}_3(2)$. In order to show that there exists more than one conjugacy class of complements of $\mathsf{V}(3, 2)$, it is enough to prove that $\mathbf{H}^1(\mathrm{GL}_3(2), \mathsf{V}(3, 2)) \neq 0$ by Theorem I.17.3 (pg. 129).

By Theorem 2.5, there exists an isomorphism $\mu : \mathrm{GL}_4(2) \to A_8$. By Theorem 6.9, which we call upon prematurely here, the transvections constitute a conjugacy class of elements of order 2 in $\mathrm{GL}_4(2)$. In A_8, there exist exactly two conjugacy classes of elements of order 2, namely the conjugacy classes of $i_1 = (1, 2)(3, 4)$ and $i_2 = (1, 2)(3, 4)(5, 6)(7, 8)$. An easy computation shows that $|\mathbf{C}_{A_8}(i_1)| = 2^5 \cdot 3$ and $|\mathbf{C}_{A_8}(i_2)| = 2^6 \cdot 3$. If t is a transvection in $\mathrm{GL}_4(2)$, then $|\mathbf{C}_{\mathrm{GL}_4(2)}(t)| = 2^6 \cdot 3$. Therefore, μ maps the conjugacy class of the transvection t in $\mathrm{GL}_4(2)$ onto the conjugacy class i_2 in A_8.

Let H be the preimage of A_7 in $\mathrm{GL}_4(2)$ under μ. Then H contains no transvections. Let K be the subgroup of H that stabilizes the hyperplane $\mathsf{H} = \langle v_1, v_2, v_3\rangle$. Since the 15 hyperplanes of $\mathsf{V}(4, 2)$ are permuted by H, we have $[H : K] \leq 15$. If $k \in K$, then

[5] Correction 11 from errata.

$$v_i k = v_i k^{\epsilon} \quad (i = 1, 2, 3) \quad \text{and} \quad v_4 k = v_4 + f(k),$$

where $f(k) \in \langle v_1, v_2, v_3 \rangle$ and $k^{\epsilon} \in \mathrm{Hom}_K(\mathsf{H}, \mathsf{H})$. Clearly, ϵ is a homomorphism from K to the group $\mathrm{GL}(\mathsf{H}) \cong \mathrm{GL}_3(2)$. If $k^{\epsilon} = 1$, then either k is a transvection or $k = 1$. Since K does not contain any transvections, the first case cannot occur. So ϵ is a monomorphism from K to $\mathrm{GL}_3(2)$. It follows that

$$168 = \frac{|H|}{15} \leq |K| \leq |\mathrm{GL}_3(2)| = 168.$$

So $|K| = 168$, and therefore $K \cong \mathrm{GL}_3(2)$.

For $k_1, k_2 \in K$, we have

$$v_4 + f(k_1 k_2) = v_4 k_1 k_2 = (v_4 + f(k_1))k_2 = v_4 + f(k_1)k_2 + f(k_2).$$

So f is a 1-cocycle in $\mathbf{Z}^1(K, \mathsf{H})$. Assume for contradiction that f is a coboundary; that is,

$$f = w\delta \in \mathbf{B}^1(K, \mathsf{H})$$

for some $w \in \mathbf{C}^0(K, \mathsf{H}) = \mathsf{H}$. Then for all $k \in K$, we have that

$$f(k) = wk - w,$$

and it follows that

$$(v_4 - w)k = v_4 + f(k) - wk = v_4 - w.$$

So $\mathsf{V} = \mathsf{H} \oplus \langle v_4 - w \rangle$ is a K-invariant decomposition of V. Since K acts on H as the entire group $\mathrm{GL}_3(2)$, the group K contains transvections, contrary to our previous observation. Therefore, f is not a coboundary, and so $\mathbf{H}^1(\mathrm{GL}_3(2), \mathsf{H}) \neq 0$.

Theorem 3.5. *Let G be a primitive permutation group of degree p^m on Ω containing a minimal and abelian normal subgroup N. Then G is permutation isomorphic to a subgroup of the group $A(m, p)$ of all affine maps of the vector space $\mathsf{V}(m, p)$ of dimension m over the field $\mathrm{GF}(p)$. The unique minimal normal subgroup N of G consists of all translations $v \mapsto v + a$. Defining*

$$H = \{g \in G \mid 0^g = 0\},$$

we have that $\mathsf{V}(m, p)$ is an irreducible H-module.

Proof. By Theorem 3.2, N is elementary abelian of order p^m. Let π be an isomorphism from N onto the additive group V^+ of $\mathsf{V} = \mathsf{V}(m, p)$. We map the set Ω bijectively onto V via the map δ defined by

$$i^{\delta} = (1^n)^{\delta} = n^{\pi},$$

where the regularity of N means the element $n \in N$ satisfying $1^n = i$ is determined uniquely. We may define an action of G on V by $(i^\delta)^g = (i^g)^\delta$. Let $m \in N$ with, say, $i^m = j$, and let $n \in N$ such that $1^n = i$. Then $1^{nm} = i^m = j$, and therefore

$$(i^\delta)^m = (i^m)^\delta = (nm)^\pi = n^\pi + m^\pi = i^\delta + m^\pi.$$

So m acts on V as the translation $v \mapsto v + m^\pi$.

Let $n \in N$ with $1^n = i$, and let $h \in G_1$ with $i^h = j$. Then

$$1^{h^{-1}nh} = i^h = j.$$

The map $h : i \mapsto j$ on Ω corresponds to the map

$$i^\delta = n^\pi \mapsto j^\delta = (n^h)^\pi$$

on V. Since $n \mapsto n^h$ is an automorphism of N, the map $n^\pi \mapsto (n^h)^\pi$ is an automorphism of V^+. That is, it is a linear map of V (note that over the prime field $\mathrm{GF}(p)$, linear maps and additive maps coincide). Since

$$1^\delta = (1^1)^\delta = 1^\pi = 0,$$

we have

$$G_1 = H = \{g \in G \mid 0^g = 0\}.$$

Assume for contradiction that W is a nonzero proper subspace of V that is fixed by H. We define

$$W = \left\{ \begin{pmatrix} v \\ v + w \end{pmatrix} \,\middle|\, w \in \mathsf{W} \right\}.$$

For $h \in H$ and $t = \binom{v}{v+w} \in W$, it follows that

$$t^h = \begin{pmatrix} v \\ v + w^h \end{pmatrix} \in W.$$

Therefore, W is a normal subgroup of $G = NH$, contradicting the minimality of N. We conclude that V is an irreducible H-module. □

The structure of transitive solvable permutation groups of prime degree is completely determined in the following theorem.

Theorem 3.6 (Galois). *Assume G is a transitive permutation group of prime degree p. Then the following are equivalent:*

a) *G has exactly one Sylow p-subgroup.*
b) *G is solvable.*
c) *G is permutation isomorphic to a subgroup of the group $A(1, p)$ of all maps $x \mapsto ax + b$ $(a \neq 0)$ over the prime field $\mathrm{GF}(p)$.*

d) *If a permutation $g \in G$ fixes at least two points, then $g = 1$.* (*This means G is a so-called Frobenius group; see Theorem V.8.2, pg. 526.*)

Proof. a) $\Rightarrow$ b): Since G is transitive of prime degree p, we have that p divides $|G|$ and $|G|$ divides $|S_p| = p!$. Since p^2 is not a divisor of $p!$, the unique Sylow p-subgroup P of G has order p. The subgroup P must then be regular since it is generated by a p-cycle. From Lemma 3.1, we conclude that $\mathbf{C}_G(P) = P$. The factor group $G/P = G/\mathbf{C}_G(P)$ is isomorphic to a subgroup of the automorphism group $\mathbf{Aut}(P)$ of P by Theorem I.4.5 (pg. 21), so it must be cyclic by Theorem I.13.19 (pg. 89). Therefore, G is solvable. (In fact, $G'' = 1$.)

b) $\Rightarrow$ c): Let N be a minimal normal subgroup of the solvable group G. Then N is abelian. By Theorems 1.3 and 1.5, N is transitive. Therefore, N is regular by Theorem I.5.13 (pg. 28), and the claim follows from Theorem 3.5.

c) $\Rightarrow$ d): This is trivial since all the nonidentity affine linear maps over a field have at most one fixed point.

d) $\Rightarrow$ a): Assume for contradiction that G has two distinct Sylow p-subgroups P_1 and P_2. Then P_1P_2 is a subset of G, and by Lemma I.2.12 a) (pg. 7) we have

$$|P_1P_2| = \frac{|P_1||P_2|}{|P_1 \cap P_2|} = p^2.$$

So certainly $p^2 \leq |G|$. Let

$$H = \{g \in G \mid 1^g = 1\} \quad \text{and} \quad K = \{g \in G \mid 1^g = 1,\ 2^g = 2\}.$$

The assumption in d) gives $K = 1$. Therefore, we have

$$|G| = [G : K] = [G : H][H : K].$$

The transitivity of G implies $[G : H] = p$ by Theorem I.5.11 (pg. 28). Furthermore, $[H : K]$ is the size of the H-orbit containing 2, so $[H : K] \leq p - 1$. Altogether we obtain the contradiction

$$p^2 \leq |G| \leq p(p - 1),$$

so G has only one Sylow p-subgroup. □

The following theorem of Burnside gives more insight into permutation groups of prime degree:

If G is a transitive permutation group of prime degree, then G is solvable or 2-transitive.

A proof of this theorem will be given in Theorem V.21.3 (pg. 645) using character theory.

Theorem 3.2 has important implications for the structure of solvable groups, to which we now turn.

Theorem 3.7. *If G is solvable, then every maximal subgroup of G has prime power index.*

Proof. Let U be a maximal subgroup of G. We consider the permutation representation of G acting on the cosets of U. The subgroup stabilizing U under this action is U itself. The kernel of the representation is $K = \bigcap_{g \in G} U^g$ by Theorem I.6.2 (pg. 29). Since U/K is a maximal subgroup of G/K, the induced permutation representation of G/K is a faithful permutation representation of G/K as a primitive permutation group by Theorem 1.4. We conclude that

$$[G : U] = [G/K : U/K]$$

is a prime power by Theorem 3.2. □

The converse of Theorem 3.7 does not hold. The maximal subgroups of the simple group of order 168, which we consider in §6 and §7, have indices 7 and 8. This is the only known nonabelian simple group whose maximal subgroups have prime power index.[6] We will show in Theorem VI.9.4, however, that if the index of every maximal subgroup of G is prime or the square of a prime, then G is solvable.

Another consequence of Theorem 3.2 is the following:

Theorem 3.8 (Ore). *Let G be solvable. Then there is at most one conjugacy class of maximal subgroups of G that contain no nontrivial normal subgroup of G. Therefore, G admits at most one faithful representation as a primitive permutation group up to permutation isomorphism.*

Proof. Suppose that U_1 and U_2 are maximal subgroups of G such that $\bigcap_{g \in G} U_i^g = 1$ for $i = 1, 2$. Then we obtain faithful primitive permutation representations of G on the cosets of the U_i. The group G has a unique minimal normal subgroup N by Theorem 3.2 d), so $G = NU_i$ and $N \cap U_i = 1$. It follows by Theorem 3.2 f) that U_1 and U_2 are conjugate in G. □

Theorem 3.9 (Ore). *Let U and V be maximal subgroups of the solvable group G. Then exactly one of the following occurs:*
a) *$G = UV$ and U, V are not conjugate in G.*
b) *$G \neq UV$ and U, V are conjugate in G.*

Proof. a) Assume $G = UV$, and assume for contradiction that $U^g = V$ for some $g \in G$. Then $g = uv$ for some $u \in U$ and $v \in V$. It follows that $V = U^g = U^v$, then $U = V^{v^{-1}} = V$ and so $G = UV = U$. This contradicts our assumption that U is a maximal subgroup of G. Therefore, it follows just from $G = UV$ that U and V are not conjugate. (That is, up to this point we have not used the fact that G is solvable nor the maximality of U and V; only $U < G$ is required for this step!)

b) Let U and V be maximal subgroups of G with $G \neq UV$. Suppose $N \trianglelefteq G$ is a normal subgroup of G contained in U. Then $N \leq V$, since otherwise we would have, by the maximality of V, that

[6] This is in fact the only example; see R. M. Guralnick, Subgroups of prime power index in a simple group, J. Algebra **81** (1983), no. 2, 304–311. *trans.*

$$G = NV \subseteq UV \neq G.$$

By the symmetry of U and V, we conclude that $\bigcap_{g\in G} U^g = \bigcap_{g\in G} V^g = K$. Applying Theorem 3.8 to G/K with the maximal subgroups U/K and V/K yields the conjugacy of U/K and V/K, so U and V are conjugate in G. □

Theorem 3.10. *Let $\mathsf{V} = \mathsf{V}(n, p^f)$ be the vector space of dimension n over the field $\mathsf{K} = \mathrm{GF}(p^f)$. Let A be an abelian group of K-linear transformations of V onto itself, and assume V is irreducible as an A-module. Then viewing A as a permutation group on the set V, the group A is permutation isomorphic to a group of linear maps over $\mathrm{GF}(p^{nf})$ in which, for all $a \in A$ and $x \in \mathsf{V}$, the maps have the form*

$$x^a = u(a)\, x \quad (u(a) \in \mathrm{GF}(p^{nf})).$$

In particular, A is cyclic and we have that $|A|$ divides $p^{nf} - 1$, where n is determined uniquely by $|A|$ to be the smallest natural number such that $p^{nf} - 1 \equiv 0 \bmod |A|$. Finally, if $A = \langle a_0 \rangle$, then $\mathrm{GF}(p^f)[u(a_0)] = \mathrm{GF}(p^{nf})$.

Proof. Let C be the commutative subring of $\mathrm{Hom}_{\mathsf{K}}(\mathsf{V}, \mathsf{V})$ generated by A and K. Since V is an irreducible C-module, $\mathrm{Hom}_C(\mathsf{V}, \mathsf{V})$ is a division ring by Schur's Lemma I.10.5 (pg. 59). Since A and C are commutative, we have $C \subseteq \mathrm{Hom}_C(\mathsf{V}, \mathsf{V})$. Applying Lemma I.2.2 (pg. 5) to the finite, multiplicatively-closed subset $C - \{0\}$ of $\mathrm{Hom}_C(\mathsf{V}, \mathsf{V})^\times$ shows that $C - \{0\}$ is a group. Therefore, C is a field. Now, since V is an irreducible C-module, $\mathsf{V} = vC$ for all $0 \neq v \in \mathsf{V}$. For $0 \neq v \in \mathsf{V}$, the map $\epsilon : C \to \mathsf{V}$ defined by $c\epsilon = vc$ for all $c \in C$ is clearly a C-homomorphism from the C-module C onto V. Since $\ker \epsilon$ is an ideal of the field C, ϵ is an isomorphism. Therefore, $p^{nf} = |\mathsf{V}| = |C|$ and hence $C \cong \mathrm{GF}(p^{nf})$. It is well known that $C^\times$ is cyclic; for example, by Exercise I.37 (pg. 45). So A is also cyclic, say $A = \langle a_0 \rangle$. Then $C = \mathrm{GF}(p^f)[u(a_0)]$, and $|A| = |a_0|$ is a divisor of $|C^\times| = p^{nf} - 1$.

It remains to prove that n is the smallest natural number such that

$$p^{nf} - 1 \equiv 0 \mod |A|.$$

We identify C with $\mathrm{GF}(p^{nf})$. Let $k > 0$ be the smallest positive integer such that $p^{kf} - 1 \equiv 0 \bmod |A|$. Then $k \mid n$. Since the cyclic group $\mathrm{GF}(p^{nf})^\times$ has exactly one subgroup of order $p^{kf} - 1$, namely $\mathrm{GF}(p^{kf})^\times$, it follows that $A \leq \mathrm{GF}(p^{kf})^\times$. Hence, $v\mathrm{GF}(p^{kf})$ is an A-submodule of V. The irreducibility of V then forces $k = n$.

Let $A = \langle a_0 \rangle$. Since $|a_0| = |A|$ and

$$p^{kf} - 1 \not\equiv 0 \mod |A|$$

for $k < n$, $u(a_0)$ does not lie in any proper subfield $\mathrm{GF}(p^{kf})$ of $\mathrm{GF}(p^{nf})$. Thus, $\mathrm{GF}(p^f)[u(a_0)] = \mathrm{GF}(p^{nf})$. □

Lemma 3.11. *Let G be a subgroup of $\mathrm{GL}_n(p^f)$, and suppose A is an abelian normal subgroup of G. We set $\mathsf{K} = \mathrm{GF}(p^f)$. When considered as a $\mathsf{K}[A]$-module, assume*

the vector space $\mathsf{V} = \mathsf{V}(n, p^f)$ *is the direct sum of* s *isomorphic irreducible* $\mathsf{K}[A]$*-modules. Let* $k = \frac{n}{s}$*. Then the group* G*, when viewed as a permutation group on* V*, is permutation isomorphic to a group of semilinear transformations over* $\mathrm{GF}(p^f)$ *that satisfy*

$$(v_1' + v_2')g = v_1'g + v_2'g \quad \textit{for} \quad v_i' \in \mathsf{V}(s, p^{kf})$$

and

$$(v'c)g = (v'g)c^g \quad \textit{for} \quad v' \in \mathsf{V}(s, p^{kf}) \quad \textit{and} \quad c \in \mathrm{GF}(p^{kf}),$$

where the map $c \mapsto c^g$ *is a field automorphism of* $\mathrm{GF}(p^{kf})$ *over* $\mathrm{GF}(p^f)$*. The group of* $\mathrm{GF}(p^{kf})$*-linear maps of* $\mathsf{V}(s, p^{kf})$ *lying in* G *is exactly* $\mathbf{C}_G(A)$*.*

Proof. Suppose

$$\mathsf{V} = \mathsf{V}(n, p^f) = \mathsf{V}_1 \oplus \cdots \oplus V_s$$

for some isomorphic irreducible $\mathsf{K}[A]$-modules V_i, and let $n = sk$. By Theorem 3.10, there exist maps $\delta_i : \mathsf{V}_i \to \mathrm{GF}(p^{kf})^+$ such that for all $a \in A$ and $v_i \in \mathsf{V}_i$ we have

$$(v_i a)^{\delta_i} = u(a)\, v_i^{\delta_i}.$$

Since the V_i are isomorphic A-modules, $u(a)$ does not depend on i. Let δ be the map from $\mathsf{V} = \mathsf{V}_1 \oplus \cdots \oplus \mathsf{V}_s$ onto $\mathsf{V}' = \mathsf{V}(s, p^{kf})$ defined by

$$(v_1, \ldots, v_s)^{\delta} = (v_1^{\delta_1}, \ldots, v_s^{\delta_s}) \quad (v_i \in \mathsf{V}_i).$$

By Theorem 3.10, A is cyclic, say $A = \langle a_0 \rangle$, and so

$$v_i^{\delta_i} a_0 = u v_i^{\delta_i},$$

where $u = u(a_0) \in \mathrm{GF}(p^{kf})$. So a_0 acts on V' as the diagonal matrix $u1_{\mathsf{V}'}$.

Now we show that every $g \in G$ induces a $\mathrm{GF}(p^{kf})$-semilinear transformation of V' via the action $v^{\delta} g = (vg)^{\delta}$. It is clear that the transformation is additive.

Let $b \in \mathrm{GF}(p^{kf})$. Since $\mathrm{GF}(p^{kf}) = \mathrm{GF}(p^f)[u]$ by Theorem 3.10, we have that

$$b = \sum_{i=0}^{k-1} b_i u^i$$

for uniquely determined $b_i \in \mathrm{GF}(p^f)$. Let $g \in G$ and write $a_0^g = a_0^r$ for some integer r. For all $v' \in \mathsf{V}'$, it follows that

$$(v'b)g = \left(\sum_{i=0}^{k-1} v' b_i a_0^i \right) g = \sum_{i=0}^{k-1} v' b_i g a_0^{ri} = \sum_{i=0}^{k-1} (v'g) b_i u^{ri} = (v'g) b^g,$$

where

$$b^g = \sum_{i=0}^{k-1} b_i u^{ri}.$$

The map $b \mapsto b^g$ is obviously linear over $\mathrm{GF}(p^f)$. Since

$$(v'g)(bc)^g = (v'bc)g = (v'bg)c^g = (v'g)b^g c^g,$$

we have $(bc)^g = b^g c^g$. Therefore, $b \mapsto b^g$ is an endomorphism of the field $\mathrm{GF}(p^{kf})$. Since $1^g = 1$, this is not the zero homomorphism, so $b \mapsto b^g$ is an automorphism of $\mathrm{GF}(p^{kf})$ over $\mathrm{GF}(p^f)$.

It is obvious that $b^g = b$ for all $b \in \mathrm{GF}(p^{kf})$ is equivalent to $u^g = u^r = u$; that is, it is equivalent to $a_0^g = a_0^r = a_0$. Therefore, $\mathbf{C}_G(A)$ is the subgroup of all $\mathrm{GF}(p^{kf})$-linear maps in G. □

The following theorem is a useful special case.

Theorem 3.12. *Let G be a primitive permutation group on $\Omega = \{0, \dots\}$ with the following properties:*
(1) *G contains a minimal normal subgroup N that is abelian.*
(2) *G_0 contains an abelian normal subgroup A such that N is a minimal normal subgroup of NA.*
Then G is permutation isomorphic to a subgroup of the group $\Gamma(p^n)$ of affine semilinear maps

$$x \mapsto x^g = ax^\alpha + b$$

over the field $\mathrm{GF}(p^n)$.

Proof. By Theorem 3.5, we may assume that G consists of affine transformations of $\mathsf{V} = \mathsf{V}(n, p)$ and N is the group of translations on V. As N is a minimal normal subgroup of NA, V is an irreducible A-module. Then the claim follows from Lemma 3.11 by setting $s = 1$. □

We now show that, as a consequence of Theorem 3.12, solvable groups cannot be highly transitive.

Theorem 3.13. *Let G be a solvable permutation group.*
a) *If G is 2-primitive, then G is permutation isomorphic to the group of all linear or the group of all affine semilinear transformations over a field K, where $\mathsf{K} = \mathrm{GF}(2)$, $\mathsf{K} = \mathrm{GF}(3)$, or $\mathsf{K} = \mathrm{GF}(2^q)$ with $2^q - 1$ (and therefore q) a prime.*
b) *If G is 3-transitive, then $G = S_3$ or $G = S_4$.*

Proof. a) By Theorem 3.5, we may assume that the set on which G acts is the vector space $\mathsf{V} = \mathsf{V}(m, p)$ of dimension m over $\mathrm{GF}(p)$ and that G consists entirely of affine transformations. The case that $p^m = 2$ is trivial, so assume that $p^m > 2$. Since the action of G_0 on $\mathsf{V} - \{0\}$ is primitive, a minimal normal subgroup A of G_0 is transitive on $\mathsf{V} - \{0\}$ by Theorem 1.5. In particular, V is an irreducible A-module. By Theorem 3.12, G consists (up to permutation isomorphism) of affine

semilinear transformations over $\mathrm{GF}(p^m)$. Since G is 2-primitive, the group $\Gamma(p^m)$ of all affine semilinear transformations over $\mathrm{GF}(p^m)$ must be 2-primitive as well. By Example 1.18 d), this forces $p^m - 1$ to be a prime. If $p^m = 3$, then G is the group of all linear transformations over $\mathrm{GF}(3)$. Otherwise, $p = 2$ and m is a prime number, say q. If $G_{0,1} \neq 1$, then it follows that

$$|G| = 2^q(2^q - 1)q,$$

and G is the group $\Gamma(2^q)$ of all affine semilinear transformations over $\mathrm{GF}(2^q)$. Now suppose that $G_{0,1} = 1$. Then $A = G_0 = \mathbf{C}_G(A)$ consists of linear maps by Lemma 3.11. So G is the group of all linear transformations over $\mathrm{GF}(2^q)$.

b) follows directly from a) and the statement in Example 1.18 d) regarding the transitivity of the affine semilinear group $\Gamma(p^m)$. □

The 2-transitive solvable groups are, up to finitely many exceptions of degree 3^2, 5^2, 7^2, 11^2, 23^2 and 3^4, always groups of affine semilinear transformations over a finite field. We will prove this in Volume 3.

Exercises

1) (Ritt [356]). Let G be a solvable primitive permutation group of degree p^m. Show that G contains no p^m-cycle except when $m = 1$ (for any p) or $p^m = 4$.
(Hint: Use Theorem 3.5 and the Jordan normal form of elements of p-power order in $\mathrm{GL}_{m+1}(p)$.)

4 Primitive groups with transitive subgroups of smaller degree

Lemma 4.1. *Assume G is transitive on Ω. Let $U \leq G$, and let Δ be an orbit of U. If U^Δ is primitive on Δ and $|\Omega| < 2|\Delta|$, then G is primitive on Ω.* (*For the meaning of U^Δ see Definition I.5.9, pg. 27.*)

Proof. Assume for contradiction that Ψ is a block for G. By conjugating Ψ if necessary, we may assume that $1 \leq |\Psi \cap \Delta|$ and $|\Psi^g \cap \Delta| \leq |\Psi \cap \Delta|$ for all $g \in G$. We define $\Lambda = \Psi \cap \Delta$.

If $\Lambda \cap \Lambda^u \neq \emptyset$ for some $u \in U$, then certainly $\Psi \cap \Psi^u \neq \emptyset$. Since Ψ is a block for G, it follows that $\Psi^u = \Psi$ and

$$\Lambda^u = (\Psi \cap \Delta)^u = \Psi^u \cap \Delta^u = \Psi \cap \Delta = \Lambda.$$

Since U^Δ is primitive on Δ, we must have $\Lambda = \Delta$ or $|\Lambda| = 1$.

If $\Lambda = \Delta$, then $\Delta \subseteq \Psi$. Since

$$\frac{|\Omega|}{2} < |\Delta| \leq |\Psi|$$

and $|\Psi|$ divides $|\Omega|$, we obtain the contradiction $\Psi = \Omega$.

If $|\Lambda| = 1$, then

$$|\Psi^g \cap \Delta| \leq |\Psi \cap \Delta| \leq 1$$

for all $g \in G$. Since $\Omega = \bigcup_{g \in G} \Psi^g$, there must be at least $|\Delta|$ distinct Ψ^g. But then

$$|\Omega| < 2|\Delta| \leq |\Psi||\Delta| \leq |\Omega|,$$

which is impossible. □

Lemma 4.2. *Assume G is transitive on Ω and that $G = \langle C, D \rangle$ is generated by two subgroups $C, D \leq G$. Suppose Γ and Δ are subsets of Ω with the following properties:*

$\Gamma^C = \Gamma$; the group C^Γ is primitive on Γ, and $C^{\Omega - \Gamma} = 1$;

$\Delta^D = \Delta$; the group D^Δ is primitive on Δ, and $D^{\Omega - \Delta} = 1$.

Then G is primitive on Ω.

Proof. By the transitivity of $G = \langle C, D \rangle$ on Ω, we have $\Omega = \Gamma \cup \Delta$ and $\Gamma \cap \Delta \neq \emptyset$. Therefore, $|\Omega| < 2|\Gamma|$ or $|\Omega| < 2|\Delta|$, and the claim follows from Lemma 4.1. □

Theorem 4.3 (C. Jordan). *Let $\Omega = \Gamma \cup \Delta$ and $\Gamma \cap \Delta = \emptyset$ with $1 < |\Gamma| < |\Omega|$. Assume that G is a permutation group on Ω and G_Δ is transitive on Γ. Then:*

a) *If G is primitive on Ω, then G is 2-transitive.*

b) *If G is primitive on Ω and G_Δ is primitive on Γ, then G is 2-primitive.*

Proof. In either case, the assumptions ensure that G is transitive on Ω. We prove the claims by induction on $|\Delta|$.

First assume that $|\Delta| = 1$, say $\Delta = \{d\}$. In case a), the subgroup G_d is transitive on $\Gamma = \Omega - \{d\}$, so G is 2-transitive by Lemma 1.8. In case b), the group G_d is primitive on Γ, so G is 2-primitive on Ω by Definition 1.14 a).

Now assume that $|\Delta| > 1$. There are two cases:

Case 1: Assume $2|\Delta| < |\Omega|$, so $2|\Gamma| > |\Omega|$. Then $\Gamma \cap \Gamma^g \neq \emptyset$ for all $g \in G$. Since $|\Delta| > 1$, there exist two distinct points a and b in Δ. Rudio's Lemma 1.6 ensures there exists some $g \in G$ such that $a \in \Delta^g$ and $b \notin \Delta^g$. By our assumption and $\Gamma \cap \Gamma^g \neq \emptyset$, we have that $H = \langle G_\Delta, G_\Delta^g \rangle$ is transitive on $\Gamma' = \Gamma \cup \Gamma^g$. Therefore, $\Delta' = \Delta \cap \Delta^g = \Omega - \Gamma'$ is exactly the set of points fixed by H. It follows from $\Gamma \subseteq \Gamma'$ that $1 < |\Gamma'|$. Furthermore, $a \in \Delta \cap \Delta^g = \Delta'$, and so $1 \leq |\Delta'|$. Since $b \in \Delta$ and $b \notin \Delta^g$, we have $|\Delta'| < |\Delta|$.

In case a), since $H \leq G_{\Delta'}$, the group $G_{\Delta'}$ is certainly transitive on Γ'. So by the induction hypothesis, G is 2-transitive.

If G_Δ is primitive on Γ, then $H = \langle G_\Delta, G_\Delta^g \rangle$ is primitive on Γ' by Lemma 4.2. So G is 2-primitive by the induction hypothesis.

Case 2: Assume $|\Omega| \leq 2|\Delta|$. Let $a, b \in \Gamma$ be distinct. By Lemma 1.6, there exists a $g \in G$ such that $a \in \Gamma^g$ and $b \notin \Gamma^g$. Now $a \in \Gamma \cap \Gamma^g \neq \emptyset$. Therefore, $H = \langle G_\Delta, G_\Delta^g \rangle$ is transitive on $\Gamma' = \Gamma \cup \Gamma^g$. It follows from $2|\Gamma| \leq |\Omega|$ and $\Gamma \cap \Gamma^g \neq \emptyset$ that

$$|\Gamma'| = |\Gamma \cup \Gamma^g| < 2|\Gamma|,$$

so that $1 < |\Gamma| \leq |\Gamma'| < |\Omega|$.

Since $b \in \Gamma$ and $b \notin \Gamma^g$, we have $\Gamma \subset \Gamma'$. With $\Delta' = \Omega - \Gamma'$, we then have $1 \leq |\Delta'| < |\Delta|$. As in case 1, the claim follows by induction. □

Theorem 4.3 can be sharpened to the following:

Theorem 4.4 (C. Jordan). *Assume G is primitive on Ω with $|\Omega| = n$. Let $\Gamma \subset \Omega$ with $1 < |\Gamma| = m < n$, and let $\Delta = \Omega - \Gamma$. Assume G_Δ is primitive on Γ. Then G is $(n - m + 1)$-primitive on Ω.*

Proof. We prove the claim by induction on $|\Delta| = n - m$. Observe that G is certainly 2-primitive by Theorem 4.3 b). So we have proved the base case $n - m = 1$. Now assume that $n - m > 1$, and let $d \in \Delta$. By the 2-primitivity of G on Ω, the subgroup G_d is primitive on $\Omega' = \Omega - \{d\}$. Furthermore, $G_\Delta \leq G_d$ and G_Δ is primitive on $\Gamma = \Omega' - \Delta'$, where $\Delta' = \Delta - \{d\}$. Since $|\Delta'| = (n-1) - m$, the induction hypothesis implies that G_d is $(n - m)$-primitive. Therefore, G is $(n - m + 1)$-primitive. □

Theorem 4.5. *Let G be primitive of degree n.*

a) *If G contains a cycle of length p a prime, then G is at least $(n - p + 1)$-primitive, so in particular $(n - p + 1)$-transitive.*

b) *If G contains a 2-cycle, then $G = S_n$.*

c) *If G contains a 3-cycle, then $G \geq A_n$.*

Proof. a) Suppose g is a p-cycle, and let $\Gamma = \{a \in \Omega \mid a^g \neq a\}$. Then $U = \langle g \rangle$ is primitive on Γ, and $U \leq G_\Delta$ if we define $\Delta = \Omega - \Gamma$. So G_Δ is primitive on Γ, and the claim follows by Theorem 4.4.

b) By a) G is $(n - 1)$-transitive, so $G = S_n$.

c) By a) G is $(n - 2)$-transitive. By Theorem 1.11, we have

$$|G| \geq n(n-1)\cdots 3,$$

so $[S_n : G] \leq 2$. We may assume that $G < S_n$. By Theorem I.3.3 (pg. 13), we conclude that $G \lhd S_n$.

Assume for contradiction that $G \neq A_n$. If $n \geq 5$, then A_n is simple by Theorem 2.4, and so $G \cap A_n = 1$. But then the product formula of Lemma I.2.12 a) (pg. 7) yields the contradiction

$$|GA_n| = |G||A_n| \leq |S_n|.$$

For $n = 3$, every 3-cycle generates the alternating group A_3. It remains to consider $n = 4$. If we define $N = G \cap A_4$, then $N \lhd S_4$ and

$$|N| = \frac{|G||A_4|}{|GA_4|} = 6.$$

By Theorem I.8.10 (pg. 41), N has exactly one Sylow 3-subgroup, which is normal in S_4 since it is characteristic in $N \trianglelefteq S_4$. This is a contradiction since S_4 has exactly four Sylow 3-subgroups. □

Theorem 4.6 (Bochert). *Let G be primitive on Ω with $|\Omega| = n$, and assume $A_n \not\leq G$. Then*

$$[S_n : G] \geq \lfloor \frac{n+1}{2} \rfloor ! .$$

Proof. a) Let $g, h \in G$, and assume for contradiction that there exists exactly one point $i \in \Omega$ such that $i^g \neq i \neq i^h$. Define $j, k \in \Omega$ by $i = j^g = k^h$.

Let $s \notin \{i, j, k\}$, so that $s \neq i$, $s^g \neq i$ and $s^h \neq i$. We show that $s^{gh} = s^{hg}$.

Since $s \neq i$, our assumption implies that $s^g = s$ or $s^h = s$. By the symmetry in g and h, we may assume that $s^g = s$. If $s^h = s$ as well, then $s^{gh} = s = s^{hg}$. If $s^h \neq s$, then since $s^h \neq i$ we certainly have $s^{hg} = s^h = s^{gh}$.

Since $i^g \neq i$, we have $i^{g^2} \neq i^g$. Then the uniqueness of i implies $i^{gh} = i^g$. In the same way, we may conclude that $i^{hg} = i^h$. If $i^h = i^g$, then $i^{h^2} \neq i^h$ and $i^{g^2} \neq i^g$ together with our assumption implies that $i^h = i^g = i$, which is nonsense. Hence, $i^{gh} \neq i^{hg}$ and therefore $ghg^{-1}h^{-1} \neq 1$. Thus, $ghg^{-1}h^{-1}$ is a cycle of length 2 or 3. But then our assumption that $A_n \not\leq G$ contradicts Theorem 4.5.

b) Let S be a subgroup of S_n with the following properties:

(1) $S \cap G = 1$.
(2) There exists a subset $\Delta \subseteq \Omega$ that is fixed setwise by S, and S^Δ is the full symmetric group on Δ.
(3) $S^{\Omega - \Delta} = 1$.
(4) S is chosen so that $k = |\Delta|$ is as large as possible.

Assume for contradiction that $2k < n$. Let $T \leq S_n$ be chosen such that $T^{\Omega-\Delta}$ is the full symmetric group on $\Omega - \Delta$ and $T^\Delta = 1$. Since

$$|\Omega - \Delta| > \frac{n}{2} > k$$

and k is chosen to be maximal, there exists some $g \in G$ such that $1 \neq g \in G \cap T$. Since $g \notin S$, the element g moves some $i \in \Omega - \Delta$ but Δ is fixed pointwise by g. Let $\Delta' = \Delta \cup \{i\}$. Choose the subgroup $U \leq S_n$ such that $U^{\Delta'}$ is the full symmetric group on Δ' and $U^{\Omega - \Delta'} = 1$. Since $|\Delta| < |\Delta'|$, there exists some h such that $1 \neq h \in G \cap U$. If $i^h = i$ then we obtain the contradiction $1 \neq h \in G \cap S = 1$. So $i^h \neq i$ but $x^h = x$ for all $x \in \Omega - \Delta' = \Omega - \Delta - \{i\}$. Thus, i is the only point such that $i^g \neq i \neq i^h$. But this is impossible by part a). Hence, we must have $n \leq 2k$.

Now we have $|S_n| \geq |GS| = |G||S| = |G|k!$, and so

$$[S_n : G] \geq k! .$$

It follows from $k \geq \frac{n}{2}$ that $k \geq \lfloor \frac{n+1}{2} \rfloor$, and the claim is proved. □

As an application, we classify all primitive permutation groups of degree 6:

Theorem 4.7. *There exist exactly four primitive groups of degree* 6 *up to permutation isomorphism; namely* S_6, A_6 *and two* 2*-transitive groups of degree* 6 *isomorphic to* S_5 *and* A_5.

Proof. a) Let G be primitive of degree 6 with $A_6 \not\leq G$. By Theorem 4.6, we conclude that $|G| \leq 120$, and G is not solvable by Theorem 3.2 a). By Exercises 34 and 35 and Theorem I.8.14, the only nonabelian simple group of order at most 120 is the alternating group A_5. Therefore, G has a composition factor isomorphic to A_5, and so $G \cong A_5$ or $|G| = 120$.

Assume $|G| = 120$, and let N be a minimal normal subgroup of G. By Theorem 1.5, the subgroup N is transitive. It follows that $6 \mid |N|$, so $|G/N| \leq 20$ and G/N is solvable. So N is not solvable, and we have $N \cong A_5$.

Let K be a Sylow 2-subgroup of N. Then $|\mathbf{N}_N(K)| = 12$. If P is a Sylow 2-subgroup of G with $K < P$, then $K = P \cap N \lhd P$. It follows that $|\mathbf{N}_G(K)| = 24j$ and $[G : \mathbf{N}_G(K)] = \frac{5}{j}$ for some j. Since $6 \nmid |K|$, K is not transitive. So $K \not\trianglelefteq G$ and $[G : \mathbf{N}_G(K)] = 5$. Therefore, G has a transitive permutation representation of degree 5 on the cosets of $\mathbf{N}_G(K)$ by Theorem I.6.2 (pg. 29). The kernel T of this representation satisfies $|T| \leq 24$. However, we have seen that every minimal normal subgroup of G has order 60. So $T = 1$ and $G \cong S_5$.

b) It remains to show that both S_5 and A_5 admit exactly one transitive permutation representation of degree 6 up to permutation isomorphism. The normalizers of the Sylow 5-subgroups of S_5 and A_5 have index 6 and yield transitive and faithful representations of degree 6 by Theorem I.6.2 (pg. 29). To show that these representations are 2-transitive, and so certainly primitive by Theorem 1.9, it suffices to observe that the elements of order 5 must be represented by 5-cycles.

Now let U be any subgroup of S_5 or A_5 of index 6. Then U has order 20 or 10. By Sylow's theorems, U has exactly one subgroup of order 5. Thus, U is the full normalizer of a Sylow 5-subgroup of S_5 or A_5, respectively. Since every transitive[7] permutation representation of degree 6 is permutation isomorphic to a representation on the cosets of a subgroup of index 6 by Remark I.6.4 (pg. 30) and conjugate subgroups yield permutation isomorphic representations (see Exercise I.23, pg. 32), the groups S_5 and A_5 admit exactly one transitive permutation representation of degree 6 up to permutation isomorphism. □

5 The symmetric and alternating groups

Theorem 5.1. *For* $n \neq 2, 4$, *the only composition series* (*chief series, characteristic series*) *of* S_n *is* $1 \lhd A_n \lhd S_n$.

Proof. It suffices to show that A_n is the unique minimal normal subgroup of S_n.

[7] Correction 12 from errata.

Assume for contradiction that there exists a minimal normal subgroup N of S_n that is distinct from A_n. Since A_n is simple for $n \neq 2, 4$ by Theorem 2.4, $N \cap A_n = 1$. This implies that

$$|NA_n| = |N||A_n| \text{ divides } |S_n|,$$

and so $|N| = 2$. But this is impossible since N is necessarily transitive by Theorem 1.5. □

Remark 5.2. Since the symmetric group S_4 is a primitive solvable permutation group, it contains a unique minimal normal subgroup K by Theorem 3.2, and K is regular and abelian. It is given by

$$K = \{1, (1,2)(3,4), (1,3)(2,4), (1,4)(2,3)\}.$$

The group K is called the Klein four group. Defining

$$H = \{h \in S_4 \mid 1^h = 1\},$$

we have $S_4 = KH$ by Theorem 3.2 b). Therefore,

$$S_4/K \cong H \cong S_3.$$

Since S_3 has only one nontrivial normal subgroup, namely A_3, the subgroups K and A_4 are the only nontrivial normal subgroups of S_4. Therefore,

$$1 \lhd K \lhd A_4 \lhd S_4$$

is the unique chief series (characteristic series) of S_4.

Theorem 5.3.

a) *For $n \neq 4$, the symmetric group S_n does not contain a subgroup of index k with $2 < k < n$. (In contrast, the Sylow 2-subgroups of S_4 have index 3.)*

b) *The number of conjugacy classes of subgroups of S_n with index n is equal to the order of the outer automorphism group $\mathbf{Aut}(S_n)/\mathbf{Inn}(S_n)$ of S_n.*

Proof. a) If $[S_n : U] = k$, then the permutation representation of S_n as a permutation group on the cosets of U yields a homomorphism from S_n to S_k. Since $k < n$, this homomorphism must have a nontrivial kernel N with $N \leq U$. However, since $2 < k$ it follows that $|N| < |A_n|$, contradicting Theorem 5.1.

b) For $1 \leq j \leq n$, we define

$$V_j = \{g \in S_n \mid j^g = j\}.$$

(1) For every subgroup $U \leq S_n$ of index n, there exists an automorphism α of S_n such that $U^\alpha = V_1$:

Let $S_n = \bigcup_i Uh_i$ be the decomposition of S_n into cosets of U with $h_1 = 1$. From Theorem 5.1 (and Remark 5.2 for $n = 4$), we have $\bigcap_{g \in S_n} U^g = 1$. Therefore, the map

α defined by $g^\alpha = \mathbf{P}(g) = \binom{i}{i'}$ for $Uh_i g = Uh_{i'}$ is an automorphism of S_n. Clearly, $1^{\mathbf{P}(g)} = 1$ is equivalent to $g \in U$. Therefore, $U^\alpha = V_1$.

(2) Let

$$B = \{\alpha \in \mathbf{Aut}(S_n) \mid V_j^\alpha \in \{V_1, \ldots, V_n\} \text{ for all } j = 1, \ldots, n\}.$$

By (1), the index $[\mathbf{Aut}(S_n) : B]$ is the number of conjugacy classes of subgroups of S_n with index n. We show that $B = \mathbf{Inn}(S_n)$:

Suppose $\alpha \in B$ and $V_i^\alpha = V_{i'}$ $(i = 1, \ldots, n)$. Let τ be the inner automorphism of S_n induced by the element $x = \binom{i}{i'}$. Then $V_i^\alpha = V_i^\tau$, and so $\mu = \alpha\tau^{-1}$ fixes every V_i. Hence, μ also fixes

$$\bigcap_{j \neq i \neq k} V_i = \langle (j, k) \rangle.$$

Since S_n is generated by its transpositions by Example I.19.7 (pg. 146), it follows that $\mu = 1$. So $\alpha = \tau$ is the inner automorphism of S_n induced by the element x. □

Lemma 5.4.

a) *For* $1 < t < n - 1$, *the inequality* $(n - t)!t! < (n - 1)!$ *always holds.*

b) *Let* $n = rs$ *with* $r \neq 1 \neq s$. *For* $4 < n$, *we have the inequality*

$$(s!)^r r! < (n - 1)!\,.$$

c) *For* $4 < n$ *and* $n \neq 6$, *we have*

$$n < \left\lfloor \frac{n+1}{2} \right\rfloor !\,.$$

Proof. a) We define $N(t) = (n - t)!t!$. Then

$$\frac{N(t)}{N(t+1)} = \frac{n-t}{t+1} \geq 1 \quad \text{for} \quad n \geq 2t + 1.$$

Furthermore, $N(n - t) = N(t)$ and $N(n - 1) = (n - 1)!$. This yields the claim.

b) If $r = 2$, then $s = \frac{n}{2} > 2$ and

$$\frac{n!}{2(s!)^2} = \frac{(s+1)\cdots 2s}{2s!} = (s+1)\frac{s+2}{2}\cdots\frac{2s}{2s} > 2(s+1) > 2s = n,$$

as desired. For $r \geq 2$, we have[8]

$$\begin{aligned} n! &= (1 \cdot 2 \cdots s)((s+1)\cdots 2s)\cdots\big(\big((r-1)s+1\big)\cdots rs\big) \\ &> s!\,(s\,s!\,2)\cdots\big((r-1)s\,s!\,r\big) = (s!)^r r! s^{r-1}(r-1)!\,. \end{aligned}$$

It therefore suffices to show for $r > 2$ that

[8] Correction 13 from errata.

$$s^{r-1}(r-1)! > n = rs.$$

For $r \geq 4$, we have $(r-1)! > r$ and so $s^{r-1}(r-1)! > sr = n$. For $r = 3$, since $s > 1$ we certainly have $2s^{r-1} = 2s^2 > 3s = n$.

c) For $6 < n$, we have $\lfloor \frac{n+1}{2} \rfloor ! > \frac{n}{2} \cdot 3 \cdot 2 = 3n$. For $n = 5$, it is easy to check that the inequality holds as well. □

Theorem 5.5.

a) *Let $n \neq 6$. Then the symmetric group S_n has exactly one conjugacy class of subgroups of index n, namely the stabilizers of a single point. Every automorphism of S_n is an inner automorphism.*

b) *In S_6, there exist exactly two conjugacy classes of subgroups of index 6, namely the stabilizers of points and a conjugacy class of 2-transitive subgroups isomorphic to S_5. We have $|\mathbf{Aut}(S_6)/\mathbf{Inn}(S_6)| = 2$.*

Proof. Assume $[S_n : U] = n$.

First assume that U is not transitive, and suppose U has an orbit of size t with $1 < t < n-1$. Lemma 5.4 a) yields the contradiction

$$(n-1)! = |U| \leq t!(n-t)! < (n-1)!.$$

For $n \leq 5$, U is necessarily not transitive since $n \nmid |U|$. Thus, we may assume in what follows that $n \geq 6$.

Now assume that U is transitive but imprimitive with r blocks of size s. Then $n = rs$. By Theorem 1.2, we have that $|U|$ divides $(s!)^r r!$. Since $n \geq 6$, Lemma 5.4 b) yields the contradiction

$$(n-1)! = |U| \leq (s!)^r r! < (n-1)!\,.$$

Finally, assume U is primitive. Since $A_n \not\leq U$, Theorem 4.6 and Lemma 5.4 c) together yield the contradiction

$$n = [S_n : U] \geq \lfloor \frac{n+1}{2} \rfloor ! > n$$

for $n > 6$. Hence, if $n \neq 6$ then U has an orbit of size 1, and so U is the stabilizer of a point. By Theorem 5.3 b), it therefore follows that $\mathbf{Aut}(S_n) = \mathbf{Inn}(S_n)$ for $n \neq 6$.

Finally, assume $n = 6$, and suppose U is not the stabilizer of a point. We have already seen that U must be primitive. By Theorem 4.7, there is exactly one conjugacy class of primitive subgroups of order 120 in S_6. By Theorem 5.3 b), we conclude that $|\mathbf{Aut}(S_6)/\mathbf{Inn}(S_6)| = 2$. □

Exercises

2) Give an outer automorphism of S_6 by explicitly writing down its action on the 2-cycles of S_6 (which generate all of S_6).

3) Let G be a transitive permutation group on Ω that can be written in the form $G = G_1 \times \cdots \times G_r$, where $(|G_i|, |G_j|) = 1$ for $i \neq j$. Let $|\Omega| = n_1 \cdots n_r$, where n_i divides $|G_i|$. Then there exist transitive, faithful representations of G_i on sets Ω_i with $|\Omega_i| = n_i$ such that G is permutation isomorphic to the permutation representation of G on $\Omega_1 \times \cdots \times \Omega_r$, where

$$(a_1, \ldots, a_r)^{(g_1,\ldots,g_r)} = (a_1^{g_1}, \ldots, a_r^{g_r}).$$

4) Let

$$\mathcal{N} = \{H \leq S_n \mid H \text{ nilpotent, transitive}\}.$$

(For the definition of a nilpotent group, see Theorem I.9.5, pg. 50.) Show that the maximal elements of $\mathcal{N}$ are conjugate.

5) (P. Hall [219]). Let S_n be the symmetric group of degree n. Let p and q be primes with $p < q \leq n$. Suppose there exists a subgroup $H \leq S_n$ such that $|H| = p^a q^b$, $p \nmid [S_n : H]$ and $q \nmid [S_n : H]$. Then $p = 2$, $q = 3$ and $n \in \{3, 4, 5, 7, 8\}$.

(Hint: The solvability of H, which will be proved in Chapter V, must be assumed.

a) If H is primitive, then $n = r^m$ is a prime power ($r = p$ or $r = q$). By Theorem 3.5, we have that $|H|$ divides $r^m |\mathrm{GL}_m(r)|$. Show that $n = 3$ or $n = 4$ by comparing with the highest power of r that divides $r^m!$.

b) Assume H is transitive but imprimitive with k blocks of size m. Then H is isomorphic to a subgroup of the wreath product $S_m \wr S_k$. Show that this only occurs for $m = 4$, $k = 2$ and $n = 8$.

c) If H is not transitive, then $n = 5$ or $n = 7$; the orbits of H have sizes 1 and 4 or 3 and 4, respectively.)

6 Linear and projective groups

In addition to the alternating groups A_n ($n \geq 5$), other important families of finite simple groups are derived from the automorphism groups of vector spaces, with or without a scalar product. In this section, we address only the simplest case: that where the vector space carries no scalar product. Most of our statements will not require the underlying field to be finite.

Definition 6.1. Let $\mathsf{V} = \mathsf{V}(n, \mathsf{K})$ be a vector space of dimension n over the field K. We denote by $\mathrm{GL}_n(\mathsf{K})$ the group of all K-linear maps from V onto itself. By the product rule for determinants, the mapping of $g \in \mathrm{GL}_n(\mathsf{K})$ to its determinant $\det g$ is an epimorphism from $\mathrm{GL}_n(\mathsf{K})$ onto the multiplicative group $\mathsf{K}^\times$ of K. We denote the kernel of this epimorphism by $\mathrm{SL}_n(\mathsf{K})$. Hence, $\mathrm{SL}_n(\mathsf{K})$ consists of the linear maps with determinant 1. The group Z of all multiples $a1$ of the identity map 1 lies in the center of $\mathrm{GL}_n(\mathsf{K})$. We denote the factor group $\mathrm{GL}_n(\mathsf{K})/Z$ by $\mathrm{PGL}_n(\mathsf{K})$. This factor group can be understood as the group of projective maps of the projective space

$\mathsf{P}(n-1,\mathsf{K})$ of dimension $n-1$ over K since it is well known that the elements of Z are the only linear maps that take every one-dimensional subspace of V to itself. Furthermore, we define

$$\mathrm{SL}_n(\mathsf{K})/(\mathrm{SL}_n(\mathsf{K}) \cap Z) = \mathrm{PSL}_n(\mathsf{K}),$$

where

$$\mathrm{SL}_n(\mathsf{K}) \cap Z = \{a1_V \mid a^n = 1\}.$$

If $\mathsf{K} = \mathrm{GF}(p^f)$ is a finite field, and we are indeed most interested in this case, we also write

$$\mathrm{GL}_n(\mathsf{K}) = \mathrm{GL}_n(p^f),\ \mathrm{SL}_n(\mathsf{K}) = \mathrm{SL}_n(p^f),\ \mathrm{PGL}_n(\mathsf{K}) = \mathrm{PGL}_n(p^f)$$

and $\mathrm{PSL}_n(\mathsf{K}) = \mathrm{PSL}_n(p^f)$.

Lemma 6.2. *We have:*
(1) $|\mathrm{GL}_n(p^f)| = (p^{nf}-1)(p^{nf}-p^f)\cdots(p^{nf}-p^{(n-1)f})$.
(2) $|\mathrm{SL}_n(p^f)| = |\mathrm{PGL}_n(p^f)| = (p^{nf}-1)(p^{nf}-p^f)\cdots(p^{nf}-p^{(n-2)f})p^{(n-1)f}$.
(3) $|\mathrm{PSL}_n(p^f)| = |\mathrm{SL}_n(p^f)|k^{-1}$, *where* $k = (n, p^f-1)$.

Proof. Let $\{v_1,\dots,v_n\}$ be a K-basis for $\mathsf{V} = \mathsf{V}(n,p^f)$. Then every linear map g of V onto itself is uniquely determined by the linearly independent vectors $v_1g,\dots,v_ng$. Therefore, we have the conditions

$$v_1g \neq 0, \quad v_2g \notin \langle v_1g\rangle, \quad \dots, \quad v_ng \notin \langle v_1g,\dots,v_{n-1}g\rangle.$$

This gives

$$|\mathrm{GL}_n(p^f)| = (p^{nf}-1)(p^{nf}-p^f)\cdots(p^{nf}-p^{(n-1)f}).$$

Claim (2) follows immediately from

$$|\mathrm{SL}_n(p^f)| = |\mathrm{PGL}_n(p^f)| = \frac{|\mathrm{GL}_n(p^f)|}{|\mathsf{K}^\times|} = \frac{|\mathrm{GL}_n(p^f)|}{p^f-1}.$$

The group $\mathrm{SL}_n(p^f) \cap Z$ consists of the elements $a1$ such that

$$\det a1 = a^n = 1.$$

Since the group $\mathsf{K}^\times$ is cyclic, the number of elements satisfying this equation is (n, p^f-1). This proves claim (3) regarding the order of $\mathrm{PSL}_n(p^f)$. □

In what follows, set $\mathsf{V} = \mathsf{V}(n,\mathsf{K})$ with $n > 1$.

Definition 6.3. Let H be a hyperplane in $\mathsf{V} = \mathsf{V}(n,\mathsf{K})$. A map $t \neq 1$ in $\mathrm{GL}_n(\mathsf{K})$ is called a transvection over the hyperplane H whenever

$$vt = v \quad \text{for all } v \in \mathsf{H}$$

and

$$vt - v \in \mathsf{H} \quad \text{for all } v \in \mathsf{V}.$$

Note that H and V/H are fixed elementwise by t.

Lemma 6.4. *Let t be a transvection over the hyperplane H, and let be μ a linear functional on V such that*

$$\mathsf{H} = \{v \in \mathsf{V} \mid \mu(v) = 0\}.$$

Then there exists a vector $0 \neq a \in \mathsf{H}$ (so $\mu(a) = 0$) such that

$$vt = v - \mu(v)a$$

for all $v \in \mathsf{V}$, and we write $t = t(a, \mu)$.

Conversely, if $\mu \neq 0$ is a linear functional on V and $0 \neq a \in \mathsf{V}$ such that $\mu(a) = 0$, then the map t defined by $vt = v - \mu(v)a$ is a transvection over the hyperplane $\mathsf{H} = \{v \in \mathsf{V} \mid \mu(v) = 0\}$.

Proof. Let $\mathsf{V} = \mathsf{H} \oplus \langle w \rangle$ with $\mu(w) = 1$. We set $a = w - wt$. Since $t \neq 1$, we have $a \neq 0$. For $v = h + cw$ with $h \in \mathsf{H}$ and $c \in \mathsf{K}$, we have $\mu(v) = c\mu(w) = c$ and

$$vt = (h + cw)t = h + c(wt) = h + c(w - a) = v - ca = v - \mu(v)a.$$

The converse is trivial. □

A simple calculation gives the following lemma.

Lemma 6.5. *Let a_1, a_2 and a be nonzero vectors in the hyperplane*

$$\mathsf{H} = \{v \in \mathsf{V} \mid \mu(v) = 0\}.$$

Then we have

$$t(a_1, \mu)t(a_2, \mu) = t(a_1 + a_2, \mu) \quad \textit{and} \quad t(a, \mu)t(-a, \mu) = 1.$$

That is, the transvections over a fixed hyperplane H together with the identity map constitute an abelian subgroup $T(\mathsf{H})$ that is isomorphic to the additive group H^+ of H. Since

$$t(a, \mu_1)t(a, \mu_2) = t(a, \mu_1 + \mu_2)$$

for $\mu_i(a) = 0$, the set

$$T_a = \{1,\, t(a, \mu) \mid 0 \neq \mu \in \mathsf{V}^* = \mathrm{Hom}_{\mathsf{K}}(\mathsf{V}, \mathsf{K}),\, \mu(a) = 0\}$$

is also an abelian subgroup of $\mathrm{GL}_n(\mathsf{K})$.

Lemma 6.6. *All transvections have determinant* 1*, so transvections lie in* $\mathrm{SL}_n(\mathsf{K})$.

Proof. Let t be a transvection over the hyperplane H. Then t acts as the identity map on H and V/H, so t has determinant 1. □

Theorem 6.7. *The transvections generate* $\mathrm{SL}_n(\mathsf{K})$.

Proof. We prove the claim by induction on n.

Let T be the subgroup of $\mathrm{SL}_n(\mathsf{K})$ generated by all the transvections of $\mathrm{GL}_n(\mathsf{K})$. Since the conjugate of a transvection is clearly again a transvection, T is a normal subgroup of $\mathrm{SL}_n(\mathsf{K})$.

Let $g \in \mathrm{SL}_n(\mathsf{K})$. Let $0 \neq v \in \mathsf{V}$ and $vg = v'$. If v and v' are linearly independent, then there exists a linear functional μ on V such that $\mu(v') = 1$ and $\mu(v' - v) = 0$. Defining $t = t(v' - v, \mu)$, we have $vgt = v$ and $gtT = gT$.

If $vg = cv$ for some $c \in \mathsf{K}$ and $w \in \mathsf{V}$ is any vector linearly independent of v, then with $t' = t(w, \mu')$ and $\mu'(v) = 1$, $\mu'(w) = 0$ we find

$$vt'g = (v - w)g \notin \langle v \rangle.$$

Then we can apply the previous construction to $t'g$. Note that $t'gT = gT$.

Hence, in each case there exists some $x \in \mathrm{SL}_n(\mathsf{K})$ such that $xT = gT$ and $v_1 x = v_1$ for some $v_1 \neq 0$. On $\overline{\mathsf{V}} = \mathsf{V}/\langle v_1 \rangle$, the element x acts as a linear map $\overline{x}$ with $\det \overline{x} = 1$. If $\dim \overline{\mathsf{V}} = n - 1 = 1$, then $\overline{x} = \overline{1}$ and x is a transvection over the hyperplane $\langle v_1 \rangle$. So assume that $n > 2$.

By our induction hypothesis, there exist transvections $\overline{t}(\overline{w}_i, \overline{\mu}_i)$ on $\overline{\mathsf{V}}$ such that $\overline{x} = \prod_i \overline{t}(\overline{w}_i, \overline{\mu}_i)$. Let $w_j \in \mathsf{V}$ be any preimages of the $\overline{w}_j$, and define linear functionals μ_j on V by $\mu_j(v_1) = 0$ and $\mu_j(v) = \overline{\mu}_j(\overline{v})$ for $\overline{v} = v + \langle v_1 \rangle$. We define

$$k = x \prod_j t(w_j, \mu_j)^{-1}.$$

Then $x^{-1}k \in T$, $v_1 k = v_1$ and $vk - v \in \langle v_1 \rangle$ for all $v \in \mathsf{V}$. Let $\{v_1, \ldots, v_n\}$ be a basis of V, and let $v_j k = v_j - a_j v_1$ ($a_j \in \mathsf{K}$). Then

$$k = \prod_{j=2}^{n} t(a_j v_1, \tau_j),$$

where the linear functionals τ_j are defined by $\tau_j(v_i) = \delta_{ij}$. Therefore, $k \in T$ and so $g \in T$ as well. □

Lemma 6.8. *Let* H *and* H' *be hyperplanes in* V*, and let* $0 \neq a \in \mathsf{H}$, $b \notin \mathsf{H}$, $0 \neq a' \in \mathsf{H}'$, $b' \notin \mathsf{H}'$. *Then there exists some* $g \in \mathrm{GL}_n(\mathsf{K})$ *such that* $\mathsf{H}g = \mathsf{H}'$, $ag = a'$ *and* $bg = b'$. *For* $n \geq 3$*, we may find such an element* g *in* $\mathrm{SL}_n(\mathsf{K})$.

Proof. Let $\{a_1, \ldots, a_n\}$ and $\{b_1, \ldots, b_n\}$ be K-bases of V such that

$$a_1 = a,\ a_n = b,\ \mathsf{H} = \langle a_1, \ldots, a_{n-1}\rangle \quad \text{and}$$
$$b_1 = a',\ b_n = b',\ \mathsf{H}' = \langle b_1, \ldots, b_{n-1}\rangle.$$

Then there exists exactly one linear map $h \in \mathrm{GL}_n(\mathsf{K})$ such that $a_i h = b_i$ for all $i = 1, \ldots, n$.

For $n \geq 3$, we define the linear map h' by $a_i h' = a_i$ for $i \neq 2$ and $a_2 h' = (\det h)^{-1} a_2$. If we set $g = h$ for $n = 2$ and $g = h'h$ for $n \geq 3$, we obtain the desired linear maps. □

Lemma 6.9. *All transvections are conjugate in* $\mathrm{GL}_n(\mathsf{K})$. *If* $n \geq 3$, *then all transvections are already conjugate in* $\mathrm{SL}_n(\mathsf{K})$. *In* $\mathrm{SL}_2(\mathsf{K})$, *the groups* $T(\mathsf{H})$ *are conjugate.*

Proof. Let $t = t(a, \mu)$ be a transvection and $b \in \mathsf{V}$ such that $\mu(b) = 1$. Let $t' = t(a', \mu')$ be another transvection and $\mu'(b') = 1$. We denote by H, respectively H', the hyperplane consisting of the zeros of μ, respectively μ'. By Lemma 6.8, there exists some $g \in \mathrm{GL}_n(\mathsf{K})$ such that $ag = a'$, $bg = b'$ and $\mathsf{H}g = \mathsf{H}'$; for $n \geq 3$ we may choose $g \in \mathrm{SL}_n(\mathsf{K})$. Defining $x' = xg \in \mathsf{H}'$ for all $x \in \mathsf{H}$, we have

$$x'g^{-1}tg = xtg = xg = x' \quad \text{and} \quad b'g^{-1}tg = btg = (b - a)g = b' - a'.$$

Therefore, $g^{-1}tg = t'$.

For $n = 2$, we may choose $g \in \mathrm{SL}_2(\mathsf{K})$ in such a way that there exists $c \in \mathsf{K}^\times$ such that $ag = ca'$ and $bg = b'$. This implies

$$g^{-1}t(a, \mu)g = t(ca', \mu').$$ □

The next theorem gives our first indication of the structure of $\mathrm{SL}_n(\mathsf{K})$.

Theorem 6.10. *For* $n \geq 3$, *as well as for* $n = 2$ *and* $|\mathsf{K}| > 3$, *we have*

$$\mathrm{GL}_n(\mathsf{K})' = \mathrm{SL}_n(\mathsf{K})' = \mathrm{SL}_n(\mathsf{K}).$$

Proof. Since

$$\mathrm{GL}_n(\mathsf{K})/\mathrm{SL}_n(\mathsf{K}) \cong \mathsf{K}^\times$$

is abelian, we have

$$\mathrm{SL}_n(\mathsf{K})' \leq \mathrm{GL}_n(\mathsf{K})' \leq \mathrm{SL}_n(\mathsf{K}).$$

To conclude, we will show that $\mathrm{SL}_n(\mathsf{K})' = \mathrm{SL}_n(\mathsf{K})$. By Theorem 6.7, it suffices to show that every transvection lies in $\mathrm{SL}_n(\mathsf{K})'$.

First assume that $n \geq 3$. If t, t' are transvections, then by Theorem 6.9 there exists some $g \in \mathrm{SL}_n(\mathsf{K})$ such that $g^{-1}tg = t'$. So all transvections lie in the same coset of $\mathrm{SL}_n(\mathsf{K})'$.

Let $\mathsf{H} = \{v \mid \mu(v) = 0\}$ be a hyperplane in V. Since $\dim \mathsf{H} = n - 1 \geq 2$, there exist vectors $a_1, a_2 \in \mathsf{H}$ satisfying

$$a_1 \neq 0 \quad \text{and} \quad 0 \neq a_2 \neq -a_1.$$

By Lemma 6.5, we have

$$t(a_1, \mu)t(a_2, \mu) = t(a_1 + a_2, \mu).$$

Since all transvections lie in the same coset of $\mathrm{SL}_n(\mathsf{K})'$, it follows that[9]

$$t(a_1, \mu)t(a_2, \mu)\mathrm{SL}_n(\mathsf{K})' = t(a_2, \mu)\mathrm{SL}_n(\mathsf{K})'.$$

So $t(a_1, \mu) \in \mathrm{SL}_n(\mathsf{K})'$, and all transvections lie in $\mathrm{SL}_n(\mathsf{K})'$.

For $n = 2$ we must proceed more carefully. Let t be a transvection over the hyperplane $\langle v_1 \rangle$. With respect to a suitable basis $\{v_1, v_2\}$ of V, the transvection t has the form

$$v_1 t = v_1, \quad v_2 t = v_1 + v_2.$$

Since $|\mathsf{K}| > 3$, there exists some $d \in \mathsf{K}$ such that $0 \neq d^2 \neq 1$. We define a linear map $g \in \mathrm{SL}_2(\mathsf{K})$ by

$$v_1 g = dv_1 \quad \text{and} \quad v_2 g = d^{-1} v_2.$$

Then we have

$$v_1 g^{-1} t g = (d^{-1} v_1) t g = v_1$$

and

$$v_2 g^{-1} t g = (dv_2) t g = d(v_1 + v_2) g = v_2 + d^2 v_1.$$

We therefore obtain $g^{-1} t g t^{-1} = t(w, \mu) \in \mathrm{SL}_2(\mathsf{K})'$, where $w = (1 - d^2)v_1 \neq 0$ and $\mu(v_1) = 0$, $\mu(v_2) = 1$.

Since $\mathrm{SL}_2(\mathsf{K})' \trianglelefteq \mathrm{GL}_2(\mathsf{K})$ and all transvections are conjugate under $\mathrm{GL}_2(\mathsf{K})$, all transvections lie in $\mathrm{SL}_2(\mathsf{K})'$. □

We will see in Theorem 6.14 that the groups $\mathrm{SL}_2(2)$ and $\mathrm{SL}_2(3)$ are solvable, so the conclusion of Theorem 6.10 certainly does not hold for these groups.

Lemma 6.11. *The group* $\mathrm{PSL}_n(\mathsf{K})$ *is* 2*-transitive on the points of the projective space* $\mathsf{P}(n-1, \mathsf{K})$ *of dimension* $n-1$ *over* K*. If* $|\mathsf{K}| = p^f$*, then* $\mathrm{PSL}_n(\mathsf{K})$ *is a permutation group of degree* $(p^{nf} - 1)/(p^f - 1)$ *on* $\mathsf{P}(n-1, \mathsf{K})$*.*

Proof. Let a_1, a_2 and a_1', a_2' be pairs of points in $\mathsf{P}(n-1, \mathsf{K})$ such that $a_1 \neq a_2$ and $a_1' \neq a_2'$. Let $a_i = \langle v_i \rangle$ and $a_i' = \langle v_i' \rangle$ for some $v_i, v_i' \in \mathsf{V} = \mathsf{V}(n, \mathsf{K})$. Then v_1 and v_2 are linearly independent, as well as v_1' and v_2'. Therefore, there exists some $g \in \mathrm{SL}_n(\mathsf{K})$ such that

[9] Correction 14 from errata.

$$v_1 g = v_1' \quad \text{and} \quad v_2 g = c v_2'$$

for some $c \in \mathsf{K}^\times$. (For $n \geq 3$, we may take $c = 1$.) Then the element g, considered as a map on the projective space $\mathsf{P}(n-1, \mathsf{K})$, takes (a_1, a_2) to (a_1', a_2'). □

Lemma 6.12. *Let G be a primitive permutation group on the possibly infinite set Ω. Suppose the following conditions are satisfied:*
(1) $G = G'$.
(2) *The stabilizer G_a of $a \in \Omega$ contains a solvable normal subgroup K with*

$$K^G = \langle K^g \mid g \in G \rangle = G.$$

(*By the transitivity of G on Ω, it is equivalent to demand that* (2) *holds for one or for all $a \in \Omega$.*)

Then G is simple.

Proof. Let $1 < N \trianglelefteq G$. Since G is primitive, N is transitive by Theorem 1.5. By Theorem 1.4, G_a is a maximal subgroup of G, and so $G = NG_a$. Since $K \trianglelefteq G_a$, it follows that

$$NK \trianglelefteq NG_a = G.$$

By condition (2), we have

$$G = \langle K^g \mid g \in G \rangle \leq NK,$$

and so $G = NK$. Since K is solvable, there exists an integer k such that $K^{(k)} = 1$. Therefore, it follows from condition (1) that

$$G = G^{(k)} = (NK)^{(k)} \leq NK^{(k)} = N.$$

Hence, G is simple. □

Main Theorem 6.13. *The group* $\mathrm{PSL}_n(\mathsf{K})$ *is simple, aside from the two cases* $\mathrm{PSL}_2(2)$ *and* $\mathrm{PSL}_2(3)$.

Proof. (Iwasawa) By Theorem 6.11, the group $G = \mathrm{PSL}_n(\mathsf{K})$ is a 2-transitive permutation group on the points of the projective space $\mathsf{P}(n-1, \mathsf{K})$. By Theorem 6.10, we have

$$(\mathrm{SL}_n(\mathsf{K})Z/Z)' = \mathrm{SL}_n(\mathsf{K})'Z/Z = \mathrm{SL}_n(\mathsf{K})Z/Z$$

for $n \geq 3$ or $|\mathsf{K}| > 3$. Then $\mathrm{PSL}_n(\mathsf{K}) = \mathrm{PSL}_n(\mathsf{K})'$ as $\mathrm{PSL}_n(\mathsf{K}) = \mathrm{SL}_n(\mathsf{K})Z/Z$.

Let $0 \neq w \in \mathsf{V}$, and let T_w be the group consisting of 1 and all transvections t of the form

$$vt = v - \mu(v)w \quad \text{with} \quad \mu(w) = 0.$$

Then T_w is an abelian group. For all $g \in \mathrm{SL}_n(\mathsf{K})$, we have $T_w^g = T_{wg}$. With the help of Theorem 6.7, we conclude that

$$\langle T_w^g \mid g \in \mathrm{SL}_n(\mathsf{K})\rangle = \langle T_{wg} \mid wg \in \mathsf{V}\rangle = \mathrm{SL}_n(\mathsf{K}).$$

Let $K \leq \mathrm{PSL}_n(\mathsf{K})$ be the subgroup consisting of the maps in T_w regarded as maps on the projective space $\mathsf{P}(n-1,\mathsf{K})$. Let $a = \langle w\rangle$ and

$$G_a = \{g \in \mathrm{PSL}_n(\mathsf{K}) \mid ag = a\}.$$

Then $K \trianglelefteq G_a$. The conditions of Lemma 6.12 are satisfied, and we conclude that $G = \mathrm{PSL}_n(\mathsf{K})$ is simple, aside from the two cases $\mathrm{PSL}_2(2)$ and $\mathrm{PSL}_2(3)$. □

Some of the groups $\mathrm{PSL}_n(\mathsf{K})$ are isomorphic to some $\mathrm{PSL}_{n'}(\mathsf{K}')$ or an alternating or symmetric group:

Theorem 6.14. *We have the following isomorphisms:*
(1) $\mathrm{PSL}_2(2) \cong \mathrm{SL}_2(2) = \mathrm{GL}_2(2) \cong S_3$.
(2) $\mathrm{PSL}_2(3) \cong A_4$.
(3) $\mathrm{PSL}_2(4) \cong \mathrm{PSL}_2(5) \cong A_5$.
(4) $\mathrm{PSL}_2(7) \cong \mathrm{PSL}_3(2)$.
(5) $\mathrm{PSL}_4(2) \cong A_8$.
(6) $\mathrm{PSL}_2(9) \cong A_6$.
(*We mention without proof that these are the only isomorphisms between groups* $\mathrm{PSL}_n(\mathsf{K})$ *and alternating or symmetric groups, see Artin [14].*)

Proof. (1) $\mathrm{GL}_2(2)$ has order 6 and admits a faithful permutation representation of degree 3, namely by acting on the three nonzero vectors in the vector space $\mathsf{V}(2,2)$. This proves (1).

(2) We have $|\mathrm{PSL}_2(3)| = 12$, and $\mathrm{PSL}_2(3)$ has a faithful permutation representation on the four points of the projective line $\mathsf{P}(1,3)$. This gives $\mathrm{PSL}_2(3) \cong A_4$.

(3) $\mathrm{PSL}_2(4)$ has order 60 and admits a faithful representation as a permutation group of degree 5 on $\mathsf{P}(1,4)$. It follows that $\mathrm{PSL}_2(4) \cong A_5$. Since $\mathrm{PSL}_2(5)$ is a simple group of order 60, the isomorphism $\mathrm{PSL}_2(5) \cong A_5$ follows from Theorem I.8.14 (pg. 43). (This isomorphism provides, up to permutation isomorphism, the representation of A_5 as a 2-transitive permutation group of degree 6 given in Theorem 4.7.)

(4) The groups $\mathrm{PSL}_2(7)$ and $\mathrm{PSL}_3(2)$ are simple by Theorem 6.13, and both have order 168 by Lemma 6.2. We will show in Theorem 6.15 that there exists only one simple group of order 168 up to isomorphism.

(5) We have already proved $\mathrm{PSL}_4(2) \cong A_8$ in Theorem 2.5.

(6) We will prove $\mathrm{PSL}_2(9) \cong A_6$ in Theorem 8.14 when we study the subgroups of $\mathrm{PSL}_2(p^f)$. □

Theorem 6.15. *If G is a simple group of order* 168*, then $G \cong \mathrm{PSL}_2(7)$.*

Proof. Let t be the number of Sylow 7-subgroups of G. Then we have $t \equiv 1 \bmod 7$, $t > 1$, and t is a divisor of 24. This forces $t = 8$. So G admits a (necessarily

faithful) transitive permutation representation of degree 8, namely on the cosets of the normalizer of a Sylow 7-subgroup. We denote this set by $\{\infty\} \cup \mathrm{GF}(7)$ via a permutation isomorphism.

Let $P = \langle g \rangle$ be a Sylow 7-subgroup of G. With an appropriate labeling of the points, we may take

$$g = \begin{pmatrix} x \\ x+1 \end{pmatrix} = (\infty)(0,1,2,3,4,5,6).$$

The unique fixed point ∞ of P is also fixed by $\mathbf{N}_G(P)$. The group $\mathbf{N}_G(P)$ of order 21 acts as a transitive permutation group of degree 7 on GF(7). Since $P \trianglelefteq \mathbf{N}_G(P)$, the group $\mathbf{N}_G(P)$ is solvable by Theorem 3.6 (pg. 171). We may write $\mathbf{N}_G(P) = \langle g, n \rangle$, where $\infty^n = \infty$ and $x^n = 2x$ for all $x \in \mathrm{GF}(7)$.

Let $H = \mathbf{N}_G(N)$ be the normalizer of the Sylow 3-subgroup $N = \langle n \rangle$ of G, and let $[G : H] = s$. If H had an element h of order 7, then $h \in \mathbf{C}_G(n)$ and nh would be an element of order 21. This is impossible since the group acts on 8 points. Therefore $7 \mid s$, furthermore $s \mid 56$ and $s \equiv 1 \bmod 3$. Hence, we must have $s = 7$ or $s = 28$.

Assume for contradiction that $[G : H] = 7$. Then G has exactly seven Sylow 3-subgroups. But $\mathbf{N}_G(P)$ already has seven Sylow 3-subgroups and is generated by them. So $\mathbf{N}_G(P) \lhd G$, contradicting the simplicity of G.

Therefore, $[G : H] = 28$ and $|H| = 6$. So there exists some $u \in H$ with $|u| = 2$.

Assume for contradiction that $nu = un$. Then $z = nu \in G$ is an element of order 6 with $\mathbf{N}_G(\langle z \rangle) = H$. The following elements are contained in G:

$8 \cdot 6 = 48$ elements of order 7 in the 8 conjugates of P;

$28 \cdot 2 = 56$ elements of order 3 in the 28 conjugates of N;

$28 \cdot 2 = 56$ elements of order 6 in the 28 conjugates of $\langle z \rangle$.

Only 8 elements remain, so these must constitute the (necessarily unique) Sylow 2-subgroup of G. But this is impossible since G is simple.

Hence, we must have $n^u = n^{-1}$. (The case $H = \mathbf{N}_G(N) = \mathbf{C}_G(N)$ can also be handled with Burnside's Transfer Theorem IV.2.6.)

Any subgroup of G that leaves fixed one of the 8 points has order 21 by the transitivity of G, so does not contain an element of order 2. So u has no fixed points. The points 0 and ∞ fixed by n must therefore be interchanged by u. It follows from $n^u = n^{-1}$ and $x^n = 2x$ that

$$(2x)^u = x^{nu} = x^{un^{-1}} = 2^{-1}x^u.$$

Since u has no fixed points, we have $1^u \neq 1$, $2^u = 2^{-1}1^u \neq 2$, so $1^u \neq 4$, and $4^u = 4^{-1}1^u \neq 4$, so $1^u \neq 2$. Then, since $u^2 = 1$, the element u has the four disjoint cycles

$$(0,\infty),\ (1,1^u),\ (2,2^{-1}1^u) \quad \text{and} \quad (4,4^{-1}1^u)$$

in common with the linear fractional transformation l defined by $x^l = x^{-1}1^u$. Hence, $l = u$. Therefore g, n and u are linear fractional transformations of $\{\infty\} \cup \mathrm{GF}(7)$. Letting $U = \langle g, n, u \rangle$, we have that $|U|$ is a multiple of $|g||n||u| = 42$. It follows from

the simplicity of G and $[G : U] \leq 4$ that $G = U$. So G is isomorphic to a subgroup of $\mathrm{PGL}_2(7)$ of index 2. Since the determinants of the maps g, n and u are squares in GF(7) ($\det u = -1^u$ is a square as -1 and 1^u are not squares since $1^u \notin \{1, 2, 4\}$!) it follows that $G \cong \mathrm{PSL}_2(7)$. □

Exercises

6) Let $\mathrm{SL}_n(\mathsf{K}) \leq G \leq \mathrm{GL}_n(\mathsf{K})$. Let N be a normal subgroup of G. If $n \geq 3$ or $|\mathsf{K}| > 3$, then $N \leq \{a1 \mid a \in \mathsf{K}^\times\}$ or $\mathrm{SL}_n(\mathsf{K}) \leq N$.
7) Show that $|\mathrm{PSL}_4(2)| = |\mathrm{PSL}_3(4)|$ but

$$\mathrm{PSL}_4(2) \ncong \mathrm{PSL}_3(4).$$

(Hint: Consider the centers of the Sylow 2-subgroups and use Theorem 7.1.)
8) Show that $\mathrm{PGL}_2(9) \ncong S_6$.
9) Determine the centralizers of the elements of order 2 in $\mathrm{PSL}_3(q)$, where $q = p^f$. (The cases $p = 2$ and $p \neq 2$ should be handled separately!)

7 Subgroups of $\mathrm{PGL}_n(p^f)$

In this section, we set $\mathsf{K} = \mathrm{GF}(p^f)$. First we describe three subgroups of $\mathrm{GL}_n(p^f)$ and $\mathrm{PGL}_n(p^f)$, respectively, for any $n > 1$.

Theorem 7.1. *Let $\{v_1, \ldots, v_n\}$ be a K-basis of $\mathsf{V} = \mathsf{V}(n, p^f)$. The group P of all linear maps x of the form*

$$v_i x = \sum_{j=1}^{i} a_{ij} v_j \quad (a_{ij} \in \mathsf{K})$$

with $a_{ii} = 1$ for $i = 1, \ldots, n$ is a Sylow p-subgroup of $\mathrm{GL}_n(p^f)$ and $\mathrm{SL}_n(p^f)$; it is isomorphic to a Sylow p-subgroup of $\mathrm{PSL}_n(p^f)$. The normalizer of P in $\mathrm{GL}_n(p^f)$ consists of all maps g of the form

$$v_i g = \sum_{j=1}^{i} a_{ij} v_j \quad (a_{ii} \neq 0).$$

(*We will thoroughly study the structure of P in Chapter III, §16.*)

Proof. It is clear that P is a subgroup of $\mathrm{SL}_n(p^f)$ of order p^k, where $k = \frac{fn(n-1)}{2}$. By Lemma 6.2, this is the highest power of p that divides the order of $\mathrm{GL}_n(p^f)$, resp. $\mathrm{SL}_n(p^f)$. Therefore, P is a Sylow p-subgroup of $\mathrm{GL}_n(p^f)$ and $\mathrm{SL}_n(p^f)$. Since

the kernel $Z = \{a1 \mid a \in \mathsf{K}^\times\}$ of the epimorphism from $\mathrm{GL}_n(p^f)$ onto $\mathrm{PGL}_n(p^f)$ has order $p^f - 1$ coprime to p, the group P is isomorphic to a Sylow p-subgroup of $\mathrm{PGL}_n(p^f)$ and $\mathrm{PSL}_n(p^f)$.

One sees easily that the subspaces $\mathsf{V}_i = \langle v_1, \ldots, v_i\rangle$ form the only P-invariant chain of subspaces of V with one-dimensional quotient spaces. If $P^g = P$, then we must have $\mathsf{V}_i g = \mathsf{V}_i$ $(i = 1, \ldots, n)$, so g has the desired form. □

Theorem 7.2. *Let $\{v_1, \ldots, v_n\}$ be a basis of $\mathsf{V} = \mathsf{V}(n, p^f)$ over $\mathsf{K} = \mathrm{GF}(p^f)$.*

a) *The maps d of the form*

$$v_i d = a_i v_i \quad (a_i \in \mathsf{K}^\times,\ i = 1, \ldots, n)$$

constitute an abelian subgroup D of $G = \mathrm{GL}_n(p^f)$ of order $(p^f - 1)^n$. If $p^f \neq 2$, then $\mathbf{C}_G(D) = D \neq 1$; the normalizer $\mathbf{N}_G(D)$ is exactly the set of so-called monomial maps g of the form

$$v_i g = b_i v_{i'} \quad (b_i \in \mathsf{K}^\times),$$

where $\binom{i}{i'}$ is a permutation in the symmetric group S_n. Furthermore, we have $\mathbf{N}_G(D)/D \cong S_n$.

b) *The group $S = D \cap \mathrm{SL}_n(p^f)$ has order $(p^f - 1)^{n-1}$. For $n \geq 3$ or $p^f > 3$, we have*

$$\mathbf{N}_{\mathrm{SL}_n(p^f)}(S) = \mathbf{N}_{\mathrm{GL}_n(p^f)}(D) \cap \mathrm{SL}_n(p^f).$$

c) *Let $n \geq 3$ or $p^f > 3$. The groups S and $\mathbf{N}_{\mathrm{SL}_n(p^f)}(S)$ correspond to subgroups U and $\mathbf{N}_{\mathrm{PSL}_n(p^f)}(U)$ of $\mathrm{PSL}_n(p^f)$ of orders*

$$\frac{(p^f - 1)^{n-1}}{k} \quad \text{and} \quad \frac{(p^f - 1)^{n-1}}{k} n!,$$

where $k = (p^f - 1, n)$.

Proof. a) Clearly, D is an abelian subgroup of $\mathrm{GL}_n(p^f)$ having the given order. It is a trivial calculation to show that for $p^f > 2$ and $i = 1, \ldots, n$, the $\langle v_i\rangle$ are the only one-dimensional subspaces of V that are D-invariant. For all $d \in D$ and $g \in \mathbf{N}_G(D)$, there exists some $c_i \in \mathsf{K}^\times$ such that

$$(v_i g)d = v_i(g d g^{-1})g = c_i v_i g.$$

So $\langle v_i g\rangle$ is D-invariant, and it follows that $v_i g = a_i v_{i'}$. The converse statement, that all monomial maps lie in $\mathbf{N}_G(D)$, is clear.

Now let $g \in \mathbf{C}_G(D)$. Since $g \in \mathbf{N}_G(D)$, we have $v_i g = a_i v_{i'}$. Assume for contradiction that $i \neq i'$ for some i. There exists some $d \in D$ such that $v_i d = v_i$ and $v_{i'} d = b v_{i'}$ for some $1 \neq b \in \mathsf{K}^\times$. But then

$$a_i v_{i'} = v_i g = v_i d g = v_i g d = a_i v_{i'} d = a_i b v_{i'},$$

a contradiction. So $\mathbf{C}_G(D) = D$.

b) We show that for $n \geq 3$ or $p^f > 3$, the $\langle v_i \rangle$ are the only one-dimensional S-invariant subspaces of V. The claim then follows as in a).

Assuming $\langle v \rangle$ is S-invariant, we may write $v = \sum_{i=1}^{n} b_i v_i$ with, say, $b_1 \neq 0$. We show that $v = b_1 v_1$. Let $s \in S$ such that

$$v_1 s = a v_1, \quad v_2 s = a^{-1} v_2, \quad v_i s = v_i \ (i \geq 3)$$

with $1 \neq a \in \mathsf{K}^\times$; for $p^f > 3$ we may demand that $a^2 \neq 1$. Then for some $c \in \mathsf{K}^\times$ we have

$$vs = cv = c \sum_{i=1}^{n} b_i v_i = a b_1 v_1 + a^{-1} b_2 v_2 + \sum_{i=3}^{n} b_i v_i.$$

This forces $a = c$ and $a b_i = b_i$ for $i \geq 3$, so $b_i = 0$. If $n \geq 3$, we are done since we may run the same argument after switching the roles of 2 and 3. For $n = 2$ and $p^f > 3$, we then have $a b_2 = c b_2 = a^{-1} b_2$, so $b_2 = 0$ since $a^2 \neq 1$.

c) The claim follows since

$$\widetilde{Z} = \{a1 \mid a^n = 1, a \in \mathsf{K}^\times\} \leq S$$

and the quotient map from $\mathrm{SL}_n(p^f)$ onto $\mathrm{PSL}_n(p^f) = \mathrm{SL}_n(p^f)/\widetilde{Z}$ is an epimorphism. □

Theorem 7.3.

a) *The group $G = \mathrm{GL}_n(p^f)$ contains a cyclic subgroup T of order $p^{nf} - 1$. We have $\mathbf{C}_G(T) = T$, and the quotient $\mathbf{N}_G(T)/T$ is cyclic of order n. If $t \in T$ and the order of t does not divide $p^{mf} - 1$ for every $m < n$ with $m \mid n$, then $\mathbf{N}_G(\langle t \rangle) = \mathbf{N}_G(T)$.*

(*In geometric investigations, the subgroup T occasionally arises under the name Singer cycle.*)

b) *Let $L = T \cap \mathrm{SL}_n(p^f)$. Then*

$$|L| = \frac{p^{nf} - 1}{p^f - 1}$$

and $\mathbf{N}_{\mathrm{SL}_n(p^f)}(L) = \mathbf{N}_{\mathrm{GL}_n(p^f)}(T) \cap \mathrm{SL}_n(p^f)$.

c) *The group T corresponds to a cyclic subgroup $\overline{T}$ of $\overline{G} = \mathrm{PGL}_n(p^f)$, where*

$$|\overline{T}| = \frac{p^{nf} - 1}{p^f - 1} \quad \text{and} \quad |\overline{T} \cap \mathrm{PSL}_n(p^f)| = \frac{p^{nf} - 1}{d(p^f - 1)}$$

with $d = (\frac{p^{nf}-1}{p^f-1}, n)$. The quotient $\mathbf{N}_{\overline{G}}(\overline{T})/\overline{T}$ is cyclic of order n. If $\bar{t} \in \overline{T}$ and $|\bar{t}|$ is not a divisor of

$$\frac{p^{mf}-1}{p^f-1}$$

for all $m < n$ with $m \mid n$, then $\mathbf{N}_{\overline{G}}(\langle \overline{t} \rangle) = \mathbf{N}_{\overline{G}}(\overline{T})$.

Proof. a) We identify $\mathrm{GF}(p^{nf})^+$, regarded as a vector space over $\mathrm{GF}(p^f)$, with $\mathsf{V} = \mathsf{V}(n, p^f)$. Then the linear maps $g \in \mathrm{GL}_n(p^f)$ of the form

$$x^g = ax \quad (a \in \mathrm{GF}(p^{nf})^\times)$$

constitute a cyclic subgroup $T \leq \mathrm{GL}_n(p^f)$ of order $p^{nf}-1$. Note that T is a regular permutation group on $\mathsf{V} - \{0\}$. By Lemma 3.1, we then have $\mathbf{C}_G(T) = T$. Since V is an irreducible T-module, we may set $s = 1$ in Lemma 3.11 to conclude that $\mathbf{N}_G(T)$ is permutation isomorphic to the group of all semilinear maps

$$x \mapsto ax^\alpha,$$

where α is a field automorphism of $\mathrm{GF}(p^{nf})$ over $\mathrm{GF}(p^f)$. The automorphism group of $\mathrm{GF}(p^{nf})$ over $\mathrm{GF}(p^f)$ is well known to be cyclic of order n. So $\mathbf{N}_G(T)/T$ is also cyclic of order n.

Let $t \in T$, and assume $|t| \nmid p^{mf}-1$ for all $m < n$ with $m \mid n$. Suppose t acts as $x^t = ax$ on $\mathrm{GF}(p^{nf})$. Since $|t| = |a| \nmid p^{mf}-1$, the element $a \in \mathrm{GF}(p^{nf})^\times$ does not lie in any of the fields $\mathrm{GF}(p^{mf})$ for $m < n$ and $m \mid n$. So $\mathrm{GF}(p^{nf}) = \mathrm{GF}(p^f)[a]$. For every nonzero $x \in \mathrm{GF}(p^{nf})$, we then have $\mathrm{GF}(p^{nf}) = x\mathrm{GF}(p^f)[a]$. Hence, $\mathrm{GF}(p^{nf})^+$ is an irreducible $\langle t \rangle$-module. So $\mathsf{V} = \mathsf{V}(n, p^f)$ is also an irreducible $\langle t \rangle$-module. By Lemma 3.11, it follows that $\mathbf{N}_G(\langle t \rangle) = \mathbf{N}_G(T)$.

b) Let $\mathrm{GF}(p^{nf})^\times = \langle a \rangle$. The $\mathrm{GF}(p^f)$-linear map g defined by

$$x^g = ax \quad (x \in \mathrm{GF}(p^{nf}))$$

has the eigenvalue a since the minimal polynomial of g clearly has a zero at a. Therefore, the conjugates of a over $\mathrm{GF}(p^f)$

$$a, a^{p^f}, \ldots, a^{p^{(n-1)f}}$$

are also eigenvalues of g. As these are pairwise distinct, they are all the eigenvalues of g. Hence, $\det g = a^k$, where

$$k = \frac{p^{nf}-1}{p^f-1}.$$

This shows that[10]

$$|T \cap \mathrm{SL}_n(p^f)| = |\langle g^{p^f-1} \rangle| = \frac{p^{nf}-1}{p^f-1}.$$

[10] Correction 15 from errata.

Since

$$|g^{p^f-1}| = \frac{p^{nf}-1}{p^f-1} = 1 + p^f + \cdots + p^{(n-1)f} \geq 1 + p^{(n-1)f} > p^{mf} - 1$$

for all $m < n$, it follows from a) that

$$\mathbf{N}_G(\langle g^{p^f-1}\rangle) = \mathbf{N}_G(T).$$

This yields the claim.

c) follows immediately from a) upon passing from $\mathrm{GL}_n(p^f)$ to $\mathrm{PGL}_n(p^f) = \mathrm{GL}_n(p^f)/Z$, where

$$Z = \{a1 \mid a \in \mathrm{GF}(p^f)^\times\} \leq T. \qquad \square$$

Theorem 7.4. *Every proper subgroup of* $\mathrm{PSL}_3(3)$ *is solvable.*

Proof. The group $G = \mathrm{PSL}_3(3)$ is a 2-transitive permutation group on the 13 points of the projective plane $\mathsf{P}(2,3)$. By Lemma 6.2, we have

$$|\mathrm{PSL}_3(3)| = 13 \cdot 3^3 \cdot 2^4.$$

Let G_a be the stabilizer of a point $a \in \mathsf{P}(2,3)$ and H the preimage of G_a in $\mathrm{SL}_3(3)$. Since

$$\{c1 \in \mathrm{SL}_3(3) \mid c \in \mathrm{GF}(3),\ c^3 = 1\} = 1,$$

we have $\mathrm{PSL}_3(3) \cong \mathrm{SL}_3(3)$ and $G_a \cong H$. With an appropriate choice of basis for $\mathsf{V}(3,3)$, the matrices in H have the form

$$g = \begin{pmatrix} a_1 & 0 & 0 \\ a_2 & & \\ a_3 & & g_1 \end{pmatrix},$$

for some $g_1 \in \mathrm{GL}_2(3)$ with $a_1 \det g_1 = 1$. The map ϵ defined by $g^\epsilon = g_1$ is an epimorphism from H onto $\mathrm{GL}_2(3)$, which is solvable by Theorem 6.14 (2). The kernel of ϵ has order 3^2, so it is abelian. Therefore, H is solvable.

Let U be a proper subgroup of $G = \mathrm{PSL}_3(3)$. We distinguish two cases:

Case 1: Assume $13 \mid |U|$. Let P be a Sylow 13-subgroup of U, and let $[U : \mathbf{N}_U(P)] = s$. Then $s \equiv 1 \bmod 13$ and $s \mid 3^3 \cdot 2^4$. By Theorem 7.3 c), we have $|\mathbf{N}_G(P)| = 3 \cdot 13$. Since G is simple, it is generated by its Sylow 13-subgroups. Then since $U < G$, we have

$$s = [U : \mathbf{N}_U(P)] < [G : \mathbf{N}_G(P)] = 3^2 \cdot 2^4.$$

This forces $s = 1$ or $s = 3^3$; since $|\mathbf{N}_U(P)|$ divides $3 \cdot 13$, we have that $|U|$ divides $3^4 \cdot 13$. Then U is solvable by Theorem I.8.11 (pg. 42).

Case 2: Assume $13 \nmid |U|$. Then U is certainly not transitive and $|U|$ divides $3^3 \cdot 2^4$. The orbits of U on $\mathsf{P}(2,3)$ have possible sizes 1, 2, 3, 4, 6, 8, 9, 12.[11] If U has an orbit of size 1, then U lies is some G_a and is then solvable. If U has an orbit Δ with $|\Delta| \le 4$, then $U^\Delta \cong U/U_\Delta$ is solvable and U_Δ has fixed points, so is solvable. So if U were not solvable, then all of its orbits would have size 6 or 8. But this is impossible since $6+6 < 13 < 6+8$. □

Remark 7.5. The nonabelian simple groups whose proper subgroups are solvable have been classified in a large and extraordinarily deep work of Thompson [436]. They are the following:
(1) $\mathrm{PSL}_2(p)$, where $p > 3$ is prime and $p^2 - 1 \not\equiv 0 \bmod 5$.
(2) $\mathrm{PSL}_2(2^q)$, where q is prime.
(3) $\mathrm{PSL}_2(3^q)$, where q is an odd prime.
(4) $\mathrm{PSL}_3(3)$.
(5) The Suzuki groups $\mathrm{Sz}(2^q)$, where q is an odd prime. (We will discuss these groups in detail in Volume 3.)

Exercises

10) Let $q = p^f$ with $p > 2$. For a prime r and integer m, we write $r^n \top m$ if $r^n \mid m$ but $r^{n+1} \nmid m$.
   a) For $q \equiv 1 \bmod 4$ and $2^s \top q-1$, show that the Sylow 2-subgroups of $\mathrm{PSL}_3(q)$ may be defined via the relations
$$a^{2^s} = b^{2^s} = c^2 = 1, \quad a^c = a^{-1}, \quad b^c = ab.$$
   b) For $q \equiv -1 \bmod 4$ and $2^s \top q^2 - 1$, show that the Sylow 2-subgroups of $\mathrm{PSL}_3(q)$ may be defined via the relations
$$a^{2^s} = b^2 = 1, \quad a^b = a^{-1+2^{s-1}}.$$
   (This is type (4) from Theorem I.14.9, pg. 96.)
11) Let $q = p^f$ be a power of the prime p, and let $r > 2$ be a prime distinct from p. Let t be the order of q modulo r, and let $q^t - 1 = ar^s$ with $r \nmid a$. Show that:
   a) $r^{s+k} \top q^{tx} - 1$, where $r^k \top x$.
   b) $r^m \top (q^n - 1)(q^{n-1} - 1) \cdots (q - 1)$, where
$$m = s\left\lfloor \tfrac{n}{t} \right\rfloor + \left\lfloor \tfrac{n}{tr} \right\rfloor + \left\lfloor \tfrac{n}{tr^2} \right\rfloor + \cdots.$$
12) Let the notation be as in Exercise 11, and let $k = \lfloor \frac{n}{t} \rfloor$.
   a) Let $\mathsf{V} = \mathsf{V}(n,q) = \mathsf{V}_1 \oplus \mathsf{V}_2$ with $\mathsf{V}_1 = \mathsf{V}(tk,q)$. Let $\mathsf{W} = \mathsf{V}(k,q^t)$. Since W and V_1 are isomorphic $\mathrm{GF}(q)$-vector spaces, there is a monomorphism τ from $\mathrm{GL}_k(q^t)$ to $\mathrm{GL}_{tk}(q)$. If R is a Sylow r-subgroup of $\mathrm{GL}_k(q^t)$, then

[11] Correction 16 from errata.

$$R^* = \{r^* \mid v_1 r^* = v_1 r^\tau \text{ for } v_1 \in \mathsf{V}_1,\ v_2 r^* = v_2 \text{ for } v_2 \in \mathsf{V}_2; r \in R\}$$

is a Sylow r-subgroup of $\mathrm{GL}_n(q)$.

b) Let $\{w_1, \dots, w_k\}$ be a basis of $\mathsf{W} = \mathsf{V}(k, q^t)$ over $\mathrm{GF}(q^t)$. Let P be a Sylow r-subgroup of the symmetric group S_k. Then

$$R = \{r \in \mathrm{GL}_k(q^t) \mid v_i r = a_i v_{ig} \text{ for } a_i \in \mathrm{GF}(q^t),\ g \in P\}$$

is a Sylow r-subgroup of $\mathrm{GL}_k(q^t)$.

(One may find a description of the Sylow 2-subgroups of $\mathrm{GL}_n(q)$ in Carter, Fong [93].)

13) Show that $\mathrm{PSL}_3(4)$ contains a subgroup isomorphic to the alternating group A_6.

(Hint: For every 3 pairwise non-collinear points a_1, a_2, a_3 in $\mathsf{P}(2,4)$, there exist exactly three other points a_4, a_5, a_6 such that $\{a_1, \dots, a_6\}$ contains no collinear triple. For $a_i' \in \{a_1, \dots, a_6\}$ $(i = 1, 2, 3)$, there exists some $g \in \mathrm{PGL}_3(4)$ such that $a_i g = a_i'$ $(i = 1, 2, 3)$.)

8 The subgroups of $\mathrm{PSL}_2(p^f)$

It is only for a handful of nonabelian groups that a complete list of subgroups is known. In this section, we will determine all the subgroups of $\mathrm{PSL}_2(p^f)$. We begin by describing some subgroups in Theorems 8.2–8.20, and then we show that these are in fact all the subgroups of $\mathrm{PSL}_2(p^f)$.

Our first lemma is trivial.

Lemma 8.1. *The group* $\mathrm{PSL}_2(p^f)$ *consists of the linear fractional transformations of the form*

$$x \mapsto \frac{ax + b}{cx + d},$$

where $a, b, c, d \in \mathrm{GF}(p^f)$ *and* $ad - bc$ *is a square in* $\mathrm{GF}(p^f)$*; by scaling the coefficients* a, b, c, d *by a common factor we may choose* $ad - bc = 1$*. The group* $\mathrm{PSL}_2(p^f)$ *has order*

$$\frac{(p^f + 1)p^f(p^f - 1)}{d},$$

where $d = (p^f - 1, 2)$*. Furthermore,* $\mathrm{PSL}_2(p^f)$ *is a 2-transitive permutation group on the* $p^f + 1$ *points of the projective line* $\mathsf{P}(1, p^f)$ *over* $\mathrm{GF}(p^f)$*. Every nonidentity element of* $\mathrm{PSL}_2(p^f)$ *fixes at most two points of* $\mathsf{P}(1, p^f)$*.*

Theorem 8.2. *Let* P *be a Sylow* p*-subgroup of* $\mathrm{PSL}_2(p^f)$*.*

a) P *is isomorphic to the additive group of* $\mathrm{GF}(p^f)$*. The elements of* P *have a common fixed point, and every nonidentity element of* P *fixes only this point.*

b) $\mathrm{PSL}_2(p^f)$ *has exactly* $p^f + 1$ *Sylow* p*-subgroups.*
c) *The intersection of any two distinct Sylow* p*-subgroups of* $\mathrm{PSL}_2(p^f)$ *is* 1.

Proof. We set $G = \mathrm{PSL}_2(p^f)$. We identify the point having homogeneous coordinates $(a, 1)$ in $\mathsf{P}(1, p^f)$ with $a \in \mathrm{GF}(p^f)$. We identify the point $(1, 0)$ with ∞.

a) The maps $g \in \mathrm{PSL}_2(p^f)$ of the form $xg = x+a$ constitute a group P isomorphic to $\mathrm{GF}(p^f)^+$, which is a Sylow p-subgroup of $\mathrm{PSL}_2(p^f)$. The point ∞ is fixed by all $g \in P$ and is the unique fixed point of all $1 \neq g \in P$.

b) Since ∞ is the only fixed point of P, the point ∞ is also fixed by all elements of $\mathbf{N}_G(P)$. So we have

$$\mathbf{N}_G(P) \leq G_\infty = \{g \in G \mid xg = a^2x + b\}.$$

Clearly, $P \trianglelefteq G_\infty$. It follows that $\mathbf{N}_G(P) = G_\infty$ and

$$[G : \mathbf{N}_G(P)] = |\mathsf{P}(1, p^f)| = p^f + 1$$

is the number of Sylow p-subgroups of $\mathrm{PSL}_2(p^f)$.

c) The elements of P^g have the unique fixed point ∞g by a). Let $1 \neq h \in P \cap P^g$. Then ∞ and ∞g are fixed by h, so we must have $\infty = \infty g$. Then $g \in G_\infty = \mathbf{N}_G(P)$ and $P = P^g$. □

Theorem 8.3.
a) *The subgroup* U *of* $G = \mathrm{PSL}_2(p^f)$ *that fixes* 0 *and* ∞ *is cyclic of order* $|U| = \frac{p^f-1}{d}$, *where* $d = (p^f - 1, 2)$.
b) *The intersection of any two distinct conjugates of* U *is* 1.
c) *For every* $1 \neq u \in U$, *the subgroup* $\mathbf{N}_G(\langle u \rangle)$ *is a dihedral group of order* $2|U|$.

Proof. a) The subgroup U consists of the linear maps u of the form $xu = a^2x$ ($a \in \mathrm{GF}(p^f)^\times$), so it has the given properties.

b) Let $x \in U \cap U^g$. The element x fixes the points 0, ∞, $0g$ and ∞g. If $U \neq U^g$, then $\{0, \infty\} \neq \{0g, \infty g\}$. So x has at least three fixed points, and it follows from Lemma 8.1 that $x = 1$.

c) Let $xu = a^2x$. If we define the element $s \in G$ by $xs = -x^{-1}$, then $\langle U, s \rangle \leq \mathbf{N}_G(\langle u \rangle)$. On the other hand, the two fixed points 0 and ∞ of u can be at most interchanged by elements of $\mathbf{N}_G(\langle u \rangle)$, so $|\mathbf{N}_G(\langle u \rangle)| \leq 2|G_{0,\infty}| = 2|U|$. Hence, we have $\mathbf{N}_G(\langle u \rangle) = U\langle s \rangle$. Since $u^s = u^{-1}$ for all $u \in U$ and $s^2 = 1$, the subgroup $\mathbf{N}_G(\langle u \rangle)$ is dihedral of order $2|U|$. □

Theorem 8.4.
a) *The group* $G = \mathrm{PSL}_2(p^f)$ *contains a cyclic subgroup* S *of order* $|S| = \frac{p^f+1}{d}$, *where* $d = (p^f - 1, 2)$.
b) *For every* $1 \neq s \in S$, *the subgroup* $\mathbf{N}_G(\langle s \rangle)$ *is a dihedral group of order* $2|S|$.
c) *The intersection of any two distinct conjugates of* S *is* 1.
d) *If* $1 \neq s \in S$, *then* s *does not fix any point of* $\mathsf{P}(1, p^f)$.

Proof. a) The subgroup $S = \overline{T} \cap \mathrm{PSL}_2(p^f)$ with the given properties was constructed in Theorem 7.3 c).

b) By the construction of S, the subgroup S has a cyclic preimage $L = \langle l \rangle$ of order $p^f + 1$ in $\mathrm{SL}_2(p^f)$. If we identify $\mathsf{V}(2, p^f)$ with $\mathrm{GF}(p^{2f})^+$, we may assume that $xl = ax$ $(a, x \in \mathrm{GF}(p^{2f}))$. Since

$$l \notin Z = \{a1 \mid a \in \mathrm{GF}(p^f)^\times\},$$

we have $a \notin \mathrm{GF}(p^f)$. Therefore, $|l| \nmid p^f - 1$ and so $\mathbf{N}_{\mathrm{SL}_2(p^f)}(L)$ consists of semilinear transformations of $\mathrm{GF}(p^{2f})$ by Theorem 7.3 a). The linear maps $g \in \mathbf{N}_{\mathrm{SL}_2(p^f)}(L)$ defined over $\mathrm{GF}(p^f)$ have the form $xg = cx$ for some $c \in \mathrm{GF}(p^f)$ and $\det g = c^{1+p^f} = 1$, so $g \in L$. Thus, $\mathbf{N}_{\mathrm{SL}_2(p^f)}(L) = \langle L, k \rangle$ for some element k of the form $xk = bx^{p^f}$. Since

$$xk^2 = b^{1+p^f} x \quad \text{with} \quad b^{1+p^f} \in \mathrm{GF}(p^f)^\times,$$

we have $k^2 \in Z$. It follows from

$$xklk = b^{1+p^f} a^{p^f} x$$

that $k^{-1}lk \equiv klk \equiv l^{p^f} \equiv l^{-1} \bmod Z$. Therefore, $\mathbf{N}_G(S)$ is a dihedral group of order $2|S|$.

Let $1 \neq s \in S$. Then $|s|$ divides $\frac{p^f+1}{d}$, and by Theorem 7.3 c) we have $\mathbf{N}_G(\langle s \rangle) = \mathbf{N}_G(S)$.

c) Assume that $1 \neq x \in S \cap S^g$. By b), it follows that

$$\mathbf{N}_G(S) = \mathbf{N}_G(\langle x \rangle) = \mathbf{N}_G(S^g).$$

If $|S| > 2$, then S is the unique cyclic subgroup of index 2 in the dihedral group $\mathbf{N}_G(S)$. It follows immediately that $S = S^g$. If $|S| = 2$, then the claim is trivially true.

d) Suppose $s \in S$ fixes a point of $\mathsf{P}(1, p^f)$. Then $|s|$ is a divisor of $|S| = \frac{p^f+1}{d}$ and of $|G_\infty| = \frac{p^f(p^f-1)}{d}$. This forces $s = 1$. □

Theorem 8.5.

a) *Let $\mathscr{P}$ be the following collection of subgroups of* $\mathrm{PSL}_2(p^f)$:

$$\mathscr{P} = \{P^x, U^x, S^x \mid x \in \mathrm{PSL}_2(p^f)\}.$$

Then every nonidentity element of $\mathrm{PSL}_2(p^f)$ *lies in exactly one of the groups in $\mathscr{P}$. (The set $\mathscr{P}$ is a so-called partition of* $\mathrm{PSL}_2(p^f)$.)

b) *Let $\mu(g)$ be the number of fixed points of the element $g \in G = \mathrm{PSL}_2(p^f)$ acting on $\mathsf{P}(1, p^f)$. We define*

$$I_j = \{g \in \mathrm{PSL}_2(p^f) \mid \mu(g) = j\}$$

for $j = 0, 1, 2$. Then

$$I_0 = \bigcup_{g\in G}(S-1)^g, \quad I_1 = \bigcup_{g\in G}(P-1)^g, \quad I_2 = \bigcup_{g\in G}(U-1)^g.$$

Proof. It follows immediately from Theorems 8.2 c), 8.3 b), 8.4 c) and the pairwise coprime orders of P, U and S that every nonidentity element of $\mathrm{PSL}_2(p^f)$ lies in at most one of the conjugates of P, U or S. Counting elements, we have

$$\begin{aligned}
&|\{1\}| + \Big|\bigcup_{g\in G}(P-1)^g\Big| + \Big|\bigcup_{g\in G}(U-1)^g\Big| + \Big|\bigcup_{g\in G}(S-1)^g\Big| \\
&= 1 + [\mathrm{PSL}_2(p^f) : \mathbf{N}_G(P)](|P|-1) + [\mathrm{PSL}_2(p^f) : \mathbf{N}_G(U)](|U|-1) \\
&\quad + [\mathrm{PSL}_2(p^f) : \mathbf{N}_G(S)](|S|-1) \\
&= 1 + (p^f+1)(p^f-1) + \frac{p^f(p^f+1)}{2}\Big(\frac{p^f-1}{d} - 1\Big) \\
&\quad + \frac{p^f(p^f-1)}{2}\Big(\frac{p^f+1}{d} - 1\Big) \\
&= \frac{(p^f+1)p^f(p^f-1)}{d} = |\mathrm{PSL}_2(p^f)|.
\end{aligned}$$

This proves a); part b) follows immediately from Theorems 8.2 a), 8.3 a) and 8.4 d).□

Remark 8.6. The groups $\mathrm{PSL}_2(p^f)$ with $p^f > 3$ and the Suzuki groups $\mathrm{Sz}(2^{2n+1})$ (considered in Volume 3) are the only simple groups with a nontrivial partition. (We will have more to say about partitions in Volume 3; see also Suzuki [408].)

Lemma 8.7. *Let* $\mathsf{K} = \mathrm{GF}(p^f)$ *and* $\mathsf{L} = \mathrm{GF}(p^{nf})$. *Then the norm map* $\mathrm{N}_{\mathsf{L}/\mathsf{K}}$ *is an epimorphism from* $\mathsf{L}^\times$ *to* $\mathsf{K}^\times$.

Proof. Clearly, $\mathrm{N}_{\mathsf{L}/\mathsf{K}}$ is a homomorphism from $\mathsf{L}^\times$ to $\mathsf{K}^\times$. For $x \in \mathsf{L}^\times$, $\mathrm{N}_{\mathsf{L}/\mathsf{K}}(x) = x^k$, where

$$k = 1 + p^f + p^{2f} + \cdots + p^{(n-1)f} = \frac{p^{nf}-1}{p^f-1}.$$

Hence, the kernel of $\mathrm{N}_{\mathsf{L}/\mathsf{K}}$ has order k. It follows that

$$|\mathrm{im}\,\mathrm{N}_{\mathsf{L}/\mathsf{K}}| = [\mathsf{L}^\times : \ker \mathrm{N}_{\mathsf{L}/\mathsf{K}}] = p^f - 1 = |\mathsf{K}^\times|.$$

So $\mathrm{N}_{\mathsf{L}/\mathsf{K}}$ maps $\mathsf{L}^\times$ onto $\mathsf{K}^\times$. □

In order to describe further subgroups of $\mathrm{PSL}_2(p^f)$, it is convenient to first discuss the unitary groups.

Lemma 8.8. *Let* $\mathsf{W} = \mathsf{V}(2, p^{2f})$ *be the vector space of dimension* 2 *over* $\mathrm{GF}(p^{2f})$. *Let* $\{w_1, w_2\}$ *be a basis for* W *over* $\mathrm{GF}(p^{2f})$. *We equip* W *with a Hermitian scalar product by defining*

$$(w, w') = x_1\overline{y}_1 + x_2\overline{y}_2,$$

where $w = x_1w_1 + x_2w_2$, $w' = y_1w_1 + y_2w_2$ *and* $\overline{x} = x^{p^f}$ *for* $x \in \mathrm{GF}(p^{2f})$. *We have* $\overline{(w, w')} = (w', w)$. *We define the special unitary group* $\mathrm{SU}_2(p^{2f})$ *to be the group of all* $\mathrm{GF}(p^{2f})$*-linear maps* g *from* W *onto itself such that* $\det g = 1$ *and* $(w, w') = (wg, w'g)$ *for all* $w, w' \in \mathsf{W}$. *We have* $\mathrm{SL}_2(p^f) \cong \mathrm{SU}_2(p^{2f})$. *With respect to the basis* $\{w_1, w_2\}$, *the group* $\mathrm{SU}_2(p^{2f})$ *consists of the matrices of the form*

$$\begin{pmatrix} a & b \\ -\overline{b} & \overline{a} \end{pmatrix},$$

where $a, b \in \mathrm{GF}(p^{2f})$ *and* $a\overline{a} + b\overline{b} = 1$.

Proof. a)[12] We first show that $\mathrm{SL}_2(p^f)$ is isomorphic to a subgroup of $\mathrm{SU}_2(p^{2f})$. By Lemma 8.7, there exists some $a \in \mathrm{GF}(p^{2f})$ such that $\mathrm{N}(a) = a\overline{a} = -1$. We would like to choose a such that $a \neq \overline{a}$. Suppose $a = \overline{a} \in \mathrm{GF}(p^f)$. Since

$$|\ker \mathrm{N}| = p^f + 1 > |\mathrm{GF}(p^f)|,$$

there exists some $b \in \ker \mathrm{N}$ such that $b \notin \mathrm{GF}(p^f)$. Then $b\overline{b} = 1$ and $b \neq \overline{b}$. It follows that $ab \notin \mathrm{GF}(p^f)$, so $ab \neq \overline{ab}$, but

$$\mathrm{N}(ab) = \mathrm{N}(a)\mathrm{N}(b) = -1.$$

Now we define

$$u_1 = w_1 + aw_2, \qquad u_2 = -aw_1 + w_2.$$

Since $a^2 \neq a\overline{a} = -1$, we have

$$\det \begin{pmatrix} 1 & a \\ -a & 1 \end{pmatrix} = 1 + a^2 \neq 0.$$

Therefore $\{u_1, u_2\}$ is a basis of W over $\mathrm{GF}(p^{2f})$. We have

$$\begin{aligned} (u_1, u_1) &= 1 + a\overline{a} = 0, \\ (u_2, u_2) &= a\overline{a} + 1 = 0, \\ (u_1, u_2) &= -\overline{a} + a = -(u_2, u_1). \end{aligned}$$

Thus, for $w = x_1u_1 + x_2u_2$ and $w' = y_1u_1 + y_2u_2$, we have

$$(w, w') = (a - \overline{a})(x_1\overline{y}_2 - x_2\overline{y}_1), \quad \text{where} \quad 0 \neq a - \overline{a} \in \mathrm{GF}(p^{2f}).$$

An easy calculation shows that the maps g of the form

[12] Correction 17 from errata runs from here until "An easy calculation shows...".

$$u_1 g = a_{11}u_1 + a_{12}u_2$$
$$u_2 g = a_{21}u_1 + a_{22}u_2,$$

where $a_{ij} \in \mathrm{GF}(p^f)$ and $a_{11}a_{22} - a_{12}a_{21} = 1$, preserve the scalar product $(_,_)$. The group of linear maps over $\mathrm{GF}(p^{2f})$ with determinant 1 that preserve $(_,_)$ is clearly $\mathrm{SU}_2(p^{2f})$. This gives rise to an embedding of $\mathrm{SL}_2(p^f)$ in $\mathrm{SU}_2(p^{2f})$.

b) Let $g \in \mathrm{SU}_2(p^{2f})$. With respect to the basis $\{w_1, w_2\}$, we may associate to g the matrix

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix}.$$

It follows from $(w_1, w_1) = (w_2, w_2) = 1$ and $(w_1, w_2) = 0$ that

$$(1) \quad a\overline{a} + b\overline{b} = 1;$$
$$(2) \quad c\overline{c} + d\overline{d} = 1;$$
$$(3) \quad a\overline{c} + b\overline{d} = 0.$$

Since $\det g = 1$, we also find that

$$(4) \quad ad - bc = 1.$$

If $a = 0$, then it follows from (1), (3) and (4) that

$$b\overline{b} = 1, \quad d = 0 \quad \text{and} \quad c = -b^{-1} = -\overline{b}.$$

If $a \neq 0$, then we obtain from (4) that

$$d = \frac{1 + bc}{a},$$

then using (3)

$$a\overline{c} + b\frac{1 + \overline{b}\overline{c}}{\overline{a}} = 0,$$

so that by (1)

$$0 = \overline{c}(a\overline{a} + b\overline{b}) + b = \overline{c} + b.$$

Finally, by (4) and (1) we conclude that

$$d = \frac{1 + bc}{a} = \frac{1 - b\overline{b}}{a} = \overline{a},$$

so g has the desired form.

The number of solutions to $a\overline{a}+b\overline{b} = 1$ with $a, b \in \mathrm{GF}(p^{2f})$ is $(p^f+1)p^f(p^f-1)$:

First, there exist exactly $p^f + 1$ solutions to

$$a\overline{a} = a^{1+p^f} = 1.$$

Furthermore, there exist $p^{2f} - p^f - 1$ elements $a \in \mathrm{GF}(p^{2f})$ such that $a\overline{a} \neq 1$; since $1 - a\overline{a} \in \mathrm{GF}(p^f)$, to every such a there exist exactly $p^f + 1$ elements b with $b\overline{b} = 1 - a\overline{a}$ by Lemma 8.7. Therefore,

$$|\mathrm{SU}_2(p^{2f})| = (p^f + 1)p^f(p^f - 1) = |\mathrm{SL}_2(p^f)|,$$

so with the help of a) we conclude that $\mathrm{SU}_2(p^{2f}) \cong \mathrm{SL}_2(p^f)$. □

Remark 8.9. Let $c \in \mathrm{GF}(p^{2f})$ be an element of order $p^f + 1$. Then

$$g = \begin{pmatrix} c & 0 \\ 0 & \overline{c} \end{pmatrix}$$

is an element of order $p^f + 1$ in $\mathrm{SU}_2(p^{2f})$. By virtue of the isomorphism in Lemma 8.8, the group $\mathrm{SL}_2(p^f)$ contains a cyclic subgroup of order $p^f + 1$. Up to conjugacy, this subgroup corresponds to the cyclic subgroup S of $\mathrm{PSL}_2(p^f)$ given in Theorem 8.4.

Theorem 8.10. *Let r be a prime number.*

a) *The Sylow r-subgroups of $\mathrm{SL}_2(p^f)$ are cyclic for $2 \neq r \neq p$, generalized quaternion for $r = 2 \neq p$, and elementary abelian for $r = p$.*

b) *For $r \neq 2$, the Sylow r-subgroups of $\mathrm{PSL}_2(p^f)$ are isomorphic to the Sylow r-subgroups of $\mathrm{SL}_2(p^f)$. For $p \neq 2$, the Sylow 2-subgroups of $\mathrm{PSL}_2(p^f)$ are dihedral groups. (Here, the elementary abelian group of type $(2, 2)$ is regarded as a dihedral group of order 4.)*

Proof. Since $|\mathrm{SL}_2(p^f)| = |\mathrm{PSL}_2(p^f)|k$ with $k = 1$ or $k = 2$, the Sylow r-subgroups of $\mathrm{SL}_2(p^f)$ and $\mathrm{PSL}_2(p^f)$ are isomorphic for $r \neq 2$. The claim regarding the Sylow p-subgroups then follows from Theorem 8.2.

Now suppose $p \neq r \neq 2$. If $r^j \nmid |\mathrm{PSL}_2(p^f)|$, then either $r^j \mid \frac{p^f+1}{k} = s$ or $r^j \mid \frac{p^f-1}{k} = u$. By Theorems 8.3 and 8.4, the group $\mathrm{PSL}_2(p^f)$ contains cyclic subgroups S and U of orders s and u, respectively. Therefore, the Sylow r-subgroups of $\mathrm{PSL}_2(p^f)$ are cyclic.

It remains to consider the case $r = 2 \neq p$. Let $2^j \nmid |\mathrm{SL}_2(p^f)|$.

First assume that $2^{j-1} \nmid p^f - 1$. Let $a \in \mathrm{GF}(p^f)^\times$ be an element of order 2^{j-1}. We define

$$g = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \quad \text{and} \quad x = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Then $g^{2^{j-1}} = 1$, $g^x = g^{-1}$ and $g^{2^{j-2}} = -1 = x^2$. Since $|\langle g, x\rangle| = 2^j$, the subgroup $\langle g, x\rangle$ is a generalized quaternion group of order 2^j (see Theorem I.14.9, pg. 96).

Now let $j \geq 3$, and assume $2^{j-1} \mid p^f + 1$. We use the fact that $\mathrm{SU}_2(p^{2f})$ is isomorphic to $\mathrm{SL}_2(p^f)$. Let $a \in \mathrm{GF}(p^{2f})^\times$ be an element of order 2^{j-1}. It follows from Lemma 8.8 that

$$g = \begin{pmatrix} a & 0 \\ 0 & \overline{a} \end{pmatrix} \quad \text{and} \quad x = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

lie in $\mathrm{SU}_2(p^{2f})$, and we have

$$g^{2^{j-1}} = 1, \quad g^{2^{j-2}} = x^2, \quad g^x = g^{-1}.$$

Hence, $\langle g, x \rangle$ is a generalized quaternion group, and it is a Sylow 2-subgroup of $\mathrm{SU}_2(p^{2f}) \cong \mathrm{SL}_2(p^f)$.

According to Theorems 8.3 and 8.4, for $p \neq 2$ the group $G = \mathrm{PSL}_2(p^f)$ contains dihedral groups $D_1 = \mathbf{N}_G(U)$ and $D_2 = \mathbf{N}_G(S)$ of orders $p^f - 1$ and $p^f + 1$, respectively. Exactly one of these numbers is divisible by 4. The corresponding group D_i $(i = 1, 2)$ contains a Sylow 2-subgroup of $\mathrm{PSL}_2(p^f)$, which is thus a dihedral group. □

Remark 8.11. It follows from Burnside's Transfer Theorem IV.2.6 that the Sylow 2-subgroups of a nonabelian simple group cannot be cyclic. (See also Exercise I.21, pg. 32.) R. Brauer and Suzuki have shown that the Sylow 2-subgroups of a simple group cannot be generalized quaternion (see Brauer, Suzuki [70]). The simple groups whose Sylow 2-subgroups are dihedral have been classified by Gorenstein and Walter. Aside from the groups $\mathrm{PSL}_2(p^f)$ with $p \neq 2$ and $p^f > 3$, only the alternating group A_7 has this property (see the series of papers by Gorenstein and Walter [180, 181, 182, 183, 184]).

In preparation for the construction of additional subgroups of $\mathrm{PSL}_2(p^f)$, we prove:

Lemma 8.12. *Let*

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

be an element of $\mathrm{SL}_2(p^f)$.
a) *If* $\operatorname{tr} g = a + d = \pm 1$, *then* $g^3 = \pm 1$.
b) *If* $\operatorname{tr} g = 0$, *then* $g^2 = -1$.

Proof. a) Let w and w^{-1} be the eigenvalues of g in $\mathrm{GF}(p^{2f})$. First assume that

$$\operatorname{tr} g = w + w^{-1} = -1.$$

It follows that $w^2 + w + 1 = 0$, so $w^3 = 1$. If the eigenvalues w and w^{-1} are distinct, then it is well known that g can be diagonalized in $\mathrm{GL}_2(p^{2f})$, and it follows that $g^3 = 1$. If $w = w^{-1}$, then $w = 1$ and $\operatorname{tr} g = 2 = -1$, so $p = 3$. Now g can be brought into lower triangular form

$$\begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}$$

and this show that $g^3 = 1$.

If $\operatorname{tr} g = 1$ then $\operatorname{tr}(-g) = -1$, and the previous paragraph demonstrates that $g^3 = -1$.

b) It follows from

$$g = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}$$

with $-a^2 - bc = 1$ that $g^2 = (a^2 + bc)1 = -1$. □

Theorem 8.13. *The group* $\mathrm{PSL}_2(p^f)$ *contains subgroups isomorphic to* A_5 *if and only if* $p = 5$ *or* $p^{2f} - 1 \equiv 0 \bmod 5$.

Proof. If $\mathrm{PSL}_2(p^f)$ contains a subgroup isomorphic to A_5, then we must have $|\mathrm{PSL}_2(p^f)| \equiv 0 \bmod 5$. This show the given conditions are necessary.

By Theorem 6.14 (3), $A_5 \cong \mathrm{PSL}_2(5)$. Since $\mathrm{PSL}_2(5)$ is isomorphic to a subgroup of $\mathrm{PSL}_2(5^f)$, the condition $p = 5$ is also sufficient. Now assume $p \neq 5$ and $p^{2f} - 1 \equiv 0 \bmod 5$.

Case 1: Assume $p^f - 1 \equiv 0 \bmod 5$, and let a be a primitive 5-th root of unity in $\mathrm{GF}(p^f)$. We define

$$g = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \quad \text{and} \quad x = \begin{pmatrix} b & c \\ d & -b \end{pmatrix}$$

with $-b^2 - cd = 1$. Then $g^5 = 1$. Since $\operatorname{tr} x = 0$, we conclude by Lemma 8.12 that $\overline{x}^2 = 1$, where $\overline{x}$ is the projective linear map induced by x. We choose b so that

$$\operatorname{tr} gx = b(a - a^{-1}) = 1,$$

and then choose c and d so the equality $-b^2 - cd = 1$ holds. Then by Lemma 8.12, $(\overline{gx})^3 = 1$ as a projective map. Consequently, we have

$$\overline{g}^5 = \overline{x}^2 = (\overline{gx})^3 = 1.$$

These are the defining relations for the alternating group A_5 by Example I.19.9 (pg. 149). So $\langle \overline{g}, \overline{x} \rangle$ is an epimorphic image of the simple group A_5. Since $\overline{g} \neq 1$, it follows that $\langle \overline{g}, \overline{x} \rangle \cong A_5$.

Case 2: Now assume that $p^f + 1 \equiv 0 \bmod 5$. By passing to the quotient by the center, Lemma 8.8 yields an isomorphism $\mathrm{PSL}_2(p^f) \cong \mathrm{PSU}_2(p^{2f})$. Let a be a primitive 5-th root of unity in $\mathrm{GF}(p^{2f})$. Since $p^f + 1 \equiv 0 \bmod 5$, we have

$$a\overline{a} = a^{1+p^f} = 1.$$

We let $b = (a - \overline{a})^{-1}$ and determine $c \in \mathrm{GF}(p^{2f})$ via the equation $c\overline{c} = 1 - b\overline{b}$; this is possible since $1 - b\overline{b} \in \mathrm{GF}(p^f)$ is the norm of some element $c \in \mathrm{GF}(p^{2f})$ by Lemma 8.7. Now setting

$$g = \begin{pmatrix} a & 0 \\ 0 & \overline{a} \end{pmatrix} \quad \text{and} \quad x = \begin{pmatrix} b & c \\ -\overline{c} & \overline{b} \end{pmatrix},$$

we have $g^5 = 1$. Since $\operatorname{tr} x = b + \overline{b} = 0$ and $\operatorname{tr} gx = b(a - \overline{a}) = 1$, we have $\overline{x}^2 = (\overline{gx})^3 = 1$ by Lemma 8.12. As in case 1, we conclude that $\langle \overline{g}, \overline{x} \rangle \cong A_5$. □

Having proved Theorem 8.13, we can deduce without difficulty the remaining isomorphism in Theorem 6.14 (6).

Theorem 8.14. *The groups* $\mathrm{PSL}_2(9)$ *and* A_6 *are isomorphic.*

Proof. By Theorem 8.13, the group $\mathrm{PSL}_2(9)$ contains a subgroup V isomorphic to A_5. The permutation representation of $\mathrm{PSL}_2(9)$ on the cosets of V is a faithful and transitive representation of $\mathrm{PSL}_2(9)$ of degree 6. By Theorem 5.1, A_6 is the only subgroup of S_6 with index 2. Since $|\mathrm{PSL}_2(9)| = |A_6| = 360$, it follows that $\mathrm{PSL}_2(9) \cong A_6$. □

Lemma 8.15. *Let D be a dihedral group of order $4m$.*[13] *If m is even, then there are exactly two conjugacy classes of subgroups of D of type $(2,2)$; if m is odd there is only one. If K is a subgroup of D of type $(2,2)$, then $\mathbf{C}_D(K) = K$.*

Proof. Let $D = \langle a, b \rangle$, where

$$a^{2m} = b^2 = 1 \quad \text{and} \quad a^b = a^{-1}.$$

The subgroups of D that have type (2,2) are those of the form

$$\langle a^m, a^i b \rangle \quad (i = 0, 1, \ldots, 2m - 1).$$

If m is odd, then these subgroups are all conjugate since they are Sylow 2-subgroups of D. If m is even, then they split into two conjugacy classes

$$\langle a^m, a^{2j} b \rangle \quad (j = 0, 1, \ldots, m - 1)$$

and

$$\langle a^m, a^{2j+1} b \rangle \quad (j = 0, 1, \ldots, m - 1).$$

Let $K = \langle a^m, a^i b \rangle$ and $x \in \mathbf{C}_D(K)$. If $x = a^j$, then it follows that

$$a^j = (a^j)^{a^i b} = (a^j)^b = a^{-j},$$

so $x \in K$. If $x = a^j b$, we have

$$a^i b = (a^i b)^{a^j b} = a^{2j-i} b.$$

Thus, $2i \equiv 2j \bmod 2m$ and so $x = a^i b$ or $x = a^{i+m} b$; in either case, $x \in K$. □

[13] Correction 18 from errata runs from here until "If m is even, then they split...".

Lemma 8.16. *Let $p \neq 2$. Then $G = \mathrm{PSL}_2(p^f)$ has exactly $\frac{|\mathrm{PSL}_2(p^f)|}{12}$ subgroups K of type $(2,2)$. If $p^{2f} - 1 \not\equiv 0 \bmod 16$, then they are all conjugate. If $p^{2f} - 1 \equiv 0 \bmod 16$, then they are distributed between two conjugacy classes of the same size. We have $\mathbf{C}_G(K) = K$, and furthermore*

$$|\mathbf{N}_G(K)| = 12 \quad \textit{for} \quad p^{2f} - 1 \not\equiv 0 \bmod 16$$
$$\textit{and}$$
$$|\mathbf{N}_G(K)| = 24 \quad \textit{for} \quad p^{2f} - 1 \equiv 0 \bmod 16.$$

Proof. Let $4m = p^f + 1$ or $4m = p^f - 1$, depending on whether $p^f + 1 \equiv 0 \bmod 4$ or $p^f - 1 \equiv 0 \bmod 4$. Let $Z = U$ if $p^f - 1 \equiv 0 \bmod 4$ (see Theorem 8.3) or $Z = S$ if $p^f + 1 \equiv 0 \bmod 4$ (see Theorem 8.4). Then Z is cyclic of order $2m$. By Theorem 8.3 or Theorem 8.4, the subgroup $\mathbf{N}_G(Z)$ is dihedral of order $4m$ and distinct conjugates of Z intersect trivially.

We count the ordered pairs of elements $x, y \in G$ that generate a group of type $(2,2)$:

Since $x^2 = 1$, it follows from the partition of $\mathrm{PSL}_2(p^f)$ in Theorem 8.5 and $(|U|, |S|) = 1$ that $x \in Z^g$ for some $g \in \mathrm{PSL}_2(p^f)$, and that every Z^g contains exactly one element of order 2. Therefore, $\mathrm{PSL}_2(p^f)$ contains exactly

$$k = [\mathrm{PSL}_2(p^f) : \mathbf{N}_G(Z)] = \frac{(p^f + 1)p^f(p^f - 1)}{8m}$$

elements of order 2, and they are all conjugate. Let x be an element of order 2 in Z. It follows from $xy = yx$ that $x = x^y \in Z \cap Z^y$, and so $y \in \mathbf{N}_G(Z)$ by Theorem 8.3 b) or Theorem 8.4 c). Since $x \neq y$, it follows that $y \notin Z$. In the dihedral group $\mathbf{N}_G(Z)$ of order $4m$, there exist exactly $2m$ elements y of the desired type, namely the elements in $\mathbf{N}_G(Z) - Z$. With this we conclude that there are

$$2km = \frac{(p^f + 1)p^f(p^f - 1)}{4}$$

pairs x, y. Since every group of type $(2,2)$ contains exactly 6 such pairs, $\mathrm{PSL}_2(p^f)$ contains exactly

$$\frac{(p^f + 1)p^f(p^f - 1)}{24} = \frac{|\mathrm{PSL}_2(p^f)|}{12}$$

subgroups of type $(2,2)$.

If $K = \langle x, y \rangle$ is a subgroup of $\mathrm{PSL}_2(p^f)$ of type $(2,2)$, then $x^g \in Z$ for some $g \in \mathrm{PSL}_2(p^f)$. As above, it follows that $y^g \in \mathbf{N}_G(Z)$, so $K^g \leq \mathbf{N}_G(Z)$. By Lemma 8.15, there exist at most two conjugacy classes of subgroups of type $(2,2)$ in $\mathrm{PSL}_2(p^f)$.

We have $\mathbf{C}_G(K) = K$:

Let $K = \langle x, y \rangle$, where $x \in Z$ and $K \leq \mathbf{N}_G(Z)$. As before, it follows from $x^g = x \in Z \cap Z^g$ that $g \in \mathbf{N}_G(Z)$ and then, with the help of Lemma 8.15,

$$g \in \mathbf{N}_G(Z) \cap \mathbf{C}_G(K) = K.$$

Thus, $\mathbf{N}_G(K)/K = \mathbf{N}_G(K)/\mathbf{C}_G(K)$ is isomorphic to a subgroup of the automorphism group of K, and $|\mathbf{N}_G(K)/K|$ is a divisor of 6.

Case 1: Assume $m \equiv 1 \bmod 2$, so $p^{2f} - 1 \not\equiv 0 \bmod 16$.

Then $4 \nmid |\mathrm{PSL}_2(p^f)|$. The subgroups of type $(2,2)$ are Sylow 2-subgroups of $\mathrm{PSL}_2(p^f)$ and are all conjugate.

That is, for $p^{2f} - 1 \not\equiv 0 \bmod 16$ there is exactly one conjugacy class of subgroups of type $(2,2)$ in $\mathrm{PSL}_2(p^f)$.

Case 2: Assume $m \equiv 0 \bmod 2$, so $p^{2f} - 1 \equiv 0 \bmod 16$.

Let K be a subgroup of type $(2,2)$ in $\mathrm{PSL}_2(p^f)$, and let $P \leq \mathrm{PSL}_2(p^f)$ be a Sylow 2-subgroup that contains K. Since $|\mathrm{PSL}_2(p^f)| \equiv 0 \bmod 8$, we have $K < P$, and $K < \mathbf{N}_P(K)$ as well by Theorem I.8.8 (pg. 41). Therefore, $|\mathbf{N}_G(K)| \equiv 0 \bmod 8$. If $\mathrm{PSL}_2(p^f)$ had only one conjugacy class of subgroups of type $(2,2)$, then we would have $|\mathrm{PSL}_2(p^f)|/12 = [\mathrm{PSL}_2(p^f) : \mathbf{N}_G(K)]$; that is, $|\mathbf{N}_G(K)| = 12$. Hence, if $m \equiv 0 \bmod 2$, then there must be exactly two conjugacy classes of subgroups of type $(2,2)$ in $\mathrm{PSL}_2(p^f)$. Let K_1 and K_2 be representatives for each of the conjugacy classes of type $(2,2)$. If we set $|\mathbf{N}_G(K_i)/K_i| = k_i$ $(i = 1, 2)$, then it follows that

$$\frac{|\mathrm{PSL}_2(p^f)|}{12} = \frac{|\mathrm{PSL}_2(p^f)|}{4k_1} + \frac{|\mathrm{PSL}_2(p^f)|}{4k_2}.$$

Since $k_i \mid 6$, this forces $k_1 = k_2 = 6$, so both conjugacy classes have the same size, and we have $|\mathbf{N}_G(K_i)| = 24$. □

Lemma 8.17.

a) *Let G be a group of order* 12 *or* 24. *If G contains a normal elementary abelian subgroup N of type* $(2,2)$ *such that $\mathbf{C}_G(N) = N$, then $G \cong A_4$ or $G \cong S_4$.*

b) *If G is a group of order* 12 *with exactly four Sylow* 3*-subgroups, then $G \cong A_4$.*

c) *If G is a group of order* 24 *with exactly four Sylow* 3*-subgroups and $\mathbf{Z}(G) = 1$, then $G \cong S_4$.*

Proof. a) We first show that N is a minimal normal subgroup of G. Let $N = \langle n_1, n_2 \rangle$, and assume for contradiction that $\langle n_1 \rangle \lhd G$. Let $g \in G$ such that $|g| = 3$. Then $n_1^g = n_1$ and $n_2^g = n_1^a n_2$ for some integer a, so $n_i^{g^2} = n_i$ for $i = 1, 2$. This yields the contradiction $g^2 \in \mathbf{C}_G(N) = N$.

Thus, N is a minimal normal subgroup of G. Then G has a faithful representation as a primitive permutation group of degree 4 by Theorem 3.3. Since $|G| = 12$ or $|G| = 24$, it follows from Remark 5.2 that either $G \cong A_4$ or $G \cong S_4$.

b) Since there are exactly 8 elements of order 3 in the four Sylow 3-subgroups of G, the group G has only one Sylow 2-subgroup, say N, and it is normal in G. Clearly, $\mathbf{C}_G(N) = N$, and then $G \cong A_4$ by a).

c) Let P be a Sylow 3-subgroup of G. Since $[G : \mathbf{N}_G(P)| = 4$, G has a representation $\mathbf{P}$ as a transitive permutation group of degree 4. We have $\ker \mathbf{P} = \bigcap_{g \in G} \mathbf{N}_G(P)^g$. Since $P \not\lhd G$, it follows that $|\ker \mathbf{P}| \leq 2$. The assumption $\mathbf{Z}(G) = 1$ then forces $\ker \mathbf{P} = 1$. Therefore, $\mathbf{P}$ is faithful and G is isomorphic to S_4. □

Theorem 8.18.

a) $\mathrm{PSL}_2(p^f)$ *has subgroups isomorphic to* S_4 *if and only if* $p^{2f} - 1 \equiv 0 \bmod 16$. *They are distributed between two conjugacy classes.*

b) $\mathrm{PSL}_2(p^f)$ *has subgroups isomorphic to* A_4 *if and only if* $p > 2$ *or* $p = 2$ *and* $f \equiv 0 \bmod 2$. *They are all conjugate if* $p^{2f} - 1 \not\equiv 0 \bmod 16$, *and they are distributed between two conjugacy classes if* $p^{2f} - 1 \equiv 0 \bmod 16$.

Proof. a) First assume that $p \neq 2$. A necessary condition for the existence of subgroups in $\mathrm{PSL}_2(p^f)$ isomorphic to S_4 is that 24 divides $|\mathrm{PSL}_2(p^f)|$. This implies that $p^{2f} - 1 \equiv 0 \bmod 16$.

If this condition is satisfied, then $G = \mathrm{PSL}_2(p^f)$ has exactly two conjugacy classes of subgroups K of type $(2, 2)$ by Lemma 8.16. We have $|\mathbf{N}_G(K)| = 24$ and $\mathbf{C}_G(K) = K$, so $\mathbf{N}_G(K) \cong S_4$ by Lemma 8.17.

If $p = 2$, then the Sylow 2-subgroups of $\mathrm{PSL}_2(2^f)$ are abelian by Theorem 8.2. However, the Sylow 2-subgroups of S_4 are nonabelian, so $\mathrm{PSL}_2(2^f)$ cannot contain subgroups isomorphic to S_4.

b) Again, first assume that $p > 2$. If $p^{2f} - 1 \equiv 0 \bmod 16$, then by Lemma 8.16 there are two conjugacy classes of subgroups K of type $(2, 2)$ in $\mathrm{PSL}_2(p^f)$. By Lemma 8.17 a), we have $\mathbf{N}_G(K) \cong S_4$ and so $\mathbf{N}_G(K)' \cong A_4$. Therefore, there exist two conjugacy classes of subgroups isomorphic to A_4.

If $p^{2f} - 1 \not\equiv 0 \bmod 16$, then by Lemma 8.16 there is one conjugacy class of subgroups of type $(2, 2)$ in $\mathrm{PSL}_2(p^f)$, and then by Lemma 8.17 a) we have $\mathbf{N}_G(K) \cong A_4$. So for $p^{2f} - 1 \not\equiv 0 \bmod 16$, there is exactly one conjugacy class of subgroups in $\mathrm{PSL}_2(p^f)$ isomorphic to A_4.

It remains to consider the case that $p = 2$. Since $\mathrm{PSL}_2(4) \cong A_5$ (see Theorem 6.14 (3)), the group $\mathrm{PSL}_2(4)$ certainly contains subgroups isomorphic to A_4. From the embedding of $\mathrm{PSL}_2(4)$ in $\mathrm{PSL}_2(2^{2m})$ (by virtue of the embedding of fields) it follows that $\mathrm{PSL}_2(2^{2m})$ also has subgroups isomorphic to A_4.

Next, we show that the condition $f \equiv 0 \bmod 2$ is necessary for the existence of subgroups in $\mathrm{PSL}_2(2^f)$ that are isomorphic to A_4. Suppose $A_4 \cong T < \mathrm{PSL}_2(2^f)$, and let $K < T$ be the normal subgroup of order 4 in T. By Theorem 8.2, we may assume (taking a conjugate if necessary) that $K = \langle k_1, k_2 \rangle$ with

$$k_i = \begin{pmatrix} 1 & 0 \\ a_i & 1 \end{pmatrix}.$$

Let $x \in T$ be an element of order 3. Since $x \in \mathbf{N}_G(K)$, it follows that x has the form

$$x = \begin{pmatrix} c & 0 \\ d & e \end{pmatrix}.$$

The equation $x^3 = 1$ (considered as a linear fractional transformation) forces $c^3 = e^3$, so $(ce^{-1})^3 = 1$. If f is odd, then $3 \nmid 2^f - 1 = |\mathrm{GF}(2^f)^\times|$, and so $c = e$. But then x has order 1 or 2, contradicting our assumption.

It remains to show that for $f \equiv 0 \bmod 2$, all the subgroups of $\mathrm{PSL}_2(2^f)$ that are isomorphic to A_4 are conjugate. Let

$$A_4 \cong \langle k_1, k_2, x\rangle < \mathrm{SL}_2(2^f),$$

where $k_i^2 = x^3 = 1$, $k_1^x = k_2$, $k_2^x = k_1 k_2$. By Theorem 8.2, we may pass to conjugates of the form

$$k_1 = \begin{pmatrix} 1 & 0 \\ x_1 & 1 \end{pmatrix} \quad \text{and} \quad k_2 = \begin{pmatrix} 1 & 0 \\ x_2 & 1 \end{pmatrix}.$$

Since

$$\begin{pmatrix} c^{-1} & 0 \\ 0 & c \end{pmatrix} \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix} \begin{pmatrix} c & 0 \\ 0 & c^{-1} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ c^2 a & 1 \end{pmatrix}$$

and every element in $\mathrm{GF}(2^f)$ is a square, we may force $x_1 = 1$. Then x has the form

$$x = \begin{pmatrix} a & 0 \\ b & a^{-1} \end{pmatrix}.$$

It follows from $k_1 x = x k_2$ that $a^2 = x_2$, so $x_2^3 = 1$. Therefore, x_2 is uniquely determined up to inversion. Since $x_2^{-1} = 1 + x_2$ in the field $\mathrm{GF}(2^f)$ of characteristic 2, we have

$$\begin{pmatrix} 1 & 0 \\ x_2^{-1} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ x_2 & 1 \end{pmatrix} \in \langle k_1, k_2\rangle.$$

Hence, for $f \equiv 0 \bmod 2$ there is exactly one group K of type $(2, 2)$ that contains

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

and can be embedded in a subgroup of $\mathrm{PSL}_2(2^f)$ isomorphic to A_4. From

$$t = \begin{pmatrix} 1 & 0 \\ y & 1 \end{pmatrix} \in \mathbf{C}_G(K) \quad \text{and} \quad txt^{-1} = \begin{pmatrix} a & 0 \\ b' & a^{-1} \end{pmatrix},$$

where $b' = b + y(a - a^{-1}) = 0$ for an appropriate value of y, it follows that all subgroups of $\mathrm{PSL}_2(2^f)$ isomorphic to A_4 are conjugate to $\langle k_1, k_2, g\rangle$, where

$$k_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad k_2 = \begin{pmatrix} 1 & 0 \\ x_2 & 1 \end{pmatrix}, \quad g = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}$$

and $x_2^3 = a^3 = 1$. □

Remark 8.19. The two conjugacy classes of subgroups isomorphic to S_4 in the simple group $\mathrm{PSL}_2(7)$ of order 168 may be interpreted geometrically. By Theorem 6.14 (4), we have $\mathrm{PSL}_2(7) \cong \mathrm{PSL}_3(2)$. The group $\mathrm{PSL}_3(2)$ is a 2-transtive permutation group on the 7 points of the projective plane $\mathsf{P}(2, 2)$ by Theorem 6.11; it is also 2-transitive on the 7 lines in this plane. The stabilizers of points, resp. lines,

constitute the two conjugacy classes of subgroups isomorphic to S_4. The group $\mathrm{PSL}_3(2) \cong \mathrm{GL}_3(2)$ admits an outer automorphism α given by $g^\alpha = (g^T)^{-1}$, where g^T is the transposed matrix of g. This automorphism interchanges the two conjugacy classes of subgroups of index 7.

Theorem 8.20.
a) *If* $m \mid f$, *then* $\mathrm{PSL}_2(p^f)$ *contains subgroups isomorphic to* $\mathrm{PSL}_2(p^m)$.
b) *If* $2m \mid f$, *then* $\mathrm{PSL}_2(p^f)$ *contains subgroups isomorphic to* $\mathrm{PGL}_2(p^m)$.

Proof. a) This claim follows immediately from $\mathrm{GF}(p^m) \subseteq \mathrm{GF}(p^f)$.

b) We certainly have $\mathrm{PGL}_2(p^m) \leq \mathrm{PGL}_2(p^f)$ by virtue of the embedding of $\mathrm{GF}(p^m)$ in $\mathrm{GF}(p^f)$. We need to show that for $f = rm \equiv 0 \bmod 2m$, the elements of $\mathrm{GF}(p^m)^\times$ are squares in $\mathrm{GF}(p^f)^\times$. This is trivial for $p = 2$ since every element is a square in this case. For $p > 2$, this follows from

$$[\mathrm{GF}(p^{rm})^\times : \mathrm{GF}(p^m)^\times] = \frac{p^{rm}-1}{p^m-1} = 1 + p^m + \cdots + p^{(r-1)m} \equiv r \equiv 0 \bmod 2. \quad \square$$

Now we will show that every subgroup of $\mathrm{PSL}_2(p^f)$ belongs to one of the classes we have described. The key to the proof is the partition of $\mathrm{PSL}_2(p^f)$ in Theorem 8.5.

We begin by proving two lemmas.

Lemma 8.21. *Let* G *be a subgroup of* $\mathrm{PSL}_2(p^f)$, *let* $P > 1$ *be a Sylow* p*-subgroup of* G *and let* $N = \mathbf{N}_G(P)$. *Then* P *is isomorphic to a subgroup of the additive group* $\mathrm{GF}(p^f)^+$ *of* $\mathrm{GF}(p^f)$. *There is exactly one point* x *on the projective line* $\mathsf{P}(1, p^f)$ *that is fixed by all elements of* P. *There exists a subgroup* Z *of* G *such that* $N = PZ$ *and* $P \cap Z = 1$. *We have* $Z = 1$ *or* Z *is a maximal cyclic subgroup of* G. *The point* x *is fixed by* Z, *and we have that* $|Z|$ *divides* $|P| - 1$ *and* $|Z|$ *divides* $p^f - 1$.

Proof. The Sylow p-subgroup P of G is contained in a Sylow p-subgroup of $\mathrm{PSL}_2(p^f)$. It therefore follows by Theorem 8.2 that P is isomorphic to a subgroup of $\mathrm{GF}(p^f)^+$ and that there is exactly one point $x \in \mathsf{P}(1, p^f)$ fixed by all of P. Hence, we may assume that the elements in P are represented by matrices of the form

$$\begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix}.$$

Then $(1, 0)$ is the fixed point of P, and we may assume the elements in N are represented by matrices of the form

$$\begin{pmatrix} 1 & 0 \\ c & b \end{pmatrix}.$$

The elements b that appear constitute a subgroup B of $\mathrm{GF}(p^f)^\times$. Let $B = \langle b \rangle$ and

$$z = \begin{pmatrix} 1 & 0 \\ c & b \end{pmatrix} \in N.$$

Suppose that $|B| = s > 1$. Then $s \mid p^f - 1$, and since

$$1 + b + \cdots + b^{s-1} = 0$$

we have $z^s = 1$. Defining $Z = \langle z \rangle$, we have $Z \cap P = 1$, $N = PZ$ and $|Z| = s \mid p^f - 1$.

Let $Z \neq 1$, and let $Z_1 = \langle z_1 \rangle$ be a maximal cyclic subgroup of G with $Z \leq Z_1$. Since $\mu(z) \geq 1$ (see Theorem 8.5), $|z| = s \mid p^f - 1$ and z lies in one of the conjugates of U in the partition from Theorem 8.5. Since every cyclic subgroup of $\mathrm{PSL}_2(p^f)$ lies in exactly one component of the partition, the subgroup Z_1 also lies in a conjugate of U. Consequently, the fixed point $(1, 0)$ of P is also fixed by z_1. By Theorem 8.2, the subgroups P and P^{z_1}, which both fix $(1, 0)$, are contained in the same Sylow p-subgroup of $\mathrm{PSL}_2(p^f)$ and therefore commute. This gives $P = P^{z_1}$, and so

$$Z_1 \leq N = \mathbf{N}_G(P).$$

It remains to show that $|Z|$ divides $|P| - 1$: If

$$1 \neq x = \begin{pmatrix} 1 & 0 \\ c & 1 \end{pmatrix} \in P,$$

then $\mathbf{C}_G(x) = P$. Therefore, $P - 1$ decomposes under the conjugation action of Z into conjugacy classes of size $|Z|$, and so $|Z|$ divides $|P| - 1$. □

Lemma 8.22. *Let G be a subgroup of $\mathrm{PSL}_2(p^f)$ with $|G| = n$. Let $Z_1, \ldots, Z_r$ be representatives of the conjugacy classes of maximal cyclic subgroups of G with order coprime to p. We let $|Z_i| = z_i$ and $|\mathbf{N}_G(Z_i)| = z_i s_i$. Let P be a Sylow p-subgroup of G and $|P| = p^m$. With a suitable labeling of the Z_i, we have $|\mathbf{N}_G(P)| = p^m$ or $|\mathbf{N}_G(P)| = p^m z_1$. Furthermore, $s_i \leq 2$, the z_i are divisors of $\frac{p^f \pm 1}{d}$ with $d = (p^f - 1, 2)$, and we have*

$$n = 1 + \frac{(p^m - 1)n}{p^m z_1} + \sum_{i=1}^{r} \frac{(z_i - 1)n}{z_i s_i}. \qquad (*)$$

If $\mathbf{N}_G(P) = P$, then $\frac{(p^m-1)n}{p^m z_1}$ should be replaced by $\frac{(p^m-1)n}{p^m}$.

Proof. By Theorem 8.5, the conjugates of P and the cyclic subgroups $U^x \cap G$ and $S^x \cap G$ ($x \in \mathrm{PSL}_2(p^f)$) that are nontrivial constitute a partition of G. Let $Z_1, \ldots, Z_r$ be representatives of the G-conjugacy classes of those $U^x \cap G$ and $S^x \cap G$ that are distinct from 1. (Note that $U^x \cap G$ and $U^y \cap G$ are not necessarily conjugate in G!) Then the Z_i are maximal cyclic subgroups of G. It follows from Theorem 8.3 c) that

$$\mathbf{N}_G(U^x \cap G) = \mathbf{N}_{\mathrm{PSL}_2(p^f)}(U^x) \cap G$$

and therefore

$$\begin{aligned} [\mathbf{N}_G(U^x \cap G) : U^x \cap G] &= [\mathbf{N}_{\mathrm{PSL}_2(p^f)}(U^x) \cap G : U^x \cap G] \\ &\leq [\mathbf{N}_{\mathrm{PSL}_2(p^f)}(U^x) : U^x] = 2. \end{aligned}$$

We proceed in the same way for S with the help of Theorem 8.4 b), and we conclude that $s_i \leq 2$. The statement regarding $\mathbf{N}_G(P)$ is in Lemma 8.21. Since

$$\Big|\bigcup_{g\in G}(P-1)^g\Big| = \frac{(p^m-1)n}{p^m z_1}$$

and

$$\Big|\bigcup_{g\in G}(Z_i-1)^g\Big| = \frac{(z_i-1)n}{z_i s_i},$$

counting the elements of G gives equation $(*)$. □

Our next concern is to examine the equation $(*)$ in Lemma 8.22 more closely. We must distinguish between different cases:

8.23 (Case 1). Let $p^m > 1$ and $\mathbf{N}_G(P) = P$. Then one of the following must occur:
(1) $n = p^m$, and G is an elementary abelian p-group.
(2) $p^m = 2$, $n = 2z_1$ with $2 \nmid z_1$, and G is a dihedral group.
(3) $p^m = 3$, and G is isomorphic to the alternating group A_4.

Proof. We have $z_i \geq 2$ and $s_i \leq 2$. We obtain from $(*)$ that

$$0 = 1 - \frac{n}{p^m} + \sum_{i=1}^{r}\frac{(z_i-1)n}{z_i s_i} > -\frac{n}{p^m} + \frac{rn}{4}.$$

It follows that $r < 4p^{-m} \leq 2$, and so $r \leq 1$.

If $r = 0$, then $n = p^m$, which is case (1). If $r = 1$, then $p^m < 4$, so $p^m = 2$ or $p^m = 3$.

The assumption $p^m = 2$ implies

$$0 = 1 - \frac{n}{2} + \frac{(z_1-1)n}{z_1 s_1},$$

so

$$\frac{n}{2} > \frac{(z_1-1)n}{z_1 s_1} \geq \frac{n}{2s_1}$$

and then $s_1 = 2$ since $s_1 \leq 2$. This gives

$$n = 2z_1 = |\mathbf{N}_G(Z_1)|,$$

and so G is a dihedral group.

Now assume $p^m = 3$. Then we have $\frac{1}{2s_1} \leq \frac{(z_1-1)}{z_1 s_1} < \frac{1}{3}$, so $z_1 = 2$ and $s_1 = 2$. This implies $n = 12$. Since the Sylow 3-subgroups of G are equal to their normalizers by assumption, G has exactly four Sylow 3-subgroups and so is isomorphic to A_4 by Lemma 8.17 b). This is case (3). □

8.24 (Case 2). Let $p^m = 1$, so p does not divide n. Then one of the following must occur:
(4) G is cyclic of order z_1 and $z_1 \Big| \frac{p^f \pm 1}{d}$.
(5) G is a dihedral group of order $2z_1$, and z_1 is as in (4).
(6) G is isomorphic to A_4.
(7) G is isomorphic to S_4.
(8) G is isomorphic to A_5.

Proof. Let $s_i = 1$ for $1 \le i \le t$ and $s_i = 2$ for $t+1 \le i \le t+s$, so $r = s+t$. We have

$$\begin{aligned} 1 &= \frac{1}{n} + \sum_{i=1}^{t} \left(1 - \frac{1}{z_i}\right) + \sum_{i=t+1}^{t+s} \left(\frac{1}{2} - \frac{1}{2z_i}\right) \\ &= \frac{1}{n} + t + \frac{s}{2} - \sum_{i=1}^{t} \frac{1}{z_i} - \sum_{i=t+1}^{t+s} \frac{1}{2z_i} \\ &\ge \frac{1}{n} + t + \frac{s}{2} - \frac{t}{2} - \frac{s}{4} = \frac{1}{n} + \frac{t}{2} + \frac{s}{4}. \end{aligned}$$

This implies

$$\frac{t}{2} + \frac{s}{4} < 1, \qquad (**)$$

so $t \le 1$ and $s \le 3$.

If $t = 1$, then we obtain $\frac{s}{4} < \frac{1}{2}$ from $(**)$, so $s \le 1$. As a consequence, we need to consider only the following possibilities:

$$(s,t) = (0,1), (1,1), (1,0), (2,0), (3,0).$$

a) If $r = s+t = 1$, then the group G is covered by the conjugates of Z_1. Therefore, $G = Z_1$ (see Exercise I.4, pg. 12), which is case (4).

b) Assume $s = t = 1$, so $s_1 = 1$ and $s_2 = 2$. Then we have

$$n = 1 + \frac{(z_1 - 1)n}{z_1} + \frac{(z_2 - 1)n}{2z_2}$$

and therefore

$$1 > \frac{z_1 - 1}{z_1} + \frac{z_2 - 1}{2z_2}.$$

If $z_1 = 2$, then $n = 2z_2$. Since $s_2 = 2$, the group $G = \mathbf{N}_G(Z_2)$ is dihedral. This is case (5).

If, however, $z_1 > 2$, then it follows that $\frac{z_1 - 1}{z_1} \ge \frac{2}{3}$, therefore $\frac{z_2 - 1}{2z_2} < \frac{1}{3}$ and $z_2 = 2$. Then there remains

$$\frac{z_1 - 1}{z_1} < \frac{3}{4},$$

so $z_1 = 3$. This implies $n = 12$. Then G has exactly $\frac{n}{z_1} = 4$ subgroups of order 3, and therefore G is isomorphic to A_4 by Lemma 8.17 b). This is case (6).

c) Let $s = 2$ and $t = 0$. Then

$$n = 1 + \frac{(z_1 - 1)n}{2z_1} + \frac{(z_2 - 1)n}{2z_2},$$

so

$$n = \frac{2z_1 z_2}{z_1 + z_2}.$$

Since $2z_1 = |\mathbf{N}_G(Z_1)|$ is a divisor of n, $\frac{z_2}{z_1+z_2}$ must be an integer. This is impossible. So this case does not occur.

d) It remains to consider the case $s = 3$, $t = 0$. We have

$$\sum_{i=1}^{3} \frac{(z_i - 1)}{z_i} = 2\left(1 - \frac{1}{n}\right) < 2.$$

Let $z_1 \geq z_2 \geq z_3$. Then $\frac{z_3-1}{z_3} < \frac{2}{3}$, so $z_3 = 2$. There remains

$$\frac{z_1 - 1}{z_1} + \frac{z_2 - 1}{z_2} < \frac{3}{2}.$$

This forces $2 \leq z_2 \leq 3$.

Here, $z_2 = 2$ implies $n = 2z_1$, and then $G = \mathbf{N}_G(Z_1)$ is a dihedral group. (At the moment, z_1 is even and there are two conjugacy classes of involutions outside of Z_1, these correspond to $z_2 = z_3 = 2$.)

On the other hand, $z_2 = 3$ implies $\frac{z_1-1}{z_1} < \frac{5}{6}$, so $3 \leq z_1 \leq 5$.

If $z_1 = z_2 = 3$, then we would have $n = 12$ and G would have two conjugacy classes of cyclic subgroups of order 3. This is impossible by Sylow's theorem.

It remains to consider $z_1 = 4$ and $z_1 = 5$. If $z_1 = 4$, then we find $n = 24$. Then G has exactly $\frac{n}{2z_2} = 4$ subgroups of order 3. If G had central elements of order 2, then it would also have elements of order 6. Then one of the z_i would be divisible by 6, which is not the case. Therefore, G is isomorphic to S_4 by Lemma 8.17 c). This is case (7).

If $z_1 = 5$, $z_2 = 3$ and $z_3 = 2$, then we find $n = 60$. Now G has exactly $\frac{n}{2z_1} = 6$ Sylow 5-subgroups, so G admits a transitive permutation representation of degree 6. This representation is faithful, since otherwise G would contain cyclic subgroups of order 10.

The next lemma finishes the proof that $G \cong A_5$. □

Lemma 8.25. *Let G be a transitive permutation group of degree 6 and order 60. Then $G \cong A_5$.*

Proof. The elements of order 5 in G leave one point fixed and permute the rest of the points in a single 5-cycle. Therefore, G is 2-transitive, so certainly primitive. As a primitive group of degree 6 and order 60, G is isomorphic to A_5 by Theorem 4.7.□

8.26 (Case 3). Now assume $p^m > 1$ and $\mathbf{N}_G(P) > P$. This leads to the following subgroups of $\mathrm{PSL}_2(p^f)$:

(9) $n = p^m z_1$, and G is a semidirect product of its elementary abelian Sylow p-subgroup with a cyclic group of order z_1.
(10) $p^m = 3$ and $G \cong A_5$.
(11) $G \cong \mathrm{PGL}_2(p^m)$ with $2m \mid f$.
(12) $G \cong \mathrm{PSL}_2(p^m)$ with $m \mid f$.

Proof. By Lemma 8.21, we have $|\mathbf{N}_G(P)| = p^m z_1$ with $1 < z_1 \leq p^m - 1$, so

$$\frac{(p^m - 1)n}{p^m z_1} \geq \frac{n}{z_1 + 1}.$$

With the help of equation $(*)$ from Lemma 8.22, we have

$$n > \frac{n}{z_1 + 1} + \frac{(z_1 - 1)n}{z_1 s_1} + \sum_{i=2}^{r} \frac{(z_i - 1)n}{z_i s_i}.$$

a) Assuming $s_1 = 1$, we obtain

$$\begin{aligned} n &> \frac{n}{z_1 + 1} + \frac{(z_1 - 1)n}{z_1} + \sum_{i=2}^{r} \frac{(z_i - 1)n}{z_i s_i} \\ &\geq \frac{(z_1^2 + z_1 - 1)n}{z_1(z_1 + 1)} + \frac{(r - 1)n}{4} \geq \frac{5n}{6} + \frac{(r - 1)n}{4}. \end{aligned}$$

It follows that $\frac{r-1}{4} < \frac{1}{6}$, so $r = 1$, and $s_1 = 1$ implies $n = p^m z_1 = |\mathbf{N}_G(P)|$. This is case (9).

b) Now assume $s_1 = 2$. This gives

$$\begin{aligned} n &> \frac{(z_1^2 + 2z_1 - 1)n}{2z_1(z_1 + 1)} + \frac{(r - 1)n}{4} \\ &> \frac{(z_1^2 + z_1)n}{2z_1(z_1 + 1)} + \frac{(r - 1)n}{4} = \frac{n}{2} + \frac{(r - 1)n}{4}, \end{aligned}$$

so $\frac{r-1}{4} < \frac{1}{2}$ and thus $r \leq 2$. Assume for contradiction that $r = 1$. Then

$$n = 1 + \frac{(p^m - 1)n}{p^m z_1} + \frac{(z_1 - 1)n}{2z_1},$$

so $2p^m z_1 = n(p^m z_1 - p^m + 2)$.

Since $p^m z_1 \mid n$, it necessarily follows that $p^m(z_1 - 1) + 2 \leq 2$, which is impossible since $z_1 > 1$. So $r = 2$. We have

$$n = 1 + \frac{(p^m - 1)n}{p^m z_1} + \frac{(z_1 - 1)n}{2z_1} + \frac{(z_2 - 1)n}{s_2 z_2}.$$

Assume for contradiction that $s_2 = 1$. Then we have

$$n = 1 + \frac{n}{z_1} - \frac{n}{p^m z_1} + \frac{n}{2} - \frac{n}{2z_1} + n - \frac{n}{z_2},$$

but since $p^m \geq 2$ and $z_2 \geq 2$, this yields the contradiction

$$0 = 1 + \frac{n}{2z_1} - \frac{n}{p^m z_1} + \frac{n}{2} - \frac{n}{z_2} \geq 1.$$

Therefore, $r = 2$ and $s_1 = s_2 = 2$.

c) (z_1, z_2) is a divisor of 2:

The $\frac{n}{2z_1}$ conjugates Z_1^x of Z_1 are permuted by the elements of Z_1. There exists an orbit of size 1, namely Z_1 itself. The orbit of some Z_1^x with $Z_1^x \neq Z_1$ has size

$$[Z_1 : Z_1 \cap \mathbf{N}_G(Z_1^x)].$$

However, $Z_1 \cap Z_1^x = 1$ for $Z_1^x \neq Z_1$, and since

$$[\mathbf{N}_G(Z_1^x) : Z_1^x] = s_1 = 2,$$

we have $|Z_1 \cap \mathbf{N}_G(Z_1^x)| \leq 2$. It follows that

$$[Z_1 : Z_1 \cap \mathbf{N}_G(Z_1^x)] = z_1 \quad \text{or} \quad [Z_1 : Z_1 \cap \mathbf{N}_G(Z_1^x)] = \frac{z_1}{2}.$$

Therefore,

$$\frac{n}{2z_1} = 1 + \frac{jz_1}{2}$$

for some integer j.

The Z_1^x are also permuted by elements of Z_2. Since $|Z_2 \cap \mathbf{N}_G(Z_1^x)| \leq 2$, the orbits have sizes z_2 or $\frac{z_2}{2}$. This gives

$$1 + \frac{jz_1}{2} = \frac{j'z_2}{2}$$

for some integer j'. It follows immediately that the greatest common divisor (z_1, z_2) of z_1 and z_2 is a divisor of 2.

d) Let c be the least common multiple of p^m, $2z_1$ and $2z_2$ (so in particular, $c = p^m z_1 z_2$ for $p = 2$). We claim $n = c$:

It follows from the equation

$$n = 1 + \frac{(p^m - 1)n}{p^m z_1} + \frac{(z_1 - 1)n}{2z_1} + \frac{(z_2 - 1)n}{2z_2}$$

that $n = 1 + \frac{jn}{c}$ for some integer j. So n and $\frac{n}{c}$ are coprime, and $n = c$.

e) Now assume that $n = p^m z_1 z_2$. This gives

$$p^m z_1 z_2 = 1 + (p^m - 1)z_2 + \frac{(z_1 - 1)p^m z_2}{2} + \frac{(z_2 - 1)p^m z_1}{2}$$
$$= 1 + (p^m - 1)z_2 + p^m z_1 z_2 - \frac{p^m(z_1 + z_2)}{2},$$

and therefore $(z_2 - z_1)p^m = 2(z_2 - 1)$.

If $p \neq 2$, it follows immediately that $z_2 = 1 + kp^m$ for some k. If $p = 2$, we see that z_1 and z_2 are odd since they are divisors of $2^{2f} - 1$, so $z_2 - z_1$ is even and we obtain the same statement. This yields

$$(z_2 - z_1)p^m = 2kp^m$$

and then, with the help of Lemma 8.21,

$$z_1 = z_2 - 2k = k(p^m - 2) + 1 \leq p^m - 1.$$

The inequality $1 < z_1 \leq p^m - 1$ forces $k = 1$, then $z_1 = p^m - 1$, $z_2 = p^m + 1$ and

$$n = (p^m + 1)p^m(p^m - 1).$$

f) Now assume that $n = 2p^m z_1 z_2$, so certainly $p \neq 2$. We then have

$$2p^m z_1 z_2 = 1 + (p^m - 1)2z_2 + (z_1 - 1)p^m z_2 + (z_2 - 1)p^m z_1,$$

so $p^m(z_2 - z_1) = 2z_2 - 1$.

If we let $k = z_2 - z_1$, then $z_2 = \frac{kp^m+1}{2}$. Since $p \neq 2$, k is odd. We then have

$$z_1 = z_2 - k = \frac{kp^m + 1}{2} - k \leq p^m - 1$$

and then, since $3 \leq p^m$, we conclude $k \leq 3$. So we need to consider the cases $k = 1$ and $k = 3$.

The case $k = 1$ yields $z_2 = \frac{p^m+1}{2}$, $z_1 = \frac{p^m-1}{2}$ and then

$$n = 2p^m z_1 z_2 = \frac{(p^m + 1)p^m(p^m - 1)}{2}.$$

If $k = 3$, we obtain

$$\frac{3p^m}{2} - \frac{5}{2} \leq p^m - 1,$$

thus $p^m = 3$. It follows that $z_1 = 2$, $z_2 = z_1 + k = 5$ and $n = 60$. Then G has exactly $\frac{n}{2z_2} = 6$ Sylow 5-subgroups, and using Lemma 8.25 we conclude as before that $G \cong A_5$.

g) It remains to clarify the structure of the groups appearing in e) and f) of orders $(p^m + 1)p^m(p^m - 1)$ and $\frac{(p^m+1)p^m(p^m-1)}{2}$. In these cases, we found that $z_1 = p^m - 1$ or $z_1 = \frac{p^m-1}{2}$. By Lemma 8.21, we have $z_1 \mid p^f - 1$. It follows that $p^m - 1 \mid 2(p^f - 1)$.

Writing $f = tm + e$ with $0 \leq e < m$, we have $p^m - 1 \mid 2(p^e - 1)$. The inequalities

$$2(p^e - 1) \leq p^{e+1} - 2 \leq p^m - 2$$

force $e = 0$, so $m \mid f$.

By our previous considerations of the current case, G has exactly $\frac{n}{p^m z_1} = p^m + 1$ Sylow p-subgroups. Each of these fixes exactly one point of the projective line $\mathsf{P}(1, p^f)$ by Lemma 8.21. The set $\mathscr{P}'$ of these $p^m + 1$ fixed points is mapped onto itself by G. We now consider G as a permutation group on $\mathscr{P}'$. Since every nonidentity element of G has at most two fixed points, G acts faithfully on $\mathscr{P}'$. Using an appropriate transformation by an element in the 3-transitive group $\mathrm{PGL}_2(p^f)$, we may assume that the points $(0, 1)$, $(1, 0)$ and $(1, 1)$ lie in $\mathscr{P}'$.

As all the Sylow p-subgroups of G are conjugate, G acts transitively on $\mathscr{P}'$. Let P be a Sylow p-subgroup of G that fixes the point ∞ with coordinates $(1, 0)$. Then $\mathbf{N}_G(P) \leq G_\infty$ and

$$[G : \mathbf{N}_G(P)] = \frac{n}{p^m z_1} = p^m + 1 = [G : G_\infty].$$

So we have $G_\infty = \mathbf{N}_G(P)$.

Since the nonidentity elements of P fix only the point ∞ in $\mathsf{P}(1, p^f)$, the group P acts regularly on $\mathscr{P}' - \{\infty\}$. So G acts 2-transitively on $\mathscr{P}'$. Therefore, there exists some element $h \in G$ such that $(0, 1)^h = \infty$ and $\infty^h = (0, 1)$.

We have $\mathbf{N}_G(P) = PZ_1$. By Theorem 8.2, the only elements of $\mathrm{PSL}_2(p^f)$ with exactly one fixed point are those of order p. Therefore, Z_1 fixes another point $x \in \mathsf{P}(1, p^f)$ that is distinct from ∞. If x does not lie in $\mathscr{P}'$, then $\mathscr{P}' - \{\infty\}$ decomposes into orbits of size z_1 under Z_1. This gives the contradiction

$$z_1 \quad \text{divides} \quad |\mathscr{P}' - \{\infty\}| = p^m.$$

So we must have $x \in \mathscr{P}'$. By the transitivity of P on $\mathscr{P}' - \{\infty\}$, we may pass to a conjugate of Z_1 in $\mathbf{N}_G(P)$ if necessary to ensure that ∞ and $(0, 1)$ are fixed by Z_1. Then the cyclic group Z_1 is generated by an element g of the form

$$g = \begin{pmatrix} 1 & 0 \\ 0 & r \end{pmatrix}.$$

Since $|g| = |r| = z_1 \mid p^m - 1$ and $m \mid f$, we have $r \in \mathrm{GF}(p^m)$. Since P acts regularly on the points of $\mathscr{P}' - \{\infty\}$, there exists exactly one $t \in P$ such that $(0, 1)^t = (1, 1)$. Then

$$t = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

We have

$$P = \left\{ \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix} \;\middle|\; x \in W \right\},$$

where W is a subgroup of the additive group $\mathrm{GF}(p^f)^+$ of order p^m. Since

$$g^{-j}tg^j = \begin{pmatrix} 1 & 0 \\ r^{-j} & 1 \end{pmatrix} \in P,$$

all r^{-j} lie in W. Thus, W contains at least

$$1 + z_1 \geq 1 + \frac{p^m - 1}{2} > p^{m-1}$$

elements of $\mathrm{GF}(p^m)$. So $W = \mathrm{GF}(p^m)^+$ and then

$$\mathscr{P}' = \{\infty, (1,1)^g \mid g \in P\} = \{(1,0), (x,1) \mid x \in \mathrm{GF}(p^m)\}.$$

The element h with $(0,1)^h = \infty$ and $\infty^h = (0,1)$ has the form

$$h = \begin{pmatrix} 0 & 1 \\ s & 0 \end{pmatrix}.$$

Since $(1,1)^h = (s,1)$, it follows that $s \in \mathrm{GF}(p^m)$. Consequently, we have

$$\langle PZ_1, h \rangle \leq \mathrm{PGL}_2(p^m).$$

Noting that $PZ_1 = G_\infty$ is a maximal subgroup of G by the 2-transitivity of G on $\mathscr{P}'$, we have $G = \langle PZ_1, h \rangle$. Since

$$|G| = (p^m + 1)p^m(p^m - 1) \quad \text{or} \quad |G| = \frac{(p^m + 1)p^m(p^m - 1)}{2},$$

the claim now follows. □

We collect the previous results together:

Main Theorem 8.27 (Dickson). *The group* $\mathrm{PSL}_2(p^f)$ *contains only the following subgroups:*

(1) *Elementary abelian p-groups.*
(2) *Cyclic groups of order z, where* $z \mid \frac{p^f \pm 1}{d}$ *and* $d = (p^f - 1, 2)$.
(3) *Dihedral groups of order* $2z$ *with z as in* (2).
(4) *Alternating groups* A_4 *for* $p > 2$ *or* $p = 2$ *and* $f \equiv 0 \bmod 2$.
(5) *Symmetric groups* S_4 *for* $p^{2f} - 1 \equiv 0 \bmod 16$.
(6) *Alternating groups* A_5 *for* $p = 5$ *or* $p^{2f} - 1 \equiv 0 \bmod 5$.
(7) *Semidirect products of elementary abelian groups of order* p^m *with cyclic groups of order t, where* $t \mid \frac{p^m - 1}{d}$ *and* $t \mid p^f - 1$.[14]
(8) *Groups* $\mathrm{PSL}_2(p^m)$ *for* $m \mid f$ *and* $\mathrm{PGL}_2(p^m)$ *for* $2m \mid f$.

We easily obtain the following result from Theorem 8.27. It was already known to Galois for $f = 1$.

[14] Correction 19 from errata.

Theorem 8.28 (Galois). *Let $m > 1$ be the degree of a representation of $\mathrm{PSL}_2(p^f)$ as a transitive permutation group. Then $m \geq p^f + 1$ unless one of the following occurs:*
(1) $p^f = 2$, $m = 2$.
(2) $p^f = 3$, $m = 3$.
(3) $p^f = 5$, $m = 5$.
(4) $p^f = 7$, $m = 7$.
(5) $p^f = 9$, $m = 6$.
(6) $p^f = 11$, $m = 11$.

Proof. The claim follows easily from Theorem 8.27 by inspecting the possible indices of proper subgroups of $\mathrm{PSL}_2(p^f)$. The subgroups of given index m are alternating groups A_4 in case (3), symmetric groups S_4 in case (4), and alternating groups A_5 in cases (5) and (6). □

Exercises

14) The linear fractional transformations s and t defined by

$$xs = -\frac{1}{x} \quad \text{and} \quad xt = x + 1$$

generate $\mathrm{PSL}_2(p)$.

15) Let $\mathrm{P\Gamma L}_2(p^f)$ be the group of semilinear fractional transformations

$$x \mapsto \frac{ax^\alpha + b}{cx^\alpha + d}$$

over $\mathrm{GF}(p^f)$, where α is an automorphism of $\mathrm{GF}(p^f)$. Show that if δ is an automorphism of $\mathrm{PGL}_2(p^f)$, then there exists an $x \in \mathrm{P\Gamma L}_2(p^f)$ such that $g^\delta = g^x$ for all $g \in \mathrm{PGL}_2(p^f)$.

(Hint: Since one can modify δ by an inner automorphism of $\mathrm{PGL}_2(p^f)$, one can assume that the stabilizers of 0, 1 and ∞ are fixed by δ.)

16) The group $\mathrm{PSL}_2(11)$ has a representation as a 2-transitive permutation group of degree 11.

17) (Lüneburg [322]). The group $\mathrm{P\Gamma L}_2(8)$ has a representation as a 2-transitive permutation group of degree 28.

(Hint: Consider the action of $\mathrm{P\Gamma L}_2(8)$ on the cosets of the normalizer in $\mathrm{P\Gamma L}_2(8)$ of a Sylow 3-subgroup of $\mathrm{PSL}_2(8)$.)

18) For $p^f > 3$, the group $\mathrm{PSL}_2(p^f)$ is generated by an element a of order $\frac{p^f+1}{d}$ with $d = (p^f - 1, 2)$ and an element b of order 2.

9 The symplectic groups

In addition to the general linear groups, the automorphism groups of vector spaces equipped with a scalar product yield important examples of finite simple groups. In this section, we consider the case of a skew-symmetric scalar product, which plays a role in numerous group theoretical considerations (see, for example, Theorem III.13.7). For most proofs in this section, the finiteness of the field K will not be necessary.

Definition 9.1.

a) Let V be a vector space of finite dimension over the field K. We say V is a symplectic space whenever V is equipped with a scalar product $(_,_)$ (a map from $\mathsf{V} \times \mathsf{V}$ to K) satisfying the following properties:

(1) $(_,_)$ is bilinear, that is

$$(a_1 v_1 + a_2 v_2, w) = a_1(v_1, w) + a_2(v_2, w),$$
$$(w, a_1 v_1 + a_2 v_2) = a_1(w, v_1) + a_2(w, v_2)$$

for all $a_i \in \mathsf{K}$ and $v_i, w \in \mathsf{V}$.

(2) $(v, v) = 0$ for all $v \in \mathsf{V}$.

From (1) and (2), it follows for all $v, w \in \mathsf{V}$ that

$$0 = (v + w, v + w) = (v, v) + (v, w) + (w, v) + (w, w) = (v, w) + (w, v),$$

thus

(3) $(v, w) = -(w, v)$.

(Note that if $\operatorname{char} \mathsf{K} \neq 2$, then one can of course conclude (2) from (3).)

b) Two symplectic spaces V and W over the same field K are said to be isomorphic if there is a linear bijection t from V to W such that

$$(v_1 t, v_2 t) = (v_1, v_2)$$

for all $v_i \in \mathsf{V}$. We call t an isometry from V to W. (Here we have used the same notation for the scalar products on V and W.)

Definition 9.2. Let V be a symplectic space.

a) If U is a subspace of V, then we define

$$\mathsf{U}^\perp = \{v \in \mathsf{V} \mid (u, v) = 0 \text{ for all } u \in \mathsf{U}\}.$$

Clearly, $\mathsf{U}^\perp$ is a subspace of V.

b) We let $\mathsf{V}^\perp = \mathbf{R}(\mathsf{V})$ and call $\mathbf{R}(\mathsf{V})$ the radical of V. If $\mathbf{R}(\mathsf{V}) = 0$, then we say V is nondegenerate; otherwise V is called degenerate. So V is nondegenerate if and only if whenever $(v, x) = 0$ for all $v \in \mathsf{V}$ we have $x = 0$. For every subspace $\mathsf{U} \subseteq \mathsf{V}$, clearly

$$\mathsf{U} \cap \mathsf{U}^\perp = \mathbf{R}(\mathsf{U}).$$

Definition 9.3. Let V be a symplectic space. We say V is the orthogonal sum of subspaces V_i $(i = 1, \ldots, m)$,

$$\mathsf{V} = \mathsf{V}_1 \perp \mathsf{V}_2 \perp \cdots \perp \mathsf{V}_m,$$

if $\mathsf{V} = \mathsf{V}_1 \oplus \cdots \oplus \mathsf{V}_m$ and whenever $v \in \mathsf{V}_i$, $w \in \mathsf{V}_j$ with $i \neq j$, we have $(v, w) = 0$. In this situation, it is clear that

$$\mathbf{R}(\mathsf{V}) = \mathbf{R}(\mathsf{V}_1) \perp \cdots \perp \mathbf{R}(\mathsf{V}_m).$$

Lemma 9.4. *Let* U *be a subspace of the symplectic space* V.
a) *We have* $\dim \mathsf{U}^\perp \geq \dim \mathsf{V} - \dim \mathsf{U}$.
b) *If* V *is nondegenerate, then the map* ϵ *from* V *to its dual space* V^* *defined by* $v^\epsilon(w) = (v, w)$ *is a bijective linear map from* V *to* V^*.
c) *If* V *is nondegenerate, then* $\dim \mathsf{U}^\perp = \dim \mathsf{V} - \dim \mathsf{U}$.
d) *If* U *is nondegenerate, then* $\mathsf{V} = \mathsf{U} \perp \mathsf{U}^\perp$.

Proof. a) Let $\{u_1, \ldots, u_k\}$ be a K-basis of U. Then $\mathsf{U}^\perp$ is the space of solutions x of the k linear equations

$$(u_i, x) = 0 \quad (i = 1, \ldots, k).$$

It is well known that therefore $\dim \mathsf{U}^\perp \geq \dim \mathsf{V} - k = \dim \mathsf{V} - \dim \mathsf{U}$.

b) Assume V is nondegenerate. We define a K-linear map ϵ from V to its dual space V^* by

$$v^\epsilon(w) = (v, w) \quad (v, w \in \mathsf{V}).$$

We have $v^\epsilon = 0$ exactly when $v \in \mathbf{R}(\mathsf{V}) = 0$. Since $\dim \mathsf{V} = \dim \mathsf{V}^*$, the map ϵ is a bijective linear map from V to V^*.

c) Let $\{u_1, \ldots, u_k\}$ be a K-basis of U, and define $f_i \in \mathsf{V}^*$ by $f_i = u_i^\epsilon$ $(i = 1, \ldots, k)$. Assume that

$$\sum_{i=1}^{k} a_i f_i = 0$$

for some $a_i \in \mathsf{K}$. Then for all $v \in \mathsf{V}$, we have

$$\sum_{i=1}^{k} a_i f_i(v) = \sum_{i=1}^{k} a_i (u_i, v) = 0.$$

It follows that

$$\sum_{i=1}^{k} a_i u_i \in \mathbf{R}(\mathsf{V}) = 0.$$

By the linear independence of the u_i, the a_i are all 0. Thus, the f_i are linearly independent elements of V^*. Since

$$\mathsf{U}^\perp = \{x \in \mathsf{V} \mid (u_i, x) = f_i(x) = 0;\ i = 1, \dots, k\},$$

we may conclude that $\dim \mathsf{U}^\perp = \dim \mathsf{V} - k$.

d) By a), we have $\dim \mathsf{U}^\perp \geq \dim \mathsf{V} - \dim \mathsf{U}$. By assumption, $\mathsf{U} \cap \mathsf{U}^\perp = \mathbf{R}(\mathsf{U}) = 0$. It follows that

$$\dim(\mathsf{U} + \mathsf{U}^\perp) = \dim \mathsf{U} + \dim \mathsf{U}^\perp \geq \dim \mathsf{V}.$$

Therefore, $\mathsf{V} = \mathsf{U} \oplus \mathsf{U}^\perp$, and so $\mathsf{V} = \mathsf{U} \perp \mathsf{U}^\perp$. □

Definition 9.5. Let V be a symplectic space.

a) If $v_1, v_2 \in \mathsf{V}$ and $(v_1, v_2) = 1$, then we call $\{v_1, v_2\}$ a hyperbolic pair.

b) Every two-dimensional symplectic space $\mathsf{H} = \langle v_1, v_2 \rangle$ that is generated by a hyperbolic pair $\{v_1, v_2\}$ is called a hyperbolic plane. Clearly, hyperbolic planes are nondegenerate.

Now we can completely describe the structure of symplectic spaces.

Theorem 9.6.

a) *Let* V *be a symplectic space of dimension* n *over* K *with* $\dim \mathbf{R}(\mathsf{V}) = r$. *Then* V *is determined up to isomorphism by the pair* n, r. *There exists an orthogonal decomposition*

$$\mathsf{V} = \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_m \perp \mathbf{R}(\mathsf{V}),$$

where the H_i *are hyperbolic planes. In particular,* $n - r = 2m$ *is even.*

b) *Let* V *be a nondegenerate symplectic space. Then* $\dim \mathsf{V} = 2m$ *is even and* V *is the orthogonal sum of* m *hyperbolic planes. Thus,* V *is uniquely determined up to isomorphism by its dimension.*

Proof. a) If $\mathsf{V} = \mathbf{R}(\mathsf{V})$, then we are done. So assume $h_1 \in \mathsf{V} - \mathbf{R}(\mathsf{V})$. Then there exists an $h_2 \in \mathsf{V}$ such that $(h_1, h_2) = 1$. Therefore, $\mathsf{H}_1 = \langle h_1, h_2 \rangle$ is a hyperbolic plane. Define $\mathsf{V}_1 = \mathsf{H}_1^\perp$. By Lemma 9.4 d), it follows immediately that $\mathsf{V} = \mathsf{H}_1 \perp \mathsf{V}_1$ since $\mathbf{R}(\mathsf{H}_1) = 0$. We also have

$$\mathbf{R}(\mathsf{V}) = \mathbf{R}(\mathsf{H}_1) \perp \mathbf{R}(\mathsf{V}_1) = \mathbf{R}(\mathsf{V}_1).$$

By induction on n, we may assume that V_1 has the desired form

$$\mathsf{V}_1 = \mathsf{H}_2 \perp \cdots \perp \mathsf{H}_m \perp \mathbf{R}(\mathsf{V})$$

for some hyperbolic planes H_i. We are finished.

b) is a special case of a). □

Definition 9.7. Let V be the nondegenerate symplectic space of dimension $2n$ over K, which is uniquely determined up to isomorphism by Theorem 9.6. We call the group

of all isometries of V the symplectic group of dimension $2n$ and write $\mathrm{Sp}_{2n}(\mathsf{K})$. If K is finite with $|\mathsf{K}| = q$ (q a prime power), then we also write $\mathrm{Sp}_{2n}(q)$.

Lemma 9.8. *Let V be a nondegenerate symplectic space. Suppose that $x_1, \dots, x_r$ are linearly independent elements of V such that $(x_i, x_j) = 0$ for all i, j. Then there exist linearly independent elements $y_1, \dots, y_r$ in V such that the $\mathsf{H}_i = \langle x_i, y_i \rangle$ are hyperbolic planes and*

$$\mathsf{V} = \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_r \perp \mathsf{V}'$$

for some subspace V' of V. (In particular, $2r \leq \dim \mathsf{V}$.)

Proof. Since V is nondegenerate, the map ϵ defined in Lemma 9.4 b) is a bijective map from V to the dual space V^*. By the linear independence of the x_i, there exists a $y_1 \in \mathsf{V}$ such that

$$x_1^\epsilon(y_1) = (x_1, y_1) = 1 \quad \text{and} \quad x_i^\epsilon(y_1) = (x_i, y_1) = 0 \quad (i = 2, \dots, r).$$

Then $\mathsf{H}_1 = \langle x_1, y_1 \rangle$ is a hyperbolic plane. By Lemma 9.4 d), it follows that $\mathsf{V} = \mathsf{H}_1 \perp \mathsf{H}_1^\perp$. We have $x_i \in \mathsf{H}_1^\perp$ for $i = 2, \dots, r$. Since

$$0 = \mathbf{R}(\mathsf{V}) = \mathbf{R}(\mathsf{H}_1) \perp \mathbf{R}(\mathsf{H}_1^\perp),$$

$\mathsf{H}_1^\perp$ is nondegenerate. The claim then follows by induction on r. □

The next theorem is our first fundamental statement about the symplectic groups $\mathrm{Sp}_{2n}(\mathsf{K})$.

Theorem 9.9 (Witt). *Let V be a nondegenerate symplectic space. Let U_1 and U_2 be subspaces of V and t an isometry from U_1 to U_2. Then there exists an isometry s of V satisfying $s = t$ on U_1.*

Proof. By Theorem 9.6, there exists an orthogonal decomposition

$$\mathsf{U}_1 = \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_m \perp \mathbf{R}(\mathsf{U}_1)$$

for some collection of hyperbolic planes H_i. This decomposition of U_1 gives rise to a decomposition of U_2,

$$\mathsf{U}_2 = \mathsf{H}_1' \perp \cdots \perp \mathsf{H}_m' \perp \mathbf{R}(\mathsf{U}_2),$$

where $\mathsf{H}_i' = \mathsf{H}_i t$ and $\mathbf{R}(\mathsf{U}_2) = \mathbf{R}(\mathsf{U}_1)t$.

We let $\mathsf{H} = \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_m$ and $\mathsf{H}' = \mathsf{H}_1' \perp \cdots \perp \mathsf{H}_m'$. Since H and H' are nondegenerate, Lemma 9.4 d) gives

$$\mathsf{V} = \mathsf{H} \perp \mathsf{L} = \mathsf{H}' \perp \mathsf{L}',$$

where $\mathsf{L} = \mathsf{H}^\perp$ and $\mathsf{L}' = \mathsf{H}'^\perp$. It follows from

$$0 = \mathbf{R}(\mathsf{V}) = \mathbf{R}(\mathsf{H}) \perp \mathbf{R}(\mathsf{L})$$

that $\mathbf{R}(\mathsf{L}) = 0$, and likewise $\mathbf{R}(\mathsf{L}') = 0$. We have

$$\mathbf{R}(\mathsf{U}_1) \subseteq \mathsf{H}^\perp = \mathsf{L} \quad \text{and} \quad \mathbf{R}(\mathsf{U}_2) \subseteq \mathsf{H}'^\perp = \mathsf{L}'.$$

Let $\{x_1, \dots, x_r\}$ be a K-basis of $\mathbf{R}(\mathsf{U}_1)$. Then $(x_i, x_j) = 0$ for all i, j. Since L is nondegenerate, Lemma 9.8 gives an orthogonal decomposition

$$\mathsf{L} = \mathsf{S}_1 \perp \cdots \perp \mathsf{S}_r \perp \mathsf{M},$$

where $\mathsf{S}_i = \langle x_i, y_i \rangle$ and $(x_i, y_i) = 1$. In the same way, there is a decomposition

$$\mathsf{L}' = \mathsf{S}'_1 \perp \cdots \perp \mathsf{S}'_r \perp \mathsf{M}',$$

where $\mathsf{S}'_i = \langle x_i t, y'_i \rangle$ and $(x_i t, y'_i) = 1$. We have $\mathbf{R}(\mathsf{M}) \subseteq \mathbf{R}(\mathsf{L}) = 0$, and likewise $\mathbf{R}(\mathsf{M}') = 0$. Now,

$$\begin{aligned} \mathsf{V} = \mathsf{H} \perp \mathsf{L} &= \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_m \perp \mathsf{S}_1 \perp \cdots \perp \mathsf{S}_r \perp \mathsf{M} \\ &= \mathsf{H}'_1 \perp \cdots \perp \mathsf{H}'_m \perp \mathsf{S}'_1 \perp \cdots \perp \mathsf{S}'_r \perp \mathsf{M}'. \end{aligned}$$

Since $\mathbf{R}(\mathsf{M}) = \mathbf{R}(\mathsf{M}') = 0$ and $\dim \mathsf{M} = \dim \mathsf{M}'$, M and M' are isomorphic by Theorem 9.6 b). That is, there exists an isometry u from M to M'. Now we define the linear transformation s of V by

$$\begin{aligned} hs &= ht \quad \text{for} \quad h \in \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_m, \\ x_i s &= x_i t \quad (1 \le i \le r), \\ y_i s &= y'_i \quad (1 \le i \le r), \\ xs &= xu \quad \text{for} \quad x \in \mathsf{M}. \end{aligned}$$

Then s is an isometry of V with $s = t$ on

$$\mathsf{U}_1 = \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_m \perp \langle x_1, \dots, x_r \rangle. \qquad \square$$

Definition 9.10. A subspace U of the symplectic space V is called isotropic if $(u, u') = 0$ for all $u, u' \in \mathsf{U}$.

Theorem 9.11. *Let V be a nondegenerate symplectic space of dimension $2n$. If U is an isotropic subspace of V, then $\dim \mathsf{U} \le n$ and U is contained in some isotropic subspace of V of dimension n.*

Proof. Let $\{u_1, \dots, u_r\}$ be a basis of U. By Lemma 9.8, there exists an orthogonal decomposition

$$\mathsf{V} = \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_r \perp \mathsf{V}'$$

for some hyperbolic planes $\mathsf{H}_i = \langle u_i, v_i \rangle$. Thus $2r \leq \dim \mathsf{V} = 2n$. By Theorem 9.6 b), we have

$$\mathsf{V} = \mathsf{V}_1 \perp \cdots \perp \mathsf{V}_n$$

for some hyperbolic planes $\mathsf{V}_i = \langle x_i, y_i \rangle$. The K-linear map t defined by $u_i t = x_i$ $(1 \leq i \leq r)$ is then an isometry from U to $\langle x_1, \ldots, x_r \rangle$. Let s be an extension of t to V, whose existence is guaranteed by Theorem 9.9. Then we have

$$\mathsf{U}s = \mathsf{U}t = \langle x_1, \ldots, x_r \rangle \subseteq \langle x_1, \ldots, x_n \rangle.$$

So U lies in the isotropic subspace $\langle x_1, \ldots, x_n \rangle s^{-1}$ of dimension n. □

Lemma 9.12. *We have* $\mathrm{Sp}_2(\mathsf{K}) \cong \mathrm{SL}_2(\mathsf{K})$.

Proof. Let $\{v_1, v_2\}$ be a hyperbolic pair in the nondegenerate two-dimensional symplectic space V, which exists by Theorem 9.6 b). Let $t \in \mathrm{GL}_2(\mathsf{K})$ with

$$\begin{aligned} v_1 t &= a_{11} v_1 + a_{12} v_2 \\ v_2 t &= a_{21} v_1 + a_{22} v_2, \end{aligned}$$

where $a_{ij} \in \mathsf{K}$. Then

$$(v_1 t, v_2 t) = a_{11} a_{22} - a_{12} a_{21},$$

and so $t \in \mathrm{Sp}_2(\mathsf{K})$ if and only if $t \in \mathrm{SL}_2(\mathsf{K})$. □

Theorem 9.13.

a) *The nondegenerate symplectic space* V *of dimension* $2n$ *over the field* $\mathsf{K} = \mathrm{GF}(q)$ *contains exactly* $(q^{2n} - 1)q^{2n-1}$ (*ordered*) *hyperbolic pairs.*

b) *We have*

$$\begin{aligned} |\mathrm{Sp}_{2n}(q)| &= (q^{2n} - 1)q^{2n-1}(q^{2n-2} - 1)q^{2n-3} \cdots (q^2 - 1)q \\ &= q^{n^2} \prod_{i=1}^{n} (q^{2i} - 1). \end{aligned}$$

Proof. a) Let $\{x, y\}$ be a hyperbolic pair in V. For x we may choose any of the $q^{2n} - 1$ nonzero vectors of V. By Lemma 9.4 c), $\langle x \rangle^\perp$ has dimension $2n - 1$. So there exist $q^{2n} - q^{2n-1}$ vectors y with $(x, y) \neq 0$. Among these there are exactly

$$\frac{q^{2n} - q^{2n-1}}{q - 1} = q^{2n-1}$$

vectors y with $(x, y) = 1$. So $(q^{2n} - 1)q^{2n-1}$ is the number of hyperbolic pairs in V.

b) We have $|\mathrm{Sp}_{2n}(q)| = (q^{2n} - 1)q^{2n-1}|\mathrm{Sp}_{2n-2}(q)|$:

Let $\{x, y\}$ and $\{x', y'\}$ be hyperbolic pairs. Then the linear map t defined by $xt = x'$ and $yt = y'$ is an isometry from $\langle x, y \rangle$ to $\langle x', y' \rangle$. By Theorem 9.9 (pg. 228),

t can be extended to an isometry of V. Thus, $\mathrm{Sp}_{2n}(q)$ is transitive on hyperbolic pairs. Using a) and Theorem I.5.11 (pg. 28), it follows that

$$|\mathrm{Sp}_{2n}(q)| = (q^{2n} - 1)q^{2n-1}|U|,$$

where U is the subgroup of $\mathrm{Sp}_{2n}(q)$ fixing a hyperbolic pair $\{x_0, y_0\}$. Defining $\mathsf{H}_0 = \langle x_0, y_0\rangle$, we have $\mathsf{V} = \mathsf{H}_0 \perp \mathsf{H}_0^\perp$ by Lemma 9.4 d). Every $u \in U$ is uniquely determined by its action on $\mathsf{H}_0^\perp$, and every isometry s from $\mathsf{H}_0^\perp$ to itself can be uniquely extended to an isometry $u \in U$ by defining

$$(h + h')u = h + h's \quad \text{for} \quad h \in \mathsf{H}_0 \quad \text{and} \quad h' \in \mathsf{H}_0^\perp.$$

So $U \cong \mathrm{Sp}_{2n-2}(q)$.

The claim in b) follows by induction on n, the base case for induction being established in Lemma 9.12. □

Definition 9.14. Let V be a nondegenerate symplectic space of dimension $2n$. Let $\mathsf{P} = \mathsf{P}(2n-1, \mathsf{K})$ be the associated projective space. We call the group of collineations of P induced by $\mathrm{Sp}_{2n}(\mathsf{K})$ the projective symplectic group and write $\mathrm{PSp}_{2n}(\mathsf{K})$. It is well known that

$$\mathrm{PSp}_{2n}(\mathsf{K}) \cong \mathrm{Sp}_{2n}(\mathsf{K})/Z,$$

where $Z = \{t \in \mathrm{Sp}_{2n}(\mathsf{K}) \mid t = a1,\ a \in \mathsf{K}^\times\}$. Since

$$(vt, v't) = a^2(v, v') = (v, v') \neq 0$$

for some $v, v' \in \mathsf{V}$, we necessarily have $a^2 = 1$. Therefore, $Z = \langle -1\rangle$ for char $\mathsf{K} \neq 2$ and $Z = 1$ for char $\mathsf{K} = 2$.

The main goal of this section is to prove that $\mathrm{PSp}_{2n}(\mathsf{K})$ is simple except when $n = 1$, $|\mathsf{K}| \leq 3$ and $n = 2$, $|\mathsf{K}| = 2$. First we consider the structure of $\mathrm{PSp}_{2n}(\mathsf{K})$ as a permutation group on the projective space P:

Theorem 9.15. *We consider* $\mathrm{PSp}_{2n}(\mathsf{K})$ *as a permutation group on the points of the projective space* P. *We have:*

a) $\mathrm{PSp}_{2n}(\mathsf{K})$ *is transitive on* P.

b) *Let* U *be the stabilizer of the point* $\langle x\rangle$ *in* $\mathrm{PSp}_{2n}(\mathsf{K})$ $(0 \neq x \in \mathsf{V})$. *Then* U *has exactly three orbits on* P, *namely*

$$\Delta_1 = \{\langle x\rangle\},$$
$$\Delta_2 = \{\langle y\rangle \mid \langle y\rangle \neq \langle x\rangle,\ (x, y) = 0\}$$
$$\Delta_3 = \{\langle y\rangle \mid (x, y) = 1\}.$$

(*For* $|\mathsf{K}| = q$, *we have* $|\Delta_2| = q + q^2 + \cdots + q^{2n-2}$ *and* $|\Delta_3| = q^{2n-1}$.)

c) $\mathrm{PSp}_{2n}(\mathsf{K})$ *is primitive on* P.

Proof. a) This claim is clear since $\mathrm{Sp}_{2n}(\mathsf{K})$ is transitive on the nonzero vectors of V by Theorem 9.9.

b) Let $\langle x\rangle \neq \langle y_i\rangle$ and $(x, y_i) = 0$ for $i = 1, 2$. Since the linear map s defined by $xs = x$ and $y_1 s = y_2$ is an isometry from $\langle x, y_1\rangle$ to $\langle x, y_2\rangle$, there exists some $t \in \mathrm{Sp}_{2n}(\mathsf{K})$ such that $xt = x$ and $y_1 t = y_2$ by Theorem 9.9. Then the transformation $t' \in \mathrm{PSp}_{2n}(\mathsf{K})$ of P induced by t satisfies $\langle y_1\rangle t' = \langle y_2\rangle$ and $t' \in U$. So Δ_2 is an orbit of U.

The same proof also shows that Δ_3 is an orbit of U. It is clear that $\mathsf{P} = \Delta_1 \cup \Delta_2 \cup \Delta_3$.

c) By Lemma 9.12, $\mathrm{PSp}_2(\mathsf{K}) \cong \mathrm{PSL}_2(\mathsf{K})$, and this group is 2-transitive on P by Lemma 8.1. So we may assume $n > 1$.

Let $a = \langle v\rangle$ be a point in P. We define

$$\Omega_a = \{\langle x\rangle \mid x \in \mathsf{V},\ (x, v) = 0\} \quad \text{and}$$
$$\Omega'_a = \{a, \langle x\rangle \mid x \in \mathsf{V},\ (x, v) = 1\}.$$

Then $\mathsf{P} = \Omega_a \cup \Omega'_a$ and $\{a\} = \Omega_a \cap \Omega'_a$.

Assume for contradiction that $\mathrm{PSp}_{2n}(\mathsf{K})$ is imprimitive, and let Λ_a be a block for $\mathrm{PSp}_{2n}(\mathsf{K})$ containing a. Let U be the stabilizer of a in $\mathrm{PSp}_{2n}(\mathsf{K})$.

If $a \neq b \in \Lambda_a \cap \Omega'_a$, then it follows from b) that

$$\Lambda_a \supseteq b^U = \Omega'_a - \{a\}.$$

In the same way, if $a \neq b \in \Lambda_a \cap \Omega_a$ then

$$\Lambda_a \supseteq b^U = \Omega_a - \{a\},$$

and so $\Lambda_a \supseteq \Omega_a$ as well.

Therefore, we have either $\Lambda_a = \Omega_a$ or $\Lambda_a = \Omega'_a$ for all $a \in \mathsf{P}$. Since the blocks are transitively permuted by $\mathrm{PSp}_{2n}(\mathsf{K})$, the same holds for every block.

Case 1: Assume $\Lambda_a = \Omega_a$ for all $a \in \mathsf{P}$. Let

$$\mathsf{V} = \langle x_1, x_2\rangle \perp \mathsf{W},$$

where $(x_1, x_2) = 1$, and let $0 \neq x_3 \in \mathsf{W}$. We let $\langle x_i\rangle = a_i \in \mathsf{P}$, and it follows that

$$\Lambda_{a_3} = \Omega_{a_3} \ni a_1 \quad \text{and likewise} \quad \Lambda_{a_3} \ni a_2.$$

Thus, $\Lambda_{a_3} = \Lambda_{a_1} = \Lambda_{a_2}$. Since $\Lambda_{a_1} = \Omega_{a_1} \not\ni a_2$, we have reached a contradiction.

(One can also use the fact that the intersection of two hyperplanes Ω_a and Ω_b in $\mathsf{P}(2n-1, q)$ is nonempty.)

Case 2: Assume $\Lambda_a = \Omega'_a$ for all $a \in \mathsf{P}$. Let the x_i $(i = 1, 2, 3)$ be chosen as in case 1. We define $y = x_1 + x_3$ and $\langle x_i\rangle = a_i$ $(i = 1, 2)$.

Now it follows that $\Lambda_{a_1} = \Omega'_{a_1} \ni a_2$, so $\Omega'_{a_1} = \Lambda_{a_1} = \Lambda_{a_2} = \Omega'_{a_2}$. Since $(x_1, y) = 0$ and $\langle a_1\rangle \neq \langle y\rangle$, we have $\langle y\rangle \notin \Omega'_{a_1}$, but it follows from $(x_2, y) = (x_2, x_1) = -1$ that $\langle y\rangle \in \Omega'_{a_2}$. This is a contradiction. Therefore, $\mathrm{PSp}_{2n}(\mathsf{K})$ is primitive on P. □

Definition 9.16. A map $t \neq 1$ in $\mathrm{Sp}_{2n}(\mathsf{K})$ is called a symplectic transvection over the hyperplane H of V whenever $ht = h$ for all $h \in \mathsf{H}$ and $vt - v \in \mathsf{H}$ for all $v \in \mathsf{V}$.

Lemma 9.17. *Let V be a nondegenerate symplectic space. Let t be a symplectic transvection over the hyperplane H, and let $\mathsf{H} = \langle w \rangle^{\perp}$ (see Lemma 9.4 c)). Then for all $v \in \mathsf{V}$, we have*

$$vt = v + c(w, v)w$$

for some $c \in \mathsf{K}$. We say t is a transvection in the direction $\langle w \rangle$. Conversely, if $0 \neq w \in \mathsf{V}$ and $0 \neq c \in \mathsf{K}$, then the map t defined by

$$vt = v + c(w, v)w$$

is a symplectic transvection over the hyperplane $\langle w \rangle^{\perp}$.

Proof. Let $x \in \mathsf{V} - \mathsf{H}$. Since $t \neq 1$, we have $0 \neq xt - x = u \in \mathsf{H}$. Since $\mathsf{V} = \langle x \rangle \oplus \mathsf{H}$, every $v \in \mathsf{V}$ may be uniquely written as $v = ax + h$ for some $a \in \mathsf{K}$ and $h \in \mathsf{H}$. Then the map μ defined by $\mu(ax + h) = a$ is a linear functional on V. Since V is nondegenerate, there exists some $s \in \mathsf{V}$ such that $\mu(v) = (s, v)$ by Lemma 9.4 b). Now we have

$$vt = ax + h + au = v + (s, v)u.$$

For all $h \in \mathsf{H}$, we have

$$(x, h) = (xt, ht) = (x + u, h) = (x, h) + (u, h).$$

This forces $u \in \mathsf{H}^{\perp}$, so $u = b_1 w$ for some $b_1 \in \mathsf{K}$ with $b_1 \neq 0$. Furthermore, for all $h \in \mathsf{H}$ we have

$$h = ht = h + b_1(s, h)w.$$

This implies $s = b_2 w \in \mathsf{H}^{\perp}$. Thus, the claim follows by setting $c = b_1 b_2$.

For the converse, it follows by a simple calculation that every map t of the form

$$vt = v + c(w, v)w$$

is a symplectic transvection over the hyperplane $\langle w \rangle^{\perp}$. □

Theorem 9.18. *The symplectic transvections generate $\mathrm{Sp}_{2n}(\mathsf{K})$.*

Proof. For $n = 1$, this is true by Theorem 6.7 since $\mathrm{Sp}_2(\mathsf{K}) = \mathrm{SL}_2(\mathsf{K})$. Now let $n > 1$. Let T be the subgroup of $\mathrm{Sp}_{2n}(\mathsf{K})$ generated by the symplectic transvections. We show that:

a) T is transitive on $\mathsf{V} - \{0\}$:

Let $0 \neq v_i \in \mathsf{V}$ $(i = 1, 2)$. If $(v_1, v_2) \neq 0$, then we determine $c \in \mathsf{K}$ by $c(v_1, v_2) = -1$. Let t be the symplectic transformation defined by

$$vt = v + c(v_1 - v_2, v)(v_1 - v_2).$$

Then

$$v_1 t = v_1 - c(v_2, v_1)(v_1 - v_2) = v_2,$$

and we are done.

So assume that $(v_1, v_2) = 0$. We seek a $w \in \mathsf{V}$ such that $(v_1, w) \neq 0 \neq (v_2, w)$. If $\langle v_1 \rangle = \langle v_2 \rangle$, then every w outside $\langle v_1 \rangle^\perp$ satisfies these conditions. So we may assume that v_1 and v_2 are linearly independent vectors satisfying $(v_1, v_2) = 0$. By Lemma 9.8, there exist $w_1, w_2 \in \mathsf{V}$ such that

$$(v_1, w_1) = (v_2, w_2) = 1 \quad \text{and} \quad (v_1, w_2) = (v_2, w_1) = 0.$$

Then $w = w_1 + w_2$ has the desired properties.

By what has already been shown, there exist symplectic transvections t_1 and t_2 such that $v_1 t_1 = w$ and $w t_2 = v_2$. Then $v_1 t_1 t_2 = v_2$ with $t_1 t_2 \in T$.

b) T is transitive on the hyperbolic pairs in V:

Let $\{x_1, y_1\}$ and $\{x_2, y_2\}$ be two hyperbolic pairs in V. By a), we may assume that $x_1 = x_2 = x$.

If $(y_1, y_2) \neq 0$, then again as in a) there exists a symplectic transvection t in the direction $y_1 - y_2$ such that $y_1 t = y_2$. In addition, $xt = x$ since

$$(x, y_1 - y_2) = (x, y_1) - (x, y_2) = 0.$$

Now suppose $(y_1, y_2) = 0$. We have the three hyperbolic pairs

$$\{x, y_1\}, \quad \{x, y_1 + x\} \quad \text{and} \quad \{x, y_2\},$$

where $-(y_1, y_1 + x) = (y_1 + x, y_2) = 1$.

Thus, there exist symplectic transvections t_1 and t_2 satisfying $xt_i = x$, $y_1 t_1 = y_1 + x$ and $(y_1 + x)t_2 = y_2$. Then $t_1 t_2$ maps the pair $\{x, y_1\}$ to $\{x, y_2\}$.

c) Now we prove the claim of Theorem 9.18 by induction on n. We have already mentioned that the statement is true for $n = 1$.

Let $\{x, y\}$ be a hyperbolic pair in V, and let $g \in \mathrm{Sp}_{2n}(\mathsf{K})$. By b), there is a $t \in T$ such that $xg = xt$ and $yg = yt$. Thus, $r = gt^{-1}$ fixes the pair $\{x, y\}$. Let $\mathsf{H} = \langle x, y \rangle$. Then $\mathsf{V} = \mathsf{H} \perp \mathsf{H}^\perp$ and $\mathsf{H}^\perp r = \mathsf{H}^\perp$. So the restriction r' of r to $\mathsf{H}^\perp$ is an isometry of $\mathsf{H}^\perp$. By induction, r' is a product $r' = \prod_i t_i'$ of symplectic transvections t_i' on $\mathsf{H}^\perp$. We define symplectic transvections t_i on V by

$$(h + h')t_i = h + h' t_i' \quad \text{for} \quad h \in \mathsf{H} \quad \text{and} \quad h' \in \mathsf{H}^\perp.$$

Then

$$r = gt^{-1} = \prod_i t_i,$$

and we have proved the claim $g = \prod_i t_i t$. □

Theorem 9.19. *Every element of* $\mathrm{Sp}_{2n}(\mathsf{K})$ *has determinant 1.*

Proof. By Theorem 9.18, it suffices to prove that $\det t = 1$ for symplectic transvections t. The symplectic transvections are, however, transvections in the sense of Definition 6.3, so they have determinant 1 by Lemma 6.6. □

Theorem 9.20. *We have* $\mathrm{Sp}_{2n}(\mathsf{K})' = \mathrm{Sp}_{2n}(\mathsf{K})$*, except for the cases* $\mathrm{Sp}_2(2)$*,* $\mathrm{Sp}_2(3)$ *and* $\mathrm{Sp}_4(2)$.

Proof. a) First assume that $|\mathsf{K}| > 3$. Then there is some $a \in \mathsf{K}$ such that $0 \neq a^2 \neq 1$. Let $0 \neq w \in \mathsf{V}$. Since the linear map s satisfying $ws = aw$ is an isometry from $\langle w \rangle$ to itself, there exists some $m \in \mathrm{Sp}_{2n}(\mathsf{K})$ such that $wm = aw$ by Theorem 9.9. Let t be a symplectic transvection of the form

$$vt = v + c(w, v)w.$$

Now we have

$$\begin{aligned} vm^{-1}tmt^{-1} &= (vm^{-1} + c(w, vm^{-1})w)mt^{-1} = (v + c(wm, v)wm)t^{-1} \\ &= v + (a^2 - 1)c(w, v)w. \end{aligned}$$

With an appropriate choice of c, we can force $(a^2 - 1)c$ to be any element of K since $a^2 \neq 1$. Thus, all symplectic transvections lie in $\mathrm{Sp}_{2n}(\mathsf{K})'$. It follows by Theorem 9.18 that $\mathrm{Sp}_{2n}(\mathsf{K}) = \mathrm{Sp}_{2n}(\mathsf{K})'$.

b) Now assume $|\mathsf{K}| = 3$. It suffices to show that $\mathrm{Sp}_4(3)' = \mathrm{Sp}_4(3)$:

Let t be a symplectic transvection in $\mathrm{Sp}_{2n}(\mathsf{K})$, and let $n > 2$. Write

$$vt = v + c(w, v)w.$$

We define $v_1 = w$. Then there is some $v_2 \in \mathsf{V}$ such that $(v_1, v_2) = 1$. We have $\mathsf{V} = \langle v_1, v_2 \rangle \perp \mathsf{W}$ by Lemma 9.4 d). Let $\langle v_3, v_4 \rangle$ be a hyperbolic plane in W. Then

$$\mathsf{V} = \mathsf{V}_1 \perp \mathsf{V}_2, \quad \text{where} \quad \mathsf{V}_1 = \langle v_1, v_2 \rangle \perp \langle v_3, v_4 \rangle.$$

Let m be the restriction of t to V_1. We have $m \in \mathrm{Sp}_4(3)'$ by our assumption. We embed the symplectic group $\mathrm{Sp}_4(3)$ acting on V_1 in $\mathrm{Sp}_{2n}(3)$ by defining

$$(w_1 + w_2)a = w_1 a + w_2$$

for $a \in \mathrm{Sp}_4(3)$ and $w_i \in \mathsf{V}_i$. It follows that

$$t \in \mathrm{Sp}_4(3)' \leq \mathrm{Sp}_{2n}(3)'.$$

It remains to show that $\mathrm{Sp}_4(3)' = \mathrm{Sp}_4(3)$. In this case, we must calculate. Let

$$\mathsf{V} = \langle v_1, v_2 \rangle \perp \langle v_3, v_4 \rangle, \quad \text{where} \quad (v_1, v_2) = (v_3, v_4) = 1.$$

Then we also obtain the following orthogonal decomposition of V into hyperbolic planes:

$$\mathsf{V} = \langle v_1, v_2 + v_3\rangle \perp \langle v_3, v_1 + v_4\rangle.$$

We define $t, m \in \mathrm{Sp}_4(3)$ by

$$v_1 t = v_2, \quad v_2 t = -v_1, \quad v_3 t = v_3, \quad v_4 t = v_4$$

and

$$v_1 m = v_2 + v_3, \quad (v_2 + v_3)m = -v_1, \quad v_3 m = v_3, \quad (v_1 + v_4)m = v_1 + v_4.$$

Note that the restriction of t^2 to $\langle v_1, v_2\rangle$ is the map -1. Since the Sylow 2-subgroups of $\mathrm{Sp}_2(3) = \mathrm{SL}_2(3)$ are quaternion by Theorem 8.10, the element -1 lies in $\mathrm{Sp}_2(3)'$. It follows that $t^2 \in \mathrm{Sp}_4(3)'$, and in the same way $m^2 \in \mathrm{Sp}_4(3)'$. A simple calculation yields

$$v_i(mt)^2 = v_i \ (i = 1, 2, 3), \quad v_4(mt)^2 = v_3 + v_4.$$

Therefore $l = (mt)^2$ is the transvection

$$vl = v + (v_3, v)v_3.$$

We have

$$l = (mt)^2 = m^2(t^m)^2[m, t] \in \mathrm{Sp}_4(3)'.$$

Since $\mathrm{Sp}_4(3)$ transitively permutes the nonzero vectors of V, every symplectic transvection is conjugate to l or l^{-1}. It follows from Theorem 9.18 that $\mathrm{Sp}_4(3)' = \mathrm{Sp}_4(3)$.

c) Now assume $|\mathsf{K}| = 2$. As in the beginning of b), it follows from $\mathrm{Sp}_6(2)' = \mathrm{Sp}_6(2)$ that $\mathrm{Sp}_{2n}(2)' = \mathrm{Sp}_{2n}(2)$ for $n > 3$.

The abelian group $\mathrm{Sp}_6(2)/\mathrm{Sp}_6(2)'$ is generated by elements of order 2, namely the images of the symplectic transvections. So $\mathrm{Sp}_6(2)/\mathrm{Sp}_6(2)'$ is a 2-group. Thus, for all $l \in \mathrm{Sp}_6(2)$ with $l^3 = 1$, we have $l \in \mathrm{Sp}_6(2)'$.

Let

$$\mathsf{V} = \langle v_1, v_2\rangle \perp \langle v_3, v_4\rangle \perp \langle v_5, v_6\rangle,$$

where $(v_1, v_2) = (v_3, v_4) = (v_5, v_6) = 1$.

Then

$$\mathsf{V} = \langle v_1, v_2 + v_4\rangle \perp \langle v_1 + v_3, v_4\rangle \perp \langle v_5, v_6\rangle$$

is also an orthogonal decomposition of V into hyperbolic planes. Let t_1 be defined by

$$v_i t_1 = v_i \text{ for } i = 1, 2, 5, 6, \quad v_3 t_1 = v_3 + v_4, \quad v_4 t_1 = v_3.$$

Then t_1 is an isometry of V of order 3 since $\mathsf{K} = \mathrm{GF}(2)$. Therefore, $t_1 \in \mathrm{Sp}_6(2)'$. Using the second decomposition of V, we obtain in the same way a $t_2 \in \mathrm{Sp}_6(2)'$ with

$$v_i t_2 = v_i \ (i = 1, 5, 6), \quad (v_2 + v_4)t_2 = v_2 + v_4, \quad (v_1 + v_3)t_2 = v_4,$$
$$v_4 t_2 = v_1 + v_3 + v_4.$$

Then $m_1 = t_1 t_2$ lies in $\mathrm{Sp}_6(2)'$, and we have

$$v_1 m_1 = v_1, \quad v_2 m_1 = v_1 + v_2 + v_3, \quad v_3 m_1 = v_3, \quad v_4 m_1 = v_1 + v_4,$$
$$v_5 m_1 = v_5, \quad v_6 m_1 = v_6.$$

Using the same process, we obtain from the decomposition

$$\mathsf{V} = \langle v_1, v_2 \rangle \perp \langle v_5, v_6 \rangle \perp \langle v_3, v_4 \rangle$$

a map m_2, where

$$v_1 m_2 = v_1, \quad v_2 m_2 = v_1 + v_2 + v_5, \quad v_3 m_2 = v_3, \quad v_4 m_2 = v_4,$$
$$v_5 m_2 = v_5, \quad v_6 m_2 = v_1 + v_6.$$

Again using the same process, we obtain from the decomposition

$$\mathsf{V} = \langle v_1, v_2 \rangle \perp \langle v_3 + v_5, v_4 \rangle \perp \langle v_5, v_4 + v_6 \rangle$$

a map m_3, where

$$v_1 m_3 = v_1, \quad v_2 m_3 = v_1 + v_2 + v_3 + v_5, \quad v_3 m_3 = v_3, \quad v_4 m_3 = v_1 + v_4,$$
$$v_5 m_3 = v_5, \quad v_6 m_3 = v_1 + v_6.$$

Let $m = m_3 m_2 m_1$. Then $m \in \mathrm{Sp}_6(2)'$ and

$$v_i m = v_i \text{ for } i \neq 2, \quad v_2 m = v_1 + v_2.$$

That is, m is a symplectic transvection in the direction $\langle v_1 \rangle$ lying in $\mathrm{Sp}_6(2)'$. If $g \in \mathrm{Sp}_6(2)$ with $v_1 g = w$, then $g^{-1} m g$ is the only symplectic transvection in the direction $\langle w \rangle$. Thus, all transvections lie in $\mathrm{Sp}_6(2)'$, and then $\mathrm{Sp}_6(2) = \mathrm{Sp}_6(2)'$ by Theorem 9.18. □

Next, we consider $\mathrm{Sp}_4(2)$, which is one of the exceptional cases in Theorem 9.20.

Theorem 9.21. *We have* $\mathrm{Sp}_4(2) \cong S_6$.

Proof. Let V be the nondegenerate symplectic space of dimension 4 over $\mathrm{GF}(2)$. We consider the subsets $\mathscr{S}$ of V of the form

$$\mathscr{S} = \{v_1, \dots, v_5 \mid v_i \in \mathsf{V},\ (v_i, v_j) = 1 \text{ for } i \neq j\}.$$

Since $\langle v_1, v_2\rangle$ is a hyperbolic plane, we have $\mathsf{V} = \langle v_1, v_2\rangle \perp \mathsf{H}$ by Lemma 9.4 d). Now H is a hyperbolic plane by Theorem 9.6 since it is a two-dimensional nondegenerate symplectic space. Let $\mathsf{H} = \langle x_3, x_4\rangle$, where $(x_3, x_4) = 1$. Then up to ordering we must have

$$v_3 = v_1 + v_2 + x_3, \quad v_4 = v_1 + v_2 + x_4, \quad v_5 = v_1 + v_2 + x_3 + x_4.$$

So every hyperbolic pair $\{v_1, v_2\}$ generates exactly one set $\mathscr{S}$, and every set $\mathscr{S}$ contains 20 hyperbolic pairs. Since V has $(2^4 - 1)2^3 = 120$ hyperbolic pairs by Theorem 9.13 a), there are exactly 6 such sets $\mathscr{S}$ in V.

Since $\mathrm{Sp}_4(2)$ transitively permutes the sets $\mathscr{S}$ by Theorem 9.9, we obtain a homomorphism from $\mathrm{Sp}_4(2)$ to the symmetric group S_6. Let K be the kernel of this homomorphism. Let

$$\mathsf{V} = \langle v_1, v_2\rangle \perp \langle x_3, x_4\rangle,$$

where $(v_1, v_2) = (x_3, x_4) = 1$. Then the sets

$$\mathscr{S} = \{v_1,\ v_2,\ v_1 + v_2 + x_3,\ v_1 + v_2 + x_4,\ v_1 + v_2 + x_3 + x_4\}$$

and

$$\mathscr{S}' = \{x_3,\ x_4,\ v_1 + x_3 + x_4,\ v_2 + x_3 + x_4,\ v_1 + v_2 + x_3 + x_4\}$$

are fixed by K. Then

$$\mathscr{S} \cap \mathscr{S}' = \{v_1 + v_2 + x_3 + x_4\}$$

is also fixed by K. But since $\mathrm{Sp}_4(2)$ is transitive on the nonzero vectors of V by Theorem 9.9, it follows that $K = 1$. So we have a monomorphism from $\mathrm{Sp}_4(2)$ into S_6. Comparing orders yields the claim $\mathrm{Sp}_4(2) \cong S_6$. □

Main Theorem 9.22. *The projective symplectic group* $\mathrm{PSp}_{2n}(\mathsf{K})$ *is simple except in the following cases:*
(1) $\mathrm{PSp}_2(2) = \mathrm{PSL}_2(2) \cong S_3$.
(2) $\mathrm{PSp}_2(3) = \mathrm{PSL}_2(3) \cong A_4$.
(3) $\mathrm{PSp}_4(2) \cong S_6$.

Proof. By Theorem 9.15, $\mathrm{PSp}_{2n}(\mathsf{K})$ is a primitive permutation group on the points of P. By Theorem 9.20, we have $\mathrm{PSp}_{2n}(\mathsf{K})' = \mathrm{PSp}_{2n}(\mathsf{K})$ except for the three exceptions listed above. Let K be the group of transvections in the direction $\langle w\rangle$, and let $\overline{K}$ be the image of K in $\mathrm{PSp}_{2n}(\mathsf{K})$. Since $\mathrm{Sp}_{2n}(\mathsf{K})$ transitively permutes the nonzero vectors by Theorem 9.9, we have by Theorem 9.18 that

$$\mathrm{Sp}_{2n}(\mathsf{K}) = \langle K^x \mid x \in \mathrm{Sp}_{2n}(\mathsf{K})\rangle.$$

Thus

$$\mathrm{PSp}_{2n}(\mathsf{K}) = \langle \overline{K}^{\overline{x}} \mid \overline{x} \in \mathrm{PSp}_{2n}(\mathsf{K}) \rangle,$$

as well. Furthermore, $\overline{K}$ is normal in the stabilizer of the point $\langle w \rangle$. The simplicity of $\mathrm{PSp}_{2n}(\mathsf{K})$ then follows by Lemma 6.12. □

The following theorem is the symplectic equivalent of Theorem 3.10.

Theorem 9.23. *Let $\mathsf{V} = \mathsf{V}(2n, q)$ be the nondegenerate symplectic space of dimension $2n$ over $\mathrm{GF}(q)$. Let C be a cyclic subgroup of $\mathrm{Sp}_{2n}(q)$, and suppose V is an irreducible C-module. Then $|C|$ divides $q^n + 1$.*

Proof. Since $C = \langle g \rangle$ acts irreducibly on V, the group C is permutation isomorphic, by Theorem 3.10, to the permutation group $\langle g \rangle$ on $\mathrm{GF}(q^{2n})$ via the action $xg = cx$ for all $x \in \mathrm{GF}(q^{2n})$, where c is a $|C|$-th root of unity in $\mathrm{GF}(q^{2n})$ and $\mathrm{GF}(q^{2n}) = \mathrm{GF}(q)[c]$. The eigenvalues of g are therefore the pairwise-distinct conjugates

$$c, c^q, c^{q^2}, \ldots, c^{q^{2n-1}}$$

of c over $\mathrm{GF}(q)$.

Let $\mathsf{K} = \mathrm{GF}(q^{2n})$, and let $\mathsf{V}' = \mathsf{V}(2n, q^{2n})$. The element g acts on V' as a diagonal matrix, say

$$\mathsf{V}' = \bigoplus_{i=0}^{2n-1} \mathsf{K}v_i \quad \text{and} \quad v_i g = c^{q^i} v_i \quad (i = 0, 1, \ldots, 2n-1).$$

As is well known, V' is also a nondegenerate symplectic space since $\det\big((v_i, v_j)\big) \neq 0$, so there is some i such that

$$0 \neq (v_0, v_i) = (v_0 g, v_i g) = c^{1+q^i}(v_0, v_i).$$

Hence, $c^{1+q^i} = 1$. It follows that $c^{1-q^{2i}} = 1$, so $c \in \mathrm{GF}(q^{2i})$. This means c is contained in the field $\mathrm{GF}(q^{2d})$, where $d = (i, n)$. Since $\mathrm{GF}(q)[c] = \mathrm{GF}(q^{2n})$, we have $d = n$ and so $n \mid i$. Then since $0 \leq i < 2n$, we deduce that $i = n$, which means either $|C| = |g|$ divides $1 + q^n$ or $i = 0$ and $c = \pm 1$. The second case is impossible since $\mathrm{GF}(q)[c] = \mathrm{GF}(q^{2n})$. □

The next theorem gives information about the relationships between different symplectic groups.

Theorem 9.24.

a) *There is a monomorphism from $\mathrm{GL}_n(\mathsf{K})$ to $\mathrm{Sp}_{2n}(\mathsf{K})$.*

b) *Let L be an extension field of K with $[\mathsf{L} : \mathsf{K}] = m$. Then there is a monomorphism from $\mathrm{Sp}_{2n}(\mathsf{L})$ to $\mathrm{Sp}_{2nm}(\mathsf{K})$.*

Proof. a) Let

$$\mathsf{V} = \langle v_1, w_1\rangle \perp \langle v_2, w_2\rangle \perp \cdots \perp \langle v_n, w_n\rangle,$$

where $(v_i, w_i) = 1$. Let g be a nonsingular linear transformation of $\langle v_1, \ldots, v_n\rangle$ with

$$v_i g = \sum_{k=1}^{n} a_{ik} v_k.$$

We seek to extend g to an isometry g' of the symplectic space V by defining

$$w_j g' = \sum_{m=1}^{n} b_{jm} w_m.$$

(This is certainly possible by Witt's Theorem 9.9!) This forces

$$\delta_{ij} = (v_i, w_j) = (v_i g, w_j g') = \sum_{k,m} a_{ik} b_{jm} (v_k, w_m) = \sum_{k} a_{ik} b_{jk}.$$

So $(b_{ij}) = (g^T)^{-1}$, where g^T is the transpose of g. Thus, the map τ defined by

$$g^\tau = \begin{pmatrix} g & 0 \\ 0 & (g^T)^{-1} \end{pmatrix}$$

is a monomorphism from $\mathrm{GL}_n(\mathsf{K})$ to $\mathrm{Sp}_{2n}(\mathsf{K})$.

b) We regard the symplectic space $\mathsf{V} = \mathsf{V}(2n, \mathsf{L})$ as a vector space $\mathsf{V}' = \mathsf{V}(2nm, \mathsf{K})$ of dimension $2nm$ over K. Let μ be a K-linear map from L to K that is not identically zero. If $(_, _)$ is the scalar product on V with values in L, then we may define a scalar product on V' by $[v, w] = (v, w)^\mu$. Clearly, $[_, _]$ is bilinear over K and $[v, v] = 0$ for all $v \in \mathsf{V}'$. If $0 \neq v \in \mathsf{V}'$, then there exists a $w \in \mathsf{V}'$ such that $(v, w) = 1$. Then for some $a \in \mathsf{L}$,

$$[v, aw] = a^\mu \neq 0.$$

Thus, V' with the scalar product $[_, _]$ is a nondegenerate symplectic space over K. The maps in $\mathrm{Sp}_{2n}(\mathsf{L})$ act on V' as K-linear transformations that preserve the scalar product $[_, _]$. This gives the desired monomorphism from $\mathrm{Sp}_{2n}(\mathsf{L})$ to $\mathrm{Sp}_{2nm}(\mathsf{K})$.□

All the simple groups that we have considered so far, namely the alternating groups A_n ($n \geq 5$) and the projective special linear groups $\mathrm{PSL}_n(p^f)$, have representations as 2-transitive permutation groups. However, this is not the case for all nonabelian simple groups. In particular, this cannot be the reason that the order of a nonabelian simple group is always even:

Theorem 9.25 (Parker). *The projective symplectic group* $\mathrm{PSp}_4(3)$ *does not have a representation as a* 2*-transitive permutation group.*

Proof. By Theorem 9.13 and Definition 9.14, we have

$$|\mathrm{Sp}_4(3)| = 2^7 \cdot 3^4 \cdot 5 \quad \text{and} \quad |\mathrm{PSp}_4(3)| = 2^6 \cdot 3^4 \cdot 5.$$

We begin with some observations regarding the structure of the Sylow subgroups of $\mathrm{PSp}_4(3)$:

a) The Sylow 3-subgroups of $\mathrm{PSp}_4(3)$ are nonabelian. Their commutator subgroups have order at most 3^2:

It is clear that the Sylow 3-subgroups of $\mathrm{PSp}_4(3)$ and $\mathrm{Sp}_4(3)$ are isomorphic. Let

$$\mathsf{V} = \mathsf{V}(4,3) = \langle v_1, v_2\rangle \perp \langle v_3, v_4\rangle,$$

where $(v_1, v_2) = (v_3, v_4) = 1$. Then the maps $t = t(a,b,c,d)$ of the form

$$\begin{aligned} v_1 t &= v_1 \\ v_2 t &= dv_1 + v_2 + (b - ac)v_3 - av_4 \\ v_3 t &= av_1 + v_3 \\ v_4 t &= bv_1 + cv_3 + v_4 \end{aligned}$$

constitute a subgroup T of $\mathrm{Sp}_4(3)$ of order 3^4. The two maps $t_1, t_2 \in T$ defined by

$$\begin{aligned} v_1 t_1 &= v_1, & v_1 t_2 &= v_1, \\ v_2 t_1 &= v_2 - v_4, & v_2 t_2 &= v_2, \\ v_3 t_1 &= v_1 + v_3, & v_3 t_2 &= v_3, \\ v_4 t_1 &= v_4, & v_4 t_2 &= v_3 + v_4 \end{aligned}$$

do not commute.

The map sending $t(a,b,c,d)$ to the pair (a,c) is a homomorphism from T onto an abelian group of type $(3,3)$ whose kernel contains T' and has order 3^2. Thus, $|T'| \leq 3^2$.

b) If R is a Sylow 2-subgroup of $\mathrm{PSp}_4(3)$, then R/R' has exponent 2:

Let Q be a Sylow 2-subgroup of $\mathrm{Sp}_2(3) \cong \mathrm{SL}_2(3)$. By Theorem 8.10 a), Q is a quaternion group of order 8. With respect to the basis $\{v_1, v_2, v_3, v_4\}$ from a), the matrices

$$\begin{pmatrix} q_1 & 0 \\ 0 & q_2 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & q_1 \\ q_2 & 0 \end{pmatrix} \quad (q_i \in Q)$$

constitute a subgroup $S \leq \mathrm{Sp}_4(3)$ of order 2^7, so S is a Sylow 2-subgroup. We have

$$\begin{pmatrix} q_1 & 0 \\ 0 & q_2 \end{pmatrix}^2 = \begin{pmatrix} q_1^2 & 0 \\ 0 & q_2^2 \end{pmatrix} = \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix} \in S'$$

and

$$\begin{pmatrix} 0 & q_1 \\ q_2 & 0 \end{pmatrix}^2 = \begin{pmatrix} q_1q_2 & 0 \\ 0 & q_2q_1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \pm 1 \end{pmatrix} \begin{pmatrix} q_1q_2 & 0 \\ 0 & (q_1q_2)^{-1} \end{pmatrix} \in S',$$

since $\begin{pmatrix} q & 0 \\ 0 & q^{-1} \end{pmatrix}$ is the commutator of $\begin{pmatrix} q^{-1} & 0 \\ 0 & 1 \end{pmatrix}$ with $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Therefore, $S^2 \leq S'$. Since $R = S/Z$ (where $Z = \langle -1 \rangle$) is a Sylow 2-subgroup of $\mathrm{PSp}_4(3) = \mathrm{Sp}_4(3)/Z$, we have $R^2 \leq R'$.

c) If P is a Sylow 5-subgroup of $G = \mathrm{PSp}_4(3)$, then $\mathbf{C}_G(P) = P$ and $|\mathbf{N}_G(P)| = 20$:

Let $\widetilde{P}$ be a Sylow 5-subgroup of $\widetilde{G} = \mathrm{Sp}_4(3)$. Since $5 \nmid 3^i - 1$ for $i < 4$, $\mathsf{V} = \mathsf{V}(4,3)$ is an irreducible $\widetilde{P}$-module by Theorem 3.10. By Schur's Lemma, $\mathbf{C}_{\widetilde{G}}(\widetilde{P})$ is then cyclic as it is a multiplicative subgroup of a finite division ring. We have that $|\mathbf{C}_{\widetilde{G}}(\widetilde{P})|$ divides $3^2 + 1$ by Theorem 9.23, so $\mathbf{C}_{\widetilde{G}}(\widetilde{P}) = \widetilde{P}Z$ with $Z = \langle -1 \rangle$, and then $\mathbf{C}_G(P) = P$. Now $|\mathbf{N}_G(P)| = 5k$, where

$$k = |\mathbf{N}_G(P)/P| = |\mathbf{N}_G(P)/\mathbf{C}_G(P)| \quad \text{divides} \quad |\mathbf{Aut}(P)| = 4.$$

So $k = 2^i$ for some $0 \leq i \leq 2$. It follows from

$$[G : \mathbf{N}_G(P)] = 2^{6-i} \cdot 3^4 \equiv 1 \bmod 5$$

that $i = 2$.

(Note: Parker proves the nonexistence of 2-transitive representations for the unitary group $\mathrm{PSU}_4(4)$, which is isomorphic to $\mathrm{PSp}_4(3)$. Since the Sylow 5-subgroups of $\mathrm{SU}_4(4)$ are reducible, the proof of c) in that case is easier to carry out.)

d) Assume for contradiction that $G = \mathrm{PSp}_4(3)$ has a 2-transitive permutation representation of degree n. By Theorem 1.11, we have that

$$n(n-1) \text{ divides } |G| = 2^6 \cdot 3^4 \cdot 5 \quad \text{and} \quad |G| \text{ divides } n!.$$

This forces n to be $9, 10, 2^4$ or 3^4.

First assume that $n = 9$ or $n = 10$. The Sylow 3-subgroups of G are then full Sylow 3-subgroups of S_9, resp. S_{10}, so contain 3-cycles. But then $A_n \leq G$ with $n = 9$ or $n = 10$ by Theorem 4.5 c), a contradiction.

e) Assume $n = 3^4$. If U is the stabilizer of one of the 3^4 points, then $[G : U] = n = 3^4$, so $|U| = 2^6 \cdot 5$. By Theorem I.8.11 (pg. 42), U is solvable. Let N be a minimal normal subgroup of U and P a Sylow 5-subgroup of U. By c), we have $|\mathbf{N}_U(P)| \leq 20$, so that $P \not\trianglelefteq U$. So $|N| = 2^k$ with $k \leq 6$. By Maschke's Theorem I.17.6 (pg. 131), N is a direct sum of irreducible P-groups N_i. Since $\mathbf{C}_G(P) = P$, none of the N_i are centralized by P. It therefore follows by Theorem 3.10 that $|N_i| = 2^4$ for all i. This forces $|N| = 2^4$. Since $|U/N| \leq 20$, U/N has only one Sylow 5-subgroup, so $NP \trianglelefteq U$ and $U = N\mathbf{N}_U(P)$ by Theorem I.7.8 (pg. 36). Let $g \in N \cap \mathbf{N}_U(P)$. For all $x \in P$, we have

$$g^{-1}x^{-1}gx \in N \cap P = 1,$$

so by c)

$$g \in \mathbf{C}_G(P) \cap N = P \cap N = 1.$$

Thus, $U = N\mathbf{N}_U(P)$ and $N \cap \mathbf{N}_U(P) = 1$. So $|\mathbf{N}_U(P)| = 20$. Since $\mathbf{C}_U(P) = P$, $\mathbf{N}_U(P)/P$ is cyclic of order 4. Therefore, every Sylow 2-subgroup of U contains a normal subgroup N with cyclic factor group of order 4. This is impossible by b).

f) Now assume $[G : U] = n = 2^4$, so $|U| = 2^2 \cdot 3^4 \cdot 5$. Again let P be a Sylow 5-subgroup of U and $|\mathbf{N}_U(P)| = 5k$. It follows from

$$[U : \mathbf{N}_U(P)] = \frac{2^2 \cdot 3^4}{k} \equiv 1 \bmod 5$$

that $k = 4$. Thus, the Sylow 2-subgroup of $\mathbf{N}_U(P)$ is a Sylow 2-subgroup of U, and the Sylow 2-subgroups of U are cyclic. By Exercise I.21 (pg. 32), there then exists a normal subgroup K of U such that $|U/K| = 4$. (Here, the Burnside Transfer Theorem IV.2.6 may be used in place of Exercise I.21.) The subgroup K has exactly one Sylow 3-subgroup by Theorem I.7.3, say T. By a), T is nonabelian. Since $|T'| \leq 3^2$, it follows that T' is centralized by P. This contradicts $\mathbf{C}_G(P) = P$ in c).

Therefore, $G = \mathrm{PSp}_4(3)$ has no representation as a 2-transitive permutation group. □

Exercises

19) Let V be a nondegenerate symplectic space of dimension $2n$ over $\mathrm{GF}(q)$. How many isotropic subspaces of dimension n does V have?

20) Let V be a nondegenerate symplectic space with

$$\mathsf{V} = \langle v_1, w_1 \rangle \perp \cdots \perp \langle v_n, w_n \rangle$$

and $(v_i, w_i) = 1$ $(i = 1, \ldots, n)$. Let t be an isometry of V such that $v_i t = v_i$ for $1 \leq i \leq n$. Then

$$w_i t = \sum_{j=1}^{n} a_{ij} v_j + w_i,$$

where $a_{ij} = a_{ji}$.

21) Let t be an element of order 2 in $\mathrm{Sp}_{2n}(\mathsf{K})$.
   a) If $\mathrm{char}\, \mathsf{K} \neq 2$, then there exists an orthogonal decomposition $\mathsf{V} = \mathsf{V}_1 \perp \mathsf{V}_2$ such that $v_1 t = v_1$ for all $v_1 \in \mathsf{V}_1$ and $v_2 t = -v_2$ for all $v_2 \in \mathsf{V}_2$.
   b) If $\mathrm{char}\, \mathsf{K} = 2$, then there exists an isotropic subspace W of dimension n such that $wt = w$ for all $w \in \mathsf{W}$.
   (Hint for part b): Let $\mathsf{F} = \{v \in \mathsf{V} \mid vt = v\}$. Show that:
      1) $\mathsf{V}(t - 1)$ is an isotropic subspace of F.

2) If $\mathsf{F} = \mathbf{R}(\mathsf{F}) \perp \mathsf{F}_0$, then $\mathsf{F} \cap \mathsf{F}_0^\perp = \mathbf{R}(\mathsf{F})$.
3) $\dim \mathsf{F}_0^\perp = \dim \mathsf{F}_0^\perp(t-1) + \dim \mathbf{R}(\mathsf{F})$.
4) $\mathbf{R}(\mathsf{F})$ is a maximal isotropic subspace of $\mathsf{F}_0^\perp$.
5) If F_1 is a maximal isotropic subspace of F_0, then $\mathbf{R}(\mathsf{F}) + \mathsf{F}_1$ is a maximal isotropic subspace of V.)

10 Unitary and orthogonal groups

Definition 10.1.

a) Let V be a finite-dimensional vector space over the field K. Let τ be an automorphism of K such that $\tau^2 = 1$, and let K_0 be the fixed field of τ. (For $\tau \neq 1$, it is well known that $[\mathsf{K} : \mathsf{K}_0] = 2$.) Suppose there is a scalar product $(_, _)$ defined on V with the following properties:
(1) $(v_1 + v_2, w) = (v_1, w) + (v_2, w)$ and
$(w, v_1 + v_2) = (w, v_1) + (w, v_2)$ for all $v_1, v_2, w \in \mathsf{V}$.
(2) $(av, w) = a(v, w)$ for all $v, w \in \mathsf{V}$ and all $a \in \mathsf{K}$.
(3) $(v, w) = (w, v)^\tau$ for all $v, w \in \mathsf{V}$.
From (2) and (3), we obtain immediately that
(4) $(v, aw) = a^\tau(v, w)$ for all $v, w \in \mathsf{V}$ and all $a \in \mathsf{K}$.
It follows from (3) that $(v, v) = (v, v)^\tau$, so (v, v) lies in the fixed field K_0 of τ for every $v \in \mathsf{V}$.

If $\tau \neq 1$, then V is called a Hermitian vector space. If $\tau = 1$, then V is called an orthogonal vector space. We say that V is nondegenerate if for every $0 \neq v \in \mathsf{V}$ there exists some $w \in \mathsf{V}$ such that $(v, w) \neq 0$.

b) The Hermitian (orthogonal) vector spaces V and W (over the same field K and with the same automorphism τ) are said to be isomorphic if there is a K-linear bijective map t from V to W such that

$$(v_1 t, v_2 t) = (v_1, v_2)$$

for all $v_i \in \mathsf{V}$. We call t an isometry from V to W. (Here we have used the same notation for the scalar products on V and W.)

c) If U_1 and U_2 are subspaces of the Hermitian (orthogonal) vector space V such that $\mathsf{V} = \mathsf{U}_1 \oplus \mathsf{U}_2$ and $(u_1, u_2) = 0$ for all $u_i \in \mathsf{U}_i$ $(i = 1, 2)$, then we write $\mathsf{V} = \mathsf{U}_1 \perp \mathsf{U}_2$ as in Definition 9.3. If U is a subspace of V, then we define

$$\mathsf{U}^\perp = \{v \in \mathsf{V} \mid (u, v) = 0 \text{ for all } u \in \mathsf{U}\}.$$

In a similar way to Lemma 9.4 one can show that if V is nondegenerate, then

$$\dim \mathsf{V} = \dim \mathsf{U} + \dim \mathsf{U}^\perp.$$

If U is nondegenerate, then $\mathsf{V} = \mathsf{U} \perp \mathsf{U}^\perp$.

d) Let V be a nondegenerate Hermitian (orthogonal) vector space over K. The group of all K-linear maps g from V onto itself with $(v_1 g, v_2 g) = (v_1, v_2)$ for all $v_i \in \mathsf{V}$ is called the unitary group $\mathrm{U}(\mathsf{V})$ (the orthogonal group $\mathrm{O}(\mathsf{V})$). The subgroup

$$\mathrm{SU}(\mathsf{V}) = \{g \in \mathrm{U}(\mathsf{V}) \mid \det g = 1\}$$

is called the special unitary group. We denote by $\mathrm{PSU}(\mathsf{V})$ the group of all collineations of the projective space P associated to V that are induced by $\mathrm{SU}(\mathsf{V})$. The groups $\mathrm{SO}(\mathsf{V})$ and $\mathrm{PSO}(\mathsf{V})$ are defined in the corresponding way.

Theorem 10.2. *Let V be a nondegenerate Hermitian (orthogonal) vector space over the field K. If V is orthogonal, assume* $\operatorname{char}\mathsf{K} \neq 2$.

a) *There exists some $v \in \mathsf{V}$ such that $(v, v) \neq 0$.*

b) *There exists a K-basis $\{v_1, \ldots, v_n\}$ of V such that $(v_i, v_j) = 0$ for $i \neq j$ and $(v_i, v_i) \neq 0$ for $i = 1, \ldots, n$. The v_i constitute a so-called orthogonal basis of V.*

c) *The determinant yields a homomorphism from the unitary group $\mathrm{U}(\mathsf{V})$ onto the group*

$$\{a \in \mathsf{K}^\times \mid aa^\tau = 1\}$$

of elements of norm 1. The determinant yields a homomorphism from the orthogonal group $\mathrm{O}(\mathsf{V})$ onto the group $\{1, -1\}$ of order 2.

Proof. a) Assume for contradiction that $(x, x) = 0$ for all $x \in \mathsf{V}$. Since V is nondegenerate, there exists a pair $v, w \in \mathsf{V}$ such that $(v, w) = 1$.

First assume that V is Hermitian. Then there exists some $a \in \mathsf{K}$ such that $a + a^\tau \neq 0$: For $\operatorname{char}\mathsf{K} = 2$ this is the requirement that $a \neq a^\tau$, which can be satisfied since $\tau \neq 1$; if $\operatorname{char}\mathsf{K} \neq 2$ and $a^\tau = -a$, then $1 + a$ satisfies our desired condition since then

$$(1 + a)^\tau = 1 - a \neq -(1 + a).$$

From this follows the contradiction

$$0 = (v + aw, v + aw) = (v, v) + a^\tau (v, w) + a(w, v) + a^{1+\tau}(w, w) = a^\tau + a.$$

Now assume that V is orthogonal. Then

$$0 = (v + w, v + w) = (v, v) + (v, w) + (w, v) + (w, w) = (v, w) + (w, v).$$

On the other hand, we have $(v, w) = (w, v)$. Since $\operatorname{char}\mathsf{K} \neq 2$, we obtain the contradiction $(v, w) = 0$.

b) By a), there exists some $v_1 \in \mathsf{V}$ such that $(v_1, v_1) = a_1 \neq 0$. Then $\langle v_1 \rangle$ is a nondegenerate subspace of V. By Definition 10.1 c), it follows that $\mathsf{V} = \langle v_1 \rangle \perp \langle v_1 \rangle^\perp$. Induction on $\dim \mathsf{V}$ yields the claim.

c) Let $\{v_1, \ldots, v_n\}$ be a K-basis of V, and let $A = (a_{ij})$ with $a_{ij} = (v_i, v_j)$. Let $g \in \mathrm{U}(\mathsf{V})$ (resp. $g \in \mathrm{O}(\mathsf{V})$) with

$$v_i g = \sum_{k=1}^{n} g_{ik} v_k.$$

We define $\tilde{g} = (g_{ik})$. If V is Hermitian, it follows that

$$a_{ij} = (v_i, v_j) = (v_i g, v_j g) = \sum_{k,m} g_{ik} g_{jm}^{\tau} a_{km},$$

so $A = \tilde{g} A \tilde{g}^{T\tau}$, where $\tilde{g}^T$ is the transposed matrix of $\tilde{g}$. If V is orthogonal, then the same considerations yield $A = \tilde{g} A \tilde{g}^T$. Since V is nondegenerate, the system of linear equations

$$\Big(\sum_{i=1}^{n} x_i v_i, v_j\Big) = \sum_{i=1}^{n} x_i a_{ij} = 0 \quad (j = 1, \ldots, n)$$

has only the trivial solution. Therefore $\det A \neq 0$. Now it follows that

$$0 \neq \det A = \det \tilde{g} A \tilde{g}^{T\tau} = \det A (\det \tilde{g})^{1+\tau},$$

so $(\det \tilde{g})^{1+\tau} = 1$ (the orthogonal case yields $(\det \tilde{g})^2 = 1$).

Let $a \in \mathsf{K}$ with $aa^{\tau} = 1$. Let $\{v_1, \ldots, v_n\}$ be an orthogonal basis of V. Then the map g defined by

$$v_1 g = a v_1, \; v_i g = v_i \quad \text{for} \quad i \geq 2$$

is an isometry of V with $\det g = a$. □

Remark 10.3. The conclusion of c) is also correct for $\operatorname{char} \mathsf{K} = 2$. However, the method of the previous proof is not suitable for orthogonal vector spaces over fields of characteristic 2. The isometries of a vector space with symmetric scalar product over a field of characteristic 2 do not yield any fundamentally new groups (see Dieudonné [115], pg. 60). If $\operatorname{char} \mathsf{K} = 2$, then one considers vector spaces V over K on which a bilinear form $[_, _]$ and a quadratic form q such that

$$q(a_1 v_1 + a_2 v_2) = a_1^2 q(v_1) + a_2^2 q(v_2) + a_1 a_2 [v_1, v_2] \quad (a_i \in \mathsf{K}, v_i \in \mathsf{V})$$

is defined.

Then $[v, v] = 0$ for all $v \in \mathsf{V}$. One demands that the vector space V is a nondegenerate symplectic space with respect to $[_, _]$. (The other cases are not interesting for finite fields.) For the case of orthogonal groups with $\operatorname{char} \mathsf{K} = 2$, in place of the determinant there is a homomorphism $\mathbf{D} : \mathrm{O}(\mathsf{V}) \to \mathsf{K}^+$ which was given by Dickson. For all $g \in \mathrm{O}(\mathsf{V})$ it turns out $\mathbf{D}(g) = 0$ or $\mathbf{D}(g) = 1$. Then one defines

$$\mathrm{SO}(\mathsf{V}) = \{g \in \mathrm{O}(\mathsf{V}) \mid \mathbf{D}(g) = 0\}.$$

See Dieudonné [114].

The structure of Hermitian vector spaces over finite fields is completely described in the following theorem.

Theorem 10.4. *Let* V *be a nondegenerate Hermitian vector space over the finite field* $\mathsf{K} = \mathrm{GF}(q^2)$.

a) *There exists a so-called orthonormal basis* $\{v_1, \dots, v_n\}$ *of* V *with* $(v_i, v_j) = \delta_{ij}$. *Thus,*

$$\Big(\sum_{i=1}^{n} x_i v_i, \sum_{i=1}^{n} y_i v_i\Big) = \sum_{i=1}^{n} x_i y_i^{\tau},$$

where τ *is the uniquely determined automorphism of order* 2 *of* K *defined by* $x^{\tau} = x^q$. *In particular, the isomorphism type of* V *is uniquely determined by* $\dim \mathsf{V} = n$.

b) *If* $\dim \mathsf{V} = 2m$ *is even, then there exists a basis* $\{w_1, \dots, w_{2m}\}$ *of* V *such that*

$$(w_{2i-1}, w_{2i}) = 1 \quad (i = 1, \dots m) \quad \textit{and} \quad (w_i, w_j) = 0 \textit{ otherwise}.$$

(*In particular,* $(w_i, w_i) = 0$ *for all* i. *Compare with Theorem 9.6 b)!*) *If* $\dim \mathsf{V} = 2m + 1$ *is odd, then there exists a further basis element* w_{2m+1} *such that*

$$(w_{2m+1}, w_{2m+1}) = 1, \ (w_{2m+1}, w_i) = 0 \quad \textit{for} \quad i \le 2m.$$

c) *If* $\dim \mathsf{V} > 1$, *then there exist so-called isotropic vectors* v *with* $0 \neq v \in \mathsf{V}$ *and* $(v, v) = 0$.

Proof. a) Let $\{v'_1, \dots, v'_n\}$ be an orthogonal basis for V, which exists by Theorem 10.2 b). Let $(v'_i, v'_i) = a_i$. Then $0 \neq a_i \in \mathsf{K}_0 = \mathrm{GF}(q)$. By Lemma 8.7, there exists some $b_i \in \mathsf{K}$ such that $a_i^{-1} = b_i^{1+\tau}$. If we define $v_i = b_i v'_i$, then $\{v_1, \dots, v_n\}$ is an orthonormal basis of V.

b) By a), we have

$$\mathsf{V} = \langle v_1, v_2\rangle \perp \langle v_3, v_4\rangle \perp \cdots \perp \langle v_{2m-1}, v_{2m}\rangle \perp \mathsf{V}',$$

where $\mathsf{V}' = 0$ if $n = 2m$ and $\mathsf{V}' = \langle v_n\rangle$ if $n = 2m + 1$. It suffices to find a new basis for each two-dimensional space $\langle v_i, v_{i+1}\rangle$ $(i = 1, 3, \dots, 2m - 1)$ with the desired properties.

First assume that $\mathrm{char}\, \mathsf{K} \neq 2$. By Lemma 8.7, there exists an $a \in \mathsf{K}$ such that $a^{1+\tau} = -1$. We define

$$w_1 = v_1 + a v_2, \quad w_2 = \frac{1}{2}(v_1 - a v_2).$$

Then $(w_1, w_1) = 1 + a^{1+\tau} = 0$, similarly $(w_2, w_2) = 0$ and

$$(w_1, w_2) = \frac{1}{2}(1 - a^{1+\tau}) = 1.$$

So the basis $\{w_1, w_2\}$ of $\langle v_1, v_2\rangle$ has the desired properties.

Now suppose that char $\mathsf{K} = 2$. The equation

$$x^2 + x + 1 = 0$$

has a solution a in the quadratic extension K of K_0. Certainly $a \neq 1$. We define

$$w_1 = v_1 + v_2 \quad \text{and} \quad w_2 = av_1 + a^\tau v_2.$$

Then we have

$$(w_1, w_1) = 0, \quad (w_2, w_2) = a^{1+\tau} + a^{\tau+1} = 0 \quad \text{and} \quad (w_1, w_2) = a^\tau + a = 1.$$

c) is a trivial consequence of b). □

The theory of unitary groups over finite fields is greatly simplified by Theorem 10.4 compared to the situation over an arbitrary field. (This is unlike the case of the symplectic groups, where it was never necessary to make assumptions about the underlying field.) By Theorem 10.4 a), there exists up to isomorphism only one non-degenerate Hermitian vector space of a given dimension over the field $\mathsf{K} = \mathrm{GF}(q^2)$ with the uniquely determined automorphism τ of order 2 defined by $x^\tau = x^q$. Therefore, we may set $\mathrm{U}(\mathsf{V}) = \mathrm{U}_n(q^2)$. The existence of isotropic vectors in V is the key to the proof that $\mathrm{PSU}_n(q^2)$ is simple when $n \geq 3$ and $(n, q) \neq (3, 2)$.

Remark 10.5.

a) The unitary group $\mathrm{SU}_2(q^2)$ is isomorphic to $\mathrm{SL}_2(q)$ by Lemma 8.8 (but see Exercise 24).

b) With respect to the orthonormal basis for the Hermitian vector space V over K from Theorem 10.4 a), the group $\mathrm{U}_n(q^2)$ consists of the matrices g such that $g^\tau = (g^T)^{-1}$, where g^T denotes the transpose of g. There exist automorphisms α and μ of $\mathrm{GL}_n(q^2)$ such that

$$g^\alpha = (g^T)^{-1} \quad \text{and} \quad g^\mu = g^\tau.$$

Therefore, $\mathrm{U}_n(q^2)$ is a subgroup of the fixed points of the automorphism $\alpha\mu$ of $\mathrm{GL}_n(q^2)$.

This is the simplest and most well-known case of a construction that has led to the discovery of new simple groups. (See Ono [340], as well as Ree [351, 352] and Steinberg [394, 396] for more on so-called twisted types.)

c) We have $\mathrm{PSU}_4(4) \cong \mathrm{PSp}_4(3)$. (An elegant proof can be found in D. G. Higman [240].)

Lemma 10.6. *If K is a finite field, then every element $a \in \mathsf{K}$ can be written as $a = x^2 + y^2$ for some $x, y \in \mathsf{K}$.*

Proof. If char $\mathsf{K} = 2$, the claim is trivial since in this case every element is itself a square. So we may assume that char $\mathsf{K} \neq 2$. Let $U = \{a^2 \mid a \in \mathsf{K}^\times\}$ and $\mathsf{K}^\times = U \cup Uk$. If $uk = x^2 + y^2$ for some $u \in U$, then for every $u' \in U$ with $u' = c^2 u$ we have

$$u'k = c^2uk = (cx)^2 + (cy)^2.$$

So it suffices to show that among the elements of the form $x^2 + y^2$ there exists an element that is not a square.

Assume for contradiction that x^2+y^2 is a square for all $x, y \in \mathsf{K}$. Then $\mathsf{L} = \{0\} \cup U$ is a subfield of K. Letting $|\mathsf{K}| = q$, we have $|\mathsf{L}| = \frac{q+1}{2}$. But $\frac{q+1}{2} \nmid q$, which contradicts Lagrange's theorem. □

Definition 10.7. A two-dimensional orthogonal vector space with a basis $\{v_1, v_2\}$ satisfying $(v_1, v_1) = (v_2, v_2) = 0$ and $(v_1, v_2) = 1$ is called a hyperbolic plane. Clearly, hyperbolic planes are nondegenerate.

Lemma 10.8. *Let* V *be a nondegenerate orthogonal space over the finite field* K *with* $\operatorname{char}\mathsf{K} \neq 2$.

a) *If* $\dim \mathsf{V} \geq 3$, *then there exists a nonzero isotropic vector* $v \in \mathsf{V}$ *with* $(v, v) = 0$.

b) *If there exists a nonzero isotropic vector in* V, *then* V *has an orthogonal decomposition* $\mathsf{V} = \mathsf{H} \perp \mathsf{H}^\perp$ *for some hyperbolic plane* H.

Proof. a) Let $\{v_1, \ldots, v_n\}$ be an orthogonal basis of V. By modifying the v_i by factors in $\mathsf{K}^\times$, we can ensure that $(v_i, v_i) = 1$ or $(v_i, v_i) = k$ for a given nonsquare $k \in \mathsf{K}$. Since $n \geq 3$, we may assume that $(v_1, v_1) = (v_2, v_2)$. By Lemma 10.6, there exist elements $x, y \in \mathsf{K}$ such that

$$x^2 + y^2 = -\frac{(v_3, v_3)}{(v_1, v_1)}.$$

Then $xv_1 + yv_2 + v_3$ is an isotropic vector.

b) Suppose $v \in \mathsf{V}$ is an isotropic vector. Since V is nondegenerate, there exists a vector w such that $(v, w) = 1$. Clearly, $\langle v, w\rangle$ has dimension 2. If we define $a = -\frac{(w,w)}{2}$, then

$$(v, av + w) = (v, w) = 1$$

and

$$(av + w, av + w) = 2a + (w, w) = 0.$$

So $\mathsf{H} = \langle v, av + w\rangle$ is a hyperbolic plane. Since H is nondegenerate, $\mathsf{V} = \mathsf{H} \perp \mathsf{H}^\perp$ by Definition 10.1 c). □

Theorem 10.9. *Let* K *be a finite field with* $\operatorname{char}\mathsf{K} \neq 2$, *and let* V *be a nondegenerate orthogonal vector space of dimension n over* K.

a) *There is an orthogonal decomposition*

$$\mathsf{V} = \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_m \perp \mathsf{V}'$$

for some hyperbolic planes H_i. *The subspace* V' *contains no nonzero isotropic vectors and* $\dim \mathsf{V}' \leq 2$. *If* $\dim \mathsf{V}' = 2$, *then* V' *has an orthogonal basis* $\{v_1, v_2\}$ *such that* $(v_2, v_2) = -k(v_1, v_1)$, *where* $k \in \mathsf{K}^\times$ *is a nonsquare.*

b) *There exists an orthogonal basis* $\{v_1, \dots, v_n\}$ *of* V *such that* $(v_i, v_i) = 1$ *for* $1 \le i \le n-1$.

Proof. a) By repeated application of Lemma 10.8 b), we obtain

$$\mathsf{V} = \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_m \perp \mathsf{V}'$$

for some hyperbolic planes H_i. The subspace V' contains no nonzero isotropic vectors. Thus, $\dim \mathsf{V}' \le 2$ by Lemma 10.8 a).

Let $\dim \mathsf{V}' = 2$, and let $\{v_1, v_2\}$ be an orthogonal basis of V'. If we had $(v_2, v_2) = -a^2(v_1, v_1)$, then $av_1 + v_2$ would be an isotropic vector in V'. So $(v_2, v_2) = -k(v_1, v_1)$ for some nonsquare k.

b) It suffices to show that every nondegenerate orthogonal space of dimension at least two contains a vector v such that $(v, v) = 1$. Let $\{v_1, \dots, v_n\}$ be an orthogonal basis of V. We may assume that $(v_1, v_1) = (v_2, v_2) = k$ is a nonsquare. By Lemma 10.6, there exist elements $x, y \in \mathsf{K}$ such that $k(x^2 + y^2) = 1$. If we define $w = xv_1 + yv_2$, then $(w, w) = 1$. □

In what follows, we restrict our attention to the orthogonal and unitary groups of dimensions 2 and 3.

Theorem 10.10. *Let* $\mathsf{K} = \mathrm{GF}(q)$ *with* $2 \nmid q$.

a) *If* V *is a hyperbolic plane over* K*, then* $\mathrm{SO}(\mathsf{V})$ *is a cyclic group of order* $q - 1$.

b) *If* V *is a two-dimensional orthogonal space without isotropic vectors, then* $\mathrm{SO}(\mathsf{V})$ *is cyclic of order* $q + 1$.

Proof. a) Let $\mathsf{V} = \langle v_1, v_2 \rangle$, where $(v_1, v_1) = (v_2, v_2) = 0$ and $(v_1, v_2) = 1$. Since

$$(x_1v_1 + x_2v_2,\ x_1v_1 + x_2v_2) = 2x_1x_2,$$

the only isotropic vectors in V are multiples of v_1 and v_2. Let $g \in \mathrm{SO}(\mathsf{V})$. If

$$v_1 g = av_2 \quad \text{and} \quad v_2 g = bv_1,$$

then it follows that

$$ab = (v_1 g, v_2 g) = (v_1, v_2) = 1;$$

on the other hand, however, $\det g = -ab = 1$. This is not possible.

Hence, $v_1 g = av_1$ and $v_2 g = bv_2$. Again it follows that $ab = 1$. Therefore, $\mathrm{SO}(\mathsf{V})$ is isomorphic to the cyclic group $\mathsf{K}^\times$ of order $q - 1$.

b) We may assume that $(v_1, v_1) = 1$, $(v_1, v_2) = 0$ and $(v_2, v_2) = -k$ for some nonsquare $k \in \mathsf{K}$. Let $\mathsf{L} = \mathsf{K}(z)$ be the quadratic extension of K with $z^2 = k$. Let ϵ be the K-linear map from V onto L defined by

$$(x_1v_1 + x_2v_2)^\epsilon = x_1 + x_2z.$$

We have

$$(x_1v_1 + x_2v_2,\, x_1v_1 + x_2v_2) = x_1^2 - kx_2^2 = \mathrm{N}(x_1 + x_2z),$$

where N is the norm of L over K. By defining

$$2(x, y) = \mathrm{N}(x + y) - \mathrm{N}(x) - \mathrm{N}(y)$$

for $x, y \in \mathsf{L}$, the vector space L becomes an orthogonal space, and ϵ is an isometry from V to L. Thus, we may study $\mathrm{SO}(\mathsf{L})$ instead of $\mathrm{SO}(\mathsf{V})$.

Let $a \in \mathsf{L}$ with $\mathrm{N}(a) = 1$. Then the map $h(a)$ defined by $xh(a) = ax$ is a K-linear map from L onto itself with

$$\mathrm{N}(ax) = \mathrm{N}(a)\mathrm{N}(x) = \mathrm{N}(x).$$

So $h(a)$ preserves the scalar product on L. If $a = y_1 + y_2z$, then

$$\det h(a) = \begin{vmatrix} y_1 & y_2 \\ ky_2 & y_1 \end{vmatrix} = \mathrm{N}(a) = 1.$$

Thus, the $h(a)$ constitute a cyclic subgroup of order $q + 1$ of $\mathrm{SO}(\mathsf{L})$.

Let $g \in \mathrm{SO}(\mathsf{L})$, and let $1g = a$. Then $\mathrm{N}(a) = \mathrm{N}(1) = 1$. If we define $t = g\,h(a)^{-1}$, then $t \in \mathrm{SO}(\mathsf{L})$ and $1t = 1$. Let $zt = b + cz$ for some $b, c \in \mathsf{K}$. Then

$$\begin{aligned} 0 = (1, z) = (1t, zt) &= \mathrm{N}(1t + zt) - \mathrm{N}(1t) - \mathrm{N}(zt) \\ &= (1 + b)^2 - kc^2 - 1 - b^2 + kc^2 = 2b. \end{aligned}$$

We conclude that $\det t = c = 1$, so $t = 1$. Hence, $g = h(a)$, and $\mathrm{SO}(\mathsf{L})$ is cyclic of order $q + 1$. □

Theorem 10.11. *Let* V *be a nondegenerate orthogonal space of dimension* 3 *over* $\mathrm{GF}(q)$ *with* $2 \nmid q$. *Let* Q *be the following set of points in the projective plane* $\mathsf{E} = \mathsf{P}(2, q)$*:*

$$\mathsf{Q} = \{\langle v\rangle \mid 0 \neq v \in \mathsf{V}, (v, v) = 0\}.$$

Then we have:

a) $|\mathsf{Q}| = q + 1$.

b) *The group* $\mathrm{PSO}(\mathsf{V})$ *acts on* Q *as a sharply* 3*-transitive permutation group that is permutation isomorphic to the permutation group* $\mathrm{PGL}_2(q)$ *on the projective line* $\mathsf{P} = \mathsf{P}(1, q)$.

Proof. a) By Theorem 10.9 a), V has a basis $\{v_1, v_2, v_3\}$ such that

$$\left(\sum_{i=1}^{3} x_iv_i, \sum_{i=1}^{3} x_iv_i\right) = 2x_1x_2 + ax_3^2$$

for some $0 \neq a \in \mathsf{K}$. Let $v = \sum_{i=1}^{3} x_i v_i$ be an isotropic vector. If $x_1 = 0$, then $x_3 = 0$ and the corresponding point $\langle v \rangle$ in E has the homogeneous coordinates $(0, 1, 0)$. If $x_1 \neq 0$, then we may normalize the homogeneous coordinates such that $x_1 = 1$ and obtain

$$x_2 = -\frac{a}{2} x_3^2.$$

Thus,

$$\mathsf{Q} = \left\{ (0,1,0), (1, -\tfrac{a}{2}k^2, k) \;\middle|\; k \in \mathrm{GF}(q) \right\},$$

and so $|\mathsf{Q}| = q + 1$.

b) Let $g \in \mathrm{SO}(\mathsf{V})$ with $v_2 g = b v_2$. Then $\langle v_2 \rangle^\perp = \langle v_2, v_3 \rangle$ is also fixed by g. Therefore,

$$\begin{aligned} v_1 g &= a_{11} v_1 + a_{12} v_2 + a_{13} v_3 \\ v_2 g &= \quad\; a_{22} v_2 \\ v_3 g &= \quad\; a_{32} v_2 + a_{33} v_3 \end{aligned}$$

for some $a_{ij} \in \mathrm{GF}(q)$. So we have

$$\begin{aligned} (v_1 g, v_1 g) &= 2a_{11}a_{12} + a_{13}^2 a = 0, \\ (v_1 g, v_2 g) &= a_{11}a_{22} = 1, \\ (v_1 g, v_3 g) &= a_{11}a_{32} + a_{13}a_{33}a = 0 \end{aligned}$$

and $\det g = a_{11}a_{22}a_{33} = a_{33} = 1$.

We set $a_{11} = x$ and $a_{13} = y$. Then

$$g = g(x,y) = \begin{pmatrix} x & \frac{-y^2 a}{2x} & y \\ 0 & x^{-1} & 0 \\ 0 & \frac{-ya}{x} & 1 \end{pmatrix}.$$

If $g \in \mathrm{SO}(\mathsf{V})$, then g induces a collineation on $\mathsf{E} = \mathsf{P}(2,q)$, which we denote by $\overline{g}$. The stabilizer U of $\langle v_2 \rangle$ in $\mathrm{PSO}(\mathsf{V})$ has order $q(q-1)$ and is sharply 2-transitive on $\mathsf{Q} - \langle v_2 \rangle$ since

$$(1, -\tfrac{a}{2}k^2, k)\,\overline{g}(x,y) = (x, *, y + k) = (1, *, x^{-1}(y + k)).$$

Let $t \in \mathrm{SO}(\mathsf{V})$ with

$$v_1 t = v_2, \quad v_2 t = v_1, \quad v_3 t = -v_3.$$

Then we have

$$(0,1,0)\,\overline{t} = (1,0,0).$$

So $\mathrm{PSO}(\mathsf{V})$ is sharply 3-transitive on Q. Since U is a maximal subgroup of $\mathrm{PSO}(\mathsf{V})$, we have $\mathrm{PSO}(\mathsf{V}) = \langle U, \bar{t} \rangle$.

Let ϵ be the map from Q onto the projective line $\mathsf{P} = \mathsf{P}(1, q)$ defined by

$$(0,1,0)^\epsilon = (0,1) \quad \text{and} \quad (1, -\tfrac{a}{2}k^2, k)^\epsilon = (1,k).$$

By defining

$$(x\overline{g})^\epsilon = x^\epsilon \overline{g}$$

for $x \in \mathsf{Q}$ and $\overline{g} \in \mathrm{PSO}(\mathsf{V})$, we give $\mathrm{PSO}(\mathsf{V})$ the structure of a permutation group on P. Then we have

$$(1,k)\,\overline{g}(x,y) = (1, *, x^{-1}(k+y))^\epsilon = (1, x^{-1}(k+y))$$

and

$$(1,k)\,\overline{t} = \big((1, -\tfrac{a}{2}k^2, k)\,\overline{t}\big)^\epsilon = \big(-\tfrac{a}{2}k^2, 1, -k\big)^\epsilon = \big(1, \tfrac{2}{ak}\big).$$

Furthermore,

$$(0,1)\,\overline{g}(x,y) = (0,1) \quad \text{and} \quad (0,1)\,\overline{t} = (1,0).$$

Thus, $\mathrm{PSO}(\mathsf{V})$ acts as linear fractional transformations on P. Since

$$|\mathrm{PSO}(\mathsf{V})| = (q^2-1)q = |\mathrm{PGL}_2(q)|,$$

the claim follows. □

The orthogonal groups of dimensions 4 to 6 over finite fields also do not yield any essentially new groups. The situation is different among the unitary groups, however. Already in dimension 3 we obtain new simple 2-transitive permutation groups:

Theorem 10.12. *Let $\mathsf{K} = \mathrm{GF}(q^2)$, and let V be the three-dimensional nondegenerate Hermitian vector space over K. We denote by $\mathsf{E} = \mathsf{P}(2, q^2)$ the projective plane associated to V. Then the following hold:*

a) *There are exactly q^3+1 points $x \in \mathsf{E}$ such that $x = \langle v \rangle$ and $(v,v) = 0$. We denote by A the set of all such points.*

b) *Let $\mathsf{V} = \langle w_1, w_2, w_3 \rangle$ with*

$$(w_2, w_2) = (w_1, w_3) = 1, \quad (w_1, w_1) = (w_3, w_3) = (w_1, w_2) = (w_2, w_3) = 0.$$

(This is a trivial modification of the basis given in Theorem 10.4 b).)

Let U be the stabilizer of $\langle w_3 \rangle$ in $G = \mathrm{PSU}_3(q^2)$. Then

$$|U| = \frac{q^3(q^2-1)}{d},$$

where $d = (q+1, 3)$.

In addition, U contains a normal subgroup R acting regularly on $\mathsf{A} - \{\langle w_3\rangle\}$ with $R' = \mathbf{Z}(R)$ and $|R'| = q$. The factor group U/R is cyclic of order $\frac{q^2-1}{d}$. With respect to the basis $\{w_1, w_2, w_3\}$, the group R consists of the collineations of the form

$$r(a,b) = \begin{pmatrix} 1 & a & b \\ 0 & 1 & -a^\tau \\ 0 & 0 & 1 \end{pmatrix}$$

such that $\mathrm{tr}(b) + \mathrm{N}(a) = 0$, where the trace and norm are constructed over $\mathsf{K}_0 = \mathrm{GF}(q)$. The map τ is the automorphism of $\mathrm{GF}(q^2)$ defined by $x^\tau = x^q$. The cyclic group $H = \{h(k) \mid k \in \mathsf{K}^\times\}$ with elements

$$h(k) = \begin{pmatrix} k^{-\tau} & & \\ & k^{\tau-1} & \\ & & k \end{pmatrix}$$

is a complement of R in U.

c) *$\mathrm{PSU}_3(q^2)$ acts on A as a 2-transitive permutation group.*

d) *$|\mathrm{PSU}_3(q^2)| = (q^3+1)q^3\frac{q^2-1}{d}$, where $d = (3, q+1)$.*

e) *When $q > 2$, there exist elements $g \neq 1$ in $\mathrm{PSU}_3(q^2)$ fixing at least three points of A.*

Proof. a) Let $\{w_1, w_2, w_3\}$ be a basis of V with

$$(w_2, w_2) = (w_1, w_3) = 1, \quad (w_1, w_1) = (w_3, w_3) = (w_1, w_2) = (w_2, w_3) = 0.$$

Then for any $w = \sum_{i=1}^{3} x_i w_i$, we have

$$(w, w) = x_1 x_3^\tau + x_1^\tau x_3 + x_2^{1+\tau}.$$

Assume $(w, w) = 0$. If $x_1 = 0$, it follows that $x_2 = 0$ and $w = x_3 w_3$. If $x_1 = 1$, then we obtain

$$0 = x_3^\tau + x_3 + x_2^{1+\tau} = \mathrm{tr}(x_3) + \mathrm{N}(x_2).$$

We set $x_2 = a$ and $x_3 = b$. Every point in A distinct from $\langle w_3\rangle$ corresponds to a vector of the form $(1, a, b)$ with $\mathrm{tr}(b) + \mathrm{N}(a) = 0$. For a given $a \in \mathsf{K}$, $\mathrm{tr}(b) = -\mathrm{N}(a)$ has exactly q solutions since every element $\mathrm{N}(a)$ in K_0 is the trace of exactly q elements in K. Therefore,

$$\mathsf{A} = \{\langle w_3\rangle, (1, a, b) \mid a, b \in \mathsf{K}, \mathrm{tr}(b) + \mathrm{N}(a) = 0\}$$

and $|\mathsf{A}| = q^3 + 1$.

b) Suppose $g \in \mathrm{SU}_3(q^2)$ with $w_3 g = k w_3$. Then $\langle w_3\rangle^\perp = \langle w_2, w_3\rangle$ is also fixed by g. So with respect to the basis $\{w_1, w_2, w_3\}$, the element g has the form

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{pmatrix}$$

with $a_{33} = k$. We have

$$1 = (w_1, w_3) = (w_1 g, w_3 g) = a_{11} a_{33}^{\tau}.$$

Therefore, $a_{11} = a_{33}^{-\tau} = k^{-\tau}$ and then

$$a_{22} = a_{11}^{-1} a_{33}^{-1} = k^{\tau-1}$$

since $\det g = 1$. Clearly, the element $h(k)$ with

$$w_1 h(k) = k^{-\tau} w_1, \quad w_2 h(k) = k^{\tau-1} w_2, \quad w_3 h(k) = k w_3$$

lies in $\mathrm{SU}_3(q^2)$. Thus, $g = h(k)r$ for some r of the form

$$r = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$$

lying in $\mathrm{SU}_3(q^2)$. With this, we have

$$0 = (w_1 r, w_2 r) = c^{\tau} + a \quad \text{and} \quad 0 = (w_1 r, w_1 r) = b^{\tau} + a^{1+\tau} + b,$$

and therefore

$$r = r(a, b) = \begin{pmatrix} 1 & a & b \\ 0 & 1 & -a^{\tau} \\ 0 & 0 & 1 \end{pmatrix},$$

where $\mathrm{tr}(b) + \mathrm{N}(a) = 0$. We have

$$r(a, b)\, r(a', b') = r(a + a', b + b' - a a'^{\tau}).$$

Thus, the $r(a, b)$ with $\mathrm{tr}(b) + \mathrm{N}(a) = 0$ constitute a group R of order q^3 with

$$\mathbf{Z}(R) = R' = \{r(0, b) \mid b \in \mathsf{K}, \mathrm{tr}(b) = 0\}.$$

We have $U = RH$, and $R \trianglelefteq U$ since

$$h(k)^{-1} r(a, b) h(k) = r(k^{2\tau-1} a, k^{\tau+1} b).$$

Now,

$$(1, 0, 0) r(a, b) = (1, a, b),$$

so R acts transitively on the q^3 points of $\mathsf{A} - \{\langle w_3\rangle\}$. Since $|R| = q^3$, the action is regular.

The $h(k)$ constitute a cyclic group of linear maps of V that is isomorphic to $\mathsf{K}^\times$. What order does the associated group of collineations of $\mathsf{E} = \mathsf{P}(2, q^2)$ have? Note that $h(k) = k1$ and induces the trivial collineation of E if and only if

$$k^{-\tau} = k^{\tau-1} = k.$$

Since $k^\tau = k^q$, this is equivalent to $k^{1+q} = k^{q-2} = 1$, so $k^{1+q} = k^3 = 1$. This equation has exactly $d = (q+1, 3)$ solutions in the cyclic group $\mathsf{K}^\times$. So the $h(k)$ constitute a cyclic group H of collineations of E of order $\frac{q^2-1}{d}$.

c) Let t be the linear transformation of V defined by

$$w_1 t = w_3, \quad w_2 t = -w_2, \quad w_3 t = w_1.$$

Then $t \in \mathrm{SU}_3(q^2)$ and

$$w_3\, t\, r(a, b) = (1, a, b).$$

So $\mathrm{PSU}_3(q^2)$ is transitive on A. Since U is transitive on $\mathsf{A} - \{\langle w_3\rangle\}$ by b), $\mathrm{PSU}_3(q^2)$ is 2-transitive on A. Clearly, the permutation representation of $\mathrm{PSU}_3(q^2)$ on A is faithful.

d) By b) and c), we have

$$|\mathrm{PSU}_3(q^2)| = |\mathsf{A}|\,|U| = (q^3+1)q^3\frac{(q^2-1)}{d}.$$

e) The map μ taking $k \in \mathrm{GF}(q^3)^\times$ to $k^{1+\tau} = \mathrm{N}(k) \in \mathrm{GF}(q)^\times$ is an epimorphism from $\mathrm{GF}(q^3)^\times$ onto $\mathrm{GF}(q)^\times$ with $|\ker\mu| = q+1$ by Lemma 8.7 (pg. 203). Thus, for $q > 2$ there exists some k with $k^{1+\tau} = 1$ but $k^{\tau-1} = k^{-2} \neq k$. Then $h(k) \neq 1$ and $h(k)$ has as fixed points the $q+1$ points of A having coordinates $(1, 0, 0)$, $(0, 0, 1)$ and $(x, 0, z)$ with

$$xz^\tau + x^\tau z = \mathrm{tr}(xz^\tau) = 0. \qquad \square$$

Theorem 10.13. *For $q > 2$, $\mathrm{PSU}_3(q^2)$ is simple.*

Proof. We show (using the notation of Theorem 10.12):

a) $U' = R$:

A calculation shows

$$[r(a, b), h(k)] = r((k^{2\tau-1} - 1)a, c)$$

for some c. For $q > 2$, there exists a $k \in \mathsf{K}^\times$ with $k^{2\tau-1} = k^{2q-1} \neq 1$. Let $r(x, y)$ be given. Then there exists an $a \in \mathsf{K}$ with $a(k^{2\tau-1} - 1) = x$. Since $r(0, z) \in R' \leq U'$, we have

$$r(x,y) = [r(a,b), h(k)]\, r(0, y-c) \in U'.$$

Since H is cyclic, it holds trivially that $U' \leq R$.

b) $H \leq G'$:

Let t be the element of $\mathrm{SU}_3(q^2)$ defined by

$$w_1 t = w_3, \quad w_2 t = -w_2, \quad w_3 t = w_1.$$

Then $t^2 = 1$ and $t \in \mathbf{N}_G(H)$ since $t\, h(k)\, t = h(k^{-q})$. Since G is 2-transitive on A, we have by Theorem 1.12 that

$$G = U \cup UtU = U \cup RHtR.$$

Let $0 \neq b \in \mathsf{K}$. It follows from

$$(0,0,1)t\, r(a,b)\, t = (b, -a, 1) \neq (0,0,1)$$

that

$$t\, r(a,b)\, t = r(a',b')\, h(k)\, t\, r(a'',b'') \in RHtR$$

for some a', b', a'', b'', k. Calculating, we find

$$\begin{pmatrix} & * \\ b & \end{pmatrix} = t\, r(a,b)\, t = r(a',b')\, h(k)\, t\, r(a'',b'') = \begin{pmatrix} & * \\ k & \end{pmatrix},$$

so that $b = k$. Since $R \leq G'$, we may conclude that $h(b)\, t \in G'$. Since this holds for all $b \in \mathsf{K}^\times$, it follows that $H \leq G'$.

c) $G' = G$:

By a) and b), we have $HR = U \leq G'$. Since U is a maximal and non-normal subgroup of the 2-transitive group G, it follows that $G' = G$.

d) G is simple:

Let $1 < N \trianglelefteq G$. The subgroup N is transitive by the 2-transitivity of G, and it follows that $G = NU$. We have

$$U''' = R'' = 1.$$

Hence, by c)

$$G = G''' = (NU)''' \leq N. \qquad \square$$

Theorem 10.14. $\mathrm{PSU}_3(4)$ *is a solvable, sharply* 2*-transitive permutation group of degree* 9.

Proof. By Theorem 10.12 c), $G = \mathrm{PSU}_3(4)$ is a 2-transitive permutation group of degree 9 and order 72. So G is sharply 2-transitive.

We show that G has exactly one Sylow 3-subgroup. Let R be the Sylow 2-subgroup of G described in Theorem 10.12 b). (In the current case, R is a quaternion group

of order 8.) Note that R is the stabilizer of the point $\langle w_3\rangle \in \mathsf{A}$. If $R^x \neq R$ and $r \in R \cap R^x$, then r fixes the points $\langle w_3\rangle$ and $\langle w_3\rangle x$. Since G is sharply 2-transitive on A, it follows that $r = 1$.

In the 9 stabilizers, there are exactly $1 + 9 \cdot 7 = 64$ elements of G. The remaining 8 elements must, together with 1, form the unique Sylow 3-subgroup of G. If one specifies an orthonormal basis of V, then the Sylow 3-subgroup of $\mathrm{PSU}_3(4)$ is generated by the collineations induced by the matrices

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \end{pmatrix}$$

with $a_i^3 = a_1 a_2 a_3 = 1$. □

(The existence of a regular normal subgroup in the sharply 2-transitive group $\mathrm{PSU}_3(4)$ is a special case of a theorem of Frobenius; see Theorem IV.2.4 and Theorem V.7.6.)

Remark 10.15. Suzuki has shown that the unitary groups are characterized by the properties given in Theorem 10.12. Namely:

a) (Suzuki [417]) Let G be a 2-transitive permutation group of degree q^3+1, where q is an odd prime power. Suppose the stabilizer G_1 has a normal subgroup R that acts regularly on the points distinct from 1 and G_1/R is cyclic of order $q^2 - 1$. Then G is permutation isomorphic to the unitary group $\mathrm{PU}_3(q^2)$, which should be interpreted as a permutation group on the set A as in Theorem 10.12.

b) (Suzuki [416]) Let G be a 2-transitive permutation group of odd degree. Suppose the stabilizer G_1 has a normal subgroup R that acts regularly on the points distinct from 1 and G_1/R has odd order (so is solvable). Furthermore, suppose that G contains no regular normal subgroups (this condition simply eliminates the uninteresting cases of semilinear groups over near fields). Then G has a normal subgroup T that is permutation isomorphic to one of the following permutation groups:
   (1) $\mathrm{PSL}_2(2^m)$ as a permutation group on the projective line $\mathsf{P}(1, 2^m)$.
   (2) A Suzuki group $\mathrm{Sz}(2^m)$ as a permutation group on the Tits ovoid (see Tits [439, 440] and also Volume 3).
   (3) $\mathrm{PSU}_3(2^m)$ as a permutation group on the set A from Theorem 10.12.
   In all cases, G is then isomorphic to a subgroup of the automorphism group $\mathbf{Aut}(T)$ of T.

c) The simple groups of Ree (see Remark 10.17 b)) are 2-transitive of degree q^3+1 with $q = 3^{2n+1}$ $(n \geq 1)$ and have order $(q^3 + 1)q^3(q - 1)$. The stabilizer G_1 contains a normal subgroup R acting regularly on the points distinct from 1, and G_1/R is cyclic of (even!) order $q - 1$. At the time of writing, a characterization of the Ree groups has not been achieved (but see Ward [451]).[15]

[15] Gorenstein suggested that the characterization of the simple Ree groups "may well be the single most difficult problem of the entire classification of simple groups". A characterization was achieved

Remark 10.16.

a) The groups $\mathrm{PSU}_n(q^2)$ are simple for $n \geq 3$ and $(n, q) \neq (3, 2)$[16]. We have

$$|\mathrm{PSU}_n(q^2)| = (q^n - (-1)^n)q^{n-1}(q^{n-1} - (-1)^{n-1})q^{n-2} \cdots (q^2 - 1)qd^{-1},$$

where $d = (n, q + 1)$. The proof of simplicity proceeds in a way similar to §9 by proving the following facts:

(1) $\mathrm{PSU}_n(q^2)$ is a primitive permutation group on the set

$$\mathsf{A} = \{\langle v\rangle \mid v \in \mathsf{V} = \mathsf{V}(n, q^2),\ (v, v) = 0\}.$$

(When $n > 3$, $\mathrm{PSU}_n(q^2)$ is not 2-transitive.)

(2) $\mathrm{PSU}_n(q^2)$ is generated by unitary transvections and $\mathrm{PSU}_n(q^2)' = \mathrm{PSU}_n(q^2)$ when $(n, q) \neq (3, 2)$.

b) Let $\mathsf{K} = \mathrm{GF}(q)$, where q is odd. If the dimension $n = 2m + 1$ is odd, then there exist two nonisomorphic classes of vector spaces over K equipped with nondegenerate symmetric scalar products, namely

$$\mathsf{V} = \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_m \perp \langle v_n\rangle$$

with hyperbolic planes H_i and either $(v_n, v_n) = 1$ or $(v_n, v_n) = k$ for some nonsquare k. One may pass from one space to the other by multiplying the scalar product by k. So their orthogonal groups are the same. We may therefore write $\mathrm{O}(\mathsf{V}) = \mathrm{O}_{2m+1}(q)$. When $n = 2m$, there exist two fundamentally different vector spaces V_ϵ $(\epsilon = \pm)$, namely

$$\mathsf{V}_+ = \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_m$$

and

$$\mathsf{V}_- = \mathsf{H}_1 \perp \cdots \perp \mathsf{H}_{m-1} \perp \langle v_{n-1}, v_n\rangle$$

with hyperbolic planes H_i and

$$(v_{n-1}, v_{n-1}) = 1, \quad (v_{n-1}, v_n) = 0, \quad (v_n, v_n) = -k,$$

where again k is a nonsquare. We denote the associated orthogonal groups by $\mathrm{O}(\mathsf{V}_\epsilon) = \mathrm{O}^\epsilon_{2m}(q)$. We have

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by Thompson, a 1970 Fields Medalist, in a series of three papers published in 1967–1977. His proof left a number theoretic problem that was solved in 1980 by Bombieri, who was awarded the Fields Medal in 1974. For a discussion of this characterization, as well as the role of 2-transitive groups in the classification, see Section 3.2 of D. Gorenstein, *Finite Simple Groups*, University Series in Mathematics, Plenum, New York, 1982. *trans.*

[16] Correction 20 from errata.

$$|\mathrm{O}_{2m+1}(q)| = 2q^{m^2}\prod_{i=1}^{m}(q^{2i}-1)$$

and[17]

$$|\mathrm{O}^{+}_{2m}(q)| = 2q^{m(m-1)}(q^m-1)\prod_{i=1}^{m-1}(q^{2i}-1),$$
$$|\mathrm{O}^{-}_{2m}(q)| = 2q^{m(m-1)}(q^m+1)\prod_{i=1}^{m-1}(q^{2i}-1).$$

In all cases, $|\mathrm{O}(\mathsf{V})/\mathrm{SO}(\mathsf{V})| = 2$. For $n \geq 3$, the quotient

$$\mathrm{SO}(\mathsf{V})/\mathrm{SO}(\mathsf{V})' \cong \mathsf{K}^{\times}/(\mathsf{K}^{\times})^2$$

has order 2. The group $\Omega(\mathsf{V}) = \mathrm{SO}(\mathsf{V})'$ is the kernel of the so-called spinor norm. The groups $\mathrm{P}\Omega(\mathsf{V}) = \Omega(\mathsf{V})/\mathbf{Z}(\Omega(\mathsf{V}))$ are simple for $n \geq 5$. However, when $n \leq 6$ one does not obtain any new groups, but rather projective and symplectic groups. The proof of the simplicity of $\mathrm{P}\Omega(\mathsf{V})$ proceeds in a fashion similar to Theorem 9.22 (Tamagawa [422]).

c) We have $|\mathrm{PSp}_{2m}(q)| = |\mathrm{P}\Omega_{2m+1}(q)|$ for all m and all odd prime powers q. We have

$$\mathrm{PSp}_2(q) = \mathrm{PSL}_2(q) \cong \mathrm{P}\Omega_3(q) \quad \text{(see Theorem 10.11)}$$

and

$$\mathrm{PSp}_4(q) \cong \mathrm{P}\Omega_5(q)$$

(the most elegant proof is in D. G. Higman [240]), but $\mathrm{PSp}_{2m}(q) \ncong \mathrm{P}\Omega_{2m+1}(q)$ for $m \geq 3$. Together with the coincidence

$$|\mathrm{PSL}_4(2)| = |\mathrm{PSL}_3(4)| \quad \text{but} \quad \mathrm{PSL}_4(2) \ncong \mathrm{PSL}_3(4)$$

(see Exercise 7 of §6), these are the only cases of nonisomorphic simple groups having the same order (see Artin [13, 14]).

d) If $\mathsf{K} = \mathrm{GF}(q)$ with $q = 2^h$, then there are two nondegenerate quadratic forms q_ϵ, namely

$$q_+\left(\sum_{i=1}^{2m} x_i v_i\right) = \sum_{i=1}^{m} x_i x_{m+i}$$

and

[17] Correction 21 from errata.

$$q_-\left(\sum_{i=1}^{2m} x_i v_i\right) = \sum_{i=1}^{m-1} x_i x_{m+i} + x_m^2 + x_m x_{2m} + a x_{2m}^2,$$

where $x^2 + x + a$ is an irreducible polynomial over K. The associated orthogonal groups $\mathrm{SO}^{\epsilon}_{2m}(q)$ have orders

$$|\mathrm{SO}^{+}_{2m}(q)| = (q^m - 1)(q^{2(m-1)} - 1)q^{2(m-1)} \cdots (q^2 - 1)q^2$$
$$|\mathrm{SO}^{-}_{2m}(q)| = (q^m + 1)(q^{2(m-1)} - 1)q^{2(m-1)} \cdots (q^2 - 1)q^2$$

and are simple when $m \geq 3$. (Recall that one constructs $\mathrm{SO}(\mathsf{V})$ via the Dickson invariant and not the determinant!) In odd dimensions, one recovers the symplectic groups. (For all of these statements, one may consult Artin [15] and Dieudonné [112, 115].)

Remark 10.17. As we noted in Remark 10.5 b), the description of the unitary[18] groups given there leads to new series of simple groups.

a) The symplectic group $\mathrm{PSp}_4(q)$ with $q = 2^{2n+1}$ and $n \geq 1$ admits an automorphism α such that the set of fixed points of α is a simple group $\mathrm{Sz}(2^{2n+1})$ of order $(q^2 + 1)q^2(q - 1)$. These groups were discovered by Suzuki. We will discuss them in detail in Volume 3. The group $\mathrm{Sz}(2^{2n+1})$ admits a representation as a 2-transitive permutation group of degree $q^2 + 1$ (see Ono [340]).

b) Let $\mathsf{K} = \mathrm{GF}(q)$ with $q = 3^{2n+1}$ and $n > 1$. Let H be the simple Chevalley group arising from the exceptional Lie algebra G_2 over K. Then H admits an automorphism α such that the group G of fixed points of α is a simple group of order $(q^3 + 1)q^3(q - 1)$. The group G has a representation as a 2-transitive permutation group of degree $q^3 + 1$. The stabilizer of a point contains a normal subgroup R that acts regularly on the rest of the points. The subgroup R is nilpotent of class 3 and has order q^3. The stabilizer of two points is cyclic of order $q - 1$. The Sylow 2-subgroups of G are elementary abelian of order 8 (see Ree [352] and Tits [439]).

c) Let $\mathsf{K} = \mathrm{GF}(q)$ with $q = 2^{2n+1}$ $(n \geq 1)$, and let H be the simple group arising from the exceptional Lie algebra F_4 over K. Then H admits an automorphism α such that the group G of fixed points of α is a simple group of order

$$q^{12}(q^6 + 1)(q^4 - 1)(q^3 + 1)(q - 1),$$

see Ree [351]. Over $\mathrm{GF}(2)$, the same construction leads to a group with a simple normal subgroup of index 2 (see Tits [442]).

d) Let $\mathsf{K} = \mathrm{GF}(q^3)$ with $q \geq 4$, and let H be the simple group arising from the Lie algebra D_4 over K. Then H admits an automorphism α such that the group of fixed points of α is a simple group of order

$$q^{12}(q^2 - 1)(q^6 - 1)(q^8 + q^4 + 1),$$

[18] Correction 22 from errata.

see Steinberg [394].

e) Let $\mathsf{K} = \mathrm{GF}(q^2)$ with $q \geq 4$, and let H be the simple group arising from the exceptional Lie algebra E_6 over K. Then H admits an automorphism α such that the group of fixed points of α is a simple group of order

$$q^{36}(q^2-1)(q^5+1)(q^6-1)(q^8-1)(q^9+1)(q^{12}-1)d^{-1},$$

where $d = (3, q+1)$ (see Steinberg [394]).

Exercises

22) Let V be a vector space over K equipped with a symmetric nondegenerate scalar product. Let H be a hyperplane in V and t a nontrivial isometry of V such that $ht = h$ for all $h \in \mathsf{H}$.

a) If $vt - v \in \mathsf{H}$ for all $v \in \mathsf{V}$, then $\mathsf{H} \cap \mathsf{H}^\perp \neq 0$ and $\mathrm{char}\,\mathsf{K} = 2$. (Thus, only in $\mathrm{char}\,\mathsf{K} = 2$ do there exist transvections that are isometries.)

b) If $\mathsf{H} \cap \mathsf{H}^\perp = 0$, then $\mathrm{char}\,\mathsf{K} \neq 2$ and t acts as a reflection on H; that is, setting $\mathsf{H}^\perp = \langle a \rangle$ we have

$$vt = v - \frac{2(v,a)}{(a,a)}a$$

for all $v \in \mathsf{V}$.

23) Let V be a vector space over K equipped with a symmetric nondegenerate scalar product. Assume $\mathrm{char}\,\mathsf{K} \neq 2$. Let H be a hyperplane in V with $\mathsf{H}^\perp = \langle s \rangle \subseteq \mathsf{H}$ and t an isometry of V such that

$$\begin{aligned} st &= s, \\ ht - h \in \langle s \rangle &\quad \text{for all } h \in \mathsf{H}, \\ vt - v \in \mathsf{H} &\quad \text{for all } v \in \mathsf{V}. \end{aligned}$$

Let $\mathsf{V} = \mathsf{H} \oplus \langle x \rangle$ with $(x, s) = 1$. Then there exists some $w \in \mathsf{H} \cap \langle x \rangle^\perp$ such that

$$\begin{aligned} ht &= h - (w,h)s \quad \text{for all } h \in \mathsf{H}, \\ xt &= x + w - \frac{(w,w)}{2}s. \end{aligned}$$

(For $\mathrm{char}\,\mathsf{K} \neq 2$, these t are the substitute for the transvections that do not exist by Exercise 22 a).)

24) If $\mathbb{R}$ and $\mathbb{C}$ are the real and complex numbers, respectively, and the Hermitian form on $\mathsf{V}(2,\mathbb{C})$ is given by $(x,y) = x_1\overline{y}_1 + x_2\overline{y}_2$, then $\mathrm{SU}_2(\mathbb{C}) \ncong \mathrm{SL}_2(\mathbb{R})$. (Compare with Lemma 8.8 (pg. 203).)

Notes for Chapter II

§1: The presentation of the material in this section largely follows Wielandt [477].

§2: Theorems 2.1–2.3 are taken from Wielandt [477]. To the best of my knowledge, Theorem 2.6 was first proved by N. Itô.

§3: The statements in Theorems 3.2, 3.7–3.9 were first given in Ore [341]. Theorem 3.6 was already given in Galois' famous letter. Theorem 3.10 and a special case of Lemma 3.11 can be found in Huppert [259]; Theorem 3.10 may have been widely known beforehand. I thank McLaughlin for sharing Example 3.4 b) with me in conversation. One may find generalizations of Exercise 1 in Huppert [258] and Itô [270].

§4: The presentation here again follows Wielandt [477].

§6: The proof of the simplicity of $\mathrm{PSL}_n(\mathsf{K})$ given here follows Iwasawa [289]. The presentation is strongly influenced by Artin [15].

§7: The results in this section are contained in Dickson [111], although we do not follow the presentation given there.

§8: Our discussion of the Diophantine equation follows Dickson [111] in some places.

§9: Our treatment of the symplectic groups largely follows Artin [15]. However, the proof of simplicity in Theorem 9.22 follows Tamagawa [422]. I thank E. C. Dade for sharing the proof of Lemma 9.23. Theorem 9.25 can be found in Parker [342], along with an alternative proof. Exercises 20 and 21 stem from Artin [15].

§10: The presentation in this section again follows Artin [15].

Chapter III
Nilpotent groups and p-groups

The Sylow theorems show that the groups of prime power order play a distinguished role in the theory of finite groups. Admittedly, the construction of a finite group from its Sylow subgroups is generally a very difficult task. In the case of solvable groups, the theory of Sylow systems developed by P. Hall and the theory of p-length developed by P. Hall and G. Higman give significant insight; we will discuss these theories in Chapter VI. For simple groups, any understanding of their construction from Sylow subgroups is still in its infancy. However, the work of Thompson on simple groups has given significant insight and guided the theory of p-groups in new directions.

One of the most difficult remaining tasks in group theory is the determination of all p-groups. The sheer number of them seems to be a fundamental impediment. G. Higman [239, 240] and Sims [386] have shown that the number of p-groups of order p^n is asymptotically

$$p^{a(n,p)n^3}, \quad \text{where} \quad a(n,p) = \frac{2}{27} + O(n^{-1/3}).$$

It therefore seems questionable whether the determination of all p-groups is a well-posed problem. It would be desirable to have a classification of p-groups according to some equivalence relation weaker than isomorphism. An attempt in this direction can be found in P. Hall [214], which was used by M. Hall and Senior [205] in the classification of all groups of order 2^6.

A systematic theory of p-groups is only possible under strong additional assumptions. By far the most general and successful approach is the theory of regular p-groups developed by P. Hall (§10). This theory considers, in addition to the commutators, the formation of p-th powers. There are numerous relationships between the characteristic subgroups

$$\Omega_i(G) = \langle g \in G \mid g^{p^i} = 1 \rangle, \quad \mho_i(G) = \langle g^{p^i} \mid g \in G \rangle$$

and the elements of the upper and lower central series. For p-groups of exponent p, however, the theory of regular p-groups yields no insight; here one is in the realm

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B. Huppert, *Finite Groups I*, Grundlehren der mathematischen Wissenschaften 364,
https://doi.org/10.1007/978-3-031-87529-8_3

of the Burnside problem (see Remark 6.7), where many questions have received no complete answer thus far. An important extension of the theory of regular p-groups is the theory of p-groups of maximal class discussed in §14.

Aside from the theory of regular p-groups, there is no entirely coherent theory for p-groups. Except for some general but very elementary statements (§7 and §8), we only study some special classes of p-groups that play a role in applications. For example, the p-groups with only small abelian normal subgroups discussed in §12 have derived new significance from Feit and Thompson's work on simple groups. In the more delicate theory of solvable groups that we develop in Volume 2, the special and extraspecial groups play a dominant role (§13).

Besides the theory of p-groups, in this chapter we consider the groups whose construction from their Sylow subgroups is particularly easy; namely, those groups that are a direct product of Sylow subgroups. This class of groups, called nilpotent (for reasons that only become apparent in their connection to Lie rings), may be characterized in numerous ways (§2–§5). This part of the theory has reached a satisfactory conclusion, and the results are of great importance for later sections, in particular the theory of solvable groups in Chapter VI.

Sections 18 and 19 contain new results regarding the number of generators and relations of p-groups, as well as their automorphisms; the module theoretic methods in §18 and the cohomological considerations in §19 have not previously been used in the theory of p-groups.

The inexperienced reader is advised to restrict themselves to the following sections on a first reading: §1–§5, §7, §8–§8.4, §13–§13.7.

1 Commutators and commutator subgroups

In this section, we develop a systematic commutator calculus. Nowhere will we require that the groups be finite. In fact, Theorem 2.11 b) will be needed for free groups in §9.

Definition 1.1. With our earlier definitions

$$a^b = b^{-1}ab \quad \text{and} \quad [a,b] = a^{-1}b^{-1}ab$$

(see Lemma I.2.14, pg. 8 and Definition I.8.1, pg. 38), we have

$$[a,b] = a^{-1}a^b = b^{-a}b.$$

In agreement with Definition I.10.8 (pg. 61), we write

$$[a,b] = a^{-1+b} = b^{-a+1}.$$

Clearly, we have

$$a^{b(c+d)} = a^{bc+bd} \quad \text{and} \quad a^{(b+c)d} = a^{bd+cd}.$$

(It should be noted, however, that the addition introduced in the exponent is not necessarily commutative!) We call $[a, b]$ the commutator of a and b. We have $[a, b] = 1$ if and only if $ab = ba$.

The following computation rules hold for commutators:

Lemma 1.2. *We have*
a) $[a, b] = [b, a]^{-1}$.
b) $[a, bc] = [a, c][a, b]^c = [a, c][a, b][[a, b], c]$.
c) $[ab, c] = [a, c]^b[b, c] = [a, c][[a, c], b][b, c]$.

Proof. a) is completely trivial.
b) We have

$$\begin{aligned}[a, bc] &= a^{-1+bc} = a^{-1+c-c+bc} = a^{-1+c}a^{(-1+b)c} \\ &= [a, c][a, b]^c = [a, c][a, b][[a, b], c].\end{aligned}$$

c) Similarly,

$$\begin{aligned}[ab, c] &= c^{-ab+1} = c^{(-a+1)b}c^{-b+1} = [a, c]^b[b, c] \\ &= [a, c][[a, c], b][b, c].\end{aligned}$$

□

The next lemma is often useful.

Lemma 1.3.
a) *If the commutator $[a, b]$ commutes with a, then for all integers n we have*

$$[a^n, b] = [a, b]^n.$$

b) *If $[a, b]$ commutes with a and b, then for all natural numbers n we have*

$$(ab)^n = a^n b^n [b, a]^{\binom{n}{2}}.$$

Proof. a) We prove the claim for $n > 0$ by induction on n. For $n = 1$, it is clearly true. We have

$$\begin{aligned}[a^n, b] &= [aa^{n-1}, b] = [a, b]^{a^{n-1}}[a^{n-1}, b] && \text{(by Lemma 1.2 c))} \\ &= [a, b][a^{n-1}, b] && \text{(since } [a, b] \text{ commutes with } a^{n-1}\text{)} \\ &= [a, b][a, b]^{n-1} && \text{(by the induction hypothesis)} \\ &= [a, b]^n.\end{aligned}$$

It follows by Lemma 1.2 c) that for $n > 0$,

$$\begin{aligned}1 &= [a^n a^{-n}, b] = [a^n, b]^{a^{-n}}[a^{-n}, b] \\ &= ([a, b]^n)^{a^{-n}}[a^{-n}, b] = [a, b]^n[a^{-n}, b],\end{aligned}$$

so that $[a^{-n}, b] = [a, b]^{-n}$.

b) For $n = 1$, the claim is trivial. For $n > 1$, we prove the claim by induction as follows:

$$\begin{aligned}
(ab)^n &= (ab)^{n-1}ab = a^{n-1}b^{n-1}[b,a]^{\binom{n-1}{2}}ab && \text{(by the induction hypothesis)}\\
&= a^{n-1}b^{n-1}ab[b,a]^{\binom{n-1}{2}} && \text{(as } a \text{ and } b \text{ commute with } [b,a] = [a,b]^{-1}\text{)}\\
&= a^{n-1}ab^{n-1}b^{-(n-1)}a^{-1}b^{n-1}ab[b,a]^{\binom{n-1}{2}}\\
&= a^n b^{n-1}[b^{n-1},a]b[b,a]^{\binom{n-1}{2}}\\
&= a^n b^{n-1}[b,a]^{n-1}b[b,a]^{\binom{n-1}{2}} && \text{(by Lemma 1.3 a))}\\
&= a^n b^n [b,a]^{\binom{n}{2}}.
\end{aligned}$$

□

Theorem 1.4 (Witt's identity). *Writing* $[[a,b],c] = [a,b,c]$, *we have*

$$[a, b^{-1}, c]^b [b, c^{-1}, a]^c [c, a^{-1}, b]^a = 1.$$

Proof. We let $u = aca^{-1}ba$. By cyclically permuting a, b, c, we construct $v = bab^{-1}cb$ and $w = cbc^{-1}ac$. We have

$$\begin{aligned}
[a, b^{-1}, c]^b &= b^{-1}[a, b^{-1}]^{-1}c^{-1}[a, b^{-1}]cb\\
&= b^{-1}[b^{-1}, a]c^{-1}[a, b^{-1}]cb\\
&= b^{-1}ba^{-1}b^{-1}ac^{-1}a^{-1}bab^{-1}cb = u^{-1}v.
\end{aligned}$$

If we calculate the other terms in Witt's identity in the same way, we obtain

$$[a, b^{-1}, c]^b [b, c^{-1}, a]^c [c, a^{-1}, b]^a = u^{-1}vv^{-1}ww^{-1}u = 1.$$ □

Definition 1.5. Let A and B be subsets of the group G. As in Definition I.8.1 (pg. 38), we define

$$[A, B] = \langle [a, b] \mid a \in A, b \in B \rangle.$$

Clearly, $[A, B]$ is a subgroup of G. The group $[G, G]$ coincides with the commutator subgroup G'. (Note that $[A, B]$ consists not only of commutators $[a, b]$, but also all products of such commutators and their inverses. Furthermore, note that $[a, b]$ and $[\{a\}, \{b\}] = \langle [a, b] \rangle$ should definitely be distinguished.)

Lemma 1.6. *If A and B are subgroups of G, then:*

a) $[A, B] = [B, A]$.

b) $[A, B] \trianglelefteq \langle A, B \rangle$.

c) *If* $A_1 \leq A$ *and* $B_1 \leq B$, *then* $[A_1, B_1] \leq [A, B]$.

d) $[A, B] \leq A$ *if and only if* $B \leq \mathbf{N}_G(A)$.

e) *If μ is a homomorphism of G, then* $[A, B]^\mu = [A^\mu, B^\mu]$.

f) *If A and B are normal (characteristic) subgroups of G, then $[A, B]$ is a normal (characteristic) subgroup of G, and $[A, B] \leq A \cap B$.*

Proof. a) Since $[b, a] = [a, b]^{-1} \in [A, B]$, we have $[B, A] \leq [A, B]$. The reverse inclusion, and thus the claim, follows in exactly the same way.

b) Let $a, a' \in A$ and $b, b' \in B$. Then by Lemma 1.2 b) and c), we have

$$[a, b]^{b'} = [a, b']^{-1}[a, bb'] \in [A, B]$$

and

$$[a, b]^{a'} = [aa', b][a', b]^{-1} \in [A, B].$$

So that $[A, B] \trianglelefteq \langle A, B \rangle$.

c) is trivial.

d) $[a, b] = a^{-1}b^{-1}ab \in A$ is equivalent to $b^{-1}ab = a^b \in A$. Thus, $[A, B] \leq A$ is equivalent to $B \leq \mathbf{N}_G(A)$.

e) We have

$$\begin{aligned}[A, B]^\mu &= \langle [a, b]^\mu \mid a \in A, b \in B \rangle = \langle [a^\mu, b^\mu] \mid a^\mu \in A^\mu, b^\mu \in B^\mu \rangle \\ &= [A^\mu, B^\mu].\end{aligned}$$

f) The fact that A and B normal (characteristic) subgroups of G implies $[A, B]$ is a normal (characteristic) subgroup of G follows from e) if we allow μ to run over all inner automorphisms (all automorphisms) of G.

If $A \leq \mathbf{N}_G(B) = G$, then $[A, B] \leq B$ from d); likewise, $[A, B] \leq A$. □

Definition 1.7. Let F be the free group in the free generators f_{ij} $(j = 1, \ldots, k;$ $i = 1, \ldots)$, and let

$$w = \prod_{m,n} f_{i_m j_n}^{b_{mn}}$$

be an element of F. Let A_j $(j = 1, \ldots, k)$ be subgroups of the group G and M the set of homomorphisms μ from F to G such that

$$f_{ij}^\mu \in A_j \quad (j = 1, \ldots, k).$$

(By Definition I.19.1, pg. 142, for any given $a_{ij} \in A_j$ there always exists a homomorphism μ from F to G with $f_{ij}^\mu = a_{ij}$.) We define

$$w(A_1, \ldots, A_k) = \langle w^\mu \mid \mu \in M \rangle = \langle \prod_{m,n} a_{i_m j_n}^{b_{mn}} \mid a_{ij} \in A_j \rangle.$$

If ϵ is a homomorphism of G, then clearly we have

$$w(A_1, \ldots, A_k)^\epsilon = w(A_1^\epsilon, \ldots, A_k^\epsilon).$$

If all the A_i are normal (characteristic) subgroups of G, then $w(A_1, \dots, A_k)$ is a normal (characteristic) subgroup of G.

For example, if $w = [f_{11}, f_{12}]$, then $w(A_1, A_2) = [A_1, A_2]$; if $w = f_{11}^n$, then $w(A_1)$ is generated by the n-th powers of the elements in A_1.

Lemma 1.8. *Let f and f' be elements of the free group F with free generating set $\{f_{ij}, f'_{ij} \mid j = 1, \dots, k; i = 1, 2, \dots\}$. Let f be a product of powers of the f_{ij}, and let f' be a product of powers of the f'_{ij}. We let $w = [f, f']$. If the A_j $(j = 1, \dots, k)$ are normal subgroups of G, then*

$$w(A_1, \dots, A_k) = [f(A_1, \dots, A_k), f'(A_1, \dots, A_k)].$$

Proof. Let M be the set of homomorphisms μ from F to G such that $f_{ij}^{\mu} \in A_j$ and $f'_{ij}{}^{\mu} \in A_j$ $(j = 1, \dots, k)$. Then for every $\mu \in M$, we have

$$w^{\mu} = [f^{\mu}, f'^{\mu}] \in [f(A_1, \dots, A_k), f'(A_1, \dots, A_k)],$$

and so

$$w(A_1, \dots, A_k) = \langle w^{\mu} \mid \mu \in M \rangle \leq [f(A_1, \dots, A_k), f'(A_1, \dots, A_k)].$$

To prove the reverse inclusion, we consider the factor group

$$\overline{G} = G/w(A_1, \dots, A_k);$$

here we use that the A_j are normal subgroups of G, so $w(A_1, \dots, A_k)$ is also normal in G. Let ϵ be the natural epimorphism from G onto $\overline{G}$ defined by

$$g^{\epsilon} = g\, w(A_1, \dots, A_k).$$

Now

$$[f^{\mu\epsilon}, f'^{\mu\epsilon}] = [f^{\mu}, f'^{\mu}]^{\epsilon} = w^{\mu\epsilon} \in w(A_1, \dots, A_k)^{\epsilon} = 1.$$

For a fixed $f^{\mu} = \prod_{n,m} f_{i_n j_m}^{a_{nm}\mu}$, the $f'^{\mu} = \prod_{s,t} f_{i_s j_t}^{\prime\, a'_{st}\mu}$ (with $a_{nm} = \pm 1, a'_{st} = \pm 1$) generate all of $f'(A_1, \dots, A_k)$, so it follows that

$$[f(A_1, \dots, A_k), f'(A_1, \dots, A_k)] \leq \ker \epsilon = w(A_1, \dots, A_k). \qquad \square$$

Definition 1.9. We recursively define

$$[a_1, \dots, a_n] = [[a_1, \dots, a_{n-1}], a_n].$$

The commutators with this special bracketing are said to be "left normed". If w is the element

$$w = [f_1, \dots, f_n]$$

in the free group F with free generators f_j, then we set $w(G) = \mathbf{K}_n(G)$. So by Definition 1.7,

$$\mathbf{K}_n(G) = \langle [g_1, \ldots, g_n] \mid g_i \in G \rangle.$$

Lemma 1.8 shows that we also obtain $\mathbf{K}_n(G)$ by recursively defining

$$\mathbf{K}_1(G) = G \quad \text{and} \quad \mathbf{K}_{i+1}(G) = [\mathbf{K}_i(G), G].$$

From the description of $\mathbf{K}_n(G)$, it follows immediately that if μ is a homomorphism of G, then $\mathbf{K}_n(G^\mu) = \mathbf{K}_n(G)^\mu$.

Lemma 1.10. *Let A, B and C be subgroups of G. Then:*
a) *If $B \leq \mathbf{N}_G(A) \cap \mathbf{N}_G(C)$, then $[AB, C] = [A, C][B, C]$.*
b) *(P. Hall) If $N \trianglelefteq G$, $[B, C, A] \leq N$ and $[C, A, B] \leq N$, then $[A, B, C] \leq N$.*
(Recall that according to Definition 1.7,

$$[A, B, C] = \langle [a, b, c] \mid a \in A, b \in B, c \in C \rangle.)$$

c) *If $A \trianglelefteq G$, then $[AB, AC] \leq [B, C]A$.*

Proof. a) Since $B \leq \mathbf{N}_G(A)$, $AB = \{ab \mid a \in A, b \in B\}$ is a subgroup of G. Thus,

$$[AB, C] = \langle [ab, c] \mid a \in A, b \in B, c \in C \rangle.$$

By Lemma 1.6 b), $[A, C]$ is a normal subgroup of $\langle A, C \rangle$. Since $B \leq \mathbf{N}_G(C)$, we have $[B, C] \leq C$ by Lemma 1.6 d). Together this implies

$$[B, C] \leq C \leq \mathbf{N}_G([A, C]).$$

Thus, $[A, C][B, C]$ is a subgroup of G. With the help of Lemma 1.2 c), we obtain

$$[ab, c] = [a, c]^b [b, c] = [a^b, c^b][b, c] \in [A, C][B, C].$$

Hence, we have shown that

$$[AB, C] \leq [A, C][B, C].$$

The reverse inclusion follows immediately from Lemma 1.6 c) since $A \leq AB$ and $B \leq AB$.

b) The Witt identity of Theorem 1.4 yields

$$[a, b^{-1}, c]^b = [c, a^{-1}, b]^{-a} [b, c^{-1}, a]^{-c} \in N^a N^c = N.$$

So $[a, b^{-1}, c] \in N$ for all $a \in A$, $b \in B$, $c \in C$, and thus $[A, B, C] \leq N$ as well.

c) Let μ be the natural epimorphism from G onto G/A with $g^\mu = gA$. Then $B^\mu = AB/A$ and $C^\mu = AC/A$. Using Lemma 1.6 e), we now obtain

$$[AB, AC]A/A = [AB/A, AC/A] = [B^\mu, C^\mu] = [B, C]^\mu = [B, C]A/A,$$

and therefore

$$[AB, AC] \leq [B, C]A. \qquad \square$$

Lemma 1.11. *Suppose the group G is generated by the subset M. Then:*
a) $\mathbf{K}_i(G) = \langle [m_1, \ldots, m_i]^g \mid m_j \in M, g \in G \rangle$.
b) $\mathbf{K}_i(G) = \langle [m_1, \ldots, m_i], \mathbf{K}_{i+1}(G) \mid m_j \in M \rangle$.
c) *If $G = \langle a, b \rangle$, then $\mathbf{K}_2(G) = G' = \langle [a, b], \mathbf{K}_3(G) \rangle$. In particular, $\mathbf{K}_2(G)/\mathbf{K}_3(G)$ is cyclic.*

Proof. a) Clearly, $[m_1, \ldots, m_i]^g \in \mathbf{K}_i(G)$.

Since $\mathbf{K}_1(G) = G = \langle M \rangle$, the claim holds for $i = 1$. We will prove the claim by induction on i. Assume for induction that

$$\mathbf{K}_i(G) = \langle [m_1, \ldots, m_i]^g \mid m_j \in M,\ g \in G \rangle.$$

We define

$$N_{i+1} = \langle [m_1, \ldots, m_{i+1}]^g \mid m_j \in M, g \in G \rangle.$$

We have

$$[[m_1, \ldots, m_i]^g, m_{i+1}^g] = [m_1, \ldots, m_{i+1}]^g \in N_{i+1}.$$

Thus, modulo the normal subgroup N_{i+1}, every generator $[m_1, \ldots, m_i]^g$ of $\mathbf{K}_i(G)$ commutes with the entire generating set $\{m_j^g \mid m_j \in M\}$ of G. So $[\mathbf{K}_i(G), G] = \mathbf{K}_{i+1}(G) \leq N_{i+1}$. The reverse inclusion is trivial.

b) follows immediately from a) upon recognizing that

$$[m_1, \ldots, m_i]^g = [m_1, \ldots, m_i][m_1, \ldots, m_i, g] \in [m_1, \ldots, m_i]\mathbf{K}_{i+1}(G).$$

c) follows from b) by setting $M = \{a, b\}$. $\square$

Now we return to an observation made in Definition I.8.1 (pg. 38):

Example 1.12. We give an example of a group G in which not all elements of G' are commutators. We define this group G via the generators a_i $(1 \leq i \leq 4)$ and the relations

$$a_i^p = [a_i, a_j, a_k] = 1$$

for all i, j, k. By Lemma 1.3 a), it follows that $[a_i, a_j]^p = 1$. We define $a_{ij} = [a_i, a_j]$. One sees easily that every element of G may be written as

$$\prod_i a_i^{x_i} \prod_{i<j} a_{ij}^{x_{ij}},$$

where the x_i and x_{ij} are integers uniquely determined mod p. Therefore $|G| = p^{10}$. If follows from Lemma 1.2 that

$$\left[\prod_i a_i^{x_i}, \prod_i a_i^{y_i}\right] = \prod_{i<j} a_{ij}^{z_{ij}}$$

with $z_{ij} = x_i y_j - y_i x_j$, so we have

$$z_{12}z_{34} - z_{13}z_{24} + z_{14}z_{23} = 0.$$

Therefore, the equations

$$z_{13} = z_{24} = 1, \quad z_{ij} = 0 \quad \text{for all remaining pairs } i, j$$

are not solvable. Thus, $a_{13}a_{24}$ is not a commutator, but lies in G'.

Exercises

1) Assume $G'' = 1$. Show that

$$[a, b, c][b, c, a][c, a, b] = 1$$

for all $a, b, c \in G$.

2) Suppose A is an abelian normal subgroup of G with cyclic factor group $G/A = \langle xA \rangle$. Then:
   a) $a \mapsto [a, x]$ is an epimorphism from A onto G'.
   b) If G is finite, then $|A| = |G'||A \cap \mathbf{Z}(G)|$.
   c) Let G be a finite group with $G'' = 1$ and G/G' cyclic. Then for every x such that $G = \langle G', x \rangle$, we have $|x| = |G/G'|$; thus, $\langle x \rangle$ is a complement of G' in G.

2 Central series and nilpotent groups

Definition 2.1.
a) A chain

$$N_1 \geq N_2 \geq \cdots \geq N_r$$

of subgroups of G is called a central series of G if $[N_i, G] \leq N_{i+1}$ for $i = 1, \ldots, r-1$ and $N_r \trianglelefteq G$. Since

$$[N_i, G] \leq N_{i+1} \leq N_i$$

for $i < r$, it follows by Lemma 1.6 d) that $N_i \trianglelefteq G$. Thus, the terms in a central series are always normal subgroups of G. The condition $[N_i, G] \leq N_{i+1}$ is clearly equivalent to

$$N_i/N_{i+1} \leq \mathbf{Z}(G/N_{i+1}).$$

b) If G has a central series with $N_1 = G$ and $N_r = 1$, then we say that G is nilpotent. (We will show in Theorem 2.3 that this definition of nilpotent agrees with the definition given in Theorem I.9.5, pg. 50 for finite groups.)

Definition 2.2.

a) The chain of subgroups

$$G = \mathbf{K}_1(G) \geq \mathbf{K}_2(G) \geq \cdots$$

given in Definition 1.9 is called the lower central series of G. (Since we have

$$[\mathbf{K}_i(G), G] = \mathbf{K}_{i+1}(G),$$

it is a central series of G.) We define

$$\mathbf{K}_\infty(G) = \bigcap_i \mathbf{K}_i(G).$$

b) The chain of subgroups

$$1 = \mathbf{Z}_0(G) \leq \mathbf{Z}_1(G) \leq \mathbf{Z}_2(G) \leq \cdots$$

with $\mathbf{Z}_1(G) = \mathbf{Z}(G)$ and $\mathbf{Z}_{i+1}(G)/\mathbf{Z}_i(G) = \mathbf{Z}(G/\mathbf{Z}_i(G))$ is called the upper central series of G. It is clearly a central series of G. We define

$$\mathbf{Z}_\infty(G) = \bigcup_i \mathbf{Z}_i(G)$$

and call $\mathbf{Z}_\infty(G)$ the hypercenter of G.

The key result of this section is Theorem 2.3, which is largely due to Burnside.

Main Theorem 2.3. *Let G be a finite group. The following conditions are equivalent:*

a) *G is nilpotent.*
b) *For all sufficiently large n, we have $\mathbf{K}_n(G) = 1$. That is, the lower central series of G terminates at 1.*
c) *If $U < G$, then $U < \mathbf{N}_G(U)$.*
d) *Every maximal subgroup of G is a normal subgroup of G.*
e) *G is the direct product of its Sylow subgroups.*
f) *Every epimorphic image of G distinct from 1 has a center distinct from 1. In particular, $\mathbf{Z}_m(G) = G$ for all large enough m. That is, the upper central series terminates at G.*
g) *If x and y are elements of G of coprime order, then $xy = yx$.*

Proof. a) ⇒ b): Let

$$G = N_1 \geq N_2 \geq \cdots \geq N_r = 1$$

be a central series of G. Then $G = \mathbf{K}_1(G) = N_1$. Assume for induction that $\mathbf{K}_i(G) \leq N_i$. Then it follows that

$$\mathbf{K}_{i+1}(G) = [\mathbf{K}_i(G), G] \leq [N_i, G] \leq N_{i+1}.$$

Thus, ultimately $\mathbf{K}_r(G) \leq N_r = 1$.

b) ⇒ c): Let $U < G$. We fix the integer i such that $\mathbf{K}_{i+1}(G) \leq U$ but $\mathbf{K}_i(G) \not\leq U$; since $\mathbf{K}_1(G) = G \not\leq U$ and $\mathbf{K}_n(G) = 1 \leq U$, there certainly exists such an i. Now

$$[\mathbf{K}_i(G), U] \leq [\mathbf{K}_i(G), G] = \mathbf{K}_{i+1}(G) \leq U,$$

so $\mathbf{K}_i(G) \leq \mathbf{N}_G(U)$ by Lemma 1.6 d). Hence, we have $U < \mathbf{N}_G(U)$.

c) ⇒ d): Let U be a maximal subgroup of G. We have $U < \mathbf{N}_G(U)$ by c), so $\mathbf{N}_G(U) = G$ and U is normal in G.

d) ⇒ e): Let P be a Sylow subgroup of G. We show that P is a normal subgroup of G. Assume for contradiction that $\mathbf{N}_G(P) < G$. Then there is a maximal subgroup U of G such that $\mathbf{N}_G(P) \leq U$. Every overgroup of the normalizer of a Sylow subgroup is self normalizing by Theorem I.7.6 (pg. 35). So we obtain $\mathbf{N}_G(U) = U$. But this contradicts our assumption that all maximal subgroups of G are normal in G. Therefore, $P \trianglelefteq G$. Then G is the direct product of its Sylow subgroups by Theorem I.9.5 (pg. 50).

e) ⇒ f): It is enough to show that $\mathbf{Z}(G) > 1$ since the condition that all Sylow subgroups of G are normal, which is equivalent to e) by Theorem I.9.5, is inherited by homomorphic images of G. Since G is the direct product

$$G = P_1 \times \cdots \times P_r$$

of its Sylow subgroups $P_i \neq 1$, we have by Lemma I.9.9 (pg. 50) that

$$\mathbf{Z}(G) = \mathbf{Z}(P_1) \times \cdots \times \mathbf{Z}(P_r).$$

By Theorem I.6.9 (pg. 31), $\mathbf{Z}(P_i) > 1$. Thus, $\mathbf{Z}(G) > 1$ as well.

f) ⇒ a): This is trivial since the assumption of f) yields the upper central series

$$1 = \mathbf{Z}_0(G) < \mathbf{Z}_1(G) < \cdots < \mathbf{Z}_m(G) = G,$$

a central series of G from 1 to G.

e) ⇒ g): Let $p_1, \ldots, p_s$ be the prime divisors of $|x|$ and $q_1, \ldots, q_t$ the prime divisors of $|y|$. Let P_i be the Sylow p_i-subgroup and Q_j the Sylow q_j-subgroup of G. By Lemma I.2.12 a) (pg. 7), the order of

$$J = \langle P_1 \times \cdots \times P_s, x \rangle$$

is only divisible by the primes $p_1, \ldots, p_s$. Since $p_i \nmid [G : J]$ $(i = 1, \ldots, s)$, it follows that

$$x \in P_1 \times \cdots \times P_s,$$

and in the same way

$$y \in Q_1 \times \cdots \times Q_t.$$

Since $[P_i, Q_j] \leq P_i \cap Q_j = 1$, we have $[x, y] = 1$.

g) $\Rightarrow$ e): Let P be a Sylow p-subgroup of G. If Q is a Sylow q-subgroup of G with $p \neq q$, then we have by g) that $Q \leq \mathbf{C}_G(P) \leq \mathbf{N}_G(P)$. Since $P \leq \mathbf{N}_G(P)$ trivially, it follows by comparison of the orders that $\mathbf{N}_G(P) = G$. An application of Theorem I.9.5 then gives e). □

Remark 2.4.

a) The following statement is an immediate consequence of Theorem 2.3 g): A finite group G is nilpotent if and only if for all $x, y \in G$, the subgroup $\langle x, y \rangle$ is nilpotent.

b) Clearly, the proof of b) from a) in Theorem 2.3 goes through without assuming that G is finite. Part a) follows trivially from b). So the condition that $\mathbf{K}_n(G) = 1$ for some natural number n characterizes the (not necessarily finite) nilpotent groups G.

Theorem 2.5.

a) *If G is nilpotent, then so is every subgroup and every homomorphic image of G.*

b) *If G_1 and G_2 are nilpotent, then $G_1 \times G_2$ is also nilpotent.*

c) *If G/M and G/N are nilpotent, then $G/(M \cap N)$ is also nilpotent.*

d) *Let G be finite and $N \trianglelefteq G$. Then G/N is nilpotent if and only if $\mathbf{K}_\infty(G) \leq N$.*

e) *If N and G/N are nilpotent, then G is not necessarily nilpotent.*

Proof. a) It follows from $\mathbf{K}_n(G) = 1$ and $U \leq G$ that $\mathbf{K}_n(U) \leq \mathbf{K}_n(G) = 1$, so U is nilpotent.

If μ is a homomorphism of G, then $\mathbf{K}_n(G)^\mu = \mathbf{K}_n(G^\mu)$ by Definition 1.9. Thus, G nilpotent implies G^μ is nilpotent as well.

b) The nilpotency of $G_1 \times G_2$ follows from the trivial observation that

$$\mathbf{K}_n(G_1 \times G_2) = \mathbf{K}_n(G_1) \times \mathbf{K}_n(G_2).$$

c) By Lemma I.9.6 (pg. 50), $G/(M \cap N)$ is isomorphic to a subgroup of $G/M \times G/N$, so is nilpotent by a) and b).

d) Assume G is finite. If G/N is nilpotent, then for all large enough n we have

$$\mathbf{K}_n(G/N) = \mathbf{K}_n(G)N/N = 1,$$

so $\mathbf{K}_n(G) \leq N$ and therefore $\mathbf{K}_\infty(G) \leq N$, as well.

Conversely, it follows from $\mathbf{K}_\infty(G) = \mathbf{K}_n(G) \leq N$ (here the finiteness of G is used!) that $\mathbf{K}_n(G/N) = 1$, and so G/N is nilpotent.

e) The alternating group A_3 and S_3/A_3 are nilpotent. However, S_3 is not nilpotent. It has three Sylow 2-subgroups, for example. □

Theorem 2.6. *Let G be nilpotent and $1 < N \trianglelefteq G$. Then*

$$[N, G] < N \quad \textit{and} \quad N \cap \mathbf{Z}(G) > 1.$$

In particular, every minimal normal subgroup of a nilpotent group lies in the center.

Proof. We define the normal subgroups N_i of G recursively by $N_1 = N$ and $N_{i+1} = [N_i, G]$. Then clearly $N_i \leq N$ and $N_i \leq \mathbf{K}_i(G)$. Since G is nilpotent, for large enough m it certainly holds that $N_m \leq \mathbf{K}_m(G) = 1$. Thus, $N_2 = [N, G] < N$, since otherwise we would have $N_i = N \neq 1$ for all i. Let n be chosen so that $N_n = 1 < N_{n-1}$. Then $[N_{n-1}, G] = N_n = 1$, so we have

$$1 < N_{n-1} \leq N \cap \mathbf{Z}(G). \qquad \square$$

Theorem 2.7. *Let G be a finite nilpotent group with central series*

$$G = \mathbf{K}_1(G) > \mathbf{K}_2(G) > \cdots > \mathbf{K}_{r+1}(G) = 1$$

and

$$1 = \mathbf{Z}_0(G) < \mathbf{Z}_1(G) < \cdots < \mathbf{Z}_c(G) = G.$$

Then $r = c$. We call $c = c(G)$ the nilpotency class of G. For every central series

$$G = N_1 \geq N_2 \geq \cdots \geq N_{s+1} = 1$$

of G we have $\mathbf{K}_i(G) \leq N_i \leq \mathbf{Z}_{s+1-i}(G)$ and $c \leq s$.

Proof. We have already proved that $\mathbf{K}_i(G) \leq N_i$ in the course of proving a) implies b) in Theorem 2.3.

It follows from $[N_s, G] \leq N_{s+1} = 1$ that $N_s \leq \mathbf{Z}_1(G)$. Assume for induction that $N_i \leq \mathbf{Z}_{s+1-i}(G)$. Then

$$[N_{i-1}, G] \leq N_i \leq \mathbf{Z}_{s+1-i}(G),$$

so

$$N_{i-1}\mathbf{Z}_{s+1-i}(G)/\mathbf{Z}_{s+1-i}(G) \leq \mathbf{Z}(G/\mathbf{Z}_{s+1-i}(G)) = \mathbf{Z}_{s+2-i}(G)/\mathbf{Z}_{s+1-i}(G)$$

and thus

$$N_{i-1} \leq \mathbf{Z}_{s+2-i}(G).$$

Finally, setting $i = 1$ gives $N_1 = G \leq \mathbf{Z}_s(G)$, thus $s \geq c$. The special case $N_i = \mathbf{K}_i(G)$ yields $r \geq c$.

We have $\mathbf{K}_1(G) = G = \mathbf{Z}_c(G)$. Assume for induction that $\mathbf{K}_i(G) \leq \mathbf{Z}_{c+1-i}(G)$. Then it follows that

$$\mathbf{K}_{i+1}(G) = [\mathbf{K}_i(G), G] \leq [\mathbf{Z}_{c+1-i}(G), G] \leq \mathbf{Z}_{c-i}(G).$$

Hence, we ultimately obtain $\mathbf{K}_{c+1}(G) \leq \mathbf{Z}_0(G) = 1$, and therefore $c \geq r$. So $c = r$, as claimed. □

Theorem 2.8 (P. Hall). *Let N and L be subgroups of the (not necessarily finite) group G. Suppose*

$$N = N_0 \trianglerighteq N_1 \trianglerighteq \cdots$$

is a chain of normal subgroups of N with $[N_i, L] \leq N_{i+1}$ for all $i = 0, 1, 2, \ldots$. We define

$$L_j = \{l \in L \mid [N_i, l] \leq N_{i+j} \text{ for all } i\}$$

(in particular, $L_1 = L$). Then the L_j are subgroups of G, and for all i, j, m we have

$$[L_j, L_m] \leq L_{j+m} \quad \text{and} \quad [N_i, \mathbf{K}_j(L)] \leq N_{i+j}.$$

Proof. Since $[N_i, L] \leq N_{i+1} \leq N_i$, we have $L \leq \mathbf{N}_G(N_i)$ for all i by Lemma 1.6 d).

For $l, l' \in L_j$ and $n_i \in N_i$, we have by Lemma 1.2 b) that

$$[n_i, ll'] = [n_i, l'][n_i, l]^{l'} \in N_{i+j} N_{i+j}^{l'} = N_{i+j}.$$

It follows from

$$1 = [n_i, ll^{-1}] = [n_i, l^{-1}][n_i, l]^{l^{-1}}$$

that

$$[n_i, l^{-1}] = ([n_i, l]^{-1})^{l^{-1}} \in N_{i+j}^{l^{-1}} = N_{i+j}.$$

Therefore, we have $ll' \in L_j$ and $l^{-1} \in L_j$, and so L_j is a subgroup of G.

Without loss of generality, we may assume that G is generated by N and L. Since $L \leq \mathbf{N}_G(N_i)$, the subgroups N_i are then normal subgroups of G. By the definition of L_j, we have

$$[N_i, L_j, L_m] \leq [N_{i+j}, L_m] \leq N_{i+j+m}$$

and

$$[L_m, N_i, L_j] \leq [N_{i+m}, L_j] \leq N_{i+m+j}.$$

By Lemma 1.10 b), it then follows that

$$[[L_j, L_m], N_i] = [L_j, L_m, N_i] \leq N_{i+j+m},$$

and so we obtain

$$[L_j, L_m] \leq L_{j+m}$$

as desired.

In particular, $[L_j, L_1] \leq L_{j+1}$. Hence, the L_j form a central series of L. By Theorem 2.7, $\mathbf{K}_j(L) \leq L_j$ and therefore

$$[N_i, \mathbf{K}_j(L)] \leq [N_i, L_j] \leq N_{i+j}. \qquad \square$$

Theorem 2.9. *Let*

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_s = 1$$

be a chain of normal subgroups of G, and let A be a group of automorphisms of G such that

$$g_i^{\alpha} g_i^{-1} \in G_{i+1}$$

for all $g_i \in G_i$, all $i = 1, 2, \ldots, s-1$ and all $\alpha \in A$. Then A is nilpotent, and the class of A is less than s.

(*The special case $s = 2$ was considered in Theorem I.4.4, pg. 20; in that case, A is abelian.*)

Proof. Let H be the semidirect product of G with A constructed as in Example I.14.4 (pg. 94). We apply Theorem 2.8 with G in place of N, with G_i in place of N_i and with A in place of L. We then obtain

$$[G, \mathbf{K}_s(A)] \leq G_s = 1,$$

thus $\mathbf{K}_s(A) = 1$. $\square$

Example 2.10. Let K be a field, and let V be the vector space of dimension n over K. Let $\{v_1, \ldots, v_n\}$ be a K-basis of V, and let T be the group of all linear maps t of V such that

$$v_1 t = v_1 \quad \text{and} \quad v_i t - v_i \in \langle v_1, \ldots, v_{i-1} \rangle \quad (i = 2, \ldots, n).$$

Then T is nilpotent with $c(T) \leq n - 1$:

We apply Theorem 2.9 with $G = \mathsf{V}^+$, $G_i = \langle v_1, \ldots, v_{n-i} \rangle$ and $A = T$. The assumptions of Theorem 2.9 are then clearly satisfied. One can easily calculate (and we will do this in §16) that the class of T is $n - 1$.

The following theorem is fundamental to the more refined theory of nilpotent groups.

Main Theorem 2.11 (P. Hall).
a) *For every central series*

$$N_1 \geq \cdots \geq N_r$$

of G we have $[N_i, \mathbf{K}_j(G)] \leq N_{i+j}$; here we define $N_k = N_r$ for $k > r$.
b) *In particular, $[\mathbf{K}_i(G), \mathbf{K}_j(G)] \leq \mathbf{K}_{i+j}(G)$. If G is nilpotent of class c, then $\mathbf{K}_i(G)$ is abelian for $2i > c$.*
c) *We have*

$$[\mathbf{K}_i(G), \mathbf{Z}_j(G)] \leq \mathbf{Z}_{j-i}(G);$$

here we define $\mathbf{Z}_{j-i}(G) = 1$ for $j \leq i$. In particular, we have

$$[\mathbf{K}_i(G), \mathbf{Z}_i(G)] = 1;$$

that is, elements of $\mathbf{K}_i(G)$ commute with elements of $\mathbf{Z}_i(G)$.

Proof. a) Since the N_i constitute a central series, we have $[N_i, G] \leq N_{i+1}$. Theorem 2.8 with G in place of L yields the claim

$$[N_i, \mathbf{K}_j(G)] \leq N_{i+j}.$$

b) The first statement is a special case of a) with $N_i = \mathbf{K}_i(G)$. The second statement is proved as follows:
We have

$$\mathbf{K}_i(G)' = [\mathbf{K}_i(G), \mathbf{K}_i(G)] \leq \mathbf{K}_{2i}(G).$$

Therefore, $\mathbf{K}_i(G)$ is abelian for $2i > c$.
c) We consider the central series

$$N_1 \geq N_2 \geq \cdots \geq N_{j+1} = 1$$

of G with $N_i = \mathbf{Z}_{j-i+1}(G)$. Then we have by a) that

$$[\mathbf{K}_i(G), \mathbf{Z}_j(G)] = [\mathbf{K}_i(G), N_1] \leq N_{i+1} = \mathbf{Z}_{j-i}(G). \qquad \square$$

The connection to the concepts introduced in §8 of Chapter I for finite groups is contained in the following theorem.

Theorem 2.12. *Let $G^{(i)}$ be the higher commutator groups of G from Definition I.8.1 (pg. 38), which are defined to be $G^{(0)} = G$ and $G^{(i+1)} = G^{(i)'}$. Then:*
a) *$G^{(i)} \leq \mathbf{K}_{2^i}(G)$; in particular, every finite nilpotent group is solvable.*
b) *If G can be generated by two elements, then in fact $G'' \leq \mathbf{K}_5(G)$.*

Proof. a) We prove the claim by induction on i. For $i = 0$, we have

$$G^{(0)} = G = \mathbf{K}_1(G).$$

So assume for induction that $G^{(i)} \leq \mathbf{K}_{2^i}(G)$. Using Theorem 2.11 b), we obtain

$$G^{(i+1)} = [G^{(i)}, G^{(i)}] \leq [\mathbf{K}_{2^i}(G), \mathbf{K}_{2^i}(G)] \leq \mathbf{K}_{2^{i+1}}(G).$$

b) By passing to the quotient $G/\mathbf{K}_5(G)$, we may assume without loss of generality that $\mathbf{K}_5(G) = 1$. Then we must show that $G'' = 1$. By Theorem 2.11,

$$[\mathbf{K}_2(G), \mathbf{K}_3(G)] \leq \mathbf{K}_5(G) = 1.$$

So $\mathbf{K}_3(G)$ lies in the center of G'. By Theorem 1.11 c), however, we have

$$G' = \mathbf{K}_2(G) = \langle [a, b], \mathbf{K}_3(G) \rangle,$$

where $\{a, b\}$ is any generating set of G. Thus, $G'/\mathbf{Z}(G')$ is cyclic. So G' is clearly abelian, and it follows that $G'' = 1$. □

Theorem 2.13.
a) *The exponent of* $\mathbf{Z}_{i+1}(G)/\mathbf{Z}_i(G)$ *divides the exponent of* $\mathbf{Z}_i(G)/\mathbf{Z}_{i-1}(G)$.
b) *The exponent of* $\mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)$ *divides the exponent of* $\mathbf{K}_{i-1}(G)/\mathbf{K}_i(G)$.

Proof. a) First assume that $i = 1$. Let $x \in \mathbf{Z}_2(G)$, so $[g, x] \in \mathbf{Z}_1(G)$ for all $g \in G$. Let n be the exponent of $\mathbf{Z}_1(G)$. By Lemma 1.3, we have

$$1 = [g, x]^n = [g, x^n].$$

So every x^n lies in $\mathbf{Z}_1(G)$ and the exponent of $\mathbf{Z}_2(G)/\mathbf{Z}_1(G)$ is a divisor of n.

The general case is handled in the same way by first passing to the quotient $G/\mathbf{Z}_{i-1}(G)$.

b) Let n be the exponent of $\mathbf{K}_{i-1}(G)/\mathbf{K}_i(G)$. By passing to the quotient $G/\mathbf{K}_{i+1}(G)$, we may assume that $\mathbf{K}_{i+1}(G) = 1$. Then $\mathbf{K}_i(G) \leq \mathbf{Z}(G)$. We have

$$\mathbf{K}_i(G) = [\mathbf{K}_{i-1}(G), G] = \langle [y, x] \mid x \in G, y \in \mathbf{K}_{i-1}(G) \rangle.$$

Since $[y, x] \in \mathbf{K}_i(G) \leq \mathbf{Z}(G)$, it follows from Lemma 1.3 that for $y \in \mathbf{K}_{i-1}(G)$ we have

$$[y, x]^n = [y^n, x] \in [\mathbf{K}_i(G), G] = \mathbf{K}_{i+1}(G) = 1.$$

Thus, the exponent of $\mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)$ is a divisor of n. □

Finally, we discuss a peculiar result of P. Hall:

Theorem 2.14 (P. Hall [218]). *Let* $\mathbf{C}_G(\mathbf{K}_i(G))$ *be the centralizer of* $\mathbf{K}_i(G)$ *in* G. *We have*

$$[\mathbf{C}_G(\mathbf{K}_{n+1}(G)), \mathbf{C}_G(\mathbf{K}_{m+1}(G))] \leq \mathbf{Z}_{m+n-1}(G);$$

where we define $\mathbf{Z}_j(G) = 1$ *for* $j \leq 0$. *If* $\mathbf{Z}_1(G) = 1$, *then for* $m = n$ *it follows, in particular, that*

$$[\mathbf{C}_G(\mathbf{K}_{n+1}(G)), \mathbf{C}_G(\mathbf{K}_{n+1}(G))] = 1;$$

hence, in this case the centralizers of terms of the lower central series of G are abelian.

Proof. For $N \trianglelefteq G$, we define $N_0 = N$ and $N_{i+1} = [N_i, G]$. Then the N_i are all normal subgroups of G contained in N. With this notation, we first prove the following:

If H and K are normal subgroups of G, then

$$[H, K]_k \leq \prod_{i=0}^{k} [H_i, K_{k-i}].$$

For $k = 0$, this is clearly true. We prove the general result by induction on k. A repeated application of Lemma 1.10 a) yields

$$\begin{aligned}[H, K]_{k+1} = [[H, K]_k, G] &\leq \left[\prod_{i=0}^{k} [H_i, K_{k-i}], G \right] \\ &= \prod_{i=0}^{k} [H_i, K_{k-i}, G].\end{aligned}$$

It follows by Lemma 1.10 b) that

$$\begin{aligned}[[H_i, K_{k-i}], G] &\leq [[K_{k-i}, G], H_i][[G, H_i], K_{k-i}] \\ &= [K_{k-i+1}, H_i][H_{i+1}, K_{k-i}].\end{aligned}$$

The claim follows by taking the product over i from 0 to k and using that $H_i K_j = K_j H_i$.

Setting $H = \mathbf{C}_G(\mathbf{K}_{n+1}(G))$ and $K = \mathbf{C}_G(\mathbf{K}_{m+1}(G))$, it follows by what we have just shown that

$$[H, K]_{n+m-1} \leq \prod_{i=0}^{n+m-1} [H_i, K_{n+m-1-i}],$$

where either $i \geq m$ or $n + m - 1 - i \geq n$. In the first case, it follows that

$$[H_i, K_{n+m-1-i}] \leq [\mathbf{K}_{m+1}(G), \mathbf{C}_G(\mathbf{K}_{m+1}(G))] = 1,$$

and in the second case,

$$[H_i, K_{n+m-1-i}] \leq [\mathbf{C}_G(\mathbf{K}_{n+1}(G)), \mathbf{K}_{n+1}(G)] = 1.$$

So $[H, K]_{n+m-1} = 1$, which means exactly that

$$[H, K] \leq \mathbf{Z}_{n+m-1}(G). \qquad \square$$

One can find an interesting application of Theorem 2.14 to infinite groups in Hall [218].

Exercises

3) Show that if G is finite, then

$$\mathbf{Z}_\infty(G) = \bigcap_{\mathbf{Z}(G/N)=1} N \quad \text{and} \quad \mathbf{K}_\infty(G) = \langle N \mid [N, G] = N \rangle.$$

4) Show that Theorem 2.3 b) and Theorem 2.3 f) are equivalent, even when G is infinite.
5) Let G be a (not necessarily finite) group with $\bigcap_i \mathbf{K}_i(G) = 1$. Show that if $x, y \in G$ are elements of finite coprime order, then $xy = yx$.
6) Construct a non-nilpotent group such that $\bigcap_i \mathbf{K}_i(G) = 1$.
7) Show that if $G = G'$, then $\mathbf{Z}_2(G) = \mathbf{Z}_1(G)$. For every n, construct a G such that $\mathbf{Z}(G) = 1$ and $G > \mathbf{K}_2(G) > \cdots > \mathbf{K}_n(G)$.
8) Let $G = \mathbf{K}_1(G) > \mathbf{K}_2(G) > \cdots > \mathbf{K}_{c+1}(G) = 1$ be the lower central series of the nilpotent group G. We write the abelian factor groups $L_i = \mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)$ additively and construct $L = \bigoplus_{i=1}^{c} L_i$. We define a multiplication on L by

$$g\mathbf{K}_{i+1}(G) \circ h\mathbf{K}_{j+1}(G) = [g, h]\mathbf{K}_{i+j+1}(G)$$

for $g\mathbf{K}_{i+1}(G) \in L_i$ and $h\mathbf{K}_{j+1}(G) \in L_j$, and then declaring that

$$\left(\sum_i a_i\right) \circ \left(\sum_j b_j\right) = \sum_{i,j} a_i \circ b_j$$

for all $a_i, b_i \in L_i$. Show that L is a Lie ring. That is, for all $a, b, c \in L$ we have:
(1) $a \circ (b + c) = a \circ b + a \circ c$ and $(b + c) \circ a = b \circ a + c \circ a$.
(2) $a \circ a = 0$ and $a \circ b = -b \circ a$.
(3) $(a \circ b) \circ c + (b \circ c) \circ a + (c \circ a) \circ b = 0$ (the Jacobi identity).

3 The Frattini subgroup

From now on, all groups are assumed to be finite unless otherwise stated.

Definition 3.1. For $G > 1$, let $\Phi(G)$ be the intersection of all maximal subgroups of G. We define $\Phi(1) = 1$. We call $\Phi(G)$ the Frattini subgroup of G. It is clear that $\Phi(G)$ is a characteristic subgroup of G.

It is important to observe that $\Phi(G)$ can be described in another way, namely:

Theorem 3.2.
a) *An element $x \in G$ lies in $\Phi(G)$ if and only if x can be omitted from every generating set of G; that is, for every subset K of G such that $G = \langle K, x\rangle$, we have $G = \langle K\rangle$.*
b) *Let $N \trianglelefteq G$. There exists some H such that $H < G$ and $G = NH$ if and only if N does not lie in $\Phi(G)$.*

Proof. a) Let $x \in \Phi(G)$ and $G = \langle K, x\rangle$. Assume for contradiction that $\langle K\rangle < G$. Then there exists a maximal subgroup M of G such that $\langle K\rangle \leq M$. Since $x \in \Phi(G) \leq M$, we have $x \in M$ as well. From this, we obtain the contraction

$$G = \langle K, x\rangle \leq M.$$

So in fact $G = \langle K\rangle$.

Now assume that for all subsets K of G such that $\langle K, x\rangle = G$, we have $\langle K\rangle = G$. Let U be an arbitrary maximal subgroup of G. Then we have $U = \langle U\rangle < G$, so by our assumption, $U \leq \langle U, x\rangle < G$ as well. Since U is a maximal subgroup of G, it follows that $U = \langle U, x\rangle$, and so $x \in U$. As this holds for all maximal subgroups of G, we see that $x \in \Phi(G)$.

b) If $N \leq \Phi(G)$ and

$$G = NH = \Phi(G)H = \langle \Phi(G), H\rangle,$$

then it follows by repeated application of a) that $G = H$. Thus, $G = NH$ and $H < G$ imply that $N \not\leq \Phi(G)$.

Conversely, assume $N \not\leq \Phi(G)$. Then there exists a maximal subgroup H of G such that $N \not\leq H$. Since $H < NH \leq G$, we then obtain $G = NH$. □

The following two lemmas are often useful when working with the Frattini subgroup.

Lemma 3.3.
a) *If $N \trianglelefteq G$, $U \leq G$ and $N \leq \Phi(U)$, then $N \leq \Phi(G)$.*
b) *If $M \trianglelefteq G$, then $\Phi(M) \leq \Phi(G)$.*
c) *The statement in b) cannot be extended to arbitrary subgroups of G; that is, $U < G$ does not always imply that $\Phi(U) \leq \Phi(G)$.*

Proof. a) Assume for contradiction that $N \not\leq \Phi(G)$. Then there exists a maximal subgroup V of G such that $N \not\leq V$. Thus, $V < VN$ and then $G = VN$ since V is maximal. Since $N \leq U$, it follows however that

$$U = U \cap NV = N(U \cap V).$$

By assumption, we have $N \leq \Phi(U)$. Therefore, we may omit all elements of N from the generation of U by Theorem 3.2 and obtain $U = U \cap V \leq V$. But then $N \leq \Phi(U) \leq V$, contradicting our assumption that $N \not\leq V$.

b) Since $\Phi(M)$ is a characteristic subgroup of the normal subgroup M of G, $\Phi(M) \trianglelefteq G$ by Theorem I.4.8 a) (pg. 22). Now we use a) with $N = \Phi(M)$ and $U = M$. This yields the claim $\Phi(M) \leq \Phi(G)$.

c) We construct a counterexample. Let G be the semidirect product of the cyclic group $\langle a \rangle$ of order 5 with the cyclic group $\langle b \rangle$ of order 4 with respect to the automorphism $a^b = a^2$. We certainly have

$$\Phi(\langle b \rangle) = \langle b^2 \rangle \neq 1.$$

The subgroup $\langle b \rangle$ has index 5 in G, so it is a maximal subgroup of G. Thus, to prove $\Phi(G) = 1$ it suffices to show that $\langle b \rangle \cap \langle b^a \rangle = 1$. Otherwise, we would have

$$b^2 = (b^2)^a \in \langle b \rangle \cap \langle b^a \rangle.$$

However, we have

$$(b^2)^a = b^2[b^2, a] = b^2 a^{-4} a = b^2 a^2 \neq b^2. \qquad \square$$

Lemma 3.4.
a) *If N is a normal subgroup of G, then*

$$\Phi(G)N/N \leq \Phi(G/N).$$

This statement can also be formulated as follows: If μ is a homomorphism of G, then $\Phi(G)^\mu \leq \Phi(G^\mu)$.
b) *If $N \trianglelefteq G$ and $N \leq \Phi(G)$, then $\Phi(G)/N = \Phi(G/N)$.*
c) *If $N \trianglelefteq G$, then it does not always follow that $\Phi(G)N/N = \Phi(G/N)$.*

Proof. a) By Theorem I.3.10 (pg. 16), the maximal subgroups of G/N have the form U/N, where U runs through the set $\mathscr{S}$ of maximal subgroups of G with $N \leq U$. Thus,

$$\Phi(G/N) = \bigcap_{U \in \mathscr{S}} U/N \geq \Phi(G)N/N.$$

b) If $N \leq \Phi(G)$, then $N \leq U$ for all maximal subgroups U of G. So

$$\Phi(G/N) = \bigcap_{U\max} U/N = \left(\bigcap_{U\max} U \right)/N = \Phi(G)/N.$$

c) We use the example from Lemma 3.3 c). We have $\Phi(G) = 1$, but $G/\langle a \rangle$ is isomorphic to a cyclic group of order 4, so it has a Frattini subgroup of order 2. $\square$

Theorem 3.5 (Gaschütz [155]). *Let D and M be normal subgroups of G such that $D \trianglelefteq M$ and $D \leq \Phi(G)$. If M/D is nilpotent, then M itself is nilpotent.*

Proof. Let P be a Sylow p-subgroup of M. Then PD/D is a Sylow p-subgroup of M/D by Lemma I.7.7 b) (pg. 35). Since M/D is nilpotent, the Sylow p-subgroup

PD/D of M/D is characteristic in M/D since it is unique. Thus, $PD/D \trianglelefteq G/D$. From this we obtain $PD \trianglelefteq G$.

Since P is a Sylow p-subgroup of the normal subgroup PD of G, it follows by Theorem I.7.8 (pg. 36) that

$$G = PD\mathbf{N}_G(P) = D\mathbf{N}_G(P).$$

Since $D \leq \Phi(G)$, we have by Theorem 3.2 b) that $G = \mathbf{N}_G(P)$, so $P \trianglelefteq G$. Since this holds for all Sylow subgroups of M, the subgroup M is nilpotent by Theorem 2.3.□

Important properties of the Frattini subgroup follow from Theorem 3.5:

Theorem 3.6 (Frattini 1885!). *The Frattini subgroup $\Phi(G)$ is always nilpotent.*

Proof. We set $M = D = \Phi(G)$ in Theorem 3.5. Certainly $M/D = 1$ is nilpotent, so $M = \Phi(G)$ is also nilpotent. □

Theorem 3.7. *If $G/\Phi(G)$ is nilpotent, then G is also nilpotent.*

Proof. This is a special case of Theorem 3.5 with $M = G$ and $D = \Phi(G)$. □

Theorem 3.8. *If p is a prime number dividing $|\Phi(G)|$, then p divides $|G/\Phi(G)|$.*

Proof. Assume for contradiction that $p \nmid |G/\Phi(G)|$, and let P be a Sylow p-subgroup of $\Phi(G)$. Since $\Phi(G)$ is nilpotent by Theorem 3.6, P is characteristic in $\Phi(G)$, and so $P \trianglelefteq G$. Since $|G/P|$ is coprime to p, there exists a complement S of P in G by the Zassenhaus Theorem I.18.1. But then using Theorem 3.2 b), we reach the contradiction

$$G = PS \leq \Phi(G)S = S < G.$$

So p is a divisor of $|G/\Phi(G)|$. □

Theorem 3.9 (Huppert [257]). *Let $\mathscr{E}$ be a group property that satisfies the following conditions:*
(1) *If $\mathscr{E}$ holds for $H/\Phi(H)$, then $\mathscr{E}$ holds for H.*
(2) *If $\mathscr{E}$ holds for H, then $\mathscr{E}$ holds for all epimorphic images of H.*
Then:
If G/N has the property $\mathscr{E}$, then there exists a subgroup U with the property $\mathscr{E}$ and with $G = NU$.

Proof. We consider the set

$$\mathscr{M} = \{U \leq G \mid G = NU\}.$$

Since $G \in \mathscr{M}$, the set $\mathscr{M}$ is nonempty. Let U be a minimal element of $\mathscr{M}$. We will show that U has the property $\mathscr{E}$.

If $U \cap N \not\leq \Phi(U)$, then there exists a maximal subgroup V of U such that $U \cap N \not\leq V$, so $U = (U \cap N)V$. It follows that

$$G = NU = NV,$$

contradicting the minimality of U. Hence, $U \cap N \leq \Phi(U)$.

Since $G = NU$, we have $G/N \cong U/(U \cap N)$. So $U/(U \cap N)$ has the property $\mathscr{E}$. Since $U \cap N \leq \Phi(U)$, the quotient $U/\Phi(U)$ is an epimorphic image of $U/(U \cap N)$. By (2), the quotient $U/\Phi(U)$ then has the property $\mathscr{E}$, and so by (1) the subgroup U does as well. □

Theorems 3.7 and 3.9 yield immediately:

Theorem 3.10. *If G/N is nilpotent, then there exists a nilpotent subgroup U of G such that $G = NU$.*

(*However, there does not necessarily exist a complement of N in G; that is, it need not be the case that $U \cong G/N$!*)

The Frattini subgroup provides a new characterization of (finite) nilpotent groups:

Theorem 3.11 (Wielandt [460]). *The group G is nilpotent if and only if $G' \leq \Phi(G)$. More generally, if N is a normal subgroup of G, then N is nilpotent if and only if $N' \leq \Phi(G)$.*

Proof. Assume $N' \leq \Phi(G)$. Certainly $N' \trianglelefteq G$ and N/N' is abelian, so nilpotent. So N is nilpotent by Theorem 3.5.

Conversely, assume N is nilpotent. By Theorem 2.3 d), all maximal subgroups of N are normal in N, so they necessarily have prime index. Hence, $N' \leq U$ for every maximal subgroup U of N. It follows that $N' \leq \Phi(N)$. Using the containment $\Phi(N) \leq \Phi(G)$ from Lemma 3.3 b), we obtain $N' \leq \Phi(G)$, as well. □

The following theorem will be used in Chapter V when we discuss the Schur multiplier and projective representations.

Theorem 3.12 (Gaschütz [155]). *We have $G' \cap \mathbf{Z}(G) \leq \Phi(G)$.*

Proof. We let $D = G' \cap \mathbf{Z}(G)$. Assume for contradiction that $D \not\leq \Phi(G)$. Then there exists a maximal subgroup U of G such that $D \not\leq U$. Thus, $G = DU$ and we can write every element $g \in G$ in the form

$$g = du \quad \text{with} \quad d \in D \quad \text{and} \quad u \in U.$$

Since $d \in D \leq \mathbf{Z}(G)$, it follows that

$$U^g = U^{du} = U^u = U.$$

So U is a normal subgroup of G. Since U is also a maximal subgroup of G, the index of U is a prime number. It follows that $G' \leq U$ and then $D \leq G' \leq U$, contradicting our assumption. □

Not much is known about which nilpotent groups can be the Frattini subgroup of a finite group G.[1] We prove:

Theorem 3.13 (Gaschütz [155]). *If $N \lhd G$ and $N \leq \Phi(G)$, then the inner automorphism group* $\mathbf{Inn}(N)$ *of N lies in the Frattini subgroup of the full automorphism group* $\mathbf{Aut}(N)$.

Proof. We associate to every element $g \in G$ an automorphism g^τ of N defined by $g^\tau : n \mapsto n^g$ for all $n \in N$. This produces an epimorphism τ from G onto a subgroup $U = G^\tau$ of the automorphism group $\mathbf{Aut}(N)$ of N. Under this homomorphism, N is mapped to the group $N^\tau = \mathbf{Inn}(N)$ of inner automorphisms of N. Since $N \leq \Phi(G)$, it follows from Lemma 3.4 a) that

$$N^\tau = \mathbf{Inn}(N) \leq \Phi(G)^\tau \leq \Phi(G^\tau) = \Phi(U).$$

On the other hand, $\mathbf{Inn}(N) \trianglelefteq \mathbf{Aut}(N)$ by Theorem I.4.2 (pg. 19). Thus, Lemma 3.3 a) gives

$$\mathbf{Inn}(N) \leq \Phi(\mathbf{Aut}(N)). \qquad \square$$

Considerably more can be said about the Frattini subgroups of p-groups:

Theorem 3.14. *Let G be a p-group. Then:*

a) $\Phi(G) = G'G^p$, *where* $G^p = \langle g^p \mid g \in G\rangle$.
So $G/\Phi(G)$ is an elementary abelian p-group, and $\Phi(G)$ is the smallest normal subgroup of G with an elementary abelian factor group.

b) *If $p = 2$, then $\Phi(G) = G^2$. In this case, the Frattini subgroup is generated by the squares of the elements of G.*

c) *If $U \leq G$, then $\Phi(U) \leq \Phi(G)$; if $N \trianglelefteq G$, then $\Phi(G)N/N = \Phi(G/N)$.*

Proof. a) Let U be a maximal subgroup of G. Since G is nilpotent by Theorem 2.3, being a p-group, we have $U \lhd G$. By the maximality of U, $|G/U| = p$. Then

$$G'G^p \leq U,$$

and therefore $G'G^p \leq \Phi(G)$, as well.

Next, we prove the following claim: If G/N is elementary abelian, then $\Phi(G) \leq N$. Let $x \in G - N$. Since G/N is elementary abelian, it is isomorphic to the additive group of a finite dimensional vector space over GF(p). Hence, there exists a basis $\{x_iN \mid i = 1, \ldots k\}$ of G/N with $x_1 = x$. Now

$$G = \langle x_1, \ldots, x_k, N\rangle > \langle x_2, \ldots, x_k, N\rangle.$$

It follows that $x_1 \notin \Phi(G)$ by Theorem 3.2 a). Thus, $\Phi(G) \leq N$.

Letting $D = G'G^p$, we have that G/D is elementary abelian. Hence, $\Phi(G) \leq D$ and so $D = G'G^p = \Phi(G)$.

[1] It is now known that the converse of Theorem 3.13 holds; see B. Eick, The converse of a theorem of W. Gaschütz on Frattini subgroups, Math. Z. **224** (1997), no. 1, 103–111. *trans.*

b) We need to show that commutators can be written as products of squares. This follows from

$$[x, y] = x^{-1}y^{-2}xx^{-2}xyxy = (y^x)^{-2}x^{-2}(xy)^2.$$

Hence, $G' \leq G^2$, and then $G^2 = \Phi(G)$ by a).

c) is a direct consequence of the description of the Frattini subgroup in a). □

The following theorem of Burnside is fundamental for the theory of p-groups.

Theorem 3.15 (Burnside's Basis Theorem).
Let G be a p-group with $|G/\Phi(G)| = p^d$. Then every minimal generating set of G contains exactly d elements, and every element of G not contained in $\Phi(G)$ belongs to at least one generating set with d elements.

(*Note that the corresponding statement for arbitrary nilpotent groups does not hold: The cyclic group of order* 6 *has a generating set consisting of one element, but it also has a minimal generating set containing two elements of orders* 2 *and* 3.)

Proof. If $G/\Phi(G) = \langle x_i\Phi(G) \mid i = 1, \ldots, d\rangle$, then it follows with the help of Theorem 3.2 a) that

$$G = \langle x_i, \Phi(G) \mid i = 1, \ldots, d\rangle = \langle x_i \mid i = 1, \ldots, d\rangle.$$

The converse clearly holds as well. Therefore, the x_i constitute a generating set of G if and only if the cosets $x_i\Phi(G)$ constitute a generating set of $G/\Phi(G)$. Hence, we need only prove the claim for elementary abelian groups.

Let G be elementary abelian of order p^d. Then G is isomorphic to the additive group of a vector space of dimension d over $\mathsf{K} = \mathrm{GF}(p)$. It is well known that every generating set of G contains a K-basis of G, that every K-basis consists of exactly d elements, and that every nonidentity element of G appears in at least one basis. □

One simple consequence of the Burnside Basis Theorem anticipates considerations we will take up in §8:

Theorem 3.16. *If the group G has a unique maximal subgroup, then G is cyclic of prime power order.*

Proof. Let U be the unique maximal subgroup of G. Then $U = \Phi(G)$. If G does not have prime power order, then every Sylow subgroup of G lies in U. The contradiction $|U| = |G|$ follows. Thus, G is a p-group for some prime p. We have $p^d = |G/\Phi(G)| = |G/U| = p$, so $d = 1$. By Theorem 3.15, the group G can be generated by one element, so is cyclic. □

Theorem 3.17 (P. Hall [207]). *Assume that the group G can be generated by d elements. Then the order of the automorphism group* $\mathbf{Aut}(G)$ *of G is a divisor of*

$$|\mathbf{Aut}(G/\Phi(G))|\,|\Phi(G)|^d.$$

Proof. We associate to every $\alpha \in \mathbf{Aut}(G)$ the automorphism $\overline{\alpha}$ of $G/\Phi(G)$ induced by α constructed according to Theorem I.4.3 (pg. 19). It is defined by

$$(g\Phi(G))^{\overline{\alpha}} = g^{\alpha}\Phi(G).$$

The map $\alpha \mapsto \overline{\alpha}$ provides a homomorphism from $\mathbf{Aut}(G)$ to $\mathbf{Aut}(G/\Phi(G))$. We call the kernel of this homomorphism B. Then B consists of exactly those automorphisms α of G such that $g^{\alpha}\Phi(G) = g\Phi(G)$ for all $g \in G$.

Let $\{r_1, \ldots, r_d\}$ be a generating set of G. We consider all ordered sets $\{r'_1, \ldots, r'_d\}$ with $r'_i = r_i f_i$ and $f_i \in \Phi(G)$. The number of these sets is clearly $|\Phi(G)|^d$, and they are all generating sets for G. The automorphisms in B permute these sets. The set of all $|\Phi(G)|^d$ ordered generating sets decomposes into orbits T_i under B. The size of T_i is the index of B_i in B by Theorem I.5.10 (pg. 27), where B_i is the subgroup of B that fixes an ordered generating set in T_i. Then all automorphisms in B_i fix some ordered generating set of G, and it follows that $B_i = 1$. Thus, we have $|T_i| = |B|$ for all i. If k is the number of the orbits T_i, then

$$k|B| = |\Phi(G)|^d.$$

It follows that

$$\begin{array}{lll} |\mathbf{Aut}(G)| & \text{divides} & |\mathbf{Aut}(G/\Phi(G))|\,|B|, \\ & \text{so} & \\ |\mathbf{Aut}(G)| & \text{divides} & |\mathbf{Aut}(G/\Phi(G))|\,|\Phi(G)|^d. \end{array}$$ □

A special case of Theorem 3.17 is particularly useful:

Theorem 3.18. *Let α be an automorphism of G such that $|\alpha|$ is coprime to $|\Phi(G)|$. Furthermore, assume that $g^{\alpha}\Phi(G) = g\Phi(G)$ for all $g \in G$; that is, α induces the identity automorphism on $G/\Phi(G)$. Then $\alpha = 1$.*

In particular, if G is a p-group and α is an automorphism of G such that $g^{\alpha}\Phi(G) = g\Phi(G)$ for all $g \in G$, then $|\alpha|$ is a power of p.

Proof. The automorphism α lies in the group B of Theorem 3.17. Thus, $|\alpha|$ divides $|B|$ and so divides $|\Phi(G)|^d$ as well. But since $|\alpha|$ is coprime to $|\Phi(G)|$, we have $\alpha = 1$. □

Later, in Theorem IV.5.12 (pg. 465), we will prove a statement that is dual to Theorem 3.18 for p-groups.

We obtain from Theorem 3.17 an upper bound for the order of the automorphism group of a p-group:

Theorem 3.19 (P. Hall [207]). *Let G be a p-group of order p^n with $|G/\Phi(G)| = p^d$. Then the order of the automorphism group $\mathbf{Aut}(G)$ is a divisor of*

$$p^{d(n-d)}(p^d - 1)(p^d - p)\cdots(p^d - p^{d-1}).$$

Proof. We have $|\Phi(G)| = p^{n-d}$. By Lemma II.6.2 (pg. 186), we have

$$|\mathbf{Aut}(G/\Phi(G))| = |\mathrm{GL}_d(p)| = (p^d - 1)(p^d - p)\cdots(p^d - p^{d-1}).$$

The claim follows by Theorem 3.17. □

It is considerably more difficult to bound the order of the automorphism group of a p-group from below. Gaschütz proved the following statement using cohomological methods: If G is a p-group with $|G| > p$, then p divides $|\mathbf{Aut}(G)/\mathbf{Inn}(G)|$ (see §19 of Chapter III). There are examples of 2-groups whose automorphism groups are again 2-groups (see Exercise I.56, pg. 100).

Exercises

9) Show that $\Phi(G_1 \times G_2) = \Phi(G_1) \times \Phi(G_2)$.
10) (Kovács, Neubüser, B.H. Neumann)
   a) Assume the finite group G is generated by n elements. Show that the following statements are equivalent:
      (1) $g \in \Phi(G)$.
      (2) If $\{g_1, \ldots, g_n\}$ is a generating set of G with n elements, then we also have $G = \langle g_1 g, g_2, \ldots, g_n\rangle$.
   b) Let n be the minimum size of a generating set of G. Then the following are equivalent:
      (1) $g \in \Phi(G)$.
      (2) If $G = \langle g, g_1, \ldots, g_n\rangle$, then $G = \langle g_1, \ldots, g_n\rangle$.
      (Hint: Use a) in the proof of b).)
11) (Gaschütz [155]) Let $\Delta(G)$ be the intersection of all non-normal, maximal subgroups of G. Show that:
   a) $\Delta(G)/\Phi(G) = \mathbf{Z}(G/\Phi(G))$.
   b) $\Delta(G)$ is nilpotent.
   c) $\Delta(G)$ contains the hypercenter $\mathbf{Z}_\infty(G)$ of G.
12) (Deskins [109]) Let $\Phi_p(G)$ be the intersection of all those maximal subgroups of G whose index is coprime to the prime p. Show that:
   a) $\Phi_p(G)$ has a normal Sylow p-subgroup with nilpotent factor group.
   b) If p and q are two distinct primes, then

$$\Phi_p(G) \cap \Phi_q(G) = \Phi(G).$$

4 The Fitting subgroup

Theorem 4.1 (Fitting [139]). *If A and B are nilpotent normal subgroups of G with nilpotency classes a and b, respectively, then AB is a nilpotent normal subgroup of G, and AB has nilpotency class at most $a + b$.*

(*It is not necessary to assume that G is finite.*)

Proof. If A, B and C are normal subgroups of G, then we have by Lemma 1.10 a) that

$$[AB, C] = [A, C][B, C] \quad \text{and} \quad [A, BC] = [A, B][A, C].$$

Let $c = a + b$. A repeated application of these identities gives

$$\mathbf{K}_{c+1}(AB) = \prod\prod T,$$

where every factor T has the form $T = [H_1, \ldots, H_{c+1}]$ with $H_i = A$ or $H_i = B$. In a given T, the subgroup A appears, say, n times and B appears m times. Then $n + m = c + 1 = a + b + 1$, so $n > a$ or $m > b$. Say $n \geq a + 1$. We define

$$T_i = [\ldots, A],$$

where the commutator ends after the i-th appearance of A. Since $A \trianglelefteq G$, we have $T_1 \leq A$ and $T_1 \trianglelefteq G$. Therefore,

$$T_2 = [T_1, B, \ldots, B, A] \leq [A, A] = \mathbf{K}_2(A).$$

By induction on i, it follows more generally since $\mathbf{K}_i(A) \trianglelefteq G$ that

$$\begin{aligned} T_{i+1} &= [T_i, B, \ldots, B, A] \leq [\mathbf{K}_i(A), B, \ldots, B, A] \leq [\mathbf{K}_i(A), A] \\ &= \mathbf{K}_{i+1}(A). \end{aligned}$$

Finally, we obtain $T_{a+1} \leq \mathbf{K}_{a+1}(A) = 1$, and therefore $T = 1$. So $\mathbf{K}_{c+1}(AB) = 1$. □

Assume in what follows that G is finite.

Theorem 4.2.

a) *The product of all nilpotent normal subgroups of G is a nilpotent normal subgroup $\mathbf{F}(G)$ of G. We call $\mathbf{F}(G)$ the Fitting subgroup of G.*

b) *If $\mathbf{C}_G(\mathbf{F}(G))$ is the centralizer of $\mathbf{F}(G)$ in G, then*

$$\mathbf{C}_G(\mathbf{F}(G))\mathbf{F}(G)/\mathbf{F}(G)$$

contains no nonidentity solvable normal subgroups of $G/\mathbf{F}(G)$. In particular, if G is solvable, then $\mathbf{C}_G(\mathbf{F}(G)) \leq \mathbf{F}(G)$.

c) *The Frattini subgroup $\Phi(G)$ of G lies in $\mathbf{F}(G)$. If G is solvable and $G \neq 1$, then in fact $\Phi(G) < \mathbf{F}(G)$.*

d) *We have* $\mathbf{F}(G/\Phi(G)) = \mathbf{F}(G)/\Phi(G)$.
e) *If N is a minimal normal subgroup of G, then* $\mathbf{F}(G) \leq \mathbf{C}_G(N)$. *In particular, if N is abelian, then* $N \leq \mathbf{Z}(\mathbf{F}(G))$.

Proof. a) is a direct consequence of Theorem 4.1 and the finiteness of G.

b) Let $N/\mathbf{F}(G)$ be an abelian normal subgroup of the quotient $G/\mathbf{F}(G)$ with $N \leq \mathbf{C}_G(\mathbf{F}(G))\mathbf{F}(G)$. Then $N = \mathbf{F}(G)C$, where $C = N \cap \mathbf{C}_G(\mathbf{F}(G))$. Let c be the class of the nilpotent group

$$N/C \cong \mathbf{F}(G)/(\mathbf{F}(G) \cap C).$$

Since $N \leq \mathbf{F}(G)\mathbf{C}_G(\mathbf{F}(G))$, it follows that

$$\mathbf{K}_{c+1}(N) \leq C \cap N' \leq C \cap \mathbf{F}(G) \leq \mathbf{Z}(\mathbf{F}(G)) \leq \mathbf{Z}(N)$$

and so $\mathbf{K}_{c+2}(N) = 1$. Thus, N is nilpotent and we have $N \leq \mathbf{F}(G)$.

We have shown that $G/\mathbf{F}(G)$ contains no nontrivial abelian normal subgroups lying in $\mathbf{C}_G(\mathbf{F}(G))\mathbf{F}(G)/\mathbf{F}(G)$. Therefore, there also do not exist any nontrivial solvable normal subgroups of $G/\mathbf{F}(G)$ lying in $\mathbf{C}_G(\mathbf{F}(G))\mathbf{F}(G)/\mathbf{F}(G)$.

c) Since $\Phi(G)$ is nilpotent by Theorem 3.6, it follows from Theorem 4.2 a) that $\Phi(G) \leq \mathbf{F}(G)$.

Assume $G \neq 1$ is solvable. Then $G/\Phi(G)$ is also solvable and distinct from 1, so it contains a nonidentity abelian normal subgroup $M/\Phi(G)$. The nilpotency of M follows from the nilpotency of $M/\Phi(G)$ and Theorem 3.5. So $M \leq \mathbf{F}(G)$, and thus certainly $\Phi(G) < \mathbf{F}(G)$.

d) Let $\mathbf{F}(G/\Phi(G)) = M/\Phi(G)$ be the Fitting subgroup of $G/\Phi(G)$. The nilpotency of M follows from the nilpotency of $M/\Phi(G)$ and Theorem 3.5. So $M \leq \mathbf{F}(G)$. The reverse inclusion $M \geq \mathbf{F}(G)$ is trivial since $\mathbf{F}(G)/\Phi(G)$ is a nilpotent normal subgroup of $G/\Phi(G)$.

e) If the minimal normal subgroup N of G is nonabelian, then $N \not\leq \mathbf{F}(G)$, so $N \cap \mathbf{F}(G) = 1$ and thus $\mathbf{F}(G) \leq \mathbf{C}_G(N)$ by Theorem I.3.13 (pg. 18). If N is abelian, then we certainly have $N \leq \mathbf{F}(G)$. So $1 < N \cap \mathbf{Z}(\mathbf{F}(G)) \trianglelefteq G$ by Theorem 2.6. The minimality of N forces $N \leq \mathbf{Z}(\mathbf{F}(G))$. □

The following characterization of the Fitting subgroup is of great importance.

Theorem 4.3. *Let*

$$G = G_0 \rhd G_1 \rhd \cdots \rhd G_s = 1$$

be an arbitrary chief series of G. Then

$$\mathbf{F}(G) = \bigcap_{i=0}^{s-1} \mathbf{C}_G(G_i/G_{i+1}).$$

(*For the definition of* $\mathbf{C}_G(G_i/G_{i+1})$, *see Theorem I.4.3, pg. 19.*)

Proof. We define

$$D = \bigcap_{i=0}^{s-1} \mathbf{C}_G(G_i/G_{i+1})$$

and $D_i = D \cap G_i$. Then

$$[D_i, D] \leq [G_i, \mathbf{C}_G(G_i/G_{i+1})] \cap D \leq G_{i+1} \cap D = D_{i+1}.$$

So the D_i constitute a central series of D from D to 1, and D is nilpotent. Since $D \trianglelefteq G$, Theorem 4.2 a) gives $D \leq \mathbf{F}(G)$.

It remains to show that $\mathbf{F}(G) \leq \mathbf{C}_G(G_i/G_{i+1})$. We have that G_i/G_{i+1} is a minimal normal subgroup of G/G_{i+1}. Furthermore, $\mathbf{F}(G)G_{i+1}/G_{i+1}$ is a nilpotent normal subgroup of G/G_{i+1}. Thus, it follows from Lemma 4.2 a) and e) that

$$\mathbf{F}(G)G_{i+1}/G_{i+1} \leq \mathbf{F}(G/G_{i+1}) \leq \mathbf{C}_{G/G_{i+1}}(G_i/G_{i+1}) = \mathbf{C}_G(G_i/G_{i+1})/G_{i+1}.$$

This gives the desired containment

$$\mathbf{F}(G) \leq \mathbf{C}_G(G_i/G_{i+1}). \qquad \square$$

Theorem 4.4. *Let A be an abelian normal subgroup of G with $A \cap \Phi(G) = 1$. Then there exists a complement of A in G.*

Proof. We choose S to be a minimal element of the nonempty set

$$\{T \leq G \mid G = AT\}.$$

We have $A \cap S \trianglelefteq S$, and $A \cap S \trianglelefteq A$ since $A' = 1$. So we have $A \cap S \trianglelefteq AS = G$.

First assume that $A \cap S \leq \Phi(S)$. By Lemma 3.3 a), it follows that

$$A \cap S \leq A \cap \Phi(G) = 1,$$

and we are done.

Therefore, we may assume that $A \cap S \not\leq \Phi(S)$. Then there exists a maximal subgroup U of S such that $S = (A \cap S)U$. It follows that $G = AS = AU$, contradicting the minimality of S. $\square$

The quotient $\mathbf{F}(G)/\Phi(G)$ can now be described in the following way:

Theorem 4.5 (Gaschütz [155]). *The factor group $\mathbf{F}(G)/\Phi(G) = \mathbf{F}(G/\Phi(G))$ is equal to the direct product of all abelian minimal normal subgroups of $G/\Phi(G)$.*

(By Theorem 4.1, every abelian normal subgroup of $G/\Phi(G)$ is contained in $\mathbf{F}(G)/\Phi(G)$.)

Proof. We have $\mathbf{F}(G/\Phi(G)) = \mathbf{F}(G)/\Phi(G)$ by Theorem 4.2 d). By passing to the quotient $G/\Phi(G)$, we may assume that $\Phi(G) = 1$. By Theorem 3.11 and Lemma 3.3 b), we have

$$\mathbf{F}(G)' \leq \Phi(\mathbf{F}(G)) \leq \Phi(G) = 1.$$

So $\mathbf{F}(G)$ is an abelian normal subgroup of G.

Let T be a largest possible direct product of minimal abelian normal subgroups of G. Assume for contradiction that $T < \mathbf{F}(G)$. Since $T \cap \Phi(G) = 1$, there exists by Theorem 4.4 a subgroup S of G such that $G = TS$ and $T \cap S = 1$. Then

$$\mathbf{F}(G) = \mathbf{F}(G) \cap TS = T(S \cap \mathbf{F}(G)) = T \times (S \cap \mathbf{F}(G)),$$

where $S \cap \mathbf{F}(G) \trianglelefteq TS = G$. If $S \cap \mathbf{F}(G) \neq 1$, then there exists a minimal abelian normal subgroup A of G with $A \leq S \cap \mathbf{F}(G)$. But then $T < A \times T \leq \mathbf{F}(G)$, contradicting the maximal choice of T. Thus, $T = \mathbf{F}(G)$. □

Theorem 4.6. *Let G be a solvable group. We define characteristic subgroups F_i and K_i of G recursively as follows:*
(1) *Let $F_0 = 1$, and let*

$$F_i/F_{i-1} = \mathbf{F}(G/F_{i-1})$$

be the Fitting subgroup of G/F_{i-1}.
(2) *Let $K_0 = G$, and let $K_i = \mathbf{K}_\infty(K_{i-1})$ be the smallest normal subgroup of K_{i-1} with nilpotent factor group.*
If $F_{n-1} < F_n = G$ and $K_{m-1} > K_m = 1$, then $n = m$ and $K_i \leq F_{n-i}$.

Proof. a) First we show that $K_{m-i} \leq F_i$.

Since $\mathbf{K}_\infty(K_{m-1}) = K_m = 1$, the subgroup K_{m-1} is a nilpotent normal subgroup of G. Thus, $K_{m-1} \leq \mathbf{F}(G) = F_1$ by Theorem 4.2.

Now assume for induction that $K_{m-i} \leq F_i$. Since

$$\mathbf{K}_\infty(K_{m-i-1}F_i/F_i) = \mathbf{K}_\infty(K_{m-i-1})F_i/F_i = K_{m-i}F_i/F_i = 1,$$

$K_{m-i-1}F_i/F_i$ is a nilpotent normal subgroup of G/F_i. It follows that

$$K_{m-i-1}F_i/F_i \leq F_{i+1}/F_i,$$

and so $K_{m-i-1} \leq F_{i+1}$. Finally, $K_0 = G \leq F_m$ and thus $n \leq m$.

b) Since F_{i+1}/F_i is nilpotent, it follows immediately by induction on i that $K_i \leq F_{n-i}$. This gives $K_n \leq F_0 = 1$, and therefore $m \leq n$ as well. □

Definition 4.7. The integer n defined in Theorem 4.6 is called the nilpotent length of G, and we write $n = n(G)$. (It is easy to show that if $U \leq G$ then $n(U) \leq n(G)$, and if μ is a homomorphism of G then $n(G^\mu) \leq n(G)$.)

5 Minimal non-nilpotent groups

In many applications, it is useful to be familiar with the non-nilpotent groups whose proper subgroups are nilpotent. In proofs of nilpotency, these groups appear as "minimal counterexamples". The following theorem is often sufficient.

Theorem 5.1 (O. Schmidt [367]). *If every proper subgroup of G is nilpotent, then G is solvable.*

Proof. Let G be a counterexample of smallest possible order. Our hypothesis is clearly inherited by all epimorphic images of G. Assume for contradiction that G has a normal subgroup N such that $1 < N \lhd G$. Then N is a nilpotent subgroup of G, being a proper subgroup, and G/N is solvable by the minimality of $|G|$. But then G is solvable as well by Theorem I.8.6 a) (pg. 40). Since this is not the case, a minimal counterexample G must be simple.

Let U_1 and U_2 be distinct maximal subgroups of G, and let $D = U_1 \cap U_2$. We choose the U_i so that $|D|$ is a large as possible.

Assume for contradiction that $1 < D$. Then the nilpotency of the U_i implies by Theorem 2.3 c) that

$$1 < D < \mathbf{N}_G(D) \cap U_i = V_i.$$

So $D \lhd V_i \leq U_i (i = 1, 2)$. Since G is simple, $\mathbf{N}_G(D) < G$. So there exists a maximal subgroup U_3 of G with $\mathbf{N}_G(D) \leq U_3$. Now we have

$$D < V_1 \leq U_1 \cap U_3,$$

thus $U_1 = U_3$ by the maximal choice of D. In the same way, it follows from

$$D < V_2 \leq U_2 \cap U_3$$

that $U_2 = U_3$. But then $U_1 = U_2$, contradicting the choice of U_1 and U_2 as distinct maximal subgroups of G.

Therefore, any two distinct maximal subgroups of G have intersection 1. Since G is simple, the maximal subgroups of G are their own normalizers in G. Each maximal subgroup U of G then has exactly $[G : U]$ conjugates in G.

Now we count the elements of G. Every nonidentity element lies in exactly one maximal subgroup of G. Letting U_i $(i = 1, \ldots, s)$ be representatives of the conjugacy classes of maximal subgroups of G, we have

$$\begin{aligned} |G| &= 1 + \sum_{i=1}^{s} (|U_i| - 1)[G : U_i] \\ &= 1 + s|G| - \sum_{i=1}^{s} [G : U_i]. \end{aligned}$$

Using $[G : U_i] \leq \frac{|G|}{2}$, we have

$$|G| \geq 1 + s|G| - \frac{s|G|}{2} = 1 + \frac{s|G|}{2},$$

and thus $s = 1$. But then

$$|G| = 1 + |G| - [G : U_1]$$

and $G = U_1$, a contradiction. So G is solvable. □

Compare this proof with Theorem I.8.11 (pg. 42)!

Theorem 5.1 is sufficient in many applications. In some situations, however, more precise knowledge of minimal non-nilpotent groups is needed. The following theorem greatly clarifies the structure of these groups.

Theorem 5.2. *If every proper subgroup of G is nilpotent but G itself is not nilpotent, then G has the following properties:*

a) *$|G| = p^a q^b$ for some prime numbers $p \neq q$; the Sylow p-subgroup is normal in G; the Sylow q-subgroups are cyclic, and for every Sylow q-subgroup Q of G we have $\Phi(Q) \leq \mathbf{Z}(G)$.*

b) *The nilpotency class of the Sylow p-subgroup P is at most two. More precisely, we have $\Phi(P) \leq \mathbf{Z}(G)$.*

c) *If $p > 2$, then P has exponent p; if $p = 2$, then P has exponent at most 4.*

Proof. a) By Theorem 5.1, G is solvable. Assume for contradiction that $|G| = \prod_{i=1}^{r} p_i^{a_i}$ for some $a_i > 0$ and $r \geq 3$. Since G is solvable, it contains a normal subgroup H of prime index, say $|G/H| = p_1$. By our assumption, H is nilpotent. The Sylow subgroups of H are therefore characteristic in H, and so they are normal in G. Let P_i be a Sylow p_i-subgroup of G. For each $i > 1$, P_i is a Sylow p_i-subgroup of H, so it is normal in G. The products $P_1 P_i$ with $i \geq 2$ are always proper subgroups of G since $r \geq 3$, so they are nilpotent. Thus, P_1 commutes with P_i if $i > 1$, and so P_1 is normal in G. But then G is nilpotent, contradicting our hypothesis. We conclude that $r = 2$.

Let $|G| = p^a q^b$. Let $H \lhd G$ and $|G/H| = q$. Since H is nilpotent, the Sylow p-subgroup P of H is a normal subgroup of G. But P is also a Sylow p-subgroup of G.

Let $Q = \langle y_1, \ldots, y_d \rangle$ be a Sylow q-subgroup of G with d minimal. If $d > 1$, then $P\langle y_i \rangle$ is a proper subgroup of G for $i = 1, \ldots, d$, so it is nilpotent. Thus, y_i commutes with all elements of P. Since this holds for all i, P and Q commute. It follows that $G = PQ = P \times Q$, and G is nilpotent. This contradicts our assumption on G, so $d = 1$ and Q is cyclic.

The group $P\Phi(Q)$ is a proper subgroup of G, so it is nilpotent. Therefore, $\Phi(Q)$ commutes with P, and $\Phi(Q)$ lies in $\mathbf{Z}(G)$.

b) Assume for contradiction that there exists a normal subgroup N of G such that $|G/N| = p^f > 1$. Then N is nilpotent, and we obtain $N = Q \times (N \cap P)$. Hence, Q is a characteristic subgroup of N, so Q is normal in G and G is nilpotent. But this is not the case. Thus, there is no normal subgroup N of G such that $|G/N| = p^f > 1$. In particular, we have

$$Q^G = \langle Q^g \mid g \in G \rangle = G.$$

The group $\Phi(P)Q$ is a proper subgroup of G, so it is nilpotent. Hence, $\Phi(P)$ commutes with Q. The same conclusion holds for all conjugates Q^g of Q. Since the Q^g generate all of G, we obtain $\Phi(P) \leq \mathbf{Z}(G)$. In particular, using Theorem 3.11 we have

$$P' \leq \Phi(P) \leq P \cap \mathbf{Z}(G) \leq \mathbf{Z}(P),$$

and hence P has class at most 2.

c) By Lemma 1.6 b), we have $[P, Q] \lhd \langle P, Q \rangle = G$.

Assume for contradiction that $[P, Q] < P$. Then $[P, Q]Q$ is a proper subgroup of G, so it is nilpotent. Therefore, $[P, Q]$ commutes with Q. It follows that

$$Q^g \leq \mathbf{C}_G([P, Q]^g) = \mathbf{C}_G([P, Q]),$$

and then $Q^G = G$ gives that $[P, Q] \leq \mathbf{Z}(G)$. But then $G/\mathbf{Z}(G)$ is nilpotent, so G is nilpotent as well, contradicting our hypothesis.

Therefore, we have

$$P = [P, Q] = \langle x^{y-1} \mid x \in P, y \in Q \rangle.$$

If $x \in P$, then $x^p \in \Phi(P) \leq \mathbf{Z}(G)$ by Theorem 3.14. So $(x^p)^y = x^p$ for all $y \in Q$. Since P has class at most 2 by b), it follows from Lemma 1.3 that

$$[x, x']^p = [x^p, x'] = 1 \quad \text{for all} \quad x, x' \in P.$$

In particular,

$$1 = [x^{-p}, x^y] = [x^{-1}, x^y]^p.$$

For $p > 2$, the divisibility of $\binom{p}{2}$ by p and Lemma 1.3 yield

$$(x^{y-1})^p = (x^y)^p x^{-p} [x^{-1}, x^y]^{\binom{p}{2}} = 1.$$

Therefore, the generators x^{y-1} of P all have order p. Since

$$(xx')^p = x^p (x')^p [x', x]^{\binom{p}{2}} = x^p (x')^p,$$

all the nonidentity elements of P have order p. So P has exponent p.

For $p = 2$, we must proceed more carefully. As above, we first obtain that $x^2 \in \mathbf{Z}(G)$ and $[x, x']^2 = 1$ for all $x, x' \in P$. This time, using Lemma 1.3 it follows that

$$(x^{y-1})^4 = (x^y)^4 x^{-4} [x^{-1}, x^y]^6 = 1.$$

Therefore, the generators x^{y-1} of P have order at most 4. It follows from

$$(xx')^4 = x^4(x')^4[x', x]^6 = x^4(x')^4$$

that the elements of P have order at most 4. □

As a first application of Theorem 5.2, we prove:

Theorem 5.3 (Itô [276]). *Assume the group G has odd order.*
a) *If every minimal subgroup of G lies in $\mathbf{Z}(G)$, then G is nilpotent.*
b) *If every minimal subgroup of G' is normal in G, then the commutator subgroup G' is nilpotent and G is solvable.*
(*The statement in b) is evidently an attempt to dualize the characterization of nilpotent groups in Theorem 2.3 d).*)

Proof. a) Let G be a minimal counterexample. The assumptions are clearly inherited by subgroups of G. Thus, all proper subgroups of G are nilpotent, but G itself is not. By Theorem 5.2, G then has the following structure: $G = PQ$ with $P \lhd G$, and P has exponent p. If $1 \neq x \in P$, then $\langle x \rangle$ is a minimal subgroup of G, so it lies in $\mathbf{Z}(G)$. Therefore, P lies in $\mathbf{Z}(G)$, so we have $G = P \times Q$ and G is nilpotent.

b) Let U be a minimal subgroup of G'. Then U has prime order, and the factor group $G/\mathbf{C}_G(U)$ is isomorphic to a subgroup of the automorphism group of the cyclic group U by Theorem I.4.5 (pg. 21), so it is abelian. Thus, we obtain $G' \leq \mathbf{C}_G(U)$. Hence, all minimal subgroups of G' lie in $\mathbf{Z}(G')$, and we may apply a) to G'. This shows that G' is nilpotent, and so G' is solvable. Therefore, G is solvable as well.□

We will see a useful generalization of Theorem 5.1 in Theorem IV.5.4 (pg. 462). There we will prove a statement, also for groups of even order, corresponding to Theorem 5.3.

Our next objective is a generalization of Sylow's theorems due to Wielandt.

Definition 5.4. Let π be a set of prime numbers. A subgroup U of G is called a π-subgroup of G if every prime divisor of $|U|$ lies in π. A subgroup H of G is called a Hall π-subgroup of G when the following hold:
(1) Every prime divisor of $|H|$ lies in π.
(2) No prime divisor of $[G : H]$ lies in π.
If H is a Hall π-subgroup of G for some set π of primes, then we call H a Hall subgroup of G.

Definition 5.5. Let π be a set of primes. We say that the π-Sylow theorem holds in G whenever the following condition is satisfied: If U is a π-subgroup of G and V is a maximal π-subgroup of G (that is, a π-subgroup of G that is not properly contained in any π-subgroup of G), then there exists some $g \in G$ such that $U^g \leq V$. It clearly suffices to verify that this condition holds for a *single* maximal π-subgroup V of G.

Theorem 5.6. *If the π-Sylow theorem holds in G, then every maximal π-subgroup of G is a Hall π-subgroup of G, all Hall π-subgroups of G are conjugate, and every π-subgroup of G lies in a Hall π-subgroup of G.*

Proof. Let p be a prime in π, let P be a Sylow p-subgroup of G and let V be a maximal π-subgroup of G. Since the π-Sylow theorem holds, there exists a $g \in G$ such that $P^g \leq V$. Thus $p \nmid [G : V]$, and V is a Hall π-subgroup of G. The rest of the theorem follows trivially. □

The content of the Sylow theorems (Theorems I.7.3 and I.7.5, pg. 34) is that the $\{p\}$-Sylow theorem holds in every group. By no means does the π-Sylow theorem hold for all sets of primes π in all groups. We will show in Chapter VI that the π-Sylow theorem holds in G for all sets of primes π if and only if G is solvable. Next, we give a few examples of groups in which the π-Sylow theorem does not hold:

Example 5.7.

a) The alternating group A_5 has no Hall π-subgroups for the sets of primes $\pi = \{2, 5\}$ and $\pi = \{3, 5\}$; otherwise A_5 would have a subgroup of index 3 or 4 and then would not be simple by Theorem I.6.6 (pg. 30).

b) The simple group $\mathrm{PSL}_2(11)$ has the following nonisomorphic Hall π-subgroups for $\pi = \{2, 3\}$:
   (1) the alternating group A_4 (Theorem II.8.18 b)),
   (2) the dihedral group of order 12 (Theorem II.8.4).

c) The simple group $\mathrm{PSL}_2(59)$ has the following nonisomorphic Hall π-subgroups for $\pi = \{2, 3, 5\}$:
   (1) the alternating group A_5 (Theorem II.8.13),
   (2) the dihedral group of order 60 (Theorem II.8.4).

d) The simple group $\mathrm{PSL}_2(7)$ has, for $\pi = \{2, 3\}$, two conjugacy classes of Hall π-subgroups isomorphic to the symmetric group S_4 (Theorem II.8.18 a)).

There are two ways to search for sufficient conditions for the π-Sylow theorem:

(1) One imposes some conditions on the entire group G. (This is what we will do in Chapter VI, where we will demand that G is solvable.)

(2) One imposes some conditions on the π-subgroups of G.

The following result of the second type has found numerous applications.

Theorem 5.8 (Wielandt [465]). *If G has a nilpotent Hall π-subgroup, then the π-Sylow theorem holds in G.*

Proof. Let H be a nilpotent Hall π-subgroup, and let M be a π-subgroup of G. We prove by induction on $|M|$ that there exists some $g \in G$ such that $M^g \leq H$.

For every proper subgroup U of M, there exists some $g \in G$ (depending on U) such that $U^g \leq H$ by the induction hypothesis. So every proper subgroup of M is nilpotent.

If M is not nilpotent, then M has a normal Sylow p-subgroup P by Theorem 5.2, and $|M/P|$ is a power of a prime q with $q \nmid |P|$.

Suppose, on the other hand, that M is nilpotent, and let q be a prime divisor of $|M|$. Then M decomposes as $M = P \times Q$, where Q is a Sylow q-subgroup of M. Again $q \nmid |P|$ and $|M/P|$ is a power of q.

In either case, the Sylow q-subgroup Q of M has a normal complement P in M.

Since H is nilpotent, there is a decomposition $H = H_1 \times H_2$ where $|H_1| = q^b$ and $q \nmid |H_2|$. By the induction hypothesis, there exists an $x \in G$ such that $P^x \leq H$; so $P^x \leq H_2$. Since $P \lhd M$ and $P^x \leq H_2$, we have

$$\mathbf{N}_G(P^x) \geq \langle M^x, H_1 \rangle.$$

Let Q be a Sylow q-subgroup of M. Since H_1 is a Sylow q-subgroup of $\mathbf{N}_G(P^x)$, there exists some $y \in \mathbf{N}_G(P^x)$ such that $Q^{xy} \leq H_1$ by Theorem I.7.5 (pg. 35). Altogether, we have

$$M^{xy} = Q^{xy} P^{xy} = Q^{xy} P^x \leq H_1 \times H_2 = H,$$

as claimed. □

Exercises

13) (Pazderski [345]) Let $n = \prod_{i=1}^{r} p_i^{a_i}$.
   a) Every group of order n is nilpotent if and only if for every $i \neq j$ we have
   $$p_i \nmid (p_j^{a_j} - 1)(p_j^{a_j - 1} - 1) \cdots (p_j - 1).$$
   b) Every group of order n is abelian (resp. cyclic) if and only if the condition in a) holds and $a_i \leq 2$ (resp. $a_i = 1$) for all i.
14) (Miller, Moreno [331], Rédei [348]) Suppose G is a nonabelian group whose proper subgroups are abelian.
   a) If $|G|$ is a prime power, then G' has prime order and G has class 2.
   b) If $|G|$ is not a prime power, then $G = PQ$, where P is an elementary abelian minimal normal subgroup of G and Q is a (non-normal) cyclic Sylow subgroup of G.
15) (MacDonald [325]) Let G be a nonabelian group. Then there always exists a pair of conjugate elements g and g^x in G such that $g \neq g^x$ and $[g, g^x] = 1$.

6 Engel groups and Engel elements

Theorem 6.1. *Let $w = [f_{i_1}, f_{i_2}, \ldots, f_{i_n}]$ be a left-normed commutator in the free group F in the free generators $f_1, \ldots, f_k$. Assume $i_1 \neq i_2$, and let all i_j be taken from the set $\{1, \ldots, k\}$. If $[g_{i_1}, g_{i_2}, \ldots, g_{i_n}] = 1$ for all g_i in the finite group G, then G is nilpotent.*

(The condition $[g_1, g_2, \ldots, g_k] = 1$ for all $g_i \in G$ is equivalent to $\mathbf{K}_k(G) = 1$ by Definition 1.9, forcing $c(G) \leq k-1$. Theorem 6.1 says that, for example, the condition $[g_1, g_2, g_2] = 1$ or the condition $[g_1, g_2, g_1, g_2] = 1$ forces G to be nilpotent.)

Proof. Let G be a minimal counterexample. Since the assumption is inherited by all subgroups and all factor groups of G, every proper subgroup of G is nilpotent and every proper factor group of G is nilpotent. Since G, however, is not itself nilpotent, Theorem 5.2 shows that G has the form $G = PQ$, where P is a normal Sylow p-subgroup of G and Q is a cyclic Sylow q-subgroup of G. If $\Phi(P) \neq 1$, then $G/\Phi(P)$ is nilpotent. Since $P \lhd G$, $\Phi(P) \leq \Phi(G)$ by Lemma 3.3 b). But then the nilpotency of G follows from the nilpotency of $G/\Phi(P)$ with the help of Theorem 3.5, contradicting our assumption.

Therefore, $\Phi(P) = 1$. By Theorem 3.14, P is then an elementary abelian p-group. Let x be an arbitrary element of P, and let y be an arbitrary element of Q. We set $g_j = yx$ for $j = i_1$ and $g_j = y$ for all remaining j. Now we analyze the condition

$$[g_{i_1}, g_{i_2}, \ldots, g_{i_n}] = 1.$$

We have

$$[g_{i_1}, g_{i_2}] = [yx, y] = x^{-1}y^{-1}y^{-1}yxy = x^{-1+y} = x^{y-1},$$

where the last equality follows from the commutativity of P. Assume for induction that

$$[g_{i_1}, g_{i_2}, \ldots, g_{i_j}] = x^{(y-1)^{j-1}}.$$

Then

$$[g_{i_1}, g_{i_2}, \ldots, g_{i_{j+1}}] = \begin{cases} [x^{(y-1)^{j-1}}, yx] & \text{for } g_{i_{j+1}} = yx \\ [x^{(y-1)^{j-1}}, y] & \text{for } g_{i_{j+1}} = y. \end{cases}$$

By the commutativity of P, we have in each case that

$$[g_{i_1}, g_{i_2}, \ldots, g_{i_{j+1}}] = [x^{(y-1)^{j-1}}, y] = x^{-(y-1)^{j-1}}y^{-1}x^{(y-1)^{j-1}}y = x^{(y-1)^j}.$$

Ultimately, we arrive at

$$[g_{i_1}, g_{i_2}, \ldots, g_{i_n}] = x^{(y-1)^{n-1}} = 1 \quad \text{for all} \quad x \in P.$$

If μ is the automorphism of P defined by $x^\mu = x^y$ and $p^m \geq n-1$, then $(\mu-1)^{p^m} = 0$ in the endomorphism ring of P. Since P has exponent p, it follows that

$$0 = (\mu - 1)^{p^m} = \sum_{i=0}^{p^m} \binom{p^m}{i} (-1)^i \mu^{p^m - i} = \mu^{p^m} + (-1)^{p^m} = \mu^{p^m} - 1.$$

So we have

$$x^{\mu^{p^m}} = x^{y^{p^m}} = x.$$

Since the order of y is coprime to p, this forces $y \in \mathbf{C}_G(P)$. Thus, P and Q commute, and G is nilpotent. □

Definition 6.2. The condition

$$[x, y, \underset{n}{\ldots}, y] = 1$$

is called the n-Engel condition. Groups in which every pair of elements satisfies an Engel condition are called Engel groups. (This name stems from the theory of Lie algebras, where Engel gave a corresponding condition which plays an important role.)

We formulate the following special case of Theorem 6.1 as a separate theorem:

Theorem 6.3 (Zorn). *If the n-Engel condition*

$$[x, y, \underset{n}{\ldots}, y] = 1$$

holds for every pair of elements x, y in the finite group G, then G is nilpotent.

It is natural to ask to what extent the nilpotency class of a group satisfying the n-Engel condition can be estimated by n. Little is known about this problem.[2] Here we will consider more closely only the 2-Engel condition $[x, y, y] = 1$. Our statements will hold for infinite groups as well since we will not use Theorem 6.1 or Theorem 6.3.

Lemma 6.4. *The following statements are equivalent:*
a) *We have $[x, y, y] = 1$ for all $x, y \in G$.*
b) *Every two conjugate elements of G commute with each other.*

Proof. By Lemma 1.2 c),

$$[x, y, y] = [[x, y], y] = [y^{-x}y, y] = [y^{-x}, y]^y.$$

So $[x, y, y] = 1$ is equivalent to $[y^{-x}, y] = 1$, which holds if and only if y^x and y commute. □

The following theorem gives precise information about the groups satisfying the 2-Engel condition.

[2] It is still a difficult open problem whether a finitely-generated n-Engel group is nilpotent. (Although it has been claimed that a counterexample exists, nothing has yet been published.) However, it is known that even if an n-Engel group is locally nilpotent, it does not have to be nilpotent. For more information about n-Engel groups and (local) nilpotency, see R. G. Burns and Y. Medvedev, A note on Engel groups and local nilpotence, J. Austral. Math. Soc. Ser. A **64** (1998), no. 1, 92–100; M. I. Golovanov, On the degree of nilpotency of 4-Engel Lie rings, Algebra i Logika **25** (1986), no. 5, 508–532, 613; H. Heineken, Engelsche Elemente der Länge drei, Illinois J. Math. **5** (1961), 681–707; G. Traustason, On 4-Engel groups, J. Algebra **178** (1995), no. 2, 414–429. *trans.*

Theorem 6.5 (F. W. Levi [317]). *If $[x, y, y] = 1$ for all $x, y \in G$, then G is nilpotent and has class at most 3. If G has no elements of order 3, then G has class at most 2.*

Proof. For a fixed $y \in G$, all elements y^x commute with y by Lemma 6.4. Thus,

$$y^G = \langle y^g \mid g \in G \rangle = A(y)$$

is an abelian normal subgroup of G. We abbreviate the left-normed commutators as follows:

$$[x_1, \ldots, x_n] = x_{1,\ldots,n}.$$

We show:

(1) $x_{\ldots,i,\ldots,i,\ldots} = 1$.

We have

$$x_{\ldots,i,\ldots,i,\ldots} = [a, x_i, b_1, \ldots, b_s, x_i, c_1, \ldots, c_t]$$

for some $a, b_k, c_j \in G$. Here,

$$[a, x_i] \in A(x_i) \trianglelefteq G,$$

and thus

$$[a, x_i, b_1, \ldots, b_s] \in A(x_i)$$

as well. Since $A(x_i)$ is abelian, it follows that

$$[a, x_i, b_1, \ldots, b_s, x_i] = 1,$$

which proves the claim.

(2) We have $x_{1,3,2} = x_{1,2,3}^{-1}$:

By Lemma 1.2 b), $[x_1, x_2x_3] = x_{1,3}\, x_{1,2}\, x_{1,2,3}$. Thus, it follows with the help of (1) that

$$\begin{aligned}[x_1, x_2x_3]^{x_2} &= x_{1,3}^{x_2}\, x_{1,2}^{x_2}\, x_{1,2,3}^{x_2} = x_{1,3}^{x_2}\, x_{1,2}\, x_{1,2,3} \\ &= x_{1,3}\, x_{1,3,2}\, x_{1,2}\, x_{1,2,3}.\end{aligned}$$

In the same way, it follows that

$$[x_1, x_2x_3]^{x_2x_3} = x_{1,3}\, x_{1,3,2}\, x_{1,2}^{x_3}\, x_{1,2,3} = x_{1,3}\, x_{1,3,2}\, x_{1,2}\, x_{1,2,3}^2$$

and by (1),

$$\begin{aligned}[x_1, x_2x_3]^{x_2x_3} &= [x_1, x_2x_3] \\ &= x_{1,3}\, x_{1,2}\, x_{1,2,3}.\end{aligned}$$

All commutators here contain x_1, so they lie in $A(x_1)$ and commute with each other. Thus, there remains only

$$x_{1,3,2}x_{1,2,3} = 1,$$

which is what we wanted to show.

(3) For all $x, y \in G$, we have $[x^{-1}, y] = [x, y^{-1}] = [x, y]^{-1}$:
By (1), we have

$$1 = [xx^{-1}, y] = [x, y]^{x^{-1}}[x^{-1}, y] = [x, y][x^{-1}, y]$$

and then

$$[x, y^{-1}] = [y^{-1}, x]^{-1} = [y, x] = [x, y]^{-1}.$$

(4) We have $x_{1,2,3} = x_{2,1,3}^{-1}$:
We have

$$\begin{aligned} x_{1,2,3} = [x_{1,2}, x_3] = [x_{2,1}^{-1}, x_3] = [x_{2,1}, x_3]^{-1} \quad &\text{(by (3))} \\ = x_{2,1,3}^{-1}. \end{aligned}$$

(5) We have $x_{1,2,3}^3 = 1$:
We have

$$[x, y^{-1}, z]^y = [x, y^{-1}, z],$$

since $[x, y^{-1}, z]$ lies in the normal abelian subgroup $A(y)$, so it commutes with y. Furthermore,

$$\begin{aligned} [x, y^{-1}, z]^y = [[x, y^{-1}], z] = [[x, y]^{-1}, z] \quad &\text{(by (3))} \\ = [x, y, z]^{-1} \quad &\text{(again by (3)).} \end{aligned}$$

Now, Witt's identity from Theorem 1.4 gives that

$$x_{1,2,3}^{-1}\, x_{2,3,1}^{-1}\, x_{3,1,2}^{-1} = 1.$$

But by (2) and (4), the elements $x_{i,j,k}$ are inverted under transposition of indices, so they are invariant under even permutations of the indices. We describe this situation by saying that the $x_{i,j,k}$ are "alternating in the indices". This yields

$$x_{1,2,3}^3 = 1.$$

In particular, if G has no elements of order 3, then all 3-fold left-normed commutators are the identity, so G has class at most 2.

(6) We have $x_{1,2,3,4} = x_{1,2,4,3}^{-1}$:

This follows from (2) upon substituting $x_{1,2}$ in place of x_1, x_3 in place of x_2, and x_4 in place of x_3.

(7) $x_{2,1,3,4} = x_{1,2,3,4}^{-1}$:
By (4), we have

$$\begin{aligned} x_{2,1,3,4} &= [x_{2,1,3}, x_4] = [x_{1,2,3}^{-1}, x_4] = [x_{1,2,3}, x_4]^{-1} \quad \text{(by (3))} \\ &= x_{1,2,3,4}^{-1}. \end{aligned}$$

(8) $x_{1,3,2,4} = x_{1,2,3,4}^{-1}$:
By (2), we have

$$\begin{aligned} x_{1,3,2,4} &= [x_{1,3,2}, x_4] = [x_{1,2,3}^{-1}, x_4] = [x_{1,2,3}, x_4]^{-1} \quad \text{(by (3))} \\ &= x_{1,2,3,4}^{-1}. \end{aligned}$$

Since the transpositions $(1,2)$, $(2,3)$ and $(3,4)$ generate the entire symmetric group S_4, the element $x_{1,2,3,4}$ is alternating in the indices.

(9) $x_{1,2,3,4}^2 = 1$:
By (2) and (4),

$$\begin{aligned} x_{1,2,3,4} &= [x_{1,2}, x_3, x_4] = [x_3, x_4, x_{1,2}] = [x_{3,4}, x_{1,2}] = [x_{1,2}, x_{3,4}]^{-1} \\ &= x_{3,4,1,2}^{-1} = x_{1,2,3,4}^{-1}. \end{aligned}$$

(10) $x_{1,2,3,4}^3 = 1$:
This follows from (5) by replacing x_1, x_2, x_3 with $x_{1,2}, x_3, x_4$.

(11) From (9) and (10), we obtain that $x_{1,2,3,4} = 1$. So all left-normed commutators of length 4 are equal to the identity, and G has class at most 3. □

As a first application of Theorem 6.5, we deduce a statement about (not necessarily finite) groups of exponent 3; namely, we give the solution to the Burnside Problem for exponent 3:

Theorem 6.6 (F.W. Levi, v. d. Waerden [318]). *Let G be a group of exponent 3. Then:*

a) $[x, y, y] = 1$ *for all* $x, y \in G$.

b) *If G can be generated by d elements, then G is finite and the order of G is a divisor of 3^k, where $k = d + \binom{d}{2} + \binom{d}{3}$.*

Proof. a) For all $x, y \in G$, we have

$$1 = (xy)^3 = xyxyxy$$

and therefore

$$xyx = y^{-1}x^{-1}y^{-1}.$$

It follows that

$$\begin{aligned} x^{-1}yxyx^{-1} &= x^{-1}yx^{-1}x^{-1}yx^{-1} \quad (\text{since } x = x^{-2}) \\ &= y^{-1}xy^{-1}y^{-1}xy^{-1} = y^{-1}xyxy^{-1} = y^{-1}y^{-1}x^{-1}y^{-1}y^{-1} \\ &= yx^{-1}y, \end{aligned}$$

and so $yxyx^{-1} = xyx^{-1}y$.

Thus, y and xyx^{-1} commute. So the conditions of Lemma 6.4 and Theorem 6.5 are satisfied, and we conclude that $[x, y, y] = 1$ and $\mathbf{K}_4(G) = 1$.

b) Let $G = \langle x_1, \ldots, x_d \rangle$. Then G/G' is an abelian group of exponent 3 that can be generated by d elements. Therefore, $|G/G'| \leq 3^d$. The factor group $G'/\mathbf{K}_3(G)$ is generated by the cosets $[x_i, x_j]\mathbf{K}_3(G)$ by Theorem 1.11 b). Since $[x_i, x_j] = [x_j, x_i]^{-1}$, it suffices to take as generators the pairs (i, j) with $i < j$. This yields $\binom{d}{2}$ generators for the abelian group $G'/\mathbf{K}_3(G)$, and it follows that

$$|G'/\mathbf{K}_3(G)| \leq 3^{\binom{d}{2}}.$$

By Theorem 1.11 b), $\mathbf{K}_3(G)/\mathbf{K}_4(G) = \mathbf{K}_3(G)$ is generated by the $[x_i, x_j, x_k]$. Since these commutators are alternating in the indices by Theorem 6.5, it suffices to take the $\binom{d}{3}$ generators $[x_i, x_j, x_k]$ with $i < j < k$, and we have

$$|\mathbf{K}_3(G)| \leq 3^{\binom{d}{3}}.$$

We now know that G is finite and has exponent 3. So its order is a power of 3, yielding the claim. □

We mention without proof that the bound for the order of G in Theorem 6.6 is sharp (see Exercise 16).

Remark 6.7. W. Burnside posed the following question in 1902:

The Burnside problem: Let G be a group of exponent m that can be generated by d elements. Is G necessarily finite?

If the answer to this question is yes, then in particular the quotient

$$B(m, d) = F/\langle f^m \mid f \in F \rangle$$

of the free group $F = \langle f_1, \ldots, f_d \rangle$ in the free generators f_i is finite, and every group G of exponent m generated by d elements is a homomorphic image of $B(m, d)$. So

$$|G| \leq |B(m, d)|.$$

The answer to the Burnside problem is known in the following cases:

(1) It is easy to see that $B(2, d)$ of exponent 2 is abelian, so

$$|B(2, d)| = 2^d.$$

(2) $|B(3, d)| = 3^{k(d)}$, where $k(d) = d + \binom{d}{2} + \binom{d}{3}$; we showed that $|B(3, d)| \leq 3^{k(d)}$ in Theorem 6.6.

(3) $|B(4,d)| < \infty$ (Sanov; see M. Hall [204], pg. 324).
The order of $B(4,d)$ is known only for $d = 2$, namely $|B(4,2)| = 2^{12}$ (see Exercise 20).
(4) $|B(6,d)| = 3^{a(d)}2^{b(d)}$, where

$$a(d) = k + \binom{k}{2} + \binom{k}{3}, \quad k = 1 + 2^d(d-1)$$

and

$$b(d) = 1 + 3^j(d-1), \quad j = d + \binom{d}{2} + \binom{d}{3}$$

(M. Hall [203], P. Hall, G. Higman [224]).[3]

We will return to the groups of exponent 6 in Theorem VI.6.7 when we discuss the p-length of solvable groups.

Nothing is known about $B(5,2)$!

The following is a closely related question:

The restricted Burnside problem: Does there exist a function $b(m,d)$ such that for every finite group G of exponent m that can be generated by d elements, we have that $|G| \leq b(m,d)$? If $B(m,d)$ is finite, then of course $|G| \leq |B(m,d)|$ and we can take $b(m,d) = |B(m,d)|$; this is clearly the best possible bound. While the answer to the Burnside problem is only known in the cases listed above, Kostrikin was able to show using considerations from the theory of Lie rings that for every prime p there exists a function $b(p,d)$ such that every finite group of exponent p generated by d elements has order at most $b(p,d)$. (See Kostrikin [311]; the Zassenhaus identity of Theorem 9.7 plays a central role in these studies.)

In preparation for a further application of Theorem 6.5, we prove the following easy lemma.

Lemma 6.8. *Let $k \geq 2$ and $g_i \in \mathbf{K}_{n_i}(G)$ $(i = 1, \ldots, k)$. Then we have*

$$[g_1^{a_1}, \ldots, g_k^{a_k}] = [g_1, \ldots, g_k]^{a_1 \cdots a_k} t$$

for some $t \in \mathbf{K}_{n+1}(G)$ with $n = \sum_{i=1}^{k} n_i$.

Proof. For $k = 2$, an application of Lemma 1.3 a) to $G/\mathbf{K}_{n+1}(G)$ yields the claim immediately since $[g_1, g_2, g_1] \in \mathbf{K}_{n+1}(G)$ and $[g_1, g_2, g_2] \in \mathbf{K}_{n+1}(G)$. Now let $k > 2$, and assume for induction that

[3] The only values of m for which it is known that $B(m,d)$ is finite (for all d) are still $m = 1, 2, 3, 4, 6$. On the other hand, Adian proved that $B(m,d)$ is infinite if $d > 1$ and $m \geq 665$ is odd. For a textbook reference, see M. Vaughan-Lee, *The restricted Burnside problem*, London Mathematical Society Monographs. New Series, 5, The Clarendon Press, Oxford University Press, New York, 1990. There are also results showing that $B(m,d)$ is infinite for $d > 1$ and sufficiently large even exponent m; see S. V. Ivanov, The free Burnside groups of sufficiently large exponents, Internat. J. Algebra Comput. **4** (1994), 1–308 and I. G. Lysenok, Infinite Burnside groups of even exponent, Izv. Math. **60** (1996), 453–654. *trans.*

$$[g_1^{a_1}, \dots, g_{k-1}^{a_{k-1}}] = [g_1, \dots, g_{k-1}]^b h,$$

where $b = a_1 \cdots a_{k-1}$ and $h \in \mathbf{K}_{n-n_k+1}(G)$. By Lemma 1.2 c), it follows that

$$[g_1^{a_1}, \dots, g_k^{a_k}] = [[g_1, \dots, g_{k-1}]^b, g_k^{a_k}][[g_1, \dots, g_{k-1}]^b, g_k^{a_k}, h][h, g_k^{a_k}].$$

By Theorem 2.11 b),

$$[h, g_k^{a_k}] \in [\mathbf{K}_{n-n_k+1}(G), \mathbf{K}_{n_k}(G)] \leq \mathbf{K}_{n+1}(G)$$

and

$$[[g_1, \dots, g_{k-1}]^b, g_k^{a_k}, h] \in [\mathbf{K}_{n-n_k}(G), \mathbf{K}_{n_k}(G), G] \leq \mathbf{K}_{n+1}(G).$$

Since

$$[[g_1, \dots, g_{k-1}], g_k] \in \mathbf{K}_n(G)$$

and $\mathbf{K}_n(G)/\mathbf{K}_{n+1}(G) \leq \mathbf{Z}(G/\mathbf{K}_{n+1}(G))$, applying Lemma 1.3 a) to $G/\mathbf{K}_{n+1}(G)$ yields

$$[[g_1, \dots, g_{k-1}]^b, g_k^{a_k}] = [g_1, \dots, g_k]^{ba_k} r$$

for some $r \in \mathbf{K}_{n+1}(G)$. □

Theorem 6.9 (Heineken [228]). *Suppose that for all x, y, z in the (not necessarily finite) group G we have*

(1) $[x, y, y, z] = 1$ *and* (2) $[x, y, z, z] = 1$.

Then G is nilpotent and has class at most 3.

Proof. In what follows, let x, y, z, u, v be arbitrary elements of G. Condition (1) implies $[x, y, y] \in \mathbf{Z}(G)$. So the 2-Engel condition holds in $G/\mathbf{Z}(G)$. Therefore, $c(G/\mathbf{Z}(G)) \leq 3$ by Theorem 6.5, so $c(G) \leq 4$ and (see step (5) in the proof of Theorem 6.5)

$$[x, y, z]^3 \in \mathbf{Z}(G).$$

Using $\mathbf{K}_5(G) = 1$ and Lemma 6.8, we have

(3) $1 = [[x, y, z]^3, u] = [x, y, z, u]^3$.

From (1) with uv in place of y and y in place of z, we obtain $1 = [x, uv, uv, y]$. We rearrange this expression with the help of Lemma 1.2 b) and c) and keep in mind that every commutator with at least four arguments (and arbitrary bracketing) lies in $\mathbf{K}_4(G) \leq \mathbf{Z}(G)$ by Theorem 2.11 b). This yields

$$\begin{aligned}
1 &= [[x,v][x,u]\cdot[x,u,v],uv,y] \\
&= [[[x,v][x,u],uv]\cdot[[x,v][x,u],uv,[x,u,v]]\cdot[x,u,v,uv],y] \\
&= [[x,v][x,u],uv,y] = [[x,v,uv][x,u,uv],y] \\
&= [[x,v,v][x,v,u][x,u,v][x,u,u],y] \\
&= [x,v,v,y][x,v,u,y][x,u,v,y][x,u,u,y].
\end{aligned}$$

By (1), the first and fourth factors are 1. There remains

(4) $[x,v,u,y] = [x,u,v,y]^{-1}$.

We proceed in a similar way with (2):

$$\begin{aligned}
1 &= [x,y,uv,uv] = [[x,y,v][x,y,u][x,y,u,v],uv] \\
&= [x,y,v,uv][x,y,u,uv] \\
&= [x,y,v,v][x,y,v,u][x,y,u,v][x,y,u,u].
\end{aligned}$$

Therefore, we have

(5) $[x,y,v,u] = [x,y,u,v]^{-1}$.

Furthermore, since $\mathbf{K}_5(G) = 1$ we have

$$\begin{aligned}
1 &= [[x,y][y,x],u,v] = [[x,y,u][y,x,u],v] \\
&= [x,y,u,v][y,x,u,v],
\end{aligned}$$

and so

(6) $[x,y,u,v] = [y,x,u,v]^{-1}$.

The relations (4)–(6) show that the commutators $[x,y,u,v]$ are alternating under permutation of the arguments.

In Witt's identity (see Theorem 1.4)

$$[x,y^{-1},z]^y[y,z^{-1},x]^z[z,x^{-1},y]^x = 1,$$

we set $z = [u,v]$. Then all commutators appearing in this expression lie in $\mathbf{K}_4(G) \leq \mathbf{Z}(G)$ by Theorem 2.11 b). Replacing y by y^{-1}, we have

$$[[x,y],[u,v]]\,[y^{-1},[u,v]^{-1},x]\,[[u,v],x^{-1},y^{-1}] = 1.$$

With the help of Lemma 6.8, it follows that

$$\begin{aligned}
[[x,y],[u,v]] &= [[u,v],x^{-1},y^{-1}]^{-1}[y^{-1},[u,v]^{-1},x]^{-1} \\
&= [u,v,x,y]^{-1}[u,v,y,x] \\
&= [u,v,x,y]^{-2}.
\end{aligned}$$

So by virtue of the alternating property of $[x,y,u,v]$,

$$\begin{aligned} 1 &= [[x,y],[u,v]]\,[[u,v],[x,y]] = [u,v,x,y]^{-2}[x,y,u,v]^{-2} \\ &= [x,y,u,v]^{-2}[x,y,u,v]^{-2} \\ &= [x,y,u,v]^{-4}. \end{aligned}$$

This relation, together with (3), yields the desired statement $[x,y,u,v] = 1$. So G has class at most 3. □

The question of whether a group G has class 1 (that is, whether G is abelian) can already be decided by examining the subgroups of G that are generated by two elements. A nontrivial analog of this fact is:

Theorem 6.10 (Heineken [228]). *Let $n > 2$. If every subgroup of G that is generated by n elements has class at most n, then G itself has class at most n.*

(*For $n = 2$, the conclusion of Theorem 6.10 no longer holds; this can be illustrated by suitable groups of exponent 3. Using Theorem 6.5 in this case, we obtain only $c(G) \leq 3$.*)

Proof. For all elements x, y, z, a_i $(i = 1, \ldots, n-3)$ in G we have

$$[x,y,y,z,a_1,\ldots,a_{n-3}] = [x,y,z,z,a_1,\ldots,a_{n-3}] = 1.$$

This implies $[x,y,y,z] \in \mathbf{Z}_{n-3}(G)$ and $[x,y,z,z] \in \mathbf{Z}_{n-3}(G)$. Then by Theorem 6.9, we have $c(G/\mathbf{Z}_{n-3}(G)) \leq 3$, and so $c(G) \leq n$. □

Remark 6.11. Heineken has proved the following theorem (Heineken [227]): If every pair of elements $x, y \in G$ satisfies the 3-Engel relation $[x,y,y,y] = 1$, and if G contains no elements of orders 2 or 5, then G is nilpotent and has class at most 4. The finiteness of G is not required in the proof. The condition that G has no elements of order 2 cannot be removed; at the time of writing, it is unknown if the condition that G contains no elements of order 5 is truly necessary.[4] The proof of this theorem of Heineken requires numerous intricate commutator calculations which we will not reproduce here.

Up to now, we have required that all elements of G satisfy an Engel condition. In what follows, we consider individual elements that satisfy an Engel condition.

Definition 6.12. An element $g \in G$ is said to be left-Engel if for large enough n and all $x \in G$, we have $[g, \underset{n}{x, \ldots, x}] = 1$.

An element $g \in G$ is said to be right-Engel if for large enough n and all $x \in G$, we have $[x, \underset{n}{g, \ldots, g}] = 1$.

We call $g \in G$ an Engel element if g is left- or right-Engel.

[4] It is now known that this is a necessary condition; see S. Bachmuth and H. Y. Mochizuki, Third Engel groups and the Macdonald-Neumann conjecture, Bull. Austral. Math. Soc. **5** (1971), 379–386. *trans.*

Lemma 6.13. *Assume G is solvable, finite and generated by Engel elements. Then G is nilpotent.*

Proof. Let G be a minimal counterexample. We show:

(1) G has exactly one minimal normal subgroup N. The subgroup N is abelian and G/N is nilpotent:

Since homomorphisms take Engel elements to Engel elements, every homomorphic image of G again satisfies our assumption. So if $1 < N \lhd G$, then G/N is nilpotent.

Assume for contradiction that G has two distinct minimal normal subgroups N_1 and N_2. Using the nilpotency of G/N_i and Theorem 2.5 c), we see that $G = G/(N_1 \cap N_2)$ is also nilpotent, contradicting our assumption regarding G. Thus, G has exactly one minimal normal subgroup N. Since G is solvable, N is abelian.

(2) There exists an Engel element $g \in G$ such that $G = \langle \mathbf{C}_G(N), g\rangle$:

Let

$$\overline{G} = G/N = \langle \overline{g}_i \mid \overline{g}_i = g_i N,\ g_i \text{ Engel}, i = 1, \ldots, k\rangle,$$

and let k be as small as possible. Then

$$\overline{U} = \langle \overline{g}_1, \ldots, \overline{g}_{k-1}\rangle < \overline{G}.$$

Since $\overline{G}$ is nilpotent, it follows from Theorem 2.3 d) that

$$\overline{U}^{\overline{G}} = \langle \overline{g}_1^{\overline{x}}, \ldots, \overline{g}_{k-1}^{\overline{x}} \mid \overline{x} \in \overline{G}\rangle < \overline{G}.$$

Let

$$W = \langle g_1^x, \ldots, g_{k-1}^x \mid x \in G\rangle.$$

Then $W \lhd G$ and W is generated by Engel elements. So W is nilpotent.

If $k = 1$, then $G = \langle N, g_1\rangle$ and we are done since $N \leq \mathbf{C}_G(N)$. If $k > 1$, then $1 < W$. Since N is the unique minimal normal subgroup of G, we have $N \leq W$. By Theorem 2.6, $N \cap \mathbf{Z}(W) > 1$ since W is nilpotent. The minimality of N forces $N \leq \mathbf{Z}(W)$, so $W \leq \mathbf{C}_G(N)$. It follows that

$$G = \langle N, g_1, \ldots, g_k\rangle = \langle W, g_k\rangle = \langle \mathbf{C}_G(N), g_k\rangle.$$

(3) First assume that $G = \langle \mathbf{C}_G(N), g\rangle$ for some right-Engel element g. Let $1 \neq n \in N$. Then there exists some $m \geq 0$ such that $[n, g, \ldots, g] = n' \neq 1$ but $[n', g] = 1$. Since $G = \langle \mathbf{C}_G(N), g\rangle$, we have $1 \neq n' \in N \cap \mathbf{Z}(G)$. Thus, $N \leq \mathbf{Z}(G)$ and G is nilpotent by virtue of (1).

(4) Now assume that $G = \langle \mathbf{C}_G(N), g\rangle$ for some left-Engel element g. Let $n \in N$ such that $n' = [g, n] \neq 1$. (If there does not exist such an n, then $N \leq \mathbf{Z}(G)$ and we are done.) For large enough m, we have

$$1 = [g, gn, \underset{m}{\ldots}, gn] = [[g, n], gn, \underset{m-1}{\ldots}, gn] = [n', g, \underset{m-1}{\ldots}, g].$$

For some i, it follows again that

$$1 \neq [n', g, \underset{i}{\ldots}, g] \in N \cap \mathbf{Z}(G),$$

and we are done. □

Lemma 6.14 (Baer [20]). *If G is finite and generated by Engel elements, then G is nilpotent.*

Proof. (If we were able to show directly that products of Engel elements are again Engel elements, then this lemma would be a trivial consequence of Theorem 6.3.)

By Lemma 6.13, it is enough to show that G is solvable. So if we let G be a minimal counterexample, then G has no nontrivial solvable normal subgroups.

For every subgroup U of G, we define

$$\mathscr{E}(U) = \{x \in U \mid x \text{ is an Engel element of } G\}.$$

We show:

(1) Assume

$$D = \langle \mathscr{E}(D) \rangle < U = \langle \mathscr{E}(U) \rangle < G.$$

Then there is an $x \in \mathscr{E}(U)$ such that $x \in \mathbf{N}_U(D) - D$:

Since $U = \langle \mathscr{E}(U) \rangle < G$, U is nilpotent. Let

$$D = U_0 \lhd U_1 \lhd \cdots \lhd U_r = U$$

be a chain from D to U; such a chain exists since if $V < U$, then $V < \mathbf{N}_U(V)$ by Theorem 2.3. Let $\mathscr{E}(D) = \mathscr{E}(U_i) \subset \mathscr{E}(U_{i+1})$, and let $x \in \mathscr{E}(U_{i+1}) - \mathscr{E}(D)$.

Since $U_i \lhd U_{i+1}$, we have $\mathscr{E}(U_i)^x = \mathscr{E}(U_i)$ and therefore

$$D^x = \langle \mathscr{E}(D)^x \rangle = \langle \mathscr{E}(U_i)^x \rangle = \langle \mathscr{E}(U_i) \rangle = \langle \mathscr{E}(D) \rangle = D.$$

So $x \in \mathbf{N}_U(D) - D$ and $x \in \mathscr{E}(U)$.

(2) If V_1 and V_2 are distinct subgroups of G that are each maximal with respect to inclusion satisfying $V_i = \langle \mathscr{E}(V_i) \rangle < G$ $(i = 1, 2)$, then $\mathscr{E}(V_1 \cap V_2) = \{1\}$:

We choose the V_i $(i = 1, 2)$ with $V_i = \langle \mathscr{E}(V_i) \rangle < G$ and $V_1 \neq V_2$ such that $\mathscr{E}(V_1 \cap V_2)$ is as large as possible. We define $S = V_1 \cap V_2$ and $D = \langle \mathscr{E}(S) \rangle = \langle \mathscr{E}(D) \rangle$. Assume for contradiction that $D > 1$. By (1), there exist elements $x, y \in \mathscr{E}(G)$ such that

$$x \in \mathbf{N}_{V_1}(D) - D \quad \text{and} \quad y \in \mathbf{N}_{V_2}(D) - D.$$

So $V_3 = \langle x, y, \mathscr{E}(S) \rangle$ is generated by Engel elements.

If $V_3 = G$, then $1 < \langle \mathscr{E}(S) \rangle = D \lhd G$. But this is impossible since D is nilpotent. Consequently,

$$\langle x, y, \mathscr{E}(S)\rangle = V_3 = \langle \mathscr{E}(V_3)\rangle < G.$$

Now it follows that

$$\mathscr{E}(V_1 \cap V_3) \supseteq \{x, \mathscr{E}(S)\} \supset \mathscr{E}(S), \quad \text{so} \quad V_1 = V_3.$$

In the same way, we have

$$\mathscr{E}(V_2 \cap V_3) \supseteq \{y, \mathscr{E}(S)\} \supset \mathscr{E}(S), \quad \text{so} \quad V_2 = V_3.$$

But this contradicts our assumption that $V_1 \neq V_2$. Thus, $\mathscr{E}(S) = \{1\}$.

(3) Let U be a subgroup of G satisfying

$$U = \langle \mathscr{E}(U)\rangle < G$$

that is as large as possible. Then U is nilpotent and therefore $U \not\trianglelefteq G$. Let $x \in \mathscr{E}(U)$ and $g \in G$ such that $y = x^g \notin U$.

Assume for contradiction that $x^y \in \mathbf{N}_G(U)$. Then $U \trianglelefteq \langle U, x^y\rangle < G$, and by the choice of U we must have $x^y \in U \cap U^y$. Since x^y is an Engel element, we have in fact that

$$1 \neq x^y \in \mathscr{E}(U \cap U^y).$$

By (2), this forces $U = U^y$. Since $U \trianglelefteq \langle U, y\rangle < G$, we have $y \in U$. But this contradicts the choice of y.

Thus, we have $x^y \notin \mathbf{N}_G(U)$ and so

$$[x, y] = x^{-1}x^y \notin \mathbf{N}_G(U).$$

Since x and y are Engel elements of the same type, for large enough n we certainly have

$$[y, x, \underset{n}{\ldots}, x] = 1.$$

Thus, there exists some $i > 1$ such that

$$[y, x, \underset{i}{\ldots}, x] \in \mathbf{N}_G(U) \quad \text{and} \quad t = [y, x, \underset{i-1}{\ldots}, x] \notin \mathbf{N}_G(U).$$

Now

$$[t, x] = (x^t)^{-1}x = [y, x, \underset{i}{\ldots}, x] \in \mathbf{N}_G(U),$$

and since $x \in U$ we also have $x^t \in \mathbf{N}_G(U)$. Since $U \trianglelefteq \langle U, x^t\rangle < G$, it follows as before that $x^t \in U \cap U^t$. By (2), $U = U^t$ and so $t \in \mathbf{N}_G(U)$, which is a contradiction. □

Step (2) in the proof of Lemma 6.14 is apparently a clever adaptation of the proof of Theorem 5.1 to the situation at hand.

We are now able to describe the Engel elements:

Theorem 6.15 (Baer [20]). *Let G be a finite group. An element $g \in G$ is a right-Engel element if and only if $g \in \mathbf{F}(G)$, the Fitting subgroup of G.*

Proof. First assume that $g \in \mathbf{F}(G)$ and $x \in G$. Then $[x, g] \in \mathbf{F}(G)$ and so for large enough n, we have

$$[x, g, \underset{n}{\ldots}, g] \in \mathbf{K}_{n-1}(\mathbf{F}(G)) = 1.$$

So every element of $\mathbf{F}(G)$ is right-Engel.

Conversely, let

$$M = \langle g \in G \mid g \text{ is a right-Engel element of } G \rangle.$$

Then $M \trianglelefteq G$, and M is nilpotent by Lemma 6.14. So $M \leq \mathbf{F}(G)$. Thus, every right-Engel element of G lies in $\mathbf{F}(G)$. □

Theorem 6.16 (Baer [20]). *Let G be a finite group. An element $g \in G$ is a left-Engel element if and only if $g \in \mathbf{Z}_\infty(G)$, the hypercenter of G.*

(*By Theorem 6.15, every left-Engel element is also right-Engel since we have* $\mathbf{Z}_\infty(G) \leq \mathbf{F}(G)$.)

Proof. a) First assume that $g \in \mathbf{Z}_\infty(G) = \mathbf{Z}_m(G)$. For all $x \in G$, it follows that

$$[g, x, \underset{m}{\ldots}, x] \in \mathbf{Z}_0(G) = 1,$$

and g is left-Engel.

b) Let

$$M = \langle g \in G \mid g \text{ is a left-Engel element of } G \rangle.$$

Clearly, $M \trianglelefteq G$. We need to show that $M \leq \mathbf{Z}_\infty(G)$. Let G be a minimal counterexample.

(1) G has exactly one minimal normal subgroup N:

If N_i $(i = 1, 2)$ are two distinct minimal normal subgroups of G, then

$$MN_i/N_i \leq \mathbf{Z}_\infty(G/N_i).$$

For large enough k, we then have

$$[M, G, \underset{k}{\ldots}, G] \leq N_1 \cap N_2 = 1,$$

and then $M \leq \mathbf{Z}_\infty(G)$, contradicting our assumption.

(2) M is a p-group:

By Lemma 6.14, M is nilpotent. Using (1), it follows that M has prime power order.

(3) There exists a $g \in G$ such that $p \nmid |g|$ and $G = \langle M, g \rangle$:

Assume for contradiction that for all $x \in G$ with order coprime to p, we have $\langle M, x\rangle < G$. Then $M \le \mathbf{Z}_\infty(\langle M, x\rangle)$. Thus, $\langle M, x\rangle$ is a nilpotent group, and since $(|M|, |x|) = 1$ and $N \le M$, it follows that $x \in \mathbf{C}_G(M) \le \mathbf{C}_G(N)$. Consequently, all elements of G with order coprime to p lie in $\mathbf{C}_G(N)$. So if P is a Sylow p-subgroup of G, we have $G = \mathbf{C}_G(N)P$. If we define $D = N \cap \mathbf{Z}(P)$, then $1 < D \le \mathbf{Z}(G)$. So $N \le \mathbf{Z}(G)$, then since $M/N \le \mathbf{Z}_\infty(G/N)$ we have $M \le \mathbf{Z}_\infty(G)$.

(4) $\Phi(M) = 1$:

Assume for contradiction that $\Phi(M) > 1$. As N is the unique minimal normal subgroup of G, we have $N \le \Phi(M)$. Since $G = \langle M, g\rangle$ and $M/N \le \mathbf{Z}_\infty(G/N)$, G/N is nilpotent. By Lemma 3.3 b), we have $N \le \Phi(M) \le \Phi(G)$. So G is nilpotent by Theorem 3.7. Then certainly $M \le G = \mathbf{Z}_\infty(G)$.

(5) $M \le \mathbf{Z}_\infty(G)$:

Let $1 \ne m \in M$, and let m be left-Engel. In the endomorphism ring of the elementary abelian p-group M, we have the identity $(\alpha - 1)^{p^k} = \alpha^{p^k} - 1$. By virtue of this identity, for large enough k we have

$$1 = [m, g, \underset{k}{\ldots}, g] = m^{(g-1)^{p^k}} = m^{g^{p^k}-1}.$$

Thus, $g^{p^k} \in \mathbf{C}_G(m)$, and in fact $g \in \mathbf{C}_G(M)$ since $p \nmid |g|$. Note that M is abelian by (4) and Theorem 3.11. Furthermore, $G = \langle M, g\rangle$ by (3), so we conclude that $m \in \mathbf{Z}(G)$ and we are done. □

Exercises

16) Construct a group G of exponent 3 that is generated by d elements and has order 3^k, where $k = d + \binom{d}{2} + \binom{d}{3}$.

17) (Baer [16]) Let $B_n = \langle b_0, b_1, \ldots, b_{p^n-1}\rangle$ be the elementary abelian group of order p^n. We express the indices i of the b_i p-adically:

$$i = \sum_{j=0}^{n-1} c_j p^j \quad \text{with} \quad 0 \le c_j \le p - 1.$$

Let π_k $(1 \le k \le n-1)$ be the permutation of $\{0, 1, \ldots, p^n - 1\}$ defined by

$$\left(\sum_{j=0}^{n-1} c_j p^j\right)\pi_k = \begin{cases} \sum\limits_{j \ne k-1} c_j p^j + (c_{k-1} + 1)p^{k-1} & \text{for } c_{k-1} \ne p - 1 \\ \sum\limits_{j \ne k-1} c_j p^j & \text{for } c_{k-1} = p - 1. \end{cases}$$

Then all the π_k have order p and commute pairwise, so they generate an elementary abelian group C_n of order p^{n-1}. Let G_n be the semidirect product of B_n with C_n. Now show:

(1) G_n has exponent p^2.

(2) The class of G_n is at least $n-1$.

(This example shows that for the exponent p^2, there is no bound on the class that is independent of the number of generators. For groups of exponent 4, it is known that $c(G) \leq 3d(G)$, see Wright [488].)

18) (Burnside [77]) Let $G = \langle a, b\rangle$ be a group of exponent 4 with $a^2 = 1$. Show that:
   a) $G' = \langle abab^{-1}, ab^{-1}ab\rangle$ is abelian.
   b) $|G'| \leq 2^3$.
   c) $|G| \leq 2^6$.

19) (Burnside [77]) Let $G = \langle a, b\rangle$ be a group of exponent 4 with $[a^2, b^2] = 1$. Show that:
   a) $\langle a^2\rangle^G$ is elementary abelian and has order at most 2^3.
   b) $|G| \leq 2^9$.
   (Hint: Show that $(a^2)^{bab} = a^2(a^2)^b(a^2)^{ba}$ and

$$\langle a^2\rangle^G = \langle a^2, (a^2)^b, (a^2)^{ba}\rangle.)$$

20) (Burnside [77]) Let $G = \langle a, b\rangle$ be a group of exponent 4. Define $c = [a^2, b^2]$ and show that:
   a) $\langle c\rangle^G = \langle c, c^a, c^b\rangle$ is elementary abelian.
   b) $|G| \leq 2^{12}$.
   c) $c(G) \leq 5$.
   (Hint: For the proof of a), show that:
   (1) $[c, a^2] = [c, b^2] = 1$.
   (2) $[c, b^{-1}, b] = 1$.
   (3) $[c, b^{-1}, a^{-1}b] = 1$, so $[c, b^{-1}] \in \mathbf{Z}(G)$.)

7 Elementary theory of p-groups

In this section, we assume until Theorem 7.12 that G is a p-group. We will prove a number of elementary and loosely related theorems about such groups. A systematic theory of a large class of p-groups follows in §10.

Lemma 7.1.
a) *If G/N is cyclic and $N \leq \mathbf{Z}(G)$, then G is abelian.*
(*Here it is not necessary to assume that G is a p-group.*)
b) *If G is nonabelian, then p^2 is a divisor of $|G/\mathbf{Z}(G)|$.*
c) *If G/G' is cyclic, then G is itself cyclic.*
d) *If G is nonabelian, then p^2 is a divisor of $|G/G'|$. (In particular, every group of order p^2 is abelian, as we already saw in Theorem I.6.10, pg. 32.)*

Proof. a) Let $G/N = \langle aN\rangle$. Then we have $G = \langle a, N\rangle$. If $N \leq \mathbf{Z}(G)$, then the elements of the generating set $\{a, N\}$ of G commute, so G is abelian.

b) If p^2 is not a divisor of $|G/\mathbf{Z}(G)|$, then $G/\mathbf{Z}(G)$ is cyclic and G is abelian by a).

c) If G/G' is cyclic, then $G/\Phi(G)$ is also cyclic since $G' \leq \Phi(G)$ by Theorem 3.11. It follows from the Burnside Basis Theorem 3.15 that $d = 1$, so G is cyclic.

d) is a trivial consequence of c). □

Theorem 7.2.

a) *If* $1 < N \trianglelefteq G$, *then* $1 < N \cap \mathbf{Z}(G)$. *In particular, every minimal normal subgroup of a* p*-group lies in the center.*

b) *If* $N \trianglelefteq G$ *and* $|N| = p^i$, *then* $N \leq \mathbf{Z}_i(G)$.

c) *All chief factors of* G *have order* p.

d) *If* M *and* N *are normal subgroups of* G *with* $|M/N| = p^m$, *then for every* k *with* $0 \leq k \leq m$ *there exists a normal subgroup* K *of* G *such that* $N \leq K \leq M$ *and* $|K/N| = p^k$.

e) *If* U *and* V *are subgroups of* G *with* $[U : V] = p^m$, *then for every* k *with* $0 \leq k \leq m$ *there exists a subgroup* W *of* G *such that* $V \leq W \leq U$ *and* $[W : V] = p^k$.

Proof. a) We have already proved a) in Theorem I.6.9 (pg. 31); it is also contained in Theorem 2.6.

b) By a), $N \cap \mathbf{Z}(G) > 1$. For $i = 1$, we have $|N| = p$ and thus $N \leq \mathbf{Z}(G)$. We prove the claim by induction on i:

Note that $N\mathbf{Z}_1(G)/\mathbf{Z}_1(G)$ is a normal subgroup of $G/\mathbf{Z}_1(G)$. If $|N| = p^i$, then because $N \cap \mathbf{Z}_1(G) > 1$ we have

$$|N\mathbf{Z}_1(G)/\mathbf{Z}_1(G)| = |N/(N \cap \mathbf{Z}_1(G))| \leq p^{i-1}.$$

It follows by our induction hypothesis that

$$N\mathbf{Z}_1(G)/\mathbf{Z}_1(G) \leq \mathbf{Z}_{i-1}(G/\mathbf{Z}_1(G)) = \mathbf{Z}_i(G)/\mathbf{Z}_1(G)$$

and consequently $N \leq \mathbf{Z}_i(G)$.

c) By the Jordan–Hölder theorem for chief series in Theorem I.11.7 (pg. 68), it suffices to prove the existence of a single chief series with factor groups of order p. By a), G has a normal subgroup N_1 of order p. Induction on the order of the group yields the claim immediately.

d) We refine the chain $1 \leq M < N \leq G$ of normal subgroups of G to a chief series of G. By part c) and the Jordan–Hölder theorem for chief series in Theorem I.11.7, every chief factor in this chief series has order p. Therefore, there exists in this chief series exactly one member K that satisfies the required properties.

e) Let $V < U \leq G$ and $[U : V] = p^m$. Let S be a maximal subgroup of U with $V \leq S$. Since U is nilpotent, Theorem 2.3 guarantees that $S \lhd U$, so $|U/S| = p$. Induction on $m - k$ yields the claim. □

(Note that this construction results in a chain

$$V = V_0 \lhd V_1 \lhd \cdots \lhd V_m = U,$$

but it does not follow that $V \lhd U$!)

Theorem 7.3. *Let A be a maximal abelian normal subgroup of G; that is, A is not properly contained in any abelian normal subgroup of G. Then:*

a) *A is equal to its centralizer in G.*

b) *If $|G| = p^n$ and $|A| = p^a$, then $2n \leq a(a+1)$.*

c) *If $|A| = p^a$, then $\mathbf{K}_a(G) \leq A$ and $\mathbf{K}_{2a}(G) = 1$.*

d) *Let A_1 and A_2 be maximal abelian normal subgroups of G with $\exp A_i = p^{a_i}$ $(i = 1, 2)$. Then $a_2 \leq 2a_1$ (and of course $a_1 \leq 2a_2$, as well).*

Proof. a) Assume for contradiction that $A < \mathbf{C}_G(A) \trianglelefteq G$. By Theorem 7.2 d), there exists a normal subgroup L of G such that $A < L \leq \mathbf{C}_G(A)$ and $|L/A| = p$. We have $A \leq \mathbf{Z}(L)$. So L is a cyclic extension of its center and is therefore abelian by Lemma 7.1 a). But this contradicts our choice of A as a maximal abelian normal subgroup of G.

b) We have $G/A = G/\mathbf{C}_G(A)$, and this is isomorphic to a subgroup of the automorphism group $\mathbf{Aut}(A)$ of A by Theorem I.4.5 (pg. 21). Letting $|A/\Phi(A)| = p^d$, Theorem 3.19 says that $|\mathbf{Aut}(A)|$ is a divisor of

$$p^{d(a-d)}(p^d - 1)(p^d - p)\cdots(p^d - p^{d-1}).$$

Thus, the highest power p^k of p that divides the order of $\mathbf{Aut}(A)$ satisfies

$$k \leq 1 + 2 + \cdots + (d-1) + d(a-d) = \frac{d(d-1)}{2} + d(a-d) \leq \frac{a(a-1)}{2}.$$

Hence,

$$|G| = p^n \leq |A|p^k \leq p^a p^{\frac{a(a-1)}{2}}$$

and then $2n \leq a(a+1)$.

c) Since $|A| = p^a$, we certainly have $A \leq \mathbf{Z}_a(G)$ by Theorem 7.2 b). It follows by Theorem 2.11 c) that

$$[A, \mathbf{K}_a(G)] \leq [\mathbf{Z}_a(G), \mathbf{K}_a(G)] = 1,$$

therefore by a),

$$\mathbf{K}_a(G) \leq \mathbf{C}_G(A) = A.$$

From $|A| = p^a$ and

$$\mathbf{K}_a(G) > \mathbf{K}_{a+1}(G) > \cdots > \mathbf{K}_{c+1}(G) = 1,$$

we conclude that $c \leq 2a - 1$.

d) Let $B = \langle A_1, A_2 \rangle$. Since

$$\mathbf{Z}(B) \leq \mathbf{C}_G(A_1) \cap \mathbf{C}_G(A_2) = A_1 \cap A_2 \leq \mathbf{Z}(B),$$

it follows that $\mathbf{Z}(B) = A_1 \cap A_2$. We then conclude from $[A_1, A_2] \leq A_1 \cap A_2$ that B has class at most 2. Using Theorem 2.13 a), we obtain

$$\exp(B/\mathbf{Z}(B)) \leq \exp \mathbf{Z}(B) \leq p^{a_1}$$

and thus

$$p^{a_2} = \exp A_2 \leq \exp B \leq p^{2a_1}. \qquad \square$$

Example 7.4. Let $\mathsf{V} = \mathsf{V}(n, p)$ be the vector space of dimension n over the field $\mathrm{GF}(p)$, and let $\{v_1, \ldots, v_n\}$ be a basis of V. We define $\mathsf{V}_0 = \{0\}$ and $\mathsf{V}_i = \langle v_1, \ldots, v_i \rangle$ for $1 \leq i \leq n$. Let T be the group of all linear maps t from V onto itself with

$$v_i t - v_i \in \mathsf{V}_{i-1}$$

for all $i = 1, \ldots, n$. (So with respect to the basis $\{v_1, \ldots, v_n\}$, the maps in T are triangular matrices with 1's on the diagonal.) It is easy to show that the maps $a \in T$ satisfying $va - v \in \mathsf{V}_1$ for all $v \in \mathsf{V}$ constitute an abelian normal subgroup A of T of order p^{n-1}. Since $\mathbf{C}_T(A) = A$, the subgroup A is a maximal abelian normal subgroup of T. We have $|A| = p^{n-1}$ and $|T| = p^{\frac{n(n-1)}{2}}$. Thus, the bound in Theorem 7.3 b) cannot be improved.

Suppose $n = 2m$. Then the $b \in T$ such that

$$vb = v \quad \text{for all} \quad v \in \mathsf{V}_m \quad \text{and} \quad vb - v \in \mathsf{V}_m \quad \text{for all} \quad v \in \mathsf{V}$$

constitute an abelian normal subgroup B of T. We have $|B| = p^{m^2}$, and this is larger than $|A|$ for $m \geq 2$. Thus, the maximal abelian normal subgroups of p-groups need not have the same order.

Lemma 7.5. *Let N be a noncyclic normal subgroup of G.*

a) *If $p > 2$, then there exists a normal subgroup K of G that lies in N and is elementary abelian of type (p, p).*

b) *If $p = 2$, then the previous statement also holds as long as $N \leq \Phi(G)$.*

Proof. We prove the claim by induction on $|G|$. The base case for induction is $|G| = p^2$ and $G = N = K$, and the statement holds.

By Theorem 7.2 d), there exists a normal subgroup T of G of order p lying in N. We consider G/T.

If N/T is cyclic, then N is abelian by Lemma 7.1 a). Since N is not cyclic by assumption, $d(N) = 2$. Then the solutions to $x^p = 1$ lying in N constitute a characteristic subgroup K of N, and K has type (p, p). Since K is characteristic in $N \trianglelefteq G$, the subgroup K is certainly normal in G by Theorem I.4.8 a) (pg. 22).

Now assume that N/T is not cyclic. By the induction hypothesis, there is a normal subgroup M of G such that $M \leq N$ and M/T is abelian of type (p, p). So M has order p^3. If M has exponent p, then we take K to be any normal subgroup of G lying in M of order p^2; such a subgroup exists by Theorem 7.2 d). So we may

assume $\exp M > p$. If M is abelian, then we conclude as above. So we may assume M is nonabelian. We have $|M/M'| \geq p^2$ by Lemma 7.1 d), so $|M'| = p$. Thus, $M' \leq \mathbf{Z}(M)$ by Theorem 7.2 b).

Now we must distinguish the cases $p > 2$ and $p = 2$:

First assume that $p > 2$. We apply Lemma 1.3 to M and obtain

$$(xy)^p = x^p y^p [y,x]^{\binom{p}{2}} = x^p y^p$$

for all $x, y \in M$. So the mapping of x onto x^p is a homomorphism ϵ from M onto a subgroup U of M. Since $\exp M > p$, we have $U \neq 1$. On the other hand, since M/M' has type (p,p), we have $U \leq M'$. Altogether it follows that $|U| = p$. Thus, the kernel K of ϵ has order p^2 and consists exactly of those elements of M that satisfy the equation $x^p = 1$. Therefore, K has type (p,p) and is a normal subgroup of G, being a characteristic subgroup of M.

Now assume that $p = 2$. Let $L \lhd G$ with $L < M$ and $|L| = 4$; such a subgroup L exists by Theorem 7.2 d). We may assume that $L = \langle l \rangle$ is cyclic; otherwise we set $L = K$ and are done. Now, for all $g \in G$ we have $l^g = l^{\pm 1}$. It follows that $[G : \mathbf{C}_G(l)] \leq 2$. Therefore

$$\mathbf{C}_G(l) \geq \Phi(G) \geq N \geq M$$

and so $l \in \mathbf{Z}(M)$. But then M is abelian, contradicting our assumption. □

Lemma 7.5 admits a range of interesting applications, to which we now turn.

Theorem 7.6. *Suppose all abelian normal subgroups of G are cyclic.*

a) *If $p > 2$, then G is itself cyclic.*

b) *If $p = 2$, then G has a cyclic normal subgroup of index 2.*

Proof. a) Assume for contradiction that G is not cyclic. Setting $N = G$ in Lemma 7.5 a), we see that G has an abelian normal subgroup of type (p,p). But this contradicts our assumption that every abelian normal subgroup of G is cyclic.

b) In the case $p = 2$, we can certainly apply Lemma 7.5 b) to the normal subgroup $\Phi(G)$ of G and conclude that $\Phi(G)$ is cyclic. Let A be a maximal abelian normal subgroup of G with $\Phi(G) \leq A$. By our assumption, A is cyclic, say $A = \langle a \rangle$. Since $\Phi(G) \leq A$, the quotient G/A is an elementary abelian 2-group by Theorem 3.14.

If G/A is cyclic, then G/A has order 2 and we are done. So assume that G/A is not cyclic. By Theorem 7.3 a), $\mathbf{C}_G(A) = A$ and so G/A is isomorphic to a subgroup of the automorphism group $\mathbf{Aut}(A)$ of the cyclic group A. Let $|A| = 2^n$. Then $\mathbf{Aut}(A)$ is isomorphic to the multiplicative group consisting of the residue classes modulo 2^n of the integers coprime to 2 by Theorem I.4.6 (pg. 21). This has order 2 for $n = 2$ and type $(2^{n-2}, 2)$ for $n > 2$ by Theorem I.13.19 (pg. 89). In the case at hand, G/A is elementary abelian and noncyclic, so $n > 2$, and G/A is necessarily of type $(2,2)$. The residue classes -1 and $1 + 2^{n-1}$ generate the unique subgroup of type $(2,2)$ in the group of residue classes modulo 2^n coprime to 2. Thus, there exist elements $b, c \in G$ such that

$$a^b = a^{-1} \quad \text{and} \quad a^c = a^{1+2^{n-1}}.$$

Since $[b,c] \in \Phi(G) \leq A$, we have $[b,c] = a^r$ for some integer r. Furthermore, $c^2 \in \Phi(G)$, so $c^2 = a^j$. Note that 2 is certainly a divisor of j, since otherwise c would generate a cyclic normal subgroup of G that is larger than the maximal abelian normal subgroup A. Now we calculate

$$(a^x c^{-1})^2 = a^x c^{-1} a^x c c^{-2} = a^x a^{x(1+2^{n-1})} a^{-j}.$$

The congruence $x(2+2^{n-1}) \equiv j \bmod 2^n$ can be solved for x since $n > 2$ means $2^2 \nmid 2+2^{n-1}$, and $2 \mid j$. We now replace c by $a^x c^{-1}$. Then

$$a^c = a^{1+2^{n-1}}$$

still holds since $c^{-1} = ca^{-j}$, and we now have $c^2 = 1$. It follows that

$$1 = [b, c^2] = [b,c][b,c]^c = a^r a^{r(1+2^{n-1})},$$

so $r(2+2^{n-1}) \equiv 0 \bmod 2^n$. Therefore, $r \equiv 0 \bmod 2^{n-1}$. As a consequence, we obtain $[b,c] \in \langle a^{2^{n-1}} \rangle$, and we already have $[a,c] = a^{2^{n-1}} \in \langle a^{2^{n-1}} \rangle$. Then since $c^2 = 1$,

$$\langle a^{2^{n-1}}, c \rangle$$

is an abelian, noncyclic normal subgroup of G, contradicting our assumption. So we must have $|G/A| = 2$. □

We will classify the p-groups ($p > 3$) whose abelian normal subgroups are generated by two elements in Theorem 12.4.

Theorem 7.7. *Suppose $\mathbf{Z}_2(G)$ is cyclic. Then:*
a) *If $p > 2$, then G is itself cyclic.*
b) *If $p = 2$, then G has a cyclic normal subgroup of index 2.*

Proof. Assume for contradiction that there exists an abelian, noncyclic normal subgroup A of G. Then by Theorem 7.2 d), there exists a normal subgroup K of G lying in $\Omega_1(A)$ of type (p,p). Then $K \leq \mathbf{Z}_2(G)$ by Theorem 7.2 b). But this contradicts our assumption that $\mathbf{Z}_2(G)$ is cyclic. So all abelian normal subgroups of G are cyclic, and the claim follows from Theorem 7.6. □

Theorem 7.8.
a) (*P. Hall*) *If $G' \cap \mathbf{Z}_2(G)$ is cyclic, then G' is also cyclic.*
b) (*Burnside*) *If $\mathbf{Z}(G')$ is cyclic, then G' is also cyclic.*
c) (*Hobby [245]*) *If $\mathbf{Z}(\Phi(G))$ is cyclic, then $\Phi(G)$ is also cyclic.*

Proof. a) Assume for contradiction that G' is not cyclic. By Theorem 3.11, we have $G' \leq \Phi(G)$. Thus, there exists by Lemma 7.5 an abelian normal subgroup $K \trianglelefteq G$ of type (p,p) with $K \leq G'$. Since $|K| = p^2$, we have $K \leq \mathbf{Z}_2(G)$ by Theorem 7.2 b).

So $K \leq G' \cap \mathbf{Z}_2(G)$. But this is impossible since $G' \cap \mathbf{Z}_2(G)$ is assumed to be cyclic. Therefore, G' is cyclic.

b) We have $[G', \mathbf{Z}_2(G)] = 1$ by Theorem 2.11 c), and it follows that

$$G' \cap \mathbf{Z}_2(G) \leq \mathbf{Z}(G').$$

So if $\mathbf{Z}(G')$ is cyclic, then $G' \cap \mathbf{Z}_2(G)$ is also cyclic and we can apply a).

c) Assume for contradiction that $\mathbf{Z}(\Phi(G))$ is cyclic but $\Phi(G)$ is not. By Lemma 7.5, G contains a normal subgroup K of type (p, p) with $K \leq \Phi(G)$. Since

$$|G/\mathbf{C}_G(K)| \quad \text{divides} \quad |\mathbf{Aut}(K)| = |\mathrm{GL}_2(p)| = (p^2 - 1)(p^2 - p),$$

it follows that $|G/\mathbf{C}_G(K)| \leq p$. So $\Phi(G) \leq \mathbf{C}_G(K)$ and then $K \leq \mathbf{Z}(\Phi(G))$. This is a contradiction. □

Theorem 7.9 (Blackburn [35]). *If the commutator subgroup G' can be generated by two elements, then G'' is cyclic and G' has class at most* 2.

(*We state the following without proof: If G and G' can each be generated by two elements, then G' is in fact abelian; see Blackburn [35].*)

Proof. First we show that G'' is cyclic. Assume for contradiction that G'' is not cyclic. Then by the Burnside Basis Theorem 3.15,

$$|G''/\Phi(G'')| = p^m \geq p^2.$$

Thus, there exists by Theorem 7.2 d) a normal subgroup $K \trianglelefteq G$ such that $\Phi(G'') \leq K \leq G''$ and $|G''/K| = p^2$. To prove that G'' is cyclic, it is enough to derive a contradiction under the assumption $K = 1$; otherwise we consider G/K.

So assume that G'' is abelian of type (p, p). The factor group $G/\mathbf{C}_G(G'')$ is isomorphic to a subgroup of the automorphism group $\mathbf{Aut}(G'')$ of G'' by Theorem I.4.5 (pg. 21). Since

$$|\mathbf{Aut}(G'')| = |\mathrm{GL}_2(p)| = (p^2 - 1)(p^2 - p),$$

it follows that $|G/\mathbf{C}_G(G'')| \leq p$. Therefore, $G' \leq \mathbf{C}_G(G'')$ and consequently $G'' \leq \mathbf{Z}(G')$. So the class of G' is at most two. By assumption, G' can be generated by two elements c and d. By Theorem 1.11 c), we have

$$G'' = \langle [c, d], \mathbf{K}_3(G') \rangle = \langle [c, d] \rangle.$$

We have proved that G'' is cyclic.

It remains to show that G' has class at most 2. (Thus far, we have only proved this under the assumption that $|G''| = p^2$!) The group $G/\mathbf{C}_G(G'')$ is isomorphic to a subgroup of the automorphism group of the cyclic group G''. So $G/\mathbf{C}_G(G'')$ is abelian and therefore $G' \leq \mathbf{C}_G(G'')$. This means $G'' \leq \mathbf{Z}(G')$, and G' has class at most 2. □

Lemma 7.10. *Suppose N is a nonabelian normal subgroup of G with $N \leq \mathbf{K}_i(G)$. Then*

$$|\mathbf{Z}(N)| \geq p^i, \quad |N| \geq p^{i+2} \quad \text{and} \quad |N/N'| \geq p^{i+1}.$$

Proof. By Theorem 2.11 c), we have $[\mathbf{K}_i(G), \mathbf{Z}_i(G)] = 1$. It follows that $N \not\leq \mathbf{Z}_i(G)$, since otherwise N would be abelian. Now $N \cap \mathbf{Z}_i(G) \leq \mathbf{Z}(N)$. For $j \leq i$, we have by Theorem 7.2 a) that

$$\begin{aligned} 1 &< \mathbf{Z}(G/\mathbf{Z}_j(G)) \cap N\mathbf{Z}_j(G)/\mathbf{Z}_j(G) \\ &= \mathbf{Z}_{j+1}(G)/\mathbf{Z}_j(G) \cap N\mathbf{Z}_j(G)/\mathbf{Z}_j(G) \\ &= (\mathbf{Z}_{j+1}(G) \cap N)\mathbf{Z}_j(G)/\mathbf{Z}_j(G). \end{aligned}$$

Therefore,

$$1 < N \cap \mathbf{Z}_1(G) < N \cap \mathbf{Z}_2(G) < \cdots < N \cap \mathbf{Z}_i(G)$$

and consequently

$$|\mathbf{Z}(N)| \geq |N \cap \mathbf{Z}_i(G)| \geq p^i.$$

It follows from Lemma 7.1 b) that $|N/\mathbf{Z}(N)| \geq p^2$, and then $|N| \geq p^{i+2}$.

Let $M \lhd G$ and $|N'/M| = p$. We define $\overline{N} = N/M$. Then we have

$$\overline{N} \trianglelefteq G/M$$

with $\overline{N}' = N'/M \neq 1$ and $\overline{N} \leq \mathbf{K}_i(G)M/M = \mathbf{K}_i(G/M)$. By what we have already shown, $|\overline{N}| \geq p^{i+2}$ and therefore $|N/N'| \geq p^{i+1}$. □

Theorem 7.11 (P. Hall [207]). *If $G^{(k)} \neq 1$, then $|G| \geq p^{2^k+k}$.*

Proof. If $i < k$, then $G^{(i)} > G^{(i+1)} > 1$. By Theorem 2.12 a), we have $G^{(i)} \leq \mathbf{K}_{2^i}(G)$. Then Lemma 7.10 yields the bound

$$|G^{(i)}/G^{(i+1)}| \geq p^{2^i+1}.$$

Let $|G| = p^n$. Since $|G^{(k)}| \geq p$, we have

$$n \geq 1 + \sum_{i=0}^{k-1}(2^i + 1) = 2^k + k.$$ □

We will show in §17 that this bound cannot be considerably improved.

If G is a nilpotent group and $U \leq G$, then a repeated application of Theorem 2.3 c) yields a chain

$$U = U_0 \lhd U_1 \lhd \cdots \lhd U_s = G.$$

If there exists such a chain, then we say that U is subnormal in G and write $U \lhd\lhd G$. In a finite nilpotent group, every subgroup is subnormal, and the finite nilpotent groups are characterized by this property by Theorem 2.3 d). The class of finite groups whose subgroups are normal is much smaller:

Theorem 7.12 (Dedekind). *Assume that G is nonabelian and every subgroup of G is normal in G. Then*

$$G = Q \times A \times B,$$

where Q is a quaternion group of order 8, A is an abelian group of odd order and B is an abelian group of exponent 1 or 2. Conversely, if G has the given structure, then every subgroup of G is normal in G. (These groups are sometimes called Hamiltonian groups.)

Proof. Since every Sylow subgroup of G is normal, G is nilpotent. In the (possibly abelian) Sylow subgroups of G, all subgroups are also normal. Therefore, we may assume that G is a p-group.

First assume that G is nonabelian. We choose elements $a, b \in G$ so that $Q = \langle a, b\rangle$ is nonabelian and $|a| + |b|$ is as small as possible. Since $\langle a\rangle \lhd G$ and $\langle b\rangle \lhd G$, we have $1 \neq [a, b] \in \langle a\rangle \cap \langle b\rangle$, and so

$$[a, b] = a^{rp^m} = b^{sp^n}$$

for some integers m, n, r, s. Here we may assume $rs \not\equiv 0 \bmod p$. It follows from $Q' = \langle [a, b]\rangle \leq \mathbf{Z}(Q)$ that Q has class 2. Now, $\langle a^p, b\rangle$ is abelian since $|a^p| + |b| < |a| + |b|$, so we have by Lemma 1.3 a) that

$$1 = [a^p, b] = [a, b]^p.$$

Consequently, $|a| = p^{m+1}$ and $|b| = p^{n+1}$. Suppose without loss of generality that $m \geq n$. Since

$$Q = \langle a, a^{-rp^{m-n}} b^s\rangle,$$

by the choice of a and b we certainly have

$$|a^{-rp^{m-n}} b^s| \geq |b| = p^{n+1}.$$

By Lemma 1.3, it follows that

$$1 \neq (a^{-rp^{m-n}} b^s)^{p^n} = a^{-rp^m} b^{sp^n} [b, a]^t$$

with $t = -rsp^{m-n}\binom{p^n}{2}$.

So we must have $t \not\equiv 0 \bmod p$. This is only possible if $m = n = 1$ and $p = 2$. Thus, $Q = \langle a, b\rangle$ is a quaternion group of order 8. In particular, it follows that G is abelian for $p > 2$.

Now suppose that G is a nonabelian 2-group in which every subgroup is normal. Then G contains a quaternion group $Q = \langle a, b\rangle$. Since $\langle a\rangle \lhd G$ and $\langle b\rangle \lhd G$, we have for every $g \in G$ that $a^g = a^{\pm 1}$ and $b^g = b^{\pm 1}$. That is, g acts on Q as one of the four inner automorphisms of Q, and we obtain $G = Q\mathbf{C}_G(Q)$.

Assume for contradiction that there exists an $x \in \mathbf{C}_G(Q)$ such that $|x| = 4$. Then

$$(ax)^b = a^{-1}x \in \langle ax\rangle.$$

This leads to the contradiction $a^2 = 1$ or $x^2 = 1$.

So $\mathbf{C}_G(Q)$ has exponent 2 and is elementary abelian. Therefore, we have $\mathbf{C}_G(Q) = \mathbf{Z}(Q) \times B$ for some B, and it follows that $G = Q \times B$. It is not difficult to verify that in the groups listed above, all subgroups are normal. □

Exercises

21) Let G be a p-group. If G has two maximal subgroups that are abelian, then at least $p + 1$ maximal subgroups of G are abelian, and G has class at most 2.
22) (Rédei [348]) Let G be a nonabelian p-group whose maximal subgroups are abelian. Then G is isomorphic to one of the following groups:
    a) The quaternion group of order 8.
    b) $G = \langle a, b\rangle$ with the defining relations

$$a^{p^m} = b^{p^n} = 1, \quad a^b = a^{1+p^{m-1}} \quad (m \geq 2, n \geq 1); \quad |G| = p^{m+n}.$$

    c) $G = \langle a, b\rangle$ with the defining relations

$$a^{p^m} = b^{p^n} = c^p = 1, \quad [a, b] = c, \quad [a, c] = [b, c] = 1; \quad |G| = p^{m+n+1}.$$

    (Hint: Choose a generating set $\{a, b\}$ of G so that $|a|+|b|$ is as small as possible.)
23) (Alperin [7]) Let G be a p-group with an abelian subgroup of index p^2. Then G has an abelian normal subgroup of index p^2.
24) Let G be a p-group. We define $\beta(g)$ for $g \in G$ by

$$p^{\beta(g)} = [G : \mathbf{C}_G(g)]$$

    and define $\beta(G) = \max \beta(g)$. Show that:
    a) $p^{\beta(G)} \leq |G'|$.
    b) (Knoche [306]) We have $\beta(G) = 1$ if and only if $|G'| = p$.
    (Hint for b):
    (1) $G' \leq \mathbf{Z}(G)$.
    (2) G' has exponent p.
    (3) If $[a, b] \neq 1 \neq [a', b']$, then there exists an $a'' \in G$ such that

$$a'' \notin \mathbf{C}_G(b) \cup \mathbf{C}_G(b').$$

Then $[a'', b] = [a, b]^i$ for some i.)

25) (Wielandt) If G is a p-group with $\beta(G) < p$, then $c(G) \leq \beta(G) + 1$. (It was conjectured that we always have $c(G) \leq \beta(G) + 1$. P. Neumann and Wiegold showed that $c(G) \leq \frac{p}{p-1}\beta(G) + 1$ for p-groups.[5])
(Hint: Prove the claim by induction on $|G|$.
a) Since the claim holds for $G/\mathbf{Z}(G)$, we have $c(G) \leq \beta(G) + 2$.
b) Assume $c(G) = \beta(G) + 2 = c$. Let $a \notin \mathbf{Z}(G)$ and $\mathbf{C}_G(a) = C$. In the chain

$$C = C\mathbf{K}_c(G) \leq C\mathbf{K}_{c-1}(G) \leq \cdots \leq C\mathbf{K}_1(G) = G,$$

there is a third equality. It follows that $a \in \mathbf{C}_G(\mathbf{K}_i(G)/\mathbf{K}_{i+2}(G))$ for some i with $2 \leq i \leq c - 1$.)

8 Counting theorems

Definition 8.1. Let $|G| = p^n$, and let $0 \leq k \leq n$. We denote by $s_k = s_k(G)$ the number of subgroups of G with order p^k. Clearly, $s_0 = s_n = 1$.

The following theorem has found numerous applications.

Theorem 8.2. *Suppose the p-group G contains only one subgroup of order p, then:*
a) *For $p > 2$, G is cyclic.*
b) *For $p = 2$, G is cyclic or a generalized quaternion group with the defining relations*

$$a^{2^{n-1}} = 1, \quad b^2 = a^{2^{n-2}}, \quad a^b = a^{-1} \quad (n \geq 3).$$

Proof. Every abelian normal subgroup of G has only one subgroup of order p, so is cyclic by the fundamental theorem of finitely generated abelian groups, Theorem I.13.12 (pg. 85). So if $p > 2$, then G is certainly cyclic by Theorem 7.6 a). If $p = 2$, then G has a cyclic normal subgroup of index 2 by Theorem 7.6 b). We classified all the 2-groups with a cyclic normal subgroup of index 2 in Theorem I.14.9 (pg. 96). Since G has only one subgroup of order 2, G cannot be a semidirect product. So by Theorem I.14.9, G is a generalized quaternion group with the relations

$$a^{2^{n-1}} = 1, \quad b^2 = a^{2^{n-2}}, \quad a^b = a^{-1}.$$

This group has a single subgroup of order 2, namely $\langle b^2 \rangle$; all elements of G of order 2 lie in $\langle a \rangle$ since $(a^i b)^2 = b^2 \neq 1$. □

[5] It is now known that this so-called class-breadth conjecture, due to Leedham-Green, Neumann and Wiegold, is false. Counterexamples can be found in W. Felsch, J. Neubüser and W. Plesken, Space groups and groups of prime-power order. IV. Counterexamples to the class-breadth conjecture, J. London Math. Soc. (2) **24** (1981), no. 1, 113–122 and B. Eick, M. F. Newman and E. A. O'Brien, The class-breadth conjecture revisited, J. Algebra **300** (2006), no. 1, 384–393. *trans.*

Theorem 8.3. *Let $|G| = p^n$. If $s_m(G) = 1$ for some m with $1 < m < n$, then G is cyclic.*

Proof. Let U be the unique subgroup of G with order p^m. By Theorem 7.2 e), there exists a subgroup V of G of order p^{m+1} with $U < V$. Then U is the unique maximal subgroup of V. By Theorem 3.16, V is cyclic and so U is cyclic as well. Every subgroup of G of order p or p^2 lies in a subgroup of order p^m by Theorem 7.2 e), so lies in U. Since U is cyclic, G has only one subgroup of order p and one subgroup of order p^2. If $p > 2$, then G is cyclic by Theorem 8.2 a). If $p = 2$, then G is cyclic by Theorem 8.2 b) since the generalized quaternion group $Q = \langle a, b\rangle$ of order 2^n has two subgroups of order 4, namely

$$\langle a^{2^{n-3}}\rangle \quad \text{and} \quad \langle b\rangle. \qquad \square$$

Theorem 8.4. *Let $|G| = p^n$. Suppose there is some m with $1 < m \leq n$ such that every subgroup of G with order p^m is cyclic. Then G is cyclic, except possibly when $p^m = 4$; in this case, G may be a generalized quaternion group.*

Proof. Since $m \geq 2$, each subgroup of order p^2 lies in a subgroup of order p^m by Theorem 7.2 e). So each subgroup of order p^2 is cyclic. Assume for contradiction that G has two distinct subgroups U_1 and U_2 of order p. Let $N \leq \mathbf{Z}(G)$ and $|N| = p$. Then at least one of the groups U_i is distinct from N, say $N \neq U_1$. Then $N \times U_1$ is a noncyclic subgroup of G with order p^2, a contradiction. Thus, G has only one subgroup of order p. If $p > 2$, then G is cyclic by Theorem 8.2 a). If $p = 2$, then G is cyclic or generalized quaternion by Theorem 8.2 b). But when $m \geq 3$, the generalized quaternion group $Q = \langle a, b\rangle$ of order 2^n has a noncyclic subgroup of order 2^m, namely

$$\langle a^{2^{n-m}}, b\rangle.$$

So G can be noncyclic only when $p^m = 4$. $\square$

For the following theorems regarding the number of subgroups of a given order in a p-group, we will need precise knowledge of this number for elementary abelian groups.

Lemma 8.5. *Let A be an elementary abelian group of order p^d. For $0 \leq k$ we define $s_k(A) = f(d, k)$. We have:*

a) $f(d, k) = f(d, d - k)$.

b) $$f(d,k) = \frac{(p^d-1)(p^d-p)\cdots(p^d-p^{k-1})}{(p^k-1)(p^k-p)\cdots(p^k-p^{k-1})} = \frac{\prod_{i=d-k+1}^{d}(p^i-1)}{\prod_{i=1}^{k}(p^i-1)}$$

for $1 \leq k \leq d$.

c) $f(d+1, k) = f(d, k) + p^{d-k+1} f(d, k-1)$.

d) $f(d,d-1) \equiv 1 \bmod p$;
$f(d,d-2) \equiv 1 \bmod p$ *for* $d \geq 2$;
$f(d,d-1) \equiv 1 + p \bmod p^2$ *for* $d \geq 2$.

e) $\prod_{i=1}^{d}(x - p^{i-1}) = \sum_{i=0}^{d}(-1)^i p^{\binom{i}{2}} f(d,i)\, x^{d-i}$.

Proof. a) Let V be the vector space of dimension of d over $\mathrm{GF}(p)$. It is well known that V is isomorphic to its dual space V^*. By the duality theory of finite-dimensional vector spaces, the map

$$\mathsf{U} \mapsto \mathsf{U}^\perp = \{f \in \mathsf{V}^* \mid f = 0 \text{ on } \mathsf{U}\}$$

is a bijective correspondence between the subspaces of V of dimension k and the subspaces of V^* of dimension $d-k$. Since the elementary abelian group A is isomorphic to the additive group of V and the subgroups of V^+ are exactly the subspaces of V, we have

$$f(d,k) = f(d,d-k).$$

b) The number of pairs $\{X,Y\}$ with $X < Y \leq A$, $|X| = p^k$ and $|Y| = p^{k+1}$ is the same as the number of pairs $\{X, Y/X\}$ with $|X| = p^k$ and $|Y/X| = p$, so is given by

$$f(d,k)f(d-k,1) = f(d,k)\frac{p^{d-k}-1}{p-1}.$$

If we instead first count the Y and the number of its subgroups of index p, then we find the same number is given by

$$f(d,k+1)f(k+1,k) = f(d,k+1)f(k+1,1) = f(d,k+1)\frac{p^{k+1}-1}{p-1}.$$

This yields the recursion formula

$$f(d,k+1) = f(d,k)\frac{p^{d-k}-1}{p^{k+1}-1}.$$

The formula in b) follows by induction on k.

c) Let $B = A \times \langle g \rangle$ be an elementary abelian group of order p^{d+1}. Let V be a subgroup of B of order p^k with $k \leq d$. There are exactly $f(d+1,k)$ such subgroups V in B. In A, there are $f(d,k)$ subgroups of order p^k. If $V \not\leq A$, then $|V \cap A| = p^{k-1}$ and $V = \langle V \cap A, ag \rangle$ for some $a \in A$. Here, V on depends only on the coset of $V \cap A$ determined by a. That is, the group V is uniquely determined by the cosets $V \cap A$ and $(V \cap A)a$, so there are exactly

$$f(d,k-1)[A : V \cap A] = f(d,k-1)p^{d-k+1}$$

subgroups V of B with $|V| = p^k$ and $V \not\leq A$. Altogether, it follows that

$$f(d+1,k) = f(d,k) + p^{d-k+1} f(d,k-1).$$

d) By a),

$$f(d,d-1) = f(d,1) = \frac{p^d - 1}{p-1} = 1 + p + \cdots + p^{d-1}.$$

So for $d \geq 2$, we always have $f(d,d-1) \equiv 1 \bmod p$ and $f(d,d-1) \equiv 1+p \bmod p^2$. Furthermore,

$$f(d,d-2) = f(d,2) = \frac{(p^d-1)(p^{d-1}-1)}{(p-1)(p^2-1)} \equiv 1 \mod p \quad \text{for} \quad d \geq 2.$$

e) We prove the claim by induction on d. For $d = 1$ the formula is

$$x - 1 = f(1,0)x - f(1,1),$$

and it is clearly correct. By the induction hypothesis,

$$\prod_{i=1}^{d+1} (x - p^{i-1}) = \left(\sum_{i=0}^{d} (-1)^i p^{\binom{i}{2}} f(d,i) x^{d-i} \right) (x - p^d).$$

The coefficient of x^{d+1} is $f(d,0) = 1$. For $1 \leq i \leq d$, the coefficient of x^{d+1-i} is

$$\begin{aligned} &(-1)^i p^{\binom{i}{2}} f(d,i) - (-1)^{i-1} p^{\binom{i-1}{2}} f(d,i-1) p^d \\ &= (-1)^i p^{\binom{i}{2}} \big(f(d,i) + p^{d-i+1} f(d,i-1) \big) = (-1)^i p^{\binom{i}{2}} f(d+1,i), \end{aligned}$$

where we used part c) in the last equality. The coefficient of x^0 is

$$-(-1)^d p^{\binom{d}{2}} p^d = (-1)^{d+1} p^{\binom{d+1}{2}}. \qquad \square$$

Theorem 8.6 (Hall's Enumeration Principle). *Let G be a p-group of order p^n with $|G/\Phi(G)| = p^d$. Let $0 \leq i \leq d$. We denote by $\mathscr{S}_i'$ the set of subgroups M of G such that $\Phi(G) \leq M \leq G$ and $|G/M| = p^i$. Let $\mathscr{S}$ be any set of proper subgroups of G. We denote by $n(M)$ the number of subgroups in $\mathscr{S}$ that lie in the group M. Then:*

$$\begin{aligned} n(G) - \sum_{M \in \mathscr{S}_1'} n(M) + p \sum_{M \in \mathscr{S}_2'} n(M) - \cdots + (-1)^i p^{\binom{i}{2}} \sum_{M \in \mathscr{S}_i'} n(M) + \cdots \\ + (-1)^d p^{\binom{d}{2}} n(\Phi(G)) = 0. \end{aligned}$$

Proof. Let $N \in \mathscr{S}_s'$ be the intersection of all those $M \in \mathscr{S}_i'$ $(i = 0, 1, 2, \dots)$ containing a given $U \in \mathscr{S}$. We certainly have $s \geq 1$ since the proper subgroup U of G lies in at least one $M \in \mathscr{S}_1'$. Therefore, the number of $M \in \mathscr{S}_i'$ with $M \geq U$ is equal to the number of $M \in \mathscr{S}_i'$ with $M \geq N$, so is equal to the number of subgroups

of index p^i in the elementary abelian group G/N of order p^s. This number is $f(s,i)$. Thus, a fixed $U \in \mathscr{S}$ is counted in the above formula with multiplicity

$$v(U) = 1 - f(s,1) + pf(s,2) - \cdots + (-1)^s p^{\binom{s}{2}}$$

since $f(s,s) = 1$ and $f(s,i) = 0$ for $i > s$. Setting $x = 1$ in Lemma 8.5 e) yields $v(U) = 0$. □

Numerous theorems can be elegantly proved using Theorem 8.6. First we must address a particular case:

Lemma 8.7. *Let $|G| = p^{n+1}$ with $n \geq 2$, let $p > 2$ and assume G is not cyclic. Assume G has a cyclic normal subgroup of index p. Then G has exactly p cyclic normal subgroups of index p and one noncyclic normal subgroup of index p.*

Proof. Let $\langle a \rangle$ be a cyclic normal subgroup of G of index p. Since G is not cyclic, G is generated by elements a and b such that

$$a^{p^n} = b^p = 1 \qquad \text{and} \qquad a^b = a \quad \text{or} \quad a^b = a^{1+p^{n-1}}$$

by Theorem I.14.9 (pg. 96). In the first case, $(ab^{-i})^p = a^p$ for $0 \leq i \leq p-1$. In the second case,

$$(ab^{-i})^p = aa^{b^i}a^{b^{2i}} \cdots a^{b^{(p-1)i}}b^{-pi} = a^t,$$

where

$$\begin{aligned} t &= 1 + (1+p^{n-1})^i + (1+p^{n-1})^{2i} + \cdots + (1+p^{n-1})^{(p-1)i} \\ &\equiv p + ip^{n-1}\frac{p(p-1)}{2} \equiv p \mod p^n \quad (\text{since } p > 2). \end{aligned}$$

So in each case, $(ab^{-i})^p = a^p$. Hence, ab^{-i} has order p^n. The p cyclic subgroups $\langle ab^{-i} \rangle$ are certainly distinct since they are already distinct modulo $\langle a^p \rangle$. So we have found p cyclic subgroups of index p in G. In addition, G contains the noncyclic subgroup $\langle a^p, b \rangle$ of index p (here it is necessary to assume that $n \geq 2$!). Since $d(G) = 2$, these $p+1$ subgroups of index p are all the maximal subgroups of G. □

Theorem 8.8. *Let $|G| = p^n$. Then:*

a) *$s_k(G) \equiv 1 \bmod p$ for $0 \leq k \leq n$.*
(*This is a special case of Theorem I.7.2, pg. 33.*)

b) (*Kulakoff*) *If $p > 2$ and G is not cyclic, then*

$$s_k(G) \equiv 1 + p \mod p^2 \quad \text{for} \quad 1 \leq k \leq n-1.$$

Proof. a) If $k = 0$ or $k = n$, then the theorem is trivial. So assume $0 < k < n$. We prove the theorem by induction on n. By the induction hypothesis, we may assume that all maximal subgroups M of G satisfy $s_k(M) \equiv 1 \bmod p$. Let $|G/\Phi(G)| = p^d$. If $\mathscr{S}_k$ is the set of subgroups of G of order p^k, then by Theorem 8.6 we have

$$s_k(G) \equiv \sum_{M \in \mathscr{S}_1'} s_k(M) \equiv \sum_{M \in \mathscr{S}_1'} 1 = f(d, d-1) \equiv 1 \mod p,$$

where the last congruence follows from Lemma 8.5 d).

b) The claim is correct for $n = 2$; in this case, G is elementary abelian of type (p, p) and has exactly $p + 1$ subgroups of order p. So assume $n > 2$. We again proceed by induction on n. Let the set $\mathscr{S}_k$ and the integer d be defined as in a). If $k = n - 1$, then we have by Lemma 8.5 d) that

$$s_{n-1}(G) = f(d, d-1) \equiv 1 + p \mod p^2$$

since $d \geq 2$. So assume $k \leq n - 2$. Theorem 8.6 yields

$$s_k(G) \equiv \sum_{M \in \mathscr{S}_1'} s_k(M) - p \sum_{M \in \mathscr{S}_2'} s_k(M) \mod p^2.$$

Since $k \leq n - 2$, we have by a) that $s_k(M) \equiv 1 \bmod p$ for $M \in \mathscr{S}_2'$. Therefore, by Lemma 8.5 d),

$$p \sum_{M \in \mathscr{S}_2'} s_k(M) \equiv p \sum_{M \in \mathscr{S}_2'} 1 = p f(d, d-2) \equiv p \mod p^2.$$

If no maximal subgroup of G is cyclic, then we may use the induction hypothesis to conclude $s_k(M) \equiv 1 + p \bmod p^2$ for all $M \in \mathscr{S}_1'$. Using Lemma 8.5 d), it follows that

$$s_k(G) \equiv (1 + p) f(d, d-1) - p \equiv (1 + p)^2 - p \equiv 1 + p \mod p^2.$$

It remains to consider the case that G contains a cyclic normal subgroup of index p. In this case, G has exactly p cyclic normal subgroups of index p and one noncyclic normal subgroup of index p by Lemma 8.7. Hence,

$$s_k(G) \equiv p + (p + 1) - p \equiv 1 + p \mod p^2. \qquad \square$$

For related results, see Theorem 11.8 and P. Hall [208].

9 The identities of P. Hall and Zassenhaus

Definition 9.1. Let $F = \langle f_1, \ldots, f_n \rangle$ be the free group in the free generators $f_1, \ldots, f_n$. We define:

a) $w(f) = j$ if f lies in $\mathbf{K}_j(F)$ but f does not lie in $\mathbf{K}_{j+1}(F)$. If f lies in all $\mathbf{K}_j(F)$ $(j = 1, 2, \ldots)$, then we define $w(f) = \infty$.

(It turns out that $\bigcap_j \mathbf{K}_j(F) = 1$, so $w(f) = \infty$ is equivalent to $f = 1$. However, we will not need to make use of this fact, which was first proved by F. W. Levi,

since we will never require the exact value of $w(f)$. We will always be able to manage with bounding $w(f)$ from below.)

b) Let J be a subset of $\{f_1, \ldots, f_n\}$ and define

$$N_{\mathsf{J}} = \langle f_j^f \mid f_j \in \mathsf{J}, f \in F \rangle$$

to be the smallest normal subgroup of F generated by the $f_j \in \mathsf{J}$ and their conjugates.

The following lemma is crucial for our calculations with the function w:

Lemma 9.2. *For all $f, f' \in F$, we have $w([f, f']) \geq w(f) + w(f')$.*

Proof. Let $w(f) \geq m$, so $f \in \mathbf{K}_m(F)$, and let $w(f') \geq m'$, so $f' \in \mathbf{K}_{m'}(F)$. By Theorem 2.11 b) (which was proved for arbitrary groups!), we have

$$[f, f'] \in [\mathbf{K}_m(F), \mathbf{K}_{m'}(F)] \leq \mathbf{K}_{m+m'}(F).$$

So $w([f, f']) \geq m + m'$. □

The main results of this section, Theorems 9.4 and 9.7, rely upon the following theorem:

Theorem 9.3 (P. Hall's "commutator collecting process"). *Let $F = \langle f_1, \ldots, f_n \rangle$ be the free group in the free generators $f_1, \ldots, f_n$. Suppose the set $\{f_1, \ldots, f_n\}$ is partitioned into pairwise disjoint subsets J_i with $i \in \mathscr{M} = \{1, \ldots, m\}$.*

a) *We order the set of subsets of $\mathscr{M}$ in the following way: We say $\mathscr{R} \ll \mathscr{S}$ if $|\mathscr{R}| < |\mathscr{S}|$; we order the sets of equal cardinality lexicographically. We say $\mathscr{R} \lessdot\!\!\lessdot \mathscr{S}$ if $\mathscr{R} \ll \mathscr{S}$ or $\mathscr{R} = \mathscr{S}$.*

b) *For any $\emptyset \neq \mathscr{R} \subseteq \mathscr{M}$, define $C_{\mathscr{R}}$ to be the set of those $f \in F$ with the following properties:*

(1) *$w(f) \geq |\mathscr{R}|$, so $f \in \mathbf{K}_{|\mathscr{R}|}(F)$.*

(2) *$f \in N_{\mathsf{J}_i}$ for all $i \in \mathscr{R}$.*

(3) *$f \in \langle f_j \mid f_j \in \bigcup_{i \in \mathscr{R}} \mathsf{J}_i \rangle$.*

With these conventions, we now describe the commutator collecting process:

Let $g = f_1 \cdots f_n$. Then

$$g = \prod_{\mathscr{R} \subseteq \mathscr{M}} h_{\mathscr{R}}$$

for some elements $h_{\mathscr{R}} \in C_{\mathscr{R}}$; the subsets $\mathscr{R}$ of $\mathscr{M}$ appear here in the total order given above.

Proof. For $\mathscr{S} \subseteq \mathscr{M}$, we show that

$$g = \prod_{\mathscr{R} \ll \mathscr{S}} h_{\mathscr{R}} t_1 \cdots t_k, \qquad (*)$$

where the product of the $h_{\mathscr{R}}$ is already ordered correctly and the t_i lie in some $C_{\mathscr{T}}$ with $\mathscr{S} \trianglelefteqq \mathscr{T}$.

If we have proved the claim for $\mathscr{S} = \mathscr{M}$, then the t_i lie in $C_{\mathscr{M}}$ and the theorem is proved.

If $\mathscr{S} = \{1\}$, then the formula $(*)$ is correct:

For $f_i \in \mathsf{J}_j$, we have $f_i \in C_{\{j\}}$, and $\{1\} \trianglelefteqq \{j\}$ for all sets $\{j\}$. Thus, the ordered product at the beginning of expression $(*)$ is empty in this case.

We prove the formula $(*)$ by induction on $\mathscr{S}$ with respect to the order we have imposed on subsets of $\mathscr{M}$, and prove the step from $\mathscr{S}$ to its successor. To this end, we bring the element $t \in C_{\mathscr{S}}$ that is furthest to the left in the expression $(*)$ one step further to the left via

$$\ldots t_i t \ldots = \ldots t\, t_i [t_i, t] \ldots$$

if such an element $t_i \in C_{\mathscr{R}}$ with $\mathscr{R} \gg \mathscr{S}$ exists. Since $t_i \in C_{\mathscr{R}}$ with $\mathscr{R} \gg \mathscr{S}$, we have by Lemma 9.2 that

(1) $w([t_i, t]) \geq w(t_i) + w(t) \geq |\mathscr{R}| + |\mathscr{S}| \geq |\mathscr{R} \cup \mathscr{S}|$.

(2) $[t_i, t] \in [N_{\mathsf{J}_j}, F] \leq N_{\mathsf{J}_j}$ for all $j \in \mathscr{R}$ and
$[t_i, t] \in [F, N_{\mathsf{J}_k}] \leq N_{\mathsf{J}_k}$ for all $k \in \mathscr{S}$, thus
$[t_i, t] \in N_{\mathsf{J}_j}$ for all $j \in \mathscr{R} \cup \mathscr{S}$.

(3) $[t_i, t] \in \langle f_j \mid f_j \in \bigcup_{k \in \mathscr{R} \cup \mathscr{S}} \mathsf{J}_k \rangle$.

So we have $[t_i, t] \in C_{\mathscr{R} \cup \mathscr{S}}$ with $\mathscr{R} \cup \mathscr{S} \trianglerighteqq \mathscr{R} \gg \mathscr{S}$. Consequently, our procedure only generates new factors in $C_{\mathscr{T}}$ with $\mathscr{S} \ll \mathscr{T}$. After finitely many steps, we have brought all the $t_i \in C_{\mathscr{S}}$ leftwards into the correct position and reached the desired form. □

Theorem 9.4 (Identity of P. Hall and Petresco). *Let F_2 be the free group in the free generators x and y. Then there exist elements $c_i \in \mathbf{K}_i(F_2)$ $(i = 2, 3, \ldots)$ such that for all $m > 0$ we have*

$$x^m y^m = (xy)^m c_2^{\binom{m}{2}} \cdots c_{m-1}^{\binom{m}{m-1}} c_m.$$

(If $\mathbf{K}_3(\langle x, y \rangle) = 1$, then we obtain the formula given in Lemma 1.3 b).)

Proof. a) We work initially in the free group F with the free generators $f_1, \ldots, f_{2m}$. Anticipating the use of Theorem 9.3, we set $\mathsf{J}_i = \{f_i, f_{i+m}\}$; then $\mathscr{M} = \{1, \ldots, m\}$. For every subset $\mathscr{R}$ of $\mathscr{M}$ we set

$$g_{\mathscr{R}} = \prod_{i \in \mathscr{R}} f_i \prod_{i \in \mathscr{R}} f_{i+m}.$$

Let $\mu_{\mathscr{R}}$ be the homomorphism from F onto the subgroup

$$F_{\mathscr{R}} = \langle f_i, f_{i+m} \mid i \in \mathscr{R} \rangle$$

defined by

$$f_i^{\mu_{\mathscr{R}}} = f_i \quad \text{for} \quad i \in \mathscr{R},$$
$$f_{i+m}^{\mu_{\mathscr{R}}} = f_{i+m} \quad \text{for} \quad i \in \mathscr{R} \quad \text{and}$$
$$f_j^{\mu_{\mathscr{R}}} = 1 \quad \text{for all other } f_j.$$

We note that:

(1) For $\mathscr{S} \subseteq \mathscr{R}$ and $h_{\mathscr{S}} \in C_{\mathscr{S}}$, we have $h_{\mathscr{S}}^{\mu_{\mathscr{R}}} = h_{\mathscr{S}}$:
Since $h_{\mathscr{S}}$ lies in $\langle f_i, f_{i+m} \mid i \in \mathscr{S} \rangle$ by the definition of $C_{\mathscr{S}}$, the claim holds.

(2) For $\mathscr{S} \not\subseteq \mathscr{R}$, we have $h_{\mathscr{S}}^{\mu_{\mathscr{R}}} = 1$:
In this case, there exists a j such that $j \in \mathscr{S}$ and $j \notin \mathscr{R}$. Then

$$h_{\mathscr{S}} \in N_{\mathsf{J}_j} = \langle f_j^f, f_{j+m}^f \mid f \in F \rangle.$$

Since $j \notin \mathscr{R}$, it follows that $h_{\mathscr{S}}^{\mu_{\mathscr{R}}} = 1$.

Setting $g = f_1 \cdots f_{2m}$, we obtain from Theorem 9.3 that

$$g^{\mu_{\mathscr{R}}} = g_{\mathscr{R}} = \prod_{\mathscr{S}} h_{\mathscr{S}}^{\mu_{\mathscr{R}}} = \prod_{\mathscr{S} \subseteq \mathscr{R}} h_{\mathscr{S}}.$$

The subsets $\mathscr{S}$ of $\mathscr{R}$ that arise here appear according to the order described in Theorem 9.3 a).

b) Let $F_2 = \langle x, y \rangle$ be the free group in the two free generators x and y, and let ϵ be the homomorphism from F onto F_2 defined by

$$f_i^{\epsilon} = x \quad \text{for} \quad i \leq m \quad \text{and} \quad f_i^{\epsilon} = y \quad \text{for} \quad m+1 \leq i \leq 2m.$$

For all subsets $\mathscr{R}$ of $\mathscr{M}$ of cardinality $|\mathscr{R}| = r$, we obtain by a) that

$$g_{\mathscr{R}}^{\epsilon} = x^r y^r = \prod_{\mathscr{S} \subseteq \mathscr{R}} h_{\mathscr{S}}^{\epsilon}.$$

We claim: If $|\mathscr{R}_1| = |\mathscr{R}_2| = r$, then $h_{\mathscr{R}_1}^{\epsilon} = h_{\mathscr{R}_2}^{\epsilon}$.

We prove the claim by induction on r. If $r = 1$, then $\mathscr{R} = \{i\}$ and so

$$g_{\{i\}}^{\epsilon} = xy = \prod_{\mathscr{S} \subseteq \{i\}} h_{\mathscr{S}}^{\epsilon} = h_{\{i\}}^{\epsilon}.$$

So $h_{\{i\}}^{\epsilon} = xy$ is independent of i. The general case is proved as follows:

We have

$$\prod_{\mathscr{S} \subset \mathscr{R}_1} h_{\mathscr{S}}^{\epsilon} h_{\mathscr{R}_1}^{\epsilon} = g_{\mathscr{R}_1}^{\epsilon} = x^r y^r = g_{\mathscr{R}_2}^{\epsilon} = \prod_{\mathscr{S} \subset \mathscr{R}_2} h_{\mathscr{S}}^{\epsilon} h_{\mathscr{R}_2}^{\epsilon}.$$

By our induction hypothesis, the $h_{\mathscr{S}}^{\epsilon}$ with $|\mathscr{S}| < r$ only depend on $|\mathscr{S}| = s$. Note that the same number of $h_{\mathscr{S}}^{\epsilon}$ appear on both sides of this equation since the number of subsets of $\mathscr{R}_i$ of cardinality s is the same for $i = 1, 2$. Furthermore, these $h_{\mathscr{S}}^{\epsilon}$ appear on both sides in the same order according to increasing $|\mathscr{S}|$. Thus,

$$h^{\epsilon}_{\mathscr{R}_1} = h^{\epsilon}_{\mathscr{R}_2}.$$

For the subsets $\mathscr{S}$ of $\mathscr{R}$ with $|\mathscr{S}| = i$, we set $c_i = h^{\epsilon}_{\mathscr{S}}$. If $|\mathscr{R}| = r$, then there are exactly $\binom{r}{i}$ subsets $\mathscr{S}$ of $\mathscr{R}$ such that $|\mathscr{S}| = i$. It follows that

$$x^r y^r = \prod_{i=1}^{r} c_i^{\binom{r}{i}}.$$

In this expression, $c_1 = xy$. Since $h_{\mathscr{S}} \in C_{\mathscr{S}} \subseteq \mathbf{K}_i(F)$ for $|\mathscr{S}| = i$, we have

$$c_i = h^{\epsilon}_{\mathscr{S}} \in \mathbf{K}_i(F^{\epsilon}) = \mathbf{K}_i(F_2). \qquad \square$$

We will see an important consequence of the identity in Theorem 9.4 for p-groups in the next section.

Lemma 9.5. *Let p be an odd prime. In the Hall–Petresco identity of Theorem 9.4 we have*

$$c_p \equiv [y, x, \underset{p-1}{\ldots}, x]^a \prod_i v_i^{a_i} \mod \mathbf{K}_{p+1}(\langle x, y\rangle),$$

where $a \equiv -1 \bmod p$ and every v_i has the form

$$v_i = [y, x, z_3, \ldots, z_p]$$

with $z_j \in \{x, y\}$ and $z_j = y$ for at least one j.

Proof. Let F be the free group in the free generators x and y. By Theorem 1.11 b), $\mathbf{K}_p(F)/\mathbf{K}_{p+1}(F)$ is generated by all $[z_1, \ldots, z_p]\mathbf{K}_{p+1}(F)$ with $z_j \in \{x, y\}$. By Lemma 6.8, we have

$$\begin{aligned}[x, y, z_3, \ldots, z_p] &= [[y, x]^{-1}, z_3, \ldots, z_p] \\ &\equiv [y, x, z_3, \ldots, z_p]^{-1} \mod \mathbf{K}_{p+1}(F).\end{aligned}$$

So that

$$c_p \equiv [y, x, \underset{p-1}{\ldots}, x]^a \prod_i v_i^{a_i} \mod \mathbf{K}_{p+1}(F),$$

where the v_i are commutators of the form $[y, x, z_3, \ldots, z_p]$ with $z_i \in \{x, y\}$ and at least one z_j is equal to y.

Let $G = Z \wr_r Z$ be the regular wreath product of two cyclic groups Z of order p constructed according to Definition I.15.6 (pg. 104). Then G has an elementary abelian normal subgroup $D = \langle w_1, \ldots, w_p\rangle$ of order p^p and $G = D\langle z\rangle$ for some z with $z^p = 1$ and $w_i^z = w_{i+1}$ ($i \bmod p$). Since $G' \leq D$, $g^p = 1$ for all $g \in G'$.

Let ϵ be the homomorphism from F to G defined by $x^{\epsilon} = z$ and $y^{\epsilon} = w_1$. Since $(c_i^{\binom{p}{i}})^{\epsilon} = 1$ for $2 \leq i \leq p-1$ and $\mathbf{K}_{p+1}(G) = 1$, the identity of Theorem 9.4 yields

$$1 = z^p w_1^p = (zw_1)^p [w_1, z, \underset{p-1}{\ldots}, z]^a \prod_i v_i^{\epsilon a_i}.$$

In this expression,

$$v_i^{\epsilon} = [w_1, z, y_3, \ldots, y_p],$$

where $y_j \in \{w_1, z\}$ and, say, $y_k = w_1$. Since $D' = 1$, it follows that

$$[w_1, z, \ldots, y_k] \in [D, y_k] = 1.$$

Thus,

$$(zw_1)^p [w_1, z, \ldots, z]^a = 1.$$

A simple computation in the wreath product G shows that

$$[w_1, z, \ldots, z] = w_1^{k_1} w_2^{k_2} \cdots w_{p-1}^{k_{p-1}} w_p$$

(for some integers k_i) and

$$(zw_1)^p = zw_1 z^{-1} z^2 w_1 z^{-2} \cdots z^p w_1 z^{-p} = w_1 w_2 \cdots w_p.$$

And it follows that $a \equiv -1 \bmod p$. □

Lemma 9.6.

a) *Let $U \neq \{1\}$ be a finite subgroup of the multiplicative group of a field* K. *Then* $\sum_{x \in U} x = 0$.

b) *We have*

$$\sum_{x \in \mathrm{GF}(p)} x^r = \begin{cases} 0 & \text{for } 0 < r < p-1 \\ -1 & \text{for } r = p-1. \end{cases}$$

Proof. a) Let $1 \neq a \in U$. It follows from $a \sum_{x \in U} x = \sum_{x \in U} x$ that $\sum_{x \in U} x = 0$.

b) Let

$$U = \{x^r \mid x \in \mathrm{GF}(p)^\times\}$$

with $0 < r \leq p-1$. Let L be the kernel of the homomorphism μ from $\mathrm{GF}(p)^\times$ onto U defined by $x^\mu = x^r$. If $U \neq \{1\}$, then by a),

$$\sum_{x \in \mathrm{GF}(p)} x^r = |L| \sum_{u \in U} u = 0.$$

If $U = \{1\}$, then $r = p-1$ and we obtain

$$\sum_{x\in \mathrm{GF}(p)} x^r = p-1 = -1. \qquad \square$$

Theorem 9.7 (The Zassenhaus identity). *Let G be a group of exponent p. Then for all $x, y \in G$ we have*

$$[y, x, \underset{p-1}{\ldots}, x] \in \mathbf{K}_{p+1}(G).$$

(*In fact, for $p = 3$, $[x, y, y] = 1$ by Theorem 6.6.*)

Proof. If $p = 2$, then G is abelian and the claim is trivial. So assume that $p > 2$. We may assume that $\mathbf{K}_{p+1}(G) = 1$ and then show $[y, x, \underset{p-1}{\ldots}, x] = 1$. By the identity of Theorem 9.4, we have

$$1 = x^p y^p = (xy)^p c_2^{\binom{p}{2}} \cdots c_{p-1}^{\binom{p}{p-1}} c_p = c_p.$$

In this expression, Lemma 9.5 says that

$$c_p = [y, x, \underset{p-1}{\ldots}, x]^a \prod_i v_i^{a_i},$$

where the v_i have the form $v_i = [y, x, z_3, \ldots, z_p]$ with $z_j \in \{x, y\}$ and $z_j = y$ for at least one j.

We replace x by x^s with $1 \le s \le p-1$. If in some v_i exactly $r(i)$ arguments are x and $p - r(i)$ arguments are y, then v_i passes over to $v_i^{s^{r(i)}}$ under this replacement by Lemma 6.8 since $\mathbf{K}_{p+1}(G) = 1$.

So we obtain for all s with $1 \le s \le p-1$ that

$$1 = [y, x, \underset{p-1}{\ldots}, x]^{as^{p-1}} \prod_i v_i^{a_i s^{r(i)}}.$$

Forming the product over s yields, by Lemma 9.6,

$$1 = [y, x, \ldots, x]^{-a} \prod_i \left(\prod_{s=1}^{p-1} v_i^{s^{r(i)}} \right)^{a_i} = [y, x, \ldots, x]^{-a}.$$

By Lemma 9.5, $a \equiv -1 \bmod p$. This leads to $[y, x, \underset{p-1}{\ldots}, x] = 1$, as claimed. $\square$

The Zassenhaus identity has important consequences for the theory of groups of exponent p, but we will not discuss them (Kostrikin [311]). We will call upon the Zassenhaus identity in §14 when we consider groups of maximal class.

10 Regular p-groups

In this section, we consider a class of groups introduced by P. Hall, the regular p-groups. (There is no connection with the concept of regular permutation groups from Definition I.5.12, pg. 28. There is, however, only one place in our discussion where this unfortunate coincidence of terminology could lead to confusion.) In these groups, the map $x \mapsto x^{p^i}$ is not necessarily an endomorphism, but it is true that

$$\Omega_i(G) = \langle g \in G \mid g^{p^i} = 1 \rangle = \{g \in G \mid g^{p^i} = 1\}$$

and

$$\mho_i(G) = \langle g^{p^i} \mid g \in G \rangle = \{g^{p^i} \mid g \in G\}.$$

For regular p-groups, there are connections between these two subgroups that are similar to those in abelian p-groups (see Chapter I, §13, Exercise 51). We will also thoroughly study the commutation relations between $\Omega_i(G)$ and $\mho_i(G)$, which are trivial in abelian p-groups. The importance of the theory developed here rests in part on the fact that p-groups of class $c(G) < p$ are always regular. The theory appeared in a major paper by P. Hall in 1933 (P. Hall [207]) and has progressed little since then.

Unless otherwise stated, all theorems in this section come from P. Hall [207, 208].

Definition 10.1. A p-group G is called regular if for all $x, y \in G$ we have

$$x^p y^p = (xy)^p \prod_i d_i^p$$

for some $d_i \in \langle x, y \rangle'$.

All abelian p-groups are regular. The property of being regular is clearly inherited by subgroups and factor groups. We will use this fact frequently in our proofs.

First, we show that the collection of regular p-groups is not exceedingly small, so the theory developed here covers a wide array of p-groups.

Theorem 10.2. *Let G be a p-group. Then:*
a) *If the class of G is smaller than p, then G is regular.*
b) *If $|G| \leq p^p$, then G is regular.*
c) *If G' is cyclic and $p > 2$, then G is regular.*
d) *If $\exp G = p$, then G is regular.*

Proof. a) Assume $c(G) < p$. By the identity of Theorem 9.4, we have

$$x^p y^p = (xy)^p c_2^{\binom{p}{2}} \cdots c_{p-1}^{\binom{p}{p-1}} c_p$$

for some $c_i \in \mathbf{K}_i(\langle x, y \rangle) \leq \langle x, y \rangle'$. Since $c_p \in \mathbf{K}_p(G) = 1$ and all binomial coefficients $\binom{p}{i}$ with $1 \leq i \leq p-1$ are divisible by p, we obtain the defining condition for regularity.

b) We may assume that $|G| > p$. Then we have $|G/\mathbf{K}_2(G)| \geq p^2$ and $|\mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)| \geq p$ for $2 \leq i \leq c(G)$. Setting $c = c(G)$, we have $|G| \geq p^{c+1}$. Thus, if $|G| \leq p^p$, then $c(G) < p$ and G is regular by a).

c) Let $H = \langle x, y\rangle$ be a subgroup of G. Since $H' \leq G'$, the subgroup H' is also cyclic. Let $H' = \langle h\rangle > 1$. Then $\mathbf{K}_3(H) < H'$, so $\mathbf{K}_3(H) \leq \langle h^p\rangle$. Applying the formula of Lemma 1.3 b) to $H/\mathbf{K}_3(H)$ yields

$$(xy)^p = x^p y^p [y,x]^{\binom{p}{2}} t$$

with $t \in \mathbf{K}_3(H) \leq \langle h^p\rangle$. Since $p > 2$, p is a divisor of $\binom{p}{2}$. Thus,

$$(xy)^p = x^p y^p h^{ap}$$

with $h \in H'$, and G is regular.

d) is trivial. □

Next, we describe the limits of the concept of regularity.

Theorem 10.3.

a) *Every regular 2-group is abelian.*

b) *If G is a regular 3-group with two generators, then G' is cyclic.*

(*Alperin has shown the following* (*Alperin [3]*): *Let G be a p-group with $p > 2$. If every subgroup of G that is generated by two elements has a cyclic commutator subgroup, then $G'' = 1$. So in particular, every regular 3-group has derived length at most 2. If $p > 3$, then the derived length of a regular p-group can become arbitrarily large, as we will show in §17.*)

c) (*Wielandt*) *There exists a regular 3-group G with $|G| = 3^5$ such that $G \times G$ is not regular. Hence, regularity is not inherited by direct products.*

d) *For every prime p, there exists an irregular group of order p^{p+1}, for example the Sylow p-subgroups of the symmetric groups S_{p^2}.*

Proof. a) Let G be a regular 2-group. For $H = \langle x, y\rangle \leq G$, we then have

$$(xy)^2 = xyxy = x^2 y[y,x]y = x^2 y^2 [y,x][y,x,y].$$

On the other hand,

$$(xy)^2 = x^2 y^2 c$$

for some $c \in (H')^2 \leq \Phi(H')$. It follows that

$$[y,x] = c[y,x,y]^{-1} \in \Phi(H')\mathbf{K}_3(H),$$

and therefore by Theorem 1.11 c),

$$H' = \langle [y,x], \mathbf{K}_3(H)\rangle \leq \Phi(H')\mathbf{K}_3(H).$$

If $1 < H'$, then $\mathbf{K}_3(H) < H'$, however, and so $\Phi(H')\mathbf{K}_3(H) < H'$ by Theorem 3.2 b). Thus, $G' = 1$ and G is abelian.

b) Assume for contradiction that G is a regular 3-group generated by two elements and G' is not cyclic. By passing to an appropriate quotient of G, we may assume that G' is elementary abelian of type $(3,3)$. Then $|G/\mathbf{C}_G(G')| \leq 3$, and we may choose one of the generating elements of G from $\mathbf{C}_G(G')$. So let $G = \langle a, b\rangle$ with $b \in \mathbf{C}_G(G')$ and $G' = \langle c\rangle \times \langle d\rangle$ with $c = [b, a]$. By Theorem 1.11 c), we have $G' = \langle c, \mathbf{K}_3(G)\rangle$. Thus, we may assume that $d \in \mathbf{K}_3(G) \leq \mathbf{Z}(G)$. Now we have

$$c^a = cd^n \text{ for some integer } n \text{ and } c^b = c.$$

Since G' has exponent 3, the regularity of G implies that

$$(ab)^3 = a^3b^3.$$

On the other hand, however,

$$\begin{aligned}(ab)^3 &= ababab = aabcabcb = a^2bac^ab^2c\\ &= a^3bcc^ab^2c = a^3b^3c^3d^n,\end{aligned}$$

and so $d^n = 1$. It follows that $\mathbf{K}_3(G) = 1$ and G' is cyclic, a contradiction.

c) The components G_i $(i = 1, 2)$ of the direct product will be given by the defining relations

$$a_i^{3^3} = b_i^{3^2} = 1, \quad a_i^{b_i} = a_i^4.$$

That is, the group G_i is a semidirect product of the cyclic group $\langle a_i\rangle$ of order 3^3 with the cyclic group $\langle b_i\rangle$ of order 3^2. We have $G_i' = \langle a_i^3\rangle$. So G_i' is cyclic and therefore G_i is regular by Theorem 10.2 c). We will construct two elements $h_1, h_2 \in G_1 \times G_2$ that do not satisfy the required condition for regularity. Namely, let

$$h_1 = a_1b_2^{-1}, \quad h_2 = b_1^{-1}a_2 \quad \text{and} \quad c = [h_1, h_2].$$

We have

$$a_i^{b_i^{-1}} = a_i^7.$$

Now we calculate

$$c = [h_1, h_2] = [a_1, b_1^{-1}][b_2^{-1}, a_2] = a_1^{-1}a_1^{b_1^{-1}}a_2^{-b_2^{-1}}a_2 = a_1^6a_2^{-6}$$

and

$$[c, h_1] = [a_1^6a_2^{-6}, a_1b_2^{-1}] = [a_2^{-6}, b_2^{-1}] = a_2^6a_2^{-6b_2^{-1}} = a_2^{-9}.$$

Now, a_2^{-9} is not a power of c. So $\langle h_1, h_2\rangle'$ is not cyclic and $\langle h_1, h_2\rangle$ is not regular by b).

d) Let $G = Z \wr_r Z$ be the regular wreath product of two cyclic groups of order p constructed according to Definition I.15.6 (pg. 104). By Theorem I.15.3 (pg. 102), G

has a faithful representation as a permutation group of degree p^2, so G is isomorphic to a Sylow p-subgroup of S_{p^2}. The group G has an elementary abelian normal subgroup $D = \langle w_1, \dots, w_p \rangle$ of order p^p, and $G = D\langle z \rangle$ for some z with $z^p = 1$ and $w_i^z = w_{i+1}$ ($i \bmod p$). So $G' \le D$ and thus $c^p = 1$ for all elements $c \in G'$. The regularity of G would require $(xy)^p = x^p y^p$. However, we have

$$(w_1 z^{-1})^p = w_1^{1+z+\cdots+z^{p-1}} z^{-p} = w_1 w_2 \cdots w_p \neq 1 = w_1^p z^{-p}.$$

Consequently, the group G is irregular. □

(We emphasize that the groups given in d) are not the only irregular groups of order p^{p+1}; see Example 10.15.)

Now we investigate the extent to which the properties of abelian p-groups carry over to regular p-groups.

Definition 10.4. In an arbitrary (not necessarily regular) p-group G we define the characteristic subgroups $\Omega_i(G)$ and $\mathfrak{V}_i(G)$ by

$$\Omega_i(G) = \langle g \in G \mid g^{p^i} = 1 \rangle \quad \text{and}$$
$$\mathfrak{V}_i(G) = \langle g^{p^i} \mid g \in G \rangle.$$

We let $|G/\mathfrak{V}_1(G)| = p^{\omega(G)}$.

Main Theorem 10.5. *Let G be a regular p-group and k a natural number.*

a) *If $x^{p^k} = y^{p^k} = 1$, then $(xy)^{p^k} = 1$ as well. Hence, the elements $x \in G$ with $x^{p^k} = 1$ constitute the characteristic subgroup $\Omega_k(G)$ of G.*

b) *For all $x, y \in G$, we have $x^{p^k} y^{p^k} = z^{p^k}$ for some $z \in G$ that depends on x and y. Hence, the p^k-th powers of elements of G constitute the characteristic subgroup $\mathfrak{V}_k(G)$ of G.*

Proof. a) We first prove the statement for $k = 1$ by induction on $|G|$.

Let $x^p = y^p = 1$. We need to show that $(xy)^p = 1$. We may assume that $G = \langle x, y \rangle$ and that G is nonabelian. Let

$$L = x^G = \langle x^g \mid g \in G \rangle,$$

and let U be a maximal subgroup of G with $x \in U$. By Theorem 2.3, $U \lhd G$ and thus $L \le U < G$. The proper subgroup L of G is generated by the conjugates x^g of x, which all have order p. Since L is also regular, it follows by the induction hypothesis that L has exponent p. We have $[x, y] = x^{-1}x^y$, where x and x^y are elements of L. Therefore, $[x, y]^p = 1$. By Theorem 1.11 a), the group G' is generated by the conjugates $[x, y]^g$ of $[x, y]$, which all have order p. Since $G' < G$, we have $c^p = 1$ for all $c \in G'$. Now the regularity of G yields

$$(xy)^p = x^p y^p \prod_i d_i^p = 1,$$

which is what we wanted to show.

For $k > 1$, we prove the claim in a) by induction on k.

If $x^{p^k} = y^{p^k} = 1$, then $x^p \in \Omega_{k-1}(G)$ and $y^p \in \Omega_{k-1}(G)$. Defining $\overline{x} = x\Omega_{k-1}(G)$ and $\overline{y} = y\Omega_{k-1}(G)$, we have $\overline{x}^p = \overline{y}^p = \overline{1}$ in the factor group $G/\Omega_{k-1}(G)$. The case $k = 1$, which we have already proved, shows that $(\overline{x}\,\overline{y})^p = \overline{1}$ as well. This means that $(xy)^p \in \Omega_{k-1}(G)$, and then $(xy)^{p^k} = 1$, as claimed.

b) Again we settle the case $k = 1$ by induction on $|G|$. We may again assume that $G = \langle x, y\rangle$ and G is nonabelian. Then $L = \langle xy, G'\rangle$ is a proper normal subgroup of G since $G' \le \Phi(G)$. By the induction hypothesis, the theorem holds for L. Since G is regular, we have

$$x^p y^p = (xy)^p \prod_i d_i^p$$

for some $d_i \in G' \le L$ and $xy \in L$. The right side of this equation is a p-th power of an element of L by the induction hypothesis, and so $x^p y^p$ is a p-th power of an element of G.

Now we handle the general case by induction on k. We have

$$x^{p^k} y^{p^k} = (x^p)^{p^{k-1}} (y^p)^{p^{k-1}} = t^{p^{k-1}}$$

for some $t \in \langle x^p, y^p\rangle$. By the $k = 1$ case, which we have already proved, $t = z^p$ for some $z \in G$. With this we have the desired result

$$x^{p^k} y^{p^k} = z^{p^k}. \qquad \square$$

(Note: We do not claim that $z = xy$ in Theorem 10.5 b)!)

Theorem 10.6. *Let G be a regular p-group. Then:*

a) *The equation $x^{p^k} = y^{p^k}$ is equivalent to $(xy^{-1})^{p^k} = 1$.*

b) $[x^{p^k}, y^{p^n}] = 1$ *is equivalent to* $[x, y]^{p^{k+n}} = 1$.

Proof. a) We again settle the case $k = 1$ by induction on $|G|$. We may assume that $G = \langle x, y\rangle$ and that G is nonabelian. We let $z = [x, y]$.

(1) It follows from $x^p = y^p$ that $[x^p, y] = 1$. Thus,

$$x^p = y^{-1}x^p y = (y^{-1}xy)^p = (xz)^p.$$

Since $\langle x, z\rangle \le \langle x, G'\rangle < G$, we obtain by the induction hypothesis that $x^p = (xz)^p$ implies $(xzx^{-1})^p = 1$, and so $z^p = 1$ as well.

By Theorem 1.11 a), the elements z^g ($g \in G$) of order p generate all of G'. So the regular group G' has exponent p by Theorem 10.5 a). The regularity condition yields

$$(xy^{-1})^p = x^p y^{-p} \prod_i d_i^p = x^p y^{-p} = 1,$$

as claimed.

(2) Conversely, we begin with $(xy^{-1})^p = 1$ and we must show that $x^p = y^p$. We have

$$(yx^{-1})^p = ((xy^{-1})^{-1})^p = 1$$

and

$$(y^{-1}x)^p = y^{-1}(xy^{-1})^p y = 1,$$

so by Theorem 10.5 a), we have $(y^{-1}xyx^{-1})^p = 1$ as well. The group G', which is generated by commutators $[y, x^{-1}]^g$ by Theorem 1.11 a), then has exponent p by Theorem 10.5 a). Now the regularity condition yields the claim

$$1 = (xy^{-1})^p = x^p y^{-p} \prod_i d_i^p = x^p y^{-p}.$$

The general case follows quickly again by induction on k. By our induction hypothesis, $x^{p^k} = y^{p^k}$ is equivalent to $(x^p y^{-p})^{p^{k-1}} = 1$, and this means that $x^p \equiv y^p \bmod \Omega_{k-1}(G)$. But by applying the $k = 1$ case, which we have already proved, to $G/\Omega_{k-1}(G)$, this is equivalent to $(xy^{-1})^p \equiv 1 \bmod \Omega_{k-1}(G)$. So we have that $(xy^{-1})^{p^k} = 1$.

b) Note that $[x^{p^k}, y^{p^n}] = x^{-p^k}(y^{-p^n}xy^{p^n})^{p^k}$. Thus, $[x^{p^k}, y^{p^n}] = 1$ is equivalent to $(y^{-p^n}xy^{p^n})^{p^k} = x^{p^k}$, and this is equivalent to $(x^{-1}y^{-p^n}xy^{p^n})^{p^k} = 1$ by a). This means

$$x^{-1}y^{-p^n}xy^{p^n} = (x^{-1}yx)^{-p^n}y^{p^n} \in \Omega_k(G).$$

Applying a) to $G/\Omega_k(G)$ shows that this is equivalent to

$$(y^{-1}x^{-1}yx)^{p^n} \in \Omega_k(G),$$

and this means

$$[y, x]^{p^{k+n}} = 1. \qquad \square$$

The following theorem reveals the close similarities between regular p-groups and abelian p-groups. In particular, we may introduce group invariants akin to those of abelian p-groups.

Theorem 10.7. *Let G be a regular p-group.*

a) *We have $|G/\Omega_k(G)| = |\mho_k(G)|$.*

b) *If we let $|\Omega_k(G)/\Omega_{k-1}(G)| = p^{\omega_k}$, then we have*

$$\omega_1 \geq \omega_2 \geq \cdots \geq \omega_\mu;$$

where p^μ is the exponent of G.

Proof. a) We consider the map $x \mapsto x^{p^k}$ from G onto $\mathfrak{V}_k(G)$. (This map is not necessarily an endomorphism of G!) By Theorem 10.6 a), $x^{p^k} = y^{p^k}$ if and only if $(xy^{-1})^{p^k} = 1$; that is, if and only if $xy^{-1} \in \Omega_k(G)$. Consequently, the elements in a fixed coset of $\Omega_k(G)$ generate the same p^k-th powers. Thus, the number $|\mathfrak{V}_k(G)|$ of p^k-th powers is equal to the index of $\Omega_k(G)$ in G.

b) We define $K = \Omega_{k+1}(G)/\Omega_{k-1}(G)$. Then

$$\Omega_1(K) = \Omega_k(G)/\Omega_{k-1}(G) :$$

Since $x^p \in \Omega_{k-1}(G)$ implies $x^{p^k} = 1$, we have $x \in \Omega_k(G)$. The converse holds as well.

The group K has exponent p^2. Thus, $\mathfrak{V}_1(K) \leq \Omega_1(K)$. Using a) gives

$$\begin{aligned} p^{\omega_{k+1}} &= |\Omega_{k+1}(G)/\Omega_k(G)| = |K/\Omega_1(K)| = |\mathfrak{V}_1(K)| \leq |\Omega_1(K)| \\ &= |\Omega_k(G)/\Omega_{k-1}(G)| = p^{\omega_k}. \end{aligned}$$

□

The following theorems connect the commutator structure of G, as expressed by the elements of the lower central series, with the power structure of G, as expressed by the characteristic subgroups $\Omega_k(G)$ and $\mathfrak{V}_k(G)$. For abelian p-groups, the following theorems are trivial.

Theorem 10.8. *Let G be a regular p-group, and let M and N be normal subgroups of G. We have:*

a) $[\mathfrak{V}_r(M), \mathfrak{V}_s(N)] = \mathfrak{V}_{r+s}([M, N])$.
b) $\mathbf{K}_j(\mathfrak{V}_r(G)) = \mathfrak{V}_{rj}(\mathbf{K}_j(G))$.
c) $[\mathfrak{V}_r(M), N, \underset{k}{\ldots}, N] = \mathfrak{V}_r([M, N, \underset{k}{\ldots}, N])$.
d) $\mathbf{K}_{j+1}(G) \leq \Omega_r(G)$ *is equivalent to* $\mathfrak{V}_r(G) \leq \mathbf{Z}_j(G)$.
e) $[M, G] \leq \Omega_r(G)$ *is equivalent to* $[M, \mathfrak{V}_r(G)] = 1$.
f) $[\Omega_r(G), \mathfrak{V}_s(G)] \leq \Omega_{r-s}(G)$, *where we define* $\Omega_k(G) = 1$ *for* $k \leq 0$. *In particular,*

$$[\Omega_r(G), \mathfrak{V}_r(G)] = 1,$$

so the p^r-th powers commute with the solutions of $x^{p^r} = 1$.
g) *For all $x, y \in G$ and all $m \geq 1$, we have*

$$(xy)^{p^m} = x^{p^m} y^{p^m} t^{p^m}$$

for some $t \in G'$.

Proof. a) We have

$$[\mathfrak{V}_r(M), \mathfrak{V}_s(N)] = \langle [m^{p^r}, n^{p^s}] \mid m \in M, n \in N \rangle$$

and

$$[m, n]^{p^{r+s}} \equiv 1 \mod \mathfrak{V}_{r+s}([M, N]).$$

By Theorem 10.6 b), it follows that

$$[m^{p^r}, n^{p^s}] \equiv 1 \mod \mho_{r+s}([M,N]).$$

So we have $[\mho_r(M), \mho_s(N)] \leq \mho_{r+s}([M,N])$.

Conversely, we have

$$[m^{p^r}, n^{p^s}] \equiv 1 \mod [\mho_r(M), \mho_s(N)],$$

so that, using Theorem 10.6 b),

$$[m,n]^{p^{r+s}} \equiv 1 \mod [\mho_r(M), \mho_s(N)].$$

Since $[M,N]$ is generated by commutators $[m,n]$ with $m \in M$ and $n \in N$, $[M,N]/[\mho_r(M), \mho_s(N)]$ is generated by elements of order at most p^{r+s}. So by the regularity of G and Theorem 10.5 a), elements of $[M,N]/[\mho_r(M), \mho_s(N)]$ have order at most p^{r+s}. We obtain

$$\mho_{r+s}([M,N]) \leq [\mho_r(M), \mho_s(N)].$$

b) The claim is trivial for $j = 1$. We prove the general case by induction on j. By the induction hypothesis and a), we have

$$\begin{aligned} \mathbf{K}_{j+1}(\mho_r(G)) &= [\mathbf{K}_j(\mho_r(G)), \mho_r(G)] = [\mho_{rj}(\mathbf{K}_j(G)), \mho_r(G)] \\ &= \mho_{rj+r}([\mathbf{K}_j(G), G]) = \mho_{r(j+1)}(\mathbf{K}_{j+1}(G)). \end{aligned}$$

c) For $k = 1$, the claim follows from a) by setting $s = 0$. For $k > 1$, we prove the claim by induction on k:

$$\begin{aligned} [\mho_r(M), N, \underset{k}{\ldots}, N] &= [\mho_r([M, N, \underset{k-1}{\ldots}, N]), N] \qquad \text{(induction hypothesis)} \\ &= \mho_r([M, N, \underset{k}{\ldots}, N]) \qquad \text{(by a)).} \end{aligned}$$

d) The following statements are clearly equivalent:

$$\begin{aligned} \mathbf{K}_{j+1}(G) &\leq \Omega_r(G); \\ \mho_r(\mathbf{K}_{j+1}(G)) &= 1; \\ [\mho_r(G), G, \underset{j}{\ldots}, G] &= 1 \qquad \text{(by c));} \\ \mho_r(G) &\leq \mathbf{Z}_j(G). \end{aligned}$$

e) By a), $[G,M] \leq \Omega_r(G)$ is equivalent to

$$1 = \mho_r([G,M]) = [\mho_r(G), M].$$

f) From $[G, \Omega_r(G)] \leq \Omega_r(G)$, it follows by setting $M = \Omega_r(G)$ in e) that

$$[\Omega_r(G), \mho_r(G)] = 1.$$

This proves the claim in f) in the special case $r = s$. In the general case we proceed as follows: For $r > s$, we have

$$\Omega_s(G/\Omega_{r-s}(G)) = \Omega_r(G)/\Omega_{r-s}(G).$$

Applying what we have already proved to $G/\Omega_{r-s}(G)$ gives

$$\begin{aligned}[\Omega_r(G)/\Omega_{r-s}(G), \mho_s(G)\Omega_{r-s}(G)/\Omega_{r-s}(G)] \\ = [\Omega_s(G/\Omega_{r-s}(G)), \mho_s(G/\Omega_{r-s}(G))] = 1.\end{aligned}$$

Thus

$$[\Omega_r(G), \mho_s(G)] \leq \Omega_{r-s}(G)$$

for $r > s$. For $r \leq s$, we have

$$[\Omega_r(G), \mho_s(G)] \leq [\Omega_r(G), \mho_r(G)] = 1.$$

g) By the definition of regularity, we have for $m = 1$ that

$$(xy)^p = x^p y^p \prod_i d_i^p$$

for some $d_i \in \langle x, y\rangle' \leq G'$. Since G' is also regular, there exists a $t \in G'$ such that $\prod_i d_i^p = t^p$ by Theorem 10.5 b).

We prove the claim for $m > 1$ by induction on m. Suppose for induction that

$$(xy)^{p^m} = x^{p^m} y^{p^m} s^{p^m}$$

for some $s \in G'$. It follows from the regularity of G that

$$(xy)^{p^{m+1}} = (x^{p^m} y^{p^m})^p s^{p^{m+1}} \prod_i k_i^p = x^{p^{m+1}} y^{p^{m+1}} \prod_j l_j^p s^{p^{m+1}} \prod_i k_i^p,$$

where $s^{p^{m+1}} \in \mho_{m+1}(G')$ and, furthermore, by a)

$$k_i \in \langle x^{p^m}, y^{p^m}, s^{p^m}\rangle' \leq [\mho_m(G), \mho_m(G)] = \mho_{2m}(G') \leq \mho_{m+1}(G')$$

and in the same way

$$l_j \in \langle x^{p^m}, y^{p^m}\rangle' \leq [\mho_m(G), \mho_m(G)] \leq \mho_{m+1}(G').$$

Then using Theorem 10.5 b), we have for some $t \in G'$ that

$$(xy)^{p^{m+1}} = x^{p^{m+1}} y^{p^{m+1}} t^{p^{m+1}}.$$

□

Lemma 10.9. *Let A be an abelian normal subgroup of an arbitrary group G.*
a) *For all $a_1, a_2 \in A$ and all $g \in G$, we have*

$$[a_1 a_2, g] = [a_1, g][a_2, g].$$

b) *For $a \in A$ and $g \in G$ and all natural numbers n, we have*

$$(ga)^n = g^n a^n \prod_{i=2}^{n} [a, g, \underset{i-1}{\ldots}, g]^{\binom{n}{i}}.$$

Proof. a) follows immediately from Lemma 1.2 c) considering that $[a_1, g] \in A$. (See Exercise 2.)

b) The claim is clearly true if $n = 1$. We prove the general case by induction on n. Suppose the formula has been proved for $n - 1$. Then using a), it follows that

$$\begin{aligned}
(ga)^n &= g^{n-1} a^{n-1} \left(\prod_{i=2}^{n-1} [a, g, \underset{i-1}{\ldots}, g]^{\binom{n-1}{i}} \right) ga \\
&= g^n a^{n-1} [a, g]^{n-1} \left(\prod_{i=2}^{n-1} [a, g, \underset{i-1}{\ldots}, g]^{\binom{n-1}{i}} \prod_{i=2}^{n-1} [a, g, \underset{i}{\ldots}, g]^{\binom{n-1}{i}} \right) a \\
&= g^n a^n \left(\prod_{i=2}^{n-1} [a, g, \underset{i-1}{\ldots}, g]^{\binom{n-1}{i} + \binom{n-1}{i-1}} \right) [a, g, \underset{n-1}{\ldots}, g] \\
&= g^n a^n \prod_{i=2}^{n} [a, g, \underset{i-1}{\ldots}, g]^{\binom{n}{i}}.
\end{aligned}$$

□

In a very special case, we can prove a converse of Theorem 10.2 a):

Theorem 10.10. *Suppose the p-group G has an abelian normal subgroup A of index p. If G is regular, then G has class at most $p - 1$.*

Proof. Let G be a minimal counterexample. Then G has class p, and we have $|\mathbf{K}_p(G)| = p$. Let $G = \langle A, g \rangle$. Since the set $\{A, g\}$ is a generating set of G, the subgroup $\mathbf{K}_p(G)$ is generated by a commutator of the form $[x_1, \ldots, x_p]$ $(\neq 1)$ with $x_i = g$ or $x_i \in A$ by Theorem 1.11 a). Then we must have $[x_1, x_2] = [a, g]^{\pm 1}$ for some $a \in A$, and we have $x_3 = \cdots = x_p = g$. By Lemma 6.8, it follows that

$$[x_1, \ldots, x_p] = [a, g, \underset{p-1}{\ldots}, g]^{\pm 1}.$$

On the other hand, $G' = [G, A]$, and since $\mho_1(G) \leq A$, Theorem 10.8 a) implies

$$\mho_1(G') = [\mho_1(G), A] = 1.$$

Consequently, the regularity of G requires that $(ga)^p = g^p a^p$ for all $a \in A$. Then by Lemma 10.9 b),

$$g^p a^p = (ga)^p = g^p a^p [a,g]^{\binom{p}{2}} \cdots [a,g,\underset{p-2}{\ldots},g]^{\binom{p}{p-1}} [a,g,\underset{p-1}{\ldots},g]$$
$$= g^p a^p [a,g,\underset{p-1}{\ldots},g].$$

This yields $[a,g,\underset{p-1}{\ldots},g] = 1$, which contradicts

$$\mathbf{K}_p(G) = \langle [a,g,\underset{p-1}{\ldots},g] \rangle \neq 1.$$

So G has class at most $p-1$. □

In preparation for a deeper regularity criterion, we prove two lemmas:

Lemma 10.11. *Let G be an irregular p-group of order p^{p+1}. If G has class p, then $|G/G'| = |G/\Phi(G)| = p^2$ and $|\mathbf{Z}(G)| = p$.*

Proof. By Theorem 10.2 a), $c(G) \geq p$. Therefore, we have

$$G = \mathbf{K}_1(G) > \mathbf{K}_2(G) > \cdots > \mathbf{K}_p(G) > \mathbf{K}_{p+1}(G) \geq 1.$$

Certainly $|G/\mathbf{K}_2(G)| \geq p^2$ by Lemma 7.1 d). Hence,

$$|\mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)| = p$$

for $2 \leq i \leq p$ and $\mathbf{K}_{p+1}(G) = 1$. The claims $c(G) = p$ and $|G/G'| = p^2$ have been proved. Since G is not cyclic and $G' \leq \Phi(G)$, we also have

$$p^2 \leq p^{d(G)} = |G/\Phi(G)| \leq |G/G'| = p^2.$$

So $|G/\Phi(G)| = p^2$.

If $|\mathbf{Z}(G)| > p$, then $|G/\mathbf{Z}(G)| \leq p^{p-1}$ and therefore

$$c(G/\mathbf{Z}(G)) \leq p-2.$$

This yields the contradiction $c(G) \leq p-1$. □

Lemma 10.12. *If $\omega(G) < p$ and H is a subgroup of G, then $\omega(H) \leq \omega(G)$.*
(*For the meaning of $\omega(G)$, see Definition 10.4.*)

Proof. Assume for contradiction that $\omega(H) > \omega(G)$. We may assume that $[G:H] = p$. Thus,

$$p^{\omega(G)} = |G/\mho_1(G)| < |H/\mho_1(H)| = \frac{1}{p}|G/\mho_1(G)|\,|\mho_1(G)/\mho_1(H)|.$$

From this it follows that $|\mho_1(G)/\mho_1(H)| \geq p^2$.

Therefore, there exists by Theorem 7.2 d) a normal subgroup L of G such that

$$\mho_1(H) \leq L < \mho_1(G)$$

and $|\mho_1(G)/L| = p^2$.

Now we consider the factor group G/L, so we may assume without loss of generality that $L = 1$. Then $\mho_1(H) = 1$, which means H has exponent p and $|H| = p^{\omega(H)}$. Furthermore, we have $|\mho_1(G)| = p^2$. Therefore, $|G| = p^{\omega(G)+2}$. Now it follows that

$$p^{\omega(H)} = |H| = \frac{|G|}{p} > p^{\omega(G)}.$$

If G were regular, then since $\Omega_1(H) \leq \Omega_1(G)$, we would have

$$p^{\omega(G)} = |G/\mho_1(G)| = |\Omega_1(G)| \geq |\Omega_1(H)| = |H/\mho_1(H)| = p^{\omega(H)},$$

which contradicts the assumption that $\omega(H) > \omega(G)$. So G is not regular. Thus, $\omega(G) + 2 \geq p + 1$ by Theorem 10.2 b), and so $\omega(G) \geq p - 1$. On the other hand, $\omega(G) < p$ by our assumption, so we must have $\omega(G) = p - 1$ and $|G| = p^{p+1}$.

Since $|G/H| = p$, it is certainly true that $G = \langle H, x\rangle$ for every $x \in G$ that does not lie in H. By Lemma 10.11, we have $|G/\Phi(G)| = p^2$. By the Burnside Basis Theorem 3.15, we thus have $G = \langle x, y\rangle$ for every $y \in H - \Phi(G)$. It follows from $|\mho_1(G)| = p^2$ and $|\mathbf{Z}(G)| = p$ that $\mathbf{Z}(G) < \mho_1(G)$. Hence, there exists some $x_1 \in G$ with $x_1^p \notin \mathbf{Z}(G)$. Since H has exponent p, certainly $x_1 \notin H$. Thus, we may use this x_1 as our x from above.

Since $G' < H$, we have $\mho_1(G') \leq \mho_1(H) = 1$. The Hall–Petresco identity of Theorem 9.4 yields

$$(xy)^p = x^p y^p \prod_{i=2}^{p-1} c_i^{\binom{p}{i}} c_p \quad \text{with} \quad c_i \in G' \quad \text{and} \quad c_p \in \mathbf{K}_p(G),$$

where

$$c_i^{\binom{p}{i}} \in \mho_1(G') = 1 \quad (2 \leq i \leq p-1)$$

and $y^p \in \mho_1(H) = 1$. Thus, we obtain $(xy)^p = x^p c_p$. Therefore, $x^p c_p \in \mho_1(G)$. Since $c_p \in \mathbf{K}_p(G) \leq \mathbf{Z}(G)$ and $x^p \notin \mathbf{Z}(G)$, $x^p c_p$ does not lie in $\mathbf{Z}(G)$ either. So from $|\mho_1(G)| = p^2$ it follows that

$$\mho_1(G) = \langle x^p, \mathbf{Z}(G)\rangle = \langle (xy)^p, \mathbf{Z}(G)\rangle.$$

However, from this we obtain

$$\mathbf{C}_G(\mho_1(G)) \geq \langle x, xy\rangle = \langle x, y\rangle = G.$$

This gives the contradiction $\mho_1(G) \leq \mathbf{Z}(G)$. □

The next theorem is the aforementioned regularity criterion.

Theorem 10.13. *If $\omega(\mathbf{K}_i(G)) \leq p - i$ for some $1 \leq i \leq p$, then G is regular.*

(*In the special case $i = 2$, the theorem says: If $\omega(G') \le p-2$, then G is regular. This encompasses Theorem 10.2 c), since $G' = \mathbf{K}_2(G)$ is cyclic there and so for $p > 2$ it is certainly true that $\omega(G') = 1 \le p-2$.*)

Proof. We prove the theorem by induction on $|G|$. If $H < G$, then $\mathbf{K}_i(H) \le \mathbf{K}_i(G)$. Since $\omega(\mathbf{K}_i(G)) \le p - i < p$, we have the inequality $\omega(\mathbf{K}_i(H)) \le \omega(\mathbf{K}_i(G))$ by Lemma 10.12. So H is regular by the induction hypothesis. Furthermore, we may assume that G can be generated by two elements; otherwise, every pair of elements of G lies in a regular subgroup, and then G is regular. Thus, we have $|G/\Phi(G)| = p^2$.

1) We prove $\mathbf{K}_p(G) \le \mathfrak{V}_1(G')$:

First assume that $i > 1$. Then $\mathbf{K}_i(G) < G$, and therefore $\mathbf{K}_i(G)$ is regular. Furthermore, we have

$$|\mathbf{K}_i(G)/\mathfrak{V}_1(\mathbf{K}_i(G))| = p^{\omega(\mathbf{K}_i(G))} \le p^{p-i}.$$

It follows that $\mathbf{K}_p(G) = [\mathbf{K}_i(G), G, \underset{p-i}{\ldots}, G] \le \mathfrak{V}_1(\mathbf{K}_i(G))$.

Now the difficult case $i = 1$: We may assume that $\mathfrak{V}_1(G') = 1$, and we must show that $\mathbf{K}_p(G) = 1$. Assume for contradiction that $\mathbf{K}_p(G) \ne 1$. Since G' has exponent p, we have $|G'| = p^{\omega(G')}$, and $\omega(G') \le \omega(G) \le p-1$ by Lemma 10.12. On the other hand, since $c(G) \ge p$ we have $|G'| \ge p^{p-1}$. Altogether we obtain $\omega(G) = p-1$ and $|G'| = p^{p-1}$.

For all $x, y \in G$, we have

$$y^{-p}x^{-1}y^p x = y^{-p}(x^{-1}yx)^p = y^{-p}(yw)^p \quad \text{with} \quad w = [y,x] \in G'.$$

Since $\langle y, w\rangle \le \langle y, G'\rangle < G$, the subgroup $\langle y, w\rangle$ is certainly regular. Thus, we have

$$(yw)^p = y^p w^p \prod_i d_i^p$$

with w^p and d_i^p in $\mathfrak{V}_1(G') = 1$. There remains only $(yw)^p = y^p$, and hence $[y^p, x] = 1$. We have shown that $\mathfrak{V}_1(G) \le \mathbf{Z}(G)$. Using Theorem 3.14 a), it follows that

$$\begin{aligned} |G/\mathfrak{V}_1(G)| = p^{\omega(G)} = p^{p-1} &= |G/\mathfrak{V}_1(G)G'|\,|\mathfrak{V}_1(G)G'/\mathfrak{V}_1(G)| \\ &= |G/\Phi(G)|\,|G'/G' \cap \mathfrak{V}_1(G)| = p^2|G'/G' \cap \mathfrak{V}_1(G)|. \end{aligned}$$

Hence, we obtain

$$|G'/G' \cap \mathfrak{V}_1(G)| = p^{p-3},$$

so that

$$\mathbf{K}_{p-1}(G) = \mathbf{K}_{2+p-3}(G) \le G' \cap \mathfrak{V}_1(G) \le \mathbf{Z}(G).$$

Thus $\mathbf{K}_p(G) = 1$, which is what we wanted to show.

2) Now we can prove the regularity criterion:

If $\mathbf{K}_p(G) = 1$, then G is regular by Theorem 10.2 a) and we are finished. So let $1 \neq r \in \mathbf{K}_p(G) \cap \mathbf{Z}(G)$. Since

$$\begin{aligned}\mho_1(\mathbf{K}_i(G/\langle r\rangle)) &= \mho_1(\mathbf{K}_i(G)/\langle r\rangle) \quad (\text{using } i \leq p)\\ &= \mho_1(\mathbf{K}_i(G))\langle r\rangle/\langle r\rangle,\end{aligned}$$

we certainly have

$$\omega(\mathbf{K}_i(G/\langle r\rangle)) \leq \omega(\mathbf{K}_i(G)) \leq p - i.$$

So $G/\langle r\rangle$ is regular by the induction hypothesis. For all $x, y \in G$ such that $G = \langle x, y\rangle$ (and we need only consider such x, y since all proper subgroups of G are regular), we have

$$(xy)^p = x^p y^p \prod_i d_i^p r^a,$$

for some $d_i \in \langle x, y\rangle'$ and some integer a. We have by a) that

$$r^a \in \mathbf{K}_p(G) \leq \mho_1(G'),$$

and so $r^a = \prod_i l_i^p$ with $l_i \in G' = \langle x, y\rangle'$.

The regularity of G is proved. □

The following theorem is a useful consequence of Theorem 10.13.

Theorem 10.14. *Let G be a p-group. If G does not have a normal subgroup N of exponent p with $|N| \geq p^{p-1}$, then G is regular.*

Proof. If $\mho_1(G) = 1$, then it is clear that G is regular. If $\mho_1(G) \neq 1$, then G has a normal subgroup M with $|M| = p$ and $M \leq \mho_1(G)$. Note that the quotient G/M also satisfies our hypothesis:

Namely, assume for contradiction that G/M has a normal subgroup K/M of exponent p and order p^{p-1}. Then K has order p^p and is regular by Theorem 10.2 b). But then

$$|\Omega_1(K)| = |K/\mho_1(K)| \geq |K/M| = p^{p-1}$$

by Theorem 10.7 a), and $\Omega_1(K) \trianglelefteq G$ as well. This contradicts our assumptions on G. Thus, G/M satisfies the hypothesis of the theorem. By induction, we may conclude that G/M is regular. Since $M \leq \mho_1(G)$, it follows that

$$p^{\omega(G)} = |G/\mho_1(G)| = |G/M\,/\mho_1(G/M)| = |\Omega_1(G/M)| \leq p^{p-2}.$$

Hence, we have $\omega(G) \leq p - 2$. The regularity of G then follows from Theorem 10.13. □

Example 10.15 (Blackburn). For every prime p, there exists an irregular p-group G with the following properties:

(1) The elements of G of order p constitute a characteristic subgroup $\Omega_1(G)$ of G with $|\Omega_1(G)| = p^{p-1}$.
(2) The p-th powers of elements of G constitute a characteristic subgroup $\mho_1(G)$ of G with $|\mho_1(G)| = p$.
(3) G has exponent p^2 and order p^{p+1}.

(A complete classification of the irregular p-groups of order p^{p+1} is still unknown. However, Blackburn's theory of the groups of maximal class, which we will discuss in §14, gives much information about these groups.)

Proof. Let

$$A = \langle a_1 \rangle \times \langle a_2 \rangle \times \cdots \times \langle a_{p-1} \rangle$$

be an abelian group with $|a_1| = p^2$ and $|a_i| = p$ for $2 \le i \le p-1$. Then the map α defined by

$$a_i^\alpha = a_i a_{i+1} \quad (1 \le i \le p-2), \quad a_{p-1}^\alpha = a_{p-1} a_1^{-p}$$

is an automorphism of A. For every $a \in A$, we have $(a^{\alpha-1})^p = 1$. Noting that

$$\binom{p-2}{i} \equiv (-1)^i (i+1) \mod p,$$

we have

$$\begin{aligned} a^{(\alpha-1)^{p-1}} &= a^{(\alpha-1)(\alpha^{p-2}+2\alpha^{p-3}+\cdots+(p-2)\alpha+p-1)} \\ &= a^{\alpha^{p-1}+\alpha^{p-2}+\cdots+\alpha+1} a^{-p}. \end{aligned}$$

We also have

$$a_1^{(\alpha-1)^{p-1}} = a_2^{(\alpha-1)^{p-2}} = \cdots = a_{p-1}^{\alpha-1} = a_1^{-p}$$

and

$$a_j^{(\alpha-1)^{p-1}} = \cdots = a_{p-1}^{(\alpha-1)^j} = (a_1^{-p})^{(\alpha-1)^{j-1}} = 1 \quad \text{for} \quad j \ge 2.$$

It then follows that

$$a^{1+\alpha+\cdots+\alpha^{p-1}} = 1$$

for all $a \in A$.

Now we consider the group $G = A\langle x \rangle$, where $a^x = a^\alpha$ for all $a \in A$ and $x^p = a_1^p$; since $(a_1^p)^x = a_1^p$, this group has order p^{p+1} by Example I.14.8 (pg. 96). Then

$$(x^j a)^p = a^{1+\alpha+\cdots+\alpha^{p-1}} x^{jp} = a_1^{jp}.$$

Hence, $\mho_1(G) = \langle a_1^p \rangle$ has order p. On the other hand, $(x^j a)^p = a_1^{jp} \neq 1$ for $j \not\equiv 0 \bmod p$. Thus, it follows that

$$|\Omega_1(G)| = |\Omega_1(A)| = p^{p-1}.$$

Since

$$|\mho_1(G)| = p \neq |G/\Omega_1(G)| = p^2,$$

the group G is not regular by Theorem 10.7 a). □

Exercises

26) [6]

27) (Grün [200]) If G_1 and G_2 are regular and $\mho_1(G_1') = 1$, then $G_1 \times G_2$ is regular.

11 Metacyclic p-groups

Definition 11.1. A group G is called metacyclic if G has a cyclic normal subgroup with cyclic factor group. It is clear that epimorphic images and subgroups of metacyclic groups are again metacyclic.

The metacyclic p-groups have a simple description:

Theorem 11.2. *If G is a metacyclic p-group, then G has defining relations of the form*

$$x^{p^a} = 1, \quad y^{p^b} = x^{p^c}, \quad x^y = x^k,$$

where $c \geq 0$, $k^{p^b} \equiv 1 \bmod p^a$ and $p^c(k-1) \equiv 0 \bmod p^a$. So $|G| = p^{a+b}$.

Proof. Let $G/\langle x \rangle$ by cyclic with $|x| = p^a$ and $|G/\langle x \rangle| = p^b$. Let $y_1\langle x \rangle$ be a generating coset of the cyclic factor group $G/\langle x \rangle$. Then $G = \langle x, y_1 \rangle$. We have

$$y_1^{p^b} = x^r$$

for some integer r. Let $r = tp^c$, where $t \not\equiv 0 \bmod p$. We solve the congruence $st \equiv 1 \bmod p^a$ and define $y = y_1^s$. Then we have

$$y^{p^b} = (y_1^s)^{p^b} = x^{sr} = x^{p^c}.$$

[6] The exercise originally given here is false. Correction 23 from errata.

Finally, we have

$$x = x^{y^{p^b}} = x^{k^{p^b}},$$

thus $k^{p^b} \equiv 1 \bmod p^a$ and

$$x^{p^c} = (x^{p^c})^y = x^{kp^c},$$

and so $p^c(k-1) \equiv 0 \bmod p^a$. □

Lemma 11.3 (Blackburn [37]). *Let G be a p-group. The group G is metacyclic if and only if $G/\Phi(G')\mathbf{K}_3(G)$ is metacyclic.*

Proof. If G is metacyclic, then clearly $G/\Phi(G')\mathbf{K}_3(G)$ is metacyclic as well. Conversely, assume $G/\Phi(G')\mathbf{K}_3(G)$ is metacyclic. We show:

(1) We have $|G'/\Phi(G')\mathbf{K}_3(G)| = p$:

Since $G/\Phi(G')\mathbf{K}_3(G)$ is metacyclic, it is generated by two elements. Furthermore,

$$\Phi(G')\mathbf{K}_3(G) \leq G' \leq \Phi(G).$$

Hence, G is itself generated by two elements by the Burnside Basis Theorem 3.15. Then $G'/\mathbf{K}_3(G)$ is cyclic by Theorem 1.11 c). From this follows the claim that $|G'/\Phi(G')\mathbf{K}_3(G)| = p$.

(2) If N is a normal subgroup of G such that $|G'/N| = p$, then $N = \Phi(G')\mathbf{K}_3(G)$:

Since N is a maximal subgroup of G', we certainly have $\Phi(G') \leq N$. Furthermore, since $|G'/N| = p$, we have $G'/N \leq \mathbf{Z}(G/N)$ by Theorem 7.2 b) and therefore $[G', G] = \mathbf{K}_3(G) \leq N$. Altogether we have $\Phi(G')\mathbf{K}_3(G) \leq N$. By (1), we have equality here.

(3) Assume $\Phi(G')\mathbf{K}_3(G) \neq 1$ (otherwise there is nothing to prove), and let K be a normal subgroup of G such that $K \leq \Phi(G')\mathbf{K}_3(G)$ and $|K| = p$. Then

$$\Phi((G/K)')\mathbf{K}_3(G/K) = \Phi(G')\mathbf{K}_3(G)/K,$$

so our hypotheses are again satisfied in G/K. We may therefore assume that G/K is metacyclic. Hence, there exists a normal subgroup L of G such that G/L and L/K are cyclic. Since $K \leq \mathbf{Z}(G)$, L is abelian. If L is cyclic, then we are done. Otherwise, $L = M \times K$ for some cyclic M of order p^a. Certainly G/G' is not cyclic, since otherwise G itself would be cyclic by Lemma 7.1 c). Since G/L is cyclic and $\mathbf{K}_3(G) < G'$, we have

$$1 < \Phi(G')\mathbf{K}_3(G) < G' < L.$$

It follows that $|L| \geq p^3$. We have $\mho_1(M) = \mho_1(L) \trianglelefteq G$. Now we define $N = \Omega_{a-1}(L) = \Omega_{a-1}(M) \times K$. Then N is a normal subgroup of G. Since M is a cyclic group of order p^a, we have $\Omega_{a-1}(M) = \mho_1(M)$. The factor group G/N is abelian since G/L is cyclic and L/N has order p, so it lies in the center of G/N.

Thus, we have $G' \leq N$. Now

$$|G'/G' \cap \mho_1(M)| = |G'\mho_1(M)/\mho_1(M)| \leq |N/\mho_1(M)| = p.$$

Then by (2),

$$G' = G' \cap \mho_1(M) \quad \text{or} \quad G' \cap \mho_1(M) = \Phi(G')\mathbf{K}_3(G).$$

In both cases we obtain

$$K \leq \Phi(G')\mathbf{K}_3(G) \leq M,$$

which is certainly not the case. So L is in fact cyclic and G is metacyclic. □

Now we may prove a useful characterization of metacyclic groups:

Theorem 11.4 (Huppert [256]). *Let G be a p-group and $p > 2$. The group G is metacyclic if and only if $|G/\mho_1(G)| \leq p^2$. If $p > 2$, the regular p-groups with $\omega(G) = 2$ are exactly those p-groups that are metacyclic but not cyclic.*

Proof. 1) Assume G is metacyclic. Then $G/\mho_1(G)$ is metacyclic of exponent p, so $|G/\mho_1(G)| \leq p^2$.

2) Assume $|G/\mho_1(G)| \leq p^2$. By Lemma 11.3, it is enough to prove that $G/\Phi(G')\mathbf{K}_3(G)$ is metacyclic. So we may assume that $\Phi(G')\mathbf{K}_3(G) = 1$. Since $\mho_1(G) \leq \Phi(G)$, we have

$$|G/\Phi(G)| \leq |G/\mho_1(G)| \leq p^2.$$

So G can be generated by two elements by the Burnside Basis Theorem 3.15. By Theorem 1.11 c), $G'/\mathbf{K}_3(G) = G'$ is therefore cyclic. Since $\Phi(G') = 1$, the subgroup G' has order at most p. Let $G = \langle x, y\rangle$, so $G' = \langle [x, y]\rangle$. We conclude from $|G/\mho_1(G)| \leq p^2$ that $G' \leq \mho_1(G)$. Since G has class at most 2, G is regular by Theorem 10.2 a). Since $G' \leq \mho_1(G)$, $[x, y]$ is a product of p-th powers, so is also a p-th power. Let

$$[x, y] = s^{p^a},$$

where s is not itself a p-th power. So s does not lie in $\mho_1(G)$. Now, $G' \leq \langle s\rangle \trianglelefteq G$. If $|G/\Phi(G)| = p$, then G is cyclic by the Burnside Basis Theorem 3.15. If $|G/\Phi(G)| = p^2$, then we have $\Phi(G) = \mho_1(G)$. Since s does not lie in $\Phi(G)$, we may generate G with s and one additional element t by Theorem 3.15. Then $G/\langle s\rangle$ is cyclic and G is metacyclic. □

Theorem 11.5 (Huppert [256]). *Let $G = \langle x\rangle\langle y\rangle$ be a p-group with $p > 2$. Then G is metacyclic. (If $p = 2$, the claim no longer holds; see Exercise 28 and Blackburn [38].)*

Proof. The factor group $G/\mho_1(G)$ is the product of two cyclic groups and has exponent p. Therefore, $|G/\mho_1(G)| \leq p^2$. By Theorem 11.4, G is metacyclic. □

Next, we would like to extend the counting considerations of §8:

Theorem 11.6 (Huppert [256]). *Let $|G| = p^n$ with $p > 3$. If G has at most $p + 1$ subgroups of order p^r for some r with $1 \le r \le n - 2$, then G is metacyclic.*

Proof. 1) First we consider the case $r = 1$. If N is a normal subgroup of G of exponent p, then N also has at most $p + 1$ subgroups of order p. So $|N| \le p^2$. Since $p > 3$, G is regular by Theorem 10.14. Thus, it follows by Theorem 10.7 a) that

$$|G/\mho_1(G)| = |\Omega_1(G)| \le p^2,$$

and then G is metacyclic by Theorem 11.4.

2) Now assume that $r \ge 2$. Since $r \le n - 2$, there exists a subgroup H of G with order p^{r+2} by Theorem 7.2 e). Note that $|H/\mho_1(H)| \le p^2$: Otherwise there would exist a normal subgroup K of H with $|H/K| = p^3$ such that H/K has exponent p. The $1 + p + p^2$ subgroups of H/K with order p would correspond to the same number of subgroups of H with order p^r, contradicting our assumption.

Hence, $|H/\mho_1(H)| \le p^2$, and H is then metacyclic by Theorem 11.4. If all the subgroups of G with order p^{r+2} are cyclic, then G is itself cyclic by Theorem 8.4 and we are done. So we may assume that there exists a noncyclic subgroup H of order p^{r+2}. Let $s_r(H)$ be the number of subgroups of H with order p^r. Since H is not cyclic, $s_r(H) > 1$ by Theorem 8.4. By Theorem 8.8 a), $s_r(H) \equiv 1 \bmod p$. Consequently,

$$1 + p \le s_r(H) \le s_r(G) \le 1 + p.$$

So every subgroup of G with order p^r lies in H. Since every subgroup of G with order p lies in at least one subgroup of order p^r, H also contains all subgroups of G with order p. Since H is regular, we have

$$|H/\mho_1(H)| = |\Omega_1(H)| = p^2.$$

So H, and thus G itself, has exactly $p + 1$ subgroups of order p. So G is metacyclic by 1). □

Remark 11.7. The group G of order 3^4 from Example 10.15 shows that the conclusion of Theorem 11.6 no longer holds if $p = 3$. Namely, G has exactly four subgroups of order 3 since $|\Omega_1(G)| = 3^2$. But G is not regular, so G is also not metacyclic.

Theorem 11.8 (Huppert). *Let G be a p-group with $p > 3$, and assume that G is not metacyclic. Let $s_1(G)$ be the number of subgroups of G with order p. Then*

$$s_1(G) \equiv 1 + p + p^2 \mod p^3.$$

Proof. We proceed with the help of Hall's Enumeration Principle, Theorem 8.6. Accordingly, we have

$$s_1(G) = \sum_{M_1 \in \mathscr{S}_1'} s_1(M_1) - p \sum_{M_2 \in \mathscr{S}_2'} s_1(M_2) + p^3 \sum_{M_3 \in \mathscr{S}_3'} s_1(M_3) + \cdots.$$

1) First we assume that no $M_1 \in \mathscr{S}_1'$ is metacyclic, and thus no $M_2 \in \mathscr{S}_2'$ is cyclic. By the induction hypothesis, we conclude

$$s_1(M_1) \equiv 1 + p + p^2 \mod p^3.$$

Furthermore, by Kulakoff's Theorem 8.8 b),

$$s_1(M_3) \equiv 1 + p \mod p^2.$$

Now first assume that $d = d(G) \geq 3$. Since

$$f(d, d-1) = 1 + p + \cdots + p^{d-1} \equiv 1 + p + p^2 \mod p^3$$

it follows that

$$\begin{aligned} s_1(G) &\equiv (1 + p + p^2) f(d, d-1) - p(1+p) f(d, d-2) \\ &\equiv 1 + p + p^2 \mod p^3. \end{aligned}$$

For $d = 2$, we conclude as follows:

Since $\Phi(G)$ is the only $M_2 \in \mathscr{S}_2'$ in this case, we have

$$\begin{aligned} s_1(G) &\equiv \sum_{M_1 \in \mathscr{S}_1'} s_1(M_1) - p s_1(M_2) \\ &\equiv (1 + p + p^2)(1 + p) - p(1 + p) \equiv 1 + p + p^2 \mod p^3. \end{aligned}$$

2) It remains to consider the case that at least one $M_1 \in \mathscr{S}_1'$ is metacyclic. Then we have by Theorem 11.4 that

$$p^{\omega(G)} = |G/\mho_1(G)| \leq |G/\mho_1(M_1)| = p|M_1/\mho_1(M_1)| \leq p^3.$$

Since $\omega(G) \leq 3$ and $p > 3$, the regularity of G follows from Theorem 10.13. (It is only here that we require the assumption that $p > 3$.) Since G is not metacyclic, $\omega(G) = 3$ by Theorem 11.4. Therefore

$$|\Omega_1(G)| = |G/\mho_1(G)| = p^3,$$

and G has exactly $1 + p + p^2$ subgroups of order p. □

Theorem 11.9.

a) (*Taussky [427]*) *Let G be a nonabelian 2-group, and assume that G/G' is elementary abelian of type $(2, 2)$. Then G is a dihedral group, a generalized quaternion group or a quasidihedral group* (*see Theorem I.14.9, pg. 96*).

b) *If $|G| = 2^n$ and G has class $n - 1$, then G is one of the groups listed in a).*

Proof. a) Since $|G/\Phi(G)| \leq |G/G'| = 4$, G can be generated by two elements by Theorem 3.15. By Theorem 1.11 c), the quotient $G'/\mathbf{K}_3(G)$ is cyclic and therefore $|G'/\Phi(G')\mathbf{K}_3(G)| = 2$. It follows that $|G/\Phi(G')\mathbf{K}_3(G)| = 2^3$. Since

$\Phi(G')\mathbf{K}_3(G) < G'$, the quotient $G/\Phi(G')\mathbf{K}_3(G)$ is nonabelian, so it does not have exponent 2. Consequently, $G/\Phi(G')\mathbf{K}_3(G)$ contains a cyclic normal subgroup of order 4 and is metacyclic. Hence, G is itself metacyclic by Lemma 11.3.

Now let L be a normal subgroup of G such that G/L and L are cyclic. Then $G' \leq L$. Therefore,

$$\exp(G/L) = |G/L| \leq \exp(G/G') = 2.$$

Hence, L has index 2. We have already classified the groups containing a cyclic normal subgroup of index 2 in Theorem I.14.9 b) (pg. 96). The isomorphism type (3) given there does not satisfy our hypothesis since in that case G/G' has type $(2^{n-1}, 2)$ with $n \geq 3$.

b) It follows immediately from our hypothesis that $|G/G'| = 4$, and we may apply a). □

We list some additional theorems without proof:

Theorem 11.10 (Blackburn [37]). *Let G be a p-group, and suppose G/G' has type (p^n, p^m). Then G' can be minimally generated by at most $(p^n-1)(p^m-1)$ elements.*

(*The Schreier estimate of Theorem I.19.11, pg. 151 for the size of a minimal generating set of a subgroup of the given index yields $d(G') \leq 1 + p^{n+m}$ in this case.*)

Theorem 11.11 (Blackburn [39]). *Let G be a p-group. Suppose that all proper subgroups of G are metacyclic, but G itself is not metacyclic. Then one of the following is true:*

a) *G is elementary abelian of order p^3.*
b) *$p > 2$ and G is the nonabelian group of exponent p and order p^3.*
c) *G is a 3-group of class 3 and order 3^4.*
d) *G is a 2-group with $|G| \leq 2^5$.*

Theorem 11.12 (Blackburn [39]). *Let $p > 2$, and let $|G| = p^n$ with $n \geq 5$. Suppose there exists an r with $3 \leq r \leq n-2$ such that all subgroups of G with order p^r can be generated by two elements. Then one of the following is true:*

a) *G is metacyclic.*
b) *G is a 3-group of maximal class (see §14).*
c) *We have $r = 3$. The elements of order p in G constitute (with 1) a nonabelian normal subgroup $\Omega_1(G)$ of G with order p^3, and $G/\Omega_1(G)$ is cyclic.*

Theorem 11.13 (Blackburn [39]). *Let $|G| = 2^n$ with $n \geq 5$. Suppose there exists an r with $5 \leq r \leq n$ such that all subgroups of G with orders 2^{r-1} and 2^r can be generated by two elements. Then G is metacyclic.*

Exercises

28) Construct an example of a 2-group that is a product of two cyclic groups, but that does not have a cyclic commutator subgroup. (Hint: Consider an appropriate

extension (not a semidirect product!) of an elementary abelian group of type $(2, 2, 2)$ by the cyclic group of order 4.)

12 Abelian normal subgroups of p-groups

We have already said something about the abelian normal subgroups of p-groups in Theorem 7.3 and Theorem 7.6. Now we take a closer look.

Theorem 12.1 (Alperin [4]). *Let G be a p-group, n a natural number and A a maximal element of the set*

$$\{A \trianglelefteq G \mid A' = 1, \exp(A) \leq p^n\}.$$

If $x \in \mathbf{C}_G(A)$ and $x^{p^n} = 1$, then $x \in A$ as long as $p^n \neq 2$.

(The conclusion of Theorem 12.1 does not hold when $p^n = 2$. The dihedral group of order 2^4 has only one abelian normal subgroup of exponent 2, namely its center, but it has 9 elements of order 2.)

Proof (Blackburn). Let $x \in \mathbf{C}_G(A) - A$ and $x^{p^n} = 1$. We choose x so that $X = \langle x^g \mid g \in G\rangle$ has smallest possible order. Let $B = \langle x, A\rangle$. Then B is abelian and $\exp B \leq p^n$. Since $A < B$, $B \not\trianglelefteq G$ by the maximality of A. Hence, there exists some $y \in G$ with $[x, y] \notin A$. We choose y so that $Y = \langle y^g \mid g \in G\rangle$ has smallest possible order. Let $z = [x, y]$. Since $x \in \mathbf{C}_G(A) \trianglelefteq G$, we have $z \in \mathbf{C}_G(A) - A$. Then

$$z = [x, y] \in [X, G] < X,$$

so we have

$$|\langle z^g \mid g \in G\rangle| \leq |[X, G]| < |X|.$$

By the choice of x, it now follows that $z^{p^n} \neq 1$.

If $z \notin \mathbf{N}_G(B)$, then $[x, z] \notin A$ and we can replace y with z. But then

$$z = [x, y] \in [G, Y] < Y,$$

contradicting the choice of y. So $z \in \mathbf{N}_G(B)$. By the choice of x, we have $x^p \in A$. Thus, $|B/A| = p$ and $B/A \trianglelefteq \langle z, B\rangle/A$. So $B/A \leq \mathbf{Z}(\langle z, B\rangle/A)$ and consequently $[z, x] \in A$. Then $[z, x]$ commutes with both z and x. Since $z \in \mathbf{C}_G(A)$ and $x^p \in A$, it follows by Lemma 1.3 a) that

$$[z, x]^p = [z, x^p] = 1.$$

On the other hand, it follows by Lemma 1.3 b) that

$$1 = (x^{p^n})^y = (x^y)^{p^n} = (xz)^{p^n} = x^{p^n} z^{p^n} [z,x]^{\binom{p^n}{2}} = z^{p^n} [z,x]^{\binom{p^n}{2}},$$

so

$$1 \neq z^{p^n} = [z,x]^{-\binom{p^n}{2}}.$$

This requires that $p \nmid \binom{p^n}{2}$, so we must have $p^n = 2$. □

The reader may compare Theorem 12.1 with Blackburn [40].

Lemma 12.2 (Thompson). *Let G be a p-group with $p > 2$. Suppose that every element of order p in G lies in $\mathbf{Z}(G)$. Then the minimum sizes of generating sets satisfy*

$$d(G) \leq d(\mathbf{Z}(G)).$$

Proof (Blackburn). We prove the claim by induction on $|G|$, and we define $Z = \Omega_1(G)$. By our hypothesis, $Z = \Omega_1(\mathbf{Z}(G)) \leq \mathbf{Z}(G)$ and so $d(Z) = d(\mathbf{Z}(G))$. Let $d(Z) = n$.

Let A/Z be an abelian normal subgroup of G/Z of exponent p that is as large as possible. Since $A' \leq Z \leq \mathbf{Z}(G)$, A has class at most 2 and A' has exponent at most p.

Let $a \in A$, let $g \in G$ and $a^g = ab$. Since $a^p \in Z$, $b \in A$ and $p > 2$ we have

$$a^p = (a^p)^g = (a^g)^p = (ab)^p = a^p b^p [b,a]^{\binom{p}{2}} = a^p b^p.$$

It follows that $b^p = 1$ and thus $b \in Z$. Hence, $A/Z \leq \mathbf{Z}(G/Z)$. By Theorem 12.1, every element of order p lying in $G/Z = \mathbf{C}_{G/Z}(A/Z)$ lies in A/Z. So $A/Z = \Omega_1(\mathbf{Z}(G/Z))$. Using the regularity of A, it follows by Theorem 10.7 a) that

$$|A/Z| \leq |A/\mho_1(A)| = |\Omega_1(A)| = |Z| = p^n,$$

and so $d(A/Z) \leq n$. The induction hypothesis now yields

$$d(G/Z) \leq d(\mathbf{Z}(G/Z)) = d(A/Z) \leq n.$$

If $Z \leq \Phi(G)$, then the claim follows since $d(G/Z) = d(G)$. If $Z \not\leq \Phi(G)$, then there exists a maximal subgroup M of G with $Z \not\leq M$. We have

$$M \cap Z = \{m \in M \mid m^p = 1\} = \Omega_1(\mathbf{Z}(M)).$$

The induction hypothesis implies

$$d(M) \leq d(\mathbf{Z}(M)) = d(\Omega_1(\mathbf{Z}(M))) = d(M \cap Z) \leq n - 1.$$

From this it follows that $d(G) \leq d(M) + 1 \leq n = d(\mathbf{Z}(G))$. □

Theorem 12.3 (Thompson). *Let G be a p-group with $p > 2$. Suppose that every abelian normal subgroup of G can be generated by k elements. Then every subgroup of G can be generated by $\frac{k(k+1)}{2}$ elements.*

Proof. We may assume that G contains a maximal abelian normal subgroup A with $d(A) = k$. We let $B = \Omega_1(A)$ and $C = \mathbf{C}_G(B)$. By Theorem 12.1, $\Omega_1(C) = B$.

Let U be a subgroup of G. Then $U/(U \cap C)$ is isomorphic to a subgroup of the automorphism group $\mathbf{Aut}(B)$ of B. So by Theorem 3.19,

$$|U/(U \cap C)| \leq p^{\frac{k(k-1)}{2}}.$$

If we can show that $d(U \cap C) \leq k$, then we will have

$$d(U) \leq d(U/U \cap C) + d(U \cap C) \leq \frac{k(k+1)}{2}.$$

So let $V \leq C = \mathbf{C}_G(B)$. Then $VB = V \times B_1$ for some B_1 and thus $d(V) \leq d(VB)$. Replacing V by VB if necessary, we may assume $B \leq V \leq C$. Then $B = \Omega_1(V) \leq \mathbf{Z}(V)$. Then Lemma 12.2 yields the desired inequality

$$d(V) \leq d(\mathbf{Z}(V)) = d(\Omega_1(\mathbf{Z}(V))) = d(B) = k. \qquad \square$$

The abelian normal subgroups whose generating sets of minimum size are as large as possible play a role in numerous investigations; for example, in Feit and Thompson's proof of the solvability of groups of odd order and in the proof of Thompson's fundamental p-nilpotency criterion (see Chapter IV, §6). The case in which all abelian normal subgroups of a p-group are cyclic was addressed in Theorem 7.6. For $p > 2$, Blackburn handled those p-groups whose abelian normal subgroups can be generated by at most two elements. Here we restrict ourselves to the case that $p > 3$ in order to take full advantage of the theory of regular p-groups.

Theorem 12.4 (Blackburn [39]). *Let G be a nonabelian p-group of order p^n with $p > 3$. Suppose that every abelian normal subgroup of G can be generated by at most two elements. Then G belongs to one of the following isomorphism classes:*
(1) *G is metacyclic.*
(2) *$G = \langle x, y, z \rangle$ with the defining relations*

$$x^p = y^p = z^{p^{n-2}} = [x, z] = [y, z] = 1, \quad y^x = yz^{p^{n-3}}.$$

So G is the central product of the nonabelian group $P = \langle x, y \rangle$ of order p^3 and exponent p with the cyclic group $Z = \langle z \rangle$ of order p^{n-2}, where P' and $\Omega_1(Z)$ are identified (in the sense of Theorem I.9.10, pg. 51). One can also describe G as the semidirect product of the abelian normal subgroup $\langle y, z \rangle$ of type (p, p^{n-2}) with the cyclic group $\langle x \rangle$ of order p with respect to the automorphism

$$y^x = yz^{p^{n-3}}, \quad z^x = z.$$

If $n = 3$, then G is the nonabelian group of exponent p and order p^3.

(3) $G = \langle x, y, z\rangle$ *with the defining relations*

$$x^p = y^p = z^{p^{n-2}} = [y, z] = 1, \quad y^x = yz^{sp^{n-3}}, \quad z^x = yz,$$

where $n \geq 4$ and s is equal to 1 *or an arbitrary quadratic nonresidue modulo p. The two nonisomorphic groups described in this way are semidirect products of the abelian normal subgroup $\mathbf{C}_G(G') = \langle y, z\rangle$ with the cyclic group $\langle x\rangle$ of order p with respect to the automorphism*

$$y^x = yz^{sp^{n-3}}, \quad z^x = yz.$$

Proof. If $n = 3$, the two isomorphism classes of groups that occur are described in (1) and (2). So we may assume that $n \geq 4$.

If $|\Omega_1(G)| \leq p^2$, then G is metacyclic by Theorem 11.6 since $p > 3$. This is case (1). So assume from now on that $|\Omega_1(G)| \geq p^3$. By our assumptions about the abelian normal subgroups of G, the subgroup $\Omega_1(G)$ is certainly not abelian. So there exists a normal subgroup A of G of type (p, p) contained in $\Omega_1(G)$ by Lemma 7.5 a). We let $N = \mathbf{C}_G(A)$. By Theorem 12.1, it follows that $A = \Omega_1(N) < \Omega_1(G)$. So $N < G$ and since

$$|G/N| \quad \text{divides} \quad |\mathbf{Aut}(A)| = (p^2 - 1)(p^2 - p),$$

we have $|G/N| = p$. Since $|\Omega_1(N)| = |A| = p^2$ and $p > 3$, N is metacyclic by Theorem 11.6. Using Theorem 11.4, it follows that $|N/\mho_1(N)| \leq p^2$. From

$$|G/\mho_1(G)| \leq |G/\mho_1(N)| \leq p^3,$$

we conclude that G is regular by Theorem 10.13 since $p > 3$. Thus,

$$|\Omega_1(G)| = |G/\mho_1(G)| = p^3.$$

Now, since $\Omega_1(G)$ is nonabelian and $|\Omega_1(N)| = p^2$, we have $\Omega_1(N) \not\leq \mathbf{Z}(\Omega_1(G))$. Assume for contradiction that $\Omega_1(N) \leq \mho_1(N)$. By Theorem 10.8 f), we have

$$[\mho_1(G), \Omega_1(G)] = 1,$$

but then

$$\Omega_1(N) \leq \mho_1(G) \cap \Omega_1(G) \leq \mathbf{Z}(\Omega_1(G)),$$

a contradiction. So $\Omega_1(N) \not\leq \mho_1(N)$. Therefore, $\mho_1(N)$ contains only one subgroup of order p, and $\mho_1(N)$ is cyclic by Theorem 8.2 a).

Let $z \in N$ with $\mho_1(N) = \langle z^p\rangle$ and $y \in \Omega_1(N)$ with $y \notin \langle z\rangle$. Then $N = \langle y, z\rangle$. Since

$$y \in \Omega_1(N) \leq \mathbf{Z}(N),$$

we have that N is abelian of type (p^{n-2}, p) with $|z| = p^{n-2}$. Let

$$x \in \Omega_1(G) - \Omega_1(N).$$

Since $|G/N| = p$, we have $G = \langle x, y, z\rangle$ with the relations

$$x^p = y^p = z^{p^{n-2}} = [y, z] = 1.$$

We have $\mho_{n-3}(N) = \langle z^{p^{n-3}}\rangle \trianglelefteq G$ and $|\Omega_1(G)/\mho_{n-3}(N)| = p^2$. Since

$$\Omega_1(G) = \langle x, y, z^{p^{n-3}}\rangle$$

is nonabelian and $z^{p^{n-3}} \in \mathbf{Z}(G)$, we have

$$1 \neq [y, x] \in \Omega_1(G)' = \mho_{n-3}(N),$$

so

$$y^x = yz^{rp^{n-3}}$$

with $r \not\equiv 0 \bmod p$.

Since $x^p = 1$, it follows from the regularity of G and Theorem 10.6 b) that

$$[z, x]^p = 1.$$

So $z^x = zw$ with

$$w = [z, x] \in \Omega_1(N) = \langle y, z^{p^{n-3}}\rangle.$$

Now we distinguish two cases:

Case 1: Assume $[z, x] = z^{sp^{n-3}} \in \mho_{n-3}(N)$. Then

$$z^x = z^{1+sp^{n-3}}$$

for some integer s. We determine t via the equation $s + tr \equiv 0 \bmod p$. Then

$$(zy^t)^x = z^{1+sp^{n-3}} y^t z^{trp^{n-3}} = zy^t.$$

Now we replace z by $z' = (zy^t)^r$. Then $(z')^x = z'$, $y^x = y(z')^{p^{n-3}}$, and the other relations still hold. So G is the group described in (2).

Case 2: Assume $[z, x] \notin \mho_{n-3}(N)$. We determine the element $y \in N$ of order p via the equation $z^x = zy$. Then we still have $N = \langle y, z\rangle = \langle y\rangle \times \langle z\rangle$ and

$$y^x = yz^{rp^{n-3}}$$

with $r \not\equiv 0 \bmod p$. Let t be a quadratic nonresidue modulo p. We define k such that

$$k^2 r = s \in \{1, t\}.$$

We have

$$y^{x^k} = y z^{k r p^{n-3}} \quad \text{and} \quad z^{x^k} = z^{1+\binom{k}{2} r p^{n-3}} y^k.$$

We define $x' = x^k$, $y' = y^k z^{\binom{k}{2} r p^{n-3}}$, $z' = z$ and obtain

$$y'^{x'} = y' z^{k^2 r p^{n-3}} = y' (z')^{s p^{n-3}} \quad \text{and} \quad (z')^{x'} = z' y'.$$

These are the two groups described in (3).

One can see that these two groups are not isomorphic in the following way: In the group $G = \langle x, y, z \rangle$ with the relations in (3) and $s = 1$, we have

$$[z, x, x] = z^{p^{n-3}}$$

with $z \in \mathbf{C}_G(G') - G'$ and $x \in G - \mathbf{C}_G(G')$.

Assume for contradiction that in the group $G = \langle x, y, z \rangle$ with the relations in (3) and $s = t$ (a quadratic nonresidue modulo p), there exist $a, b \in G$ such that

$$[a, b, b] = a^{p^{n-3}}$$

with $a \in \mathbf{C}_G(G') - G'$ and $b \in G - \mathbf{C}_G(G')$. Then

$$a = z^{m_1} y^{m_2} \quad \text{and} \quad b = c x^{m_3} \quad \text{with} \quad c \in \mathbf{C}_G(G')$$

and some integers m_1, m_2, m_3. But this gives

$$[a, b, b] = a^{t m_3^2 p^{n-3}} \neq a^{p^{n-3}}. \qquad \square$$

Remark 12.5.

a) For $p = 3$, Blackburn [36, 39] classified the groups treated in Theorem 12.4. Besides the isomorphism classes given in Theorem 12.4, there exist 3-groups of maximal class (see §14).

b) For $p = 2$, no complete classification of the groups in question exists. J. Thompson proved the following in [436]:

Let P be a nonabelian 2-group in which every abelian normal subgroup can be generated by at most two elements. Suppose there exists a group A of automorphisms of P such that $2 \nmid |A|$ and $P = \langle x^{-1} x^\alpha \mid x \in P, \alpha \in A \rangle$. Then one of the following must occur:

(1) P is a quaternion group of order 8.

(2) P is the central product of a quaternion group of order 8 with a dihedral group of order 8, where the centers are identified.

(3) P is a special p-group (in the sense of Definition 13.1) with $|P/\mathbf{Z}(P)| = 2^4$ and $|\mathbf{Z}(P)| = 2^2$.

We are now in a position to classify all groups of order p^4 for $p > 3$:

Theorem 12.6. *For $p > 3$, there exist exactly 15 isomorphism classes of groups of order p^4, namely:*

1–5) *The five abelian groups of types*

$$(p^4),\ (p^3, p),\ (p^2, p^2),\ (p^2, p, p),\ (p, p, p, p).$$

6) *A metacyclic group $G = \langle a, b\rangle$ of exponent p^3 with*

$$a^{p^3} = b^p = 1, \quad a^b = a^{1+p^2}.$$

(*This is the only nonabelian group of exponent p^3 and order p^4.*)

7) *A metacyclic group $G = \langle a, b\rangle$ of exponent p^2 with*

$$a^{p^2} = b^{p^2} = 1, \quad a^b = a^{1+p}.$$

8) *A central product $G = P \circ Z$ of the nonabelian group P of order p^3 and exponent p with the cyclic group Z of order p^2, where P' and $\Omega_1(Z)$ are identified.*

9–10) *The two groups $G = \langle x, y, z\rangle$ from Theorem 12.4 (3) with*

$$x^p = y^p = z^{p^2} = [y, z] = 1, \quad y^x = yz^{sp}, \quad z^x = zy,$$

where s is either 1 or a quadratic nonresidue modulo p.

11) *$G = P \times \langle a\rangle$ with $a^p = 1$, where P is the nonabelian group of exponent p and order p^3.*

12) *A semidirect product $G = A\langle x\rangle$ of the abelian normal subgroup $A = \langle a_1, a_2, a_3\rangle$ of type (p, p, p) with the cyclic group $\langle x\rangle$ of order p by means of the automorphism*

$$a_1^x = a_1, \quad a_2^x = a_1a_2, \quad a_3^x = a_2a_3.$$

(*This group has exponent p.*)

13)–15) *The (nonsplit) extensions $G = A\langle x\rangle$ of the abelian normal subgroup A of type (p, p, p), say $A = \langle a_1, a_2, a_3\rangle$, with the cyclic group of order p with $x^p = a_1$ by means of the following automorphisms:*

13) $a_1^x = a_1$, $a_2^x = a_1a_2$, $a_3^x = a_2a_3$.

14) $a_1^x = a_1$, $a_2^x = a_1a_2$, $a_3^x = a_3$.

Then $G = \langle a_2, x\rangle \times \langle a_3\rangle$, where $\langle a_2, x\rangle$ is the nonabelian group of order p^3 and exponent p^2.

15) $a_1^x = a_1$, $a_2^x = a_2$, $a_3^x = a_2a_3$.

Proof. After accounting for the abelian groups in 1) – 5), we distinguish three cases:

Case 1: Every abelian normal subgroup of G has at most two generators. Then G is either metacyclic or one of the groups given in 8) – 10) by Theorem 12.4.

Assume G is nonabelian and metacyclic of order p^4. First assume that $\exp G = p^3$. Then there is an $a \in G$ with $|a| = p^3$. So $\langle a\rangle \lhd G$. By Theorem I.14.9 a) (pg. 96), G is then the group in 6).

Now assume G is nonabelian and metacyclic of exponent p^2. Let $G = \langle a, b\rangle$ with $\langle a\rangle \lhd G$. It follows from

$$p^4 = |G| = \frac{|\langle a\rangle||\langle b\rangle|}{|\langle a\rangle \cap \langle b\rangle|} \leq \frac{p^4}{|\langle a\rangle \cap \langle b\rangle|}$$

that $|a| = |b| = p^2$ and $\langle a\rangle \cap \langle b\rangle = 1$. Let $a^b = a^k$. From $b^{p^2} = 1$ we obtain $k^{p^2} \equiv 1 \bmod p^2$, so $k \equiv 1 \bmod p$. Since $G' \neq 1$, we have $k \not\equiv 1 \bmod p^2$. By modifying b if necessary, we can force $k = 1 + p$ and obtain type 7).

Case 2: G has an abelian normal subgroup A of type (p, p, p) and $\exp G = p$. Then G is a semidirect product $G = A\langle x\rangle$. With an appropriate choice of basis $\{a_1, a_2, a_3\}$ of A, the automorphism of A induced by x is in Jordan normal form, so has the form

$$a_1^x = a_1, \quad a_2^x = a_1a_2, \quad a_3^x = a_3$$

or

$$a_1^x = a_1, \quad a_2^x = a_1a_2, \quad a_3^x = a_2a_3.$$

These are the groups in 11) and 12).

Case 3: G has an abelian normal subgroup A of type (p, p, p) and $\exp G = p^2$. Let $x \in G$ with $|x| = p^2$, and let $a_1 = x^p$. Then $G = A\langle x\rangle$. There are only two possibilities for the automorphism of A induced by x:

a) A is a directly indecomposable G-module. Since in this case the set of points fixed by x is $\langle a_1\rangle$, x has Jordan normal form

$$a_1^x = a_1, \quad a_2^x = a_1a_2, \quad a_3^x = a_2a_3.$$

This gives the group in 13).

b) The $\langle x\rangle$-module A, written multiplicatively, is a direct product of indecomposable modules of dimensions 1 and 2, say $A = \langle b_1, b_2, b_3\rangle$ with

$$b_1^x = b_1, \quad b_2^x = b_2, \quad b_3^x = b_2b_3.$$

Since $(x^p)^x = a_1^x = a_1$, we have $a_1 = b_1^m b_2^n$ for some integers m and n.

First assume that $m = 0$, so $n \neq 0$. We set $a_1 = b_2^n$, $a_2 = b_3^n$, $a_3 = b_1$. Then we obtain the relations

$$a_1^x = a_1, \quad a_2^x = a_1a_2, \quad a_3^x = a_3, \quad x^p = a_1.$$

This is the group in 14).

Now assume $m \neq 0$. We set $a_1 = b_1^m b_2^n$, $a_2 = b_2$, $a_3 = b_3$ and obtain the relations

$$a_1^x = a_1, \quad a_2^x = a_2, \quad a_3^x = a_2a_3, \quad x^p = a_1.$$

This is type 15).

It remains to show that the groups in this list are pairwise nonisomorphic.

Groups 1)–5) are nonisomorphic abelian groups. The groups 6)–10) are nonabelian groups whose abelian normal subgroups are generated by two elements; the groups 6) and 7) are metacyclic of exponent p^3, resp. p^2, so are not isomorphic; the group in 8) is minimally generated by 3 elements and has commutator subgroup of order p; the two groups in 9) and 10) are not isomorphic by Theorem 12.4; they have commutator subgroups of order p^2 and are not metacyclic since they contain subgroups of order p^3 and exponent p, namely $\langle x, y, z^{p^{n-3}}\rangle$ in the notation of Theorem 12.4. The groups in 11) and 12) are nonabelian groups of exponent p containing abelian normal subgroups of type (p, p, p); in 11), we have $|G'| = p$ and in 12), $|G'| = p^2$. The groups in 13)–15) are nonabelian groups of exponent p^2 containing abelian normal subgroups of type (p, p, p); we have

$$\begin{aligned} G' = \langle a_1, a_2\rangle > \mho_1(G) \quad &\text{in 13),} \\ G' = \langle a_1\rangle = \mho_1(G) \quad &\text{in 14) and} \\ G' \cap \mho_1(G) = \langle a_2\rangle \cap \langle a_1\rangle = 1 \quad &\text{in 15).} \end{aligned}$$

□

Exercises

29) Show that if we set $p = 3$ in Theorem 12.6 and replace the group in 13), which is isomorphic to 12) when $p = 3$, by the irregular group of order 3^4 in Example 10.15, we obtain a complete list of the nonisomorphic groups of order 3^4.

30) Classify the nonabelian groups of order 2^4 and exponent 4. (There are 5 isomorphism classes.)
(Hint: G has an abelian normal subgroup A of type $(4, 2)$.
Letting $G = A\langle x\rangle$, distinguish three cases:
(1) There is an $a \in A$ such that $|a| = 4$ and $a^x = a$. Then $G = D \curlyvee Z$, where D is a dihedral group of order 8, Z is a cyclic group of order 4, and D' is identified with $\Omega_1(Z)$.
(2) There is an $a \in A$ such that $|a| = 4$ and $a^x = a^{-1}$. Then $G = H \times \langle y\rangle$ with $y^2 = 1$, or G is a metacyclic group of exponent 4.
(3) If cases (1) and (2) do not occur, then $G = \langle a_1, a_2, x\rangle$, where $a_1^4 = a_2^2 = x^2 = [a_1, a_2] = [a_2, x] = 1$, $a_1^x = a_1 a_2$.)

31) (Alperin [7]). Let H be a group of exponent $p > 2$ and class 2 with

$$H = \langle x_1, \ldots, x_p, y_1, \ldots, y_p\rangle$$

and the further defining relations

$$[x_i, y_i] = 1 \quad (i = 1, \ldots, p).$$

Show that

a) $H' = \langle [x_i, x_j], [y_i, y_j], [x_i, y_j] \mid i \neq j;\ i, j = 1, \ldots, p \rangle$ and therefore

$$|H'| = p^{2p^2-2p} \quad \text{and} \quad |H| = p^{2p^2}.$$

b) Let $G = \langle H, z \rangle$ with $x_i^z = x_{i+1}$, $y_i^z = y_{i+1}$ ($i \bmod p$), and $z^p = 1$. Then G has an abelian subgroup of order p^{2p^2-2p+2}, but G has no abelian normal subgroup of this order.

13 Special and extraspecial p-groups

Definition 13.1. Let G be a p-group.

a) G is called special when it satisfies one of the following conditions:
   (1) G is an elementary abelian p-group.
   (2) $\Phi(G) = G' = \mathbf{Z}(G)$ is elementary abelian.
   (In particular, a special group has class at most 2.)

b) G is called extraspecial when G is a nonabelian special p-group and $|\mathbf{Z}(G)| = p$.

We will prove two theorems about automorphism groups. First we make a definition.

Definition 13.2. Let G be an A-group equipped with the automorphisms $g \mapsto g^a$ for $g \in G$ and $a \in A$. We define the subgroups $[G, A]$ and $\mathbf{C}_G(A)$ of G by

$$[G, A] = \langle g^a g^{-1} \mid g \in G, a \in A \rangle$$

and

$$\mathbf{C}_G(A) = \{g \in G \mid g^a = g \text{ for all } a \in A\}.$$

If we consider G and A as subgroups inside the semidirect product GA, then $[G, A]$ is simply the commutator group of G and A as in Definition 1.5. It therefore follows from Lemma 1.6 b) that $[G, A] \trianglelefteq G$ and $[G, A]^a = [G, A]$ for all $a \in A$. It holds trivially that $\mathbf{C}_G(A)^a = \mathbf{C}_G(A)$ for all $a \in A$, but $\mathbf{C}_G(A)$ need not be a normal subgroup of G.

Lemma 13.3. *Let G be an A-group.*

a) *If $[G, A] = G$ and N is a normal subgroup of G such that $[N, A] = 1$, then $N \leq \mathbf{Z}(G)$.*

b) *If $(|G|, |A|) = 1$, then $[G, A] = [[G, A], A]$.*

Proof. a) We construct the semidirect product $H = GA$. Since $[N, G] \leq N$, it follows by our assumptions and Lemma 1.10 b) that

$$[G, N] = [[G, A], N] \leq [[A, N], G]^H [[N, G], A]^H = 1.$$

So $N \leq \mathbf{Z}(G)$.

b) For all $g \in G$ and all $a \in A$, we clearly have

$$[G, A]g^a = [G, A]g.$$

Since $(|G|, |A|) = 1$, it follows by Theorem I.18.6 (pg. 140) that $G = [G, A]\mathbf{C}_G(A)$. In the same way,

$$[G, A] = [[G, A], A](\mathbf{C}_G(A) \cap [G, A]).$$

So $G = [[G, A], A]\mathbf{C}_G(A)$. If $g = g_1 g_2$ with $g_1 \in [[G, A], A]$ and $g_2 \in \mathbf{C}_G(A)$, then it follows for all $a \in A$ that

$$g^a g^{-1} = g_1^a g_2^a g_2^{-1} g_1^{-1} = g_1^a g_1^{-1} \in [[G, A], A].$$

This shows that $[G, A] \leq [[G, A], A]$; the reverse inclusion is trivial. □

The following theorem is useful in a wide variety of applications.

Theorem 13.4 (Zassenhaus).

a) *If A is an abelian normal subgroup of G with $(|A|, |G/A|) = 1$, then*

$$A = [A, G] \times (A \cap \mathbf{Z}(G)).$$

b) *If A is an abelian $\overline{G}$-group with $(|A|, |\overline{G}|) = 1$, then $A = [A, \overline{G}] \times \mathbf{C}_A(\overline{G})$ in the notation of Definition 13.2.*

Proof. a) We let $\overline{G} = G/A$ and $|\overline{G}| = m$. By defining $a^{Ag} = a^g$, A becomes a $\overline{G}$-group. For all $a \in A$, we have

$$a^m = \prod_{\overline{g} \in \overline{G}} a^{\overline{1}-\overline{g}} \prod_{\overline{g} \in \overline{G}} a^{\overline{g}}$$

with $a^{\overline{1}-\overline{g}} \in [A, G]$ and $\prod_{\overline{g} \in \overline{G}} a^{\overline{g}} \in A \cap \mathbf{Z}(G)$.

Since $(m, |A|) = 1$, it follows that

$$A = \{a^m \mid a \in A\} = [A, G](A \cap \mathbf{Z}(G)).$$

The map μ defined by

$$a^\mu = \prod_{\overline{g} \in \overline{G}} a^{\overline{g}}$$

is a homomorphism from A to $A \cap \mathbf{Z}(G)$. Since

$$(a^{\overline{1}-\overline{g}})^\mu = 1,$$

$[A, G]$ lies in the kernel of μ. For $a \in [A, G] \cap (A \cap \mathbf{Z}(G))$, it therefore follows that

$$1 = a^{\mu} = a^{m},$$

and since $(m, |A|) = 1$, in fact $a = 1$. Consequently,

$$[A, G] \cap (A \cap \mathbf{Z}(G)) = 1$$

and therefore

$$A = [A, G] \times (A \cap \mathbf{Z}(G)).$$

b) The claim follows immediately by applying a) to the semidirect product $A\overline{G}$.□

In numerous applications, the following two theorems bring special groups into consideration:

Theorem 13.5 (Hall–Higman reduction). *Let Q be a p-subgroup of G, let $B \trianglelefteq A \leq \mathbf{N}_G(Q)$ with $p \nmid |A|$ and $B \not\leq \mathbf{C}_G(Q)$. Let P be a minimal element of the set*

$$\mathscr{P} = \{V \leq Q \mid A \leq \mathbf{N}_G(V),\ B \not\leq \mathbf{C}_G(V)\}.$$

Then

a) *$P/\Phi(P)$ is an irreducible A-group with $[P, B] = P$.*

b) *$[\Phi(P), B] = 1$.*

c) *P is a special p-group.*

Proof. a) Let

$$P/\Phi(P) = \overline{P} = \bigtimes_i \overline{P}_i$$

be a direct decomposition of $\overline{P}$ into irreducible A-modules $\overline{P}_i = P_i/\Phi(P)$, which exists by Maschke's Theorem I.17.7 (pg. 132) since $(|P|, |A|) = 1$. Since $[P, B] \neq 1$ and $(|P|, |B|) = 1$, we also have $[\overline{P}, B] \neq 1$ by Theorem 3.18. Suppose that, say, $[\overline{P}_1, B] \neq 1$. Then $P_1 \in \mathscr{P}$, and so $P = P_1$ by the minimality of P. Hence, $P/\Phi(P)$ is an irreducible A-module.

Since $B \trianglelefteq A$, $[P, B]$ is invariant under A. By Lemma 13.3 b),

$$1 \neq [P, B] = [P, B, B].$$

It follows that $[P, B] \in \mathscr{P}$ and thus $P = [P, B]$.

b) As P is a minimal element of $\mathscr{P}$, we have $\Phi(P) \notin \mathscr{P}$ and so $[\Phi(P), B] = 1$.

c) Since $P = [P, B]$ and $[\Phi(P), B] = 1$, it follows by Lemma 13.3 a) that

$$\Phi(P) \leq \mathbf{Z}(P).$$

Thus, we obtain from Lemma 1.3 a) that for all $x, y \in P$,

$$1 = [x^p, y] = [x, y]^p.$$

So P' has exponent at most p.

We show that $P' = \Phi(P)$:

Let $\widetilde{P} = P/P'$. An application of Theorem 13.4 b) yields with the help of a) that

$$[\widetilde{P}, B] = \widetilde{P} = [\widetilde{P}, B] \times \mathbf{C}_{\widetilde{P}}(B).$$

Thus, $\mathbf{C}_{\widetilde{P}}(B) = 1$. By b), $\Phi(P) \leq \mathbf{C}_P(B)$. Hence, $\Phi(P) \leq P'$. The reverse inclusion $P' \leq \Phi(P)$ holds by Theorem 3.11.

If $P' = \Phi(P) = 1$, then P is elementary abelian. If, on the other hand, $1 \neq P' = \Phi(P)$, then it follows immediately from $\Phi(P) \leq \mathbf{Z}(P)$ and the irreducibility of $P/\Phi(P)$ as an A-module that $\mathbf{Z}(P) = \Phi(P)$. We have seen already that $P' = \Phi(P)$ has exponent p. So P is a special p-group. □

Theorem 13.6 (Thompson). *Let $G = PA$ be a semidirect product of the p-group $P \neq 1$ with A (where $P \lhd G$), and assume G has the following properties:*
(1) $(|P|, |A|) = 1$.
(2) $[P, A] = P$.
(3) *A centralizes all abelian characteristic subgroups of P.*
Then P is a (necessarily nonabelian) special p-group.

Proof. a) P has class 2:

Assume for contradiction that $c = c(P) > 2$. We let $T = \mathbf{K}_{c-1}(P)$. Since $2c - 2 \geq c + 1$, T is abelian by Theorem 2.11 b). So T is an abelian characteristic subgroup of P, and then $A \leq \mathbf{C}_G(T)$ by our assumption. For $t \in T$, $a \in A$ and all $x \in P$, we have $[x, t] \in \mathbf{K}_c(P) \leq T \leq \mathbf{C}_G(A)$, and then

$$[x, t] = [x, t]^a = [x^a, t^a] = [x^a, t].$$

We may conclude from

$$[x^a, t, x^{-1}] \in [P, \mathbf{K}_{c-1}(P), P] \leq \mathbf{K}_{c+1}(P) = 1$$

with the help of Lemma 1.2 c) and Lemma 1.3 b) that

$$[x^a x^{-1}, t] = [x^a, t][x^a, t, x^{-1}][x^{-1}, t] = [x^a, t][x, t]^{-1} = 1.$$

Then by our assumption (2), it follows that

$$\mathbf{C}_G(T) \geq \langle x^a x^{-1} \mid x \in P, a \in A \rangle = [P, A] = P.$$

Thus

$$1 = [P, T] = [P, \mathbf{K}_{c-1}(P)] = \mathbf{K}_c(P),$$

contradicting that $c(P) = c$.

b) P/P' and P' have exponent p:

Let $\exp P/P' = p^m$. Then for all $x, y \in P$, we have $x^{p^m} \in P' \leq \mathbf{Z}(P)$ and so by Lemma 1.3 a),

$$[x, y]^{p^m} = [x^{p^m}, y] = 1.$$

We conclude that $\exp P' \leq p^m$.

Assume for contradiction that $m > 1$. Since $2m - 2 \geq m$, we have

$$[x^{p^{m-1}}, y^{p^{m-1}}] = [x, y]^{p^{2m-2}} = 1.$$

Therefore,

$$V = \mathfrak{V}_{m-1}(P) = \langle x^{p^{m-1}} \mid x \in P \rangle$$

is abelian. It follows by our assumption (3) that $A \leq \mathbf{C}_G(V)$. We may give $\widetilde{P} = P/P'$ the structure of an A-group by defining $(xP')^a = x^a P'$. By Theorem 13.4 b), we have

$$\widetilde{P} = [\widetilde{P}, A] \times \mathbf{C}_{\widetilde{P}}(A),$$

and by assumption (2), we have $\widetilde{P} = [\widetilde{P}, A]$. So $\mathbf{C}_{\widetilde{P}}(A) = 1$ and therefore $V = \mathfrak{V}_{m-1}(P) \leq P'$. However, this contradicts $\exp P/P' = p^m$. So $\exp P/P' = p$ and $\exp P' \leq p$. By (2) and (3), P is nonabelian, and consequently $\exp P' = p$.

c) P is a special p-group:

By a) and b), we have that $\Phi(P) = P' \leq \mathbf{Z}(P)$. It remains to show that $\mathbf{Z}(P) \leq P'$. Again let $\widetilde{P} = P/P'$. Since

$$[\widetilde{P}, A] = \widetilde{P} = [\widetilde{P}, A] \times \mathbf{C}_{\widetilde{P}}(A),$$

we have $\mathbf{C}_{\widetilde{P}}(A) = 1$. On the other hand, $A \leq \mathbf{C}_G(\mathbf{Z}(P))$ by our assumption (3). It therefore follows that $\mathbf{Z}(P) \leq P'$. □

The following theorem describes the structure of extraspecial groups.

Theorem 13.7. *Let G be a nonabelian p-group, let $G/\mathbf{Z}(G)$ be elementary abelian and let $\mathbf{Z}(G)$ be cyclic. (The additional property $|\mathbf{Z}(G)| = p$ satisfied by extraspecial p-groups is not required here.) Then:*

a) $|G'| = p$.

b) *Letting $G' = \langle c \rangle$, we may give $G/\mathbf{Z}(G)$ the structure of a nondegenerate symplectic vector space over* $\mathrm{GF}(p)$ *by defining*

$$f(\overline{x}, \overline{y}) = a \quad \text{for} \quad \overline{x} = x\mathbf{Z}(G),\ \overline{y} = y\mathbf{Z}(G) \text{ and } [x, y] = c^a.$$

c) $|G/\mathbf{Z}(G)|$ *is a square.*

d) G *is a central product (with identified centers) of nonabelian groups G_i with $|G_i/\mathbf{Z}(G_i)| = p^2$ and $\mathbf{Z}(G_i) = \mathbf{Z}(G)$.*

e) *If $|G/\mathbf{Z}(G)| = p^{2m}$, then all maximal abelian normal subgroups of G have order $p^m|\mathbf{Z}(G)|$.*

f) *For every maximal abelian normal subgroup A_1 of G, there exists a maximal abelian normal subgroup A_2 of G such that*

$$G = A_1A_2 \quad \textit{and} \quad A_1 \cap A_2 = \mathbf{Z}(G).$$

Proof. a) Since G has class 2, it follows from $x^p \in \mathbf{Z}(G)$ and Lemma 1.3 a) that for all $x, y \in G$, we have

$$1 = [x^p, y] = [x, y]^p.$$

Since G' is cyclic, we obtain $|G'| = p$.

b) Clearly, f is a well-defined function taking values in GF(p). As $[x, y] = [y, x]^{-1}$ and $[x, x] = 1$, $f(\overline{x}, \overline{y}) = -f(\overline{y}, \overline{x})$ and $f(\overline{x}, \overline{x}) = 0$. Since G has class 2, we have by Lemma 1.2 that

$$[x_1x_2, y] = [x_1, y][x_2, y] \quad \text{and} \quad [x, y_1y_2] = [x, y_1][x, y_2].$$

So f is bilinear and $G/\mathbf{Z}(G)$, considered as a vector space over GF(p) with the scalar product f, is a symplectic vector space over GF(p) in the sense of Chapter II, §9.

If $f(\overline{x}, \overline{y}) = 0$ for all $\overline{y} \in G/\mathbf{Z}(G)$, then $[x, y] = 1$ for all $y \in G$, and so $x \in \mathbf{Z}(G)$. Hence, the symplectic space $G/\mathbf{Z}(G)$ is nondegenerate.

c) By Theorem II.9.6 b) (pg. 227), $\dim G/\mathbf{Z}(G) = 2m$ is even. So

$$|G/\mathbf{Z}(G)| = p^{2m}$$

is a square.

d) By Theorem II.9.6 a), $G/\mathbf{Z}(G)$ has a basis $\{\overline{x}_1, \ldots, \overline{x}_m, \overline{y}_1, \ldots, \overline{y}_m\}$ such that

$$f(\overline{x}_i, \overline{y}_i) = 1 \quad \text{and} \quad f(\overline{x}_i, \overline{x}_j) = f(\overline{y}_i, \overline{y}_j) = f(\overline{x}_i, \overline{y}_j) = 0 \ \text{ for } i \neq j.$$

Let $\overline{x}_i = x_i\mathbf{Z}(G)$ and $\overline{y}_j = y_j\mathbf{Z}(G)$. The groups $G_i = \langle x_i, y_i, \mathbf{Z}(G)\rangle$ are nonabelian since $[x_i, y_i] = c$. They have order $p^2|\mathbf{Z}(G)|$ with $G_i' = \langle c\rangle \leq \mathbf{Z}(G)$. Also, $[G_i, G_j] = 1$ and $G_i \cap G_j = \mathbf{Z}(G)$ for $i \neq j$. Hence, G is a central product of the subgroups G_i, where the centers are identified.

e) Let A be a maximal abelian normal subgroup of G. Then $\mathbf{Z}(G) \leq A$. Hence, $A/\mathbf{Z}(G)$ is a subspace of the symplectic vector space $G/\mathbf{Z}(G)$ satisfying $f(\overline{a}_1, \overline{a}_2) = 0$ for all $a_i \in A$. That is, $A/\mathbf{Z}(G)$ is a maximal isotropic subspace of $G/\mathbf{Z}(G)$. Then by Theorem II.9.11 (pg. 229), $\dim A/\mathbf{Z}(G) = m$ and so $|A| = p^m|\mathbf{Z}(G)|$.

f) By e), we have $A_1 = \langle x_1, \ldots, x_m, \mathbf{Z}(G)\rangle$ for some $x_i \in G$. Defining $\overline{x}_i = x_i\mathbf{Z}(G)$, we have

$$1 = [x_i, x_j] = c^{f(\overline{x}_i, \overline{x}_j)}$$

for $i, j = 1, \ldots, m$. Thus, $\langle \overline{x}_1, \ldots, \overline{x}_m\rangle$ is a maximal isotropic subspace of the symplectic vector space $G/\mathbf{Z}(G)$. Then by Lemma II.9.8 (pg. 228), there exist elements $\overline{y}_1, \ldots, \overline{y}_m$ in $G/\mathbf{Z}(G)$ such that $f(\overline{y}_i, \overline{y}_j) = 0$ and $f(\overline{x}_i, \overline{y}_j) = \delta_{ij}$. Let

$\overline{y}_i = y_i\mathbf{Z}(G)$ and $A_2 = \langle y_1, \ldots, y_m, \mathbf{Z}(G)\rangle$. Then A_2 is an abelian normal subgroup of G since $G' \le \mathbf{Z}(G) \le A_2$, and clearly $G = A_1A_2$ with $A_1 \cap A_2 = \mathbf{Z}(G)$. Since $|A_2| = p^m|\mathbf{Z}(G)|$, A_2 is a maximal abelian normal subgroup of G by e). □

For $p > 2$, the extraspecial p-groups that appear in applications most often have exponent p. An extraspecial p-group G of exponent $p > 2$ is uniquely determined by its order $|G| = p^{1+2m}$; namely, G is defined by the generators $x_1, \ldots, x_m, y_1, \ldots, y_m$ and the relations

$$\begin{aligned} x_i^p = y_i^p = [x_i, y_i]^p &= 1, \\ [x_i, x_j] = [y_i, y_j] &= 1, \\ [x_i, y_j] &= 1 \quad (\text{for } i \ne j), \\ [x_i, y_i, x_j] = [x_i, y_i, y_j] &= 1. \end{aligned}$$

The situation is somewhat more complicated for extraspecial 2-groups:

Theorem 13.8. *Let G be an extraspecial 2-group of order 2^{1+2m}.*

a) *There are two isomorphism classes of such groups, namely:*
   (1) *The central product $G_{m,1}$ of m dihedral groups of order 8 with identified centers.*
   (2) *The central product $G_{m,2}$ of $m-1$ dihedral groups of order 8 and a quaternion group with identified centers.*

b) *$G_{m,1}$ contains maximal abelian normal subgroups of types $(4, 2, \underset{m-1}{\ldots}, 2)$ and $(2, \underset{m+1}{\ldots}, 2)$. On the other hand, all maximal abelian normal subgroups of $G_{m,2}$ have type $(4, 2, \underset{m-1}{\ldots}, 2)$.*

c) *On the vector space $G/\mathbf{Z}(G)$ over* GF(2)*, we define a function q taking values in* GF(2) *as follows: Let $\mathbf{Z}(G) = \langle c\rangle$. If $\overline{g} = g\mathbf{Z}(G) \in G/\mathbf{Z}(G)$ and $g^2 = c^a$ $(a = 0, 1)$, then we set $q(\overline{g}) = a$. Then q is a quadratic form on $G/\mathbf{Z}(G)$. (See Remark II.10.3, pg. 246.)*

Proof. a) By Theorem 13.7 d), we have $G = G_1 \cdots G_m$ with $[G_i, G_j] = 1$ and $G_i \cap G_j = \mathbf{Z}(G)$ for $i \ne j$. Here, the G_i are nonabelian groups of order 8, so are either dihedral groups D or quaternion groups Q. First we show that $D \curlyvee D \cong Q \curlyvee Q$, where both central products have identified centers.

Let $G = Q_1Q_2$ be the central product of the quaternion groups $Q_1 = \langle q_1, q_2\rangle$ and $Q_2 = \langle q_3, q_4\rangle$ with identified centers. We define $D_1 = \langle d_1, d_2\rangle$ and $D_2 = \langle d_3, d_4\rangle$, where

$$d_1 = q_1q_4, \quad d_2 = q_2q_4, \quad d_3 = q_1q_2q_3, \quad d_4 = q_1q_2q_3q_4.$$

Then $d_i^2 = 1$ $(i = 1, 2, 3, 4)$, $[d_1, d_2] = [d_3, d_4] \ne 1$ and $[D_1, D_2] = 1$. Since $\exp G = 4$, the subgroups D_1 and D_2 are dihedral groups of order 8 by Example I.19.5 (pg. 144). Furthermore, $\mathbf{Z}(D_1) = \mathbf{Z}(D_2) = D_1 \cap D_2$. Thus, $Q \curlyvee Q \cong D \curlyvee D$ is proved.

By Theorem 13.7, there exist at most the two given types $G_{m,1}$ and $G_{m,2}$ of extraspecial 2-groups of order 2^{1+2m}. Once we prove b), it will follow that these two types are not isomorphic.

b) First we prove that every extraspecial 2-group G of order 2^{1+2m} contains an abelian normal subgroup of type $(4, 2, \underset{m-1}{\ldots}, 2)$:

Since G is nonabelian, there exists an $a \in G$ such that $|a| = 4$. Since $G' = \langle a^2 \rangle$, we have $\langle a \rangle \lhd G$. Let A be a maximal abelian normal subgroup of G with $a \in A$. By Theorem 13.7 e), we have $|A| = 2^{m+1}$. Since $|a| = 4$ and $\exp G/\mathbf{Z}(G) = 2$, A is an abelian normal subgroup of G of type $(4, 2, \underset{m-1}{\ldots}, 2)$.

Let $G_{m,1} = D_1 \cdots D_m$ be the central product of the dihedral groups $D_i = \langle a_i, b_i \rangle$ with identified centers, where $a_i^2 = 1$ for $i = 1, \ldots, m$. Then it is clear that $\langle \mathbf{Z}(G), a_1, \ldots, a_m \rangle$ is a maximal abelian normal subgroup of $G_{m,1}$ of type $(2, \ldots, 2)$.

Now let $G_{m,2} = D_1 \cdots D_{m-1} Q$, and let A be a maximal abelian normal subgroup of $G_{m,2}$. The elements $a \in A$ may be decomposed into products $a = a_1 a_2$ with $a_1 \in D_1 \cdots D_{m-1}$ and $a_2 \in Q$, where a_1 and a_2 are determined only up to a factor lying in $\mathbf{Z}(G)$. We define A_i $(i = 1, 2)$ to be the subgroup generated by all the a_i. Then A_1, resp. A_2, are abelian normal subgroups of $D_1 \cdots D_{m-1}$, resp. Q. Since $A \leq A_1 A_2$ and $A_1 \cap A_2 = \mathbf{Z}(G)$, it follows with the help of Theorem 13.7 e) that

$$2^{m+1} = |A| \leq |A_1 A_2| = \frac{|A_1||A_2|}{2} \leq \frac{2^m 2^2}{2} = 2^{m+1}.$$

So $A = A_1 A_2$, and A_2 is a maximal abelian normal subgroup of Q. Therefore, A_2 is cyclic of order 4 and consequently $\exp A = 4$.

c) The function q is clearly well defined on $G/\mathbf{Z}(G)$. Since $(xy)^2 = x^2 y^2 [y, x]$, it follows that

$$q(\overline{xy}) = q(\overline{x}) + q(\overline{y}) + f(\overline{y}, \overline{x}),$$

where f is the skew-symmetric bilinear form of Theorem 13.7 b). So q is a quadratic form over GF(2). □

Remark 13.9.

a) If $G_{m,1} = D_1 \cdots D_m$, where the $D_i = \langle a_i, b_i \rangle$ are dihedral groups with $a_i^2 = b_i^2 = 1$, then

$$q(\overline{a}_1^{x_1} \overline{b}_1^{y_1} \cdots \overline{a}_m^{x_m} \overline{b}_m^{y_m}) = x_1 y_1 + \cdots + x_m y_m.$$

If $G_{m,2} = D_1 \cdots D_{m-1} Q$, where $D_i = \langle a_i, b_i \rangle$ $(i = 1, \ldots, m-1)$ as before and $Q = \langle a_m, b_m \rangle$, then

$$q(\overline{a}_1^{x_1} \overline{b}_1^{y_1} \cdots \overline{a}_m^{x_m} \overline{b}_m^{y_m}) = x_1 y_1 + \cdots + x_{m-1} y_{m-1} + x_m^2 + x_m y_m + y_m^2.$$

These are the normal forms of both of the possible nondegenerate quadratic forms over GF(2) (see Dieudonné [115]).

b) Let G be an extraspecial 2-group of order 2^{1+2m}. By Theorem 13.8 c), the automorphism group $\mathbf{Aut}(G)$ acts on $G/\mathbf{Z}(G)$ as a subgroup of the orthogonal group $\mathrm{O}_{2m}(2)$ associated to the quadratic form q. Note that the quadratic form q and

the bilinear form f on $G/\mathbf{Z}(G)$ determine a system of defining relations for G:

$$[x, y] = c^{f(\overline{x},\overline{y})}, \quad x^2 = c^{q(\overline{x})} \quad (x, y \in G).$$

It is then not difficult to see that every orthogonal map of $G/\mathbf{Z}(G)$ preserving q may be extended to an automorphism of G. That is, $\mathbf{Aut}(G)$ acts on $G/\mathbf{Z}(G)$ as the full orthogonal group associated to q.

Theorem 13.10 (P. Hall). *Let G be a p-group. Assume that every abelian characteristic subgroup of G is cyclic, but G itself is not cyclic.*

a) *If $p > 2$, then G is the central product of an extraspecial p-group P of exponent p and the cyclic group $\mathbf{Z}(G)$, where $\mathbf{Z}(P)$ and $\Omega_1(\mathbf{Z}(G))$ are identified.*

b) *If $p = 2$, then one of the following occurs:*

(1) *G is extraspecial.*

(2) *G is a dihedral group, a generalized quaternion group or a quasidihedral group (see Theorem I.14.9 b), pg. 96).*

(3) *$G = P \circ Q$; where P is extraspecial, Q is cyclic, a dihedral group, a generalized quaternion group or a quasidihedral group and $\mathbf{Z}(P)$ is identified with $\Omega_1(\mathbf{Z}(Q))$.*

Proof. Note that $\mathbf{Z}(\Phi(G))$ is cyclic, being a characteristic abelian subgroup of G. Then $\Phi(G)$ is also cyclic by Theorem 7.8 c).

a) Assume $p > 2$. We show:

(1) G is regular:

Since $G' \leq \Phi(G)$, we have that G' is cyclic. The regularity of G then follows from Theorem 10.2 c) since $p > 2$.

(2) $G/\Omega_1(G)$ is cyclic:

Since $\mho_1(G)$ is cyclic, being a subgroup of $\Phi(G)$, it follows from Theorem 10.7 a) that

$$|\Omega_1(G/\Omega_1(G))| = |\Omega_2(G)/\Omega_1(G)| = |\mho_1(G)/\mho_2(G)| \leq p.$$

So $G/\Omega_1(G)$ has at most one subgroup of order p, and then it is cyclic by Theorem 8.2 a) since $p > 2$.

(3) $\Omega_1(G)$ is extraspecial:

Since G is not cyclic, it follows from Theorem 8.2 a) that $\Omega_1(G)$ is also not cyclic. So it follows from our assumptions that $\Omega_1(G)' \neq 1$. Since $\Phi(\Omega_1(G)) \leq \Phi(G)$, we have that $\Phi(\Omega_1(G))$ is cyclic of exponent p. Hence, $\Omega_1(G)' = \Phi(\Omega_1(G))$. Since $\mathbf{Z}(\Omega_1(G))$ is cyclic, being an abelian characteristic subgroup of G, it follows that $\Omega_1(G)' = \Phi(\Omega_1(G)) = \mathbf{Z}(\Omega_1(G))$ and $|\Omega_1(G)'| = p$. So $\Omega_1(G)$ is extraspecial.

(4) Assume for contradiction that $\Omega_1(G)\mathbf{Z}(G) < G$. Since $G/\Omega_1(G)$ is cyclic by (2), we have $G' \leq \Omega_1(G) \cap \Phi(G)$. It follows that $|G'| = p$. Hence, for all $x, y \in G$ we have

$$[x^p, y] = [x, y]^p = 1$$

by Lemma 1.3 a). Thus, $\mho_1(G) \leq \mathbf{Z}(G)$ and so $G/\mathbf{Z}(G)$ is elementary abelian. Furthermore, $\Phi(G) = \mho_1(G)G' \leq \mathbf{Z}(G)$. It then follows from Theorem 13.7 that $|G/\mathbf{Z}(G)| = p^{2k}$ for some k. As $G/\Omega_1(G)$ is cyclic, we have

$$|G/\Omega_1(G)\big/\Phi(G/\Omega_1(G))| = |G/\Phi(G)\Omega_1(G)| = p.$$

It follows from

$$\Omega_1(G)\Phi(G) \leq \Omega_1(G)\mathbf{Z}(G) < G$$

that $\Omega_1(G)\Phi(G) = \Omega_1(G)\mathbf{Z}(G)$.

Furthermore, since $\Omega_1(G)$ is extraspecial, we have

$$1 < |\Omega_1(G) \cap \mathbf{Z}(G)| \leq |\mathbf{Z}(\Omega_1(G))| = p,$$

so that $\Omega_1(G) \cap \mathbf{Z}(G) = \mathbf{Z}(\Omega_1(G))$. By Theorem 13.7, $|\Omega_1(G)/\mathbf{Z}(\Omega_1(G))| = p^{2m}$ for some m. But then we obtain the contradiction

$$\begin{aligned} p^{2k} &= |G/\mathbf{Z}(G)| = p|\Omega_1(G)\mathbf{Z}(G)/\mathbf{Z}(G)| = p|\Omega_1(G)/\Omega_1(G) \cap \mathbf{Z}(G)| \\ &= p|\Omega_1(G)/\mathbf{Z}(\Omega_1(G))| = p^{1+2m}. \end{aligned}$$

So we must have $G = \Omega_1(G)\mathbf{Z}(G)$ and $\Omega_1(G) \cap \mathbf{Z}(G) = \mathbf{Z}(\Omega_1(G))$.

b) Now assume $p = 2$. We show:

(1) $|G/\mathbf{C}_G(\Phi(G))| \leq 2$:

Since $\Phi(G)$ is cyclic, there exists an $a \in G$ such that $\Phi(G) = \langle a^2 \rangle$ (see Theorem 3.14 b)). Then $\langle a \rangle \trianglelefteq G$. For every $g \in G$, we have $a^g = a^k$ for some integer k. Letting $|a| = 2^r$, it follows from $g^2 \in \Phi(G)$ that $k^2 \equiv 1 \bmod 2^r$. By Theorem I.13.19 (pg. 89), k must be one of the values $1, -1, 1+2^{r-1}$ or $-1+2^{r-1}$. So $|G/\mathbf{C}_G(a)| \leq 4$ and $|G/\mathbf{C}_G(a^2)| \leq 2$.

(2) We let $U = \mathbf{C}_G(\Phi(G))$. Then $|U'| \leq 2$ and

$$\{u \in U \mid u^4 = 1\}$$

is the subgroup $\Omega_2(U)$ of U:

Since $\Phi(U) \leq \Phi(G) \leq \mathbf{Z}(U)$, the subgroup U has class at most 2. So for all $u_1, u_2 \in U$, we have $u_1^2 \in \Phi(U) \leq \mathbf{Z}(U)$ and then

$$1 = [u_1^2, u_2] = [u_1, u_2]^2$$

by Lemma 1.3 a). Now, U' is cyclic since $U' \leq \mathbf{Z}(U)$, so $|U'| \leq 2$. The formula

$$(u_1u_2)^4 = u_1^4u_2^4[u_2, u_1]^6 = u_1^4u_2^4$$

from Lemma 1.3 b) shows that $\{u \in U \mid u^4 = 1\}$ is a subgroup of U.

(3) If $\Omega_2(U)$ is abelian, then G is a dihedral group, a generalized quaternion group or a quasidihedral group:

Assume $\Omega_2(U)$ is abelian. Since $\Omega_2(U)$ is also a characteristic subgroup of G, $\Omega_2(U)$ is cyclic. Hence, U has only one subgroup of order 4. So U is cyclic by Theorem 8.3. It follows from

$$\Phi(G) = \langle a^2 \rangle < \langle a \rangle \leq U$$

and $a \notin \Phi(G)$ that $U = \langle a \rangle$. Let $G = \langle a, b \rangle$. Since $b^2 \in U$ but $b \notin U = \mathbf{C}_G(a^2)$, we have either $a^b = a^{-1}$ or $a^b = a^{-1+2^{r-1}}$. It follows by Theorem I.14.9 (pg. 96) that G is one of the specified groups.

So assume in what follows that $\Omega_2(U)$ is nonabelian.

(4) If $\exp G = 4$, then $G = P\mathbf{Z}(G)$ for some extraspecial 2-group P, and we have $P \cap \mathbf{Z}(G) = \mathbf{Z}(P)$:

In this case, $|\Phi(G)| = 2$. If $|\mathbf{Z}(G)| = 2$, then we have $1 < G' = \Phi(G) = \mathbf{Z}(G)$, and G is extraspecial. Suppose $|\mathbf{Z}(G)| = 4$, so $\mathbf{Z}(G) > \Phi(G)$. Then there exists a maximal subgroup P of G satisfying $G = P\mathbf{Z}(G)$ and $P \cap \mathbf{Z}(G) = \Phi(G)$. It follows that $G' = P'$ and $\mathbf{Z}(P) = P \cap \mathbf{Z}(G)$. So

$$\mathbf{Z}(P) = \Phi(G) = G' = P' \leq \Phi(P) \leq \Phi(G).$$

Hence, $\mathbf{Z}(P) = P' = \Phi(P)$ and P is extraspecial.

Assume in what follows that $\exp G > 4$. Since $\Phi(G) = \langle a^2 \rangle$, we have $|a| > 4$.

(5) We have $U = \Omega_2(U)\mathbf{Z}(U)$ and $\Omega_2(U) \cap \mathbf{Z}(U) = \mathbf{Z}(\Omega_2(U))$:

Assume as before that $\Phi(G) = \langle a^2 \rangle$, so $a \in U = \mathbf{C}_G(\Phi(G))$. For all $u \in U$, we have $u^2 \in \Phi(G) = \langle a^2 \rangle$, so $u^4 = a^{4k}$ for some integer k. It follows by (2) that

$$(ua^{-k})^4 = u^4 a^{-4k} [a^{-k}, u]^6 = 1,$$

and so $ua^{-k} \in \Omega_2(U)$. Hence, $U = \Omega_2(U)\langle a \rangle$. Since $\Phi(U) \leq \mathbf{Z}(U)$ and $\Omega_2(U)' \neq 1$, we have by Theorem 13.7 that

$$|U/\mathbf{Z}(U)| = 2^{2m} \quad \text{and} \quad |\Omega_2(U)/\mathbf{Z}(\Omega_2(U))| = 2^{2n}$$

for some integers m and n. Since $a^2 \in \mathbf{Z}(U)$ and $|a| > 4$, we have

$$\Omega_2(\langle a \rangle) \leq \Omega_2(U) \cap \mathbf{Z}(U) \leq \mathbf{Z}(\Omega_2(U)).$$

Since $\Omega_2(\langle a \rangle)$ and $\mathbf{Z}(\Omega_2(U))$ are cyclic of order 4, it follows that

$$\Omega_2(U) \cap \langle a \rangle = \Omega_2(\langle a \rangle) = \Omega_2(U) \cap \mathbf{Z}(U) = \mathbf{Z}(\Omega_2(U)).$$

Suppose that $\mathbf{Z}(U) = \langle a^2 \rangle$. Then we obtain the contradiction

$$\begin{aligned} 2^{2m} = |U/\mathbf{Z}(U)| &= \frac{|\Omega_2(U)\langle a \rangle|}{|\langle a^2 \rangle|} = 2|\Omega_2(U)/\Omega_2(U) \cap \langle a \rangle| \\ &= 2|\Omega_2(U)/\mathbf{Z}(\Omega_2(U))| = 2^{1+2n}. \end{aligned}$$

So $\langle a^2\rangle < \mathbf{Z}(U)$. Since $\Omega_2(U) \cap \mathbf{Z}(U) = \Omega_2(U) \cap \langle a\rangle$, it follows by comparing orders that $U = \Omega_2(U)\mathbf{Z}(U)$.

By an appropriate choice of a, we may thus obtain $\mathbf{Z}(U) = \langle a\rangle$ and $\Phi(G) = \langle a^2\rangle$.

(6) If $U = G$, then $G = P\mathbf{Z}(G)$ for some extraspecial 2-group P:

It is clear that $\Omega_2(G)$ again satisfies the assumptions of the theorem.

By (4), we have $\Omega_2(G) = P\mathbf{Z}(\Omega_2(G))$ for an extraspecial group P. By (5), we have $G = \Omega_2(G)\mathbf{Z}(G)$ and $\Omega_2(G) \cap \mathbf{Z}(G) = \mathbf{Z}(\Omega_2(G))$. So we have $G = P\mathbf{Z}(G)$. It follows from $1 < P \cap \mathbf{Z}(G) \leq \mathbf{Z}(P)$ and $|\mathbf{Z}(P)| = p$ that $P \cap \mathbf{Z}(G) = \mathbf{Z}(P)$.

Assume in what follows that $U < G$, so that $|G/U| = 2$ by (1).

(7) G is one of the groups listed in (3):

Let $b \in G - U$, so $b \notin U = \mathbf{C}_G(a^2)$. Since $b^2 \in \Phi(G) = \langle a^2\rangle$, we have $a^b = a^{-1}$ or $a^b = a^{-1+2^{r-1}}$. Thus, $Q = \langle a, b\rangle$ is by Theorem I.14.9 (pg. 96) either a dihedral group, a generalized quaternion group, or a quasidihedral group. Since $U = \Omega_2(U)\langle a\rangle$, we have

$$G = \langle U, b\rangle = \langle \Omega_2(U), a, b\rangle = \Omega_2(U)Q.$$

Let $u \in \Omega_2(U)$. Then $u^2 \in \Phi(U) \leq \mathbf{Z}(U) = \langle a\rangle$ and $|u^2| \leq 2$, hence $u^2 \in \mathbf{Z}(G)$. We have

$$a^u = a \quad \text{and} \quad b^u = a^i b$$

for some i. Since $u^2 \in \mathbf{Z}(G)$, we have $b = b^{u^2} = a^{2i}b$, so $a^{2i} = 1$. It follows that $|\Omega_2(U)/\Omega_2(U) \cap \mathbf{C}_G(Q)| \leq 2$. Since

$$a^{2^{r-2}} \in \Omega_2(U), \quad \text{but} \quad a^{2^{r-2}} \notin \mathbf{C}_G(Q),$$

we have $|\Omega_2(U)/\Omega_2(U) \cap \mathbf{C}_G(Q)| = 2$. We let $P = \Omega_2(U) \cap \mathbf{C}_G(Q)$. It follows from $\Omega_2(U) = P\langle a^{2^{r-2}}\rangle$ that $U = \Omega_2(U)\langle a\rangle = P\langle a\rangle$ and

$$G = \Omega_2(U)Q = PQ.$$

We have

$$1 < \Omega_2(U)' = P' \leq \Phi(P) \leq P \cap \Phi(U) \leq P \cap \mathbf{Z}(U) = P \cap \langle a\rangle = \mathbf{Z}(P).$$

The group $\mathbf{Z}(P) = P \cap \mathbf{Z}(U) = P \cap \langle a\rangle$ has order 2 since $a^{2^{r-2}} \notin P$. So P is extraspecial. Furthermore, we have

$$P \cap Q \leq \mathbf{C}_G(Q) \cap Q = \mathbf{Z}(Q).$$

Since $|\mathbf{Z}(Q)| = 2$, it follows that $P \cap Q = \mathbf{Z}(P) = \mathbf{Z}(Q) = \mathbf{Z}(G)$. □

Exercises

32) If G is an extraspecial p-group of exponent p, then $\mathbf{Z}(G)$ is the only nontrivial characteristic subgroup of G.
33) Let $G = PA$ be the semidirect product of P with A. Assume $(|P|, |A|) = 1$, $P = [P, A]$, and suppose every characteristic abelian subgroup of the p-group P is cyclic. Then P is either cyclic or extraspecial.
34) Let Z be a cyclic 2-group with $|Z| \geq 4$, let Q be the quaternion group and D the dihedral group of order 8. Then $Q \curlyvee Z \cong D \curlyvee Z$, where $\Omega_1(Z)$ and Q', resp. D', are identified.

14 The p-groups of maximal class

Definition 14.1. Let G be a p-group of order p^n with $n \geq 2$. Since $|G/\mathbf{K}_2(G)| \geq p^2$, the nilpotency class $c(G)$ of G is at most $n-1$. We say that G is a p-group of maximal class if G has order p^n and nilpotency class $n-1$.

The theory of p-groups of maximal class (in fact, a considerably larger family of p-groups) was developed in Blackburn [36]. That paper is the most significant contribution to the theory of p-groups since the fundamental work of P. Hall [207, 208]. The presentation given here follows unpublished notes of Blackburn.

Lemma 14.2. *Let G be a p-group of maximal class of order p^n.*
a) *We have $|G/\mathbf{K}_2(G)| = p^2$ and $|\mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)| = p$ for $2 \leq i \leq n-1$.*
b) *For $i \geq 2$, $\mathbf{K}_i(G)$ is the only normal subgroup of G of index p^i.*
c) *If $N \lhd G$ and $|G/N| \geq p^2$, then G/N is a p-group of maximal class.*
d) *We have $\mathbf{Z}_i(G) = \mathbf{K}_{n-i}(G)$ for $0 \leq i \leq n-1$.*

Proof. a) The claim follows immediately from

$$|G| = p^n = \prod_{i=1}^{n-1} |\mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)|$$

and the inequalities

$$|G/\mathbf{K}_2(G)| \geq p^2 \quad \text{and} \quad |\mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)| \geq p \quad \text{for} \quad 2 \leq i \leq n-1.$$

b) Let $N \lhd G$ with $|G/N| = p^i$. Then G/N has class at most $i-1$. It follows that

$$1 = \mathbf{K}_i(G/N) = \mathbf{K}_i(G)N/N,$$

so $\mathbf{K}_i(G) \leq N$. Since $|G/\mathbf{K}_i(G)| = p^i$, this forces $N = \mathbf{K}_i(G)$.

c) Assume $|G/N| = p^i \geq p^2$. By b), $N = \mathbf{K}_i(G)$. So G/N has class $i-1$ and is therefore a p-group of maximal class.

d) Since $c(G) = n - 1$, we have $\mathbf{Z}_{n-1}(G) = G$. It follows from

$$1 = \mathbf{Z}_0(G) < \mathbf{Z}_1(G) < \cdots < \mathbf{Z}_{n-1}(G) = G$$

and $|G/\mathbf{Z}_{n-2}(G)| \leq p^2$ that $|\mathbf{Z}_i(G)| = p^i$ for $i \leq n-2$. So $\mathbf{Z}_i(G) = \mathbf{K}_{n-i}(G)$ by b). □

The 2-groups of maximal class were all described in Theorem 11.9 b). So we may assume in what follows that $p > 2$.

All nonabelian p-groups of order p^3 clearly have maximal class. For $p > 3$, the p-groups of maximal class and order p^4 can be easily extracted from Theorem 12.6: They are the groups given in 9), 10), 12) and 13). (For $p = 3$, one should replace the group in 13) with the group given in Exercise 29.) Therefore, in our study of p-groups of maximal class, we assume $n \geq 5$ from now on.

Next, we define a characteristic subgroup that is fundamental to the description of p-groups of maximal class.

Definition 14.3. Let G be a p-group of maximal class with $|G| = p^n$. We define the characteristic subgroup G_1 by

$$G_1 = \mathbf{C}_G(\mathbf{K}_2(G)/\mathbf{K}_4(G)).$$

For $2 \leq i \leq n$, we define $G_i = \mathbf{K}_i(G)$. (So $G_n = 1$.)

Lemma 14.4. *Let G be a p-group of maximal class with $|G| = p^n$ and $n \geq 4$. Then:*
a) $|G/G_1| = p$.
b) $G_1/G_3 \not\leq \mathbf{Z}(G/G_3)$.

Proof. a) As $|G_2/G_4| = p^2$, we have $p^2 \nmid |\mathbf{Aut}(G_2/G_4)|$. Since

$$|G/G_1| \quad \text{divides} \quad |\mathbf{Aut}(G_2/G_4)|,$$

it follows that $|G/G_1| \leq p$. On the other hand, since $[G_2, G] = G_3 \not\leq G_4$, we have

$$G_1 = \mathbf{C}_G(G_2/G_4) < G.$$

So $|G/G_1| = p$.

b) Assume for contradiction that $G_1/G_3 \leq \mathbf{Z}(G/G_3)$. Then G/G_3 is a cyclic extension of a subgroup of its center, so is abelian. This yields the contradiction $G' = G_2 \leq G_3$. □

Definition 14.5. Let G be a p-group of maximal class with $|G| = p^n$ and $n \geq 5$. We say G is an exceptional group whenever there exists an i with $3 \leq i \leq n-2$ such that $G_1 \neq \mathbf{C}_G(G_i/G_{i+2})$. In this case, $[G_i, G_1] \leq G_{i+1}$ but $[G_i, G_1] \not\leq G_{i+2}$. So it follows by Lemma 14.2 b) that $[G_i, G_1] = G_{i+1}$. Thus, G is not an exceptional group if and only if we have $G_1 = \mathbf{C}_G(G_i/G_{i+2})$, which is equivalent to $[G_i, G_1] \leq G_{i+2}$, for all $2 \leq i \leq n-2$. If G is not an exceptional group, then clearly no factor group of G is an exceptional group.

Next, we state the main theorems.

Main Theorem 14.6. *Let G be a p-group of maximal class with $|G| = p^n$ and $n \geq 5$. Then:*

a) *G/G_{n-1} is not an exceptional group; that is, we have $G_1 = \mathbf{C}_G(G_i/G_{i+2})$ for $2 \leq i \leq n-3$.*

b) *If G is an exceptional group, then $p > 3$. Furthermore, we have $6 \leq n \leq p+1$ and n is even.*

Main Theorem 14.7. *Let G be a p-group of maximal class with $|G| = p^n$ and $n \geq 5$. Then the following are equivalent:*

a) *G is not an exceptional group; that is, $[G_i, G_1] \leq G_{i+2}$ for $i = 2, \ldots, n-2$.*

b) *We have $[G_i, G_j] \leq G_{i+j+1}$ for all i, j with $i + j > 2$. (For $k \geq n$, we define $G_k = 1$.)*

c) *The abelian factor groups G_i/G_{i+2} (with $2 \leq i \leq n-2$) of order p^2 are pairwise isomorphic as G-groups.*

We prove the main theorems in a series of intermediate steps.

Lemma 14.8. *Let G be a p-group of maximal class with $|G| = p^n$ and $n \geq 4$. Assume G is not an exceptional group. We choose elements s and s_1 from G such that*

$$G = \langle G_1, s \rangle \quad \text{and} \quad G_1 = \langle G_2, s_1 \rangle.$$

We define the elements $s_i \in G$ $(i \geq 2)$ recursively by $s_i = [s_{i-1}, s]$. Then we have $G_i = \langle G_{i+1}, s_i \rangle$ for $2 \leq i \leq n-1$.

Proof. We have

$$G = \langle G_1, s \rangle = \langle G_2, s_1, s \rangle.$$

Since $G_2 = G' \leq \Phi(G)$ (see Theorem 3.11), we have $G = \langle s_1, s \rangle$. So by Theorem 1.11 b),

$$G_2 = \mathbf{K}_2(G) = \langle [s_1, s], \mathbf{K}_3(G) \rangle = \langle s_2, G_3 \rangle.$$

Assume for induction that $G_{i-1} = \langle G_i, s_{i-1} \rangle$. Since $G = \langle s, s_1 \rangle$, it follows that

$$G_i = [G_{i-1}, G] = \langle G_{i+1}, [s_{i-1}, s], [s_{i-1}, s_1] \rangle.$$

As G is not an exceptional group, we have

$$[s_{i-1}, s_1] \in [G_{i-1}, G_1] \leq G_{i+1}$$

for $i \leq n-1$. Since $[s_{i-1}, s] = s_i$, the claim is proved. □

Lemma 14.9. *Let G be a p-group of maximal class with $|G| = p^n$ and $n \geq 5$. Suppose that for all i, j with $2 < i + j \leq n-2$, we have $[G_i, G_j] \leq G_{i+j+1}$. Let*

$s, s_1 \in G$ such that $G = \langle G_1, s\rangle$ and $G_1 = \langle G_2, s_1\rangle$. For $2 \le i \le n-2$, let s_i be defined recursively by $s_i = [s_{i-1}, s]$. Then

$$[s_1, s_{n-2}] = [s_2, s_{n-3}]^{-1} = \cdots = [s_i, s_{n-i-1}]^{(-1)^{i-1}} = \cdots = [s_{n-2}, s_1]^{(-1)^{n-1}}.$$

Proof. By our assumption,

$$[G_i, G_j] \le G_{i+j+1} \quad \text{for} \quad 2 < i + j \le n-2. \tag{$*$}$$

Let $2 \le i \le n-2$. Then we have

$$\begin{aligned}[s_i, s_{n-i-1}] &= s_i^{-1} s_i^{s_{n-i-1}} = s_i^{-1}[s_{i-1}, s]^{s_{n-i-1}} \\ &= s_i^{-1}[s_{i-1}^{s_{n-i-1}}, s^{s_{n-i-1}}].\end{aligned}$$

Now by $(*)$,

$$[s_{i-1}, s_{n-i-1}] \in [G_{i-1}, G_{n-i-1}] \le G_{n-1} = \mathbf{Z}(G).$$

So there remains

$$\begin{aligned}[s_i, s_{n-i-1}] &= s_i^{-1}[s_{i-1}, s[s, s_{n-i-1}]] = s_i^{-1}[s_{i-1}, s s_{n-i}^{-1}] \\ &= s_i^{-1}[s_{i-1}, s_{n-i}^{-1}][s_{i-1}, s]^{s_{n-i}^{-1}}\end{aligned}$$

by Lemma 1.2 b). In this expression,

$$[s_{i-1}, s_{n-i}^{-1}] \in [G_{i-1}, G_{n-i}] \le G_{n-1} = \mathbf{Z}(G),$$

and so by Lemma 1.3 a),

$$[s_{i-1}, s_{n-i}^{-1}] = [s_{i-1}, s_{n-i}]^{-1} \in \mathbf{Z}(G).$$

Furthermore, we have

$$[s_{i-1}, s] = s_i$$

and since $2 \le i \le n-2$,

$$[s_i, s_{n-i}] \in [G_i, G_{n-i}] = [\mathbf{K}_i(G), \mathbf{K}_{n-i}(G)] \le \mathbf{K}_n(G) = 1.$$

So there remains

$$[s_i, s_{n-i-1}] = s_i^{-1}[s_{i-1}, s_{n-i}]^{-1} s_i = [s_{i-1}, s_{n-i}]^{-1}$$

for $2 \le i \le n-2$. □

14.10 (Proof of Theorem 14.7). We prove the claim by induction on $|G|$.

a) $\Rightarrow$ b): Since G is not an exceptional group, we have

(1) $[G_1, G_i] \le G_{i+2}$ for all $i \ge 2$.

Since G/G_{n-1} is also not an exceptional group, applying the induction hypothesis to G/G_{n-1} yields

(2) $[G_i, G_j] \leq G_{i+j+1}$ for $2 < i + j \leq n - 2$.

Hence, the assumptions of Lemma 14.9 are satisfied. It follows from (1) that

$$[s_1, s_{n-2}] \in [G_1, G_{n-2}] \leq G_n = 1,$$

and so by Lemma 14.9,

$$1 = [s_1, s_{n-2}] = [s_i, s_{n-i-1}] \quad \text{for} \quad i \leq n - 2.$$

By Lemma 14.8,

$$G_i = \langle G_{i+1}, s_i \rangle \quad \text{and} \quad G_{n-i-1} = \langle G_{n-i}, s_{n-i-1} \rangle.$$

Now we have

$$\begin{aligned} [G_{n-i-1}, G_{i+1}] &\leq G_n = 1, \\ [s_{n-i-1}, s_i] &= 1 \qquad \text{and} \\ [G_{n-i}, s_i] &\leq G_n = 1. \end{aligned}$$

So $[G_{n-i-1}, G_i] = 1$, and hence $[G_i, G_j] \leq G_{i+j+1}$ for $i + j = n - 1$. For $i + j \geq n$ and $i, j \geq 2$, the claim in Theorem 14.7 b) follows from Theorem 2.11 b) since

$$[G_i, G_j] = [\mathbf{K}_i(G), \mathbf{K}_j(G)] \leq \mathbf{K}_{i+j}(G) \leq \mathbf{K}_n(G) = 1.$$

b) $\Rightarrow$ c): Let $t_i \in G - \mathbf{C}_G(G_i/G_{i+2})$ for $2 \leq i \leq n - 2$. The map τ_i defined by

$$(xG_{i+2})^{\tau_i} = [x, t_i]G_{i+3}$$

for all $x \in G_i$ is a homomorphism from G_i/G_{i+2} to G_{i+1}/G_{i+3} for $2 \leq i \leq n - 3$:

For the purpose of the proof, we may assume $G_{i+3} = 1$. Since $[G_{i+2}, t_i] = 1$, the map τ_i is well defined. For $x_1, x_2 \in G_i$, we have by Lemma 1.2 c) and

$$\begin{aligned} [x_1, t_i, x_2] \in [G_i, G, G_i] = 1 \qquad &\text{that} \\ (x_1x_2G_{i+2})^{\tau_i} = [x_1x_2, t_i] &= [x_1, t_i][x_1, t_i, x_2][x_2, t_i] \\ &= [x_1, t_i][x_2, t_i] = (x_1G_{i+2})^{\tau_i}(x_2G_{i+2})^{\tau_i}. \end{aligned}$$

Next, we claim that τ_i is in fact a G-homomorphism:

For $x \in G_i$ and $g \in G$, we want to prove that

$$[x^g, t_i] = [x^g, t_i^g].$$

It clearly suffices to prove that $t_i^g t_i^{-1} \in \mathbf{C}_G(G_i)$. By our assumption in b), we have for all $x \in G_i$ and $i \leq n - 3$ that

$$[x, t_i^g t_i^{-1}] \in [G_i, G_2] \leq G_{i+3} = 1.$$

Now, since τ_i is a G-homomorphism from G_i/G_{i+2} to G_{i+1}, $\ker \tau_i$ and $\operatorname{im} \tau_i$ are normal subgroups of G. Since $t_i \notin \mathbf{C}_G(G_i/G_{i+2})$, we have $\operatorname{im} \tau_i \not\leq G_{i+2}$. So $\operatorname{im} \tau_i = G_{i+1}$ and consequently $\ker \tau_i = G_{i+2}$. That is, τ_i is a G-isomorphism from G_i/G_{i+2} to G_{i+1}/G_{i+3}.

c) $\Rightarrow$ a): As the G_i/G_{i+2} (for $2 \leq i \leq n-2$) are isomorphic G-groups, they all have the same centralizer, namely G_1. Thus, G is not an exceptional group. □

Theorem 14.11. *Let G be a p-group of maximal class. If $G'' = 1$, then G is not an exceptional group.*

Proof. We prove the claim by induction on $|G| = p^n$. We may assume that $n \geq 5$ and that G/G_{n-1} is not an exceptional group. So $[G_1, G_i] \leq G_{i+2}$ for $2 \leq i \leq n-3$. It follows from Lemma 14.9 that

$$[s_1, s_{n-2}] = [s_2, s_{n-3}]^{-1} \in [G_2, G_2] = 1.$$

This implies $G_1 = \langle s_1, G_2 \rangle \leq \mathbf{C}_G(G_{n-2})$, and so G is not an exceptional group. □

Next, we prove part of Theorem 14.6:

Theorem 14.12. *Let G be a p-group of maximal class with $|G| = p^n$ and $5 \leq n \leq p+2$. Then G/G_{n-1} is not an exceptional group. If G is an exceptional group, then n is even.*

Proof. We prove the claim by induction on n.

a) Assume $n = 5$. By Theorem 7.11, $G'' = 1$ and so G is not an exceptional group by Theorem 14.11.

b) Now assume $6 \leq n \leq p+2$.

Case 1: First assume that G/G_{n-1} is not an exceptional group. By Theorem 14.7, $[G_i, G_j] \leq G_{i+j+1}$ for $2 < i+j \leq n-2$. Then by Lemma 14.9,

$$[s_1, s_{n-2}] = [s_1, s_{n-2}]^{(-1)^n}.$$

If n is odd, then $[s_1, s_{n-2}] = 1$ since $p > 2$, and then $[G_1, G_{n-2}] = 1$ as well. As G/G_{n-1} is not an exceptional group, we have $G_1 = \mathbf{C}_G(G_i/G_{i+2})$ for $2 \leq i \leq n-2$, and thus G is not an exceptional group.

Case 2: Now assume that G/G_{n-1} is an exceptional group. Since $n \geq 6$, applying Theorem 14.12 to G/G_{n-1} shows that G/G_{n-2} is not an exceptional group and that n is odd. We again choose $s, s_1 \in G$ such that $G = \langle G_1, s \rangle$ and $G_1 = \langle G_2, s_1 \rangle$. We define $s_i \in G_i$ for $2 \leq i \leq n-3$ by

(1) $s_i = [s_{i-1}, s]$,

just as we did in Lemma 14.8. An application of Theorem 14.7 to G/G_{n-2} yields

(2) $[s_i, s_j] \in [G_i, G_j] \leq G_{i+j+1}$ for $2 < i+j \leq n-3$.

Since G/G_{n-2} is not an exceptional group, applying Lemma 14.9 to G/G_{n-1} for $2 \leq i \leq n-3$ yields

(3) $[s_1, s_{n-3}] \equiv [s_i, s_{n-i-2}]^{(-1)^{i-1}} \bmod G_{n-1}$.

As G/G_{n-1} is an exceptional group but G/G_{n-2} is not, we have $[G_1, G_i] \leq G_{i+2}$ for $i \leq n-4$ and $[G_1, G_{n-3}] \not\leq G_{n-1}$, but $[G_1, G_{n-3}] \leq G_{n-2}$. By Lemma 14.2 b), this forces $[G_1, G_{n-3}] = G_{n-2}$. Now we set

(4) $s_{n-2} = [s_{n-3}, s_1]$.

We have $G_1 = \langle G_2, s_1\rangle$. Since G/G_{n-2} is not an exceptional group, we have $G_{n-3} = \langle G_{n-2}, s_{n-3}\rangle$ by Lemma 14.8. Thus, it follows that

(5) $G_{n-2} = [G_1, G_{n-3}] = \langle G_{n-1}, s_{n-2}\rangle$.

From (3) and (4) we obtain

(6) $[s_i, s_{n-i-2}] \equiv s_{n-2}^{(-1)^i} \bmod G_{n-1}$ $(i = 2, \ldots, n-3)$.

In particular,

$$[s_2, s_{n-4}] \equiv s_{n-2} \bmod G_{n-1},$$

and then, since $G_{n-1} = \mathbf{Z}(G)$,

(7) $[s_{n-2}, s_1] = [[s_2, s_{n-4}], s_1] = [s_2, s_{n-4}]^{-1}[s_2, s_{n-4}]^{s_1}$.

In this expression, we have

$$[s_2, s_{n-4}]^{s_1} = [s_2^{s_1}, s_{n-4}^{s_1}] = [s_2[s_2, s_1], s_{n-4}[s_{n-4}, s_1]].$$

Since $n > 5$, we have by (2) that $[s_2, s_1] \in G_4$. Therefore, $[s_2, s_1]$ commutes with the element $s_{n-4}^{s_1} \in G_{n-4}$. So by Lemma 1.2 c),

$$[s_2, s_{n-4}]^{s_1} = [s_2, s_{n-4}[s_{n-4}, s_1]]^{[s_2, s_1]}.$$

Since $[s_2, s_1] \in G_4$, $[s_2, G_{n-4}] \leq G_{n-2}$ and $[G_4, G_{n-2}] = 1$, it follows that

$$[s_2, s_{n-4}]^{s_1} = [s_2, s_{n-4}[s_{n-4}, s_1]].$$

By (2), $[s_{n-4}, s_1] \in G_{n-2}$. Since $[G_2, G_{n-2}] = 1$, it then follows by Lemma 1.3 b) that

$$[s_2, s_{n-4}]^{s_1} = [s_2, s_{n-4}]^{[s_{n-4}, s_1]} = [s_2, s_{n-4}].$$

So from (7), it follows that $[s_{n-2}, s_1] = 1$. Therefore, $s_{n-2} \in \mathbf{C}_G(G_1)$. Then by (5) we have

$$G_{n-2} = \langle G_{n-1}, s_{n-2}\rangle \leq \mathbf{C}_G(G_1).$$

Hence, we have

(8) $[G_1, G_{n-2}] = 1$.

Since $G_{n-2} \not\leq G_{n-1} = \mathbf{Z}(G)$, certainly $s_{n-2} \notin \mathbf{Z}(G)$. So $\mathbf{C}_G(s_{n-2}) = G_1$ and $[s_{n-2}, s] \neq 1$. We set

(9) $s_{n-1} = [s_{n-2}, s]$.

Then $G_{n-1} = \langle s_{n-1}\rangle$.

Next, we prove by induction on i that

(10) $[s_i, s_{n-i-1}] = s_{n-1}^{(-1)^{i-1}(i-1)} \quad (i = 2, \dots, n-3)$.
For $i = 2$,

$$[s_2, s_{n-3}] = s_2^{-1} s_2^{s_{n-3}} = s_2^{-1}[s_1, s]^{s_{n-3}} = s_2^{-1}[s_1^{s_{n-3}}, s^{s_{n-3}}]$$
$$= s_2^{-1}[s_1[s_1, s_{n-3}], s[s, s_{n-3}]].$$

Here, $[s, s_{n-3}] \in G_{n-2}$, and so $[s, s_{n-3}]$ commutes with every element of G_1 by (8). So by (4),

$$[s_2, s_{n-3}] = s_2^{-1}[s_1 s_{n-2}^{-1}, s] = s_2^{-1}[s_1, s]^{s_{n-2}^{-1}}[s_{n-2}^{-1}, s].$$

Furthermore, since $[s_1, s] = s_2 \in \mathbf{C}_G(G_{n-2})$ and $[s_{n-2}, s] \in G_{n-1} = \mathbf{Z}(G)$, it follows that

$$[s_2, s_{n-3}] = s_2^{-1} s_2 [s_{n-2}, s]^{-1} = s_{n-1}^{-1}.$$

This is the claim in (10) for $i = 2$.
Assume for induction that $i > 2$ and
(11) $[s_{i-1}, s_{n-i}] = s_{n-1}^{(-1)^i (i-2)}$.
Since $2 < i \le n-3$, we have by (1) that

$$[s_i, s_{n-i-1}] = s_i^{-1}[s_{i-1}, s]^{s_{n-i-1}} = s_i^{-1}[s_{i-1}^{s_{n-i-1}}, s^{s_{n-i-1}}]$$
$$= s_i^{-1}[s_{i-1}[s_{i-1}, s_{n-i-1}], s[s, s_{n-i-1}]]$$
$$= s_i^{-1}[s_{i-1}[s_{i-1}, s_{n-i-1}], s s_{n-i}^{-1}].$$

By (3) (with $i-1$ in place of i) and (4),

$$[s_{i-1}, s_{n-i-1}] \equiv [s_1, s_{n-3}]^{(-1)^i} \equiv s_{n-2}^{(-1)^{i-1}} \bmod G_{n-1}.$$

Since $G_{n-1} = \mathbf{Z}(G)$, using Lemma 1.2 b) we have

$$[s_i, s_{n-i-1}] = s_i^{-1}[s_{i-1} s_{n-2}^{(-1)^{i-1}}, s\, s_{n-i}^{-1}]$$
$$= s_i^{-1}[s_{i-1} s_{n-2}^{(-1)^{i-1}}, s_{n-i}^{-1}][s_{i-1} s_{n-2}^{(-1)^{i-1}}, s]^{s_{n-i}^{-1}}$$
$$= s_i^{-1}[s_{i-1} s_{n-2}^{(-1)^{i-1}}, s_{n-i}^{-1}][s_{i-1} s_{n-2}^{(-1)^{i-1}}, s]$$

since $[\mathbf{K}_i(G), s_{n-i}] \le \mathbf{K}_n(G) = 1$. Taking into consideration that

$$[s_{i-1}, s_{n-i}^{-1}] = [s_{i-1}, s_{n-i}]^{-1} \in G_{n-1} = \mathbf{Z}(G),$$
$$[s_{n-2}^{(-1)^{i-1}}, s_{n-i}^{-1}] \in [G_{n-2}, G_2] = 1 \quad (\text{since } i \le n-3),$$
$$[s_{i-1}, s] = s_i \quad \text{and}$$
$$[s_{n-2}^{(-1)^{i-1}}, s] = s_{n-1}^{(-1)^{i-1}} \quad (\text{by (9) and Lemma 1.3 a)}),$$

it follows from (11) that

$$[s_i, s_{n-i-1}] = s_i^{-1}[s_{i-1}, s_{n-i}]^{-1} s_i s_{n-1}^{(-1)^{i-1}} = s_{n-1}^{(-1)^{i-1}(i-2)} s_{n-1}^{(-1)^{i-1}}$$
$$= s_{n-1}^{(-1)^{i-1}(i-1)}.$$

This proves the claim in (10).

In (10), we set $i = \frac{n-1}{2}$; this makes sense since n is odd; it is allowed since $n \geq 5$ means that $\frac{n-1}{2} \leq n-3$. This yields

$$s_{n-1}^{\frac{n-3}{2}} = 1.$$

Since $s_{n-1} \neq 1$, it follows that $n - 3 \equiv 0 \bmod p$, contradicting our assumption that $n \leq p+2$. □

We have proved Theorem 14.6 for $n \leq p+2$. We must postpone for now the proof of Theorem 14.6 for $n > p+2$ and develop some more theory.

Lemma 14.13. *Let G be a p-group of maximal class with $|G| = p^n$ and $n \geq 4$. Assume $s \in G$ and $s \notin \mathbf{C}_G(G_i/G_{i+2})$ for all i with $2 \leq i \leq n-2$. (In particular, $s \notin \mathbf{C}_G(G_2/G_4) = G_1$ and therefore $G = \langle G_1, s\rangle$.) Then:*

a) $\mathbf{C}_G(s) = \langle G_{n-1}, s\rangle = \langle s\rangle G_{n-1}$.

b) $s^p \in G_{n-1}$, *so* $|s| \leq p^2$ *and* $|\mathbf{C}_G(s)| = p^2$.

c) $\{s^g \mid g \in G\} = sG_2$.

d) *If* $sG_2 = \overline{s}G_2$, *then* $s^p = \overline{s}^p$.

(*Note: Once Theorem 14.6 a) is proved, we will know that* $G_1 = \mathbf{C}_G(G_i/G_{i+2})$ *for* $2 \leq i \leq n-3$. *Since G has exactly $p+1$ subgroups of index p, there exists some $U \leq G$ such that $[G : U] = p$ and $G_1 \neq U \neq \mathbf{C}_G(G_{n-2})$. If $U = \langle G_2, s\rangle$, then s has the properties described in Lemma 14.13.*)

Proof. a) Since $G = \langle G_1, s\rangle = \langle s\rangle G_1$, every $g \in G$ has the form $g = s^a x$ for some $x \in G_1$. So to prove a), it suffices to show that

$$G_1 \cap \mathbf{C}_G(s) = G_{n-1}.$$

Let $z \in G_1 \cap \mathbf{C}_G(s)$. Then there is some $j \geq 1$ with $z \in G_j - G_{j+1}$.

Assume for contradiction that $j = 1$. Then $G_1 = \langle G_2, z\rangle$. Furthermore, we have $G = \langle G_1, s\rangle$ and $z \in \mathbf{C}_G(s)$. Hence,

$$G = \langle G_2, z, s\rangle = \langle \Phi(G), z, s\rangle = \langle z, s\rangle.$$

Since $z \in \mathbf{C}_G(s)$, G is then abelian. This is a contradiction.

Now assume for contradiction that $2 \leq j \leq n-2$. Then $G_j = \langle G_{j+1}, z\rangle$, $[s, z] = 1$ and $[s, G_{j+1}] \leq G_{j+2}$. But this yields $[s, G_j] \leq G_{j+2}$, contradicting our choice of s.

We conclude that $z \in G_{n-1}$ and $G_1 \cap \mathbf{C}_G(s) = G_{n-1}$.

b) We have $s^p \in G_1 \cap \mathbf{C}_G(s) = G_{n-1}$, and therefore $|s| \leq p^2$. Since $s \notin G_{n-1}$, it then follows from a) that

$$|\mathbf{C}_G(s)| = |\langle s\rangle G_{n-1}| = p|G_{n-1}| = p^2.$$

c) Since $|\mathbf{C}_G(s)| = p^2$ by b), it follows that $|\{s^g \mid g \in G\}| = [G : \mathbf{C}_G(s)| = p^{n-2}$. On the other hand, since $s^g = s[s, g] \in sG_2$, we have

$$\{s^g \mid g \in G\} \subseteq sG_2.$$

Since $|sG_2| = p^{n-2}$, the claim follows.

d) Let $sG_2 = \overline{s}G_2$. By c), there exists an $x \in G$ such that $\overline{s} = s^x$. By b), $s^p \in G_{n-1} = \mathbf{Z}(G)$. This yields

$$\overline{s}^p = (s^x)^p = (s^p)^x = s^p. \qquad \square$$

Lemma 14.14. *Let G be a p-group of maximal class with $|G| = p^n$ and $5 \leq n \leq p+1$. Then both G/G_{n-1} and G_2 have exponent p.*

Proof. a) By Theorem 14.12, G/G_{n-1} is not an exceptional group. Then

$$G_1 = \mathbf{C}_G(G_i/G_{i+2}) \quad \text{for} \quad 2 \leq i \leq n-3,$$

but we could have $G_1 \neq \mathbf{C}_G(G_{n-2})$. Since $p > 2$, G has exactly $p + 1 \geq 4$ maximal subgroups. So there exist two distinct maximal subgroups $\langle s, G_2\rangle$ and $\langle \overline{s}, G_2\rangle$ of G that are themselves distinct from all $\mathbf{C}_G(G_i/G_{i+2})$ $(2 \leq i \leq n-2)$. Hence,

$$s, \overline{s} \notin \mathbf{C}_G(G_i/G_{i+2}) \quad \text{for} \quad 2 \leq i \leq n-2.$$

We have $G = \langle s, \overline{s}, G_2\rangle = \langle s, \overline{s}\rangle$. As the assumptions of Lemma 14.13 are satisfied, we have $s^p \in G_{n-1}$ and $\overline{s}^p \in G_{n-1}$. As a consequence, we see that G/G_{n-1} can be generated by two elements that have order at most p. Now, G/G_{n-1} is regular by Theorem 10.2 b) since

$$|G/G_{n-1}| = p^{n-1} \leq p^p.$$

So G/G_{n-1} has exponent p by Theorem 10.5 a).

b) Since $|G_1| = p^{n-1} \leq p^p$, G_1 is also regular by Theorem 10.2 b). It follows from a) that

$$\mho_1(G_1) \leq \mho_1(G) \leq G_{n-1}.$$

Then by Theorem 10.7 a),

$$|\Omega_1(G_1)| = |G_1/\mho_1(G_1)| \geq |G_1/G_{n-1}| = p^{n-2}.$$

From this it follows that $G_2 \leq \Omega_1(G_1)$, and so G_2 has exponent p. $\square$

Lemma 14.15. *Let G be a p-group of maximal class, let $|G| = p^n$ with $n > p + 1$ and $G_1 = \langle G_2, s_1\rangle$. Then $s_1^p \in G_p - G_{p+1}$.*

(*The assumption that $n > p + 1$ cannot be removed, as shown by the regular wreath product of two cyclic groups of order p.*)

Proof. Applying Lemma 14.14 to G/G_{p+1} shows that $s_1^p \in G_p$.

Now let $G = \langle G_1, s\rangle$ and $s_i = [s_{i-1}, s]$ for $i = 2, \ldots, p$. Since $p > 2$, $p+2$ is odd and so G/G_{p+2} is not an exceptional group by Theorem 14.12. Thus, $G_p = \langle G_{p+1}, s_p\rangle$ by Lemma 14.8, and so $s_p \notin G_{p+1}$. The Zassenhaus identity (Theorem 9.7) implies

$$s_p = [s_1, \underset{p-1}{s, \ldots, s}] \in G_{p+1}\mho_1(G).$$

Since $s_p \notin G_{p+1}$, we have $\mho_1(G) \not\leq G_{p+1}$. Hence, there exists an $x \in G$ such that $x^p \notin G_{p+1}$.

Assume for contradiction that $x \notin G_1$. As G/G_{p+2} is not an exceptional group, we obtain $x^p \in G_{p+1}$ by Lemma 14.13 b), but this contradicts our choice of x.

So $x \in G_1 = \langle G_2, s_1\rangle$. By Lemma 14.14, we have $\mho_1(G_2) \leq G_{p+1}$. If $s_1^p \in G_{p+1}$, then it follows from the regularity of G_1/G_{p+1} that G_1/G_{p+1} has exponent p. But since $x \in G_1$, we then obtain the contradiction $x^p \in G_{p+1}$. Hence, $s_1^p \notin G_{p+1}$. □

Theorem 14.16. *Let G be a p-group of maximal class with $|G| = p^n$ and $n > p+1$. Then*

$$\mho_1(G_i) = G_{i+p-1} \quad \textit{for} \quad 1 \leq i \leq n-p+1.$$

Also, G_1 is a regular p-group with $\Omega_1(G_1) = G_{n-p+1}$ and $|G_1/\mho_1(G_1)| = p^{p-1}$.

Proof. By Lemma 14.2 b), $\mho_1(G_1) = G_k$ for some k. Applying Lemma 14.14 to G/G_{p+1} shows that $\mho_1(G_1) \leq G_p$. This means $p \leq k$. If $p < k$, then it would follow that $s_1^p \in \mho_1(G_1) \leq G_{p+1}$, contradicting Lemma 14.15. So $\mho_1(G_1) = G_p$. From

$$|G_1/\mho_1(G_1)| = |G_1/G_p| = p^{p-1}$$

we conclude that G_1 is regular by Theorem 10.13. Therefore,

$$|\Omega_1(G_1)| = |G_1/\mho_1(G_1)| = p^{p-1}$$

by Theorem 10.7, and then $\Omega_1(G_1) = G_{n-p+1}$ by Lemma 14.2 b).

Let $1 < i \leq n-p+1$. Then G_i is regular, being a subgroup of G_1. Since $i \leq n-p+1$, we have $\Omega_1(G_1) = G_{n-p+1} \leq G_i$, and thus $\Omega_1(G_1) = \Omega_1(G_i)$. This yields

$$|G_i/\mho_1(G_i)| = |\Omega_1(G_i)| = p^{p-1},$$

and consequently $|G/\mho_1(G_i)| = p^{i+p-1}$. Finally, we may conclude

$$\mho_1(G_i) = G_{i+p-1}$$

by Lemma 14.2 b). □

We extract a few consequences from Theorem 14.16.

Theorem 14.17. *If G is a 3-group of maximal class, then $G'' = 1$. In addition, G_1 is metacyclic with $c(G_1) \leq 2$. (In particular, every 3-group of maximal class is not an exceptional group by Theorem 14.11.)*

Proof. If $|G| \leq 3^4$, then $|G'| \leq 3^2$ and thus $G'' = 1$. So assume that $|G| = 3^n$ with $n > 4$. By Theorem 14.16, $|G_1/\mho_1(G_1)| = 3^2$. Hence, G_1 is a metacyclic group by Theorem 11.4. This means G_1' is cyclic, and $G_1' = G_k$ for some k by Lemma 14.2 b). It follows from Theorem 14.16 that $\mho_1(G_{n-2}) = G_n = 1$. Consequently, G_{n-2} has exponent 3 and is not cyclic since $|G_{n-2}| = 3^2$. So $k \geq n-1$. This forces $|G_1'| \leq 3$ and $G_1' \leq \mathbf{Z}(G_1)$. With the help of Lemma 1.3 a), it now follows that for all $x, y \in G_1$,

$$[x^3, y] = [x, y]^3 = 1.$$

So we have $\mho_1(G_1) \leq \mathbf{Z}(G_1)$. We conclude that every maximal subgroup of G_1 is abelian, being a cyclic extension of $\mho_1(G_1) = \Phi(G_1)$. This conclusion applies, in particular, to $G_2 = G'$. □

Now we complete the proof of Theorem 14.6:

14.18 (Proof of Theorem 14.6 for $n > p + 2$).

Proof. We must show that if G is a p-group of maximal class with $|G| = p^n$ and $n \geq p+2$, then G is not an exceptional group. We prove the claim by induction on n. For $n = p + 2$, the claim holds by Theorem 14.12. So assume that $n > p + 2$. Since $n - 1 \geq p+2$, we conclude that G/G_{n-1} is not an exceptional group by the induction hypothesis. Assume as before that

$$G = \langle G_1, s\rangle, \quad G_1 = \langle G_2, s_1\rangle \quad \text{and} \quad s_{i+1} = [s_i, s] \text{ for } 1 \leq i \leq n-2.$$

Then $G_i = \langle G_{i+1}, s_i\rangle$ for $i = 1, 2, \ldots, n-2$ by Lemma 14.8. Lemma 14.15 gives

$$s_1^p \in G_p - G_{p+1}.$$

So there exists some k with $0 < k < p$ and some $x \in G_{p+1}$ such that

$$s_1^p = x s_p^{-k} \in G_p = G_{p+1}\langle s_p\rangle.$$

It follows that

$$[s_1^p s_p^k, s_{n-p-1}] \in [G_{p+1}, G_{n-p-1}] \leq G_n = 1.$$

Then by Lemma 1.2 c),

$$1 = [s_1^p, s_{n-p-1}]^{s_p^k}[s_p^k, s_{n-p-1}].$$

Since

$$[s_1^p, s_{n-p-1}] \in [G_p, G_{n-p-1}] \leq G_{n-1} = \mathbf{Z}(G)$$

and $[s_p, s_{n-p-1}] \in G_{n-1}$, it follows that

$$[s_{n-p-1}, s_p]^k = [s_1^p, s_{n-p-1}] = s_1^{-p}(s_1^p)^{s_{n-p-1}} = s_1^{-p}(s_1^{s_{n-p-1}})^p$$
$$= s_1^{-p}(s_1 y)^p,$$

where $y = [s_1, s_{n-p-1}]$. Since G/G_{n-1} is not an exceptional group and we have $n-p-1 \leq n-3$, it follows from Theorem 14.7 that

$$y = [s_1, s_{n-p-1}] \in [G_1, G_{n-p-1}] \leq G_{n-p+1}.$$

By Theorem 14.16, G_1 is regular and $\Omega_1(G_1) = G_{n-p+1}$ has exponent p. Since $y \in G_{n-p+1}$, we also have $\langle s_1, y\rangle' \leq G_{n-p+1}$. Then the regularity of G_1 yields

$$(s_1 y)^p = s_1^p y^p = s_1^p.$$

Hence, we obtain that $[s_{n-p-1}, s_p]^k = 1$. Since $k \not\equiv 0 \bmod p$, it follows that $[s_{n-p-1}, s_p] = 1$. Since G/G_{n-1} is not an exceptional group, we obtain from Theorem 14.7 and Lemma 14.9 that

$$[s_1, s_{n-2}] = [s_p, s_{n-p-1}]^{(-1)^{p-1}} = 1.$$

We have $G_1 = \langle G_2, s_1\rangle$. Applying Lemma 14.8 to G/G_{n-1}, we may conclude that $G_{n-2} = \langle G_{n-1}, s_{n-2}\rangle$. It follows that $[G_1, G_{n-2}] = 1$. Since G/G_{n-1} is not an exceptional group, we have

$$[G_1, G_i] \leq G_{i+2}$$

for $i \leq n-3$. Hence, G is itself not an exceptional group. □

Theorem 14.17 admits an extension to all prime numbers:

Theorem 14.19 (Alperin [1]). *Let G be a p-group of maximal class. Then the derived length $k(G)$ of G is bounded above by a function $k(p)$ of p that is independent of G.*

(The exact value of $k(p)$ is only known for $p \leq 5$; namely, $k(3) = 2$ and $k(5) = 3$. Not even an estimate of $k(p)$ is known for $p > 5$. It is conjectured that even the nilpotency class of G' is bounded above by a function of p that is independent of G.[7] *For $p = 3$, we proved in Theorem 14.17 that $c(G_1) \leq 2$; Blackburn proved $c(G_1) \leq 3$ for $p = 5$ (Blackburn [36]).)*

[7] This follows from the proofs of the so-called coclass conjectures. The conjectures were stated in a famous article by Leedham-Green and Newman in 1980, with final proofs due to Shalev (1994) and Leedham-Green (1994). A summary of the proofs appears in C. R. Leedham-Green and S. McKay, *The structure of groups of prime power order*, London Mathematical Society Monographs. New Series, 27, Oxford Univ. Press, Oxford, 2002. For a survey article on the coclass conjectures, see H. Dietrich and B. Eick, The origins of coclass theory, Arch. Math. (Basel) **121** (2023), no. 5-6, 495–510. *trans.*

Proof. By Theorem 14.6 and the remark in Lemma 14.13, there exists an $s \in G$ such that $|\mathbf{C}_G(s)| = p^2$ and $s^p \in G_{n-1} = \mathbf{Z}(G)$. Thus, the inner automorphism α of G defined by $g^\alpha = g^s$ for all $g \in G$ has order p and has exactly p^2 fixed points.

Now we use without proof the following theorem of Alperin [1]: Let G be a p-group and α an automorphism of G of order p. Suppose α fixes exactly p^m elements of G. Then the derived length $k(G)$ of G is bounded above by a function $t(p, m)$ of p and m. For groups of maximal class, this yields

$$k(G) \leq t(p, 2) = k(p). \qquad \square$$

Theorem 14.20. *For $p > 2$, the Sylow p-subgroup of the symmetric group S_{p^2} is not the homomorphic image of a p-group of maximal class of order p^n with $n > p + 1$.*

Proof. Let G be a p-group of maximal class with $|G| = p^n$ and $n > p + 1$. By Theorem 14.16, we have $|G_1/\mathfrak{V}_1(G_1)| = p^{p-1}$. Let T be the Sylow p-subgroup of S_{p^2}, and assume for contradiction that $G/K \cong T$. Since $|G/K| = p^{p+1}$, we must have $K = G_{p+1}$ by Lemma 14.2 b). Then

$$T_1 = \mathbf{C}_T(T_2/T_4) \cong G_1/G_{p+1},$$

and therefore by Theorem 14.16

$$|T_1/\mathfrak{V}_1(T_1)| = |G_1/\mathfrak{V}_1(G_1)G_{p+1}| = |G_1/G_p| = p^{p-1}.$$

Let Z be a regular permutation group of degree p. Then the regular wreath product $T = Z \wr_r Z$, regarded as a permutation group of degree p^2 according to Theorem I.15.3 (pg. 102), is a Sylow p-subgroup of the symmetric group S_{p^2}.

The group $T = Z \wr_r Z$ contains the elementary abelian normal subgroup $D = Z \wr 1$ of index p. Since $T_2 = T' \leq D$, we have

$$D \leq \mathbf{C}_T(T_2) \leq \mathbf{C}_T(T_2/T_4) = T_1,$$

so $D = T_1$. But then $|T_1/\mathfrak{V}_1(T_1)| = |T_1| = p^p$, contradicting our observation above.$\square$

Theorem 14.21. *Let G be a p-group of maximal class with $|G| \geq p^{p+1}$. Then G is irregular.*

Proof. The group G has a factor group G/N of maximal class and order p^{p+1}. It clearly suffices to show that G/N is irregular. For this purpose, we may assume that $N = 1$.

Let $G_1 = \mathbf{C}_G(G_2/G_4)$. Since G/G_p is not an exceptional group by Theorem 14.6 a), we have

$$G_1 = \mathbf{C}_G(G_i/G_{i+2}) \quad \text{for} \quad 2 \leq i \leq p - 2.$$

Choose an element $s \in G$ such that $s \notin G_1$ and $s \notin \mathbf{C}_G(G_{p-1})$. Then $G = \langle G_1, s \rangle$. We choose s_1 such that $G_1 = \langle G_2, s_1 \rangle$ and set $s_i = [s_{i-1}, s]$ for $2 \leq i \leq p$.

Since G/G_p is not an exceptional group by Theorem 14.6 a), it follows that $G_i = \langle G_{i+1}, s_i \rangle$ for $2 \le i \le p-1$ by applying Lemma 14.8 to G/G_p. In particular, $G_{p-1} = \langle G_p, s_{p-1} \rangle$. Since $[s, G_p] = 1$ and $s \notin \mathbf{C}_G(G_{p-1})$,

$$s_p = [s_{p-1}, s] = [s_1, s, \underset{p-1}{\ldots}, s] \neq 1.$$

The identity of Theorem 9.4 gives

$$s^p s_1^p = (ss_1)^p c_2^{\binom{p}{2}} \cdots c_{p-1}^{\binom{p}{p-1}} c_p.$$

By Lemma 14.14, $\mho_1(G_2) = 1$ and so

$$c_i^{\binom{p}{i}} \in \mho_1(G') = 1$$

for $2 \le i \le p-1$. Since $G_{p+1} = 1$, it follows from Lemma 9.5 that

$$c_p = [s_1, s, \underset{p-1}{\ldots}, s]^a \prod_i v_i^{a_i}$$

with $a \equiv -1 \bmod p$. In this expression, the v_i have the form $v_i = [s_1, s, x_3, \ldots, x_p]$ with $x_j \in \{s_1, s\}$ and $x_j = s_1$ for at least one j. Let

$$v_i = [s_1, s, \underset{k}{\ldots}, s, s_1, \ldots].$$

Then for $k \le p-3$,

$$[s_1, s, \underset{k}{\ldots}, s, s_1] \in [G_{k+1}, G_1] \le G_{k+3}$$

and therefore

$$v_i \in [G_{k+3}, G, \underset{p-k-2}{\ldots}, G] \le G_{p+1} = 1.$$

So there remains only

$$s^p s_1^p = (ss_1)^p s_p^a [s_1, s, \underset{p-2}{\ldots}, s, s_1]^b.$$

Assume for contradiction that G is regular. Then $s^p s_1^p = (ss_1)^p$ since $\mho_1(G') = 1$. So $s_p^a = t^{-b}$, where $t = [s_1, s, \underset{p-2}{\ldots}, s, s_1]$. Since $G_{p+1} = 1$, we have by Lemma 6.8 that

$$[s_1^2, s, \ldots, s, s_1^2] = [s_1, s, \ldots, s, s_1]^4 \quad \text{and} \quad [s_1^2, s, \ldots, s] = [s_1, s, \ldots, s]^2.$$

Replacing s_1 by s_1^2, which is possible since $p > 2$, then yields

$$(ss_1^2)^p = s^p s_1^{2p} = (ss_1^2)^p s_p^{2a} t^{4b}.$$

It follows that

$$s_p^{2a} = t^{-4b} = s_p^{4a}.$$

Hence, $s_p^{2a} = 1$. But since $s_p \neq 1$ and $a \equiv -1 \bmod p$, this is a contradiction. We conclude that G is irregular. □

Theorem 14.22. *Let G be a p-group of maximal class.*

a) *If G is not an exceptional group, then every maximal subgroup of G that is distinct from G_1 is a p-group of maximal class.*

b) *If $|G| > p^{p+1}$, then every maximal subgroup of G that is distinct from G_1 is irregular. Furthermore, G_1 is the only maximal subgroup of G that is regular.*

Proof. a) We need to show that every maximal subgroup U of G distinct from G_1 has class $n-2$.

Let $G = \langle G_1, s\rangle$, let $G_1 = \langle G_2, s_1\rangle$ and $s_{i+1} = [s_i, s]$. Then every maximal subgroup U of G distinct from G_1 has the form $U = \langle G_2, ss_1^i\rangle$ for some i with $0 < i \leq p-1$.

For $1 \leq j \leq n-3$, we prove the formula

$$[s_2, \underset{j}{ss_1^i, \ldots, ss_1^i}] \equiv s_{j+2} \bmod G_{j+3}$$

by induction on j. For $j = 1$, the statement is clearly true since $[s_2, s_1] \in G_4$. Suppose

$$[s_2, \underset{j-1}{ss_1^i, \ldots, ss_1^i}] = s_{j+1} l$$

for some $l \in G_{j+2}$ and $j \leq n-3$. Then

$$\begin{aligned}[s_{j+1} l, ss_1^i] &= [s_{j+1}, ss_1^i]^l [l, ss_1^i] \equiv [s_{j+1}, ss_1^i] \\ &\equiv [s_{j+1}, s_1^i][s_{j+1}, s][s_{j+1}, s, s_1^i] \equiv [s_{j+1}, s_1^i] s_{j+2} \mod G_{j+3}.\end{aligned}$$

By assumption, G is not an exceptional group. Thus,

$$[s_{j+1}, s_1^i] \in [G_{j+1}, G_1] \leq G_{j+3}$$

since $j+1 \leq n-2$. This proves our intermediate claim.

In particular, since $G_n = 1$ we now have

$$s_{n-1} = [s_2, \underset{n-3}{ss_1^i, \ldots, ss_1^i}] \in \mathbf{K}_{n-2}(U).$$

Since G/G_{n-1} is not an exceptional group by Theorem 14.6 a), it follows from Lemma 14.8 that $G_{n-2} = \langle G_{n-1}, s_{n-2}\rangle$.

Since $s \notin G_1 = \mathbf{C}_G(G_{n-2})$, we have $[s_{n-2}, s] = s_{n-1} \neq 1$. Hence, U has class at least $n-2$. Since $|U| = p^{n-1}$, it is a trivial observation that U has class at most $n-2$.

b) Now assume $|G| > p^{p+1}$. Then G is not an exceptional group by Theorem 14.6 b). If U is a maximal subgroup of G with $U \neq G_1$, then U is a p-group of maximal class with $|U| \geq p^{p+1}$ by a). This means U is irregular by Theorem 14.21. On the other hand, G_1 is regular by Theorem 14.16. □

The p-groups of maximal class can be characterized by the property that they contain large conjugacy classes:

Theorem 14.23. *Let G be a p-group with $|G| = p^n$ and $n \geq 3$. The group G is a p-group of maximal class if and only if there exists an element $s \in G$ such that*

$$|\{s^g \mid g \in G\}| = p^{n-2}.$$

(*This statement also holds for $p = 2$.*)

Proof. 1) Let $s \in G$ with

$$|\{s^g \mid g \in G\}| = p^{n-2} = [G : \mathbf{C}_G(s)].$$

For $n = 3$, G is not abelian, so G has class 2. We prove the claim for $n > 3$ by induction on n.

Since $\mathbf{Z}(G) \leq \mathbf{C}_G(s)$ and $s \notin \mathbf{Z}(G)$, we have $|\mathbf{Z}(G)| = p$. We let $Z = \mathbf{Z}(G)$. Now, $\mathbf{C}_{G/Z}(sZ) = L/Z$, where

$$L = \{g \in G \mid s^g = sz \text{ with } z = [s, g] \in Z\}.$$

The map $g \mapsto [s, g]$ is then a homomorphism from L to Z with kernel $\mathbf{C}_G(s)$. It follows that

$$|L| \leq p|\mathbf{C}_G(s)| = p^3.$$

If we define $\overline{G} = G/Z$ and $\overline{s} = sZ$, then

$$|\{\overline{s}^{\overline{g}} \mid \overline{g} \in \overline{G}\}| = [\overline{G} : \mathbf{C}_{\overline{G}}(\overline{s})] = [G : L] \geq p^{n-3}.$$

On the other hand, since

$$\{\overline{s}^{\overline{g}} \mid \overline{g} \in \overline{G}\} \subseteq \overline{s}\overline{G}',$$

it follows that

$$|\{\overline{s}^{\overline{g}} \mid \overline{g} \in \overline{G}\}| \leq |\overline{G}'| \leq p^{n-3}.$$

So we have

$$|\{\overline{s}^{\overline{g}} \mid \overline{g} \in \overline{G}\}| = p^{n-3}.$$

Therefore, by the induction hypothesis,

$$1 \neq \mathbf{K}_{n-2}(G/Z) = \mathbf{K}_{n-2}(G)Z/Z.$$

So $\mathbf{K}_{n-2}(G) \not\leq Z = \mathbf{Z}(G)$, and consequently $\mathbf{K}_{n-1}(G) \neq 1$. Thus, G is a group of maximal class.

2) Now let G be a p-group of maximal class.

If $p > 2$, then by Lemma 14.13 there exists an $s \in G$ such that $|\mathbf{C}_G(s)| = p^2$. Then

$$|\{s^g \mid g \in G\}| = [G : \mathbf{C}_G(s)] = p^{n-2}.$$

If $p = 2$, then $G = \langle a, b\rangle$ with $a^{2^{n-1}} = 1$ and $a^b = a^{-1+2^{n-2}}$ or $a^b = a^{-1}$ by Theorem 11.9. Then

$$|\{b^g \mid g \in G\}| = |\{ba^{2j} \mid j = 0, 1, \dots\}| = 2^{n-2}.$$

The claim follows by setting $s = b$. □

Next, we will show by examples that the statement of Theorem 14.6 b) cannot be improved.

Example 14.24. For every $p > 3$ and every r with $6 \leq 2r \leq p + 1$, there exists an exceptional group of order p^{2r}:

Let A be an elementary abelian group of order p^r with $3 \leq r \leq \frac{p+1}{2}$. Let $\{a_1, \dots, a_r\}$ be a basis of A. Let B be the group of automorphisms b of A such that

$$a_r^b = a_r \quad \text{and} \quad a^b a^{-1} \in \langle a_r\rangle$$

for all $a \in A$. Then B is elementary abelian of order p^{r-1}. We show:

(1) The elements $b_j \in B$ with

$$a_i^{b_j} = a_i a_r^{(-1)^{r+i-1}\binom{r-j-1}{i-1}} \quad (i = 1, \dots, r-1;\ j = 1, \dots, r-1)$$
$$a_r^{b_j} = a_r$$

are a basis for B:

We have $a_i^{b_j} = a_i$ for $i > r - j$ and $a_i^{b_{r-i}} = a_i a_r^{(-1)^{r+i-1}}$. Let $b \in B$, and let $a_i^b = a_i$ for $i > k$ and $a_k^b = a_k a_r^x$ for some integer x. We define

$$c = b b_{r-k}^{(-1)^{r+k}x},$$

so that $a_i^c = a_i$ for $i \geq k$. From this the claim follows easily.

(2) Let H be the semidirect product of A with B. Then H admits an automorphism a such that

$$a_r^a = a_r, \quad a_{r-1}^a = a_{r-1}, \quad a_i^a = a_i a_{i+1}^{-1}\ (1 \leq i \leq r-2),$$
$$b_{r-1}^a = b_{r-1} a_1^{-1}, \quad b_i^a = b_i b_{i+1}^{-1}\ (1 \leq i \leq r-2).$$

One can confirm by direct calculation that a is an automorphism of H with order p. Let G be the semidirect product of H with $\langle a \rangle$.

(3) We have $|G| = p^{2r}$, G has class $2r-1$ and G is an exceptional group:
It follows immediately from

$$[b_i, a] = b_{i+1}^{-1} \ (1 \le i \le r-2), \quad [b_{r-1}, a] = a_1^{-1},$$
$$[a_i, a] = a_{i+1}^{-1} \ (1 \le i \le r-2), \quad [a_{r-1}, b_1] = a_r$$

that

$$\begin{aligned} \mathbf{K}_i(G) &= \langle b_i, \ldots, b_{r-1}, a_1, \ldots, a_r \rangle \quad (2 \le i \le r-1), \\ \mathbf{K}_r(G) &= \langle a_1, \ldots, a_r \rangle, \\ \mathbf{K}_{r+j}(G) &= \langle a_{j+1}, \ldots, a_r \rangle \quad (0 \le j \le r-1). \end{aligned}$$

Hence, G has class $2r-1$. That is, G is a group of maximal class. Since

$$b_2^a = b_2 b_3^{-1} \not\equiv b_2 \bmod \mathbf{K}_4(G),$$

we have $a \notin \mathbf{C}_G(\mathbf{K}_2(G)/\mathbf{K}_4(G)) = G_1$. On the other hand, $a_{r-1}^a = a_{r-1}$ and therefore $a \in \mathbf{C}_G(G_{2r-2})$. So $G_1 \ne \mathbf{C}_G(G_{2r-2})$ and G is an exceptional group.

Exercises

35) Let G be a group of maximal class with $|G| = p^{2m}$. Show that:
   a) If G is not an exceptional group, then G_{m-1} is abelian.
   b) If G is an exceptional group, then G_m is the unique maximal abelian normal subgroup of G.

15 The Sylow p-subgroups of the symmetric groups S_{p^n}

In this section, we study the Sylow p-subgroups of the symmetric groups S_{p^n}. We first prove two easy lemmas regarding the calculation of the lower central series:

Lemma 15.1. *Let A be a normal subgroup of G. We define the normal subgroups A_i of G recursively by $A_1 = A$ and $A_{i+1} = [A_i, G]$. Then $[\mathbf{K}_i(G), A] \le A_{i+1}$.*

Proof. The A_i constitute a central series of G, in the sense of Definition 2.1, beginning with $A = A_1$. Therefore, $[\mathbf{K}_i(G), A] \le A_{i+1}$ by Theorem 2.11 a). □

Lemma 15.2. *Let $G = LA$ with $A \lhd G$. Then $\mathbf{K}_i(G) = \mathbf{K}_i(L)A_i$, where the normal subgroups A_i of G are defined recursively by $A_1 = A$ and $A_{i+1} = [A_i, G]$.*

Proof. We prove the lemma by induction on i. The claim is clearly true for $i = 1$ since

$$\mathbf{K}_1(G) = G = LA = \mathbf{K}_1(L)A_1.$$

We show that:
(1) $\mathbf{K}_{i+1}(L)A_{i+1} \leq \mathbf{K}_{i+1}(G)$:
By the induction hypothesis,

$$A_{i+1} = [A_i, G] \leq [\mathbf{K}_i(G), G] = \mathbf{K}_{i+1}(G).$$

Since $\mathbf{K}_{i+1}(L) \leq \mathbf{K}_{i+1}(G)$ and $A_{i+1} \trianglelefteq G$, we also have

$$\mathbf{K}_{i+1}(L)A_{i+1} \leq \mathbf{K}_{i+1}(G).$$

(2) $\mathbf{K}_{i+1}(L)A_{i+1} \trianglelefteq G$:
Certainly $A_{i+1} \trianglelefteq G$ and $\mathbf{K}_{i+1}(L) \trianglelefteq L$. So to prove (2), it suffices to show that

$$\mathbf{K}_{i+1}(L)^a \leq \mathbf{K}_{i+1}(L)A_{i+1}$$

for all $a \in A$. Let $l \in \mathbf{K}_{i+1}(L)$. Then it follows by Lemma 15.1 that

$$l^a = l[l, a] \in \mathbf{K}_{i+1}(L)[\mathbf{K}_{i+1}(L), A] \leq \mathbf{K}_{i+1}(L)A_{i+1}.$$

(3) $\mathbf{K}_{i+1}(G) \leq \mathbf{K}_{i+1}(L)A_{i+1}$:
By the induction hypothesis, we have

$$\mathbf{K}_{i+1}(G) = [\mathbf{K}_i(G), G] = [\mathbf{K}_i(L)A_i, LA].$$

We let $N = \mathbf{K}_{i+1}(L)A_{i+1}$. We have

$$[\mathbf{K}_i(L), L] = \mathbf{K}_{i+1}(L) \leq N,$$

and furthermore, by Lemma 15.1,

$$\begin{aligned} [\mathbf{K}_i(L), A] &\leq A_{i+1} \leq N \quad \text{and} \\ [A_i, AL] &= [A_i, G] = A_{i+1} \leq N. \end{aligned}$$

Using (2), we conclude that $\mathbf{K}_{i+1}(G) \leq \mathbf{K}_{i+1}(L)A_{i+1}$. Then together with (1), the claim follows. □

With the help of the wreath product, we can describe the Sylow p-subgroups of the symmetric group of degree p^n:

Theorem 15.3 (Kaloujnine [296]). *Let Z be the cyclic group of order p. We define*

$$P_n = Z \wr \overset{n}{\cdots} \wr Z$$

to be the n-fold iterated wreath product of Z, where Z is regarded as a regular permutation group. By Theorem I.15.3 (pg. 102), P_n is a permutation group of degree p^n. We claim that:

a) *P_n is a Sylow p-subgroup of the symmetric group S_{p^n} of degree p^n.*
b) *P_n has exponent p^n and can be generated by elements of order p. In particular, P_n is not regular for $n > 1$.*
c) *The minimum number of generators $d(P_n)$ of P_n is n.*
d) *P_n has derived length n.*
e) *P_n has nilpotency class p^{n-1}.*

Proof. a) P_n is a permutation group of degree p^n and order p^t, where $t = 1 + p + p^2 + \cdots + p^{n-1}$. It is not difficult to calculate that p^t is the highest power of p that divides $(p^n)!$. So P_n is a Sylow p-subgroup of the symmetric group S_{p^n} since it has the correct order.

b) Every Sylow p-subgroup of S_{p^n} contains cycles of length p^n, but longer cycles obviously cannot appear in the cycle decomposition of a permutation in S_{p^n}. So $\exp P_n = p^n$.

Assume for induction that P_{n-1} can be generated by elements of order p. By Theorem I.15.4 (pg. 103),

$$P_n = Z \wr (Z \wr \cdots \wr Z) = Z \wr P_{n-1}.$$

So P_n can be generated by elements of order p. The observation that P_n is irregular follows with the help of Theorem 10.5 a).

c) We always denote by $d(G)$ the minimum number of generators of a p-group G (see Theorem 3.15). Assume for induction that $d(P_{n-1}) = n - 1$, and let

$$P_{n-1} = \langle x_1, \ldots, x_{n-1} \rangle.$$

So $\{x_1, \ldots, x_{n-1}\}$ is a generating set of P_{n-1} with exactly $n - 1$ elements. Then by Theorem I.15.2 (pg. 101), we have $P_n = AP_{n-1}$ with

$$A = Z \wr 1 = \langle a_1 \rangle \times \cdots \times \langle a_{p^{n-1}} \rangle;$$

In this expression, we regard 1 as the trivial permutation group on p^{n-1} points, and P_{n-1} is identified with a subgroup of $P_n = Z \wr P_{n-1}$. The a_i are transitively permuted under conjugation by elements of P_{n-1}. So we obtain $P_n = \langle a_1, x_1, \ldots, x_{n-1} \rangle$. Note that a_1 is certainly not redundant in this generating set since $P_{n-1} < P_n$. In addition, none of the x_j are redundant, since otherwise we would have

$$P_{n-1} \cong P_n/A = \langle x_i A \mid i = 1, \ldots, n-1,\ i \neq j \rangle,$$

contradicting the induction hypothesis $d(P_{n-1}) = n - 1$. So $d(P_n) = n$.

d) We have $P_n = P_{n-1} \wr Z$, where $Z = \langle z \rangle$ is regarded as a cyclic, transitive permutation group of degree p. Let $D_n = P_{n-1} \wr 1$ (where 1 is regarded as the trivial permutation group on p points). Since $P_n/D_n \cong Z$, we have

$$P_n' \leq D_n \cong P_{n-1} \times \cdots \times P_{n-1}.$$

It follows by a simple induction on n that P_n has derived length at most n.

The derived length of P_n is at least n: In the direct product

$$D_n = P_{n-1} \wr 1 \cong P_{n-1} \times \cdots \times P_{n-1},$$

we choose an element x_1 from the first factor. Then

$$[x_1, z] = x_1^{-1} x_1^z \in P_n'.$$

Thus, the projection of elements of P_n' onto their first component in the direct product above is an epimorphism from P_n' onto P_{n-1}. Consequently, the derived length of P_n satisfies $k(P_n) \geq 1 + k(P_{n-1})$. An easy induction yields $k(P_n) \geq n$.

(In fact, this calculation shows the following more general fact: If A is abelian and $G \neq 1$, then $k(G \wr A) = k(G) + 1$.)

e) We have $P_n = Z \wr P_{n-1} = AP_{n-1}$, where $A = Z \wr 1$ is an elementary abelian normal subgroup of P_n of order $p^{p^{n-1}}$. Assume for induction that $c(P_{n-1}) = p^{n-2}$. By Lemma 15.2,

$$\mathbf{K}_i(P_n) = A_i \mathbf{K}_i(P_{n-1}).$$

Since

$$\mathbf{K}_{p^{n-2}+1}(P_{n-1}) = 1,$$

it follows for $i > p^{n-2}$ that $\mathbf{K}_i(P_n) = A_i$. We show that $A_{p^{n-1}} \neq 1 = A_{p^{n-1}+1}$:

We define r by

$$A = A_1 > A_2 > \cdots > A_r = 1.$$

Since $|A| = p^{p^{n-1}}$, we have $r \leq p^{n-1} + 1$.

Furthermore, r can be bounded from below in the following way: Let

$$A = \langle a_1 \rangle \times \cdots \times \langle a_{p^{n-1}} \rangle,$$

and let s be the p^{n-1}-cycle in the permutation group P_{n-1} defined by $a_i^s = a_{i+1}$ ($i \bmod p^{n-1}$). Then we have

$$[a_1, s] = a_1^{-1} a_1^s = a_1^{-1} a_2 \in A_2,$$
$$[a_1, s, s] = a_1 a_2^{-2} a_3 \in A_3,$$

and finally

$$[a_1, s, s, \underset{p^{n-1}-1}{\ldots}, s] = a_1^{\pm 1} \cdots a_{p^{n-1}} \in A_{p^{n-1}}.$$

Hence, $A_{p^{n-1}} \neq 1$ and then $r \geq p^{n-1} + 1$. This, when combined with what we have already shown, means $r = p^{n-1} + 1$ and $c(P_n) = p^{n-1}$. □

We end this section with a remark about the abelian normal subgroups of P_n:

Theorem 15.4.

a) *For $p > 2$, all abelian normal subgroups of P_n lie in the elementary abelian normal subgroup $A = Z \wr 1$ of order $p^{p^{n-1}}$ (where 1 is regarded as a trivial permutation group on p^{n-1} points). In particular, all abelian normal subgroups of P_n have exponent p.*

b) *For $p = 2$, P_n also has abelian normal subgroups of type $(4, 4, \ldots, 4)$ of order $2^{2^{n-1}}$.*

(*Compare with Theorem 7.3 d).*)

Proof. a) We have $P_n = Z \wr P_{n-1}$. Clearly

$$A = Z \wr 1 = \langle a_1 \rangle \times \cdots \times \langle a_{p^{n-1}} \rangle$$

is an elementary abelian normal subgroup of P_n with order $p^{p^{n-1}}$. Let B be any abelian normal subgroup of P_n. Note that AB has class at most 2 since

$$[A, B] \leq A \cap B \leq \mathbf{Z}(AB).$$

Hence, for all $a \in A$ and all $t \in AB$, we have

$$[a, t, t] = 1.$$

We have $P_n = AP_{n-1}$, and therefore $AB = AT$, where $T = AB \cap P_{n-1}$. The group P_{n-1} permutes the p^{n-1} direct factors $\langle a_i \rangle$ of A under conjugation. Let $1 \neq t = (i_1, i_2, \ldots, i_p, \ldots) \cdots \in T$. Then

$$1 = [a_{i_1}, t, t] = a_{i_1} a_{i_2}^{-2} a_{i_3}.$$

Since $p > 2$, this is a contradiction. So $T = 1$ and therefore $B \leq A$.

b) Let $p = 2$. By Theorem I.15.4 (pg. 103), we have

$$P_n = (Z \wr Z) \wr (Z \wr \cdots \wr Z) = D \wr P_{n-2},$$

where $D = Z \wr Z$ is clearly a dihedral group of order 8. The normal subgroup $D \wr 1$ of P_n is a direct product of 2^{n-2} groups isomorphic to D. Thus, it clearly has a characteristic subgroup B of type $(4, \ldots, 4)$ and order $2^{2^{n-1}}$. Then B is the desired abelian normal subgroup of P_n. □

Exercises

36) Let G be a Sylow p-subgroup of the symmetric group of degree p^2. Show that there exist two maximal subgroups U_1 and U_2 of G such that $U_1 \cup U_2$ is exactly the set of all elements of G with order 1 or p.

16 The Sylow p-subgroups of the linear groups $\mathrm{GL}_n(p^f)$

Definition 16.1.

a) Let K be the finite field $\mathrm{GF}(q)$ with $q = p^f$, and let V be the vector space of dimension n over K with a basis $\{v_1, \ldots, v_n\}$. We define $\mathsf{V}_0 = 0$ and $\mathsf{V}_i = \langle v_1, \ldots, v_i \rangle$. Let $G = T(n, q)$ be the group of all K-linear maps g from V onto itself with

$$vg - v \in \mathsf{V}_{i-1} \quad \text{for all } v \in \mathsf{V}_i$$

and all $i = 1, \ldots, n$.

If $\{e_{jk} \mid j, k = 1, \ldots, n\}$ is the usual basis of the full matrix ring of degree n over K satisfying

$$v_i e_{jk} = \delta_{ij} v_k$$

(that is, $e_{ij} e_{km} = \delta_{jk} e_{im}$, where δ_{jk} is the Kronecker delta symbol), then G consists of the triangular matrices of the form

$$1 + \sum_{k<j} a_{jk} e_{jk} \quad (a_{jk} \in \mathsf{K}).$$

Clearly, $|G| = q^{\frac{n(n-1)}{2}}$. By Theorem II.7.1 (pg. 194), G is a Sylow p-subgroup of $\mathrm{GL}_n(q)$ and $\mathrm{SL}_n(q)$.

b) Let

$$H_i = \{g \in G \mid vg - v \in \mathsf{V}_{j-i} \text{ for all } v \in \mathsf{V}_j \text{ and all } j\}.$$

(In this expression, we let $\mathsf{V}_k = 0$ if $k \leq 0$.) The H_i are clearly normal subgroups of G and consist of the matrices

$$1 + \sum_{j-k \geq i} a_{jk} e_{jk}.$$

We have $H_1 = G$ and $H_n = 1$.

Lemma 16.2. *The group H_i is generated by the elements $1 + a e_{jk}$ with $j - k \geq i$ and $a \in \mathsf{K}$.*

Proof. We order the matrix coefficients lexicographically; that is,

$$a_{21}, a_{31}, a_{32}, a_{41}, \ldots, a_{n,n-1}.$$

Let $g \in H_i$, and let a_{jk} be the first nonzero coefficient in g. Then $j - k \geq i$. Let $h = 1 - a_{jk}e_{jk}$. A simple calculation shows that in the product gh, all the coefficients before a_{jk} remain zero and the coefficient with indices (j, k) is also zero. The product gh again lies in H_i, so we may conclude using induction. □

Next, we calculate the lower central series of G:

Theorem 16.3.
a) *We have* $[H_i, H_j] = H_{i+j}$.
b) *We have* $\mathbf{K}_i(G) = H_i$, $G^{(i)} = \mathbf{K}_{2^i}(G)$ *and* $[\mathbf{K}_i(G), \mathbf{K}_j(G)] = \mathbf{K}_{i+j}(G)$. *In particular,* $G = T(n, q)$ *has nilpotency class* $n - 1$.

Proof. a) We consider the semidirect product of the elementary abelian group V with G. We apply Theorem 2.8 with V in place of N, with V_{n-i} in place of N_i, and with G in place of L. Then the H_i appear in place of the L_i in Theorem 2.8. We obtain

$$[H_i, H_j] \leq H_{i+j}.$$

We prove the reverse inclusion as follows: By Lemma 16.2, the subgroup H_{i+j} is generated by the elements $1 + ae_{km}$ with $k - m \geq i + j$ and $a \in \mathsf{K}$. We choose a t such that $k - i \geq t \geq m + j$. Then $k - t \geq i$ and $t - m \geq j$. By a direct calculation, we obtain

$$1 + ae_{km} = [1 + ae_{kt}, 1 + e_{tm}] \in [H_i, H_j].$$

Therefore, $H_{i+j} \leq [H_i, H_j]$.
b) follows easily by induction. □

Theorem 16.4. *We have* $\mathbf{Z}_i(G) = H_{n-i} = \mathbf{K}_{n-i}(G)$. *That is, the upper and lower central series of* G *coincide.*

Proof. a) First assume $i = 1$, and let

$$z = 1 + \sum_{j<i} a_{ij}e_{ij} \in \mathbf{Z}(G).$$

The relation

$$(1 + e_{km})z = z(1 + e_{km}) \quad (1 \leq m < k)$$

shows that only a_{n1} can be nonzero. Thus, $\mathbf{Z}(G) \leq H_{n-1}$. Conversely, one sees immediately that H_{n-1} lies in $\mathbf{Z}(G)$. We conclude that $H_{n-1} = \mathbf{Z}(G)$.

b) The subspace $\mathsf{V}_{n-1} = \langle v_1, \ldots, v_{n-1}\rangle$ is fixed setwise by G. Let g^ϵ be the restriction of $g \in G$ to V_{n-1}. Then ϵ is a homomorphism from $G = T(n,q)$ onto $T(n-1,q)$. The kernel N_1 of ϵ consists exactly of those matrices of the form

$$1 + \sum_{j<n} a_{nj} e_{nj}.$$

The subspace $\mathsf{V}_1 = \langle v_1\rangle$ is also fixed by G. Let g^τ be the linear transformation of the quotient space V/V_1 induced by $g \in G$. The map τ is also a homomorphism from $T(n,q)$ onto $T(n-1,q)$. The kernel N_2 of τ consists exactly of those matrices of the form

$$1 + \sum_{j>1} a_{j1} e_{j1}.$$

We have $N_1 \cap N_2 = H_{n-1}$ and $N_1 H_i \cap N_2 H_i = H_i$ for $i \le n-1$.

c) Assume for induction that the claim is true for $T(n-1,q)$. Furthermore, suppose it has been shown that $\mathbf{Z}_i(G) = H_{n-i}$.

The statement that $g \in \mathbf{Z}_{i+1}(G)$ is equivalent to $[g,x] \in \mathbf{Z}_i(G) = H_{n-i}$ for all $x \in G$. Since $i \ge 1$, it follows that

$$[g^\epsilon, x^\epsilon] \in H^\epsilon_{n-i} = H^\epsilon_{(n-1)-(i-1)} = \mathbf{Z}_{i-1}(G^\epsilon).$$

By the induction hypothesis, we obtain

$$g^\epsilon \in \mathbf{Z}_i(G^\epsilon) = H^\epsilon_{n-i-1}.$$

Hence, $g \in N_1 H_{n-i-1}$.

By applying the homomorphism τ, we obtain in the same way that $g \in N_2 H_{n-i-1}$, and then

$$g \in N_1 H_{n-i-1} \cap N_2 H_{n-i-1} = H_{n-i-1}.$$

Thus, $\mathbf{Z}_{i+1}(G) \le H_{n-i-1}$. However, by Theorem 2.7 we have

$$H_{n-i-1} = \mathbf{K}_{n-i-1}(G) \le \mathbf{Z}_{i+1}(G),$$

and the claim is proved. □

The power structure of $G = T(n,q)$ is described in the following theorem.

Theorem 16.5.
a) *We have* $\mho_i(G) = H_{p^i}$.
b) *G is generated by its elements of order p, so it is regular for $n \le p$ and irregular for $n > p$.*

Proof. a) First we show that $\mho_1(H_i) \le H_{ip}$.

Let $g = 1 + h \in H_i$, so h has the form

$$h = \sum_{j-k \geq i} a_{jk} e_{jk}.$$

Then $g^p = 1 + h^p$, and h^p has the form $h^p = \sum_{j-k \geq ip} b_{jk} e_{jk}$ since from

$$\mathsf{V}_j h \subseteq \mathsf{V}_{j-i} \quad \text{it follows that} \quad \mathsf{V}_j h^p \subseteq \mathsf{V}_{j-ip}.$$

Hence, we have proved that $\mathfrak{V}_1(H_i) \leq H_{ip}$. It follows by an easy induction on i that $\mathfrak{V}_i(G) \leq H_{p^i}$.

Let $g = 1 + ae_{km} \in H_{p^i}$ ($a \in \mathsf{K}$), so $k - m \geq p^i$. We show that g lies in $\mathfrak{V}_i(G)$.

Let x be the linear map of V defined by

$$v_k x = v_{m+p^i-1}, \quad v_{j+m} x = v_{j+m-1} \quad \text{for } 1 < j \leq p^i - 1,$$
$$v_{m+1} x = a v_m \quad \text{and} \quad v_s x = 0 \text{ otherwise.}$$

Then

$$x^{p^i} = a e_{km}.$$

If we let $h = 1 + x$, then it follows that

$$h^{p^i} = 1 + x^{p^i} = g.$$

Since these g generate all of H_{p^i} by Lemma 16.2, we have

$$H_{p^i} \leq \mathfrak{V}_i(G).$$

b) By Lemma 16.2, the elements $g = 1 + ae_{jk}$ ($j > k$, $a \in \mathsf{K}$) generate all of $G = H_1$. Since $e_{jk}^2 = 0$, we have $g^p = 1 + a^p e_{jk}^p = 1$. Thus, G is regular if and only if it has exponent p; that is, G is regular if and only if $n \leq p$. □

Theorem 16.6.

a) *If N is a normal subgroup of $T(n, q)$, then*

$$\langle v(n-1) \mid n \in N, v \in \mathsf{V} \rangle = \mathsf{V}_k$$

for some k with $0 \leq k \leq n-1$.

b) *If A is an abelian normal subgroup of $T(n, q)$ and*

$$\langle v(a-1) \mid a \in A, v \in \mathsf{V} \rangle = \mathsf{V}_k,$$

then $|A| \leq q^{k(n-k)}$.

c) *For every k with $0 \leq k \leq n-1$, the set*

$$A_k = \{1 + x \mid \mathsf{V}x \subseteq \mathsf{V}_k, \mathsf{V}_k x = 0\}$$

is an elementary abelian normal subgroup of $T(n, q)$ with $|A_k| = q^{k(n-k)}$.

d) *If A is an abelian normal subgroup of $T(n,q)$, then $|A| \leq q^{m^2}$ for $n = 2m$ and $|A| \leq q^{m(m+1)}$ for $n = 2m+1$.*

(*For the determination of the abelian normal subgroups of largest possible order, see Exercises 37–40.*)

Proof. a) Let

$$\mathsf{W} = \langle v(n-1) \mid n \in N, v \in \mathsf{V} \rangle$$

and $g \in T(n,q)$. Since

$$v(n-1)g = vg(n^g - 1) \in \mathsf{W},$$

the subspace W of V is G-invariant. Then W is clearly one of the V_k.

b) We prove the claim by induction on k. If $k = 0$, then $A = 1$ and the claim is trivial. Assume from now on that $k > 0$.

Let τ be the homomorphism from $T(n,q)$ onto $T(n-1,q)$ defined in part b) of the proof of Theorem 16.4 by

$$(v + \mathsf{V}_1)g^\tau = vg + \mathsf{V}_1.$$

Again, let $\ker\tau = N_2$. Then $A^\tau = AN_2/N_2$ is an abelian normal subgroup of $T(n-1,q)$ with

$$\langle \overline{v}(\overline{a} - \overline{1}) \mid \overline{v} \in \mathsf{V}/\mathsf{V}_1, \overline{a} \in A^\tau \rangle = \langle \overline{v}_2, \ldots, \overline{v}_k \rangle.$$

It follows from the induction hypothesis that

$$|A^\tau| = |A/(A \cap N_2)| \leq q^{(k-1)(n-k)}.$$

Let $1 + x \in A \cap N_2$. Since A is abelian, it follows from $\mathsf{V}x \subseteq \mathsf{V}_1$ that

$$\begin{aligned} \mathsf{V}_k x &= \langle v(a-1)x \mid a \in A, v \in \mathsf{V} \rangle = \langle vx(a-1) \mid a \in A, v \in \mathsf{V} \rangle \\ &\subseteq \langle v_1(a-1) \mid a \in A, v_1 \in \mathsf{V}_1 \rangle = 0. \end{aligned}$$

Thus,

$$|A \cap N_2| \leq |\mathrm{Hom}_{\mathsf{K}}(\mathsf{V}/\mathsf{V}_k, \mathsf{V}_1)| = q^{n-k}.$$

This yields the claim $|A| \leq q^{k(n-k)}$.

c) follows by an easy calculation, and d) follows from b). □

Exercises

37) We denote the transpose of the matrix x by x^T. Suppose there exists an element $l \in \mathrm{GL}_n(q)$ such that $l^{-1}xl = (x^T)^{-1}$ for all $x \in T(n,q)$. Then $n \leq 2$. (Hint: Consider the subspaces V_i, which are invariant under the action of $T(n,q)$.)

38) Let A be an abelian normal subgroup of $T(n,q)$ of order q^{n-1}. Suppose the group A consists of all the matrices of the form

$$a(\mathfrak{a}, f(\mathfrak{a}), z) = \begin{pmatrix} 1 & & & & & 0 \\ \mathfrak{a} & 1 & & & & \\ & 0 & 1 & & & \\ & & \ddots & \ddots & & \\ & & & 0 & 1 & \\ z & f(\mathfrak{a}) & & & & 1 \end{pmatrix},$$

where $\mathfrak{a}$ is a column vector of length $n-2$ and f is a map of column vectors to row vectors. Then $n \leq 4$.

39) The abelian normal subgroups of $T(3,q)$ with order q^2 have the form

$$\left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & 1 \end{pmatrix} \;\middle|\; a, b \in \mathsf{K} \right\}$$

or

$$\left\{ \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & xa & 1 \end{pmatrix} \;\middle|\; a, b \in \mathsf{K} \right\}$$

for some fixed $x \in \mathsf{K} = \mathrm{GF}(q)$. (If $x = 0$, we obtain the normal subgroup N_2.)

40) Let $n \geq 4$, and let A be an abelian normal subgroup of $T(n,q)$ with

$$\langle v(a-1) \mid a \in A, v \in \mathsf{V} \rangle = \mathsf{V}_k$$

and $|A| = q^{k(n-k)}$. Then $A = \{1 + x \mid \mathsf{V}x \subseteq \mathsf{V}_k,\ \mathsf{V}_k x = 0\}$.

41) Let $f_{i_1}^{m_1} \cdots f_{i_n}^{m_n}$ be an element of the free group F in the free generators f_i $(i = 1, \ldots, k)$, and assume that $i_j \neq i_{j+1}$. Show that for every prime p there exists a p-group G and elements $g_i \in G$ $(i = 1, \ldots, k)$ with

$$g_{i_1}^{m_1} \cdots g_{i_n}^{m_n} \neq 1.$$

That is, there does not exist a relation that holds in all p-groups for all elements.

(Hint following P. Hall: Let $m_1 \cdots m_n = p^t m$ with $p \nmid m$. Let G be the group of triangular matrices

$$g = 1 + \sum_{\substack{i,j=0 \\ i>j}}^{n} a_{ij} e_{ij}$$

of degree $n+1$ over the ring $\mathbb{Z}/p^{t+1}\mathbb{Z}$ with diagonal coefficients 1. We define

$$g_j = \prod_{i_s=j} (1 + e_{s-1,s}).$$

Then

$$g_j^t = 1 + t \sum_{i_s=j} e_{s-1,s}$$

and

$$g_{i_1}^{m_1} \cdots g_{i_n}^{m_n} = 1 + p^t m e_{0n} + \cdots \neq 1.)$$

17 Binary p-adic groups

Let $p > 2$ be a prime, and let R be the ring of p-adic integers.

Definition 17.1. Let h, k and m by integers with $h \geq 0$, $k > 0$, $m > 0$ and $h+k \geq m$. Let $M_{h,k,m}$ be the set of all 2×2 matrices

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

over R with $ad - bc = 1$, $b \equiv 0 \bmod p^h$, $c \equiv 0 \bmod p^k$ and $d \equiv 1 \bmod p^m$. Then since $h + k \geq m$, we also have

$$a = \frac{1 + bc}{d} \equiv 1 \bmod p^m.$$

One easily calculates that $M_{h,k,m}$ is a group for the given h, k, m. All of these groups lie in the group $M_{0,1,1}$, which we denote in what follows by G (with p fixed). Of course, these groups are all infinite.

The following lemma gives an overview of generators for $M_{h,k,m}$.

Lemma 17.2.
a) *We define*

$$b(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \quad c(x) = \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix}, \quad d(x) = \begin{pmatrix} x^{-1} & 0 \\ 0 & x \end{pmatrix}$$

and consider the following subgroups of G:

$$B_h = \{b(x) \mid x \equiv 0 \bmod p^h\},$$
$$C_k = \{c(x) \mid x \equiv 0 \bmod p^k\},$$
$$D_m = \{d(x) \mid x \equiv 1 \bmod p^m\}.$$

The factor groups

$$B_h/B_{h+n}, \quad C_k/C_{k+n}, \quad D_m/D_{m+n}$$

are all cyclic of order p^n.

b) *Every element* $x \in M_{h,k,m}$ *can be uniquely written in the form* $x = bcd$ *with* $b \in B_h$, $c \in C_k$ *and* $d \in D_m$. *So we have*

$$M_{h,k,m} = B_h C_k D_m.$$

Likewise,

$$M_{h,k,m} = C_k D_m B_h = D_m B_h C_k.$$

(*However, the factors* B_h, C_k *and* D_m *cannot be exchanged pairwise since* $B_h C_k \neq C_k B_h$, *for example.*)

c) *We have* $[G : M_{h,k,m}] = p^{h+k+m-2}$.

Proof. a) The claims follow immediately from

$$\begin{aligned} B_h/B_{h+n} &= \langle b(p^h)B_{h+n}\rangle, \\ C_k/C_{k+n} &= \langle c(p^k)C_{k+n}\rangle, \\ D_m/D_{m+n} &= \langle d(1+p^m)D_{m+n}\rangle. \end{aligned}$$

b) Let

$$x = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in M_{h,k,m}.$$

One can easily calculate that

$$x = b(\beta_1)c(\gamma_1)d(\delta_1) = c(\gamma_2)d(\delta_2)b(\beta_2) = d(\delta_3)b(\beta_3)c(\gamma_3),$$

where $\beta_1 = \frac{\beta}{\delta}$, $\gamma_1 = \gamma\delta$, $\delta_1 = \delta$; $\beta_2 = \frac{\beta}{\alpha}$, $\gamma_2 = \frac{\gamma}{\alpha}$, $\delta_2 = \frac{1}{\alpha}$; $\beta_3 = \beta\delta$, $\gamma_3 = \frac{\gamma}{\delta}$, $\delta_3 = \delta$.

It follows from $\beta \equiv 0 \bmod p^h$ that $\beta_i \equiv 0 \bmod p^h$ for $i = 1, 2, 3$. In the same way, $\gamma_i \equiv 0 \bmod p^k$ and $\delta_i \equiv 1 \bmod p^m$ for $i = 1, 2, 3$.

Conversely, if the $\beta_i, \gamma_i, \delta_i$ are determined by these congruences, then

$$b(\beta_1)c(\gamma_1)d(\delta_1) \in M_{h,k,m}$$

and likewise for $i = 2, 3$.

Now we address the uniqueness:

If $b(\beta)c(\gamma)d(\delta) = b(\beta')c(\gamma')d(\delta')$, then

$$\begin{pmatrix} 1+\gamma(\beta-\beta') & \beta-\beta' \\ \gamma & 1 \end{pmatrix} = b(\beta-\beta')c(\gamma) = c(\gamma')d\left(\frac{\delta'}{\delta}\right) = \begin{pmatrix} \frac{\delta}{\delta'} & 0 \\ \frac{\gamma'\delta}{\delta'} & \frac{\delta'}{\delta} \end{pmatrix},$$

so $\beta = \beta'$, $\gamma = \gamma'$ and $\delta = \delta'$.

The uniqueness for the other decompositions follows in the same way.

c) We have the following identities:

$$\begin{aligned} M_{h,k,m}D_1 &= B_hC_kD_1 = M_{h,k,1}, \\ M_{h,k,1}C_1 &= D_1B_hC_kC_1 = D_1B_hC_1 = M_{h,1,1} \\ M_{h,1,1}B_0 &= C_1D_1B_hB_0 = C_1D_1B_0 = M_{0,1,1} = G. \end{aligned}$$

Furthermore, since the decompositions in b) yield unique expressions,

$$D_1 \cap M_{h,k,m} = D_m, \quad C_1 \cap M_{h,k,1} = C_k \quad \text{and} \quad B_0 \cap M_{h,1,1} = B_h.$$

With this, we obtain

$$\begin{aligned} [G : M_{h,k,m}] &= [G : M_{h,1,1}][M_{h,1,1} : M_{h,k,1}][M_{h,k,1} : M_{h,k,m}] \\ &= [M_{h,1,1}B_0 : M_{h,1,1}][M_{h,k,1}C_1 : M_{h,k,1}][M_{h,k,m}D_1 : M_{h,k,m}] \\ &= [B_0 : M_{h,1,1} \cap B_0][C_1 : M_{h,k,1} \cap C_1][D_1 : M_{h,k,m} \cap D_1] \\ &= [B_0 : B_h][C_1 : C_k][D_1 : D_m] = p^h p^{k-1} p^{m-1} = p^{h+k+m-2}. \quad \square \end{aligned}$$

Lemma 17.3. *We have:*

a) $[B_h, D_m] = B_{h+m}$.

b) $[C_k, D_m] = C_{k+m}$.

c) $[B_h, C_k] \leq B_{2h+k}C_{h+2k}D_{h+k}$.

Proof. By direct calculation, we have

(1) $[b(\beta), d(\delta)] = b(\beta(\delta^2 - 1))$

(2) $[c(\gamma), d(\delta)] = c(\gamma(\delta^{-2} - 1))$

(3) $[b(\beta), c(\gamma)] = b(\beta^2\gamma\epsilon^{-1})\, c(-\beta\gamma^2\epsilon)\, d(\epsilon)$

with $\epsilon = 1 - \beta\gamma \equiv 1 \bmod p^{h+k}$. Since $\delta \equiv 1 \bmod p$ and $p > 2$, the element $\delta^{\pm 1} + 1$ is a unit in the p-adic ring R. Now, from (1) we obtain

$$[B_h, D_m] = B_{h+m}$$

and from (2)

$$[C_k, D_m] = C_{k+m}.$$

Equation (3) yields immediately that

$$[B_h, C_k] \leq B_{2h+k}C_{h+2k}D_{h+k}. \quad \square$$

Next, we determine the lower central series of G.

Theorem 17.4. *We have:*

(1) $\mathbf{K}_{2i}(G) = M_{i,i+1,i}$.

(2) $\mathbf{K}_{2i+1}(G) = M_{i,i+1,i+1}$.

Proof. a) For all i, we have

$$M_{i,i+1,i} = B_i C_{i+1} D_i \trianglelefteq G = B_0 C_1 D_1 \quad \text{and}$$
$$M_{i,i+1,i+1} = B_i C_{i+1} D_{i+1} \trianglelefteq G :$$

By Lemma 17.3, we have

$$\begin{aligned}
[B_i, C_1] &\le B_{2i+1} C_{i+2} D_{i+1} \le M_{i,i+1,i+1},\\
[B_i, D_1] &= B_{i+1} \le M_{i,i+1,i+1},\\
[C_{i+1}, B_0] &\le B_{i+1} C_{2i+2} D_{i+1} \le M_{i,i+1,i+1},\\
[C_{i+1}, D_1] &= C_{i+2} \le M_{i,i+1,i+1},\\
[D_i, B_0] &= B_i \le M_{i,i+1,i+1} \quad \text{(and likewise for } D_{i+1}),\\
[D_i, C_1] &= C_{i+1} \le M_{i,i+1,i} \quad \text{(and likewise for } D_{i+1}).
\end{aligned}$$

These calculations show that $M_{i,i+1,i}$ and $M_{i,i+1,i+1}$ are normal in G.

b) Assume for induction that

$$\mathbf{K}_{2i+1}(G) = M_{i,i+1,i+1} = B_i C_{i+1} D_{i+1};$$

this is clearly true for $i = 0$. Then

$$\mathbf{K}_{2i+2}(G) = [B_i C_{i+1} D_{i+1}, B_0 C_1 D_1],$$

where we have

$$\begin{aligned}
[B_i, C_1] &\le B_{2i+1} C_{i+2} D_{i+1} \le M_{i+1,i+2,i+1},\\
[B_i, D_1] &= B_{i+1} \le M_{i+1,i+2,i+1},\\
[C_{i+1}, B_0] &\le B_{i+1} C_{2i+2} D_{i+1} \le M_{i+1,i+2,i+1},\\
[C_{i+1}, D_1] &= C_{i+2} \le M_{i+1,i+2,i+1},\\
[D_{i+1}, B_0] &= B_{i+1} \le M_{i+1,i+2,i+1},\\
[D_{i+1}, C_1] &= C_{i+2} \le M_{i+1,i+2,i+1}.
\end{aligned}$$

So it follows that

$$B_{i+1} C_{i+2} \le \mathbf{K}_{2i+2}(G) \le M_{i+1,i+2,i+1}.$$

Let $\gamma \equiv 0 \bmod p^{i+1}$. With $\epsilon = 1 - \gamma$, we have

$$[b(1), c(\gamma)] = b(\gamma\epsilon^{-1}) c(-\gamma^2\epsilon) d(\epsilon) \in \mathbf{K}_{2i+2}(G).$$

Since $b(\gamma\epsilon^{-1}) \in B_{i+1} \le \mathbf{K}_{2i+2}(G)$ and

$$c(-\gamma^2\epsilon) \in C_{2i+2} \le \mathbf{K}_{2i+2}(G),$$

it follows that $d(1-\gamma) \in \mathbf{K}_{2i+2}(G)$. By an appropriate choice of γ, we may write any element of D_{i+1} in the form $d(1-\gamma)$. So $D_{i+1} \leq \mathbf{K}_{2i+2}(G)$. Altogether we obtain

$$\mathbf{K}_{2i+2}(G) = M_{i+1,i+2,i+1}.$$

In the same way, one shows that

$$\mathbf{K}_{2i+3}(G) = [M_{i+1,i+2,i+1}, M_{0,1,1}] = M_{i+1,i+2,i+2}. \qquad \square$$

Lemma 17.5. *We have* $M'_{h,k,m} = M_{h+m,k+m,h+k}$.

Proof. As in Theorem 17.4 a), one easily shows that

$$M_{h+m,k+m,h+k} \trianglelefteq M_{h,k,m}.$$

Furthermore, since $h + k \geq m$,

$$\begin{aligned}
[B_h, C_k] &\leq B_{2h+k} C_{h+2k} D_{h+k} \leq M_{h+m,k+m,h+k}, \\
[B_h, D_m] &= B_{h+m} \leq M'_{h,k,m} \cap M_{h+m,k+m,h+k}, \\
[C_k, D_m] &= C_{k+m} \leq M'_{h,k,m} \cap M_{h+m,k+m,h+k}.
\end{aligned}$$

It follows that

$$B_{h+m} C_{k+m} \subseteq M'_{h,k,m} \leq M_{h+m,k+m,h+k}.$$

Let $\beta \equiv 0 \bmod p^h$, $\gamma \equiv 0 \bmod p^k$ and $\epsilon = 1 - \beta\gamma$. Since

$$[b(\beta), c(\gamma)] = b(\beta^2 \gamma \epsilon^{-1})\, c(-\beta\gamma^2\epsilon)\, d(\epsilon),$$

where

$$b(\beta^2\gamma\epsilon^{-1}) \in B_{2h+k} \leq B_{h+m} \leq M'_{h,k,m}$$

and

$$c(-\beta\gamma^2\epsilon) \in C_{h+2k} \leq C_{k+m} \leq M'_{h,k,m},$$

we have $d(1-\beta\gamma) \in M'_{h,k,m}$. By varying β and γ, $d(1-\beta\gamma)$ runs through all elements of D_{h+k}. Therefore, $D_{h+k} \leq M'_{h,k,m}$, and we are done. $\square$

Now we can determine the higher commutator subgroups of G:

Theorem 17.6. *We have* $G^{(2s)} = M_{t-1,t,t}$ *and* $G^{(2s+1)} = M_{2t-1,2t,2t-1}$, *where* $t = t(s) = \frac{1}{3}(2^{2s+1}+1)$ $(s = 0, 1, \ldots)$.

Proof. The claim follows from Lemma 17.5 by an easy induction argument. $\square$

Theorem 17.7. *For* $p > 2$ *and every* $k = 0, 1, \ldots$, *there exists a group* H *of order* $p^{2^{k+1}-1}$ *with* $H^{(k)} \neq 1$.

(*This shows that the bound in Theorem 7.11 has the correct order of magnitude.*)

Proof. We let $L = G/G^{(k+1)}$ with $G = M_{0,1,1}$. Then

$$L^{(k)} = G^{(k)}/G^{(k+1)} \neq 1.$$

It follows from Lemma 17.2 c) and Theorem 17.6, setting $t = t(s)$, that

$$|G^{(2s)}/G^{(2s+1)}| = [G : M_{2t-1,2t,2t-1}]/[G : M_{t-1,t,t}] = p^{3t-1} = p^{2^{2s+1}}$$

and

$$|G^{(2s+1)}/G^{(2s+2)}| = p^{6t-2} = p^{2^{2s+2}}.$$

Therefore, for all i with $i \leq k$ we have

$$|L^{(i)}/L^{(i+1)}| = p^{2^{i+1}}.$$

So $|L| = p^n$ with $n = 2^{k+2} - 2$ and $|L^{(k)}| = p^{2^{k+1}}$.

Let N be a normal subgroup of L with $|L^{(k)}/N| = p$. We let $H = L/N$. Then $H^{(k)} \neq 1$ and

$$|H| = p^{2^{k+1}-1}. \qquad \square$$

Theorem 17.8. *Let $s \geq 1$, and let $t = t(s)$ be defined as in Theorem 17.6. The group*

$$H = G/M_{t-1,t,t}$$

has the following properties when $p > 3$:

a) *H is regular with $|H/\mho_1(H)| = p^3$.*

b) *We have*

$$\mho_1(H) = M_{1,2,2}/M_{t-1,t,t} = \mathbf{K}_3(H) < H' = M_{1,2,1}/M_{t-1,t,t}.$$

c) *$\Omega_1(H) = M_{t-2,t-1,t-1}/M_{t-1,t,t}$ is abelian.*
So $\Omega_1(H)$ and $H/\mho_1(H)$ are not isomorphic (but they have the same order, as in every regular p-group).

d) *H has derived length $2s$. So for $p > 3$, there exist regular p-groups with $\omega(H) = 3$ and arbitrarily large derived length.*
Regular 3-groups are metabelian by the remark in Theorem 10.3 b).

Proof. Since $\mho_1(B_h) = B_{h+1}$ and the corresponding identities hold for C_k and D_m, we certainly have

$$\mho_1(H) \geq M_{1,2,2}/M_{t-1,t,t}.$$

It follows from $|G/M_{1,2,2}| = p^3$ that $\omega(H) \leq 3$. Hence, H is regular for $p > 3$ by Theorem 10.13.

The group $G/M_{1,2,2}$ of order p^3 is regular and is generated by its elements of order p. So $G/M_{1,2,2}$ has exponent p, and we obtain

$$\mho_1(H) = M_{1,2,2}/M_{t-1,t,t}$$

and $\omega(H) = 3$.

It remains to prove c): We have

$$\Omega_1(H) = M_{t-2,t-1,t-1}/M_{t-1,t,t}$$

since this group has exponent p and the correct order

$$|H/\mho_1(H)| = p^3.$$

Since $t \geq 3$, the commutativity of $\Omega_1(H)$ follows from

$$M'_{t-2,t-1,t-1} = M_{2t-3,2t-2,2t-3} \leq M_{t-1,t,t}.$$

The claim in d) follows from Theorem 17.6. □

In this context, the following theorem is worth mentioning.

Theorem 17.9 (Blackburn [39]). *Let $p > 3$, and let G be a p-group with $\mho_1(G) = \mathbf{K}_3(G)$ and $|G/\mho_1(G)| = p^3$. We define the nonnegative integers f_i by*

$$|\mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)| = p^{f_i}.$$

Then $(f_1, f_2, \dots) = (2, 1, 2, 1, \dots, 1, 2, 1, f)$ with $0 \leq f \leq 2$. If i or j is odd, then

$$[\mathbf{K}_i(G), \mathbf{K}_j(G)] = \mathbf{K}_{i+j}(G).$$

In particular,

$$G^{(s)} = \mathbf{K}_{j(s)}(G)$$

with $j(s) = \frac{1}{6}(2^{s+3} + (-1)^s - 3)$.

Proof. a) We first prove by induction on i that

$$\mho_1(\mathbf{K}_i(G)) = \mathbf{K}_{i+2}(G) :$$

Since $[G : \mho_1(G)] = p^3 < p^p$, the group G is regular by Theorem 10.13.

The claim clearly holds for $i = 1$. Thus, it follows immediately from the induction hypothesis

$$\mho_1(\mathbf{K}_i(G)) = \mathbf{K}_{i+2}(G)$$

and Theorem 10.8 a) that

$$\begin{aligned}\mho_1(\mathbf{K}_{i+1}(G)) &= \mho_1([\mathbf{K}_i(G), G]) = [\mho_1(\mathbf{K}_i(G)), G] = [\mathbf{K}_{i+2}(G), G] \\ &= \mathbf{K}_{i+3}(G).\end{aligned}$$

b) If $i > 1$ is odd, then using induction with the help of Theorem 10.8 a) and part a), we have

$$\begin{aligned}[\mathbf{K}_i(G), \mathbf{K}_j(G)] &= [\mho_1(\mathbf{K}_{i-2}(G)), \mathbf{K}_j(G)] = \mho_1([\mathbf{K}_{i-2}(G), \mathbf{K}_j(G)]) \\ &= \mho_1(\mathbf{K}_{i+j-2}(G)) = \mathbf{K}_{i+j}(G).\end{aligned}$$

c) We have

$$\begin{aligned}|\mathbf{K}_i(G)/\mathbf{K}_{i+2}(G)| &= |\mathbf{K}_i(G)/\mho_1(\mathbf{K}_i(G))| = |\Omega_1(\mathbf{K}_i(G))| \\ &\leq |\Omega_1(G)| = |G/\mho_1(G)| = p^3.\end{aligned}$$

Assume that

$$|\mathbf{K}_i(G)/\mathbf{K}_{i+2}(G)| \leq p^2.$$

We prove that $\mathbf{K}_{i+2}(G) = 1$ by induction on i. We may assume for contradiction that $|\mathbf{K}_{i+2}(G)| = p$. Then

$$|\mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)| = |\mathbf{K}_{i+1}(G)/\mathbf{K}_{i+2}(G)| = |\mathbf{K}_{i+2}(G)| = p.$$

First assume $i = 2$. Then $|G| = p^5$ and

$$(f_1, f_2, f_3, f_4) = (2, 1, 1, 1).$$

So G is a p-group of maximal class. Since $p > 3$, Lemma 14.14 yields the contradiction

$$\mathbf{K}_3(G) = \mho_1(G) \leq \mathbf{K}_4(G).$$

Now assume $i > 2$. The induction hypothesis implies that for $j < i$,

$$\begin{aligned}&|\mathbf{K}_j(G)/\mathbf{K}_{j+2}(G)| = p^3 \quad \text{and} \\ &|\mathbf{K}_j(G)/\mathbf{K}_{j+1}(G)| = \begin{cases} p^2 & \text{for } j \equiv 1 \bmod 2 \\ p & \text{for } j \equiv 0 \bmod 2. \end{cases}.\end{aligned}$$

Case 1: Assume i is odd. Then we have

$$|\mathbf{K}_i(G)/\mathbf{K}_{i+2}(G)| \leq p^2$$

and

$$p^3 = |\mathbf{K}_{i-1}(G)/\mathbf{K}_{i+1}(G)| = p|\mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)|.$$

It follows that

$$|\mathbf{K}_i(G)/\mathbf{K}_{i+1}(G)| = |\mathbf{K}_i(G)/\mathbf{K}_{i+2}(G)|$$

and therefore $\mathbf{K}_{i+1}(G) = \mathbf{K}_{i+2}(G) = 1$.

Case 2: Assume $i = 2r$ is even and $2 \leq r$. Then

$$(f_1, \ldots, f_{2r+2}) = (2, 1, 2, 1, \ldots, 2, 1, 1, 1).$$

By a), we have

$$\begin{aligned} p^3 = |\mathbf{K}_{2r-1}(G)/\mathbf{K}_{2r+1}(G)| &= |\mathbf{K}_{2r-1}(G)/\mathfrak{V}_1(\mathbf{K}_{2r-1}(G))| \\ &= |\Omega_1(\mathbf{K}_{2r-1}(G))| \end{aligned}$$

and

$$p^2 = |\mathbf{K}_{2r}(G)/\mathbf{K}_{2r+2}(G)| = |\Omega_1(\mathbf{K}_{2r}(G))|.$$

Consequently, there exists an element x with $|x| = p$ and $x \in \mathbf{K}_{2r-1}(G) - \mathbf{K}_{2r}(G)$. Using a), we obtain

$$\mathfrak{V}_{r-2}(\mathbf{K}_3(G)) = \mathbf{K}_{2r-1}(G) \quad \text{and} \quad \mathfrak{V}_{r-2}(\mathbf{K}_4(G)) = \mathbf{K}_{2r}(G).$$

Thus, there exists an $a \in \mathbf{K}_3(G) - \mathbf{K}_4(G)$ with $x = a^{p^{r-2}}$.

Let $\mathbf{K}_3(G) = \mathfrak{V}_1(G) = \langle \mathbf{K}_4(G), a, b \rangle$.

If $c \in \mathbf{K}_4(G)$, then by Theorem 10.8 g) we have

$$(a^i b^j c)^{p^{r-2}} = a^{ip^{r-2}} b^{jp^{r-2}} c^{p^{r-2}} t^{p^{r-2}}$$

for some $t \in \mathbf{K}_3(G)' \leq \mathbf{K}_4(G)$. Hence,

$$\begin{aligned} \mathbf{K}_{2r-1}(G) = \mathfrak{V}_{r-2}(\mathbf{K}_3(G)) &= \langle a^{p^{r-2}}, b^{p^{r-2}}, \mathfrak{V}_{r-2}(\mathbf{K}_4(G)) \rangle \\ &= \langle a^{p^{r-2}}, b^{p^{r-2}}, \mathbf{K}_{2r}(G) \rangle. \end{aligned}$$

Using b), it follows that

$$\begin{aligned} \mathbf{K}_{2r+2}(G) &= [\mathbf{K}_3(G), \mathbf{K}_{2r-1}(G)] \\ &= \langle [a^{p^{r-2}}, b], [a, b^{p^{r-2}}], [\mathbf{K}_4(G), \mathbf{K}_{2r-1}(G)], [\mathbf{K}_3(G), \mathbf{K}_{2r}(G)] \rangle. \end{aligned}$$

Since $\mathbf{K}_{2r+3}(G) = 1$, there remains only

$$\mathbf{K}_{2r+2}(G) = \langle [a^{p^{r-2}}, b], [a, b^{p^{r-2}}] \rangle.$$

Here, $a^{p^{r-2}} = x \in \Omega_1(G)$ and $b \in \mathfrak{V}_1(G)$, so by Theorem 10.8 f),

$$[a^{p^{r-2}}, b] \in [\Omega_1(G), \mathfrak{V}_1(G)] = 1.$$

Now since $[a^{p^{r-2}}, b] = 1$, it follows from Theorem 10.6 b) that $[a, b^{p^{r-2}}] = 1$, as well. But then $\mathbf{K}_{2r+2}(G) = 1$, contradicting our assumption.

d) We prove by induction on s that $G^{(s)} = \mathbf{K}_{j(s)}(G)$, where $j(s) = \frac{1}{6}(2^{s+3} + (-1)^s - 3)$. For $s = 1$, the claim holds since $j(1) = 2$. Now assume for induction that $G^{(i)} = \mathbf{K}_{j(i)}(G)$ for some $i \geq 1$. If $j(i)$ is odd, then i is even and it follows immediately from what we have already proved that

$$G^{(i+1)} = [\mathbf{K}_{j(i)}(G), \mathbf{K}_{j(i)}(G)] = \mathbf{K}_{2j(i)}(G) = \mathbf{K}_{j(i+1)}(G).$$

Now suppose that i is odd, so $j(i)$ is even.[8] Then

$$\mathbf{K}_{2j(i)+1}(G) = [\mathbf{K}_{j(i)}(G), \mathbf{K}_{j(i)+1}(G)] \leq \mathbf{K}_{j(i)}(G)'.$$

Since $|\mathbf{K}_{j(i)}(G)/\mathbf{K}_{j(i)+1}(G)| = p$, the quotient $\mathbf{K}_{j(i)}(G)/\mathbf{K}_{2j(i)+1}(G)$ is abelian, as it is a cyclic extension of the central subgroup $\mathbf{K}_{j(i)+1}(G)/\mathbf{K}_{2j(i)+1}(G)$. So $\mathbf{K}_{j(i)}(G)' \leq \mathbf{K}_{2j(i)+1}(G)$ and finally

$$\mathbf{K}_{j(i)}(G)' = \mathbf{K}_{2j(i)+1}(G) = \mathbf{K}_{j(i+1)}(G). \qquad \square$$

18 Generators and relations in p-groups

The primary objective of this section is to prove the following theorem of Golod and Shafarevich.

Theorem 18.1. *Let G be a p-group. Let d be the minimum size of a generating set for G, and let $G = \langle g_1, \ldots, g_d \rangle$. If G can be defined by r relations between the g_i (in the sense of Chapter I, §19), then*

$$r > \frac{d^2}{4}.$$

Theorem 18.1 plays a crucial role in disproving the so-called class field tower conjecture. Before proving Theorem 18.1, we must equip ourselves with a few theorems regarding group rings that are also important in other applications. If H is a (not necessarily finite) group and S is a commutative ring with unity, then we denote by $S[H]$ the group ring of H over S as described in Definition I.16.6 (pg. 112).

Lemma 18.2. *Let α be an epimorphism from H to G.*

a) *The map $\overline{\alpha}$ defined by*

$$\left(\sum_{h \in H} a_h h \right) \overline{\alpha} = \sum_{h \in H} a_h h^{\alpha}$$

[8] Note that i is odd if and only if $j(i)$ is even. *trans.*

is a ring homomorphism from the group ring $S[H]$ of H over S onto $S[G]$ with

$$\ker \overline{\alpha} = \left\{ \sum_{h\in H} a_h h \;\middle|\; \sum_{h\in H} a_h h^\alpha = 0 \right\}.$$

b) *If $\ker \alpha = R$ and $H = \bigcup_{i\in I} Rh_i$ is the decomposition of H into cosets of R, then $\ker \overline{\alpha}$ is a free S-module in the free generators $(t-1)h_i$ with $1 \neq t \in R$ and $i \in I$.*

Proof. a) is trivial.

b) If

$$\sum_{h\in H} a_h h = \sum_{\substack{t\in R\\ i\in I}} a_{th_i} t h_i \in \ker \overline{\alpha},$$

then

$$0 = \sum_{i\in I} \left(\sum_{t\in R} a_{th_i} \right) h_i^\alpha.$$

Since the h_i^α are free generators of the free S-module $S[G]$, it follows that

$$\sum_{t\in R} a_{th_i} = 0.$$

Consequently, we have

$$\sum_{h\in H} a_h h = \sum_{\substack{t\in R\\ i\in I}} a_{th_i} (t-1) h_i. \qquad \square$$

Setting $G = 1$ in Lemma 18.2 yields the following important special case:

Definition 18.3. Let $S[H]$ be the group ring of H over S. The map ϵ defined by

$$\left(\sum_{h\in H} a_h h \right)\epsilon = \sum_{h\in H} a_h$$

is a ring homomorphism from $S[H]$ onto S. We define I_H by

$$I_H = \ker \epsilon = \left\{ \sum_{h\in H} a_h h \;\middle|\; \sum_{h\in H} a_h = 0 \right\}.$$

The two-sided ideal I_H of $S[H]$ is called the augmentation ideal of $S[H]$.

Lemma 18.4. *Let F be a free group in the free generators f_i $(i \in I)$ and $\mathfrak{m}$ a right F-module. Given a collection of elements $m_i \in \mathfrak{m}$, there exists a 1-cocycle $\varphi \in \mathbf{Z}^1(F, \mathfrak{m})$ with $\varphi(f_i) = m_i$ for all $i \in I$.*

Proof. We construct the set

$$T = \{(f, m) \mid f \in F, m \in \mathfrak{m}\}$$

and define a multiplication by

$$(f, m)(f', m') = (ff', mf' + m').$$

It is easy to verify that T is a group. Since F is free in the generators f_i, there exists by Definition I.19.1 (pg. 142) a homomorphism μ from F to T with

$$f_i^\mu = (f_i, m_i).$$

Then clearly $f^\mu = (f, \varphi(f))$ for some map φ from F to $\mathfrak{m}$. We have

$$\begin{aligned}(ff', \varphi(ff')) &= (ff')^\mu = f^\mu (f')^\mu = (f, \varphi(f))(f', \varphi(f')) \\ &= (ff', \varphi(f)f' + \varphi(f')).\end{aligned}$$

Therefore, $\varphi \in \mathbf{Z}^1(F, \mathfrak{m})$ and $\varphi(f_i) = m_i$. □

Lemma 18.5. *Let F be a free group in the free generators f_i $(i \in I)$. Then I_F is a free right $S[F]$-module in the free generators $f_i - 1$.*

Proof. By Lemma 18.2, the collection of the $f - 1$ with $f \in F$ is a generating set for I_F as an S-module. It follows from

$$xy - 1 = (x - 1)y + (y - 1) \quad \text{and} \quad x^{-1} - 1 = -(x - 1)x^{-1}$$

that the $f_i - 1$ generate all of I_F as a right $S[F]$-module.

It remains to show that if

$$\sum_j (f_j - 1)c_j = 0 \quad \text{with} \quad c_j \in S[F],$$

then all the c_j are zero.

For every i, there exists a 1-cocycle $\varphi \in \mathbf{Z}^1(F, S[F])$ such that

$$\varphi(f_j) = \begin{cases} 1 & \text{for } j = i \\ 0 & \text{for } j \neq i \end{cases}$$

by Lemma 18.4. We extend φ to an S-homomorphism from $S[F]$ to $S[F]$. Then

$$\begin{aligned}\varphi(1) &= \varphi(1)1 + \varphi(1), \quad \text{so} \quad \varphi(1) = 0, \\ \varphi(f - 1) &= \varphi(f) - \varphi(1) = \varphi(f), \\ \varphi((f - 1)x) &= \varphi(fx) - \varphi(x) = \varphi(f)x + \varphi(x) - \varphi(x) = \varphi(f)x \\ &= \varphi(f - 1)x\end{aligned}$$

for all $f, x \in F$. So in general, we have

$$\varphi(ab) = \varphi(a)b$$

for all $a \in I_F$ and $b \in S[F]$. Therefore, φ is an $S[F]$-homomorphism of the right $S[F]$-module I_F. Now it follows that

$$0 = \varphi\left(\sum_j (f_j - 1)c_j\right) = \sum_j \varphi(f_j - 1)c_j = c_i. \qquad \square$$

In the rest of this section, we fix the following notation:

Let $G = \langle g_1, \ldots, g_d \rangle$ be a p-group with $|G/\Phi(G)| = p^d$, and let F be a free group in the d free generators f_i $(i = 1, \ldots, d)$ with $F/R = G$. Let α be the natural homomorphism from F onto G, so $\ker \alpha = R$. Let $t_1, \ldots, t_r$ be a collection of defining relations for G in terms of the generators $g_1, \ldots, g_d$. Let $\mathsf{K} = \mathrm{GF}(p)$, and let $\overline{\alpha}$ be the ring homomorphism from $\mathsf{K}[F]$ onto $\mathsf{K}[G]$ constructed as in Lemma 18.2. We let $\mathfrak{n} = \ker \overline{\alpha}$. Furthermore, let

$$I_F = \left\{ \sum_{f \in F} a_f f \;\middle|\; \sum_{f \in F} a_f = 0 \right\}$$

and

$$I_G = \left\{ \sum_{g \in G} a_g g \;\middle|\; \sum_{g \in G} a_g = 0 \right\}$$

be the augmentation ideals of $\mathsf{K}[F]$ and $\mathsf{K}[G]$, respectively.

The relationship between what we have developed regarding group rings and our stated problem is made clear by the following two lemmas.

Lemma 18.6. *We let $\widetilde{R} = R/R'R^p$. The elementary abelian p-group $\widetilde{R}$ becomes a right $\mathsf{K}[F]$-module, respectively right $\mathsf{K}[G]$-module, in a natural way by defining*

$$(tR'R^p)^f = t^f R'R^p, \quad \textit{respectively} \quad (tR'R^p)^g = t^f R'R^p,$$

for $g = Rf \in G$ and $t \in R$.

Let r' be the minimum size of a generating set for $\widetilde{R}$ as a right $\mathsf{K}[F]$-module. (By Theorem I.19.10, pg. 151, R and thus $\widetilde{R}$ is finitely generated even as a group, so r' is finite.) Then $r \geq r'$.

Proof. If the elements $t_1, \ldots, t_r$ of F are defining relations for G in the generators g_i, then

$$R = \langle t_i^f \mid f \in F, i = 1, \ldots, r \rangle.$$

So $\widetilde{R} = R/R'R^p$ is generated as a right $\mathsf{K}[F]$-module by the $\widetilde{t_i} = t_i R'R^p$. Hence, $r \geq r'$. $\square$

To prove Theorem 18.1, we will show that $r' > \dfrac{d^2}{4}$.

Lemma 18.7. *The right* $\mathsf{K}[G]$*-modules* $\widetilde{R} = R/R'R^p$ *and* $\mathfrak{n}/I_F\mathfrak{n}$ *are isomorphic.* (*Recall that* $\mathfrak{n} = \ker\overline{\alpha}$. *The product* $I_F\mathfrak{n}$ *of the ideals* I_F *and* $\mathfrak{n}$ *should be understood in the usual way.*)

Proof. For $t, t' \in R$, we have

$$tt' - 1 = (t-1) + (t'-1) + (t-1)(t'-1),$$

so that

(1) $tt' - 1 \equiv (t-1) + (t'-1) \mod I_F\mathfrak{n}$.

For $t \in R$ and $f \in F$, we have

$$t^f - 1 = f^{-1}tf - 1 = (t-1)f + (1-f)(t^f - 1),$$

so that

(2) $t^f - 1 \equiv (t-1)f \mod I_F\mathfrak{n}$.

a) We define the map μ from R to $\mathfrak{n}/I_F\mathfrak{n}$ by

$$t^\mu = t - 1 + I_F\mathfrak{n}.$$

By (1), the map μ is a homomorphism from R to the additive group of $\mathfrak{n}/I_F\mathfrak{n}$. Since K has characteristic p, we have $(R'R^p)^\mu = 0$. So μ induces a homomorphism ϵ from $\widetilde{R} = R/R'R^p$ to $\mathfrak{n}/I_F\mathfrak{n}$ defined by

$$(tR'R^p)^\epsilon = t^\mu = t - 1 + I_F\mathfrak{n}.$$

By (2), we have

$$\begin{aligned}(t^f R'R^p)^\epsilon = (t^f)^\mu = t^f - 1 + I_F\mathfrak{n} &= (t-1)f + I_F\mathfrak{n} \\ &= (tR'R^p)^\epsilon f.\end{aligned}$$

That is, ϵ is in fact a $\mathsf{K}[G]$-homomorphism from $\widetilde{R}$ to $\mathfrak{n}/I_F\mathfrak{n}$.

b) Let V be a transversal for R in F, so that $F = \bigcup_{v\in V} vR$. Then by Lemma 18.2 b), $\{v(t-1) \mid 1 \neq t \in R, v \in V\}$ is a K-basis of $\mathfrak{n}$. We define a K-linear map τ from $\mathfrak{n}$ to $\widetilde{R} = R/R'R^p$ by

$$(v(t-1))^\tau = tR'R^p.$$

If $f \in F$ and $fv = v't'$, then

$$\begin{aligned}(fv(t-1))^\tau = (v't'(t-1))^\tau &= (v'(t't-1) - v'(t'-1))^\tau \\ &= t't(t')^{-1}R'R^p = tR'R^p = (v(t-1))^\tau.\end{aligned}$$

It follows that

$$((f-1)v(t-1))^\tau = 1,$$

and so $I_F\mathfrak{n} \subseteq \ker\tau$. Hence, τ induces a K-linear map δ from $\mathfrak{n}/I_F\mathfrak{n}$ to $R/R'R^p$ defined by

$$(v(t-1)+I_F\mathfrak{n})^\delta = tR'R^p.$$

By a), $(tR'R^p)^\epsilon = t-1+I_F\mathfrak{n}$.

Therefore,

$$(tR'R^p)^{\epsilon\,\delta} = (t-1+I_F\mathfrak{n})^\delta = tR'R^p$$

for all $t \in R$. Hence, ϵ is a monomorphism from $R/R'R^p$ to $\mathfrak{n}/I_F\mathfrak{n}$. Since

$$v(t-1) \equiv t-1 \bmod I_F\mathfrak{n},$$

the map ϵ is also an epimorphism from $R/R'R^p$ onto $\mathfrak{n}/I_F\mathfrak{n}$. So ϵ is an isomorphism. □

Lemma 18.8. *We have* $\mathfrak{n} \subseteq I_F^2$.

Proof. By Theorem 3.14 a) (pg. 288), we have

$$\begin{aligned} p^d &= |G/\Phi(G)| = |G/G'G^p| = |(F/R)\big/(F'F^pR/R)| \\ &= |F/F'F^pR| \le |F/F'F^p| = p^d. \end{aligned}$$

Therefore, we must have equality throughout, and so $R \le F'F^p$.

Applying Lemma 18.7 with $G = 1$ and $R = F$, we obtain

$$F/F'F^p = \widetilde{F} \cong I_F/I_F^2.$$

The isomorphism is given by

$$\widetilde{f} = fF'F^p \mapsto f-1+I_F^2.$$

Let $t \in R \le F'F^p$. Then $\widetilde{t} = 1$ and $t-1 \in I_F^2$. Since $\mathfrak{n}$ is generated by the $t-1$ as a right $\mathsf{K}[F]$-module by Lemma 18.2 b), $\mathfrak{n} \subseteq I_F^2$. □

18.9. Proof of Theorem 18.1 (Roquette).

The homomorphism α from F onto G induces the homomorphism $\overline{\alpha}$ from $\mathsf{K}[F]$ onto $\mathsf{K}[G]$ with kernel $\mathfrak{n}$. The induced map clearly takes I_F onto I_G. Since $\mathfrak{n} \subseteq I_F$,

$$(1) \qquad 0 \longrightarrow \mathfrak{n} \underset{\iota}{\longrightarrow} I_F \underset{\overline{\alpha}}{\longrightarrow} I_G \longrightarrow 0$$

is an exact sequence of right $\mathsf{K}[F]$-modules.

In this exact sequence, ι is the inclusion map and $\ker\overline{\alpha} = \operatorname{im}\iota$. As $\mathfrak{n}$ is a two-sided ideal, we have $I_F\mathfrak{n} \subseteq \mathfrak{n} = \ker\overline{\alpha}$. Therefore, the sequence

$$(2) \qquad 0 \longrightarrow \mathfrak{n}/I_F\mathfrak{n} \longrightarrow I_F/I_F\mathfrak{n} \underset{\varphi}{\longrightarrow} I_G \longrightarrow 0$$

is also exact, where φ is defined by

$$(a + I_F\mathfrak{n})\varphi = a\overline{\alpha} \quad \text{for} \quad a \in I_F.$$

By Lemma 18.5, I_F is a free right $\mathsf{K}[F]$-module of rank d, say

$$I_F = \bigoplus_{i=1}^{d} a_i\mathsf{K}[F].$$

So $I_F\mathfrak{n} = \bigoplus_{i=1}^{d} a_i\mathfrak{n}$, and then

$$I_F/I_F\mathfrak{n} \cong \bigoplus_{i=1}^{d} a_i\mathsf{K}[F]/a_i\mathfrak{n}.$$

The map μ_i sending $x \mapsto a_i x + a_i\mathfrak{n}$ is a module homomorphism from $\mathsf{K}[F]$ onto $a_i\mathsf{K}[F]/a_i\mathfrak{n}$. Since the a_i constitute a free generating set of I_F, we have $\ker\mu_i = \mathfrak{n}$. Hence,

$$a_i\mathsf{K}[F]/a_i\mathfrak{n} \cong \mathsf{K}[F]/\mathfrak{n} \cong \mathsf{K}[G].$$

Therefore, $I_F/I_F\mathfrak{n}$ is a free right $\mathsf{K}[G]$-module of rank d.

By Lemma 18.7, we have $\mathfrak{n}/I_F\mathfrak{n} \cong \widetilde{R}$. We define

$$\mathfrak{m}_0 = \mathfrak{n}/I_F\mathfrak{n} \quad \text{and} \quad \mathfrak{m}_1 = I_F/I_F\mathfrak{n}.$$

The group $\widetilde{R}$ is generated by r' elements as a $\mathsf{K}[G]$-module. Let $\mathfrak{f}$ be a free $\mathsf{K}[G]$-module of rank r', and let ψ be an epimorphism from $\mathfrak{f}$ onto $\mathfrak{m}_0$. Then the sequence

$$(3) \qquad \mathfrak{f} \underset{\psi}{\longrightarrow} \mathfrak{m}_1 \underset{\varphi}{\longrightarrow} I_G \longrightarrow 0$$

is exact since $\mathfrak{f}\psi = \mathfrak{m}_0 = \ker\varphi$. By Lemma 18.8, we have

$$\mathfrak{f}\psi = \mathfrak{m}_0 = \mathfrak{n}/I_F\mathfrak{n} \subseteq I_F^2/I_F\mathfrak{n} = \mathfrak{m}_1 I_F = \mathfrak{m}_1 I_G,$$

where the last equality follows from $\mathfrak{m}_1\mathfrak{n} = 0$.

The homomorphism φ induces a map

$$\mathfrak{m}_1 I_G^n \to (\mathfrak{m}_1 I_G^n)\varphi = (\mathfrak{m}_1\varphi)I_G^n = I_G^{n+1}$$

with kernel $\mathfrak{m}_1 I_G^n \cap \mathfrak{m}_0$.

Let $\mathfrak{b}_n = \{f \in \mathfrak{f} \mid f\psi \in \mathfrak{m}_1 I_G^n \cap \mathfrak{m}_0\}$.

We obtain two further exact sequences, namely

$$(4) \qquad 0 \longrightarrow \mathfrak{b}_n \underset{\psi}{\longrightarrow} \mathfrak{m}_1 I_G^n \longrightarrow I_G^{n+1} \longrightarrow 0$$

and, from (4),

$$(5) \qquad 0 \longrightarrow \mathfrak{b}_n/\mathfrak{b}_{n+1} \longrightarrow \mathfrak{m}_1 I_G^n/\mathfrak{m}_1 I_G^{n+1} \longrightarrow I_G^{n+1}/I_G^{n+2} \longrightarrow 0.$$

We consider the dimensions of the modules appearing in these sequences as K-vector spaces and set

$$d_n = \dim_{\mathsf{K}} I_G^n/I_G^{n+1} \quad (n \geq 0).$$

Since $I_G^n \subseteq \mathsf{K}[G]$, d_n is certainly finite, and only finitely many of the d_n are nonzero. Furthermore,

$$d_0 = \dim_{\mathsf{K}} \mathsf{K}[G]/I_G = 1.$$

Since $\mathfrak{m}_1$ is a free $\mathsf{K}[G]$-module of rank d, we have

$$\dim_{\mathsf{K}} \mathfrak{m}_1 I_G^n/\mathfrak{m}_1 I_G^{n+1} = d\, d_n.$$

From (5), we now obtain

$$(6) \qquad \dim_{\mathsf{K}} \mathfrak{b}_n/\mathfrak{b}_{n+1} = d\, d_n - d_{n+1}.$$

We have

$$(\mathfrak{f} I_G^n)\psi = (\mathfrak{f}\psi) I_G^n \subseteq \mathfrak{m}_1 I_G^{n+1},$$

so

$$\mathfrak{f} I_G^n \subseteq \mathfrak{b}_{n+1} \subseteq \mathfrak{b}_n \subseteq \mathfrak{f}.$$

In particular, $\mathfrak{b}_0/\mathfrak{b}_1 = 0$. We let

$$\dim_{\mathsf{K}} \mathfrak{b}_n/\mathfrak{b}_{n+1} = s_{n-1} \quad (n \geq 1).$$

Since $\mathfrak{f}$ is a free $\mathsf{K}[G]$-module of rank r', it follows from

$$\mathfrak{f} I_G^n \subseteq \mathfrak{b}_{n+1} \subseteq \mathfrak{b}_n \subseteq \mathfrak{f}$$

that

$$(7) \qquad \sum_{i=0}^{n-1} s_i = \sum_{i=1}^{n} \dim_{\mathsf{K}} \mathfrak{b}_i/\mathfrak{b}_{i+1} \leq \dim_{\mathsf{K}} \mathfrak{f}/\mathfrak{f} I_G^n = \sum_{i=0}^{n-1} \dim_{\mathsf{K}} \mathfrak{f} I_G^i/\mathfrak{f} I_G^{i+1}$$
$$= r' \sum_{i=0}^{n-1} d_i.$$

We define generating functions $p(t) = \sum\limits_n d_n t^n$ and $s(t) = \sum\limits_n s_n t^n$ in the indeterminate t. It follows from (7) that

$$\frac{s(t)}{1-t} = \sum_n (s_0 + \cdots + s_n)t^n \le r' \sum_n (d_0 + \cdots + d_n)t^n = r' \frac{p(t)}{1-t}$$

for $0 < t < 1$, so $s(t) \le r' p(t)$. Multiplying the equations

$$s_{n-1} = d\, d_n - d_{n+1}$$

in (6) with t^{n+1} and summing over n, we obtain

$$t^2 s(t) + p(t) - 1 = dt\, p(t)$$

since $d_0 = 1$. Therefore, for $0 < t < 1$,

$$1 = t^2 s(t) + p(t) - dt\, p(t) \le p(t)(1 - dt + r' t^2). \tag{8}$$

Hence,

$$r' t^2 - dt + 1 > 0 \quad \text{for} \quad 0 < t < 1. \tag{9}$$

Letting $t \to 1$, we obtain from (9) the inequality $d < r' + 1$. Therefore,

$$t_0 = \frac{d}{2r'} \le \frac{d}{r'+1} < 1.$$

Plugging t_0 into (9) yields the claim $r' > \dfrac{d^2}{4}$. □

Example 18.10.

a) (Kostrikin) Let G be the group generated by the $2t$ elements $x_1, \ldots, x_t, y_1, \ldots, y_t$ and the $\frac{3}{2}(t^2 + t)$ relations

$$\begin{aligned} x_i^p = y_i^p &= 1 \quad (i = 1, \ldots, t) \\ [x_i, y_j] &= 1 \quad (i, j = 1, \ldots, t) \\ [x_i, x_j] &= [y_i, y_j] \quad (i < j;\ i, j = 1, \ldots, t). \end{aligned}$$

Since

$$[x_i, x_j, x_k] = [y_i, y_j, x_k] = 1$$

and

$$[y_i, y_j, y_k] = [x_i, x_j, y_k] = 1,$$

G has class at most 2. If $p > 2$, then G has exponent p, and it follows that $|G| \le p^s$ with $s = 2t + \frac{t(t-1)}{2}$. Hence, G is a finite p-group with $d = 2t$ generators and r relations with

$$r \leq \frac{3}{8}d^2 + \frac{3}{4}d.$$

The elementary abelian group of order p^{2t} is clearly a homomorphic image of G. So $d(G) = 2t$.

Therefore, it is not always true that $r \geq \frac{d(d-1)}{2}$, as is sometimes conjectured.

b) (Mennicke [330]) Let $G = \langle a, b, c\rangle$ with the defining relations

$$(1) \qquad bab^{-1} = a^{1+t}, \quad cbc^{-1} = b^{1+t}, \quad aca^{-1} = c^{1+t}$$

for some positive integer t. We set $n = 1 + t$ and show:

$$(2) \qquad b^x a^y b^{-x} = a^{yn^x} :$$

This follows immediately by a repeated application of (1).

(3) Letting $k = n^t - 1$, we have $nk + n = n^{1+t} = n^n$. We show that

$$c^{nk} b^{nk} a^{nk} = 1 :$$

By (2),

$$a^{nk} a^n = a^{n^n} = b^n a b^{-n}.$$

It follows that

$$a^{nk} = b^n a b^{-n} a^{-n},$$

and we obtain corresponding relations by cyclically permuting the a, b, c. Hence,

$$\begin{aligned} c^{nk} b^{nk} a^{nk} &= a^n c a^{-n} c^{-n} c^n b c^{-n} b^{-n} b^n a b^{-n} a^{-n} \\ &= a^n (c a^{-n} b c^{-n} a b^{-n}) a^{-n} \\ &= a^n (c b a^{-1} b^{-1} b a c^{-1} a^{-1} a c b^{-1} c^{-1}) a^{-n} = 1. \end{aligned}$$

Next, we show that

$$(4) \qquad a^{nk} b^{n^{nk}} a^{-nk} = b :$$

By (2), we have

$$b = c^{-nk} b^{n^{nk}} c^{nk}.$$

Using (3) to write $c^{nk} = a^{-nk} b^{-nk}$, it follows that

$$b = b^{nk} (a^{nk} b^{n^{nk}} a^{-nk}) b^{-nk} = a^{nk} b^{n^{nk}} a^{-nk}.$$

Now we show that

(5) $b^{n^{nk}-1} = a^{tk}$:

Using (1), it follows from (4) that

$$b = ba^k b^{-1} b^{n^{nk}} a^{-tk} a^{-k},$$

so

$$a^{tk} = b^{n^{nk}-1}.$$

(6) Conjugating (5) by b^{-1} yields

$$a^{tk} = b^{n^{nk}-1} = ba^{tk}b^{-1} = a^{ntk} = a^{tk}a^{t^2k}.$$

We conclude that $a^{t^2k} = 1$. In the same way, $b^{t^2k} = c^{t^2k} = 1$. Since every element of G can be written in the form $a^x b^y c^z$ by (1), the group G is finite. Two applications of the product formula to the decomposition $G = ABC$ show that $|G|$ divides t^6k^3. In particular, if we set $t = 2$, then $k = 8$ and G is a 2-group defined by $d = 3$ generators and $r = 3$ relations.

19 Automorphisms of p-groups

The goal of this section is to prove the following theorem:

Theorem 19.1 (Gaschütz [165]). *If G is a p-group with $|G| > p$, then p divides $|\mathbf{Aut}(G)/\mathbf{Inn}(G)|$.*

We make some preparations before proceeding with the proof.

Lemma 19.2. *Let G be a p-group.*

a) *If $|G/N| = p$ and $\mathbf{Z}(N) \leq \mathbf{Z}(G)$, then there exists an outer automorphism of G with order p.*

b) *If A is a maximal abelian normal subgroup of G and $\mathbf{H}^1(G/A, A) \neq 1$, then G admits an outer automorphism of p-power order.*

Proof. a) Let τ be a nontrivial homomorphism from G to $\mathbf{Z}(N)$ with $\ker \tau = N$. We define a map α from G to itself by $g^\alpha = gg^\tau$. Since $\ker \tau = N$, we have

$$(g^{1+\tau})^{1-\tau} = g^{1-\tau^2} = g.$$

So α is a map from G onto itself. It follows immediately from $g^\tau \in \mathbf{Z}(N) \leq \mathbf{Z}(G)$ that α is an automorphism of G. Clearly, $|\alpha| = p$.

Assume for contradiction that $g^\alpha = g^x$ for some $x \in G$. Since $n^\alpha = n$ for all $n \in N$, it follows that $x \in \mathbf{C}_G(N)$. If $\mathbf{C}_G(N) \leq N$, then we obtain

$$x \in \mathbf{C}_G(N) = \mathbf{Z}(N) \leq \mathbf{Z}(G),$$

contradicting $\alpha \neq 1$. If $\mathbf{C}_G(N) \not\leq N$, then we have $G = N\mathbf{C}_G(N)$. Then

$$\mathbf{C}_G(N)/\mathbf{Z}(N) \cong G/N$$

is cyclic, and since $\mathbf{Z}(N) \leq \mathbf{Z}(G)$, this means $\mathbf{C}_G(N)$ is abelian. It now follows from $G = N\mathbf{C}_G(N)$ that

$$x \in \mathbf{C}_G(N) \leq \mathbf{Z}(G),$$

again contradicting $\alpha \neq 1$.

b) Let $f \in \mathbf{Z}^1(G/A, A) - \mathbf{B}^1(G/A, A)$. By Theorem I.17.1 (pg. 127), the map α defined by

$$g^\alpha = gf(gA)$$

is an automorphism of G with $|\alpha|$ dividing $|A|$. Assume for contradiction that $g^\alpha = g^x$ for some $x \in G$. Since $a^\alpha = a$ for $a \in A$ and $\mathbf{C}_G(A) = A$ (see Theorem 7.3 a)), it follows that $x \in A$. But then Theorem I.17.1 yields a contradiction since $f \notin \mathbf{B}^1(G/A, A)$. □

Lemma 19.3.

a) *If G is an abelian p-group with $|G| > p$, then G admits an outer automorphism of order p.*

b) *If G is a nonabelian 2-group with a cyclic normal subgroup of index 2, then G admits an outer automorphism of 2-power order.*

Proof. a) Let $G = \langle g_1\rangle \times \langle g_2\rangle \times \cdots \times \langle g_n\rangle$, where $|g_i| = p^{a_i}$ and $a_1 \geq a_2 \geq \cdots \geq a_n$. If $a_1 > 1$, then α defined by

$$g_1^\alpha = g_1^{1+p^{a_1-1}}, \quad g_i^\alpha = g_i \quad (i \geq 2)$$

is an (outer, clearly) automorphism of G of order p. If $a_1 = \cdots = a_n = 1$, then α defined by

$$g_1^\alpha = g_1 g_2, \quad g_i^\alpha = g_i \quad (i \geq 2)$$

is an automorphism with the desired properties.

b) Let $\langle a\rangle$ be the cyclic normal subgroup of G with index 2.

First suppose that $\langle a\rangle$ has a complement $\langle b\rangle$ in G. If $|a| > 4$, then $\langle a\rangle$ admits an automorphism β of order 2 with $a^\beta \neq a^b$. Then α defined by

$$a^\alpha = a^\beta, \quad b^\alpha = b$$

is an automorphism of G, and α has order 2. Clearly, α is an outer automorphism of G.

If $|a| = 4$, then G is a dihedral group of order 8. Then α defined by

$$a^\alpha = a, \quad b^\alpha = ab$$

is an automorphism of G, and α has order 2. Since $b^\alpha b^{-1} = a \notin G'$, the map α is certainly not an inner automorphism.

It remains to consider the case that $\langle a \rangle$ does not have a complement in G. Then G has only one subgroup of order 2, so G must be a generalized quaternion group with the defining relations

$$a^{2^n} = 1, \quad b^2 = a^{2^{n-1}}, \quad a^b = a^{-1}.$$

Now α defined by

$$a^\alpha = a, \quad b^\alpha = ab$$

is an outer automorphism of G, and α has order 2^n. □

We will require a few results from cohomology theory:

Theorem 19.4 (Herbrandt).

a) *Let* $\mathfrak{m}$ *be a finite abelian group, written additively, and let* φ *and* ψ *be endomorphisms of* $\mathfrak{m}$ *such that* $\varphi\psi = \psi\varphi = 0$. *Then*

$$[\ker\varphi : \operatorname{im}\psi] = [\ker\psi : \operatorname{im}\varphi].$$

b) *Let* G *be a finite cyclic group, and let* $\mathfrak{m}$ *be a* G*-module with* $|\mathfrak{m}| < \infty$. *Then*

$$|\mathbf{H}^1(G, \mathfrak{m})| = |\mathbf{H}^2(G, \mathfrak{m})|.$$

Proof. a) Since $\varphi\psi = \psi\varphi = 0$, we have $\operatorname{im}\varphi \leq \ker\psi$ and $\operatorname{im}\psi \leq \ker\varphi$. Therefore,

$$\begin{aligned} |\mathfrak{m}| &= [\mathfrak{m} : \ker\varphi][\ker\varphi : \operatorname{im}\psi]|\operatorname{im}\psi| \\ &= |\operatorname{im}\varphi|[\ker\varphi : \operatorname{im}\psi][\mathfrak{m} : \ker\psi]. \end{aligned}$$

The claim now follows immediately.

b) Let $G = \langle g \rangle$ with $|G| = n$. We define the element N of the group ring to be

$$N = \sum_{i=0}^{n-1} g^i.$$

By Theorem I.16.10 (pg. 116), we have

$$\mathbf{H}^1(G, \mathfrak{m}) \cong \{m \in \mathfrak{m} \mid mN = 0\}/\{m(1-g) \mid m \in \mathfrak{m}\}$$

and

$$\mathbf{H}^2(G, \mathfrak{m}) \cong \{m \in \mathfrak{m} \mid mg = m\}/\{mN \mid m \in \mathfrak{m}\}.$$

We define endomorphisms φ and ψ of $\mathfrak{m}$ by $m\varphi = mN$ and $m\psi = m(1-g)$. Then $\varphi\psi = \psi\varphi = 0$ and

$$\mathbf{H}^1(G,\mathfrak{m}) \cong \ker\varphi/\operatorname{im}\psi, \quad \mathbf{H}^2(G,\mathfrak{m}) \cong \ker\psi/\operatorname{im}\varphi.$$

The claim now follows from a). □

Theorem 19.5 (Gaschütz [163]). *Let G be a p-group, and let A be an abelian p-group, written multiplicatively. Suppose A is also a G-module. If $\mathbf{H}^1(G,A) = 1$, then $\mathbf{H}^1(U,A) = 1$ and $\mathbf{H}^2(U,A) = 1$ for all subgroups U of G.*

(In fact, $\mathbf{H}^i(U,A) = 1$ for all $i \geq 1$, but we will not need this.)

Proof. The claim is very easy to prove using the Hochschild–Serre exact sequence, which relates the cohomology groups of G to the cohomology groups of G/N and N (see Hochschild, Serre [246] or MacLane [327], pg. 354). The following elementary proof is due to W. Gaschütz. An elementary cohomological proof can be found in Hoechsmann [247].

We prove the claim by induction on $|G|$. Let N be a maximal subgroup of G. We define

$$A_N = \{a \in A \mid a^n = a \text{ for all } n \in N\}.$$

Then $(A_N)^g = A_N$ for all $g \in G$. The G-module A_N may be given the structure of a G/N-module via the well-defined action $a^{Ng} = a^g$. We show:

(1) $\mathbf{H}^1(G/N, A_N) = 1$:

Let $f \in \mathbf{Z}^1(G/N, A_N)$. Then we define the map $\widetilde{f} : G \to A$ by $\widetilde{f}(g) = f(gN)$. (This map $\widetilde{f}$ is the so-called inflation of f.) We have

$$\begin{aligned}(\widetilde{f}\delta)(g_1,g_2) &= \widetilde{f}(g_1)^{g_2}\widetilde{f}(g_2)\widetilde{f}(g_1g_2)^{-1} = f(g_1N)^{g_2N}f(g_2N)f(g_1g_2N)^{-1}\\ &= (f\delta)(g_1N, g_2N) = 1.\end{aligned}$$

So $\widetilde{f} \in \mathbf{Z}^1(G,A)$. Since $\mathbf{H}^1(G,A) = 1$, it follows that $\widetilde{f} \in \mathbf{B}^1(G,A)$. Therefore, we have $f(gN) = \widetilde{f}(g) = a^g a^{-1}$ for some $a \in A$. In particular, for $n \in N$ we have

$$1 = \widetilde{f}(1) = f(N) = f(nN) = a^n a^{-1}.$$

So $a \in A_N$, and consequently $f \in \mathbf{B}^1(G/N, A_N)$. Therefore, $\mathbf{H}^1(G/N, A_N) = 1$.

(2) $\mathbf{H}^2(G/N, A_N) = 1$:

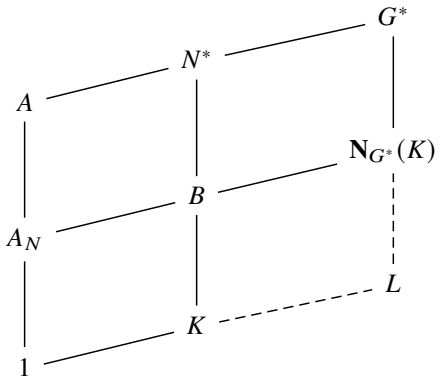
Since G/N is cyclic, it follows from (1) and Theorem 19.4 b) that

$$|\mathbf{H}^2(G/N, A_N)| = |\mathbf{H}^1(G/N, A_N)| = 1.$$

(3) Let G^* be any extension of A by G, and let N^* be the extension of A by N contained in G^*. Then $G^*/N^* \cong G/N$. Suppose K is a complement of A in N^*, and suppose $G^* = A\mathbf{N}_{G^*}(K)$. We claim that K lies in a complement L of A in G^*:

We define $D = A \cap \mathbf{N}_{G^*}(K)$. Since $A_N \leq \mathbf{Z}(N^*)$, clearly $A_N \leq D$. On the other hand, since $D \leq \mathbf{N}_{G^*}(K)$ we have $[D, N^*] = [D, AK] = [D,K] \leq A \cap K = 1$. So we have $A \cap \mathbf{N}_{G^*}(K) = A_N$.

Let $B = \mathbf{N}_{G^*}(K) \cap N^*$. By the Dedekind identity, $B = B \cap AK = (B \cap A)K = A_N K$.



It follows that $B/K \cong A_N$ as $\mathbf{N}_{G^*}(K)$-modules. Since $B \leq N^*$, B/K and A_N are isomorphic $\mathbf{N}_{G^*}(K)/B$-modules. By assumption, we have

$$G^* = A\mathbf{N}_{G^*}(K) = N^*\mathbf{N}_{G^*}(K).$$

By virtue of the isomorphisms

$$G/N \cong G^*/N^* \cong \mathbf{N}_{G^*}(K)/\mathbf{N}_{N^*}(K) = \mathbf{N}_{G^*}(K)/B,$$

it follows with the help of (2) that

$$\mathbf{H}^2(\mathbf{N}_{G^*}(K)/B, B/K) \cong \mathbf{H}^2(G/N, A_N) = 1.$$

Therefore, B/K has a complement L/K in $\mathbf{N}_{G^*}(K)/K$ by Theorem I.17.2 c) (pg. 128). Then we have

$$L \cap A \leq \mathbf{N}_{G^*}(K) \cap A = A_N \leq B,$$

so

$$L \cap A \leq L \cap B \cap A = K \cap A = 1.$$

Furthermore, we have

$$\begin{aligned} |L/K| &= |\mathbf{N}_{G^*}(K)/B| = |\mathbf{N}_{G^*}(K)/\mathbf{N}_{G^*}(K) \cap N^*| \\ &= |\mathbf{N}_{G^*}(K)N^*/N^*| = |G^*/N^*| = p, \end{aligned}$$

so then

$$|L| = |K|p = |N^*/A|\,|G/N| = |N|\,|G/N| = |G|.$$

Hence, L is a complement of A in G^*.

(4) $\mathbf{H}^1(U, A) = 1$ for all $U \leq G$:

We may clearly assume that $U = N$ is a maximal subgroup of G. By (1) and (2), we have

$$\mathbf{H}^1(G/N, A_N) = \mathbf{H}^2(G/N, A_N) = 1.$$

Now let G^* be the semidirect product of A with G, and let $\mathscr{L}_1, \ldots, \mathscr{L}_s$ be the A-conjugacy classes of complements of A in N^*. By Theorem I.17.3 (pg. 129), $s = |\mathbf{H}^1(N, A)|$ and s is a power of p. Since $\mathbf{H}^1(G, A) = 1$, all complements of A in G^* are A-conjugate. Therefore, the complements of A in N^* lying in exactly one of the classes $\mathscr{L}_i$, say $\mathscr{L}_1$, may be extended to complements of A in G^*.

We give G^*/N^* the structure of a permutation group on $\{\mathscr{L}_1, \ldots, \mathscr{L}_s\}$ by means of the well-defined action

$$\{K \mid K \in \mathscr{L}_i\}^{N^* g^*} = \{K^{g^*} \mid K \in \mathscr{L}_i\}.$$

We now show that if $\mathscr{L}_i$ is fixed by G^*/N^*, then $i = 1$:

Let $K \in \mathscr{L}_i$, and assume $\mathscr{L}_i$ is fixed by G^*/N^*. For every $g^* \in G^*$, there then exists an $a \in A$ such that $K^{g^*} = K^a$. So we have $G^* = \mathbf{N}_{G^*}(K)A$. Then by (3), K lies in a complement L of A in G^*, so $i = 1$. By partitioning $\{\mathscr{L}_1, \ldots, \mathscr{L}_s\}$ into G^*/N^*-orbits, we then have

$$s = |\mathbf{H}^1(N, A)| \equiv 1 \mod p.$$

Since s is a power of p, it follows that $s = 1$ and $\mathbf{H}^1(N, A) = 1$.

(5) $\mathbf{H}^2(U, A) = 1$ for all $U \leq G$:

If $U < G$, then $\mathbf{H}^1(U, A) = 1$ and it follows immediately by our induction hypothesis that $\mathbf{H}^2(U, A) = 1$. Thus, it remains only to show that $\mathbf{H}^2(G, A) = 1$. That is, we need to show that every extension $\widetilde{G}$ of A by G is a semidirect product. Let $|G/N| = p$, and let $\widetilde{N}$ be the preimage of N in $\widetilde{G}$. Then $\mathbf{H}^2(N, A) = 1$ by induction. So $\widetilde{N}$ is a semidirect product of A with a complement K. Since $\mathbf{H}^1(N, A) = 1$ (see (4)), all complements of A in $\widetilde{N}$ are A-conjugate. So $\widetilde{G} = A\mathbf{N}_{\widetilde{G}}(K)$. Then by (3), there exists a complement of A in $\widetilde{G}$. □

19.6 (Proof of Theorem 19.1).

Suppose G is a counterexample to Theorem 19.1. By Lemma 19.3 a), G is not abelian. If all abelian normal subgroups of G are cyclic, then either G is cyclic or G contains a cyclic normal subgroup of index 2 by Theorem 7.6. Both cases are impossible by Lemma 19.3. So let A be a maximal abelian normal subgroup of G that is not cyclic. By Lemma 19.2 b), $\mathbf{H}^1(G/A, A) = 1$. With the help of Theorem 19.5, it follows that $\mathbf{H}^2(G/A, A) = 1$, and then A has a complement in G by Theorem I.17.2 (pg. 128). Therefore, there exists a maximal subgroup N of G with $A \not\leq N$. We let $B = A \cap N$, so $G = AN$ and $|G/N| = |A/B| = p$.

It follows from Lemma 19.2 a) that $\mathbf{Z}(N) \not\leq \mathbf{Z}(G)$, so $\mathbf{Z}(N) \not\leq A$ since $G = AN$. Choose $h \in \mathbf{Z}(N)$ such that $|\langle h, A\rangle/A| = p$. Since $B = A \cap N$ and $h \in \mathbf{Z}(N)$, we have $B \leq \mathbf{C}_A(h)$. Since A is a maximal abelian normal subgroup, $A = \mathbf{C}_G(A)$ by

Theorem 7.3 a) (pg. 319). Thus, $h \notin A = \mathbf{C}_G(A)$ and consequently $B \leq \mathbf{C}_A(h) < A$. Together with $|A/B| = p$, it follows that $B = \mathbf{C}_A(h)$.

From $\mathbf{H}^1(G/A, A) = 1$ and Theorem 19.5, it follows upon letting $\overline{h} = hA$ that

$$\mathbf{H}^1(\langle \overline{h} \rangle, A) = 1.$$

Then $\mathbf{H}^2(\langle \overline{h} \rangle, A) = 1$ by Theorem 19.4 b). It then follows from Theorem I.16.10 (pg. 116) that

$$\{a \in A \mid a^h = a\} = \{a^{1+h+\cdots+h^{p-1}} \mid a \in A\}.$$

Hence,

$$B = \mathbf{C}_A(h) = \{a^{1+h+\cdots+h^{p-1}} \mid a \in A\}.$$

Let $A = \langle B, a_0 \rangle$. Since $B^p = \Phi(B)$, we have

$$B = \langle B^p, a_0^{1+h+\cdots+h^{p-1}} \rangle = \langle a_0^{1+h+\cdots+h^{p-1}} \rangle.$$

We set $b_0 = a_0^{1+h+\cdots+h^{p-1}}$. Since $A = \langle a_0, b_0 \rangle$ is not cyclic, we may choose a_0 to lie in $A - B$ with $a_0^p = 1$. Since $B = A \cap N \trianglelefteq G$, we have $a_0^h = a_0 b_1$ with $b_1^p = 1$. Therefore,

$$b_0 = a_0^{1+h+\cdots+h^{p-1}} = a_0^p b_1^{\frac{p(p-1)}{2}} = b_1^{\frac{p(p-1)}{2}}.$$

For $p \neq 2$, it follows that $b_0 = 1$, contradicting $|A| \geq p^2$. For $p = 2$, we have $b_1 = b_0$, so $|A| = 4$ and then $|G| = 8$ since $|G/\mathbf{C}_G(A)|$ divides $|\mathbf{Aut}(A)|$. So G is the dihedral group of order 8, which admits an outer automorphism of 2-power order by Lemma 19.3 b). □

Notes for Chapter III

§1: The systematic development of the commutator calculus, in particular the fundamental Lemma 1.10 b), is due to P. Hall [207].

§2: The fundamental Theorems 2.11 and 2.12 regarding nilpotent groups and the relationships between central series were proved by P. Hall [207], partly sharpening earlier results of Burnside [82, 83]. Theorem 2.13 is due to Blackburn [38].

§3: The theory of the Frattini subgroup described in this section mostly follows Gaschütz [155] until Theorem 3.7. The proof given here of Burnside's Basis Theorem 3.15 follows P. Hall [207].

§5: Theorem 5.1 is important in many applications and was proved by O. J. Schmidt [367]. Another proof was given by Iwasawa [288] which is easier to generalize (see Theorem IV.5.4). The elementary proof of Theorem 5.1 given here stems, to the best of my knowledge, from W. Gaschütz. One can find all the results in The-

orem 5.2 in Rédei [350], where the minimal non-nilpotent groups are completely classified. Theorem 5.3 is a special case of results of Itô [276]. The fundamental Theorem 5.8 is contained in Wielandt [465]. A somewhat more general formulation can be found in Wielandt [473].

§6: A more general formulation of Theorem 6.5 is contained in Kappe [298].

§7: The theorems in this section stem in large part from Burnside [82, 83] and P. Hall [207]. Theorem 7.6 is contained in Roquette [358]. Theorem 7.7 a) is a lemma from Feit, M. Hall, Thompson [128]. The proof of Theorem 7.12 follows the presentation in Zassenhaus [496]. Exercise 25 was communicated to me in a conversation with H. Wielandt.

§8: Hall's Enumeration Principle is contained in P. Hall [207].

§9: The derivation of the identity in Theorem 9.4 follows P. Hall [223], which itself uses ideas from Petresco [346]. The proof of the Zassenhaus identity in Theorem 9.7 follows the presentation in M. Hall [204].

§10: The theory of regular p-groups is the main content of P. Hall [207, 208], where nearly all the theorems of §10 can be found. The commutativity of regular 2-groups appears to be "folklore". Theorem 10.3 b) stems from Alperin [3]. The counterexample in Theorem 10.3 c) was communicated to me by H. Wielandt. One may consult Grün [200] and Weichsel [452] for results regarding the regularity of direct products. Example 10.15 was communicated to me in a conversation with Blackburn.

§12: Lemma 12.2 and Theorem 12.3 stem from unpublished notes of J. Thompson. The classification of all groups of order p^4 can be found in O. Hölder [251].

§13: Theorem 13.4 can be found in Zassenhaus [496], pg. 172. The fundamental Theorem 13.5 is contained in Hall, Higman [224]. Theorem 13.6 is in Thompson [436]. Theorem 13.8 is an unpublished result of P. Hall. One may also consult Gorenstein [176].

§14: The entire contents of §14 stem from Blackburn [36]. However, the presentation follows unpublished notes of Blackburn.

§15: The treatment given here partially follows Weir [454]. One may also consult Alperin [7]. Theorem 15.3 first appeared in Kaloujnine [296]. Exercise 36 is an observation of P. Hall.

§16: The results of this section are partially contained in Grün [193] and Weir [453]. Exercises 37–40 reproduce a result of I. Schur [372] and are partly related to conversations I had with E. C. Dade. Exercise 41 was communicated to me in a letter from P. Hall.

§17: The treatment of the binary p-adic groups follows unpublished notes of Blackburn and P. Hall. One may also consult P. Hall [222].

§18: The formulation of the main result of Golod and Shafarevich [173] follows unpublished notes of Roquette. Example 18.10 a) was given by Kostrikin in Oberwolfach.

§19: The main theorems in this section are due to Gaschütz [163, 165]. The elementary proof of Theorem 19.5 was communicated to me in a letter from Gaschütz.

Chapter IV
Transfer and p-nilpotent groups

One of the most important problems in group theory is proving that a given group is not simple, or even more ambitiously, showing that the group is solvable. The elementary considerations described in Chapter I, §8 and the subsequent exercises are not widely applicable. However, there are two powerful techniques available: In this chapter, we consider the method of monomial representations and transfer. An approach using character theory is discussed in Chapter V and Volume 2. The two approaches can be compared as follows:

1) The method of transfer is concerned with a homomorphism $\mathbf{V}$ from the group G to the commutator factor group H/H' of a subgroup H of G. Hence, the technique is limited to proving the existence of abelian (or somewhat more generally, solvable) factor groups of G. (This is the reason that the proof of Theorem 2.4 cannot be extended to a full proof of Frobenius' Theorem, which we will prove with group characters in Theorem V.7.6.)

The case in which H is a Sylow p-subgroup of G has been studied extensively. The transfer into a Sylow p-subgroup always yields the largest abelian p-factor group of G; this is the statement of Theorem 3.3. If the structure of the Sylow p-subgroup P of G is not too wild, then the largest abelian p-factor groups of G and $\mathbf{N}_G(P)$ are isomorphic. A sufficient condition for this is the commutativity of P (see Theorem 2.6, due to Burnside); but already the regularity of P (in the sense of Chapter III, §10) suffices, as Wielandt has shown (see Theorem 8.1). The general case requires, in addition to knowledge about $\mathbf{N}_G(P)$, information regarding the embedding of P in G, perhaps in the form of a statement about the intersection of the conjugates of P. The most useful formulation is the theorem of Grün (see Theorem 3.7), which has found numerous applications.

Thus, the method of transfer is always applicable when a prime p has been identified and it is suspected that G has a nontrivial p-factor group. We address situations of this type in §5. Calculating the transfer to a Sylow p-subgroup P only requires information about the embedding of P in G. Herein lies the strength and weakness of this technique. The many complex relationships between the Sylow subgroups for different primes are almost completely ignored. For example, the famous theorem of Burnside, Theorem V.7.3, that all groups of order $p^a q^b$ are

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B. Huppert, *Finite Groups I*, Grundlehren der mathematischen Wissenschaften 364,
https://doi.org/10.1007/978-3-031-87529-8_4

solvable for any primes p and q has not yet admitted a proof using transfer. Here the method of group characters is clearly superior. (A character-free proof was given by Thompson for $p \neq 2 \neq q$, but the proof is not easy.) The method of transfer already breaks down in the face of the following question (although admittedly all other known methods do as well): Does there exist a nonabelian simple group whose Sylow subgroups are all self-normalizing?[1] By Theorem 8.1, due to Wielandt, all the Sylow subgroups must be irregular in this situation, and the embedding of the Sylow subgroups must be somewhat complicated by Grün's theorem. But beyond this, it appears that transfer has nothing to say. The proof of Burnside's famous conjecture that all groups of odd order are solvable also requires character theory at crucial points, although transfer does play a minor role as well (see Feit, Thompson [130]).[2]

2) In its original form, developed by Burnside, the study of transfer begins with monomial representations of G; that is, homomorphisms $\mu : g \mapsto (a_{ij}(g))$ from G to the group of so-called monomial matrices, which are the matrices over $\mathbb{C}$ with only one nonzero entry in each row and column. The transfer is then (up to a sign) the map $\mathbf{V}$ defined by $\mathbf{V}(g) = \det(a_{ij}(g))$. The monomial representations have many similarities with the permutation representations we discussed in Chapter I, §6. In particular, the permutation representation of G on the cosets of a subgroup admits an easy generalization; see Theorem 1.4.

From the viewpoint of general representation theory, the monomial representations are particularly simple representations; namely, those induced from a one-dimensional representation of a subgroup. We will return to this in Chapter V, §18. This algorithm for constructing monomial representations, which relies on information about subgroups, greatly simplifies the treatment of these representations. However, it is not known for which groups this procedure gives rise to irreducible representations (but see Theorems V.18.4–V.18.10 for more in this direction).

1 Monomial representations and transfer

Definition 1.1. Let G be a (not necessarily finite) group, and let n be a natural number. Let S_n be the symmetric group of degree n, considered as a permutation group of degree n. We define $\mathbf{M}_n(G) = G \wr S_n$ and refer to the wreath product $\mathbf{M}_n(G)$ as the monomial group of degree n over G. By Definition I.15.1 (pg. 101), we have

$$\mathbf{M}_n(G) = \{(f, h) \mid h \in S_n,\ f : \{1, \ldots, n\} \longrightarrow G\},$$

[1] It is now known that if all nonidentity Sylow subgroups of a finite group are self-normalizing, then the group must have prime power order, as conjectured by Zassenhaus. See Theorem X.8.15 of B. Huppert and N. Blackburn, *Finite Groups III*, Grundlehren der Mathematischen Wissenschaften, 243, Springer, Berlin, 1982. *trans.*

[2] Glauberman has shown that if $p \geq 7$ and P is a Sylow p-subgroup of G with $\mathbf{N}_G(P) = P$, then G has a normal subgroup of index p. (Footnote in original; see Theorem X.8.13 of B. Huppert and N. Blackburn, *Finite Groups III*, Grundlehren der Mathematischen Wissenschaften, 243, Springer, Berlin, 1982. *trans.*)

and the multiplication is given by the rule

$$(f_1, h_1)(f_2, h_2) = (g, h_1 h_2)$$

with $g(i) = f_1(i) f_2(i^{h_1})$.

Lemma 1.2. *Let G be abelian. Then the map δ defined by*

$$\delta((f, h)) = \prod_{i=1}^{n} f(i)$$

is a homomorphism from $\mathbf{M}_n(G)$ to G. (If G is a subgroup of the multiplicative group of complex numbers and one writes the elements of $\mathbf{M}_n(G)$ as permutation matrices with coefficients in G, then δ is (up to a sign) the determinant.)

Proof. We have

$$\begin{aligned}\delta((f_1, h_1)(f_2, h_2)) &= \delta((g, h_1 h_2)) = \prod_{i=1}^{n} g(i) = \prod_{i=1}^{n} f_1(i) f_2(i^{h_1}) \\ &= \prod_{i=1}^{n} f_1(i) \prod_{i=1}^{n} f_2(i) = \delta((f_1, h_1))\delta((f_2, h_2)).\end{aligned}$$

□

Lemma 1.3. *If α is a homomorphism from G to $\overline{G}$, then the map $\overline{\alpha}$ defined by*

$$(f, h)^{\overline{\alpha}} = (g, h) \quad \text{with} \quad g(i) = f(i)^{\alpha}$$

is a homomorphism from $\mathbf{M}_n(G)$ to $\mathbf{M}_n(\overline{G})$.

Proof. This follows immediately from the multiplication rule in $\mathbf{M}_n(G)$. Note that this result was given already in Lemma I.15.7 (pg. 104). □

Main Theorem 1.4. *Let G be a (not necessarily finite) group, and let H be a subgroup of G with finite index $[G : H] = n$. Let $G = \bigcup_{i=1}^{n} Hr_i$ be the decomposition of G into cosets of H. For every $g \in G$, we define a permutation $\mathbf{P}(g)$ of $\{1, \ldots, n\}$ and elements $h_i(g) \in H$ $(i = 1, \ldots, n)$ by*

$$r_i\, g = h_i(g)\, r_{i^{\mathbf{P}(g)}}.$$

Then:

a) *The map $\mathbf{M}$ from G to $\mathbf{M}_n(H)$ defined by $\mathbf{M}(g) = (f, \mathbf{P}(g))$ with $f(i) = h_i(g)$ is a monomorphism. We say $\mathbf{M}$ is a monomial representation of G over H.*

b) *Let $H/\overline{H}$ be an abelian factor group of H. The map $\mathbf{V}$ from G to $H/\overline{H}$ defined by $\mathbf{V}(g) = \prod_{i=1}^{n} h_i(g)\overline{H}$ is a homomorphism. We refer to $\mathbf{V}$ as the transfer map from G to $H/\overline{H}$. If $\overline{H} = H'$, then we refer to $\mathbf{V}$ (somewhat imprecisely) as the transfer map from G to H and occasionally write $\mathbf{V}_{G \to H}$.*

(*The map* $\mathbf{P}$ *defined by* $Hr_ig = Hr_{i^{\mathbf{P}(g)}}$ *is the permutation representation of* G *considered in Theorem I.6.2, pg. 29; so monomial representations are a refinement of permutation representations.*)

Proof. a) It follows from

$$\begin{aligned} h_i(gg')r_{i^{\mathbf{P}(gg')}} &= r_i(gg') = (r_ig)g' = h_i(g)r_{i^{\mathbf{P}(g)}}g' \\ &= h_i(g)h_{i^{\mathbf{P}(g)}}(g')r_{i^{\mathbf{P}(g)\mathbf{P}(g')}} \end{aligned}$$

that $\mathbf{P}(gg') = \mathbf{P}(g)\mathbf{P}(g')$ and $h_i(gg') = h_i(g)h_{i^{\mathbf{P}(g)}}(g')$. So we have that $\mathbf{M}(gg') = (f, \mathbf{P}(g)\mathbf{P}(g'))$, where

$$f(i) = h_i(gg') = h_i(g)h_{i^{\mathbf{P}(g)}}(g').$$

Therefore, $\mathbf{M}(gg') = \mathbf{M}(g)\mathbf{M}(g')$.

If $\mathbf{M}(g) = 1$, then $r_ig = r_i$ for all i and so $g = 1$. Hence, $\mathbf{M}$ is a monomorphism from G to $\mathbf{M}_n(H)$.

b) Let α be the epimorphism from H to $H/\overline{H}$ defined by $h^\alpha = h\overline{H}$. Using the notation in Lemma 1.2 and Lemma 1.3, for every $g \in G$ we define

$$\mathbf{V}(g) = \delta(\mathbf{M}(g)^{\overline{\alpha}}).$$

By Lemma 1.2, Lemma 1.3 and Theorem 1.4 a), $\mathbf{V}$ is a homomorphism from G to $H/\overline{H}$ with

$$\mathbf{V}(g) = \delta((f, \mathbf{P}(g))^{\overline{\alpha}}) = \prod_{i=1}^{n} f(i)^\alpha = \prod_{i=1}^{n} h_i(g)\overline{H}. \qquad \square$$

Lemma 1.5. *The transfer from* G *to* $H/\overline{H}$ *is independent of the choice of representatives for the cosets of* H.[3]

Proof. Let $G = \bigcup_{i=1}^{n} Hr_i = \bigcup_{i=1}^{n} Hs_i$ with $Hr_i = Hs_i$. Then $s_i = t_ir_i$ for some $t_i \in H$. The permutation $\sigma = \mathbf{P}(g)$ of the cosets of H is independent of the choice of representatives since

$$Hs_{i^\sigma} = Hs_ig = Hr_ig = Hr_{i^\sigma}.$$

Let

$$r_ig = h_i(g)r_{i^\sigma} \quad \text{and} \quad s_ig = k_i(g)s_{i^\sigma}.$$

With $\sigma = \mathbf{P}(g)$, we have

$$k_i(g)t_{i^\sigma}r_{i^\sigma} = k_i(g)s_{i^\sigma} = s_ig = t_ir_ig = t_ih_i(g)r_{i^\sigma}.$$

[3] Correction 24 from errata.

Hence, $k_i(g) = t_i h_i(g)(t_{i^\sigma})^{-1}$. Since $t_i \in H$, it follows that

$$\prod_{i=1}^{n} k_i(g)\overline{H} = \prod_{i=1}^{n} t_i h_i(g)(t_{i^\sigma})^{-1}\overline{H} = \prod_{i=1}^{n} h_i(g)\overline{H}. \qquad \square$$

Theorem 1.6. *Let $K \le H \le G$, and assume that the index $[G : K]$ is finite. Letting $\mathbf{V}_{G\to H}(g) = hH'$, we have $\mathbf{V}_{G\to K}(g) = \mathbf{V}_{H\to K}(h)$.*

Proof. Let $G = \bigcup_{i=1}^{n} Hr_i$ be the decomposition of G into cosets of H, and let $H = \bigcup_{j=1}^{m} Ks_j$ be the decomposition of H into cosets of K. Then $G = \bigcup_{i,j} Ks_j r_i$ is the decomposition of G into cosets of K. Let

$$r_i g = h_i(g) r_{i'} \quad \text{and} \quad s_j h = k_j(h) s_{j'}.$$

Then

$$\mathbf{V}_{G\to H}(g) = \prod_{i=1}^{n} h_i(g)H' = hH'.$$

We have

$$s_j r_i g = s_j h_i(g) r_{i'} = k_j(h_i(g)) s_{j'} r_{i'},$$

and therefore

$$\begin{aligned}\mathbf{V}_{G\to K}(g) &= \prod_{i=1}^{n}\prod_{j=1}^{m} k_j(h_i(g))K' = \prod_{i=1}^{n} \mathbf{V}_{H\to K}(h_i(g)) \\ &= \mathbf{V}_{H\to K}\Big(\prod_{i=1}^{n} h_i(g)\Big).\end{aligned}$$

The element $\mathbf{V}_{H\to K}(h)$ is independent of the choice of h in the coset hH' since the homomorphism $\mathbf{V}_{H\to K}$ from H to the abelian group K/K' certainly takes H' to 1. Then it follows from $\mathbf{V}_{G\to H}(g) = hH' = \prod_{i=1}^{n} h_i(g)H'$ that $\mathbf{V}_{G\to K}(g) = \mathbf{V}_{H\to K}(h)$.$\square$

The following lemma describes a choice of coset representatives that is often convenient for evaluating the transfer map.

Lemma 1.7. *Assume $[G : H]$ is finite, and let $g \in G$. Let the element g act on the cosets of H in G as a permutation with cycles*

$$(Hs_j, Hs_j g, \ldots, Hs_j g^{f_j - 1})$$

for $j = 1, \ldots, t$.

(*We may assume that* $s_1 = 1$.) *Then*

$$\mathbf{V}_{G\to H}(g) = \prod_{j=1}^{t} s_j g^{f_j} s_j^{-1} H'.$$

Proof. The $r_{ij} = s_j g^i$ with $0 \le i < f_j$ and $1 \le j \le t$ are a transversal for the cosets of H in G. For $i < f_j - 1$, we have

$$r_{ij} g = s_j g^{i+1} = r_{i+1,j},$$

so that the corresponding $h_{ij}(g)$ equals 1. Furthermore, we have

$$r_{f_j-1,j}\, g = s_j g^{f_j} = h_{f_j-1,j}(g)\, s_j \in H s_j.$$

Thus,

$$h_{f_j-1,j}(g) = s_j g^{f_j} s_j^{-1},$$

and it follows that

$$\mathbf{V}_{G\to H}(g) = \prod_{j=1}^{t} s_j g^{f_j} s_j^{-1} H'. \qquad \square$$

Exercises

1) (Wielandt) If G is a finite group, define the characteristic subgroup $\Psi(G)$ of G to be the subgroup generated by all minimal subgroups of G. (So $\Psi(G)$ is a dual construction to the Frattini subgroup $\Phi(G)$.) Show that if H is an arbitrary finite group, then there exists a finite group G such that $G/\Psi(G) \cong H$. (That is, the nilpotency of $\Phi(G)$ does not correspond to any statement about $G/\Psi(G)$; see Gaschütz [164].)

   (Hint: Let $U_1, \ldots, U_k$ be the minimal subgroups of H. Let $n_i = [H : U_i]$ and $p_i = |U_i|$. Construct $H_i \le \mathbf{M}_{n_i}(U_i)$ with $H_i \cong H$ following Theorem 1.4. Let V_i be the cyclic group of order p_i^2, let α_i be a homomorphism from V_i onto U_i and let $\overline{\alpha}_i$ be the homomorphism from $\mathbf{M}_{n_i}(V_i)$ onto $\mathbf{M}_{n_i}(U_i)$ given in Lemma 1.3. Let G_i be the full $\overline{\alpha}_i$-preimage of H_i in $\mathbf{M}_{n_i}(V_i)$, and let $N_i = G_i \cap \ker \overline{\alpha}_i$. Following Theorem I.9.11 (pg. 52), construct $G = G_1 \curlywedge \cdots \curlywedge G_k$ with identified factor group H. Now show that $\Psi(G) = N_1 \times \cdots \times N_k$ (under the natural identification) and $G/\Psi(G) \cong H$.)

2 First applications of transfer

Lemma 2.1. *Let $H < G$ with $[G : H] = n$, and let $x \in \mathbf{Z}(G)$. Then we have $\mathbf{V}_{G\to H}(x) = x^n H'$.*

Proof. Since $x \in \mathbf{Z}(G)$, we have by Lemma 1.7 that

$$\mathbf{V}(x) = \prod_{j=1}^{t} s_j x^{f_j} s_j^{-1} H' = \prod_{j=1}^{t} x^{f_j} H' = x^n H'. \qquad \square$$

Theorem 2.2. *Let G be a finite group and $H < G$. Assume $[H : H']$ and $[G : H]$ are coprime. Then*

$$H \cap G' \cap \mathbf{Z}(G) \leq H'.$$

In particular, if H is an abelian Hall subgroup of G, then

$$H \cap G' \cap \mathbf{Z}(G) = 1.$$

Proof. Let $g \in H \cap G' \cap \mathbf{Z}(G)$. By Lemma 2.1, we have

$$\mathbf{V}_{G\to H}(g) = g^{[G:H]} H'.$$

Since $\mathbf{V}_{G\to H}$ is a homomorphism from G to the abelian group H/H', clearly $\mathbf{V}_{G\to H}(g) = H'$ for $g \in G'$. Thus, for $g \in G' \cap \mathbf{Z}(G)$ we have

$$\mathbf{V}_{G\to H}(g) = g^{[G:H]} H' = H'.$$

Since $([G : H], [H : H']) = 1$ and $g \in H$, it follows that $g \in H'$, as claimed. $\square$

Theorem 2.2 will be useful in Chapter VI, §15 when we discuss the theory of solvable groups with abelian Sylow subgroups. Next, we prove a theorem concerning not necessarily finite groups which will be needed in our discussion of the Schur multiplier in Chapter V, §23.

Theorem 2.3 (I. Schur). *Let G be a (not necessarily finite) group, and suppose that $|G/\mathbf{Z}(G)| < \infty$. Then $|G'| < \infty$.*

Proof. Since

$$G'/(G' \cap \mathbf{Z}(G)) \cong G'\mathbf{Z}(G)/\mathbf{Z}(G) \leq G/\mathbf{Z}(G),$$

we have $|G'/(G' \cap \mathbf{Z}(G))| < \infty$. Let $G = \bigcup_{i=1}^{n} \mathbf{Z}(G) r_i$. For all $z_i, z_j \in \mathbf{Z}(G)$, we have $[z_i r_i, z_j r_j] = [r_i, r_j]$. It follows that

$$G' = \langle [r_i, r_j] \mid i, j = 1, \ldots, n \rangle,$$

and so G' is finitely generated. Since $|G'/(G' \cap \mathbf{Z}(G))| < \infty$, the group $G' \cap \mathbf{Z}(G)$ is also finitely generated by Theorem I.19.10 (pg. 151). To prove the claim, it now suffices to show that the abelian group $G' \cap \mathbf{Z}(G)$ is a torsion group, since then $|G' \cap \mathbf{Z}(G)| < \infty$. Let $\mathbf{V}$ be the transfer map from G to $\mathbf{Z}(G)$. Using Lemma 2.1 with $\mathbf{Z}(G)$ in place of H, it follows for $g \in G' \cap \mathbf{Z}(G)$ that

$$1 = \mathbf{V}_{G \to \mathbf{Z}(G)}(g) = g^n.$$

Thus, $G' \cap \mathbf{Z}(G)$ is a torsion group. □

In the remainder of this section, all groups are assumed to be finite. The next theorem is a special case of a fundamental theorem of Frobenius (see Theorem V.7.6, pg. 526). Since there is currently no known proof of Frobenius' theorem without the use of character theory, the following elementary proof of a special, but useful, case is of interest.

Theorem 2.4. *Let H be a solvable subgroup of G such that $H \cap H^t = 1$ for all $t \in G - H$. (In particular, if $H \neq 1$ then $\mathbf{N}_G(H) = H$.) Then the set*

$$F = G - \bigcup_{g \in G} (H - 1)^g$$

is a normal subgroup of G with $G = FH$ and $F \cap H = 1$.

Proof. (Shaw [381]) If $H = 1$, then $F = G$ and the claim is trivial. So we may assume that $H \neq 1$.

It follows immediately from the definition of F that all inner automorphisms of G map F onto itself. The difficulty lies in proving that F is actually a subgroup of G. We assume for the moment that this has already been proved. From our hypothesis, it follows that $H^t \cap H^{t'} = 1$ if $t't^{-1} \in G - H$. Therefore,

$$|F| = |G| - [G : H](|H| - 1) = [G : H].$$

Since $F \cap H = 1$ and $F \leq G$, we have $|FH| = |F||H| = |G|$ and therefore $G = FH$.

To prove that F is a subgroup of G, we construct the transfer map $\mathbf{V} = \mathbf{V}_{G \to H}$ from G to H and show:

a) We have $\mathbf{V}(h) = hH'$ for all $h \in H$; in particular, $H \cap K = H'$, where $K = \ker \mathbf{V}$:

By Lemma 1.7,

$$\mathbf{V}(h) = \prod_i s_i h^{f_i} s_i^{-1} H',$$

where $s_1 = 1$ and $s_i h^{f_i} s_i^{-1} \in H$. Since $h \in H$, we have $Hs_1 h = Hh = H$, so $f_1 = 1$. On the other hand, our hypothesis yields for $i > 1$ that

$$s_i h^{f_i} s_i^{-1} \in H \cap s_i H s_i^{-1} = 1.$$

So $\mathbf{V}(h) = hH'$.

b) We have $G = KH$:

Let $g \in G$. There exists an $h \in H$ such that $\mathbf{V}(g) = hH'$. By a), $\mathbf{V}(h) = hH'$, so we have $\mathbf{V}(gh^{-1}) = H'$ and consequently $gh^{-1} \in \ker \mathbf{V} = K$.

c) Since $K \cap H = H'$ and $H \neq 1$ is solvable, we have $K < G$. We claim that our assumptions are again satisfied by the group K and the subgroup H': If $k \in K - H'$, then in fact $k \in G - H$ since $K \cap H = H'$. Thus,

$$H' \cap H'^k \leq H \cap H^k = 1.$$

By induction, we may assume that the theorem has been proved for K. So

$$N = K - \bigcup_{k \in K} (H' - 1)^k$$

is a normal subgroup of K satisfying $K = NH'$, $N \cap H' = 1$ and $|N| = [K : H']$. For all $g = hk \in HK = G$, we have

$$N \cap H^g = N \cap H^k = N \cap (K \cap H^k) = N \cap H'^k = 1.$$

So we have

$$N \subseteq G - \bigcup_{g \in G} (H - 1)^g = F.$$

It follows from

$$|N| = [K : H'] = [K : K \cap H] = [KH : H] = [G : H] = |F|$$

that $N = F$. So F is a subgroup of G. □

Lemma 2.5. *Suppose K and L are subsets of the Sylow p-subgroup P of G with $K^x = K$ and $L^x = L$ for all $x \in P$. If there exists an element $g \in G$ such that $K^g = L$, then there also exists a $w \in \mathbf{N}_G(P)$ such that $K^w = L$.*

Proof. We have $P \leq \mathbf{N}_G(L)$ and $P^g \leq \mathbf{N}_G(K^g) = \mathbf{N}_G(L)$. So P and P^g are Sylow p-subgroups of $\mathbf{N}_G(L)$ and thus conjugate in $\mathbf{N}_G(L)$. That is, there exists an element $y \in \mathbf{N}_G(L)$ such that $P = P^{gy}$. Now, $gy \in \mathbf{N}_G(P)$ and $K^{gy} = L^y = L$, so the element $w = gy$ satisfies the claim. □

The following theorem of Burnside is a special case of more general theorems that we will see later (see Theorems 3.7, 5.8, 6.2 and 8.1).

Main Theorem 2.6 (Burnside). *Let P be a Sylow p-subgroup of G, and suppose that $P \leq \mathbf{Z}(\mathbf{N}_G(P))$. (That is, $\mathbf{C}_G(P) = \mathbf{N}_G(P)$, and so P is abelian.) Then there exists a normal subgroup N of G such that $G/N \cong P$.*

Proof. Let N be the kernel of the transfer map $\mathbf{V} = \mathbf{V}_{G \to P}$ from G to the abelian group P. By the Homomorphism Theorem I.3.8 (pg. 14), we have $G/N \cong \mathbf{V}(G)$. If we can show that $\mathbf{V}(P) = P$, then $G/N \cong P$ and we are done.

For all $x \in P$, we have by Lemma 1.7 that

$$\mathbf{V}(x) = \prod_j s_j x^{f_j} s_j^{-1}.$$

In this expression, x^{f_j} and $s_j x^{f_j} s_j^{-1}$ lie in P. We set $K = \{x^{f_j}\}$ and $L = \{s_j x^{f_j} s_j^{-1}\}$. Since P is abelian, the sets K and L are invariant under conjugation by elements of P, and they are conjugate by s_j^{-1}. By Lemma 2.5, there exists a $w \in \mathbf{N}_G(P)$ such that $K^w = L$. In fact, since $K \subseteq P \leq \mathbf{Z}(\mathbf{N}_G(P))$, we have $K = K^w = L$ and so $x^{f_j} = s_j x^{f_j} s_j^{-1}$. Letting $n = [G : P]$, this yields

$$\mathbf{V}(x) = \prod_j x^{f_j} = x^n.$$

Consequently, $\mathbf{V}(P)$ contains all x^n. Since $(n, p) = 1$, this means that $\mathbf{V}(P)$ contains all $x \in P$. □

Theorem 2.7. *Let p be the smallest prime divisor of $|G|$, and suppose that P is an abelian Sylow p-subgroup of G. Assume that all invariants of P are pairwise distinct. (See Theorem I.13.12, pg. 85 for the definition of the invariants of a finitely generated abelian group.) Then G has a normal subgroup N such that $G/N \cong P$.*

Proof. If we show that the order of the automorphism group $\mathbf{Aut}(P)$ of P is only divisible by primes less than or equal to p, then it follows immediately from $|\mathbf{N}_G(P)/\mathbf{C}_G(P)|$ dividing $|\mathbf{Aut}(P)|$ and $P \leq \mathbf{C}_G(P)$ that $\mathbf{N}_G(P) = \mathbf{C}_G(P)$. Then we obtain the claim from Theorem 2.6.

Let $P = Z_1 \times \cdots \times Z_k$ for some cyclic groups $Z_i = \langle z_i \rangle$ with $|Z_i| = p^{a_i}$ and $a_1 < a_2 < \cdots < a_k$. We define $P_i = \mho_1(P)\Omega_i(P)$. (For the notation, see Exercise I.51, pg. 91.) Then,

$$\mho_1(P) = P_0 \leq P_1 \leq \cdots \leq P_{a_k} = P.$$

If $a_j \leq i < a_{j+1}$, then we have

$$P_i = \langle z_1 \rangle \times \cdots \times \langle z_j \rangle \times \langle z_{j+1}^p \rangle \times \cdots \times \langle z_k^p \rangle.$$

Therefore, $|P_{i+1}/P_i| \leq p$ for all i.

Suppose α is an automorphism of P of order $|\alpha|$ not divisible by p, and let $\overline{\alpha}$ be the automorphism of $\overline{P} = P/\mho_1(P)$ induced by α. The subgroups $\overline{P_i} = P_i/\mho_1(P)$ of $\overline{P}$ are fixed by $\overline{\alpha}$. By Maschke's Theorem I.17.7 (pg. 132), $\overline{P}$ is a direct product of $\overline{\alpha}$-invariant subgroups of order p. Thus, $|\overline{\alpha}|$ is a divisor of $p - 1$. Using that $\mho_1(P) = \Phi(P)$ and applying Theorem III.3.18 (pg. 290), we conclude that $|\mathbf{Aut}(P)|$ is only divisible by primes q with $q = p$ or $q \mid p - 1$, that is, $q \leq p$. □

We formulate an important special case of Theorem 2.7:

Theorem 2.8. *Let p be the smallest prime divisor of $|G|$. Suppose G has a cyclic Sylow p-subgroup P. Then there exists a normal subgroup N of G such that $G/N \cong P$.*

(*In particular, the Sylow* 2*-subgroups of a nonabelian simple group can never be cyclic.*)

Theorem 2.8 and an easy induction on the group order yield the following generalization of Theorem I.8.13 (pg. 43). For another proof, see Exercise I.40 (pg. 46).

Theorem 2.9 (O. Hölder). *If all Sylow subgroups of G are cyclic, then G is solvable.*

Theorem 2.10 (Zassenhaus). *If G'/G'' and G''/G''' are cyclic, then $G'' = G'''$.*

Proof. By passing to the quotient G/G''', we may assume that $G''' = 1$; then we must show that $G'' = 1$.

Since G'' is cyclic, $\mathbf{Aut}(G'')$ is abelian. Then $G/\mathbf{C}_G(G'')$ is also abelian, as it is isomorphic to a subgroup of $\mathbf{Aut}(G'')$. Therefore, $G' \leq \mathbf{C}_G(G'')$ and $G'' \leq \mathbf{Z}(G')$. As G'/G'' is cyclic, we have $G' = \langle G'', a\rangle$ for some $a \in G'$. Hence, G' is abelian and $G'' = 1$. □

The structure of groups whose Sylow subgroups are all cyclic can now be completely classified:

Theorem 2.11 (Zassenhaus). *Suppose that all the Sylow subgroups of G are cyclic. Then G is generated by elements a and b with the following defining relations:*

$$a^m = b^n = 1, \quad b^{-1}ab = a^r,$$

where $r^n \equiv 1 \bmod m$, $(m, n(r-1)) = 1$ and $|G| = mn$. The commutator subgroup $G' = \langle a\rangle$ and the factor group G/G' are cyclic. Moreover, $|G'| = m$ and $|G/G'| = n$. Conversely, if a group satisfies the given relations, then all of its Sylow subgroups are cyclic.

Proof. a) Let G be a group whose Sylow subgroups are all cyclic. By Theorem 2.9, the group G is solvable. Since all Sylow subgroups of the abelian groups G'/G'' and G''/G''' are cyclic, the groups G'/G'' and G''/G''' are cyclic. It follows from Theorem 2.10 that $G'' = G'''$, which means $G'' = 1$ since G is solvable.

Let $G' = \langle a\rangle$ and $G/G' = \langle bG'\rangle$, let $|a| = m$ and $|bG'| = n$. Then $a^m = 1$; moreover, $b^n = a^s \in \langle a\rangle$ and $b^{-1}ab = a^r \in \langle a\rangle$ for some natural numbers r and s.

We claim that the commutator $[a, b] = a^{r-1}$ generates all of $G' = \langle a\rangle$: Since $N = \langle a^{r-1}\rangle$ is a normal subgroup of G containing $[a, b]$, the quotient G/N is abelian and thus $G' \leq N$. The reverse conclusion is obvious. Hence, $r - 1$ is coprime to m. It follows from

$$a^{r^n} = b^{-n}ab^n = a^{-s}\, a\, a^s = a$$

that $r^n \equiv 1 \bmod m$. Furthermore, we have

$$a^s = b^n = b^{-1}b^n b = b^{-1}a^s b = (b^{-1}ab)^s = a^{rs},$$

and therefore $s(r-1) \equiv 0 \bmod m$. Since $(r-1, m) = 1$, this forces $s \equiv 0 \bmod m$. So $b^n = a^s = 1$.

It remains to show that m and n are coprime. Assume for contradiction that p is a common divisor of $m = pm'$ and $n = pn'$. Then since $\langle a^{m'} \rangle \trianglelefteq G$, the set $\langle a^{m'} \rangle \langle b^{n'} \rangle$ is certainly a subgroup of G and has type (p, p). But this is impossible since every p-subgroup of G lies in a Sylow p-subgroup of G by Theorem I.7.5 (pg. 35) and the latter are cyclic by our hypothesis. Thus, m and n are coprime.

b) Now assume that $G = \langle a, b \rangle$ is defined by the relations in the statement of the theorem. Since $(m, n) = 1$, a Sylow p-subgroup of G lies either in $\langle a \rangle$ or $\langle b \rangle$ for each prime p. Thus, all Sylow subgroups of G are cyclic. (The condition that $(m, r - 1) = 1$ is only necessary if one wants to force $G' = \langle a \rangle$.) □

Remark 2.12. Let G be a group such that $|G/G'| < \infty$ and G'/G'' is finitely generated. Then $\mathbf{V}_{G \to G'}(G) = G''/G''$.[4] This so-called principal ideal theorem plays an important role in class field theory; it follows from this theorem that all ideals in the ring of integers of an algebraic number field become principal ideals in the ring of integers of the Hilbert class field. The principal ideal theorem was first proved by Furtwängler. For easier proofs, see Zassenhaus [496], pg. 176 and also Witt [483]. This theorem has not yet found an application in group theory.

Exercises

2) Show that all finite division rings are commutative. (Hint, following Brandis [45]: Let K be a minimal counterexample with center Z, let $|\mathsf{Z}| = q$ and $|\mathsf{K}| = q^n$. Show that:
   a) $\mathsf{K}^\times$ contains no subgroup of type (p, p) and no quaternion subgroup.
   b) All the Sylow subgroups of $\mathsf{K}^\times = G$ are cyclic.
   c) Letting $G' = \langle a \rangle$ and $G = \langle a \rangle \langle b \rangle$ with $\langle a \rangle \cap \langle b \rangle = 1$, we have that $\mathbf{Z}(G) = \mathsf{Z}^\times = \langle b^s \rangle$ for some integer s.
   d) There exist natural numbers m and t such that

$$|\langle a \rangle| = \frac{q^m - 1}{q - 1} \quad \text{and} \quad |\langle b \rangle| = q^t - 1.$$

   From this, conclude that $n = 1$.)

3 Grün's theorems

In this section, all groups are assumed to be finite.

Definition 3.1. Let P be a Sylow p-subgroup of G. The group G is said to be p-normal if for every $g \in G$ such that $\mathbf{Z}(P)^g \leq P$, we have $\mathbf{Z}(P)^g = \mathbf{Z}(P)$. In

[4] Correction 25 from errata.

particular, G is p-normal whenever the Sylow p-subgroups of G are abelian or whenever distinct Sylow p-subgroups intersect trivially.

Definition 3.2. The p-commutator subgroup $G'(p)$ of G is defined to be the smallest normal subgroup M of G with abelian p-factor group G/M. (By Theorem I.9.7, pg. 50, $G'(p)$ is the intersection of all normal subgroups N of G with abelian p-factor group G/N.)

Theorem 3.3. *Let P be a Sylow p-subgroup of G. We have*

$$\mathbf{V}_{G\to P}(G) = \mathbf{V}_{G\to P}(P) \cong P/(P \cap G') \cong G/G'(p).$$

Proof. The quotient $G/G'(p)$ is a p-group, so $G = G'(p)P$ and

$$G/G'(p) \cong P/(P \cap G'(p)).$$

Since $G' \leq G'(p)$, we have $P \cap G' \leq P \cap G'(p)$. On the other hand, we have

$$G/G' = PG'/G' \times G'(p)/G'.$$

This yields

$$P \cap G'(p) \leq PG' \cap G'(p) = G',$$

so that $P \cap G'(p) \leq P \cap G'$. We have proved that $P \cap G'(p) = P \cap G'$ and $G/G'(p) \cong P/(P \cap G')$.

Under the homomorphism $\mathbf{V} = \mathbf{V}_{G\to P}$ from G to the abelian p-group P/P', the group $G'(p)$ is mapped onto 1. Consequently, we have $\mathbf{V}(G) = \mathbf{V}(G'(p)P) = \mathbf{V}(P)$. Now, for some s_i depending on g, we have

$$\mathbf{V}(g) = \prod_i s_i g^{f_i} s_i^{-1} P'$$

by Lemma 1.7. In particular, if we let $n = [G : P]$ then it follows for $g = x \in P$ that

$$\mathbf{V}(x) = \prod_i x^{f_i}[x^{f_i}, s_i^{-1}]P' = x^n t P'$$

for some $t \in G'$. If $\mathbf{V}(x) = P'$, then $x^n \in G'$, so $(n, p) = 1$ then implies $x \in P \cap G'$. It is trivial to see that $\mathbf{V}(P \cap G') = 1$. Therefore, $\ker \mathbf{V} \cap P = P \cap G'$ and $\mathbf{V}(P) \cong P/(P \cap G')$. □

The next theorem describes the kernel of the transfer map to a Sylow p-subgroup.

Theorem 3.4 (Grün's first theorem). *Let P be a Sylow p-subgroup of G. Then*

$$P \cap G' = \langle P \cap \mathbf{N}_G(P)',\ P \cap P'^g \mid g \in G \rangle$$

is the kernel of the homomorphism $\mathbf{V}_{G\to P}$ from P onto $\mathbf{V}_{G\to P}(P)$.

Proof. We let

$$D = \langle P \cap \mathbf{N}_G(P)',\ P \cap P'^g \mid g \in G \rangle.$$

Clearly, $P' \leq D \leq P \cap G'$ and so $D \trianglelefteq P$.

Assume for contradiction that $D < P \cap G'$. Let u be an element of smallest possible order such that $u \in P \cap G'$ but $u \notin D$. Note that if $t > 0$ and $(u^{p^t})^g \in P$, then $(u^{p^t})^g \in P \cap G'$ and so $(u^{p^t})^g \in D$ since $|u^{p^t}| < |u|$.

We compute $\mathbf{V}_{G\to P}(u)$. For the computation, we choose the following representatives for the cosets of P in G:

Let $G = \bigcup_i Pg_iP$ be the decomposition of G into double cosets of P (see Theorem I.2.19, pg. 11). For a fixed $g = g_i$, the distinct cosets Pgx with $x \in P$ constitute an orbit of P with respect to right multiplication. Therefore, $|PgP|/|P|$ is a power p^s of p. The element u induces a permutation $\widetilde{u}$ on the cosets of P lying in PgP. Let p^{m_i} with $m_1 \leq m_2 \leq \cdots \leq m_r$ be the cycle lengths of $\widetilde{u}$. We have $\sum_{i=1}^{r} p^{m_i} = p^s$. Now choose g such that

$$(Pg, Pgu, \ldots, Pgu^{p^{m_1}-1})$$

is a cycle of $\widetilde{u}$ of length p^{m_1}. Then $Pgu^{p^{m_1}} = Pg$ and therefore

$$gu^{p^{m_1}}g^{-1} \in P.$$

To continue the computation, we distinguish two cases:

Case 1: Assume $p^s > 1$. For all $i = 1, \ldots, r$, let $x_i \in P$ such that

$$(Pgx_i, Pgx_iu, \ldots, Pgx_iu^{p^{m_i}-1})$$

is a cycle of $\widetilde{u}$ of length p^{m_i} acting on the cosets of P in PgP. By Lemma 1.7, each of these cycles contributes to $\mathbf{V}(u)$ an element

$$t_i = gx_iu^{p^{m_i}}x_i^{-1}g^{-1} \in P.$$

We have

$$gu^{-p^{m_i}}g^{-1}t_i = g[u^{p^{m_i}}, x_i^{-1}]g^{-1} \in gP'g^{-1}.$$

Since $m_1 \leq m_i$, the element $u^{p^{m_i}}$ is a power of $u^{p^{m_1}}$. Hence,

$$gu^{p^{m_i}}g^{-1} \in \langle gu^{p^{m_1}}g^{-1} \rangle \leq P.$$

Then we also have

$$gu^{-p^{m_i}}g^{-1}t_i \in P,$$

and it follows that

$$gu^{-p^{m_i}}g^{-1}t_i \in P \cap gP'g^{-1} \leq D.$$

Consequently, the contribution to $\mathbf{V}(u)$ arising from the cycles in PgP is

$$\prod_i t_i \equiv \prod_i gu^{p^{m_i}}g^{-1} = gu^{p^s}g^{-1} \bmod D.$$

Since $t_i \in P$ and $D \leq P$, it follows that $gu^{p^s}g^{-1} \in P$. By our initial remark, we have $gu^{p^s}g^{-1} \in D$ since $s > 0$.

Case 2: Now assume that $p^s = 1$, which means $Pgx = Pg$ for all $x \in P$. This is equivalent to $g \in \mathbf{N}_G(P)$. The contribution to $\mathbf{V}(u)$ from this cycle is now $gug^{-1} = u[u, g^{-1}]$, where $[u, g^{-1}] \in P \cap \mathbf{N}_G(P)' \leq D$.

Altogether, we find that $\mathbf{V}(u) = u^k tP'$ for some $t \in D$ and $k = [\mathbf{N}_G(P) : P]$. Since $u \in P \cap G'$, we have $\mathbf{V}(u) = P'$. Then it follows from $P' \leq D$ that $u^k \in D$. However, $k \not\equiv 0 \bmod p$ then implies that $u \in D$, a contradiction. □

Next, we give an application of Theorem 3.4.

Theorem 3.5 (Wong [484]). *Suppose a Sylow 2-subgroup P of G is the group of type* (3) *in Theorem I.14.9 (pg. 96) with generators a and b satisfying the defining relations*

$$a^{2^n} = b^2 = 1, \quad a^b = a^{1+2^{n-1}} \quad (n > 2).$$

Then G has a normal subgroup N such that $G/N \cong P$.

Proof. The commutator subgroup $P' = \langle a^{2^{n-1}} \rangle$ of P is cyclic of order 2. The elements $a^{2^{n-1}}$, $a^{2^{n-1}}b$ and b are the only elements of order 2 in P.

a) First, we show that the automorphism group $\mathbf{Aut}(P)$ of P is a 2-group:

If α is an automorphism of P, then $b^\alpha = b$ or $b^\alpha = a^{2^{n-1}}b$. Clearly, the map α defined on generators by $a^\alpha = a$ and $b^\alpha = a^{2^{n-1}}b$ is an automorphism of P. Thus, there exists a subgroup B of $\mathbf{Aut}(P)$ such that $[\mathbf{Aut}(P) : B] = 2$ and $b^\alpha = b$ for all $\alpha \in B$. For $\alpha \in B$, we have

$$\begin{aligned} a^\alpha &= a^x \quad \text{with} \quad x \not\equiv 0 \bmod 2 \quad \text{or} \\ a^\alpha &= a^x b \quad \text{with} \quad x \not\equiv 0 \bmod 2, \end{aligned}$$

and these are all automorphisms of P. Consequently, $|B| = 2^{2(n-1)}$ and we have $|\mathbf{Aut}(P)| = 2^{2n-1}$.

b) By a) and the Zassenhaus Theorem I.18.1 (pg. 135), we have $\mathbf{N}_G(P) = P \times Q$, where $2 \nmid |Q|$. By Theorem 3.4, we have

$$\begin{aligned} P \cap G' &= \langle \mathbf{N}_G(P)' \cap P, P \cap P'^g \mid g \in G \rangle \\ &= \langle P', P \cap P'^g \mid g \in G \rangle. \end{aligned}$$

Since $|P'| = 2$ and all the elements of order 2 in P lie in

$$\Omega_1(P) = \langle a^{2^{n-1}}, b\rangle,$$

it follows that $P \cap G' \leq \Omega_1(P)$.

Let $|G/G'| = 2^m k$ with $2 \nmid k$. Then $m \geq n-1$. Let $G' \leq T \leq G$ with $|T/G'| = k$, and let $S = \langle a^2, b\rangle T$. Then $[G : S] = 2$ and the group

$$P \cap S = \langle a^2, b\rangle (P \cap T) = \langle a^2, b\rangle (P \cap G') = \langle a^2, b\rangle$$

is abelian. Since $|a^2| = 2^{n-1} > 2 = |b|$, the group S has a normal subgroup N such that $|S/N| = 2^n$ by Theorem 2.7. Since N is a characteristic subgroup of S, it follows that $N \lhd G$ and $|G/N| = 2^{n+1} = |P|$. □

Remark 3.6. By Theorem II.8.10 (pg. 206), the groups $\mathrm{PSL}_2(p^f)$ with $p \neq 2$ and $p^f > 3$ are examples of simple groups with dihedral Sylow 2-subgroups. Gorenstein and Walter [180]–[184] showed that the only simple groups with dihedral Sylow 2-subgroups are the groups $\mathrm{PSL}_2(p^f)$ ($p \neq 2$) and the alternating group A_7. The Sylow 2-subgroups of the simple groups $\mathrm{PSL}_3(p^f)$ with $p^f \equiv -1 \bmod 4$ are quasidihedral with defining relations of the form

$$a^{2^n} = b^2 = 1, \quad a^b = a^{-1+2^{n-1}} \quad \text{(See Exercise II.10, pg. 199).}$$

The Mathieu group M_{11}, which we will address in Volume 3, also has a Sylow 2-subgroup of this type. Finally, the Sylow 2-subgroups of $\mathrm{SL}_2(p^f)$ with $p \neq 2$ are generalized quaternion groups by Theorem II.8.10 (pg. 206), but $\mathrm{SL}_2(p^f)$ does not have a normal subgroup of odd order with 2-power index. However, $\mathrm{SL}_2(p^f)$ is not simple since $|\mathbf{Z}(\mathrm{SL}_2(p^f))| = 2$. R. Brauer and Suzuki have shown that the Sylow 2-subgroups of a simple group cannot be generalized quaternion groups (Brauer, Suzuki [70], Suzuki [403], Brauer [59]). We will return to this point in Chapter V, §22.

Theorem 3.7 (Grün's second theorem). *Suppose G is p-normal, and let P be a Sylow p-subgroup of G. Then*

$$G/G'(p) \cong \mathbf{N}_G(\mathbf{Z}(P))/\mathbf{N}_G(\mathbf{Z}(P))'(p).$$

Proof. We know from Theorem 3.3 that

$$G/G'(p) \cong P/(P \cap G').$$

Similarly, applying Theorem 3.3 to $\mathbf{N}_G(\mathbf{Z}(P))$ yields

$$\mathbf{N}_G(\mathbf{Z}(P))/\mathbf{N}_G(\mathbf{Z}(P))'(p) \cong P/(P \cap \mathbf{N}_G(\mathbf{Z}(P))').$$

We show that $P \cap G' = P \cap \mathbf{N}_G(\mathbf{Z}(P))'$.

Clearly, $P \cap \mathbf{N}_G(\mathbf{Z}(P))' \leq P \cap G'$. So it remains to show that

$$P \cap G' \leq P \cap \mathbf{N}_G(\mathbf{Z}(P))'.$$

By Theorem 3.4, we have

$$P \cap G' = \langle P \cap \mathbf{N}_G(P)', P \cap P'^g \mid g \in G \rangle.$$

Since $\mathbf{N}_G(P) \leq \mathbf{N}_G(\mathbf{Z}(P))$, it holds trivially that

$$P \cap \mathbf{N}_G(P)' \leq P \cap \mathbf{N}_G(\mathbf{Z}(P))'.$$

To finish, we show that

$$P \cap P'^g \leq P \cap \mathbf{N}_G(\mathbf{Z}(P))' :$$

We define $T = P \cap P'^g$. Clearly, $\mathbf{Z}(P) \leq \mathbf{N}_G(T)$ and $\mathbf{Z}(P)^g \leq \mathbf{N}_G(T)$ as well. Therefore, there exist Sylow p-subgroups P_1 and P_2 of $\mathbf{N}_G(T)$ such that $\mathbf{Z}(P) \leq P_1$ and $\mathbf{Z}(P)^g \leq P_2$. By Sylow's theorem, there exists an $s \in \mathbf{N}_G(T)$ such that $P_1 = P_2^s$. The group P_1 lies in a Sylow p-subgroup P^* of G. Now we have

$$\mathbf{Z}(P) \leq P_1 \leq P^* \quad \text{and} \quad \mathbf{Z}(P)^{gs} \leq P_2^s = P_1 \leq P^*.$$

By the p-normality of G, it follows that $\mathbf{Z}(P) = \mathbf{Z}(P^*) = \mathbf{Z}(P)^{gs}$, so $gs \in \mathbf{N}_G(\mathbf{Z}(P))$. Since $s \in \mathbf{N}_G(T)$, we have

$$T = T^s = P^s \cap P'^{gs}.$$

Moreover, it follows from $P \leq \mathbf{N}_G(\mathbf{Z}(P))$ that

$$P'^{gs} \leq \mathbf{N}_G(\mathbf{Z}(P))'^{gs} = \mathbf{N}_G(\mathbf{Z}(P))'.$$

Thus, $T = P \cap P'^g = P^s \cap P'^{gs} \leq P \cap \mathbf{N}_G(\mathbf{Z}(P))'$. □

For our first application of Grün's second theorem, we prove a decomposition theorem:

Theorem 3.8 (Gaschütz [154]). *Suppose N is a normal subgroup of G with the following properties:*
(1) *G/N is a p-group.*
(2) *N has abelian Sylow p-subgroups.*
(3) *N has no nontrivial p-factor groups.*
Then N has a complement in G.

Proof. Let P be a Sylow p-subgroup of G. Then $Q = P \cap N$ is a Sylow p-subgroup of N by Lemma I.7.7 (pg. 35). So Q is abelian. By the Zassenhaus Theorem I.18.1 (pg. 135), there exists an S such that $\mathbf{N}_N(Q) = QS$ and $Q \cap S = 1$.

a) We first show that $Q \cap \mathbf{Z}(QS) = 1$:

If $[Q, S] < Q$, then $[Q, S]S \lhd QS$ and $\mathbf{N}_N(Q) = QS$ has a nontrivial p-factor group $QS/[Q, S]S$. Since Q is abelian, N is p-normal. It follows from Theorem 3.7 that

$$N/N'(p) \cong \mathbf{N}_N(Q)/\mathbf{N}_N(Q)'(p) \neq 1,$$

but this contradicts condition (3) of our hypothesis. Therefore, $Q = [Q, S]$. By Theorem III.13.4 (pg. 370), we have

$$[Q, S] = Q = (Q \cap \mathbf{Z}(QS)) \times [Q, S].$$

It follows that $Q \cap \mathbf{Z}(QS) = 1$.

b) We let $T = \mathbf{N}_G(Q)$. We claim that $\mathbf{N}_G(Q) = Q\mathbf{N}_T(S)$ and $Q \cap \mathbf{N}_T(S) = 1$:

Let $g \in \mathbf{N}_G(Q)$. Then S^g is a complement of Q in QS because

$$\mathbf{N}_N(Q) = QS \trianglelefteq \mathbf{N}_G(Q).$$

Since Q is solvable, we may apply the conjugacy part of the Zassenhaus Theorem I.18.2 (pg. 136) to conclude that there exists an $x \in \mathbf{N}_N(Q) = QS$ such that $S^g = S^x$. So $gx^{-1} \in \mathbf{N}_T(S)$. It follows that

$$\mathbf{N}_G(Q) = (QS)\mathbf{N}_T(S) = Q\mathbf{N}_T(S).$$

We let $L = \mathbf{N}_T(S)$ and $D = Q \cap L$. Since $D \leq Q \trianglelefteq QS$ and $D \leq L \leq \mathbf{N}_G(S)$, we have $[D, S] \leq Q \cap S = 1$. So $D \leq Q \cap \mathbf{Z}(QS)$, and it follows from a) that $D = 1$.

c) Now we have

$$N \cap \mathbf{N}_T(S) = \mathbf{N}_N(Q) \cap \mathbf{N}_T(S) = QS \cap L = S(Q \cap L) = S.$$

Hence,

$$[\mathbf{N}_T(S) : S] = [\mathbf{N}_T(S) : \mathbf{N}_T(S) \cap N]$$

is a power of p, being a divisor of $|G/N|$. The Zassenhaus Theorem I.18.1 (pg. 135) yields

$$L = \mathbf{N}_T(S) = SR$$

for some p-group R. Now we have by Theorem I.7.8 (pg. 36) that

$$G = N\mathbf{N}_G(Q) = N(QL) = NL = NSR = NR$$

and

$$N \cap R = N \cap \mathbf{N}_G(Q) \cap R = QS \cap R.$$

Since Q is the unique Sylow p-subgroup of $QS = \mathbf{N}_N(Q)$, it follows that

$$N \cap R \leq Q \cap R \leq Q \cap L = 1.$$

That is, R is a complement of N in G. □

We will see further applications of Theorem 3.7 in §5.

4 p-nilpotent groups

Definition 4.1. A group G is said to be p-nilpotent if G contains a normal subgroup K of order coprime to p such that G/K is a p-group. This subgroup K is unique if it exists, as it is the subgroup of G consisting of all elements of order coprime to p. So K is a characteristic subgroup of G. If P is a Sylow p-subgroup of G, then $G = KP$, $K \cap P = 1$ and $G/K \cong P$. If G is p-nilpotent, then we refer to K as the normal p-complement of G. (If G is p-nilpotent for all primes p, then clearly G is nilpotent in the sense of Chapter III.)

Lemma 4.2. *If G is p-nilpotent, then all subgroups and factor groups of G are also p-nilpotent.*

Proof. a) Let $U \leq G$, and let K be the normal p-complement of G. Then $K \cap U \trianglelefteq U$, $|K \cap U|$ is coprime to p, and

$$|U/(K \cap U)| = |KU/K|$$

is a power of p, as it is a divisor of $|G/K|$. So $K \cap U$ is a normal p-complement of U, and U is p-nilpotent.

b) Let $N \trianglelefteq G$, and again let K be the normal p-complement of G. Then $KN/N \trianglelefteq G/N$ and

$$|KN/N| = |K/(K \cap N)|$$

is coprime to p. Furthermore,

$$[G/N : KN/N] = |G/KN|$$

is a power of p. Hence, KN/N is a normal p-complement of G/N. □

Lemma 4.3. *Assume G is p-nilpotent. If N is a minimal normal subgroup of G and p divides $|N|$, then $N \leq \mathbf{Z}(G)$.*

Proof. Let K be the normal p-complement of G. Since p divides $|N|$ and p does not divide $|K|$, it follows that $K \cap N < N$. The minimality of N forces $K \cap N = 1$ and $[K, N] \leq K \cap N = 1$. Hence, $|N| = |NK/K|$ is a power of p. Let P be a Sylow p-subgroup of G. By Theorem I.6.9 (pg. 31), we have $1 < N \cap \mathbf{Z}(P)$ since $1 < N \trianglelefteq P$. It follows from $[K, N] = 1$ that $N \cap \mathbf{Z}(P) \trianglelefteq KP = G$. By the minimality of N, we have $N \leq \mathbf{Z}(P)$, and then $N \leq \mathbf{Z}(G)$, as well. □

Now we can give a characterization of p-nilpotent groups that corresponds to the characterization of nilpotent groups by central series in Theorem III.2.3 (pg. 274):

Theorem 4.4. *The following are equivalent:*
a) *G is p-nilpotent.*

b) *If*

$$1 = G_0 \lhd G_1 \lhd \cdots \lhd G_r = G$$

is an arbitrary chief series of G, then for all i such that $p \mid |G_i/G_{i-1}|$, we have

$$G_i/G_{i-1} \leq \mathbf{Z}(G/G_{i-1}).$$

c) *There exists a chief series*

$$1 = G_0 \lhd G_1 \lhd \cdots \lhd G_r = G$$

of G such that for each i, either $p \nmid |G_i/G_{i-1}|$ or $G_i/G_{i-1} \leq \mathbf{Z}(G/G_{i-1})$.

Proof. a) $\Rightarrow$ b): By Lemma 4.2, the quotient G/G_{i-1} is again p-nilpotent. Therefore, b) follows immediately upon applying Lemma 4.3 to the minimal normal subgroup G_i/G_{i-1} of G/G_{i-1}.

b) $\Rightarrow$ c): This is trivial.

c) $\Rightarrow$ a): Clearly, G/G_1 also satisfies the hypothesis of c). Using induction, we may assume that G/G_1 is p-nilpotent. Let L/G_1 be the normal p-complement of G/G_1. Then

$$[G/G_1 : L/G_1] = |G/L|$$

is a power of p and $|L/G_1|$ is coprime to p. If $p \nmid |G_1|$, then $p \nmid |L|$ and L is the normal p-complement of G.

Assume $p \mid |G_1|$. By our assumption in c), we have $G_1 \leq \mathbf{Z}(G)$ and so $|G_1| = p$. By the Zassenhaus Theorem I.18.1 (pg. 135), it follows that $L = G_1 \times K$ for some subgroup K of order coprime to p. Since K is characteristic in L, it follows that $K \trianglelefteq G$ and $|G/K| = p|G/L|$ is a power of p. Therefore, G is p-nilpotent. □

Next, we note the following theorem.

Theorem 4.5. *If G is p-nilpotent, then G is p-normal.*

Proof. Let K be the normal p-complement of G and P a Sylow p-subgroup of G. Then $\mathbf{Z}(G/K) = \mathbf{Z}(P)K/K$. So for all $g \in G$, we have

$$\mathbf{Z}(P)K = (\mathbf{Z}(P)K)^g = \mathbf{Z}(P)^g K.$$

If $\mathbf{Z}(P)^g \leq P$, then it follows that

$$\mathbf{Z}(P)^g = K\mathbf{Z}(P)^g \cap P = K\mathbf{Z}(P) \cap P = \mathbf{Z}(P),$$

and G is p-normal. □

We prove an interesting connection between p-nilpotency and the Frattini subgroup:

Theorem 4.6 (Roquette [359]). *Let $N \trianglelefteq G$, and let S be a subgroup of G with the following properties:*
(1) $N \cap S \leq \Phi(S)$.
(2) $(|N \cap S|, [G : S]) = 1$.
(*Note that condition* (2) *is satisfied if S is a Hall subgroup of G.*)

Let π be the set of all prime divisors of $|N \cap S|$. Then there exists a normal subgroup M of N with $M \cap S = 1$ such that N/M is a π-group and $|N \cap S|$ divides $|N/M|$.

Proof. Let M be the smallest normal subgroup of N subject to the condition that N/M is a π-group. The subgroup M is clearly characteristic in N, so M is normal in G. If $M \cap S = 1$, then

$$|N \cap S| = |(N \cap S)/(M \cap S)| = |(N \cap S)M/M|,$$

and this is a divisor of $|N/M|$. So we are done in this case.

Now assume for contradiction that $Q = M \cap S \neq 1$. By condition (1) of our hypothesis, we have $Q \leq \Phi(S)$. By the minimality of M, the subgroup M certainly has no nontrivial π-factor groups. We consider the group $L = MS$.

Since $Q \leq \Phi(S)$, the subgroup Q is nilpotent by Theorem III.3.6 (pg. 286). Thus, there exists a characteristic subgroup T of Q such that Q/T is a nonidentity elementary abelian p-group for some $p \in \pi$. It follows from $Q = M \cap S \trianglelefteq S$ that $T \trianglelefteq S$.

We define $G_1 = L$, $N_1 = M$ and $G_2 = S/T$, $N_2 = Q/T$. Then

$$G_1/N_1 = L/M = MS/M \cong S/Q \cong G_2/N_2.$$

We have homomorphisms μ_i $(i = 1, 2)$ from G_i onto S/Q defined by

$$(ms)^{\mu_1} = sQ = (sT)^{\mu_2} \quad (\text{for } m \in M, s \in S).$$

(Note that μ_1 is well defined since $Q = M \cap S$!)

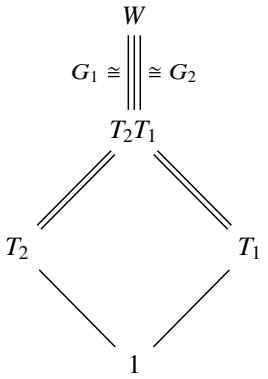
Now we construct the group

$$W = G_1 \leftthreetimes G_2 = \{(ms, sT) \mid m \in M, s \in S\}$$

according to Theorem I.9.11 (pg. 52) and define

$$T_1 = \{(m, T) \mid m \in M = N_1\} \quad \text{and} \quad T_2 = \{(1, qT) \mid q \in Q\}.$$

Then $T_i \cong N_i$ $(i = 1, 2)$, $T_i \trianglelefteq W$ and $T_1 \cap T_2 = 1$.



We first show that T_2 has a complement in W:

By Theorem I.17.4 (pg. 130), it suffices to show that the abelian p-group T_2 has a complement in a Sylow p-subgroup of W. The map $\epsilon : S \to W$ defined by

$$s^\epsilon = (s, sT)$$

for all $s \in S$ is clearly a monomorphism from S to W. If $s \in S$ such that

$$(s, sT) \in S^\epsilon \cap T_2,$$

then $s = 1$. Hence, S^ϵ is a complement of T_2 in $S^\epsilon T_2$. By condition (2) of our hypothesis, we have

$$[W : S^\epsilon T_2] = [G_1 : S] \not\equiv 0 \bmod p.$$

So $S^\epsilon T_2$ contains a Sylow p-subgroup of W. Then T_2 has a complement in this Sylow p-subgroup since T_2 has a complement in $S^\epsilon T_2$, namely S^ϵ.

By what we have just shown, T_2 has a complement R in W. Then the map α_1 defined by

$$r^{\alpha_1} = (g_1, g_2)^{\alpha_1} = g_1 \quad \text{for} \quad r = (g_1, g_2) \in R$$

is an isomorphism from R to G_1 since $R \cap \ker \alpha_1 = R \cap T_2 = 1$. Let δ be the inverse of α_1. Then δ is a monomorphism from G_1 to W defined by

$$g_1^\delta = (g_1, g_1^\tau),$$

where τ is a homomorphism from G_1 to G_2. For $m \in M = N_1$, we have

$$m^\delta = (m, m^\tau) \in W.$$

So $m^{\mu_1} = Q = (m^\tau)^{\mu_2}$ and consequently

$$m^\tau \in \ker \mu_2 = N_2 = Q/T.$$

Since $\tau|_M$ is a homomorphism from M to the p-group Q/T and M has no nontrivial p-factor groups, it follows that $m^\tau = T$ for all $m \in M$. Thus,

$$M^\delta = \{(m, m^\tau) \mid m \in M\} = \{(m, T) \mid m \in M\} = T_1.$$

Define a homomorphism α_2 from W onto G_2 by

$$(g_1, g_2)^{\alpha_2} = g_2 \quad (g_i \in G_i).$$

Since $W = G_1^\delta T_2$ and $G_1^\delta \cap T_2 = 1$, applying the homomorphism α_2 we find

$$G_2 = W^{\alpha_2} = G_1^{\delta\alpha_2} T_2^{\alpha_2} = G_1^{\delta\alpha_2} N_2.$$

Now, since

$$\ker \alpha_2 = T_1 = M^\delta \leq G_1^\delta$$

and $G_1^\delta \cap T_2 = 1$, it also follows that $G_1^{\delta\alpha_2} \cap N_2 = 1$. Hence, $G_1^{\delta\alpha_2}$ is a complement of $N_2 = Q/T$ in $G_2 = S/T$. In particular, if Q/T is not the identity subgroup, then a maximal subgroup containing the complement of Q/T in S/T does not contain Q/T, which means $Q/T \not\leq \Phi(S/T)$. However, we have by Lemma III.3.4 (pg. 285) that

$$Q/T \leq \Phi(S)/T \leq \Phi(S/T).$$

So Q/T is trivial, contradicting our assumption that $Q \neq T$. Thus, $Q = M \cap S = 1$.□

The following theorem is an important special case of Theorem 4.6.

Theorem 4.7 (J. Tate [425]). *Let P be a Sylow p-subgroup of G, and suppose N is a normal subgroup of G with $N \cap P \leq \Phi(P)$. Then N is p-nilpotent.*

Proof. We apply Theorem 4.6 with P in place of S. Then N contains a normal subgroup M such that $|N \cap P|$ divides $|N/M| = p^a$. Since $N \cap P$ is a Sylow p-subgroup of N, the subgroup M is the normal p-complement of N. □

We can now give a new characterization of nilpotent groups:

Theorem 4.8. *The finite group G is nilpotent if and only if for every prime p there exists at least one Sylow p-subgroup P of G such that $P \cap G' \leq \Phi(P)$. Moreover, we then have $P \cap G' = P'$.*

Proof. If G is nilpotent, then it follows immediately from the direct decomposition

$$G = P_1 \times \cdots \times P_r$$

of G into the direct product of its Sylow subgroups P_i that $P_i \cap G' = P_i'$. By Theorem III.3.11 (pg. 287), we have $P_i' \leq \Phi(P_i)$.

Conversely, assume that for every prime p there is a Sylow p-subgroup P of G such that $P \cap G' \leq \Phi(P)$. Let K/G' be the normal p-complement of the abelian

group G/G'. Then $P \cap G'$ is a Sylow p-subgroup of G' and a Sylow p-subgroup of K. By our hypothesis,

$$P \cap K = P \cap G' \leq \Phi(P),$$

and so K is p-nilpotent by Theorem 4.7. The quotient G/K is a p-group, so G is also p-nilpotent. Since this conclusion holds for all p, the group G is nilpotent. □

Next, we prove another criterion for p-nilpotency.

Theorem 4.9. *Let P be a Sylow p-subgroup of G. The following are equivalent:*
a) *G is p-nilpotent.*
b) *If x_1 and x_2 are elements of P and there exists an element $g \in G$ such that $x_1^g = x_2$, then there exists an element $x \in P$ such that $x_1^x = x_2$.*

Proof. a) ⇒ b): Let G be p-nilpotent, and let K be the normal p-complement of G. Then $G = PK$. Let $g = xk$ with $x \in P$ and $k \in K$. Since $K \trianglelefteq G$, we have

$$x_1^{-x} x_2 = x_1^{-x} x_1^g = x_1^{-x} x_1^{xk} = [x_1^x, k] \in K \cap P = 1.$$

Thus, $x_1^x = x_2$.

b) ⇒ a): Let $\mathbf{V}$ be the transfer map from G to P/P'. By Lemma 1.7, we have for all $x \in P$ that

$$\mathbf{V}(x) = \prod_j s_j x^{f_j} s_j^{-1} P'.$$

In this expression, each $s_j x^{f_j} s_j^{-1}$ lies in P. By our assumption in b), there exist elements $x_j \in P$ such that

$$s_j x^{f_j} s_j^{-1} = x_j x^{f_j} x_j^{-1}.$$

It follows that

$$\mathbf{V}(x) = \prod_j x^{f_j} P' = x^{[G:P]} P'.$$

Thus, $\mathbf{V}(P) = P/P'$ and $\ker \mathbf{V} \cap P = P'$. Since $P' \leq \Phi(P)$ by Theorem III.3.11 (pg. 287), the kernel $\ker \mathbf{V}$ is p-nilpotent by Theorem 4.7. Since $G/\ker \mathbf{V}$ is a p-group, G is itself p-nilpotent. □

We plan to include another application of Theorem 4.7 when we discuss the theory of subnormal subgroups in later volumes.

5 Minimal non-p-nilpotent groups

In applications, the case where G is not p-normal frequently poses some difficulties. The following theorem often makes it easier to proceed.

Theorem 5.1 (Burnside). *Let P and P^* be Sylow p-subgroups of G. Let $1 < M \lhd P$ and $M < P^*$, but $M \not\lhd P^*$. Then there exists an element $g \in G$ of prime power order q^b with $q \neq p$ such that:*

a) $g \notin \mathbf{N}_G(M)$.

b) $J = MM^g \cdots M^{g^{q^b-1}}$ *is a p-group.*

c) *We have $g \in \mathbf{N}_G(J)$, but $g \notin \mathbf{C}_G(J)$.*

Proof. We choose P^* from among the Sylow p-subgroups of G such that

(1) $M < P^*$,

(2) $M \not\lhd P^*$ and

(3) $[P^* : \mathbf{N}_{P^*}(M)]$ is as small as possible.

We define $S = \mathbf{N}_{P^*}(M) = P^* \cap \mathbf{N}_G(M)$.

1) By replacing P by a conjugate P^y with $M \lhd P^y$ if necessary, we may assume that $S \leq P$:

Since $M \lhd P$, the group P is a Sylow p-subgroup of $\mathbf{N}_G(M)$. Since S is a p-subgroup of $\mathbf{N}_G(M)$, there exists a Sylow p-subgroup P^y of $\mathbf{N}_G(M)$ such that $S \leq P^y$. Clearly, $M \lhd P^y$. Thus, we may replace P by P^y.

2) We define $T = \mathbf{N}_P(S)$ and $T^* = \mathbf{N}_{P^*}(S)$. Then $S < T$ and $S < T^*$:

Since $S = \mathbf{N}_{P^*}(M)$ and $M \not\lhd P^*$, we have $S < P^*$. As P^* is nilpotent, it follows from Theorem III.2.3 (pg. 274) that $S < \mathbf{N}_{P^*}(S) = T^*$. In the same way, $S < T = \mathbf{N}_P(S)$ since $|S| < |P^*| = |P|$ and $S \leq P$ implies $S < P$.

3) If $T \leq V \leq G$ and $|V| = p^a$, then $M \trianglelefteq V$:

For some Sylow p-subgroup P^{**} of G, we have

$$\mathbf{N}_{P^*}(M) = S < T \leq V \leq P^{**}.$$

Since $T = \mathbf{N}_P(S) \leq P \leq \mathbf{N}_G(M)$, we also have $T \leq \mathbf{N}_{P^{**}}(M)$. It follows that

$$[P^{**} : \mathbf{N}_{P^{**}}(M)] \leq [P^{**} : T] < [P^* : S] = [P^* : \mathbf{N}_{P^*}(M)].$$

By the choice of P^*, we have $M \lhd P^{**}$ and thus $M \trianglelefteq V$, as claimed.

4) Concluding the proof:

It follows from $S \lhd T$ and $S \lhd T^*$ that $S \lhd \langle T, T^* \rangle = L$. Let R be a Sylow p-subgroup of L with $T \leq R$. By 3), $M \trianglelefteq R$. Since $\mathbf{N}_{P^*}(M) = S < T^*$, we have $T^* \not\leq \mathbf{N}_G(M)$, which means $L \not\leq \mathbf{N}_G(M)$. Thus, there exists a prime $q \neq p$ and a Sylow q-subgroup Q of L with $Q \not\leq \mathbf{N}_G(M)$. Choose $g \in Q$ such that $g \notin \mathbf{N}_G(M)$. From $S \lhd L$ and $M \lhd S$, we deduce that

$$M^{g^i} \lhd S^{g^i} = S$$

for all i. Hence,

$$J = MM^g \cdots M^{g^{q^b}-1} \quad (\text{where } q^b = |g|)$$

is a p-group normalized by g. Since $M^g \neq M$, we certainly have $g \notin \mathbf{C}_G(J)$. □

Lemma 5.2. *Let P be a Sylow p-subgroup of G, and let U be a characteristic subgroup of P. If $U^g \trianglelefteq P$ for some $g \in G$, then $U = U^g$.*

Proof. Let $U \trianglelefteq P$ and $U^g \trianglelefteq P$. Then P and $P^{g^{-1}}$ lie in $\mathbf{N}_G(U)$. By Sylow's theorem, there exists an $h \in \mathbf{N}_G(U)$ such that $P^{g^{-1}} = P^h$. Since U is characteristic in P, it follows that $hg \in \mathbf{N}_G(P) \leq \mathbf{N}_G(U)$, so $g \in \mathbf{N}_G(U)$. This proves that $U = U^g$. □

Lemma 5.3. *Suppose G is not p-normal. Then there exists a p-subgroup $J \leq G$ and an element $g \in G$ with the following properties:*
(1) $g \in \mathbf{N}_G(J) - \mathbf{C}_G(J)$.
(2) *$|g|$ is a power of a prime q with $q \neq p$.*

Proof. Since G is not p-normal, there exists an element $h \in G$ such that $\mathbf{Z}(P)^h \leq P$ but $\mathbf{Z}(P)^h \neq \mathbf{Z}(P)$. This means $\mathbf{Z}(P)^h \not\trianglelefteq P$ by Lemma 5.2. The claim then follows by setting $M = \mathbf{Z}(P)$ and $P^* = P^{h^{-1}}$ in Theorem 5.1. □

Using Grün's second theorem together with Lemma 5.3 is a powerful tool. As a typical application, we prove a useful generalization of Theorem III.5.2 (pg. 297):

Theorem 5.4 (N. Itô [270]). *Suppose that every proper subgroup of G is p-nilpotent, but G itself is not p-nilpotent. Then:*
(1) *Every proper subgroup of G is nilpotent.*
(2) *$|G| = p^a q^b$ for some prime q.*
(3) *G has a normal Sylow p-subgroup P. Moreover, if $p > 2$, then P has exponent p; if $p = 2$, then P has exponent at most 4.*
(4) *The Sylow q-subgroups of G are cyclic.*

Proof. a) G has a normal Sylow p-subgroup:

First assume that G is not p-normal. Then by Lemma 5.3, there exists a subgroup $J\langle g\rangle$ of G with $g \in \mathbf{N}_G(J) - \mathbf{C}_G(J)$, $|J| = p^a$ and $|g| = q^b$. If $J\langle g\rangle < G$, then $J\langle g\rangle$ is p-nilpotent, and it follows that $\langle g\rangle \lhd J\langle g\rangle$. Then $[J, \langle g\rangle] \leq J \cap \langle g\rangle = 1$ and $g \in \mathbf{C}_G(J)$. This is a contradiction. Therefore, $J\langle g\rangle = G$ and J is the normal Sylow p-subgroup of G.

Now assume G is p-normal, and let P be a Sylow p-subgroup of G. If $\mathbf{N}_G(\mathbf{Z}(P)) < G$, then $\mathbf{N}_G(\mathbf{Z}(P))$ is p-nilpotent. Consequently,

$$G/G'(p) \cong \mathbf{N}_G(\mathbf{Z}(P))/\mathbf{N}_G(\mathbf{Z}(P))'(p) \neq 1$$

by Theorem 3.7. In particular, $G'(p) < G$ and therefore $G'(p)$ is p-nilpotent. But then G is p-nilpotent since the normal p-complement of $G'(p)$, being a characteristic subgroup of $G'(p)$, is also a normal p-complement of G. This contradicts our assumption that G is not p-nilpotent.

Therefore, we must have $\mathbf{N}_G(\mathbf{Z}(P)) = G$. If $\mathbf{Z}(P) = P$, then G has a normal Sylow p-subgroup. If $\mathbf{Z}(P) < P$ and $G/\mathbf{Z}(P)$ is not p-nilpotent, then $P/\mathbf{Z}(P) \trianglelefteq G/\mathbf{Z}(P)$

by induction. Again, we conclude that $P \trianglelefteq G$. Finally, assume that $G/\mathbf{Z}(P)$ is p-nilpotent and $\mathbf{Z}(P) < P$. Let $M/\mathbf{Z}(P)$ be the normal p-complement of $G/\mathbf{Z}(P)$. Since p divides $|G/\mathbf{Z}(P)|$, we have $M < G$ and M is p-nilpotent. The normal p-complement of M is a normal p-complement of G. This contradicts our assumption that G is not p-nilpotent.

We have proved that G has a normal Sylow p-subgroup.

b) Every proper subgroup of G is nilpotent:

By the Zassenhaus Theorem I.18.1 (pg. 135), P has a complement Q in G. Since G is not p-nilpotent, we have $Q \not\leq \mathbf{C}_G(P)$. Let $g \in Q$ be an element of prime power order q^b with $g \notin \mathbf{C}_G(P)$. Then $P\langle g\rangle$ is not p-nilpotent, so $G = P\langle g\rangle$. In particular, $|G| = p^a q^b$ and G is q-nilpotent. If $U < G$, then U is p-nilpotent and q-nilpotent, so U is nilpotent.

c) Statements (2)–(4) now follow immediately from Theorem III.5.2. □

Theorem 5.4 has numerous consequences. We give a few of them here.

Theorem 5.5.

a) *Let p be an odd prime. If every subgroup of G of order p lies in $\mathbf{Z}(G)$, then G is p-nilpotent.*

b) *If all elements of order 2 and 4 lie in $\mathbf{Z}(G)$, then G is 2-nilpotent.*

Proof. Let G be a minimal counterexample. Since the hypothesis is inherited by the subgroups of G, every proper subgroup of G is p-nilpotent but G itself is not. So G has a normal Sylow p-subgroup P by Theorem 5.4. Furthermore, if $p > 2$, then P has exponent p by Theorem 5.4. It follows that $P \leq \mathbf{Z}(G)$. If $p = 2$, then P has exponent at most 4 by Theorem 5.4 and again $P \leq \mathbf{Z}(G)$. Therefore, G is p-nilpotent. □

Example 5.6. If we only assume that all elements of order 2 lie in $\mathbf{Z}(G)$, then it does not follow that G is 2-nilpotent. The following construction provides an example: Let $Q = \langle a, b\rangle$ be the quaternion group of order 8 with relations

$$a^4 = 1, \quad a^2 = b^2 = [a, b].$$

Then Q admits an automorphism α of order 3 defined by

$$a^\alpha = b, \quad b^\alpha = ab.$$

Let G be the semidirect product of Q with $\langle\alpha\rangle$. Then G is not 2-nilpotent, but its unique subgroup $\langle a^2\rangle$ of order 2 lies in $\mathbf{Z}(G)$. □

By Theorem III.2.3 (pg. 274), a finite group is nilpotent if and only if all maximal subgroups are normal. The dual condition, that all minimal subgroups are normal, does not lead to a particularly important class of groups. We prove:

Theorem 5.7 (Gaschütz, N. Itô). *Suppose that every minimal subgroup of G is normal in G. Then G is solvable and the commutator subgroup G' has a normal Sylow 2-subgroup with nilpotent factor group.*

Proof. Let U be a subgroup of G' of prime order p. Then $U \trianglelefteq G$, and $G/\mathbf{C}_G(U)$ is abelian. So $G' \leq \mathbf{C}_G(U)$. The assumptions of Theorem 5.5 are now satisfied in G' for every odd prime p. Therefore, G' is p-nilpotent for every odd prime p, which gives the claim. □

Theorem 5.8 (Frobenius).
a) *If $\mathbf{N}_G(U)/\mathbf{C}_G(U)$ is a p-group for every p-subgroup U of G, then G is p-nilpotent.*
b) *If $\mathbf{N}_G(U)$ is p-nilpotent for every p-subgroup $U \neq 1$ of G, then G is p-nilpotent.*

Proof. a) Let V be a subgroup of G, and suppose that U is a nonidentity p-subgroup of V. Then

$$\begin{aligned} |\mathbf{N}_V(U)/\mathbf{C}_V(U)| &= |(\mathbf{N}_G(U) \cap V)/(\mathbf{C}_G(U) \cap V)| \\ &= |(\mathbf{N}_G(U) \cap V)\mathbf{C}_G(U)/\mathbf{C}_G(U)| \end{aligned}$$

is a power of p, being a divisor of $|\mathbf{N}_G(U)/\mathbf{C}_G(U)|$. Hence, the hypothesis is inherited by all subgroups of G.

Let G be a minimal counterexample. Then all proper subgroups of G are p-nilpotent, but G itself is not. By Theorem 5.4, the group G has a normal Sylow p-subgroup $P \neq 1$. By our hypothesis,

$$\mathbf{N}_G(P)/\mathbf{C}_G(P) = G/\mathbf{C}_G(P)$$

is a p-group, so $G = \mathbf{C}_G(P)P$. By the Zassenhaus Theorem I.18.1, the subgroup $\mathbf{C}_G(P) \cap P = \mathbf{Z}(P)$ of $\mathbf{C}_G(P)$ has a complement S in $\mathbf{C}_G(P)$. Then $G = P \times S$ is p-nilpotent.

b) Let U be a nonidentity p-subgroup of G, and let K be the normal p-complement of the p-nilpotent group $\mathbf{N}_G(U)$. Then

$$[U, K] \leq U \cap K = 1,$$

so $K \leq \mathbf{C}_G(U)$. Therefore, $\mathbf{N}_G(U)/\mathbf{C}_G(U)$ is a p-group, and the claim follows from a). □

Theorem 5.9 (Frobenius). *Let P be a Sylow p-subgroup of G. Assume every subgroup of P can be generated by at most d elements, and suppose $|G|$ is coprime to*

$$(p^d - 1)(p^{d-1} - 1) \cdots (p^2 - 1)(p - 1).$$

Then G is p-nilpotent.

Proof. Let $U \leq P$. Since U can be generated by d elements, we have that

$$|\mathbf{N}_G(U)/\mathbf{C}_G(U)| \text{ divides } |\mathbf{Aut}(U)| \text{ divides } p^k(p^d - 1)(p^{d-1} - 1) \cdots (p - 1)$$

for some k by Theorem III.3.19 (pg. 290). By our hypothesis, $|\mathbf{N}_G(U)/\mathbf{C}_G(U)|$ is a power of p, and we may apply Theorem 5.8 a). □

Note that we easily obtain Theorem 2.8 again from Theorem 5.9. The next result is another consequence of Theorem 5.9.

Theorem 5.10. *If the Sylow p-subgroups of G are metacyclic and $|G|$ is coprime to $p^2 - 1$, then G is p-nilpotent.* (*See also Theorem 8.5.*)

Proof. Let P be a metacyclic Sylow p-subgroup of G. Let Z be a cyclic normal subgroup of P with cyclic factor group P/Z. If U is a subgroup of P, then $U \cap Z$ and $U/(U \cap Z) \cong UZ/Z$ are cyclic. Thus, U may be generated by two elements. Applying Theorem 5.9 with $d = 2$ yields the claim. □

As a special case of Theorem 5.10, we have:

Theorem 5.11. *Assume that the Sylow 2-subgroups of G are metacyclic. If $|G|$ is coprime to 3, then G is 2-nilpotent.*

The following theorem is the dual statement of Theorem III.3.18 (pg. 290).

Theorem 5.12. *Let P be a p-group, and let α be an automorphism of P whose order is coprime to p.*
a) *For $p > 2$, suppose that α fixes all elements of P of order p.*
b) *For $p = 2$, suppose that α fixes all elements of P of orders 2 and 4.*
Then $\alpha = 1$.

Proof. Let P be a minimal counterexample. If $U < P$ and $U^\alpha = U$, then $\alpha|_U = 1$ by the minimality of P. We may clearly assume that α has prime order q with $q \neq p$.

Let $G = P\langle y\rangle$ be the semidirect product of P with the cyclic group $\langle y\rangle$ of order q defined by $x^y = x^\alpha$ for all $x \in P$. Since $\alpha \neq 1$, G is not p-nilpotent.

We show that every proper subgroup V of G is p-nilpotent. If V is a p-group, this is trivial. If q divides $|V|$, then $\langle y\rangle \leq V^g$ for some $g \in G$. Then

$$V^g = (V^g \cap P)\langle y\rangle.$$

By the remark in the first paragraph, α acts as the identity on $V^g \cap P$, so V^g is p-nilpotent. By Theorem 5.4, P has exponent p for $p > 2$ and exponent at most 4 for $p = 2$. This means that α acts trivially on all of P. □

6 Thompson's normal p-complement theorem

Definition 6.1. Let P be a p-group. We define

$$m(P) = \max\{d(A) \mid A \leq P,\ A' = 1\}$$

and

$$\mathbf{J}(P) = \langle A \leq P \mid A' = 1,\ d(A) = m(P)\rangle.$$

(Recall that $d(A)$ is the minimum size of a generating set for A.)

It is clear that $\mathbf{J}(P)$ is a characteristic subgroup of P. It is also easy to see that if $\mathbf{J}(P) \le K \le P$, then $\mathbf{J}(P) = \mathbf{J}(K)$.

For applications, it is important to be able to deduce the p-nilpotency of G from the p-nilpotency of the smallest possible collection of subgroups of G. Theorem 5.8 is a statement of this type. The best known criterion in this direction is the following:

Main Theorem 6.2 (Thompson [432]). *Let $p > 2$, and let P be a Sylow p-subgroup of G. If $\mathbf{C}_G(\mathbf{Z}(P))$ and $\mathbf{N}_G(\mathbf{J}(P))$ are p-nilpotent, then G is p-nilpotent.*

Proof. Suppose G is a minimal counterexample.

a) Let

$$\mathscr{H} = \{H \mid 1 < H \le G,\ H \text{ a } p\text{-group},\ \mathbf{N}_G(H) \text{ not } p\text{-nilpotent}\}.$$

We define a partial order $\ll$ on $\mathscr{H}$ by saying $H_1 \ll H_2$ if

(1) $|\mathbf{N}_G(H_1)|_p < |\mathbf{N}_G(H_2)|_p$ or

(2) $|\mathbf{N}_G(H_1)|_p = |\mathbf{N}_G(H_2)|_p$ and $|H_1| < |H_2|$.

(The highest power of p dividing $|T|$ is denoted by $|T|_p$.) We claim that $\mathscr{H} \neq \emptyset$ and $P \notin \mathscr{H}$:

If $\mathscr{H} = \emptyset$, then $\mathbf{N}_G(H)$ is p-nilpotent for every p-group H with $1 < H \le G$. But then G is also p-nilpotent by Theorem 5.8 b), contradicting our choice of G as a counterexample. So $\mathscr{H} \neq \emptyset$. Since $\mathbf{J}(P)$ is characteristic in P, we have $\mathbf{N}_G(P) \le \mathbf{N}_G(\mathbf{J}(P))$. Then $\mathbf{N}_G(P)$ is p-nilpotent since $\mathbf{N}_G(\mathbf{J}(P))$ is p-nilpotent by assumption. This means $P \notin \mathscr{H}$.

Now we fix an element $H \in \mathscr{H}$ that is maximal with respect to the order $\ll$.

b) Letting $N = \mathbf{N}_G(H)$, the quotient N/H is p-nilpotent:

Let $\overline{P}$ be a Sylow p-subgroup of N. Then certainly $H \le \overline{P}$. By passing to a conjugate if necessary, we may assume that $\overline{P} \le P$.

Let $P_1/H = \mathbf{Z}(\overline{P}/H)$ and $P_2/H = \mathbf{J}(\overline{P}/H)$. Then $\overline{P} \le \mathbf{N}_G(P_i)$ $(i = 1, 2)$ and

$$|\mathbf{N}_G(P_i)|_p \ge |\overline{P}| = |\mathbf{N}_G(H)|_p.$$

Since $H < P$, we have $H < \mathbf{N}_G(H) \cap P = \overline{P}$ by Theorem III.2.3 (pg. 274). Then $1 < \mathbf{Z}(\overline{P}/H) = P_1/H$ and $1 < \mathbf{J}(\overline{P}/H) = P_2/H$ as well. Thus, $|H| < |P_i|$. If $P_i \in \mathscr{H}$, then we would have $H \ll P_i$, contradicting the maximality of H in $\mathscr{H}$. Therefore, $P_i \notin \mathscr{H}$ and $\mathbf{N}_G(P_i)$ is p-nilpotent.

We have $\mathbf{C}_{N/H}(P_1/H) \le \mathbf{N}_G(P_1)/H$, so $\mathbf{C}_{N/H}(\mathbf{Z}(\overline{P}/H))$ is p-nilpotent. Moreover, $\mathbf{N}_{N/H}(P_2/H) \le \mathbf{N}_G(P_2)/H$, so $\mathbf{N}_{N/H}(\mathbf{J}(\overline{P}/H))$ is p-nilpotent. By the minimality of G, the quotient N/H is p-nilpotent.

c) We have $N = \mathbf{N}_G(H) = G$:

Since $H < P$, we have $\mathbf{Z}(P) \le \mathbf{N}_P(H) = \overline{P}$, and $\mathbf{Z}(P) \le \mathbf{Z}(\overline{P})$ as well. It follows from $\mathbf{C}_N(\mathbf{Z}(\overline{P})) \le \mathbf{C}_N(\mathbf{Z}(P))$ that $\mathbf{C}_N(\mathbf{Z}(\overline{P}))$ is p-nilpotent.

Assume for contradiction that $\mathbf{N}_G(\mathbf{J}(\overline{P}))$ is not p-nilpotent. Then $\mathbf{J}(\overline{P}) \in \mathscr{H}$ and $H \not\ll \mathbf{J}(\overline{P})$ by the maximality of H. This means

$$|\overline{P}| = |\mathbf{N}_G(H)|_p \ge |\mathbf{N}_G(\mathbf{J}(\overline{P}))|_p.$$

If $\overline{P} < P$, then there exists a P^* such that $\overline{P} \lhd P^* \leq P$ by Theorem III.2.3 (pg. 274). But then $P^* \leq \mathbf{N}_G(\mathbf{J}(\overline{P}))$ and

$$|\mathbf{N}_G(\mathbf{J}(\overline{P}))|_p \geq |P^*| > |\overline{P}| \geq |\mathbf{N}_G(\mathbf{J}(\overline{P}))|_p.$$

This is impossible. Thus, $\overline{P} = P$ and $\mathbf{N}_G(\mathbf{J}(\overline{P})) = \mathbf{N}_G(\mathbf{J}(P))$ is p-nilpotent.

Assume for contradiction that $N < G$. Then by the minimality of G, we deduce from the p-nilpotency of $\mathbf{C}_N(\mathbf{Z}(\overline{P}))$ and $\mathbf{N}_N(\mathbf{J}(\overline{P}))$ that N is p-nilpotent. But this contradicts that $H \in \mathscr{H}$. Hence, $N = G$.

d) G contains no nontrivial normal subgroup of order coprime to p:

Assume for contradiction that there exists a subgroup T of G such that $1 < T \lhd G$ and $p \nmid |T|$. Then we have $\mathbf{J}(PT/T) = \mathbf{J}(P)T/T$. As $\mathbf{J}(P)$ is a Sylow p-subgroup of $\mathbf{J}(P)T$, if $g \in \mathbf{N}_G(\mathbf{J}(P)T)$ then $\mathbf{J}(P)^g = \mathbf{J}(P)^t$ for some $t \in T$ by Sylow's theorem. Hence,

$$\mathbf{N}_G(\mathbf{J}(P)T) = \mathbf{N}_G(\mathbf{J}(P))T.$$

Then

$$\mathbf{N}_{G/T}(\mathbf{J}(P)T/T) = \mathbf{N}_G(\mathbf{J}(P)T)/T = \mathbf{N}_G(\mathbf{J}(P))T/T,$$

and this group is p-nilpotent.

In the same way, we have

$$\mathbf{C}_{G/T}(\mathbf{Z}(P)T/T) \leq \mathbf{N}_{G/T}(\mathbf{Z}(P)T/T) = \mathbf{N}_G(\mathbf{Z}(P))T/T.$$

Let $a \in \mathbf{N}_G(\mathbf{Z}(P))$ such that $(zT)^a = z^aT = zT$ for all $z \in \mathbf{Z}(P)$. Then we have $z^{-1}z^a \in T \cap \mathbf{Z}(P) = 1$, so $a \in \mathbf{C}_G(\mathbf{Z}(P))$. This proves that

$$\mathbf{C}_{G/T}(\mathbf{Z}(P)T/T) = \mathbf{C}_G(\mathbf{Z}(P))T/T,$$

and this group is p-nilpotent. Now, G/T satisfies the hypothesis of the theorem and $|G/T| < |G|$, so G/T is p-nilpotent. But since $p \nmid |T|$, this implies G itself is p-nilpotent, contradicting our choice of G as a counterexample.

e) H is the maximal normal p-subgroup of G:

Assume for contradiction that there exists a p-subgroup H_1 of G such that $H < H_1 \lhd G$. Then $\mathbf{N}_G(H_1) = G$ is not p-nilpotent, so $H_1 \in \mathscr{H}$. Since $|H| < |H_1|$ and $|\mathbf{N}_G(H)|_p = |\mathbf{N}_G(H_1)|_p = |P|$, we have $H \ll H_1$. But this contradicts the maximality of H in $\mathscr{H}$.

f) We have $G = PQ$, where Q is an elementary abelian q-group with $q \neq p$; furthermore, P is a maximal subgroup of G and QH/H is a minimal normal subgroup of G/H; in particular, G is solvable:

Note that if $P \leq U < G$, then U satisfies the assumptions of the theorem. So U is p-nilpotent.

Recall that G/H is p-nilpotent by b) and c), so let K/H be the normal p-complement of G/H. By the Zassenhaus theorem, H has a complement S in K.

Then

$$1 < H \lhd HS \lhd G$$

is a normal series of G. Let HT/H with $1 < T \leq S$ be a minimal normal subgroup of G/H. If $T < S$, then $P(HT) = PT < G$. Then by the remark of the last paragraph, PT is p-nilpotent. This means $T \lhd PT$. Since $HT \cap S = T \lhd S$, it follows that $T \lhd PS = G$, contradicting d). Thus, $S = T$ and HS/H is a minimal normal subgroup of G/H. In particular, S is characteristically simple.

Let $Q \neq 1$ be a Sylow q-subgroup of S for some prime $q \neq p$. By the Frattini argument of Theorem I.7.8 (pg. 36), we have $G = HS\mathbf{N}_G(Q)$. So $H\mathbf{N}_G(Q)$ contains a Sylow p-subgroup of G. If $H\mathbf{N}_G(Q) < G$, then $H\mathbf{N}_G(Q)$ is p-nilpotent by our previous remark, which means $[H, Q] = 1$. Now, for all $s \in S$ it follows that

$$Q^s \leq \mathbf{C}_G(H^s) = \mathbf{C}_G(H).$$

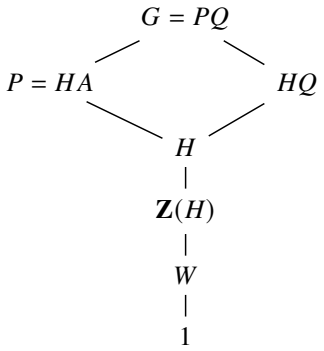
Since S is characteristically simple, we then have

$$S = \langle Q^s \mid s \in S \rangle \leq \mathbf{C}_G(H).$$

But this implies $HS = H \times S \lhd G$, and then $S \lhd G$, contradicting d).

Hence, we have $H\mathbf{N}_G(Q) = G$, which means $HQ \lhd G$. Since HS/H is a minimal normal subgroup of G/H, the subgroup $S = Q$ is elementary abelian.

Let U be a subgroup of G such that $P < U \leq G$. Then $U = PQ_1$ with $1 < Q_1 = Q \cap U$. Now, $HQ_1 = HQ \cap U$ is normalized by Q and U, so HQ_1 is normalized by $\langle U, Q \rangle = PQ = G$. This forces $Q_1 = Q$ and $U = G$. That is, P is a maximal subgroup of G.



g) There exists an abelian subgroup A of P such that $d(A) = m(P)$ and $A \not\leq H$. If A is chosen to be minimal with respect to the conditions $d(A) = m(P)$ and $A \not\leq H$, then $A/(A \cap H)$ is elementary abelian:

If $\mathbf{J}(P) \leq H$, then $G = \mathbf{N}_G(H) \leq \mathbf{N}_G(\mathbf{J}(H)) = \mathbf{N}_G(\mathbf{J}(P))$. But then G is p-nilpotent, a contradiction. So $\mathbf{J}(P) \not\leq H$. Thus, there exists an abelian subgroup A of P with $d(A) = m(P)$ and $A \not\leq H$. Now choose A to be minimal with respect to

these properties. Let p^m be the exponent of A. Then $d(\Omega_{m-1}(A)) = d(A) = m(P)$ and $\Omega_{m-1}(A) < A$. So $\Omega_{m-1}(A) \leq A \cap H$, and $A/(A \cap H)$ has exponent p.

h) We have $\mathbf{C}_G(H) \leq H$ and $\mathbf{C}_{G/H}(QH/H) = QH/H$:

By d) and e), the Fitting subgroup of G is H. Since G is solvable, we have $\mathbf{C}_G(H) \leq H$ by Theorem III.4.2 (pg. 292). Clearly, QH/H is the Fitting subgroup of G/H, which means $\mathbf{C}_{G/H}(QH/H) \leq QH/H$.

In what follows, we use the 'bar' notation to denote passing to the quotient $\overline{G} = G/H$. In particular, $\overline{A} = AH/H$ for the subgroup A that we chose in g).

i) We have $\overline{Q} = [\overline{Q}, \overline{A}]$; moreover, $\overline{Q} \cap \mathbf{Z}(\overline{QA}) = \overline{1}$ and $G = HQA$:

By Theorem III.13.4 (pg. 370), we have $\overline{Q} = \overline{Q}_1 \times \overline{Q}_2$, where $\overline{Q}_1 = \overline{Q} \cap \mathbf{Z}(\overline{QA})$ and $\overline{Q}_2 = [\overline{Q}, \overline{A}]$. If $\overline{Q}_2 = \overline{1}$, then using h) we obtain the contradiction $\overline{A} \leq \mathbf{C}_{\overline{G}}(\overline{Q}) = \overline{Q}$. Hence, $\overline{Q}_2 \neq \overline{1}$.

Write $\overline{Q}_2 = Q_2H/H$ for some $Q_2 \leq Q$. Then $Q_2 \neq 1$. We define $K = HQ_2A$. Note that K is a subgroup of G. Moreover, $P_0 = HA$ is a Sylow p-subgroup of K and $A \leq \mathbf{J}(P_0)$. Let $\mathbf{N}_K(\mathbf{J}(P_0)) = Q_2^*P_0$, where $Q_2^* \leq Q_2$. Since $HQ_2 \lhd HQA$, we have

$$[Q_2^*, A] \leq \mathbf{J}(P_0) \cap HQ_2 \leq H.$$

So $\overline{A}$ acts trivially on $\overline{Q_2^*}$, and it follows that $\overline{Q_2^*} \leq \overline{Q}_1 \cap \overline{Q}_2 = \overline{1}$. Hence, $Q_2^* = 1$ and consequently $\mathbf{N}_K(\mathbf{J}(P_0))$ is p-nilpotent, being a p-group.

By h), we have $\mathbf{Z}(P) \leq \mathbf{C}_G(H) \leq H$. So $\mathbf{Z}(P) \leq HA = P_0$, which means $\mathbf{Z}(P) \leq \mathbf{Z}(P_0)$ and $\mathbf{C}_G(\mathbf{Z}(P_0)) \leq \mathbf{C}_G(\mathbf{Z}(P))$. Hence, $\mathbf{N}_K(\mathbf{J}(P_0))$ and $\mathbf{C}_K(\mathbf{Z}(P_0))$ are both p-nilpotent. If $K < G$, then K is p-nilpotent and Q_2 is the normal p-complement of K. This yields

$$[Q_2, H] \leq Q_2 \cap H = 1.$$

But this means $Q_2 \leq \mathbf{C}_G(H) \leq H$ by h). So $Q_2 = 1$, which contradicts our assumption above that $Q_2 \neq 1$. Therefore, $G = K = HQ_2A$. It follows that $Q_2 = Q$, so $\overline{Q} = [\overline{Q}, \overline{A}]$ and $\overline{Q} \cap \mathbf{Z}(\overline{QA}) = \overline{1}$.

j) Letting $m = m(P) = d(A)$, we have

$$|A/(A \cap H)| = |G/HQ| = p$$

and $d(A \cap H) \geq m - 1$:

Recall that $\overline{G} = G/H = \overline{QA}$ by i), and we have that $\overline{Q}$ is a minimal normal subgroup of $\overline{G}$ by f). By h), we have $\mathbf{C}_{\overline{G}}(\overline{Q}) = \overline{Q}$. So $\overline{A}$ acts faithfully and irreducibly on $\overline{Q}$. By Theorem II.3.10 (pg. 174), this means $\overline{A}$ is cyclic. Since

$$\overline{A} = AH/H \cong A/(A \cap H)$$

has exponent p by g), it follows that $|A/(A \cap H)| = p$. Since $d(A) = m$, we must have $d(A \cap H) \geq m - 1$.

k) Let $W = \langle \mathbf{Z}(P)^g \mid g \in G \rangle$. We have $W \leq \mathbf{Z}(H)$:

Since $\mathbf{Z}(P) \leq \mathbf{C}_G(H) \leq H \lhd G$, we have $\mathbf{C}_G(H) = \mathbf{Z}(H)$; therefore, $\mathbf{Z}(P) \leq \mathbf{Z}(H) \lhd G$ and consequently $W \leq \mathbf{Z}(H)$.

l) $\mathbf{C}_G(\mathbf{Z}(P)) = P$:

By f), the subgroup P is a maximal subgroup of G. Since $\mathbf{C}_G(\mathbf{Z}(P))$ is p-nilpotent, but G is not, $\mathbf{C}_G(\mathbf{Z}(P))$ is a proper subgroup of G. Hence, we may conclude from $P \leq \mathbf{C}_G(\mathbf{Z}(P)) < G$ that $P = \mathbf{C}_G(\mathbf{Z}(P))$.

m) We have $W = [W, Q] \times (W \cap \mathbf{Z}(WQ))$ and $[W, Q] \neq 1$:

The first claim follows by applying Theorem III.13.4 (pg. 370) to the group WQ containing the abelian normal subgroup W. Since $Q \not\leq \mathbf{C}_G(\mathbf{Z}(P)) = P$, we certainly have $Q \not\leq \mathbf{C}_G(W)$. Hence, $[W, Q] \neq 1$.

n) We define $W_1 = [W, Q]$, $V = \Omega_1(W_1)$ and $V_0 = V \cap A$. Then $V \lhd G$; moreover, $V = [V, Q]$, $V \cap \mathbf{Z}(VQ) = 1$, and $d(V/V_0) \leq 1$:

To prove that $V = \Omega_1(W_1) \lhd G$, it suffices to show that $W_1 \lhd G$. Since $W \leq \mathbf{Z}(H)$ by k), we have

$$W_1 = [W, Q] = [W, HQ].$$

It then follows from $W \lhd G$ and $HQ \lhd G$ that $W_1 = [W, HQ] \lhd G$. By Theorem III.13.4 (pg. 370), we have

$$V = [V, Q] \times (V \cap \mathbf{Z}(VQ))$$

and

$$V \cap \mathbf{Z}(VQ) \leq W_1 \cap \mathbf{Z}(WQ) = 1.$$

This proves that $V = [V, Q]$ and $V \cap \mathbf{Z}(VQ) = 1$.

We have

$$V_0 = V \cap A = (V \cap H) \cap A = V \cap A_0,$$

where $A_0 = A \cap H$. Since $V \leq W \leq \mathbf{Z}(H)$ and $A_0 \leq H$, the group $\langle A_0, V \rangle$ is abelian and V is elementary abelian. This means

$$d(\langle A_0, V \rangle) = d(A_0) + d(V/V_0).$$

On the other hand,

$$d(\langle A_0, V \rangle) \leq m \leq d(A_0) + 1$$

by j). Together these inequalities imply $d(V/V_0) \leq 1$.

o) $d(V) \leq 2$:

Write $A = \langle a, A \cap H \rangle$. Then $\mathbf{C}_V(a) \geq A \cap V$, so $[V : \mathbf{C}_V(a)] \leq p$ by n). Since

$$P = HA = \langle H, a \rangle \not\lhd G,$$

there exists some $x \in G$ such that $a^x \notin P$. Recalling that P is a maximal subgroup of G by f), it follows that

$$G = \langle P, a^x \rangle = \langle H, a, a^x \rangle.$$

From $V^x = V$, we obtain $[V : \mathbf{C}_V(a^x)] \leq p$; hence,

$$[V : \mathbf{C}_V(\langle a, a^x \rangle)] \leq p^2.$$

Since $G = \langle H, a, a^x \rangle$ and $V \leq \mathbf{Z}(H)$ by k), it follows by n) that

$$\mathbf{C}_V(\langle a, a^x \rangle) \leq V \cap \mathbf{Z}(G) \leq V \cap \mathbf{Z}(VQ) = 1.$$

Thus, $|V| \leq p^2$.

p) $\mathbf{C}_G(V) = H$:

Since $V \leq \mathbf{Z}(H)$, we certainly have $H \leq \mathbf{C}_G(V) \trianglelefteq G$. If $H < \mathbf{C}_G(V)$, then $\mathbf{C}_G(V)$ is not a p-group by e). So $\mathbf{C}_G(V) \cap Q \neq 1$. Since $\overline{Q}$ is a minimal normal subgroup of $\overline{G}$, we have $Q \leq \mathbf{C}_G(V)$ and $[V, Q] = 1$. But this contradicts $[V, Q] = V$ from n). Hence, $\mathbf{C}_G(V) = H$.

q) $|V| = p^2$:

Assume for contradiction that $|V| = p$. Then $G/\mathbf{C}_G(V) = G/H$ is cyclic of order coprime to p. But this implies $H = P$, contradicting $H \neq P$ from a).

r) Concluding the proof:

By p) and q), we have that

$$\overline{G} = G/H = G/\mathbf{C}_G(V) = \overline{QA}$$

is isomorphic to a subgroup of $\mathrm{GL}_2(p)$. The eigenvalues of an arbitrary $\overline{x} \in \overline{Q}$ lie in $\mathrm{GF}(p^2)$. Since $p \nmid |\overline{x}|$, all such $\overline{x}$ can therefore be diagonalized in $\mathrm{GL}_2(p^2)$. It is a well-known fact that since $\overline{Q}$ is abelian, all elements of $\overline{Q}$ can be simultaneously diagonalized. Let $\mathsf{V} = \mathsf{V}(2, p^2) = \langle v_1, v_2 \rangle$ with

$$v_i \overline{x} = a_i(\overline{x}) v_i \quad (i = 1, 2).$$

Since $[\overline{Q}, \overline{A}] = \overline{Q}$ by i), not all $\overline{x} \in \overline{Q}$ are proportional to the identity matrix. Hence, the vectors v_i $(i = 1, 2)$ are uniquely determined up to scalar multiples.

Let $\overline{1} \neq \overline{a} \in \overline{A} \leq \mathbf{N}_{\overline{G}}(\overline{Q})$. Then

$$v_i \overline{a}\, \overline{x} = v_i \overline{x}^{\overline{a}^{-1}} \overline{a} = a_i(\overline{x}^{\overline{a}^{-1}}) v_i \overline{a} \quad (i = 1, 2).$$

Since $\overline{a} \notin \mathbf{C}_{\overline{G}}(\overline{Q}) = \overline{Q}$, the element $\overline{a}$ is not a diagonal matrix with respect to the basis $\{v_1, v_2\}$. Hence,

$$v_1 \overline{a} = b_1 v_2 \quad \text{and} \quad v_2 \overline{a} = b_2 v_1.$$

This means $\overline{a}^2 \in \mathbf{C}_{\overline{G}}(\overline{Q})$. But since $p \neq 2$ (and here we use the assumption $p \neq 2$ for the first time!), we obtain the contradiction $\overline{a} \in \mathbf{C}_{\overline{G}}(\overline{Q})$. □

Example 6.3. Let G be the symmetric group S_4, and let P be a Sylow 2-subgroup of G. Then $P = \mathbf{J}(P)$ and $\mathbf{C}_G(\mathbf{Z}(P)) = \mathbf{N}_G(\mathbf{J}(P)) = P$ is 2-nilpotent, but G itself is not 2-nilpotent. This example shows that the condition $p \neq 2$ in Theorem 6.2 cannot be removed, even though it is used only in the last line of the proof.

We will discuss important applications of Thompson's theorem in the next section and, above all, in Chapter V, §8.

7 Nilpotent subgroups

To begin this section, we recall Definition III.5.4 (pg. 299):

Definition 7.1. Let π be a set of prime numbers. A subgroup U of G is called a π-subgroup if π contains all the primes dividing $|U|$. A subgroup H of G is called a Hall π-subgroup of G if every prime divisor of $|H|$ lies in π and no prime divisor of $[G : H]$ lies in π.

The following lemma is proved in the same way as Lemma I.7.7 (pg. 35):

Lemma 7.2. *Suppose H is a Hall π-subgroup of G and $N \trianglelefteq G$. Then:*
a) *$N \cap H$ is a Hall π-subgroup of N.*
b) *HN/N is a Hall π-subgroup of G/N.*

Theorem 7.3 (Wielandt [471]). *Suppose H is a nilpotent Hall π-subgroup of G, but not a Sylow subgroup of G. Assume that for every prime p dividing $|H|$ and Sylow p-subgroup H_p of H, we have $\mathbf{N}_G(H_p) = H$. Then there exists a normal subgroup K of G such that $G = KH$ and $K \cap H = 1$. (So G is p-nilpotent for all p in π.)*

Proof. Let P be a Sylow p-subgroup of H, and suppose that x_1 and x_2 are elements of P with $x_2 = x_1^g$ for some $g \in G$. Since H is not a Sylow subgroup of G, there exists a Sylow q-subgroup Q of H with $1 \neq Q \neq P$. We have $x_i \in \mathbf{C}_G(Q)$ $(i = 1, 2)$. Now, Q and Q^g are both Sylow q-subgroups of $\mathbf{C}_G(x_2)$, so there exists an $x \in \mathbf{C}_G(x_2)$ such that $Q^{gx} = Q$. It follows that $gx \in \mathbf{N}_G(Q) = H$ and

$$x_1^{gx} = x_2^x = x_2.$$

Since H is nilpotent, there exists a $y \in P$ with $x_1^y = x_2$. So G is p-nilpotent by Theorem 4.9. Let $K(p)$ be the normal p-complement of G. Letting $K = \bigcap_{p \in \pi} K(p)$, we have $|G/K| = |H|$, so $G = KH$ and $K \cap H = 1$. □

In Chapter VI, §4, we will study products of pairwise permutable nilpotent groups, and Theorem 7.3 will be very useful. Here we give the following application of Theorem 7.3:

Theorem 7.4 (Deskins, Janko, Thompson). *Let H be a maximal subgroup of G. Assume that H is nilpotent and the Sylow 2-subgroup of H has class at most 2. Then G is solvable.*

Proof. Let G be a minimal counterexample. We proceed in a series of steps.

a) H is a Hall subgroup of G:

Assume for contradiction that p is a prime such that $1 < H_p < G_p$ for some Sylow p-subgroup G_p of G. By Theorem III.2.3 (pg. 274), we have

$$H_p \lhd \mathbf{N}_G(H_p) \cap G_p,$$

and so $\mathbf{N}_G(H_p) > H$. By the maximality of H, we have $H_p \lhd G$. Then G/H_p and its nilpotent maximal subgroup H/H_p satisfy the assumptions of the theorem, so G/H_p is solvable. But then G is itself solvable, contradicting the choice of G as a counterexample. We conclude that H is a Hall subgroup of G.

b) G does not contain a normal subgroup K such that $G = KH$ and $K \cap H = 1$:

Assume for contradiction that $G = KH$ and $K \cap H = 1$. Let q be a prime with $K_q \neq 1$. By the Frattini argument of Theorem I.7.8 (pg. 36), we have $G = K\mathbf{N}_G(K_q)$, and then

$$|H| = |G/K| = |\mathbf{N}_G(K_q)/(\mathbf{N}_G(K_q) \cap K)| = |\mathbf{N}_G(K_q)/\mathbf{N}_K(K_q)|.$$

Since H is a Hall subgroup of G by a), $|H|$ and $|\mathbf{N}_K(K_q)|$ are coprime. So by the Zassenhaus Theorem I.18.1 (pg. 135), there exists a subgroup V of $\mathbf{N}_G(K_q)$ so that $|V| = |\mathbf{N}_G(K_q)/\mathbf{N}_K(K_q)| = |H|$. Since the Hall subgroup H is nilpotent, it follows from Wielandt's Theorem III.5.8 (pg. 300) that H and V are conjugate in G. That is, there exists a $g \in G$ such that

$$H = V^g \leq \mathbf{N}_G(K_q^g).$$

The maximality of H implies $H < HK_q^g = G$. But then $K = K_q \lhd G$ and G is solvable, contradicting our choice of G.

c) H is a Sylow subgroup of G:

Assume for contradiction that H is not a Sylow subgroup of G. We certainly have $H \leq \mathbf{N}_G(H_p)$ for all p. If there exists a p with $1 \neq H_p$ and $H < \mathbf{N}_G(H_p)$, then it follows immediately from the maximality of H that $H_p \lhd G$. But then we obtain the contradiction that G is solvable, just like we did in a). Hence, $\mathbf{N}_G(H_p) = H$ for all primes p dividing $|H|$. It then follows from Theorem 7.3 that G contains a normal subgroup $K \lhd G$ so that $G = KH$ and $K \cap H = 1$. But this is impossible by b). Therefore, H is a Sylow subgroup of G.

d) H is a Sylow 2-subgroup of G:

Assume for contradiction that $|H| = p^n$ for some $p > 2$. Let $U = \mathbf{J}(H)$ or $U = \mathbf{Z}(H)$ (the group $\mathbf{J}(H)$ is defined in Definition 6.1). Then clearly $H \leq \mathbf{N}_G(U)$. If $H < \mathbf{N}_G(U)$, then it follows that $U \lhd G$ and G is solvable. Hence, $\mathbf{N}_G(U) = H$ is p-nilpotent (in fact, it is nilpotent). This means $\mathbf{N}_G(\mathbf{J}(H))$ and $\mathbf{C}_G(\mathbf{Z}(H))$ are p-nilpotent, the latter being a subgroup of $\mathbf{N}_G(\mathbf{Z}(H))$. Since $p > 2$, we may apply

Thompson's Theorem 6.2 to conclude that G is p-nilpotent. But this contradicts b), so $p = 2$.

e) H has nilpotency class at least 3:

Assume for contradiction that H has class at most 2. (Note that in the following argument, it will only be necessary to assume that H is a Sylow p-subgroup of G for some prime p. We will not make use of the fact that $p = 2$.) We set $Z = \mathbf{Z}(H)$. Since G does not contain a nontrivial normal p-subgroup, $\mathbf{N}_G(Z) = H$. Since G has class at most 2, we have $H' \leq \mathbf{Z}(H) = Z$.

Assume for contradiction that G is not p-normal. Then there exists a $g \in G$ such that $Z^g \leq H$, but $Z^g \neq Z$. Since $H' \leq Z$, we have

$$H' \leq \langle Z, Z^g \rangle \leq H.$$

We set $J = \langle Z, Z^g \rangle$. Note that $J \trianglelefteq H$.

Since $Z^g \leq H$, we have $Z \leq \mathbf{N}_G(Z^g) = H^g$ and then

$$H'^g \leq \langle Z, Z^g \rangle = J \trianglelefteq H^g$$

as above. Now, $H \neq H^g$, so the maximality of H together with what we have shown implies

$$1 < J \triangleleft \langle H, H^g \rangle = G.$$

This contradicts the fact that G contains no nontrivial normal p-subgroup.

Hence, G is p-normal. By Grün's Theorem 3.7 (pg. 286), it follows that

$$G/G'(p) \cong \mathbf{N}_G(Z)/\mathbf{N}_G(Z)'(p) = H/H' \neq 1.$$

Since $G/G'(p)$ is an abelian p-group, we have $H' \leq H \cap G'(p)$. Then from

$$H/H' \cong G/G'(p) = G'(p)H/G'(p) \cong H/(H \cap G'(p))$$

we conclude that $H' = H \cap G'(p)$.

If $H' = 1$, then G is p-nilpotent, which contradicts b). If $H' \neq 1$, then $\mathbf{N}_G(H') = H$ and so

$$\mathbf{N}_{G'(p)}(H') = H \cap G'(p) = H'.$$

Since H' is abelian, $G'(p)$ is p-nilpotent by Burnside's Theorem 2.6 (pg. 445). But then G is p-nilpotent, contradicting what we showed in b). This is our final contradiction and the theorem is proved. □

Theorem 7.4 cannot be improved, as the following remark shows.

Remark 7.5. A Sylow 2-subgroup of the simple group $\mathrm{PSL}_2(17)$ is the dihedral group D of order 16 by Theorem II.8.10 (pg. 206). Moreover, D is a maximal subgroup of $\mathrm{PSL}_2(17)$ by Theorem II.8.27 (pg. 223). The group D has class 3.

Related to this example is the following conjecture: Let G be a nonabelian simple group. Suppose P is a Sylow p-subgroup of G that is also a maximal subgroup of G. Then $p = 2$ (this was proved in step d) of the proof of Theorem 7.4) and P is a dihedral group. Moreover, $G \cong \mathrm{PSL}_2(q)$, where q is a prime of the form $q = 2^n \pm 1$ or $q = 9$.[5]

Thompson proved the following theorem (Thompson [430]): Let G be a simple group containing a nilpotent maximal subgroup P. Suppose that the Sylow 2-subgroup of P is a dihedral group or a generalized quaternion group. Then $G \cong \mathrm{PSL}_2(q)$, where q is a prime of the form $q = 2^n \pm 1$ or $q = 9$.

8 Groups with regular Sylow subgroups

In this section, we show that transfer to a regular Sylow subgroup is a particularly favorable situation. (For the definition of a regular p-group, see Chapter III, §10.) Our objective is to prove the following theorem:

Theorem 8.1 (Wielandt). *Let P be a regular Sylow p-subgroup of G. Let $\mathbf{O}^p(N)$ be the smallest normal subgroup of $N = \mathbf{N}_G(P)$ with p-power index, and let $\mathbf{O}^p(G)$ be the smallest normal subgroup of G with p-power index. Then we have*

$$N/\mathbf{O}^p(N) \cong G/\mathbf{O}^p(G).$$

(*Note that in this theorem, we compare the p-factor groups of $\mathbf{N}_G(P)$ with those of G. In contrast, in Grün's Theorem 3.7 we compare the p-factor groups of $\mathbf{N}_G(\mathbf{Z}(P))$ with those of G.*)

Before proceeding with the proof of Theorem 8.1, we prove two lemmas.

Lemma 8.2. *Let $\mathbf{M}$ be a homomorphism from G to the monomial group $\mathbf{M}_n(H)$, where $\mathbf{M}(g) = (f, \mathbf{P}(g))$. Let T be an orbit of $\{\mathbf{P}(g) \mid g \in G\}$ on $\{1, \ldots, n\}$, and let $|T| = k$. Then the map*

$$g \mapsto \mathbf{M}_T(g) = (f|_T, \mathbf{P}(g)|_T)$$

is a homomorphism from G to $\mathbf{M}_k(H)$.

Proof. The claim follows immediately from the multiplication rule in the monomial groups. □

Now we come to the most important lemma:

[5] This conjecture was proved in B. Baumann, Endliche nichtauflösbare Gruppen mit einer nilpotenten maximalen Untergruppe, J. Algebra **38** (1976), no. 1, 119–135. For more recent results in a similar direction, see R. M. Guralnick, G. Malle and G. Navarro Ortega, Self-normalizing Sylow subgroups, Proc. Amer. Math. Soc. **132** (2004), no. 4, 973–979; as well as B. Baumeister, T. C. Burness, R. M. Guralnick, H. P. Tong-Viet, On the maximal overgroups of Sylow subgroups of finite groups, Adv. Math. **444** (2024), Paper No. 109632. *trans.*

Lemma 8.3. *Let P be a regular Sylow p-subgroup of G, and let $N = \mathbf{N}_G(P)$. Let $\overline{N}$ be a normal subgroup of N with abelian p-factor group $N/\overline{N}$. Let $G = \bigcup_{i=1}^{n} Nr_i$ be the decomposition of G into cosets of N with $r_1 = 1$. For all $g \in G$, we write $r_i g = n_i(g) r_{i^{\mathbf{P}(g)}}$ with $n_i(g) \in N$. Let $\mathbf{M}$ be the homomorphism from G to $\mathbf{M}_n(N/\overline{N})$ constructed according to Theorem 1.4 and Lemma 1.3, so*

$$\mathbf{M}(g) = (g', \mathbf{P}(g)) \quad \text{and} \quad g'(i) = n_i(g)\overline{N}.$$

Finally, let $x \in P$ with $\mathbf{M}(x) = (h, \mathbf{P}(x))$. We denote the transfer map from G to $N/\overline{N}$ by $\mathbf{V}$. Then $\mathbf{V}(x) = h(1)^n$, and for all i such that $i^{\mathbf{P}(x)} = i$, we have $h(i) = h(1)$.

Proof. We prove the claims by induction on $|x|$. For $x = 1$, the claim is clearly true.

For every $g \in G$, we define the subset $\mathsf{F}(g)$ of $\{1, \ldots, n\}$ by

$$\mathsf{F}(g) = \{i \mid i^{\mathbf{P}(g)} = i\}.$$

Then $1 \in \mathsf{F}(g)$ is equivalent to $Ng = N$, or $g \in N$.

a) There is a decomposition of $\Omega = \{2, \ldots, n\}$ into disjoint subsets Ω_{ij} of size p with the following properties: If $x \in P$ and $\Omega_{ij} \cap \mathsf{F}(x) \neq \emptyset$, then $\Omega_{ij} \subseteq \mathsf{F}(x)$. Moreover, there exists a $y_i \in P$ such that $\mathbf{P}(y_i)|_{\Omega_{ij}}$ is a p-cycle on Ω_{ij}:

Let $\Omega = \bigcup_{i=1}^{t} \Lambda_i$ be the decomposition of Ω into orbits of $\mathbf{P}(P) = \{\mathbf{P}(x) \mid x \in P\}$. Assume for contradiction there is a $j \in \Omega$ that is fixed by $\mathbf{P}(P)$. Then $Nr_jP = Nr_j$, so $r_jPr_j^{-1} \leq N$. Since P is the only Sylow p-subgroup of $N = \mathbf{N}_G(P)$, we have $r_jPr_j^{-1} = P$. Then $r_j \in N = \mathbf{N}_G(P)$ and so $j = 1$, but $1 \notin \Omega$. So none of the Λ_i have size 1.

For $x \in P$, we define $\mathbf{P}_i(x) = \mathbf{P}(x)|_{\Lambda_i}$ and $\mathbf{P}_i(P) = \{\mathbf{P}_i(x) \mid x \in P\}$. Choose $y_i \in P$ such that $1 \neq \mathbf{P}_i(y_i) \in \mathbf{Z}(\mathbf{P}_i(P))$ and $\mathbf{P}_i(y_i)^p = 1$. Then on Λ_i, $\mathbf{P}_i(y_i)$ is a product of p-cycles on some subsets Ω_{ij} $(j = 1, \ldots, s_i)$, and $\mathbf{P}_i(y_i)$ has no fixed points. These Ω_{ij} $(i = 1, \ldots, t;\ j = 1, \ldots, s_i)$ form the desired decomposition of Ω.

Let $k \in \Lambda_i \cap \mathsf{F}(x)$. Then k lies in one of the sets Ω_{ij}. Since $\mathbf{P}_i(x)$ commutes with $\mathbf{P}_i(y_i) \in \mathbf{Z}(\mathbf{P}_i(P))$, all points in Ω_{ij} are fixed by $\mathbf{P}(x)$. So we have $\Omega_{ij} \subseteq \mathsf{F}(x)$.

b) Let $x \in P$ and $\mathbf{M}(x) = (h, \mathbf{P}(x))$. For all $i \in \mathsf{F}(x)$, we have $h(i)^p = h(1)^p$:

Let $i \in \mathsf{F}(x)$. Then $i \in \mathsf{F}(x^p)$ as well. Furthermore, $\mathbf{M}(x^p) = (f, \mathbf{P}(x^p))$, where $f(1) = h(1)^p$ and $f(i) = h(i)^p$. Thus, $h(i)^p = h(1)^p$ by induction.

c) Suppose $\Omega_{ij} \subseteq \mathsf{F}(x)$. Then $\prod_{l \in \Omega_{ij}} h(l) = h(1)^p$:

We consider the subgroup $\langle x, y \rangle$ of P, where $y = y_i$ from a). Using the construction in Lemma 8.2, we also consider the monomial representation $\mathbf{M}^* = \mathbf{M}_{\Omega_{ij}}$ stemming from the orbit Ω_{ij} of this subgroup. By a), we may assume that $\Omega_{ij} = \{2, \ldots, p+1\}$ and $\mathbf{P}^*(y) = (2, \ldots, p+1)$. Then $\mathbf{M}^*(y) = (f, \mathbf{P}^*(y))$ for some map f from $\{2, \ldots, p+1\}$ to $N/\overline{N}$. Let $\mathbf{M}(x) = (h, \mathbf{P}(x))$ as in b). Since $\Omega_{ij} \subseteq \mathsf{F}(x)$, we have by b) that

$$\mathbf{M}^*(x) = (h|_{\Omega_{ij}}, 1) \quad \text{with} \quad h(l)^p = h(1)^p \quad \text{for} \quad l = 2, \ldots, p+1.$$

If we define elements $a_1 = (m_1, 1)$ and $a_2 = (m_2, 1)$ with $m_1(l) = h(1)$ and $m_2(l) = h(l)h(1)^{-1}$ for all $l = 2, \ldots, p+1$, then $\mathbf{M}^*(x) = (m_1, 1)(m_2, 1) = a_1 a_2$. Since $a_1 = (m_1, 1) \in \mathbf{Z}(\mathbf{M}^*(N/\overline{N}))$, we have

$$[\mathbf{M}^*(y), \mathbf{M}^*(x)] = [\mathbf{M}^*(y), a_2].$$

Note that $a_2^p = 1$ since $m_2(l)^p = h(l)^p h(1)^{-p} = \overline{N}$. The regularity of P implies that $\langle \mathbf{M}^*(x), \mathbf{M}^*(y) \rangle$ is regular. Since $a_2^p = 1$, we have by Theorem III.10.6 b) (pg. 343) that

$$[\mathbf{M}^*(y), \mathbf{M}^*(x)]^p = 1.$$

Hence, the commutator subgroup $\langle \mathbf{M}^*(x), \mathbf{M}^*(y) \rangle'$ of $\langle \mathbf{M}^*(x), \mathbf{M}^*(y) \rangle$, which is generated by the conjugates of $[\mathbf{M}^*(y), \mathbf{M}^*(x)]$ in $\langle \mathbf{M}^*(x), \mathbf{M}^*(y) \rangle$, has exponent p as well. Then the regularity of $\langle \mathbf{M}^*(x), \mathbf{M}^*(y) \rangle$ implies

$$(\mathbf{M}^*(x)\mathbf{M}^*(y))^p = \mathbf{M}^*(x)^p \mathbf{M}^*(y)^p.$$

By b), we have

$$\mathbf{M}^*(x)^p = (u, 1) \quad \text{with} \quad u(k) = h(k) = h(1)^p \quad \text{for} \quad 2 \leq k \leq p+1,$$

furthermore,

$$\mathbf{M}^*(y)^p = (v, 1) \quad \text{with} \quad v(k) = f(2) \cdots f(p+1) \quad \text{for} \quad 2 \leq k \leq p+1$$

and

$$(\mathbf{M}^*(x)\mathbf{M}^*(y))^p = (w, 1) \quad \text{with} \quad w(k) = \prod_{l=2}^{p+1} f(l) \prod_{l=2}^{p+1} h(l).$$

This yields

$$\prod_{l=2}^{p+1} h(l) = h(1)^p.$$

d) If $x \in P$ and Δ is a p^k-cycle of $\mathbf{P}(x)$ with $k > 0$, then

$$\prod_{i \in \Delta} h(i) = h(1)^{p^k} :$$

Let

$$\mathbf{M}(x^{p^k}) = (f, \mathbf{P}(x^{p^k})).$$

For $i \in \Delta$, we have $i \in \mathsf{F}(x^{p^k})$ and $f(i) = \prod_{j\in\Delta} h(j)$. Furthermore, it is clear that $f(1) = h(1)^{p^k}$. Since $|x^{p^k}| < |x|$, our claim holds for x^{p^k}, and so

$$h(1)^{p^k} = f(1) = f(i) = \prod_{j\in\Delta} h(j).$$

e) $\mathbf{V}(x) = h(1)^n$:

We decompose $\{1, \dots, n\}$ into cycles with respect to the action of $\mathbf{P}(x)$. By d), every cycle of length p^k with $k > 0$ contributes exactly $h(1)^{p^k}$ to $\mathbf{V}(x)$. Following a), we group the fixed points of $\mathbf{P}(x)$ not equal to 1 into sets Ω_{ij} of size p. By b) and c), we have

$$\prod_{l\in\Omega_{ij}} h(l) = h(1)^p.$$

This yields the claim $\mathbf{V}(x) = h(1)^n$.

f) It remains to show that $h(1) = h(i)$ for $i \in \mathsf{F}(x)$:

Let $i^{\mathbf{P}(x)} = i \neq 1$. By the transitivity of $\{\mathbf{P}(g) \mid g \in G\}$ acting on $\{1, \dots, n\}$, there is a $g \in G$ such that $i^{\mathbf{P}(g)} = 1$. Now we have

$$\mathbf{M}(g)^{-1}\mathbf{M}(x)\mathbf{M}(g) = \mathbf{M}(x^g) = (k, \mathbf{P}(x^g)),$$

where $k(1) = h(1^{\mathbf{P}(g)^{-1}}) = h(i)$ and $1^{\mathbf{P}(x^g)} = 1$. This forces $x^g \in N$ by the remark at the beginning of the proof. Then in fact $x^g \in P$ since $|x^g| = |x|$ is a power of p. By e), it now follows that

$$h(1)^n = \mathbf{V}(x) = \mathbf{V}(x^g) = h(i)^n.$$

Since $(n, p) = 1$, we conclude that $h(1) = h(i)$ for $i \in \mathsf{F}(x)$. □

8.4 (Proof of Theorem 8.1).

a) If $N/\overline{N}$ is an abelian p-factor group of N, then there exists a normal subgroup $\overline{G}$ of G with $G/\overline{G} \cong N/\overline{N}$ and $\overline{G} \cap N = \overline{N}$:

Let $\mathbf{V}$ be the transfer map from G to $N/\overline{N}$ constructed according to Lemma 8.3. Since $N/\overline{N}$ is a p-group, we certainly have $\mathbf{V}(G) = \mathbf{V}(P)$ and $N = P\overline{N}$.

For $x \in P$ with $\mathbf{M}(x) = (h, \mathbf{P}(x))$, we have $\mathbf{V}(x) = h(1)^n$ by Lemma 8.3. Now assume that $\mathbf{V}(x) = h(1)^n = \overline{N}$. Then $h(1) = \overline{N}$ as well because $(p, n) = 1$. Since $h(1) = n_1(x)\overline{N}$ and $r_1 = 1$, this is equivalent to $x = n_1(x) \in \overline{N}$. Hence, $\ker \mathbf{V} \cap P = \overline{N} \cap P$ and

$$\mathbf{V}(P) \cong P/(\overline{N} \cap P) \cong P\overline{N}/\overline{N} = N/\overline{N}.$$

Letting $\overline{G} = \ker \mathbf{V}$, we have $G/\overline{G} \cong N/\overline{N}$ and

$$\overline{G} \cap N = \ker \mathbf{V} \cap \overline{N}P = \overline{N}(\ker \mathbf{V} \cap P) = \overline{N}.$$

b) Now let $N/\mathbf{O}^p(N)$ be the largest p-factor group of N. If $N/\mathbf{O}^p(N)$ is cyclic, then we are done by a). Otherwise, there exist two distinct N_i with $\mathbf{O}^p(N) < N_i \lhd N$ and $|N/N_i| = p$ $(i = 1, 2)$. Then $P_i = N_i \cap P$ is a Sylow p-subgroup of N_i. If $\mathbf{N}_G(P_i) = G$ for $i = 1, 2$, then $P = P_1P_2 \trianglelefteq G$. In this case, $N = \mathbf{N}_G(P) = G$ and the theorem is trivial.

So we may assume that, say, $H = \mathbf{N}_G(P_1) < G$. By a), there exists some $G_1 \lhd G$ such that $G/G_1 \cong N/N_1$ and $G_1 \cap N = N_1$. So $|G/G_1| = p$. Since $P_1 = N_1 \cap P \lhd N = \mathbf{N}_G(P)$, we have $N \leq H = \mathbf{N}_G(P_1)$. Applying the induction hypothesis to $H = \mathbf{N}_G(P_1)$, we conclude that the maximal p-factor group $H/\mathbf{O}^p(H)$ of H is isomorphic to $N/\mathbf{O}^p(N)$; moreover, $\mathbf{O}^p(N) = N \cap \mathbf{O}^p(H)$. We define $H_1 = \mathbf{N}_{G_1}(P_1)$. It follows from $|G/G_1| = p$ that

$$\mathbf{O}^p(H) \leq H \cap G_1 = H_1 \trianglelefteq H,$$

and hence $H_1/\mathbf{O}^p(H)$ is the maximal p-factor group of H_1. Since $H_1 = \mathbf{N}_{G_1}(P_1)$, the maximal p-factor group $G_1/\mathbf{O}^p(G)$ of G_1 is isomorphic to $H_1/\mathbf{O}^p(H)$ by induction, and $\mathbf{O}^p(H) = H_1 \cap \mathbf{O}^p(G)$. Now, we have

$$|G/\mathbf{O}^p(G)| = p|G_1/\mathbf{O}^p(G)| = p|H_1/\mathbf{O}^p(H)| = |H/\mathbf{O}^p(H)| = |N/\mathbf{O}^p(N)|.$$

It follows from $G = \mathbf{O}^p(G)N$ that

$$G/\mathbf{O}^p(G) \cong N/(N \cap \mathbf{O}^p(G)).$$

Since $\mathbf{O}^p(N) \leq N \cap \mathbf{O}^p(G)$, we conclude that $\mathbf{O}^p(N) = N \cap \mathbf{O}^p(G)$. □

In preparation for an application of Theorem 8.1, we prove a lemma.

Lemma 8.5. *Assume N contains a normal Sylow p-subgroup P with $p > 2$. Furthermore, assume P is metacyclic and nonabelian. Then N has nontrivial p-factor groups.*

Proof. By Theorem III.10.2 c) (pg. 339), P is regular.

a) P contains a characteristic subgroup of index p: Let p^m be the exponent of P. If $|\mho_{m-1}(P)| = p$, then $\Omega_{m-1}(P)$ is a characteristic subgroup of P of index p since

$$p = |\mho_{m-1}(P)| = |P/\Omega_{m-1}(P)| \quad \text{(see Theorem III.10.7, pg. 344)}.$$

So assume $|\mho_{m-1}(P)| = p^2$. Then $\mho_{m-1}(P)$ has type (p, p) and thus $\mho_{m-1}(P) \not\leq P'$. We let

$$\Omega_{m-1}(P/P') = U/P'.$$

By Theorem III.10.7 (pg. 344), we have

$$\begin{aligned}[P : U] &= [P/P' : U/P'] = [P/P' : \Omega_{m-1}(P/P')] \\ &= |\mho_{m-1}(P/P')| = |\mho_{m-1}(P)P'/P'| \\ &= |\mho_{m-1}(P)/\mho_{m-1}(P) \cap P'| = p.\end{aligned}$$

Thus, U is the desired characteristic subgroup of index p.

b) By the Zassenhaus theorem, P has a complement S in N. Let P_1 be the characteristic subgroup of P having index p that we established in a). Since $p \nmid |S|$, $P/\Phi(P)$ is a completely reducible S-module by Maschke's theorem (see Theorem I.17.7), say

$$P/\Phi(P) = P_1/\Phi(P) \times P_2/\Phi(P).$$

Let $P_i = \langle x_i, \Phi(P)\rangle$ and

$$x_i^s \equiv x_i^{a_i(s)} \mod \Phi(P)$$

for $i = 1, 2$ and $s \in S$. Since $\Phi(P) = \mho_1(P)$ (see Theorem III.11.4, pg. 356) and

$$[P, \mho_1(P)] = \mho_1(P') \quad \text{(see Theorem III.10.8 a), pg. 345)}$$

we have

$$[x_1, x_2]^s \equiv [x_1, x_2]^{a_1(s)a_2(s)} \mod \mho_1(P').$$

Letting $\Omega_1(P') = \langle k\rangle$, we have

$$k^s = k^{a_1(s)a_2(s)}$$

for all $s \in S$.

Since P is regular, it follows from Theorem III.10.8 g) (pg. 345) that

$$(xy)^{p^{m-1}} = x^{p^{m-1}} y^{p^{m-1}} t^{p^{m-1}}$$

for some $t \in P'$. By Theorem III.11.4 (pg. 356), we have $P' \leq \mho_1(P)$ and thus $t^{p^{m-1}} = 1$. Hence, the map μ from P onto $\mho_{m-1}(P)$ defined by $x^\mu = x^{p^{m-1}}$ is an S-homomorphism whose kernel contains $\mho_1(P) = \Phi(P)$.

First assume that $\Omega_1(P') \leq \mho_{m-1}(P) = P^\mu$. Then $\Omega_1(P')$ is isomorphic as an S-module to a submodule of $P/\mho_1(P)$. This forces

$$a_1(s)a_2(s) \equiv a_i(s) \mod p$$

for $i = 1$ or $i = 2$ and all $s \in S$. So $a_j(s) \equiv 1 \bmod p$ for $j \neq i$. Then $N' \leq P_i S$, and N has a nontrivial p-factor group $N/P_i S$.

Finally, assume $\Omega_1(P') \not\leq \mho_{m-1}(P)$. Then all the assumptions of the theorem hold in $N/\mho_{m-1}(P)$ with the nonabelian, metacyclic Sylow p-subgroup $P/\mho_{m-1}(P)$. By induction, $N/\mho_{m-1}(P)$ has a nontrivial p-factor group, so N does as well. □

Theorem 8.6 (Huppert [261]). *Let $p > 2$. Assume that the Sylow p-subgroups of G are metacyclic and nonabelian. Then G has a nontrivial p-factor group. In particular, G is not simple.*

Proof. Let $N = \mathbf{N}_G(P)$. By Lemma 8.5, N has a nontrivial p-factor group. Then since the metacyclic p-group P is regular, G itself has a nontrivial p-factor group by Theorem 8.1. □

Remark 8.7. Theorem 8.6 cannot be improved to include $p = 2$. By Theorem II.8.10 (pg. 206), the Sylow 2-subgroups of $\mathrm{PSL}_2(7)$ are dihedral groups of order 8, and $\mathrm{PSL}_2(7)$ is simple. The Sylow 2-subgroups of $\mathrm{SL}_2(5)$ are quaternion groups of order 8, but $\mathrm{SL}_2(5)$ has no nontrivial 2-factor group. All known statements regarding groups with metacyclic Sylow 2-subgroups are deep and have not admitted a proof without the use of group characters at the time of writing.

The assumption that P is nonabelian cannot be removed from Theorem 8.6, as illustrated by the simple groups $\mathrm{PSL}_2(p^2)$ with elementary abelian Sylow p-subgroups of type (p, p).

Example 8.8 (Wielandt [462]). We illustrate with an example that the assumption that P is regular in Theorem 8.1 cannot be removed.

Let $Z = \langle z \rangle$ be the cyclic group of order $p > 2$. Let G be the following subgroup of the monomial group $\mathbf{M}_{p+1}(Z)$:

$$G = \left\{ (f, x) \,\middle|\, x \in S_{p+1},\ f(i) \in Z,\ \prod_{i=0}^{p} f(i) = 1 \right\}.$$

Then $|G| = p^p(p+1)!$ and

$$D = \left\{ (f, 1) \,\middle|\, \prod_{i=0}^{p} f(i) = 1 \right\}$$

is an elementary abelian normal subgroup of G of order p^p. Let $s = (h, r)$ with $r = (0)(1, \ldots, p)$ and $h(i) = 1$ for all i. Then $P = \langle D, s \rangle$ is a Sylow p-subgroup of G. The normalizer $N = \mathbf{N}_G(P)$ only contains elements (f, x) with $0^x = 0$. Hence, the map defined by

$$(f, x) \mapsto f(0)$$

is a homomorphism from N onto Z. So N has a nontrivial p-factor group.

Assume for contradiction there is some $\overline{G} \lhd G$ with $|G/\overline{G}| = p^a > 1$. Then

$$\overline{G}D/D \trianglelefteq G/D.$$

Since G/D is isomorphic to the symmetric group S_{p+1}, which has no nontrivial p-factor groups when $p > 2$, we have $G = \overline{G}D$. So the quotient

$$G/\overline{G} \cong D/(\overline{G} \cap D)$$

is abelian. Let[6]

[6] Correction 26 from errata.

$$t = (u, (0, 1)) \quad \text{with } u(i) = 1 \text{ for all } i \text{ and}$$
$$d = (v, 1) \quad \text{with } v(0) = 1,\ v(i) = z \text{ for all } i \geq 1.$$

Then $t, d \in G$ and

$$[t, d] = (g, 1) \quad \text{with} \quad g(0) = z^{-1},\ g(1) = z,\ g(i) = 1 \quad \text{for } i \geq 2.$$

The conjugates of $[t, d]$ generate all of D. So $D \leq G' \leq \overline{G}$, which yields the contradiction $G = \overline{G}D = \overline{G}$. Thus, G does not have a nontrivial p-factor group.

Notes for Chapter IV

§1: Exercise 1 was communicated to me in a conversation with H. Wielandt; the same result was proved in Gaschütz [164] using different methods.

§2: See Grün [195] for Theorem 2.4. The hint for Exercise 2 is a modification of the proof in Brandis [45].

§3: The fundamental theorems of Grün are contained in Grün [194]. Theorem 3.5 was proved by other means (namely those of D. G. Higman [232]) in Wong [484].

§4: A proof of Theorem 4.7 using cohomological methods was given in Tate [425].

§5: Much of Theorem 5.4 is in Itô [270], Theorem 5.5 a) in Huppert [262]. Theorem 5.7 can be found in Itô [276], but it was proved at the same time by W. Gaschütz. Theorem 5.12 is in Huppert [262].

§6: The fundamental Theorem 6.2 was first proved (in a somewhat weaker form) in Thompson [428]. The proof given here follows Thompson [432].

§7: Theorem 7.4 is partially given in Thompson [428] and Deskins [110]. The complete result is contained in Janko [290].

§8: Theorem 8.1 is a very special case of the results in Wielandt [462].

Chapter V
Representation theory

The objects of study in this chapter are the so-called representations; that is, homomorphisms from a given group G to the general linear group $\mathrm{GL}_n(\mathsf{K})$ ($n = 1, 2, \dots$) over a given field K. In the first phase of the theory (Frobenius, Burnside and I. Schur, approximately 1890–1910), the field K was usually the complex numbers. However, much of the theory can be established in the same way over algebraically closed fields of positive characteristic, as long as the characteristic of the field does not divide the order of the group. Namely, it is this condition that establishes the fundamental semisimplicity of the group ring $\mathsf{K}[G]$. We do not address the case in which the characteristic of the field divides the group order, so-called modular representation theory (see Curtis, Reiner [102], Chapter XII). One can remove the assumption that the field is algebraically closed with a certain amount of success (see §11–§14); this is important in some applications to solvable groups, where representations of the group over prime fields arise via automorphisms of its chief factors.

The modern algebraic approach to representation theory stems from E. Noether, in which the main theorems are an application of the theory of semisimple algebras and their representations (§2–§4). It is of fundamental importance to the applicability of representation theory that one can largely avoid calculating with the linear maps $\mathbf{D}(g)$ and instead calculate with the character χ of the representation $\mathbf{D}$; here the map χ from G to K is defined by $\chi(g) = \operatorname{tr}\mathbf{D}(g)$. Using the tensor product (Kronecker product) of representations, the set of generalized characters inherits a ring structure (§10). The two cornerstones of the theory were already known to Frobenius:

(1) The orthogonality relations of Theorem 5.8, which show that the characters of irreducible representations form an orthonormal basis for the space of class functions on the group (equipped with a suitable scalar product).

(2) The method of induced characters, which associates to every character χ of a subgroup U of G a character χ^G of G. For calculations with induced characters, the Frobenius reciprocity described in Theorem 16.5 is of fundamental importance. Frobenius reciprocity relates, for a subgroup $U \leq G$, the induction map $\chi \mapsto \chi^G$ (χ a character of U) and the restriction map $\psi \mapsto \psi_U$ (ψ a character of G). More detailed knowledge is available when U is a normal subgroup of G (§17). There

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B. Huppert, *Finite Groups I*, Grundlehren der mathematischen Wissenschaften 364,
https://doi.org/10.1007/978-3-031-87529-8_5

is not yet a completely satisfactory definition of characters that is independent of representations. However, the generalized characters of G can be recognized as the class functions on G whose restrictions to certain "small" subgroups of G are generalized characters of these subgroups (§19). This is occasionally useful in the construction of special characters of G that yield information about the structure of G (see the proof of Application 19.12 and Theorem 19.13).

The theory of characters yields deep statements about the existence of normal subgroups, which arise as kernels of representations. Representation theory is the only effective method for proving the existence of nonsolvable factor groups; compare this to the remarks at the beginning of Chapter IV. In this chapter, we treat the classical theorems of Burnside (Theorem 7.3) and Frobenius (Theorem 7.6), the theorems of Burnside and Itô regarding permutation groups of prime degree (Theorem 21.3 and Theorem 21.6), as well as the theorem of Brauer and Suzuki (Theorem 22.9) proving the nonsimplicity of groups whose Sylow 2-subgroups are generalized quaternion. In §8, we prove some theorems regarding Frobenius groups that are important in numerous applications. Many more theorems have been proved in which character theory plays a crucial role.[1] Some of these results, which are based on the theory of so-called exceptional characters, will be addressed in Volume 3.

1 Algebras and their representations

In the first four sections of this chapter, we will develop the structure theory of algebras only as far as is necessary for representation theory. Although the entire theory holds for rings satisfying the minimal condition for right ideals, we restrict our attention to algebras of finite dimension.

Definition 1.1. Let K be a field. An algebra over K is a set $\mathfrak{a}$ with the following properties:

(1) With respect to an operation $+$, the set $\mathfrak{a}$ is a vector space of finite dimension over K. The dimension of an algebra $\mathfrak{a}$ over K always refers to the dimension of $\mathfrak{a}$ over K as a vector space.

(2) The set $\mathfrak{a}$ is equipped with an associative and two-sided distributive multiplication; that is, for all $a, b, c \in \mathfrak{a}$ we have

$$a(bc) = (ab)c, \quad a(b+c) = ab + ac, \quad (b+c)a = ba + ca.$$

[1] The character theory of finite groups has advanced considerably in light of the classification of finite simple groups and the Deligne–Lusztig theory of representations of finite groups of Lie type, particularly with respect to local-global conjectures. For an introductory survey, see G. Malle, Local-global conjectures in the representation theory of finite groups, in *Representation theory—current trends and perspectives*, 519–539, EMS Ser. Congr. Rep, Eur. Math. Soc., Zürich, 2017. For a textbook reference, see G. Navarro Ortega, *Character theory and the McKay conjecture*, Cambridge Studies in Advanced Mathematics, 175, Cambridge Univ. Press, Cambridge, 2018. Much progress has been made since the publication of these works. *trans.*

(3) For all $a, b \in \mathfrak{a}$ and all $k \in \mathsf{K}$ we have

$$(ka)b = a(kb) = k(ab).$$

(4) There exists an element $1 \in \mathfrak{a}$ such that $1a = a1 = a$ for all $a \in \mathfrak{a}$.

Definition 1.2. A right ideal $\mathfrak{r}$ of an algebra $\mathfrak{a}$ over K is a subset $\mathfrak{r}$ of $\mathfrak{a}$ with the following properties:
(1) $\mathfrak{r}$ is a K-vector subspace of the K-vector space $\mathfrak{a}$.
(2) For $a \in \mathfrak{a}$ and $r \in \mathfrak{r}$, we have $ra \in \mathfrak{r}$.

We define left ideals of $\mathfrak{a}$ in the corresponding fashion. If $\mathfrak{i}$ is a right ideal and a left ideal, we say that $\mathfrak{i}$ is a two-sided ideal of $\mathfrak{a}$.

A straightforward calculation proves the following theorem.

Theorem 1.3. *Let $\mathfrak{r}_1$ and $\mathfrak{r}_2$ be right ideals of $\mathfrak{a}$. We define*

$$\mathfrak{r}_1 + \mathfrak{r}_2 = \{r_1 + r_2 \mid r_i \in \mathfrak{r}_i\}$$

and

$$\mathfrak{r}_1\mathfrak{r}_2 = \Big\{ \sum_{i=1}^{n} a_i b_i \,\Big|\, a_i \in \mathfrak{r}_1,\ b_i \in \mathfrak{r}_2,\ n = 1, 2, 3, \dots \Big\}.$$

Then $\mathfrak{r}_1 + \mathfrak{r}_2$ and $\mathfrak{r}_1\mathfrak{r}_2$ are right ideals of $\mathfrak{a}$. In fact, the set $\mathfrak{r}_1\mathfrak{r}_2$ is already a right ideal when $\mathfrak{r}_2$ is a right ideal and $\mathfrak{r}_1$ is any subset of $\mathfrak{a}$. The following associative and distributive laws hold for the operations on ideals introduced above:

$$\begin{aligned} \mathfrak{r}_1 + (\mathfrak{r}_2 + \mathfrak{r}_3) &= (\mathfrak{r}_1 + \mathfrak{r}_2) + \mathfrak{r}_3, \\ \mathfrak{r}_1(\mathfrak{r}_2\mathfrak{r}_3) &= (\mathfrak{r}_1\mathfrak{r}_2)\mathfrak{r}_3, \\ \mathfrak{r}_1(\mathfrak{r}_2 + \mathfrak{r}_3) &= \mathfrak{r}_1\mathfrak{r}_2 + \mathfrak{r}_1\mathfrak{r}_3, \\ (\mathfrak{r}_1 + \mathfrak{r}_2)\mathfrak{r}_3 &= \mathfrak{r}_1\mathfrak{r}_3 + \mathfrak{r}_2\mathfrak{r}_3. \end{aligned}$$

If $\mathfrak{i}_1$ and $\mathfrak{i}_2$ are two-sided ideals, then $\mathfrak{i}_1 + \mathfrak{i}_2$ and $\mathfrak{i}_1\mathfrak{i}_2$ are also two-sided ideals.

Definition 1.4. Let $\mathfrak{a}_1$ and $\mathfrak{a}_2$ be algebras over K. A map μ from $\mathfrak{a}_1$ to $\mathfrak{a}_2$ is said to be a homomorphism if for all $a, b \in \mathfrak{a}_1$ and all $k \in \mathsf{K}$ we have

$$\begin{aligned} (a + b)\mu &= a\mu + b\mu, \\ (ab)\mu &= (a\mu)(b\mu), \\ (ka)\mu &= k(a\mu). \end{aligned}$$

In particular, μ is a linear map from the K-vector space $\mathfrak{a}_1$ to the K-vector space $\mathfrak{a}_2$. If μ is a homomorphism and a bijective map from $\mathfrak{a}_1$ to $\mathfrak{a}_2$, then μ is said to be an isomorphism; we then say that $\mathfrak{a}_1$ and $\mathfrak{a}_2$ are isomorphic and write $\mathfrak{a}_1 \cong \mathfrak{a}_2$.

Definition 1.5. Let $\mathfrak{i}$ be a two-sided ideal of the algebra $\mathfrak{a}$, and let $\mathfrak{a}/\mathfrak{i}$ be the quotient vector space of the vector space $\mathfrak{a}$ by the subspace $\mathfrak{i}$. Then $\mathfrak{a}/\mathfrak{i}$ becomes an algebra over K upon defining

(1) $(a + \mathfrak{i}) + (b + \mathfrak{i}) = (a + b) + \mathfrak{i}$.
(2) $k(a + \mathfrak{i}) = ka + \mathfrak{i}$.
(3) $(a + \mathfrak{i})(b + \mathfrak{i}) = ab + \mathfrak{i}$.

(The assignment in (3) is well defined since $\mathfrak{i}$ is a two-sided ideal of $\mathfrak{a}$.) We call $\mathfrak{a}/\mathfrak{i}$ the quotient algebra of $\mathfrak{a}$ by $\mathfrak{i}$. It is clear that

$$\dim_{\mathsf{K}} \mathfrak{a}/\mathfrak{i} = \dim_{\mathsf{K}} \mathfrak{a} - \dim_{\mathsf{K}} \mathfrak{i}.$$

The following theorem is also trivial.

Homomorphism Theorem 1.6. *Let μ be an epimorphism from the algebra $\mathfrak{a}_1$ to the algebra $\mathfrak{a}_2$. Then*

$$\mathfrak{k} = \{a \in \mathfrak{a}_1 \mid a\mu = 0\}$$

is a two-sided ideal of $\mathfrak{a}_1$. We call $\mathfrak{k}$ the kernel of μ. The map τ defined by

$$(a + \mathfrak{k})\tau = a\mu$$

is an isomorphism from $\mathfrak{a}_1/\mathfrak{k}$ to $\mathfrak{a}_2$.

Conversely, if $\mathfrak{k}$ is a two-sided ideal of $\mathfrak{a}_1$, then the map ν defined by $a\nu = a + \mathfrak{k}$ is an epimorphism from $\mathfrak{a}_1$ to $\mathfrak{a}_1/\mathfrak{k}$ with kernel $\mathfrak{k}$.

Definition 1.7. Let $\mathfrak{a}$ be an algebra over K. A right $\mathfrak{a}$-module V is defined to be a finite-dimensional vector space V over K equipped with a right multiplication by elements of $\mathfrak{a}$ with the following properties:

(1) $v(a_1 + a_2) = va_1 + va_2$,
(2) $v(a_1a_2) = (va_1)a_2$,
(3) $v1 = v$,
(4) $(v_1 + v_2)a = v_1a + v_2a$,
(5) $(kv)a = v(ka) = k(va)$

for all $v_1, v_2, v \in \mathsf{V}$, for all $a_1, a_2, a \in \mathfrak{a}$ and for all $k \in \mathsf{K}$. Note that this definition is more restricted than the definition of a module over a ring given in Chapter I, §10.

Properties (4) and (5) mean that the mapping of v to va is a K-linear map from V to itself; properties (1), (2) and (5) mean that the mapping of $a \in \mathfrak{a}$ to the endomorphism ϵ_a of V defined by $v\epsilon_a = va$ is a homomorphism from $\mathfrak{a}$ to the K-algebra $\mathrm{Hom}_{\mathsf{K}}(\mathsf{V}, \mathsf{V})$.

Left $\mathfrak{a}$-modules are defined in the corresponding fashion. We will consider right modules almost exclusively and often refer to them simply as $\mathfrak{a}$-modules. (In our treatment of the tensor product of modules in §9, however, we will need to consider both right modules and left modules.)

The concepts of submodule, quotient module, direct sum of modules, homomorphisms of modules and isomorphisms of modules are to be understood in the sense of Chapter I, §10. Module homomorphisms are always K-linear maps since

$$(kv)\mu = (v(k1))\mu = (v\mu)(k1) = k(v\mu).$$

If W_1 and W_2 are submodules of V, then

$$\mathsf{W}_1 + \mathsf{W}_2 = \{w_1 + w_2 \mid w_i \in \mathsf{W}_i\}$$

is also a submodule of V. If W is a subset of V and $\mathfrak{b}$ is a subset of the algebra $\mathfrak{a}$, then we define

$$\mathsf{W}\mathfrak{b} = \Big\{ \sum_i w_i b_i \,\Big|\, w_i \in \mathsf{W},\ b_i \in \mathfrak{b} \Big\}.$$

If W or $\mathfrak{b}$ is a K-vector space, then $\mathsf{W}\mathfrak{b}$ is a K-vector subspace of V. Every submodule W of V is a vector space over K since

$$w(k1) = k(w1) = kw \in \mathsf{W}$$

for all $w \in \mathsf{W}$ and all $k \in \mathsf{K}$. Thus, the maximal and minimal conditions for submodules (see Definition I.10.4) are always satisfied since every nonempty set of subspaces of a finite-dimensional vector space V always contains elements of maximal and minimal dimension.

Example 1.8.

a) Let V be a vector space of dimension n over K with basis $\{v_1, \dots, v_n\}$. Then $\mathrm{Hom}_{\mathsf{K}}(\mathsf{V}, \mathsf{V})$ is an algebra over K of dimension n^2. A K-basis of $\mathrm{Hom}_{\mathsf{K}}(\mathsf{V}, \mathsf{V})$ is given by the linear maps e_{ij} $(i, j = 1, \dots, n)$ defined by

$$v_k e_{ij} = \delta_{ik} v_j \quad (\delta_{ik} \text{ is the Kronecker delta}).$$

We have $e_{ij}e_{ks} = \delta_{jk}e_{is}$. It is well known that $\mathrm{Hom}_{\mathsf{K}}(\mathsf{V}, \mathsf{V})$ is isomorphic to the full matrix ring of degree n over K.

b) Let G be a finite group, and let K be a field. Let $\mathsf{K}[G]$ be the group ring of G over K as described in Definition I.16.6 (pg. 112). It consists of the sums

$$\sum_{g \in G} a_g g \quad (a_g \in \mathsf{K}),$$

with

$$\Big(\sum_{g \in G} a_g g \Big) + \Big(\sum_{g \in G} b_g g \Big) = \sum_{g \in G} (a_g + b_g) g,$$

$$\Big(\sum_{g \in G} a_g g \Big)\Big(\sum_{g \in G} b_g g \Big) = \sum_{g \in G} \Big(\sum_{x \in G} a_x b_{x^{-1}g} \Big) g,$$

$$k \Big(\sum_{g \in G} a_g g \Big) = \sum_{g \in G} (k a_g) g.$$

The elements $g \in G$ constitute a K-basis of $\mathsf{K}[G]$. Hence, $\dim_{\mathsf{K}} \mathsf{K}[G] = |G|$.

Definition 1.9. A representation of an algebra $\mathfrak{a}$ over K on the module V is an algebra homomorphism $\mathbf{D}$ from $\mathfrak{a}$ to the algebra $\mathrm{Hom}_{\mathsf{K}}(\mathsf{V},\mathsf{V})$ of endomorphisms of the finite-dimensional vector space V over K. Hence, we have

(1) $\mathbf{D}(a_1 + a_2) = \mathbf{D}(a_1) + \mathbf{D}(a_2)$,
(2) $\mathbf{D}(a_1 a_2) = \mathbf{D}(a_1)\mathbf{D}(a_2)$,
(3) $\mathbf{D}(ka) = k\mathbf{D}(a)$

for all $a_1, a_2, a \in \mathfrak{a}$ and all $k \in \mathsf{K}$. Furthermore, we demand that

(4) $\mathbf{D}(1) = 1_{\mathsf{V}}$.

The vector space V becomes a right $\mathfrak{a}$-module upon defining $va = v\mathbf{D}(a)$. We call V the representation module of $\mathbf{D}$. When we refer to V as an $\mathfrak{a}$-module, we always mean this module structure on V. If V is an irreducible $\mathfrak{a}$-module (see Definition I.10.4), then we say that $\mathbf{D}$ is an irreducible representation. (We recall that irreducible modules are always distinct from 0 by Definition I.10.4!) The representation $\mathbf{D}$ is said to be faithful and V a faithful $\mathfrak{a}$-module when $\mathbf{D}$ is a monomorphism from $\mathfrak{a}$ to $\mathrm{Hom}_{\mathsf{K}}(\mathsf{V},\mathsf{V})$. The dimension of V over K is called the degree of the representation $\mathbf{D}$.

Definition 1.10. Two representations $\mathbf{D}_1$, resp. $\mathbf{D}_2$, of $\mathfrak{a}$ on the modules V_1, resp. V_2, are said to be equivalent when V_1 and V_2 are isomorphic $\mathfrak{a}$-modules; that is, when there exists a bijective K-linear map τ from V_1 to V_2 such that

$$(v\mathbf{D}_1(a))\tau = (v\tau)\mathbf{D}_2(a)$$

for all $a \in \mathfrak{a}$ and all $v \in \mathsf{V}_1$.

Definition 1.11.

a) A representation $\mathbf{D}$ of a group G over the field K is a group homomorphism $\mathbf{D}$ from G to the group $\mathrm{GL}(\mathsf{V})$ of invertible linear maps from a finite-dimensional vector space $\mathsf{V} \neq 0$ over K to itself. By defining

$$\mathbf{D}\Big(\sum_{g\in G} a_g g\Big) = \sum_{g\in G} a_g \mathbf{D}(g),$$

the map $\mathbf{D}$ becomes a representation of the group ring $\mathsf{K}[G]$ on V. Conversely, every representation $\mathbf{D}$ of $\mathsf{K}[G]$ on a vector space $\mathsf{V} \neq 0$ yields a representation of G over K by restricting $\mathbf{D}$ to G; the linear maps $\mathbf{D}(g)$ are invertible since

$$\mathbf{D}(g)\mathbf{D}(g^{-1}) = \mathbf{D}(gg^{-1}) = \mathbf{D}(1) = 1_{\mathsf{V}}.$$

Hence, we need not distinguish between representations of G over K and representations of $\mathsf{K}[G]$. We say that a representation of G is irreducible when the corresponding representation of $\mathsf{K}[G]$ is irreducible. Two representations of G are said to be equivalent when the corresponding representations of $\mathsf{K}[G]$ are equivalent.

b) We say that a representation $\mathbf{D}$ of G is faithful when $\mathbf{D}$ is a monomorphism from G to $\mathrm{GL}(\mathsf{V})$; if G has a faithful representation on V, we say V is a faithful G-module. However, $\mathbf{D}$ need not necessarily be a faithful representation of the group ring $\mathsf{K}[G]$ in this case! For example, let $G = \{1, g\}$ be the cyclic group of order 2. Then the homomorphism $\mathbf{D}$ defined by $\mathbf{D}(1) = 1$ and $\mathbf{D}(g) = -1$ over any field K with $\mathrm{char}\,\mathsf{K} \neq 2$ is a faithful representation of G, but the corresponding representation of the group ring has a nontrivial kernel since $\mathbf{D}(1 + g) = 0$.

Definition 1.12. Let $\mathfrak{a}$ be an algebra over K. We give $\mathfrak{a}$ the structure of a right $\mathfrak{a}$-module in a natural way by means of its right multiplication (see Example I.10.12). The corresponding representation $\mathbf{R}$ defined by

$$a\mathbf{R}(b) = ab$$

is called the regular representation of $\mathfrak{a}$. The submodules of the right $\mathfrak{a}$-module $\mathfrak{a}$ are exactly the right ideals of $\mathfrak{a}$.

Theorem 1.13. *Let V be an irreducible $\mathfrak{a}$-module. Then there exists a maximal right ideal $\mathfrak{r}$ of $\mathfrak{a}$ such that V and $\mathfrak{a}/\mathfrak{r}$ are isomorphic as right $\mathfrak{a}$-modules.*

Proof. Let $0 \neq v \in \mathsf{V}$. Then

$$v\mathfrak{a} = \{va \mid a \in \mathfrak{a}\}$$

is a submodule of V. It is nonzero since $v1 = v \neq 0$. Since the module V is irreducible, we have $v\mathfrak{a} = \mathsf{V}$. The map μ from $\mathfrak{a}$ to V defined by $a\mu = va$ is a module epimorphism since

$$\begin{aligned}(a_1 + a_2)\mu = v(a_1 + a_2) = va_1 + va_2 = a_1\mu + a_2\mu,\\ (ka)\mu = v(ka) = k(va) = k(a\mu),\\ (a_1a_2)\mu = v(a_1a_2) = (va_1)a_2 = (a_1\mu)a_2.\end{aligned}$$

If $\mathfrak{r}$ is the kernel of μ, then $\mathfrak{r}$ is a right ideal of $\mathfrak{a}$. By the homomorphism theorem for modules, we have $\mathfrak{a}/\mathfrak{r} \cong \mathsf{V}$. Since V is irreducible, $\mathfrak{r}$ is a maximal right ideal of $\mathfrak{a}$.□

Example 1.14. Every group G admits the trivial representation $\mathbf{D}$ defined by $\mathbf{D}(g) = 1$ for all $g \in G$. This corresponds to the representation $\mathbf{D}$ of the group ring $\mathsf{K}[G]$ of G defined by

$$\mathbf{D}\left(\sum_{g\in G} a_g g\right) = \sum_{g\in G} a_g.$$

The kernel of $\mathbf{D}$ is the two-sided ideal

$$\mathfrak{i} = \left\{\sum_{g\in G} a_g g \;\middle|\; \sum_{g\in G} a_g = 0\right\}$$

of $\mathsf{K}[G]$, the so-called augmentation ideal (see Chapter III, §18).

Exercises

1) a) Let $\mathbf{D}$ be a representation of the finite group G on the vector space V over the complex numbers. Show that there exists a Hermitian positive definite scalar product $(_,_)$ on V such that $(v_1\mathbf{D}(g), v_2\mathbf{D}(g)) = (v_1, v_2)$ for all $g \in G$ and all $v_i \in \mathsf{V}$.
   b) Using a), deduce a new proof of Maschke's Theorem I.17.7 for the special case of the complex numbers.
2) Let $\mathfrak{a}$ be an algebra over K, and let $\mathfrak{m}$ be a right $\mathfrak{a}$-module in the general sense of Chapter I, §10. Show that, upon defining $km = m(k1)$, $\mathfrak{m}$ is a finite-dimensional vector space over K (so an $\mathfrak{a}$-module in the sense of Definition 1.7) if and only if $\mathfrak{m}$ is finitely generated as an $\mathfrak{a}$-module.

2 The Jacobson radical

Definition 2.1. Let $\mathfrak{a}$ be an algebra. We define

$$\mathbf{J}(\mathfrak{a}) = \{a \in \mathfrak{a} \mid \mathsf{V}a = 0 \text{ for all irreducible right } \mathfrak{a}\text{-modules } \mathsf{V}\}$$

and call $\mathbf{J}(\mathfrak{a})$ the Jacobson radical of $\mathfrak{a}$. In what follows, we often refer to the Jacobson radical simply as the radical. It is clear that $\mathbf{J}(\mathfrak{a})$ is the intersection of the kernels of all irreducible representations of $\mathfrak{a}$, so it is a two-sided ideal of $\mathfrak{a}$.

Theorem 2.2. *Let $\mathfrak{a}$ be an algebra. The following statements are equivalent for an element $x \in \mathfrak{a}$:*
a) *The element x lies in every maximal right ideal of $\mathfrak{a}$.*
b) *For all $a \in \mathfrak{a}$, $1 - xa$ has a right inverse in $\mathfrak{a}$.*
c) *The element x lies in the radical $\mathbf{J}(\mathfrak{a})$ of $\mathfrak{a}$.*

Proof. a) $\Rightarrow$ b): Assume for contradiction that $(1 - xa)\mathfrak{a} \subset \mathfrak{a}$. Then there exists a maximal right ideal $\mathfrak{r}$ of $\mathfrak{a}$ such that $(1 - xa)\mathfrak{a} \subseteq \mathfrak{r}$ (since $\mathfrak{a}$ is finite dimensional, this is trivial; the conclusion can also be obtained more generally for rings with unity using Zorn's Lemma). In particular, it follows that $1 - xa = b \in \mathfrak{r}$. Then since $x \in \mathfrak{r}$, we also have $1 = xa + b \in \mathfrak{r}$. But this contradicts that $\mathfrak{r} \subset \mathfrak{a}$. So $(1 - xa)\mathfrak{a} = \mathfrak{a}$, and thus there exists some $c \in \mathfrak{a}$ with $(1 - xa)c = 1$.

b) $\Rightarrow$ c): Let V be an irreducible right $\mathfrak{a}$-module. Assume for contradiction that $\mathsf{V}x \neq 0$. Then there exists some $v \in \mathsf{V}$ with $vx \neq 0$. Since $vx\mathfrak{a}$ is a nonzero submodule of V and V is irreducible, it follows that $vx\mathfrak{a} = \mathsf{V}$. In particular, there exists some $a \in \mathfrak{a}$ such that $vxa = v$; that is, $v(1 - xa) = 0$. By our assumption, there exists some $c \in \mathfrak{a}$ such that $(1 - xa)c = 1$. From this follows the contradiction

$$0 = v(1 - xa)c = v.$$

So in fact $Vx = 0$, and therefore x lies in the radical $\mathbf{J}(\mathfrak{a})$.

c) $\Rightarrow$ a): Let $x \in \mathbf{J}(\mathfrak{a})$, and let $\mathfrak{r}$ be a maximal right ideal of $\mathfrak{a}$. Then $\mathfrak{a}/\mathfrak{r}$ is an irreducible right $\mathfrak{a}$-module, and therefore $(\mathfrak{a}/\mathfrak{r})x = 0$, so $\mathfrak{a}x \subseteq \mathfrak{r}$. In particular, it follows that $x = 1x \in \mathfrak{r}$. □

Theorem 2.2 a) proves that $\mathbf{J}(\mathfrak{a})$ is the intersection of the maximal right ideals of $\mathfrak{a}$. It is easy to show that $\mathbf{J}(\mathfrak{a})$ is also the intersection of the maximal left ideals of $\mathfrak{a}$, but we will not need this.

While the two-sided ideals correspond to normal subgroups, as they arise as kernels of homomorphisms, the one-sided ideals have no analog in groups. Thus, there is no construction corresponding to the radical $\mathbf{J}(\mathfrak{a})$ in groups. Although the description of the radical in Theorem 2.2 a) resembles that of the Frattini subgroup in Definition III.3.1 (pg. 283), the construction of $\mathbf{J}(\mathfrak{a})$ in Definition 2.1 is more closely related to the characterization of the Fitting subgroup in Theorem III.4.3 (pg. 293) as the centralizer of all chief factors.

Definition 2.3.

a) An ideal $\mathfrak{n}$ of $\mathfrak{a}$ is called a nil ideal if for every $x \in \mathfrak{n}$ there exists a natural number $n(x)$ such that $x^{n(x)} = 0$.

b) An ideal $\mathfrak{n}$ of $\mathfrak{a}$ is called nilpotent if there exists a natural number n such that $\mathfrak{n}^n = 0$. Clearly, it is enough that for all $x_1, \ldots, x_n$ in $\mathfrak{n}$ we have $x_1 \cdots x_n = 0$. (It is also clear that nilpotent ideals are always nil ideals.)

Theorem 2.2 makes no use of the fact that $\mathfrak{a}$ is an algebra, but rather holds more generally for arbitrary rings with unity. Part a) of Theorem 2.4 below also holds for rings satisfying the minimal condition for right ideals. However, we restrict ourselves to a proof for algebras.

Theorem 2.4.

a) *The radical* $\mathbf{J}(\mathfrak{a})$ *of an algebra* $\mathfrak{a}$ *is always nilpotent.*

b) *Every right nil ideal of* $\mathfrak{a}$ *lies in the radical* $\mathbf{J}(\mathfrak{a})$. *So every right nil ideal is nilpotent.*

Proof. a) Let

$$\mathfrak{a} = \mathfrak{a}_0 \supset \mathfrak{a}_1 \supset \cdots \supset \mathfrak{a}_r = 0$$

be a composition series of $\mathfrak{a}$ as a right $\mathfrak{a}$-module; such a composition series exists by Lemma I.11.2 (pg. 65) since $\mathfrak{a}$ satisfies the maximal and minimal condition for right ideals. Since $\mathfrak{a}_i/\mathfrak{a}_{i+1}$ is irreducible, we have $\mathfrak{a}_i\mathbf{J}(\mathfrak{a}) \subseteq \mathfrak{a}_{i+1}$ for all i by Definition 2.1. It follows that

$$\mathbf{J}(\mathfrak{a})^r = \mathfrak{a}\mathbf{J}(\mathfrak{a})^r \subseteq \mathfrak{a}_r = 0,$$

and so $\mathbf{J}(\mathfrak{a})$ is nilpotent.

b) Let $\mathfrak{n}$ be a right nil ideal of $\mathfrak{a}$. If $x \in \mathfrak{n}$ and $a \in \mathfrak{a}$, then xa lies in $\mathfrak{n}$. So we have $(xa)^k = 0$ for some integer k. It then follows that

$$(1 - xa)(1 + xa + (xa)^2 + \cdots + (xa)^{k-1}) = 1.$$

Hence, x lies in $\mathbf{J}(\mathfrak{a})$ by Theorem 2.2. □

Definition 2.5. An algebra $\mathfrak{a}$ is called semisimple if $\mathbf{J}(\mathfrak{a}) = 0$.

The next lemma gives a sufficient condition for semisimplicity.

Lemma 2.6. *Let $\mathfrak{a}$ be an algebra over K with K-basis $\{a_1, \ldots, a_n\}$, and let $\mathbf{D}$ be a representation of $\mathfrak{a}$ on a module V. We define*

$$b_{ij} = \operatorname{tr} \mathbf{D}(a_i a_j) \quad \text{and} \quad \Delta = \det(b_{ij}),$$

where $\operatorname{tr} \mathbf{D}(s)$ is the trace of the linear map $\mathbf{D}(s)$. If $\Delta \neq 0$, then $\mathfrak{a}$ is semisimple.

Proof. If some $a'_i = \sum_{j=1}^{n} c_{ij} a_j$ with $c_{ij} \in \mathsf{K}$ and $\det(c_{ij}) \neq 0$ constitute another basis of $\mathfrak{a}$, then we have

$$(\operatorname{tr} \mathbf{D}(a'_i a'_j)) = (c_{ik})(\operatorname{tr} \mathbf{D}(a_i a_j))(c_{ik})^T.$$

Therefore, the assertion that $\Delta \neq 0$ is independent of the basis.

Assume for contradiction that $\mathbf{J}(\mathfrak{a}) \neq 0$. Let $\{a'_1, \ldots, a'_n\}$ be a K-basis of $\mathfrak{a}$ with $a'_1 \in \mathbf{J}(\mathfrak{a})$. Then $a'_1 a'_i \in \mathbf{J}(\mathfrak{a})$ for all $i = 1, \ldots, n$. If

$$\mathsf{V} = \mathsf{V}_0 \supset \mathsf{V}_1 \supset \cdots \supset \mathsf{V}_r = 0$$

is a composition series of V as an $\mathfrak{a}$-module, then $\mathsf{V}_j(a'_1 a'_i) \subseteq \mathsf{V}_{j+1}$ by Definition 2.1 since $a'_1 a'_i \in \mathbf{J}(\mathfrak{a})$. Therefore, $\operatorname{tr} \mathbf{D}(a'_1 a'_i) = 0$ for $i = 1, \ldots, n$. By our previous remark, we obtain the contradiction $\Delta = 0$. □

With this lemma, it is easy to determine which group rings are semisimple:

Theorem 2.7. *Let $\mathfrak{a} = \mathsf{K}[G]$ be the group ring of a finite group G of order n over the field K. Then $\mathfrak{a}$ is semisimple if and only if $\operatorname{char} \mathsf{K}$ does not divide n. This condition is always satisfied when $\operatorname{char} \mathsf{K} = 0$.*

Proof. 1) Let $\mathbf{R}$ be the regular representation of $\mathfrak{a}$ on the right $\mathfrak{a}$-module $\mathfrak{a}$. We choose the group elements $g_1, \ldots, g_n$ as a basis for $\mathfrak{a}$. Then $\operatorname{tr} \mathbf{R}(g) = 0$ for $g \neq 1$ and $\operatorname{tr} \mathbf{R}(1) = n$. Hence,

$$\operatorname{tr} \mathbf{R}(g_i g_j) = \begin{cases} 0 & \text{for } g_i g_j \neq 1 \\ n & \text{for } g_i g_j = 1. \end{cases}$$

Therefore, it follows that $\det(\operatorname{tr} \mathbf{R}(g_i g_j)) = \pm n^n$, and this is nonzero if $\operatorname{char} \mathsf{K}$ does not divide n. So the given condition is sufficient to guarantee the semisimplicity of $\mathfrak{a}$ by Lemma 2.6.

2) The condition is also necessary: Namely, assume that char $\mathsf{K} = p$ is a divisor of n. We let $i = \sum_{g \in G} g$. Then $gi = i = ig$ for all $g \in G$. It follows that $i^2 = ni = 0$. Since i commutes with all elements of $\mathfrak{a}$, we have for all $a \in \mathfrak{a}$ that

$$(ia)^2 = i^2 a^2 = 0.$$

Hence, $i\mathfrak{a}$ is a nil ideal of $\mathfrak{a}$. By Theorem 2.4, we have $0 \neq i\mathfrak{a} \subseteq \mathbf{J}(\mathfrak{a})$ and $\mathfrak{a}$ is not semisimple. □

Exercises

3) Without using the minimal condition, show that $\mathbf{J}(\mathfrak{a})$ is the intersection of the maximal left ideals of $\mathfrak{a}$. (Hint: Let $\mathbf{J}'(\mathfrak{a})$ be the intersection of the maximal left ideals of $\mathfrak{a}$. Show:
   (1) $\mathbf{J}'(\mathfrak{a}) = \{x \in \mathfrak{a} \mid$ For every $a \in \mathfrak{a}$, there exists some $z \in \mathfrak{a}$ such that $(1 - z)(1 - ax) = 1\}$.
   (2) If $y \in \mathbf{J}'(\mathfrak{a})$, then $1 - ya$ has a right inverse for all $a \in \mathfrak{a}$.)
4) Let K be a field, and let $f \in \mathsf{K}[x]$. Describe $\mathbf{J}(\mathsf{K}[x]/(f))$ and show that the algebra $\mathsf{K}[x]/(f)$ is semisimple if and only if f is not divisible by the square of an irreducible polynomial in $\mathsf{K}[x]$.

3 Completely reducible modules and semisimple algebras

In this section, $\mathfrak{a}$ is always an algebra over K. By Definition 1.7, all $\mathfrak{a}$-modules are finite-dimensional vector spaces over K.

Definition 3.1. A right $\mathfrak{a}$-module V is said to be completely reducible if V is the direct sum of (necessarily finitely many, since $\dim \mathsf{V} < \infty$) irreducible right $\mathfrak{a}$-modules. It is convenient to declare the 0-module to be completely reducible. We say that a representation is completely reducible whenever the corresponding representation module V is completely reducible.

Theorem 3.2. *The following are equivalent:*
a) V *is a sum of irreducible modules.*
b) *If* W *is a submodule of* V*, then there exists a submodule* U *of* V *such that* $\mathsf{V} = \mathsf{W} \oplus \mathsf{U}$.
c) V *is completely reducible.*

Proof. a) ⇒ b): Let T be a submodule of V satisfying $\mathsf{W} \cap \mathsf{T} = 0$ that is as large as possible. Assume for contradiction that $\mathsf{W} + \mathsf{T} \subset \mathsf{V}$. By the decomposition of $\mathsf{V} = \sum_i \mathsf{V}_i$ into irreducibles V_i, there exists an i with $\mathsf{V}_i \not\subseteq \mathsf{W} + \mathsf{T}$. Since the V_i are irreducible, it follows that $\mathsf{V}_i \cap (\mathsf{W} + \mathsf{T}) = 0$.

Suppose that

$$w = t + v \in \mathsf{W} \cap (\mathsf{T} + \mathsf{V}_i)$$

for some $w \in \mathsf{W}$, $t \in \mathsf{T}$ and $v \in \mathsf{V}_i$. Then

$$v = w - t \in \mathsf{V}_i \cap (\mathsf{W} + \mathsf{T}) = 0,$$

so $w = t \in \mathsf{W} \cap \mathsf{T} = 0$. Hence, $\mathsf{W} \cap (\mathsf{T} + \mathsf{V}_i) = 0$. But this contradicts our choice of T since $\mathsf{T} \subset \mathsf{T} + \mathsf{V}_i$. Therefore, $\mathsf{V} = \mathsf{W} + \mathsf{T}$ and $\mathsf{W} \cap \mathsf{T} = 0$, so $\mathsf{V} = \mathsf{W} \oplus \mathsf{T}$.

b) $\Rightarrow$ c): Let W be a submodule of V that is a direct sum of finitely many irreducible modules and is as large as possible. If $\mathsf{W} \subset \mathsf{V}$, then $\mathsf{V} = \mathsf{W} \oplus \mathsf{T}$ with $\mathsf{T} \neq 0$. Let S be an irreducible submodule of T; there certainly exists such a submodule since $\dim \mathsf{T} < \infty$. Then $\mathsf{W} \subset \mathsf{W} \oplus \mathsf{S}$, and $\mathsf{W} \oplus \mathsf{S}$ is a direct sum of finitely many irreducible modules, contradicting our choice of W. So $\mathsf{V} = \mathsf{W}$, and hence c) follows from b).

c) $\Rightarrow$ a): This is trivial. □

Theorem 3.2 also holds without assuming that all modules are vector spaces of finite dimension; the proof in this case requires Zorn's Lemma.

Main Theorem 3.3.

a) *If $\mathfrak{a}$ is a semisimple algebra, then $\mathfrak{a}$ is completely reducible as a right $\mathfrak{a}$-module. We have $\mathfrak{a} = \bigoplus_{i=1}^{n} \mathfrak{r}_i$ for some minimal right ideals $\mathfrak{r}_i = e_i\mathfrak{a}$. The e_i are pairwise orthogonal idempotents; that is,*

$$e_i^2 = e_i \neq 0, \quad \textit{and} \quad e_i e_j = 0 \quad \textit{for} \quad i \neq j.$$

Furthermore, $1 = \sum_{i=1}^{n} e_i$. *(This decomposition of $\mathfrak{a}$ is not unique in general.)*

b) *If $\mathfrak{a}$ is completely reducible as an $\mathfrak{a}$-module, then $\mathfrak{a}$ is semisimple.*

Proof. a) We have $\mathbf{J}(\mathfrak{a}) = 0$, and $\mathbf{J}(\mathfrak{a})$ is the intersection of the maximal right ideals of $\mathfrak{a}$ by Theorem 2.2. Since $\mathfrak{a}$ satisfies the minimal condition for right ideals, there exist finitely many maximal right ideals $\mathfrak{m}_i$ with $\bigcap_{i=1}^{n} \mathfrak{m}_i = 0$. Let these ideals be chosen so that their number n is minimal; that is, $\bigcap_{j \neq i} \mathfrak{m}_j \neq 0$ for all $i = 1, \ldots, n$. We define $\mathfrak{r}_i = \bigcap_{j \neq i} \mathfrak{m}_j$. Then

$$\mathfrak{r}_i \cap \mathfrak{m}_i = \bigcap_{j=1}^{n} \mathfrak{m}_j = 0,$$

but $\mathfrak{r}_i \not\subseteq \mathfrak{m}_i$. Since $\mathfrak{m}_i$ is maximal, we have that $\mathfrak{a} = \mathfrak{r}_i + \mathfrak{m}_i$, so $\mathfrak{a} = \mathfrak{r}_i \oplus \mathfrak{m}_i$ and $\mathfrak{a}/\mathfrak{m}_i \cong \mathfrak{r}_i$. As $\mathfrak{a}/\mathfrak{m}_i$ is an irreducible $\mathfrak{a}$-module, $\mathfrak{r}_i$ is a minimal right ideal.

We have $\mathfrak{a} = \bigoplus_{i=1}^{n} \mathfrak{r}_i$:

Since $\mathfrak{a} = \mathfrak{r}_j \oplus \mathfrak{m}_j$, we have for all $a \in \mathfrak{a}$ a unique decomposition

$$a = r_j + m_j \quad \text{with} \quad r_j \in \mathfrak{r}_j \quad \text{and} \quad m_j \in \mathfrak{m}_j.$$

Since $\mathfrak{r}_i = \bigcap_{k \neq i} \mathfrak{m}_k \subseteq \mathfrak{m}_j$ for $i \neq j$, it follows that

$$a - \sum_{i=1}^{n} r_i = (a - r_j) - \sum_{i \neq j} r_i \in \mathfrak{m}_j,$$

and thus $a - \sum_{i=1}^{n} r_i \in \bigcap_{j=1}^{n} \mathfrak{m}_j = 0$.

The decomposition $a = \sum_{i=1}^{n} x_i$ with $x_i \in \mathfrak{r}_i$ is unique:

If $\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i'$ with $x_i, x_i' \in \mathfrak{r}_i$, then

$$x_i - x_i' = -\sum_{j \neq i} (x_j - x_j') \in \mathfrak{r}_i \cap \mathfrak{m}_i = 0,$$

and thus $x_i = x_i'$. So $\mathfrak{a} = \bigoplus_{i=1}^{n} \mathfrak{r}_i$.

Let $1 = \sum_{i=1}^{n} e_i$ with $e_i \in \mathfrak{r}_i$ be the decomposition of the unity element 1 of $\mathfrak{a}$. Then

$$e_j = \sum_{i=1}^{n} e_i e_j$$

with $e_i e_j \in \mathfrak{r}_i$. By the uniqueness of the decomposition, it follows that $e_i e_j = 0$ for $i \neq j$ and $e_j^2 = e_j$. We have $e_i \neq 0$ for all i: Otherwise, $e_j = 0$ yields the contradiction

$$\mathfrak{a} = 1\mathfrak{a} = \sum_{i=1}^{n} e_i \mathfrak{a} \subseteq \bigoplus_{i \neq j} \mathfrak{r}_i \subset \mathfrak{a}.$$

Hence, $e_i \mathfrak{a}$ is a nonzero submodule of the irreducible right module $\mathfrak{r}_i$, so $e_i \mathfrak{a} = \mathfrak{r}_i$.

b) Let $\mathfrak{a} = \bigoplus_{i=1}^{n} \mathfrak{r}_i$, where the $\mathfrak{r}_i$ are irreducible right ideals. By Definition 2.1, we have $\mathfrak{r}_i \mathbf{J}(\mathfrak{a}) = 0$ for all i. It follows that

$$\mathbf{J}(\mathfrak{a}) = \mathfrak{a}\mathbf{J}(\mathfrak{a}) = 0,$$

and $\mathfrak{a}$ is semisimple. □

Theorem 3.4. *If $\mathfrak{a}$ is a semisimple algebra, then every $\mathfrak{a}$-module is completely reducible.*

Proof. Let $\mathfrak{a} = \bigoplus_{i=1}^{n} \mathfrak{r}_i$, where the $\mathfrak{r}_i$ are irreducible right ideals. Let $\mathsf{V} = \bigoplus_{j=1}^{m} \mathsf{K}v_j$ be a right $\mathfrak{a}$-module. Then

$$\mathsf{V} = \mathsf{V}\mathfrak{a} = \sum_{j,i} v_j \mathfrak{r}_i .$$

The map $r \mapsto v_j r$ from $\mathfrak{r}_i$ onto $v_j \mathfrak{r}_i$ is an $\mathfrak{a}$-homomorphism. So $v_j \mathfrak{r}_i$ is either irreducible or the 0-module. Therefore, V is a sum of irreducible modules, so V is completely reducible by Theorem 3.2. □

Theorems 3.4 and 2.7 together yield a new proof of Maschke's Theorem I.17.7.

Lemma 3.5. *Let $\mathfrak{a}$ be a semisimple algebra, and let $\mathfrak{r} \neq 0$ be a right ideal of $\mathfrak{a}$. Then there exists an idempotent element $e \in \mathfrak{a}$ with $\mathfrak{r} = e\mathfrak{a}$, and $\mathfrak{a} = e\mathfrak{a} \oplus (1-e)\mathfrak{a}$ is a direct decomposition of $\mathfrak{a}$ as a right $\mathfrak{a}$-module (a so-called Pierce decomposition).*

Proof. By Theorem 3.3, $\mathfrak{a}$ is completely reducible as a right $\mathfrak{a}$-module. Therefore, there exists a direct decomposition $\mathfrak{a} = \mathfrak{r} \oplus \mathfrak{s}$ for some right ideal $\mathfrak{s}$ of $\mathfrak{a}$ by Theorem 3.2. Let $1 = e + e'$ with $e \in \mathfrak{r}$ and $e' \in \mathfrak{s}$. It follows from $e = e^2 + e'e$ that $e^2 = e$, so e is idempotent. We therefore have $\mathfrak{a} = e\mathfrak{a} \oplus e'\mathfrak{a}$ with $e\mathfrak{a} \subseteq \mathfrak{r}$ and $e'\mathfrak{a} \subseteq \mathfrak{s}$. Thus, it follows that $\mathfrak{r} = e\mathfrak{a}$. □

Lemma 3.6. *Let $\mathfrak{a}$ be a semisimple algebra, and let $\mathfrak{b}$ be a two-sided ideal of $\mathfrak{a}$. Let $\mathfrak{a} = \mathfrak{b} \oplus \mathfrak{c}$ for some right ideal $\mathfrak{c}$ of $\mathfrak{a}$. (There exists such a decomposition by Theorem 3.3 and Theorem 3.2.) Then $\mathfrak{c}$ is a two-sided ideal.*

Proof. Since $\mathfrak{b}$ is a left ideal and $\mathfrak{c}$ is a right ideal, we have

$$\mathfrak{c}\mathfrak{b} \subseteq \mathfrak{b} \cap \mathfrak{c} = 0.$$

It follows that

$$(\mathfrak{b}\mathfrak{c})^2 = \mathfrak{b}\mathfrak{c}\mathfrak{b}\mathfrak{c} = 0.$$

Therefore, $\mathfrak{b}\mathfrak{c}$ is a nilpotent right ideal of $\mathfrak{a}$. Since $\mathfrak{a}$ is semisimple, $\mathfrak{b}\mathfrak{c} = 0$ by Theorem 2.4 b) and so

$$\mathfrak{a}\mathfrak{c} = (\mathfrak{b} + \mathfrak{c})\mathfrak{c} = \mathfrak{c}^2 \subseteq \mathfrak{c}.$$

So $\mathfrak{c}$ is a two-sided ideal of $\mathfrak{a}$. □

Definition 3.7. An algebra $\mathfrak{a} \neq 0$ is said to be simple if 0 and $\mathfrak{a}$ are the only two-sided ideals of $\mathfrak{a}$. (Clearly, every simple algebra is semisimple.)

Theorem 3.8. *Let $\mathfrak{a}$ be a semisimple algebra. Then $\mathfrak{a}$ is the direct sum of all the minimal two-sided ideals of $\mathfrak{a}$, say $\mathfrak{a} = \bigoplus_{i=1}^{k} \mathfrak{a}_i$. For $i \neq j$, we have $\mathfrak{a}_i \mathfrak{a}_j = 0$. The $\mathfrak{a}_i$ are simple algebras. Every two-sided ideal of $\mathfrak{a}$ is a direct sum of some of the $\mathfrak{a}_i$. In particular, every minimal two-sided ideal of $\mathfrak{a}$ is one of the $\mathfrak{a}_i$.*

Proof. Let $\mathfrak{a}_1$ be a minimal two-sided ideal of $\mathfrak{a}$. By Lemma 3.6, there exists a decomposition $\mathfrak{a} = \mathfrak{a}_1 \oplus \mathfrak{a}_1'$ for some two-sided ideal $\mathfrak{a}_1'$. We have $\mathfrak{a}_1\mathfrak{a}_1' \subseteq \mathfrak{a}_1 \cap \mathfrak{a}_1' = 0$. Now, $\mathfrak{a}_1$ is a simple algebra and $\mathfrak{a}_1'$ is a semisimple algebra because the two-sided ideals of $\mathfrak{a}_1$, resp. $\mathfrak{a}_1'$, are also two-sided ideals of $\mathfrak{a}$ since $\mathfrak{a}_1'\mathfrak{a}_1 = \mathfrak{a}_1\mathfrak{a}_1' = 0$; if $\mathfrak{n}$ is the radical of $\mathfrak{a}_1'$, then $\mathfrak{n}$ is also a nilpotent right ideal of $\mathfrak{a}$. The semisimplicity of $\mathfrak{a}$ then yields $\mathfrak{n} = 0$. By induction on the dimension of $\mathfrak{a}$, we obtain a decomposition $\mathfrak{a} = \bigoplus_{i=1}^{k} \mathfrak{a}_i$ for some minimal two-sided ideals $\mathfrak{a}_i$ of $\mathfrak{a}$ with $\mathfrak{a}_i\mathfrak{a}_j = 0$ for $i \neq j$.

Let $\mathfrak{b} \neq 0$ be a two-sided ideal of $\mathfrak{a}$. We have

$$\mathfrak{b} = \mathfrak{a}\mathfrak{b} = \sum_{i=1}^{k} \mathfrak{a}_i\mathfrak{b}.$$

If $\mathfrak{a}_i\mathfrak{b} \neq 0$, then it is a two-sided ideal of $\mathfrak{a}$ satisfying $0 \subset \mathfrak{a}_i\mathfrak{b} \subseteq \mathfrak{a}_i$. By the minimality of $\mathfrak{a}_i$, it follows that $\mathfrak{a}_i\mathfrak{b} = \mathfrak{a}_i$. So

$$\mathfrak{b} = \sum_{i'} \mathfrak{a}_{i'},$$

where the sum is restricted to those i' such that $\mathfrak{a}_{i'}\mathfrak{b} \neq 0$. □

Exercises

5) Submodules and homomorphic images of completely reducible modules are again completely reducible.
6) Assume $\mathfrak{m}/\mathfrak{m}_1$ and $\mathfrak{m}/\mathfrak{m}_2$ are completely reducible. Then $\mathfrak{m}/(\mathfrak{m}_1 \cap \mathfrak{m}_2)$ is also completely reducible.
7) Let $\mathfrak{r}$ be a right ideal of $\mathfrak{a}$. Then $\mathfrak{a}/\mathfrak{r}$ is a completely reducible right $\mathfrak{a}$-module if and only if $\mathbf{J}(\mathfrak{a}) \subseteq \mathfrak{r}$.
8) Let $\mathfrak{a}$ be a semisimple algebra. For all two-sided ideals $\mathfrak{i}$, $\mathfrak{j}$, $\mathfrak{k}$ of $\mathfrak{a}$, prove the relations

$$\mathfrak{i}\mathfrak{j} = \mathfrak{i} \cap \mathfrak{j} \quad \text{and} \quad (\mathfrak{i} + \mathfrak{j}) \cap \mathfrak{k} = (\mathfrak{i} \cap \mathfrak{k}) + (\mathfrak{j} \cap \mathfrak{k}).$$

9) Let $\mathfrak{a}$ be semisimple. For every right ideal $\mathfrak{r}$ of $\mathfrak{a}$, we define the left ideal $\mathbf{L}(\mathfrak{r})$ of $\mathfrak{a}$ by $\mathbf{L}(\mathfrak{r}) = \{x \in \mathfrak{a} \mid x\mathfrak{r} = 0\}$. In the same way, for every left ideal $\mathfrak{l}$ we define the right ideal $\mathbf{R}(\mathfrak{l}) = \{x \in \mathfrak{a} \mid \mathfrak{l}x = 0\}$. Show that:
   a) If $\mathfrak{r} = e\mathfrak{a}$ with $e^2 = e$, then $\mathbf{L}(\mathfrak{r}) = \mathfrak{a}(1 - e)$.
   b) $\mathbf{RL}(\mathfrak{r}) = \mathfrak{r}$ for all right ideals $\mathfrak{r}$ and $\mathbf{LR}(\mathfrak{l}) = \mathfrak{l}$ for all left ideals $\mathfrak{l}$ of $\mathfrak{a}$.
   c) $\mathbf{L}(\mathfrak{r}_1 + \mathfrak{r}_2) = \mathbf{L}(\mathfrak{r}_1) \cap \mathbf{L}(\mathfrak{r}_2)$ and $\mathbf{L}(\mathfrak{r}_1 \cap \mathfrak{r}_2) = \mathbf{L}(\mathfrak{r}_1) + \mathbf{L}(\mathfrak{r}_2)$.
10) Let $N \trianglelefteq G$, and let ν be the homomorphism from G onto $\overline{G} = G/N$ defined by $g^\nu = gN$.
   a) The map μ defined by

$$\Big(\sum_{g\in G} a_g g\Big)\mu = \sum_{g\in G} a_g g^\nu$$

is an algebra homomorphism from $\mathsf{K}[G]$ onto $\mathsf{K}[\overline{G}]$ with kernel

$$\mathfrak{k} = \Big\{ \sum_{g\in G} a_g g \;\Big|\; \sum_{g\in G} a_g g^\nu = 0 \Big\}.$$

b) If $\operatorname{char}\mathsf{K} \nmid |N|$, then

$$e = \frac{1}{|N|}\sum_{x\in N} x$$

is an idempotent in $\mathsf{K}[G]$. Furthermore, $\mathsf{K}[G] = e\mathsf{K}[G] \oplus (1-e)\mathsf{K}[G]$ with $e\mathsf{K}[G] \cong \mathsf{K}[\overline{G}]$ and $(1-e)\mathsf{K}[G] = \mathfrak{k}$.

11) Let $Z = \langle z\rangle$ be the cyclic group of order n, let $\operatorname{char}\mathsf{K} \nmid n$ and let ϵ be a primitive n-th root of unity in K. We define

$$e_i = \frac{1}{n}\sum_{j=1}^{n} \epsilon^{-ij} z^j.$$

Then $\mathsf{K}[Z] = \bigoplus_{i=0}^{n-1} \mathsf{K}e_i$, and the e_i are pairwise orthogonal idempotents with $e_i z = \epsilon^i e_i$.

4 Wedderburn's theorems

Theorem 4.1.

a) *If $\mathfrak{a}$ is a simple algebra, then there exists exactly one irreducible $\mathfrak{a}$-module up to isomorphism, and this module is faithful.*

b) *If the algebra $\mathfrak{a}$ has an irreducible, faithful $\mathfrak{a}$-module, then $\mathfrak{a}$ is simple.*

Proof. a) Let $\mathfrak{r}$ be a maximal right ideal of $\mathfrak{a}$. Then $\mathfrak{a}/\mathfrak{r}$ is an irreducible right $\mathfrak{a}$-module. The kernel $\mathfrak{k}$ of this representation of $\mathfrak{a}$ on $\mathfrak{a}/\mathfrak{r}$ is a two-sided ideal of $\mathfrak{a}$. Since $\mathfrak{a}$ is simple, we must have $\mathfrak{k} = 0$ or $\mathfrak{k} = \mathfrak{a}$. Since $(\mathfrak{a}/\mathfrak{r})1 \neq 0$, the second case is impossible. Thus, $\mathfrak{a}$ has a faithful irreducible representation on $\mathfrak{a}/\mathfrak{r}$.

If V is any irreducible right $\mathfrak{a}$-module, then by Theorem 1.13, V is isomorphic as a right $\mathfrak{a}$-module to a quotient module $\mathfrak{a}/\mathfrak{r}$ of $\mathfrak{a}$ by some maximal right ideal $\mathfrak{r}$ of $\mathfrak{a}$. By Theorem 3.3, $\mathfrak{a} = \bigoplus_{i=1}^{n} \mathfrak{r}_i$ for some irreducible right ideals $\mathfrak{r}_i$. By the Jordan–Hölder Theorem I.11.8, V is then isomorphic to one of the minimal right ideals $\mathfrak{r}_i$. So we must show that in a simple algebra $\mathfrak{a}$, all minimal right ideals are isomorphic as right $\mathfrak{a}$-modules.

For every $a \in \mathfrak{r}_j$, the map $x \mapsto ax$ is an $\mathfrak{a}$-module homomorphism from $\mathfrak{r}_i$ to $\mathfrak{r}_j$. If $\mathfrak{r}_i$ and $\mathfrak{r}_j$ are nonisomorphic $\mathfrak{a}$-modules, it follows that $\mathfrak{r}_j\mathfrak{r}_i = 0$.

Let $\mathfrak{b} = \bigoplus_{i=1}^{s} \mathfrak{r}_i$ be the sum of the $\mathfrak{r}_i$ that are isomorphic to $\mathfrak{r}_1$ (with some suitable labeling). Then

$$\mathfrak{r}_j\mathfrak{b} \subseteq \mathfrak{r}_j \subseteq \mathfrak{b}$$

for $j \leq s$ and, by the remark of the last paragraph, $\mathfrak{r}_j\mathfrak{b} = 0$ for $j > s$. It follows that $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{b}$, and so $\mathfrak{b}$ is a two-sided ideal of $\mathfrak{a}$. Since $\mathfrak{a}$ is simple, $\mathfrak{a} = \mathfrak{b}$ and hence all the $\mathfrak{r}_i$ are isomorphic.

b) Assume $\mathfrak{a}$ is faithfully represented on the irreducible right module V. Since $\mathsf{V}\mathbf{J}(\mathfrak{a}) = 0$ by Definition 2.1, we must have $\mathbf{J}(\mathfrak{a}) = 0$. That is, $\mathfrak{a}$ is semisimple. Let $\mathfrak{a} = \bigoplus_{i=1}^{k} \mathfrak{a}_i$ be the decomposition of $\mathfrak{a}$ into minimal two-sided ideals $\mathfrak{a}_i$ according to Theorem 3.8. Since

$$(\mathsf{V}\mathfrak{a}_1)\mathfrak{a} = \mathsf{V}(\mathfrak{a}_1\mathfrak{a}) = \mathsf{V}\mathfrak{a}_1,$$

the set $\mathsf{V}\mathfrak{a}_1$ is a submodule of V. As V is a faithful $\mathfrak{a}$-module, we have $\mathsf{V}\mathfrak{a}_1 \neq 0$. It then follows from the irreducibility of V that $\mathsf{V} = \mathsf{V}\mathfrak{a}_1$.

Assume for contradiction that $k > 1$. Then

$$\mathsf{V}\mathfrak{a}_2 = (\mathsf{V}\mathfrak{a}_1)\mathfrak{a}_2 = \mathsf{V}(\mathfrak{a}_1\mathfrak{a}_2) = 0,$$

contradicting that V is a faithful $\mathfrak{a}$-module. Therefore, we have $k = 1$ and $\mathfrak{a}$ is simple. □

Lemma 4.2 (Jacobson). *Let $\mathfrak{a}$ be an algebra, and let V be an irreducible right $\mathfrak{a}$-module. Let $\mathfrak{c} = \mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$. (By Schur's Lemma I.10.5, pg. 59, $\mathfrak{c}$ is a division ring.) Let n be a natural number. If $v_1, \ldots, v_n$ are elements of V that are linearly independent over $\mathfrak{c}$ and $v'_1, \ldots, v'_n$ are arbitrary elements of V, then there exists an $a \in \mathfrak{a}$ such that $v_i a = v'_i$ for $i = 1, \ldots, n$.*

Proof. It is enough to prove that there exist elements $a_i \in \mathfrak{a}$ $(i = 1, \ldots, n)$ such that $v_j a_i = \delta_{ij} v'_j$, because then $a = \sum_{i=1}^{n} a_i$ has the desired property.

a) For $n = 1$, the claim is trivial: Since $v_1 \neq 0$, we have $v_1\mathfrak{a} = \mathsf{V}$ by the irreducibility of V. Hence, $v_1 a = v'_1$ for some $a \in \mathfrak{a}$.

b) We prove the general claim by induction on n. Define the right $\mathfrak{a}$-module W by $\mathsf{W} = \mathsf{V} \oplus \overset{n}{\cdots} \oplus \mathsf{V}$. We let $w = (v_1, \ldots, v_n)$ and consider the submodule

$$w\mathfrak{a} = \{(v_1 a, \ldots, v_n a) \mid a \in \mathfrak{a}\}$$

of W for this fixed w. By our induction hypothesis, we have

$$\mathsf{V} \oplus \overset{n-1}{\cdots} \oplus \mathsf{V} = \{(v_1 a, \ldots, v_{n-1} a) \mid a \in \mathfrak{a}\}.$$

Assume for contradiction that whenever $v_1a = \cdots = v_{n-1}a = 0$, we also have $v_na = 0$. Then the map μ from $\mathsf{V} \oplus \underset{n-1}{\cdots} \oplus \mathsf{V}$ to V defined by

$$(v_1a, \ldots, v_{n-1}a)\mu = v_na$$

is clearly a well-defined $\mathfrak{a}$-homomorphism. Define homomorphisms π_i from V to V by

$$v\pi_i = (0, \ldots, 0, \underset{i}{v}, 0, \ldots, 0)\mu \quad (i = 1, \ldots, n-1).$$

Then the π_i lie in $\mathfrak{c} = \mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$. It follows that

$$v_n = (v_1, \ldots, v_{n-1})\mu = \sum_{i=1}^{n-1} v_i\pi_i,$$

contradicting the linear independence of the v_i over $\mathfrak{c}$.

Therefore, there exists some $a \in \mathfrak{a}$ such that $v_1a = \cdots = v_{n-1}a = 0 \neq v_na$. By a), there exists some $b \in \mathfrak{a}$ such that $(v_na)b = v'_n$. Then $v_i(ab) = \delta_{in}v'_n$, as desired. □

Lemma 4.3 (I. Schur). *Let $\mathfrak{a}$ be an algebra over the algebraically closed field K, and let V be an irreducible $\mathfrak{a}$-module that is finite dimensional as a vector space over K. Then any map in $\mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$ is simply multiplication by an element of K; so $\mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$ is isomorphic to K.*

Proof. Since V is irreducible, we have by Schur's Lemma I.10.5 (pg. 59) that $\mathfrak{c} = \mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$ is a division ring. Let $c \in \mathfrak{c}$. Then c is a linear map of the finite-dimensional vector space V onto itself, so c has an eigenvalue ϵ in the algebraically closed field K. Let 1_{V} be the identity map on V. Now, $c - \epsilon 1_{\mathsf{V}} \in \mathfrak{c}$ and $c - \epsilon 1_{\mathsf{V}}$ is not invertible. Since $\mathfrak{c}$ is a division ring, it follows that $c = \epsilon 1_{\mathsf{V}}$. □

Main Theorem 4.4 (Wedderburn).

a) *Let $\mathfrak{a}$ be a simple algebra over K, and let V be an irreducible $\mathfrak{a}$-module. Then $\mathfrak{a}$ is isomorphic to the full matrix ring $(\mathfrak{c})_n$ over the division ring $\mathfrak{c}$, where $\mathfrak{c}$ is anti-isomorphic to $\mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$, itself an algebra over K. If K is algebraically closed, then $\mathfrak{a}$ is isomorphic to the full matrix ring over K.*

b) *Let $\mathfrak{a}$ be the full matrix ring of degree n over the division ring $\mathfrak{c}$. Then $\mathfrak{a}$ is simple.*

c) *Let $\mathfrak{c}$ and $\mathfrak{d}$ be division rings. If $(\mathfrak{c})_n \cong (\mathfrak{d})_m$, then $\mathfrak{c} \cong \mathfrak{d}$ and $n = m$.*

Proof. a) By Theorem 4.1, V is a faithful right $\mathfrak{a}$-module. We let $\mathfrak{c}^* = \mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$. By Schur's Lemma I.10.5 (pg. 59), $\mathfrak{c}^*$ is a division ring.

We define a new multiplication on the additive abelian group $\mathfrak{c}^+$ of $\mathfrak{c}^*$ by $c_1 \circ c_2 = c_2c_1$. With this multiplication, the set $\mathfrak{c}^+$ clearly becomes a division ring $\mathfrak{c}$ that is anti-isomorphic to the division ring $\mathfrak{c}^*$. By defining $cv = vc$, V becomes a left $\mathfrak{c}$-module since

$$(c_1 \circ c_2)v = (c_2c_1)v = v(c_2c_1) = (vc_2)c_1 = c_1(c_2v).$$

The vector space V has finite dimension over K, so it certainly has finite dimension over $\mathfrak{c}$. Let $\{v_1, \ldots, v_n\}$ be a $\mathfrak{c}$-basis of V. If $a \in \mathfrak{a}$, then

$$v_i a = \sum_{j=1}^{n} a_{ij} v_j \quad \text{with} \quad a_{ij} \in \mathfrak{c}$$

defines a matrix (a_{ij}). The mapping of a to (a_{ij}) yields a homomorphism from $\mathfrak{a}$ to the full matrix ring of degree n over $\mathfrak{c}$ since

$$\begin{aligned} v_i(ab) = (v_i a)b &= \left(\sum_{j=1}^{n} a_{ij} v_j \right) b = \sum_{j=1}^{n} a_{ij}(v_j b) \\ &= \sum_{j=1}^{n} a_{ij} \left(\sum_{k=1}^{n} b_{jk} v_k \right) = \sum_{k=1}^{n} \left(\sum_{j=1}^{n} a_{ij} \circ b_{jk} \right) v_k \end{aligned}$$

and

$$v_i(a+b) = v_i a + v_i b = \sum_{j=1}^{n} (a_{ij} + b_{ij}) v_j .$$

By Lemma 4.2, for any given elements $a_{ij} \in \mathfrak{c}$ there exists an $a \in \mathfrak{a}$ such that

$$v_i a = \sum_{i=1}^{n} a_{ij} v_j \quad (i = 1, \ldots, n).$$

Hence, the map we have constructed is an epimorphism from $\mathfrak{a}$ onto the full matrix ring of degree n over $\mathfrak{c}$. If $a_{ij} = 0$ for all $i, j = 1, \ldots, n$, then $v_i a = 0$ for $i = 1, \ldots, n$, and so $\mathsf{V}a = 0$. Since $\mathfrak{a}$ is represented faithfully on V, this forces $a = 0$. Therefore, $\mathfrak{a}$ is isomorphic to the full matrix ring of degree n over $\mathfrak{c}$. If K is algebraically closed, then $\mathfrak{c}$ is isomorphic to K by Lemma 4.3.

b) Let $\mathfrak{a} = (\mathfrak{c})_n$ be the full matrix ring over the division ring $\mathfrak{c}$.

Let $\mathsf{V} = \underbrace{\mathfrak{c} \oplus \cdots \oplus \mathfrak{c}}_{n}$. Defining

$$(c_1, \ldots, c_n)a = \left(\sum_{j=1}^{n} c_j a_{j1}, \ldots, \sum_{j=1}^{n} c_j a_{jn} \right)$$

for $a = (a_{ij})$, V becomes a right $\mathfrak{a}$-module. If $0 \neq v \in \mathsf{V}$, then clearly $v\mathfrak{a} = \mathsf{V}$. So V is an irreducible $\mathfrak{a}$-module. Clearly, V is also a faithful $\mathfrak{a}$-module. So $\mathfrak{a}$ is simple by Theorem 4.1 b).

c) Since $\mathfrak{a}$ is simple, the $\mathfrak{a}$-module V described in b) is the unique (up to isomorphism) irreducible $\mathfrak{a}$-module by Lemma 4.2. We compute $\mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$.

Let $\alpha \in \mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$. We define $v_i = (0, \ldots, 0, \underset{i}{1}, 0, \ldots, 0)$ and

$$v_i \alpha = (c_{i1}, \ldots, c_{in}).$$

Let $e_{jk} \in \mathfrak{a}$ with $v_i e_{jk} = \delta_{ij} v_k$. Then it follows that

$$\begin{aligned}(\delta_{ij} v_k)\alpha &= \delta_{ij}(c_{k1}, \dots, c_{kn}) = (v_i e_{jk})\alpha = (v_i \alpha) e_{jk} \\ &= (c_{i1}, \dots, c_{in}) e_{jk} = (0, \dots, 0, \underset{k}{c_{ij}}, 0, \dots, 0).\end{aligned}$$

This yields $c_{ij} = 0$ for $i \neq j$ and $c_{ii} = c_{kk}$ for all $i, k = 1, \dots, n$. We let $c_{ii} = c$. So

$$v_i \alpha = (0, \dots, 0, \underset{i}{c}, 0, \dots, 0).$$

Letting $a = (a_{ij})$ with $a_{1j} = x_j$ and $a_{ij} = 0$ for $i > 1$, we have

$$(x_1, \dots, x_n)\alpha = (v_1 a)\alpha = (v_1 \alpha) a = (cx_1, \dots, cx_n).$$

Now the map $\alpha \mapsto c$ is clearly an isomorphism from $\operatorname{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$ to $\mathfrak{c}^*$. So $\mathfrak{c}$ is uniquely determined by $\mathfrak{a}$ since $\mathfrak{c} \cong \operatorname{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})^*$. Then n is also uniquely determined by $n = \dim_{\mathfrak{c}^*} \mathsf{V}$. □

We now sum up our results.

Theorem 4.5. *If $\mathfrak{a}$ is a semisimple algebra, then $\mathfrak{a} = \bigoplus_{i=1}^{k} \mathfrak{a}_i$, where the $\mathfrak{a}_i$ are full matrix rings of degree n_i over division rings $\mathfrak{c}_i$. If $i \neq j$, then $\mathfrak{a}_i \mathfrak{a}_j = 0$. The algebra $\mathfrak{a}$ has exactly k inequivalent irreducible representations $\mathbf{D}_i$ $(i = 1, \dots, k)$. Moreover, $\mathbf{D}_i$ is a faithful representation of $\mathfrak{a}_i$ with kernel $\ker \mathbf{D}_i = \bigoplus_{j \neq i} \mathfrak{a}_j$. We also have*

$$\deg \mathbf{D}_i = n_i \dim_{\mathsf{K}} \mathfrak{c}_i \quad \textit{and} \quad \dim_{\mathsf{K}} \mathfrak{a} = \sum_{i=1}^{k} n_i^2 \dim_{\mathsf{K}} \mathfrak{c}_i.$$

In particular, if K is algebraically closed, then all the division rings $\mathfrak{c}_i$ are isomorphic to K, $\deg \mathbf{D}_i = n_i$ and $\dim_{\mathsf{K}} \mathfrak{a} = \sum_{i=1}^{k} n_i^2$.

Proof. By Theorem 4.1 b), every irreducible representation $\mathbf{D}_i$ of $\mathfrak{a}$ has kernel $\mathfrak{k}_i$ with simple quotient algebra $\mathfrak{a}/\mathfrak{k}_i$. By Theorem 3.8, $\mathfrak{k}_i$ has the form $\mathfrak{k}_i = \bigoplus_{j \neq i} \mathfrak{a}_j$ for some labeling of the $\mathfrak{a}_j$. So by Theorem 4.1 a), $\mathbf{D}_i$ is the unique (up to equivalence) irreducible faithful representation of $\mathfrak{a}_i$. Conversely, the (unique, by Theorem 4.1 a)) irreducible representation of $\mathfrak{a}_i$ gives rise to an irreducible representation $\mathbf{D}_i$ of $\mathfrak{a}$ with kernel $\ker \mathbf{D}_i = \bigoplus_{j \neq i} \mathfrak{a}_j$. The $\mathbf{D}_i$ are inequivalent since they have different kernels. The simple ring $\mathfrak{a}_i$ is a full matrix ring, say of degree n_i, over a division ring $\mathfrak{c}_i$ by Theorem 4.4. The irreducible $\mathfrak{a}_i$-module is then obtained from the vector space V_i of dimension n_i over $\mathfrak{c}_i$. We have $\deg \mathbf{D}_i = \dim_{\mathsf{K}} \mathsf{V}_i = n_i \dim_{\mathsf{K}} \mathfrak{c}_i$. The remaining claims are trivially true. □

5 Group characters

First, we clarify the structure of group rings.

Main Theorem 5.1. *Let h be the number of conjugacy classes of G, and let K be an algebraically closed field with* $\operatorname{char}\mathsf{K} \nmid |G|$. *Then the group ring $\mathsf{K}[G]$ is the direct sum of h full matrix rings $\mathfrak{a}_i$ $(i = 1, \dots, h)$ over K. The group G has exactly h inequivalent irreducible representations $\mathbf{D}_i$ $(i = 1, \dots, h)$ over K, where for every $a \in \mathsf{K}[G]$, the image $\mathbf{D}_i(a)$ is given by the i-th component of a in the direct decomposition* $\mathsf{K}[G] = \bigoplus_{i=1}^{h} \mathfrak{a}_i$. *Letting* $\deg \mathbf{D}_i = n_i$, *we have* $|G| = \sum_{i=1}^{h} n_i^2$.

Proof. By Theorem 2.7, the algebra $\mathsf{K}[G]$ is semisimple since $\operatorname{char}\mathsf{K} \nmid |G|$. So we have by Theorem 4.5 that

$$\mathsf{K}[G] = \bigoplus_{i=1}^{k} \mathfrak{a}_i,$$

where the $\mathfrak{a}_i$ are full matrix rings over the algebraically closed field K. We must prove that $h = k$.

If $\mathfrak{a}$ is an algebra, then we define the center $\mathbf{Z}(\mathfrak{a})$ of $\mathfrak{a}$ to be

$$\mathbf{Z}(\mathfrak{a}) = \{x \in \mathfrak{a} \mid xa = ax \text{ for all } a \in \mathfrak{a}\}.$$

Clearly

$$\mathbf{Z}(\mathsf{K}[G]) = \bigoplus_{i=1}^{k} \mathbf{Z}(\mathfrak{a}_i).$$

One computes easily that the center $\mathbf{Z}(\mathfrak{a}_i)$ of the full matrix ring $\mathfrak{a}_i$ over K consists of the multiples $c1_{n_i}$ $(c \in \mathsf{K})$ of the identity matrix 1_{n_i}. This yields

$$\dim_{\mathsf{K}} \mathbf{Z}(\mathsf{K}[G]) = \sum_{i=1}^{k} \dim_{\mathsf{K}} \mathbf{Z}(\mathfrak{a}_i) = k.$$

On the other hand, if $z = \sum_{g \in G} a_g g \in \mathbf{Z}(\mathsf{K}[G])$ and $x \in G$, then

$$\sum_{g \in G} a_g g = z = x^{-1} z x = \sum_{g \in G} a_g x^{-1} g x.$$

Therefore, a_g depends only on the conjugacy class of g in G. Let $K_1, \dots, K_h$ be the conjugacy classes of G. We let

$$k_i = \sum_{g \in K_i} g.$$

Then every element in $\mathbf{Z}(\mathsf{K}[G])$ has the form $z = \sum_{i=1}^{h} b_i k_i$ with $b_i \in \mathsf{K}$. Conversely, it is clear that linear combinations of the k_i lie in $\mathbf{Z}(\mathsf{K}[G])$. Since the k_i are linearly independent over K, they constitute a K-basis of $\mathbf{Z}(\mathsf{K}[G])$. Hence, $\dim_{\mathsf{K}} \mathbf{Z}(\mathsf{K}[G]) = h$ and thus $h = k$. The remaining claims are contained in Theorem 4.5. □

The following generalization of Theorem 5.1 lies deeper and stems from R. Brauer: Let K be an algebraically closed field of characteristic p. The number of inequivalent irreducible representations of $\mathsf{K}[G]$ is exactly the number of conjugacy classes of G that consist of elements with order coprime to p (see Curtis, Reiner [102], pg. 596).

Definition 5.2. Let $\mathfrak{a}$ be an algebra, and let V be an $\mathfrak{a}$-module.

a) If a composition series of V has exactly s composition factors that are isomorphic to a given irreducible $\mathfrak{a}$-module W, then we say that W appears with multiplicity s in V. (This definition does not depend on the chosen composition series of V by the Jordan–Hölder Theorem I.11.8, pg. 68.) If V is a completely reducible $\mathfrak{a}$-module, then the isomorphism type of V is clearly determined by the multiplicities of the irreducible constituents of V.

b) Let $\mathbf{D}$ be a representation of $\mathsf{K}[G]$ on the $\mathsf{K}[G]$-module V. If a composition series of V has exactly s composition factors that are isomorphic to the irreducible module V' with irreducible representation $\mathbf{D}'$, then we say that the irreducible representation $\mathbf{D}'$ appears in $\mathbf{D}$ with multiplicity s. If V is completely reducible, then $\mathbf{D}$ is determined up to equivalence by the multiplicities of the irreducible constituents.

Theorem 5.3. *Let* K *be an algebraically closed field with* $\operatorname{char}\mathsf{K} \nmid |G|$. *The regular representation of* $\mathsf{K}[G]$ (*see Definition 1.12*) *contains the irreducible representation* $\mathbf{D}_i$ *with multiplicity* $n_i = \deg \mathbf{D}_i$. (*This shows once again the relation*

$$|G| = \dim \mathsf{K}[G] = \sum_{i=1}^{h} n_i^2$$

from Theorem 5.1.)

Proof. We have $\mathsf{K}[G] = \bigoplus_{i=1}^{h} \mathfrak{a}_i$, where $\mathfrak{a}_i$ is a full matrix ring of degree n_i over K. The simple ring $\mathfrak{a}_i$ has exactly one irreducible faithful representation, namely the representation $\mathbf{D}_i$ of degree n_i. Since $\dim \mathfrak{a}_i = n_i^2$, the regular representation of $\mathfrak{a}_i$ contains the representation $\mathbf{D}_i$ exactly n_i times. □

Definition 5.4.

a) Let f be a map from G to K. We say that f is a class function if $f(g^{g'}) = f(g)$ for all $g, g' \in G$. The collection $\mathsf{C}(G, \mathsf{K})$ of class functions on G is clearly a vector space of dimension h over K. Let $K_1, \ldots, K_h$ be the conjugacy classes of G. A K-basis of $\mathsf{C}(G, \mathsf{K})$ is given by the functions f_i defined by

$$f_i(x) = \begin{cases} 1 & \text{for } x \in K_i, \\ 0 & \text{for } x \notin K_i. \end{cases}$$

b) If $\mathbf{D}$ is a representation of $\mathsf{K}[G]$, then we define the character χ of $\mathbf{D}$ by $\chi(g) = \operatorname{tr}\mathbf{D}(g)$. If $\mathbf{D}$ is an irreducible representation of $\mathsf{K}[G]$, then we say χ is an irreducible character of G over K.

We occasionally refer to the kernel of $\mathbf{D}$ also as the kernel of χ.

If $\mathbf{D}$ is a representation of $\mathsf{K}[G]$ with character χ on the K-vector space V and V contains the irreducible $\mathsf{K}[G]$-module V_i corresponding to the irreducible representation $\mathbf{D}_i$ with character χ_i with multiplicity t_i, then clearly

$$\chi = \sum_i t_i \chi_i.$$

(One makes use of a basis of V derived from a composition series.)

If $\mathbf{D}$ is a representation of $\mathsf{K}[G]$ and L is an extension field of K, then $\mathbf{D}$ induces a representation $\widetilde{\mathbf{D}}$ of $\mathsf{L}[G]$ by defining

$$\widetilde{\mathbf{D}}\Big(\sum_{g\in G} a_g g\Big) = \sum_{g\in G} a_g \mathbf{D}(g) \quad (\text{for } a_g \in \mathsf{L}).$$

However, the irreducibility of $\mathbf{D}$ does not imply that $\widetilde{\mathbf{D}}$ is irreducible.

Theorem 5.5. *Let χ be the character of the representation $\mathbf{D}$ of $\mathsf{K}[G]$. Then χ is a class function in $\mathsf{C}(G, \mathsf{K})$, and χ depends only on the equivalence class of $\mathbf{D}$.*

Proof. Since

$$\chi(g^{g'}) = \operatorname{tr}\mathbf{D}(g^{g'}) = \operatorname{tr}\left(\mathbf{D}(g')^{-1}\mathbf{D}(g)\mathbf{D}(g')\right) = \operatorname{tr}\mathbf{D}(g) = \chi(g),$$

the character χ is a class function on G. If $\mathbf{D}$ and $\mathbf{D}'$ are equivalent representations, then there is a linear map t from the representation module V of $\mathbf{D}$ onto the representation module V' of $\mathbf{D}'$ with $t^{-1}\mathbf{D}(g)t = \mathbf{D}'(g)$ for all $g \in G$. Then

$$\operatorname{tr}\mathbf{D}(g) = \operatorname{tr}\mathbf{D}'(g). \qquad \square$$

5.6 (Notation). We fix the following notation: h is the class number of G and $|G|$ is the order of G;

$$K_i = \{g_i^x \mid x \in G\} \quad (i = 1, \ldots, h)$$

are the conjugacy classes of G, $K_1 = \{1\}$, $|K_i| = h_i$ and $k_i = \sum_{g\in K_i} g$. Furthermore, let $\mathbf{D}_1, \ldots, \mathbf{D}_h$ be irreducible, pairwise inequivalent representations of G over the algebraically closed field K with $\operatorname{char}\mathsf{K} \nmid |G|$. (By Theorem 5.1, every irreducible representation of $\mathsf{K}[G]$ is equivalent to one of the $\mathbf{D}_i$.) Let $\mathbf{D}_1$ be the trivial representation of $\mathsf{K}[G]$ defined by $\mathbf{D}_1(g) = 1$ for all $g \in G$. We denote by χ_i the character of

$\mathbf{D}_i$ and let $\deg \mathbf{D}_i = n_i$. We often denote by 1 the character χ_1 of $\mathbf{D}_1$ with $\chi_1(g) = 1$ for all $g \in G$.

Theorem 5.7. *Let* $\mathbf{D}$ *and* $\mathbf{D}'$ *be irreducible representations of* $\mathsf{K}[G]$ *on the* $\mathsf{K}[G]$*-modules* V *and* V'*. Let* $\mathbf{D}(g)$ *be the matrix* $(a_{ij}(g))$ *with respect to some basis of* V*, and let* $\mathbf{D}'(g)$ *be the matrix* $(a'_{ij}(g))$ *with respect to some basis of* V'*.*

a) *If* $\mathbf{D}$ *is not equivalent to* $\mathbf{D}'$*, then*

$$\sum_{g\in G} a_{ij}(g)\, a'_{kl}(g^{-1}) = 0$$

for all i, j, k, l*.*

b) *If* K *is algebraically closed and* $\operatorname{char}\mathsf{K} \nmid |G|$*, then*

$$\sum_{g\in G} a_{ij}(g)\, a_{kl}(g^{-1}) = \delta_{jk}\delta_{il}\frac{|G|}{\deg \mathbf{D}}.$$

Proof. Let $x \in \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}, \mathsf{V}')$. With respect to the chosen bases of V and V', let x be the rectangular matrix (x_{rs}). We construct the sum

$$t(x) = \sum_{g\in G} \mathbf{D}(g)\, x\, \mathbf{D}'(g^{-1}).$$

For all $g' \in G$, we have

$$\mathbf{D}(g')t(x) = \sum_{g\in G} \mathbf{D}(g'g)\, x\, \mathbf{D}'((g'g)^{-1})\, \mathbf{D}'(g') = t(x)\mathbf{D}'(g').$$

So $t(x) \in \operatorname{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{V}')$.

a) Assume V is not isomorphic to V'. Since V and V' are irreducible, $t(x) = 0$ for all x. We choose $x = (x_{rs})$ with $x_{rs} = \delta_{rj}\delta_{sk}$. Then

$$0 = \sum_{g\in G}\sum_{r,s} a_{ir}(g) x_{rs} a'_{sl}(g^{-1}) = \sum_{g\in G} a_{ij}(g) a'_{kl}(g^{-1}).$$

b) Now assume K is algebraically closed. By Lemma 4.3,

$$\operatorname{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{V}) = \{a1 \mid a \in \mathsf{K}\}.$$

So $t(x) = a(x)1$ for some $a(x) \in \mathsf{K}$. Furthermore,

$$\operatorname{tr} t(x) = \sum_{g\in G} \operatorname{tr}\left(\mathbf{D}(g)\, x \mathbf{D}(g)^{-1}\right) = |G| \operatorname{tr} x.$$

As in a), we choose $x = (x_{rs})$ with $x_{rs} = \delta_{rj}\delta_{sk}$. Then

$$\operatorname{tr} x = \begin{cases} 0 & \text{for } j \neq k \\ 1 & \text{for } j = k. \end{cases}$$

Therefore, we obtain

$$|G|\,\delta_{jk} = |G| \operatorname{tr} x = \operatorname{tr} t(x) = \operatorname{tr}(a(x)1) = a(x) \deg \mathbf{D}.$$

The special case $j = k$ shows that the characteristic of K does not divide $\deg \mathbf{D}$; that is, we may divide by $\deg \mathbf{D}$ in K. This yields

$$\sum_{g \in G} \mathbf{D}(g)\, x\, \mathbf{D}(g^{-1}) = \frac{|G|\,\delta_{jk}}{\deg \mathbf{D}}\, 1$$

and therefore

$$\sum_{g \in G} a_{ij}(g) a_{kl}(g^{-1}) = \begin{cases} 0 & \text{for } j \neq k \text{ or } i \neq l. \\ \dfrac{|G|}{\deg \mathbf{D}} & \text{for } j = k,\ i = l. \end{cases}$$ □

The following orthogonality relations are of fundamental importance.

Main Theorem 5.8 (Frobenius). *Let χ and χ' be characters of the irreducible representations $\mathbf{D}$ and $\mathbf{D}'$ of $\mathsf{K}[G]$.*

a) (1) *If $\mathbf{D}$ is not equivalent to $\mathbf{D}'$, then over any field K we have*

$$\sum_{g \in G} \chi(g) \chi'(g^{-1}) = 0.$$

(2) *If K is algebraically closed and* $\operatorname{char} \mathsf{K} \nmid |G|$, *then*

$$\sum_{g \in G} \chi(g) \chi(g^{-1}) = |G|.$$

(3) *If* $\operatorname{char} \mathsf{K} = 0$, *then*

$$\sum_{g \in G} \chi(g) \chi(g^{-1})$$

is a positive integer that is divisible by $|G|$.

b) *Let K be algebraically closed, and assume* $\operatorname{char} \mathsf{K} \nmid |G|$. *Using Notation 5.6, we have*

$$h_j \sum_{k=1}^{h} \chi_k(g_i) \chi_k(g_j^{-1}) = \delta_{ij} |G|.$$

Proof. a) Let $\mathbf{D}_i$ be representations on the vector spaces V_i with irreducible characters χ_i. With respect to some basis of V_i, let $\mathbf{D}_i(g)$ be the matrix $(a_{jk}^{(i)}(g))$. Then

$$\sum_{g\in G}\chi_i(g)\chi_j(g^{-1}) = \sum_{g\in G}\sum_k a_{kk}^{(i)}(g)\sum_m a_{mm}^{(j)}(g^{-1})$$
$$= \sum_{k,m}\sum_{g\in G} a_{kk}^{(i)}(g)a_{mm}^{(j)}(g^{-1}).$$

For $i \neq j$, it follows immediately from Theorem 5.7 a) that

$$\sum_{g\in G}\chi_i(g)\chi_j(g^{-1}) = 0.$$

If K is algebraically closed and $\operatorname{char}\mathsf{K} \nmid |G|$, then we obtain from Theorem 5.7 b) that

$$\sum_{g\in G}\chi_i(g)\chi_i(g^{-1}) = \sum_{k,m}\sum_{g\in G} a_{kk}(g)a_{mm}(g^{-1})$$
$$= \sum_{k,m}\delta_{km}\frac{|G|}{\deg \mathbf{D}_i} = |G|.$$

Now assume that $\operatorname{char}\mathsf{K} = 0$, but K is not necessarily algebraically closed. Let L be an algebraically closed extension field of K, and let $\psi_1, \ldots, \psi_h$ be the characters of the irreducible representations of $\mathsf{L}[G]$. Let

$$\chi_i = \sum_{j=1}^{h} a_{ij}\psi_j$$

be the decomposition of the characters χ_i into irreducible characters of $\mathsf{L}[G]$ for some nonnegative integers a_{ij}. By (2), it follows that

$$\sum_{g\in G}\chi_i(g)\chi_i(g^{-1}) = |G|\sum_{j=1}^{h} a_{ij}^2.$$

b) We define the h by h square matrices $A = (a_{ik})$ and $B = (b_{kj})$ by $a_{ik} = \chi_k(g_i)$ and $b_{kj} = \frac{h_j}{|G|}\chi_k(g_j^{-1})$. The result in a) implies that $BA = 1$. So $AB = 1$ as well. This means

$$h_j\sum_k \chi_k(g_i)\chi_k(g_j^{-1}) = |G|\,\delta_{ij}. \qquad \square$$

Theorem 5.9.
a) *Let* K *be algebraically closed with* $\operatorname{char}\mathsf{K} \nmid |G|$. *Upon defining*

$$(f_1, f_2)_G = \frac{1}{|G|}\sum_{g\in G} f_1(g)f_2(g^{-1}),$$

the K*-vector space* $\mathsf{C}(G,\mathsf{K})$ *of class functions on* G *becomes a vector space with a symmetric, nondegenerate scalar product. The characters* χ_i $(i = 1,\dots,h)$ *of the irreducible representations of* G *constitute an orthonormal basis of* $\mathsf{C}(G,\mathsf{K})$.

(*If the group* G *is clear from the context, then we often write* $(_,_)$ *instead of* $(_,_)_G$.)

b) *Let* $\mathbb{C}$ *be the field of complex numbers. Then* $\chi(g^{-1}) = \overline{\chi(g)}$ *for all characters* χ *of* G. *The values* $\chi(g)$ *of the characters are algebraic integers in the field generated by the primitive* $|G|$*-th roots of unity over* $\mathbb{Q}$. *If* $f \neq 0$ *is a linear combination of the* χ_i *with real coefficients, then* $(f,f) > 0$. *Hence,* $(_,_)$ *is a positive definite scalar product on the vector space of real linear combinations of the* χ_i $(i = 1,\dots,h)$.

Proof. a) Since $\dim \mathsf{C}(G,\mathsf{K}) = h$, the χ_i constitute an orthonormal basis of $\mathsf{C}(G,\mathsf{K})$ by Theorem 5.8 a). It is well known that the scalar product must then be nondegenerate on $\mathsf{C}(G,\mathsf{K})$.

b) Let $\mathbb{C}$ be the field of complex numbers. If $\mathbf{D}$ is a representation of G over $\mathbb{C}$ with character χ, then

$$\chi(g) = \operatorname{tr}\mathbf{D}(g) = \sum_i \epsilon_i,$$

where the ϵ_i are the eigenvalues of $\mathbf{D}(g)$. Since $\mathbf{D}(g)^{|G|} = \mathbf{D}(g^{|G|}) = 1$, the ϵ_i are $|G|$-th roots of unity. Therefore, $\chi(g)$ is an algebraic integer. We also have $\epsilon_i^{-1} = \overline{\epsilon}_i$ and thus

$$\chi(g^{-1}) = \sum_i \epsilon_i^{-1} = \sum_i \overline{\epsilon}_i = \overline{\chi(g)}.$$

If $f = \sum_{i=1}^{h} c_i\chi_i$ with real c_i, then

$$(f,f) = \sum_{i=1}^{h} c_i^2.$$

This yields the final claim. □

Theorem 5.10. *Assume* $\operatorname{char}\mathsf{K} = 0$.

a) *The equivalence class of a representation* $\mathbf{D}$ *is uniquely determined by its character* χ. *Namely, the representation* $\mathbf{D}$ *contains the irreducible representation* $\mathbf{D}_i$ *having character* χ_i *with multiplicity* $(\chi,\chi_i)/(\chi_i,\chi_i)$.

b) *If* χ *is a character of* G *with* $(\chi,\chi) = 1$, *then* χ *is irreducible.*

Proof. a) Let V be a representation module for $\mathbf{D}$. Since V is completely reducible by Theorem 3.4, the isomorphism type of V is determined by the multiplicities t_i of the irreducible modules V_i with representations $\mathbf{D}_i$ appearing in the direct decomposition of V by the Jordan–Hölder Theorem I.11.8 (pg. 68). Clearly, $\chi(g) = \sum_{j=1}^{h} t_j\chi_j(g)$. By

Theorem 5.8 a), it follows that

$$(\chi, \chi_i) = \sum_{j=1}^{h} t_j (\chi_j, \chi_i) = t_i (\chi_i, \chi_i).$$

Since char $\mathsf{K} = 0$, $(\chi_i, \chi_i) > 0$ by Theorem 5.8 a), (3); so $\mathbf{D}$ is uniquely determined up to equivalence by the character χ.

b) Let $\chi = \sum_{i=1}^{h} t_i \chi_i$ for some natural numbers t_i. Then

$$1 = (\chi, \chi) = \sum_{i=1}^{h} t_i^2 (\chi_i, \chi_i).$$

Since char $\mathsf{K} = 0$ and $(\chi_i, \chi_i) > 0$, there exists exactly one i such that $t_i = 1$ and all other t_j are 0. So $\chi = \chi_i$ is an irreducible character of G. □

Lemma 5.11. *Let* K *be an algebraically closed field of characteristic* 0. *Then the quantities*

$$a_{ik} = \frac{h_i \chi_k(g_i)}{n_k} \quad (i, k = 1, \ldots, h)$$

are algebraic integers.

Proof. Since the elements $k_i = \sum_{x \in K_i} x$ constitute a K-basis of the center $\mathbf{Z}(\mathsf{K}[G])$ of $\mathsf{K}[G]$, we have

$$k_i k_j = \sum_{m=1}^{h} c_{ijm} k_m.$$

Clearly, the c_{ijm} are integers. We obtain

(1) $\mathbf{D}_l(k_i)\mathbf{D}_l(k_j) = \sum_{m=1}^{h} c_{ijm} \mathbf{D}_l(k_m)$.

By Schur's Lemma 4.3, we have $\mathbf{D}_l(k_i) = a_{il} 1_{n_l}$ for some $a_{il} \in \mathsf{K}$. It follows that

$$a_{il} n_l = \operatorname{tr} \mathbf{D}_l(k_i) = \operatorname{tr} \sum_{x \in K_i} \mathbf{D}_l(x) = |K_i| \operatorname{tr} \mathbf{D}_l(g_i) = h_i \chi_l(g_i).$$

Hence, the a_{il} have the form given in the statement of the theorem. From (1), we obtain

(2) $a_{il} a_{jl} = \sum_{m=1}^{h} c_{ijm} a_{ml}$.

For fixed i and l, we consider this as a system of linear equations $a_{il} v = tv$ for the matrix $t = (c_{ijm})_{j,m}$ and the vector $v = (a_{1l}, \ldots, a_{hl})$. Here, $a_{1l} = h_1 = 1$. Therefore, a_{il} is an eigenvalue of the integer matrix t, so it is an algebraic integer. □

We establish an important relation regarding the degrees n_i of the irreducible representations:

Theorem 5.12. *If* $\mathbf{D}$ *is an irreducible representation of* G *over the algebraically closed field* K *of characteristic* 0*, then the degree of* $\mathbf{D}$ *divides the order of* G*.*

Proof. The quantities a_{ik} defined in Theorem 5.11 are algebraic integers. Since $\chi_k(g_i^{-1})$ is also an algebraic integer, being a sum of $|G|$-th roots of unity, we obtain with the help of Theorem 5.8 a) the rational number

$$\sum_{i=1}^{h} a_{ik}\chi_k(g_i^{-1}) = \frac{1}{n_k}\sum_{g\in G}\chi_k(g)\chi_k(g^{-1}) = \frac{|G|}{n_k},$$

which is also an algebraic integer. It is well known that this means $\frac{|G|}{n_k}$ must be an integer. So n_k divides $|G|$. □

Theorem 5.12 holds under the weaker assumption that K is an algebraically closed field with $\operatorname{char}\mathsf{K} \nmid |G|$. However, we will first be able to prove this in Theorem 12.11.

Let K be an algebraically closed field of characteristic 0, and let $\mathfrak{a} = \bigoplus_{i=1}^{h}(\mathsf{K})_{n_i}$ be a semisimple algebra over K. When is $\mathfrak{a}$ the group ring of a finite group? By Theorem 5.12, it is necessary that $n_1 = 1$ and every n_i is a divisor of $\dim_{\mathsf{K}}\mathfrak{a} = \sum_{i=1}^{h} n_i^2$. However, there are easy examples which show that these conditions are not sufficient. One sees easily that if $n_1 = \cdots = n_6 = 1$ and $n_7 = 2$, then the algebra $\mathfrak{a}$ is not a group ring; since $\dim_{\mathsf{K}}\mathfrak{a} = 10$, the algebra $\mathfrak{a}$ would have to be the group ring of the dihedral group of order 10. But this group has two irreducible representations of degree 2 (see Lemma 21.5, pg. 649). No further relations between the n_i are known besides those given in Theorem 5.1 and Theorem 5.12.

If K is an algebraically closed field with $\operatorname{char}\mathsf{K}$ dividing $|G|$, then the conclusion of Theorem 5.12 no longer holds in general; we give an example of this in Example 5.13. At the time of writing, the following conjecture is open[2]: Let K be an algebraically closed field of characteristic p, and let $\mathbf{D}$ be an irreducible representation of $\mathsf{K}[G]$. If p^k is a divisor of the degree of $\mathbf{D}$, then p^k is also a divisor of $|G|$.

Example 5.13 (R. Brauer, Nesbitt [68]). We let $G = \mathrm{SL}_2(p)$. Let K be a field of characteristic p and V_m the K-vector space consisting of the homogeneous polynomials of degree m in the indeterminates x and y. Upon defining

$$x^i y^{m-i} g = (a_{11}x + a_{12}y)^i (a_{21}x + a_{22}y)^{m-i}$$

[2] This conjecture is now known to be false. John Thackray constructed a counterexample for the MacLaughlin simple group and $p = 2$; see J. Thackray, *Modular representations of some finite groups*. Ph.D. Thesis, Cambridge, 1981. Dave Benson found another counterexample for $p = 2$ in the symmetric group S_{15}; see D. J. Benson, Some remarks on the decomposition numbers for the symmetric groups, in *The Arcata Conference on Representations of Finite Groups (Arcata, Calif., 1986)*, 381–394, Proc. Sympos. Pure Math., 47, Part 1, Amer. Math. Soc., Providence, RI. *trans.*

for $g = (a_{ij})$ $(i, j = 1, 2)$, the vector space V_m becomes a $\mathsf{K}[G]$-module. We first show that V_m is irreducible for $0 \le m \le p-2$.

Let W be a submodule of V_m and

$$0 \neq f = \sum_{j=0}^{m} a_j x^j y^{m-j} \in \mathsf{W}.$$

For all $k \in \mathrm{GF}(p)$, we define $s(k) = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ and $t(k) = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$.

Let $a_n \neq 0 = a_{n+1} = \cdots = a_m$. Then

$$fs(k) = \sum_{j=0}^{n} a_j (x + ky)^j y^{m-j} = \sum_{j=0}^{n} f_j(x, y) k^j \in \mathsf{W},$$

where $f_n(x, y) = a_n y^m$. We let $\mathrm{GF}(p)^\times = L$. Since $m \le p-2$, it follows from the formula

$$\sum_{k \in L} k^i = \begin{cases} -1 & \text{for } i \equiv 0 \bmod (p-1) \\ 0 & \text{otherwise} \end{cases} \qquad \text{(see Lemma III.9.6, pg. 337)}$$

that

$$\sum_{k \in L} k^{-i} fs(k) = -f_i(x, y) \in \mathsf{W}.$$

In particular, $y^m \in \mathsf{W}$. In the same way, we find

$$\sum_{k \in L} k^{-j} y^m t(k) = \sum_{k \in L} k^{-j} (kx + y)^m = -\binom{m}{j} x^j y^{m-j} \in \mathsf{W}.$$

Since $\binom{m}{j}$ is not divisible by p, it follows that $x^j y^{m-j} \in \mathsf{W}$ for all j with $0 \le j \le m$. Hence, $\mathsf{W} = \mathsf{V}_m$ and V_m is an irreducible $\mathsf{K}[G]$-module.

In particular, $\mathrm{SL}_2(p)$ has an irreducible representation of degree $p-2$ over the field K of characteristic p. For $p \equiv 1 \bmod 3$, we have that $|\mathrm{SL}_2(p)| = p(p^2-1)$ is coprime to $p-2$, and in this case the conclusion of Theorem 5.12 does not hold. (It is possible to show that V_m is actually irreducible for $0 \le m \le p-1$, and that these V_m are all the irreducible modules for $\mathrm{SL}_2(p)$ over K; see Brauer, Nesbitt [68].)

Theorem 5.14 (Burnside). *Let* $\mathbf{D}$ *be an irreducible representation of* G *of degree* m *over an algebraically closed field* K *of arbitrary characteristic. Then there exist exactly* m^2 K*-linearly independent maps among the* $\mathbf{D}(g)$ *with* $g \in G$.

Proof. We also denote by $\mathbf{D}$ the representation of the group ring $\mathsf{K}[G]$ induced by the representation $\mathbf{D}$ of G. Since $\mathbf{D}$ is irreducible, the radical $\mathbf{J}(\mathsf{K}[G])$ lies in the kernel of $\mathbf{D}$ by Definition 2.1. So $\mathbf{D}$ induces an irreducible representation $\widetilde{\mathbf{D}}$ on the semisimple algebra $\mathfrak{a} = \mathsf{K}[G]/\mathbf{J}(\mathsf{K}[G])$. Let $\mathfrak{a} = \bigoplus_i \mathfrak{a}_i$ for some full matrix rings $\mathfrak{a}_i$

of degree n_i over K according to Theorem 4.5. We may assume that $\widetilde{\mathbf{D}}$ is a faithful representation of $\mathfrak{a}_j$. Then the elements

$$\widetilde{\mathbf{D}}(g + \mathbf{J}(\mathsf{K}[G])) = \mathbf{D}(g)$$

for all $g \in G$ span $\mathfrak{a}_j$ over K. Since $\dim_{\mathsf{K}} \mathfrak{a}_j = m^2$, the claim follows. □

The unity elements in the simple summands of $\mathsf{K}[G]$ can be explicitly calculated using the characters:

Theorem 5.15. *Let* K *be an algebraically closed field, and assume that* $\operatorname{char} \mathsf{K} \nmid |G|$.

a) *Let* $\mathbf{D}$ *be an irreducible representation of* $\mathsf{K}[G]$ *with character* χ. *By Theorem 5.1, there corresponds to* $\mathbf{D}$ *exactly one simple direct summand* $\mathfrak{a}$ *of* $\mathsf{K}[G]$ *that is faithfully represented by* $\mathbf{D}$. *Let* e *be the unity element of* $\mathfrak{a}$. *Then*

$$e = \frac{\chi(1)}{|G|} \sum_{g \in G} \chi(g^{-1}) g.$$

b) *Let* K_0 *be the smallest subfield of* K *that contains all* $\chi(g)$ *for* $g \in G$. *If* m *is the degree of* $\mathbf{D}$, *then* $m\chi$ *is the character of a representation of* $\mathsf{K}_0[G]$.

Proof. a) Let $\mathbf{D}_j$ $(j = 1, \ldots, h)$ be the irreducible representations of $\mathsf{K}[G]$, and let $\mathbf{D} = \mathbf{D}_k$. Then $\mathbf{D}_j(e) = 0$ for $j \neq k$ and $\mathbf{D}_k(e) = 1_m$ with $m = \deg \mathbf{D}$ because $\mathfrak{a}$ is faithfully represented by $\mathbf{D}_k$ and lies in the kernel of $\mathbf{D}_j$ when $j \neq k$. Let

$$e = \sum_{g \in G} a(g)\, g.$$

Since $e \in \mathbf{Z}(\mathsf{K}[G])$, we have $a(g) = a(g^{g'})$ for all $g, g' \in G$.

Then we have

$$m\, \delta_{jk} = \operatorname{tr} \mathbf{D}_j(e) = \sum_{g \in G} a(g)\, \chi_j(g).$$

By the orthogonality relations of Theorem 5.8 b), it follows that

$$\begin{aligned} m\, \chi(g_i^{-1}) &= \sum_{j=1}^{h} m\, \delta_{jk}\, \chi_j(g_i^{-1}) = \sum_{g \in G} a(g) \sum_{j=1}^{h} \chi_j(g) \chi_j(g_i^{-1}) \\ &= \sum_{g \in K_i} \frac{|G|}{h_i} a(g) = |G|\, a(g). \end{aligned}$$

So

$$e = \sum_{g \in G} a(g)\, g = \frac{m}{|G|} \sum_{g \in G} \chi(g^{-1}) g.$$

b) By a), the element e lies in $\mathsf{K}_0[G]$. Therefore, $e\mathsf{K}_0[G]$ is a $\mathsf{K}_0[G]$-module. The group G has equivalent representations on $e\mathsf{K}_0[G]$ and $e\mathsf{K}[G] = (\mathsf{K})_m$. Since $(\mathsf{K})_m$ is the simple direct summand of $\mathsf{K}[G]$ with representation $\mathbf{D}$ of degree m, G has a representation on $(\mathsf{K})_m$ with character $m\,\chi$. □

The representation $\mathbf{D}$ need not itself be equivalent to a representation of $\mathsf{K}_0[G]$ in general, where K_0 is a field containing all the $\chi(g)$. We will address the natural questions that arise here in §14.

Next, we prove two simple observations regarding representations over fields of nonzero characteristic.

Theorem 5.16. *Let G be a p-group, and let K be a field of characteristic p.*

a) *There exists only one irreducible representation of G over K, namely the trivial representation.*

b) *If $|G| = p^m$, then the radical of $\mathsf{K}[G]$ has dimension $p^m - 1$. We have*

$$\mathbf{J}(\mathsf{K}[G]) = \Big\{ \sum_{g\in G} a_g\, g \;\Big|\; \sum_{g\in G} a_g = 0 \Big\}.$$

c) *If G is a transitive permutation group on the points $\{1, \dots, m\}$ and we give the m-dimensional K-vector space $\mathsf{V} = \langle v_1, \dots, v_m\rangle$ the structure of a G-module by defining $v_i g = v_{ig}$, then V is a directly indecomposable G-module. In particular, $\mathsf{K}[G]$ is a directly indecomposable $\mathsf{K}[G]$-module.*

Proof. a) Let $\mathbf{D}$ be an irreducible representation of $\mathsf{K}[G]$ on the module V. We may assume that $\mathbf{D}$ is a faithful representation of G; otherwise, we consider the faithful representation that $\mathbf{D}$ induces on a quotient of G. We must show that $G = 1$.

Assume for contradiction that $G \neq 1$. Let z be an element of order p in $\mathbf{Z}(G)$. We define

$$\mathsf{W} = \{v \in \mathsf{V} \mid v\mathbf{D}(z) = v\}.$$

The subspace W of V is left invariant by G since for all $w \in \mathsf{W}$ and $g \in G$ we have

$$(w\mathbf{D}(g))\mathbf{D}(z) = w\mathbf{D}(z)\mathbf{D}(g) = w\mathbf{D}(g).$$

We claim that $\mathsf{W} \neq 0$: Since $\operatorname{char}\mathsf{K} = p$, we have for all $v \neq 0$ in V that

$$0 = v\mathbf{D}(z)^p - v = v(\mathbf{D}(z) - 1)^p.$$

Therefore, there exists an $i \geq 0$ such that

$$w = v(\mathbf{D}(z) - 1)^i \neq 0, \quad \text{but} \quad w(\mathbf{D}(z) - 1) = v(\mathbf{D}(z) - 1)^{i+1} = 0.$$

Then $0 \neq w \in \mathsf{W}$.

It follows from the irreducibility of $\mathbf{D}$ that $\mathsf{V} = \mathsf{W}$, so $\mathbf{D}(z) = 1$. But this contradicts our assumption that G is faithfully represented by $\mathbf{D}$.

b) Since the trivial representation of G is the only irreducible representation of G, resp. $\mathsf{K}[G]$, we have by Definition 2.1 that

$$\mathbf{J}(\mathsf{K}[G]) = \left\{ \sum_{g\in G} a_g g \;\middle|\; \sum_{g\in G} a_g = 0 \right\}.$$

c) Let W be a maximal submodule of V. Since V/W is an irreducible $\mathsf{K}[G]$-module, G has the trivial representation on V/W by a). So

$$v_i g - v_i \in \mathsf{W} \text{ for all } i.$$

Hence, it follows that

$$\langle v_i - v_1 \mid i = 2, \ldots, m\rangle \subseteq \mathsf{W}.$$

Therefore, $\langle v_i - v_1 \mid i = 2, \ldots, m\rangle$ is the unique maximal submodule of V. Clearly, V is then directly indecomposable. □

The next theorem is a useful consequence of Theorem 5.16.

Theorem 5.17. *Let* $\mathbf{D}$ *be an irreducible representation of* G *over a field* K *of characteristic* p*. If* N *is a normal subgroup of* G *with* p*-power order, then* N *lies in the kernel of* $\mathbf{D}$*.*

(*One can also formulate this statement as follows: The elements* $n - 1$ *with* $n \in N$ *lie in the radical* $\mathbf{J}(\mathsf{K}[G])$ *of the group ring* $\mathsf{K}[G]$*.*)

Proof. Let V be the irreducible $\mathsf{K}[G]$-module corresponding to $\mathbf{D}$ and define

$$\mathsf{W} = \{v \in \mathsf{V} \mid v\,\mathbf{D}(n) = v \text{ for all } n \in N\}.$$

Since N admits only the trivial irreducible representation over K by Theorem 5.16, we have $\mathsf{W} \neq 0$. The subspace W is G-invariant since for $w \in \mathsf{W}$ and $n \in N$ we have

$$(w\,\mathbf{D}(g))\mathbf{D}(n) = w\,\mathbf{D}(n^{g^{-1}})\mathbf{D}(g) = w\,\mathbf{D}(g).$$

It follows from the irreducibility of V that $\mathsf{V} = \mathsf{W}$, so $\mathbf{D}(n) = 1$ for all $n \in N$. □

We conclude this section by noting a trivial lemma that will be useful later when we calculate with some examples.

Lemma 5.18. *Let* G *be a finite group with generators* $g_1, \ldots, g_k$ *and defining relations* $r_j(g_1, \ldots, g_k) = 1$ $(j = 1, \ldots, m)$*. Let* V *be a vector space of finite dimension over* K*, and let* τ *be a map from* $\{g_1, \ldots, g_k\}$ *to* $\mathrm{GL}(\mathsf{V})$ *with* $r_j(\tau(g_1), \ldots, \tau(g_k)) = 1_{\mathsf{V}}$ *for* $j = 1, \ldots, m$*. Then there exists exactly one representation* $\mathbf{D}$ *of* G *on* V *with*

$$\mathbf{D}(g_i) = \tau(g_i) \quad (i = 1, \ldots, k).$$

Exercises

12) Let char $\mathsf{K} = 0$, and let $N \trianglelefteq G$. Let the idempotent e_N of $\mathsf{K}[G]$ be defined as in Exercise 10 b). We define $\mathbf{S}(N) = \mathsf{K}[G]e_N$. For $N_i \trianglelefteq G$ $(i = 1, 2)$, we then have $e_{N_1N_2} = e_{N_1}e_{N_2}$ and

$$\mathbf{S}(N_1N_2) = \mathbf{S}(N_1)\mathbf{S}(N_2) = \mathbf{S}(N_1) \cap \mathbf{S}(N_2).$$

13) Let S be the product of all minimal normal subgroups of G. (The group S is called the socle of G.) Show:
   a) If $N \trianglelefteq G$ and $N \leq S$, then there exists a normal subgroup M of G such that $S = N \times M$.
   b) The lattice of normal subgroups of G lying in S admits a duality map ϵ such that $(N_1N_2)^\epsilon = N_1^\epsilon \cap N_2^\epsilon$, $(N_1 \cap N_2)^\epsilon = N_1^\epsilon N_2^\epsilon$ and $|N^\epsilon| = [S : N]$.

   (Directly decompose S into characteristically simple, characteristic subgroups of G.)

14) (Gaschütz [156]) Let G be a finite group, and let K be an algebraically closed field with char $\mathsf{K} \nmid |G|$. There exists a faithful irreducible representation of G over K if and only if the socle S of G is generated by a single conjugacy class of G.

   (Hint: Let $N_1, \ldots, N_s$ be the minimal normal subgroups of G. Prove that the following statements are equivalent:
   a) G has a faithful irreducible representation.
   b) $\mathsf{K}[G] \neq \sum_{i=1}^{s} \mathbf{S}(N_i)$.
   c) $|S| > \sum_{k=1}^{s} (-1)^k \sum_{i_1 < \cdots < i_k} [S : N_{i_1} \cdots N_{i_k}]$.
   d) If $M_1, \ldots, M_s$ are the normal subgroups of G that are maximal in S, then

$$|S| - \sum_{k=1}^{s} (-1)^k \sum_{i_1 < \cdots < i_k} |M_{i_1} \cap \cdots \cap M_{i_k}| = \Big|S - \bigcup_{i=1}^{s} M_i\Big| > 0.$$

   e) There exists an $s \in S$ such that $S = \langle s^g \mid g \in G \rangle$.)

15) Let G be a p-group, and let K be an algebraically closed field with char $\mathsf{K} \nmid |G|$. Prove the following statement without using Exercise 14: G has a faithful, irreducible representation over K if and only if $\mathbf{Z}(G)$ is cyclic.

6 Characters of abelian groups

With an application from §23 in mind, we do not require that all abelian groups be finite in this section.

Theorem 6.1. *Let A be a (not necessarily finite) abelian group, and let K be an algebraically closed field.*

a) *Every irreducible representation of A with finite degree over K has degree* 1.
b) *If A is finite and* $\operatorname{char}\mathsf{K} \nmid |A|$, *then A has exactly $|A|$ distinct irreducible representations over K.* (*For representations of degree* 1, *being distinct and being inequivalent are clearly the same.*)

Proof. a) Let V be an irreducible A-module, and let $\mathbf{D}$ be a representation of A on V. By Schur's Lemma 4.3, $\operatorname{Hom}_A(\mathsf{V},\mathsf{V})$ consists of the maps of the form $c1_{\mathsf{V}}$ with $c \in \mathsf{K}$. Since A is abelian, $\mathbf{D}(a)$ lies in $\operatorname{Hom}_A(\mathsf{V},\mathsf{V})$ for all $a \in A$. Hence, $\mathbf{D}(a) = c(a)1_{\mathsf{V}}$ for all $a \in A$. It follows from the irreducibility of V that $\dim \mathsf{V} = 1$.

b) Since A is abelian, $|A|$ is the class number of A. As $\operatorname{char}\mathsf{K} \nmid |A|$, we may apply Theorem 5.1 and conclude that A has exactly $|A|$ irreducible representations over K. □

Definition 6.2. Let A be a (not necessarily finite) abelian group, and let K be a field. The set $\hat{A} = \operatorname{Hom}(A, \mathsf{K}^\times)$ admits the structure of a group upon defining

$$(\chi_1\chi_2)(a) = \chi_1(a)\chi_2(a)$$

for all $\chi_i \in \hat{A}$ and $a \in A$. We call $\hat{A}$ the character group of A over K. The elements of $\hat{A}$ are called the characters of A over K. (By Theorem 6.1 a), the elements of $\hat{A}$ are exactly the characters of irreducible finite-dimensional representations of A over K when K is algebraically closed. We will always interpret the characters of an abelian group A as elements of $\hat{A}$.)

Theorem 6.3. *Let K be an algebraically closed field, and let A be a* (*not necessarily finite*) *abelian group.*

a) *If $B \leq A$ and $\mu \in \operatorname{Hom}(B, \mathsf{K}^\times)$, then there exists a $\chi \in \operatorname{Hom}(A, \mathsf{K}^\times)$ such that the restriction χ_B of χ to B agrees with μ. Thus, the restriction map $\chi \mapsto \chi_B$ is an epimorphism from $\hat{A} = \operatorname{Hom}(A, \mathsf{K}^\times)$ to $\hat{B} = \operatorname{Hom}(B, \mathsf{K}^\times)$ with kernel*

$$B^\perp = \{\chi \in \hat{A} \mid \chi(b) = 1 \text{ for all } b \in B\}.$$

We have $\hat{B} \cong \hat{A}/B^\perp$.

b) *Let*

$$K = \{a \in A \mid \chi(a) = 1 \text{ for all } \chi \in \hat{A}\}.$$

If $\operatorname{char}\mathsf{K} = 0$, *then $K = 1$. If* $\operatorname{char}\mathsf{K} = p$, *then*

$$K = \{a \in A \mid |a| \text{ is a power of } p\}.$$

c) *The map τ from A to $\hat{\hat{A}}$ defined by $a^\tau(\chi) = \chi(a)$ for $a \in A$ and $\chi \in \hat{A}$ is a homomorphism from A to the character group $\hat{\hat{A}}$ of $\hat{A}$. The kernel of τ is the group K described in b).*

Proof. a) Since K is algebraically closed, the group $\mathsf{K}^\times$ is divisible. So every homomorphism from B to $\mathsf{K}^\times$ extends to a homomorphism from A to $\mathsf{K}^\times$ by Theorem I.13.17 (pg. 87). The remaining claims in part a) now follow easily.

b) If $\operatorname{char} \mathsf{K} = p$ and $a \in A$ with $a^{p^m} = 1$, then for every character χ of A we have

$$1 = \chi(a^{p^m}) = \chi(a)^{p^m}.$$

Since 1 is the only p-th root of unity in K, this forces $\chi(a) = 1$.

Let $1 \neq a \in A$, and assume that $|a|$ is not a power of the characteristic of K. It is enough to exhibit a character χ of the cyclic group $\langle a \rangle$ such that $\chi(a) \neq 1$ since χ extends to a character of A by a).

First assume that $|a| = \infty$. Then we choose some $1 \neq s \in \mathsf{K}^\times$ and define $\chi(a^i) = s^i$. Clearly, χ is a character of $\langle a \rangle$ satisfying $\chi(a) \neq 1$.

Now assume that $|a|$ is finite and not a power of the characteristic of K. Let m be a divisor of $|a|$ such that $m > 1$ and $\operatorname{char} \mathsf{K} \nmid m$. Since K is algebraically closed, there exists a primitive m-th root of unity ϵ in K. Defining $\chi(a^i) = \epsilon^i$, χ is again a character of $\langle a \rangle$ satisfying $\chi(a) \neq 1$.

c) Since

$$a^\tau(\chi_1\chi_2) = (\chi_1\chi_2)(a) = \chi_1(a)\chi_2(a) = a^\tau(\chi_1)a^\tau(\chi_2),$$

a^τ is a character of $\hat{A}$. The map τ is a homomorphism from A to $\hat{\hat{A}}$ since

$$(a_1a_2)^\tau(\chi) = \chi(a_1a_2) = \chi(a_1)\chi(a_2) = a_1^\tau(\chi)a_2^\tau(\chi).$$

Clearly

$$\ker \tau = \{a \in A \mid a^\tau(\chi) = \chi(a) = 1 \text{ for all } \chi \in \hat{A}\}. \qquad \square$$

Theorem 6.4 (Duality theorem for finite abelian groups). *Let A be a finite abelian group, and let K be an algebraically closed field with $\operatorname{char} \mathsf{K} \nmid |A|$.*

a) *The groups A and $\hat{A} = \operatorname{Hom}(A, \mathsf{K}^\times)$ are isomorphic.*

b) *The map τ defined by $a^\tau(\chi) = \chi(a)$ is an isomorphism from A to $\hat{\hat{A}}$.*

c) *For $B \leq A$, we define*

$$B^\perp = \{\chi \in \hat{A} \mid \chi(b) = 1 \text{ for all } b \in B\}.$$

Then $B^\perp \cong \widehat{A/B}$.

d) *For all $B \leq A$, we have $B^{\perp\perp} = B^\tau$.*

e) *The map $B \mapsto B^\perp$ is a duality map from the lattice of subgroups of A onto the lattice of subgroups of $\hat{A}$; that is, we have*

$$(B_1B_2)^\perp = B_1^\perp \cap B_2^\perp \quad \text{and} \quad (B_1 \cap B_2)^\perp = B_1^\perp B_2^\perp$$

for all $B_i \leq A$, and $B \mapsto B^\perp$ is a bijection.

Proof. a) By Theorem 6.1 b), $|A|$ is the number of distinct irreducible characters of A. So $|A| = |\hat{A}|$.

Let $A = A_1 \times \cdots \times A_n$ be a direct decomposition of A into cyclic groups $A_i = \langle a_i \rangle$. Let $|A_i| = n_i$, and let ϵ_i be a primitive n_i-th root of unity in the algebraically closed

field K; there exists such an ϵ_i since n_i divides $|A|$ and char $\mathsf{K} \nmid |A|$. For every element $a = a_1^{y_1} \cdots a_n^{y_n} \in A$, we define a character χ_a of A by

$$\chi_a(a_1^{x_1} \cdots a_n^{x_n}) = \epsilon_1^{x_1 y_1} \cdots \epsilon_n^{x_n y_n}.$$

Clearly, $\chi_{aa'} = \chi_a \chi_{a'}$. If

$$a = a_1^{y_1} \cdots a_n^{y_n} \neq a' = a_1^{y_1'} \cdots a_n^{y_n'},$$

then there exists some i such that $y_i \not\equiv y_i' \bmod n_i$. Then

$$\chi_a(a_i) = \epsilon_i^{y_i} \neq \epsilon_i^{y_i'} = \chi_{a'}(a_i).$$

So, all characters χ_a are pairwise distinct. Thus, the map $a \mapsto \chi_a$ is an isomorphism from A onto $\hat{A}$.

b) By Theorem 6.3 c), τ is a homomorphism from A to $\hat{\hat{A}}$ with kernel K. Since we assume that char $\mathsf{K} \nmid |A|$, we have $K = 1$. Furthermore, we have by a) that $|\hat{\hat{A}}| = |\hat{A}| = |A|$. So τ is an isomorphism from A to $\hat{\hat{A}}$.

c) If $\chi \in B^\perp$, then we may define a character χ' on A/B by

$$\chi'(aB) = \chi(a).$$

This assignment is clearly well defined since $\chi \in B^\perp$. Then $\chi \mapsto \chi'$ is a monomorphism from $B^\perp$ to $\widehat{A/B}$. By Theorem 6.3 a), we have $\hat{B} \cong \hat{A}/B^\perp$. Therefore, it follows by a) that

$$|B^\perp| = \frac{|\hat{A}|}{|\hat{B}|} = \frac{|A|}{|B|} = |A/B| = |\widehat{A/B}|,$$

and so $B^\perp \cong \widehat{A/B}$.

d) The inclusion $B^\tau \leq B^{\perp\perp}$ holds trivially. Using c), we obtain $|B^\perp| = [A : B]$ and therefore

$$|B^{\perp\perp}| = [\hat{A} : B^\perp] = |B| = |B^\tau|.$$

This shows that $B^\tau = B^{\perp\perp}$.

e) The map $B \mapsto B^\perp$ is a bijection:

If $B_1^\perp = B_2^\perp$, then it follows from d) that

$$B_1^\tau = B_1^{\perp\perp} = B_2^{\perp\perp} = B_2^\tau.$$

So $B_1 = B_2$ by b). Since A and $\hat{A}$ are isomorphic by a), they have the same number of subgroups. Thus, $B \mapsto B^\perp$ is also a surjection to the lattice of subgroups of $\hat{A}$.

The relations $(B_1 B_2)^\perp = B_1^\perp \cap B_2^\perp$ and $(B_1 \cap B_2)^\perp \geq B_1^\perp B_2^\perp$ are trivial. By the relations already proved in e) and c), we have

$$|B_1^\perp B_2^\perp| = \frac{|B_1^\perp||B_2^\perp|}{|B_1^\perp \cap B_2^\perp|} = \frac{|B_1^\perp||B_2^\perp|}{|(B_1B_2)^\perp|}$$
$$= \frac{|A|}{|B_1|}\frac{|A|}{|B_2|}\frac{|B_1B_2|}{|A|} = \frac{|A|}{|B_1 \cap B_2|} = |(B_1 \cap B_2)^\perp|.$$

So $(B_1 \cap B_2)^\perp = B_1^\perp B_2^\perp$, as well. □

Finally, we note the following for later use:

Theorem 6.5. *Let* K *be an algebraically closed field, and assume that* $\operatorname{char}\mathsf{K} \nmid |G|$. *The representations of* G *of degree* 1 *correspond bijectively to the irreducible representations of* G/G' *over* K*; their number is* $|G/G'|$. (*We also refer to the characters of the one-dimensional representations of* G *as linear characters.*)

Proof. Every representation of G of degree 1 clearly has the commutator subgroup G' in its kernel, so it corresponds bijectively to a representation of G/G' of degree 1. By Theorem 6.1 b), all irreducible representations of G/G' over K have degree 1, and their number is $|G/G'|$. □

Exercises

16) Let A be a finite abelian p-group. Prove that

$$\Omega_i(A)^\perp = \mho_i(\hat{A}) \quad \text{and} \quad \mho_i(A)^\perp = \Omega_i(\hat{A}).$$

(See Exercise I.51, pg. 91, for the notation.)

17) Let A be a finite abelian p-group, and let α be an automorphism of A that fixes every element of order p in A. Then $|\alpha|$ is a power of p. (This is a special case of Theorem IV.5.12, pg. 465.) (Hint: Define an automorphism $\hat{\alpha}$ of $\hat{A}$ by $\chi^{\hat{\alpha}}(a) = \chi(a^\alpha)$ for $a \in A$ and $\chi \in \hat{A}$. Show that $\hat{\alpha}$ induces the identity automorphism on $\hat{A}/\Phi(\hat{A})$.)

7 Theorems of Burnside, Wielandt and Frobenius

As a first application of the theory of group characters, we prove the famous theorems of Burnside and Frobenius in this section. Both theorems have found numerous applications, some of which we will discuss in later chapters.

Lemma 7.1. *Let* $\mathbf{D}$ *be an irreducible representation of* G *over the complex numbers with* $\deg \mathbf{D} = n$. *We denote the character of* $\mathbf{D}$ *by* χ. *Let* $K = \{k^g \mid g \in G\}$ *be a conjugacy class of* G *with size* $|K| = m$ *satisfying* $(m, n) = 1$. *Then* $\chi(k) = 0$ *or* $\mathbf{D}(k) = c1_n$ *for some* $c \in \mathbb{C}$.

Proof. By Theorem 5.11, the quantity $\frac{m}{n}\chi(k)$ is an algebraic integer. Since $(m,n)=1$, there exist integers a and b such that $ma+nb=1$. Then

$$c = a\frac{m}{n}\chi(k) + b\chi(k) = \frac{\chi(k)}{n}$$

is also an algebraic integer. Let $\epsilon_1, \ldots, \epsilon_n$ be the eigenvalues of $\mathbf{D}(k)$. Then

$$|c| = \frac{1}{n}\Big|\sum_{i=1}^{n}\epsilon_i\Big| \leq \frac{1}{n}\sum_{i=1}^{n}|\epsilon_i| = 1.$$

If the ϵ_i are not all equal, then it is well known that $|c|<1$. If α is an automorphism of the field generated by the $|G|$-th roots of unity over $\mathbb{Q}$, then

$$|c^\alpha| = \frac{1}{n}\Big|\sum_{i=1}^{n}\epsilon_i^\alpha\Big| < 1.$$

It follows that $|\mathrm{N}(c)| = |\prod_\alpha c^\alpha| < 1$. Since c is an algebraic integer, the norm $\mathrm{N}(c)$ of c is an integer. So $\mathrm{N}(c)=0$. Then $c=0$ and $\chi(k)=0$, as well.

Now assume that all ϵ_i are equal, so $c=\epsilon_i$. The restriction of $\mathbf{D}$ to $\langle k\rangle$ is a representation of the cyclic group $\langle k\rangle$. The irreducible constituents of this representation are one-dimensional by Theorem 6.1. Since the representation is completely reducible by Theorem 3.4, $\mathbf{D}(k)$ can be brought into diagonal form, which means $\mathbf{D}(k) = c1_n$. □

The following theorem is an important consequence of Lemma 7.1.

Theorem 7.2 (Burnside). *If G contains a conjugacy class $K \neq \{1\}$ of prime power size, then G is not a simple group of composite order.*

Proof. Let

$$K = \{k^g \mid g \in G\} \neq \{1\}$$

with $|K| = p^e$. If $e=0$, then $1 \neq k \in \mathbf{Z}(G)$ and we are done. So, assume that $e>0$. Then p is a divisor of $|G|$.

If $\chi_j(k) \neq 0$ and $p \nmid n_j$ for some $j \geq 2$, then $\mathbf{D}_j(k) = c1_{n_j}$ by Lemma 7.1. Then G is not simple of composite order, since either the kernel of $\mathbf{D}_j$ is a nontrivial normal subgroup of G (note that $j>1$) or $\mathbf{D}_j$ is faithful and $1 \neq k \in \mathbf{Z}(G)$.

So assume that $\chi_j(k)=0$ for all $j>1$ with $p \nmid n_j$. Since k has trace 0 in the regular representation, Theorem 5.3 yields

$$0 = \sum_{j=1}^{h} n_j\chi_j(k) = 1 + p\sum_{j=2}^{h}\frac{n_j}{p}\chi_j(k) = 1 + pd.$$

But then $d = -\frac{1}{p} = \sum_{j=2}^{h}\frac{n_j}{p}\chi_j(k)$ is an algebraic integer, a contradiction. □

Now there follows a famous criterion for solvability:

Main Theorem 7.3 (Burnside [80]). *If $|G| = p^a q^b$ for some primes p and q, then G is solvable.*

Proof. Let P be a Sylow p-subgroup of G. By Theorem I.6.9 (pg. 31), there exists some $1 \neq x \in \mathbf{Z}(P)$. Then $P \leq \mathbf{C}_G(x)$ and therefore

$$|\{x^g \mid g \in G\}| = [G : \mathbf{C}_G(x)] = q^{b'}$$

with $b' \leq b$. By Theorem 7.2, G is not simple of composite order. So, there exists a normal subgroup N of G such that $1 < N < G$, and we may apply induction to conclude that N and G/N are solvable. Therefore, G is itself solvable. □

Proofs of Theorems 7.2 and 7.3 without the use of group characters are not known at the time of writing.[3] (But see Theorem I.8.11 for the special case $|G| = p^a q$.) J. Thompson proved Theorem 7.3 for $p \neq 2 \neq q$ without character theory. However, many tools from the theory of p-groups and solvable groups were required. His proof is significantly more complicated than the above proof due to Burnside.

We will discuss some important generalizations of Theorem 7.3 in Chapter VI (see Theorem VI.1.10 and Theorem VI.4.3).

Lemma 7.4. *Let $\mathbf{D}$ be a representation of G over the complex numbers, and let χ be the character of $\mathbf{D}$. Then*

$$\ker \mathbf{D} = \{g \in G \mid \chi(g) = \chi(1)\}.$$

Proof. Let $\deg \mathbf{D} = n$. If $\mathbf{D}(g) = 1$, then clearly $\chi(g) = n = \chi(1)$.

Conversely, assume that $\chi(g) = n$. Letting ϵ_i $(i = 1, \ldots, n)$ be the eigenvalues of $\mathbf{D}(g)$, we have

$$n = \chi(g) = \Big|\sum_{i=1}^{n} \epsilon_i\Big| \leq \sum_{i=1}^{n} |\epsilon_i| = n.$$

It follows that all ϵ_i are equal to 1. Since $\mathbf{D}$ restricts to a completely reducible representation of $\langle g \rangle$, we have $\mathbf{D}(g) = 1$. □

Theorem 7.5 (Wielandt [470]). *Let $H^* \trianglelefteq H < G$. Suppose that $H \cap H^g \leq H^*$ for all $g \in G - H$. We define*

$$G^* = G - \bigcup_{g \in G} (H - H^*)^g.$$

Then:

[3] In the early 1970s, D. Goldschmidt, H. Bender and H. Matsuyama proved Burnside's $p^a q^b$-theorem using purely group-theoretic methods. For a textbook reference, see I. M. Isaacs, *Finite group theory*, Graduate Studies in Mathematics, 92, Amer. Math. Soc., Providence, RI, 2008. *trans.*

(1) $G^* \trianglelefteq G$.
(2) $G^* \cap H = H^*$.
(3) $G^*H = G$.

Proof. Clearly, G^* is a subset of G that is mapped onto itself by all inner automorphisms of G. However, it is in no way trivial that G^* is a subgroup of G.

Suppose that $h^* = h^g \in H^* \cap H^g$ for some $g \in G$. If $g \in G - H$, then

$$h = h^{*g^{-1}} \in H \cap H^{*g^{-1}} \leq H^*.$$

On the other hand, if $g \in H$ then $h^g \in H^* \cap H^g \leq H^{*g}$ since $H^* \trianglelefteq H$, and so again $h \in H^*$. That is, $H^* \cap (H - H^*)^g = \emptyset$ for all $g \in G$, and it follows that $H^* \subseteq G^*$.

a) We first show that if ψ is a nontrivial irreducible character of H over the complex numbers with $\ker \psi \geq H^*$, then the map

$$\tau(g) = \begin{cases} \psi(1) & \text{for } g \in G^* \\ \psi(g^x) & \text{for } g^x \in (H - H^*) \end{cases}$$

is an irreducible character of G satisfying $\tau(h) = \psi(h)$ for all $h \in H$.

We proceed with the proof in a series of steps.

(1) τ is well defined:

Let $g^x \in H - H^*$ and $g^y \in H - H^*$. Then

$$g^x \in (H - H^*) \cap (H - H^*)^{y^{-1}x}.$$

It follows that $y^{-1}x \in H$ by our assumption. Since ψ is a character of H, it is a class function on H, and we have

$$\psi(g^x) = \psi(g^{yy^{-1}x}) = \psi(g^y).$$

(2) τ is a class function on G:

If $g \in G^*$ and $t \in G$, then $g^t \in G^*$ as well, so $\tau(g^t) = \psi(1) = \tau(g)$. Now suppose

$$g = h^{x^{-1}} \in (H - H^*)^{x^{-1}}.$$

Then

$$g^t = h^{x^{-1}t} \in (H - H^*)^{x^{-1}t},$$

so $\tau(g^t) = \psi(h) = \tau(g)$.

(3) Let χ_i $(i = 1, \ldots, h(G))$ be the irreducible characters of G. We define $\mu = \tau - \psi(1)\chi_1$, where χ_1 is the trivial character of G. Then $\mu = \sum_i a_i \chi_i$ for some integers a_i and $a_1 = -\psi(1)$:

We have

$$\mu(g) = \begin{cases} 0 & \text{for } g \in G^* \\ \psi(g^x) - \psi(1) & \text{for } g^x \in H - H^*. \end{cases}$$

Let $G = \bigcup_{i=1}^{m} Hr_i$ be the decomposition of G into cosets of H. Using the disjoint decomposition

$$G - G^* = \bigcup_{i=1}^{m} (H - H^*)^{r_i}$$

as well as the fact that $H^* \subseteq G^*$ and $\mu = 0$ on G^*, we have

$$\begin{aligned} a_i = (\mu, \chi_i)_G &= \frac{1}{|G|} \sum_{g \in G} \mu(g)\chi_i(g^{-1}) = \frac{1}{|G|}\frac{|G|}{|H|} \sum_{h \in H - H^*} \mu(h)\chi_i(h^{-1}) \\ &= \frac{1}{|H|} \sum_{h \in H} \mu(h)\chi_i(h^{-1}) = (\mu, \chi_i)_H. \end{aligned}$$

Note that a_i is an integer by the orthogonality relations: Letting ψ_1 be the trivial character of H, $\psi - \psi(1)\psi_1$ is a linear combination of the irreducible characters of H with integer coefficients. The same is true of the restriction of χ_i to H. In particular, since $\psi \neq \psi_1$ we have

$$a_1 = (\mu, \chi_1)_G = (\psi - \psi(1)\psi_1, \psi_1)_H = -\psi(1).$$

(4) We have

$$(\mu, \mu)_G = \sum_i a_i^2 = \psi(1)^2 + 1 = a_1^2 + 1 :$$

By a calculation similar to that in (3), we have

$$\begin{aligned} (\mu, \mu)_G &= \frac{1}{|G|} \sum_{g \in G} \mu(g)\mu(g^{-1}) = \frac{1}{|G|}\frac{|G|}{|H|} \sum_{h \in H - H^*} \mu(h)\mu(h^{-1}) \\ &= \frac{1}{|H|} \sum_{h \in H} \mu(h)\mu(h^{-1}) \\ &= (\mu, \mu)_H = (\psi - \psi(1)\psi_1, \psi - \psi(1)\psi_1)_H = 1 + \psi(1)^2. \end{aligned}$$

So, there is exactly one $i \geq 2$ with $a_i = \pm 1$, all the other a_j with $j \geq 2$ are 0. Hence,

$$\mu = \tau - \psi(1)\chi_1 = \pm\chi_i - \psi(1)\chi_1.$$

It follows that $\tau = \pm\chi_i$. Since

$$\pm\chi_i(1) = \tau(1) = \psi(1) > 0,$$

the class function $\tau = \chi_i$ is in fact an irreducible character of G.

(5) $\tau(h) = \psi(h)$ for all $h \in H$:

If $h \in H - H^*$, this follows immediately from the definition of τ. If $h \in H^* \leq \ker \psi$, then $\tau(h) = \psi(1) = \psi(h)$.

b) The regular representation $\mathbf{R}$ of H/H^* gives rise to a representation $\mathbf{F}$ of H with kernel H^* by defining $\mathbf{F}(h) = \mathbf{R}(hH^*)$. Let ψ be the character of $\mathbf{F}$ and decompose ψ into a linear combination

$$\psi = \sum_i m_i \psi_i$$

of irreducible characters ψ_i of H with $H^* \leq \ker \psi_i$. Clearly,

$$\psi(h) = \begin{cases} 0 & \text{for } h \in H - H^* \\ |H/H^*| & \text{for } h \in H^*. \end{cases}$$

Using the construction in a), we associate to each ψ_i $(i \geq 2)$ an irreducible character τ_i of G. Then

$$\chi = m_1 \chi_1 + \sum_{i \geq 2} m_i \tau_i$$

is a character of G of degree $\chi(1) = \psi(1) = |H/H^*|$. Let $\mathbf{D}$ be a representation of G with $\operatorname{tr} \mathbf{D}(g) = \chi(g)$. We let $K = \ker \mathbf{D}$. By Lemma 7.4, we have

$$K = \{g \in G \mid \chi(g) = \chi(1) = |H/H^*|\}.$$

For $g \in G^*$,

$$\chi(g) = m_1 + \sum_{i \geq 2} m_i \tau_i(g) = \sum_i m_i \psi_i(1) = \psi(1) = |H/H^*|$$

by the construction of the τ_i. So $G^* \subseteq K$. Furthermore,

$$K \cap H = \{h \in H \mid \chi(h) = \chi(1) = \psi(1)\} = \ker \mathbf{F} = H^*.$$

We have

$$|G^*| = |G| - [G : H](|H| - |H^*|) = [G : H]|H^*|.$$

Since $G^* \subseteq K$, it now follows that

$$|KH| = \frac{|K||H|}{|H^*|} \geq \frac{|G^*||H|}{|H^*|} = |G|.$$

Therefore, $G = KH$ and $G^* = K$. Claims (1)–(3) have been proved. □

The following special case of Theorem 7.5, in which $H^* = 1$, was already known to Frobenius; it has found many applications.

Main Theorem 7.6 (Frobenius). *Let $1 < H < G$, and suppose that $H \cap H^g = 1$ for all $g \in G - H$. Then $F = G - \bigcup_{g \in G} (H-1)^g$ is a normal subgroup of G with $G = FH$ and $F \cap H = 1$.*

Currently, no character-free proof of Theorem 7.6 is known. Using transfer, we proved Theorem 7.6 for G solvable in Theorem IV.2.4 (pg. 444) (see also Exercise 18).

8 Frobenius groups

In this section, we consider more closely the structure of the groups appearing in Theorem 7.6.

Definition 8.1. Let $1 < H < G$, and suppose that $H \cap H^g = 1$ for all $g \in G - H$. (In particular, $H = \mathbf{N}_G(H)$.) By Theorem 7.6, $F = G - \bigcup_{g \in G} (H-1)^g$ is a normal subgroup of G, and G is a semidirect product of F with H. We call G a Frobenius group with Frobenius complement H and Frobenius kernel F. Letting $G = \bigcup_{i=1}^{n} Hg_i$, we say that the decomposition of G

$$G = F \cup H^{g_1} \cup \cdots \cup H^{g_n}$$

into subgroups of G that have pairwise intersection 1 is the Frobenius partition of G. (Designating this partition as "the" Frobenius partition is justified; namely, we will prove in Theorem 8.17 that every Frobenius group admits exactly one Frobenius partition.)

In the remainder of this section, we use the notation introduced in Definition 8.1. First, we translate the statement of Theorem 7.6 into the language of permutation groups:

Theorem 8.2.

a) *Let G be a transitive, nonregular permutation group. Assume that every nonidentity element of G has at most one fixed point. Let H be the stabilizer of a point. Then G is a Frobenius group with Frobenius complement H. The fixed point free permutations in G together with the identity constitute the Frobenius kernel F of G, and F is regular.*

b) *Let G be a Frobenius group with Frobenius complement H. Then the permutation representation of G on the cosets of H yields an isomorphism from G to a transitive, nonregular permutation group of degree $[G : H]$ in which every nonidentity element has at most one fixed point.*

Proof. a) Let $H = G_1$ be the stabilizer of the point 1, and let $G = \bigcup_{i=1}^{n} Hg_i$ be the decomposition of G into cosets of H with $1^{g_i} = i$. By our assumption, we have for all $h \in H$ and $i > 1$ that

$$1 = H \cap G_i = H \cap H^{hg_i}.$$

Since G is not regular, we have $1 < H < G$. Hence, the assumptions of Theorem 7.6 are satisfied. Therefore,

$$F = G - \bigcup_{i=1}^{n} (H-1)^{g_i}$$

is a normal subgroup of G with $F \cap H = 1$. Since

$$[F : F \cap H] = |F| = [G : H] = n,$$

the subgroup F is also transitive, so it is regular.

b) Let G be a Frobenius group with Frobenius complement H. By Theorem I.6.2 (pg. 29), the kernel of the permutation representation of G on the cosets of H is $\bigcap_{g \in G} H^g = 1$. If $G = \bigcup_{i=1}^{n} Hg_i$ is the decomposition of G into cosets of H, then H^{g_i} is the stabilizer of the point Hg_i. Since

$$H^{g_j} \cap H^{g_i} = (H \cap H^{g_i g_j^{-1}})^{g_j} = 1 \quad (\text{for } i \neq j),$$

every nonidentity element of G fixes at most one of the cosets Hg_i. □

Theorem 8.3. *Let G be a Frobenius group with Frobenius complement H. Then $|H|$ is a divisor of $[G : H] - 1$. That is, H and the Frobenius kernel F of G are Hall subgroups of G. In particular, F is a characteristic subgroup of G.*

Proof. We represent G as a permutation group of degree $n = [G : H]$ on the set $\Omega = \{1, \ldots, n\}$ according to Theorem 8.2 b). Let $1 \neq i \in \Omega$. The orbit of i under $H = G_1$ has size

$$|i^{G_1}| = [G_1 : G_1 \cap G_i] = |G_1| = |H|.$$

Hence, $\Omega - \{1\}$ decomposes under H into orbits of size $|H|$, and $|H|$ is thus a divisor of $n - 1$. Since $|G| = |F||H|$ and $|F| = n$, F is a normal Hall subgroup of G. By Theorem I.4.9 b) (pg. 22), the subgroup F is therefore a characteristic subgroup of G. □

Definition 8.4.

a) Let α be an automorphism of G. We say α is fixed point free if α fixes no nonidentity element of G.

b) If A is a subgroup of the automorphism group $\mathbf{Aut}(G)$ of G, then we say A is fixed point free if every nonidentity automorphism in A is a fixed point free automorphism of G.

In our further considerations, we will use the following description of Frobenius groups multiple times.

Theorem 8.5. *The following are equivalent:*

a) *G is a Frobenius group with Frobenius complement H and Frobenius kernel F.*

b) *We have $G = FH$ for some $H < G$ and $F \lhd G$. The map μ from H to $\mathbf{Aut}(F)$ defined by $f^{h^\mu} = h^{-1}fh$ for $h \in H$ and $f \in F$ is an isomorphism from H to a fixed point free group of automorphisms of F.*

Proof. a) $\Rightarrow$ b): By Theorem 7.6, we have $G = FH$ with $F \lhd G$ and $F \cap H = 1$. Let $h \in H$ and $f \in F$ such that $f^h = f$. Then

$$fhf^{-1} = h \in H \cap H^{f^{-1}}.$$

If $f \neq 1$, then $f^{-1} \in G - H$, so it follows by our assumptions that $H \cap H^{f^{-1}} = 1$, so $h = 1$. If $h \neq 1$, then we must have $f = 1$. Thus, H acts on F as a fixed point free group of automorphisms.

b) $\Rightarrow$ a): Let $x \in F \cap H$. Then $x = x^x$, so $x = 1$ by the assumption in b). So $F \cap H = 1$, and $G = FH$ is a semidirect product of F with H.

Let $h_1 \in H \cap H^g$ with $g \in G - H$. Then $g = hf$ for some $h \in H$ and $1 \neq f \in F$. It follows that $H^g = H^f$ and

$$h_1 = h_2^f \in H \cap H^f$$

for some $h_2 \in H$. Since $F \lhd G$, we have

$$h_1 h_2^{-1} = [f, h_2^{-1}] \in H \cap F = 1,$$

so that $h_1 = h_2$. Then $f^{h_1} = f$, and it follows by our assumption that $h_1 = 1$ since $f \neq 1$. So $H \cap H^g = 1$, and G is a Frobenius group with Frobenius complement H.□

Example 8.6.

a) Let G be the group consisting of all maps g on the set $\mathsf{K} = \mathrm{GF}(p^f)$ with $p^f > 2$ of the form $xg = ax + b$ $(a \neq 0)$. Every nonidentity element of G clearly has at most one fixed point. Since G is transitive but not regular, G is a Frobenius group with Frobenius complement

$$H = \left\{ \begin{pmatrix} x \\ ax \end{pmatrix} \middle| \, a \in \mathsf{K}^\times \right\}$$

by Theorem 8.2 a). The group

$$F = \left\{ \begin{pmatrix} x \\ x + b \end{pmatrix} \middle| \, b \in \mathsf{K} \right\}$$

is the Frobenius kernel of G.

b) (Zassenhaus [494]) Let q be a prime power and m a natural number with $m \mid q - 1$. Let $\mathsf{K} = \mathrm{GF}(q^m)$ and $\mathsf{K}^\times = \langle a \rangle$. We define a subgroup H of the group of semilinear maps on K in the following way: Let

$$V = \left\{ \begin{pmatrix} x \\ vx \end{pmatrix} \Big| \, v \in (\mathsf{K}^\times)^m \right\},$$

and let $H = \bigcup_{i=0}^{m-1} h_i V$ with $h_0 = 1$ and

$$xh_i = a^{1+q+\cdots+q^{i-1}} x^{q^i}$$

for $1 \le i \le m-1$ and $x \in \mathsf{K}$. We have

$$xh_ih_j = a^{1+q+\cdots+q^{j-1}} (a^{1+q+\cdots+q^{i-1}} x^{q^i})^{q^j} = a^{1+q+\cdots+q^{i+j-1}} x^{q^{i+j}}.$$

So $h_ih_j = h_{i+j}$ for $i+j \le m-1$; if $i+j = m+k$ with $0 \le k \le m-2$, then since $a^{q^m} = a$ we have

$$xh_ih_j = a^{1+q+\cdots+q^{m-1}} a^{1+q+\cdots+q^{k-1}} x^{q^k} = xh_ku,$$

where u is defined by $xu = a^{1+q+\cdots+q^{m-1}} x$. Since $q \equiv 1 \bmod m$, we have

$$a^{1+q+\cdots+q^{m-1}} \in (\mathsf{K}^\times)^m,$$

so $u \in V$. Hence, H is a group.

H acts regularly on $\mathsf{K}^\times$:

Clearly, V acts fixed point freely on $\mathsf{K}^\times$. Assume for contradiction that $x \in \mathsf{K}^\times$ and

$$x = xh_iu = va^{1+q+\cdots+q^{i-1}} x^{q^i}$$

for some i with $0 < i \le m-1$ and some $v \in (\mathsf{K}^\times)^m$. Since $q^i \equiv 1 \bmod m$, this leads to

$$a^{1+q+\cdots+q^{i-1}} = v^{-1} x^{1-q^i} \in (\mathsf{K}^\times)^m.$$

But since a is a generator of $\mathsf{K}^\times$, this yields the contradiction

$$1 + q + \cdots + q^{i-1} \equiv i \equiv 0 \mod m.$$

We construct the semidirect product G of H with $F = \{\left(\begin{smallmatrix} x \\ x+b \end{smallmatrix}\right) \mid b \in \mathsf{K}\}$ inside the group $\Gamma(q^m)$ of all affine semilinear maps from K to itself. Then G is a permutation group on K, where F is a regular normal subgroup of G and $G_0 = H$. Since H acts regularly on $\mathsf{K}^\times$, G is a Frobenius group by Theorem 8.2 a). Clearly, G is sharply 2-transitive on K.

(This example plays a role in the construction of finite near-fields in Zassenhaus [494].)

c) (N. Itô) Working over the field $\mathsf{K} = \mathrm{GF}(p^f)$, we consider the group

$$F = \{(k_{ij}) \mid k_{ij} \in \mathsf{K}, k_{ii} = 1, k_{ij} = 0 \text{ for } j > i; i, j = 1, \ldots, n\}$$

of all lower triangular matrices with diagonal elements 1. By Theorem III.16.3 (pg. 405), F is a nilpotent group of class $n-1$. Let q be a prime divisor of p^f-1 with $q \geq n$. Then there exist pairwise distinct elements $d_1, \ldots, d_n$ in K with $d_i^q = 1$. We construct the diagonal matrix

$$h = \begin{pmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_n \end{pmatrix}$$

and we claim: For $0 < m < q$, h^m acts on F as a fixed point free automorphism via conjugation. Namely, let $f = (k_{ij})$ and suppose $h^{-m} f h^m = f$. This forces $d_i^{-m} k_{ij} d_j^m = k_{ij}$ for all i, j. Since $(m, q) = 1$, we have $d_i^m \neq d_j^m$ for $i \neq j$. So $k_{ij} = 0$ for $i \neq j$ and therefore $f = 1$. By Theorem 8.5, the semidirect product $G = FH$ with $H = \langle h \rangle$ is then a Frobenius group with Frobenius complement H and Frobenius kernel F.

The groups we constructed in Example 8.6 have very special properties: The Frobenius kernels F were nilpotent and the Frobenius complements H were cyclic or metacyclic. We will show that there are strong constraints on the subgroups F and H in a Frobenius group. Our main objective is the following theorem.

Main Theorem 8.7. *Let G be a Frobenius group with Frobenius complement H and Frobenius kernel F.*

a) (*Thompson*) *F is nilpotent.*

b) (*Burnside*) *The Sylow p-subgroups of H are cyclic if $p > 2$, and they are cyclic or generalized quaternion if $p = 2$.*

Remark 8.8. Let G be a Frobenius group with Frobenius complement H and Frobenius kernel F.

a) Example 8.6 c) shows that there is no upper bound for the nilpotency class $c(F)$ of the Frobenius kernel. However, the class $c(F)$ can be bounded if we have some knowledge of the prime divisors of $|H|$. Namely,
(1) If 2 divides $|H|$, then F is abelian (see Theorem 8.18).
(2) If 3 divides $|H|$, then $c(F) \leq 2$ (Burnside [84], pg. 90).
(3) If p is an odd prime divisor of $|H|$, then

$$c(F) \leq \frac{(p-1)^{2^{p-1}-1} - 1}{p-2}$$

(Kostrikin, Kreknin [312], Kreknin [314], G. Higman [238]).
G. Higman has conjectured that $c(F) \leq \frac{p^2-1}{4}$ is the best possible bound; this has been proved for $p \leq 5$ (see G. Higman [238]).

b) The conjecture of Burnside [84] (pg. 336) that H is nilpotent is not correct. A Frobenius complement H need not necessarily even be solvable, as the following example shows:

For $p^2 \equiv 1 \bmod 5$, the group $\mathrm{PSL}_2(p)$ contains a subgroup isomorphic to the alternating group A_5 by Theorem II.8.13 (pg. 208). This gives rise to a nonsolvable subgroup H of order 120 inside $\mathrm{SL}_2(p)$. We show that H is a fixed point free automorphism group of the elementary abelian group $\mathsf{V} = \mathsf{V}(2, p)$ of type (p, p), written additively:

Suppose the element $h \in H$ fixes some nonzero vector $v_1 \in \mathsf{V}$. Since $\det h = 1$, the two eigenvalues of h are both 1. So there exists a basis $\{v_1, v_2\}$ of V such that

$$v_1 h = v_1 \quad \text{and} \quad v_2 h = a v_1 + v_2$$

for some $a \in \mathrm{GF}(p)$. It follows that $h^p = 1$. Since $p \nmid |H|$, this forces $h = 1$.

Before proving Theorem 8.7 a), we first prove a few useful theorems regarding fixed point free automorphisms.

Theorem 8.9. *Let α be a fixed point free automorphism of G with $|\alpha| = n$.*
a) *If $(m, n) = 1$, then α^m is also a fixed point free automorphism of G.*
b) *The map μ defined by $g^\mu = g^{-1} g^\alpha$ is a bijection from G onto itself (however, it is not an automorphism of G in general).*
c) *If g and g^α are conjugate in G, then $g = 1$.*
d) *For all $g \in G$, we have $g g^\alpha g^{\alpha^2} \cdots g^{\alpha^{n-1}} = 1$.*

Proof. a) Since $(m, n) = 1$, α is a power of α^m, which means that the only fixed point of α^m is also 1.

b) Suppose $g_1, g_2 \in G$ and

$$g_1^{-1} g_1^\alpha = g_1^\mu = g_2^\mu = g_2^{-1} g_2^\alpha.$$

It follows that $g_1 g_2^{-1} = (g_1 g_2^{-1})^\alpha$, which means $g_1 = g_2$ since α is fixed point free. Hence, μ is an injective map from G to itself. Since G is finite, μ is also a surjective map.

c) Assume $g^\alpha = x^{-1} g x$ for some $g, x \in G$. By b), there exists some $y \in G$ such that $x = y^{-1} y^\alpha$. It follows that

$$g^\alpha = y^{-\alpha} y g y^{-1} y^\alpha,$$

so $y g y^{-1} = (y g y^{-1})^\alpha$. This forces $y g y^{-1} = 1$, so $g = 1$.

d) We let $h = g g^\alpha \cdots g^{\alpha^{n-1}}$. Since $\alpha^n = 1$, we have

$$h^\alpha = g^\alpha g^{\alpha^2} \cdots g^{\alpha^{n-1}} g^{\alpha^n} = g^{-1} h g.$$

Then it follows from c) that $h = 1$. □

Theorem 8.10. *Let α be a fixed point free automorphism of G, and let N be a normal subgroup of G such that $N^\alpha = N$. Then α induces a fixed point free automorphism of G/N.*

Proof. Clearly, α restricts to a fixed point free automorphism of N. Then we have $N = \{n^{-1}n^\alpha \mid n \in N\}$ by Theorem 8.9 b).

Let $g \in G$ with $gN = g^\alpha N$. Then $g^{-1}g^\alpha \in N$, so $g^{-1}g^\alpha = n^{-1}n^\alpha$ for some $n \in N$. It follows that $(gn^{-1})^\alpha = gn^{-1}$, so $g = n$. The automorphism $\overline{\alpha}$ of G/N induced by α defined by $(gN)^{\overline{\alpha}} = g^\alpha N$ is therefore fixed point free. □

If α is a fixed point free automorphism of G, then it is easy to show that $(|G|, |\alpha|) = 1$.[4] Thus, Theorem 8.10 is also an easy consequence of Theorem I.18.6 (pg. 140).

Theorem 8.11. *Let α be a fixed point free automorphism of G. If G contains Hall π-subgroups and all Hall π-subgroups of G are conjugate, then there is exactly one Hall π-subgroup H of G such that $H^\alpha = H$.*

In particular, for every prime p there is exactly one Sylow p-subgroup P of G such that $P^\alpha = P$.

Proof. Let H be an arbitrary Hall π-subgroup of G. By our assumption, $H^\alpha = H^g$ for some $g \in G$. By Theorem 8.9 b), we have $g = x^{-1}x^\alpha$ for some $x \in G$. Then

$$(xHx^{-1})^\alpha = x^\alpha H^\alpha x^{-\alpha} = x^\alpha g^{-1} H g x^{-\alpha} = xHx^{-1},$$

and so xHx^{-1} is an α-invariant Hall π-subgroup of G.

Let H be a Hall π-subgroup of G with $H^\alpha = H$. Suppose that $(H^g)^\alpha = H^g$ for some $g \in G$. Then

$$H^g = (H^g)^\alpha = (H^\alpha)^{g^\alpha} = H^{g^\alpha},$$

so $gg^{-\alpha} \in \mathbf{N}_G(H)$. Since α fixes H, it also fixes $\mathbf{N}_G(H)$. So α restricts to a fixed point free automorphism of $\mathbf{N}_G(H)$, which means $gg^{-\alpha} = n^{-1}n^\alpha$ for some $n \in \mathbf{N}_G(H)$ by Theorem 8.9 b). It follows that $ng = (ng)^\alpha$, so $ng = 1$. Hence, $H^g = H^{n^{-1}} = H$, and H is therefore the unique α-invariant Hall π-subgroup of G. □

Lemma 8.12. *Let V be a vector space of arbitrary dimension over the field K, and let A be a finite group of linear maps from V onto itself. Assume that A is fixed point free; that is, if $1 \neq a \in A$ and $0 \neq v \in \mathsf{V}$ then $va \neq v$.*

a) *If $|A| = pq$ for some (not necessarily distinct) primes p and q, then A is cyclic.*

b) *The Sylow p-subgroups of A are cyclic for $p > 2$, and they are cyclic or generalized quaternion groups for $p = 2$.*

Proof. a) Suppose $p = q$, and assume for contradiction that A is not cyclic. Then A has exactly $p + 1$ subgroups A_i ($i = 1, \ldots, p + 1$) of order p. Then $A = \bigcup_{i=1}^{p+1} A_i$ and $A_i \cap A_j = 1$ for $i \neq j$.

[4] This statement is not true; for example, $G = C_3 \times (C_2 \times C_2)$ admits a fixed point free automorphism of order 6. If, however, $\langle \alpha \rangle$ is fixed point free, then it is easy to see that $(|G|, |\alpha|) = 1$ by considering the orbits of $\langle \alpha \rangle$ on G as in the proof of Theorem 8.3. It also follows that Theorem 8.10 cannot be deduced from Theorem I.18.6. *trans.*

Now suppose $p > q$, and assume for contradiction that A is not cyclic. By Theorem I.8.10 (pg. 41), A has exactly one Sylow p-subgroup A_1 and p distinct Sylow q-subgroups $A_2, \ldots, A_{p+1}$. Again, $A = \bigcup_{i=1}^{p+1} A_i$ and $A_i \cap A_j = 1$ for $i \neq j$.

In what follows, we handle the cases $p = q$ and $p > q$ together.

Let $0 \neq v \in \mathsf{V}$, and let $1 \neq a' \in A$. Then

$$\left(\sum_{a \in A} va\right)a' = \sum_{a \in A} va,$$

so that $\sum_{a \in A} va = 0$ by our assumption. In the same way, we find that $\sum_{a \in A_i} va = 0$. Therefore, we obtain

$$0 = \sum_{a \in A} va = -pv + \sum_{i=1}^{p+1} \sum_{a \in A_i} va = -pv.$$

Hence, $\operatorname{char} \mathsf{K} = p$. Let $x \in A$ with $|x| = p$. Since $\operatorname{char} \mathsf{K} = p$, we have

$$0 = v(x^p - 1) = v(x - 1)^p.$$

So there exists some i with $0 \leq i < p$ such that $v(x-1)^i \neq 0$ and $v(x-1)^{i+1} = 0$. But then letting $w = v(x-1)^i$, we obtain the contradiction $wx = w \neq 0$.

b) By a), the Sylow p subgroups of A do not contain a subgroup of type (p, p), so they contain a unique subgroup of order p. The claim then follows by Theorem III.8.2 (pg. 526). □

Main Theorem 8.13 (Hughes, Kegel, Thompson). *Let p be a prime number. If G admits an automorphism α such that $\alpha^p = 1$ and*

$$gg^{\alpha} \cdots g^{\alpha^{p-1}} = 1$$

for all $g \in G$, then G is nilpotent.

Proof. Let G be a minimal counterexample to the theorem. We construct the semidirect product $S = GH$ of G with the cyclic group $H = \langle h \rangle$ of order p by defining conjugation of $g \in G$ by h to be $g^h = g^{\alpha}$. We show:

1) If $1 < N \lhd G$ and $N^{\alpha} = N$, then N and G/N are nilpotent:

The automorphism α of G induces an automorphism $\overline{\alpha}$ of G/N with $\overline{\alpha}^p = 1$ by defining $(gN)^{\overline{\alpha}} = g^{\alpha}N$. Then we have

$$(gN)^{1+\overline{\alpha}+\cdots+\overline{\alpha}^{p-1}} = g^{1+\alpha+\cdots+\alpha^{p-1}}N = N$$

for all $g \in G$. Since $|G/N| < |G|$, the quotient G/N is nilpotent. (It is possible that $\overline{\alpha} = 1$, but that does not interfere with our conclusion.) Clearly, N also satisfies our assumptions, so N is also nilpotent.

2) We have $\mathbf{Z}(G) = 1$:

If $1 < \mathbf{Z}(G) \leq G$, then $G/\mathbf{Z}(G)$ is nilpotent because of 1), and then G is itself nilpotent. This contradicts that G is a counterexample.

3) G is not solvable, so G is characteristically simple by 1):

Assume for contradiction that G is solvable. Then we show:

(3a) There exists a normal subgroup Q of G with $Q^\alpha = Q$ and $1 < Q \lhd G$. The groups Q and G/Q are elementary abelian:

Let G/Q be a chief factor of S. Then G/Q is an elementary abelian r-group for some prime number r. By 1), Q is nilpotent. Let $Q = Q_1 \times \cdots \times Q_k$ be the direct decomposition of Q into Sylow subgroups, where $Q_i \neq 1$ for all i. By 1), the factor groups G/Q_i are nilpotent. If $k > 1$, then it follows from Theorem III.2.5 c) (pg. 276) that $G = G/(Q_1 \cap Q_2)$ is nilpotent, contradicting our hypothesis. So $k = 1$ and Q is a q-group for some prime q. Assume for contradiction that $\Phi(Q) \neq 1$. Since $Q \lhd S$, we have $\Phi(Q) \lhd S$ and so $G/\Phi(Q)$ is nilpotent by 1). Since $Q \lhd G$, we have $\Phi(Q) \leq \Phi(G)$ (see Lemma III.3.3 b), pg. 284). But then G is nilpotent by Theorem III.3.7 (pg. 286), contradicting our choice of G. Therefore, $\Phi(Q) = 1$ and Q is an elementary-abelian q-group.

(3b) We have $|G/Q| = p$:

Assume for contradiction that $|G/Q| = r^a$ for some prime r distinct from p. We define $\overline{S} = S/Q$ and $\overline{G} = G/Q$. Since Q is abelian, we may give Q the structure of an $\overline{S}$-module by defining $x^{Qs} = x^s$ for all $x \in Q$ and $s \in S$.

Assume for contradiction that $\overline{g}^{\overline{h}} = \overline{g}$ for some $\overline{1} \neq \overline{g} \in \overline{G}$. Then

$$\overline{1} = \overline{g}^{\overline{1}+\overline{h}+\cdots+\overline{h}^{p-1}} = \overline{g}^p,$$

contradicting our assumption that $r \neq p$. Thus, the conjugation action of $\overline{h}$ on $\overline{G}$ is a fixed point free automorphism of order p. By Theorem 8.5, this means $\overline{S}$ is a Frobenius group with Frobenius kernel $\overline{G}$.

We now compute $m(x) = \prod_{\overline{s} \in \overline{S}} x^{\overline{s}}$ for $x \in Q$ in two different ways. (Note that $m(x)$ is well defined and independent of the order of the factors since Q is commutative.) Let

$$\overline{S} = \overline{G} \cup \bigcup_{i=1}^{r^a} \overline{H}_i$$

be the Frobenius partition of $\overline{S}$, where the subgroups $\overline{H}_i$ are the Sylow p-subgroups of $\overline{S}$. Since the $\overline{H}_i$ are conjugate to $\overline{H}$ in $\overline{S}$, we have for all $x \in Q$ and every i that

$$\prod_{\overline{h} \in \overline{H}_i} x^{\overline{h}} = 1.$$

This yields

$$m(x) = \prod_{\overline{g}\in\overline{G}} x^{\overline{g}} \prod_{i=1}^{r^a} \prod_{\overline{1}\neq\overline{h}\in\overline{H}_i} x^{\overline{h}} = x^{-r^a} \prod_{\overline{g}\in\overline{G}} x^{\overline{g}}.$$

Now we use the decomposition $\overline{S} = \bigcup_{i=0}^{p-1} \overline{G}\,\overline{h}^i$ of $\overline{S}$ into cosets of $\overline{G}$. This gives

$$m(x) = \left(\prod_{\overline{g}\in\overline{G}} x^{\overline{g}}\right)^{\overline{1}+\overline{h}+\cdots+\overline{h}^{p-1}} = \left(\prod_{\overline{g}\in\overline{G}} x^{\overline{g}}\right)^{1+\alpha+\cdots+\alpha^{p-1}} = 1.$$

Putting our two calculations of $m(x)$ together, we conclude that

$$x^{r^a} = \prod_{\overline{g}\in\overline{G}} x^{\overline{g}}.$$

Since $r \neq q$ (otherwise G is nilpotent) and Q is commutative, this implies that $Q \leq \mathbf{Z}(G)$, contradicting 2). So in fact $p = r$. Then $|S/Q|$ is a power of p, which means $|G/Q| = p$ since G/Q is a chief factor of S/Q.

(3c) Let P^* be a Sylow p-subgroup of S with $h \in P^*$. We define $P = G \cap P^*$. By (3b), we have $|P| = p$, so $P = \langle y\rangle$ for some y. We claim that h and y induce fixed point free automorphisms of Q:

Assume for contradiction that $1 \neq x \in Q$ and $x^h = x$. Then

$$1 = x^{1+h+\cdots+h^{p-1}} = x^p.$$

But then $q = p$ and G is nilpotent, contradicting the choice of G. If $x \in Q$ such that $x^y = x$, then it follows that $x \in \mathbf{Z}(QP) = \mathbf{Z}(G) = 1$.

(3d) The elementary abelian p-group $\langle h, y\rangle$ cannot be a group of fixed point free automorphisms of Q by Lemma 8.12. Since h and y act fixed point freely on Q by (3c), this means there exists an element $k = h^i y^j$ with $i \not\equiv 0 \not\equiv j \bmod p$ that centralizes some $1 \neq x \in Q$. By replacing h with h^i and y with y^j, we may take $k = hy$. Now it follows that

$$\begin{aligned} 1 &= (xy^{-1})^{1+\alpha+\cdots+\alpha^{p-1}} = (xy^{-1})^{1+h+\cdots+h^{p-1}} \\ &= xy^{-1}x^h y^{-1}x^{h^2}y^{-1}\cdots x^{h^{p-1}}y^{-1} \\ &= xy^{-1}x^h yy^{-2}x^{h^2}y^2\cdots y^{-(p-1)}x^{h^{p-1}}y^{p-1} \\ &= x^{1+hy+(hy)^2+\cdots+(hy)^{p-1}} = x^p. \end{aligned}$$

But this forces the contradiction $q = p$. Therefore, G is not solvable.

4) Now we handle the general case:

Let G be a non-nilpotent group admitting an automorphism α with the given properties. Since G is not a 2-group, $|G|$ is divisible by an odd prime q. If $p = 2$, then $g^\alpha = g^{-1}$ for all $g \in G$ and G is abelian. So $p \neq 2$.

First assume that p divides $|G|$, and let $1 + kp$ be the number of Sylow p-subgroups of G. These are permuted by α, and the $\langle\alpha\rangle$-orbits have sizes 1 and p. So there must exist an orbit of size 1, and so there exists a Sylow p-subgroup P of G such that $P^\alpha = P$.

Now assume that p does not divide $|G|$. If $g \in G$ is fixed by α, then it follows that

$$1 = g^{1+\alpha+\cdots+\alpha^{p-1}} = g^p,$$

and so $g = 1$. Hence, α is a fixed point free automorphism of G with order p. Thus, for every prime divisor q of $|G|$, there exists a Sylow q-subgroup Q of G such that $Q^\alpha = Q$ by Theorem 8.11.

Now let Q be some Sylow subgroup of G for an odd prime (possibly the prime p) dividing $|G|$ such that $Q^\alpha = Q$. Let $\mathbf{J}(Q)$ be the characteristic subgroup of Q defined in Definition IV.6.1 (pg. 465). Since $Q^\alpha = Q$ and $\mathbf{J}(Q) \not\trianglelefteq G$, we have $\mathbf{N}_G(\mathbf{J}(Q))^\alpha = \mathbf{N}_G(\mathbf{J}(Q)) < G$. So $\mathbf{N}_G(\mathbf{J}(Q))$ is nilpotent; in particular, it is q-nilpotent. In the same way, $\mathbf{C}_G(\mathbf{Z}(Q))$ is q-nilpotent. By Thompson's Theorem IV.6.2 (pg. 466), we conclude that G itself is q-nilpotent. The normal q-complement of G is characteristic in G, so it is invariant under α. But this contradicts the observation in 1) that G is characteristically simple. □

Theorem 8.13 has an interesting application in the theory of groups with partitions (see Baer [28, 29, 30], Kegel [300]). We will return to this in Volume 3. The most important application of Theorem 8.13 is the following theorem of Thompson, from which Theorem 8.7 a) immediately follows:

Main Theorem 8.14 (Thompson). *If the group G admits a fixed point free automorphism of prime order, then G is nilpotent.*

Proof. The claim follows immediately from Theorem 8.9 d) and Theorem 8.13. □

The following theorem contains Theorem 8.7 b).

Theorem 8.15. *Let A be a fixed point free group of automorphisms of G.*

a) *The Sylow p-subgroups of A are cyclic for $p > 2$, and they are cyclic or generalized quaternion for $p = 2$.*

b) *If p_1 and p_2 are primes, then every subgroup of A of order p_1p_2 is cyclic.*

Proof. We may assume that $A \neq 1$. Then A contains elements of prime order, which act as fixed point free automorphisms of G of the same prime order. So G is nilpotent by Theorem 8.14. (One can also easily avoid applying Thompson's deep Theorem 8.14 here by using Theorem 8.11 to prove the claim instead.)

Let Q be a Sylow subgroup of the nilpotent group G. By Theorem 8.10, every $1 \neq \alpha \in A$ induces a fixed point free automorphism of $Q/\Phi(Q)$. So A is isomorphic to a fixed point free automorphism group of the elementary abelian group $Q/\Phi(Q)$. If we consider $Q/\Phi(Q)$ as a vector space over a prime field, then applying Lemma 8.12 yields the claim. □

Now we give a few consequences of the main theorems.

Theorem 8.16. *Let G be a Frobenius group with Frobenius complement H and Frobenius kernel F. If $N \trianglelefteq G$, then $N \leq F$ or $F < N$.*

Proof. Assume $N \not\leq F$. Then there exists an $n \in N$ such that $n \notin F$. By the Frobenius partition of G, the element n must lie in a conjugate of H. By replacing n with a conjugate if necessary, we may force $n \in H$. As n is a fixed point free automorphism of F, it follows by Theorem 8.9 b) that

$$F = \{f^{-1}f^n = [f, n] \mid f \in F\} \leq N. \qquad \square$$

Theorem 8.17. *Every Frobenius group has exactly one Frobenius partition.*

Proof. Let G be a Frobenius group, and suppose G has Frobenius complements H_1 and H_2 with corresponding Frobenius kernels F_1 and F_2. By Theorem 8.16, we must have $F_1 \leq F_2$ or $F_2 < F_1$. Assume that, say, $F_1 \leq F_2$.

Assume for contradiction that $F_1 < F_2$. Since $G = F_1H_1$ and $F_1 < F_2$, it follows that $F_2 = F_1(F_2 \cap H_1)$. The groups F_1 and $F_2 \cap H_1$ have coprime order since $(|F_1|, |H_1|) = 1$ by Theorem 8.3. Since F_2 is nilpotent by Theorem 8.7 a), the subgroups F_1 and $F_2 \cap H_1$ commute. But this is impossible since the nonidentity subgroup $F_2 \cap H_1$ acts on F_1 as a group of fixed point free automorphisms.

So we have $F_1 = F_2$. Then H_1 and H_2 are conjugate in G by the Zassenhaus Theorem I.18.2 (pg. 136), being complements of the nilpotent normal Hall subgroup F_1.$\square$

Theorem 8.18. *Let G be a Frobenius group with Frobenius complement H and Frobenius kernel F.*

a) *If $|H|$ is even, then H contains exactly one element i of order 2. We have $f^i = f^{-1}$ for all $f \in F$. In particular, F is abelian in this case.*

b) *If $|H|$ is odd, then H is metacyclic. If x is an element of H with prime order, then $\langle x \rangle \trianglelefteq H$ and $x \in H'$ or $x \in \mathbf{Z}(H)$.*

c) *We always have $\mathbf{Z}(H) \neq 1$.*

Proof. a) Let $i \in H$ with $|i| = 2$. By Theorem 8.9 b),

$$F = \{f^{-1}f^i \mid f \in F\}.$$

We have

$$(f^{-1}f^i)^i = f^{-i}f^{i^2} = f^{-i}f = (f^{-1}f^i)^{-1}.$$

So $f^i = f^{-1}$ for all $f \in F$, and F is abelian. If i_1 and i_2 are involutions in H, then

$$i_2^{-1}i_1 \in H \cap \mathbf{C}_G(F) = 1.$$

So H contains a unique involution.

b) Assume $|H|$ is odd. By Theorem 8.15, all the Sylow subgroups of H are cyclic. We determined the structure of groups whose Sylow subgroups are cyclic in Theorem IV.2.11 (pg. 447): We have $H = \langle a \rangle \langle b \rangle$ with $H' = \langle a \rangle$ and $(|a|, |b|) = 1$.

Let $x \in H$ with $|x| = p$. If p divides $|a|$, then $\langle x \rangle$ is a characteristic subgroup of H', so $\langle x \rangle$ is normal in H. Now assume that p divides $|b|$. By passing to a conjugate of $\langle x \rangle$ if necessary, we may assume that $\langle x \rangle \leq \langle b \rangle$. Let $Q = \langle y \rangle$ be a Sylow q-subgroup of H' for some prime q dividing $|a|$. By Theorem 8.15 b), the subgroup $\Omega_1(Q) \leq Q$ is centralized by x. Let $|y| = q^m$ and $y^x = y^k$ for some integers m and k. Since

$$(y^{q^{m-1}})^x = y^{q^{m-1}},$$

we have $k \equiv 1 \bmod q$, so $k^{q^m} \equiv 1 \bmod q^m$. Since $k^p \equiv 1 \bmod q^m$ and $p \neq q$, it follows that $k \equiv 1 \bmod q^m$. That is, $y^x = y$. Since this holds for all prime divisors q of $|a|$, we conclude that $x \in \mathbf{Z}(H)$.

c) If $|H|$ is even, then $1 \neq i \in \mathbf{Z}(H)$ by a). If $|H|$ is odd, then H is metacyclic by b). Since $(|H'|, |H/H'|) = 1$ and $H' < H$, we have by b) that p divides $|\mathbf{Z}(H)|$ for every prime divisor p of $|H/H'|$. □

We will discuss the many applications of the theory of Frobenius groups to the theory of permutation groups in Volume 3. Here we mention only the following easy result:

Theorem 8.19. *Let G be a primitive, nonregular permutation group. Assume that every nonidentity element of G has at most one fixed point. Then G is a Frobenius group with an elementary abelian Frobenius kernel.*

Proof. By Theorem 8.2 a), G is a Frobenius group with Frobenius complement $H = G_1$. Letting F be the Frobenius kernel of G, we have $G = FG_1$ and $F \cap G_1 = 1$. Since G is primitive, G_1 is a maximal subgroup of G by Theorem II.1.4 (pg. 155). So F is a minimal normal subgroup of G, which means it is characteristically simple. Since F is nilpotent by Theorem 8.7 a), F is elementary abelian. □

We will describe the irreducible representations of Frobenius groups in Theorem 16.13.

Exercises

18) Let G be a Frobenius group with Frobenius complement H. Give an elementary proof for the existence of the Frobenius kernel F in the following special cases:
   a) $[G : H]$ is a prime power.
   b) $[G : H] = n$ and $|H| = n - 1$ or $|H| = \frac{n-1}{2}$. In fact, the Frobenius kernel F is abelian in these cases.

   (Hint for b): For $1 \neq f \in G - \bigcup_{g \in G} (H - 1)^g$, prove that $\mathbf{C}_G(f) \cap H = 1$ and $|\mathbf{C}_G(f)| \geq n$.)

9 Tensor products of modules and algebras

The concept of the tensor product of modules plays an important role in every part of algebra. We introduce the tensor product in full generality here, but we develop the theory only as far as we will need it for our purposes.

Definition 9.1. Let R be a ring with unity, let $\mathfrak{v}$ be a right R-module, and let $\mathfrak{w}$ be a left R-module in the general sense of Chapter I, §10.

a) A map μ from the product set $\mathfrak{v} \times \mathfrak{w}$ to the abelian group $\mathfrak{s}$ (written additively) is said to be R-balanced if for all $v_1, v_2, v \in \mathfrak{v}$, all $w_1, w_2, w \in \mathfrak{w}$, and all $r \in R$ we have

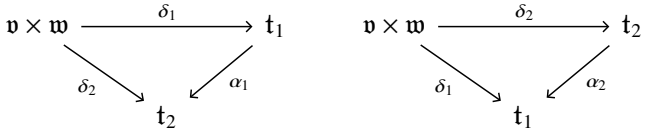
$$(v_1 + v_2, w)\mu = (v_1, w)\mu + (v_2, w)\mu$$
$$(v, w_1 + w_2)\mu = (v, w_1)\mu + (v, w_2)\mu$$
$$(vr, w)\mu = (v, rw)\mu.$$

b) An abelian group $\mathfrak{t}$ (written additively) is called the tensor product of $\mathfrak{v}$ and $\mathfrak{w}$ over R if:
   (1) There exists an R-balanced map δ from $\mathfrak{v} \times \mathfrak{w}$ to $\mathfrak{t}$ such that $\mathfrak{t}$ is generated by the $(v, w)\delta$.
   (2) If μ is an R-balanced map from $\mathfrak{v} \times \mathfrak{w}$ to some abelian group $\mathfrak{s}$, then there exists a group homomorphism α from $\mathfrak{t}$ to $\mathfrak{s}$ such that $\mu = \delta\alpha$.
   We write $\mathfrak{t} = \mathfrak{v} \otimes_R \mathfrak{w}$.

The justification for calling this construction "the" tensor product is provided by the following theorem.

Theorem 9.2. *There exists at most one tensor product $\mathfrak{v} \otimes_R \mathfrak{w}$ up to isomorphism.*

Proof. Let $\mathfrak{t}_1$ and $\mathfrak{t}_2$ be tensor products of $\mathfrak{v}$ and $\mathfrak{w}$ with R-balanced maps δ_1 and δ_2. Then we have the "commutative diagrams"



with $\delta_2 = \delta_1\alpha_1$ and $\delta_1 = \delta_2\alpha_2$. It follows that $\delta_2 = \delta_2\alpha_2\alpha_1$ and $\delta_1 = \delta_1\alpha_1\alpha_2$. Hence, $\alpha_2\alpha_1 = 1$ on the image of δ_2 and $\alpha_1\alpha_2 = 1$ on the image of δ_1. Since the images of $\mathfrak{v} \times \mathfrak{w}$ under δ_i generate all of $\mathfrak{t}_i$, it follows that $\alpha_2\alpha_1 = 1$ on $\mathfrak{t}_2$ and $\alpha_1\alpha_2 = 1$ on $\mathfrak{t}_1$. Therefore, α_1 is an isomorphism from $\mathfrak{t}_1$ to $\mathfrak{t}_2$. □

Theorem 9.3. *Let $\mathfrak{v}$ be a right R-module, and let $\mathfrak{w}$ be a left R-module. Then there exists a tensor product $\mathfrak{v} \otimes_R \mathfrak{w}$.*

Proof. Let $\mathfrak{f}$ be the free abelian group (written additively) in the free generators $f(v, w)$ with $v \in \mathfrak{v}$ and $w \in \mathfrak{w}$. Let the subgroup $\mathfrak{d}$ of $\mathfrak{f}$ be generated by elements of the form

$$f(v_1 + v_2, w) - f(v_1, w) - f(v_2, w),$$
$$f(v, w_1 + w_2) - f(v, w_1) - f(v, w_2),$$
$$f(vr, w) - f(v, rw)$$

with $v, v_1, v_2 \in \mathfrak{v}$, $w, w_1, w_2 \in \mathfrak{w}$ and $r \in R$.

We define $\mathfrak{t} = \mathfrak{f}/\mathfrak{d}$, and we show that $\mathfrak{t}$ is a tensor product of $\mathfrak{v}$ and $\mathfrak{w}$ with R-balanced map δ defined by $(v, w)\delta = f(v, w) + \mathfrak{d}$.

It is clear that the $(v, w)\delta$ generate all of $\mathfrak{t}$, so condition (1) of Definition 9.1 is satisfied.

Let μ be an R-balanced map from $\mathfrak{v} \times \mathfrak{w}$ to the abelian group $\mathfrak{s}$. We define the homomorphism τ from the free abelian group $\mathfrak{f}$ to $\mathfrak{s}$ by

$$f(v, w)\tau = (v, w)\mu.$$

Since μ is R-balanced, we have

$$\begin{aligned}(f(v_1 + v_2, w) - f(v_1, w) - f(v_2, w))\tau \\ = (v_1 + v_2, w)\mu - (v_1, w)\mu - (v_2, w)\mu = 0,\\ (f(v, w_1 + w_2) - f(v, w_1) - f(v, w_2))\tau \\ = (v, w_1 + w_2)\mu - (v, w_1)\mu - (v, w_2)\mu = 0,\\ (f(vr, w) - f(v, rw))\tau = (vr, w)\mu - (v, rw)\mu = 0.\end{aligned}$$

So $\mathfrak{d}$ is mapped to 0 by τ, and defining

$$(f(v, w) + \mathfrak{d})\alpha = (v, w)\mu$$

yields the desired homomorphism α from $\mathfrak{t}$ to $\mathfrak{s}$. □

In what follows, we always write $(v, w)\delta = v \otimes w$. Then the following relations hold:

$$(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w,$$
$$v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2,$$
$$vr \otimes w = v \otimes rw.$$

Determining the structure of the tensor product can be very difficult. The following lemma is occasionally useful in this regard.

Lemma 9.4. *Let* $\mathfrak{v} = \bigoplus_{i \in I} \mathfrak{v}_i$ *be a direct decomposition of the right* R*-module* $\mathfrak{v}$*, and let* $\mathfrak{w} = \bigoplus_{j \in J} \mathfrak{w}_j$ *be a direct decomposition of the left* R*-module* $\mathfrak{w}$*. Then*

$$\mathfrak{v} \otimes_R \mathfrak{w} \cong \bigoplus_{i,j} \mathfrak{v}_i \otimes_R \mathfrak{w}_j.$$

Proof. Let $\mathfrak{t}$ be the tensor product of $\mathfrak{v}$ and $\mathfrak{w}$. The map π from $\mathfrak{v} \times \mathfrak{w}$ to $\mathfrak{t}' = \bigoplus_{i,j} \mathfrak{v}_i \otimes_R \mathfrak{w}_j$ defined by

$$\Big(\sum_i v_i, \sum_j w_j\Big)\pi = \sum_{i,j} v_i \otimes w_j \quad (v_i \in \mathfrak{v}_i, w_j \in \mathfrak{w}_j)$$

is clearly R-balanced. So there exists a map α in $\mathrm{Hom}(\mathfrak{t}, \mathfrak{t}')$ such that

$$\Big(\Big(\sum_i v_i\Big) \otimes' \Big(\sum_j w_j\Big)\Big)\alpha = \sum_{i,j} v_i \otimes w_j.$$

(For the moment, we denote the tensor product in $\mathfrak{v} \otimes_R \mathfrak{w}$ using $\otimes'$ and use $\otimes$ for the images in the $\mathfrak{v}_i \otimes_R \mathfrak{w}_j$.)

Let τ_{ij} be the R-balanced map from $\mathfrak{v}_i \times \mathfrak{w}_j$ to $\mathfrak{v} \otimes'_R \mathfrak{w}$ defined by

$$(v_i, w_j)\tau_{ij} = v_i \otimes' w_j.$$

(The tensor product on the right-hand side is constructed in $\mathfrak{v} \otimes'_R \mathfrak{w}$.) Then there exist μ_{ij} in $\mathrm{Hom}(\mathfrak{v}_i \otimes_R \mathfrak{w}_j, \mathfrak{t})$ such that the following diagram commutes:

$$\begin{array}{ccc} \mathfrak{v}_i \times \mathfrak{w}_j & \xrightarrow{\ \delta_{ij}\ } & \mathfrak{v}_i \otimes_R \mathfrak{w}_j \\ & {\scriptstyle \tau_{ij}} \searrow \quad \swarrow {\scriptstyle \mu_{ij}} & \\ & \mathfrak{t} & \end{array}$$

Therefore, $(v_i \otimes w_j)\mu_{ij} = v_i \otimes' w_j$.

Now we define a group homomorphism μ from $\mathfrak{t}'$ to $\mathfrak{t}$ by requiring that the restriction of μ to the direct summand $\mathfrak{v}_i \otimes_R \mathfrak{w}_j$ of $\mathfrak{t}'$ is exactly μ_{ij}. Then

$$\begin{aligned} \Big(\Big(\sum_i v_i\Big) \otimes' \Big(\sum_j w_j\Big)\Big)\alpha\mu &= \Big(\sum_{i,j} v_i \otimes w_j\Big)\mu = \sum_{i,j} v_i \otimes' w_j \\ &= \Big(\sum_i v_i\Big) \otimes' \Big(\sum_j w_j\Big). \end{aligned}$$

So $\alpha\mu = 1$ on $\mathfrak{t}$. It similarly follows that $\mu\alpha = 1$ on $\mathfrak{t}'$, and therefore $\mathfrak{t} \cong \mathfrak{t}'$. □

We note the following trivial consequence of Lemma 9.4.

Lemma 9.5. *Let $\mathfrak{v}$ be a right R-module, and let $\mathfrak{w}$ be a free left R-module in the free generators w_i with $i \in I$. Then the abelian group $\mathfrak{v} \otimes_R \mathfrak{w}$ is isomorphic to the direct sum of $|I|$ groups, each of which is isomorphic to the additive group $\mathfrak{v}^+$ of $\mathfrak{v}$. More precisely, we have*

$$\mathfrak{v} \otimes_R \mathfrak{w} = \bigoplus_{i \in I} \mathfrak{v}_i \quad \textit{with} \quad \mathfrak{v}_i = \{v \otimes w_i \mid v \in \mathfrak{v}\} \cong \mathfrak{v}^+.$$

Proof. Let $\mathfrak{w} = \bigoplus_{i\in I} Rw_i$. By Lemma 9.4, $\mathfrak{v} \otimes_R \mathfrak{w} \cong \bigoplus_{i\in I} \mathfrak{v} \otimes_R Rw_i$. Thus, it only remains to show that

$$\mathfrak{v} \otimes_R Rw_i = \{v \otimes w_i \mid v \in \mathfrak{v}\} \cong \mathfrak{v}^+.$$

Let μ be the R-balanced map from $\mathfrak{v} \times Rw_i$ to $\mathfrak{v}^+$ defined by $(v, rw_i)\mu = vr$. By Definition 9.1, there exists a map ψ in $\mathrm{Hom}(\mathfrak{v} \otimes_R Rw_i, \mathfrak{v}^+)$ such that $(v \otimes rw_i)\psi = vr$. Furthermore, let π be the homomorphism from $\mathfrak{v}^+$ to $\mathfrak{v} \otimes_R Rw_i$ defined by $v\pi = v \otimes w_i$. Then $\psi\pi = 1$ on $\mathfrak{v} \otimes_R Rw_i$ and $\pi\psi = 1$ on $\mathfrak{v}^+$. So ψ is an isomorphism from $\mathfrak{v} \otimes_R Rw_i$ to $\mathfrak{v}^+$. □

Theorem 9.6 (Functoriality of the tensor product). *Let $\mathfrak{v}$ and $\mathfrak{v}'$ be right R-modules, and let $\alpha \in \mathrm{Hom}_R(\mathfrak{v}, \mathfrak{v}')$. Furthermore, let $\mathfrak{w}$ and $\mathfrak{w}'$ be left R-modules, and let $\beta \in \mathrm{Hom}_R(\mathfrak{w}, \mathfrak{w}')$. Then there exists a group homomorphism $\mu \in \mathrm{Hom}(\mathfrak{v} \otimes_R \mathfrak{w}, \mathfrak{v}' \otimes_R \mathfrak{w}')$ such that*

$$(v \otimes w)\mu = v\alpha \otimes w\beta$$

for all $v \in \mathfrak{v}$ and all $w \in \mathfrak{w}$. We write $\mu = \alpha \otimes \beta$.

If $\alpha' : \mathfrak{v}' \to \mathfrak{v}''$ and $\beta' : \mathfrak{w}' \to \mathfrak{w}''$ are further R-homomorphisms, then $(\alpha \otimes \beta)(\alpha' \otimes \beta') = \alpha\alpha' \otimes \beta\beta'$.

Proof. The map π from $\mathfrak{v} \times \mathfrak{w}$ to $\mathfrak{v}' \otimes_R \mathfrak{w}'$ defined by $(v, w)\pi = v\alpha \otimes w\beta$ is clearly R-balanced. So there exists a group homomorphism μ from $\mathfrak{v} \otimes_R \mathfrak{w}$ to $\mathfrak{v}' \otimes_R \mathfrak{w}'$ with $(v \otimes w)\mu = v\alpha \otimes w\beta$. We have

$$\begin{aligned}(v \otimes w)(\alpha \otimes \beta)(\alpha' \otimes \beta') &= (v\alpha \otimes w\beta)(\alpha' \otimes \beta') = v\alpha\alpha' \otimes w\beta\beta' \\ &= (v \otimes w)(\alpha\alpha' \otimes \beta\beta').\end{aligned}$$

Since the $v \otimes w$ generate all of $\mathfrak{v} \otimes_R \mathfrak{w}$, the claim follows. □

Tensor products of R-modules are simply abelian groups, so they have no R-module structure. However, they can be made into R-modules under some additional assumptions.

Definition 9.7. Let R and S be rings. The set $\mathfrak{m}$ is called an (R, S)-module whenever $\mathfrak{m}$ is a left R-module and a right S-module with respect to the same operation $+$, and in addition $r(ms) = (rm)s$ for all $r \in R$, all $s \in S$ and all $m \in \mathfrak{m}$.

Theorem 9.8. *Let $\mathfrak{v}$ be a right R-module, and let $\mathfrak{w}$ be an (R, S)-module. Then the tensor product $\mathfrak{v} \otimes_R \mathfrak{w}$ admits the structure of a right S-module upon defining*

$$(v \otimes w)s = v \otimes ws.$$

Proof. The map π from $\mathfrak{v} \times \mathfrak{w}$ to $\mathfrak{v} \otimes_R \mathfrak{w}$ defined by

$$(v, w)\pi = v \otimes ws$$

is R-balanced: Note that the additive conditions are satisfied and we have

$$(vr, w)\pi = vr \otimes ws = v \otimes r(ws) = v \otimes (rw)s = (v, rw)\pi.$$

So there exists a homomorphism μ_s from $\mathfrak{v} \otimes_R \mathfrak{w}$ to itself with $(v \otimes w)\mu_s = v \otimes ws$. It is then easy to verify that upon defining

$$(v \otimes w)s = (v \otimes w)\mu_s,$$

the tensor product $\mathfrak{v} \otimes_R \mathfrak{w}$ satisfies all the axioms for a right S-module. □

Remark 9.9. In the special case of a commutative ring R, we may always give a right R-module $\mathfrak{m}$ the structure of an (R, R)-module by defining $rm = mr$. Since R is commutative, we have

$$(r_1r_2)m = m(r_1r_2) = m(r_2r_1) = (mr_2)r_1 = r_1(r_2m)$$

and

$$r_1(mr_2) = (mr_2)r_1 = m(r_2r_1) = m(r_1r_2) = (mr_1)r_2 = (r_1m)r_2.$$

Hence, tensor products of R-modules over a commutative ring R can always be given the structure of an R-module using the construction in Theorem 9.8.

The following associativity property of the tensor product will be useful when we discuss induced representations in §16.

Theorem 9.10. *Let R, S and T be rings, let $\mathfrak{a}$ be a right R-module, let $\mathfrak{b}$ be an (R, S)-module, and let $\mathfrak{c}$ be an (S, T)-module. Then the map μ defined by*

$$((a \otimes b) \otimes c)\mu = a \otimes (b \otimes c)$$

is a T-isomorphism from $(\mathfrak{a} \otimes_R \mathfrak{b}) \otimes_S \mathfrak{c}$ to $\mathfrak{a} \otimes_R (\mathfrak{b} \otimes_S \mathfrak{c})$.

Proof. The map π defined by

$$(a \otimes b, c)\pi = a \otimes (b \otimes c)$$

is an S-balanced map:

It is easy to show that π is well defined and additive in both arguments. We also have

$$((a \otimes b)s, c)\pi = (a \otimes bs, c)\pi = a \otimes (bs \otimes c) = a \otimes (b \otimes sc) = (a \otimes b, sc)\pi.$$

So there exists a homomorphism μ from $(\mathfrak{a} \otimes_R \mathfrak{b}) \otimes_S \mathfrak{c}$ to $\mathfrak{a} \otimes_R (\mathfrak{b} \otimes_S \mathfrak{c})$ defined by

$$((a \otimes b) \otimes c)\mu = a \otimes (b \otimes c).$$

It is clear that μ is in fact a T-epimorphism. In the same way, one constructs a map τ that is inverse to μ by defining

$$(a \otimes (b \otimes c))\tau = (a \otimes b) \otimes c,$$

and this yields the claim. □

We require a few more theorems regarding tensor products of vector spaces and algebras.

Theorem 9.11. *Let* V *and* W *be vector spaces over* K.

a) *The tensor product* $\mathsf{V} \otimes_{\mathsf{K}} \mathsf{W}$ *is a vector space over* K. *If* $\{v_i \mid i \in I\}$ *is a* K*-basis of* V *and* $\{w_j \mid j \in J\}$ *is a* K*-basis of* W, *then*

$$\{v_i \otimes w_j \mid i \in I, j \in J\}$$

is a K*-basis of* $\mathsf{V} \otimes_{\mathsf{K}} \mathsf{W}$.

b) *If* α, *resp.* β, *are* K*-linear maps from* V, *resp.* W, *to itself, then the ordered pair* (α, β) *gives rise to a* K*-linear map* $\gamma = \alpha \otimes \beta$ *from* $\mathsf{V} \otimes_{\mathsf{K}} \mathsf{W}$ *to itself with the following properties:*

(1) $(v \otimes w)(\alpha \otimes \beta) = v\alpha \otimes w\beta$ *for all* $v \in \mathsf{V}$ *and all* $w \in \mathsf{W}$.

(2) $(\alpha_1 \otimes \beta_1)(\alpha_2 \otimes \beta_2) = \alpha_1\alpha_2 \otimes \beta_1\beta_2$,
$(\alpha_1 + \alpha_2) \otimes \beta = \alpha_1 \otimes \beta + \alpha_2 \otimes \beta$,
$\alpha \otimes (\beta_1 + \beta_2) = \alpha \otimes \beta_1 + \alpha \otimes \beta_2$,
$k\alpha \otimes \beta = \alpha \otimes k\beta = k(\alpha \otimes \beta)$
for all $\alpha, \alpha_1, \alpha_2 \in \mathrm{Hom}_{\mathsf{K}}(\mathsf{V}, \mathsf{V})$, *all* $\beta, \beta_1, \beta_2 \in \mathrm{Hom}_{\mathsf{K}}(\mathsf{W}, \mathsf{W})$ *and all* $k \in \mathsf{K}$.

(3) *If* $v_i\alpha = \sum_{s \in I} a_{is} v_s$ *and* $w_j\beta = \sum_{t \in J} b_{jt} w_t$, *then*

$$(v_i \otimes w_j)(\alpha \otimes \beta) = \sum_{s \in I, t \in J} a_{is} b_{jt} (v_s \otimes w_t).$$

(*Here, the sums over* I *and* J *contain only finitely many nonzero terms*.)

Proof. a) Let T be a vector space over K with K-basis $\{t_{ij} \mid i \in I, j \in J\}$. The map δ defined by

$$\Big(\sum_{i \in I} a_i v_i, \sum_{j \in J} b_j w_j\Big)\delta = \sum_{i \in I, j \in J} a_i b_j t_{ij}$$

is a K-balanced map, and the $(kv_i, w_j)\delta = kt_{ij}$ generate T as an abelian group.

If μ is a K-balanced map from $\mathsf{V} \times \mathsf{W}$ to the abelian group $\mathfrak{s}$, then we define a group homomorphism α from T to $\mathfrak{s}$ by

$$\Big(\sum_{i \in I, j \in J} a_{ij} t_{ij}\Big)\alpha = \sum_{i \in I, j \in J} (a_{ij} v_i, w_j)\mu.$$

Since the t_{ij} constitute a K-basis of T, the map α is well defined. We have $\mu = \delta\alpha$. So the vector space T is a tensor product of V and W. In what follows, we write $v_i \otimes w_j$ in place of $t_{ij} = (v_i, w_j)\delta$.

b) Given maps α and β as in the statement of part b), we obtain the homomorphism $\gamma = \alpha \otimes \beta$ from $\mathsf{V} \otimes_{\mathsf{K}} \mathsf{W}$ to itself by Theorem 9.6. It is then easy to compute that γ satisfies the properties (1)–(3). □

Lemma 9.12. *Let $\mathfrak{a}_i$ $(i = 1, 2)$ be algebras of not necessarily finite dimension over K. Let V_i be a vector space of arbitrary dimension over K that is also a right $\mathfrak{a}_i$-module in the sense of Chapter I, §10. Then the map τ defined by*

$$(v_1 \otimes v_2)((a_1 \otimes a_2)\tau) = v_1 a_1 \otimes v_2 a_2$$

is a K-linear map from the K-vector space $\mathfrak{a}_1 \otimes_{\mathsf{K}} \mathfrak{a}_2$ to

$$\mathrm{Hom}_{\mathsf{K}}(\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2, \mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2).$$

Proof. For every $a_i \in \mathfrak{a}_i$, we define a K-linear map $a_i\delta_i$ from V_i to itself by $v_i(a_i\delta_i) = v_i a_i$ $(i = 1, 2)$. Furthermore, we define a map μ from $\mathfrak{a}_1 \times \mathfrak{a}_2$ to $\mathrm{Hom}_{\mathsf{K}}(\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2, \mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2)$ according to Theorem 9.11 b) by

$$(a_1, a_2)\mu = (a_1\delta_1) \otimes (a_2\delta_2).$$

It is clear that μ is K-balanced. Therefore, there exists a group homomorphism τ from $\mathfrak{a}_1 \otimes_{\mathsf{K}} \mathfrak{a}_2$ to $\mathrm{Hom}_{\mathsf{K}}(\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2, \mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2)$ that satisfies

$$(a_1 \otimes a_2)\tau = (a_1\delta_1) \otimes (a_2\delta_2).$$

Since

$$\begin{aligned}(k(a_1 \otimes a_2))\tau &= (ka_1 \otimes a_2)\tau = (ka_1)\delta_1 \otimes a_2\delta_2 = k(a_1\delta_1) \otimes a_2\delta_2 \\ &= k((a_1\delta_1) \otimes (a_2\delta_2)) = k(a_1 \otimes a_2)\tau\end{aligned}$$

for $k \in \mathsf{K}$, the map τ is a K-linear map. □

Theorem 9.13. *Let $\mathfrak{a}_1$ and $\mathfrak{a}_2$ be algebras of not necessarily finite dimension over K.*

a) *The tensor product $\mathfrak{a}_1 \otimes_{\mathsf{K}} \mathfrak{a}_2$ becomes an algebra over K upon defining*

$$(a_1 \otimes a_2)(a_1' \otimes a_2') = a_1 a_1' \otimes a_2 a_2'$$

for all $a_i, a_i' \in \mathfrak{a}_i$.

b) *Let V_i be a vector space of arbitrary dimension over K that is also a right $\mathfrak{a}_i$-module. Then $\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2$ admits the structure of an $\mathfrak{a}_1 \otimes_{\mathsf{K}} \mathfrak{a}_2$-module upon defining*

$$(v_1 \otimes v_2)(a_1 \otimes a_2) = v_1 a_1 \otimes v_2 a_2.$$

Proof. a) By Lemma 9.12, there exists a K-linear map τ from $\mathfrak{a}_1 \otimes_{\mathsf{K}} \mathfrak{a}_2$ to $\mathrm{Hom}_{\mathsf{K}}(\mathfrak{a}_1 \otimes_{\mathsf{K}} \mathfrak{a}_2, \mathfrak{a}_1 \otimes_{\mathsf{K}} \mathfrak{a}_2)$ with

$$(a_1 \otimes a_2)((a_1' \otimes a_2')\tau) = a_1 a_1' \otimes a_2 a_2'.$$

So the multiplication on $\mathfrak{a}_1 \otimes_{\mathsf{K}} \mathfrak{a}_2$ given above is well defined. It is easy to verify that $\mathfrak{a}_1 \otimes_{\mathsf{K}} \mathfrak{a}_2$ with this multiplication is an algebra over K.

b) follows from Lemma 9.12 in a way similar to a). □

Theorem 9.14. *Let V_1 and V_2 be vector spaces of finite dimension over K. Then the map τ defined by*

$$(v_1 \otimes v_2)((a_1 \otimes a_2)\tau) = v_1 a_1 \otimes v_2 a_2$$

for all $v_i \in \mathsf{V}_i$ and $a_i \in \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_i, \mathsf{V}_i)$ is an algebra isomorphism from

$$\operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_1, \mathsf{V}_1) \otimes_{\mathsf{K}} \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_2, \mathsf{V}_2) \quad \textit{to} \quad \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2, \mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2).$$

Proof. Letting $\dim_{\mathsf{K}} \mathsf{V}_i = n_i$, we have by Theorem 9.11 a) that

$$\dim_{\mathsf{K}} \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2, \mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2) = (\dim_{\mathsf{K}} \mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2)^2 = (n_1 n_2)^2$$

and

$$\begin{aligned}&\dim_{\mathsf{K}}(\operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_1, \mathsf{V}_1) \otimes_{\mathsf{K}} \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_2, \mathsf{V}_2)) \\ &\quad = \dim_{\mathsf{K}} \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_1, \mathsf{V}_1) \dim_{\mathsf{K}} \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_2, \mathsf{V}_2) = n_1^2 n_2^2.\end{aligned}$$

Thus, it suffices by Lemma 9.12 to prove that the K-linear map τ is an epimorphism. Let

$$\{v_1^{(i)}, \ldots, v_{n_i}^{(i)}\} \quad (i = 1, 2)$$

be a K-basis of V_i $(i = 1, 2)$, and let $e_{jk}^{(i)} \in \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_i, \mathsf{V}_i)$ be defined by

$$v_m^{(i)} e_{jk}^{(i)} = \delta_{mj} v_k^{(i)}.$$

Then

$$(v_s^{(1)} \otimes v_t^{(2)})((e_{jk}^{(1)} \otimes e_{lm}^{(2)})\tau) = \delta_{sj}\delta_{tl}\, v_k^{(1)} \otimes v_m^{(2)}.$$

Therefore, the $(e_{jk}^{(1)} \otimes e_{lm}^{(2)})\tau$ constitute a K-basis of $\operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2, \mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2)$. □

10 Tensor products of representations

Lemma 10.1. *Let $a \in \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_1, \mathsf{V}_1)$ and $b \in \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_2, \mathsf{V}_2)$ with $\dim \mathsf{V}_k < \infty$ $(k = 1, 2)$. Let the eigenvalues of a in some extension field L of K be ϵ_i $(i = 1, \ldots, n)$, and let δ_j $(j = 1, \ldots, m)$ be the eigenvalues of b. Then the $\epsilon_i \delta_j$ are all the eigenvalues of $a \otimes b$ with the correct multiplicities. In particular, we have $\operatorname{tr}(a \otimes b) = \operatorname{tr}(a) \operatorname{tr}(b)$ and $\det(a \otimes b) = (\det a)^m (\det b)^n$.*

Proof. We denote again by a and b the matrices of a and b with respect to some arbitrary but fixed basis of V_1, resp. V_2. We also denote again by $a \otimes b$ the matrix of $a \otimes b$ with respect to the corresponding basis of $\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2$.

It is well known that over a large enough extension field L of K there exist nonsingular matrices s and t such that

$$a' = s^{-1}as = \begin{pmatrix} \epsilon_1 & & \\ & \ddots & \\ * & & \epsilon_n \end{pmatrix} \quad \text{and} \quad b' = t^{-1}bt = \begin{pmatrix} \delta_1 & & \\ & \ddots & \\ * & & \delta_m \end{pmatrix}$$

are triangular matrices. Therefore,

$$a' \otimes b' = s^{-1}as \otimes t^{-1}bt = (s \otimes t)^{-1}(a \otimes b)(s \otimes t)$$

is a triangular matrix with diagonal elements $\epsilon_i \delta_j$. Hence, the $\epsilon_i \delta_j$ are the eigenvalues of $a' \otimes b'$, so they are also the eigenvalues of $a \otimes b$, and they occur with the correct multiplicities. Therefore,

$$\operatorname{tr}(a \otimes b) = \sum_{i,j} \epsilon_i \delta_j = \sum_i \epsilon_i \sum_j \delta_j = \operatorname{tr}(a) \operatorname{tr}(b)$$

and

$$\det(a \otimes b) = \prod_{i,j} (\epsilon_i \delta_j) = \Big(\prod_i \epsilon_i \Big)^m \Big(\prod_j \delta_j \Big)^n = (\det a)^m (\det b)^n. \qquad \square$$

Lemma 10.2. *We have the following algebra isomorphism of group rings:*

$$\mathsf{K}[G_1] \otimes_{\mathsf{K}} \mathsf{K}[G_2] \cong \mathsf{K}[G_1 \times G_2].$$

(*Here,* $\mathsf{K}[G_1] \otimes_{\mathsf{K}} \mathsf{K}[G_2]$ *is an algebra over* K *constructed according to Theorem 9.13.*)

Proof. By Theorem 9.11 a),

$$\{g_1 \otimes g_2 \mid g_i \in G_i, i = 1, 2\}$$

is a K-basis of the algebra $\mathsf{K}[G_1] \otimes_{\mathsf{K}} \mathsf{K}[G_2]$. Therefore, the map τ defined by

$$\Big(\sum_{g_i \in G_i} a_{g_1, g_2} \, g_1 \otimes g_2 \Big) \tau = \sum_{g_1 g_2 \in G_1 \times G_2} a_{g_1, g_2} \, g_1 g_2$$

is an algebra isomorphism from $\mathsf{K}[G_1] \otimes_{\mathsf{K}} \mathsf{K}[G_2]$ to $\mathsf{K}[G_1 \times G_2]$. $\square$

Now we classify the representations of direct products.

Theorem 10.3.

a) *Let* $\mathbf{D}_i$ *be a representation of* G_i *on the* K*-vector space* V_i $(i = 1, 2)$. *Then* $\mathbf{D}$ *defined by*

$$\mathbf{D}(g_1g_2) = \mathbf{D}_1(g_1) \otimes \mathbf{D}_2(g_2) \quad (\textit{for } g_i \in G_i)$$

is a representation of $G_1 \times G_2$ *on* $\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2$.

b) *If* K *is algebraically closed and* V_i *is an irreducible* $\mathsf{K}[G_i]$*-module* $(i = 1, 2)$*, then* $\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2$ *is an irreducible* $\mathsf{K}[G_1 \times G_2]$*-module.*

c) *Assume* K *is algebraically closed and* $\operatorname{char} \mathsf{K} \nmid |G_i|$ *for* $i = 1, 2$. *Let* $\mathbf{D}_1, \ldots, \mathbf{D}_h$ *be all the irreducible representations of* G_1 *over* K *up to equivalence, and let* $\mathbf{H}_1, \ldots, \mathbf{H}_k$ *be all the irreducible representations of* G_2 *over* K *up to equivalence. Then the* $\mathbf{F}_{ij}$ *defined by*

$$\mathbf{F}_{ij}(g_1g_2) = \mathbf{D}_i(g_1) \otimes \mathbf{H}_j(g_2) \quad (i = 1, \ldots, h; j = 1, \ldots, k)$$

are all the irreducible representations of $G_1 \times G_2$ *over* K *up to equivalence.*

Proof. a) By Theorem 9.13 b), $\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2$ becomes a $\mathsf{K}[G_1] \otimes_{\mathsf{K}} \mathsf{K}[G_2]$-module upon defining

$$(v_1 \otimes v_2)(g_1 \otimes g_2) = v_1\mathbf{D}_1(g_1) \otimes v_2\mathbf{D}_2(g_2).$$

By the isomorphism of Lemma 10.2, $\mathbf{D}$ is then a representation of $G_1 \times G_2$ on $\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2$.

b) If V_i is an irreducible $\mathsf{K}[G_i]$-module, then there are $(\dim \mathsf{V}_i)^2$ linearly independent maps among the $\mathbf{D}_i(g)$ with $g \in G_i$ by Theorem 5.14. Therefore,

$$\Big\{ \sum_{g \in G_i} a_g \, \mathbf{D}_i(g) \mid a_g \in \mathsf{K} \Big\} = \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_i, \mathsf{V}_i) \quad (i = 1, 2).$$

It follows by Theorem 9.14 that

$$\Big\{ \sum_{g_i \in G_i} a_{g_1,g_2} \, \mathbf{D}_1(g_1) \otimes \mathbf{D}_2(g_2) \mid a_{g_1,g_2} \in \mathsf{K} \Big\} = \operatorname{Hom}_{\mathsf{K}}(\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2, \mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2).$$

So $\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2$ is an irreducible $\mathsf{K}[G_1 \times G_2]$-module.

c) Let χ_i be the character of $\mathbf{D}_i$, and let ψ_j be the character of $\mathbf{H}_j$. By Lemma 10.1, ψ_{ij} defined by

$$\psi_{ij}(g_1g_2) = \chi_i(g_1)\psi_j(g_2) \quad (g_k \in G_k)$$

is the character of $\mathbf{F}_{ij}$. By the orthogonality relations of Theorem 5.8 a), we have

$$\begin{aligned} \sum_{g_1g_2 \in G_1 \times G_2} & \psi_{ij}(g_1g_2)\psi_{i'j'}(g_1^{-1}g_2^{-1}) \\ &= \sum_{g_1 \in G_1} \chi_i(g_1)\chi_{i'}(g_1^{-1}) \sum_{g_2 \in G_2} \psi_j(g_2)\psi_{j'}(g_2^{-1}) \\ &= \delta_{ii'}|G_1|\delta_{jj'}|G_2| \end{aligned}$$

$$= \begin{cases} |G_1 \times G_2| & \text{for } i = i', j = j' \\ 0 & \text{otherwise.} \end{cases}$$

Therefore, the ψ_{ij} are pairwise distinct irreducible characters of $G_1 \times G_2$. (Note that we use that $\mathbf{F}_{ij}$ is irreducible from b); if the characteristic of K is nonzero, then we may not apply Theorem 5.10 b).) The fact that the $\mathbf{F}_{ij}$ are all the irreducible representations of $\mathsf{K}[G_1 \times G_2]$ up to equivalence follows either from

$$\sum_{i,j} (\deg \mathbf{F}_{ij})^2 = \sum_i (\deg \mathbf{D}_i)^2 \sum_j (\deg \mathbf{H}_j)^2 = |G_1||G_2| = |G_1 \times G_2|$$

or the easily confirmed formula $h(G_1 \times G_2) = h(G_1)h(G_2)$ for the class numbers.□

Theorem 10.3 c) can be proved for a field K of arbitrary characteristic in the same way: If char $\mathsf{K} = p$, then one uses that the number of equivalence classes of irreducible representations of $\mathsf{K}[G]$ is equal to the number of conjugacy classes $h_p(G)$ of G consisting of elements of order coprime to p (see pg. 504). It is easy to see that $h_p(G_1 \times G_2) = h_p(G_1)h_p(G_2)$.

Definition 10.4. Let $\mathbf{D}_1$ and $\mathbf{D}_2$ be representations of G over K. By Theorem 10.3, the map $\mathbf{D}$ defined by

$$\mathbf{D}(g_1 g_2) = \mathbf{D}_1(g_1) \otimes \mathbf{D}_2(g_2)$$

is a representation of $G \times G$ over K. By composing $\mathbf{D}$ with the monomorphism ϵ from G to $G \times G$ defined by

$$g^{\epsilon} = (g, g),$$

we obtain a representation $\mathbf{D}_1 \otimes \mathbf{D}_2$ of G satisfying

$$(\mathbf{D}_1 \otimes \mathbf{D}_2)(g) = \mathbf{D}_1(g) \otimes \mathbf{D}_2(g).$$

We call $\mathbf{D}_1 \otimes \mathbf{D}_2$ the tensor product of $\mathbf{D}_1$ and $\mathbf{D}_2$. (In the literature, this is often referred to as the Kronecker product.) Note that if $\mathbf{D}_1$ and $\mathbf{D}_2$ are irreducible representations of G over an algebraically closed field, then the representation $\mathbf{D}$ of $G \times G$ is irreducible, but the restriction $\mathbf{D}_1 \otimes \mathbf{D}_2$ of $\mathbf{D}$ to the diagonal $\{(g, g) \mid g \in G\}$ is not necessarily irreducible.

Definition 10.5. Let $\chi_1, \ldots, \chi_h$ be the irreducible characters of G over an arbitrary field K. A class function of the form

$$\sum_{i=1}^{h} a_i \chi_i$$

with integer a_i is called a generalized character of G over K. (Clearly, the $\sum_{i=1}^{h} a_i\chi_i$ with $a_i \geq 0$ and $\sum_{i=1}^{h} a_i > 0$ are exactly the characters of representations of G.)

Theorem 10.6. *The generalized characters of G over K constitute a commutative ring with respect to the operations*

$$(\tau_1 + \tau_2)(g) = \tau_1(g) + \tau_2(g)$$
$$(\tau_1\tau_2)(g) = \tau_1(g)\tau_2(g).$$

More precisely:

$$\chi_i\chi_j = \sum_{k=1}^{h} c_{ijk}\chi_k$$

for some nonnegative integers c_{ijk}.

Proof. Note that $\chi_i\chi_j$ is the character of the representation $\mathbf{D}_i \otimes \mathbf{D}_j$ of G. The decomposition of $\chi_i\chi_j$ into irreducible characters of G yields the given formula. It follows immediately that the generalized characters form a ring. □

Remark 10.7. Let $\mathbf{D}$ be an irreducible representation of G over K, and let $\mathbf{D}'$ be a representation of degree 1 over K. Then $\mathbf{D} \otimes \mathbf{D}'$ is an irreducible representation of G. It is entirely possible, however, that $\mathbf{D} \otimes \mathbf{D}'$ is equivalent to $\mathbf{D}$ even when $\mathbf{D}'$ is not the trivial representation of G.

Proof. Letting $\mathbf{D}'(g) = a(g) \in \mathsf{K}$, we have $(\mathbf{D} \otimes \mathbf{D}')(g) = a(g)\mathbf{D}(g)$. Therefore, the subspaces of the representation module that are invariant under the $\mathbf{D}(g)$ $(g \in G)$ and under the $(\mathbf{D} \otimes \mathbf{D}')(g)$ coincide. In particular, the irreducibility of $\mathbf{D}$ implies $\mathbf{D} \otimes \mathbf{D}'$ is also irreducible. □

Theorem 10.8 (R. Brauer [56]). *Let χ be the character of a faithful representation $\mathbf{D}$ of G over the field $\mathbb{C}$ of complex numbers. Suppose that χ takes exactly r distinct values on G. Then every irreducible character of G appears as a constituent of at least one of the characters $\chi^0 = 1, \chi, \chi^2, \ldots, \chi^{r-1}$. That is, all the irreducible representations of G can be found by decomposing the representations $\mathbf{D} \otimes \cdots \otimes \mathbf{D}$ for $i \leq r - 1$.* (*Admittedly, this decomposition is rarely feasible in practice.*)

Proof. Let $a_j = \chi(g_j)$ $(j = 1, \ldots, r)$ be the distinct values that χ takes on G, let $a_1 = \chi(1)$ and let

$$A_j = \{g \in G \mid \chi(g) = \chi(g_j) = a_j\}.$$

Since $\mathbf{D}$ is faithful,

$$A_1 = \ker \mathbf{D} = 1$$

by Lemma 7.4. Assume for contradiction that χ_k is an irreducible character of G that does not appear as a constituent of any χ^i for $0 \le i \le r-1$. By Theorem 5.9,

$$0 = |G|(\chi^i, \chi_k) = \sum_{j=1}^{r} \chi^i(g_j) \sum_{g \in A_j} \chi_k(g^{-1}).$$

It is well known that the Vandermonde determinant

$$\det(\chi^i(g_j)) = \det(a_j^i)$$

is nonzero. It follows that

$$\sum_{g \in A_j} \chi_k(g^{-1}) = 0$$

for all $j = 1, \ldots, r$. In particular, this yields the contradiction

$$0 = \sum_{g \in A_1} \chi_k(g^{-1}) = \chi_k(1). \qquad \square$$

Exercises

19) Let the complex vector space V be an irreducible G-module with representation $\mathbf{D}$ of G.
   a) If the character χ of $\mathbf{D}$ takes only real values on G, then there exists (up to scalar multiple) exactly one bilinear form f on V such that $f(v_1 g, v_2 g) = f(v_1, v_2)$ for all $g \in G$ and all $v_i \in \mathsf{V}$. The bilinear form f is nondegenerate.
   b) The bilinear form f in a) is either symmetric or antisymmetric.
   c) If $\deg \mathbf{D}$ is odd, then f is symmetric.
20) Give an example to show that K being algebraically closed in Theorem 10.3 b) is a necessary condition.
21) Let $G = G_1 \curlyvee G_2$ be the central product of G_1 and G_2 with identified central subgroup (see Theorem I.9.10, pg. 51). Describe all the irreducible representations of G over an algebraically closed field of characteristic 0.

11 Splitting fields

In this section, we investigate the following question: Over which not necessarily algebraically closed fields do the main results of §5 still hold? By an algebra over K we always mean an algebra of finite dimension over K.

Lemma 11.1.

a) *Let* $\mathfrak{a}$ *be an algebra over* K*, and let* L *be an extension field of* K*. Then* $\mathfrak{a}_\mathsf{L} = \mathsf{L} \otimes_\mathsf{K} \mathfrak{a}$ *becomes an algebra over* L *upon defining*

$$(y_1 \otimes a_1)(y_2 \otimes a_2) = y_1 y_2 \otimes a_1 a_2$$

and

$$y_1(y_2 \otimes a) = y_1 y_2 \otimes a$$

for all $a_1, a_2, a \in \mathfrak{a}$ *and all* $y_1, y_2 \in \mathsf{L}$*. If* $\{a_1, \ldots, a_n\}$ *is a* K*-basis of* $\mathfrak{a}$ *satisfying*

$$a_i a_j = \sum_{s=1}^{n} k_{ijs} a_s \quad (k_{ijs} \in \mathsf{K}),$$

then the $a'_i = 1 \otimes a_i$ $(i = 1, \ldots, n)$ *constitute an* L*-basis of* $\mathfrak{a}_\mathsf{L}$ *satisfying*

$$a'_i a'_j = \sum_{s=1}^{n} k_{ijs} a'_s.$$

In particular, $\dim_\mathsf{L} \mathfrak{a}_\mathsf{L} = \dim_\mathsf{K} \mathfrak{a}$*. We call* $\mathfrak{a}_\mathsf{L}$ *the base field extension of* $\mathfrak{a}$ *by* L*.*

b) *If* V *is a right* $\mathfrak{a}$*-module, then* $\mathsf{V}_\mathsf{L} = \mathsf{L} \otimes_\mathsf{K} \mathsf{V}$ *becomes a right* $\mathfrak{a}_\mathsf{L}$*-module upon defining*

$$(y_1 \otimes v)(y_2 \otimes a) = y_1 y_2 \otimes va.$$

If $\{v_1, \ldots, v_m\}$ *is a* K*-basis of* V*, then* $\{1 \otimes v_i \mid i = 1, \ldots, m\}$ *is an* L*-basis of* V_L*. In particular,* $\dim_\mathsf{K} \mathsf{V} = \dim_\mathsf{L} \mathsf{V}_\mathsf{L}$*.*

Proof. a) By Theorem 9.13 a), $\mathfrak{a}_\mathsf{L} = \mathsf{L} \otimes_\mathsf{K} \mathfrak{a}$ is an algebra over K. By defining

$$y_1(y_2 \otimes a) = (y_1 \otimes 1)(y_2 \otimes a) = y_1 y_2 \otimes a,$$

$\mathfrak{a}_\mathsf{L}$ becomes an algebra over L. Let $\{a_1, \ldots, a_n\}$ be a K-basis of $\mathfrak{a}$. Since $\mathfrak{a} = \bigoplus_{i=1}^{n} \mathsf{K} a_i$, it follows by Lemma 9.5 that

$$\mathsf{L} \otimes_\mathsf{K} \mathfrak{a} = \bigoplus_{i=1}^{n} \mathfrak{a}_i,$$

where $\mathfrak{a}_i = \{l \otimes a_i = l(1 \otimes a_i) \mid l \in \mathsf{L}\}$. So $\{1 \otimes a_i \mid i = 1, \ldots, n\}$ is an L-basis of $\mathfrak{a}_\mathsf{L}$.

b) follows in the same way as a) with the help of Theorem 9.13 b). □

Definition 11.2.

a) Let $\mathfrak{a}$ be an algebra over K. An extension field L of K is called a splitting field for $\mathfrak{a}$ if the algebra $\mathfrak{a}_\mathsf{L}$ is a direct sum of full matrix rings over L.

b) A field L is called a splitting field for the group G if the group ring $\mathsf{L}[G]$ is a direct sum of full matrix rings over L.

By Theorem 5.1, every algebraically closed field L with $\operatorname{char}\mathsf{L} \nmid |G|$ is a splitting field for G. The splitting fields for G are in no way uniquely determined up to isomorphism.

Lemma 11.3.
a) *If* L *is an extension field of* K*, then*

$$\mathsf{K}[G]_{\mathsf{L}} \cong \mathsf{L}[G].$$

In particular, L *is a splitting field for* $\mathsf{K}[G]$ *if and only if* L *is a splitting field for* G*.*
b) *If* K *is a splitting field for the algebra* $\mathfrak{a}$*, then every field containing* K *is also a splitting field for* $\mathfrak{a}$*.*

Proof. a) The algebra $\mathsf{K}[G]$ has a K-basis $\{g \mid g \in G\}$. By Lemma 11.1, $\mathsf{K}[G]_{\mathsf{L}}$ has an L-basis $\{1 \otimes g \mid g \in G\}$ satisfying

$$(1 \otimes g_1)(1 \otimes g_2) = 1 \otimes g_1 g_2.$$

So $\mathsf{K}[G]_{\mathsf{L}}$ and $\mathsf{L}[G]$ are isomorphic as L-algebras.

b) Since K is a splitting field for $\mathfrak{a}$, we have $\mathfrak{a}_{\mathsf{K}} = \bigoplus_i \mathfrak{a}_i$, where the $\mathfrak{a}_i$ are full matrix rings over K. With the help of Lemma 9.4, we obtain

$$\mathfrak{a}_{\mathsf{L}} \cong \bigoplus_i (\mathfrak{a}_i)_{\mathsf{L}}.$$

So we need only show that if $\mathfrak{b}$ is a full matrix ring over K, then $\mathfrak{b}_{\mathsf{L}}$ is a full matrix ring over L.

As a full matrix ring over K, $\mathfrak{b}$ has a K-basis $\{e_{ij} \mid i, j = 1, \dots, n\}$ satisfying $e_{ij}e_{km} = \delta_{jk}e_{im}$. By Lemma 11.1, $\mathfrak{b}_{\mathsf{L}}$ then has an L-basis $\{e'_{ij} \mid i, j = 1, \dots, n\}$ satisfying $e'_{ij}e'_{km} = \delta_{jk}e'_{im}$. So $\mathfrak{b}_{\mathsf{L}}$ is a full matrix ring over L. □

Theorem 11.4. *Assume* $\operatorname{char}\mathsf{K} \nmid |G|$*. Then the following are equivalent:*
a) K *is a splitting field for* G*.*
b) *If* h *is the number of conjugacy classes of* G*, then there exist at least* h *irreducible, pairwise inequivalent representations* $\mathbf{D}_i$ *of* $\mathsf{K}[G]$ *such that* $\sum_{i=1}^{h} n_i^2 \leq |G|$*, where* $\deg \mathbf{D}_i = n_i$*.*

Proof. a) ⇒ b): Let K be a splitting field for G. Then

$$\mathsf{K}[G] = \bigoplus_{i=1}^{k} \mathfrak{a}_i,$$

where the $\mathfrak{a}_i$ are full matrix rings over K. As in the proof of Theorem 5.1, it follows that

$$h = \dim_{\mathsf{K}} \mathbf{Z}(\mathsf{K}[G]) = \dim_{\mathsf{K}} \bigoplus_{i=1}^{k} \mathbf{Z}(\mathfrak{a}_i) = k.$$

The projections onto the $\mathfrak{a}_i$ yield irreducible representations of $\mathsf{K}[G]$. These representations are inequivalent since they have different kernels. If $\mathfrak{a}_i$ is a full matrix ring of degree n_i over K, then

$$\sum_{i=1}^{h} n_i^2 = \sum_{i=1}^{h} \dim_{\mathsf{K}} \mathfrak{a}_i = \dim_{\mathsf{K}} \mathsf{K}[G] = |G|.$$

b) $\Rightarrow$ a): Let $\mathsf{K}[G] = \bigoplus_{i=1}^{k} \mathfrak{a}_i$ be the decomposition of the semisimple algebra $\mathsf{K}[G]$ into simple algebras $\mathfrak{a}_i$. This decomposition exists by Theorem 2.7 and Theorem 4.5. Let $\mathfrak{a}_i$ be a full matrix ring of degree m_i over the division ring $\mathfrak{c}_i$. Then $\mathsf{K}[G]$ has exactly k irreducible, pairwise inequivalent representations $\mathbf{D}_1, \ldots, \mathbf{D}_k$ by Theorem 4.5, and $\deg \mathbf{D}_i = m_i \dim_{\mathsf{K}} \mathfrak{c}_i$. By our assumption in b), it now follows that $k \geq h$. On the other hand, each matrix ring $\mathfrak{a}_i$ contributes at least one dimension to the dimension of the center $\mathbf{Z}(\mathsf{K}[G])$ by virtue of the K-multiples of its identity element. This yields $h = \dim_{\mathsf{K}} \mathbf{Z}(\mathsf{K}[G]) \geq k$. So $h = k$ and $n_i = \deg \mathbf{D}_i = m_i \dim_{\mathsf{K}} \mathfrak{c}_i$. It now follows by the second assumption in b) that

$$\sum_{i=1}^{h} m_i^2 (\dim_{\mathsf{K}} \mathfrak{c}_i)^2 = \sum_{i=1}^{h} n_i^2 \leq |G| = \sum_{i=1}^{h} m_i^2 \dim_{\mathsf{K}} \mathfrak{c}_i.$$

This forces $\dim_{\mathsf{K}} \mathfrak{c}_i = 1$, so $\mathfrak{c}_i = \mathsf{K}$ for all i. Therefore, K is a splitting field for G. □

Example 11.5. Let $\mathbb{Q}$ be the field of rational numbers, and let $Q = \langle a, b \rangle$ be the quaternion group of order 8. Then Q has the following representations of degree 1 over $\mathbb{Q}$:

$$\begin{aligned}
&\mathbf{D}_1 \quad \text{defined by} \quad \mathbf{D}_1(a) = \mathbf{D}_1(b) = 1,\\
&\mathbf{D}_2 \quad \text{defined by} \quad \mathbf{D}_2(a) = 1,\ \mathbf{D}_2(b) = -1,\\
&\mathbf{D}_3 \quad \text{defined by} \quad \mathbf{D}_3(a) = -1,\ \mathbf{D}_3(b) = 1,\\
&\mathbf{D}_4 \quad \text{defined by} \quad \mathbf{D}_4(a) = \mathbf{D}_4(b) = -1.
\end{aligned}$$

Let $\mathfrak{c}$ be the division ring of quaternions with the $\mathbb{Q}$-basis $\{e_0, e_1, e_2, e_3\}$, where $e_0 = 1$ and the multiplication satisfies

$$\begin{aligned}
&e_i^2 = -1\ (i \geq 1),\ e_0 e_i = e_i e_0 = e_i,\ e_1 e_2 = -e_2 e_1 = e_3,\\
&e_2 e_3 = -e_3 e_2 = e_1,\ e_3 e_1 = -e_1 e_3 = e_2.
\end{aligned}$$

The $\mathbb{Q}$-linear map ϵ defined by $a\epsilon = e_1$, $b\epsilon = e_2$ yields an algebra homomorphism from $\mathbb{Q}[Q]$ onto $\mathfrak{c}$. Letting $\mathfrak{k}$ be the kernel of ϵ, we have

$$\mathbb{Q}[Q] \cong \mathfrak{c} \oplus \mathfrak{k}$$

by Theorem 3.8. Therefore, $\mathbb{Q}[Q]$ has an irreducible representation of degree 4 on $\mathfrak{c}$. The Wedderburn decomposition of $\mathbb{Q}[Q]$ then has the form

$$\mathbb{Q}[Q] = \mathfrak{c} \oplus \mathbb{Q} \oplus \mathbb{Q} \oplus \mathbb{Q} \oplus \mathbb{Q}.$$

So $\mathbb{Q}$ is not a splitting field for Q. The assumption of Theorem 11.4 b) regarding the number of irreducible representations is now satisfied since $h(Q) = 5$, but

$$\sum_{i=1}^{5} n_i^2 = 4^2 + 4 > |Q|.$$

Let $\mathbb{C}$ be the field of complex numbers. It is easy to verify that $\mathfrak{c} \otimes_{\mathbb{Q}} \mathbb{C}$ is a full matrix ring of degree 2 over $\mathbb{C}$. So the Wedderburn decomposition of $\mathbb{C}[Q]$ is

$$\mathbb{C}[Q] = (\mathbb{C})_2 \oplus \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C}.$$

Now we have

$$\sum_{i=1}^{5} n_i^2 = 8 = |Q|.$$

Definition 11.6. A $\mathfrak{p}$-adic field is a field K of characteristic 0 with the following properties:
(1) K admits a non-Archimedean, additive valuation v that takes discrete values.
(2) K is complete with respect to v.
(3) If $I = \{x \in \mathsf{K} \mid v(x) \geq 0\}$ is the valuation ring of K and $\mathfrak{p} = \{x \in \mathsf{K} \mid v(x) > 0\}$ is the valuation ideal of K, then $I/\mathfrak{p} = \overline{\mathsf{K}}$ is a field of characteristic p.
(See Hasse [225] for more about these concepts.)

Theorem 11.7.
a) *Let G be a group of order $|G|$ and P a prime field with* $\operatorname{char} \mathsf{P} \nmid |G|$. *Then G has splitting fields that are finite Galois extensions of P.*
b) *For any prime p, there exists a $\mathfrak{p}$-adic splitting field K for G whose residue field has characteristic p.*

Proof. a) Let L be the algebraic closure of P. By Theorem 5.1, L is a splitting field for G, so there exist h irreducible, pairwise inequivalent representations $\mathbf{D}_i$ $(i = 1, \ldots, h)$ of $\mathsf{L}[G]$. Letting $\deg \mathbf{D}_i = n_i$, we have by Theorem 5.1 that

$$|G| = \sum_{i=1}^{h} n_i^2.$$

The finitely many coefficients of the matrices $\mathbf{D}_i(g)$ (with $g \in G$; $i = 1, \ldots, h$) are algebraic over P, so they generate a finite algebraic extension K of P. Now, the $\mathbf{D}_i$ yield h representations $\widetilde{\mathbf{D}}_i$ of $\mathsf{K}[G]$ by defining

$$\widetilde{\mathbf{D}}_i\Big(\sum_{g\in G} a_g g\Big) = \sum_{g\in G} a_g \mathbf{D}_i(g) \quad (\text{with } a_g \in \mathsf{K}).$$

These representations $\widetilde{\mathbf{D}}_i$ are clearly irreducible and pairwise inequivalent. Furthermore, we retain the relation $|G| = \sum_{i=1}^{h} n_i^2$ on degrees. Therefore, K is a splitting field for G by Theorem 11.4.

Since K is a finite algebraic extension of P, there always exists an extension field M of K that is a finite algebraic Galois extension of P. By Lemma 11.3 b), M is then a Galois splitting field for G.

b) Let K be a splitting field for G that is a finite algebraic extension of the rational numbers. Let $\mathfrak{p}$ be a prime ideal in K with $\mathfrak{p} \mid (p)$, and let $\mathsf{K}_\mathfrak{p}$ be the completion of K with respect to the valuation of K associated to $\mathfrak{p}$. Then it is well known that $\mathsf{K}_\mathfrak{p}$ is a $\mathfrak{p}$-adic field and the residue field has characteristic p. By Lemma 11.3 b), $\mathsf{K}_\mathfrak{p}$ is a splitting field for G since it is an extension of K. □

Definition 11.8. Let $\mathbf{D}$ be a representation of G on the $\mathsf{K}[G]$-module V. We say that $\mathbf{D}$ is absolutely irreducible if for all extension fields L of K, $\mathsf{V}_\mathsf{L} = \mathsf{L} \otimes_\mathsf{K} \mathsf{V}$ is always an irreducible $\mathsf{L}[G]$-module. If χ is the character of an absolutely irreducible representation of G, then χ is called an absolutely irreducible character of G.

Lemma 11.9. *Let* V *be a* $\mathsf{K}[G]$*-module, and let* L *be an extension field of* K*. Then* $\mathrm{Hom}_{\mathsf{L}[G]}(\mathsf{V}_\mathsf{L}, \mathsf{V}_\mathsf{L})$ *and* $\mathsf{L} \otimes_\mathsf{K} \mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{V})$ *are isomorphic as rings.*

Proof. If $\alpha \in \mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{V})$, then clearly $1 \otimes \alpha \in \mathrm{Hom}_{\mathsf{L}[G]}(\mathsf{V}_\mathsf{L}, \mathsf{V}_\mathsf{L})$. We show that every element $\beta \in \mathrm{Hom}_{\mathsf{L}[G]}(\mathsf{V}_\mathsf{L}, \mathsf{V}_\mathsf{L})$ has the form $\beta = \sum_i l_i \otimes \alpha_i$ for some $\alpha_i \in \mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{V})$ and $l_i \in \mathsf{L}$. Let $\{l_i \mid i \in I\}$ be a K-basis of L. Then $\mathsf{L} = \bigoplus_{i\in I} l_i\mathsf{K}$ is a free right K-module. It follows by Lemma 9.5 that

$$\mathsf{V}_\mathsf{L} = \mathsf{L} \otimes_\mathsf{K} \mathsf{V} = \bigoplus_{i\in I} l_i \otimes \mathsf{V}.$$

Hence, every element $w \in \mathsf{V}_\mathsf{L}$ can be uniquely written as $w = \sum_{i\in I} l_i \otimes v_i$ for some $v_i \in \mathsf{V}$, where only finitely many of the v_i are nonzero.

Let $\beta \in \mathrm{Hom}_{\mathsf{L}[G]}(\mathsf{V}_\mathsf{L}, \mathsf{V}_\mathsf{L})$. We define maps α_i by

$$(1 \otimes v)\beta = \sum_{i\in I} l_i \otimes v\alpha_i$$

for all $v \in \mathsf{V}$. Clearly, $\alpha_i \in \mathrm{Hom}_\mathsf{K}(\mathsf{V}, \mathsf{V})$. For all $g \in G$, we have

$$\sum_{i\in I} l_i \otimes (vg)\alpha_i = (1\otimes vg)\beta = (1\otimes v)g\beta = (1\otimes v)\beta g$$
$$= \sum_{i\in I} l_i \otimes (v\alpha_i)g.$$

By the uniqueness of the decomposition, it follows that $(vg)\alpha_i = (v\alpha_i)g$, so we have $\alpha_i \in \mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V},\mathsf{V})$. □

Lemma 11.10. *Let* $\mathbf{D}$ *be a representation of* G *on the* $\mathsf{K}[G]$*-module* V*, and assume that* $\mathrm{char}\,\mathsf{K} \nmid |G|$*. Then the following are equivalent:*

a) $\mathbf{D}$ *is absolutely irreducible.*

b) $\mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V},\mathsf{V})$ *consists only of the* K*-multiples of the identity map* 1_{V} *on* V*.*

Proof. a) ⇒ b): Let $\mathbf{D}$ be absolutely irreducible, and let L be an algebraically closed extension of K. Since V_{L} is an irreducible $\mathsf{L}[G]$-module, Schur's Lemma 4.3 implies that

$$\mathrm{Hom}_{\mathsf{L}[G]}(\mathsf{V}_{\mathsf{L}},\mathsf{V}_{\mathsf{L}}) = \{a1 \mid a \in \mathsf{L}\}.$$

So $\dim_{\mathsf{L}} \mathrm{Hom}_{\mathsf{L}[G]}(\mathsf{V}_{\mathsf{L}},\mathsf{V}_{\mathsf{L}}) = 1$. Then $\dim_{\mathsf{K}} \mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V},\mathsf{V}) = 1$ by Lemma 11.9, so $\mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V},\mathsf{V}) = \{a1 \mid a \in \mathsf{K}\}$.

b) ⇒ a): Now we have that $\dim_{\mathsf{K}} \mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V},\mathsf{V}) = 1$. It follows from Lemma 11.9 that $\dim_{\mathsf{L}} \mathrm{Hom}_{\mathsf{L}[G]}(\mathsf{V}_{\mathsf{L}},\mathsf{V}_{\mathsf{L}}) = 1$, so $\mathrm{Hom}_{\mathsf{L}[G]}(\mathsf{V}_{\mathsf{L}},\mathsf{V}_{\mathsf{L}}) = \{a1 \mid a \in \mathsf{L}\}$.

Assume for contradiction that V_{L} is reducible. Since $\mathsf{L}[G]$ is a semisimple algebra (here we require that $\mathrm{char}\,\mathsf{K} \nmid |G|$), V_{L} is completely reducible by Theorem 3.4 (pg. 495). Let $\mathsf{V}_{\mathsf{L}} = \mathsf{V}_1 \oplus \mathsf{V}_2$ be a nontrivial direct decomposition of V_{L} into $\mathsf{L}[G]$-submodules. The projections π_i from V_{L} onto V_i $(i = 1, 2)$ lie in $\mathrm{Hom}_{\mathsf{L}[G]}(\mathsf{V}_{\mathsf{L}},\mathsf{V}_{\mathsf{L}})$. Clearly, $\pi_1\pi_2 = 0$. This is impossible, however, since $\mathrm{Hom}_{\mathsf{L}[G]}(\mathsf{V}_{\mathsf{L}},\mathsf{V}_{\mathsf{L}})$ is a field. □

Theorem 11.11. *Let* K *be a splitting field for* G*, and let* $\mathbf{D}_i$ $(i = 1,\ldots,h)$ *be the irreducible representations of* $\mathsf{K}[G]$ *(up to equivalence). Assume that* $\mathrm{char}\,\mathsf{K} \nmid |G|$*. Let* V_i *be a* $\mathsf{K}[G]$*-module corresponding to* $\mathbf{D}_i$*. Then:*

a) *The* $\mathbf{D}_i$ *are absolutely irreducible.*

b) *For every extension field* L *of* K*, the* $\mathsf{L}\otimes\mathsf{V}_i$ *are a complete system of irreducible, pairwise nonisomorphic* $\mathsf{L}[G]$*-modules.*

Proof. a) By the definition of splitting fields, for each fixed i the $\mathbf{D}_i(g)$ with $g \in G$ span a full matrix ring over K. So

$$\mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V}_i,\mathsf{V}_i) = \{x \in \mathrm{Hom}_{\mathsf{K}}(\mathsf{V}_i,\mathsf{V}_i) \mid \mathbf{D}_i(g)x = x\mathbf{D}_i(g) \text{ for all } g \in G\}$$
$$= \{a1 \mid a \in \mathsf{K}\}.$$

Therefore, $\mathbf{D}_i$ is absolutely irreducible by Lemma 11.10.

b) Since the $\mathbf{D}_i$ are absolutely irreducible, the $\mathsf{L}\otimes_{\mathsf{K}}\mathsf{V}_i$ are irreducible $\mathsf{L}[G]$-modules. We must show that the corresponding representations $\widetilde{\mathbf{D}}_i$ of $\mathsf{L}[G]$ are pairwise inequivalent. It suffices to show that they have distinct kernels. Let

$$\mathsf{K}[G] = \bigoplus_{i=1}^{h} \mathfrak{a}_i \quad \text{with} \quad \ker \mathbf{D}_i = \bigoplus_{j \neq i} \mathfrak{a}_j.$$

Then $\mathsf{L}[G] = \bigoplus_{i=1}^{h} (\mathfrak{a}_i)_{\mathsf{L}}$ and $\ker \widetilde{\mathbf{D}}_i \supseteq \bigoplus_{j \neq i} (\mathfrak{a}_j)_{\mathsf{L}}$.

(Since the $(\mathfrak{a}_i)_{\mathsf{L}}$ are simple, being full matrix rings over L, we have in fact that $\ker \widetilde{\mathbf{D}}_i = \bigoplus_{j \neq i} (\mathfrak{a}_j)_{\mathsf{L}}$.) So the $\widetilde{\mathbf{D}}_i$ are pairwise inequivalent representations of $\mathsf{L}[G]$. □

Theorem 11.11 means that we may always restrict our study of the absolutely irreducible representations of G to splitting fields for G. Field extensions of a splitting field yield no new irreducible representations and preserve irreducibility. In §19 we will prove the following deep theorem of R. Brauer: If G is a group of exponent m, then the field generated by the m-th roots of unity over the rational numbers is a splitting field for G.

Exercises

22) Give an example of a reducible representation of $\mathsf{K}[G]$ on a vector space V over K such that $\mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{V})$ is a division ring, but not a field. (It is necessary that char K divides $|G|$.)

12 Integral representations and reduction modulo $\mathfrak{p}$

Definition 12.1. Let I be an integral domain, and let K be the field of fractions of I. We say that the representation $\mathbf{D}$ of G on the $\mathsf{K}[G]$-module V can be made integral if there exists a K-basis $\{v_1, \ldots, v_n\}$ of V such that for all $g \in G$,

$$v_i \mathbf{D}(g) = \sum_{j=1}^{n} a_{ij}(g) v_j$$

with $a_{ij}(g) \in I$ for all $i, j = 1, \ldots, n$.

Theorem 12.2. *Let I be an integral domain, and let K be the field of fractions of I. Suppose that every finitely generated ideal of I is a principal ideal. Let V be a $\mathsf{K}[G]$-module, and let $\mathbf{D}$ be the representation of G on V. Then $\mathbf{D}$ can be made integral.*

Proof. Let $\{v_1, \ldots, v_n\}$ be a K-basis of V and

$$v_i \mathbf{D}(g) = \sum_{j=1}^{n} b_{ij}(g) v_j.$$

There exists a $0 \neq c \in I$ such that all $cb_{ij}(g)$ lie in I. We define

$$\mathsf{W} = \left\{ \sum_{g \in G} \sum_{j=1}^{n} a_{g,j} v_j \mathbf{D}(g) \,\middle|\, a_{g,j} \in I \right\}.$$

It is clear that W is a finitely generated submodule of the free I-module

$$\mathsf{F} = \left\{ \sum_{j=1}^{n} \frac{c_j}{c} v_j \,\middle|\, c_j \in I \right\}$$

in the free generators $\frac{1}{c} v_j$. By Theorem I.13.4 b) (pg. 78), W is therefore a free I-module with free generators $w_1, \ldots, w_s$ ($s \leq n$). Since $v_i \in \mathsf{W}$, the w_j span all of V over K. Hence, $s = n$ and the w_j constitute a K-basis of V. Since $w_i \mathbf{D}(g) \in \mathsf{W}$, we have relations of the form

$$w_i \mathbf{D}(g) = \sum_{j=1}^{n} a_{ij}(g) w_j$$

with $a_{ij}(g) \in I$ for all $g \in G$. So $\mathbf{D}$ can be made integral. □

Two special cases of Theorem 12.2 are of particular interest for us. To handle the first case, we need two observations regarding algebraic number fields.

Lemma 12.3. *Let K be an algebraic number field, let I be the ring of integers of K and let $\mathfrak{i}$ be an ideal in I. Then there exists a finite algebraic extension K' of K with the following properties: If I' is the ring of integers of K', then the ideal $\mathfrak{i}I'$ is a principal ideal in I'.*

Proof. Since the order of the ideal class group of K is finite (see Hasse [225], pg. 542), there exists a natural number n such that $\mathfrak{i}^n$ is a principal ideal in I. Let $\mathfrak{i}^n = aI$ with $a \in I$. We construct the field extension $\mathsf{K}' = \mathsf{K}(b)$ of K using any solution b of $b^n = a$. If I' is the ring of algebraic integers of K', then

$$(\mathfrak{i}I')^n = aI' = (bI')^n.$$

Since the group of ideals of I' is a free abelian group (see Hasse [225], pg. 251), we have $\mathfrak{i}I' = bI'$. □

Lemma 12.4. *Let J be the ring of all algebraic integers. Then every finitely generated ideal of J is a principal ideal.*

Proof. Let $\mathfrak{i} = \sum_{i=1}^{m} Ja_i$ be an ideal in J. Since the a_i are algebraic integers, they lie in a finite algebraic extension K' of the rationals and generate an ideal $\mathfrak{i}' = \sum_{i=1}^{m} I'a_i$ in the ring I' consisting of the algebraic integers of K'. By Lemma 12.3, there exists an

algebraic extension K'' of K' such that $\mathfrak{i}'I'' = bI''$, where I'' is the ring of algebraic integers in K''. It follows that

$$\mathfrak{i} = J\mathfrak{i}' = J(I''\mathfrak{i}') = J(bI'') = Jb. \qquad \square$$

Theorem 12.5. *Let* $\mathbf{D}$ *be a representation of* G *on the* $\mathsf{K}[G]$*-module* V.

a) *If* K *is a* $\mathfrak{p}$*-adic field, then* $\mathbf{D}$ *can be made integral in the ring* I *of* $\mathfrak{p}$*-adic integers of* K.

b) *If* K *is an algebraic number field, then there is a finite algebraic extension* L *of* K *over which the representation of* G *on* V_L *can be made integral.*

Proof. a) The claim follows immediately from Theorem 12.2 since the ring I of $\mathfrak{p}$-adic integers is a principal ideal domain.

b) Let A be the field of all algebraic numbers. By Lemma 12.4 and Theorem 12.2, V_A has an A-basis $\{v_1, \ldots, v_n\}$ such that

$$v_i\mathbf{D}(g) = \sum_{j=1}^{n} a_{ij}(g)v_j,$$

where the $a_{ij}(g)$ are algebraic integers. The finitely many $a_{ij}(g)$ lie in a finite algebraic extension L of K for which the claim holds. $\square$

As an application of Theorem 12.5, we prove a stronger version of Theorem 5.12.

Theorem 12.6. *Let* $\mathbf{D}$ *be an absolutely irreducible representation of* G *over a field* K *of characteristic* 0. *Then the degree of* $\mathbf{D}$ *divides the index* $[G : \mathbf{Z}(G)]$ *of the center of* G.

Proof. (Gaschütz) Let $\mathbf{D}$ be a representation of $\mathsf{K}[G]$. Since $\mathbf{D}$ is absolutely irreducible, we may consider $\mathbf{D}$ as an irreducible representation of $\mathsf{L}[G]$, where L is an algebraically closed extension of K. Let A be the algebraic closure of the prime field of L in L. By Theorem 5.1, A is a splitting field for G. By Theorem 11.11, the representation module corresponding to the representation $\mathbf{D}$ of $\mathsf{L}[G]$ arises (up to isomorphism) by extending the base field of an irreducible $\mathsf{A}[G]$-module. So let $\mathbf{D}$ be a representation of $\mathsf{A}[G]$ on the $\mathsf{A}[G]$-module M. By Theorem 12.5 b), $\mathbf{D}$ can be made integral. Hence, the matrices $\mathbf{D}(g)$, with respect to some basis of M, have algebraic integers as entries.

Let $G = \bigcup_{j=1}^{m} \mathbf{Z}(G)r_j$. Let x be an integer matrix with $\operatorname{tr} x = 1$, and then let

$$y = \sum_{j=1}^{m} \mathbf{D}(r_j)\, x\, \mathbf{D}(r_j^{-1}).$$

So $\operatorname{tr} y = m \operatorname{tr} x = m$. For every $g \in G$, we have $gr_j = r_{j'}z_{j'}$ for some $z_{j'} \in \mathbf{Z}(G)$. Since $\mathbf{D}$ is absolutely irreducible, Lemma 11.10 now implies that $\mathbf{D}(z_{j'}) = a_{j'}1$ with $a_{j'} \in \mathsf{A}$. For all $g \in G$, we then have

$$\mathbf{D}(g)y = \sum_{j=1}^{m} \mathbf{D}(gr_j)\, x\, \mathbf{D}(r_j^{-1}) = \sum_{j=1}^{m} \mathbf{D}(r_{j'})\, x\, a_{j'}\, \mathbf{D}(r_j^{-1})$$
$$= \sum_{j=1}^{m} \mathbf{D}(r_{j'})\, x\, \mathbf{D}(r_{j'}^{-1} g) = y\mathbf{D}(g).$$

So we have $y \in \mathrm{Hom}_{\mathsf{A}[G]}(\mathsf{M}, \mathsf{M})$. By Lemma 11.10, $y = k1$ for some $k \in \mathsf{A}$. Since the entries of the matrix x and the entries of the matrices $\mathbf{D}(r_j^{\pm 1})$ lie in the ring I consisting of the algebraic integers of A, it follows that $k \in I$. Letting n be the degree of $\mathbf{D}$, we then have

$$m = \operatorname{tr} y = \operatorname{tr}(k1) = kn.$$

Thus, $k = \frac{m}{n}$ is a rational number and an algebraic integer. It is well known that this means k is an integer. This shows that $n = \deg \mathbf{D}$ divides $m = [G : \mathbf{Z}(G)]$. □

If one prefers a proof of Theorem 12.6 that avoids using Lemma 12.3 and Lemma 12.4 regarding algebraic numbers, then one can also construct a proof using $\mathfrak{p}$-adic fields. We will prove another strengthening of Theorem 5.12 in Theorem 17.10.

Definition 12.7. Let $\mathbf{D}$ be a representation of a finite group G over a $\mathfrak{p}$-adic field K. With respect to an appropriate basis of the representation module of $\mathbf{D}$, we may associate to $\mathbf{D}$ the matrices $\mathbf{D}(g) = (a_{ij}(g))$ with $a_{ij}(g) \in I$ by Theorem 12.5 a). We define $\overline{a} = a + \mathfrak{p}$ for $a \in I$. The map $\overline{\mathbf{D}}$ defined by $\overline{\mathbf{D}}(g) = (\overline{a_{ij}(g)})$ clearly yields a representation of G over the residue field $\overline{\mathsf{K}} = I/\mathfrak{p}$ of K. We say that $\overline{\mathbf{D}}$ is the reduction of $\mathbf{D}$ modulo $\mathfrak{p}$. (We will address the extent to which $\overline{\mathbf{D}}$ is uniquely determined by $\mathbf{D}$ in Theorem 12.9.)

Theorem 12.8 (R. Brauer, Nesbitt [68]). *Let $\mathbf{D}$ be an absolutely irreducible representation of G over a $\mathfrak{p}$-adic field K. Let p be the characteristic of the residue field $\overline{\mathsf{K}}$ of K, and let p^a be the largest power of p dividing $|G|$. If $p^a \mid \deg \mathbf{D}$, then every reduction of $\mathbf{D}$ modulo $\mathfrak{p}$ is irreducible over $\overline{\mathsf{K}}$.*

Proof. Assume for contradiction that $\overline{\mathbf{D}}$, a reduction modulo $\mathfrak{p}$ of the integral representation $\mathbf{D}$, is reducible. Then there exists a matrix $\bar{t}$ with entries in $\overline{\mathsf{K}}$ such that

$$\bar{t}^{-1}\overline{\mathbf{D}}(g)\bar{t} = \begin{pmatrix} \overline{\mathbf{D}}_1(g) & 0 \\ * & \overline{\mathbf{D}}_2(g) \end{pmatrix}$$

for all $g \in G$. Let $t = (t_{ij})$ with $t_{ij} \in I$ and $\bar{t} = (\bar{t}_{ij})$. Since $\det \bar{t} \neq 0$, this means $\det t \notin \mathfrak{p}$. So $\det t$ is a unit in I, which means t^{-1} is also a matrix with coefficients in I. Now we have

$$t^{-1}\mathbf{D}(g)t = (a_{ij}(g))$$

for some $a_{ij}(g) \in I$, and $\overline{a_{1n}(g)} = 0$ for all $g \in G$, where $n = \deg \mathbf{D}$. Since $\mathbf{D}$ is absolutely irreducible, it follows from Theorem 5.7 b) that

$$\frac{|G|}{\deg \mathbf{D}} = \sum_{g \in G} a_{1n}(g) a_{n1}(g^{-1}) \equiv 0 \bmod \mathfrak{p},$$

contradicting that $p^a \mid \deg \mathbf{D}$. So $\overline{\mathbf{D}}$ is irreducible. □

Theorem 12.8 plays an important role in modular representation theory. We concern ourselves here only with the case that the characteristic of the residue field $\overline{\mathsf{K}}$ of K does not divide $|G|$. The following theorem then describes the situation completely:

Main Theorem 12.9. *Let K be a $\mathfrak{p}$-adic splitting field for G with residue field $\overline{\mathsf{K}}$ of characteristic p. Assume $p \nmid |G|$, and let $\mathbf{D}_1, \ldots, \mathbf{D}_h$ be the irreducible representations of $\mathsf{K}[G]$. Let $\overline{\mathbf{D}}_i$ be some reduction of $\mathbf{D}_i$ modulo $\mathfrak{p}$. Then the $\overline{\mathbf{D}}_1, \ldots, \overline{\mathbf{D}}_h$ are all the irreducible representations of $\overline{\mathsf{K}}[G]$ up to equivalence, and $\overline{\mathsf{K}}$ is a splitting field for G. In particular, the reduction $\overline{\mathbf{D}}_i$ of $\mathbf{D}_i$ modulo $\mathfrak{p}$ is uniquely determined up to equivalence.*

Proof. Since $p \nmid |G|$, the $\overline{\mathbf{D}}_i$ are irreducible by Theorem 12.8.

Assume for contradiction that $\overline{\mathbf{D}}_i$ is equivalent to $\overline{\mathbf{D}}_j$ for some $i \neq j$. Then

$$\overline{\chi_i(g)} = \operatorname{tr} \overline{\mathbf{D}}_i(g) = \operatorname{tr} \overline{\mathbf{D}}_j(g) = \overline{\chi_j(g)}.$$

Since $(\chi_i, \chi_j) = 0$, we have

$$|G| = \sum_{g \in G} (\chi_i(g) - \chi_j(g)) \chi_i(g^{-1}) \equiv 0 \mod \mathfrak{p},$$

which contradicts our assumption that $p \nmid |G|$. Hence, the $\overline{\mathbf{D}}_1, \ldots, \overline{\mathbf{D}}_h$ are pairwise inequivalent representations of $\overline{\mathsf{K}}[G]$. We have

$$\sum_{i=1}^{h} (\deg \overline{\mathbf{D}}_i)^2 = \sum_{i=1}^{h} (\deg \mathbf{D}_i)^2 = |G|.$$

Since the conditions of Theorem 11.4 b) are satisfied over $\overline{\mathsf{K}}$, the field $\overline{\mathsf{K}}$ is a splitting field for G and the $\overline{\mathbf{D}}_i$ are all the absolutely irreducible representations of G over $\overline{\mathsf{K}}$ up to equivalence.

Let $\overline{\mathbf{D}}_i$ and $\widetilde{\mathbf{D}}_i$ be two reductions of $\mathbf{D}_i$ modulo $\mathfrak{p}$. By what we have already shown, the representations $\overline{\mathbf{D}}_i$ and $\widetilde{\mathbf{D}}_i$ are irreducible and are not equivalent to any $\overline{\mathbf{D}}_j$ with $j \neq i$. Therefore, $\overline{\mathbf{D}}_i$ is equivalent to $\widetilde{\mathbf{D}}_i$. □

If char $\overline{\mathsf{K}}$ divides $|G|$, then most of the previous statements no longer hold. The reductions modulo $\mathfrak{p}$ are no longer determined up to equivalence, but their irreducible constituents are still uniquely determined (see R. Brauer [53], pg. 443). Furthermore, the reductions of irreducible representations of $\mathsf{K}[G]$ modulo $\mathfrak{p}$ can be reducible. It is possible that not all representations of $\overline{\mathsf{K}}[G]$ can be obtained by reducing representations of $\mathsf{K}[G]$ modulo $\mathfrak{p}$. We give some examples of these phenomena:

Example 12.10.

a) Let $G = \langle a, b\rangle$ be the dihedral group of order 6 with relations $a^3 = b^2 = 1$, $a^b = a^{-1}$. If K is a $\mathfrak{p}$-adic field with char $\overline{\mathsf{K}} = 3$ and ϵ is a primitive third root of unity in K, then the homomorphism $\mathbf{D}$ defined by

$$\mathbf{D}(a) = \begin{pmatrix} \epsilon & 0 \\ 0 & \epsilon^{-1} \end{pmatrix}, \quad \mathbf{D}(b) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

is clearly a representation of G over K. It is easy to see with the help of Lemma 11.10 that $\mathbf{D}$ is absolutely irreducible. Since $\overline{\epsilon} = 1$, $\overline{\mathbf{D}}$ is a representation of the abelian group $G/\langle a\rangle$, so it is clearly reducible.

b) By virtue of the isomorphism

$$\mathrm{SL}_2(4) = \mathrm{PSL}_2(4) \cong A_5$$

from Theorem II.6.14 (pg. 192), the alternating group A_5 has a faithful representation $\mathbf{F}$ of degree 2 over GF(4). This representation is absolutely irreducible, since otherwise A_5 would be isomorphic to a group of triangular matrices over an extension of GF(4) and would then be solvable.

We show that A_5 does not have a faithful irreducible representation of degree 2 over a field of characteristic 0. This means that $\mathbf{F}$ is not the reduction modulo $\mathfrak{p}$ of a representation of A_5 over a $\mathfrak{p}$-adic field.

Assume for contradiction that $\mathbf{D}$ is a faithful representation of A_5 of degree 2. Let P be a Sylow 2-subgroup of A_5. Then the restriction of $\mathbf{D}$ to P is completely reducible, and it follows (with respect to some basis) that

$$\{\mathbf{D}(x) \mid x \in P\} = \left\{ \begin{pmatrix} \epsilon_1 & 0 \\ 0 & \epsilon_2 \end{pmatrix} \;\middle|\; \epsilon_i = \pm 1 \right\}.$$

So there exists some $x_0 \in P$ such that $\mathbf{D}(x_0) = -1$. But since $\mathbf{D}$ is faithful, we obtain the contradiction $x_0 \in \mathbf{Z}(A_5)$.

c) Let $G = \langle g\rangle$ be the cyclic group of order 2, let K be a $\mathfrak{p}$-adic field with char $\overline{\mathsf{K}} = 2$ and let $\mathsf{V} = \langle v_1, v_2\rangle$ be the $\mathsf{K}[G]$-module with

$$v_1\mathbf{D}(g) = v_1 \quad \text{and} \quad v_2\mathbf{D}(g) = -v_2.$$

The representation is integral with respect to the basis $\{v_1, v_2\}$, and it is clear that we obtain

$$\overline{\mathbf{D}}(g) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

upon reducing modulo $\mathfrak{p}$. If $\{w_1, w_2\}$ is the basis defined by $w_1 = v_1 + \frac{1}{2}v_2$ and $w_2 = v_2$, then

$$w_1\mathbf{D}(g) = w_1 - w_2 \quad \text{and} \quad w_2\mathbf{D}(g) = -w_2.$$

The reduction of $\mathbf{D}$ modulo $\mathfrak{p}$ with respect to this basis is

$$\widetilde{\mathbf{D}}(g) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Clearly, $\overline{\mathbf{D}}$ and $\widetilde{\mathbf{D}}$ are not equivalent.

Theorem 12.11.

a) *Let* K_1 *and* K_2 *be splitting fields for* G *with* $\operatorname{char}\mathsf{K}_i \nmid |G|$. *Then the degrees of the irreducible representations of* $\mathsf{K}_1[G]$ *and* $\mathsf{K}_2[G]$ *are the same.*

b) *If* K *is a splitting field for* G *with* $\operatorname{char}\mathsf{K} \nmid |G|$, *then the degree of every irreducible representation of* $\mathsf{K}[G]$ *divides* $[G : \mathbf{Z}(G)]$.

Proof. a) First, we assume that K_1 and K_2 have the same characteristic. Then there exists a field L in which K_1 and K_2 are isomorphically embedded. (For example, one may choose L to be an algebraically closed field of sufficiently large transcendental degree.) By Lemma 11.3 b), L is a splitting field for G. By Theorem 11.11, all irreducible representations of $\mathsf{L}[G]$ are obtained by base field extension of irreducible representations of $\mathsf{K}_i[G]$ $(i = 1, 2)$. The degrees remain unchanged under base field extension.

Now assume that $\operatorname{char}\mathsf{K}_1 = p$. It is well known that there exists a discrete valuation field K whose residue field $\overline{\mathsf{K}}$ is isomorphic to K_1. Let L be a splitting field for G that is a finite algebraic extension of K. It is well known that L is again a discrete valuation field and that the residue field $\overline{\mathsf{L}}$ of L is an extension of $\overline{\mathsf{K}}$. Since the reduction modulo $\mathfrak{p}$ of all irreducible representations of $\mathsf{K}[G]$ are absolutely irreducible, the irreducible representation of $\mathsf{K}[G]$ remain irreducible under base field extension to L, so they are absolutely irreducible. Hence, K is a splitting field for G, and we may apply Theorem 12.9. The degrees of the irreducible representations of G over K_1 are the same as those over a splitting field of characteristic 0. By the first part of the proof, these degrees coincide for all fields we consider.

b) The claim follows immediately from Theorem 12.6 and part a). □

Exercises

23) Let $\mathbf{D}$ be a representation of G on $\overline{\mathsf{V}}$ over a splitting field $\overline{\mathsf{K}}$ of G with $\operatorname{char}\overline{\mathsf{K}} = p$ not dividing $|G|$. Let K be a $\mathfrak{p}$-adic field with residue field $I/\mathfrak{p} = \overline{\mathsf{K}}$ and with $\mathfrak{p} = \pi I$. Let $\mathbf{F}_1(g)$ be matrices with entries in I such that $\overline{\mathbf{F}_1(g)} = \mathbf{D}(g)$. (The reduction map modulo $\mathfrak{p}$ on matrices is also denoted with the bar notation.) Prove the following:

a) Suppose we have already found matrices $\mathbf{F}_i(g)$ (for $g \in G$) and $\mathbf{B}(g, g')$ with entries in I that satisfy

$$\mathbf{F}_i(g)\mathbf{F}_i(g') = \mathbf{F}_i(gg') + \pi^i \mathbf{B}(g, g')$$

for all $g, g' \in G$. Then f defined by $f(g, g') = \mathbf{D}(gg')^{-1}\overline{\mathbf{B}(g, g')}$ is a 2-cocycle contained in $\mathbf{Z}^2(G, \mathrm{Hom}_{\overline{\mathsf{K}}}(\overline{\mathsf{V}}, \overline{\mathsf{V}}))$. Show that there exists some $\mathbf{F}_{i+1}(g) = \mathbf{F}_i(g) + \pi^i \mathbf{A}(g)$ such that for all $g, g' \in G$ we have

$$\mathbf{F}_{i+1}(g)\,\mathbf{F}_{i+1}(g') \equiv \mathbf{F}_{i+1}(gg') \mod (\pi^{i+1}).$$

b) Construct a representation $\mathbf{F}$ of G over K by defining

$$\mathbf{F}(g) = \lim_{i \to \infty} \mathbf{F}_i(g).$$

Prove that irreducibility and inequivalence are preserved using this procedure. Give a new proof of Theorem 12.9.

(The procedure described here is an analog of Hensel's Lemma in valuation theory.)

24) Let G be a finite group of matrices whose coefficients lie in the ring I of $\mathfrak{p}$-adic integers of the $\mathfrak{p}$-adic field K. Let $I/\mathfrak{p} = \mathrm{GF}(p^f)$ be the residue field of I. We define $\overline{a} = a + \mathfrak{p}$ for $a \in I$. Then the map $(a_{ij}) \mapsto (\overline{a_{ij}})$ is a homomorphism from G to $\mathrm{GL}_n(p^f)$ whose kernel is a p-group. If $pI = \mathfrak{p}^e$ and $e < p - 1$, then the map is an isomorphism.

13 Algebraically conjugate characters

In this section, K is always a field of characteristic 0 and L is a splitting field for G that is a finite algebraic Galois extension of K, except in Lemma 13.2 and Theorem 13.3. We say that χ is a character of G over K when there exists a representation $\mathbf{D}$ of $\mathsf{K}[G]$ such that $\mathrm{tr}\,\mathbf{D}(g) = \chi(g)$ for all $g \in G$; the character χ is said to be irreducible over K when $\mathbf{D}$ is an irreducible representation of $\mathsf{K}[G]$.

Theorem 13.1. *Let L be a Galois extension of K. Assume L is a splitting field for G.*

a) *If χ is a character of G over L and α is a field automorphism of L over K, then χ^α defined by*

$$\chi^\alpha(g) = \chi(g)^\alpha$$

is a character of G over L. We say that χ^α is conjugate to χ over K. If χ is irreducible, then χ^α is also irreducible.

b) *If χ is a character of G over L with $\chi(g) \in \mathsf{K}$ for all $g \in G$, then*

$$(\chi, \tau) = (\chi, \tau^\alpha)$$

for all characters τ of G over L and all automorphisms α of L over K. (Note that $\chi(g) \in \mathsf{K}$ for all $g \in G$ in no way implies that χ is a character of G over K!)

c) *Suppose the splitting field L of G contains the field $\mathbb{Q}(\epsilon)$ generated by the primitive $|G|$-th roots of unity over the field $\mathbb{Q}$ of rational numbers. If $(|G|, m) = 1$, then it is well known that there exists an automorphism α_m of $\mathbb{Q}(\epsilon)$ over $\mathbb{Q}$ such*

that $\epsilon^{\alpha_m} = \epsilon^m$. *Then*

$$\chi^{\alpha_m}(g) = \chi(g^m)$$

for all characters χ *of* G *over* L *and all* $g \in G$.

Proof. a) Let $\mathbf{D}$ be a representation of $\mathsf{L}[G]$ on a vector space V over L with $\operatorname{tr}\mathbf{D}(g) = \chi(g)$. Let $\{v_1, \dots, v_n\}$ be an L-basis of V and

$$v_i\mathbf{D}(g) = \sum_{j=1}^{n} a_{ij}(g)v_j.$$

We define $\mathbf{D}^\alpha(g)$ by

$$v_i\mathbf{D}^\alpha(g) = \sum_{j=1}^{n} a_{ij}(g)^\alpha v_j.$$

Since $\mathbf{D}^\alpha(g_1)\mathbf{D}^\alpha(g_2) = \mathbf{D}^\alpha(g_1g_2)$, $\mathbf{D}^\alpha$ is a representation of G over L with

$$\operatorname{tr}\mathbf{D}^\alpha(g) = \sum_{i=1}^{n} a_{ii}(g)^\alpha = \left(\sum_{i=1}^{n} a_{ii}(g)\right)^\alpha = \chi^\alpha(g).$$

If χ is irreducible over L, then it follows from

$$\sum_{g\in G} \chi(g)\chi(g^{-1}) = |G|$$

that

$$\sum_{g\in G} \chi^\alpha(g)\chi^\alpha(g^{-1}) = |G|$$

as well. Therefore, χ^α is irreducible by Theorem 5.10 b) since $\operatorname{char}\mathsf{K} = 0$.

b) Since (χ, τ) is always an integer, it is fixed by every automorphism. We have

$$\begin{aligned}(\chi,\tau) = (\chi,\tau)^\alpha &= \left(\frac{1}{|G|}\sum_{g\in G}\chi(g)\tau(g^{-1})\right)^\alpha = \frac{1}{|G|}\sum_{g\in G}\chi(g)\tau(g^{-1})^\alpha \\ &= (\chi,\tau^\alpha).\end{aligned}$$

c) It is well known that the automorphism α_m of $\mathbb{Q}(\epsilon)$ over $\mathbb{Q}$ defined by $\epsilon^{\alpha_m} = \epsilon^m$ can be extended to the field L containing $\mathbb{Q}(\epsilon)$. Let $\mathbf{D}$ be a representation of G over L with $\operatorname{tr}\mathbf{D}(g) = \chi(g)$, and let ϵ^{n_i} be the eigenvalues of $\mathbf{D}(g)$. These eigenvalues are $|G|$-th roots of unity since

$$\mathbf{D}(g)^{|G|} = \mathbf{D}(g^{|G|}) = \mathbf{D}(1) = 1.$$

Then $\mathbf{D}(g^m)$ and $\mathbf{D}^{\alpha_m}(g)$ have eigenvalues ϵ^{mn_i}. Therefore, it follows that

$$\chi^{\alpha_m}(g) = \operatorname{tr}\mathbf{D}^{\alpha_m}(g) = \sum_i \epsilon^{mn_i} = \operatorname{tr}\mathbf{D}(g^m) = \chi(g^m). \qquad \square$$

Lemma 13.2. *Let* L *be a Galois extension of* K*, and let* V *be a right* $\mathsf{K}[G]$*-module. We let* $\mathsf{V}' = \mathsf{L} \otimes_{\mathsf{K}} \mathsf{V}$*. Then* V' *is a right* $\mathsf{L}[G]$*-module. Let* H *be the Galois group of* L *over* K*, and let* $\{v_1, \ldots, v_n\}$ *be a* K*-basis of* V*. By defining*

$$\left(\sum_{i=1}^{n}(l_i \otimes v_i)\right)h = \sum_{i=1}^{n}(l_i h \otimes v_i)$$

for all $l_i \in \mathsf{L}$*, the vector space* V' *becomes an* H*-module. If* W *is an* $\mathsf{L}[G]$*-submodule of* V' *with* $\mathsf{W}h = \mathsf{W}$ *for all* $h \in H$*, then there exists a* $\mathsf{K}[G]$*-submodule* U *of* V *such that* $\mathsf{W} = \mathsf{L} \otimes_{\mathsf{K}} \mathsf{U}$*.*

Proof. Let $\{w_1, \ldots, w_r\}$ be an L-basis of W. It is well known that this set can be extended to an L-basis of V' by adding an appropriate subset of the L-basis $\{v_i' = 1 \otimes v_i \mid i = 1, \ldots, n\}$ of V'. So after re-indexing if necessary, we have that

$$\{w_1, \ldots, w_r, v_{r+1}', \ldots, v_n'\}$$

is an L-basis of V'. Then we have

$$v_i' = \sum_{j=r+1}^{n} v_j' l_{ij} + w_i' \quad (1 \le i \le r)$$

for some $w_i' \in \mathsf{W}$ and $l_{ij} \in \mathsf{L}$. Since the $v_1', \ldots, v_n'$ are linearly independent over L, the w_i' are also linearly independent over L, so the w_i' constitute an L-basis of W. Now, for all $h \in H$ we have $w_i'h \in \mathsf{W}$, and so for some $a_{ki} \in \mathsf{L}$ we have

$$\begin{aligned} w_i'h &= v_i' - \sum_{j=r+1}^{n} v_j'(l_{ij}h) \\ &= \sum_{k=1}^{r} w_k' a_{ki} = \sum_{k=1}^{r} v_k' a_{ki} - \sum_{k=1}^{r}\sum_{j=r+1}^{n} v_j' l_{kj} a_{ki}. \end{aligned}$$

Comparing coefficients, we have $a_{ki} = \delta_{ki}$ $(i, k = 1, \ldots, r)$. Thus, $w_i'h = w_i'$. Letting $w_i' = \sum_{j=1}^{n} v_j' b_{ij}$ for some $b_{ij} \in \mathsf{L}$, it follows that $b_{ij}h = b_{ij}$ for all $h \in H$, so $b_{ij} \in \mathsf{K}$. We define

$$u_i = \sum_{j=1}^{n} v_j b_{ij}.$$

Then $\mathsf{U} = \langle u_1, \ldots, u_r \rangle$ is a K-subspace of V such that

$$\mathsf{L} \otimes_{\mathsf{K}} \mathsf{U} = \langle w'_1, \ldots, w'_r \rangle = \mathsf{W}.$$

Since

$$\mathsf{U} = \langle v \in \mathsf{V} \mid 1 \otimes v \in \mathsf{W} \rangle,$$

U is a $\mathsf{K}[G]$-module. □

Theorem 13.3. *Let* V *be an irreducible* $\mathsf{K}[G]$*-module, and let* L *be a Galois extension of* K*. Then the* $\mathsf{L}[G]$*-module* $\mathsf{V}' = \mathsf{L} \otimes_{\mathsf{K}} \mathsf{V}$ *is a direct sum of irreducible* $\mathsf{L}[G]$*-modules that arise from one another by automorphisms of* L *over* K*.*

(*No assumption regarding the characteristic of* K *is necessary here.*)

Proof. Let M be an irreducible $\mathsf{L}[G]$-submodule of $\mathsf{V}' = \mathsf{L} \otimes_{\mathsf{K}} \mathsf{V}$. By Lemma 13.2, V' is a module for the Galois group H of L over K. In addition to M, all the $\mathsf{M}h$ with $h \in H$ are irreducible $\mathsf{L}[G]$-modules. Let $\mathsf{T} = \sum_{h \in H} \mathsf{M}h$. By Theorem 3.2, T is a completely reducible $\mathsf{L}[G]$-module, so it is a direct sum of some of the $\mathsf{M}h$. Clearly, $\mathsf{T}h = \mathsf{T}$ for all $h \in H$. So it follows by Lemma 13.2 that $\mathsf{T} = \mathsf{L} \otimes_{\mathsf{K}} \mathsf{U}$, where U is a $\mathsf{K}[G]$-submodule of V. Since V is irreducible, we conclude that $\mathsf{U} = \mathsf{V}$ and $\mathsf{V}' = \mathsf{L} \otimes_{\mathsf{K}} \mathsf{V} = \bigoplus_{i=1}^{n} \mathsf{M}h_i$ for some $h_i \in H$. □

Theorem 13.4. *Let* G *be a permutation group on the set* Ω*. For every* $g \in G$*, we denote the number of fixed points of* g *on* Ω *by* $\pi(g)$*. If* G *has exactly* t *orbits on* Ω*, then*

$$\sum_{g \in G} \pi(g) = t|G|.$$

Proof. First assume that $t = 1$, so G acts transitively on $\Omega = \{1, \ldots, n\}$. Then

$$\sum_{g \in G} \pi(g) = \sum_{i=1}^{n} |\{g \in G \mid i^g = i\}| = \sum_{i=1}^{n} |G_i| = n|G_1| = |G|.$$

For the general case, let $\Omega = \bigcup_{i=1}^{t} \Omega_i$ be the decomposition of Ω into orbits Ω_i of G. If $\pi_i(g)$ is the number of fixed points of g on Ω_i, then by what we have already shown,

$$\sum_{g \in G} \pi_i(g) = |G|.$$

Since $\pi(g) = \sum_{i=1}^{t} \pi_i(g)$, it follows that $\sum_{g \in G} \pi(g) = t|G|$. □

Theorem 13.5 (R. Brauer [47]). *Suppose the group* A *is represented*

(1) *as a permutation group on the absolutely irreducible characters of* G *over a splitting field for* G *of characteristic* 0 *with action* $\chi \mapsto \chi^a$ *for all* $a \in A$*.*

(2) *as a permutation group on the set of elements of G that preserves the partition into conjugacy classes; that is, we demand for all $g \in G$ and all $a \in A$ that*

$$\{g^{xa} \mid x \in G\} = \{g^{ax} \mid x \in G\}.$$

(*Thus, the group A can be considered a permutation group on the set of conjugacy classes of G.*) *Assume that for all absolutely irreducible characters χ of G, all $g \in G$ and all $a \in A$ we have*

$$\chi^a(g) = \chi(g^a).$$

Then:

a) *The number of conjugacy classes fixed by $a \in A$ is equal to the number of irreducible characters of G fixed by a.*
b) *The number t_1 of orbits of A on the set of absolutely irreducible characters of G is equal to the number t_2 of orbits of A on the set of conjugacy classes of G.*

Proof. Let $\chi_1, \ldots, \chi_h$ be the absolutely irreducible characters of G, and let $g_1, \ldots, g_h$ be representatives of the conjugacy classes of G. We consider the matrix $t = (\chi_i(g_j))$. By our assumption, the map $\chi_i(g_j) \mapsto \chi_i^a(g_j)$ is a permutation $\mathbf{P}(a)$ on the rows of t and $\chi_i(g_j) \mapsto \chi_i(g_j^a)$ is a permutation $\mathbf{Q}(a)$ on the columns of t. If we express $\mathbf{P}(a)$ and $\mathbf{Q}(a)$ as permutation matrices with h rows and columns, then we have

$$\mathbf{P}(a)(\chi_i(g_j)) = (\chi_i^a(g_j)) = (\chi_i(g_j^a)) = (\chi_i(g_j))\mathbf{Q}(a)^T.$$

The matrix t is nonsingular since the χ_i $(i = 1, \ldots, h)$ are a basis of the space of class functions on G by Theorem 5.9, so they are linearly independent. Hence, the inverse t^{-1} of t exists. Therefore, we have for the fixed points that

$$\pi(\mathbf{P}(a)) = \operatorname{tr} \mathbf{P}(a) = \operatorname{tr} t^{-1}\mathbf{P}(a)t = \operatorname{tr} \mathbf{Q}(a)^T = \pi(\mathbf{Q}(a)).$$

We have proved a). By Theorem 13.4, we have

$$t_1|A| = \sum_{a \in A} \pi(\mathbf{P}(a)) = \sum_{a \in A} \pi(\mathbf{Q}(a)) = t_2|A|,$$

and so $t_1 = t_2$ since $\operatorname{char} \mathsf{K} = 0$.

Now we give a few applications of these considerations.

Definition 13.6. Let χ be a character of G over the complex numbers.

a) We call χ a rational character if $\chi(g)$ is a rational number for all $g \in G$.
b) We call χ a real character of G if $\chi(g)$ is a real number for all $g \in G$. (Since $\chi(g^{-1}) = \overline{\chi(g)}$, this is equivalent to $\chi(g) = \chi(g^{-1})$ for all $g \in G$.)
c) We call $g \in G$ a real element of the group G if g and g^{-1} are conjugate in G. (Since the irreducible characters are a basis for the space of class functions, the element g is real if and only if $\chi(g) = \chi(g^{-1}) = \overline{\chi(g)}$ for all irreducible χ; that

is, g is real if and only if $\chi(g)$ is real for all irreducible characters χ of G.) It is clear that if $g \in G$ is real, then all conjugates of g are also real. We refer to the conjugacy classes consisting of real elements as real classes.

Theorem 13.7.

a) *The number of absolutely irreducible real characters of G is equal to the number of real classes.*

b) *All absolutely irreducible characters of G are rational if and only if for every m such that $(m, |G|) = 1$ and for all $g \in G$, the element g is conjugate to g^m in G.*

c) *All irreducible characters of the symmetric group S_n are rational. (In fact, it can be shown that the field $\mathbb{Q}$ of rational numbers is a splitting field for S_n; this follows from a close study of the idempotent elements in $\mathbb{Q}[S_n]$; see Curtis, Reiner [102], §28.)*

Proof. a) Let $A = \{1, z\}$ be the cyclic group of order 2. We represent A as a permutation group on G via the action $g^1 = g$ and $g^z = g^{-1}$ for all $g \in G$. Furthermore, we represent A as a permutation group on the absolutely irreducible characters of G via the action $\chi^1 = \chi$ and $\chi^z = \overline{\chi}$. Then $\chi^a(g) = \chi(g^a)$ for all $a \in A$. The conjugacy classes fixed by z are exactly the real classes, and the characters fixed by z are exactly the real characters. By Theorem 13.5 a), the number of real conjugacy classes of G is equal to the number of real absolutely irreducible characters of G.

b) Let A be the group of residue classes modulo $|G|$ that are coprime to $|G|$. If a is the residue class of some m with $(m, |G|) = 1$, then we define $g^a = g^m$ and $\chi^a = \chi^{\alpha_m}$, where α_m is the automorphism of the field $\mathbb{Q}(\epsilon)$ generated by the $|G|$-th roots of unity defined by $\epsilon^{\alpha_m} = \epsilon^m$. These assignments are clearly well defined. Then by Theorem 13.1 c), we have

$$\chi^a(g) = \chi(g)^{\alpha_m} = \chi(g^m) = \chi(g^a).$$

All absolutely irreducible characters of G are rational if and only if the permutation group $\{\mathbf{P}(a) \mid a \in A\}$ from Theorem 13.5 has exactly h orbits, where h is the number of conjugacy classes of G. By Theorem 13.5 b), this is equivalent to $\{\mathbf{Q}(a) \mid a \in A\}$ having exactly h orbits; that is, $\chi_i(g^m) = \chi_i(g)$ for all absolutely irreducible characters χ_i of G and all m coprime to $|G|$. Since the χ_i are a basis for the space of class functions on G, this is equivalent to the conjugacy of g and g^m in G.

c) Let $g \in S_n$. For all integers m such that $(m, n!) = 1$, g and g^m have the same cycle type, so they are conjugate in S_n. The claim then follows from b). □

We note a simple consequence.

Theorem 13.8 (Burnside). *Let G be a group of odd order. Then the trivial character χ_1 defined by $\chi_1(g) = 1$ for all $g \in G$ is the only real, absolutely irreducible character of G.*

Proof. Suppose that $K = \{x^g \mid g \in G\}$ is a real conjugacy class of G in the sense of Definition 13.6 c). Then $x^g = x^{-1}$ for some $g \in G$. It follows that $x^{g^2} = x$, so $g^2 \in \mathbf{C}_G(x)$. Since the order $|g|$ of g is odd, it follows that $g \in \mathbf{C}_G(x)$ as well. Thus, $x^{-1} = x^g = x$ and so $x^2 = 1$, which implies $x = 1$. We conclude that $\{1\}$ is the only real conjugacy class of G. Therefore, there is only one real, absolutely irreducible character of G by Theorem 13.7 a), namely χ_1. □

14 The Schur index

For the entirety of this section, let G be a finite group and K a field with $\operatorname{char} \mathsf{K} \nmid |G|$. Let L be an extension field of K that is a splitting field for G, and let $\mathbf{D}$ be an irreducible representation of $\mathsf{L}[G]$ on the L-vector space V. We will answer the following questions:

(1) What are the irreducible $\mathsf{K}[G]$-modules W whose base field extension $\mathsf{W}_\mathsf{L} = \mathsf{L} \otimes_\mathsf{K} \mathsf{W}$ contains the irreducible $\mathsf{L}[G]$-module V as a composition factor?

(2) Let F be an extension field of K with $\mathsf{F} \subseteq \mathsf{L}$. When does there exist an $\mathsf{F}[G]$-module U such that $\mathsf{L} \otimes_\mathsf{F} \mathsf{U}$ is an $\mathsf{L}[G]$-module isomorphic to V? If there exists such a U, then with an appropriate choice of L-basis for V, the matrices $\mathbf{D}(g)$ (with $g \in G$) have entries in the field F. In this situation, we say that $\mathbf{D}$ can be realized over F.

(3) We denote by $k\mathbf{D}$ the representation of G on the module $\mathsf{V} \oplus \overset{k}{\cdots} \oplus \mathsf{V}$. For which k can the representation $k\mathbf{D}$ be realized over K?

A proper treatment of these questions requires some preparation using the theory of algebras.

Definition 14.1. Let $\mathfrak{a}$ be an algebra over the field K with $\dim_\mathsf{K} \mathfrak{a} < \infty$. We say that $\mathfrak{a}$ is a central simple algebra if $\mathbf{Z}(\mathfrak{a}) = \mathsf{K}$ and $\mathfrak{a}$ is a simple algebra over K; that is, 0 and $\mathfrak{a}$ are the only two-sided ideal of $\mathfrak{a}$. (Here we always identify the field K with the subring $\{k1 \mid k \in \mathsf{K}\}$ isomorphic to K which consists of the K-multiples of the unity element 1 of $\mathfrak{a}$.)

Lemma 14.2. *Let* $\mathfrak{a}$ *and* $\mathfrak{b}$ *be algebras over* K (*not necessarily of finite dimension*).

a) *We have* $\mathbf{Z}(\mathfrak{a} \otimes_\mathsf{K} \mathfrak{b}) = \mathbf{Z}(\mathfrak{a}) \otimes_\mathsf{K} \mathbf{Z}(\mathfrak{b})$.

b) *If* $\mathfrak{a}$ *is a simple algebra* (*of arbitrary dimension*) *and* $\mathbf{b}$ *is central simple, then* $\mathfrak{a} \otimes_\mathsf{K} \mathfrak{b}$ *is simple.*

Proof. Let $\{a_i \mid i \in I\}$ be a K-basis of $\mathfrak{a}$, and let $\{b_j \mid j \in J\}$ be a K-basis of $\mathfrak{b}$. We first show that every element of $\mathfrak{a} \otimes_\mathsf{K} \mathfrak{b}$ can be written as $\sum\limits_j x_j \otimes b_j$ for unique $x_j \in \mathfrak{a}$ and $x_j \neq 0$ for only finitely many j. It is clear that every element can be written in such a way. Now suppose that

$$\sum_j x_j \otimes b_j = \sum_j x'_j \otimes b_j.$$

We have that $x_j = \sum_i c_{ji} a_i$ and $x'_j = \sum_i c'_{ji} a_i$ for suitable $c_{ji}, c'_{ji} \in \mathsf{K}$. Since $\{a_i \otimes b_j \mid i \in I, j \in J\}$ is a K-basis for $\mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b}$, it follows by comparing coefficients that $c_{ji} = c'_{ji}$ for all i, j, and then $x_j = x'_j$ for all $j \in J$.

a) Let $\{a_i \mid i \in I'\}$ be a K-basis of $\mathbf{Z}(\mathfrak{a})$, and let $\{a_i \mid i \in I\}$ be a K-basis of $\mathfrak{a}$ with $I' \subseteq I$. Furthermore, let $\{b_j \mid j \in J'\}$ be a K-basis of $\mathbf{Z}(\mathfrak{b})$, and let $\{b_j \mid j \in J\}$ be a K-basis of $\mathfrak{b}$ with $J' \subseteq J$. It is clear that

$$\mathbf{Z}(\mathfrak{a}) \otimes_{\mathsf{K}} \mathbf{Z}(\mathfrak{b}) = \langle a_i \otimes b_j \mid i \in I', j \in J' \rangle \subseteq \mathbf{Z}(\mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b}).$$

Let

$$z = \sum_{i,j} k_{ij}\, a_i \otimes b_j \in \mathbf{Z}(\mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b}),$$

where $k_{ij} \in \mathsf{K}$. For all $a \in \mathfrak{a}$, we have

$$0 = z(a \otimes 1) - (a \otimes 1)z = \sum_{i,j} k_{ij}(a_i a - a a_i) \otimes b_j.$$

Since every element of $\mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b}$ can be written in the form $\sum_j x_j \otimes b_j$ for unique $x_j \in \mathfrak{a}$ by our previous remark, we have

$$a\big(\sum_i k_{ij} a_i\big) = \big(\sum_i k_{ij} a_i\big) a.$$

Therefore, $\sum_i k_{ij} a_i \in \mathbf{Z}(\mathfrak{a})$ and consequently $k_{ij} = 0$ for $i \notin I'$ and all $j \in J$. In an analogous way, we obtain $k_{ij} = 0$ for $j \notin J'$ and all $i \in I$. Therefore,

$$z = \sum_{i \in I'} \sum_{j \in J'} k_{ij}\, a_i \otimes b_j \in \mathbf{Z}(\mathfrak{a}) \otimes_{\mathsf{K}} \mathbf{Z}(\mathfrak{b}).$$

So $\mathbf{Z}(\mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b}) = \mathbf{Z}(\mathfrak{a}) \otimes_{\mathsf{K}} \mathbf{Z}(\mathfrak{b})$.

b) Again, let $\{a_i \mid i \in I\}$ be a K-basis of $\mathfrak{a}$, and let $\mathfrak{i} \neq 0$ be a two-sided ideal in $\mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b}$. If $z \in \mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b}$, then

$$z = \sum_j a_j \otimes z_j$$

for some unique $z_j \in \mathfrak{b}$. We denote by $l(z)$ the number of nonzero z_j, and we choose a z with $0 \neq z \in \mathfrak{i}$ so that $l(z)$ is minimal. Let, say, $z_1 \neq 0$. By the simplicity of $\mathfrak{b}$, we have $\mathfrak{b} z_1 \mathfrak{b} = \mathfrak{b}$. So there exist $y_i, y'_i \in \mathfrak{b}$ such that $\sum_i y_i z_1 y'_i = 1$. We define

$$z' = \sum_i (1 \otimes y_i) z (1 \otimes y'_i).$$

Then

$$z' = \sum_j a_j \otimes w_j \in \mathfrak{i}$$

for some $w_j \in \mathfrak{b}$ with $w_1 = 1$ and $l(z') \leq l(z)$. For every $b \in \mathfrak{b}$, we have

$$(1 \otimes b)z' - z'(1 \otimes b) = \sum_j a_j \otimes (bw_j - w_j b) \in \mathfrak{i}$$

and

$$l((1 \otimes b)z' - z'(1 \otimes b)) \leq l(z') - 1 < l(z).$$

By our choice of z, this forces

$$(1 \otimes b)z' - z'(1 \otimes b) = 0,$$

and therefore $bw_j = w_j b$ for all $j = 1, \ldots, n$ and all $b \in \mathfrak{b}$. So $w_j \in \mathbf{Z}(\mathfrak{b}) = \mathsf{K}$ and $z' = \sum\limits_j a_j \otimes w_j = (\sum\limits_j w_j a_j) \otimes 1 = w \otimes 1$ with $w \in \mathfrak{a}$. Since $\mathfrak{a}$ is simple, we have $\mathfrak{a}w\mathfrak{a} = \mathfrak{a}$. Therefore,

$$\mathfrak{i} \supseteq (\mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b})z'(\mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b}) = \mathfrak{a}w\mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b} = \mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b},$$

and so $\mathfrak{a} \otimes_{\mathsf{K}} \mathfrak{b}$ is simple. □

Theorem 14.3. *Let $\mathfrak{a}$ be a central simple algebra over K. Then every algebraically closed extension field of K is a splitting field for $\mathfrak{a}$ and $\dim_{\mathsf{K}} \mathfrak{a}$ is the square of a natural number.*

Proof. Let L be an algebraically closed extension of K. By Lemma 14.2, $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}$ is simple. By Wedderburn's Theorem 4.4, it follows that $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a} \cong (\mathsf{L})_n$. So L is a splitting field for $\mathfrak{a}$, and

$$\dim_{\mathsf{K}} \mathfrak{a} = \dim_{\mathsf{L}}(\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}) = \dim_{\mathsf{L}}(\mathsf{L})_n = n^2$$

is a square. □

We emphasize that not every semisimple algebra has splitting fields. A necessary and sufficient condition for the existence of splitting fields is that the simple direct summands of the center are separable extensions of the ground field.

Lemma 14.4. *Let $\mathfrak{a}$ be a simple algebra over K, so $\mathfrak{a} = (\mathfrak{c})_n$ for some division ring $\mathfrak{c}$, and let V be a right $\mathfrak{a}$-module. We let $\mathfrak{b} = \mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$. Then:*

a) *If $\mathfrak{c}^*$ is the division ring anti-isomorphic to $\mathfrak{c}$, then $\mathfrak{b} \cong (\mathfrak{c}^*)_k$ for some k. In particular, $\mathfrak{b}$ is simple.*
b) $(\dim_{\mathsf{K}} \mathsf{V})^2 = \dim_{\mathsf{K}} \mathfrak{a} \dim_{\mathsf{K}} \mathfrak{b}$.

Proof. Let W be the unique (up to isomorphism) irreducible $\mathfrak{a}$-module determined by Theorem 4.1. Then W is isomorphic to the n-dimensional vector space over $\mathfrak{c}$, so

$\dim_{\mathsf{K}} \mathsf{W} = n \dim_{\mathsf{K}} \mathfrak{c}$. By Theorem 4.4, $\mathfrak{c}^* \cong \mathrm{Hom}_{\mathfrak{a}}(\mathsf{W}, \mathsf{W})$. Since $\mathfrak{a}$ is semisimple, V is completely reducible by Theorem 3.4, say

$$\mathsf{V} \cong \mathsf{W} \oplus \underset{k}{\cdots} \oplus \mathsf{W}.$$

With respect to an appropriate K-basis of V, every $a \in \mathfrak{a}$ can be written in the form

$$\begin{pmatrix} f(a) & 0 & \cdots & 0 \\ 0 & f(a) & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & f(a) \end{pmatrix}.$$

Let $b \in \mathfrak{b} = \mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$ and write the matrix corresponding to b in the form

$$\begin{pmatrix} d_{11}(b) & \cdots & d_{1k}(b) \\ \vdots & & \vdots \\ d_{k1}(b) & \cdots & d_{kk}(b) \end{pmatrix}.$$

Since $ab = ba$, we have $f(a)d_{ij}(b) = d_{ij}(b)f(a)$ for all $a \in \mathfrak{a}$ and all $i, j = 1, \ldots, k$. This forces $d_{ij}(b) \in \mathrm{Hom}_{\mathfrak{a}}(\mathsf{W}, \mathsf{W}) \cong \mathfrak{c}^*$. It is trivial that $b_{ij} \in \mathrm{Hom}_{\mathfrak{a}}(\mathsf{W}, \mathsf{W})$ is also sufficient for $b = (b_{ij}) \in \mathrm{Hom}_{\mathfrak{a}}(\mathsf{V}, \mathsf{V})$. Hence, $\mathfrak{b} \cong (\mathfrak{c}^*)_k$ and so $\mathfrak{b}$ is simple by Theorem 4.4 b). Furthermore, it follows that

$$\dim_{\mathsf{K}} \mathfrak{a} \dim_{\mathsf{K}} \mathfrak{b} = n^2(\dim_{\mathsf{K}} \mathfrak{c})k^2(\dim_{\mathsf{K}} \mathfrak{c}^*) = (nk \dim_{\mathsf{K}} \mathfrak{c})^2 = (\dim_{\mathsf{K}} \mathsf{V})^2. \qquad \square$$

Theorem 14.5. *Let $\mathfrak{a}$ be a central simple algebra over K, and let $\mathfrak{b}$ be a simple subalgebra of $\mathfrak{a}$. Assume that the unity element of $\mathfrak{b}$ is also the unity element of $\mathfrak{a}$. We define*

$$\mathbf{C}_{\mathfrak{a}}(\mathfrak{b}) = \{a \in \mathfrak{a} \mid ab = ba \text{ for all } b \in \mathfrak{b}\}.$$

Then:

a) $\mathbf{C}_{\mathfrak{a}}(\mathfrak{b})$ *is simple.*

b) *Let $\mathfrak{a}^*$ be the K-algebra anti-isomorphic to $\mathfrak{a}$. Then $\mathbf{C}_{\mathfrak{a}}(\mathfrak{b})$ and $\mathfrak{a}^* \otimes_{\mathsf{K}} \mathfrak{b}$ are full matrix rings over anti-isomorphic division rings.*

c) $\dim_{\mathsf{K}} \mathfrak{a} = \dim_{\mathsf{K}} \mathfrak{b} \dim_{\mathsf{K}} \mathbf{C}_{\mathfrak{a}}(\mathfrak{b})$.

d) $\mathbf{C}_{\mathfrak{a}}(\mathbf{C}_{\mathfrak{a}}(\mathfrak{b})) = \mathfrak{b}$.

Proof. By Lemma 14.2 b), $\mathfrak{a}^* \otimes_{\mathsf{K}} \mathfrak{b}$ is simple. It is easy to check that upon defining

$$x(a \otimes b) = axb$$

for all $a, x \in \mathfrak{a}$ and $b \in \mathfrak{b}$, the algebra $\mathfrak{a}$ becomes a right $\mathfrak{a}^* \otimes_{\mathsf{K}} \mathfrak{b}$-module. Let $\varphi \in \mathrm{Hom}_{\mathfrak{a}^* \otimes \mathfrak{b}}(\mathfrak{a}, \mathfrak{a})$, and let $c = 1\varphi \in \mathfrak{a}$. Then

$$a\varphi = (1(a \otimes 1))\varphi = (1\varphi)(a \otimes 1) = ac.$$

That is, φ is uniquely determined by $c = 1\varphi$. For all $b \in \mathfrak{b}$, we have

$$bc = (1b)\varphi = (1(1 \otimes b))\varphi = (1\varphi)(1 \otimes b) = cb.$$

This means $c \in \mathbf{C}_{\mathfrak{a}}(\mathfrak{b})$. Conversely, it is clear that for every $c \in \mathbf{C}_{\mathfrak{a}}(\mathfrak{b})$, the map φ defined by $x\varphi = xc$ is an element of $\mathrm{Hom}_{\mathfrak{a}^* \otimes \mathfrak{b}}(\mathfrak{a}, \mathfrak{a})$. Thus, $\mathrm{Hom}_{\mathfrak{a}^* \otimes \mathfrak{b}}(\mathfrak{a}, \mathfrak{a}) \cong \mathbf{C}_{\mathfrak{a}}(\mathfrak{b})$.

Now we apply Lemma 14.4 with $\mathfrak{a}^* \otimes_{\mathsf{K}} \mathfrak{b}$ in place of $\mathfrak{a}$, $\mathfrak{a}$ in place of V, and $\mathbf{C}_{\mathfrak{a}}(\mathfrak{b})$ in place of $\mathfrak{b}$. Letting $\mathfrak{a}^* \otimes_{\mathsf{K}} \mathfrak{b} = (\mathfrak{c})_n$ for some division ring $\mathfrak{c}$, it follows that $\mathbf{C}_{\mathfrak{a}}(\mathfrak{b}) \cong (\mathfrak{c}^*)_k$ (for some k) and

$$(\dim_{\mathsf{K}} \mathfrak{a})^2 = \dim_{\mathsf{K}}(\mathfrak{a}^* \otimes_{\mathsf{K}} \mathfrak{b}) \dim_{\mathsf{K}} \mathbf{C}_{\mathfrak{a}}(\mathfrak{b}) = \dim_{\mathsf{K}} \mathfrak{a} \dim_{\mathsf{K}} \mathfrak{b} \dim_{\mathsf{K}} \mathbf{C}_{\mathfrak{a}}(\mathfrak{b}).$$

With this, the claims in a)–c) are proved.

Finally, we have $\mathfrak{b} \subseteq \mathbf{C}_{\mathfrak{a}}(\mathbf{C}_{\mathfrak{a}}(\mathfrak{b}))$, and it follows from c) that

$$\dim_{\mathsf{K}} \mathbf{C}_{\mathfrak{a}}(\mathbf{C}_{\mathfrak{a}}(\mathfrak{b})) = \frac{\dim_{\mathsf{K}} \mathfrak{a}}{\dim_{\mathsf{K}} \mathbf{C}_{\mathfrak{a}}(\mathfrak{b})} = \dim_{\mathsf{K}} \mathfrak{b}.$$

Therefore, $\mathfrak{b} = \mathbf{C}_{\mathfrak{a}}(\mathbf{C}_{\mathfrak{a}}(\mathfrak{b}))$. □

Now we prove an important theorem regarding the splitting fields of central simple algebras.

Theorem 14.6. *Let $\mathfrak{c}$ be a division ring, and let $\mathfrak{a} = (\mathfrak{c})_n$ be a central simple algebra over K. By Theorem 14.3, $\dim_{\mathsf{K}} \mathfrak{c} = m^2$ for some m.*

a) *If M is a maximal subfield of the division ring $\mathfrak{c}$, then $\dim_{\mathsf{K}} \mathsf{M} = m$, and M is a splitting field for $\mathfrak{c}$ and for $\mathfrak{a}$.*

b) *If L is a splitting field for $\mathfrak{c}$ and $[\mathsf{L} : \mathsf{K}] < \infty$, then m is a divisor of $[\mathsf{L} : \mathsf{K}]$.*

c) *Let L be a splitting field for $\mathfrak{a}$, and let V be the unique (up to isomorphism) irreducible $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}$-module. Then $\mathsf{V} \oplus \overset{k}{\cdots} \oplus \mathsf{V}$ is isomorphic to a base field extension $\mathsf{L} \otimes_{\mathsf{K}} \mathsf{U}$ of an $\mathfrak{a}$-module U if and only if m is a divisor of k.*

Proof. a) Let M be a maximal subfield of the division ring $\mathfrak{c}$. It is clear that $\mathsf{M} \subseteq \mathbf{C}_{\mathfrak{c}}(\mathsf{M})$. If $\mathsf{M} \subset \mathbf{C}_{\mathfrak{c}}(\mathsf{M})$ and $c \in \mathbf{C}_{\mathfrak{c}}(\mathsf{M}) - \mathsf{M}$, then $\mathsf{M}(c)$ is a field properly containing M, contradicting our choice of M. So $\mathsf{M} = \mathbf{C}_{\mathfrak{c}}(\mathsf{M})$. It follows by Theorem 14.5 c) that

$$m^2 = \dim_{\mathsf{K}} \mathfrak{c} = (\dim_{\mathsf{K}} \mathsf{M})^2.$$

By Theorem 14.5 b), $\mathbf{C}_{\mathfrak{c}}(\mathsf{M}) = \mathsf{M}$ and $\mathfrak{c}^* \otimes_{\mathsf{K}} \mathsf{M}$ are full matrix rings over anti-isomorphic division rings $\mathfrak{d}$ and $\mathfrak{d}^*$. Since M is a field, it contains no zero divisors, so $\mathfrak{d} = \mathsf{M}$ and $\mathfrak{c}^* \otimes_{\mathsf{K}} \mathsf{M} = (\mathsf{M})_k$. From this we conclude that $\mathfrak{c} \otimes_{\mathsf{K}} \mathsf{M} \cong (\mathsf{M})_k^* \cong (\mathsf{M})_k$ (by virtue of the map $x \mapsto x^*$), and so M is a splitting field for $\mathfrak{c}$. It is easy to verify that

$$\mathfrak{a} \otimes_{\mathsf{K}} \mathsf{M} = (\mathfrak{c})_n \otimes_{\mathsf{K}} \mathsf{M} \cong (\mathfrak{c} \otimes_{\mathsf{K}} \mathsf{M})_n \cong (\mathsf{M})_{nk},$$

and so M is also a splitting field for $\mathfrak{a}$.

b) Let L be a splitting field for $\mathfrak{a}$. A comparison of dimensions shows that $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a} \cong (\mathsf{L})_{mn}$. The simple L-algebra $(\mathsf{L})_{mn}$ has exactly one irreducible module V, and $\dim_{\mathsf{L}} \mathsf{V} = mn$. Therefore, $\dim_{\mathsf{K}} \mathsf{V} = mn[\mathsf{L} : \mathsf{K}]$. By virtue of the isomorphic embedding $a \mapsto 1 \otimes a$ of $\mathfrak{a}$ in $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}$, the K-vector space V inherits the structure of an $\mathfrak{a}$-module. The central simple algebra $\mathfrak{a}$ has exactly one irreducible representation, so V is a direct sum of t isomorphic, irreducible $\mathfrak{a}$-modules (for some t). Since $\mathfrak{a} \cong (\mathfrak{c})_n$, the irreducible $\mathfrak{a}$-module has dimension $n \dim_{\mathsf{K}} \mathfrak{c} = nm^2$. It then follows that

$$mn[\mathsf{L} : \mathsf{K}] = tnm^2.$$

So $[\mathsf{L} : \mathsf{K}] = tm$ is divisible by m.

c) Again let $\mathfrak{a} = (\mathfrak{c})_n$ and $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a} = (\mathsf{L})_{mn}$. Let W be the irreducible $\mathfrak{a}$-module, and let V be the irreducible $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}$-module. Since $\dim_{\mathsf{K}} \mathfrak{c} = m^2$, we have $\dim_{\mathsf{K}} \mathsf{W} = nm^2$ and $\dim_{\mathsf{L}} \mathsf{V} = mn$. Therefore, $\mathsf{L} \otimes_{\mathsf{K}} \mathsf{W}$ is the direct sum of m copies of the $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}$-module V.

Now suppose that U is an $\mathfrak{a}$-module such that $\mathsf{L} \otimes_{\mathsf{K}} \mathsf{U} \cong \mathsf{V} \oplus \underset{k}{\cdots} \oplus \mathsf{V}$. Then $\mathsf{U} \cong \mathsf{W} \oplus \underset{s}{\cdots} \oplus \mathsf{W}$ for some s. Comparing dimensions shows that

$$snm^2 = \dim_{\mathsf{K}} \mathsf{U} = \dim_{\mathsf{L}} (\mathsf{L} \otimes_{\mathsf{K}} \mathsf{U}) = kmn.$$

So m is a divisor of $k = sm$. □

Lemma 14.7. *Assume* $\operatorname{char} \mathsf{K} \nmid |G|$, *and let* L *be a splitting field for* G *with* $\mathsf{K} \subseteq \mathsf{L}$. *Let* $\mathbf{D}$ *be an irreducible representation of* $\mathsf{L}[G]$, *and let* χ *be the character of* $\mathbf{D}$.

a) *Let* $\mathsf{K}[G] = \bigoplus_{i=1}^{k} \mathfrak{a}_i$ *be the direct decomposition of* $\mathsf{K}[G]$ *into two-sided, directly indecomposable ideals* $\mathfrak{a}_i$. *Then there exists exactly one* $\mathfrak{a}_j$ *such that* $\mathbf{D}(\mathfrak{a}_j) \neq 0$ *and* $\mathbf{D}(\mathfrak{a}_i) = 0$ *for* $i \neq j$. (*Here, we have embedded* $\mathsf{K}[G]$ *in* $\mathsf{L}[G]$ *in the natural way. We occasionally specify* $\mathfrak{a}_j$ *by saying that* $\chi(\mathfrak{a}_j) \neq 0$.)

b) *If* $\chi(\mathfrak{a}_j) \neq 0$, *then*

$$\mathbf{Z}(\mathfrak{a}_j) \cong \mathsf{K}(\chi(g) \mid g \in G).$$

For the sake of brevity, we write $\mathsf{K}(\chi) = \mathsf{K}(\chi(g) \mid g \in G)$.

Proof. a) Let $\mathsf{L}[G] = \bigoplus_{j=1}^{h} (\mathsf{L})_{n_j}$ be the decomposition of $\mathsf{L}[G]$ into simple algebras. It follows from

$$\mathsf{L}[G] \cong \mathsf{L} \otimes_{\mathsf{K}} \mathsf{K}[G] \cong \bigoplus_{i=1}^{k} \mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}_i$$

and the uniqueness statement in Theorem 3.8 that $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}_i$ is a direct sum of some of the $(\mathsf{L})_{n_j}$. If the representation $\mathbf{D}$ corresponds to $(\mathsf{L})_{n_s}$ and $(\mathsf{L})_{n_s}$ is a direct summand

of $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}_j$, then $\mathbf{D}(\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}_i) = 0$ for $i \neq j$ and $\mathbf{D}(\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}_j) \neq 0$. The claim follows upon embedding $\mathfrak{a}_j$ isomorphically in $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}_j$ via the map $a \mapsto 1 \otimes a$.

b) The map $\mathbf{D}$ is an algebra isomorphism from the simple K-algebra $\mathfrak{a} = \mathfrak{a}_j$ to the K-algebra $(\mathsf{L})_n$ with $n = \deg \mathbf{D}$. We define $\mathfrak{b} = \mathbf{D}(\mathfrak{a})$ and calculate $\mathbf{Z}(\mathfrak{b})$.

Let $g \in G$, let k_g be the sum of the conjugates of g, and let h_g be the number of conjugates of g in G. Since $k_g \in \mathbf{Z}(\mathsf{L}[G])$, it follows by Schur's Lemma that

$$\mathbf{D}(k_g) = \omega(g) 1_n \in \mathbf{Z}(\mathfrak{b})$$

for some $\omega(g) \in \mathsf{L}$. Therefore,

$$n\omega(g) = \operatorname{tr} \mathbf{D}(k_g) = h_g \chi(g).$$

Since $\operatorname{char} \mathsf{K} \nmid |G|$ and $h_g = [G : \mathbf{C}_G(g)]$, we have $\operatorname{char} \mathsf{K} \nmid h_g$. So we obtain

$$\chi(g) 1_n = \frac{n}{h_g} \mathbf{D}(k_g) \in \mathbf{Z}(\mathfrak{b}),$$

and thus $\mathsf{K}(\chi) 1_n \subseteq \mathbf{Z}(\mathfrak{b})$.

Conversely, assume that $a = \sum_{g \in G} a_g g \in \mathfrak{a}$ (with $a_g \in \mathsf{K}$) and $\mathbf{D}(a) \in \mathbf{Z}(\mathfrak{b})$. Since $\mathbf{D}(g) \in \mathfrak{b}$ for all $g \in G$ and

$$\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{b} = \left\{ \sum_{g \in G} c_g \mathbf{D}(g) \;\middle|\; c_g \in \mathsf{L} \right\} = (\mathsf{L})_n$$

(see Theorem 5.14), we have

$$\mathbf{Z}(\mathfrak{b}) \subseteq \mathbf{Z}((\mathsf{L})_n) = \{c 1_n \mid c \in \mathsf{L}\}.$$

It therefore follows that

$$\mathbf{D}(a) = \sum_{g \in G} a_g \mathbf{D}(g) = c 1_n$$

for some $c \in \mathsf{L}$. Consequently,

$$nc = \operatorname{tr} \mathbf{D}(a) = \sum_{g \in G} a_g \chi(g) \in \mathsf{K}(\chi).$$

By Theorem 12.11, $n = \deg \mathbf{D}$ is a divisor of $|G|$, so n is coprime to $\operatorname{char} \mathsf{K}$. We now see that $c \in \mathsf{K}(\chi)$ and $\mathbf{Z}(\mathfrak{b}) \subseteq \mathsf{K}(\chi) 1_n$. □

Definition 14.8. Let K be a field with $\operatorname{char} \mathsf{K} \nmid |G|$, and let L be an extension field of K that is a splitting field for G. Let χ be the character of the irreducible representation $\mathbf{D}$ of $\mathsf{L}[G]$, and let $\mathfrak{a}$ be the simple direct summand of $\mathsf{K}[G]$ with $\chi(\mathfrak{a}) \neq 0$, which is uniquely determined by Lemma 14.7 a). By Wedderburn's Theorem (see Theorem 4.4), we have $\mathfrak{a} = (\mathfrak{c})_n$ for some unique (up to isomorphism)

division ring $\mathfrak{c}$. It follows from Lemma 14.7 b) that

$$\mathbf{Z}(\mathfrak{a}) \cong \mathbf{Z}(\mathfrak{c}) = \mathsf{K}(\chi).$$

By Theorem 14.3, $\dim_{\mathsf{K}(\chi)} \mathfrak{c}$ is a square, say s^2. We call s the Schur index of χ over K and write $s = s_{\mathsf{K}}(\chi)$.

Theorem 14.9. *Let χ be an absolutely irreducible character of G over a splitting field for $\mathsf{K}[G]$.*

a) *Let $\mathfrak{a}$ be the simple direct summand of $\mathsf{K}[G]$ specified by $\chi(\mathfrak{a}) \neq 0$, and let F be an extension field of $\mathsf{K}(\chi)$. The representation $\mathbf{D}$ with character χ can be realized over F if and only if F is a splitting field for the central simple $\mathsf{K}(\chi)$-algebra $\mathfrak{a}$.*

b) *The representation $\mathbf{D}$ is realizable over $\mathsf{K}(\chi)$ if and only if $s_{\mathsf{K}}(\chi) = 1$.*

Proof. a) Let F be an extension field of $\mathsf{K}(\chi)$, and let L be an extension of F that is a splitting field for $\mathsf{K}[G]$. By Lemma 14.7 b) and Lemma 14.2, the algebra $\mathsf{F} \otimes_{\mathsf{K}(\chi)} \mathfrak{a}$ is central simple over F. Hence, $\mathsf{F} \otimes_{\mathsf{K}(\chi)} \mathfrak{a} \cong (\mathfrak{c})_n$ for some division ring $\mathfrak{c}$ with $\mathbf{Z}(\mathfrak{c}) = \mathsf{F}$. Let $\dim_{\mathsf{F}} \mathfrak{c} = m^2$. The representation $\mathbf{D}$ yields an irreducible representation of the central simple L-algebra $\mathsf{L} \otimes_{\mathsf{K}(\chi)} \mathfrak{a}$. By Theorem 14.6 c), the representation $k\mathbf{D}$ is realizable over F if and only if $m \mid k$. So $\mathbf{D}$ is realizable over F if and only if $m \mid 1$; that is, when F is a splitting field for $\mathfrak{c}$ and for $\mathfrak{a}$.

b) is a consequence of a). □

Theorem 14.10. *If K is a finite field, then we always have $s_{\mathsf{K}}(\chi) = 1$. That is, an absolutely irreducible representation with character χ can always be realized over $\mathsf{K}(\chi)$ when K is a finite field.*

Proof. Let $\mathfrak{a}$ be the simple direct summand of $\mathsf{K}[G]$ specified by $\chi(\mathfrak{a}) \neq 0$, and let $\mathfrak{a} = (\mathfrak{c})_n$ for some division ring $\mathfrak{c}$. By a theorem of Wedderburn, the finite division ring $\mathfrak{c}$ is commutative (see Exercise I.22, pg. 32). Using Lemma 14.7 b), it then follows that $\mathfrak{c} = \mathbf{Z}(\mathfrak{c}) = \mathsf{K}(\chi)$ and thus $s_{\mathsf{K}}(\chi) = 1$. By Theorem 14.9 b), the representation with character χ can be realized over $\mathsf{K}(\chi)$. □

Now we can answer the questions posed at the beginning of this section:

Theorem 14.11. *Let $\mathbf{D}$ be an absolutely irreducible representation of G with character χ.*

a) *If $\mathbf{D}$ is realizable over the extension field L of $\mathsf{K}(\chi)$ and $[\mathsf{L} : \mathsf{K}(\chi)] < \infty$, then the Schur index $s_{\mathsf{K}}(\chi)$ is a divisor of $[\mathsf{L} : \mathsf{K}(\chi)]$.*

b) *There exists an extension field L of $\mathsf{K}(\chi)$ with $[\mathsf{L} : \mathsf{K}(\chi)] = s_{\mathsf{K}}(\chi)$ such that $\mathbf{D}$ is realizable over L.*

c) *The representation $k\mathbf{D}$ is realizable over $\mathsf{K}(\chi)$ if and only if $s_{\mathsf{K}}(\chi)$ is a divisor of k.*

d) *$s_{\mathsf{K}}(\chi)$ is a divisor of $\deg \mathbf{D}$.*

Proof. a) Let $\mathfrak{a}$ be the simple direct summand of $\mathsf{K}[G]$ specified by $\chi(\mathfrak{a}) \neq 0$. Since $\mathbf{D}$ is realizable over L, the field L is a splitting field for the central simple

$\mathsf{K}(\chi)$-algebra $\mathfrak{a}$ by Theorem 14.9 a). Let $\mathfrak{a} = (\mathfrak{c})_n$ for some division ring $\mathfrak{c}$. By Definition 14.8, $\dim_{\mathsf{K}(\chi)} \mathfrak{c} = s_{\mathsf{K}}(\chi)^2$. The claim that $s_{\mathsf{K}}(\chi)$ divides $[\mathsf{L} : \mathsf{K}(\chi)]$ now follows from Theorem 14.6 b).

b) By Theorem 14.6 a), every maximal subfield L of $\mathfrak{c}$ is a splitting field for $\mathfrak{a}$ with $[\mathsf{L} : \mathsf{K}(\chi)] = s_{\mathsf{K}}(\chi)$. Then $\mathbf{D}$ can be realized over L by Theorem 14.9 a).

c) The claim follows immediately from Theorem 14.6 c).

d) Since $\mathfrak{a} = (\mathfrak{c})_n$, we have

$$\dim_{\mathsf{K}(\chi)} \mathfrak{a} = n^2 \dim_{\mathsf{K}(\chi)} \mathfrak{c} = n^2 s_{\mathsf{K}}(\chi)^2.$$

On the other hand, $\mathsf{L} \otimes_{\mathsf{K}(\chi)} \mathfrak{a} \cong (\mathsf{L})_k$ with $k = \deg \mathbf{D}$. It follows that $n^2 s_{\mathsf{K}}(\chi)^2 = k^2$ and hence $s_{\mathsf{K}}(\chi) \mid k$. □

Theorem 14.12. *Let V be an irreducible $\mathsf{K}[G]$-module, let L be a splitting field for $\mathsf{K}[G]$, and let $\mathbf{D}$ be an absolutely irreducible constituent of the representation $\mathbf{F}$ of $\mathsf{L}[G]$ on V_{L}. Then $\mathbf{F}$ is completely reducible, the absolutely irreducible constituents of $\mathbf{F}$ are all the representations $\mathbf{D}^{\alpha}$ algebraically conjugate to $\mathbf{D}$ over K, and every equivalence class appears with the same multiplicity s. If χ is the character of $\mathbf{D}$, then $s = s_{\mathsf{K}}(\chi)$ is the Schur index of χ over K. If φ is the character of $\mathbf{F}$, then*

$$\varphi = s_{\mathsf{K}}(\chi) \sum_{\alpha \in A} \chi^{\alpha},$$

where α runs through the Galois group A of $\mathsf{K}(\chi)$ over K.

Proof. Let $\mathfrak{a} = \mathsf{K}[G]e_0$ with $e_0^2 = e_0$ be the simple direct summand of $\mathsf{K}[G]$ faithfully represented on V, and let $\mathfrak{b}$ be a simple direct summand of $\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a}$. Then $\mathfrak{b} = \mathsf{L}[G]e_{\chi}$, where

$$e_{\chi} = \frac{\chi(1)}{|G|} \sum_{g \in G} \chi(g^{-1})g$$

is the central idempotent of $\mathsf{L}[G]$ corresponding to the absolutely irreducible character χ of G (see Theorem 5.15 a)).

Since $\mathsf{K}(\chi)$ is a subfield of the field generated over K by the $|G|$-th roots of unity, the field $\mathsf{K}(\chi)$ is Galois over K. Let A be the Galois group of $\mathsf{K}(\chi)$ over K. Then $e = \sum_{\alpha \in A} e_{\chi^{\alpha}}$ is a central idempotent in $\mathsf{K}[G]$. Since $e_{\chi^{\alpha}} e_0 = e_{\chi^{\alpha}}$, it follows that $ee_0 = e$. Hence,

$$\mathsf{K}[G]e \subseteq \mathsf{K}[G]e_0 = \mathfrak{a}.$$

Therefore, $\mathfrak{a} = \mathsf{K}[G]e$ since $\mathfrak{a}$ is simple. This implies

$$\mathsf{L} \otimes_{\mathsf{K}} \mathfrak{a} = \mathsf{L}[G]e = \bigoplus_{\alpha \in A} \mathsf{L}[G]e_{\chi^{\alpha}}. \qquad (*)$$

If m is the degree of the representation $\mathbf{D}$ affording χ, then the irreducible $\mathsf{L}[G]$-module affording χ^α appears with multiplicity m on the right-hand side of $(*)$.

Since $\mathbf{D}(e) = \mathbf{D}(e_\chi) \neq 0$, we have $\mathbf{D}(\mathfrak{a}) \neq 0$. Then $\mathfrak{a} = (\mathfrak{c})_n$, where $\mathfrak{c}$ is a division ring with $\mathbf{Z}(\mathfrak{c}) = \mathsf{K}(\chi)$ by Lemma 14.7 b). By Definition 14.8, $\dim_{\mathsf{K}(\chi)} \mathfrak{c} = s_\mathsf{K}(\chi)^2$. Comparing dimensions in $(*)$ yields

$$n^2 s_\mathsf{K}(\chi)^2 [\mathsf{K}(\chi) : \mathsf{K}] = \dim_\mathsf{K} \mathfrak{a} = \dim_\mathsf{L} \mathsf{L} \otimes_\mathsf{K} \mathfrak{a} = |A| m^2 = [\mathsf{K}(\chi) : \mathsf{K}] m^2,$$

and so $n s_\mathsf{K}(\chi) = m$. The $\mathsf{K}[G]$-module $\mathfrak{a}$ is a direct sum of n irreducible $\mathsf{K}[G]$-modules isomorphic to V. Therefore, $\mathsf{L} \otimes_\mathsf{K} \mathfrak{a}$ is a direct sum of n $\mathsf{L}[G]$-modules isomorphic to V_L. Then by the Jordan–Hölder theorem, an irreducible $\mathsf{L}[G]$-module corresponding to the algebraically conjugate character χ^α (with $\alpha \in A$) appears with multiplicity $\frac{m}{n} = s_\mathsf{K}(\chi)$ in V_L. □

Theorem 14.13. *Assume* $\operatorname{char} \mathsf{K} = 0$*, and let* χ *be an absolutely irreducible character of* G *over the extension field* L *of* K*.*

a) *There exists exactly one irreducible character* ψ *of* G *over* K *with* $(\psi, \chi)_G \neq 0$*; namely,* $\psi = s_\mathsf{K}(\chi) \sum_{\alpha \in A} \chi^\alpha$*, where* A *is the Galois group of* $\mathsf{K}(\chi)$ *over* K*.*

b) *If* τ *is a character of* G *over* K*, then* $(\tau, \chi)_G$ *is divisible by the Schur index* $s_\mathsf{K}(\chi)$*.*

Proof. a) Let ψ be an irreducible character of G over K with $(\psi, \chi)_G \neq 0$. By Theorem 14.12, we have

$$\psi = s_\mathsf{K}(\lambda) \sum_{\alpha \in A} \lambda^\alpha$$

for some absolutely irreducible character λ of G. (In this expression, α now runs over the Galois group A of $\mathsf{K}(\lambda)$ over K.) Since $(\psi, \chi)_G \neq 0$, χ is one of the λ^α. This yields the claim in a).

b) Let $\tau = \sum_i \psi_i$ be the decomposition of τ into irreducible characters ψ_i over K. If $(\psi_i, \chi)_G \neq 0$, then it follows by a) that

$$\psi_i = s_\mathsf{K}(\chi) \sum_{\alpha \in A} \chi^\alpha.$$

So $(\tau, \chi)_G = \sum_i (\psi_i, \chi)_G$ is divisible by $s_\mathsf{K}(\chi)$. □

Remark 14.14. Let G be a finite group, and let L be a splitting field for G that is Galois over K. Let χ be an absolutely irreducible character of G taking values in K, and let $\mathbf{D}$ be a representation of $\mathsf{L}[G]$ on the L-vector space V affording the character χ. We again denote the Galois group of L over K by A. For every $\alpha \in A$, the representations $\mathbf{D}$ and $\mathbf{D}^\alpha$ have the same character, so they are equivalent. Therefore, there exist nonsingular linear transformations $a(\alpha)$ of V such that

$$\mathbf{D}(g)^\alpha = a(\alpha) \mathbf{D}(g) a(\alpha)^{-1}$$

for all $g \in G$ and all $\alpha \in A$. For $\alpha_1, \alpha_2 \in A$, it follows that

$$a(\alpha_1\alpha_2)\mathbf{D}(g)a(\alpha_1\alpha_2)^{-1} = \mathbf{D}(g)^{\alpha_1\alpha_2} = (a(\alpha_1)\mathbf{D}(g)a(\alpha_1)^{-1})^{\alpha_2}$$
$$= a(\alpha_1)^{\alpha_2}a(\alpha_2)\mathbf{D}(g)a(\alpha_2)^{-1}a(\alpha_1)^{-\alpha_2}.$$

Therefore, Schur's Lemma implies that

$$a(\alpha_1\alpha_2) = c(\alpha_1,\alpha_2)a(\alpha_1)^{\alpha_2}a(\alpha_2) \quad \text{with} \quad c(\alpha_1,\alpha_2) \in \mathsf{L}^\times.$$

The obvious equality $a((\alpha_1\alpha_2)\alpha_3) = a(\alpha_1(\alpha_2\alpha_3))$ yields the 2-cocyle relation

$$c(\alpha_1,\alpha_2)^{\alpha_3}c(\alpha_1\alpha_2,\alpha_3) = c(\alpha_1,\alpha_2\alpha_3)c(\alpha_2,\alpha_3).$$

Hence, $c \in \mathbf{Z}^2(A,\mathsf{L}^\times)$ by Chapter I, §17 (pg. 127), where $\mathsf{L}^\times$ is to be understood as an A-module in the natural way. Changing $a(\alpha)$ by a factor $b(\alpha) \in \mathsf{L}^\times$ changes c by the coboundary $b\delta$ of b. So the representation $\mathbf{D}$ corresponds to an element $c\mathbf{B}^2(A,\mathsf{L}^\times)$ of the cohomology group $\mathbf{H}^2(A,\mathsf{L}^\times)$.

Suppose that $c = b\delta$ is a coboundary. Then we may modify the $a(\alpha)$ by scalar factors so that $a(\alpha_1\alpha_2) = a(\alpha_1)^{\alpha_2}a(\alpha_2)$. Then a is a 1-cocyle in $\mathbf{Z}^1(A,\mathrm{GL}(\mathsf{V}))$; the nonabelian, multiplicative group $\mathrm{GL}(\mathsf{V})$ is to be understood here as an A-group in the natural way. An extension of the well-known Hilbert's Theorem 90 says $\mathbf{H}^1(A,\mathrm{GL}(\mathsf{V})) = 1$ (see Serre [380], pg. 159). Therefore, there exists some $d \in \mathrm{GL}(\mathsf{V})$ such that $a(\alpha) = d^\alpha d^{-1}$ for all $\alpha \in A$. It follows that

$$(d^{-1}\mathbf{D}(g)d)^\alpha = d^{-\alpha}a(\alpha)\mathbf{D}(g)a(\alpha)^{-1}d^\alpha = d^{-1}\mathbf{D}(g)d.$$

So $d^{-1}\mathbf{D}(g)d$ is a representation of G over K.

Conversely, it is easy to see that if $\mathbf{D}$ is equivalent to a representation of G over K, then the 2-cocycle c constructed above is a coboundary.

Exercises

25) Let χ be an absolutely irreducible character of G afforded by the representation $\mathbf{D}$ with $s_{\mathsf{K}}(\chi) = \deg \mathbf{D}$. Show:
   a) $G/\ker \mathbf{D}$ is isomorphic to a subgroup of the multiplicative group of a division ring.
   b) The Sylow p-subgroups of $G/\ker \mathbf{D}$ are cyclic for $p > 2$ and cyclic or generalized quaternion for $p = 2$.

26) Let $\mathbf{D}$ be a representation of $\mathsf{L}[G]$ with character χ, and let K be a subfield of L with $[\mathsf{L} : \mathsf{K}] < \infty$. Construct a representation $\mathbf{F}$ of $\mathsf{K}[G]$ with $\operatorname{tr}\mathbf{F}(g) = [\mathsf{L} : \mathsf{K}(\chi)]\chi(g)$. Now give new proofs for some of the theorems in this section.

15 The class number

Very little is known about the number of conjugacy classes of a finite group. Since this so-called class number is equal to the number of irreducible representations over the complex numbers, statements regarding the class number can be proved using characters. We begin with an easy theorem.

Theorem 15.1 (Burnside). *Let h be the class number of G. If the order $|G|$ of G is odd, then $|G| \equiv h \bmod 16$.*

Proof. Let $n_1, \ldots, n_h$ be the degrees of the inequivalent, irreducible representations of G over the complex numbers. By Theorem 5.1, we have $|G| = \sum_{i=1}^{h} n_i^2$. By Theorem 5.12, the n_i are divisors of $|G|$, so are also odd. Let $n_i = 1 + 2r_i$. By Theorem 13.8, $\chi_i \neq \overline{\chi}_i$ for $i > 1$ as χ_1 is the only real irreducible character of G. So we may enumerate the irreducible characters of G with an appropriate labeling as

$$\chi_1, \chi_2, \ldots, \chi_s, \overline{\chi}_2, \ldots, \overline{\chi}_s$$

with $s = \frac{h-1}{2}$. It now follows that

$$|G| = \sum_{i=1}^{h} n_i^2 = 1 + 2\sum_{i=2}^{s}(1 + 2r_i)^2 = h + 8\sum_{i=2}^{s} r_i(r_i + 1) \equiv h \mod 16. \qquad \square$$

For a strengthening of Theorem 15.1, see Hirsch [243].

The following result regarding the class number of p-groups lies somewhat deeper.

Theorem 15.2 (P. Hall). *Let G be a p-group of order p^{e+2n}, where $n \geq 0$ and $e = 0$ or $e = 1$. Then the class number h of G has the form*

$$h = p^e + (p^2 - 1)(n + k(p - 1))$$

for some nonnegative integer k. In particular, we always have

$$h \geq p^e + (p^2 - 1)n.$$

Proof. a) Let $a_m = a_m(G)$ be the number of conjugacy classes K of G with $|K| = p^m$. Then

$$|G| = \sum_m a_m p^m \quad \text{and} \quad h = \sum_m a_m.$$

Let $b_m = b_m(G)$ be the number of inequivalent, irreducible representations of G over the complex numbers of degree p^m. Since the degrees of the irreducible representations of G over the complex numbers are powers of p by Theorem 5.12, it follows by Theorem 5.1 that

$$|G| = \sum_m b_m p^{2m} \quad \text{and} \quad h = \sum_m b_m.$$

We have

$$h - p^e = p^e(p^{2n} - 1) + h - |G|,$$

and so

$$\begin{aligned}\frac{h - p^e}{p^2 - 1} &= p^e \frac{p^{2n} - 1}{p^2 - 1} + \sum_m b_m \frac{1 - p^{2m}}{p^2 - 1} \\ &= p^e(1 + p^2 + \cdots + p^{2n-2}) - \sum_m b_m(1 + p^2 + \cdots + p^{2m-2}) \\ &\equiv n - \sum_m m b_m \mod (p - 1).\end{aligned}$$

b) We prove by induction on $|G|$ that $\sum_m m b_m \equiv 0 \bmod (p-1)$:

Let $N \lhd G$ with $|N| = p$. We define $G^* = G/N$ and $b_m^* = b_m(G/N)$. Then $b_m - b_m^*$ is the number of inequivalent, irreducible representations of G of degree p^m that are nontrivial on $N = \langle z \rangle$. Let $\mathbf{D}$ be such a representation. Since $z \in \mathbf{Z}(G)$ (see Theorem III.7.2 b), pg. 318), Schur's Lemma 4.3 implies that $\mathbf{D}(z) = \epsilon 1$ for some primitive p-th root of unity ϵ. Let χ be the character of $\mathbf{D}$, and let L be a Galois splitting field for G. Let A be the Galois group of L over the field $\mathbb{Q}$ of rational numbers, and let

$$A_\chi = \{\alpha \in A \mid \chi^\alpha = \chi\}$$

and

$$A_\epsilon = \{\alpha \in A \mid \epsilon^\alpha = \epsilon\}.$$

Since $\chi(z) = p^m \epsilon$, we certainly have $A_\epsilon \geq A_\chi$. The number of characters that are conjugate to χ by an element of A is $[A : A_\chi]$, and this is a multiple of $|A/A_\epsilon|$. It is well known that A/A_ϵ is isomorphic to the Galois group of $\mathbb{Q}(\epsilon)$ over $\mathbb{Q}$ and therefore

$$|A/A_\epsilon| = [\mathbb{Q}(\epsilon) : \mathbb{Q}] = p - 1.$$

So $[A : A_\chi] \equiv 0 \bmod (p-1)$. Hence, the $b_m - b_m^*$ characters under consideration are partitioned into A-conjugacy classes, and the size of each class is a multiple of $p - 1$. This yields the congruence

$$b_m - b_m^* \equiv 0 \mod (p - 1)$$

for all m, and therefore

$$\sum_m m b_m \equiv \sum_m m b_m^* \mod (p-1).$$

We have $\sum_m m b_m^* \equiv 0 \bmod (p-1)$ by our induction hypothesis, so it follows that

$$\sum_m m b_m \equiv 0 \mod (p-1).$$

Therefore

$$\frac{h-p^e}{p^2-1} = n + k(p-1)$$

for some integer k.

c) We prove by induction on $|G|$ that $k \geq 0$:

Let $x \in G$, and let $x^* = xN \in G/N$, where N is the same as in b). We define

$$p^{\mu(x)} = [G : \mathbf{C}_G(x)] = |\{x^g \mid g \in G\}|.$$

Let $\mathbf{C}_G^*(x) = \{g \in G \mid x^g N = xN\}$, so $\mathbf{C}_G(x) \leq \mathbf{C}_G^*(x)$ and $\mathbf{C}_{G^*}(x^*) = \mathbf{C}_G^*(x)/N$.

If $g \in \mathbf{C}_G^*(x)$, then $x^g = x\,t(g)$ for some $t(g) \in N \leq \mathbf{Z}(G)$. If $g' \in \mathbf{C}_G^*(x)$ as well, then it follows that

$$x\,t(gg') = x^{gg'} = x^{g'} t(g) = x\,t(g)\,t(g').$$

So the map t sending g to $t(g)$ is a homomorphism from $\mathbf{C}_G^*(x)$ to N with kernel $\mathbf{C}_G(x)$. Therefore, $[\mathbf{C}_G^*(x) : \mathbf{C}_G(x)]$ is equal to 1 or p.

Case 1: Assume that $\mathbf{C}_G^*(x) = \mathbf{C}_G(x)$. Then we have

$$p^{\mu(x)} = [G : \mathbf{C}_G(x)] = [G/N : \mathbf{C}_G(x)/N] = [G^* : \mathbf{C}_{G^*}(x^*)] = p^{\mu(x^*)},$$

so $\mu(x) = \mu(x^*)$.

Let y' and y'' be distinct elements of N. Then we have that xy' and xy'' are not conjugate in G: Otherwise, if

$$x^g y' = (xy')^g = xy''$$

for some $g \in G$, then

$$x^g = xy''y'^{-1}$$

and therefore $g \in \mathbf{C}_G^*(x) = \mathbf{C}_G(x)$, so $y' = y''$.

Hence, the preimage in G of the conjugacy class

$$\{x^{*g^*} \mid g^* \in G^*\}$$

of size $p^{\mu(x^*)}$ in G^* splits into p distinct classes

$$\{x^g z^i \mid g \in G\} \quad (i = 0, 1, \ldots, p-1)$$

of identical size $p^{\mu(x)}$.

Case 2: Now assume that $|\mathbf{C}^*_G(x)| = p|\mathbf{C}_G(x)|$, so $\mu(x^*) = \mu(x) - 1$. We consider the set

$$L = \{x^g y \mid g \in G, y \in N\}$$

consisting of the preimages of $K^* = \{x^{*g^*} \mid g^* \in G^*\}$. We have

$$K = \{x^g \mid g \in G\} \subseteq L$$

and

$$|K| = p^{\mu(x)} \leq |L| = p|K^*| = p\, p^{\mu(x^*)} = p^{\mu(x)} = |K|.$$

So $K = L$. That is, passing to preimages in G takes K^* to the conjugacy class K of G.

Let c_m be the number of conjugacy classes of size p^m in G^* that split into p classes of size p^m in G; and let d_m be the number of classes of size p^m in G^* whose preimage is a single class of size p^{m+1} in G. Then

$$a^*_m = a_m(G/N) = c_m + d_m \quad (m = 0, 1, 2, \ldots)$$

and

$$\begin{aligned} a_m &= pc_m + d_{m-1} = pc_m + a^*_{m-1} - c_{m-1} \\ &\vdots \\ a_1 &= pc_1 + a^*_0 - c_0 \\ a_0 &= pc_0. \end{aligned}$$

It follows that

$$\sum_m a_m = \sum_m a^*_m + (p-1)\sum_m c_m.$$

Hence, we have for the class numbers h^* and h of G^*, resp. G, that

$$\begin{aligned} h - h^* &= \sum_m (a_m - a^*_m) = (p-1)\sum_m c_m \\ &= p^e + (p^2-1)(n + k(p-1)) - p^{e^*} - (p^2-1)(n^* + k^*(p-1)). \end{aligned}$$

Therefore,

$$p^{1-e} + (p^2-1)(k - k^*) = \sum_m c_m \geq 0.$$

Since $p^{1-e} < p^2 - 1$, we now see that $k \geq k^*$. Since $k^* \geq 0$ by the induction hypothesis, it therefore follows that $k \geq 0$. □

16 Induced representations

Definition 16.1.

a) Let H be a subgroup of G, and let $G = \bigcup_{i=1}^{n} Hr_i$ be the decomposition of G into right cosets of H. Let V be a right $\mathsf{K}[H]$-module. The group ring $\mathsf{K}[G]$ is a right $\mathsf{K}[G]$-module, and since $\mathsf{K}[G] = \bigoplus_{i=1}^{n} \mathsf{K}[H]r_i$, it is also a free left $\mathsf{K}[H]$-module. We construct the tensor product

$$\mathsf{V}^G = \mathsf{V} \otimes_{\mathsf{K}[H]} \mathsf{K}[G]$$

over $\mathsf{K}[H]$. By Theorem 9.8, V^G is a right $\mathsf{K}[G]$-module. We call V^G the $\mathsf{K}[G]$-module induced by V. By Lemma 9.5, the module V^G decomposes into a direct sum of K-vector spaces

$$\mathsf{V}^G = \bigoplus_{i=1}^{n} \mathsf{V}_i \quad \text{with} \quad \mathsf{V}_i = \{v \otimes r_i \mid v \in \mathsf{V}\}.$$

Using Theorem 9.8, the action of $g \in G$ on V^G is described as follows: If $r_i g = h_i r_{i'}$ for some $h_i \in H$, then

$$(v \otimes r_i)g = v \otimes r_i g = vh_i \otimes r_{i'}.$$

If $\mathbf{D}$ is the representation of H on V, then we denote the representation of G on V^G by $\mathbf{D}^G$ and we say that $\mathbf{D}^G$ is the representation of G induced by $\mathbf{D}$.

b) Let τ be a class function on the subgroup H of G. We define

$$\tau_0(g) = \begin{cases} \tau(g) & \text{for } g \in H \\ 0 & \text{for } g \notin H \end{cases}$$

and define the function τ^G on G by

$$\tau^G(g) = \sum_{i=1}^{n} \tau_0(r_i g r_i^{-1}).$$

If $r_i' = h_i r_i$ for some $h_i \in H$, then we have

$$\sum_{i=1}^{n} \tau_0(r_i' g r_i'^{-1}) = \sum_{i=1}^{n} \tau_0(h_i r_i g r_i^{-1} h_i^{-1}) = \sum_{i=1}^{n} \tau_0(r_i g r_i^{-1}).$$

So τ^G is independent of the choice of coset representatives. Therefore, for all $x \in G$ we have

$$\tau^G(g^x) = \sum_{i=1}^{n} \tau_0(r_i x^{-1} g x r_i^{-1}) = \tau^G(g).$$

So τ^G is a class function on G. The map $\tau \mapsto \tau^G$ is clearly a K-linear map from the space of class functions on H to the space of class functions on G.

Since $\tau_0(h r_i g r_i^{-1} h^{-1}) = \tau_0(r_i g r_i^{-1})$ for all $h \in H$, when $\operatorname{char} \mathsf{K} \nmid |H|$ we have

$$\tau^G(g) = \frac{1}{|H|} \sum_{x \in G} \tau_0(g^x).$$

If we start with the trivial representation $\mathbf{D}_1$ of H, then it is easy to see that $\mathbf{D}_1^G$ is the permutation representation of G on the cosets of H, written using matrices.

Lemma 16.2.

a) *Let $H \leq G$, and let V_1, V_2 be $\mathsf{K}[H]$-modules. Then*

$$\mathsf{V}_1^G \oplus \mathsf{V}_2^G \cong (\mathsf{V}_1 \oplus \mathsf{V}_2)^G.$$

b) *Induction of modules is transitive in the following sense: If H and K are subgroups of G with $K \leq H \leq G$ and V is a $\mathsf{K}[K]$-module, then $\mathsf{V}^G \cong (\mathsf{V}^H)^G$.*

c) *Let $H_i \leq G_i$ $(i = 1, 2)$, and let $\mathsf{V}_i = \mathsf{W}_i^{G_i}$ be a $\mathsf{K}[G_i]$-module that is induced by a $\mathsf{K}[H_i]$-module W_i. If we give $\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2$ the structure of a $\mathsf{K}[G_1 \times G_2]$-module according to Theorem 10.3, and if we give $\mathsf{W}_1 \otimes_{\mathsf{K}} \mathsf{W}_2$ the structure of a $\mathsf{K}[H_1 \times H_2]$-module, then the $\mathsf{K}[G_1 \times G_2]$-modules $\mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2$ and $(\mathsf{W}_1 \otimes_{\mathsf{K}} \mathsf{W}_2)^{G_1 \times G_2}$ are isomorphic.*

Proof. a) The claim follows immediately from the isomorphism

$$(\mathsf{V}_1 \oplus \mathsf{V}_2) \otimes_{\mathsf{K}[H]} \mathsf{K}[G] \cong (\mathsf{V}_1 \otimes_{\mathsf{K}[H]} \mathsf{K}[G]) \oplus (\mathsf{V}_2 \otimes_{\mathsf{K}[H]} \mathsf{K}[G])$$

in Lemma 9.4, which is clearly an isomorphism of right $\mathsf{K}[G]$-modules.

b) This follows from the associativity

$$\begin{aligned}(\mathsf{V}^H)^G &= (\mathsf{V} \otimes_{\mathsf{K}[K]} \mathsf{K}[H]) \otimes_{\mathsf{K}[H]} \mathsf{K}[G] \cong \mathsf{V} \otimes_{\mathsf{K}[K]} (\mathsf{K}[H] \otimes_{\mathsf{K}[H]} \mathsf{K}[G]) \\ &= \mathsf{V} \otimes_{\mathsf{K}[K]} \mathsf{K}[G] = \mathsf{V}^G\end{aligned}$$

in Theorem 9.10, which is again an isomorphism of right $\mathsf{K}[G]$-modules.

c) Let $G_i = \bigcup_j H_i g_{ji}$ $(i = 1, 2)$ be the decomposition of G_i into cosets of H_i. Then

$$\mathsf{V}_i = \mathsf{W}_i^{G_i} = \bigoplus_j \mathsf{W}_i \otimes g_{ji}$$

by Definition 16.1. Clearly, the decomposition of $G_1 \times G_2$ into cosets of $H_1 \times H_2$ is

$$G_1 \times G_2 = \bigcup_{j,k} (H_1 \times H_2)(g_{j1}, g_{k2}).$$

Therefore,

$$(\mathsf{W}_1 \otimes_{\mathsf{K}} \mathsf{W}_2)^{G_1 \times G_2} = \bigoplus_{j,k} (\mathsf{W}_1 \otimes_{\mathsf{K}} \mathsf{W}_2) \otimes (g_{j1}, g_{k2}).$$

It is easy to compute that the K-linear map ϵ defined by

$$((w_1 \otimes w_2) \otimes (g_{j1}, g_{k2}))\epsilon = (w_1 \otimes g_{j1}) \otimes (w_2 \otimes g_{k2})$$

for $w_i \in \mathsf{W}_i$ $(i = 1, 2)$ is a $\mathsf{K}[G_1 \times G_2]$-isomorphism from $(\mathsf{W}_1 \otimes_{\mathsf{K}} \mathsf{W}_2)^{G_1 \times G_2}$ to $\mathsf{W}_1^{G_1} \otimes_{\mathsf{K}} \mathsf{W}_2^{G_2} = \mathsf{V}_1 \otimes_{\mathsf{K}} \mathsf{V}_2$. □

Theorem 16.3. *If χ is the character of a representation $\mathbf{D}$ of H, then χ^G is the character of the induced representation $\mathbf{D}^G$.*

Proof. Let $\mathbf{D}$ be a representation of H on the $\mathsf{K}[H]$-module V, and let χ be the character of $\mathbf{D}$. By Definition 16.1, we have $\mathsf{V}^G = \bigoplus_{i=1}^{n} \mathsf{V}_i$ with $\mathsf{V}_i = \{v \otimes r_i \mid v \in \mathsf{V}\}$ and

$$(v \otimes r_i)g = vh_i \otimes r_{i'} \quad \text{for} \quad r_i g = h_i r_{i'}.$$

It follows that

$$\operatorname{tr} \mathbf{D}^G(g) = \sum_{i=i'} \chi(h_i) = \sum_{r_i g r_i^{-1} \in H} \chi(r_i g r_i^{-1}) = \sum_{i=1}^{n} \chi_0(r_i g r_i^{-1}) = \chi^G(g). \qquad \square$$

Definition 16.4. Let H be a subgroup of G. If $\mathbf{D}$ is a representation of G on the $\mathsf{K}[G]$-module V, then denote by V_H the $\mathsf{K}[H]$-module obtained by restricting the domain of operators, and denote by $\mathbf{D}_H$ the representation of H on V_H. If τ is a class function on G, then we denote the restriction of τ to H by τ_H.

In what follows, we consider only representations over an algebraically closed field K with $\operatorname{char} \mathsf{K} \nmid |G|$.

It is important to be able to decompose induced characters into their irreducible constituents. This problem is partially solved by the following fundamental theorem.

Theorem 16.5 (Frobenius reciprocity).

a) *If τ is a class function on G and ψ is a class function on the subgroup H of G, then*

$$(\psi^G, \tau)_G = (\psi, \tau_H)_H.$$

b) *Let χ_i $(i = 1, \ldots, h)$ be the irreducible characters of G over K. If*

$$(\chi_j)_H = \sum_{k=1}^{h'} d_{jk}\psi_k$$

is the decomposition of $(\chi_j)_H$ *into irreducible characters* ψ_k *of* H *(where* h' *is the class number of* H*), then*

$$\psi_i^G = \sum_{j=1}^{h} d_{ji}\chi_j$$

is the decomposition of ψ_i^G *into irreducible characters of* G.

Proof. a) By Definition 16.1, we have

$$(\psi^G, \tau)_G = \frac{1}{|G|}\sum_{g\in G}\psi^G(g)\tau(g^{-1}) = \frac{1}{|G|\,|H|}\sum_{g\in G}\sum_{t^{-1}gt\in H}\psi(t^{-1}gt)\tau(t^{-1}g^{-1}t).$$

For each fixed $x \in H$, the equation $t^{-1}gt = x$ has exactly $|G|$ pairs of solutions $\{t, g\} = \{t, txt^{-1}\}$. It follows that

$$(\psi^G, \tau)_G = \frac{1}{|H|}\sum_{x\in H}\psi(x)\tau(x^{-1}) = (\psi, \tau_H)_H.$$

b) The claim in b) follows immediately from

$$(\psi_i^G, \chi_j)_G = (\psi_i, (\chi_j)_H)_H. \qquad \square$$

If we specialize Theorem 16.5 b) to the case that $H = 1$ and $\psi = \psi_1$, the trivial character of H, then ψ^G is the character of the regular representation of G. For the absolutely irreducible characters χ_i of G, it then follows that

$$(\psi_1^G, \chi_i)_G = (\psi_1, \chi_i(1)\psi_1)_1 = \chi_i(1) = n_i.$$

This is exactly the content of Theorem 5.3.

Frobenius reciprocity has a module-theoretic generalization which holds over any field K:

Theorem 16.6 (Nakayama). *Let* K *be any field. Let* H *be a subgroup of* G*, let* V *be a* $\mathsf{K}[G]$*-module and let* W *be a* $\mathsf{K}[H]$*-module. Then we have the following* K*-isomorphisms:*

a) $\mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{W}^G) \cong \mathrm{Hom}_{\mathsf{K}[H]}(\mathsf{V}_H, \mathsf{W})$.

b) $\mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{W}^G, \mathsf{V}) \cong \mathrm{Hom}_{\mathsf{K}[H]}(\mathsf{W}, \mathsf{V}_H)$.

Proof. Let $G = \bigcup_{i=1}^{n} Hr_i$ with $r_1 = 1$ be the decomposition of G into cosets of H.

a) Let $\alpha \in \mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{W}^G)$. If $v \in \mathsf{V}$, then $v\alpha$ can be written in the form $v\alpha = \sum_{i=1}^{n} w_i \otimes r_i$ for some uniquely determined $w_i \in \mathsf{W}$. We define the maps

$\alpha_i \in \mathrm{Hom}_{\mathsf{K}}(\mathsf{V}, \mathsf{W})$ by

$$v\alpha = \sum_{i=1}^{n} v\alpha_i \otimes r_i$$

for all $v \in \mathsf{V}$. For all $h \in H$, we have

$$\begin{aligned}(vh)\alpha &= \sum_{i=1}^{n} (vh)\alpha_i \otimes r_i = (vh)\alpha_1 \otimes 1 + \cdots \\ &= (v\alpha)h = \sum_{i=1}^{n} v\alpha_i \otimes r_i h = (v\alpha_1)h \otimes 1 + \cdots .\end{aligned}$$

A comparison of the components with $i = 1$ shows that $(vh)\alpha_1 = (v\alpha_1)h$, so $\alpha_1 \in \mathrm{Hom}_{\mathsf{K}[H]}(\mathsf{V}_H, \mathsf{W})$. Furthermore,

$$\begin{aligned}(v\alpha)r_j^{-1} &= \sum_{i=1}^{n} v\alpha_i \otimes r_i r_j^{-1} = v\alpha_j \otimes 1 + \cdots \\ &= (vr_j^{-1})\alpha = \sum_{i=1}^{n} (vr_j^{-1})\alpha_i \otimes r_i = (vr_j^{-1})\alpha_1 \otimes 1 + \cdots .\end{aligned}$$

It follows that $v\alpha_j = (vr_j^{-1})\alpha_1$. So α is uniquely determined by α_1.

Conversely, assume that $\alpha_1 \in \mathrm{Hom}_{\mathsf{K}[H]}(\mathsf{V}_H, \mathsf{W})$. Then we define the linear map $\alpha \in \mathrm{Hom}_{\mathsf{K}}(\mathsf{V}, \mathsf{W}^G)$ by

$$v\alpha = \sum_{i=1}^{n} (vr_i^{-1})\alpha_1 \otimes r_i.$$

For $g \in G$ and $r_i g = h_i r_{i'} \in Hr_{i'}$, we have

$$\begin{aligned}(v\alpha)g &= \sum_{i=1}^{n} (vr_i^{-1})\alpha_1 \otimes h_i r_{i'} = \sum_{i=1}^{n} (vr_i^{-1}h_i)\alpha_1 \otimes r_{i'} \\ &= \sum_{i'=1}^{n} (vgr_{i'}^{-1})\alpha_1 \otimes r_{i'} = (vg)\alpha.\end{aligned}$$

So $\alpha \in \mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{W}^G)$. The map $\alpha \mapsto \alpha_1$ is thus a bijective and clearly K-linear map from $\mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{W}^G)$ to $\mathrm{Hom}_{\mathsf{K}[H]}(\mathsf{V}_H, \mathsf{W})$.

b) Let $\beta \in \mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{W}^G, \mathsf{V})$. We define $\beta_1 \in \mathrm{Hom}_{\mathsf{K}}(\mathsf{W}, \mathsf{V})$ by $(w \otimes 1)\beta = w\beta_1$ for all $w \in \mathsf{W}$. It is clear that β_1 lies in $\mathrm{Hom}_{\mathsf{K}[H]}(\mathsf{W}, \mathsf{V}_H)$. We have

$$(w \otimes r_i)\beta = (w \otimes 1)r_i\beta = (w \otimes 1)\beta r_i = w\beta_1 r_i.$$

So β is uniquely determined by β_1. Conversely, it is easy to verify that each $\beta_1 \in \mathrm{Hom}_{\mathsf{K}[H]}(\mathsf{W}, \mathsf{V}_H)$ gives rise to a $\beta \in \mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{W}^G, \mathsf{V})$ by defining

$$\left(\sum_{i=1}^{n} w_i \otimes r_i\right)\beta = \sum_{i=1}^{n} (w_i\beta_1)r_i. \qquad \square$$

Remark 16.7. Let K be an algebraically closed field of arbitrary characteristic, and let $H \leq G$. Let V be an irreducible $\mathsf{K}[G]$-module, and let W be an irreducible $\mathsf{K}[H]$-module. It is easy to show that $\dim_{\mathsf{K}} \mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{W}^G)$ is the multiplicity of V in the largest completely reducible $\mathsf{K}[G]$-submodule (the so-called socle) of W^G. Furthermore, $\dim_{\mathsf{K}} \mathrm{Hom}_{\mathsf{K}[H]}(\mathsf{V}_H, \mathsf{W})$ is the multiplicity of W in the largest completely reducible $\mathsf{K}[H]$-quotient module of V_H. By Theorem 16.6 a), these multiplicities are equal. (If all modules are completely reducible, then the Frobenius reciprocity of Theorem 16.5 follows again from this.) Theorem 16.6 b) yields an analogous statement: The multiplicity of V in the largest completely reducible $\mathsf{K}[G]$-quotient module of W^G is equal to the multiplicity of W in the largest completely reducible $\mathsf{K}[H]$-submodule of V_H.

Lemma 16.8. *If τ is a class function on G and ψ is a class function on the subgroup H of G, then $\psi^G\tau = (\psi\tau_H)^G$. Hence, the class functions on G that are induced by class functions on H constitute an ideal in the ring of class functions on G.*

Proof. We have

$$\begin{aligned}\psi^G(g)\tau(g) &= \frac{1}{|H|}\sum_{t\in G}\psi_0(t^{-1}gt)\tau(g) = \frac{1}{|H|}\sum_{t\in G}\psi_0(t^{-1}gt)\tau(t^{-1}gt)\\ &= (\psi\tau_H)^G(g).\end{aligned}$$

Theorem 16.9 (Mackey [326]). *Let H and K be subgroups of G, and let V be a $\mathsf{K}[H]$-module. Let $G = \bigcup_{i=1}^{k} Ha_iK$ be the decomposition of G into double cosets of H and K. Then:*

a) $(\mathsf{V}^G)_K \cong \bigoplus_{i=1}^{k}((\mathsf{V}\otimes a_i)_{H^{a_i}\cap K})^K$.

b) *If χ is the character of the representation of H on V, then*

$$(\chi^G)_K = \sum_{i=1}^{k}((\chi^{a_i^{-1}})_{H^{a_i}\cap K})^K.$$

Here, $\chi^{a_i^{-1}}$ is defined by

$$\chi^{a_i^{-1}}(g) = \chi(g^{a_i^{-1}})$$

for $g^{a_i^{-1}} \in H$, so in particular for $g \in H^{a_i} \cap K$.
(*No assumption on the field K is necessary here.*)

Proof. a) Let $G = \bigcup_{i=1}^{n} Hr_i$ be the decomposition of G into cosets of H. Then $\mathsf{V}^G = \bigoplus_{i=1}^{n} \mathsf{V}_i$, where $\mathsf{V}_i = \{v \otimes r_i \mid v \in \mathsf{V}\}$. For brevity, we write this as $\mathsf{V}_i = \mathsf{V} \otimes r_i$. If $G = \bigcup_{i=1}^{k} Ha_iK$, then the set

$$\{Hr_j \mid r_j \in Ha_iK\}$$

constitutes an orbit of K as a permutation group on $\{Hr_j \mid j = 1, \dots, n\}$ by right multiplication. Hence, $\mathsf{V}^G = \bigoplus_{i=1}^{k} \mathsf{W}_i$ with

$$\mathsf{W}_i = \bigoplus_{r_j \in Ha_iK} \mathsf{V} \otimes r_j$$

is a decomposition of V^G as a $\mathsf{K}[K]$-module. Now we fix $a = a_i$ and let $\mathsf{W} = \mathsf{W}_i$.

The relation $Hay_1 = Hay_2$ with $y_i \in K$ is equivalent to

$$y_1 y_2^{-1} \in H^a \cap K = \widetilde{K}.$$

Using the coset decomposition $K = \bigcup_{i=1}^{m} \widetilde{K} y_i$, we obtain the disjoint decomposition $HaK = \bigcup_{i=1}^{m} Hay_i$. Now we have

$$\mathsf{W} = \bigoplus_{i=1}^{m} \mathsf{V} \otimes ay_i = \bigoplus_{i=1}^{m} (\mathsf{V} \otimes a) y_i.$$

Here, $\mathsf{V} \otimes a$ is an H^a-module, so in particular it is a $\widetilde{K}$-module, where again $\widetilde{K} = H^a \cap K$. By Definition 16.1, we have the isomorphism $\mathsf{W} \cong ((\mathsf{V} \otimes a)_{\widetilde{K}})^K$ as $\mathsf{K}[K]$-modules.

b) For $\widetilde{x} = a_i^{-1} h a_i \in H^{a_i} \cap K$, we have

$$(v \otimes a_i)\widetilde{x} = v \otimes ha_i = vh \otimes a_i = v(a_i \widetilde{x} a_i^{-1}) \otimes a_i.$$

Therefore, the character of $H^{a_i} \cap K$ on $\mathsf{V} \otimes a_i$ is given by $(\chi^{a_i^{-1}})_{H^{a_i} \cap K}$. □

We note the following special case, which can also be proved easily without the use of Theorem 16.9.

Theorem 16.10. *Let $N \lhd G$, let $G = \bigcup_{i=1}^{k} Ng_i$ be the decomposition of G into cosets of N, and let V be a $\mathsf{K}[N]$-module. Then*

$$(\mathsf{V}^G)_N \cong \bigoplus_{i=1}^{k} \mathsf{V}^{g_i}.$$

Here, $\mathsf{V}^{g_i} = \mathsf{V} \otimes g_i^{-1}$ *is the* $\mathsf{K}[N]$*-module with*

$$(v \otimes g_i^{-1})n = v \otimes n^{g_i} g_i^{-1} = v n^{g_i} \otimes g_i^{-1}$$

for all $n \in N$*. If* χ *is the character of* N *on* V*, then* χ^{g_i} *defined by* $\chi^{g_i}(n) = \chi(n^{g_i})$ *for all* $n \in N$ *is the character of* N *on* V^{g_i}*. We have*

$$(\chi^G)_N = \sum_{i=1}^{k} \chi^{g_i}.$$

(*No assumption on the field* K *is necessary here.*)

Proof. We set $H = K = N$ in Theorem 16.9. Since

$$G = \bigcup_{i=1}^{k} g_i^{-1} N = \bigcup_{i=1}^{k} N g_i^{-1}$$

and

$$N g_i^{-1} N = N(g_i^{-1} N g_i) g_i^{-1} = N g_i^{-1},$$

we have that $G = \bigcup_{i=1}^{k} N g_i^{-1}$ is the decomposition of G into double cosets of the pair N, N. So by Theorem 16.9 a),

$$(\mathsf{V}^G)_N = \bigoplus_{i=1}^{k} \mathsf{V}^{g_i}. \qquad \square$$

Definition 16.11. Let $\mathbf{D}$ be a representation of G on the K-vector space V. It is well known that for every K-endomorphism a of V, there is an associated K-endomorphism a^* of the space V^* dual to V. We define the representation $\mathbf{D}^*$ of G on V^* that is dual to $\mathbf{D}$ by $\mathbf{D}^*(g) = \mathbf{D}(g^{-1})^*$. Since

$$\mathbf{D}^*(g_1 g_2) = \mathbf{D}(g_2^{-1} g_1^{-1})^* = \mathbf{D}(g_1^{-1})^* \mathbf{D}(g_2^{-1})^* = \mathbf{D}^*(g_1)\mathbf{D}^*(g_2),$$

the map $\mathbf{D}^*$ is a representation of G. If we denote the character of $\mathbf{D}$ by χ and that of $\mathbf{D}^*$ by χ^*, then clearly $\chi^*(g) = \chi(g^{-1})$. (In particular, if K is the field of complex numbers, then $\chi^*(g) = \overline{\chi(g)}$.)

We give a criterion for the irreducibility of an induced representation:

Theorem 16.12 (Mackey [326]). *Let* $H \leq G$*, and let* $\mathbf{D}$ *be an irreducible representation of* H *over an algebraically closed field* K *of characteristic* 0*. For* $g \in G$

and $h \in H$, we define the representation $\mathbf{D}^{g^{-1}}$ of H^g by $\mathbf{D}^{g^{-1}}(h^g) = \mathbf{D}(h)$. Let $G = \bigcup_{i=1}^{m} Ha_iH$ be the decomposition of G into double cosets of H, where we choose $a_1 \in H$. Then $\mathbf{D}^G$ is an irreducible representation of G if and only if for all $i = 2, \ldots, m$ the representations $\mathbf{D}_{H\cap H^{a_i}}$ and $(\mathbf{D}^{a_i^{-1}})_{H\cap H^{a_i}}$ have no irreducible constituents in common.

Proof. Let χ be the character of $\mathbf{D}$. By Theorem 5.10 b), the irreducibility of $\mathbf{D}^G$ is equivalent to

$$\frac{1}{|G|}\sum_{g\in G}\chi^G(g)\chi^G(g^{-1}) = (\chi^G, \chi^G)_G = 1 \tag{1}$$

since $\mathrm{char}\,\mathsf{K} = 0$.

Denoting by χ^* the character of the representation $\mathbf{D}^*$ of H that is dual to the representation $\mathbf{D}$, we have

$$\chi^{*G}(g) = \frac{1}{|H|}\sum_{g^x\in H}\chi^*(g^x) = \frac{1}{|H|}\sum_{g^x\in H}\chi(g^{-x}) = \chi^G(g^{-1}).$$

So the formula in (1) can be written as

$$(\chi^G\chi^{*G}, 1)_G = 1. \tag{2}$$

We denote by Δ the monomorphism from G to $G \times G$ defined by $g\Delta = (g, g)$. We consider the subgroups $H \times H$ and $G\Delta$ of $G \times G$. Let V be a $\mathsf{K}[H]$-module for the representation $\mathbf{D}$, and let V^* be the dual vector space, which is a $\mathsf{K}[H]$-module for $\mathbf{D}^*$. Then $\mathsf{V}^G \otimes \mathsf{V}^{*G}$ is a $\mathsf{K}[G \times G]$-module by Definition 10.4. The character φ of $G\Delta$ on $(\mathsf{V}^G \otimes \mathsf{V}^{*G})_{G\Delta}$ is clearly given by

$$\varphi((g, g)) = \chi^G(g)\chi^{*G}(g).$$

Therefore, formula (2) implies

$$(\varphi, 1)_{G\Delta} = 1. \tag{3}$$

By Lemma 16.2 c), we have $\mathsf{V}^G \otimes \mathsf{V}^{*G} \cong (\mathsf{V} \otimes \mathsf{V}^*)^{G\times G}$, where $\mathsf{V} \otimes \mathsf{V}^*$ is to be understood as a $\mathsf{K}[H \times H]$-module. If $G = \bigcup_{i=1}^{m} Ha_iH$ is the decomposition of G into double cosets of H, then it is easy to see that

$$G \times G = \bigcup_{i=1}^{m}(H \times H)(a_i, 1)(G\Delta)$$

is the decomposition of $G \times G$ into double cosets of $H \times H$ and $G\Delta$. Now, keeping in mind that

$$(\mathsf{V} \otimes \mathsf{V}^*) \otimes (a_i, 1) = (\mathsf{V} \otimes a_i) \otimes \mathsf{V}^*$$

and

$$(H \times H)^{(a_i,1)} \cap G\Delta = \{(h, h) \mid h \in H \cap H^{a_i}\} = (H \cap H^{a_i})\Delta,$$

we obtain with the help of Theorem 16.9 the isomorphism

$$((\mathsf{V} \otimes \mathsf{V}^*)^{G\times G})_{G\Delta} \cong \bigoplus_{i=1}^{m} \left(((\mathsf{V} \otimes a_i) \otimes \mathsf{V}^*)_{(H\cap H^{a_i})\Delta}\right)^{G\Delta}.$$

The character of $G\Delta$ on $\left(((\mathsf{V} \otimes a_i) \otimes \mathsf{V}^*)_{(H\cap H^{a_i})\Delta}\right)^{G\Delta}$ is $((\chi^{a_i^{-1}}\chi^*)_{(H\cap H^{a_i})\Delta})^{G\Delta}$. Therefore, it follows by Frobenius reciprocity that

$$\begin{aligned}(\varphi, 1)_{G\Delta} &= \sum_{i=1}^{m} \left(((\chi^{a_i^{-1}}\chi^*)_{(H\cap H^{a_i})\Delta})^{G\Delta}, 1\right)_{G\Delta} \\ &= \sum_{i=1}^{m} \left((\chi^{a_i^{-1}}\chi^*)_{H\cap H^{a_i}}, 1\right)_{H\cap H^{a_i}} \\ &= \sum_{i=1}^{m} \left((\chi^{a_i^{-1}})_{H\cap H^{a_i}}, \chi_{H\cap H^{a_i}}\right)_{H\cap H^{a_i}}.\end{aligned}$$

Since χ is irreducible on H and $a_1 \in H$, we have

$$((\chi^{a_1^{-1}})_H, \chi_H)_H = (\chi_H, \chi_H)_H = 1.$$

Thus, $(\chi^G, \chi^G)_G = 1$ if and only if

$$\left((\chi^{a_i^{-1}})_{H\cap H^{a_i}}, \chi_{H\cap H^{a_i}}\right)_{H\cap H^{a_i}} = 0$$

for $i = 2, \ldots, m$. □

As important applications, we determine the representations of Frobenius groups and extraspecial p-groups:

Theorem 16.13. *Let $G = FH$ be a Frobenius group with Frobenius kernel F and Frobenius complement H (see Definition 8.1). Let $h(H)$ be the class number of H, and let $h(F)$ be the class number of F. Let K be an algebraically closed field with* $\operatorname{char}\mathsf{K} \nmid |G|$. *Then G has exactly the following irreducible characters over K:*

(1) *$h(H)$ irreducible characters μ_i that contain F in the kernel; if η_i are the irreducible characters of H over K ($i = 1, \ldots, h(H)$), then the μ_i are given by*

$$\mu_i(xy) = \eta_i(y) \quad \textit{for} \quad x \in F \quad \textit{and} \quad y \in H.$$

(2) $\dfrac{h(F) - 1}{|H|}$ *irreducible characters that arise in the following way:*

If ψ is a nontrivial irreducible character of F, then ψ^G is an irreducible character of G. Furthermore, we have $\psi_i^G = \psi_j^G$ if and only if there exists some $y \in H$ such that $\psi_i = \psi_j^y$. We have $(\psi_i^G)_F = \sum_{y \in H} \psi_i^y$.

If V is an irreducible $\mathsf{K}[G]$-module on which F acts nontrivially, then V_H is a direct sum of regular $\mathsf{K}[H]$-modules; therefore, V_H contains the module corresponding to the trivial representation of H.

Proof. a) First assume that $\operatorname{char} \mathsf{K} = 0$. Let ψ_i $(i = 2, \ldots, h(F))$ be the nontrivial irreducible characters of F over K. Every $1 \neq y \in H$ induces a permutation on the conjugacy classes of F with respect to the conjugation map $K \mapsto K^y$. Under this permutation action, only the class 1 is fixed by Theorem 8.9 c). Therefore, the permutation induced by $1 \neq y \in H$ on the irreducible characters of F defined by $\psi \mapsto \psi^y$, where $\psi^y(x) = \psi(x^y)$, also has only one fixed point by Theorem 13.5 a); namely, the trivial character ψ_1 of F. So if $\psi_i \neq \psi_1$ and $y \neq y'$, then $\psi_i^y \neq \psi_i^{y'}$.

Since $G = FH$ and $F \cap H = 1$, we have by Theorem 16.5 and Theorem 16.10 that

$$(\psi_i^G, \psi_j^G)_G = (\psi_i, (\psi_j^G)_F)_F = \sum_{y \in H} (\psi_i, \psi_j^y)_F$$
$$= \begin{cases} 1 & \text{if } \psi_i = \psi_j^y \text{ for some } y \in H, \\ 0 & \text{otherwise.} \end{cases}$$

If $\psi_i = \psi_j^y$, then it clearly follows that $\psi_i^G = \psi_j^G$. So the given $\dfrac{h(F) - 1}{|H|}$ characters ψ_i^G are pairwise distinct and irreducible by Theorem 5.10. Furthermore, since $F \leq \ker \mu_i$, we have by Theorem 16.5 that

$$(\psi_i^G, \mu_j)_G = (\psi_i, (\mu_j)_F)_F = \mu_j(1)(\psi_i, \psi_1)_F = 0$$

for $i \neq 1$.

We have proved that the characters of G given in (1) and (2) are irreducible and pairwise distinct. The sum of the squared degrees of these characters is

$$\sum_i \mu_i(1)^2 + \frac{1}{|H|} \sum_{j \neq 1} \psi_j^G(1)^2 = \sum_i \eta_i(1)^2 + \frac{1}{|H|} \sum_{j \neq 1} [G : F]^2 \psi_j(1)^2$$
$$= |H| + |H|(|F| - 1) = |H||F| = |G|.$$

Hence, we have found all the irreducible characters of G over K.

b) Now let K be an algebraically closed field of characteristic p with $p \nmid |G|$. By Theorem 11.7, the irreducible representations of G over K can be realized in a finite splitting field L of G. By Theorem 12.9, the irreducible representations of G over L arise by reduction modulo $\mathfrak{p}$ of the irreducible representations of G over some $\mathfrak{p}$-adic field. So we only need to show that reduction modulo $\mathfrak{p}$ commutes with the construction of induced representations. This follows immediately, however, from the definition of the induced representation.

c) Let V be an irreducible $\mathsf{K}[G]$-module on which the representation of F is not a multiple of the trivial representation. Then $\mathsf{V} = \mathsf{W}^G$ for some irreducible $\mathsf{K}[F]$-module W. Since $G = FH$ and $F \cap H = 1$, it follows from the Mackey formula in Theorem 16.9 that

$$\mathsf{V}_H = (\mathsf{W}^G)_H = (\mathsf{W}_{F\cap H})^H = (\mathsf{W}_1)^H.$$

So V_H is a multiple of the regular $\mathsf{K}[H]$-module. □

Theorem 16.14. *Let G be an extraspecial p-group of order p^{1+2m} in the sense of Definition III.13.1 (pg. 369), and let K be an algebraically closed field with* $\operatorname{char}\mathsf{K} \neq p$. *Then G has exactly the following irreducible representations over K (up to equivalence):*

(1) *p^{2m} representations of degree* 1.

(2) *$p-1$ faithful representations of degree p^m, which arise in the following way: Let A be a maximal abelian normal subgroup of G, and let $\lambda \neq 1$ be a linear character of $\mathbf{Z}(G)$. If μ is an extension of λ to A, then μ^G is an irreducible character of G. In this way, we obtain exactly one irreducible character χ of G for each linear character $\lambda \neq 1$ of $\mathbf{Z}(G)$, and $\chi_{\mathbf{Z}(G)} = \chi(1)\lambda$.*

Proof. Since $|G/G'| = p^{2m}$, the group G has exactly p^{2m} characters of degree 1 by Theorem 6.5.

Let λ be a nontrivial linear character of $\mathbf{Z}(G)$. By Theorem 6.3 a), λ can be extended to a character μ of the maximal abelian normal subgroup A of G. By Theorem III.13.7 (pg. 373), we have $|G/A| = p^m$. It follows by Theorem 16.3 that

$$\mu^G(g) = \frac{1}{|A|}\sum_{x\in G}\mu_0(g^x).$$

If $g \notin A$, then $g^x \notin A$, so $\mu^G(g) = 0$. If $g \in A$, then since $[g,x] \in G' = \mathbf{Z}(G)$, we have

$$\mu^G(g) = \frac{1}{|A|}\sum_{x\in G}\mu(g[g,x]) = \frac{\mu(g)}{|A|}\sum_{x\in G}\lambda([g,x]).$$

The group G has nilpotency class 2, so the map $x \mapsto [g,x]$ is a homomorphism from G to $G' = \mathbf{Z}(G)$ by the formula in Lemma III.1.2 (pg. 267). If $g \in \mathbf{Z}(G)$, then since $|A| = p^{m+1}$ we have

$$\mu^G(g) = \frac{\lambda(g)}{|A|}|G| = p^m\lambda(g) = \mu^G(1)\lambda(g).$$

If $g \notin \mathbf{Z}(G)$, then since $\lambda \neq 1$ we find with the help of the orthogonality relations that

$$\mu^G(g) = \frac{\mu(g)}{|A|}p^{2m}\sum_{y\in\mathbf{Z}(G)}\lambda(y) = \mu(g)p^m(\lambda,1)_{\mathbf{Z}(G)} = 0.$$

So μ^G is completely determined by λ; in particular, μ^G is independent of A and the choice of extension μ of λ to A. For char $\mathsf{K} = 0$, the irreducibility of μ^G now follows from

$$(\mu^G, \mu^G)_G = \frac{1}{|G|} \sum_{g \in \mathbf{Z}(G)} \mu^G(g)\mu^G(g^{-1}) = \frac{1}{|G|} \sum_{g \in \mathbf{Z}(G)} p^{2m}\lambda(g)\lambda(g^{-1})$$
$$= \frac{p^{1+2m}}{|G|} = 1.$$

For char $\mathsf{K} = q$ with $q \neq p$, we pass from the representations of G over a $\mathfrak{q}$-adic field to the representations of G over K by a reduction modulo $\mathfrak{q}$ using Theorem 12.9 and obtain the desired result.

It remains to show that all the irreducible representations of G are enumerated in (1) and (2). This follows immediately by calculating the sum of the squared degrees. □

Exercises

27) Let $H \leq G$, and let r be a right ideal in $\mathsf{K}[H] \subseteq \mathsf{K}[G]$. Show that $\mathfrak{r}\mathsf{K}[G]$ and $\mathfrak{r}^G$ are isomorphic $\mathsf{K}[G]$-modules.
28) Let $H \leq G$. If V is a $\mathsf{K}[G]$-module and W is a $\mathsf{K}[H]$-module, then $\mathsf{W}^G \otimes_{\mathsf{K}} \mathsf{V} = (\mathsf{W} \otimes_{\mathsf{K}[H]} \mathsf{K}[G]) \otimes_{\mathsf{K}} \mathsf{V}$ and $(\mathsf{W} \otimes_{\mathsf{K}} \mathsf{V}_H)^G = (\mathsf{W} \otimes_{\mathsf{K}} \mathsf{V}_H) \otimes_{\mathsf{K}[H]} \mathsf{K}[G]$ are isomorphic as $\mathsf{K}[G]$-modules. (These isomorphisms underlie Lemma 16.8.)
29) Find all absolutely irreducible representations of the group of affine linear maps $x \mapsto ax + b$ (with $x, a, b \in \mathrm{GF}(p^f)$) over the field of complex numbers.

17 Restriction of irreducible representations to normal subgroups

Definition 17.1. Let N be a normal subgroup of G, and let $\mathbf{D}$ be a representation of N. For every $g \in G$, the representation $\mathbf{D}^g$ defined by $\mathbf{D}^g(n) = \mathbf{D}(n^g)$ for $n \in N$ is clearly a representation of N. We say that $\mathbf{D}^g$ is a representation of N that is conjugate to $\mathbf{D}$ with respect to G.

Theorem 17.2. *Let N be a normal subgroup of G.*

a) *If $\mathbf{D}$ is an irreducible representation of $\mathsf{K}[N]$, then $\mathbf{D}^g$ is an irreducible representation of $\mathsf{K}[N]$ for every $g \in G$.*

b) *Assume that K is algebraically closed and* $\mathrm{char}\,\mathsf{K} \nmid |G|$. *The number of G-conjugacy classes of absolutely irreducible representations of $\mathsf{K}[N]$ is exactly the number of conjugacy classes of G lying in N.*

Proof. a) The claim is clear since the sets of linear maps $\{\mathbf{D}(n) \mid n \in N\}$ and $\{\mathbf{D}^g(n) = \mathbf{D}(n^g) \mid n \in N\}$ are the same.

b) Note that G is represented as a permutation group on N with respect to the action $n \mapsto n^g$. Since

$$(n^x)^g = (n^g)^{g^{-1}xg}$$

and $g^{-1}xg \in N$ for all $x \in N$, the group G is also a permutation group on the conjugacy classes of N. Let χ_i $(i = 1, \ldots, h)$ be the absolutely irreducible characters of N over K. With respect to the action

$$\chi_i^g(n) = \chi_i(n^g),$$

the group G is represented as a permutation group on $\{\chi_i \mid i = 1, \ldots, h\}$. If $\operatorname{char} \mathsf{K} = 0$, then the number of orbits of G on $\{\chi_i \mid i = 1, \ldots h\}$ is equal to the number of orbits of G on the conjugacy classes of N by Theorem 13.5. This is exactly the claim.

If $\operatorname{char} \mathsf{K} = p$ and $p \nmid |G|$, then the irreducible characters χ_i of N over K arise as a reduction modulo $\mathfrak{p}$ of the irreducible characters τ_i of N over a $\mathfrak{p}$-adic splitting field for N by Theorem 12.9. The reduction modulo $\mathfrak{p}$ is given by $\chi_i(n) = \overline{\tau}_i(n)$, where the bar denotes the map to the residue field of the $\mathfrak{p}$-adic field. Each element $g \in G$ permutes the χ_i in the same way that it permutes the $\overline{\tau}_i$ since

$$\chi_i^g(n) = \chi_i(n^g) = \overline{\tau}_i(n^g) = \overline{\tau}_i^g(n). \qquad \square$$

The following theorem is of fundamental importance. The theorem requires neither that K be algebraically closed nor that $\operatorname{char} \mathsf{K} \nmid |G|$.

Main Theorem 17.3 (Clifford [96]). *Let V be an irreducible $\mathsf{K}[G]$-module, and let N be a normal subgroup of G.*

a) *If W is an irreducible $\mathsf{K}[N]$-submodule of V, then $\mathsf{V} = \sum_{g \in G} \mathsf{W}g$. Each $\mathsf{W}g$ is an irreducible $\mathsf{K}[N]$-module, and V is therefore a completely reducible $\mathsf{K}[N]$-module.*

b) *We define the "homogeneous components" V_i of the $\mathsf{K}[N]$-module V in the following way: Let $\mathsf{W}_1, \ldots, \mathsf{W}_t$ be representatives of the isomorphism types of the irreducible $\mathsf{K}[N]$-submodules of V. We define*

$$\mathsf{V}_i = \sum_{\substack{\mathsf{W} \subseteq \mathsf{V} \\ \mathsf{W} \cong \mathsf{W}_i}} \mathsf{W} \qquad (i = 1, \ldots, t).$$

Then V_i is a direct sum of $\mathsf{K}[N]$-submodules of V, each of which is isomorphic to W_i. We have $\mathsf{V} = \bigoplus_{i=1}^{t} \mathsf{V}_i$.

c) *Let $\mathbf{F}_i$ be the irreducible representation of N on W_i. Then N has the representation $\mathbf{F}_i^{g^{-1}}$ on $\mathsf{W}_i g$.*

d) *The homogeneous components* V_i *of the* $\mathsf{K}[N]$*-module* V *are transitively permuted under right multiplication by elements of* G.

e) *For every* j*, we have*

$$\{g \in G \mid \mathsf{V}_j g = \mathsf{V}_j\} = \{g \in G \mid \mathbf{F}_j^g \text{ is equivalent to } \mathbf{F}_j\}.$$

If we denote this group by U_j*, then* V_j *is an irreducible* $\mathsf{K}[U_j]$*-module, and* $\mathsf{V} \cong \mathsf{V}_j \otimes_{\mathsf{K}[U_j]} \mathsf{K}[G] = \mathsf{V}_j^G$.

f) *Let* $\mathbf{D}$ *be the representation of* G *on* V*. The irreducible constituents of* $\mathbf{D}_N$ *are exactly the* G*-conjugates* $\mathbf{F}^g$ *of an irreducible representation* $\mathbf{F}$ *of* N*, and they all appear with the same multiplicity* e.

g) *If* χ *is the character of* $\mathbf{D}$ *and* φ *is the character of an irreducible constituent* $\mathbf{F}$ *of* $\mathbf{D}_N$*, then*

$$\chi_N = e \sum_{i=1}^{t} \varphi^{g_i},$$

where the g_i *are coset representatives of the subgroup*

$$U = \{g \in G \mid \mathbf{F}^g \text{ is equivalent to } \mathbf{F}\}.$$

Proof. a) Let W be an irreducible $\mathsf{K}[N]$-submodule of V. By the irreducibility of V as a $\mathsf{K}[G]$-module, we clearly have $\mathsf{V} = \sum_{g \in G} \mathsf{W}g$. For $w \in \mathsf{W}$, $n \in N$ and $g \in G$, we have

$$(wg)n = (wgng^{-1})g \in \mathsf{W}g.$$

So $\mathsf{W}g$ is a $\mathsf{K}[N]$-module, and $\mathsf{W}g$ is clearly irreducible. Hence, V is a completely reducible $\mathsf{K}[N]$-module by Theorem 3.2 (pg. 493).

b) We first show that V_i is a direct sum of modules isomorphic to W_i:

Let V_i' be a submodule of V_i that is a direct sum of modules isomorphic to W_i and as large as possible. If $\mathsf{V}_i' \subset \mathsf{V}_i$, then there exists a submodule W_i' of V_i isomorphic to W_i with $\mathsf{W}_i' \not\subseteq \mathsf{V}_i'$. It follows from the irreducibility of W_i' that $\mathsf{V}_i' \cap \mathsf{W}_i' = 0$, and then $\mathsf{V}_i' \subset \mathsf{V}_i' \oplus \mathsf{W}_i' \subseteq \mathsf{V}_i$, contradicting our choice of V_i'. So $\mathsf{V}_i = \mathsf{V}_i'$.

Therefore, all composition factors of V_i are isomorphic to W_i. In particular, it follows that

$$(\mathsf{V}_1 + \cdots + \mathsf{V}_k) \cap \mathsf{V}_{k+1} = 0$$

for $1 \leq k \leq t-1$, and so $\mathsf{V} = \bigoplus_{i=1}^{t} \mathsf{V}_i$.

c) We make the K-vector space W_i into a $\mathsf{K}[N]$-module W_i' by defining

$$w \circ n = wgng^{-1} = w\mathbf{F}_i(gng^{-1}) = w\mathbf{F}_i^{g^{-1}}(n)$$

for $w \in \mathsf{W}_i'$ and $n \in N$. So N has the representation $\mathbf{F}_i^{g^{-1}}$ on W_i'. The K-linear map τ from $\mathsf{W}_i g$ to W_i' defined by $(wg)\tau = w$ is a $\mathsf{K}[N]$-isomorphism from $\mathsf{W}_i g$ to W_i' since

$$(wgn)\tau = (wgng^{-1}g)\tau = wgng^{-1} = w \circ n = (wg\tau) \circ n.$$

d) By c), N has the irreducible representation $\mathbf{F}_i^{g^{-1}}$ on the irreducible constituents of $\mathsf{V}_i g$. So $\mathsf{V}_i g$ lies in a homogeneous component V_j of V. If $\mathsf{V}_i g \subset \mathsf{V}_j$, then we would have $\mathsf{V}_i \subset \mathsf{V}_j g^{-1}$, where the $\mathsf{K}[N]$-composition factors of $\mathsf{V}_j g^{-1}$ are isomorphic; namely, they have representation $\mathbf{F}_j^g$. This contradicts the construction of V_i as a homogeneous component of V. So $\mathsf{V}_i g$ is one of the V_j, and the V_i are permuted by G. Of course, since V is irreducible as a $\mathsf{K}[G]$-module, the V_i are transitively permuted.

e) Let $g \in G$, let $n \in N$, and let $w \in \mathsf{W}_j$. If $\mathsf{V}_j g = \mathsf{V}_j$, then it follows that $\mathsf{W}_j g \subseteq \mathsf{V}_j g = \mathsf{V}_j$. Since N has the irreducible representation $\mathbf{F}_j^{g^{-1}}$ on $\mathsf{W}_j g$, $\mathbf{F}_j$ is equivalent to $\mathbf{F}_j^{g^{-1}}$, so $\mathbf{F}_j^g$ is equivalent to $\mathbf{F}_j$. Conversely, if $\mathbf{F}_j^g$ is equivalent to $\mathbf{F}_j$, then $\mathsf{W}_j g$ lies in V_j, and we obtain $\mathsf{V}_j g = \mathsf{V}_j$.

Let $G = \bigcup_{i=1}^{t} U g_i$ be the decomposition of G into cosets of $U = U_1$. Then $\mathsf{V} = \bigoplus_{i=1}^{t} \mathsf{V}_1 g_i$. The map ϵ defined by

$$\left(\sum_{i=1}^{t} v_i g_i \right) \epsilon = \sum_{i=1}^{t} v_i \otimes g_i$$

for $v_i \in \mathsf{V}_1$ is clearly a $\mathsf{K}[G]$-isomorphism from V onto $\mathsf{V}_1 \otimes_{\mathsf{K}[U]} \mathsf{K}[G]$. If V_1' were a nontrivial $\mathsf{K}[U]$-submodule of V_1, then $\mathsf{V}_1'^G$ would be a nontrivial $\mathsf{K}[G]$-submodule of $\mathsf{V}_1^G \cong \mathsf{V}$. But this is impossible since V is irreducible.

f) Suppose V_1 is the e-fold direct sum of $\mathsf{K}[N]$-modules isomorphic to W_1. By d), for every i there exists some $g \in G$ such that $\mathsf{V}_i = \mathsf{V}_1 g$. Therefore, V_i is an e-fold direct sum of $\mathsf{K}[N]$-submodules isomorphic to $\mathsf{W}_1 g$ by c).

g) is a trivial consequence of f). □

Definition 17.4. A map $\mathbf{P}$ from G to the general linear group $\mathrm{GL}_n(\mathsf{K})$ on the vector space $\mathsf{V} = \mathsf{V}(n, \mathsf{K})$ is called a projective representation of G of degree n over K if for all $g, g' \in G$ we have

$$\mathbf{P}(gg') = c(g, g')\mathbf{P}(g)\mathbf{P}(g')$$

with $0 \neq c(g, g') \in \mathsf{K}$. We may clearly consider $\mathbf{P}$ as a homomorphism from G to the projective general linear group $\mathrm{PGL}_n(\mathsf{K})$. We say that $\mathbf{P}$ is irreducible if no nontrivial subspace of V is fixed setwise by all the linear maps $\mathbf{P}(g)$ with $g \in G$.

The projective representations $\mathbf{P}_1$ and $\mathbf{P}_2$ on the vector spaces V_1 and V_2 are said to be equivalent if there exists a bijective linear map t from V_1 to V_2 and a map b

from G to $\mathsf{K}^\times$ such that

$$\mathbf{P}_1(g)\, t = b(g)\, t\, \mathbf{P}_2(g)$$

for all $g \in G$.

Theorem 17.5 (Clifford [96]). *Let* K *be an algebraically closed field of arbitrary characteristic, and let* V *be an irreducible* $\mathsf{K}[G]$*-module. Let* $N \lhd G$*, and suppose* V_N *is the direct sum of* s *isomorphic irreducible* $\mathsf{K}[N]$*-modules. If* $\mathbf{D}$ *is the representation of* G *on* V*, then*

$$\mathbf{D}(g) = \mathbf{P}_1(g) \otimes \mathbf{P}_2(g),$$

where the $\mathbf{P}_i$ *are irreducible projective representations of* G*, and* $\mathbf{P}_1$ *is a projective representation of* G/N *of degree* s*.*

Proof. Let $\mathsf{V} = \mathsf{W}_1 \oplus \cdots \oplus \mathsf{W}_s$ for some isomorphic irreducible $\mathsf{K}[N]$-modules W_i. If we choose a K-basis for V that respects this decomposition, then for all $n \in N$ we obtain matrices of the form

$$\mathbf{D}(n) = \begin{pmatrix} \mathbf{F}(n) & 0 & \dots & 0 \\ 0 & \mathbf{F}(n) & \dots & 0 \\ \vdots & & & \\ 0 & 0 & \dots & \mathbf{F}(n) \end{pmatrix}.$$

With respect to the same basis, let $\mathbf{D}(g) = (\mathbf{D}_{ij}(g))$ $(i, j = 1, \dots, s)$. Since

$$\mathbf{D}(n)\mathbf{D}(g) = \mathbf{D}(g)\mathbf{D}(n^g),$$

it follows that

$$\mathbf{F}(n)\mathbf{D}_{ij}(g) = \mathbf{D}_{ij}(g)\mathbf{F}(n^g) = \mathbf{D}_{ij}(g)\mathbf{F}^g(n).$$

By our assumption, the representations $\mathbf{F}$ and $\mathbf{F}^g$ of N are equivalent since they are irreducible constituents of $\mathbf{D}_N$. So there exists a nonsingular linear map $\mathbf{P}_2(g)$ such that

$$\mathbf{F}(n)\mathbf{P}_2(g) = \mathbf{P}_2(g)\mathbf{F}^g(n).$$

For $m, n \in N$, we have

$$\mathbf{F}^n(m) = \mathbf{F}(m^n) = \mathbf{F}(n)^{-1}\mathbf{F}(m)\mathbf{F}(n).$$

We may therefore choose $\mathbf{P}_2(n) = \mathbf{F}(n)$ for $n \in N$. It follows that

$$\mathbf{F}(n)\mathbf{D}_{ij}(g)\mathbf{P}_2(g)^{-1} = \mathbf{D}_{ij}(g)\mathbf{F}^g(n)\mathbf{P}_2(g)^{-1} = \mathbf{D}_{ij}(g)\mathbf{P}_2(g)^{-1}\mathbf{F}(n).$$

If we consider $\mathbf{D}_{ij}(g)\mathbf{P}_2(g)^{-1}$ as a K-linear map from W_1 onto itself, then

$$\mathbf{D}_{ij}(g)\mathbf{P}_2(g)^{-1} \in \mathrm{Hom}_{\mathsf{K}[N]}(\mathsf{W}_1, \mathsf{W}_1).$$

Since K is algebraically closed, we have by the strengthened Schur's Lemma 4.3 that

$$\mathrm{Hom}_{\mathsf{K}[N]}(\mathsf{W}_1, \mathsf{W}_1) = \{a1_{\mathsf{W}_1} \mid a \in \mathsf{K}\}.$$

It follows that $\mathbf{D}_{ij}(g) = a_{ij}(g)\mathbf{P}_2(g)$ for some $a_{ij}(g) \in \mathsf{K}$. Defining $\mathbf{P}_1(g)$ to be $\mathbf{P}_1(g) = (a_{ij}(g))$, we have for all $g \in G$ that

$$\mathbf{D}(g) = \mathbf{P}_1(g) \otimes \mathbf{P}_2(g).$$

We first show that $\mathbf{P}_2$ is a projective representation of G:

Namely, for all $g, g' \in G$ we have

$$\begin{aligned}\mathbf{F}(n)\mathbf{P}_2(g)\mathbf{P}_2(g')\mathbf{P}_2(gg')^{-1} &= \mathbf{P}_2(g)\mathbf{F}^g(n)\mathbf{P}_2(g')\mathbf{P}_2(gg')^{-1}\\ = \mathbf{P}_2(g)\mathbf{P}_2(g')\mathbf{F}^{gg'}(n)\mathbf{P}_2(gg')^{-1} &= \mathbf{P}_2(g)\mathbf{P}_2(g')\mathbf{P}_2(gg')^{-1}\mathbf{F}(n).\end{aligned}$$

It follows that

$$\mathbf{P}_2(g)\mathbf{P}_2(g')\mathbf{P}_2(gg')^{-1} \in \mathrm{Hom}_{\mathsf{K}[N]}(\mathsf{W}_1, \mathsf{W}_1),$$

and therefore, as above,

$$\mathbf{P}_2(gg') = c(g, g')\mathbf{P}_2(g)\mathbf{P}_2(g')$$

with $c(g, g') \in \mathsf{K}^\times$.

For all $g, g' \in G$, we now have

$$\begin{aligned}\mathbf{D}(gg') &= \mathbf{P}_1(gg') \otimes \mathbf{P}_2(gg') = \mathbf{P}_1(gg') \otimes c(g, g')\mathbf{P}_2(g)\mathbf{P}_2(g')\\ &= c(g, g')\mathbf{P}_1(gg') \otimes \mathbf{P}_2(g)\mathbf{P}_2(g'),\end{aligned}$$

but on the other hand,

$$\mathbf{D}(gg') = \mathbf{D}(g)\mathbf{D}(g') = \mathbf{P}_1(g)\mathbf{P}_1(g') \otimes \mathbf{P}_2(g)\mathbf{P}_2(g').$$

Since not all the entries of the nonsingular matrix $\mathbf{P}_2(g)\mathbf{P}_2(g')$ vanish, it follows that

$$\mathbf{P}_1(gg') = c(g, g')^{-1}\mathbf{P}_1(g)\mathbf{P}_1(g').$$

Hence, $\mathbf{P}_1$ is also a projective representation of G.

For $n \in N$, $\mathbf{P}_2(n) = \mathbf{F}(n)$ and therefore

$$1 \otimes \mathbf{F}(n) = \mathbf{D}(n) = \mathbf{P}_1(n) \otimes \mathbf{P}_2(n) = \mathbf{P}_1(n) \otimes \mathbf{F}(n).$$

This forces $\mathbf{P}_1(n) = 1$. So $\mathbf{P}_1$ is a projective representation of G/N.

The irreducibility of $\mathbf{P}_i$ follows trivially from the irreducibility of $\mathbf{D}$. □

In the remainder of this section, we restrict ourselves to the case that K is an algebraically closed field of characteristic 0.

Definition 17.6. Let N be a normal subgroup of G.

a) Let χ be an irreducible character of G. By Theorem 17.3 g), the restriction of χ to N has the form

$$\chi_N = e \sum_{i=1}^{t} \varphi^{g_i}$$

for some irreducible character φ of N and pairwise distinct φ^{g_i}. We call e the ramification index of χ with respect to N.

b) Let φ be an irreducible character of N. We define

$$G_\varphi = \{g \in G \mid \varphi^g = \varphi\}$$

and call G_φ the inertia group of φ. Clearly, $N \leq G_\varphi \leq G$, and $[G : G_\varphi]$ is the number of distinct G-conjugates φ^g of φ.

Theorem 17.7. *Let $N \trianglelefteq G$, and let φ be an irreducible character of N over an algebraically closed field K of characteristic 0.*

a) *If $G = \bigcup_{i=1}^{t} G_\varphi g_i$ is the decomposition of G into cosets of G_φ, then*

$$(\varphi^G)_N = [G_\varphi : N] \sum_{i=1}^{t} \varphi^{g_i}.$$

b) *The induced character φ^G is an irreducible character of G if and only if $G_\varphi = N$.*

Proof. a) It follows from Theorem 16.3 that for all $n \in N$, we have

$$\varphi^G(n) = \frac{1}{|N|} \sum_{g \in G} \varphi(n^g) = \frac{1}{|N|} \sum_{g \in G} \varphi^g(n) = [G_\varphi : N] \sum_{i=1}^{t} \varphi^{g_i}(n).$$

b) Since $\varphi^G(g) = 0$ for all $g \in G - N$, it follows from a) that

$$(\varphi^G, \varphi^G)_G = \frac{1}{[G : N]} (\varphi^G, \varphi^G)_N = \frac{[G_\varphi : N]^2}{[G : N]} [G : G_\varphi] = [G_\varphi : N].$$

So φ^G is an irreducible character of G if and only if $G_\varphi = N$. □

For the proof of Theorem 17.9, we require a lemma whose proof we defer to Theorem 24.3, where we examine projective representations more closely:

Lemma 17.8. *Let $\mathbf{P}$ be an irreducible projective representation of G over an algebraically closed field K of characteristic 0. Then the degree of $\mathbf{P}$ is a divisor of $|G|$.*

Theorem 17.9. *Let N be a normal subgroup of G. Let K be an algebraically closed field of characteristic 0, and let V be an irreducible $\mathsf{K}[G]$-module. Suppose V decomposes as a $\mathsf{K}[N]$-module into a direct sum of s irreducible constituents. Then s divides $|G/N|$.*

Proof. Using Theorem 17.3, we decompose V as $\mathsf{V} = \bigoplus_{i=1}^{t} \mathsf{V}_i$, where each $\mathsf{K}[N]$-module V_i is a direct sum of e isomorphic irreducible $\mathsf{K}[N]$-modules. By Theorem 17.3, the V_i are transitively permuted by G. Upon defining

$$U = \{g \in G \mid \mathsf{V}_1 g = \mathsf{V}_1\},$$

we have $t = [G : U]$. By Theorem 17.3 e), V_1 is an irreducible $\mathsf{K}[U]$-module that decomposes as a $\mathsf{K}[N]$-module into e isomorphic irreducible constituents. By Theorem 17.5, the representation $\mathbf{F}$ of U on V_1 can be written as a tensor product

$$\mathbf{F}(u) = \mathbf{P}_1(u) \otimes \mathbf{P}_2(u)$$

for all $u \in U$, where $\mathbf{P}_1$ is an irreducible projective representation of U/N of degree e. By Lemma 17.8, we have that e divides $|U/N|$. Therefore, $s = et$ is a divisor of $[U : N][G : U] = |G/N|$. □

As a consequence of Theorem 17.9, we prove a stronger version of Theorem 12.6:

Theorem 17.10 (Itô [271]). *Let A be an abelian normal subgroup of G, and let K be an algebraically closed field with $\operatorname{char}\mathsf{K} \nmid |G|$. Then the degree of every irreducible representation of $\mathsf{K}[G]$ divides $|G/A|$.*

Proof. Let $\mathbf{D}$ be an irreducible representation of $\mathsf{K}[G]$ of degree n, and first assume that $\operatorname{char}\mathsf{K} = 0$. By Theorem 17.3, the restriction $\mathbf{D}_A$ of $\mathbf{D}$ to A decomposes into n one-dimensional constituents. By Theorem 17.9, we therefore have that n divides $|G/A|$. For the general case that $\operatorname{char}\mathsf{K} \nmid |G|$, the claim follows from Theorem 12.11. □

Theorem 17.11. *Let K be an algebraically closed field of characteristic 0. Let $N \trianglelefteq G$, and let φ be an irreducible character of N. Furthermore, let G_φ be the inertia group of φ, and let $\varphi^{G_\varphi} = \sum_i \psi_i$ be the decomposition of φ^{G_φ} into irreducible characters ψ_i of G_φ. Then:*

a) *The ψ_i^G are irreducible characters of G.*
b) *If χ is an irreducible character of G with $(\chi_N, \varphi)_N > 0$, then $\chi = \psi_i^G$ for some i.*
c) *If $\psi_i \neq \psi_j$, then $\psi_i^G \neq \psi_j^G$.*

Proof. a) By Theorem 17.7 a), we have $(\varphi^{G_\varphi})_N = [G_\varphi : N]\varphi$. Therefore, $(\psi_i)_N = m_i\varphi$ for some m_i. Let χ be an irreducible character of G with $(\chi, \psi_i^G)_G \neq 0$. Then

$$\chi(1) \leq \psi_i^G(1) = [G : G_\varphi]\psi_i(1).$$

It follows from Frobenius reciprocity that

$$(\chi_{G_\varphi}, \psi_i)_{G_\varphi} = (\chi, \psi_i^G)_G \neq 0 \quad \text{and therefore} \quad (\chi_N, \varphi)_N \geq m_i.$$

Let $G = \bigcup_{j=1}^{t} G_\varphi g_j$ be the decomposition of G into cosets of G_φ. Then

$$(\chi_N, \varphi^{g_j})_N = ((\chi^{g_j^{-1}})_N, \varphi)_N = (\chi_N, \varphi)_N \geq m_i.$$

The φ^{g_j} $(j = 1, \ldots, [G : G_\varphi])$ are pairwise distinct and appear in χ_N with multiplicity at least m_i. This yields

$$\chi(1) \geq m_i[G : G_\varphi]\varphi(1) = [G : G_\varphi]\psi_i(1) = \psi_i^G(1).$$

Hence, $\chi = \psi_i^G$ is an irreducible character of G.

b) By Lemma 16.2,

$$\varphi^G = (\varphi^{G_\varphi})^G = \sum_i \psi_i^G.$$

Since

$$0 < (\chi_N, \varphi)_N = (\chi, \varphi^G)_G = \sum_i (\chi, \psi_i^G)_G,$$

it follows from a) that χ is one of the ψ_i^G.

c) We again write $(\psi_i)_N = m_i\varphi$. Then

$$(\varphi^{G_\varphi}, \psi_i)_{G_\varphi} = (\varphi, (\psi_i)_N)_N = m_i.$$

If $\psi_1, \ldots, \psi_k$ are the pairwise distinct characters among the ψ_i, then

$$\varphi^{G_\varphi} = \sum_{i=1}^{k} m_i\psi_i,$$

so

$$\varphi^G = \sum_{i=1}^{k} m_i\psi_i^G$$

and

$$[G_\varphi : N]\varphi(1) = \varphi^{G_\varphi}(1) = \sum_{i=1}^{k} m_i^2\varphi(1).$$

By Theorem 17.7, we have

$$(\varphi^G)_N = [G_\varphi : N] \sum_{j=1}^{t} \varphi^{g_j}$$

with $t = [G : G_\varphi]$. Assume for contradiction that $\psi_1^G = \psi_2^G$. Then

$$\begin{aligned}(m_1 + m_2)^2 + m_3^2 + \cdots + m_k^2 &\le (\varphi^G, \varphi^G)_G = (\varphi, (\varphi^G)_N)_N \\ &= (\varphi, [G_\varphi : N] \sum_{j=1}^{t} \varphi^{g_j})_N = [G_\varphi : N] = \sum_{i=1}^{k} m_i^2.\end{aligned}$$

This is clearly a contradiction. So the ψ_i^G $(i = 1, \ldots, k)$ are distinct. □

The following question often arises: Let $N \trianglelefteq G$, and let φ be an irreducible character of N. When does there exist a (necessarily irreducible) character χ of G such that $\chi_N = \varphi$? By Theorem 17.3 g), a necessary condition is $G_\varphi = G$. In the following theorem, we give two sufficient conditions.

Theorem 17.12. *Let* K *be an algebraically closed field of characteristic* 0. *Let* $N \trianglelefteq G$, *and let* φ *be an irreducible character of* N *with* $G_\varphi = G$. *Suppose one of the following two conditions is satisfied:*

(1) $(|N|, |G/N|) = 1$.

(2) *If* $\mathbf{P}$ *is a projective representation of* G/N, *then* $\mathbf{P}$ *is equivalent* (*in the sense of Definition 17.4*) *to an ordinary representation of* G/N. (*This condition is satisfied whenever* G/N *is cyclic; we will discuss other situations in which this condition is satisfied in §25.*)

Then:

a) *There exists an irreducible character* χ *of* G *with* $\chi_N = \varphi$.

b) *If the* $\psi_1, \ldots, \psi_m$ *are all the irreducible characters of* G/N, *then the* $\chi\psi_i$ $(i = 1, \ldots, m)$ *are irreducible and pairwise distinct characters of* G.

c) *We have* $\varphi^G = \sum_{i=1}^{m} \psi_i(1)\chi\psi_i$.

d) *If* λ *is an irreducible character of* G *with* $(\lambda_N, \varphi)_N > 0$, *then* $\lambda = \chi\psi_i$ *for some* i. *If* $\lambda_N = \varphi$, *then* $\lambda = \chi\psi_i$ *for some character* ψ_i *of* G/N *of degree* 1.

Proof. a) **Case 1:** First assume that condition (1) is satisfied. By the Zassenhaus Theorem I.18.1 (pg. 135), there exists a complement H of N in G. Let $\mathbf{F}$ be a representation of N with character φ. Since $G_\varphi = G$, we have $\varphi^h = \varphi$ for all $h \in H$. Therefore, the representations $\mathbf{F}^h$ and $\mathbf{F}$ are equivalent. Hence,

$$\mathbf{F}(n^h) = \mathbf{F}^h(n) = \mathbf{D}(h)^{-1}\mathbf{F}(n)\mathbf{D}(h)$$

for some nonsingular $\mathbf{D}(h)$. By Schur's Lemma, the $\mathbf{D}(h)$ are uniquely determined up to scalar multiples. We choose the scalar factors from the algebraically closed field K such that $\det \mathbf{D}(h) = 1$ for all $h \in H$. It follows from

$$\begin{aligned}\mathbf{D}(h_1h_2)^{-1}\mathbf{F}(n)\mathbf{D}(h_1h_2) &= \mathbf{F}(n^{h_1h_2}) \\ &= \mathbf{D}(h_2)^{-1}\mathbf{D}(h_1)^{-1}\mathbf{F}(n)\mathbf{D}(h_1)\mathbf{D}(h_2)\end{aligned}$$

that

$$\mathbf{D}(h_1h_2) = c(h_1, h_2)\mathbf{D}(h_1)\mathbf{D}(h_2) \qquad (*)$$

for some $c(h_1, h_2) \in \mathsf{K}^\times$. The trivial relation

$$(\mathbf{D}(h_1)\mathbf{D}(h_2))\mathbf{D}(h_3) = \mathbf{D}(h_1)(\mathbf{D}(h_2)\mathbf{D}(h_3))$$

yields

$$c(h_1, h_2)c(h_1h_2, h_3) = c(h_1, h_2h_3)c(h_2, h_3).$$

So c is a 2-cocycle in $\mathbf{Z}^2(H, \mathsf{K}^\times)$, where $\mathsf{K}^\times$ is to be understood as a trivial H-module with $xh = x$ for all $x \in \mathsf{K}^\times$ and all $h \in H$ (see Chapter I, §17, pg. 127). If we let $d = \varphi(1)$, then upon applying the determinant map to $(*)$, we see that $c(h_1, h_2)^d = 1$ for all $h_1, h_2 \in H$ since $\det \mathbf{D}(h) = 1$ for all $h \in H$; so, in particular, $c^d \in \mathbf{B}^2(H, \mathsf{K}^\times)$. By Theorem I.16.19 (pg. 126), the exponent of $\mathbf{H}^2(H, \mathsf{K}^\times) \cong \mathbf{Z}^2(H, \mathsf{K}^\times)/\mathbf{B}^2(H, \mathsf{K}^\times)$ is a divisor of $|H|$. On the other hand, $d = \varphi(1)$ is a divisor of $|N|$ by Theorem 5.12, so d is coprime to $|H|$. Hence, $c \in \mathbf{B}^2(H, \mathsf{K}^\times)$. Therefore, there exists some $b \in \mathbf{C}^1(H, \mathsf{K}^\times)$ such that

$$c(h_1, h_2) = b(h_1)b(h_2)b(h_1h_2)^{-1}$$

for all $h_1, h_2 \in H$. We now define $\mathbf{F}'$ on $G = NH$ by

$$\mathbf{F}'(nh) = \mathbf{F}(n)b(h)\mathbf{D}(h)$$

for all $n \in N$ and $h \in H$. Then

$$\begin{aligned}\mathbf{F}'(h_1h_2) &= b(h_1h_2)\mathbf{D}(h_1h_2) = b(h_1h_2)c(h_1, h_2)\mathbf{D}(h_1)\mathbf{D}(h_2)\\ &= b(h_1)\mathbf{D}(h_1)b(h_2)\mathbf{D}(h_2) = \mathbf{F}'(h_1)\mathbf{F}'(h_2)\end{aligned}$$

and

$$\mathbf{F}'(h)^{-1}\mathbf{F}'(n)\mathbf{F}'(h) = \mathbf{F}'(n^h).$$

So $\mathbf{F}'$ is an extension of the representation $\mathbf{F}$ to G, and $\chi = \operatorname{tr}\mathbf{F}'$ is the desired character of G with $\chi_N = \varphi$.

Case 2: Assume condition (2) is satisfied. Let $\mathbf{D}$ be any irreducible representation of G such that the representation $\mathbf{F}$ with character φ is a constituent of $\mathbf{D}_N$. (Since $(\chi, \varphi^G)_G = (\chi_N, \varphi)_N$, this holds for every irreducible constituent $\mathbf{D}$ of $\mathbf{F}^G$.) Since $\varphi^g = \varphi$ for all $g \in G$, it follows from Theorem 17.3 g) that $\mathbf{D}_N$ decomposes into equivalent irreducible representations of $\mathsf{K}[N]$. Therefore, $\mathbf{D}(g) = \mathbf{P}_1(g) \otimes \mathbf{P}_2(g)$ by Theorem 17.5, where $\mathbf{P}_1$ is a projective representation of G/N. By assumption (2), we may modify the $\mathbf{P}_1(g)$ by scalar factors so that $\mathbf{P}_1$ is an ordinary representation of G/N. Then $\mathbf{P}_2$ is also an ordinary representation of G. For all $n \in N$, we have

$$1 \otimes \mathbf{F}(n) = \mathbf{D}(n) = 1 \otimes \mathbf{P}_2(n).$$

So $\mathbf{P}_2$ is the desired extension of $\mathbf{F}$ to G.

b) Let $\mathbf{L}_1, \ldots, \mathbf{L}_m$ be all the irreducible representations (up to equivalence) of G/N. We define $\psi_i = \operatorname{tr} \mathbf{L}_i$. If $\mathbf{D}$ is the representation of G with the character χ determined in a), then $\psi_i \chi$ is the character of $\mathbf{L}_i \otimes \mathbf{D}$. Suppose there exists some matrix u such that

$$(\mathbf{L}_i(g) \otimes \mathbf{D}(g))u = u(\mathbf{L}_j(g) \otimes \mathbf{D}(g))$$

for all $g \in G$. It follows for $g = n \in N$ that

$$(1 \otimes \mathbf{D}(n))u = u(1 \otimes \mathbf{D}(n)).$$

If we decompose u into suitable submatrices u_{st}, then we find that $\mathbf{D}(n)u_{st} = u_{st}\mathbf{D}(n)$ for all $n \in N$ and all s, t. Since $\mathbf{D}_N$ is irreducible, it follows from Schur's Lemma that $u_{st} = x_{st}1$ for some $x_{st} \in \mathsf{K}$. So $u = a \otimes 1$, where $a = (x_{st})$, and then

$$\mathbf{L}_i(g)a = a\mathbf{L}_j(g)$$

for all $g \in G$. If $i = j$, then Schur's Lemma forces $a = x1$, so $u = x1$. Hence, if V is the representation module of $\mathbf{L}_i \otimes \mathbf{D}$, then $\operatorname{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{V}) = \{x1 \mid x \in \mathsf{K}\}$. So V is irreducible by Lemma 11.10.

If $i \neq j$, then $\mathbf{L}_i$ and $\mathbf{L}_j$ are not equivalent. It therefore follows from $\mathbf{L}_i(g)a = a\mathbf{L}_j(g)$ for all $g \in G$ that $a = 0$. Thus, $\mathbf{L}_i \otimes \mathbf{D}$ and $\mathbf{L}_j \otimes \mathbf{D}$ are not equivalent when $i \neq j$.

c) We have

$$(\varphi^G, \chi\psi_i)_G = (\varphi, \chi_N(\psi_i)_N)_N = \psi_i(1)(\varphi, \varphi)_N = \psi_i(1).$$

It follows from

$$\sum_{i=1}^{m} \psi_i(1)(\chi\psi_i)(1) = \chi(1) \sum_{i=1}^{m} \psi_i(1)^2 = \varphi(1)|G/N| = \varphi^G(1)$$

that

$$\varphi^G = \sum_{i=1}^{m} \psi_i(1)\chi\psi_i.$$

d) The claim follows immediately from c) and

$$(\lambda_N, \varphi)_N = (\lambda, \varphi^G)_G.$$ □

There is no general procedure for calculating the extension of a representation of a normal subgroup to a representation of the whole group. We carry out this calculation for a special case that is important in applications:

Theorem 17.13. *Let P be an extraspecial p-group of order p^{1+2m}, and let H be a cyclic subgroup of the automorphism group of P with $p \nmid |H|$. We assume that $\mathbf{Z}(P) = P'$ is centralized by H, but that H acts fixed point freely on $P/\mathbf{Z}(P)$. Let $G = PH$ be the semidirect product of P with H, and let K be an algebraically closed field with* $\operatorname{char}\mathsf{K} \nmid |G|$.

a) *G has exactly the following irreducible characters over K:*

(1) *The irreducible characters of the Frobenius group $G/\mathbf{Z}(P)$ described in Theorem 16.13.*

(2) *For every linear character $\lambda \neq 1$ of $\mathbf{Z}(P)$ and every linear character μ of H, there exists an irreducible character $\chi_{\lambda\mu}$ of G of degree p^m defined by*

$$\chi_{\lambda\mu}(g) = \begin{cases} p^m\lambda(g) & \text{for } g \in \mathbf{Z}(P) \\ 0 & \text{for } g \in P - \mathbf{Z}(P) \\ \delta\lambda(x)\mu(h) & \text{for } g = xh^y \text{ with } x \in \mathbf{Z}(P), y \in G, 1 \neq h \in H, \end{cases}$$

where $\delta = \pm 1$ and $|H|$ is a divisor of $p^m - \delta$.

b) *If ρ is the regular character of H, then*

$$(\chi_{\lambda\mu})_H = \frac{p^m - \delta}{|H|}\rho + \delta\mu.$$

In particular, $(\chi_{\lambda\mu})_H$ contains the trivial character of H except when $\delta = -1$, $\mu = 1$ and $|H| = p^m + 1$.

Proof. (Dade) Let $\lambda \neq 1$ be a linear character of $\mathbf{Z}(P)$. By Theorem 16.14, there exists an irreducible, faithful representation $\mathbf{D}$ of P of degree p^m with $\mathbf{D}(z) = \lambda(z)1$ for $z \in \mathbf{Z}(P)$. If φ is the character of $\mathbf{D}$, then we have by Theorem 16.14 that

$$\varphi(x) = \begin{cases} p^m\lambda(x) & \text{for } x \in \mathbf{Z}(P) \\ 0 & \text{for } x \in P - \mathbf{Z}(P). \end{cases}$$

Since $\mathbf{Z}(P) \leq \mathbf{Z}(G)$, we have $\varphi^g = \varphi$ for all $g \in G$. So $\mathbf{D}$ admits an extension to $G = PH$ by Theorem 17.12, which we also denote by $\mathbf{D}$. We calculate $\mathbf{D}(h)$ for $h \in H$:

Let $P = \bigcup_{i=1}^{p^{2m}} \mathbf{Z}(P)x_i$. By Theorem 5.14, the $\mathbf{D}(x)$ with $x \in P$ span the full matrix ring $(\mathsf{K})_{p^m}$ over K. Since $\mathbf{D}(zx_i) = \lambda(z)\mathbf{D}(x_i)$ for $z \in \mathbf{Z}(P)$, the set $\{\mathbf{D}(x_i) \mid i = 1, \ldots, p^{2m}\}$ is a K-basis of $(\mathsf{K})_{p^m}$. We equip $(\mathsf{K})_{p^m}$ with the structure of a right $\mathsf{K}[H]$-module by defining

$$u\mathbf{L}(h) = \mathbf{D}(h)^{-1}u\mathbf{D}(h)$$

for all $h \in H$ and $u \in (\mathsf{K})_{p^m}$. We determine the decomposition of $(\mathsf{K})_{p^m}$ into irreducible $\mathsf{K}[H]$-modules in two ways:

1) $(\mathsf{K})_{p^m}$ decomposes as a $\mathsf{K}[H]$-module into the direct sum of the trivial module and $\frac{p^{2m}-1}{|H|}$ copies of the regular $\mathsf{K}[H]$-module:

Let $H = \langle h_0 \rangle$. By our assumption, H acts fixed point freely on $P/\mathbf{Z}(P)$. The orbits of H on $P/\mathbf{Z}(P)$ thus have the form

$$\{\mathbf{Z}(P)\}, \{x_j\mathbf{Z}(P), x_j^{h_0}\mathbf{Z}(P), \ldots, x_j^{h_0^{|H|-1}}\mathbf{Z}(P)\},$$

where $x_j \in P$ and $j = 1, \ldots, \frac{p^{2m}-1}{|H|}$ upon choosing a suitable labeling. Since

$$\mathbf{D}(x_j^{h_0^i})\mathbf{L}(h_0) = \mathbf{D}(x_j^{h_0^{i+1}}),$$

the representation of H on the K-subspaces

$$\langle \mathbf{D}(x_j^{h_0^i}) \mid i = 0, 1, \ldots, |H| - 1 \rangle$$

of $(\mathsf{K})_{p^m}$ is the regular representation.

2) Let ϵ be a primitive $|H|$-th root of unity in K. Since $\mathbf{D}_H$ is completely reducible, we have

$$\mathbf{D}(h_0) = \begin{pmatrix} 1_{n_0} & & & & \\ & \epsilon 1_{n_1} & & & \\ & & \epsilon^2 1_{n_2} & & \\ & & & \ddots & \\ & & & & \epsilon^{|H|-1} 1_{n_{|H|-1}} \end{pmatrix}$$

for some n_j. Here, we have denoted the $n_i \times n_i$ identity matrix by 1_{n_i}; if $n_i = 0$, then this matrix is omitted. Clearly

$$p^m = \sum_{i=0}^{|H|-1} n_i. \tag{I}$$

Now we decompose every matrix $u \in (\mathsf{K})_{p^m}$ into submatrices u_{ij} according to the decomposition of $\mathbf{D}(h_0)$ above, so u_{ij} is an $n_i \times n_j$ matrix and $i, j = 0, 1, \ldots, |H| - 1$. We have

$$(u_{ij})\mathbf{L}(h_0) = \mathbf{D}(h_0)^{-1}(u_{ij})\mathbf{D}(h_0) = (\epsilon^{j-i}u_{ij}).$$

The K-subspaces

$$\mathsf{B}_{ij} = \{(u_{kl}) \mid u_{kl} = 0 \text{ for } (k, l) \neq (i, j)\}$$

are therefore $\mathsf{K}[H]$-invariant, direct summands of $(\mathsf{K})_{p^m}$. On B_{ij}, $\mathbf{L}(h_0)$ has eigenvalue ϵ^{j-i} with multiplicity $\dim_{\mathsf{K}} \mathsf{B}_{ij} = n_i n_j$.

Now we compare the results of 1) and 2): By 1), $\mathbf{L}(h_0)$ has eigenvalue 1 with multiplicity

$$1 + \frac{p^{2m} - 1}{|H|}.$$

On B_{ij}, $\mathbf{L}(h_0)$ has eigenvalue 1 exactly when $i = j$. It follows that

$$1 + \frac{p^{2m} - 1}{|H|} = \sum_{i=0}^{|H|-1} \dim \mathsf{B}_{ii} = \sum_{i=0}^{|H|-1} n_i^2. \tag{II}$$

Analogously, our determination of the multiplicity of ϵ^k ($k = 1, \ldots, |H| - 1$) as an eigenvalue of $\mathbf{L}(h_0)$ yields the relation

$$\frac{p^{2m} - 1}{|H|} = \sum_{i=0}^{|H|-1} n_i n_{i+k}. \tag{III}$$

(Here, we define $n_{|H|+s} = n_s$.) It follows from (II) and (III) that

$$2 = 2 \sum_{i=0}^{|H|-1} n_i^2 - 2 \sum_{i=0}^{|H|-1} n_i n_{i+k} = \sum_{i=0}^{|H|-1} (n_i - n_{i+k})^2. \tag{IV}$$

For $k = 1$, the relation in (IV) shows that there exist exactly two indices j such that $n_j \neq n_{j+1}$. Say

$$n_0 = \cdots = n_j \neq n_{j+1} = \cdots = n_{j+l} \neq n_{j+l+1} = \cdots = n_{|H|-1} = n_0$$

with $l \geq 1$ and $0 \leq j < j + l \leq |H| - 1$. If $l > 1$ and $|H| - l > 1$, then setting $k = 2$ in relation (IV) we obtain the contradiction

$$\begin{aligned} 2 &= \sum_{i=0}^{|H|-1} (n_i - n_{i+2})^2 \\ &\geq (n_{j-1} - n_{j+1})^2 + (n_j - n_{j+2})^2 + (n_{j+l-1} - n_{j+l+1})^2 \geq 3. \end{aligned}$$

So either $l = 1$ or $|H| - l = 1$. Hence, $|H| - 1$ of the n_i have the same value, say d, and the remaining n_j has the value $d + \delta$ with $\delta = \pm 1$. It then follows from (I) that

$$p^m = \sum_{i=0}^{|H|-1} n_i = |H|d + \delta.$$

In particular, $|H|$ is a divisor of $p^m - \delta$.

Let χ be the character of the representation $\mathbf{D}$ of G. By our calculation,

$$\chi(h_0^i) = d \sum_{s=0}^{|H|-1} \epsilon^{si} + \delta\epsilon^{ji}.$$

If we denote by ρ the regular character of H and denote by μ_0 the linear character of H with $\mu_0(h_0^i) = \epsilon^{ji}$, then it follows that

$$\chi_H = d\rho + \delta\mu_0.$$

The factor group $G/\mathbf{Z}(P)$ is a Frobenius group with Frobenius kernel $P/\mathbf{Z}(P)$. Therefore, if $P = \bigcup_i \mathbf{Z}(P)x_i$, then

$$G = P \cup \bigcup_i \mathbf{Z}(P)(H-1)^{x_i}$$

is a disjoint decomposition of G. Since $\rho(h) = 0$ for $1 \neq h \in H$, the character χ of G is completely determined by specifying

$$\chi(g) = \begin{cases} p^m\lambda(g) & \text{for } g \in \mathbf{Z}(P) \\ 0 & \text{for } g \in P - \mathbf{Z}(P) \\ \delta\lambda(x)\mu_0(h) & \text{for } g = xh^y \text{ with } x \in \mathbf{Z}(P),\ y \in G, \text{ and } 1 \neq h \in H. \end{cases}$$

If μ is a linear character of H, then we define $\chi_{\lambda\mu} = \chi\mu_0^{-1}\mu$. By Theorem 17.12, the $\chi_{\lambda\mu}$ are irreducible, pairwise distinct characters of G. The sum of squared degrees of the irreducible characters of G that we have considered so far is the sum of $|G/\mathbf{Z}(P)| = p^{2m}|H|$ (from the characters with $\mathbf{Z}(P)$ in the kernel) and $p^{2m}(p-1)|H|$ (from the $(p-1)|H|$ characters $\chi_{\lambda\mu}$ with $1 \neq \lambda \in \widehat{\mathbf{Z}(P)}$ and $\mu \in \widehat{H}$). Together, this yields $p^{1+2m}|H| = |G|$, and so we have determined all the irreducible characters of G. It is clear that the claim in b) is also satisfied. □

When analyzing solvable groups, one can often force the assumptions of Theorem 17.13 to hold with the help of the Hall–Higman reduction theorem (see Theorem III.13.5, pg. 371). We will return to this in Volume 2.

Exercises

30) (N. Itô [272]) Let G be a solvable group, and let p be a prime number. The following are equivalent:
   a) $p \nmid \chi(1)$ for every irreducible character χ of G over the complex numbers.
   b) G has a normal, abelian Sylow p-subgroup.[5]

31) (Burnside [79]) Let G be a solvable group with a faithful, irreducible representation $\mathbf{D}$ over the complex numbers of prime degree p. Suppose that $\det \mathbf{D}(g) = 1$ for all $g \in G$. Then one of the following occurs:

[5] It follows from the classification of finite simple groups that this key result in character theory, where it is referred to as the Itô–Michler theorem, holds in any finite group. See G. Navarro Ortega, *Character theory and the McKay conjecture*, Cambridge Studies in Advanced Mathematics, 175, Cambridge Univ. Press, Cambridge, 2018. *trans.*

(1) **D** is induced from a 1-dimensional representation of a subgroup of G. In this case, there exists an abelian normal subgroup A of G such that $|G/A|$ divides $p(p-1)$.
(2) The Fitting subgroup $\mathbf{F}(G)$ of G is a nonabelian group of order p^3, and it is the Sylow p-subgroup of G if $p > 3$.

(Hint: First show that in case (2), all abelian normal subgroups of G lie in $\mathbf{Z}(G)$ and $\mathbf{F}(G)$ has class 2.)

32) Let A be an abelian subgroup of G. Suppose there exists a chain

$$A = N_0 \lhd N_1 \lhd \cdots \lhd N_s = G;$$

that is, A is subnormal in G. If χ is an absolutely irreducible character of G over the complex numbers, then $\chi(1)$ divides $[G : A]$.

18 Monomial representations

Definition 18.1.
a) A representation **D** of G is said to be monomial if **D** is induced from a one-dimensional representation of a subgroup of G.
b) We call G an M-group if every irreducible representation of G over an algebraically closed field K with char $\mathsf{K} \nmid |G|$ is monomial.

For the rest of this section, let K be an algebraically closed field with char $\mathsf{K} \nmid |G|$. The construction of monomial representations is very easy:

Theorem 18.2. *Let $G = \bigcup_{i=1}^{n} Ur_i$ be the decomposition of G into cosets of U, and suppose $L \trianglelefteq U$ such that $U/L = \langle zL \rangle$ is cyclic of order m. Let ϵ be a primitive m-th root of unity, and let* **F** *be the one-dimensional representation of U with kernel L and $\mathbf{F}(z) = \epsilon$. Then $\mathsf{V} = \mathsf{K}e$ with*

$$e = \frac{1}{|U|}\left(\sum_{l \in L} l\right)\left(\sum_{j=1}^{m} \epsilon^{-j} z^j\right)$$

is a representation module for **F**, *the element $e \in \mathsf{K}[U]$ is idempotent, and $\mathsf{V}^G \cong e\mathsf{K}[G]$ is a representation module for the induced representation $\mathbf{F}^G$. The kernel of $\mathbf{F}^G$ is $\bigcap_{g \in G} L^g$. The set $\{er_1, \ldots, er_n\}$ is a K-basis of $e\mathsf{K}[G]$, and for all $g \in G$ we have*

$$(er_i)g = \epsilon^{j_i(g)} er_{i'}$$

for some integers $j_i(g)$ and permutation $\mathbf{P}(g) = \binom{i}{i'}$ on $\{1, \ldots, n\}$.

Proof. We have $e(lz^j) = \epsilon^j e$. So $\mathsf{K}e$ is a representation module for $\mathbf{F}$, and $e^2 = e$. It is clear that $\{er_1, \dots, er_n\}$ is a K-basis of $e\mathsf{K}[G]$.

For $g \in G$ and

$$r_i g = l_i z^{j_i(g)} r_{i'}$$

with $l_i \in L$, we have

$$(er_i)g = el_i z^{j_i(g)} r_{i'} = \epsilon^{j_i(g)} er_{i'}.$$

So under multiplication by elements of G, the er_i are permuted and multiplied by factors $\epsilon^{j_i(g)}$, which are m-th roots of unity. The $\mathsf{K}[G]$-module $e\mathsf{K}[G]$ is clearly isomorphic to V^G (see also Exercise 27 on pg. 598).

The relations $(er_i)g = er_i$ for $i = 1, \dots, n$ are equivalent to $r_i g r_i^{-1} \in L$, so are equivalent to $g \in \bigcap_{i=1}^{n} r_i L r_i^{-1} = \bigcap_{x \in G} L^x$. □

A criterion for the irreducibility of the representations we just constructed follows easily from Theorem 16.12 (see Shoda [382]).

Lemma 18.3. *Let $\mathscr{M}$ be a class of finite groups with the following properties:*
(1) *If G lies in $\mathscr{M}$, then every subgroup and every homomorphic image of G also lies in $\mathscr{M}$.*
(2) *If G lies in $\mathscr{M}$ and G is nonabelian, then G contains an abelian normal subgroup A such that $A \not\leq \mathbf{Z}(G)$.*
Then every group in $\mathscr{M}$ is an M-group.

Proof. We prove the claim by induction on $|G|$. Let $\mathbf{D}$ be an irreducible representation of G over an algebraically closed field K with $\operatorname{char} \mathsf{K} \nmid |G|$. If $\mathbf{D}$ is not faithful, then $\mathbf{D}$ is a representation of a quotient G/N of G. Since $G/N \in \mathscr{M}$ by (1), $\mathbf{D}$ is monomial by the induction hypothesis. So let $\mathbf{D}$ be a faithful representation of G.

If G is abelian, then $\mathbf{D}$ is one-dimensional, so it is certainly monomial. If G is nonabelian, then there exists an abelian normal subgroup A of G with $A \not\leq \mathbf{Z}(G)$ by (2). The irreducible constituents of the restriction $\mathbf{D}_A$ of $\mathbf{D}$ to A are all one-dimensional. Since $\mathbf{D}$ is faithful and $A \not\leq \mathbf{Z}(G)$, these representations are not all equivalent. Therefore, we have by Theorem 17.3 e) that $\mathbf{D} = \mathbf{H}^G$, where $\mathbf{H}$ is an irreducible representation of a proper subgroup U of G. By (1), U lies in $\mathscr{M}$. So $\mathbf{H} = \mathbf{F}^U$ for some one-dimensional representation $\mathbf{F}$ of a subgroup V of U by our induction hypothesis. Then $\mathbf{D} = (\mathbf{F}^U)^G = \mathbf{F}^G$ by Lemma 16.2 b), and $\mathbf{D}$ is monomial. □

The following theorem yields a useful criterion.

Theorem 18.4 (Huppert [255]). *Suppose the group G has a normal subgroup N with the following properties:*
(a) *N is solvable, and all the Sylow subgroups of N are abelian.*

(b) *G/N is supersolvable; that is, there exists a chief series*

$$1 = G_0/N \lhd G_1/N \lhd \cdots \lhd G_r/N = G/N$$

of G/N such that all factor groups G_{i+1}/G_i have prime order (see Chapter VI, §9).

Then G is an M-group.

Proof. We show that assumptions (1) and (2) of Lemma 18.3 are satisfied.

(1) If $U \leq G$, then $U \cap N$ is solvable and all of its Sylow subgroups are abelian. Furthermore, U possesses the normal series

$$U \cap N = U \cap G_0 \trianglelefteq U \cap G_1 \trianglelefteq \cdots \trianglelefteq U \cap G_r = U,$$

where

$$|(U \cap G_{i+1})/(U \cap G_i)| = |G_i(U \cap G_{i+1})/G_i|,$$

and this is either 1 or a prime number as it is a divisor of $|G_{i+1}/G_i|$. So every subgroup of G again satisfies conditions (a) and (b).

If K is a normal subgroup of G, then NK/K is again solvable with abelian Sylow subgroups, being a homomorphic image of N. In addition, G/NK is again supersolvable, as it is a homomorphic image of G/N.

So condition (1) of Lemma 18.3 is satisfied.

(2) First assume that N is nonabelian, and let A be an abelian normal subgroup of G lying in N that is as large as possible. If $A \not\leq \mathbf{Z}(G)$, then we are done. Otherwise, $A \leq \mathbf{Z}(G)$, so in particular $A \leq \mathbf{Z}(N)$. Then let B/A be a minimal normal subgroup of G/A lying in N/A. Since N is solvable, we have $B' \leq A \leq \mathbf{Z}(B)$. So B is nilpotent. Since B is a subgroup of N, and all the Sylow subgroups of N are abelian by assumption, B is itself abelian. But this contradicts our choice of A.

It remains to consider the case that N is abelian and G is nonabelian. If N does not lie in $\mathbf{Z}(G)$, then we are done. Otherwise, let $G_i \leq \mathbf{Z}(G)$ but $G_{i+1} \not\leq \mathbf{Z}(G)$. Then G_{i+1} is abelian since it is a cyclic extension of $G_i \leq \mathbf{Z}(G_{i+1})$, and we are done. □

We emphasize three special cases of Theorem 18.4:

Theorem 18.5.
a) *Every supersolvable group is an M-group.*
b) *Every nilpotent group is an M-group.*
c) *Every solvable group with abelian Sylow subgroups is an M-group.*

Proof. a) is the special case $N = 1$ in Theorem 18.4, and c) is the special case $N = G$ in Theorem 18.4. The claim in b) follows from a) since every p-group is supersolvable by Theorem III.7.2 c) (pg. 318), and thus every nilpotent group is supersolvable. □

Theorem 18.6. *Let G be an M-group, and let $1 = m_1 < m_2 < \cdots < m_k$ be the distinct degrees of the irreducible representations of G over an algebraically closed field K with* $\operatorname{char}\mathsf{K} \nmid |G|$.

a) *If* $\mathbf{D}$ *is an irreducible representation of* G *over* K *with* $\deg \mathbf{D} \leq m_i$*, then the* i*-th commutator subgroup* $G^{(i)}$ *of* G *lies in the kernel of* $\mathbf{D}$.
b) (*Taketa [421]*) *Every* M*-group is solvable.*

Proof. a) If $\deg \mathbf{D} = 1$, then this is trivial. We prove the claim by induction on $\deg \mathbf{D}$.

The representation $\mathbf{D}$ is monomial. For some choice of basis $\{v_1, \ldots, v_n\}$ of the representation module for $\mathbf{D}$, we have

$$v_j \mathbf{D}(g) = \epsilon_j(g) v_{jg}$$

by Theorem 18.2. The map π sending $\mathbf{D}(g)$ to the permutation $\mathbf{P}(g) = \left(\begin{smallmatrix} j \\ jg \end{smallmatrix}\right)$ is a homomorphism from $\mathbf{D}(G)$ to the symmetric group S_n. The kernel of π consists of diagonal matrices, so it is abelian. As a permutation representation of G, $\mathbf{P}$ is reducible, and it contains the trivial representation, in particular (see Theorem 13.4). The irreducible constituents of $\mathbf{P}$ can only be those irreducible representations $\mathbf{D}_j$ of G with $\deg \mathbf{D}_j \leq m_{i-1}$, and these representations contain $G^{(i-1)}$ in the kernel by our induction hypothesis. Since $\mathbf{P}$ is completely reducible, it follows that $\mathbf{P}(G^{(i-1)}) = 1$. Hence, $\mathbf{D}(G^{(i-1)}) \leq \ker \pi$ and so $\mathbf{D}(G^{(i)}) = 1$.

b) For every irreducible representation $\mathbf{D}$ of G, we have $\mathbf{D}(G^{(m_k)}) = 1$ by a). Since the regular representation of G is faithful and completely reducible, it follows that $G^{(m_k)} = 1$ and G is solvable. □

However, the solvability of G is by no means sufficient to guarantee that G is an M-group, as the following example shows.

Example 18.7. Let $Q = \langle a, b \rangle$ be the quaternion group of order 8, and let $G = QH$ be the semidirect product of Q with the cyclic group $H = \langle h \rangle$ of order 3 with $a^h = b$ and $b^h = ab$. We define a faithful irreducible representation $\mathbf{D}$ of Q over the field $\mathbb{C}$ of complex numbers by

$$\mathbf{D}(a) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \quad \text{and} \quad \mathbf{D}(b) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

where $i^2 = -1$. Note that $\mathbf{D}^g$ is equivalent to $\mathbf{D}$ for all $g \in G$. By Theorem 17.12, $\mathbf{D}$ admits an extension to $G = QH$, which we denote again by $\mathbf{D}$. Incidentally, a simple calculation shows that we can choose

$$\mathbf{D}(h) = \frac{i-1}{2} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}.$$

The representation $\mathbf{D}$ of G is not monomial: $\mathbf{D}$ cannot be induced from a linear representation of a subgroup of G because $|G/G'| = |G/Q| = 3$, so G has no subgroup of index 2. (See also Theorem 17.13.)

Theorem 18.8. *If* G_1 *and* G_2 *are both* M*-groups, then* $G_1 \times G_2$ *is also an* M*-group.*

Proof. If $\mathbf{D}$ is an irreducible representation of G, then $\mathbf{D} = \mathbf{D}_1 \otimes \mathbf{D}_2$ for some irreducible representations $\mathbf{D}_i$ of G_i $(i = 1, 2)$ by Theorem 10.3. Since G_i is an

M-group, $\mathbf{D}_i$ is induced from a linear representation $\mathbf{F}_i$ of a subgroup U_i of G_i. By Lemma 16.2 c) (which was formulated there for modules), $\mathbf{D} = \mathbf{D}_1 \otimes \mathbf{D}_2 = \mathbf{F}_1^{G_1} \otimes \mathbf{F}_2^{G_2}$ is equivalent to the representation $(\mathbf{F}_1 \otimes \mathbf{F}_2)^{G_1 \times G_2}$, which is induced from the linear representation $\mathbf{F}_1 \otimes \mathbf{F}_2$ of $U_1 \times U_2$. So all irreducible representations of $G_1 \times G_2$ are monomial, and hence $G_1 \times G_2$ is an M-group. □

Lemma 18.9. *Let* $\mathbf{D}$ *be an irreducible representation of* G *on the* $\mathsf{K}[G]$*-module* V*. Suppose that with respect to some basis* $\{v_1, \dots, v_n\}$ *of* V*, the* $\mathbf{D}(g)$ *have the form*

$$v_i \mathbf{D}(g) = a_i(g) v_{ig} \quad (0 \neq a_i(g) \in \mathsf{K}).$$

Then $\mathbf{D}$ *is monomial.*

Proof. Let $U = \{g \in G \mid 1^g = 1\}$, and let $G = \bigcup_{i=1}^{m} U r_i$ be the decomposition of G into cosets of U with $1^{r_i} = i$. Let $\langle v \rangle$ be the one-dimensional $\mathsf{K}[U]$-module such that

$$vu = a_1(u)v.$$

The $\mathsf{K}[G]$-module $\langle v \rangle^G$ induced from $\langle v \rangle$ has the basis $\{v \otimes r_i \mid i = 1, \dots, m\}$. Letting $r_i g = u_i r_{i'}$ for some $u_i \in U$, we have

$$(v \otimes r_i)g = v u_i \otimes r_{i'} = a_1(u_i) v \otimes r_{i'}.$$

Since V is irreducible, the vectors

$$w_i = v_1 \mathbf{D}(r_i) = a_1(r_i) v_i \quad (i = 1, \dots, m)$$

span all of V. So $n = m$. We have

$$w_i \mathbf{D}(g) = v_1 \mathbf{D}(r_i g) = v_1 \mathbf{D}(u_i) \mathbf{D}(r_{i'}) = a_1(u_i) w_{i'}.$$

So the map μ defined by $(v \otimes r_i)\mu = v_1 \mathbf{D}(r_i) = w_i$ is a module isomorphism from $\langle v \rangle^G$ to V, and hence $\mathbf{D}$ is monomial. □

Lemma 18.10 (Dade). *Let* G *be an* M*-group, and let* Z *be a cyclic group of prime order. Then the regular wreath product* $G \wr_r Z$ *is an* M*-group.*

Proof. Let $|Z| = p$, and let $\mathbf{D}$ be an irreducible representation of the wreath product

$$H = G \wr_r Z = \{(f, z) \mid z \in Z, f : \{1, \dots, p\} \to G\}$$

over the algebraically closed field K with $\operatorname{char} \mathsf{K} \nmid |H|$. As in §15 of Chapter I, we define

$$D = \{(f, 1) \mid f : \{1, \dots, p\} \to G\}.$$

Then $D \lhd H$ and $D \cong \underbrace{G \times \cdots \times G}_{p}$. We distinguish three cases:

Case 1: The restriction $\mathbf{D}_D$ of $\mathbf{D}$ to D is irreducible. Then by Theorem 10.3, we have

$$\mathbf{D}((f,1)) = \mathbf{F}_1(f(1)) \otimes \cdots \otimes \mathbf{F}_p(f(p))$$

for some irreducible representations $\mathbf{F}_i$ of G. Let χ be the character of $\mathbf{D}$, and let ψ_i be the character of $\mathbf{F}_i$. We let $f(i) = g_i$. For all $z \in Z$, we then have

$$\begin{aligned}\psi_1(g_1)\cdots\psi_p(g_p) &= \chi((f,1)) = \chi((f,1)^{z^{-1}}) \\ &= \psi_1(g_{1^z})\cdots\psi_p(g_{p^z}).\end{aligned}$$

Letting $1^z = i$, it follows with $g_j = 1$ for $j \neq i$ that

$$\psi_i(g_i)\prod_{j\neq i}\psi_j(1) = \psi_1(g_i)\prod_{j\neq 1}\psi_j(1).$$

So $\psi_i = k_i\psi_1$ with $k_i = \frac{\psi_i(1)}{\psi_1(1)} \neq 0$. (Note that $\mathrm{char}\,\mathsf{K} \nmid \psi_1(1)$.) Since ψ_1 and ψ_i are irreducible, it follows that

$$(\psi_i, \psi_1)_G = k_i(\psi_1, \psi_1)_G = k_i.$$

Since $(\psi_i, \psi_1)_G$ can only be 0 or 1 by the orthogonality relations, it follows that $k_i = 1$ and $\psi_i = \psi_1$. Hence, all the $\mathbf{F}_i$ are equivalent. Therefore, using an appropriate basis for the representation module, we have

$$\mathbf{D}((f,1)) = \mathbf{F}(f(1)) \otimes \cdots \otimes \mathbf{F}(f(p)).$$

Correspondingly, the representation module W of $\mathbf{D}$ has the form

$$\mathsf{W} = \mathsf{V} \underset{p}{\otimes \cdots \otimes} \mathsf{V},$$

where V is an irreducible $\mathsf{K}[G]$-module for the representation $\mathbf{F}$.

Let $t = (e, z)$, where $e(i) = 1$ for $i = 1, \ldots, p$ and $z = (1, 2, \ldots, p)$. We define the K-linear map a on $\mathsf{V} \underset{p}{\otimes \cdots \otimes} \mathsf{V}$ by

$$(v_1 \otimes \cdots \otimes v_p)a = v_p \otimes v_1 \otimes \cdots \otimes v_{p-1}$$

for $v_i \in \mathsf{V}$. Then

$$\begin{aligned}(v_1 \otimes \cdots \otimes v_p)a^{-1}\mathbf{D}((f,1))a &= v_1\mathbf{F}(f(p)) \otimes v_2\mathbf{F}(f(1)) \otimes \cdots \otimes v_p\mathbf{F}(f(p-1)) \\ &= (v_1 \otimes \cdots \otimes v_p)\mathbf{D}(t)^{-1}\mathbf{D}((f,1))\mathbf{D}(t).\end{aligned}$$

Since $\mathsf{W} = \mathsf{V} \underset{p}{\otimes \cdots \otimes} \mathsf{V}$ is irreducible as a $\mathsf{K}[D]$-module, it follows by Schur's Lemma that

$$a\mathbf{D}(t)^{-1} \in \mathrm{Hom}_{\mathsf{K}[D]}(\mathsf{W}, \mathsf{W}) = \{k1_{\mathsf{W}} \mid k \in \mathsf{K}^\times\}.$$

Hence, $\mathbf{D}(t) = ka$ for some $k \in \mathsf{K}^\times$.

By assumption, the representation $\mathbf{F}$ of G is monomial. Let $\{w_1, \dots, w_n\}$ be a K-basis of V such that

$$w_i\mathbf{F}(g) = a_i(g)w_{ig}.$$

Then the $w_{i_1} \otimes \cdots \otimes w_{i_p}$ with $1 \le i_j \le n$ constitute a K-basis of $\mathsf{W} = \underset{p}{\mathsf{V} \otimes \cdots \otimes \mathsf{V}}$. Letting $f(i) = g_i$, we have

$$(w_{i_1} \otimes \cdots \otimes w_{i_p})\mathbf{D}((f, 1)) = a_{i_1}(g_1) \cdots a_{i_p}(g_p) w_{i_1^{g_1}} \otimes \cdots \otimes w_{i_p^{g_p}}$$

and

$$(w_{i_1} \otimes \cdots \otimes w_{i_p})\mathbf{D}(t) = k\, w_{i_p} \otimes w_{i_1} \otimes \cdots \otimes w_{i_{p-1}}.$$

Therefore, $\mathbf{D}$ is a monomial representation of H by Lemma 18.9.

Case 2: Assume that $\mathbf{D}_D$ is reducible, but not all the irreducible constituents are equivalent. Then by Theorem 17.3 d), we have

$$\mathsf{W} = \mathsf{V}_1 \oplus \cdots \oplus \mathsf{V}_s,$$

where the homogeneous components V_i are permuted by H. We define

$$U = \{h \in H \mid \mathsf{V}_1\mathbf{D}(h) = \mathsf{V}_1\}.$$

By the irreducibility of $\mathbf{D}$ we have $U < H$, and it follows from $D \le U < H$ and $|H/D| = p$ that $D = U$. By Theorem 17.3 e), there exists a representation $\mathbf{X}$ of D such that $\mathbf{X}^H = \mathbf{D}$. Since G is an M-group, D is also an M-group by Theorem 18.8. Hence, there exists a one-dimensional representation $\mathbf{Y}$ of a subgroup of D such that $\mathbf{X} = \mathbf{Y}^D$. Then it follows by Lemma 16.2 b) that

$$\mathbf{D} = \mathbf{X}^H = (\mathbf{Y}^D)^H = \mathbf{Y}^H.$$

Case 3: Finally, assume that $\mathbf{D}_D$ is reducible and contains only one irreducible representation of D, and this with multiplicity s. By Theorem 17.5, we have

$$\mathbf{D}(h) = \mathbf{P}_1(h) \otimes \mathbf{P}_2(h),$$

where $\mathbf{P}_1$ is an irreducible projective representation of $H/D \cong Z$ of degrees s. Upon modifying $\mathbf{P}_1$ by appropriate scalar factors (and modifying $\mathbf{P}_2$ correspondingly), we may force $\mathbf{P}_1(t)^i = \mathbf{P}_1(t^i)$ and $\mathbf{P}_1(t)^p = 1$. Then $\mathbf{P}_1$ is an irreducible ordinary representation of the cyclic group Z. So $s = \deg \mathbf{P}_1 = 1$. Hence, $\mathbf{D}_D$ is irreducible and we are finished by Case 1. □

Theorem 18.11 (Dade). *Every solvable group can be embedded in an M-group.*

Proof. We prove the claim by induction on the group order. Let G be solvable, and let $N \lhd G$ with $|G/N| = p$. By our induction hypothesis, there exists an M-group H with $N \leq H$ (up to isomorphism). We let $Z = G/N$. By Theorem I.15.9 and Lemma I.15.8 b) (pg. 105), we have

$$G \leq N \wr_r Z \leq H \wr_r Z.$$

By Lemma 18.10, $H \wr_r Z$ is an M-group. □

Exercises

33) (Itô [273]) Show that the wreath product $G = Z \wr_r S_4$ is an M-group, where $|Z| = 2$ and S_4 is considered as a permutation group of degree 4.
(Hint: Show that:
   a) $G/\mathbf{Z}(G)$ is an M-group.
   b) G has a normal subgroup N with $|G/N| = 12$ such that G/N is a direct product of a group of order 2 with a dihedral group of order 6.
   c) G has a faithful, irreducible and monomial representation $\mathbf{D}$ of degree 4.
   d) Letting $\mathbf{D}_i$ $(i = 1, \ldots, 6)$ be the irreducible representations of G/N, the representations $\mathbf{D} \otimes \mathbf{D}_i$ are all the faithful, irreducible representations of G.)

34) (Shoda [382]) Let G be a metabelian group, so $G'' = 1$. If A is a maximal abelian normal subgroup of G with $G' \leq A$, and if $n \neq 1$ is the degree of a faithful[6] irreducible representation of G over the field of complex numbers, then $n = |G/A|$.

19 Brauer's theorems

All fields in this section are assumed to be subfields of the complex numbers.

Definition 19.1. The group F is called a p-elementary group if $F = \langle z \rangle \times P$ for some p-group P. By enlarging P if necessary, we may clearly assume that $p \nmid |z|$. We say that F is an elementary group if F is a p-elementary group for some prime number p. Subgroups of p-elementary groups are clearly again p-elementary. Elementary groups are always nilpotent, so they are M-groups by Theorem 18.5 b). (The reader should not confuse elementary groups with elementary abelian groups.)

Definition 19.2. Let R be a subring of the field $\mathbb{C}$ of complex numbers with $1 \in R$, and let $\chi_1, \ldots, \chi_h$ be the irreducible characters of G over $\mathbb{C}$. We define the character ring $\mathsf{X}_R(G)$ of G over R to be the set

[6] Correction 27 from errata.

$$\mathsf{X}_R(G) = \Big\{ \sum_{i=1}^{h} r_i \chi_i \mid r_i \in R \Big\},$$

where the ring operations are carried out pointwise. If $\mathbb{Z}$ is the ring of integers, then $\mathsf{X}_{\mathbb{Z}}(G)$ is the ring of generalized characters of G. We define

$$\mathsf{V}_R(G) = \Big\{ \sum_i r_i \psi_i^G \mid r_i \in R, \psi_i \text{ is an irreducible character of an elementary subgroup } F_i \text{ of } G \Big\}$$

and

$$\mathsf{U}_R(G) = \big\{ \chi \mid \chi \text{ is a class function on } G; \chi_F \in \mathsf{X}_R(F) \text{ for all elementary subgroups } F \text{ of } G \big\}.$$

Main Theorem 19.3 (Brauer).

a) *If χ is a character of G, then $\chi = \sum_i a_i \varphi_i^G$, where $a_i \in \mathbb{Z}$ and the φ_i are linear characters of some elementary subgroups of G. (Note that the a_i could be negative.)*

b) *Let μ be a class function on G. Then $\mu \in \mathsf{X}_R(G)$ if and only if for every elementary subgroup F of G, the restriction μ_F of μ to F lies in $\mathsf{X}_R(F)$.*

We prove Theorem 19.3 in a series of lemmas.

Lemma 19.4. *We have $\mathsf{V}_R(G) \subseteq \mathsf{X}_R(G) \subseteq \mathsf{U}_R(G)$, and $\mathsf{V}_R(G)$ is an ideal in the ring $\mathsf{U}_R(G)$.*

Proof. Clearly, $\mathsf{U}_R(G)$ is a ring with $\mathsf{X}_R(G) \subseteq \mathsf{U}_R(G)$. Let $\sum_i r_i \psi_i^G \in \mathsf{V}_R(G)$ and $\varphi \in \mathsf{U}_R(G)$. By Lemma 16.8, we have

$$\Big(\sum_i r_i \psi_i^G \Big) \varphi = \sum_i r_i (\psi_i \varphi_{F_i})^G \in \mathsf{V}_R(G).$$

So $\mathsf{V}_R(G)$ is an ideal in $\mathsf{U}_R(G)$. □

Lemma 19.5. *Let $F = A \times B$ be an elementary subgroup of G with $(|A|, |B|) = 1$, and assume A is abelian. Let m be the exponent of G, and let $S = \mathbb{Z}[\epsilon]$, where ϵ is a primitive m-th root of unity. We claim:*

For every $a \in A$ there exists some $\Phi = \Phi_a \in \mathsf{V}_S(G)$ with the following properties:

(1) $\Phi(g) \in \mathbb{Z}$ *for all* $g \in G$.

(2) $\Phi(g) = 0$ *if g is not conjugate to an element of aB.*

(3) $\Phi(a) = [\mathbf{C}_G(a) : B]$.

Proof. Let $\tau_1, \ldots, \tau_n$ be the distinct irreducible characters of A. The values $\tau_i(a)$ are m-th roots of unity, so they lie in $S = \mathbb{Z}[\epsilon]$. We define characters μ_i of $F = A \times B$ by $\mu_i(ab) = \tau_i(a)$ for $a \in A$ and $b \in B$.

Let a be a fixed element of A. We then define

$$\psi = \sum_{i=1}^{n} \overline{\mu_i(a)}\mu_i.$$

Since $\overline{\mu_i(a)} \in S$, we clearly have $\psi \in \mathsf{X}_S(F)$. For $x \in A$ and $y \in B$, we have

$$\psi(xy) = \sum_{i=1}^{n} \overline{\mu_i(a)}\mu_i(xy) = \sum_{i=1}^{n} \overline{\tau_i(a)}\tau_i(x).$$

The orthogonality relations imply

$$\psi(xy) = \begin{cases} |A| & \text{for } x = a \\ 0 & \text{for } x \neq a. \end{cases}$$

Letting $\Phi = \psi^G$, we have

$$\Phi = \psi^G \in \mathsf{V}_S(G)$$

and

$$\Phi(g) = \frac{1}{|F|} \sum_{t^{-1}gt \in F} \psi(t^{-1}gt).$$

If $t^{-1}gt \notin aB$, then $\psi(t^{-1}gt) = 0$. So

$$\Phi(g) = \frac{1}{|F|} \sum_{t^{-1}gt \in aB} \psi(t^{-1}gt) = \frac{|A||T|}{|F|},$$

where $T = \{t \in G \mid t^{-1}gt \in aB\}$. If $t \in T$, then clearly $tB \subseteq T$. Hence, T consists of full cosets of B, and it follows that $|T| = s|B|$ for some natural number s. Therefore,

$$\Phi(g) = \frac{|A||B|s}{|F|} = s \in \mathbb{Z}.$$

With this, (1) and (2) are proved. We have

$$\Phi(a) = \frac{1}{|F|} \sum_{t^{-1}at \in aB} \psi(t^{-1}at).$$

Since $(|A|, |B|) = 1$, if $t^{-1}at = ab$ then $b = 1$ and $t \in \mathbf{C}_G(a)$. Hence, we obtain

$$\Phi(a) = \frac{|A||\mathbf{C}_G(a)|}{|F|} = [\mathbf{C}_G(a) : B],$$

and (3) is proved, as well. □

Lemma 19.6. *For every $g \in G$, there exists a unique decomposition $g = g_1 g_2$ with the following properties:*
(1) *$|g_1|$ is a power of p.*
(2) *$|g_2|$ is coprime to p.*
(3) $g_1 g_2 = g_2 g_1 = g$.

Proof. By Theorem I.13.12 (pg. 85), the cyclic group $\langle g \rangle$ admits a direct decomposition $\langle g \rangle = Z_1 \times Z_2$, where $|Z_1| = p^n$ and $p \nmid |Z_2|$. Let $g = g_1 g_2$ with $g_i \in Z_i$. Then the g_i have the properties (1)–(3).

Suppose $g = g_1' g_2'$ with $|g_1'| = p^{n'}$, $|g_2'| = m' \not\equiv 0 \bmod p$ and $g_1' g_2' = g_2' g_1'$. Then $\langle g \rangle = \langle g_1' \rangle \times \langle g_2' \rangle$, and so $g_i' \in \langle g \rangle$. Therefore, $g_i = g_i'$ and the decomposition is unique. □

Definition 19.7. An element $g \in G$ is called a p'-element if its order $|g|$ is coprime to p; an element $g \in G$ is called a p-element if its order $|g|$ is a power of p. If $g = g_1 g_2$ is the decomposition in Lemma 19.6, then we call g_1 the p-part of g and g_2 the p'-part of g.

Lemma 19.8. *Let $S = \mathbb{Z}[\epsilon]$ as in Lemma 19.5. Then for every prime number p there exists an $\alpha \in \mathsf{V}_S(G)$ such that $\alpha(g) \in \mathbb{Z}$ and $\alpha(g) \equiv 1 \bmod p$ for all $g \in G$.*

Proof. Let C_i $(i = 1, \dots, k)$ be the conjugacy classes of G that consist of p'-elements. We first show:

a) For every i, there exists a $\tau_i \in \mathsf{V}_S(G)$ with the following properties:
(1) $\tau_i(g) \in \mathbb{Z}$ for all $g \in G$.
(2) $\tau_i(g) = 0$ if the p'-part of g does not lie in C_i.
(3) $\tau_i(g) \equiv 1 \bmod p$ if the p'-part of g lies in C_i.

Let $C_i = \{a_i^g \mid g \in G\}$, and let B_i be a Sylow p-subgroup of $\mathbf{C}_G(a_i)$. Then $F_i = \langle a_i \rangle \times B_i$ is a p-elementary subgroup of G. By Lemma 19.5, there exists a $\Phi_i \in \mathsf{V}_S(G)$ such that
(i) $\Phi_i(g) \in \mathbb{Z}$ for all $g \in G$.
(ii) $\Phi_i(g) = 0$ if g is not conjugate to an element of $a_i B_i$.
(iii) $\Phi_i(a_i) = [\mathbf{C}_G(a_i) : B_i] \not\equiv 0 \bmod p$.

We choose $m_i \in \mathbb{Z}$ such that $m_i \Phi_i(a_i) \equiv 1 \bmod p$, and we let $\tau_i = m_i \Phi_i$. It is clear that (1) is satisfied.

Let $g = g_1 g_2$ be the decomposition of g into its p-part g_1 and its p'-part g_2 according to Lemma 19.6, and assume that $g_2 \notin C_i$. If we had

$$t^{-1} g t = a_i b \in a_i B_i$$

for some $t \in G$, then since $a_i b = b a_i$, it would immediately follow from the uniqueness of the decomposition in Lemma 19.6 that $g_2 = t a_i t^{-1} \in C_i$. But this contradicts our assumption that $g_2 \notin C_i$.

Hence, if $g_2 \notin C_i$, then it follows that g is not conjugate to any element of $a_i B_i$. Then $\Phi_i(g) = 0$ by (ii), and (2) holds.

It remains to prove (3):

Let $g^t = a_i b$, where b is a p-element in $\mathbf{C}_G(a_i)$. We have already shown that $\tau_i(a_i) = m_i\Phi_i(a_i) \equiv 1 \bmod p$. So it suffices to prove that $\tau_i(a_i) \equiv \tau_i(a_i b) \bmod p$.

Let $|b| = p^l$, and let $H = \langle a_i b\rangle$. Since $\tau_i \in \mathsf{V}_S(G) \subseteq \mathsf{X}_S(G)$, we have $(\tau_i)_H \in \mathsf{X}_S(H)$. Therefore, we have for each fixed i a decomposition

$$(\tau_i)_H = \sum_j r_j\psi_j,$$

where $r_j \in S$ and the ψ_j are linear characters of H over the complex numbers. Let $\mathfrak{p}$ be a prime ideal dividing the principal ideal (p) in the ring $\mathbb{Z}[\epsilon]$. Then it is well known that $\mathbb{Z}[\epsilon]/\mathfrak{p}$ is a finite field of characteristic p, say $|\mathbb{Z}[\epsilon]/\mathfrak{p}| = p^f$. Let the natural number n be a multiple of f with $p^n \geq p^l = |b|$. Then $x^{p^n} \equiv x \bmod \mathfrak{p}$ for all $x \in \mathbb{Z}[\epsilon]$. It then follows that

$$\begin{aligned}\tau_i(a_i b) \equiv \tau_i(a_i b)^{p^n} &\equiv \sum_j r_j^{p^n}\psi_j(a_i^{p^n} b^{p^n}) = \sum_j r_j^{p^n}\psi_j(a_i)^{p^n}\\ &\equiv \Big(\sum_j r_j\psi_j(a_i)\Big)^{p^n} = \tau_i(a_i)^{p^n} \equiv \tau_i(a_i) \mod \mathfrak{p}.\end{aligned}$$

Since τ_i only takes values in $\mathbb{Z}$ on G by (1), it follows from $\mathbb{Z} \cap \mathfrak{p} = p\mathbb{Z}$ that $\tau_i(a_i b) \equiv \tau_i(a_i) \bmod p$.

With this, the claim in (3) of a) is proved.

b) We define $\alpha = \sum_{i=1}^{k} \tau_i$. By a), $\alpha(g) \in \mathbb{Z}$ for all $g \in G$. If the p'-part of g lies in C_i, then $\tau_i(g) \equiv 1 \bmod p$ and $\tau_j(g) = 0$ for $j \neq i$. This implies that $\alpha(g) \equiv 1 \bmod p$, as desired. □

Lemma 19.9. *If α is a class function on G with $\alpha(g) \in |G|S$ for all $g \in G$, then $\alpha \in \mathsf{V}_S(G)$.*

Proof. Let $g_1, \ldots, g_h$ be representatives of the conjugacy classes of G. By Lemma 19.5 (with $A = \langle g_i\rangle$ and $B = 1$), there exists a $\Phi_i \in \mathsf{V}_S(G)$ such that $\Phi_i(g_j) = 0$ for $i \neq j$ and $\Phi_i(g_i) = |\mathbf{C}_G(g_i)|$. By our assumption, $\alpha(g_i) = |G|r_i$ for some $r_i \in S$. It follows that

$$\alpha = \sum_{i=1}^{h} \frac{|G|}{|\mathbf{C}_G(g_i)|} r_i\Phi_i \in \mathsf{V}_S(G). \qquad \square$$

19.10 (Proof of Theorem 19.3). Every elementary subgroup of G is nilpotent, so is an M-group by Theorem 18.5. If ψ_i is an irreducible character of the elementary subgroup F_i of G, then $\psi_i = \varphi_i^{F_i}$ for some linear character φ_i of a subgroup of F_i, which is necessarily again an elementary subgroup. To prove Theorem 19.3, it suffices to prove that $\mathsf{V}_R(G) = \mathsf{X}_R(G) = \mathsf{U}_R(G)$. By Lemma 19.4, we have $\mathsf{V}_R(G) \subseteq \mathsf{X}_R(G) \subseteq \mathsf{U}_R(G)$, and $\mathsf{V}_R(G)$ is an ideal in $\mathsf{U}_R(G)$. If $1 \in \mathsf{V}_{\mathbb{Z}}(G)$, then since $\mathsf{V}_{\mathbb{Z}}(G) \subseteq \mathsf{V}_R(G)$, the claim $\mathsf{V}_R(G) = \mathsf{X}_R(G) = \mathsf{U}_R(G)$ follows. We prove the following:

a) Let $|G| = p^n m$ with $p \nmid m$, and let $S = \mathbb{Z}[\epsilon]$ as before. If we denote by m the constant class function that takes the value m everywhere, then $m \in \mathsf{V}_S(G)$:

By Lemma 19.8, there exists an $\alpha \in \mathsf{V}_S(G)$ with $\alpha(g) \in \mathbb{Z}$ and $\alpha(g) \equiv 1 \bmod p$ for all $g \in G$. Then

$$\alpha(g)^{p^n} \equiv 1 \bmod p^n.$$

We define the class function β on G by $\alpha^{p^n} = 1 + \beta$. Then $m\beta(g) \in |G|S$ for all $g \in G$. Therefore, $m\beta \in \mathsf{V}_S(G)$ by Lemma 19.9. Hence,

$$m = m\alpha^{p^n} - m\beta \in \mathsf{V}_S(G),$$

as well.

b) We have $1 \in \mathsf{V}_S(G)$:

Let $|G| = \prod_{i=1}^{s} p_i^{a_i}$, and let $m_j = \prod_{i \neq j} p_i^{a_i}$. Since the integers m_i are coprime, we have $1 = \sum_{i=1}^{s} a_i m_i$ for some $a_i \in \mathbb{Z}$. By a), we have $m_i \in \mathsf{V}_S(G)$. So $1 \in \mathsf{V}_S(G)$, as well.

c) We have $1 \in \mathsf{V}_{\mathbb{Z}}(G)$:

There exists a $\mathbb{Z}$-basis $\{\epsilon_1, \ldots, \epsilon_t\}$ of $\mathbb{Z}[\epsilon]$ with $\epsilon_1 = 1$. (In fact, the ϵ_i can be taken to be powers of ϵ; see Hasse [225], pg. 507.) Then the ϵ_i, when viewed as constant class functions in $\mathsf{X}_S(G)$, are linearly independent over $\mathsf{X}_{\mathbb{Z}}(G)$:

Let $\chi_1, \ldots, \chi_h$ be the absolutely irreducible complex characters of G, and suppose that

$$\sum_{i=1}^{t} \epsilon_i \sum_{j=1}^{h} a_{ij} \chi_j = 0$$

with $a_{ij} \in \mathbb{Z}$. Since the χ_j are linearly independent over the complex numbers, we have

$$\sum_{i=1}^{t} a_{ij} \epsilon_i = 0 \qquad (j = 1, \ldots, h),$$

and then $a_{ij} = 0$ by the linear independence of the ϵ_i over $\mathbb{Z}$.

By b), we have $1 \in \mathsf{V}_S(G)$, so $1 = \sum_{i=1}^{k} \psi_i^G$ for some $\psi_i \in \mathsf{X}_S(F_i)$, where the F_i are elementary subgroups of G. Let

$$\psi_i = \sum_{j=1}^{t} \epsilon_j \psi_{ij}$$

with $\psi_{ij} \in \mathsf{X}_{\mathbb{Z}}(F_i)$. Since $\psi_{ij}^G \in \mathsf{X}_{\mathbb{Z}}(G)$ and $\epsilon_1 = 1$, it then follows from

$$1 = \sum_{j=1}^{t} \epsilon_j \sum_{i=1}^{k} \psi_{ij}^G$$

and our previous observation that

$$1 = \sum_{i=1}^{k} \psi_{i1}^G \in \mathsf{V}_{\mathbb{Z}}(G). \qquad \square$$

The most important group theoretical application of Theorem 19.3 a) is Brauer's theorem regarding the splitting field of group rings:

Main Theorem 19.11 (R. Brauer). *Let G be a finite group of exponent m, and let ϵ be a primitive m-th root of unity. Then $\mathbb{Q}(\epsilon)$ is a splitting field for G.*

Proof. Let $\mathbb{Q}$ be the field of rational numbers, let $\mathsf{F} = \mathbb{Q}(\epsilon)$ with ϵ a primitive m-th root of unity, and let $\mathbb{C}$ be the field of complex numbers. Let V be an irreducible $\mathbb{C}[G]$-module with character μ. We must show that μ is a character over F. By Theorem 19.3 a),

$$\mu = \sum_i a_i \varphi_i^G \qquad (\text{with } a_i \in \mathbb{Z}),$$

where the φ_i are linear characters of elementary subgroups F_i of G. Clearly, the φ_i are characters of representations that can be realized over $\mathsf{F} = \mathbb{Q}(\epsilon)$. So the φ_i^G are also characters of representations of G over F.

Let $\tau_1 = \sum\limits_{a_i>0} a_i \varphi_i^G$ and $\tau_2 = -\sum\limits_{a_i<0} a_i \varphi_i^G$.

Then τ_1 and τ_2 are characters of representations of $\mathsf{F}[G]$. If W_1 and W_2 are the corresponding $\mathsf{F}[G]$-modules, then it follows by comparing characters that

$$(\mathsf{W}_1)_{\mathbb{C}} \cong (\mathsf{W}_2)_{\mathbb{C}} \oplus \mathsf{V},$$

where $(\mathsf{W}_i)_{\mathbb{C}} = \mathbb{C} \otimes_{\mathsf{F}} \mathsf{W}_i$. Among all $\mathsf{F}[G]$-modules with the property that $(\mathsf{V}_1)_{\mathbb{C}} \cong (\mathsf{V}_2)_{\mathbb{C}} \oplus \mathsf{V}$, we choose a pair $(\mathsf{V}_1, \mathsf{V}_2)$ such that V_2 has dimension as small as possible. Assume for contradiction that $\mathsf{V}_2 \neq 0$. Let U be an irreducible $\mathsf{F}[G]$-submodule of V_2. If χ is the character of U and μ_i is the character of V_i $(i = 1, 2)$, then $(\mu_2, \chi) > 0$ by the orthogonality relation of Theorem 5.8 a). Then $(\mu_1, \chi) > 0$ as well, and so U is a composition factor of V_1. Then by Maschke's Theorem, we have direct decompositions

$$\mathsf{V}_1 \cong \mathsf{U} \oplus \mathsf{V}_1' \quad \text{and} \quad \mathsf{V}_2 \cong \mathsf{U} \oplus \mathsf{V}_2'.$$

It follows that $\mathsf{U}_{\mathbb{C}} \oplus (\mathsf{V}_1')_{\mathbb{C}} \cong \mathsf{U}_{\mathbb{C}} \oplus (\mathsf{V}_2')_{\mathbb{C}} \oplus \mathsf{V}$, and then by the Krull–Remak–Schmidt Theorem I.12.4 (pg. 73), we have $(\mathsf{V}_1')_{\mathbb{C}} \cong (\mathsf{V}_2')_{\mathbb{C}} \oplus \mathsf{V}$.

But this contradicts the choice of V_2. So $\mathsf{V}_2 = 0$ and $\mathsf{V} = (\mathsf{V}_1)_{\mathbb{C}}$. Since V_1 is an $\mathsf{F}[G]$-module, this proves the claim. $\square$

The question of which fields a given absolutely irreducible representation can be realized over has been much studied. We mention without proof three results:

(1) (Fong [142]) Let $|G| = p^n m$ with $p \nmid m$, and let $\mathbf{D}$ be an absolutely irreducible representation of G over the field of complex numbers with character χ. Let ϵ be a primitive m-th root of unity. If $p > 2$, then $\mathbf{D}$ can be realized over the field $\mathbb{Q}(\epsilon, \chi(g) \mid g \in G)$; if $p = 2$, then $\mathbf{D}$ can be realized over $\mathbb{Q}(\epsilon, \sqrt{-1}, \chi(g) \mid g \in G)$. (For $|G| = p^n$, this result can already be found in Roquette [358].)

(2) (Solomon [388]) Let $|G| = \prod_{i=1}^{r} p_i^{a_i}$ be the prime factorization of $|G|$. Let $k = p_1 \cdots p_r$, and let ρ be a primitive k-th root of unity. Let $\mathbf{D}$ be an absolutely irreducible representation of G over the complex numbers with character χ. If $|G|$ is odd, then $\mathbf{D}$ can be realized over $\mathbb{Q}(\rho, \chi(g) \mid g \in G)$; if $|G|$ is even, then $\mathbf{D}$ can be realized over $\mathbb{Q}(\rho, \sqrt{-1}, \chi(g) \mid g \in G)$.

(3) (Speiser [391]) If $\mathbf{D}$ is an absolutely irreducible representation of G with odd degree over the complex numbers and real character χ, then $\mathbf{D}$ can be realized over $\mathbb{Q}(\chi(g) \mid g \in G)$. (See also Feit [126], §16.)

Theorem 19.3 b) is sometimes useful for recognizing a class function as a character. As an application of Theorem 19.3 b), we give a new proof of Frobenius' Theorem 7.6:

19.12 (Application). Let G be a Frobenius group with Frobenius complement H; that is, $H \cap H^g = 1$ for all $g \in G - H$. We let $[G : H] = n$ and show that $F = G - \bigcup_{g \in G} (H - 1)^g$ is a normal subgroup of G.

a) $|H|$ and n are coprime:

Assume for contradiction that p is a common prime divisor of $|H|$ and n. Let P be a Sylow p-subgroup of H, and let P^* be a Sylow p-subgroup of G with $P < P^*$. Then $P < \mathbf{N}_{P^*}(P)$. If $x \in \mathbf{N}_{P^*}(P) - P$, then it follows that

$$1 \neq P = P^x \leq H \cap H^x$$

and $x \notin H$, contradicting our assumption. So $|H|$ and n are coprime.

b) If U is a nilpotent subgroup of G, then $|U|$ divides n or U lies in a conjugate of H:

Suppose p is a common prime divisor of $|U|$ and $|H|$, and let P be the Sylow p-subgroup of U. By a), we may assume that $P \leq H$ by passing to a conjugate of U if necessary. Then for all $u \in U$, we have

$$1 \neq P = P^u \leq H \cap H^u,$$

so $u \in H$ and thus $U \leq H$.

c) Let ψ be an irreducible character of H. Then τ defined by

$$\tau(g) = \begin{cases} \psi(h) & \text{for } g = h^x \in H^x \\ \psi(1) & \text{otherwise} \end{cases}$$

is an irreducible character of G:

Since $H \cap H^x = 1$ for $x \notin H$, it is easy to see that τ is a well-defined class function on G. We have

$$(\tau, \tau)_G = \frac{1}{|G|}\left\{n\,\psi(1)^2 + n \sum_{1 \neq h \in H} |\psi(h)|^2\right\} = (\psi, \psi)_H = 1.$$

To prove that τ is a generalized character of G, we show that for every nilpotent subgroup U of G, the restriction τ_U is a character of U. By b), only the following two cases are possible:

(1) $U \leq H^x$ for some $x \in G$: Then $\tau(u) = \psi(u^{x^{-1}})$, so τ_U is a character of U.

(2) $|U|$ divides n: By a), we then have $U \cap H^x = 1$ for all $x \in G$. So $\tau(u) = \psi(1)$ for all $u \in U$.

Now by Theorem 19.3 b), τ is a generalized character of G. Since $(\tau, \tau)_G = 1$ and $\tau(1) = \psi(1) > 0$, τ is actually an irreducible character of G.

d) Let $\psi_1, \ldots, \psi_h$ be the irreducible characters of H. For every ψ_i, we construct an irreducible character τ_i of G according to the procedure in c). As in part b) of the proof of Theorem 7.5, one sees that $\bigcap_i \ker \tau_i$ is the desired Frobenius kernel of G. □

The proof method of Application 19.12 can also be used in more general situations to prove the existence of normal subgroups (see R. Brauer [57], Dade [103], Suzuki [413]).

As a second application of Theorem 19.3 b), we prove another theorem of Frobenius.

Theorem 19.13 (Frobenius [151]). *Let G be a group of order $|G|$, let n be a natural number, and let L be a union of conjugacy classes of G. We define the class function $\theta_{G,L}$ on G by*

$$\theta_{G,L}(g) = \begin{cases} \frac{|G|}{(|G|,n)} & \text{for } g^n \in L \\ 0 & \text{for } g^n \notin L. \end{cases}$$

Let ϵ be a primitive $|G|$-th root of unity, and let $S = \mathbb{Z}[\epsilon]$. Then $\theta_{G,L}$ lies in $\mathsf{X}_S(G)$.

Proof (R. Brauer [52]). The claim will be reduced in a series of steps.

a) We may assume that G is an elementary group:

Let F be an arbitrary elementary subgroup of G. For $f \in F$, we have

$$\theta_{G,L}(f) = \begin{cases} \frac{|G|}{(|G|,n)} & \text{for } f^n \in L \cap F \\ 0 & \text{for } f^n \notin L \cap F. \end{cases}$$

So $(\theta_{G,L})_F = c\theta_{F,F\cap L}$ with $c = \frac{|G|}{(|G|,n)} \frac{(n,|F|)}{|F|} \in \mathbb{Z}$. If we can show that $\theta_{F,F\cap L}$ lies in $\mathsf{X}_S(F)$ for all elementary subgroups F of G, then it follows by Theorem 19.3 b) that $\theta_{G,L} \in \mathsf{X}_S(G)$.

b) We may further assume that G is a p-group:

Let G be elementary, and let $G = G_1 \times G_2$ with $(|G_1|, |G_2|) = 1$. Then

$$L_i = \{l_i \mid l_1 l_2 \in L, l_1 \in G_1, l_2 \in G_2\}$$

is a union of conjugacy classes of G_i. It follows that

$$\theta_{G,L}(g_1 g_2) = \theta_{G_1,L_1}(g_1)\theta_{G_2,L_2}(g_2).$$

Suppose that

$$\theta_{G_i,L_i} = \sum_j a_{ij}\chi_{ij} \quad (i = 1, 2)$$

for some irreducible characters χ_{ij} of G_i with $a_{ij} \in S$. Then

$$\theta_{G,L}(g_1 g_2) = \theta_{G_1,L_1}(g_1)\theta_{G_2,L_2}(g_2) = \sum_{j,k} a_{1j}a_{2k}\psi_{jk}(g_1 g_2)$$

with $\psi_{jk}(g_1 g_2) = \chi_{1j}(g_1)\chi_{2k}(g_2)$. By Theorem 10.3, ψ_{jk} is an irreducible character of $G_1 \times G_2$. So $\theta_{G,L}$ lies in $\mathsf{X}_S(G_1 \times G_2)$.

c) We may assume that 1 does not lie in L: If L_1 and L_2 are unions of conjugacy classes with $L_1 \cap L_2 = \emptyset$, then clearly

$$\theta_{G,L_1 \cup L_2} = \theta_{G,L_1} + \theta_{G,L_2}.$$

If $K_1 = 1, K_2, \ldots, K_h$ are the conjugacy classes of G, then

$$\sum_{i=1}^{h} \theta_{G,K_i}(g) = \frac{|G|}{(|G|, n)}.$$

So if the θ_{G,K_i} lie in $\mathsf{X}_S(G)$ for $i > 1$, then θ_{G,K_1}, and hence every $\theta_{G,L}$, does as well.

d) Now let G be a p-group, and let K be a conjugacy class of G with $K \neq 1$. We must show that for every irreducible character χ of G, we have $(\theta_{G,K}, \chi)_G \in S$. We let $\theta = \theta_{G,K}$. Since G is an M-group, χ is induced by a linear character λ of a subgroup H of G. It follows from Frobenius reciprocity that

$$(\theta, \chi)_G = (\theta, \lambda^G)_G = (\theta_H, \lambda)_H = \frac{|G|}{|H|}\frac{1}{(|G|, n)} \sum_{h^n \in H \cap K} \lambda(h^{-1}).$$

Let $n = p^b m$ with $p \nmid m$. If $p^b \geq |H|$, then $h^n = 1 \notin K$ for all $h \in H$ and so $(\theta, \chi)_G = 0$. So assume that $p^b < |H|$, and then $(|G|, n) = p^b$. Let t be a natural number with $mt \equiv 1 \bmod |G|$. We define the subset L of H by

$$L = \{h^t \mid h \in H \cap K\}.$$

Since $p \nmid t$ and $1 \notin K$, we have $1 \notin L$. Clearly, L is a union of conjugacy classes of H. If $h^n \in H \cap K$, then $h^{p^b} = h^{nt} \in L$. Conversely, if $h^{p^b} = (h')^t \in L$ with $h' \in H \cap K$, then $h^n = h^{p^b m} = (h')^{tm} = h' \in H \cap K$. So

$$p^{-b} \sum_{h^n \in H \cap K} \lambda(h^{-1}) = p^{-b} \sum_{h^{p^b} \in L} \lambda(h^{-1}).$$

This reduces our proof to verifying the following claim:

Let H be a p-group of order p^a, let L be a union of conjugacy classes of H with $1 \notin L$, and let λ be a linear character of H. If $b \leq a$, then

$$p^{-b} \sum_{h^{p^b} \in L} \lambda(h^{-1}) \in S.$$

We define an equivalence relation $\sim$ on the set $A = \{h \in H \mid h^{p^b} \in L\}$ by letting $h_1 \sim h_2$ if $\langle h_1 \rangle = \langle h_2 \rangle$ and $h_1^{p^b} = h_2^{p^b}$. Let $A = A_1 \cup \cdots \cup A_t$ be the decomposition of A into equivalence classes with respect to $\sim$. We show that $p^{-b} \sum_{h \in A_i} \lambda(h^{-1})$ lies in S.

Let $h \in A_i$. Since $h^{p^b} \in L$ and $1 \notin L$, $h^{p^b} \neq 1$. So $|h| = p^c > p^b$. Suppose $h^j \in A_i$. This forces

$$h^{jp^b} = h^{p^b} \quad \text{and} \quad \langle h \rangle = \langle h^j \rangle.$$

These equations are satisfied if and only if j has the form $j = 1 + mp^{c-b}$. If we let $\lambda(h^{p^{c-b}}) = \rho$, then ρ is a (not necessarily primitive) p^b-th root of unity. It follows that

$$\sum_{h^j \in A_i} \lambda(h^{-j}) = \lambda(h^{-1}) \sum_{m=0}^{p^b-1} \lambda(h^{mp^{c-b}}) = \lambda(h^{-1}) \sum_{m=0}^{p^b-1} \rho^m.$$

If $\rho = 1$, then

$$p^{-b} \sum_{h^j \in A_i} \lambda(h^{-j}) = \lambda(h^{-1}) \in \mathbb{Z}[\epsilon] = S.$$

If ρ is a primitive p^d-th root of unity with $0 < d \leq b$, then

$$p^{-b} \sum_{h^j \in A_i} \lambda(h^{-j}) = p^{-d} \lambda(h^{-1}) \sum_{m=0}^{p^d-1} \rho^m = 0.$$

With this, our claim is verified in all cases. □

For an elementary proof of Theorem 19.13, see Solomon [389]; see also Frobenius [151]. The reader may compare the following theorem to Exercise I.38 (pg. 45).

Theorem 19.14. *Let K be a conjugacy class of G, and let n be a divisor of $|G|$. Then the number t of elements $g \in G$ with $g^n \in K$ is a multiple of n.*

Proof. We define $\theta_{G,K}$ as in Theorem 19.13. Then

$$(\theta_{G,K}, 1)_G = \frac{1}{n}|\{g \in G \mid g^n \in K\}| = \frac{t}{n}$$

is a rational number lying in $\mathbb{Z}[\epsilon]$ by Theorem 19.13, so it is an algebraic integer. It is well known that this implies $\frac{t}{n}$ is an integer. □

Remark 19.15. Let n be a divisor of $|G|$, and suppose that

$$|\{g \in G \mid g^n = 1\}| = n.$$

It was conjectured already by Frobenius that in this situation the set

$$\{g \in G \mid g^n = 1\}$$

is a (necessarily characteristic) subgroup of G. If G is solvable, then this is correct (see M. Hall [204], pg. 145). For nonsolvable groups, this conjecture has not been proved at the time of writing.

If G is a Frobenius group with Frobenius complement H and $[G : H] = n$, then $(n, |H|) = 1$ (see part a) of the proof of Application 19.12). The number of solutions of $g^n = 1$ with $g \in G$ is a multiple of n by Theorem 19.14, and since $(n, |H|) = 1$, all such g lie in $G - \bigcup_{g \in G} (H - 1)^g$. It follows that

$$\begin{aligned} n \leq |\{g \in G \mid g^n = 1\}| &\leq |G - \bigcup_{g \in G} (H-1)^g| \\ &= |G| - [G : H](|H| - 1) = [G : H] = n. \end{aligned}$$

If the conjecture in Remark 19.15 were true, it would follow that

$$\{g \in G \mid g^n = 1\}$$

is a subgroup of G. This would be a new proof for the existence of the Frobenius kernel of G.

20 Characters of permutation groups

Definition 20.1. Let G be a permutation group on the set $\Omega = \{1, \ldots, n\}$, and let V be a vector space of dimension n over the field K with a basis $\{v_1, \ldots, v_n\}$. We give V the structure of a $\mathsf{K}[G]$-module by defining $v_i g = v_{ig}$ for $g \in G$. If π is the character of the representation of G on V, then it is clear that $\pi(g)$ is the number of

fixed points of g on Ω when char $\mathsf{K} = 0$. We call π the permutation character of the permutation group G.

Theorem 20.2. *Let G be a permutation group on $\Omega = \{1, \dots, n\}$, and let π be the permutation character of G over the field K of characteristic 0.*

a) *If t is the number of orbits of G on Ω, then* $\sum_{g \in G} \pi(g) = t|G|$.

b) *G is transitive on Ω if and only if $(\pi, 1)_G = 1$; that is, if and only if the trivial character is a constituent of π with multiplicity* 1.

c) *If G is transitive on Ω and s is the number of orbits of the stabilizer G_1 on Ω (including the orbit $\{1\}$), then*

$$\sum_{g \in G} \pi(g)^2 = s|G|.$$

d) *Assume G is transitive on Ω. Then G is 2-transitive on Ω if and only if $\pi = 1 + \chi$ for some nontrivial absolutely irreducible character χ of G.*

Proof. We have already proved a) in Theorem 13.4, and b) follows trivially from a).

c) If G is transitive on Ω, then the stabilizers G_i ($i = 1, \dots, n$) are conjugate in G, and thus for every i the set Ω decomposes into exactly s orbits under G_i. Using a), it now follows that

$$\sum_{g \in G} \pi(g)^2 = \sum_{i=1}^{n} \sum_{g \in G_i} \pi(g) = \sum_{i=1}^{n} s|G_i| = s|G|.$$

d) Let $\pi = \sum_{i=1}^{h} a_i \chi_i$ be the decomposition of π into absolutely irreducible characters χ_i of G, where $\chi_1 = 1$. Since G is transitive, we have by b) that

$$1 = (\pi, 1)_G = a_1.$$

By c), the 2-transitivity of G is equivalent to

$$2 = (\pi, \pi)_G = 1 + \sum_{i=2}^{h} a_i^2.$$

This forces $\pi = 1 + \chi$ for some absolutely irreducible character $\chi \neq 1$ of G. □

If char $\mathsf{K} \neq 0$, then some of the conclusions in Theorem 20.2 are no longer true:

Example 20.3. Let G be a permutation group on $\Omega = \{1, \dots, n\}$, and let K be a field of characteristic p. Let V be a vector space of dimension n over K with basis $\{v_1, \dots, v_n\}$. We give V the structure of a $\mathsf{K}[G]$-module as in Definition 20.1.

We define $w = \sum_{i=1}^{n} v_i$ and $\mathsf{W} = \langle w \rangle$. Since $wg = w$ for all $g \in G$, the representation of G on W is the trivial representation. We define a subspace U of codimension 1 in V by

$$\mathsf{U} = \langle v_i - v_n \mid i = 1, \ldots, n-1 \rangle.$$

Since

$$(v_i - v_n)g = (v_{ig} - v_n) - (v_{ng} - v_n) \in \mathsf{U},$$

the subspace U is a $\mathsf{K}[G]$-submodule of V. Furthermore, since

$$v_i g - v_i = (v_{ig} - v_n) - (v_i - v_n) \in \mathsf{U},$$

the representation of G on V/U is the trivial representation.

If char K is a divisor of n, then

$$w = \sum_{i=1}^{n} v_i = \sum_{i=1}^{n} (v_i - v_n) \in \mathsf{U},$$

and so $\mathsf{W} \subseteq \mathsf{U}$. This means V contains a $\mathsf{K}[G]$-composition factor corresponding to the trivial representation with multiplicity at least 2, even when G is multiply transitive. That is, the conclusion of Theorem 20.2 b) is no longer correct. On the other hand, if char K is not a divisor of n, then $w \notin \mathsf{U}$ and therefore $\mathsf{V} = \mathsf{U} \oplus \mathsf{W}$.

In the rest of this section, all fields under consideration are assumed to have characteristic 0. Multiple transitivity can be read off from the permutation character:

Theorem 20.4. *Let G be a permutation group on Ω with permutation character π. We define integers a_{nk} by*

$$x(x-1)(x-2)\cdots(x-k+1) = \sum_{n=1}^{k} a_{nk} x^n;$$

in particular, $a_{kk} = 1$. Furthermore, we define integers $\tau(k)$ recursively by defining $\tau(1) = 1$ and

$$\tau(k) = 1 - \sum_{n=1}^{k-1} a_{nk} \tau(n).$$

(*So we have $\tau(1) = 1$, $\tau(2) = 2$, $\tau(3) = 5$, $\tau(4) = 15$.*)

We claim:

a) *G is k-transitive on Ω if and only if*

$$\sum_{g \in G} \pi(g)(\pi(g) - 1) \cdots (\pi(g) - k + 1) = |G|.$$

b) *G is k-transitive on Ω if and only if for all i with $1 \le i \le k$ we have*

$$\sum_{g \in G} \pi(g)^i = |G| \tau(i).$$

Proof. a) Let

$$\Omega^{[k]} = \{(a_1, \ldots, a_k) \mid a_i \in \Omega, \text{ all } a_i \text{ are pairwise distinct}\}.$$

Upon defining

$$(a_1, \ldots, a_k)^{g^{[k]}} = (a_1^g, \ldots, a_k^g),$$

we obtain a permutation group $G^{[k]}$ on $\Omega^{[k]}$. By Definition II.1.10 (pg. 158), G is k-transitive on Ω if and only if $G^{[k]}$ is transitive on $\Omega^{[k]}$. Since

$$\pi(g^{[k]}) = \pi(g)(\pi(g) - 1) \cdots (\pi(g) - k + 1),$$

it follows from Theorem 20.2 b) that

$$\sum_{g \in G} \pi(g)(\pi(g) - 1) \cdots (\pi(g) - k + 1) = |G|$$

is equivalent to the k-transitivity of G on Ω.

b) For $k = 1$, the claim coincides with a). We now prove the claim by induction on k.

First assume that

$$\sum_{g \in G} \pi(g)^i = |G|\tau(i) \quad \text{for} \quad 1 \le i \le k.$$

It then follows that

$$\begin{aligned}\sum_{g \in G} \pi(g)(\pi(g) - 1) \cdots (\pi(g) - k + 1) &= \sum_{n=1}^{k} a_{nk} \sum_{g \in G} \pi(g)^n \\ &= |G| \sum_{n=1}^{k} a_{nk}\tau(n) = |G|.\end{aligned}$$

By a), G is therefore k-transitive on Ω.

Conversely, assume that G is k-transitive on Ω, and assume for induction that

$$\sum_{g \in G} \pi(g)^i = |G|\tau(i)$$

for $1 \le i \le k - 1$. Then by a),

$$\begin{aligned}|G| &= \sum_{g \in G} \pi(g)(\pi(g) - 1) \cdots (\pi(g) - k + 1) = \sum_{n=1}^{k} a_{nk} \sum_{g \in G} \pi(g)^n \\ &= \sum_{g \in G} \pi(g)^k + |G| \sum_{n=1}^{k-1} a_{nk}\tau(n),\end{aligned}$$

so

$$\sum_{g\in G}\pi(g)^k = |G|(1-\sum_{n=1}^{k-1}a_{nk}\tau(n)) = |G|\tau(k). \qquad \square$$

Theorem 20.5. *Let $\delta(g)$ be the number of 2-cycles in the cycle decomposition of the permutation g. Then:*

a) *If G is 2-transitive, then* $\sum_{g\in G}\delta(g) = \frac{|G|}{2}$.

b) *If G is 3-transitive, then* $\sum_{g\in G}\delta(g)\pi(g) = \frac{|G|}{2}$.

c) *If G is 4-transitive, then*

$$\sum_{g\in G}\delta(g)^2 = \frac{3|G|}{4} \quad \textit{and} \quad \sum_{g\in G}\pi(g)^2\delta(g) = |G|.$$

Proof. In the following proof, $\{i,j\}$ is always a set (an unordered pair) and (i,j) is an ordered pair.

a) We consider G as a permutation group on the set

$$\Delta = \{\{i,j\} \mid i,j\in\Omega, i\neq j\}$$

by defining

$$\{i,j\}^g = \{i^g, j^g\}.$$

Since G is 2-transitive on Ω, G is transitive on Δ. If

$$\{i,j\}^g = \{i,j\} \quad \text{then either} \quad i^g = i,\ j^g = j \quad \text{or} \quad i^g = j,\ j^g = i.$$

The permutation character α of G on Δ is therefore

$$\alpha = \frac{\pi(\pi-1)}{2} + \delta.$$

It follows from Theorem 20.4 a) that

$$|G| = \sum_{g\in G}\alpha(g) = \frac{1}{2}\sum_{g\in G}\pi(g)(\pi(g)-1) + \sum_{g\in G}\delta(g) = \frac{|G|}{2} + \sum_{g\in G}\delta(g),$$

and this yields the claim in a).

b) Let

$$\Gamma = \{(i,\{j,k\}) \mid i,j,k\in\Omega,\ i,j,k \text{ pairwise distinct}\}.$$

Let G act as a permutation group on Γ in the natural way. Since G is now 3-transitive on Ω, G is transitive on Γ. If

$$(i, \{j, k\})^g = (i, \{j, k\}),$$

then either $i^g = i$, $j^g = j$, $k^g = k$ or $i^g = i$, $j^g = k$, $k^g = j$. So G has the character

$$\rho = \pi\left(\frac{(\pi - 1)(\pi - 2)}{2} + \delta\right)$$

on Γ. It follows from Theorem 20.4 a) that

$$\begin{aligned} |G| &= \sum_{g \in G} \rho(g) = \frac{1}{2} \sum_{g \in G} \pi(g)(\pi(g) - 1)(\pi(g) - 2) + \sum_{g \in G} \pi(g)\delta(g) \\ &= \frac{|G|}{2} + \sum_{g \in G} \pi(g)\delta(g). \end{aligned}$$

c) We consider G as a permutation group on the sets

$$\Gamma_1 = \{\{(i, j), (k, m)\} \mid i, j, k, m \in \Omega, \text{ all pairwise distinct}\}$$

and

$$\Gamma_2 = \{(\{i, j\}, \{k, m\}) \mid i, j, k, m \in \Omega, \text{ all pairwise distinct}\}.$$

By the 4-transitivity of G on Ω, the group G is transitive on Γ_1 and Γ_2. If

$$\{(i, j), (k, m)\}^g = \{(i, j), (k, m)\},$$

then either $g = (i)(j)(k)(m) \cdots$ or $g = (i, k)(j, m) \cdots$. If $g = (i, k)(j, m) \cdots$, then the following four elements of Γ_1 are the only elements containing i, j, k, m that are fixed by g:

$$\{(i, j), (k, m)\}, \quad \{(j, i), (m, k)\}, \quad \{(i, m), (k, j)\}, \quad \{(m, i), (j, k)\}.$$

Hence, the character π_1 of G on Γ_1 is given by

$$\pi_1 = \frac{\pi(\pi - 1)(\pi - 2)(\pi - 3)}{2} + \frac{4\delta(\delta - 1)}{2}.$$

It therefore follows by Theorem 20.4 that

$$|G| = \sum_{g \in G} \pi_1(g) = \frac{|G|}{2} + 2 \sum_{g \in G} \delta(g)^2 - 2 \sum_{g \in G} \delta(g),$$

and it follows from a) that

$$\sum_{g \in G} \delta(g)^2 = \frac{3|G|}{4}.$$

In a similar way, the character π_2 of G on Γ_2 is given by

$$\pi_2 = \delta(\delta - 1) + 2\delta\frac{\pi(\pi-1)}{2} + \frac{\pi(\pi-1)}{2}\frac{(\pi-2)(\pi-3)}{2}.$$

Using Theorem 20.4 and parts a) and b) of Theorem 20.5, which we have already proved, we see that

$$\begin{aligned}|G| &= \sum_{g\in G}\pi_2(g)\\ &= \sum_{g\in G}\delta(g)^2 - \sum_{g\in G}\delta(g) + \sum_{g\in G}\delta(g)\pi(g)^2 - \sum_{g\in G}\delta(g)\pi(g) + \frac{|G|}{4}\\ &= \frac{3|G|}{4} - \frac{|G|}{2} + \sum_{g\in G}\delta(g)\pi(g)^2 - \frac{|G|}{2} + \frac{|G|}{4}.\end{aligned}$$

So $\sum\limits_{g\in G}\delta(g)\pi(g)^2 = |G|$. □

Theorem 20.6. *Let G be a permutation group of degree n.*
a) *For $n \geq 4$, the class functions*

$$\lambda = \frac{(\pi-1)(\pi-2)}{2} - \delta \quad \textit{and} \quad \rho = \frac{\pi(\pi-3)}{2} + \delta$$

are characters of G.
b) *If G is 4-transitive, then λ and ρ are irreducible over every field K of characteristic 0.*

Proof. a) Let $\mathbf{P}$ be the representation of G on the vector space $\mathsf{V} = \langle v_1, \ldots, v_n\rangle$ with $v_i g = v_{ig}$. The representation $\mathbf{P}\otimes\mathbf{P}$ of G on $\mathsf{V}\otimes_{\mathsf{K}}\mathsf{V}$ has the character π^2. The set

$$\{v_i \otimes v_j \mid i, j = 1, \ldots, n\}$$

is a basis of $\mathsf{V}\otimes_{\mathsf{K}}\mathsf{V}$ with

$$(v_i \otimes v_j)g = v_{ig} \otimes v_{jg}.$$

The $\mathsf{K}[G]$-module $\mathsf{V}\otimes_{\mathsf{K}}\mathsf{V}$ admits the direct decomposition

$$\mathsf{V}\otimes_{\mathsf{K}}\mathsf{V} = \mathsf{D}\oplus\mathsf{S}\oplus\mathsf{A},$$

where

$$\begin{aligned}\mathsf{D} &= \langle v_i \otimes v_i \mid i = 1, \ldots, n\rangle\\ \mathsf{S} &= \langle v_i \otimes v_j + v_j \otimes v_i \mid i \neq j;\ i, j = 1, \ldots, n\rangle\\ \mathsf{A} &= \langle v_i \otimes v_j - v_j \otimes v_i \mid i \neq j;\ i, j = 1, \ldots, n\rangle.\end{aligned}$$

(This is a matter of the well-known decomposition of the tensor product space $\mathsf{V} \otimes_{\mathsf{K}} \mathsf{V}$ into symmetric and antisymmetric parts. Since char $\mathsf{K} \neq 2$, this really is a direct decomposition of $\mathsf{V} \otimes_{\mathsf{K}} \mathsf{V}$ as a K-vector space.)

The character of G on S is clearly $\sigma = \frac{\pi(\pi-1)}{2} + \delta$. Since $\mathsf{D} \cong \mathsf{V}$ (as $\mathsf{K}[G]$-modules), the character of G on D is π. Hence, the character α of G on A is determined by the relation

$$\pi^2 = \pi + \sigma + \alpha,$$

and is given by $\alpha = \frac{\pi(\pi-1)}{2} - \delta$. We define $s_i = \sum_{j \neq i} (v_i \otimes v_j + v_j \otimes v_i)$ and

$$\mathsf{S}_1 = \langle s_i \mid i = 1, \ldots, n \rangle.$$

If $\sum_{i=1}^{n} a_i s_i = 0$ for some $a_i \in \mathsf{K}$, then it follows by considering the coefficients of $v_j \otimes v_k$ $(j \neq k)$ that $a_j + a_k = 0$. Since $n \geq 3$, it furthermore follows that $a_j = -a_k = a_l$ for all l. So all the a_j are equal and then $2a_j = 0$, which means $a_j = 0$. Note that

$$s_i g = \sum_{j \neq i} (v_{ig} \otimes v_{jg} + v_{jg} \otimes v_{ig}) = s_{ig},$$

so S_1 is isomorphic to V as a $\mathsf{K}[G]$-module. Since $n \geq 4$, we have $\mathsf{S}_1 \subset \mathsf{S}$. The character of the representation of G on S/S_1 is

$$\rho = \sigma - \pi = \frac{\pi(\pi-1)}{2} + \delta - \pi = \frac{\pi(\pi-3)}{2} + \delta.$$

We define $t_i = \sum_{j \neq i} (v_i \otimes v_j - v_j \otimes v_i)$ and

$$\mathsf{A}_1 = \langle t_i \mid i = 1, \ldots, n \rangle.$$

Since $t_i g = t_{ig}$, A_1 is a $\mathsf{K}[G]$-submodule of A, and the map μ from V to A_1 defined by $v_i \mu = t_i$ is a $\mathsf{K}[G]$-epimorphism. Suppose that

$$v = \sum_{i=1}^{n} c_i v_i \in \ker \mu,$$

so $\sum_{i=1}^{n} c_i t_i = 0$. Consideration of the coefficient of $v_j \otimes v_k$ shows that $c_j - c_k = 0$. So

$$\ker \mu = \langle v_1 + \cdots + v_n \rangle,$$

and G has the character $\pi - 1$ on A_1. The character of G on A/A_1 is therefore

$$\lambda = \alpha - (\pi - 1) = \frac{(\pi - 1)(\pi - 2)}{2} - \delta.$$

b) Now assume that G is 4-transitive. Using the relations

$$\sum_{g \in G} \pi(g) = |G|, \quad \sum_{g \in G} \pi(g)^2 = 2|G|, \quad \sum_{g \in G} \pi(g)^3 = 5|G|, \quad \sum_{g \in G} \pi(g)^4 = 15|G|$$

from Theorem 20.4 b) and

$$\sum_{g \in G} \delta(g) = \sum_{g \in G} \pi(g)\delta(g) = \frac{|G|}{2}, \quad \sum_{g \in G} \delta(g)^2 = \frac{3|G|}{4}, \quad \sum_{g \in G} \pi(g)^2\delta(g) = |G|$$

from Theorem 20.5, it follows from a simple calculation that

$$(\lambda, \lambda)_G = (\rho, \rho)_G = 1.$$

So λ and ρ are absolutely irreducible characters. □

Theorem 20.6 is a special case of a more general theorem of Frobenius: There exists a subset Λ_k of the set of irreducible characters of S_n, which can be described explicitly, with the following properties: If H is a k-transitive subgroup of S_n, then the characters in Λ_k remain irreducible and distinct upon restriction to H. This theorem is useful in the calculation of characters of multiply-transitive groups (see Frobenius [150]) and was used with great success by N. Itô in his analysis of permutation groups of prime degree (see Itô [282, 284, 285]).

We will use Theorem 20.2 and Theorems 20.4–20.6 in our study of permutation groups of prime degree in §21. Next, we use Theorem 20.2 to determine the irreducible characters of the alternating and symmetric groups A_4, S_4 and A_5.

Example 20.7. Let K be an algebraically closed field of characteristic 0. We generate the alternating group A_4 with the elements $t = (1, 2, 3)$ and $s = (1, 2)(3, 4)$. Since $|A_4/A_4'| = 3$, there are exactly three representations of degree 1 by Theorem 6.5. If ϵ is a primitive third root of unity, then these representations are given by

$$\begin{aligned} &\mathbf{D}_1, \quad \text{where} \quad \mathbf{D}_1(s) = \mathbf{D}_1(t) = 1, \\ &\mathbf{D}_2, \quad \text{where} \quad \mathbf{D}_2(s) = 1, \ \mathbf{D}_2(t) = \epsilon, \\ &\mathbf{D}_3, \quad \text{where} \quad \mathbf{D}_3(s) = 1, \ \mathbf{D}_3(t) = \epsilon^2. \end{aligned}$$

By Theorem 20.2 d), the 2-transitive permutation representation of A_4 of degree 4 gives rise to an irreducible representation of degree 3 in the following way: Let $\mathsf{V} = \langle v_1, v_2, v_3, v_4 \rangle$ and $v_i \mathbf{D}(g) = v_{ig}$. Then $v_1 + v_2 + v_3 + v_4$ spans a one-dimensional subspace V_1 on which A_4 acts trivially. The vector space

$$\mathsf{V}_2 = \langle v_2 - v_1, v_3 - v_1, v_4 - v_1 \rangle$$

is an A_4-invariant complement of V_1 in V. On V_2, the group A_4 has a representation $\mathbf{D}_4$ defined by

$$\mathbf{D}_4(s) = \begin{pmatrix} -1 & -1 & -1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{D}_4(t) = \begin{pmatrix} -1 & -1 & -1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which is irreducible by Theorem 20.2 d). (Incidentally, the representation $\mathbf{D}_4$ is equivalent to the representation of A_4 as the symmetry group of rotations of the tetrahedron.) Since

$$3 + 3^2 = 12 = |A_4|,$$

we have found all the irreducible representations of A_4.

Example 20.8. We determine all the irreducible representations of the symmetric group S_4. Let V be the Klein four-group. Since $S_4/V \cong S_3$ and S_3 is isomorphic to the dihedral group of order 6, there exist exactly three irreducible representations $\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_3$ of S_4 whose kernel contains V. The degrees of these representations are 1, 1, 2. By Theorem 20.2 d), the 2-transitive permutation representation of S_4 of degree 4 gives rise to an irreducible representation $\mathbf{D}_4$ of degree 3. By Remark 10.7, $\mathbf{D}_5 = \mathbf{D}_4 \otimes \mathbf{D}_2$ is irreducible of degree 3, and $\mathbf{D}_5$ is not equivalent to $\mathbf{D}_4$ since

$$\operatorname{tr}\mathbf{D}_5((1,2)) = \operatorname{tr}\mathbf{D}_4((1,2)) \operatorname{tr}\mathbf{D}_2((1,2)) = -\operatorname{tr}\mathbf{D}_4((1,2)) \neq \operatorname{tr}\mathbf{D}_4((1,2)).$$

We have now found all the irreducible representations of S_4 since

$$1 + 1 + 2^2 + 2 \cdot 3^2 = 24 = |S_4|.$$

(The representation $\mathbf{D}_5$ is equivalent to the representation of S_4 as the symmetry group of rotations of the octahedron.)

Example 20.9. For an example with a simple group, we choose the alternating group A_5. Our methods yield the following irreducible representations:

(1) The trivial representation $\mathbf{D}_1$.

(2) An irreducible representation $\mathbf{D}_2$ of degree 4 arising from the 2-transitive permutation representation of degree 5.

(3) Since $A_5 \cong \mathrm{PSL}_2(5)$ (see Theorem II.6.14, pg. 192), A_5 has a 2-transitive permutation representation of degree 6. This yields an irreducible representation $\mathbf{D}_3$ of degree 5.

(4) With $v = (1,2,3,4,5)$ and $s = (1,2)(3,4)$, we have $A_5 = \langle v, s\rangle$ with defining relations

$$v^5 = s^2 = (vs)^3 = 1$$

(see Example I.19.9, pg. 149). We define $\mathbf{D}_4$ by

$$\mathbf{D}_4(v) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \epsilon^{-1} & 0 \\ 0 & 0 & \epsilon \end{pmatrix}, \quad \mathbf{D}_4(s) = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 & 2 \\ 1 & b & a \\ 1 & a & b \end{pmatrix},$$

where ϵ is a 5-th root of unity and

$$a = \epsilon + \epsilon^{-1} = \frac{-1+\sqrt{5}}{2}, \quad b = \epsilon^2 + \epsilon^{-2} = \frac{-1-\sqrt{5}}{2}.$$

It is easy to verify that $\mathbf{D}_4(v)$ and $\mathbf{D}_4(s)$ satisfy the defining relations of A_5. So according to Lemma 5.18, we obtain a representation $\mathbf{D}_4$ of A_5 of degree 3. The representation $\mathbf{D}_4$ is irreducible since no one-dimensional subspace is fixed by $\mathbf{D}_4(v)$ and $\mathbf{D}_4(s)$.

(Up to equivalence, $\mathbf{D}_4$ is the representation of A_5 as the symmetry group of rotations of the icosahedron.)

(5) The entries of the matrices appearing in the representation $\mathbf{D}_4$ all lie in the field $\mathbb{Q}(\epsilon)$ of the 5-th roots of unity. Applying the automorphism α of $\mathbb{Q}(\epsilon)$ defined by $\epsilon^\alpha = \epsilon^2$ to the entries of these matrices yields another irreducible representation $\mathbf{D}_5$ of A_5. Since

$$\operatorname{tr}\mathbf{D}_4(v) = 1 + \epsilon + \epsilon^{-1} = \frac{1+\sqrt{5}}{2} \neq \frac{1-\sqrt{5}}{2} = 1 + \epsilon^2 + \epsilon^{-2} = \operatorname{tr}\mathbf{D}_5(v),$$

the representations $\mathbf{D}_4$ and $\mathbf{D}_5$ are not equivalent. We have found all the irreducible representations of A_5 since

$$1 + 4^2 + 5^2 + 3^2 + 3^2 = 60 = |A_5|.$$

Example 20.10 (Wielandt). Let H_1 and H_2 be nonisomorphic groups that have the same number of elements of every order. (For example, one may choose $|H_1| = |H_2| = p^3$ with $p > 2$, H_1 elementary abelian and H_2 nonabelian of exponent p.) We embed H_1 and H_2 in a symmetric group S_n with $n \geq |H_i|$ such that H_i has the regular representation on one orbit and all other orbits have size 1. Let $\mathbf{P}_i$ $(i = 1, 2)$ be the permutation representation of S_n on the cosets of H_i. It is clear that $\mathbf{P}_1$ and $\mathbf{P}_2$ are not permutation isomorphic, namely because the stabilizers are isomorphic to H_1, resp. H_2, so are not conjugate in S_n. If π_i is the character of $\mathbf{P}_i$, then we have by Exercise I.24 (pg. 33) that

$$\pi_i(g) = \frac{[S_n : H_i]|g^{S_n} \cap H_i|}{|g^{S_n}|}.$$

Let $h_1 \in g^{S_n} \cap H_1$, and let $|h_1| = m$. Then the cycle decomposition of h_1 contains exactly $\frac{|H_1|}{m}$ cycles of length m and $n - |H_1|$ cycles of length 1. Since all elements of order m in H_1 have the same cycle type, these elements all lie in the same conjugacy class of S_n. Hence, it follows by our assumption that

$$\begin{aligned} |g^{S_n} \cap H_1| &= |\{h_1 \in H_1 \mid |h_1| = m\}| \\ &= |\{h_2 \in H_2 \mid |h_2| = m\}| = |g^{S_n} \cap H_2|. \end{aligned}$$

Therefore, $\pi_1(g) = \pi_2(g)$ for all $g \in S_n$. Hence, $\mathbf{P}_1$ and $\mathbf{P}_2$ are permutation representations of S_n that are equivalent as linear representations since their characters coincide, but which are not permutation isomorphic.

If one does not require the representations $\mathbf{P}_i$ to be transitive, a much simpler example exists: Consider the representations of the Klein four-group on the points, resp. lines, of the projective plane with 7 points; the orbits on the set of points have sizes 4, 1, 1, 1, while the orbits on the set of lines, however, have sizes 2, 2, 2, 1. Thompson has given examples of p-groups which admit permutation representations that are equivalent as representations but are not permutation isomorphic.

Exercises

35) Let $\delta_k(g)$ be the number of k-cycles in the cycle decomposition of the permutation g. If G is k-transitive, then $\sum_{g \in G} \delta_k(g) = \frac{|G|}{k}$.

36) Let G be a transitive permutation group of degree n. We construct the $\mathsf{K}[G]$-module V according to Definition 20.1. Show:
   a) V is always isomorphic to a submodule of $\mathsf{K}[G]$.
   b) V is isomorphic to a direct summand of $\mathsf{K}[G]$ if and only if char K does not divide the order of the stabilizer G_1.
   c) If G is 2-transitive and char $\mathsf{K} \mid n$, then V is directly indecomposable. If G is 2-transitive and char $\mathsf{K} \nmid n$, then $\mathsf{V} = \mathsf{V}_1 \oplus \mathsf{V}_2$; here, G acts trivially on V_1 and V_2 is directly indecomposable.
   (Hint: Calculate the idempotents in $\mathrm{Hom}_{\mathsf{K}[G]}(\mathsf{V}, \mathsf{V})$; see also Tsuzuku [444].)

37) (Burnside [84]) Show that if k is the degree of a faithful complex representation of S_n, then $k \geq n - 1$.
   (Hint: Let $G = S_n$, $L = G_1$, $H = G_{1,2}$ and $J = \langle H, (1,2) \rangle$. Let χ_1 be the trivial character of G, and let χ_2 be the irreducible character of degree $n - 1$ arising from the 2-transitive permutation representation of S_n of degree n according to Theorem 20.2 d). Show:
   a) If ψ is an irreducible character of G with $(\psi_L, 1_L)_L > 0$, then $\psi = \chi_1$ or $\psi = \chi_2$.
   b) We have $((\chi_2)_J, 1_J)_J = 1$ and $((\chi_2)_H, 1_H)_H = 2$.
   c) Let $\tau = (1_J)^G$. Then $\tau = \chi_1 + \chi_2 + \chi_3$ for some irreducible character χ_3 of G of degree $\frac{n(n-3)}{2}$.
   d) Let $\mu = (1_H)^G$. Then $(\mu_H, 1_H)_H = 7$, $(\mu, \mu)_G = 7$, $(\mu, \chi_2)_G = 2$, $(\mu, \chi_1)_G = (\mu, \chi_3)_G = 1$. Therefore, there exists an irreducible character χ_4 of G of degree $\frac{n(n-3)}{2} + 1$ with $\mu = \chi_1 + 2\chi_2 + \chi_3 + \chi_4$.
   e) Let ψ be an irreducible character of G. Then $(\psi_H, 1_H)_H > 0$ if and only if $\psi = \chi_i$ with $1 \leq i \leq 4$.
   f) Let $\mathbf{D}$ be a faithful irreducible representation of G with $\deg \mathbf{D} = k \leq n - 2$. Consider $\mathbf{D}((1,2))$ and $\mathbf{D}(H)$, and show that $\mathbf{D}_H$ or $(\mathbf{D} \otimes \mathbf{D}_{-1})_H$ contains

the trivial representation of H, where $\mathbf{D}_{-1}$ is the nontrivial representation with kernel A_n.)

21 Permutation groups of prime degree

Many deep theorems regarding transitive permutation groups of prime degree can be proved with the help of character theory. We begin by collecting some elementary results.

Theorem 21.1. *Let G be a transitive permutation group of prime degree $p > 2$, and let $P = \langle g \rangle$ be a Sylow p-subgroup of G.*

a) *We have $\mathbf{N}_G(P) = \langle g, h \rangle$, where $|h|$ is a divisor of $p-1$ and h has exactly one fixed point. Every nonidentity element of $\mathbf{N}_G(P)$ has at most one fixed point. (So $\mathbf{N}_G(P)$ is a Frobenius group with Frobenius kernel P.)*

b) *We have $|G| = pd(1+kp)$ with $d \mid (p-1)$ and k a nonnegative integer.*

c) *If $d = 1$, then $G = P$.*

d) *If $d = 2$ and $p \equiv -1 \bmod 4$, then G is a dihedral group of order $2p$.*

e) *If $|G| \neq p$, then G' is a (possibly abelian) simple group. The factor group G/G' is cyclic, and $|G/G'|$ divides $p-1$.*

Proof. a) Since $P \trianglelefteq \mathbf{N}_G(P)$, the normalizer $\mathbf{N}_G(P)$ is a transitive permutation group of degree p with a normal Sylow p-subgroup. Therefore, $\mathbf{N}_G(P)$ is permutation isomorphic to a subgroup of

$$\left\{ \begin{pmatrix} x \\ ax+b \end{pmatrix} \;\middle|\; x, a, b \in \mathrm{GF}(p),\ a \neq 0 \right\}$$

by Theorem II.3.6 (pg. 171). The claims follow.

b) We let $|\mathbf{N}_G(P)| = pd$. By a), we have $d \mid (p-1)$. The index $[G : \mathbf{N}_G(P)]$ is the number of Sylow p-subgroups of G. By Sylow's Theorem I.7.3 (pg. 34), we have $[G : \mathbf{N}_G(P)] \equiv 1 \bmod p$ and so $[G : \mathbf{N}_G(P)] = 1 + kp$ for some $k \geq 0$.

c) Assume $d = 1$, so $\mathbf{N}_G(P) = P$. By Burnside's Transfer Theorem IV.2.6 (pg. 445), there exists a normal subgroup K of G such that $G = KP$ and $K \cap P = 1$. Since $p \nmid |K|$, the subgroup K is intransitive. The transitive permutation group G of prime degree is primitive by Theorem II.1.3 (pg. 155). Since every nontrivial normal subgroup of a primitive group is transitive by Theorem II.1.5 (pg. 156), we have $K = 1$ and $G = P$.

d) Assume $|\mathbf{N}_G(P)| = 2p$. Then there exists some $x \in \mathbf{N}_G(P)$ with $|x| = 2$. By a), x has exactly one fixed point and is therefore the product of $\frac{p-1}{2} = t$ cycles of length 2. Since $p \equiv -1 \bmod 4$, t is odd, and therefore $\mathrm{sgn}(x) = (-1)^t = -1$. Hence, the intersection $N = G \cap A_p$ of G with the alternating group A_p has index 2 in G. Since $x \notin N$, we have $P \leq \mathbf{N}_N(P) < \mathbf{N}_G(P)$. Our assumption $|\mathbf{N}_G(P)| = 2p$ now forces $|\mathbf{N}_N(P)| = p$. By c), we therefore have $N = P \lhd G$. One now sees easily that G is a dihedral group of order $2p$.

e) Let N be a minimal normal subgroup of G. Since G is primitive, N is transitive. So p divides $|N|$, and hence the Sylow p-subgroup P of G lies in N. By Theorem I.7.8 (pg. 36), $G = N\mathbf{N}_G(P)$. By a), we have $\mathbf{N}_G(P) = PQ$ for some cyclic group Q with $|Q|$ dividing $p - 1$. Since $P \leq N$, it follows that

$$G/N \cong \mathbf{N}_G(P)/(\mathbf{N}_G(P) \cap N) = PQ/P(Q \cap N) \cong Q/(Q \cap N).$$

So G/N is cyclic since it is a homomorphic image of Q, and $|G/N|$ divides $p - 1$. It follows that $G' \leq N$, and by the minimality of N this means $G' = 1$ or $G' = N$. In the first case, since $\mathbf{C}_G(P) = P$ we must then have $G = P$, which is impossible by our assumption. In the second case, G' is characteristically simple since G' is a minimal normal subgroup of G. By Theorem I.9.12 (pg. 53), G' is then a direct product of isomorphic simple groups. Since $p^2 \nmid |G'|$, G' is simple. □

If follows easily from Theorem 21.1 e) that every two p-cycles generate a simple subgroup of the symmetric group S_p. It seems, therefore, as if there must be many simple, transitive permutation groups of prime degree. However, very few transitive groups of prime degree are known; we list them here:

Example 21.2. The following groups are the only known transitive permutation groups of prime degree p:[7]

a) Solvable groups; these were all described in Theorem II.3.6 (pg. 171).
b) The symmetric group S_p and the alternating group A_p.
c) The sharply 4-transitive Mathieu group M_{11} of degree 11.
d) The 4-transitive Mathieu group M_{23} of degree 23 (see Chapter II, pg. 163).
e) The group $\mathrm{P\Gamma L}_n(q^f)$ of collineations of the projective space $\mathsf{P}(n, q^f)$ (including the field automorphisms!) has a 2-transitive representation on the

$$s = \frac{q^{nf} - 1}{q^f - 1}$$

points of $\mathsf{P}(n, q^f)$. For some pairs n, q^f the number s is prime. (So, for example, $\mathrm{PSL}_3(2)$ has a representation as a permutation group of degree 7, $\mathrm{PSL}_3(3)$ has a representation of degree 13 and $\mathrm{PSL}_3(5)$ has a representation of degree 31.)
f) The group $\mathrm{PSL}_2(11)$ and its representation of degree 11 (see Theorem II.8.28, case (6)).[8]

N. Itô showed in [280] that all the transitive representations of prime degree of the groups $\mathrm{PSL}_n(q^f)$ are listed here.

The following theorem is fundamental to the theory of permutation groups of prime degree.

Theorem 21.3 (Burnside). *Let G be a transitive permutation group of prime degree. If G is not 2-transitive, then G is solvable.*

[7] It is now known by the classification of finite simple groups that these are all the transitive permutation groups of prime degree; see R. M. Guralnick, Subgroups of prime power index in a simple group, J. Algebra **81** (1983), no. 2, 304–311. *trans.*

[8] Correction 28 from errata.

Proof. Let **D** be the complex linear representation of G of degree p constructed according to Definition 20.1, and let π be the character of **D**. By the transitivity of G, we have by Theorem 20.2 b) that $(\pi, 1)_G = 1$, so $\pi = 1 + \psi$ with $(\psi, 1)_G = 0$. Since G is not 2-transitive by our assumption, ψ is not irreducible by Theorem 20.2 d). Let $\psi = \sum_{i=1}^{t} \psi_i$ for some irreducible characters ψ_i of G with $t \geq 2$. We now show:

a) If $P = \langle x \rangle$ is a Sylow p-subgroup of G, then the restriction π_P of π to P decomposes into exactly p distinct linear characters of P:

Since x is a p-cycle, π_P is the character of the regular representation of P. By Theorem 5.3, π_P thus contains the p irreducible characters of P as constituents, each exactly once.

b) The ψ_i in the decomposition $\pi = 1 + \sum_{i=1}^{t} \psi_i$ are pairwise distinct:

It follows from a) that $(\psi_i)_P \neq (\psi_j)_P$ for $i \neq j$.

c) The ψ_i are algebraically conjugate over the field of $\frac{|G|}{p}$-th roots of unity:

If ρ is a primitive $|G|$-th root of unity, then $\epsilon = \rho^{\frac{|G|}{p}}$ is a primitive p-th root of unity and $\mu = \rho^p$ is a primitive $\frac{|G|}{p}$-th root of unity. Since $|G|$ divides $p!$, we have $p^2 \nmid |G|$ and thus $\mathbb{Q}(\rho) = \mathbb{Q}(\epsilon, \mu)$. It is well known that the Galois group A of $\mathbb{Q}(\rho)$ over $\mathbb{Q}$ consists of the maps $\alpha(n)$ defined by $\rho^{\alpha(n)} = \rho^n$, where $(|G|, n) = 1$. Since $(p, \frac{|G|}{p}) = 1$, for a given j with $1 \leq j \leq p-1$ there exists an n such that

$$n \equiv j \bmod p \quad \text{and} \quad n \equiv 1 \bmod \frac{|G|}{p}.$$

Then $(|G|, n) = 1$. Letting $\alpha = \alpha(n)$, we have

$$\mu^\alpha = \rho^{pn} = \rho^p = \mu \quad \text{and} \quad \epsilon^\alpha = \rho^{|G|n/p} = \rho^{|G|j/p} = \epsilon^j.$$

In particular, α fixes the field $\mathbb{Q}(\mu)$ elementwise.

Let

$$(\psi_i)_P = \sum_{j=1}^{s_i} \lambda_{ij}$$

be the decomposition of $(\psi_i)_P$ into linear characters λ_{ij} of P. Let $\lambda_{ij}(x) = \epsilon^{n_{ij}}$. With a suitable labeling, we have $\lambda_{11}(x) = \epsilon$. Let α be the automorphism of $\mathbb{Q}(\rho)$ over $\mathbb{Q}(\mu)$ defined by $\epsilon^\alpha = \epsilon^{n_{ij}}$. Since all the $\psi_1(g)$ lie in the field $\mathbb{Q}(\rho)$ of primitive $|G|$-th roots of unity, ψ_1^α is a character that is algebraically conjugate to ψ_1. Then the character λ_{ij} appears in the expression

$$(\psi_1^\alpha)_P = \sum_{j=1}^{s_1} \lambda_{1j}^\alpha.$$

Since $\psi = \pi - 1$ assumes only rational values on G, it follows by b) that

$$(\psi, \psi_1^\alpha)_G = (\psi^\alpha, \psi_1^\alpha)_G = (\psi, \psi_1)_G^\alpha = 1.$$

Thus, every ψ_1^α appears in ψ. As α runs through the automorphisms of $\mathbb{Q}(\rho)$ over $\mathbb{Q}(\mu)$, we have seen that every nontrivial linear character of P appears in $(\psi_1^\alpha)_P$. So ψ is the sum of some ψ_1^α, and all ψ_i are algebraically conjugate over $\mathbb{Q}(\mu)$.

d) If $g \in G$ and $|g| \neq p$, then $\psi_1(g) = \cdots = \psi_t(g)$, and this character value is a nonnegative integer satisfying $\psi_1(g) \leq \psi_1(1)$:

It follows from $|g| \neq p$ that $p \nmid |g|$, and therefore $|g|$ divides $\frac{|G|}{p}$. Hence, all the character values $\psi_i(g)$ lie in the field $\mathbb{Q}(\mu)$ of $\frac{|G|}{p}$-th roots of unity. By c), for a given i there exists an automorphism α_i of $\mathbb{Q}(\rho)$ over $\mathbb{Q}(\mu)$ such that $\psi_i = \psi_1^{\alpha_i}$. It follows that

$$\psi_i(g) = \psi_1^{\alpha_i}(g) = \psi_1(g)^{\alpha_i} = \psi_1(g)$$

for $i = 1, 2, \ldots, t$. Since

$$\pi(g) = 1 + \sum_{i=1}^{t} \psi_i(g) = 1 + t\psi_1(g)$$

is an integer, $\psi_1(g)$ is rational. Since $\psi_1(g)$ is also an algebraic integer, $\psi_1(g)$ is in fact an integer. It follows from $\pi(g) = 1 + t\psi_1(g) \geq 0$ that $\psi_1(g) \geq 0$. It is trivial that $\psi_1(g) \leq \psi_1(1)$.

e) For $0 < i \leq s = \psi_1(1)$, let n_i be the number of elements $g \in G$ such that $|g| \neq p$ and $\psi_1(g) = i$. If N is the number of Sylow p-subgroups of G, then

$$(1)\ \sum_{i=1}^{s} i n_i = Ns \quad \text{and} \quad (2)\ \sum_{i=1}^{s} i^2 n_i = Ns^2 :$$

We have

$$0 = p(\psi_1, 1)_P = \sum_{j=0}^{p-1} \psi_1(x^j).$$

It therefore follows from d) that

$$\begin{aligned} 0 = |G|(\psi_1, 1)_G &= \sum_{g \in G} \psi_1(g) = \sum_{|g| \neq p} \psi_1(g) + \sum_{|g| = p} \psi_1(g) \\ &= \sum_{i=1}^{s} i n_i + N \sum_{j=0}^{p-1} \psi_1(x^j) - N\psi_1(1) = \sum_{i=1}^{s} i n_i - Ns. \end{aligned}$$

Furthermore,

$$0 = p(\psi_1, \psi_2)_P = \sum_{j=0}^{p-1} \psi_1(x^j)\psi_2(x^{-j}).$$

Hence, it follows from b) and d) that

$$\begin{aligned} 0 = |G|(\psi_1, \psi_2)_G &= \sum_{i=1}^{s} i^2 n_i + N \sum_{j=0}^{p-1} \psi_1(x^j)\psi_2(x^{-j}) - N\psi_1(1)\psi_2(1) \\ &= \sum_{i=1}^{s} i^2 n_i - Ns^2. \end{aligned}$$

f) G is solvable:
We multiply (1) by s and subtract (2) from the result. This yields

$$0 = (s-1)n_1 + 2(s-2)n_2 + \cdots + i(s-i)n_i + \cdots + (s-1)n_{s-1}.$$

So $n_1 = n_2 = \cdots = n_{s-1} = 0$.
Suppose $g \in G$ with $\pi(g) \geq 2$. Then $|g| \neq p$, and thus by d)

$$\pi(g) = 1 + t\psi_1(g)$$

with $\psi_1(g) \neq 0$. So $\psi_1(g) = s$ and then $\pi(g) = 1 + ts = p$, which means $g = 1$. Hence, every nonidentity element of G has at most one fixed point. By Theorem II.3.6 (pg. 171), G is then solvable. □

An elementary proof of Theorem 21.3 (that is, without the use of character theory) is given in Schur [374]. Schur's proof relies on complicated calculations in the polynomial ring over GF(p). Burnside's proof has the advantage that its method is applicable to analyzing primitive permutation groups of degree $2p$ (see Wielandt [466]).

We give a nice application of Theorem 21.3 and Theorem II.3.6 (pg. 171) to the theory of equations of prime degree:

Theorem 21.4. *Let f be an irreducible polynomial of prime degree p over the field $\mathbb{Q}$ of rational numbers. Let K be a splitting field for f, and let $f = \prod_{i=1}^{p}(x - a_i)$ be the factorization of f in $\mathsf{K}[x]$. Then:*

a) *(Galois) The polynomial f is solvable by radicals if and only if $\mathsf{K} = \mathbb{Q}(a_i, a_j)$ for all pairs i, j with $i \neq j$.*
b) *If f is not solvable by radicals, then the polynomial $\frac{f}{x-a_1}$ is irreducible in $\mathbb{Q}(a_1)[x]$.*

Proof. Let G be the Galois group of K over $\mathbb{Q}$. If $\alpha \in G$, then it is clear that $f(a_i^\alpha) = f(a_i)^\alpha = 0$. So G is a permutation group on $\{a_1, \ldots, a_p\}$. Let $\{a_1, \ldots, a_s\}$ be an orbit of G. Then all the coefficients of the polynomial $g = \prod_{i=1}^{s}(x - a_i)$ are fixed

by G, so the coefficients lie in $\mathbb{Q}$. Since $g(a_1) = 0$, the polynomial g divides f. Hence, as f is irreducible, we have $g = f$ and $s = p$. So G is transitive on $\{a_1, \ldots, a_p\}$.

a) It is well known that f is solvable by radicals if and only if G is solvable. By Theorem II.3.6, the group G is solvable if and only if $G_{a_i,a_j} = 1$ for all $i \neq j$. The subfield of K that is fixed elementwise by G_{a_i,a_j} is $\mathbb{Q}(a_i, a_j)$ since G_{a_i,a_j} is clearly the entire group fixing $\mathbb{Q}(a_i, a_j)$ elementwise. We have $G_{a_i,a_j} = 1$ if and only if the fixed field $\mathbb{Q}(a_i, a_j)$ of G_{a_i,a_j} is equal to the fixed field of 1, which is of course K.

b) If G is not solvable, then G is 2-transitive by Theorem 21.3. So the Galois group G_{a_1} of K over $\mathbb{Q}(a_1)$ is transitive on $\{a_2, \ldots, a_p\}$. Therefore, the polynomial $\prod_{i=2}^{p}(x - a_i)$ is irreducible in $\mathbb{Q}(a_1)[x]$. □

In order to treat the permutation groups of prime degree p with $|\mathbf{N}_G(P)| = 2p$, we will need the characters of the dihedral group of order $2p$. We prove the following more general lemma.

Lemma 21.5. *Let $D = \langle a, b\rangle$ be a dihedral group of order $2n$ with defining relations $a^n = b^2 = 1$, $a^b = a^{-1}$. Let ϵ be a primitive n-th root of unity. The dihedral group D has exactly the following irreducible representations over the complex numbers, up to equivalence:*

a) *For odd n:*

$$\begin{aligned} &\mathbf{D}_1 \quad \textit{with} \quad \mathbf{D}_1(a) = \mathbf{D}_1(b) = 1;\\ &\mathbf{D}_2 \quad \textit{with} \quad \mathbf{D}_2(a) = 1,\ \mathbf{D}_2(b) = -1; \end{aligned}$$

and furthermore, $\frac{n-1}{2}$ representations $\mathbf{F}_i$ $(1 \leq i \leq \frac{n-1}{2})$ of degree 2 with

$$\mathbf{F}_i(a) = \begin{pmatrix} \epsilon^i & 0 \\ 0 & \epsilon^{-i} \end{pmatrix} \quad \textit{and} \quad \mathbf{F}_i(b) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

b) *For even n:*

$$\begin{aligned} &\mathbf{D}_1 \quad \textit{with} \quad \mathbf{D}_1(a) = \mathbf{D}_1(b) = 1;\\ &\mathbf{D}_2 \quad \textit{with} \quad \mathbf{D}_2(a) = 1,\ \mathbf{D}_2(b) = -1;\\ &\mathbf{D}_3 \quad \textit{with} \quad \mathbf{D}_3(a) = -1,\ \mathbf{D}_3(b) = 1;\\ &\mathbf{D}_4 \quad \textit{with} \quad \mathbf{D}_4(a) = \mathbf{D}_4(b) = -1; \end{aligned}$$

and furthermore, the representations $\mathbf{F}_i$ defined in a) with $1 \leq i \leq \frac{n-2}{2}$.

Proof. By Theorem 6.5, $|D/D'|$ is the number of linear representations of D. Since $D' = \langle a^2\rangle$, we have $|D/D'| = 2$ when $2 \nmid n$ and $|D/D'| = 4$ when $2 \mid n$. The given representations $\mathbf{D}_i$ are clearly all the representations of D of degree 1. Since

$$\mathbf{F}_i(a)^n = \mathbf{F}_i(b)^2 = 1 \quad \text{and} \quad \mathbf{F}_i(b)^{-1}\mathbf{F}_i(a)\mathbf{F}_i(b) = \mathbf{F}_i(a)^{-1},$$

the $\mathbf{F}_i$ defined above are representations of D by Lemma 5.18.

Now assume that $\mathbf{F}_i$ is reducible. By Maschke's Theorem, $\mathbf{F}_i$ is completely reducible, so it decomposes into two representations of degree 1. Thus, D' lies in the kernel of $\mathbf{F}_i$, and it follows that $\epsilon^{2i} = 1$. Hence, $\mathbf{F}_i$ is irreducible if $2i \not\equiv 0 \bmod n$. If $i \not\equiv \pm j \bmod n$, then $\mathbf{F}_i(a)$ and $\mathbf{F}_j(a)$ have different eigenvalues, so $\mathbf{F}_i$ and $\mathbf{F}_j$ are not equivalent.

Upon calculating the sum of the squared degrees, we see that we have listed all the irreducible representations of D up to equivalence. □

Theorem 21.6 (N. Itô [279]). *Let G be a transitive, nonsolvable permutation group of prime degree p, and let P be a Sylow p-subgroup of G. Furthermore, assume that $|\mathbf{N}_G(P)| = 2p$. Then G is sharply 3-transitive.*

(In Volume 3, we will show that G is the group $\mathrm{PSL}_2(2^m)$ on the 2^m+1 points of the projective line over $\mathrm{GF}(2^m)$. In particular, p is a Fermat prime. See Zassenhaus [493] and Huppert [264].)

Proof. Since G is not solvable, G is 2-transitive by Theorem 21.3. If π is the permutation character of G, then $\pi = 1 + \psi$ for some irreducible character ψ of degree $p-1$ by Theorem 20.2 d).

We again let $P = \langle x \rangle$. Since $|\mathbf{N}_G(P)| = 2p$, the normalizer $\mathbf{N}_G(P)$ is a dihedral group, say $\mathbf{N}_G(P) = \langle x, y \rangle$ with $x^p = y^2 = 1$ and $x^y = x^{-1}$. By Lemma 21.5, $\mathbf{N}_G(P)$ has the following irreducible characters: The trivial character and the character λ defined by $\lambda(x) = 1$, $\lambda(y) = -1$; furthermore, there exist $s = \frac{p-1}{2}$ irreducible characters φ_i $(1 \le i \le s)$ of degree 2. If ϵ is a primitive p-th root of unity, then

$$\varphi_i(x^j) = \epsilon^{ij} + \epsilon^{-ij} \quad \text{and} \quad \varphi_i(x^j y) = 0.$$

We show:

a) $(\varphi_i, \varphi_j)_P = 2\delta_{ij}$ and $\psi_P = \sum_{i=1}^{s} (\varphi_i)_P$:

This follows immediately from the formulas for the φ_i.

In steps b) to f), we construct irreducible characters ρ_j of G. Itô deduced the existence of these characters using Brauer's theory (see Itô [279]), but we proceed in a more elementary fashion:

b) The functions ρ_j $(j = 1, \dots, \frac{p-1}{2})$ defined by

$$\rho_j(g) = \begin{cases} \psi(g) - 1 & \text{if } |g| \neq p \\ 1 + \sum\limits_{\substack{i=1 \\ i \neq j}}^{s} \varphi_i(x^k) & \text{if } g = (x^k)^a \text{ for some } a \in G \text{ and } k \in \mathbb{Z} \end{cases}$$

are pairwise distinct, well-defined class functions on G with $\rho_j(1) = p - 2$:

We first show that ρ_j is well defined:

Suppose $1 \neq g \in G$ and $g = (x^k)^a = (x^m)^b$ for some $a, b \in G$ and $k, m \in \mathbb{Z}$. Then $(x^k)^{ab^{-1}} = x^m \in P$, and therefore $ab^{-1} \in \mathbf{N}_G(P)$. Since φ_i is a class function on $\mathbf{N}_G(P)$, it follows that $\varphi_i(x^k) = \varphi_i((x^k)^{ab^{-1}}) = \varphi_i(x^m)$. So ρ_j is well defined.

ρ_j is a class function on G:

If $|g| \neq p$ and $a \in G$, then

$$\rho_j(g) = \psi(g) - 1 = \psi(g^a) - 1 = \rho_j(g^a).$$

Now suppose $|g| = p$, say $g = (x^k)^b$ for some $b \in G$ and $k \in \mathbb{Z}$. Then for all $a \in G$, $g^a = (x^k)^{ba}$, and so $\rho_j(g^a) = \rho_j(g)$.

We have $\rho_j(1) = p - 2$. A consideration of $(\rho_j)_P$ shows that all the ρ_j are pairwise distinct.

c) $(\rho_j, \rho_j)_P = p - 2$:

The restriction

$$(\rho_j)_P = 1 + \sum_{\substack{i=1 \\ i \neq j}}^{s} (\varphi_i)_P$$

contains exactly $1 + 2(s-1) = p - 2$ pairwise distinct irreducible characters of P. It therefore follows by the orthogonality relations that $(\rho_j, \rho_j)_P = p - 2$.

d) $(\rho_j, \rho_j)_G = 1$:

Since ψ is a nontrivial irreducible character of G,

$$\sum_{g \in G} (\psi(g) - 1)^2 = |G|(\psi - 1, \psi - 1)_G = 2|G|.$$

We have

$$\begin{aligned} |G|(\rho_j, \rho_j)_G &= \sum_{|g| \neq p} (\psi(g) - 1)^2 + \sum_{|g| = p} |\rho_j(g)|^2 \\ &= \sum_{g \in G} (\psi(g) - 1)^2 - \sum_{|g| = p} (\psi(g) - 1)^2 + \sum_{|g| = p} |\rho_j(g)|^2. \end{aligned}$$

Since the $[G : \mathbf{N}_G(P)] = \frac{|G|}{2p}$ Sylow p-subgroups of G intersect pairwise trivially, it follows from c) and the fact that $\psi(x^i) = \pi(x^i) - 1 = -1$ (for $x^i \neq 1$) that

$$\begin{aligned} |G|(\rho_j, \rho_j)_G &= 2|G| - \frac{|G|}{2p} \sum_{i=1}^{p-1} (\psi(x^i) - 1)^2 + \frac{|G|}{2p} \sum_{i=1}^{p-1} |\rho_j(x^i)|^2 \\ &= 2|G| - \frac{2|G|(p-1)}{p} + \frac{|G|}{2}(\rho_j, \rho_j)_P - \frac{|G|}{2p}\rho_j(1)^2 \\ &= 2|G| - \frac{2|G|(p-1)}{p} + \frac{|G|(p-2)}{2} - \frac{|G|(p-2)^2}{2p} = |G|. \end{aligned}$$

e) ρ_j is a generalized character of G:

By Theorem 19.3 b) (with $\mathbb{Z}$ in place of R), it is enough to show that $(\rho_j)_F$ is a generalized character of F for every elementary subgroup F of G.

First assume that $p \nmid |F|$. By the definition of ρ_j, we then have $(\rho_j)_F = (\psi - 1)_F$, and this is clearly a generalized character of F.

If p divides $|F|$, then F contains a Sylow p-subgroup of G, say P. Since F is nilpotent and the regular permutation group P coincides with its centralizer by

Lemma II.3.1 (pg. 167), it follows that $F = P$. Then

$$(\rho_j)_P = 1 + \sum_{i \neq j} (\varphi_i)_P$$

is a character of P.

f) ρ_j is an irreducible character of G:

Let $\chi_1, \ldots, \chi_h$ be the irreducible characters of G. By e), we have $\rho_j = \sum_{i=1}^{h} c_{ij}\chi_i$ for some integers c_{ij}. It follows by d) that

$$1 = (\rho_j, \rho_j)_G = \sum_{i=1}^{h} c_{ij}^2.$$

Thus, we have $\rho_j = \pm\chi_{j'}$, and then $\rho_j(1) = p - 2 > 0$ implies that $\rho_j = \chi_{j'}$.

g) If G_1 is the stabilizer of the point 1, then $(\rho_j)_{G_1}$ is an irreducible character of G_1:

The elements of G_1 do not have order p. Therefore, by the definition of ρ_j we have for all $g \in G_1$ and all j that

$$\rho_j(g) = \psi(g) - 1 = \rho_1(g).$$

Assume for contradiction that $(\rho_1)_{G_1}$ is reducible. Let τ be an irreducible character of G_1 that occurs as a constituent of $(\rho_1)_{G_1}$ and satisfies

$$\tau(1) \leq \frac{\rho_1(1)}{2} = \frac{p-2}{2}.$$

Let $(\rho_1, \tau)_{G_1} = m > 0$. Then $(\rho_j, \tau)_{G_1} = (\rho_1, \tau)_{G_1} = m$ for all $j = 1, \ldots, s$. It follows by Frobenius reciprocity that

$$(\rho_j, \tau^G)_G = (\rho_j, \tau)_{G_1} = m.$$

Furthermore, we have

$$(\psi, \tau^G)_G = (\psi, \tau)_{G_1} = (\rho_1 + 1, \tau)_{G_1} \geq m.$$

So τ^G contains each of the pairwise distinct irreducible characters $\psi, \rho_1, \ldots, \rho_s$ as a constituent with multiplicity at least m. Hence, it follows that

$$\tau^G(1) \geq m(\psi(1) + s\rho_1(1)) = m\left(p - 1 + \frac{p-1}{2}(p-2)\right) = \frac{mp(p-1)}{2}.$$

On the other hand,

$$\tau^G(1) = [G : G_1]\tau(1) = p\tau(1) \leq \frac{p(p-2)}{2}.$$

This is a contradiction. So $(\rho_j)_{G_1}$ is an irreducible character of G_1.

h) G is 3-transitive:

The permutation character of G_1 on the set $\{2, \dots, p\}$ is

$$(\pi - 1)_{G_1} = \psi_{G_1} = 1 + (\rho_1)_{G_1}.$$

Since G is 2-transitive, G_1 is transitive on $\{2, \dots, p\}$. It follows from the irreducibility of $(\rho_1)_{G_1}$ and Theorem 20.2 d) that G_1 is 2-transitive on $\{2, \dots, p\}$. So G is 3-transitive.

i) We denote by $\delta(g)$ the number of 2-cycles in the cycle decomposition of the permutation g. Then the function λ defined by

$$\lambda = \frac{(\pi - 1)(\pi - 2)}{2} - \delta$$

is a character of G with $(\lambda, \psi)_G = (\lambda, 1)_G = 0$:

By Theorem 20.6 a), λ is a character of G. Since G is 3-transitive, we have by Theorem 20.4 b) the relations

$$\sum_{g \in G} \pi(g) = |G|, \quad \sum_{g \in G} \pi(g)^2 = 2|G|, \quad \sum_{g \in G} \pi(g)^3 = 5|G|.$$

Furthermore, by Theorem 20.5,

$$\sum_{g \in G} \delta(g) = \sum_{g \in G} \pi(g)\delta(g) = \frac{|G|}{2}.$$

It then follows that

$$\begin{aligned} |G|(\lambda, \psi)_G &= |G|(\lambda, \pi - 1)_G \\ &= \frac{1}{2} \sum_{g \in G} (\pi(g)^3 - 4\pi(g)^2 + 5\pi(g) - 2 - 2\pi(g)\delta(g) + 2\delta(g)) = 0 \end{aligned}$$

and

$$|G|(\lambda, 1)_G = \frac{1}{2} \sum_{g \in G} (\pi(g)^2 - 3\pi(g) + 2 - 2\delta(g)) = 0.$$

j) $(\lambda, \rho_j)_G = 1$ for $j = 1, \dots, s$:

For $1 \neq x^k \in P$, we have $\pi(x^k) = \delta(x^k) = 0$ and $\lambda(x^k) = 1$. It follows that

$$\begin{aligned} |G|(\lambda, \rho_j)_G = &\sum_{g \in G} \lambda(g)(\psi(g) - 1) - \frac{|G|}{2p} \sum_{k=1}^{p-1} \lambda(x^k)(\psi(x^k) - 1) \\ &+ \frac{|G|}{2p} \sum_{k=1}^{p-1} \lambda(x^k)\rho_j(x^k) \end{aligned}$$

$$= |G|(\lambda, \psi - 1)_G - \frac{|G|}{2p}\sum_{k=1}^{p-1}(-2) + \frac{|G|}{2p}\sum_{k=0}^{p-1}\rho_j(x^k) - \frac{|G|}{2p}\rho_j(1)$$
$$= 0 + \frac{|G|(p-1)}{p} + \frac{|G|}{2}(1, \rho_j)_P - \frac{|G|}{2p}(p-2) \quad \text{(with the help of i))}$$
$$= |G|.$$

k) $\lambda = \sum_{j=1}^{s} \rho_j$:

By j), $\lambda = \sum_{j=1}^{s} \rho_j + \mu$ for some character μ. Comparing degrees on each side, we see that $\mu(1) = 0$, so $\mu = 0$.

l) G is sharply 3-transitive:

Let $g \in G$, and suppose $\pi(g) \geq 3$. Then $|g| \neq p$, so $\rho_j(g) = \pi(g) - 2$. It follows by k) that

$$\frac{(\pi(g)-1)(\pi(g)-2)}{2} - \delta(g) = \lambda(g) = \sum_{j=1}^{s}\rho_j(g) = \frac{p-1}{2}(\pi(g)-2),$$

and therefore

$$\frac{(\pi(g)-2)(p-\pi(g))}{2} + \delta(g) = 0.$$

If $\pi(g) \geq 3$, then since $\delta(g) \geq 0$ and $\pi(g) \leq p$, it follows that $\pi(g) = p$; that is, $g = 1$. Hence, every nonidentity element of G has at most two fixed points. By h), G is 3-transitive, and so G is sharply 3-transitive. □

H. Wielandt has proved a number of unpublished theorems regarding transitive groups of prime degree using completely different, namely ring theoretic, methods. We collect the following together without proofs:

Remark 21.7. Let G be a transitive permutation group of prime degree p, and let P be a Sylow p-subgroup of G.

a) (Wielandt) If $|\mathbf{N}_G(P)| = p(p-1)$ and G is not solvable, then G is 3-transitive. Do we in fact have $G = S_p$ in this situation?

b) (Wielandt) The outer automorphism group $\mathbf{Aut}(G)/\mathbf{Inn}(G)$ is cyclic and its order divides $p-1$. (Consequently, the Schreier Conjecture is correct for simple groups with a subgroup of prime index.)

c) (Wielandt) How many conjugacy classes of subgroups of index p can G have? In all known cases, there are at most two such conjugacy classes. If $|\mathbf{N}_G(P)|$ is even, then G has exactly one conjugacy class of subgroups of index p, the point stabilizers.

d) (N. Itô) Assume G is not solvable. If G contains an odd permutation, then is it true that $G = S_p$?

e) (Frobenius [149]) Assume G has exactly $p+1$ Sylow p-subgroups. Then one of the following occurs:

(1) $p = 5$, $G = A_5$ or $G = S_5$.
(2) $p = 7$, $G = \mathrm{PSL}_3(2)$ as a permutation group on the 7 points of the projective plane over GF(2).
(3) $p = 11$, $G = \mathrm{PSL}_2(11)$ in the 2-transitive permutation representation of degree 11 on the cosets of a subgroup A_5 (see Theorem II.8.28, pg. 224).

With the help of modular representation theory, R. Brauer [49] has proved stronger statements regarding the number of Sylow p-subgroups in a transitive permutation group of prime degree p.

Exercises

38) There exist only the following transitive permutation groups of degree 7, up to permutation isomorphism: solvable groups; S_7; A_7; the group $\mathrm{PSL}_3(2)$ on the 7 points of the projective plane over GF(2).
39) Let G be 3-transitive of prime degree p with $p > 5$. If $p - 2$ is a prime number, then G is 4-transitive. (Considerably deeper statements in this direction can be found in Itô [284, 285].)

22 Involutions

Definition 22.1. Let G be a group of even order. We say that an element of G is an involution if it has order 2. In this section, we always use m to denote the number of involutions in G, and we use r to denote the number of conjugacy classes of involutions.

Lemma 22.2 (R. Brauer, Fowler [66]). *Let $K_0, \ldots, K_{h-1}$ be the conjugacy classes of G, where $K_i = \{g_i^g \mid g \in G\}$, and order them in the following way:*
(1) $K_0 = \{1\}$;
(2) $K_1, \ldots, K_r$ *are the conjugacy classes of involutions;*
(3) $K_{r+1}, \ldots, K_s$ *are the conjugacy classes of real elements that are not involutions.* (*By Definition 13.6, an element $x \in G$ is real if x and x^{-1} are conjugate in G.*)

We define $k_i = \sum_{g \in K_i} g$ and $k = \sum_{i=1}^{r} k_i$. Let $k^2 = \sum_{i=0}^{h-1} a_i k_i$. The natural numbers a_i can be computed in the following way:
a) $a_0 = m$.
b) *If $g_i^2 \neq 1$ (so $r < i \leq h - 1$), then a_i is the number of involutions x such that $g_i^x = g_i^{-1}$; in particular, $a_i = 0$ for nonreal elements g_i, so for $i > s$. Letting $c_i = |\mathbf{C}_G(g_i)|$, we have $a_i \leq c_i$ for $r < i \leq h - 1$.*
c) *If g_i is an involution (so $1 \leq i \leq r$) and m_i is the number of involutions in $\mathbf{C}_G(g_i)$, then $a_i = m_i - 1$. Again letting $c_i = |\mathbf{C}_G(g_i)|$, we have $a_i \leq c_i - 2$.*

Proof. Clearly, a_i is the number of ordered pairs $x, y \in J = \bigcup_{i=1}^{r} K_i$ with $xy = g_i$. The claim in a) is therefore trivial.

If $g_i = xy$ for some $x, y \in J$, it follows that

$$g_i^{-1} = yx = x^{-1}xyx = (xy)^x = g_i^x.$$

Conversely, if $g_i^x = g_i^{-1}$ for some $x \in J$, then we define $y = xg_i$. Then

$$y^2 = g_i^x g_i = g_i^{-1} g_i = 1.$$

If g_i is not an involution, then $y = xg_i \neq 1$, so $y \in J$. So in this case, a_i is the number of involutions $x \in G$ such that $g_i^x = g_i^{-1}$.

We define the subgroup $\mathbf{C}_G^*(g_i)$ of G by

$$\mathbf{C}_G^*(g_i) = \{x \in G \mid g_i^x = g_i^{\pm 1}\}.$$

For a real element g_i with $g_i^2 \neq 1$, there exists an $h \in G$ with $g_i^h = g_i^{-1} \neq g_i$. Then clearly $\mathbf{C}_G^*(g_i) = \mathbf{C}_G(g_i) \cup \mathbf{C}_G(g_i)h$, so

$$|\mathbf{C}_G^*(g_i)| = 2|\mathbf{C}_G(g_i)| = 2c_i.$$

With the already proven part of b), it therefore follows that

$$a_i \leq |\mathbf{C}_G^*(g_i) - \mathbf{C}_G(g_i)| = c_i.$$

If g_i is an involution, then only the $x \in J$ with $g_i^x = g_i$ and $x \neq g_i$ lead to a decomposition $g_i = xy$ with $x, y \in J$. So in this case, $a_i = m_i - 1$, where m_i is the number of involutions in $\mathbf{C}_G(g_i)$. □

It is also natural to consider the element $\sum_{|g|=p} g$ of the group ring for odd primes p. But this has only been carried out successfully[9] in Feit, Thompson [131] with $p = 3$, where the description of groups $G = \langle a_1, a_2 \rangle$ with $a_1^3 = a_2^3 = (a_1 a_2)^3 = 1$ from Example I.19.6 (pg. 145) was used.

Lemma 22.3. *Let G be a group of even order with m involutions. Let m_i be the number of involutions in $\mathbf{C}_G(g_i)$, where the g_i $(i = 1, \ldots, r)$ are representatives of the classes $K_1, \ldots, K_r$ of involutions. We let $m' = \max_{i=1,\ldots,r} m_i$. Then*

[9] This element plays an important role in the block theory of finite groups. See, for example, Theorem 2.9 of R. Knörr, Virtually irreducible lattices, Proc. London Math. Soc. (3) **59** (1989), no. 1, 99–132; J. C. Murray, Blocks of defect zero and products of elements of order p, J. Algebra **214** (1999), no. 2, 385–399; and G. R. Robinson, Involutions, weights and p-local structure, Algebra Number Theory **5** (2011), no. 8, 1063–1068. For a textbook introduction to block theory, see G. Navarro Ortega, *Characters and blocks of finite groups*, London Mathematical Society Lecture Note Series, 250, Cambridge Univ. Press, Cambridge, 1998. Knörr's result is proved there as Theorem 5.16. *trans.*

$$m^2 \leq |G|u + (m' - u)m - u;$$

where $u = 0$ *or* $u = |\mathbf{C}_G(g)|$ *for some real element* $g \in G$ *with* $g^2 \neq 1$.

Proof. We again let $|\mathbf{C}_G(g_i)| = c_i$. Applying the trivial representation to the expression $k^2 = \sum_{i=0}^{h-1} a_i k_i$ yields

$$m^2 = \sum_{i=0}^{h-1} a_i |K_i| = \sum_{i=0}^{h-1} a_i \frac{|G|}{c_i}. \tag{1}$$

The following equations hold trivially:

$$m = \sum_{i=1}^{r} \frac{|G|}{c_i} \tag{2}$$

and

$$1 + m + \sum_{i=r+1}^{s} \frac{|G|}{c_i} \leq |G|. \tag{3}$$

By Lemma 22.2, $a_0 = m$; furthermore, $a_i = m_i - 1$ for $1 \leq i \leq r$ and $a_i \leq c_i$ for $i = r + 1, \ldots, h - 1$. Therefore, it follows from (1) with the help of (2) that

$$\begin{aligned} m^2 &\leq m + \sum_{i=1}^{r} (m_i - 1)\frac{|G|}{c_i} + (s - r)|G| \\ &\leq m + (m' - 1)\sum_{i=1}^{r} \frac{|G|}{c_i} + (s - r)|G| = m'm + (s - r)|G|. \end{aligned} \tag{4}$$

If $r = s$, then we set $u = 0$. Then the claim is given in (4). If $r < s$, then let $u = \max_{i=r+1,\ldots,s} |\mathbf{C}_G(g_i)|$. It then follows from (3) that

$$(s - r)\frac{|G|}{u} \leq \sum_{i=r+1}^{s} \frac{|G|}{c_i} \leq |G| - m - 1.$$

Therefore, we obtain from (4) that

$$m^2 \leq mm' + (s - r)|G| \leq mm' + u(|G| - m - 1) = |G|u + (m' - u)m - u,$$

where $u = |\mathbf{C}_G(g_j)|$ for some j with $r + 1 \leq j \leq s$. □

The elementary considerations in Lemma 22.2 and Lemma 22.3 have an interesting application:

Theorem 22.4 (R. Brauer, Fowler [66]). *Let* G *be a group of even order* $|G| > 2$. *Then there exists a proper subgroup* $U < G$ *such that* $|U|^3 > |G|$.

Proof. Let m be the number of involutions in G, and let $t = \frac{|G|}{m}$. Then it follows from Lemma 22.3 that

$$\frac{|G|^2}{t^2} \leq |G|u + (m' - u)\frac{|G|}{t},$$

so

$$|G| \leq t^2 u + (m' - u)t = ut(t-1) + m't.$$

Here, $m' = \max\limits_{i=1,\ldots,r} m_i = m_j$ for some j, and therefore

$$m' = m_j \leq |\mathbf{C}_G(g_j)| - 1 = c_j - 1.$$

It follows from

$$\frac{|G|}{t} = m = \sum_{i=1}^{r} \frac{|G|}{c_i}$$

that $t \leq c_i$ for $i = 1, \ldots, r$.

Hence,

$$|G| \leq uc_j(c_j - 1) + (c_j - 1)c_j = (u+1)c_j(c_j - 1). \tag{1}$$

If $c_j \leq u$, then $|G| < u^3$. Here, $u = |\mathbf{C}_G(g)|$ for some real $g \in G$ with $g^2 \neq 1$. So there is some $h \in G$ such that $g^h = g^{-1} \neq g$, and thus $\mathbf{C}_G(g) < G$. In this case we set $U = \mathbf{C}_G(g)$.

If $u < c_j < |G|$, then it follows from (1) that $|G| < c_j^3$, and we set $U = \mathbf{C}_G(g_j)$.

Finally, assume $|G| = c_j$. Then g_j is a central involution of G. First assume that $\frac{|G|}{2}$ is even and $|G| > 4$. By induction, there exists a subgroup $U/\langle g_j \rangle$ of $G/\langle g_j \rangle$ such that $U < G$ and

$$\frac{|U|^3}{8} = |U/\langle g_j \rangle|^3 > |G/\langle g_j \rangle| = \frac{|G|}{2}.$$

Then $|U|^3 > 4|G| > |G|$. If $|G| = 4$, then G has a subgroup of order 2. If $\frac{|G|}{2}$ is odd, then G has a subgroup U of index 2 by Theorem I.6.7 (pg. 30) (or by Theorem IV.2.8, pg. 446). Then the claim is trivially true. □

If $|G|$ is odd and not a prime number, then there exists a subgroup $U < G$ with $|U|^2 \geq |G|$. The only known proof of this statement uses the deep fact that groups of odd order are solvable, and the statement is trivially true for solvable groups by the existence of Sylow complements (see Theorem VI.1.8, pg. 701).

The following lemma establishes a link between our current considerations and character theory:

Lemma 22.5. *Again let k be the sum of involutions of G, and let $k^2 = \sum_{i=0}^{h-1} a_i k_i$ as in Lemma 22.2. If χ is a character of G over the complex numbers $\mathbb{C}$, then*

$$\chi(k) = \sum_{i=1}^{r} h_i \chi(g_i),$$

where $h_i = |K_i| = [G : \mathbf{C}_G(g_i)]$.

If $\chi_1, \ldots, \chi_h$ are the irreducible characters of G over the complex numbers, then

$$a_i = \frac{1}{|G|} \sum_{j=1}^{h} \frac{\chi_j(k)^2 \chi_j(g_i^{-1})}{\chi_j(1)}.$$

Proof. Let $\mathbf{D}_j$ be an irreducible representation with character χ_j. As $k \in \mathbf{Z}(\mathbb{C}[G])$, we have by Schur's Lemma that $\mathbf{D}_j(k) = b_j 1_{n_j}$ for $n_j = \chi_j(1)$ and some $b_j \in \mathbb{C}$. It follows that

$$b_j \chi_j(1) = \operatorname{tr} \mathbf{D}_j(k) = \chi_j(k).$$

Furthermore, we obtain from $\mathbf{D}_j(k^2) = b_j^2 1_{n_j}$ that

$$\chi_j(k^2) = b_j^2 \chi_j(1) = \frac{\chi_j(k)^2}{\chi_j(1)}.$$

Since $k^2 = \sum_{i=0}^{h-1} a_i k_i$, it then follows that

$$\frac{\chi_j(k)^2}{\chi_j(1)} = \chi_j(k^2) = \sum_{i=0}^{h-1} a_i \chi_j(k_i) = \sum_{i=0}^{h-1} a_i h_i \chi_j(g_i).$$

The orthogonality relations yield

$$a_i |G| = \sum_{l=0}^{h-1} a_l h_l \sum_{j=1}^{h} \chi_j(g_l) \chi_j(g_i^{-1}) = \sum_{j=1}^{h} \frac{\chi_j(k)^2}{\chi_j(1)} \chi_j(g_i^{-1}). \qquad \square$$

Definition 22.6. Let D be a nonempty subset of G. We call D a TI-set (trivial intersection set) if $D \cap D^g \subseteq \{1\}$ for all $g \in G$ with $D \neq D^g$ and D lies in its normalizer $\mathbf{N}_G(D)$.

The following lemma will play a central role in our treatment of exceptional characters in Volume 3.

Lemma 22.7. *Let D be a TI-set in G. We let $L = \mathbf{N}_G(D)$.*

a) *If φ is a class function on L such that $\varphi = 0$ on $L - D$, then $\varphi^G(d) = \varphi(d)$ for all $1 \neq d \in D$.*

b) *If φ and ψ are class functions on L such that $\varphi = \psi = 0$ on $L - D$ and $\varphi(1) = 0$, then $(\varphi^G, \psi^G)_G = (\varphi, \psi)_L$. So the map $\varphi \mapsto \varphi^G$ is an isometry from the subspace $\{\varphi \mid \varphi = 0$ on $L - D$ and $1\}$ of $\mathsf{C}(L, \mathbb{C})$ to the space $\mathsf{C}(G, \mathbb{C})$ of class functions on G.*

Proof. a) Let $G = \bigcup_{i=1}^{n} Lr_i$ be the decomposition of G into cosets of L with $r_1 = 1$. For $d \in D$, we have by Theorem 16.3 that

$$\varphi^G(d) = \sum_{d^{r_i} \in L} \varphi(d^{r_i}).$$

If $d^{r_i} \in L - D$, then $\varphi(d^{r_i}) = 0$. If $1 \neq d^{r_i} \in D$, then $D \cap D^{r_i} \not\subseteq 1$, so $r_i \in \mathbf{N}_G(D) = L$ and $r_i = r_1 = 1$. So we obtain $\varphi^G(d) = \varphi(d)$.

b) By Frobenius reciprocity,

$$(\varphi^G, \psi^G)_G = (\varphi, (\psi^G)_L)_L.$$

We define a class function μ on L by $\mu = (\psi^G)_L - \psi$. By a), $\mu(d) = 0$ for $1 \neq d \in D$. Since φ is 0 on $L - D$ and 1, it follows that

$$(\varphi, \mu)_L = \frac{1}{|L|} \sum_{l \in L} \varphi(l)\overline{\mu(l)} = 0.$$

Therefore,

$$(\varphi^G, \psi^G)_G = (\varphi, \psi + \mu)_L = (\varphi, \psi)_L. \qquad \square$$

Lemma 22.8. *Let $D \subseteq L \leq G$. Assume that for every involution x of G such that $d^x = d^{-1}$ for some $1 \neq d \in D$, we have that the involution x lies in L. Let $\chi_1, \ldots, \chi_h$ be the irreducible characters of G over the complex numbers, and let $\psi_1, \ldots, \psi_{h'}$ be those of L. Let τ be a class function on L such that $\tau = 0$ on $L - D$ and $\tau(1) = 0$. We define*

$$T_G(\chi_i) = \sum_{j=1}^{r} \frac{\chi_i(g_j)}{|\mathbf{C}_G(g_j)|},$$

where the sum runs over the representatives g_j of the conjugacy classes of involutions. Then

$$|G| \sum_{i=1}^{h} \frac{T_G(\chi_i)^2}{\chi_i(1)} (\chi_i, \tau^G)_G = |L| \sum_{j=1}^{h'} \frac{T_L(\psi_j)^2}{\psi_j(1)} (\psi_j, \tau)_L.$$

Proof. By Lemma 22.5, $k^2 = \sum_{i=0}^{h-1} a_i k_i$, where

$$a_i = \frac{1}{|G|} \sum_{j=1}^{h} \frac{\chi_j(k)^2}{\chi_j(1)} \chi_j(g_i^{-1}).$$

It is clear by Lemma 22.2 that the integers a_i are unchanged upon replacing g_i by g_i^{-1}. So we also have

$$a_i = \frac{1}{|G|} \sum_{j=1}^{h} \frac{\chi_j(k)^2}{\chi_j(1)} \chi_j(g_i).$$

Upon using

$$\chi_j(k) = \sum_{m=1}^{r} h_m \chi_j(g_m) = |G| \sum_{m=1}^{r} \frac{\chi_j(g_m)}{|\mathbf{C}_G(g_m)|} = |G| T_G(\chi_j),$$

it follows that

$$a_i = |G| \sum_{j=1}^{h} \frac{T_G(\chi_j)^2}{\chi_j(1)} \chi_j(g_i).$$

Let $1 \neq g_i \in D$. If x is an involution in G such that $g_i^x = g_i^{-1}$, then x lies in L by our assumption. If k' is the sum of the involutions in L, then the coefficient of g_i in k'^2 is uniquely determined by the number of involutions $x \in L$ such that $g_i^x = g_i^{-1}$ by Lemma 22.2, so it is exactly a_i. Therefore, we may also compute a_i in L, and we obtain as before that

$$a_i = |L| \sum_{j=1}^{h'} \frac{T_L(\psi_j)^2}{\psi_j(1)} \psi_j(g_i).$$

Hence, the class function

$$\mu = |G| \sum_{j=1}^{h} \frac{T_G(\chi_j)^2}{\chi_j(1)} (\chi_j)_L - |L| \sum_{j=1}^{h'} \frac{T_L(\psi_j)^2}{\psi_j(1)} \psi_j$$

vanishes on $D - 1$. If $\tau = 0$ on $L - D$ and 1, then it follows that $(\mu, \tau)_L = 0$. With the help of Frobenius reciprocity, this yields

$$\begin{aligned}(\mu, \tau)_L + |L| \sum_{j=1}^{h'} \frac{T_L(\psi_j)^2}{\psi_j(1)} (\psi_j, \tau)_L &= |G| \sum_{j=1}^{h} \frac{T_G(\chi_j)^2}{\chi_j(1)} ((\chi_j)_L, \tau)_L \\ &= |G| \sum_{j=1}^{h} \frac{T_G(\chi_j)^2}{\chi_j(1)} (\chi_j, \tau^G)_G. \qquad \square\end{aligned}$$

Now we come to the most important application of the results in this section:

Theorem 22.9 (R. Brauer, Suzuki). *Suppose the Sylow 2-subgroups of G are generalized quaternion of order 2^{n+1} with $n > 2$. If N is the largest normal subgroup of G with odd order, then G/N has a normal subgroup of order 2. In particular, G is not simple.*

(*The claim also holds when the Sylow 2-subgroups of G are quaternion groups of order 8; however, the only known proof of this statement uses modular character theory, see R. Brauer [59].*)

Proof (Suzuki [403]). Let $Q = \langle a, b\rangle$ be a Sylow 2-subgroup of G with relations

$$a^{2^n} = 1, \quad b^2 = a^{2^{n-1}}, \quad a^b = a^{-1}.$$

Then $x = a^{2^{n-1}}$ is the only involution in Q. Defining $S = \langle a^{2^{n-2}}\rangle$, we have that $S = \Omega_2(Q')$ is a characteristic subgroup of Q since $n > 2$. (This is not correct for $n = 2$! In the quaternion group of order 8, the three cyclic subgroups of order 4 are transitively permuted by an automorphism of order 3.)

We show:

a) All involutions of G are conjugate:

If y is an involution in G, then by Sylow's Theorem there exists some $g \in G$ such that $y^g \in Q$; that is, $y^g = x$.

b) $\mathbf{N}_G(S)$ is 2-nilpotent:

Since $\mathbf{N}_G(S)/\mathbf{C}_G(S)$ is isomorphic to a subgroup of the automorphism group $\mathbf{Aut}(S)$ of S, we have

$$|\mathbf{N}_G(S)/\mathbf{C}_G(S)| \leq |\mathbf{Aut}(S)| = 2.$$

Also, $Q \leq \mathbf{N}_G(S)$ because $S \lhd Q$. Since $a \in \mathbf{C}_G(S)$ and $b \notin \mathbf{C}_G(S)$, it follows that $\langle a\rangle$ is a Sylow 2-subgroup of $\mathbf{C}_G(S)$. So $\mathbf{C}_G(S)$ has cyclic Sylow 2-subgroups, and thus is 2-nilpotent by Burnside's Theorem IV.2.8 (pg. 446). Let M be the normal 2-complement of $\mathbf{C}_G(S)$. Then $\mathbf{C}_G(S) = M\langle a\rangle$ and $\mathbf{N}_G(S) = MQ$. Since M is characteristic in $\mathbf{C}_G(S)$, we have $M \lhd \mathbf{N}_G(S)$. So $\mathbf{N}_G(S)$ is 2-nilpotent.

c) S is the only subgroup of order 4 in $\mathbf{C}_G(S)$:

Since $S \lhd \mathbf{C}_G(S)$, S lies in every Sylow 2-subgroup of $\mathbf{C}_G(S)$. The claim then follows because the Sylow 2-subgroups of $\mathbf{C}_G(S)$ are cyclic.

d) We define $U = \mathbf{C}_G(S) = M\langle a\rangle$, $L = \mathbf{N}_G(S)$, $V = M\langle x\rangle$ and $D = U - V$. Then D is a TI-set with $L = \mathbf{N}_G(D)$:

For $l \in \mathbf{N}_G(S)$, we have

$$D^l = U^l - V^l = U - V = D,$$

so $L \leq \mathbf{N}_G(D)$.

Suppose $g \in G$ such that $D \cap D^g \neq \emptyset$. Then there exist elements $d, d' \in D = U - V$ with $d^g = d'$. We let $\overline{d} = dM$. Since

$$\overline{d} \in U/M - V/M = \langle aM\rangle - \langle xM\rangle,$$

we have that 4 divides $|\overline{d}|$. So 4 is also a divisor of $|d|$. Let $f = d^m$ such that $|f| = 4$. Then $f \in U$, but certainly $f \notin V$ since $4 \nmid |V|$. Hence, $f \in U - V = D$. Setting $f' = d'^m$, it follows analogously that $f' \in D$. Therefore, using c) we have

$$S^g = \langle f \rangle^g = \langle f' \rangle = S,$$

so $g \in \mathbf{N}_G(S) = L$. Hence, $\mathbf{N}_G(D) = L$ and $D \cap D^g = \emptyset$ for $g \notin L$.

e) D and L satisfy the assumptions of Lemma 22.8:

If $1 \neq d \in D$ and $d^g = d^{-1}$, then it follows that $D \cap D^g \neq \emptyset$ since $d^{-1} \in D$. Then by d), $g \in L = \mathbf{N}_G(D)$.

f) Let η_1 be the trivial character of L, and let η_2 be the unique nontrivial irreducible character of L with $\ker \eta_2 = U$. The quotient group

$$L/V = MQ/M\langle x \rangle \cong Q/\langle x \rangle$$

is a dihedral group of order 2^n with $n \geq 3$, so it has an irreducible character φ of degree 2 by Lemma 21.5. This corresponds to a character η_3 of L. Defining $\tau = \eta_1 + \eta_2 - \eta_3$, we have $\tau^G = \chi_1 + \chi_2 - \chi_3$ for some irreducible, pairwise distinct characters χ_1, χ_2, χ_3 of G. Here, χ_1 is the trivial character of G:

By Lemma 21.5, $\varphi = 0$ on $Q - \langle a \rangle$. So $\eta_3 = 0$ on

$$QM - \langle a \rangle M = L - U.$$

We have $\eta_2(l) = 1$ for $l \in U$ and $\eta_2(l) = -1$ for $l \in L - U$. So for $l \in L - U$, we have

$$\tau(l) = \eta_1(l) + \eta_2(l) - \eta_3(l) = 0.$$

Since $V = M\langle x \rangle \leq \ker \eta_3$, we have for $l \in V$ that

$$\tau(l) = 1 + 1 - 2 = 0.$$

So $\tau = 0$ on $(L - U) \cup V = L - (U - V) = L - D$. By d), the assumptions of Lemma 22.7 are satisfied. Therefore, it follows that

$$(\tau^G, \tau^G)_G = (\tau, \tau)_L = 3.$$

Furthermore,

$$(\tau^G, \chi_1)_G = (\tau, \eta_1)_L = 1.$$

So $\tau^G = \chi_1 \pm \chi_2 \pm \chi_3$ for some pairwise distinct, irreducible characters χ_i of G. It follows from $\tau(1) = 0$ that

$$0 = \tau^G(1) = 1 \pm \chi_2(1) \pm \chi_3(1).$$

This forces $\tau = \chi_1 + \chi_2 - \chi_3$ for some labeling of the χ_i.

g) We have $1 + \chi_2(x) - \chi_3(x) = 0$:

If $x^g \in L$, then $x^g \in V = M\langle x \rangle$ and therefore $\tau(x^g) = 0$. Hence, it follows that

$$1 + \chi_2(x) - \chi_3(x) = \tau^G(x) = \frac{1}{|L|} \sum_{x^g \in L} \tau(x^g) = 0.$$

h) In the notation of Lemma 22.8, we have

$$\sum_{i=1}^{h'} \frac{T_L(\psi_i)^2}{\psi_i(1)} (\psi_i, \tau)_L = 0 :$$

It is clear that x is the only involution in $L = \mathbf{C}_L(x)$, since otherwise L would contain a subgroup of type $(2, 2)$. Therefore,

$$T_L(\psi_i) = \frac{\psi_i(x)}{|L|}$$

and

$$\begin{aligned}\sum_{i=1}^{h'} \frac{T_L(\psi_i)^2}{\psi_i(1)} (\psi_i, \tau)_L &= \frac{T_L(\eta_1)^2}{\eta_1(1)} + \frac{T_L(\eta_2)^2}{\eta_2(1)} - \frac{T_L(\eta_3)^2}{\eta_3(1)} \\ &= \frac{1}{|L|^2} \left(\frac{\eta_1(x)^2}{\eta_1(1)} + \frac{\eta_2(x)^2}{\eta_2(1)} - \frac{\eta_3(x)^2}{\eta_3(1)} \right) \\ &= \frac{1}{|L|^2} \left(1 + 1 - \frac{2^2}{2} \right) = 0.\end{aligned}$$

i) There exists a character χ of G with $\chi(1) > 1$ that contains x in the kernel: Since G has only one conjugacy class of involutions by a), we have

$$T_G(\chi_i) = \frac{\chi_i(x)}{|\mathbf{C}_G(x)|}.$$

By e), the assumptions of Lemma 22.8 are satisfied. Therefore, it follows by f) and h) that

$$\begin{aligned}0 = \sum_{i=1}^{h} \frac{T_G(\chi_i)^2}{\chi_i(1)} (\chi_i, \tau^G)_G &= T_G(\chi_1)^2 + \frac{T_G(\chi_2)^2}{\chi_2(1)} - \frac{T_G(\chi_3)^2}{\chi_3(1)} \\ &= \frac{1}{|\mathbf{C}_G(x)|^2} \left(1 + \frac{\chi_2(x)^2}{\chi_2(1)} - \frac{\chi_3(x)^2}{\chi_3(1)} \right).\end{aligned}$$

Furthermore,

$$0 = \tau^G(1) = 1 + \chi_2(1) - \chi_3(1) \tag{1}$$

and by g),

$$0 = 1 + \chi_2(x) - \chi_3(x). \tag{2}$$

Altogether, this gives

$$\begin{aligned} 0 &= \chi_2(1)\chi_3(1) + \chi_2(x)^2\chi_3(1) - \chi_3(x)^2\chi_2(1) \\ &= \chi_2(1)(1+\chi_2(1)) + \chi_2(x)^2(1+\chi_2(1)) - (1+\chi_2(x))^2\chi_2(1) \\ &= (\chi_2(1) - \chi_2(x))^2. \end{aligned}$$

So it follows that $\chi_2(x) = \chi_2(1)$. Using (1) and (2), we then obtain that $\chi_3(x) = \chi_3(1)$. Hence, x lies in the kernel of χ_3 by Lemma 7.4. As $\chi_3(1) = 1 + \chi_2(1)$, we have $\chi_3(1) > 1$.

j) Conclusion:

Let J be the subgroup generated by all involutions in G. Then $Q_0 = J \cap Q$ is a Sylow 2-subgroup of J. Since Q_0 contains a unique involution, Q_0 is either cyclic or generalized quaternion.

Assume for contradiction that Q_0 is not cyclic. Then Q_0 contains an element of the form $a^j b$. Since $|\mathbf{C}_Q(a^j b)| = |\langle a^j b\rangle| = 4$, the element $a^j b$ has exactly 2^{n-1} conjugates in Q, which all lie in Q_0 since $Q_0 \trianglelefteq Q$. It follows from this that $1 + 2^{n-1} < |Q_0|$, so $|Q/Q_0| \leq 2$. Since

$$QJ/J \cong Q/(Q \cap J) = Q/Q_0,$$

we have that QJ/J is abelian. Applying i) to QJ shows that QJ has an irreducible, nonlinear character χ whose kernel contains x. Since all involutions of QJ are conjugate to x, it follows that

$$J = \langle x^g \mid g \in QJ\rangle \leq \ker \chi.$$

Therefore, χ gives rise to an irreducible character of the abelian group QJ/J with $\chi(1) > 1$. This is a contradiction.

So Q_0 is cyclic. By Theorem IV.2.8 (pg. 446), J is 2-nilpotent. If K is the normal 2-complement of J, then it follows that $J = K\langle x\rangle \lhd G$. □

Glauberman has considerably generalized Theorem 22.9. We state his results here without proof:

Theorem 22.10 (Glauberman [170]). *Let G be a group that has no nonidentity normal subgroup of odd order, and let x be an element in a Sylow 2-subgroup P of G. Then $x \in \mathbf{Z}(G)$ if and only if*

$$\{x^g \mid g \in G\} \cap P = \{x\}.$$

The proof of Theorem 22.10 uses the full version of Theorem 22.9 for $|Q| \geq 8$, which we have not proved here, in addition to many results from modular representation theory. Theorem 22.10 easily yields the following generalization of Theorem 22.9:

Theorem 22.11 (Glauberman [170]). *Suppose the Sylow 2-subgroups of G are direct products of a generalized quaternion group with a 2-group whose involutions lie in its center. Then G is not simple.*

In particular, the Sylow 2-subgroup of a finite simple group is never a direct product of generalized quaternion groups.

Lemma 22.8 has been useful in other applications, as well; see Itô [283] and Janko [293].

Exercises

40) (Suzuki [409], Brauer [60]) Assume $|G| = 4n$ with $2 \nmid n$, and suppose that G has no nontrivial quotients that are 2-groups. Let x be an involution in G, and let $H = \mathbf{C}_G(x)$. Show:
   a) H has a normal 2-complement U.
   b) Let $\eta_1 = 1, \eta_2, \eta_3, \eta_4$ with $\eta_2(x) = 1$ be the irreducible characters of H/U. If we define $\tau = \eta_1 + \eta_2 - \eta_3 - \eta_4$, then $\tau^G = \chi_1 + \epsilon_2\chi_2 + \epsilon_3\chi_3 + \epsilon_4\chi_4$ for some irreducible χ_i and $\epsilon_i = \pm 1$.
   (If P is a Sylow 2-subgroup of H and $\mathbf{C}_G(P) = V \times P$, then set $D = Vx$.)
   c) We have

$$|G|\left\{1 + \sum_{i=2}^{4} \epsilon_i \frac{\chi_i(x)^2}{\chi_i(1)}\right\} = 2^5[U : V]^2|U|.$$

23 The Schur multiplier and representation groups

In the next three sections, we discuss the theory of projective representations of finite groups. This theory was developed by I. Schur in three large works [371, 373, 375].

Definition 23.1. Let K be an algebraically closed field of characteristic 0, which we fix once and for all. Let G be a finite group. We give the multiplicative group $\mathsf{K}^\times$ of K the structure of a G-module by defining $kg = k$ for all $k \in \mathsf{K}^\times$ and all $g \in G$. We call the cohomology group $\mathbf{H}^2(G, \mathsf{K}^\times)$ the Schur multiplier of G. (We will show that $\mathbf{H}^2(G, \mathsf{K}^\times)$ is independent of K as long as K satisfies the conditions given above. As we will see, it only depends on the group of $|G|$-th roots of unity lying in K.) We adopt the notation for cohomology theory defined in Chapter I, §16.

Lemma 23.2. *For every 2-cocycle $c \in \mathbf{Z}^2(G, \mathsf{K}^\times)$, there exists a 2-cocycle c' with $c'c^{-1} \in \mathbf{B}^2(G, \mathsf{K}^\times)$ such that $c'(g, g')$ is a $|G|$-th root of unity in K for all $g, g' \in G$. In particular, the Schur multiplier $\mathbf{H}^2(G, \mathsf{K}^\times)$ is finite.*

Proof. Let $c \in \mathbf{Z}^2(G, \mathsf{K}^\times)$. It follows from Theorem I.16.19 a) that there is some $b \in \mathbf{C}^1(G, \mathsf{K}^\times)$ such that

$$c^{|G|} = b\delta \in \mathbf{B}^2(G, \mathsf{K}^\times).$$

Since K is algebraically closed, there exists some $d \in \mathbf{C}^1(G, \mathsf{K}^\times)$ such that $d(g)^{|G|} = b(g)$ for all $g \in G$. Then with $c' = c(d\delta)^{-1}$, we have

$$c' \equiv c \mod \mathbf{B}^2(G, \mathsf{K}^\times),$$

and the $c'(g, g')$ are $|G|$-th roots of unity. It is clear that there are at most $|G|^{|G|^2}$ such c'. So $\mathbf{H}^2(G, \mathsf{K}^\times)$ is finite. □

Lemma 23.3. *Let G be a finite group, and let H be a not necessarily finite group with an abelian normal subgroup A such that $A \leq \mathbf{Z}(H)$ and $H/A \cong G$. Let $H = \bigcup_{g \in G} Ar(g)$ be the decomposition of H into cosets of A, and let*

$$r(g_1)r(g_2) = a(g_1, g_2)r(g_1g_2).$$

Then:

a) *For α in the character group $\hat{A} = \mathrm{Hom}(A, \mathsf{K}^\times)$ of A, we define $\alpha^\mu \in \mathbf{Z}^2(G, \mathsf{K}^\times)$ by*

$$\alpha^\mu(g_1, g_2) = \alpha(a(g_1, g_2))$$

and we define $\alpha^\epsilon \in \mathbf{H}^2(G, \mathsf{K}^\times)$ by

$$\alpha^\epsilon = \alpha^\mu \mathbf{B}^2(G, \mathsf{K}^\times).$$

Then ϵ is a homomorphism from $\hat{A}$ to $\mathbf{H}^2(G, \mathsf{K}^\times)$ with kernel $(A \cap H')^\perp$.

b) *We have $\hat{A}^\epsilon \cong A \cap H'$. In particular, $A \cap H'$ is finite.*

Proof. a) It follows immediately from the relation

$$a(g_1, g_2)a(g_1g_2, g_3) = a(g_1, g_2g_3)a(g_2, g_3)$$

(see Theorem I.14.1, pg. 92) that μ is a homomorphism from $\hat{A}$ to $\mathbf{Z}^2(G, \mathsf{K}^\times)$. When does α^μ lie in $\mathbf{B}^2(G, \mathsf{K}^\times)$?

First assume that $\alpha^\mu = b\delta \in \mathbf{B}^2(G, \mathsf{K}^\times)$ for some $b \in \mathbf{C}^1(G, \mathsf{K}^\times)$. Then

$$\alpha(a(g_1, g_2)) = \alpha^\mu(g_1, g_2) = b(g_1)b(g_2)b(g_1g_2)^{-1}.$$

We define the map τ from H to $\mathsf{K}^\times$ by $\tau(ar(g)) = \alpha(a)b(g)$. Since

$$\begin{aligned}\tau(a_1r(g_1)a_2r(g_2)) &= \tau(a_1a_2a(g_1, g_2)r(g_1g_2)) \\ &= \alpha(a_1)\alpha(a_2)\alpha(a(g_1, g_2))b(g_1g_2) \\ &= \alpha(a_1)\alpha(a_2)b(g_1)b(g_2) = \tau(a_1r(g_1))\tau(a_2r(g_2)),\end{aligned}$$

we have that τ is a homomorphism from H to $\mathsf{K}^\times$. So $\tau = 1$ on H'. We have

$$1 = \tau(r(1)^{-1}r(1)) = \alpha(r(1)^{-1})b(1).$$

Therefore, we have for all $a \in A$ that

$$\tau(a) = \alpha(ar(1)^{-1})b(1) = \alpha(a)\alpha(r(1)^{-1})b(1) = \alpha(a).$$

In particular, it follows for $a \in A \cap H'$ that

$$1 = \tau(a) = \alpha(a).$$

We have shown that if $\alpha^\mu \in \mathbf{B}^2(G, \mathsf{K}^\times)$, then $\alpha \in (A \cap H')^\perp$.

Conversely, assume that $\alpha \in (A \cap H')^\perp$. For $a \in A$ and $h \in H'$, we define a function β on AH' by

$$\beta(ah) = \alpha(a).$$

Since $\alpha \in (A \cap H')^\perp$, β is well defined. It is clear that β is a homomorphism from AH' to $\mathsf{K}^\times$ with $\ker\beta \geq H'$. By the divisibility of $\mathsf{K}^\times$, we may extend β to a homomorphism γ from H to $\mathsf{K}^\times$ by Theorem I.13.17 (pg. 87). Now we have

$$\gamma(r(g_1))\gamma(r(g_2)) = \gamma(a(g_1, g_2)r(g_1g_2)) = \alpha(a(g_1, g_2))\gamma(r(g_1g_2)).$$

Defining $b(g) = \gamma(r(g))$, it follows that

$$\alpha^\mu(g_1, g_2) = \alpha(a(g_1, g_2)) = b(g_1)b(g_2)b(g_1g_2)^{-1}.$$

So α^μ lies in $\mathbf{B}^2(G, \mathsf{K}^\times)$.

b) Using Theorem 6.3 a) (pg. 517), it follows from a) that

$$\hat{A}^\epsilon \cong \hat{A}/(A \cap H')^\perp \cong \widehat{A \cap H'}.$$

Since $\hat{A}^\epsilon$ is finite by Lemma 23.2, being a subgroup of $\mathbf{H}^2(G, \mathsf{K}^\times)$, the group $\widehat{A \cap H'}$ is also finite. It then follows by Theorem 6.3 and Theorem 6.4 that $A \cap H' \cong \widehat{A \cap H'}$.□

Definition 23.4. A group K is called a representation group for the finite group G when the following conditions are satisfied: There exists a subgroup A of K such that
(1) $A \leq K' \cap \mathbf{Z}(K)$,
(2) $K/A \cong G$,
(3) $|A| = |\mathbf{H}^2(G, \mathsf{K}^\times)|$.

Main Theorem 23.5 (I. Schur [373]). *Let G be a finite group, and let F be a free group in n free generators with $F/R \cong G$. We define*

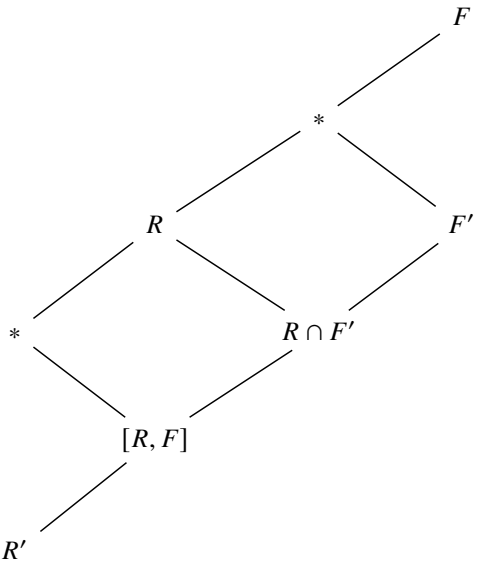
$$H = F/[R, F] \quad \textit{and} \quad J = R/[R, F].$$

Then:

a) *J is a finitely generated abelian subgroup of* $\mathbf{Z}(H)$ *and*

$$T = (R \cap F')/[R, F]$$

is the torsion group of J.
b) *J has rank n.*
c) *If* K *is algebraically closed of characteristic* 0*, then* $\mathbf{H}^2(G, \mathsf{K}^\times) \cong T$*; this proves, in particular, that* $\mathbf{H}^2(G, \mathsf{K}^\times)$ *is independent of* K*, as already remarked in Definition 23.1.*
d) *If $J = S \times T$, then H/S is a representation group for G.*
e) *If L is a representation group for G with $L/B \cong G$ and $B \leq L' \cap \mathbf{Z}(L)$, then L is a homomorphic image of $F/[R, F]$, and we have $B \cong \mathbf{H}^2(G, \mathsf{K}^\times)$.*



Proof. a) Since $J \leq \mathbf{Z}(H)$, we have $|H/\mathbf{Z}(H)| \leq |H/J| = |G| < \infty$. So H' is finite by Theorem IV.2.3 (pg. 443). Therefore,

$$H' \cap J = (F' \cap R)/[R, F]$$

lies in the torsion group T of J. As a subgroup of the free abelian group F/F' of rank n, the group $RF'/F' \cong R/(R \cap F')$ is a free abelian group by Theorem I.13.4 a) (pg. 78). Therefore, the torsion group T of $J = R/[R, F]$ lies in $(R \cap F')/[R, F]$.

Since $|F/R| = |G| < \infty$, the group R is finitely generated by Theorem I.19.10 (pg. 151). So $J = R/[R, F]$ is also finitely generated.

b) We have by the proof of a) that RF'/F' is a free abelian subgroup of F/F' with

$$[F/F' : RF'/F'] \leq |F/R| = |G| < \infty.$$

Therefore, by Theorem I.13.14 a) (pg. 85) the factor group

$$RF'/F' \cong R/(R \cap F') \cong J/T$$

has the same rank n as F/F'. So J also has rank n.

c) Let $H = \bigcup_{g \in G} Jr(g)$ be the decomposition of H into cosets of J with

$$r(g_1)r(g_2) = j(g_1, g_2)r(g_1g_2)$$

and $j(g_1, g_2) \in J$. With the map ϵ from Lemma 23.3, we have

$$(R \cap F')/[R, F] = J \cap H' \cong \hat{J}^{\epsilon}.$$

We show that ϵ is an epimorphism from the character group $\hat{J}$ of J onto $\mathbf{H}^2(G, \mathsf{K}^\times)$.

Let $c \in \mathbf{Z}^2(G, \mathsf{K}^\times)$. We equip the set

$$G(c) = \{(g, k) \mid g \in G, k \in \mathsf{K}^\times\}$$

with the multiplication

$$(g_1, k_1)(g_2, k_2) = (g_1g_2, c(g_1, g_2)c(1, 1)^{-1}k_1k_2).$$

The associativity of multiplication follows from the relation

$$1 = (c\delta)(g_1, g_2, g_3) = c(g_1, g_2)c(g_1, g_2g_3)^{-1}c(g_1g_2, g_3)c(g_2, g_3)^{-1},$$

and we have

$$c(g, 1) = c(1, g) = c(1, 1)$$

for all $g \in G$. Therefore, we have that $(1, 1)$ is the identity element of $G(c)$ and $(g^{-1}, k^{-1}c(g, g^{-1})^{-1}c(1, 1))$ is the inverse of (g, k). So $G(c)$ is a group. The map $k \mapsto (1, k)$ is an isomorphism from $\mathsf{K}^\times$ onto the central subgroup $\{(1, k) \mid k \in \mathsf{K}^\times\}$ of $G(c)$.

Let π be the homomorphism from F onto $G \cong F/R$, and let $\{f_1, \ldots, f_n\}$ be a system of free generators of F. Then the map τ defined on the generators by

$$f_i^\tau = (f_i^\pi, 1)$$

extends to a homomorphism $f \mapsto f^\tau = (f^\pi, b(f))$ from F to $G(c)$. Since $[R, F]^\tau = 1$, the map σ defined by

$$(f[R, F])^\sigma = f^\tau = (f^\pi, b(f))$$

is a homomorphism from $H = F/[R, F]$ to $G(c)$.

We define the map α from $J = R/[R, F]$ to $\mathsf{K}^\times$ by

$$j^{\sigma} = (1, \alpha(j)).$$

Then α is a character of J. It follows from

$$r(g_1)r(g_2) = j(g_1, g_2)r(g_1g_2)$$

that

$$\begin{aligned}
&\big(g_1g_2, c(g_1, g_2)c(1, 1)^{-1}b(r(g_1))b(r(g_2))\big) \\
&\qquad = \big(g_1, b(r(g_1))\big)\big(g_2, b(r(g_2))\big) \\
&\qquad = r(g_1)^{\sigma}r(g_2)^{\sigma} = j(g_1, g_2)^{\sigma}r(g_1g_2)^{\sigma} \\
&\qquad = \big(1, \alpha(j(g_1, g_2))\big)\big(g_1g_2, b(r(g_1g_2))\big) \\
&\qquad = \big(g_1g_2, c(1, g_1g_2)c(1, 1)^{-1}\alpha(j(g_1, g_2))b(r(g_1g_2))\big).
\end{aligned}$$

Since

$$c(1, 1) = c(1, g_1g_2),$$

it follows upon defining $d(g) = b(r(g))c(1, 1)^{-1}$ that

$$\alpha^{\mu}(g_1, g_2) = \alpha(j(g_1, g_2)) = c(g_1, g_2)d(g_1)d(g_2)d(g_1g_2)^{-1}.$$

Therefore,

$$\alpha^{\epsilon} = \alpha^{\mu}\mathbf{B}^2(G, \mathsf{K}^{\times}) = c\mathbf{B}^2(G, \mathsf{K}^{\times}),$$

and hence ϵ is an epimorphism from $\hat{J}$ to $\mathbf{H}^2(G, \mathsf{K}^{\times})$.

d) Let $J = S \times T$. We define $K = H/S$ and $A = J/S$. Then:

(1) $A \leq \mathbf{Z}(K)$ and

$$A = J/S = TS/S \leq H'S/S = K',$$

so $A \leq K' \cap \mathbf{Z}(K)$.

(2) $K/A \cong H/J \cong G$.

(3) $A \cong T \cong \mathbf{H}^2(G, \mathsf{K}^{\times})$.

So K is a representation group for G.

e) Since $B \leq L' \cap \mathbf{Z}(L)$, we have by Theorem III.3.12 (pg. 287) that $B \leq \Phi(L)$. Let $F = \langle f_1, \ldots, f_n \rangle$, and therefore $G = \langle g_1, \ldots, g_n \rangle$ with $g_i = f_i^{\pi}$. Let τ be the epimorphism from L onto G, and let $l_i \in L$ with $l_i^{\tau} = g_i$ $(i = 1, \ldots, n)$. Then

$$L = \langle l_1, \ldots, l_n, B \rangle = \langle l_1, \ldots, l_n, \Phi(L) \rangle = \langle l_1, \ldots, l_n \rangle.$$

Therefore, there is an epimorphism σ from F onto L with $f_i^{\sigma} = l_i$. Clearly, $\sigma\tau = \pi$. We have that $f^{\sigma} \in B$ is equivalent to $f^{\sigma\tau} = f^{\pi} = 1$; that is, $f \in R$. Furthermore, we have

$$[R, F]^\sigma = [R^\sigma, F^\sigma] = [B, L] = 1.$$

So σ induces an epimorphism $\overline{\sigma}$ from $F/[R, F]$ to L with

$$T^{\overline{\sigma}} = \big((F' \cap R)/[R, F]\big)^{\overline{\sigma}} = F'^\sigma \cap R^\sigma = L' \cap B = B.$$

Since $|B| = |\mathbf{H}^2(G, \mathsf{K}^\times)| = |T|$, it follows that $B \cong T \cong \mathbf{H}^2(G, \mathsf{K}^\times)$. □

The torsion-free part S in the decomposition $J = T \times S$ is not uniquely determined. Therefore, the isomorphism type of a representation group for G is not uniquely determined, in general. It is easy to see that both nonabelian groups of order p^3 are representation groups for the abelian group of type (p, p). However, we will show that K', $K/\mathbf{Z}(K)$, and the group ring $\mathsf{K}[K]$ are uniquely determined by G.

Theorem 23.6. *Let K_1 and K_2 be representation groups for G.*

a) (*I. Schur [373]*) *Then $K_1' \cong K_2'$. In particular, if $G = G'$ then all representation groups for G are isomorphic.*

b) (*Gaschütz, Neubüser, Yen [167]*) *We have*

$$K_1/\mathbf{Z}(K_1) \cong K_2/\mathbf{Z}(K_2).$$

Proof. a) Let K be a representation group for G, and let $H = F/[R, F]$ as in Theorem 23.5. Let $A \leq \mathbf{Z}(K) \cap K'$ with $K/A \cong G$. (So by Theorem 23.5 e), $A \cong \mathbf{H}^2(G, \mathsf{K}^\times)$.) Since $A \leq K'$, we have

$$|K'| = |G'||A| = |G'||\mathbf{H}^2(G, \mathsf{K}^\times)|.$$

By Theorem 23.5 e), there is an epimorphism μ from H to K. So $|K'| = |H'^\mu| \leq |H'|$. We show that $|K'| = |H'|$. Then $K' \cong H'$ and we will be done.

In the notation of Theorem 23.5, we have $H/J \cong G$, so

$$|G'| = |H'J/J| = |H'/H' \cap J|.$$

Furthermore, we have by Theorem 23.5 that

$$H' \cap J \cong \mathbf{H}^2(G, \mathsf{K}^\times) \cong A.$$

So we obtain $|H'| = |G'||A| = |K'|$, as desired.

b) Again let F be a free group with $F/R \cong G$. As in Theorem 23.5, we let $H = F/[R, F]$, $J = R/[R, F]$ and $T = J \cap H'$. By Theorem 23.5 e), we have $K_i \cong H/S_i$, where S_i is a complement of the torsion group T of J. It is enough to prove that $\mathbf{Z}(H/S_i) = \mathbf{Z}(H)/S_i$, since it then follows that

$$K_i/\mathbf{Z}(K_i) \cong H/S_i \Big/ \mathbf{Z}(H/S_i) \cong H/\mathbf{Z}(H)$$

for $i = 1, 2$. Let $xS_i \in \mathbf{Z}(H/S_i)$. Then for all $h \in H$ we have

$$[x, h] \in S_i \cap H' = S_i \cap (H' \cap J) = S_i \cap T = 1,$$

so $x \in \mathbf{Z}(H)$. Hence, $\mathbf{Z}(H/S_i) \leq \mathbf{Z}(H)/S_i$ and it then follows trivially that $\mathbf{Z}(H/S_i) = \mathbf{Z}(H)/S_i$. □

Definition 23.7. Let $c \in \mathbf{Z}^2(G, \mathsf{K}^\times)$. We define the twisted group algebra $\mathsf{K}[G, c]$ with cocycle c by $\mathsf{K}[G, c] = \bigoplus_{g \in G} \mathsf{K}e(g)$ with multiplication

$$e(g_1)e(g_2) = c(g_1, g_2)e(g_1g_2).$$

Note that $\mathsf{K}[G, c]$ is an associative algebra over K due to the cocycle relation

$$c(g_1, g_2)c(g_1g_2, g_3) = c(g_1, g_2g_3)c(g_2, g_3).$$

A change of basis to $e'(g) = b(g)e(g)$ for some $b(g) \in \mathsf{K}^\times$ changes c by a coboundary in $\mathbf{B}^2(G, \mathsf{K}^\times)$, and vice versa. So if $c' \equiv c \bmod \mathbf{B}^2(G, \mathsf{K}^\times)$, then $\mathsf{K}[G, c] \cong \mathsf{K}[G, c']$. Therefore, for $\overline{c} = c\mathbf{B}^2(G, \mathsf{K}^\times)$ in $\mathbf{H}^2(G, \mathsf{K}^\times)$, we may define $\mathsf{K}[G, \overline{c}]$ by $\mathsf{K}[G, \overline{c}] = \mathsf{K}[G, c]$.

Theorem 23.8. *Let K be a representation group for G. Then*

$$\mathsf{K}[K] \cong \bigoplus_{\overline{c}} \mathsf{K}[G, \overline{c}];$$

where the summation runs over all $\overline{c}$ in $\mathbf{H}^2(G, \mathsf{K}^\times)$. In particular, the isomorphism type of $\mathsf{K}[K]$ is uniquely determined by G.

Proof. Let $A \leq K' \cap \mathbf{Z}(K)$ with $K/A \cong G$, and let $K = \bigcup_{g \in G} Ar(g)$. Let $\mathsf{K}[A] = \bigoplus_{\alpha \in \hat{A}} \mathsf{K}e_\alpha$ be the decomposition of the commutative algebra $\mathsf{K}[A]$ into simple rings $\mathsf{K}e_\alpha$ according to Theorem 5.1. The e_α are pairwise orthogonal idempotents. Since the characters of A are all the irreducible representations of A, we have for every $x \in \mathsf{K}[A]$ that

$$x = \sum_{\alpha \in \hat{A}} \alpha(x)e_\alpha.$$

We embed $\mathsf{K}[A]$ inside $\mathsf{K}[K]$ in the natural way. Since $e_\alpha \in \mathbf{Z}(\mathsf{K}[K])$, we obtain

$$\mathsf{K}[K] = \bigoplus_{g \in G} \mathsf{K}[A]r(g) = \bigoplus_{\alpha \in \hat{A}} \left(\bigoplus_{g \in G} \mathsf{K}e_\alpha r(g) \right).$$

Here, we have

$$(e_{\alpha_1} r(g_1))(e_{\alpha_2} r(g_2)) = e_{\alpha_1} e_{\alpha_2} a(g_1, g_2) r(g_1g_2)$$

for some $a(g_1, g_2) \in A$. For $\alpha_1 \neq \alpha_2$, this is 0 since the e_α are orthogonal. It follows from

$$a(g_1, g_2) = \sum_{\beta \in \hat{A}} \beta(a(g_1, g_2)) e_\beta$$

that

$$(e_\alpha r(g_1))(e_\alpha r(g_2)) = \alpha(a(g_1, g_2)) e_\alpha r(g_1 g_2).$$

Hence, $\bigoplus_{g \in G} \mathsf{K} e_\alpha r(g)$ is the twisted group algebra with cocycle c defined by $c(g_1, g_2) = \alpha(a(g_1, g_2))$. Since K is a representation group for G, the map ϵ from Lemma 23.3 is an isomorphism from $\hat{A}$ onto $\mathbf{H}^2(G, \mathsf{K}^\times)$. Hence, each element $\overline{c} \in \mathbf{H}^2(G, \mathsf{K}^\times)$ appears in the direct decomposition of $\mathsf{K}[K]$ exactly once. □

By Theorem I.16.19 (pg. 126), the exponent of $\mathbf{H}^2(G, \mathsf{K}^\times)$ is a divisor of $|G|$. We can now strengthen this statement:

Theorem 23.9.

a) (*I. Schur*) *If e is the exponent of $\mathbf{H}^2(G, \mathsf{K}^\times)$, then e^2 is a divisor of $|G|$.*

b) *The exponent of $\mathbf{H}^2(G, \mathsf{K}^\times)$ divides the index of every cyclic normal subgroup of G.*

Proof. Let K be a representation group for G with $K/A \cong G$ and $A \leq K' \cap \mathbf{Z}(K)$.

a) Since $A \leq \mathbf{Z}(K)$, the restriction χ_A of an irreducible character χ of K to A has the form $\chi_A = n\varphi$ with $n = \chi(1)$ and for some linear character φ of A by Schur's Lemma. By the Frobenius Reciprocity of Theorem 16.5, we then have

$$(\varphi^K, \chi)_K = (\varphi, \chi_A)_A = \chi(1).$$

If ρ is an irreducible character of K with $\rho_A = m\beta$ and $\beta \neq \varphi$, then

$$(\varphi^K, \rho)_K = (\varphi, \rho_A)_A = 0.$$

Let φ be a one-dimensional character of A, and let τ_i $(i = 1, \ldots, s)$ be all the irreducible characters of K satisfying $(\tau_i)_A = \tau_i(1)\varphi$. Then

$$\varphi^K = \sum_{i=1}^{s} \tau_i(1)\tau_i$$

and

$$\sum_{i=1}^{s} \tau_i(1)^2 = \varphi^K(1) = |K/A| = |G|.$$

Let $\mathbf{D}$ be an irreducible representation of K such that $\mathbf{D}(a) = \varphi(a)1_n$ (with $n = \deg \mathbf{D}$) for all $a \in A$. Then the map α defined by $\alpha(x) = \det \mathbf{D}(x)$ is a one-dimensional representation of K. Since $A \leq K'$, we have

$$1 = \det \mathbf{D}(a) = \varphi(a)^n$$

for all $a \in A$. Hence, $\varphi^n = 1$. It follows from $(\tau_i)_A = n_i\varphi$ with $n_i = \tau_i(1)$ that $\varphi^{n_i} = 1$. So the order $|\varphi|$ of φ is a divisor of n_i, and therefore $|\varphi|^2$ divides

$$\sum_{i=1}^{s} n_i^2 = |G|.$$

The claim then follows since $\mathbf{H}^2(G, \mathsf{K}^\times) \cong A \cong \hat{A}$.

b) Let Z/A be a cyclic normal subgroup of the group $K/A \cong G$. Then Z is abelian as it is a cyclic extension of the central subgroup A. For every irreducible character φ of A, there is an irreducible character χ of K with $|\varphi|$ dividing $\chi(1)$ by what we showed in a). By Itô's Theorem 17.9, $\chi(1)$ is a divisor of $|K/Z|$. So the exponent of $\hat{A}$ is a divisor of $|K/Z|$. The claim follows since $\mathbf{H}^2(G, \mathsf{K}^\times) \cong \hat{A}$. □

For arbitrary cohomology groups, the conclusion of Theorem 23.9 a) is not true. For example, if L is a Galois extension of the $\mathfrak{p}$-adic field K of degree $[\mathsf{L} : \mathsf{K}] = |G|$ with Galois group G, and we equip $\mathsf{L}^\times$ with the structure of a G-module (written multiplicatively) by applying the automorphisms in G to the elements of $\mathsf{L}^\times$, then $\mathbf{H}^2(G, \mathsf{L}^\times)$ is cyclic of order $|G|$.

It is easy to estimate the size of the Schur multiplier of a p-group:

Theorem 23.10 (Green [187]). *Let G be a p-group of order p^n. Then $\mathbf{H}^2(G, \mathsf{K}^\times)$ is a p-group with $|\mathbf{H}^2(G, \mathsf{K}^\times)| \leq p^{n(n-1)/2}$.*

(*This bound is already obtained by abelian p-groups, as we will see in Theorem 25.11.*)

Proof. a) We first prove the following statement by induction on n, following J. Wiegold [457]: If G is a group with $|G/\mathbf{Z}(G)| = p^n$, then $|G'| \leq p^{n(n-1)/2}$.

The claim holds for abelian groups. So assume $n \geq 2$ and $x \in \mathbf{Z}_2(G) - \mathbf{Z}(G)$. Since

$$[x, g_1][x, g_2] = [x, g_1 g_2]$$

(see Lemma III.1.2 b), pg. 267), we have that

$$N = \{[x, g] \mid g \in G\}$$

is a normal subgroup of G lying in $\mathbf{Z}(G)$. It follows from

$$\mathbf{C}_G(x) \geq \langle \mathbf{Z}(G), x \rangle > \mathbf{Z}(G)$$

that

$$|N| = [G : \mathbf{C}_G(x)] < |G/\mathbf{Z}(G)| = p^n.$$

We define $\overline{G} = G/N$. We have

$$\mathbf{Z}(\overline{G}) \geq \langle \mathbf{Z}(G)/N, xN \rangle > \mathbf{Z}(G)/N,$$

and therefore $|\overline{G}/\mathbf{Z}(\overline{G})| \leq p^{n-1}$. Hence, we have $|\overline{G'}| \leq p^{(n-1)(n-2)/2}$ by our induction hypothesis. Then

$$|G'| = |N||\overline{G'}| \leq p^{n-1} p^{(n-1)(n-2)/2} = p^{n(n-1)/2}$$

and the claim is proved.

b) Let K be a representation group for G with $K/A \cong G$, $A \leq \mathbf{Z}(K) \cap K'$ and $A \cong \mathbf{H}^2(G, \mathsf{K}^\times)$. Since

$$|K/\mathbf{Z}(K)| \leq |K/A| = |G| = p^n,$$

it follows immediately with the help of a) that

$$|\mathbf{H}^2(G, \mathsf{K}^\times)| = |A| \leq |K'| \leq p^{n(n-1)/2},$$

which is what we wanted to show. □

Theorem 23.10 has been used to deduce lower bounds for the size of the automorphism group of a finite group of a given order; see Green [187].

Exercises

41) (Gaschütz, Neubüser, Yen [167]) Using the method of Theorem 23.10, show:
a) If G is a p-group with $|G/\Phi(G)| = p^d$ and $|G'| = p^h$, then

$$|\mathbf{H}^2(G, \mathsf{K}^\times)| \leq p^{(d-1)h} |\mathbf{H}^2(G/G', \mathsf{K}^\times)|.$$

b) Deduce from a) and Theorem 25.11 that if $|G| = p^n$, then[10]

$$|\mathbf{H}^2(G, \mathsf{K}^\times)| \leq p^{(n-h)(n-h-1)/2} p^{(d-1)h}.$$

24 Projective representations

We recall Definition 17.4:

Definition 24.1.
a) A map $\mathbf{P}$ from G to the general linear group $\mathrm{GL}_n(\mathsf{K})$ acting on the vector space $\mathsf{V} = \mathsf{V}(n, \mathsf{K})$ is called a projective representation of G of degree n over K with factor system c if for all $g_1, g_2 \in G$ we have

$$\mathbf{P}(g_1)\mathbf{P}(g_2) = c(g_1, g_2)\mathbf{P}(g_1 g_2).$$

[10] Correction 29 from errata.

b) Two projective representations $\mathbf{P}_1$ and $\mathbf{P}_2$ on V_1 and V_2 are said to be equivalent if there exists a bijective linear map t from V_1 to V_2 such that

$$\mathbf{P}_1(g)\, t = b(g)\, t\, \mathbf{P}_2(g) \quad (\text{with } b(g) \in \mathsf{K}^\times)$$

for all $g \in G$.

c) A projective representation $\mathbf{P}$ of G is said to be irreducible if no nontrivial subspace of V is mapped onto itself by all $\mathbf{P}(g)$ with $g \in G$.

Lemma 24.2.

a) *The factor system for a projective representation of G is a 2-cocycle in $\mathbf{Z}^2(G, \mathsf{K}^\times)$.*

b) *Let $\mathbf{P}$ be a projective representation of G with factor system c. The representations equivalent to $\mathbf{P}$ yield all the factor systems in $c\mathbf{B}^2(G, \mathsf{K}^\times)$. So the projective representations of G that are equivalent to an ordinary representation are exactly those with factor systems in $\mathbf{B}^2(G, \mathsf{K}^\times)$.*

c) *Every 2-cocycle in $\mathbf{Z}^2(G, \mathsf{K}^\times)$ is the factor system of some projective representation of G.*

Proof. a) We have

$$\begin{aligned}(\mathbf{P}(g_1)\mathbf{P}(g_2))\mathbf{P}(g_3) &= c(g_1, g_2)\mathbf{P}(g_1g_2)\mathbf{P}(g_3)\\ &= c(g_1, g_2)c(g_1g_2, g_3)\mathbf{P}(g_1g_2g_3)\\ &= \mathbf{P}(g_1)(\mathbf{P}(g_2)\mathbf{P}(g_3)) = c(g_2, g_3)\mathbf{P}(g_1)\mathbf{P}(g_2g_3)\\ &= c(g_1, g_2g_3)c(g_2, g_3)\mathbf{P}(g_1g_2g_3).\end{aligned}$$

b) It follows from $\mathbf{P}'(g) = b(g)t^{-1}\mathbf{P}(g)t$ that

$$\begin{aligned}c'(g_1, g_2)\mathbf{P}'(g_1g_2) &= \mathbf{P}'(g_1)\mathbf{P}'(g_2) = b(g_1)b(g_2)t^{-1}\mathbf{P}(g_1)\mathbf{P}(g_2)t\\ &= b(g_1)b(g_2)c(g_1, g_2)b(g_1g_2)^{-1}\mathbf{P}'(g_1g_2).\end{aligned}$$

So $c'c^{-1} = b\delta \in \mathbf{B}^2(G, \mathsf{K}^\times)$.

c) Let $\mathsf{V} = \mathsf{K}[G]$ be the group ring of G, viewed as a vector space over K. We define the linear map $\mathbf{P}(g)$ on V by

$$x\mathbf{P}(g) = c(x, g)xg$$

for $x \in G$. Then

$$\begin{aligned}x\mathbf{P}(g_1)\mathbf{P}(g_2) &= c(x, g_1)c(xg_1, g_2)xg_1g_2\\ &= c(x, g_1g_2)c(g_1, g_2)xg_1g_2 = c(g_1, g_2)x\mathbf{P}(g_1g_2).\end{aligned}$$

Hence, $\mathbf{P}(g_1)\mathbf{P}(g_2) = c(g_1, g_2)\mathbf{P}(g_1g_2)$, and so $\mathbf{P}$ is a projective representation of G with factor system c. □

Main Theorem 24.3. *Let* K *be an algebraically closed field of characteristic* 0*, and let* K *be a representation group for* G *with* $K/A \cong G$ *and* $A \leq K' \cap \mathbf{Z}(K)$*. Let* $K = \bigcup_{g \in G} Ar(g)$ *be the decomposition of* K *into cosets of* A*, and let*

$$r(g_1)r(g_2) = a(g_1, g_2)r(g_1g_2).$$

a) *If* $\mathbf{P}$ *is an irreducible projective representation of* G*, then there exists a projective representation* $\mathbf{P}'$ *of* G *equivalent to* $\mathbf{P}$ *and an irreducible representation* $\mathbf{D}$ *of* K *such that* $\mathbf{P}'(g) = \mathbf{D}(r(g))$ *for all* $g \in G$*.*

b) *If* $\mathbf{D}$ *is an irreducible representation of* K*, then* $\mathbf{P}$ *defined by* $\mathbf{P}(g) = \mathbf{D}(r(g))$ *is an irreducible projective representation of* G*.*

c) *The degree of every irreducible projective representation of* G *divides the order of* G*. (This proves Lemma 17.8.)*

Hence, the representation group is a solution to the following universal problem: Let α_n *be the natural homomorphism from* $\mathrm{GL}_n(\mathsf{K})$ *to the projective group* $\mathrm{PGL}_n(\mathsf{K})$*. We seek a pair* (K, β) *consisting of a finite group* K *and an epimorphism* β *from* K *onto* G *with the following universal property: If* $\mathbf{P}$ *is a homomorphism from* G *to* $\mathrm{PGL}_n(\mathsf{K})$ *(that is, a projective representation of* G*), then there exists a homomorphism* $\mathbf{D}$ *from* K *to* $\mathrm{GL}_n(\mathsf{K})$ *(that is, an ordinary representation of* K*) such that the following diagram commutes:*

$$\begin{array}{ccc} K & \xrightarrow{\mathbf{D}} & \mathrm{GL}_n(\mathsf{K}) \\ \beta \downarrow & & \downarrow \alpha_n \\ G & \xrightarrow{\mathbf{P}} & \mathrm{PGL}_n(\mathsf{K}) \end{array}$$

Proof. a) Let $\mathbf{P}$ be an irreducible projective representation of G with factor system c. Since K is a representation group for G, there exists by Lemma 23.3 and Theorem 23.5 e) some α in the character group $\hat{A}$ of A such that $\alpha^{\epsilon} = c\mathbf{B}^2(G, \mathsf{K}^\times)$, where ϵ is the homomorphism from $\hat{A}$ to $\mathbf{H}^2(G, \mathsf{K}^\times)$ established in Lemma 23.3. So there exists some 2-cocycle c' equivalent to c that satisfies

$$c'(g_1, g_2) = \alpha(a(g_1, g_2)).$$

Let $\mathbf{P}'$ be a projective representation of G equivalent to $\mathbf{P}$ with factor system c'. By Theorem 23.8, $\mathsf{K}[K]$ has a direct summand $\bigoplus_{g \in G} \mathsf{K}e_\alpha r(g)$ that is isomorphic to the twisted group algebra $\mathsf{K}[G, c']$. Clearly, the K-linear map $\mathbf{D}$ defined by

$$\mathbf{D}(e_\alpha r(g)) = \mathbf{P}'(g)$$

is an irreducible representation of the algebra $\mathsf{K}[G, c']$. We extend $\mathbf{D}$ to $\mathsf{K}[K]$ by defining $\mathbf{D}(x) = \mathbf{D}(e_\alpha x)$. Then $\mathbf{D}(r(g)) = \mathbf{P}'(g)$.

b) Let $\mathbf{D}$ be an irreducible representation of K. Since $A \leq \mathbf{Z}(K)$, we have $\mathbf{D}(a) = c(a)1$ for all $a \in A$ by Schur's Lemma 4.3. Therefore,

$$\mathbf{D}(r(g_1))\mathbf{D}(r(g_2)) = \mathbf{D}(r(g_1)r(g_2)) = \mathbf{D}(a(g_1, g_2)r(g_1g_2))$$
$$= c(a(g_1, g_2))\mathbf{D}(r(g_1g_2)).$$

So $\mathbf{P}$ defined by $\mathbf{P}(g) = \mathbf{D}(r(g))$ is a projective representation of G, and it is clearly irreducible.

c) By Theorem 12.6, the degree n of every irreducible representation of $\mathsf{K}[K]$ divides the index of $\mathbf{Z}(K)$. Since $A \leq \mathbf{Z}(K)$, it follows that n divides $|K/A| = |G|$. The claim then follows from a). □

Theorem 24.4. *Every irreducible projective representation of G is equivalent to an ordinary representation of G if and only if $\mathbf{H}^2(G, \mathsf{K}^\times) = 1$.* (*Schur referred to such groups as closed groups.*)

Proof. The claim is a trivial consequence of Lemma 24.2 b). □

In which fields can the irreducible projective representations of G be realized? To answer this question, we prove a lemma that also has independent interest:

Lemma 24.5 (Alperin, Tzee-Nan Kuo [12]). *Let G be a finite group.*
a) $\exp(G' \cap \mathbf{Z}(G)) \exp(G/(G' \cap \mathbf{Z}(G)))$ *is a divisor of* $|G/(G' \cap \mathbf{Z}(G))|$.
b) $\exp G$ *is a divisor of* $|G/(G' \cap \mathbf{Z}(G))|$.

Proof. a) We let $H = G/(G' \cap \mathbf{Z}(G))$ and $\exp H = e$. By Lemma 23.3, $G' \cap \mathbf{Z}(G)$ is isomorphic to a subgroup of $\mathbf{H}^2(H, \mathbb{C}^\times)$. Defining $e' = \exp \mathbf{H}^2(H, \mathbb{C}^\times)$, it suffices to prove that ee' divides $|H|$.

For every prime p, let H_p be a Sylow p-subgroup of H and $\exp H_p = e_p$. So $|H| = \prod_p |H_p|$ and $e = \prod_p e_p$. By Theorem I.16.21 (pg. 127), the Sylow p-subgroup of $\mathbf{H}^2(H, \mathbb{C}^\times)$ is isomorphic to a subgroup of $\mathbf{H}^2(H_p, \mathbb{C}^\times)$. Defining $e'_p = \exp \mathbf{H}^2(H_p, \mathbb{C}^\times)$, it suffices to prove that $e_p e'_p$ divides $|H_p|$.

Since $e_p = \exp H_p$, there exists a cyclic subgroup Z of H_p with $|Z| = e_p$. Let r be the restriction map from $\mathbf{H}^2(H_p, \mathbb{C}^\times)$ to $\mathbf{H}^2(Z, \mathbb{C}^\times)$ and i the corestriction map from $\mathbf{H}^2(Z, \mathbb{C}^\times)$ to $\mathbf{H}^2(H_p, \mathbb{C}^\times)$. By Theorem I.16.18 (pg. 124), the composition ri on $\mathbf{H}^2(H_p, \mathbb{C}^\times)$ is multiplication by $[H_p : Z] = |H_p|/e_p$. Since Z is cyclic, $\mathbf{H}^2(Z, \mathbb{C}^\times) = 1$ by Theorem I.16.10 (pg. 116). Therefore,

$$\mathbf{H}^2(H_p, \mathbb{C}^\times)ri = 1i = 1,$$

and hence the exponent of $\mathbf{H}^2(H_p, \mathbb{C}^\times)$ divides $|H_p|/e_p$.

b) Since $\exp G$ divides $\exp(G' \cap \mathbf{Z}(G)) \exp(G/(G' \cap \mathbf{Z}(G)))$, part b) is a trivial consequence of a). □

The reader may find a proof of Lemma 24.5 that does not use cohomology theory in Alperin, Tzee-Nan Kuo [12].

Theorem 24.6 (Reynolds [355]). *Let G be a finite group. In every equivalence class of projective irreducible representations of G over the complex numbers $\mathbb{C}$, there exists at least one that can be realized over the algebraic number field generated by the primitive $|G|$-th roots of unity over $\mathbb{Q}$.*

Proof. By Theorem 24.3, it is enough to show that all ordinary representations (up to equivalence) of a representation group K for G can be realized over the field K generated by the primitive $|G|$-th roots of unity over the rationals. By Lemma 24.5 b), $\exp K$ is a divisor of $|K/(K' \cap \mathbf{Z}(K))|$, which divides $|G|$. Therefore, every representation of K can be realized over K by Theorem 19.11. □

25 Calculating the Schur multiplier

We have two methods for calculating the Schur multiplier of a given group. The first rests on the formula

$$\mathbf{H}^2(G, \mathsf{K}^\times) \cong (R \cap F')/[R, F]$$

from Theorem 23.5 c). This method is applicable if a sufficiently manageable expression of G as a quotient $G \cong F/R$ is available. The second method relates the Sylow p-subgroup of $\mathbf{H}^2(G, \mathsf{K}^\times)$ to the cohomology group $\mathbf{H}^2(P, \mathsf{K}^\times)$ of a Sylow p-subgroup P of G using the restriction map of Definition I.16.15. This method is restricted in large part to proving that $\mathbf{H}^2(G, \mathsf{K}^\times) = 1$.

Theorem 25.1. *Let P be a Sylow p-subgroup of G. Then the Sylow p-subgroup of $\mathbf{H}^2(G, \mathsf{K}^\times)$ is isomorphic to a subgroup of $\mathbf{H}^2(P, \mathsf{K}^\times)$.*

Proof. The claim follows immediately from Theorem I.16.21 (pg. 127). A proof without the use of cohomology theory can be given as follows:

Let K be a representation group for G with $K/A \cong G$ and $A \leq K' \cap \mathbf{Z}(K)$. Furthermore, let $A \leq L \leq K$ such that $P \cong L/A$. We consider the transfer $\mathbf{V}$ from K to L/L'. Note that $[G : P] = m$ is coprime to $|P|$. Since $A \leq K' \cap \mathbf{Z}(K)$, we have by Lemma IV.2.1 (pg. 443) that for all $a \in A$

$$L' = \mathbf{V}(a) = a^m L'.$$

Therefore, $a^m \in A \cap L'$. If a lies in the Sylow p-subgroup of A, then for a suitable integer x we have

$$a = a^{mx} \in A \cap L'.$$

So the Sylow p-subgroup of A is a subgroup of $A \cap L'$. By Lemma 23.3, $A \cap L'$ is isomorphic to a subgroup of $\mathbf{H}^2(P, \mathsf{K}^\times)$. The claim then follows since $A \cong \mathbf{H}^2(G, \mathsf{K}^\times)$. □

Easy cases can be handled with the following theorem.

Theorem 25.2. *Let G be a finite group.*

a) *If G can be generated by n elements and r defining relations in these elements, then $\mathbf{H}^2(G, \mathsf{K}^\times)$ can be generated by $r - n$ elements. (If $r - n \leq 0$, then $\mathbf{H}^2(G, \mathsf{K}^\times) = 1$.)*

b) *If the group G is defined by n generators and r relations, then $r \geq n$.*
(*Situations in which $r = n$ are only known for $n \leq 3$; see Mennicke [330] and Example III.18.10 b), pg. 427.*)
c) *If G is a p-group with $d(G) \geq 4$, then $\mathbf{H}^2(G, \mathsf{K}^\times) \neq 1$.*

Proof. Let G be generated by elements $g_1, \ldots, g_n$ with relations t_i $(i = 1, \ldots, r)$. Let F be the free group in the free generators f_i $(i = 1, \ldots, n)$. Then $G \cong F/R$ with $R = \langle t_i^f \mid i = 1, \ldots, r; f \in F \rangle$. The abelian group $J = R/[R, F]$ is then generated by the elements $t_i[R, F]$ $(i = 1, \ldots, r)$. Let $J = T \times S$ according to Theorem 23.5, where S is a free abelian group of rank n. If d is the minimum number of generators of T, then $d + n \leq r$, so $d \leq r - n$ and $n \leq r$. Since $T \cong \mathbf{H}^2(G, \mathsf{K}^\times)$ (see Theorem 23.5 c)), the claims in a) and b) follow.

Now let G be a p-group, and let $n = d(G)$ be the minimum number of generators of G. Then T is also a p-group, and we obtain

$$R/R^p[R, F] \cong J/J^p \cong T/T^p \times S/S^p.$$

This yields for the number of generators

$$d(R/R^p[R, F]) = d(T/T^p) + d(S/S^p) = d(\mathbf{H}^2(G, \mathsf{K}^\times)) + n.$$

Let r' be the minimum number of generators of R/R^pR' as a G-module. By Theorem III.18.1 and Lemma III.18.6 (pg. 422), we have $r' > \frac{n^2}{4}$. If we prove that $r' \leq d(R/R^p[R, F])$, then the claim follows for $n \geq 4$ from

$$d(\mathbf{H}^2(G, \mathsf{K}^\times)) \geq r' - n > \frac{n^2}{4} - n \geq 0.$$

Clearly, $R/R^p[R, F]$ is the largest quotient module of R/R^pR' on which G acts trivially.

To show that $r' \leq d(R/R^p[R, F])$, we use additive notation and prove the following more general fact:

Let $\mathsf{K} = \mathrm{GF}(p)$. If $\mathsf{M} = \sum_{i=1}^{k} m_i \mathsf{K}[G]$ is a right $\mathsf{K}[G]$-module and k is chosen to be minimal, then

$$\mathsf{M}/\mathsf{M}\mathbf{J}(\mathsf{K}[G]) = \bigoplus_{i=1}^{k} m_i \mathsf{K}[G]/\mathbf{J}(\mathsf{K}[G]).$$

Since the elements $g - 1$ with $g \in G$ span the radical $\mathbf{J}(\mathsf{K}[G])$ of $\mathsf{K}[G]$ as a K-vector space by Theorem 5.16 b), $\mathsf{M}/\mathsf{M}\mathbf{J}(\mathsf{K}[G])$ is the largest quotient module of M on which G acts trivially.

By Theorem 5.16 b), $\mathsf{K}[G]/\mathbf{J}(\mathsf{K}[G]) \cong \mathsf{K}$. Assume for contradiction that the sum $\sum_{i=1}^{k} m_i \mathsf{K}[G]/\mathbf{J}(\mathsf{K}[G])$ is not direct. With a suitable labeling of the m_i, we then have

$$\mathsf{M}/\mathsf{M}\mathbf{J}(\mathsf{K}[G]) = \sum_{i=2}^{k} m_i \mathsf{K}[G]/\mathbf{J}(\mathsf{K}[G]).$$

Define the $\mathsf{K}[G]$-submodule M' by $\mathsf{M}' = \sum_{i=2}^{k} m_i \mathsf{K}[G]$, so that $\mathsf{M} = \mathsf{M}' + \mathsf{M}\mathbf{J}(\mathsf{K}[G])$. We now show that it follows from this that $\mathsf{M} = \mathsf{M}'$, contradicting our choice of k (Nakayama's Lemma):

Since $\mathbf{J}(\mathsf{K}[G])$ is nilpotent, there exists some integer t such that $\mathbf{J}(\mathsf{K}[G])^t = 0$. Assume for induction that $\mathsf{M} = \mathsf{M}' + \mathsf{M}\mathbf{J}(\mathsf{K}[G])^i$, then

$$\mathsf{M} = \mathsf{M}' + (\mathsf{M}' + \mathsf{M}\mathbf{J}(\mathsf{K}[G]))\mathbf{J}(\mathsf{K}[G])^i = \mathsf{M}' + \mathsf{M}\mathbf{J}(\mathsf{K}[G])^{i+1}.$$

Finally, we obtain

$$\mathsf{M} = \mathsf{M}' + \mathsf{M}\mathbf{J}(\mathsf{K}[G])^t = \mathsf{M}'. \qquad \square$$

Theorem 25.3.

a) *If G is cyclic or generalized quaternion, then $\mathbf{H}^2(G, \mathsf{K}^\times) = 1$.*

b) *If all the Sylow subgroups of G are cyclic or generalized quaternion, then $\mathbf{H}^2(G, \mathsf{K}^\times) = 1$.*

Proof. a) Since a cyclic group G of order $|G|$ can be defined by the single relation $x^{|G|} = 1$, the claim regarding cyclic groups follows immediately from Theorem 25.2 with $n = r = 1$.

The generalized quaternion groups can be defined by the relations

$$a^{2^m} = 1,\ a^{2^{m-1}} = b^2,\ a^b = a^{-1} \quad (m \geq 2).$$

The first relation follows from the other two since

$$a^{2^{m-1}} = b^2 = (b^2)^b = (a^{2^{m-1}})^b = a^{-2^{m-1}}.$$

Then the claim follows from Theorem 25.2 with $n = r = 2$.

b) follows immediately from a) and Theorem 25.1. $\square$

Lemma 25.4. *Suppose a Sylow p-subgroup P of G is elementary abelian. We have $\mathbf{N}_G(P) = PS$ with $p \nmid |S|$, and S has a representation, say $\mathbf{D}$, on P. If the tensor product $\mathbf{D} \otimes \mathbf{D}$ does not contain the identity representation of S, then p is not a divisor of $|\mathbf{H}^2(G, \mathsf{K}^\times)|$.*

Proof. Assume for contradiction that p is a divisor of $|\mathbf{H}^2(G, \mathsf{K}^\times)|$. Then there exists a homomorphic image $\overline{K}$ of a representation group for G with $\overline{K}/\overline{A} \cong G$, where $\overline{A} \leq \overline{K}' \cap \mathbf{Z}(\overline{K})$ and $|\overline{A}| = p$. Let $\overline{P}/\overline{A} \cong P$. As in the proof of Theorem 25.1, it follows by the transfer map from $\overline{K}$ to $\overline{P}/\overline{P}'$ that $\overline{A} \leq \overline{P}'$. Since $\overline{P}/\overline{A} \cong P$ is abelian, we have $\overline{A} = \overline{P}'$. So $\overline{P}$ has class 2. Therefore, we have by Lemma III.1.2 (pg. 267) that for all $\overline{x}_i$ in $\overline{P}$,

$$[\overline{x}_1\overline{x}_2, \overline{x}_3] = [\overline{x}_1, \overline{x}_3][\overline{x}_2, \overline{x}_3] \quad \text{and} \quad [\overline{x}_1, \overline{x}_2\overline{x}_3] = [\overline{x}_1, \overline{x}_2][\overline{x}_1, \overline{x}_3].$$

Let $\overline{A} = \langle \overline{a} \rangle$. If the $x_i = \overline{x}_i \overline{A}$ $(i = 1, 2)$ are elements in P with $[\overline{x}_1, \overline{x}_2] = \overline{a}^n$, then the map f defined by $f(x_1, x_2) = n$ is a balanced map from $P \times P$ onto $\mathrm{GF}(p)^+$ by the given formulas (see Definition 9.1). Therefore, there exists a homomorphism π from the tensor product $P \otimes_{\mathbb{Z}} P$ onto $\mathrm{GF}(p)^+$ with

$$(x_1 \otimes x_2)\pi = f(x_1, x_2).$$

Now we consider P as an S-module using the conjugation action $x \mapsto x^s$ for $x \in P$ and $s \in S$, and we consider $\mathrm{GF}(p)^+$ as a trivial S-module. For every $s = \overline{s}\overline{A}$ in S we have

$$\big((x_1 \otimes x_2)^s\big)\pi = (x_1^s \otimes x_2^s)\pi = m,$$

where

$$\overline{a}^m = [\overline{x}_1^{\overline{s}}, \overline{x}_2^{\overline{s}}] = [\overline{x}_1, \overline{x}_2]^{\overline{s}} = [\overline{x}_1, \overline{x}_2] = \overline{a}^n.$$

So we have

$$\big((x_1 \otimes x_2)^s\big)\pi = \big((x_1 \otimes x_2)\pi\big)^s = (x_1 \otimes x_2)\pi.$$

Therefore, π is an S-homomorphism from $P \otimes P$ onto the trivial S-module $\mathrm{GF}(p)^+$, and hence $\mathbf{D} \otimes \mathbf{D}$ contains the trivial representation of S. This contradicts our assumption. □

Theorem 25.5. *The Schur multiplier of* $\mathrm{SL}_2(p^f)$ *is trivial when* $p^f \neq 2^2, 3^2$.

Proof. By Theorem II.8.10 (pg. 206), the Sylow q-subgroups of $G = \mathrm{SL}_2(p^f)$ for $q \neq p$ are either cyclic groups or generalized quaternion groups. Therefore, it follows from Theorem 25.1 and Theorem 25.3 a) that $\mathbf{H}^2(G, \mathsf{K}^\times)$ is a p-group. We will show that the assumptions of Lemma 25.4 are satisfied when $p^f \neq 2^2, 3^2$. We have that

$$P = \left\{ x(a) = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix} \;\middle|\; a \in \mathrm{GF}(p^f) \right\}$$

is a Sylow p-subgroup of $\mathrm{SL}_2(p^f)$. By Theorem II.7.1 (pg. 194), a complement S of P in $\mathbf{N}_G(P)$ is given by the cyclic group $\langle s \rangle$ with

$$s = \begin{pmatrix} b & 0 \\ 0 & b^{-1} \end{pmatrix},$$

where b is a generator of $\mathrm{GF}(p^f)^\times$. We have $x(a)^s = x(b^2 a)$. The representation of s on P, as well as the S-module $\mathrm{GF}(p^f)^+$ isomorphic to P, is described by $x \mapsto x\tilde{s} = b^2 x$. We regard $\tilde{s}$ as a linear map of the vector space $\mathrm{GF}(p^f)$ of dimension f over $\mathrm{GF}(p)$. If m is the minimal polynomial of $\tilde{s}$, then

$$0 = 1m(\tilde{s}) = m(b^2).$$

Therefore, the elements $b^2, b^{2p}, \ldots, b^{2p^{f-1}}$ are conjugate to b^2 over $\mathrm{GF}(p)$, and are roots of m. Since $\deg m \leq f$, these are all the roots. When does $\tilde{s} \otimes \tilde{s}$ have eigenvalue 1? By Lemma 10.1, the eigenvalues of $\tilde{s} \otimes \tilde{s}$ are

$$b^{2(p^i+p^j)}$$

with $0 \leq i \leq j \leq f-1$. Assume that $b^{2(p^i+p^j)} = 1$. Then

$$2(p^i + p^j) \equiv 0 \mod (p^f - 1), \quad \text{so} \quad 2(p^{j-i} + 1) \equiv 0 \mod (p^f - 1).$$

However, if $p^f > 4$ then

$$2(p^{j-i} + 1) \leq 2(p^{f-1} + 1) < 2(p^f - 1).$$

Hence, we must have $2(p^{j-i}+1) = p^f - 1$, and therefore either $j - i = 0$ and $p^f = 5$, or $j - i > 0$ and $2 \equiv -1 \bmod p^{j-i}$. In the latter case, $p^{j-i} = 3$ and then $p^f \leq 3^2$ since

$$2(p^{j-i} + 1) = 8 \equiv 0 \mod (p^f - 1).$$

If $p^f = 3$ or $p^f = 5$, then $\mathbf{H}^2(G, \mathsf{K}^\times) = 1$ by Theorem 25.1 since the Sylow p-subgroups of $\mathrm{SL}_2(p^f)$ are cyclic in these cases. □

Before we handle the projective special linear groups $\mathrm{PSL}_2(p^f)$ in a similar way, we determine the Schur multipliers of the dihedral groups.

Theorem 25.6. *Let D be a dihedral group of order $2n$. Then the Schur multiplier of D has order 2 for even n and order 1 for odd n.*

Proof. If n is odd, then all the Sylow subgroups of D are cyclic, and so $\mathbf{H}^2(D, \mathsf{K}^\times) = 1$ by Theorem 25.3 b).

The dihedral group D is generated by elements a and b with relations

$$a^n = b^2 = 1, \quad a^b = a^{-1}.$$

Therefore, $\mathbf{H}^2(D, \mathsf{K}^\times)$ is cyclic by Theorem 25.2 a), and it then follows from Theorem 23.9 b) that $|\mathbf{H}^2(D, \mathsf{K}^\times)| \leq 2$.

It remains to show that the Schur multiplier of D has order 2 when n is even. To this end, we construct a representation group for D:

Let D_1 be the dihedral group of order $4n$ with generators a_1 and b_1 and relations

$$a_1^{2n} = b_1^2 = 1, \quad a_1^{b_1} = a_1^{-1}.$$

Then $D_1/\langle a_1^n \rangle \cong D$, and since $2 \mid n$ we have $\langle a_1^n \rangle \leq D_1' \cap \mathbf{Z}(D_1)$. It now follows from Lemma 23.3 that $|\mathbf{H}^2(D, \mathsf{K}^\times)| \geq 2$. So $|\mathbf{H}^2(D, \mathsf{K}^\times)| = 2$, and D_1 is a representation group for D. □

Theorem 25.7. *The group* $\mathrm{SL}_2(p^f)$ *is a representation group for* $\mathrm{PSL}_2(p^f)$ *when* $p^f \neq 2^2, 3^2$. *The group* $\mathrm{SL}_2(p^f)$ *is the unique representation group for* $\mathrm{PSL}_2(p^f)$ *when* $p^f > 4$ *and* $p^f \neq 3^2$*; furthermore,* $\mathrm{SL}_2(5)$ *is the unique representation group for* $\mathrm{PSL}_2(4)$. *We have*

$$|\mathbf{H}^2(\mathrm{PSL}_2(p^f), \mathsf{K}^\times)| = \begin{cases} 2 & \text{for } p > 2 \text{ and } p^f \neq 3^2 \\ 1 & \text{for } p = 2 \text{ and } p^f \neq 2^2 \\ 2 & \text{for } p^f = 2^2 \\ 6 & \text{for } p^f = 3^2. \end{cases}$$

Proof. a) Let $G = \mathrm{PSL}_2(p^f)$ and write

$$|\mathbf{H}^2(G, \mathsf{K}^\times)| = \prod_q q^{n_q},$$

where the integers q are prime. Since the Sylow q-subgroups of G are cyclic for $2 \neq q \neq p$ by Theorem II.8.10 (pg. 206), we have $n_q = 0$ for these q by Theorem 25.1 and Theorem 25.3. As the Sylow 2-subgroups of $\mathrm{PSL}_2(p^f)$ are dihedral for $p \neq 2$, we obtain the bound $n_2 \leq 1$ for $p \neq 2$ by Theorem 25.1 and Theorem 25.6. In the same way as in Theorem 25.5, it follows from Lemma 25.4 that $n_p = 0$ for $p^f \neq 2^2, 3^2$. Altogether, we have for $p^f \neq 2^2, 3^2$ that

$$|\mathbf{H}^2(\mathrm{PSL}_2(p^f), \mathsf{K}^\times)| \leq 2 \quad \text{and} \quad |\mathbf{H}^2(\mathrm{PSL}_2(2^f), \mathsf{K}^\times)| = 1.$$

Since

$$-1 \in \mathrm{SL}_2(p^f)' \cap \mathbf{Z}(\mathrm{SL}_2(p^f)),$$

we obtain with the help of Lemma 23.3 that $|\mathbf{H}^2(\mathrm{PSL}_2(p^f), \mathsf{K}^\times)| = 2$ for $p^f \neq 3^2$ and $p > 2$. Therefore, $\mathrm{SL}_2(p^f)$ is a representation group for $\mathrm{PSL}_2(p^f)$. For $p^f > 4$ and $p^f \neq 3^2$, we have by Theorem 23.6 that $\mathrm{SL}_2(p^f)$ is the unique representation group for $\mathrm{PSL}_2(p^f)$ since $\mathrm{PSL}_2(p^f)' = \mathrm{PSL}_2(p^f)$ by Theorem II.6.13.

b) We now consider the case $p^f = 3^2$ that we excluded in a). By Theorem II.8.14 (pg. 209), $G = \mathrm{PSL}_2(9)$ is isomorphic to the alternating group A_6. Since

$$-1 \in \mathrm{SL}_2(9)' \cap \mathbf{Z}(\mathrm{SL}_2(9)),$$

it follows from Lemma 23.3 that 2 is a divisor of $|\mathbf{H}^2(\mathrm{PSL}_2(9), \mathsf{K}^\times)|$. By Theorem 25.1 and Theorem 25.6, we have $|\mathbf{H}^2(\mathrm{PSL}_2(9), \mathsf{K}^\times)| = 2 \cdot 3^n$ for some integer n.

A Sylow 3-subgroup P of $G \cong A_6$ has type $(3, 3)$, so it can be generated by elements x and y subject to the three relations

$$x^3 = y^3 = [x, y] = 1.$$

So $\mathbf{H}^2(P, \mathsf{K}^\times)$ is cyclic by Theorem 25.2. It then follows from Theorem 23.9 that

$$|\mathbf{H}^2(P, \mathsf{K}^\times)|^2 \text{ divides } 9, \text{ so } |\mathbf{H}^2(P, \mathsf{K}^\times)| \text{ divides } 3.$$

In particular, this shows that $|\mathbf{H}^2(G, \mathsf{K}^\times)|$ divides 6 by Theorem 25.1.

We show that $|\mathbf{H}^2(G, \mathsf{K}^\times)| = 6$. Assume for contradiction that $|\mathbf{H}^2(G, \mathsf{K}^\times)| = 2$. Then $\mathrm{SL}_2(9)$ is a representation group for $\mathrm{PSL}_2(9) \cong A_6$. Therefore, every irreducible projective representation of A_6 is induced by an ordinary representation of $\mathrm{SL}_2(9)$.

The alternating group A_6 admits a three-dimensional projective representation $\mathbf{P}$, the so-called Valentiner representation. If we define

$$a_1 = (1,3,2),\ a_2 = (1,2)(3,4),\ a_3 = (1,2)(4,5),\ \text{and } a_4 = (1,2)(5,6),$$

then $\mathbf{P}$ is given by[11]

$$\mathbf{P}(a_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \epsilon & 0 \\ 0 & 0 & \epsilon^2 \end{pmatrix} \qquad \mathbf{P}(a_2) = \begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}$$

$$\mathbf{P}(a_3) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & \overline{\rho} \\ 0 & \rho & 0 \end{pmatrix} \qquad \mathbf{P}(a_4) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

where ϵ is a primitive third root of unity and $\rho = \frac{-1+i\sqrt{15}}{4}$.

(The easiest way to see this is to check that the defining relations

$$a_1^3 = a_i^2 = 1\ (2 \le i \le 4), \quad (a_i a_{i+1})^3 = 1\ (1 \le i \le 3),$$
$$(a_i a_j)^2 = 1\ (i+1 < j)$$

from Example I.19.8 (pg. 147) are satisfied.)

Let $\mathbf{D}$ be an ordinary representation of $\mathrm{SL}_2(9)$ of degree 3 corresponding to $\mathbf{P}$. If $\mathbf{D}$ is trivial on $Z = \langle -1 \rangle$, then $\mathbf{D}$ is a representation of A_6 of degree 3. If $\mathbf{D}(-1) = -1$, then $\mathbf{D}'$ defined by $\mathbf{D}'(g) = \mathbf{D}(g) \det \mathbf{D}(g)$ is an ordinary representation of $\mathrm{SL}_2(9)$ that is trivial on Z. Then $\mathbf{D}'$ yields a projective representation of A_6 that is equivalent to $\mathbf{P}$. So in either case there exists an ordinary representation of A_6 of degree 3, and it is faithful since A_6 is simple.

The elements $x = (1,2,3)$ and $y = (4,5,6)$ generate a Sylow 3-subgroup P of A_6. The element $z = (1,2)(4,5)$ lies in $\mathbf{N}_G(P)$ because $x^z = x^{-1}$ and $y^z = y^{-1}$. Let $T = \langle P, z \rangle$. Since $T' = P$, the group T has exactly two one-dimensional representations. Each of the four subgroups U_i of P of order 3 is normal in T, so gives rise to a two-dimensional faithful representation of the dihedral group T/U_i. These are all the irreducible representations of T since the sum of squared degrees is exactly $|T| = 18$.

Hence, the three-dimensional representation $\mathbf{D}$ of A_6 described above is reducible on T. Therefore, the restriction of $\mathbf{D}$ to $T' = P$ certainly contains the trivial representation of P. We can simultaneously diagonalize the matrices $\mathbf{D}(h)$ with $h \in P$.

[11] Correction 30 from errata.

Since $\det \mathbf{D}(g) = 1$ for all elements g in the simple group A_6, we obtain matrices of the form

$$\begin{pmatrix} \epsilon & 0 & 0 \\ 0 & \epsilon^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is clear that this cannot yield a faithful representation of the noncyclic group P.

So the assumption that $|\mathbf{H}^2(G, \mathsf{K}^\times)| = 2$ must be discarded, and the Schur multiplier of $\mathrm{PSL}_2(9) \cong A_6$ has order 6. (Incidentally, our considerations also show that the Schur multiplier of $\mathrm{SL}_2(9)$ has order 3. For the construction of the representation groups for $\mathrm{PSL}_2(9)$ and $\mathrm{SL}_2(9)$, we refer the reader to Schur [373].)

c) Since $\mathrm{SL}_2(4) \cong \mathrm{PSL}_2(4) \cong \mathrm{PSL}_2(5)$ (see Theorem II.6.14, pg. 192), the group $\mathrm{SL}_2(5)$ is the unique representation group for $\mathrm{PSL}_2(4)$ by a). □

It is possible to determine the Schur multipliers of the simple Suzuki and Ree groups using a generalization of this proof method (see Alperin, Gorenstein [10]).

Definition 25.8. Let G and H be arbitrary groups. We define the tensor product $G \otimes H$ as the group with generators $t(g, h)$ $(g \in G, h \in H)$ and the defining relations

$$t(g_1 g_2, h) = t(g_1, h)t(g_2, h) \quad \text{and} \quad t(g, h_1 h_2) = t(g, h_1)t(g, h_2).$$

Since

$$\begin{aligned} t(g_1, h_1)t(g_2, h_1)t(g_1, h_2)t(g_2, h_2) &= t(g_1 g_2, h_1)t(g_1 g_2, h_2) \\ &= t(g_1 g_2, h_1 h_2) = t(g_1, h_1 h_2)t(g_2, h_1 h_2) \\ &= t(g_1, h_1)t(g_1, h_2)t(g_2, h_1)t(g_2, h_2), \end{aligned}$$

we have $t(g_2, h_1)t(g_1, h_2) = t(g_1, h_2)t(g_2, h_1)$ for all $g_1, g_2 \in G$ and all $h_1, h_2 \in H$. That is, the tensor product $G \otimes H$ is an abelian group.

For abelian groups, this definition coincides with the definition of the tensor product over the ring of integers given in §9. The structure of the tensor product of finite groups is clarified by the following lemma.

Lemma 25.9. *We have*

a) $G \otimes H \cong G/G' \otimes H/H'$.

b) $(G_1 \times G_2) \otimes H \cong (G_1 \otimes H) \times (G_2 \otimes H)$.

c) *We denote the cyclic group of order k by Z_k. If*

$$G/G' \cong Z_{a_1} \times \cdots \times Z_{a_n} \quad \text{and} \quad H/H' \cong Z_{b_1} \times \cdots \times Z_{b_m},$$

then

$$G \otimes H \cong \bigtimes_{i,j} Z_{(a_i, b_j)},$$

where (a_i, b_j) is the greatest common divisor of a_i and b_j.

Proof. a) Since the tensor product is an abelian group, we have

$$t(g_1g_2, h) = t(g_1, h)t(g_2, h) = t(g_2, h)t(g_1, h) = t(g_2g_1, h).$$

Therefore, $t(x, h) = 1$ for all $x \in G'$. In the same way, $t(g, y) = 1$ for all $y \in H'$. This yields

$$G \otimes H \cong G/G' \otimes H/H'.$$

b) By a) and Lemma 9.4,

$$\begin{aligned}(G_1 \times G_2) \otimes H &\cong (G_1/G_1' \times G_2/G_2') \otimes H/H' \\ &\cong (G_1/G_1' \otimes H/H') \times (G_2/G_2' \otimes H/H') \\ &\cong (G_1 \otimes H) \times (G_2 \otimes H).\end{aligned}$$

c) The statement in b) reduces our task to determining the tensor product $T = Z_a \otimes Z_b$ of cyclic groups Z_a and Z_b of orders a and b. Let $Z_a = \langle x \rangle$, and let $Z_b = \langle y \rangle$. It follows from

$$t(x^m, y^n) = t(x, y)^{mn}$$

that T is generated by $t(x, y)$. Since

$$t(x, y)^a = t(x^a, y) = 1 \quad \text{and} \quad t(x, y)^b = t(x, y^b) = 1,$$

the order of $t(x, y)$ is a divisor of (a, b). Conversely, if $Z = \langle z \rangle$ is a cyclic group of order (a, b), then the map ϵ defined by

$$t(x^m, y^n)^\epsilon = z^{mn}$$

yields a homomorphism from $Z_a \otimes Z_b$ onto Z. So T is cyclic of order (a, b). □

Theorem 25.10. *If G and H are finite groups, then*

$$\mathbf{H}^2(G \times H, \mathsf{K}^\times) \cong \mathbf{H}^2(G, \mathsf{K}^\times) \times \mathbf{H}^2(H, \mathsf{K}^\times) \times (G \otimes H).$$

In particular, if $(|G/G'|, |H/H'|) = 1$, we have

$$\mathbf{H}^2(G \times H, \mathsf{K}^\times) \cong \mathbf{H}^2(G, \mathsf{K}^\times) \times \mathbf{H}^2(H, \mathsf{K}^\times).$$

(*Note: We have $\mathbf{B}^1(G, \mathsf{K}^\times) = 1$ and therefore $\mathbf{H}^1(G, \mathsf{K}^\times) = \mathbf{Z}^1(G, \mathsf{K}^\times)$. The relation for the 1-cocycles in this case is*

$$1 = (b\delta)(g_1, g_2) = b(g_1)b(g_2)b(g_1g_2)^{-1}.$$

So $\mathbf{Z}^1(G, \mathsf{K}^\times)$ is the character group of G/G', and it follows by Theorem 6.4 a) (pg. 518) that $\mathbf{H}^1(G, \mathsf{K}^\times) \cong \widehat{G/G'} \cong G/G'$. Hence, the formula above can also be written as

$$\mathbf{H}^2(G \times H, \mathsf{K}^\times) \cong \mathbf{H}^2(G, \mathsf{K}^\times) \times \mathbf{H}^2(H, \mathsf{K}^\times) \times (\mathbf{H}^1(G, \mathsf{K}^\times) \otimes \mathbf{H}^1(H, \mathsf{K}^\times)).$$

This is a special case of the so-called Künneth relation; see MacLane [327], pg. 166.)

Proof. a) Let F be the free group in the $|G| + |H|$ free generators $f(g)$ $(g \in G)$ and $f(h)$ $(h \in H)$. Let R be the normal subgroup of F generated by the elements

$$f(g_1)f(g_2)f(g_1g_2)^{-1},\ f(h_1)f(h_2)f(h_1h_2)^{-1},\ f(g)f(h)f(g)^{-1}f(h)^{-1}.$$

Then clearly $F/R \cong G \times H$. We consider the groups $F/[R, F]$ and $J = R/[R, F]$ that we constructed in Theorem 23.5, and we define

$$\overline{f}(g) = f(g)[R, F] \quad \text{and} \quad \overline{f}(h) = f(h)[R, F].$$

Then J is generated by the elements

$$l(g_1, g_2) = \overline{f}(g_1)\overline{f}(g_2)\overline{f}(g_1g_2)^{-1},\ l(h_1, h_2) = \overline{f}(h_1)\overline{f}(h_2)\overline{f}(h_1h_2)^{-1}$$

and

$$k(g, h) = \overline{f}(g)\overline{f}(h)\overline{f}(g)^{-1}\overline{f}(h)^{-1}.$$

It follows from the associativity relation

$$(\overline{f}(g_1)\overline{f}(g_2))\overline{f}(g_3) = \overline{f}(g_1)(\overline{f}(g_2)\overline{f}(g_3))$$

that

$$l(g_1, g_2)l(g_1g_2, g_3) = l(g_1, g_2g_3)l(g_2, g_3), \tag{1}$$

and similarly

$$l(h_1, h_2)l(h_1h_2, h_3) = l(h_1, h_2h_3)l(h_2, h_3). \tag{2}$$

Moreover, we have

$$\begin{aligned} l(g_1, g_2)k(g_1g_2, h)\overline{f}(h)\overline{f}(g_1g_2) &= l(g_1, g_2)\overline{f}(g_1g_2)\overline{f}(h) \\ &= (\overline{f}(g_1)\overline{f}(g_2))\overline{f}(h) = \overline{f}(g_1)(\overline{f}(g_2)\overline{f}(h)) \\ &= \overline{f}(g_1)k(g_2, h)\overline{f}(h)\overline{f}(g_2) = k(g_1, h)k(g_2, h)\overline{f}(h)\overline{f}(g_1)\overline{f}(g_2) \\ &= k(g_1, h)k(g_2, h)l(g_1, g_2)\overline{f}(h)\overline{f}(g_1g_2). \end{aligned}$$

Therefore,

$$k(g_1g_2, h) = k(g_1, h)k(g_2, h). \tag{3}$$

In the same way, we obtain

$$k(g, h_1h_2) = k(g, h_1)k(g, h_2). \tag{4}$$

Let the abelian group $J(G)$ be defined by the generators $j(g_1, g_2)$ $(g_i \in G)$ and the relations

$$j(g_1, g_2)j(g_1g_2, g_3) = j(g_1, g_2g_3)j(g_2, g_3); \tag{1′}$$

analogously, let $J(H)$ be defined by the generators $j'(h_1, h_2)$ $(h_i \in H)$ and the relations

$$j'(h_1, h_2)j'(h_1h_2, h_3) = j'(h_1, h_2h_3)j'(h_2, h_3). \tag{2′}$$

Finally, the tensor product $G \otimes H$ is defined by the generators $t(g, h)$ $(g \in G, h \in H)$ and the relations

$$t(g_1g_2, h) = t(g_1, h)t(g_2, h), \tag{3′}$$
$$t(g, h_1h_2) = t(g, h_1)t(g, h_2). \tag{4′}$$

So J is a homomorphic image of

$$J(G) \times J(H) \times (G \otimes H).$$

b) Let $S = J(G) \times J(H) \times (G \otimes H)$. We construct the central extension U of S by $G \times H$ with the multiplication rule

$$\begin{aligned}&(g_1h_1, s_1)(g_2h_2, s_2)\\ &\quad = \big(g_1g_2h_1h_2,\ s_1s_2\, j(1,1)^{-1} j(g_1, g_2)\, j'(1,1)^{-1} j'(h_1, h_2)\, t(g_2, h_1)\big).\end{aligned}$$

By the relations in (1′)–(4′), this multiplication is associative. Keeping in mind that

$$j(g, 1) = j(1, g) = j(1, 1) \quad \text{and} \quad t(g, 1) = t(1, h) = t(1, 1) = 1,$$

it follows that

$$\begin{aligned}(g_1, j(1,1))(g_2, j(1,1)) &= (g_1g_2, j(1,1))(1, j(g_1, g_2)),\\ (h_1, j'(1,1))(h_2, j'(1,1)) &= (h_1h_2, j'(1,1))(1, j'(h_1, h_2))\end{aligned}$$

and

$$(h, j'(1,1))(g, j(1,1)) = (g, j(1,1))(h, j'(1,1))(1, t(g, h)).$$

So U is generated by the $(g, j(1,1))$ and $(h, j'(1,1))$. The map π defined by

$$f(g)^\pi = (g, j(1,1)) \quad \text{and} \quad f(h)^\pi = (h, j'(1,1))$$

can be extended to a homomorphism π from the free group F with free generating set $\{f(g), f(h) \mid g \in G, h \in H\}$ onto U. Under this map, we have

$$[R, F]^\pi = [S, U] = 1.$$

So U is a homomorphic image of $F/[R, F]$, and

$$S = J(G) \times J(H) \times (G \otimes H)$$

is a homomorphic image of $J = R/[R, F]$.

c) The isomorphism

$$J \cong J(G) \times J(H) \times (G \otimes H)$$

follows from what we have proved in a) and b) and Theorem I.13.14 e) (pg. 85). The same considerations applied to the central extension of $J(G)$ by G show that $J(G)$ is exactly the group arising from the generation of G by the $f(g)$ $(g \in G)$ with the relations

$$f(g_1)f(g_2)f(g_1g_2)^{-1}$$

according to Theorem 23.5. Now, momentarily denote the torsion group of an abelian group A by $\mathbf{T}(A)$. Since $G \otimes H$ is a torsion group by Lemma 25.9 c), it follows by Theorem 23.5 that

$$\begin{aligned}\mathbf{H}^2(G \times H, \mathsf{K}^\times) &\cong \mathbf{T}(J) \cong \mathbf{T}(J(G)) \times \mathbf{T}(J(H)) \times (G \otimes H)\\ &\cong \mathbf{H}^2(G, \mathsf{K}^\times) \times \mathbf{H}^2(H, \mathsf{K}^\times) \times (G \otimes H).\end{aligned}$$ □

We obtain the following theorem from Theorem 25.10, Lemma 25.9 c), and Theorem 25.3 a).

Theorem 25.11. *If G is an abelian p-group of type $(p^{a_1}, \dots, p^{a_k})$ and $a_1 \leq \cdots \leq a_k$, then the Schur multiplier $\mathbf{H}^2(G, \mathsf{K}^\times)$ has type*

$$(\underbrace{p^{a_1}, \dots, p^{a_1}}_{k-1}, \underbrace{p^{a_2}, \dots, p^{a_2}}_{k-2}, \dots, p^{a_{k-2}}, p^{a_{k-2}}, p^{a_{k-1}}).$$

Theorem 25.12. *The Schur multiplier of S_n $(n \geq 4)$ has order 2. (By Theorem 25.3 b), S_2 and S_3 have trivial Schur multiplier.)*

Proof. a) We first show that $|\mathbf{H}^2(S_n, \mathsf{K}^\times)| \leq 2$:

By Example I.19.7 (pg. 146), we can generate the group S_n by the elements $x_i = (i, i+1)$ $(1 \leq i \leq n-1)$ subject to the defining relations

$$x_i^2 = 1 \quad (1 \leq i \leq n-1) \tag{1}$$

$$(x_ix_{i+1})^3 = 1 \quad (1 \leq i \leq n-2) \tag{2}$$

$$x_ix_j = x_jx_i \quad (i+1 < j). \tag{3}$$

Then the group $H = F/[R, F]$ from Theorem 23.5 is generated by elements h_i $(1 \leq i \leq n-1)$ with $h_iJ = x_i$, and $J = R/[R, F]$ is generated by the elements

$$j_i = h_i^2 \quad (1 \le i \le n-1) \tag{1'}$$
$$k_i = (h_i h_{i+1})^3 \quad (1 \le i \le n-2) \tag{2'}$$
$$l_{ij} = [h_i, h_j] \quad (i+1 < j). \tag{3'}$$

For $|i-j| > 1$ and $|i'-j'| > 1$ and $i, j, i', j' < n$, there exists an element $s \in S_n$ of the form

$$s = \begin{pmatrix} i & i+1 & j & j+1 & \cdots \\ i' & i'+1 & j' & j'+1 & \cdots \end{pmatrix}.$$

Let $t \in H$ with $tJ = s$. Then it follows that $h_i^t = h_{i'} a_i$ and $h_j^t = h_{j'} a_j$ for some $a_i, a_j \in J$. Since $J \le \mathbf{Z}(H)$, we have

$$l_{ij} = l_{ij}^t = [h_i^t, h_j^t] = [h_{i'}, h_{j'}] = l_{i'j'}.$$

Hence, all the l_{ij} are equal. We let $l_{ij} = l$.

We obtain from $h_i^2 = j_i \in J$ that

$$h_i^2 = (h_j^{-1} h_i h_j)^2 = (h_i l)^2 = h_i^2 l^2,$$

so $l^2 = 1$. The relation (2′) yields

$$h_i h_{i+1} h_i = k_i h_{i+1}^{-1} h_i^{-1} h_{i+1}^{-1},$$

and then by squaring,

$$j_i^2 j_{i+1} = h_i^4 h_{i+1}^2 = (h_i h_{i+1} h_i)^2 = k_i^2 h_{i+1}^{-4} h_i^{-2} = k_i^2 j_{i+1}^{-2} j_i^{-1}.$$

So $k_i^2 = j_i^3 j_{i+1}^3$ for $1 \le i \le n-2$.

Let $U = \langle j_i \mid i = 1, \ldots, n-1 \rangle$. Then J/U is generated by the elements Ul and Uk_i $(1 \le i \le n-2)$, which have order at most 2 since $l^2 = 1$ and $k_i^2 = j_i^3 j_{i+1}^3 \in U$. Therefore, $|J/U|$ divides 2^{n-1}. By Theorem 23.5 b), $r(J) = n-1$. Since U is generated by $n-1$ elements, it now follows by Theorem I.13.14 c) that U is a free abelian group. If T is the torsion group of J, then $T \cap U = 1$ and hence $|T|$ divides $|J/U|$, so $|T|$ divides 2^{n-1}.

Let $V = \langle j_1, k_1, \ldots, k_{n-2} \rangle$. Since $k_i^2 = j_i^3 j_{i+1}^3$, it follows that $j_i^3 \in V$ for all $i = 2, \ldots, n-2$. As above, we see that

$$|J/V| \text{ divides } 2 \cdot 3^{n-3}.$$

The group V is a free abelian group, again by Theorem I.13.14 c), and this means $|T|$ divides $2 \cdot 3^{n-3}$.

Altogether, we obtain

$$|T| = |\mathbf{H}^2(S_n, \mathsf{K}^\times)| \le 2.$$

b) The proof that the Schur multiplier of S_n ($n \geq 4$) has order at least 2 is rather involved. Unfortunately, the representation groups of S_n (there exist exactly two nonisomorphic such groups for $n > 3$ and $n \neq 6$) are not among the groups we have seen so far, so they must be constructed.

(For $n = 4$, $\mathrm{GL}_2(3)$ is a representation group for S_4 since $\mathrm{PGL}_2(3) \cong S_4$, as is easy to check; see Theorem II.6.14, pg. 192.)

We construct a representation group for S_{2n}:

Let r, s, t be the following 2×2 matrices over the complex numbers ($i^2 = -1$):

$$r = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad s = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \quad t = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

One can verify immediately that $r^2 = s^2 = t^2 = 1$, $rs = -sr$, $rt = -tr$, $st = -ts$. We define

$$x_{2i} = \frac{1}{\sqrt{2}} 1 \otimes \cdots \otimes \underset{i-1}{1} \otimes (1 \otimes \underset{i,i+1}{s} + r \otimes t) \otimes \underset{i+2}{t} \otimes \cdots \otimes \underset{n}{t} \quad (1 \leq i \leq n-1)$$

$$x_{2i+1} = \frac{1}{\sqrt{2}} 1 \otimes \cdots \otimes 1 \otimes \underset{i+1}{(r+s)} \otimes t \otimes \cdots \otimes \underset{n}{t} \quad (0 \leq i \leq n-1).$$

(The tensor products of linear maps are constructed in the sense of Theorem 9.11.)

One obtains the following relations by a simple calculation:

$$x_i^2 = 1 \quad (1 \leq i \leq 2n-1)$$
$$(x_i x_{i+1})^3 = -1 \quad (1 \leq i \leq 2n-2)$$
$$x_i x_j = -x_j x_i \quad \text{for} \quad i+1 < j.$$

So the x_i ($1 \leq i \leq 2n-1$) generate a group K with $-1 \in K' \cap \mathbf{Z}(K)$. Modulo $\langle -1 \rangle$, the x_i satisfy the defining relations of S_{2n}. So $K/\langle -1 \rangle$ is a homomorphic image of S_{2n}. We may assume that $2n > 4$ since the special case $2n = 4$ is settled by the representation group $\mathrm{GL}_2(3)$ for S_4. Then the proper homomorphic images of S_{2n} only have orders 1 or 2. But K contains the element $x_1 x_2$ of order 6. So $K/\langle -1 \rangle \cong S_{2n}$. Therefore, we have by Lemma 23.3 that

$$|\mathbf{H}^2(S_{2n}, \mathsf{K}^\times)| \geq 2.$$

In the same way, it follows for $n \geq 3$ that $|\mathbf{H}^2(S_{2n-1}, \mathsf{K}^\times)| \geq 2$ as well by considering the group $\langle x_1, \ldots, x_{2n-2} \rangle$. □

Remark 25.13. The nature of the projective representations of S_{2n} we have constructed is not at all clarified by the above calculations. We will say more about this now, but without giving full proofs.

Let $\mathsf{V} = \langle v_1, \ldots, v_{2n} \rangle$ be a vector space of dimension $2n$ over the complex numbers $\mathbb{C}$ equipped with the bilinear scalar product defined by $(v_i, v_j) = \delta_{ij}$. Let T be the free associative $\mathbb{C}$-algebra in the free generators x_i ($i = 1, \ldots, 2n$). (T is also called the tensor algebra over V.) Let I be the two-sided ideal in T that is generated

by the elements

$$x_i x_j + x_j x_i \quad (i \neq j) \quad \text{and} \quad x_i^2 - 1.$$

We refer to the algebra $\mathbf{C}(\mathsf{V}) = T/I$ as the Clifford algebra over V. It is not difficult to show that $\mathbf{C}(\mathsf{V})$ is an algebra of dimension 2^{2n} over $\mathbb{C}$, and $\mathbf{C}(\mathsf{V})$ has a $\mathbb{C}$-basis consisting of the element 1 and the products

$$y_{i_1} \cdots y_{i_s}$$

with $1 \leq i_1 < \cdots < i_s \leq 2n$, where $y_i = x_i + I$. The algebra $\mathbf{C}(\mathsf{V})$ is a full matrix ring of degree 2^n over $\mathbb{C}$. (If V has odd dimension $2n+1$, the algebra $\mathbf{C}(\mathsf{V})$ is a direct sum of two full matrix rings of degree 2^n over $\mathbb{C}$. For the theory of Clifford algebras, see Artin [15], pg. 186–193 and Bourbaki [44], Algèbre, chap. 9, pg. 139–150.)

The linear map π defined by $v_i\pi = y_i$ $(1 \leq i \leq 2n)$ is an injective linear map of V into $\mathbf{C}(\mathsf{V})$ with the property that

$$(v\pi)(v'\pi) + (v'\pi)(v\pi) = 2(v, v')1$$

for all $v, v' \in \mathsf{V}$. The following universal property follows easily from the construction of $\mathbf{C}(\mathsf{V})$:

Let $\mathfrak{a}$ be any algebra over $\mathbb{C}$. Suppose μ is a $\mathbb{C}$-linear map from V to $\mathfrak{a}$ such that

$$(v\mu)(v'\mu) + (v'\mu)(v\mu) = 2(v, v')1$$

for all $v, v' \in \mathsf{V}$. Then there exists an algebra homomorphism α from $\mathbf{C}(\mathsf{V})$ to $\mathfrak{a}$ such that $v\mu = v\pi\alpha$ for all $v \in \mathsf{V}$.

In particular, let τ be an orthogonal map of V onto itself. Then

$$(v\tau\pi)(v'\tau\pi) + (v'\tau\pi)(v\tau\pi) = 2(v\tau, v'\tau)1 = 2(v, v')1.$$

So there exists an algebra homomorphism α from $\mathbf{C}(\mathsf{V})$ to itself such that $(v\tau)\pi = v\pi\alpha$. It is easy to see that α is an automorphism of $\mathbf{C}(\mathsf{V})$. Since $\mathbf{C}(\mathsf{V})$ is a full matrix ring, every automorphism of $\mathbf{C}(\mathsf{V})$ is inner by the Skolem–Noether Theorem (see v. d. Waerden [448], pg. 253). Therefore, there exists some $\mathbf{P}(\tau)$ in $\mathbf{C}(\mathsf{V})$ such that

$$(v\tau)\pi = \mathbf{P}(\tau)^{-1} v\pi \mathbf{P}(\tau),$$

and $\mathbf{P}(\tau)$ is uniquely determined by this relation up to scalar factors. We have

$$\mathbf{P}(\tau_1\tau_2) = c(\tau_1, \tau_2)\mathbf{P}(\tau_1)\mathbf{P}(\tau_2)$$

for some scalar factor $c(\tau_1, \tau_2) \neq 0$. So $\mathbf{P}$ is a projective representation of the orthogonal group $\mathrm{O}(\mathsf{V})$, the so-called spin representation. The natural embedding of S_{2n} in $\mathrm{O}(\mathsf{V})$ by letting S_{2n} permute the basis vectors v_i yields the projective representation of S_{2n} of degree 2^n given by the construction of the representation

group above. (The smallest possible degree of a projective representation of S_n that is not an ordinary representation is[12] $2^{\lfloor \frac{n-1}{2} \rfloor}$, as one can see by consulting Schur [375].)

Notes for Chapter V

§2: Definition 2.1 of the radical and Theorem 2.2 originate from N. Jacobson.

§4: The proof of Jacobson's Lemma 4.2 given here follows Bourbaki [43], pg. 39.

§7: Theorem 7.5 follows Wielandt [470], where a somewhat more general theorem is proved; for a special case that can be handled with transfer, see Grün [195].

§8: Lemma 8.12 can be found in Zassenhaus [494], but was essentially already known to Burnside (see Burnside [84], pg. 336). The proof of Theorem 8.13 is a combination of Hughes, Thompson [254] and Kegel [300].

§10: Theorem 10.8 can be found in a slightly weaker form in Burnside [84]; the proof given here stems from Brauer [56].

§12: Theorem 12.9 is proved in modular representation theory using the relationship between the Cartan matrix and the decomposition matrix; see Curtis, Reiner [102], pg. 593. A direct proof of Theorem 12.11 without reducing modulo $\mathfrak{p}$ is not known.

§13: Lemma 13.2 follows v. d. Waerden [448], and Theorem 13.3 follows Powell [347].

§14: For this section, one may consult Reiner [354].

§15: Theorem 15.2 was shared with me in a letter from P. Hall.

§16: The importance of induced characters was already known to Frobenius; see Frobenius [146]. The determination of characters of Frobenius groups in Theorem 16.13 partly follows Feit [124].

§17: Our treatment of Theorems 17.12 and 17.13 is based in large part on lectures of E. C. Dade.

§18: Lemma 18.10 and the proof of Theorem 18.11 using this lemma were shared with me in a conversation with E. C. Dade.

§19: The proof of Theorem 19.3 follows Brauer, Tate [72].

§20: For the questions posed in this section, see also Tsuzuku [443, 444] and Tamaschke [423, 424]. Example 20.10 is due to a letter from H. Wielandt.

§21: The proof of Theorem 21.3 given here is the original proof in Burnside [84]. The proof of Theorem 21.6 is an elementary (however, therefore longer) version of the proof in Itô [279].

§22: Definition 22.1 through Lemma 22.5 follows Brauer, Fowler [66]; Definition 22.6 through Theorem 22.9 follows Suzuki [403], with an addition to part j) of the proof of Theorem 22.9 from Feit [126].

§23: A generalization of the method for calculating higher cohomology groups can be found in Gruenberg [192]. Twisted group algebras were studied in Conlon [97, 98].

[12] Correction 31 from errata.

§25: More examples pertaining to Theorem 25.2 a) can be found in B. H. Neumann [335]. The consequence of the Golod–Shafarevich theorem for Theorem 25.2 c) goes back to a remark of M. Rosen in Math. Reviews 31, Nr. 4841 (1966). The calculation of the Schur multiplier for the groups $\mathrm{SL}_2(p^f)$ and $\mathrm{PSL}_2(p^f)$ is already in Schur [373], as well as Theorem 25.10. A determination of the cohomology groups of an extension G of a normal subgroup N with factor group G/N from N and G/N was partially carried out by Lyndon [324], but a complete result was not obtained. A thorough analysis of the representation groups for the symmetric and alternating groups can be found in Schur [375].

Chapter VI
Solvable groups

There are two important sets of problems in the theory of solvable groups, which overlap and complement each other in many ways:

1) The "arithmetic" structure of the group. The object of study here is the construction of the group from its Sylow subgroups and Hall subgroups.

2) The "normal" structure of the group. In this direction, the focus is on the chief factors and the lattice of normal subgroups. The goal is to construct the group, in the sense of extension theory, from abelian or nilpotent groups. We plan to discuss the lattice of so-called subnormal subgroups (these are the elements in a composition series) in later volumes; this theory yields particularly sharp statements when some of the composition factors are nonsolvable. However, it has not made a substantial contribution to the theory of solvable groups at the time of writing.

The theory of Sylow systems of solvable groups (§1, §2) developed by P. Hall is the foundation of the arithmetic theory. The system normalizers provide a clear connection between the arithmetic structure and the normal structure (see the Covering-Avoidance Theorem 11.10). In particular, there is a rich theory for solvable groups whose Sylow subgroups are all abelian (§14). It is of fundamental importance for a number of solvability criteria that solvable groups are characterized by the existence of Sylow systems (Theorem 1.10), a fact discovered by P. Hall in 1937. A broad generalization of this theorem was proved by Kegel and Wielandt: Products of pairwise permutable nilpotent groups are always solvable (§4).

When analyzing the normal structure, one considers first and foremost the automorphisms induced by the group on its chief factors. Since the chief factors of solvable groups are elementary abelian, the action of a solvable group on its chief factors gives rise to representations of the group. We then have a natural opportunity to apply representation theory, although the representations are over prime fields and not (as often assumed in Chapter V) over algebraically closed fields of characteristic 0. The case of supersolvable groups (§9), in which all chief factors are cyclic, is particularly manageable in this regard; these groups are an important generalization of nilpotent groups, and their structure is determined to a large extent by the structure of their Sylow subgroups.

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B. Huppert, *Finite Groups I*, Grundlehren der mathematischen Wissenschaften 364,
https://doi.org/10.1007/978-3-031-87529-8_6

The theory of p-length (§6) developed by P. Hall and G. Higman proceeds by considering the automorphisms induced by the group on its p-factor groups. Significant insight into group structure can already be obtained from the elementary results of the theory described in this chapter. The concept of p-length has done much to organize the theory of solvable groups since its introduction in 1956. We will prove some deeper theorems regarding p-length in Volume 2.

Gaschütz's theory of formations (§7) provides a link between these two approaches to solvable groups. The theory has yielded numerous generalizations of older results and uniform proofs. It also gives an approach to Carter subgroups. These subgroups were discovered by R. Carter in 1960, who showed that every solvable group contains exactly one conjugacy class of nilpotent, self-normalizing subgroups. The significance of these subgroups is not yet completely clear, although numerous attempts have been made to connect the Carter subgroups to the structure of the whole group. There are numerous open problems in this area, some of which lead to the study of fixed point free automorphisms. We will return to this in Volume 2. Some of the known relationships between Carter subgroups, normal series and Sylow systems are illustrated in §12, §13 and §15.

The reader unfamiliar with the subject is recommended to begin with §1, §2 and §5–§9.

1 Hall subgroups of solvable groups

Our goal in this section is to prove a strengthening of the Sylow theorems for solvable groups due to P. Hall. We will also prove that these strengthened Sylow theorems characterize solvable groups.

Definition 1.1.

a) A nonempty set π of natural numbers is called an arithmetic set if it has the following properties:
(1) If $a \in \pi$ and $b \mid a$, then $b \in \pi$.
(2) If $a, b \in \pi$, then $ab \in \pi$.
Clearly,

$$\pi = \left\{ \prod_i p_i^{a_i} \mid p_i \text{ prime numbers in } \pi \right\}.$$

That is, the arithmetic sets are exactly those subsemigroups of the multiplicative semigroup of the natural numbers; the latter is generated by the prime numbers. We occasionally write

$$\pi = \langle p_i \mid p_i \in \pi,\ p_i \text{ prime} \rangle.$$

b) If π is an arithmetic set, then we may define

$$\pi' = \Big\{ \prod_i q_i^{a_i} \mid q_i \text{ prime}, q_i \notin \pi \Big\}.$$

c) If π and μ are arithmetic sets, then let

$$\pi\mu = \Big\{ ab \mid a \in \pi, b \in \mu \Big\}.$$

If π and μ are arithmetic sets, then it is clear that $\pi\mu$ and $\pi \cap \mu$ are also arithmetic sets. We always have that

$$(\pi\mu)' = \pi' \cap \mu' \quad \text{and} \quad (\pi \cap \mu)' = \pi'\mu'.$$

Definition 1.2. Let π be an arithmetic set.

a) A group G is said to be a π-group if $|G| \in \pi$; that is, the prime decomposition of $|G|$ consists only of primes lying in π.

b) A subgroup H of G is called a Hall π-subgroup of G if $|H| \in \pi$ and $[G : H] \in \pi'$. We will often then write $H = G_\pi$. If we are considering more than one Hall π-subgroup of G, then we will write G_π, G_π^* or something of the sort. (This definition coincides with those given in Definition III.5.4, pg. 299, and Definition IV.7.1, pg. 472.)

Every Sylow p-subgroup of G is a Hall π-subgroup of G for the arithmetic set $\pi = \langle p \rangle$; in this case, we write G_p instead of G_π. We also write $G_{p'}$ instead of $G_{\pi'}$ and call $G_{p'}$ a p-complement of G. (So $G_{p'}$ is not, say, a Sylow subgroup of G for a prime p'.) Occasionally, we will refer to Sylow complements without explicitly mentioning a prime.

Definition 1.3.

a) A group G is called π-separable if every chief factor of G is either a π-group or a π'-group.

b) A group G is called π-solvable if G is π-separable and every chief π-factor of G is solvable.

By the Jordan–Hölder theorem for chief series (Theorem I.11.7, pg. 68), it is enough that these conditions are satisfied by a single chief series of G.

Since every group of odd order is solvable by Feit, Thompson [132], the concepts of π-separable and π-solvable coincide if $2 \notin \pi$.

Lemma 1.4. *If G is π-solvable or π-separable, then every subgroup and every factor group of G has the same property.*

Proof. This is trivial for factor groups since their chief factors are a subset of the set of chief factors of G.

Let $U < G$, and let

$$G = G_0 \rhd G_1 \rhd \cdots \rhd G_s = 1$$

be a chief series of G. We consider the normal subgroups $U_i = G_i \cap U$ of U. We have

$$U_i/U_{i+1} = (U \cap G_i)/(U \cap G_{i+1}) \cong G_{i+1}(U \cap G_i)/G_{i+1} \leq G_i/G_{i+1}.$$

So if G_i/G_{i+1} is a π-group, π'-group or π-solvable, then U_i/U_{i+1} is as well. The claim then follows by refining the normal series of U consisting of the U_i to a chief series of U. □

Both of the following elementary lemmas regarding Hall subgroups will be used repeatedly.

Lemma 1.5. *Let π and μ be arithmetic sets with $\pi \cap \mu = \{1\}$. If $G_{\pi'}$ and $G_{\mu'}$ are arbitrary Hall subgroups of G, then $G_{\pi'} \cap G_{\mu'}$ is a Hall $(\pi\mu)'$-subgroup of G.*

Proof. Since $[G : G_{\pi'}] \in \pi$, $[G : G_{\mu'}] \in \mu$ and $\pi \cap \mu = \{1\}$, we have $([G : G_{\pi'}], [G : G_{\mu'}]) = 1$. It then follows from Lemma I.2.13 b) (pg. 8) that

$$[G : G_{\pi'} \cap G_{\mu'}] = [G : G_{\pi'}][G : G_{\mu'}] \in \pi\mu.$$

Furthermore, it holds trivially that

$$|G_{\pi'} \cap G_{\mu'}| \in \pi' \cap \mu' = (\pi\mu)'.$$

So $G_{\pi'} \cap G_{\mu'}$ is a Hall subgroup of G for the arithmetic set $(\pi\mu)'$. □

The following is Lemma IV.7.2 (pg. 472):

Lemma 1.6. *If N is a normal subgroup of G, then $G_\pi N/N$ is a Hall π-subgroup of G/N and $N \cap G_\pi$ is a Hall π-subgroup of N.*

Now we come to our first main theorem:

Main Theorem 1.7 (P. Hall [206], Čunihin [100]).

a) *If G is π-separable, then there exist Hall π-subgroups and Hall π'-subgroups in G.*

b) *If G is π-solvable, then all Hall π-subgroups of G are conjugate in G; all Hall π'-subgroups of G are also conjugate.*

c) *If G is π-solvable and U is a π-subgroup of G, then U is contained in a Hall π-subgroup of G; the corresponding statement also holds for π'-subgroups.*

(*If one assumes the conjugacy part of the Zassenhaus Theorem I.18.3, pg. 137, which always holds by the Feit–Thompson Theorem, then it suffices to assume that G is π-separable in Theorem 1.7 b) and Theorem 1.7 c).*)

Proof. We prove the claims by induction on $|G|$. Let N be a minimal normal subgroup of G. Since G is π-separable, we have $|N| \in \pi$ or $|N| \in \pi'$.

a) By the induction hypothesis, there exists a Hall π-subgroup H/N of G/N. Then $|H/N| \in \pi$ and

$$[G : H] = [G/N : H/N] \in \pi'.$$

If $|N| \in \pi$, then $|H| \in \pi$ as well and we are done. If $|N| \in \pi'$, then $|H/N|$ and $|N|$ are coprime. By the Zassenhaus Theorem I.18.1 (pg. 135), there exists a subgroup V of H such that $|V| = |H/N|$. So $|V| \in \pi$ and

$$[G : V] = [G : H][H : V] = [G : H]|N| \in \pi'.$$

Therefore, V is a Hall π-subgroup of G.

For Hall π'-subgroups, the same proof works since π'-separability is equivalent to π-separability.

b) Let V and W be Hall μ-subgroups of G, and let $\mu = \pi$ or $\mu = \pi'$. By Lemma 1.6, VN/N and WN/N are Hall μ-subgroups of G/N, so they are conjugate in G/N by induction, say by an element gN. Then $(VN)^g = WN$, and therefore $V^g \leq WN$.

If $|N| \in \mu$, then we have

$$|WN| = \frac{|W||N|}{|W \cap N|} \in \mu.$$

The subgroup W is a maximal μ-subgroup of G, as it is a Hall μ-subgroup, so $W = WN \geq V^g$. Then $V^g = W$ since their orders are equal.

If $|N| \in \mu'$, then $|V^g| = |W|$ is coprime to $|N|$. If $\mu = \pi'$, then $|N| \in \pi$, and then N is solvable since G is π-solvable. If $\mu = \pi$, then $|WN/N| \in \pi$ and WN/N is π-solvable by Lemma 1.4. Hence, WN/N is solvable. In any case, either N or WN/N is solvable. By the Zassenhaus Theorem I.18.2 (pg. 136), all complements of N are therefore conjugate in WN. In particular, this holds for W and V^g. So V and W are conjugate in G.

c) If U is a μ-subgroup of G, then UN/N is a μ-subgroup of G/N. So UN/N lies in a Hall μ-subgroup S/N of G/N by our induction hypothesis. By Lemma 1.6, every $G_\mu N/N$ is a Hall μ-subgroup of G/N, so is conjugate to S/N by b). Hence, $U \leq G_\mu^g N$ for some $g \in G$. If $|N| \in \mu$, then it follows that $G_\mu^g N = G_\mu^g$, and thus $U \leq G_\mu^g$. In any case, we have

$$UN = UN \cap G_\mu^g N = VN,$$

where $V = UN \cap G_\mu^g \leq G_\mu^g$. If $|N| \in \mu'$, then $U \cap N = 1$ and $V \cap N \leq G_\mu^g \cap N = 1$. So

$$|U| = [UN : N] = [VN : N] = |V| \in \mu.$$

So either N (for $\mu' = \pi$) or UN/N (for $\mu = \pi$) is solvable. By the Zassenhaus Theorem I.18.2 (pg. 136), both complements U and V of N are therefore conjugate in UN. This gives $U^h = V \leq G_\mu^g$ for some $h \in G$. □

We state the following theorem as an important special case.

Main Theorem 1.8 (P. Hall [206]). *If G is solvable, then G has Hall π-subgroups for all π. Every two Hall π-subgroups of G are conjugate in G, and every π-subgroup of G is contained in a Hall π-subgroup of G.*

(*Thus, the π-Sylow theorem holds in G, in the sense of Definition III.5.5, pg. 299.*)

The converse of Theorem 1.8 is of fundamental importance. In preparation for its proof, we first prove a theorem of Wielandt that has independent interest.

Theorem 1.9 (Wielandt [474]). *If the group G contains three solvable subgroups H_i $(i = 1, 2, 3)$ whose indices are pairwise coprime, then G is solvable.*

Proof. Let $D = H_1 \cap H_2$. Since the indices of H_1 and H_2 are coprime, we have by Lemma I.2.13 b) (pg. 8) that $G = H_1H_2$ and therefore $[G : H_2] = [H_1 : D]$. If $H_1 = 1$, then $G = H_2$ is solvable. So assume $H_1 \neq 1$, and let N be a minimal normal subgroup of H_1. By Theorem I.9.13 (pg. 54), N has prime power order, say $|N| = p^n$. At least one of the indices $[G : H_2]$, $[G : H_3]$ is not divisible by p. Assume without loss of generality that p does not divide $[G : H_2]$. Since $N \trianglelefteq H_1$, ND is a subgroup of H_1. Therefore, $[ND : D] = [N : N \cap D]$ is a divisor of $|N|$ and also of $[H_1 : D] = [G : H_2]$. Since $(|N|, [G : H_2]) = 1$, it follows that $[ND : D] = 1$, so $N \leq D$.

Let $g = h_1h_2$ with $h_i \in H_i$ be an arbitrary element of $G = H_1H_2$. Since $N \trianglelefteq H_1$, we have

$$N^g = N^{h_1h_2} = N^{h_2} \leq D^{h_2} \leq H_2.$$

Hence, $T = \langle N^g \mid g \in G \rangle \leq H_2$. Therefore, T is a nonidentity solvable normal subgroup of G. The factor group G/T contains the solvable subgroups H_iT/T $(i = 1, 2, 3)$ with

$$[G/T : H_iT/T] = [G : H_iT],$$

and these integers are pairwise coprime. So G/T satisfies the hypotheses of the theorem, and we may then assume that G/T is solvable by induction. Then G is also solvable by Theorem I.8.6 a) (pg. 40). □

Now the converse of Theorem 1.8 follows easily:

Main Theorem 1.10 (P. Hall [210]). *Let $|G| = \prod_{i=1}^{r} p_i^{a_i}$ be the prime decomposition of $|G|$. If G has subgroups $U_1, \ldots, U_r$ such that $[G : U_i] = p_i^{a_i}$, then G is solvable.*

Proof. We prove the theorem by induction on r. For $r = 1$, G is a group of prime power order, so is solvable by Theorem I.8.9 (pg. 41). For $r = 2$, G is solvable by Burnside's Theorem V.7.3 (pg. 522). So assume in what follows that $r \geq 3$.

We define $V_{ij} = U_i \cap U_j$ for $i \neq j$. By Lemma 1.5, we have

$$[G : V_{ij}] = p_i^{a_i} p_j^{a_j}$$

and therefore

$$[U_i : V_{ij}] = p_j^{a_j}.$$

Therefore, the U_i $(i = 1, \ldots, r)$ satisfy the hypothesis, so they are solvable by induction. The indices $[G : U_i] = p_i^{a_i}$ are pairwise coprime. So G is solvable by Theorem 1.9. □

Let G be a finite group with $|G| = \prod_{i=1}^{r} p_i^{a_i}$. At the time of writing, it is an open question whether the existence of subgroups U_{ij} $(i, j = 1, \ldots, r; i \neq j)$ with $|U_{ij}| = p_i^{a_i} p_j^{a_j}$ already forces G to be solvable.

Is it possible to characterize an interesting class of solvable groups by strengthening the hypothesis of Theorem 1.10? The following is a natural condition to consider:

(U): For every divisor d of $|G|$, there exists a subgroup U of G such that $|U| = d$.

By Theorem 1.10, the solvability of G follows from condition (U). However, not much more follows, as the following observation shows.

Theorem 1.11 (McLain [329]). *If G is any solvable group, then there exists an abelian group A such that $G \times A$ has property* (U).

Proof. Let $|G| = \prod_{i=1}^{r} p_i^{a_i}$, and let A be an abelian group of order $|A| = \prod_{i=1}^{r} p_i^{a_i - 1}$. Then $G \times A$ has property (U):

Let $d = p_1^{d_1} \cdots p_r^{d_r}$ be a divisor of $|G \times A|$, and assume that $a_i \leq d_i$ for $i = 1, \ldots, s$ and $d_i < a_i$ for $i = s+1, \ldots, r$ after reordering if necessary. Let H be a Hall π-subgroup of G for the arithmetic set $\pi = \langle p_1, \ldots, p_s \rangle$, and let B be a subgroup of A of order $p_1^{d_1 - a_1} \cdots p_s^{d_s - a_s} p_{s+1}^{d_{s+1}} \cdots p_r^{d_r}$. Then $H \times B$ is a subgroup of $G \times A$ of the desired order d. □

See also Exercise 19 (pg. 762).

Exercises

1) Prove the existence of p-complements in solvable groups without using the Zassenhaus theorem.

   (Hint: Let G be a minimal counterexample. Then G has a minimal normal subgroup N with $|N| = p^n$ and $p \nmid |G/N|$. Let M/N be a minimal normal subgroup of G/N. Then $|M/N| = q^m$ for some prime $q \neq p$. If Q is a Sylow q-subgroup of M, then $\mathbf{N}_G(Q)$ is a p-complement of G.)

2) (P. Hall [211]) We say a group G is complementable if for every subgroup U of G, there exists a subgroup V of G such that $G = UV$ and $U \cap V = 1$. Prove the following:
   a) Every complementable group is solvable.
   b) The property of being complementable is inherited by subgroups, factor groups and direct products.
   c) If G is complementable, then all chief factors of G are cyclic and all Sylow subgroups of G are elementary abelian.
   d) G is complementable if and only if G is isomorphic to a subgroup of a direct product of groups with square-free orders.

3) (Čunihin [100]) Let π be an arithmetic set. Suppose the order of each chief factor of G contains at most one prime in π. Prove the following:
   a) G contains solvable Hall π-subgroups.

b) All Hall π-subgroups of G are conjugate.

(The proof that every π-subgroup of G is contained in a Hall π-subgroup of G is considerably more difficult; see P. Hall [219].)

2 Sylow systems of solvable groups

In this section, we begin our treatment of a theory of solvable groups, developed by P. Hall, that takes Theorem 1.8 as its foundation.

Definition 2.1. Let $|G| = \prod_{i=1}^{r} p_i^{a_i}$ be the prime decomposition of $|G|$.

a) A set $\mathscr{K} = \{G_{p_1'}, \ldots, G_{p_r'}\}$ of Sylow complements of G is a called a complement system of G. (By Theorem 1.8 and Theorem 1.10, G has a complement system if and only if G is solvable.)

b) A set $\mathscr{S} = \{G_{p_1}, \ldots, G_{p_r}\}$ of Sylow subgroups of G is called a Sylow system of G if $G_{p_i} G_{p_j} = G_{p_j} G_{p_i}$ for all i, j.

Theorem 2.2. *There is a one-to-one correspondence between the Sylow systems and the complement systems of a solvable group G given by*

$$\mathscr{S} = \left\{ G_p \;\middle|\; p \text{ divides } |G| \right\} \mapsto \mathscr{K}_{\mathscr{S}} = \left\{ \prod_{q \neq p} G_q \;\middle|\; p \text{ divides } |G| \right\}$$

and

$$\mathscr{K} = \left\{ G_{p'} \;\middle|\; p \text{ divides } |G| \right\} \mapsto \mathscr{S}_{\mathscr{K}} = \left\{ \bigcap_{q \neq p} G_{q'} \;\middle|\; p \text{ divides } |G| \right\}.$$

Proof. Let $\mathscr{S} = \{G_p \mid p \text{ divides } |G|\}$ be a Sylow system of G. Since the G_q are pairwise permutable, the groups $G_{p'}^* = \prod_{q \neq p} G_q$ are a complement system $\mathscr{K}_{\mathscr{S}}$ of G.

Conversely, assume that $\mathscr{K} = \{G_{p'} \mid p \text{ divides } |G|\}$ is a complement system of G. We let $G_p^* = \bigcap_{q \neq p} G_{q'}$. Applying Lemma 1.5 repeatedly, we see that $|G_p^*| = p^{a_p}$ with $p^{a_p} \nmid |G|$. For $i \neq j$, we have

$$G_{p_i}^* G_{p_j}^* = \Big(\bigcap_{q \neq p_i} G_{q'} \Big) \Big(\bigcap_{q \neq p_j} G_{q'} \Big) \subseteq \bigcap_{q \neq p_i, p_j} G_{q'} = \bigcap_{q \in \langle p_i, p_j \rangle'} G_{q'}.$$

The right-hand group above is a Hall $\langle p_i, p_j \rangle$-subgroup of G by Lemma 1.5. Therefore, it follows by comparing orders that

$$G_{p_i}^* G_{p_j}^* = \bigcap_{q \in \langle p_i, p_j \rangle'} G_{q'} = G_{p_j}^* G_{p_i}^*,$$

and so the G_p^* form a Sylow system $\mathscr{S}_{\mathscr{K}}$ of G.

It is clear that $(\mathscr{K}_{\mathscr{S}})_{\mathscr{K}} = \mathscr{K}$ and $(\mathscr{S}_{\mathscr{K}})_{\mathscr{S}} = \mathscr{S}$. □

The following theorem is a direct consequence of Theorem 1.10 and Theorem 2.2.

Theorem 2.3 (P. Hall). *The group G is solvable if and only if G has Sylow systems (complement systems).*

We will discuss a broad generalization of Theorem 2.3 in §4: Namely, it is already enough to ensure that the group G is solvable if G is a product of pairwise permutable nilpotent groups. Theorem 2.3 will play a decisive role in the proof of this deep theorem.

There is a notable conjugacy theorem for complement systems and Sylow systems:

Theorem 2.4 (P. Hall [212]). *Assume G is solvable. Let $\mathscr{K} = \{G_{p_i'} \mid i = 1, \ldots, r\}$ and $\mathscr{K}^* = \{G^*_{p_i'} \mid i = 1, \ldots, r\}$ be complement systems of G. Then there exists an element $g \in G$ such that $G^g_{p_i'} = G^*_{p_i'}$ for all $i = 1, \ldots, r$.*

Proof. By Theorem 1.7 b), all p_i-complements of G are conjugate in G. Therefore, $[G : \mathbf{N}_G(G_{p_i'})]$ is the number of distinct p_i-complements of G. Since $G_{p_i'} \leq \mathbf{N}_G(G_{p_i'})$, the index $[G : \mathbf{N}_G(G_{p_i'})]$ is a power of p_i. So the number of complement systems of G is

$$k = \prod_{i=1}^{r} [G : \mathbf{N}_G(G_{p_i'})].$$

As the indices $[G : \mathbf{N}_G(G_{p_i'})]$ are pairwise coprime, Lemma I.2.13 b) (pg. 8) implies that

$$k = \prod_{i=1}^{r} [G : \mathbf{N}_G(G_{p_i'})] = [G : \bigcap_{i=1}^{r} \mathbf{N}_G(G_{p_i'})].$$

Therefore, the total number of complement systems of G is equal to the number of conjugates of the fixed complement system $\{G_{p_i'} \mid i = 1, \ldots, r\}$. So all the complement systems of G are conjugate. By Theorem 2.2, this is also true for Sylow systems. □

The following theorem gives information regarding the relationship between the Sylow systems of (normal) subgroups and the Sylow systems of the whole group.

Theorem 2.5.

a) *Assume G is solvable, and let U be a subgroup of G. For every Sylow system $\mathscr{S}_U = \{U_p \mid p \text{ divides } |U|\}$ of U, there exists a Sylow system $\mathscr{S}_G = \{G_p \mid p \text{ divides } |G|\}$ of G such that $U_p = U \cap G_p$ for all p.*

b) *If N is a normal subgroup of G and $\{G_p \mid p \text{ divides } |G|\}$ is a Sylow system of G, then $\{N \cap G_p \mid p \text{ divides } |N|\}$ is a Sylow system of N.*

Proof. a) Let $\mathscr{K}_U = \{U_{p'} \mid p \text{ divides } |U|\}$ be the complement system of U constructed from $\mathscr{S}_U$ according to Theorem 2.2. By Theorem 1.7 c), every $U_{p'}$ lies in at least one p-complement $G_{p'}$ of G. Let $\mathscr{K}_G$ be the complement system of G

consisting of these $G_{p'}$, and let $\mathscr{S}_G = \{G_p \mid p \text{ divides } |G|\}$ with $G_p = \bigcap_{q \neq p} G_{q'}$ be the corresponding Sylow system of G. Then $U_{p'} = U \cap G_{p'}$. By Theorem 2.2, we have

$$U_p = \bigcap_{q \neq p} U_{q'} = U \cap \bigcap_{q \neq p} G_{q'} = U \cap G_p.$$

b) By Lemma I.7.7 a) (pg. 35), $N \cap G_p = N_p$ is a Sylow p-subgroup of N. We have $N_p N_q \subseteq N \cap G_p G_q$. As $N \cap G_p G_q$ is a $\langle p, q\rangle$-subgroup of N, we have that $|N \cap G_p G_q|$ divides $|N_p||N_q|$. It follows that

$$N_p N_q = N \cap G_p G_q = N_q N_p,$$

and so the collection of the N_p is a Sylow system of N. □

One proves the following as in Theorem I.7.6 and Lemma I.7.7 c) (pg. 35).

Lemma 2.6. *Assume G is π-solvable. Then:*
a) *Every subgroup of G containing the normalizer $\mathbf{N}_G(G_\pi)$ of a Hall π-subgroup G_π of G is equal to its normalizer in G.*
b) *If $N \trianglelefteq G$ and G_π is a Hall π-subgroup of G, then*

$$\mathbf{N}_{G/N}(G_\pi N/N) = \mathbf{N}_G(G_\pi)N/N;$$

the same holds for π'.

Theorem 2.7 (P. Hall [212]). *Assume G is solvable, and let $|G| = \prod_{i=1}^{r} p_i^{a_i}$. The order of the automorphism group $\mathbf{Aut}(G)$ of G divides*[1]

$$|G| \prod_{i=1}^{r} \psi(p_i^{d_i}) p_i^{d_i(a_i - d_i)},$$

where

$$\psi(p^a) = (p^a - 1)(p^a - p) \cdots (p^a - p^{a-1})$$

and d_i is the minimum number of generators of a Sylow p_i-subgroup of G.

Proof. Let $\mathscr{S}$ be a Sylow system of G. Every automorphism α of G maps the Sylow system $\mathscr{S}$ to another Sylow system $\mathscr{S}^\alpha$ of G. By Theorem 2.4, $\mathscr{S}^\alpha$ is conjugate to $\mathscr{S}$ in G, so $\mathscr{S}$ can be mapped to $\mathscr{S}^\alpha$ by an inner automorphism of G. Thus, if we denote the group of all inner automorphisms of G by $\mathbf{Inn}(G)$ and the subgroup of $\mathbf{Aut}(G)$ fixing setwise each of the Sylow subgroups in $\mathscr{S}$ by B, we have $\mathbf{Aut}(G) = \mathbf{Inn}(G)B$. It follows that

[1] Correction 32 from errata.

$$|\mathbf{Aut}(G)| = |B|[\mathbf{Inn}(G) : \mathbf{Inn}(G) \cap B].$$

The second factor in this product is a divisor of $|G|$. The group B is completely described by its action on the Sylow subgroups G_{p_i} in $\mathscr{S}$. Therefore, it is isomorphic to a subgroup of the direct product of the automorphism groups of the G_{p_i}. That is, $|B|$ divides $\prod_{i=1}^{r} |\mathbf{Aut}(G_{p_i})|$. We recall from Theorem III.3.19 (pg. 290) the fact that $|\mathbf{Aut}(G_{p_i})|$ divides $\psi(p_i^{d_i}) p_i^{d_i(a_i - d_i)}$, which yields the claim. □

3 Groups with many Sylow systems

We have seen that the existence of Sylow systems is equivalent to solvability. By strengthening the permutability requirement for the Sylow subgroups, we can identify special classes of solvable groups. We treat such a situation in this section. More examples are considered in Exercise 6 and Theorem 6.11.

It is clear that the groups of order $p^a q^b$ (p and q primes) have the property that every Sylow p-subgroup permutes with every Sylow q-subgroup. That is, every two Sylow subgroups for distinct primes are permutable. We describe all groups with this property:

Theorem 3.1 (Huppert [265]). *The following properties of a group G are equivalent:*

a) *The group G is solvable; if H/K is a chief factor of G of p-power order, then $|G/\mathbf{C}_G(H/K)| = p^a q^b$ for some prime q (depending on H/K); that is, the order of the automorphism group induced by G on every chief factor of p-power order is divisible by at most one prime distinct from p (and is possibly divisible by p, as well).*

b) *Every two Sylow subgroups of G for distinct primes are permutable.*

c) *The group G is contained in the smallest class $\mathscr{M}$ of finite groups with the following properties:*
   (1) *$\mathscr{M}$ is closed under the formation of subgroups, factor groups and direct products.*
   (2) *$\mathscr{M}$ contains every group whose order is divisible by at most two distinct primes.*

d) *G is the homomorphic image of a subgroup of a group H having the form $H = H_1 \times \cdots \times H_k$, where $|H_i|$ is divisible by at most two distinct primes.*

If G satisfies any of the equivalent conditions given in Theorem 3.1, we call G a group with many Sylow systems.

The following condition is clearly equivalent to b):

If $|G| = \prod_{i=1}^{r} p_i^{a_i}$ and $\{G_{p_i} \mid i = 1, \ldots, r\}$ is a Sylow system of G, then for all $g_i \in G$, the collection $\{G_{p_i}^{g_i} \mid i = 1, \ldots, r\}$ is also a Sylow system of G. One can describe this situation in terms of the equality

$$\prod_{i=1}^{r}[G : \mathbf{N}_G(G_{p_i})] = [G : \bigcap_{i=1}^{r}\mathbf{N}_G(G_{p_i})].$$

However, this equation does not seem to be a suitable starting point for the analysis.

Theorem 3.1 does not contain an explicit description of the groups in question, but rather ties their description to the groups of order $p^a q^b$. Given the potential complexity of the groups of order $p^a q^b$, we cannot expect much more than this.

We now prove two lemmas in preparation for the proof of Theorem 3.1.

Lemma 3.2. *Properties a) and b) in Theorem 3.1 are inherited by subgroups, factor groups and direct products.*

Proof. It is trivial to see that the properties are inherited by factor groups and direct products.

a) Let U be a subgroup of G, and let H/K be a chief factor of G of p-power order. Let $g \in \mathbf{C}_G(H/K) \cap U$ and $u \in H \cap U$. Then $u^g = uk$ for some $k \in K \cap U$. Consequently, we have

$$\mathbf{C}_G(H/K) \cap U \leq \mathbf{C}_U((H \cap U)/(K \cap U)).$$

Therefore, $[U : \mathbf{C}_U((H \cap U)/(K \cap U))]$ divides

$$[U : \mathbf{C}_G(H/K) \cap U] = [U\mathbf{C}_G(H/K) : \mathbf{C}_G(H/K)],$$

and this has the form $p^a q^b$ as it is a divisor of $|G/\mathbf{C}_G(H/K)|$. Upon refining the normal factor $(H \cap U)/(K \cap U)$ of U to a chief factor of U, we see that property a) of Theorem 3.1 certainly holds for this chief factor. Since we may construct a chief series of U by refining the intersection of a chief series of G with U, we see that this property holds for all chief factors in some chief series of U. So property a) holds for all chief factors in any chief series of U as the chief factors are isomorphic as U-groups by the Jordan–Hölder Theorem for chief series (Theorem I.11.5, pg. 67). Hence, property a) is inherited by subgroups.

b) Let U be a subgroup of G, and let U_p and U_q be Sylow subgroups of U for distinct primes p and q. Let G_p be a Sylow p-subgroup of G containing U_p, and let G_q be a Sylow q-subgroup of G containing U_q. Since G satisfies property b), $G_p G_q \cap U = V$ is a subgroup of U. We have $U_p U_q \subseteq V$, so

$$|U_p U_q| = |U_p||U_q| \leq |V|.$$

Since the order of V has the form $p^{a'} q^{b'}$ and divides $|U|$, we also have that $|V|$ divides $|U_p||U_q|$. It follows that $V = U_p U_q$, so property b) is satisfied in U. □

Lemma 3.3. *Assume G is a solvable group that satisfies property b) of Theorem 3.1. If G admits a faithful representation as a primitive permutation group, then the order of G is divisible by at most two distinct primes.*

Proof. Since G is primitive, we have a decomposition $G = NH$ with $N \cap H = 1$ by Theorem II.3.2 (pg. 167). Here, N is a minimal normal subgroup of G and H is a

maximal subgroup of G. We may assume that $H \neq 1$. Let $|N| = p^n$, and let M be a minimal normal subgroup of H. Then M has prime power order, say q^m, and $q \neq p$ by Theorem II.3.2 e). Since $M \not\lhd G$, we have $\mathbf{N}_G(M) = H$.

Assume for contradiction that there exists a prime divisor r of $|G|$ that is distinct from both p and q. Let R be a Sylow r-subgroup of G. Then the subgroup NMR of G again has property b) by Lemma 3.2. Therefore, MR and M^xR are subgroups of G for every $x \in N$. Since $NM \lhd G$, it follows that

$$MR \cap NM = M \lhd MR,$$

and similarly $M^x \lhd M^xR$. Therefore, we obtain for every $y \in R$ that

$$M^{yxy^{-1}x^{-1}} = (M^x)^{y^{-1}x^{-1}} = M^{xx^{-1}} = M.$$

Hence, we have

$$[R, N] \leq \mathbf{N}_G(M) \cap N = H \cap N = 1.$$

So R lies in the centralizer $\mathbf{C}_G(N)$ of N. By Lemma II.3.2 c) (pg. 708), we have $N = \mathbf{C}_G(N)$. Hence, $R \leq R \cap N = 1$, contradicting our assumption that r divides $|G|$. □

Lemma 3.3 already gives some insight into the structure of groups with property b):

Theorem 3.4. *If G has property b) from Theorem 3.1, then $G/\Phi(G)$ is isomorphic to a subgroup of a direct product of groups of orders $p_i^{a_i}q_i^{b_i}$, where the p_i, q_i are (not necessarily distinct) primes.*

Proof. Using Lemma 3.3, the claim follows immediately from the embedding of $G/\Phi(G)$ in the direct product of primitive groups $G/\bigcap_{g\in G} M^g$, where M is a maximal subgroup of G. □

3.5 (Proof of Theorem 3.1). We prove the claim by induction on $|G|$, and we assume for induction that the equivalence of a)–d) holds for every smaller group.

a) $\Rightarrow$ b): Let P and Q be Sylow subgroups of G for the distinct primes p and q, and let N be a minimal normal subgroup of G. Since G/N has property a), PN/N and QN/N are permutable by our induction hypothesis. Hence, NPQ is a subgroup of G. Since property a) is inherited from G by the subgroup NPQ according to Lemma 3.2, for the purpose of proving the permutability of P and Q we may clearly assume that $G = NPQ$.

If $|N| = r^c$ with $r = p$ or $r = q$, then it follows that $NPQ = PQ$, and we are finished. So we may assume that G has no nontrivial normal subgroup whose order is a power of p or a power of q. In particular, N has a complement H in $G = NPQ$ by the Zassenhaus theorem. If $N < \mathbf{C}_G(N)$, then $\mathbf{C}_G(N) = N \times K$, where $1 < K = \mathbf{C}_G(N) \cap H \lhd G$ and $|K| = p^{a'}q^{b'}$. Then K contains a minimal normal

subgroup of G whose order is a power of p or q. But we have already eliminated this possibility. So $\mathbf{C}_G(N) = N$, and therefore

$$|G/\mathbf{C}_G(N)| = |PQ| = p^a q^b.$$

But since $|N| = r^c$ with $p \neq r \neq q$, this contradicts property a).

b) $\Rightarrow$ c): We must show that every group with property b) belongs to $\mathscr{M}$. Assume for contradiction that G is a counterexample of smallest possible order. We prove:

(1) G has only one minimal normal subgroup:

Let M and N be two distinct minimal normal subgroups of G. Then $M \cap N = 1$. The factor groups G/M and G/N have property b), so they lie in $\mathscr{M}$. Since $G = G/(M \cap N)$ is isomorphic to a subgroup of $G/M \times G/N$, the group G also lies in $\mathscr{M}$. This contradicts the choice of G as a counterexample.

(2) The Fitting subgroup $\mathbf{F}(G)$ of G is a p-group:

This follows immediately from (1), since otherwise the Sylow subgroups of $\mathbf{F}(G)$ would contain distinct minimal normal subgroups of G.

(3) If q and r are two distinct primes, each not equal to p, then every Sylow q-subgroup of G commutes with every Sylow r-subgroup of G:

Let Q be a Sylow q-subgroup of G, and let R be a Sylow r-subgroup of G. We consider the factor group $\overline{G} = G/\Phi(\mathbf{F}(G))$ and its subgroups

$$\overline{F} = \mathbf{F}(G)/\Phi(\mathbf{F}(G)),\ \overline{Q} = Q\Phi(\mathbf{F}(G))/\Phi(\mathbf{F}(G)),\ \overline{R} = R\Phi(\mathbf{F}(G))/\Phi(\mathbf{F}(G)).$$

Since $|\overline{F}| = p^n$ and $q \neq p \neq r$, we have $\overline{Q}\,\overline{R} \cong QR$, and it is enough to show that $[\overline{Q}, \overline{R}] = \overline{1}$.

By Theorem III.4.2 d) (pg. 292), $\overline{F}$ is the Fitting subgroup of $\overline{G}$. Since $(|\overline{F}|, |\overline{Q}\,\overline{R}|) = 1$, $\overline{F}$ is a completely reducible $\overline{Q}\,\overline{R}$-module by Maschke's Theorem I.17.7 (pg. 132). So $\overline{F}$ directly decomposes into irreducible constituents $\overline{F}_i$ as $\overline{F} = \bigtimes_{i=1}^{s} \overline{F}_i$.

If $s > 1$, then each $\overline{F}_i\,\overline{Q}\,\overline{R}$ satisfies condition b), so it satisfies a) as well by our induction hypothesis. This forces $\overline{Q} \leq \mathbf{C}_{\overline{G}}(\overline{F}_i)$ or $\overline{R} \leq \mathbf{C}_{\overline{G}}(\overline{F}_i)$, so $[\overline{Q}, \overline{R}] \leq \mathbf{C}_{\overline{G}}(\overline{F}_i)$ for all i. By Theorem III.4.2 b) (pg. 292), we have $\mathbf{C}_{\overline{G}}(\overline{F}) = \overline{F}$. It follows that

$$[\overline{Q}, \overline{R}] \leq \mathbf{C}_{\overline{G}}(\overline{F}) \cap \overline{Q}\,\overline{R} \leq \overline{F} \cap \overline{Q}\,\overline{R} = \overline{1}.$$

However, if $s = 1$, then $\overline{F}$ is an irreducible $\overline{Q}\,\overline{R}$-module, and $\overline{Q}\,\overline{R}$ is a maximal subgroup of $\overline{F}\,\overline{Q}\,\overline{R}$. Since $\mathbf{C}_{\overline{G}}(\overline{F}) = \overline{F}$, the group $\overline{Q}\,\overline{R}$ contains no nontrivial normal subgroup of $\overline{F}\,\overline{Q}\,\overline{R}$. Therefore, $\overline{F}\,\overline{Q}\,\overline{R}$ has a faithful representation as a primitive permutation group on the cosets of $\overline{Q}\,\overline{R}$. However, since

$$|\overline{F}\,\overline{Q}\,\overline{R}| = p^a q^b r^c$$

is divisible by three distinct primes p, q, r, we obtain a contradiction to Lemma 3.3.

(4) Let Q be a Sylow q-subgroup of G with $q \neq p$. Then the order of the normal closure

$$Q^* = Q^G = \langle Q^g \mid g \in G \rangle$$

of Q in G has the form $p^a q^b$. If q and r are two distinct primes, each not equal to p, then the normal closures Q^* and R^* of the Sylow q-subgroups, resp. Sylow r-subgroups, commute:

The conclusion that Q^* and R^* commute follows immediately from (3).

Assume for contradiction that r is a prime divisor of $|Q^*|$ with $p \neq r \neq q$. Let $\widetilde{R}$ be a Sylow r-subgroup of Q^*. Then we have

$$\widetilde{R} \leq Q^* \cap R^* \leq \mathbf{Z}(Q^*).$$

Since Q^* is a normal subgroup of G, the group G has a nontrivial normal subgroup of r-power order. But this contradicts (2). So in fact $|Q^*| = p^a q^b$.

(5) Let P be a Sylow p-subgroup of G, and let the Q_i^* be the normal closures of the Sylow q_i-subgroups of G with $q_i \neq p$ $(i = 1, \ldots, s)$. Since $Q_i^* \trianglelefteq G$, the subgroup P acts on Q_i^* via automorphisms

$$y_i \mapsto y_i^x \quad (y_i \in Q_i^*,\ x \in P).$$

Let $y_i \mapsto \widetilde{y}_i$ be an isomorphism from Q_i^* to a group $\widetilde{Q}_i$, and let $x \mapsto \widetilde{x}$ be an isomorphism from P to $\widetilde{P}$. We construct a homomorphic image of $\widetilde{P}$ in the automorphism group of $\widetilde{Q}_i$ by defining

$$\widetilde{Q}_i^{\widetilde{x}} = \widetilde{Q_i^x}.$$

With these automorphisms, we construct the semidirect product $\widetilde{G}$ of $\widetilde{Q}_1 \times \cdots \times \widetilde{Q}_s$ with $\widetilde{P}$. Let

$$\widetilde{H}_i = \widetilde{Q}_1 \times \cdots \times \widetilde{Q}_{i-1} \times \widetilde{Q}_{i+1} \times \cdots \times \widetilde{Q}_s.$$

We have $|\widetilde{G}/\widetilde{H}_i| = p^a q_i^{b_i}$. Since $\widetilde{G}$ is isomorphic to a subgroup of the direct product of the groups $\widetilde{G}/\widetilde{H}_i$, the group $\widetilde{G}$ lies in $\mathscr{M}$.

The map μ from $\widetilde{G}$ onto G defined by

$$(\widetilde{y}_1 \cdots \widetilde{y}_s \widetilde{x})^\mu = y_1 \cdots y_s x \quad (\text{for } \widetilde{y}_i \in \widetilde{Q}_i,\ \widetilde{x} \in \widetilde{P})$$

is a homomorphism from $\widetilde{G}$ onto G; namely, for

$$\widetilde{g} = \widetilde{y}_1 \cdots \widetilde{y}_s \widetilde{x} \quad \text{and} \quad \widetilde{g}' = \widetilde{y}_1' \cdots \widetilde{y}_s' \widetilde{x}'$$

we have

$$\widetilde{g}\,\widetilde{g}' = \widetilde{y}_1 (\widetilde{y}_1')^{\widetilde{x}^{-1}} \cdots \widetilde{y}_s (\widetilde{y}_s')^{\widetilde{x}^{-1}} \widetilde{x}\widetilde{x}',$$

and since $[Q_i^*, Q_j^*] = 1$ for $i \neq j$ we also have

$$\begin{aligned}(\widetilde{g}\,\widetilde{g}')^{\mu} &= y_1(y_1')^{x^{-1}} \cdots y_s(y_s')^{x^{-1}} xx' \\ &= (y_1 \cdots y_s x)(y_1' \cdots y_s' x') = \widetilde{g}^{\mu}\widetilde{g}'^{\mu}.\end{aligned}$$

Hence, G lies in $\mathscr{M}$, as it is a homomorphic image of $\widetilde{G}$. This contradicts our assumption.

c) $\Rightarrow$ a): This is trivial since property a) holds for groups whose order is divisible by at most two distinct primes, and this property is inherited by subgroups, factor groups and direct products by Lemma 3.2, so it holds for every group in $\mathscr{M}$.

c) $\Leftrightarrow$ d): We must show that the set $\mathscr{M}$ of groups given in c) contains only the groups described in d).

Let

$$\mathscr{T} = \{G \mid G = G_1 \times \cdots \times G_k,\ k = 1, 2, \ldots,\ |G_i| = p_i^{a_i} q_i^{b_i}\},$$

and let $\mathscr{S}$ be the set of groups that are a homomorphic image of a subgroup of some group in $\mathscr{T}$. We have $\mathscr{S} \subseteq \mathscr{M}$ trivially. We must show that $\mathscr{S} = \mathscr{M}$.

It is clear that $\mathscr{S}$ is closed under homomorphic images. In fact, $\mathscr{S}$ is also closed under taking subgroups and direct products:

Let $S_1 \leq S_2 \in \mathscr{S}$. Then there exists a group $G \in \mathscr{T}$, a subgroup U_2 of G and a homomorphism α such that $S_2 = U_2^{\alpha}$. If U_1 is the preimage of S_1 in U_2 with respect to α, then $S_1 = U_1^{\alpha}$ with $U_1 \leq G$. So S_1 lies in $\mathscr{S}$.

Let S_1 and S_2 be elements of $\mathscr{S}$ with $S_i = U_i^{\alpha_i}$ and $U_i \leq G_i \in \mathscr{T}$. Then

$$U_1 \times U_2 \leq G_1 \times G_2 \in \mathscr{T},$$

and $S_1 \times S_2$ is a homomorphic image of $U_1 \times U_2$. Therefore, $S_1 \times S_2$ lies in $\mathscr{S}$. So the set $\mathscr{S}$ coincides with $\mathscr{M}$. □

Exercises

4) a) Let p be a prime with $p \equiv 1 \bmod 3$, and let $P = \langle x_1, x_2, x_3, x_4 \rangle$ be the group of class 2 and exponent p defined by

$$[x_1, x_2] = [x_3, x_4] = c \neq 1 \quad \text{and} \quad [x_i, x_j] = 1 \text{ otherwise.}$$

(P is the central product of two nonabelian groups of order p^3 and exponent p, where the centers are identified.) Let H_1 be the semidirect product of P with the cyclic group $Q = \langle y_1, y_2 \rangle$ of order 6 constructed with the following automorphisms:

$$x_1^{y_1} = x_1^{-1},\ x_2^{y_1} = x_2^{-1},\ x_3^{y_1} = x_3,\ x_4^{y_1} = x_4,$$
$$x_1^{y_2} = x_1,\ x_2^{y_2} = x_2,\ x_3^{y_2} = x_3^{k},\ x_4^{y_2} = (x_4^{k})^{-1},$$

where $k \not\equiv 1 \equiv k^3 \bmod p$ and $y_1^2 = y_2^3 = 1$. Then H_1 is a group with many Sylow systems, but it is not a subgroup of a direct product of groups whose orders are each divisible by at most two distinct primes.

b) Let $H_2 = \langle x_1, x_2, y\rangle$ be the group of order p_1p_2q with the defining relations

$$x_1^{p_1} = x_2^{p_2} = y^q = [x_1, x_2] = 1, \quad x_i^{y} = x_i^{k_i},$$

where $p_1 \neq p_2$ and $k_i \not\equiv 1 \equiv k_i^q \bmod p_i$. Then H_2 is a group with many Sylow systems, but it is not a homomorphic image of a direct product of groups whose orders are each divisible by at most two distinct primes.

c) $H_1 \times H_2$ is a group with many Sylow systems, but it is neither a subgroup nor a homomorphic image of a direct product of groups whose orders are each divisible by at most two distinct primes.

5) (Huppert [263]) Let P be a p-group, and let α be a nonidentity automorphism of P with order coprime to p that maps every subgroup of P onto itself. Then P is abelian, and there exists a natural number k such that $x^\alpha = x^k$ for all $x \in P$.

6) (Huppert [263]) Let $\{P_1, \ldots, P_r\}$ be a Sylow system of the solvable group G. Show that the following two statements are equivalent:
   a) For $i \neq j$, every subgroup of P_i permutes with every subgroup of P_j.
   b) If $\mathbf{K}_\infty(G)$ is the smallest normal subgroup of G with nilpotent factor group, then:
      (1) $\mathbf{K}_\infty(G)$ is an abelian Hall subgroup of G.
      (2) For every $g \in G$, there exists a natural number $n(g)$ such that for all $x \in \mathbf{K}_\infty(G)$, $x^g = x^{n(g)}$.

   (Hint: To prove that b) follows from a), show the following:
   (i) If p is the largest prime divisor of $|G|$, then G has a normal Sylow p-subgroup P.
   (ii) Let Q be a complement of P. In the critical case that Q does not centralize P, consider the automorphisms induced by Q on P and $P/\Phi(P)$, and use Exercise 5 and Theorem III.3.18, pg. 290.)

4 Products of nilpotent groups

Definition 4.1. If the G_i $(i = 1, \ldots, n)$ are subgroups of G such that $G_iG_j = G_jG_i$ for all i, j and

$$G = G_1 \cdots G_n = \{g_1g_2 \cdots g_n \mid g_i \in G_i\},$$

then we say that G is the product of pairwise permutable subgroups G_i.

The question of how one can construct from given groups G_i $(i = 1, 2)$ a group G that is the product H_1H_2 of two subgroups H_i with $G_i \cong H_i$ has been thoroughly studied, mostly under the additional condition that $H_1 \cap H_2 = 1$. One often refers to this as the Zappa–Szép product of H_1 and H_2 (see Szép [420] and Zappa [491]). This question leads to complicated functional equations whose solutions are not fully understood even when H_1 and H_2 are cyclic groups. More tractable are questions of the following type: What can we say about the structure of G if we have a decomposition $G = G_1 \cdots G_n$ of G into pairwise permutable subgroups G_i? In particular, we are interested to find assumptions regarding the G_i which ensure that G is solvable. It is certainly not enough to simply assume that the G_i are solvable to force G to be solvable, as the following example shows:

Example 4.2. Let G be the symmetric group S_5, let $Z = \langle (1,2)(3,4,5) \rangle$, and let L be the normalizer of a Sylow 5-subgroup of G. With a suitable labeling of the points, $L = \langle x, y \rangle$ with $x = (1,2,3,4,5)$ and $y = (1,2,4,3)$. Since $Z \cap L = 1$, we have $|ZL| = |Z||L| = |G|$. So $G = ZL$ with Z and L solvable, but G is not solvable. (The groups Z and L are in fact supersolvable in the sense of §9.)

However, we do have the following beautiful theorem.

Main Theorem 4.3 (Kegel, Wielandt). *Every product of pairwise permutable nilpotent groups is solvable.*

It is clear that Theorem 4.3 is a vast generalization of Theorem 2.3. In this section, we will prove Theorem 4.3 in a long chain of lemmas. First, we address a special case that is remarkably easy to handle:

Theorem 4.4 (N. Itô [275]). *If $G = AB$ for some abelian subgroups A and B of G, then $G'' = 1$. (Here, G need not be finite.)*

Proof. a) We have

$$G' = [A, B] = \langle [a, b] \mid a \in A, b \in B \rangle :$$

By Lemma III.1.6 b) (pg. 268), we have $[A, B] \trianglelefteq \langle A, B \rangle = G$. The quotient $G/[A, B]$ is abelian as it is generated by pairwise commuting elements $a[A, B]$ and $b[A, B]$ with $a \in A$ and $b \in B$. So $G' \leq [A, B]$. Of course, the reverse inclusion $[A, B] \leq G'$ holds, so $G' = [A, B]$.

b) Now we show that every two commutators $[a, b]$ and $[a', b']$ with $a, a' \in A$ and $b, b' \in B$ commute:

Let $b^{a'} = a''b^*$ with $a'' \in A$ and $b^* \in B$, and let $a^{b'} = b''a^*$ with $a^* \in A$ and $b'' \in B$. Then

$$\begin{aligned}[a, b]^{a'b'} &= [a^{a'}, b^{a'}]^{b'} = [a, a''b^*]^{b'} = [a, b^*]^{b'} \\ &= [a^{b'}, b^{*b'}] = [b''a^*, b^*] = [a^*, b^*].\end{aligned}$$

In the same way, we obtain

$$[a,b]^{b'a'} = [b''a^*, b]^{a'} = [a^*, b]^{a'} = [a^*, a''b^*] = [a^*, b^*].$$

Altogether, this gives

$$[a,b]^{a'b'a'^{-1}b'^{-1}} = [a,b].$$

So $[a,b]$ and $[a'^{-1}, b'^{-1}]$ commute. □

It is not known whether the method of proving Theorem 4.4 can be extended to a product of nilpotent groups A and B or to a product of more than two abelian groups. Such a generalization would be interesting for multiple reasons. For example, it would allow one to remove the assumption that G is finite in Theorem 4.3. Furthermore, it might yield a proof of the following conjecture, which is stronger than Theorem 4.3: Suppose $G = G_1 \cdots G_n$ for pairwise permutable nilpotent subgroups G_i of G. Then the derived length $k(G)$ of the solvable group G is bounded above by $k(G) \leq \sum_{i=1}^{n} c(G_i)$, where $c(G_i)$ is the nilpotency class of the nilpotent group G_i.[2] (See also Theorem 14.16, pg. 800.)

Lemma 4.5. *Suppose $G = G_1G_2$. For any $g, g' \in G$, we always have $G = G_1^g G_2^{g'}$, and there exists some $h \in G$ such that $G_1^g = G_1^h$ and $G_2^{g'} = G_2^h$.*

Proof. Let $gg'^{-1} = g_1g_2$ with $g_i \in G_i$, and let $h = g_1^{-1}g$. Then $g = g_1h$ and $g' = g_2^{-1}h$. It follows that

$$G_1^g = G_1^h \quad \text{and} \quad G_2^{g'} = G_2^h.$$

Therefore,

$$G_1^g G_2^{g'} = G_1^h G_2^h = G^h = G.$$ □

If G has a decomposition $G = G_1G_2$, then for any homomorphism μ, the image G^μ inherits a decomposition $G^\mu = G_1^\mu G_2^\mu$. However, if $G_1 \cap G_2 = 1$, it does not always follow that $G_1^\mu \cap G_2^\mu = 1$. In order to study groups of the form $G = G_1G_2$, it is important to understand decompositions of certain subgroups of G in addition to the decomposition of homomorphic images. In particular, we are interested in subgroups H of the form $H = H_1H_2$ with $H_i \leq G_i$ $(i = 1, 2)$. To this end, the following theorem is helpful in many situations.

[2] Pennington proved that if a finite group G has the form $G = G_1G_2$, where G_1 and G_2 are nilpotent of classes c_1 and c_2, then the $(c_1 + c_2)$-th term $G^{(c_1+c_2)}$ of the derived series is nilpotent; see E. A. Pennington, On products of finite nilpotent groups, Math. Z. **134** (1973), 81–83. However, counterexamples were constructed showing that $G^{(c_1+c_2)}$ need not be 1, as conjectured in the text; see J. Cossey and S. E. Stonehewer, On the derived length of finite dinilpotent groups, Bull. London Math. Soc. **30** (1998), no. 3, 247–250. It is still unknown whether the derived length of G is bounded in terms of c_1 and c_2; see Problem 14.43 in the Kourovka Notebook. *trans.*

Theorem 4.6. *Suppose $G = G_1G_2$, and assume that the π-Sylow theorem holds in G, G_1 and G_2 in the sense of Definition III.5.5 (pg. 299). Then there exist Hall π-subgroups H, H_1 and H_2 of G, G_1 and G_2 such that $H = H_1H_2$.*

Proof. Let H^*, H_1^*, H_2^* be Hall π-subgroups of G, G_1, G_2. Since the π-Sylow theorem holds in G, there exist $g_i \in G$ such that $H_i^{*g_i} \leq H^*$ $(i = 1, 2)$. Now we choose some $g \in G$ such that

$$G_i^{g_i g} = G_i,$$

where such a $g \in G$ exists by Lemma 4.5. We let $H_i^{*g_i g} = H_i$ and $H^{*g} = H$. Then H, H_1, H_2 are again Hall π-subgroups of G, G_1, G_2. We have

$$H_i = H_i^{*g_i g} \leq H^{*g} = H.$$

So H_1H_2 is certainly a subset of H.

Let $|G_i| = h_ia_i$ with $|H_i| = h_i$, let $|G| = ha$ with $|H| = h$, and finally let $|G_1 \cap G_2| = h'a'$ with $h' \in \pi$ and $a' \in \pi'$. Then we have

$$ha = |G| = |G_1G_2| = \frac{|G_1||G_2|}{|G_1 \cap G_2|} = \frac{h_1h_2}{h'}\frac{a_1a_2}{a'},$$

so $h = \frac{h_1h_2}{h'}$. Therefore, it follows from $H_1 \cap H_2 \leq G_1 \cap G_2$ that $|H_1 \cap H_2| \leq h'$ and then

$$|H_1H_2| = \frac{|H_1||H_2|}{|H_1 \cap H_2|} \geq \frac{h_1h_2}{h'} = h = |H|.$$

So $H = H_1H_2$. □

The following theorems are important special cases.

Theorem 4.7. *Suppose $G = G_1G_2$. Then for every prime p, there exist Sylow p-subgroups P, P_1, P_2 of G, G_1, G_2 with $P = P_1P_2$.*

Proof. This follows immediately from Theorem 4.6 since the p-Sylow theorems hold in every finite group. □

Theorem 4.8. *Suppose $G = G_1G_2$ with G_1 and G_2 nilpotent.*

a) *If P_1 and P_2 are the Sylow p-subgroups of G_1, resp. G_2, then $P = P_1P_2$ is a Sylow p-subgroup of G.*

b) *If G is solvable, H_1 is the Hall π-subgroup of G_1 and H_2 is the Hall π-subgroup of G_2, then $H = H_1H_2$ is a Hall π-subgroup of G.*

(*By Theorem 4.3, the solvability of G follows from the hypothesis of Theorem 4.8, but we will only prove Theorem 4.3 later in this section.*)

Proof. a) The claim follows immediately from Theorem 4.7 since the nilpotent groups G_i $(i = 1, 2)$ each have only one Sylow p-subgroup.

b) This follows from Theorem 4.6 since the π-Sylow theorem holds in the solvable group G by Theorem 1.8. □

Lemma 4.9 (Wielandt [464]). *Suppose $G = G_1 \cdots G_n$ with nilpotent, pairwise permutable subgroups G_i. If every product $G_i G_j$ is solvable, then so is G itself.*

Proof. For p a prime, let K_i be the unique p-complement of the nilpotent group G_i. Since $G_i G_j$ is solvable by our hypothesis, $K_i K_j$ is a p-complement of $G_i G_j$ by Theorem 4.8 b). So the K_i are pairwise permutable. Therefore, $K = K_1 \cdots K_n$ is a subgroup of G. Calculating the order of K using the product formula from Lemma I.2.12 a) (pg. 7) shows that K is a p-complement of G. So G has a p-complement for every prime p, and hence G is solvable by Theorem 1.10. □

It is unknown if the conclusion of Lemma 4.9 holds without assuming the G_i are nilpotent.

Lemma 4.9 reduces the proof of Theorem 4.3 to the case of two factors. In preparation for a second reduction, we prove the following lemma:

Lemma 4.10 (Kegel [301]). *Let A and B be subgroups of G with $G \neq AB$ and $AB^g = B^g A$ for all $g \in G$. Then either A or B is contained in a proper normal subgroup of G.*

Proof. Let A_1 be a subgroup of G that is maximal with respect to the following properties:
(1) $A_1 B^g = B^g A_1$ for all $g \in G$.
(2) $A \leq A_1$.
(3) $G \neq A_1 B$.

If $A_1 \lhd G$, then we are done. So assume that $\mathbf{N}_G(A_1) < G$. If $h \notin \mathbf{N}_G(A_1)$, then

$$A_1^h B^{gh} = B^{gh} A_1^h.$$

So A_1^h permutes with all conjugates of B. Then we also have

$$\langle A_1, A_1^h \rangle B^g = B^g \langle A, A_1^h \rangle$$

for all $g \in G$. Since $h \notin \mathbf{N}_G(A_1)$, we have $A_1 < \langle A_1, A_1^h \rangle$. The maximality of A_1 therefore yields[3]

$$G = \langle A_1, A_1^h \rangle B.$$

We have $B^g \leq \mathbf{N}_G(A_1)$ for all $g \in G$:

Indeed, if we had $x \in B^g$ and $x \notin \mathbf{N}_G(A_1)$, then we would obtain a contradiction from

$$G = \langle A_1, A_1^x \rangle B^g = A_1 B^g < G.$$

This now yields $B^G = \langle B^g \mid g \in G \rangle \trianglelefteq G$ and $B^G \leq \mathbf{N}_G(A_1) < G$. So B^G is a proper normal subgroup of G containing B. □

[3] Correction 33 from errata.

Theorem 4.11 (Kegel [301]). *If every product of two nilpotent groups with coprime orders is solvable, then every product of two nilpotent groups is solvable.*

Proof. Let $G = AB$ with A and B nilpotent groups. Assume for contradiction that G is not solvable. We choose G to be a counterexample of minimum order. By our hypothesis, the orders $|A|$ and $|B|$ are not coprime. Let p be a common prime divisor of $|A|$ and $|B|$. By Theorem 4.8 a), the Sylow p-subgroups A_p and B_p of A and B are permutable. Furthermore, $G = AB^g$ for all $g \in G$ by Lemma 4.5. So we also have $A_p B_p^g = B_p^g A_p$. Since the nonsolvable group G is not a p-group, $G \neq A_p B_p$. By Lemma 4.10, G is not simple. Let K be a minimal normal subgroup of G. The factor group

$$G/K = (AK/K)(BK/K)$$

is solvable by the minimal choice of G. So K is not solvable.

If $AK < G$, then we have by the Dedekind identity that

$$AK = A(AK \cap B),$$

and so AK is solvable, as it is the product of nilpotent groups A and $AK \cap B$. But then K is also solvable, contradicting the observation of the last paragraph. Hence, $G = AK$; in the same way, we conclude that $G = BK$. Now,

$$G/K \cong A/(A \cap K)$$

is nilpotent. The preimage $KA_p = KB_p$ of the Sylow p-subgroup of G/K is therefore a normal subgroup of G.

The assumptions of Lemma 4.10 are again satisfied for the subgroups A_p and B_p of $KA_p = KB_p$, for certainly $KA_p \neq A_p B_p$ since K is not solvable. By Lemma 4.10, there then exists a proper normal subgroup L of $KA_p = KB_p$ with $A_p \leq L$ or $B_p \leq L$. Assume without loss of generality that $A_p \leq L$. Then we may choose for L the normal closure

$$A_p^{A_p K} = A_p^K$$

of A_p in KA_p. We have

$$L = DA_p$$

with $D = K \cap L$. Since $A_p \leq L < A_p K$, we have $D = K \cap L < K$. It now follows that

$$L^A = (A_p^K)^A = (A_p^A)^K = A_p^K = L$$

and $L^K = L$, so $L \lhd KA = G$. Therefore, $K \cap L = D \lhd G$ as well. Since K is a minimal normal subgroup of G and $D < K$, we must have $D = 1$. Hence, $L = A_p \lhd G$. Then G/A_p is again a product of two nilpotent groups, so it is solvable since $A_p \neq 1$. Since A_p is solvable, G is also solvable, a contradiction. □

By Lemma 4.9 and Theorem 4.11, it suffices to prove the following claim in order to prove Theorem 4.3: If $G = AB$ with A and B nilpotent groups of coprime order, then G is solvable. This step of the proof of Theorem 4.3 is the most difficult, and we lay the groundwork by proving additional lemmas.

The next lemma identifies subgroups that are again products of two nilpotent groups that will be crucial in the proof.

Lemma 4.12 (Wielandt). *Suppose $G = AB$. We say a subgroup S of G is factorized if $S = (S \cap A)(S \cap B)$.*

a) *Let $H \leq G$. Suppose that whenever $H^a = H^b$ for some $a \in A$ and $b \in B$, then it always follows that $H^a = H^b = H$. Then $\mathbf{N}_G(H)$ is factorized.*

b) *A relation $\mathscr{R}$ on pairs of subgroups of G is said to be invariant if whenever $C\mathscr{R}D$, we always have $C^g\mathscr{R}D^g$ for all $g \in G$. (Some examples of invariant relations are $C \leq D$, $C \trianglelefteq D$, $C \leq \mathbf{N}_G(D)$ and $CD = DC$.)*

Let $C_1, \ldots, C_k$ be subgroups of G. Suppose that for every i with $1 \leq i \leq k$, at least one of the inclusions $A \leq \mathbf{N}_G(C_i)$ or $B \leq \mathbf{N}_G(C_i)$ holds. Let $\mathscr{R}_1, \cdots, \mathscr{R}_k$ be invariant relations on the subgroups of G. Finally, suppose that H is a subgroup of G such that $H\mathscr{R}_iC_i$ for all $i = 1, \ldots, k$ and no distinct conjugate H^g of H satisfies these relations. Then $\mathbf{N}_G(H)$ is factorized.

c) *Let $A^* \trianglelefteq A$ and $B^* \trianglelefteq B$. Then $\mathbf{N}_G(\langle A^*, B^*\rangle)$ is factorized.*

Proof. a) Let $x = ab^{-1} \in \mathbf{N}_G(H)$ with $a \in A$ and $b \in B$. Then $H^a = H^b$, so $H^a = H = H^b$ by our hypothesis. Hence, $a \in \mathbf{N}_G(H)$ and $b \in \mathbf{N}_G(H)$, and therefore

$$\mathbf{N}_G(H) \leq (\mathbf{N}_G(H) \cap A)(\mathbf{N}_G(H) \cap B).$$

The reverse inclusion is trivial.

b) Let $a \in A$ and $b \in B$. If $A \leq \mathbf{N}_G(C_i)$, then since $H\mathscr{R}_iC_i$, we also have $H^a\mathscr{R}_iC_i$. If $B \leq \mathbf{N}_G(C_i)$, then we obtain in the same way that $H^b\mathscr{R}_iC_i$. If $H^a = H^b$, then this group satisfies all relations $H^a\mathscr{R}_iC_i$. By our assumption, we then have $H^a = H = H^b$, so $\mathbf{N}_G(H)$ is factorized by a).

c) We set $C_1 = A^*$ and $C_2 = B^*$ in b). We choose both invariant relations $\mathscr{R}_i$ to be the inclusion $\geq$. The group $H = \langle A^*, B^*\rangle$ then satisfies both relations $H \geq A^*$ and $H \geq B^*$, and no distinct conjugate of H satisfies both relations. Thus, $\mathbf{N}_G(H)$ is factorized by b). □

Lemma 4.13. *Suppose $G = AB$, and let $A^* \trianglelefteq A$ and $B^* \trianglelefteq B$. Then the relation*

$$A^{*g}B^{*g'} = B^{*g'}A^{*g}$$

holds either for every pair or for no pair g, g' of elements of G. If G is solvable, then this relation always holds whenever A^* and B^* are Hall subgroups of G.

Proof. Let $g, g' \in G$, and let $g'g^{-1} = ba^{-1}$ for some $a \in A$ and $b \in B$. Then each of the following relations are equivalent:

$$\begin{aligned} A^{*g}B^{*g'} &= B^{*g'}A^{*g}, \\ A^*B^{*g'g^{-1}} &= B^{*g'g^{-1}}A^*, \\ A^*B^{*ba^{-1}} &= B^{*ba^{-1}}A^*, \\ A^*B^* &= B^*A^*. \end{aligned}$$

If G is solvable, and if A^* and B^* are Hall subgroups of G, then the relation $A^{*g}B^{*g'} = B^{*g'}A^{*g}$ holds for at least one pair $g, g' \in G$ by Theorem 2.3; namely, it holds when A^{*g} and $B^{*g'}$ lie in a Hall system of G. So A^{*g} and $B^{*g'}$ are permutable for all g, g'. □

Now we turn to the remaining and most difficult part of the proof of Theorem 4.3:

Theorem 4.14 (Wielandt [471]). *Suppose $G = AB$, where A and B are nilpotent groups of coprime order. Then G is solvable, and every Sylow subgroup of A permutes with every Sylow subgroup of B.*

Proof. Let G be a counterexample of minimum order. Let p_i $(i = 1, \ldots, s)$ be the prime divisors of $|A|$, and let q_j $(j = 1, \ldots, t)$ be the prime divisors of $|B|$. Furthermore, let P_i be the Sylow p_i-subgroup of A, and let Q_j be the Sylow q_j-subgroup of B. If G is solvable, then it follows immediately from Theorem 4.8 b) that P_iQ_j is a Hall $\{p_i, q_j\}$-subgroup of G. So G is not solvable. We prove:

(1) G is simple:

Let $1 < N \trianglelefteq G$. The factor group

$$G/N = (NA/N)(NB/N)$$

satisfies our assumptions, so it is solvable. If $NA < G$, then $NA = (NA \cap B)A$. Then NA again satisfies the assumptions, so it is likewise solvable. In particular, N is then solvable. Together with the solvability of G/N, this forces G to be solvable, contradicting the observation above. So we have $G = NA$; in the same way, it follows that $G = NB$ as well. This means $|G/N|$ divides $|A|$ and $|G/N|$ divides $|B|$. Since $|A|$ and $|B|$ are coprime, we conclude that $|G/N| = 1$. So $G = N$ and G is simple.

(2) We have $s \geq 2$ and $t \geq 2$:

Assume without loss of generality that $s = 1$ and $1 \neq b \in \mathbf{Z}(B)$. Then $\mathbf{C}_G(b) \geq B$, so $[G : \mathbf{C}_G(b)] = p_1^c$ for some integer c. By Burnside's Theorem V.7.2 (pg. 521), G is not simple, contradicting (1).

(3) Let $A^* \trianglelefteq P_i$ and $B^* \trianglelefteq Q_j$ with $1 < \langle A^*, B^* \rangle = H < G$. Then there exists a normal subgroup $N \neq 1$ of H such that $N \leq P_i$ or $N \leq Q_j$:

Since A is nilpotent, it follows from $A^* \trianglelefteq P_i$ that $A^* \lhd A$ as well. Likewise, we have $B^* \lhd B$. We now define $\widetilde{G} = \mathbf{N}_G(H)$. By Lemma 4.12 c), $\widetilde{G}$ is factorized. As H is a nontrivial, proper normal subgroup of $\widetilde{G}$, we must have $\widetilde{G} < G$ since G is simple. So $\widetilde{G}$ is solvable. It follows from the decomposition

$$\widetilde{G} = (\widetilde{G} \cap A)(\widetilde{G} \cap B)$$

that $\widetilde{P}_i = P_i \cap \widetilde{G}$ and $\widetilde{Q}_j = Q_j \cap \widetilde{G}$ are Sylow subgroups of $\widetilde{G}$. By Lemma 4.13, they are permutable. Therefore, we have $H \leq \widetilde{P}_i \widetilde{Q}_j$. Since $\widetilde{G}$ is solvable, H is also solvable. Let N be a nontrivial normal subgroup of H of prime power order. Then clearly $N \leq \widetilde{P}_i \leq P_i$ or $N \leq \widetilde{Q}_j \leq Q_j$.

(4) For all i, j, we have $G = \langle P_i, Q_j \rangle$:

Assume for contradiction that $\langle P_i, Q_j \rangle = H < G$. By (3), there then exists an N such that $1 < N \trianglelefteq \langle P_i, Q_j \rangle = H$ with $N \leq P_i$ or $N \leq Q_j$. Assume without loss of generality that $N \leq P_i$. Then $N \lhd A$, and we see by (1) that

$$G = Q_j^G = Q_j^{BA} = Q_j^A \leq \langle A, Q_j \rangle \leq \mathbf{N}_G(N).$$

So $N \lhd G$, contradicting the simplicity of G established in (1).

(5) If $1 \neq a \in \mathbf{Z}(A)$, then $\langle a, Q_j \rangle = G$ for all j:

We may assume that a has prime order, say $a \in P_i$. Assume for contradiction that $\langle a, Q_j \rangle = H < G$. Letting $A^* = \langle a \rangle$ and $B^* = Q_j$ in (3), there then exists some subgroup N with $1 < N \trianglelefteq H$ such that $N \leq P_i$ or $N \leq Q_j$. If $N \leq P_i$, then we always have $P_k \leq \mathbf{N}_G(N)$ for $k \neq i$; such a k certainly exists since $s \geq 2$ by (2). It then follows by (4) that

$$\mathbf{N}_G(N) \geq \langle P_k, Q_j \rangle = G.$$

But since $1 < N \leq H < G$, this contradicts (1).

On the other hand, if $N \leq Q_j$ then $N \lhd B$, and we obtain

$$G = \langle a \rangle^G = \langle a \rangle^{AB} = \langle a \rangle^B \leq \langle a, B \rangle \leq \mathbf{N}_G(N).$$

So again $1 < N \lhd G$, contradicting (1).

(6) If $1 \neq a \in \mathbf{Z}(A)$ and $1 \neq b \in \mathbf{Z}(B)$, then $G = \langle a, b \rangle$:

We may assume that a and b have prime order, and we assume that, say, $a \in P_i$ and $b \in Q_j$. Assume for contradiction that $\langle a, b \rangle < G$. Letting $A^* = \langle a \rangle$ and $B^* = \langle b \rangle$ in (3), there exists a subgroup N with $1 < N \trianglelefteq \langle a, b \rangle$ and $N \leq P_i$ or $N \leq Q_j$. Assume without loss of generality that $N \leq Q_j$. Now we choose a Q_k with $k \neq j$; this is possible since $t \geq 2$ by (2). Then we have $Q_k \leq \mathbf{N}_G(N)$, and so by (5)

$$\mathbf{N}_G(N) \geq \langle a, Q_k \rangle = G.$$

But this contradicts (1).

(7) For $1 \neq a \in \mathbf{Z}(A)$, we have $\mathbf{C}_G(a) = A$:

Assume for contradiction that $A < \mathbf{C}_G(a)$. Then $\mathbf{C}_G(a) = AB^*$ with $1 < B^* < B$. We now choose a b with $1 \neq b \in B^* \cap Q_j$ (for an appropriate j) and then choose a Q_k with $k \neq j$. It now follows from (5) that

$$\mathbf{C}_G(b) \geq \langle a, Q_k \rangle = G,$$

contradicting (1).

(8) If $g \in G$ and $P_i \leq A^g$ for some i, then $A^g = A$:

Since $P_i \leq A^g$, we have $P_i = P_i^g$. By (7), $A = \mathbf{C}_G(\mathbf{Z}(P_i))$. It follows that

$$A^g = \mathbf{C}_G(\mathbf{Z}(P_i^g)) = \mathbf{C}_G(\mathbf{Z}(P_i)) = A.$$

(9) If $g \in G$ and $A \cap \mathbf{Z}(A^g) > 1$, then $A = A^g$:

Let $a \in A \cap \mathbf{Z}(A^g)$ be an element of prime order, say $a \in P_i$. It follows by (7) that $\mathbf{C}_G(a) = A^g$. Let $k \neq i$. Then $P_k \leq \mathbf{C}_G(a) = A^g$, and the claim $A = A^g$ follows by (8).

(10) If $\mathbf{Z}(A) \leq \widetilde{G} \leq G$, then $\widetilde{P}_i = P_i \cap \widetilde{G}$ is a Sylow p_i-subgroup of $\widetilde{G}$:

Let $\widetilde{P}$ be a Sylow p_i-subgroup of $\widetilde{G}$ with $\widetilde{P}_i = P_i \cap \widetilde{G} \leq \widetilde{P}$. For an appropriate $g \in G$, we then have

$$1 < \mathbf{Z}(P_i) \leq P_i \cap \mathbf{Z}(A) \leq P_i \cap \widetilde{G} = \widetilde{P}_i \leq \widetilde{P} \leq P_i^g \leq A^g.$$

So we have $A^g \cap \mathbf{Z}(A) \neq 1$, and therefore $A^g = A$ by (9). Then $P_i^g = P_i$. It now follows that

$$\widetilde{P}_i \leq \widetilde{P} \leq P_i \cap \widetilde{G} = \widetilde{P}_i,$$

as claimed.

(11) Let $\mathbf{Z}(A) \leq \widetilde{G} \leq G$. We again let $P_i \cap \widetilde{G} = \widetilde{P}_i$ and $A \cap \widetilde{G} = \widetilde{A}$. If $\widetilde{G} \cap \mathbf{N}_G(\widetilde{P}_i) = \widetilde{A}$ for all i, then we have $\widetilde{G} \leq A$:

We have $\widetilde{A} = \widetilde{P}_1 \times \cdots \times \widetilde{P}_s$. Since $\mathbf{Z}(A) \leq \widetilde{G}$, it follows by (10) that $\widetilde{A}$ is a nilpotent Hall subgroup of $\widetilde{G}$. Since $\widetilde{G} \cap \mathbf{N}_G(\widetilde{P}_i) = \widetilde{A}$ and $1 < \mathbf{Z}(P_i) < \mathbf{Z}(A) \leq \widetilde{A}$ for all i, the assumptions of Theorem IV.7.3 (pg. 472) are satisfied in $\widetilde{G}$. Therefore, there exists a normal subgroup N of $\widetilde{G}$ such that $\widetilde{G} = N\widetilde{A}$ and $N \cap \widetilde{A} = 1$. Here, $|N|$ is a divisor of $|B|$. By Theorem III.5.8 (pg. 300), there exists a $g \in G$ such that $N^g \leq B$. Let $g = ab$ for some $a \in A$ and $b \in B$. Then we have $N^{ab} \leq B$, so $N^a \leq B$, as well. This yields $\mathbf{Z}(B) \leq \mathbf{N}_G(N^a)$. Since $\mathbf{Z}(A) \leq \widetilde{G} \leq \mathbf{N}_G(N)$, we also have

$$\mathbf{Z}(A) = \mathbf{Z}(A)^a \leq \mathbf{N}_G(N^a).$$

Together with (6), it now follows that

$$\mathbf{N}_G(N^a) \geq \langle \mathbf{Z}(A), \mathbf{Z}(B) \rangle = G.$$

Since $|N|$ divides $|B|$, N is a proper subgroup of G, and so $N = 1$ by (1). Therefore, $\widetilde{G} = \widetilde{A} \leq A$, as claimed.

(12) We define $A^* = A \cap \mathbf{N}_G(B)$ and $B^* = B \cap \mathbf{N}_G(A)$, and we fix this notation from now on. Then A^* contains at most one subgroup of order p for every prime p:

If $a \in A^*$, then $\mathbf{Z}(B)^a = \mathbf{Z}(B)$. If $b^a = b$ with $1 \neq b \in \mathbf{Z}(B)$, then it follows by (7) that $a \in \mathbf{C}_G(b) = B$, so $a = 1$. Hence, every $1 \neq a \in A^*$ induces a fixed point free automorphism on $\mathbf{Z}(B)$. The claim now follows from Theorem V.8.15 (pg. 536).

(13) If $A^* \cap A^g > 1$, then $A^g = A$:

Let $1 \neq a \in A^* \cap A^g$. We may clearly assume that a has prime order, say $a \in P_i$. We define invariant relations $\mathscr{R}_j$ $(j = 1, 2)$ by

$$H\mathscr{R}_1 Y \iff H \leq Y$$
$$H\mathscr{R}_2 Y \iff H \leq \mathbf{N}_G(Y),$$

and we set $C_1 = A$ and $C_2 = B$. Then for $i = 1, 2$, it holds trivially that $A \leq \mathbf{N}_G(C_i)$ or $B \leq \mathbf{N}_G(C_i)$. Let $H = \langle a \rangle$. Since

$$H \leq A^* = A \cap \mathbf{N}_G(B),$$

the relations $H\mathscr{R}_i C_i$ $(i = 1, 2)$ hold. No distinct conjugate H^g of H satisfies both relations because H is the unique subgroup of $A^* = A \cap \mathbf{N}_G(B)$ of order p_i. By Lemma 4.12 b), $\widetilde{G} = \mathbf{N}_G(H)$ is factorized, say

$$\widetilde{G} = (\widetilde{P}_1 \times \cdots \times \widetilde{P}_s)(\widetilde{Q}_1 \times \cdots \times \widetilde{Q}_t).$$

Since $s \geq 2$, there is some k with $P_k \neq P_i$. As $a \in P_i$, we have $P_k \leq \mathbf{N}_G(H)$, so $P_k = \widetilde{P}_k$. We have $\widetilde{G} < G$ since G is simple. So $\widetilde{G}$ is solvable, and P_k permutes with all $\widetilde{Q}_j$ $(j = 1, \ldots, t)$. Let N be a nontrivial normal subgroup of $P_k\widetilde{Q}_j$ of prime power order. Then $N \leq P_k$ or $N \leq \widetilde{Q}_j$. If $N \leq \widetilde{Q}_j$, then we choose some $Q_m \neq Q_j$, which is possible by (2). But then we obtain a contradiction to (1) since by (4),

$$\mathbf{N}_G(N) \geq \langle P_k, Q_m \rangle = G.$$

So $N \leq P_k$. Since $N \trianglelefteq P_k$, there exists an $n \in N$ with $1 \neq n \in N \cap \mathbf{Z}(P_k)$. It now follows that

$$\langle n \rangle^{\widetilde{Q}_j} \leq N \leq P_k \leq A.$$

So for every $y \in \widetilde{Q}_j$ we have

$$\langle n \rangle^y \leq A \cap \mathbf{Z}(P_k)^y \leq A \cap \mathbf{Z}(A)^y.$$

It therefore follows from (9) that $y \in \mathbf{N}_G(A)$, so $\widetilde{Q}_j \leq \mathbf{N}_G(A)$. As this holds for all j, we have $\widetilde{G} \cap B \leq \mathbf{N}_G(A)$. Then

$$\widetilde{G} = (\widetilde{G} \cap A)(\widetilde{G} \cap B) \leq \mathbf{N}_G(A).$$

Since $a \in A^* \cap A^g$, it furthermore follows that

$$\mathbf{Z}(A)^g \leq \mathbf{N}_G(\langle a \rangle) = \widetilde{G} \leq \mathbf{N}_G(A).$$

Finally, since $|A|$ and $|\mathbf{N}_G(A)/A|$ are coprime, this forces $\mathbf{Z}(A)^g \leq A$, therefore $A = A^g$ by (9).

(14) If $a \in A$ and $B^* \cap \mathbf{C}_G(a) > 1$, then $a \in A^*$:

We have

$$1 < B^* \cap \mathbf{C}_G(a) = (B^* \cap \mathbf{C}_G(a))^a \leq B^* \cap (B^*)^a \leq B^* \cap B^a.$$

By (13) (with B in place of A), it follows that $B^a = B$, so $a \in A \cap \mathbf{N}_G(B) = A^*$.

(15) If $\mathbf{Z}(A) \le \widetilde{G} \not\le A$, then there exists some $a \in A$ such that $B^* \cap \widetilde{G}^a > 1$:

By (10), $A \cap \widetilde{G} = \widetilde{A}$ is a nilpotent Hall subgroup of $\widetilde{G}$ with Sylow subgroups $\widetilde{P}_i = P_i \cap \widetilde{G}$. Since $\widetilde{G} \not\le A$, there exists by (11) at least one i such that $\widetilde{A} < \widetilde{G} \cap \mathbf{N}_G(\widetilde{P}_i)$. As $\widetilde{A}$ is a Hall subgroup of $\widetilde{G}$ for the prime divisors of $|A|$, there exists a prime divisor q of $|B|$ such that q divides $|\widetilde{G} \cap \mathbf{N}_G(\widetilde{P}_i)|$.

Let $x \in \widetilde{G} \cap \mathbf{N}_G(\widetilde{P}_i)$ with $|x| = q$. Since $\mathbf{Z}(A) \le \widetilde{G}$, we now have

$$\mathbf{Z}(P_i) \le \widetilde{P}_i = \widetilde{P}_i^x \le A^x.$$

So $\mathbf{Z}(A) \cap A^x > 1$, and it follows by (9) that $A = A^x$. By Theorem III.5.8 (pg. 300), there exists some $g \in G$ such that $x^g \in B$. Since $G = AB$, the element g can be chosen to lie in A. So there exists an $a \in A$ such that $x^a \in B$. This yields

$$x^a \in B \cap \mathbf{N}_G(A) \cap \widetilde{G}^a = B^* \cap \widetilde{G}^a.$$

So $B^* \cap \widetilde{G}^a > 1$.

(16) If $A^g \ne A$, then $A^g \cap A = 1$:

Let $\widetilde{G} = \langle \mathbf{Z}(A), \mathbf{Z}(A)^g \rangle$. Then $\widetilde{G} \le \mathbf{C}_G(A \cap A^g)$. By (9), we have $\mathbf{Z}(A)^g \not\le A$, so $\widetilde{G} \not\le A$ as well. By (15), there then exists an $a \in A$ such that $B^* \cap \widetilde{G}^a > 1$. It now follows from $\widetilde{G}^a \le \mathbf{C}_G(A \cap A^{ga})$ that $B^* \cap \mathbf{C}_G(A \cap A^{ga}) > 1$, and therefore $A \cap A^{ga} \le A^* \cap A^{ga}$ by (14). Since $A^g \ne A$, we also have $A^{ga} \ne A$. By (13), we therefore have $A^* \cap A^{ga} = 1$, and then $A \cap A^{ga} = 1 = A \cap A^g$.

(17) Now we quickly obtain the desired contradiction:

Assume without loss of generality that $|B| < |A|$. Then

$$|G| = |A|\,|B| < |A|^2.$$

By (16), we have $A^g \cap A = 1$ for every element g not lying in $\mathbf{N}_G(A)$. But then we obtain the contradiction

$$|G| < |A|^2 = |A||A^g| = |AA^g| \le |G|. \qquad \square$$

We will study the properties of products of pairwise permutable cyclic groups in §10.

Exercises

7) (Itô [275]) Suppose $G = AB$, where A and B are abelian groups.
   a) If $A \ne B$, then G contains a normal subgroup $N \lhd G$ with $A \le N$ or $B \le N$.
   b) If $G \ne 1$, then there exists a nontrivial normal subgroup M of G with $M \le A$ or $M \le B$.

      (Hint: In the critical case $A \cap G' \ne 1$ in the proof of b), consider the groups $C = \mathbf{C}_G(A \cap G')$ and $\mathbf{Z}(C)$. Decompose $\mathbf{Z}(C) = A_1 B_1$ with $A_1 \le A$ and $B_1 \le B$.)

8) (Kegel [303]) Let $U \le V \le G$, and suppose $U^g V = VU^g$ for all $g \in G$. Then there exists an element W of a composition series of G (a so-called subnormal subgroup of G) such that $U \le W \le V$. (Use Lemma 4.10.)
9) (Kegel [303]) Let $\mathscr{A}$, $\mathscr{B}$ and $\mathscr{C}$ be three conjugacy classes of subgroups of G. For every pair X, Y of representatives of any two distinct classes, suppose that $XY = YX$ and that all these products constitute a single conjugacy class $\mathscr{K}$ of G. Then every two groups in $\mathscr{K}$ are permutable; in particular, every $K \in \mathscr{K}$ is subnormal in G.
   (Hint: Take $A \in \mathscr{A}$ and $B \in \mathscr{B}$, and let $K = AB$. Show that there exists a $C \in \mathscr{C}$ such that $K = AC = BC$.)
10) (Kegel [303]) Suppose $G = AB = BC = CA$.
   a) If A, B and C have normal Sylow p-subgroups, then so does G.
   b) If A, B and C are nilpotent, then so is G.
11) (Doerk [116]) Let G_1, G_2 and G_3 be abelian subgroups of G with pairwise coprime indices. Then G is abelian.

5 Chief series

Definition 5.1. If H and K are normal subgroups of G with $K < H$, then we define $\mathbf{Aut}_G(H/K)$ to be the group of automorphisms induced by G on H/K via the action

$$hK \mapsto h^g K \quad (h \in H, g \in G).$$

By Theorem I.4.3 (pg. 19), we have

$$\mathbf{Aut}_G(H/K) \cong G/\mathbf{C}_G(H/K),$$

where $\mathbf{C}_G(H/K)$ is defined by

$$\mathbf{C}_G(H/K) = \{g \in G \mid h^g K = hK \text{ for all } h \in H\}.$$

Definition 5.2. Let

$$1 = G_0 \lhd \cdots \lhd G_s = G$$

be a chief series of G. By the definition of a chief series, the chief factors G_{i+1}/G_i are characteristically simple; in particular, the solvable chief factors are elementary abelian.

a) If G ($\neq 1$) is solvable and $|G_{i+1}/G_i| = p_i^{n_i}$, then we define the rank of G to be $r(G) = \max n_i$. We set $r(1) = 0$.
b) If G ($\neq 1$) is p-solvable and $p^{n_i} \mid\!\mid |G_{i+1}/G_i|$, then we define the p-rank of G to be $r_p(G) = \max n_i$. We again set $r_p(1) = 0$.
c) If H/K is a chief factor of order divisible by p, then we call H/K a p-chief factor. If G is p-solvable, then every p-chief factor is an elementary abelian

p-group. (If G is not solvable, then a chief factor can of course be a p-chief factor for multiple primes p.)

By the Jordan–Hölder Theorem I.11.7 (pg. 68) for chief series, it does not matter which chief series of G is used to calculate the rank or the p-rank.

Lemma 5.3. *Assume G is solvable. Then:*

a) *For all normal subgroups N of G, we have $r(G/N) \leq r(G)$.*

b) *For all subgroups U of G, we have $r(U) \leq r(G)$.*

c) $r(G_1 \times G_2) = \max\{r(G_1), r(G_2)\}$.

d) *If N_1 and N_2 are normal subgroups of G, then*

$$r(G/(N_1 \cap N_2)) = \max\{r(G/N_1), r(G/N_2)\}.$$

Analogous statements hold for the p-rank.

Proof. a) The statement is trivial since the chief factors of G/N are isomorphic to chief factors of G.

b) Let

$$1 = G_0 \lhd G_1 \lhd \cdots \lhd G_s = G$$

be a chief series of G. We consider the normal subgroups $U_i = G_i \cap U$ of U. We have

$$U_{i+1}/U_i \cong (G_{i+1} \cap U)G_i/G_i \leq G_{i+1}/G_i.$$

A refinement of the series consisting of the U_i to a chief series of U yields the claim.

c) is trivial.

d) Using the embedding of $G/(N_1 \cap N_2)$ into $G/N_1 \times G/N_2$ (see Lemma I.9.6, pg. 50) and parts b) and c), we see immediately that

$$r(G/(N_1 \cap N_2)) \leq \max\{r(G/N_1), r(G/N_2)\}.$$

The reverse inequality

$$r(G/N_i) \leq r(G/(N_1 \cap N_2))$$

follows from a). □

Theorem 5.4. *Let*

$$1 = G_0 \lhd G_1 \lhd \cdots \lhd G_s = G$$

be a chief series of G. Then:

a) $\bigcap_{i=0}^{s-1} \mathbf{C}_G(G_{i+1}/G_i)$ *is the Fitting subgroup $\mathbf{F}(G)$ of G.*

b) *The group $\mathbf{F}_p(G)$ defined by*

$$\mathbf{F}_p(G) = \bigcap_{p||G_{i+1}/G_i|} \mathbf{C}_G(G_{i+1}/G_i)$$

is p-nilpotent and is the product of all p-nilpotent normal subgroups of G.
(*For the definition of p-nilpotency, see Definition IV.4.1, pg. 455.*)

Proof. a) has already been proved in Theorem III.4.3 (pg. 293).

b) We let $T = \bigcap_{p||G_{i+1}/G_i|} \mathbf{C}_G(G_{i+1}/G_i)$. Clearly, T is a normal subgroup of G.

(1) We first show that T is p-nilpotent:

We let $T_i = T \cap G_i$ and consider the normal series

$$1 = T_0 \trianglelefteq T_1 \trianglelefteq \cdots \trianglelefteq T_s = T$$

of T. If p divides $|T_{i+1}/T_i|$, then p divides $|G_{i+1}/G_i|$ as well. Since $T \leq \mathbf{C}_G(G_{i+1}/G_i)$, we then have

$$[T, T_{i+1}] \leq T \cap [\mathbf{C}_G(G_{i+1}/G_i), G_{i+1}] \leq T \cap G_i = T_i.$$

So if p divides $|T_{i+1}/T_i|$, then $T_{i+1}/T_i \leq \mathbf{Z}(T/T_i)$. If we refine the series consisting of the T_i to a chief series of T, then all its chief factors of order divisible by p are central. By Theorem IV.4.4 (pg. 455), T is then p-nilpotent.

(2) Let M be a p-nilpotent normal subgroup of G. We show the following: If H/K is a chief factor of G whose order is divisible by p, then H/K is centralized by M:

In the proof of this fact, we may clearly assume that $K = 1$. Since H is then a minimal normal subgroup of G, either $H \cap M = 1$ or $H \leq M$. In the first case, we obtain

$$[H, M] \leq H \cap M = 1,$$

and we are done. So assume that $H \leq M$. The subgroup M is p-nilpotent, so H is p-nilpotent, as well. Since H is a characteristically simple, p-nilpotent group of order divisible by p, H is a p-group. Let L be a minimal normal subgroup of the p-nilpotent group M that lies in H. Since p divides $|L|$, it now follows from Lemma IV.4.3 (pg. 455) that

$$1 < L \leq H \cap \mathbf{Z}(M) \trianglelefteq G.$$

Since H is a minimal normal subgroup of G, this forces $H \leq \mathbf{Z}(M)$.

(3) It follows from (2) that $M \leq T$ for every p-nilpotent normal subgroup M of G. So T is the uniquely determined largest p-nilpotent normal subgroup of G. □

The importance of Theorem 5.4 for the theory of solvable groups lies in the following observation: If the chief factor G_{i+1}/G_i of G is elementary abelian of order $p_i^{n_i}$, then the group $\mathbf{Aut}_G(G_{i+1}/G_i)$ described in Definition 5.1 is isomorphic to a solvable subgroup of the linear group $\mathrm{GL}_{n_i}(p_i)$ that is irreducible over $\mathrm{GF}(p_i)$. Since

$$G/\bigcap_i \mathbf{C}_G(G_{i+1}/G_i) = G/\mathbf{F}(G)$$

is isomorphic to a subgroup of the direct product of the $\mathbf{Aut}_G(G_{i+1}/G_i)$, the structure of G is largely determined (up to the nilpotent normal subgroup $\mathbf{F}(G)$) by the $\mathbf{Aut}_G(G_{i+1}/G_i)$. In this way, the structure of solvable groups can be traced back to the structure of solvable groups of linear transformations over prime fields. We will make much use of this observation in what follows.

Exercises

12) (Baer [18]) Assume G is solvable, and let N be a minimal normal subgroup of G. Then there exists a subgroup S of G such that $G = \mathbf{C}_G(N)S$ and $N \cap S = 1$.
(Hint: Choose S as small as possible with $G = \mathbf{C}_G(N)S$. If $N \cap S \neq 1$, then show that $N \leq S$. In the case $S < G$, use induction. For $S = G$, we have $\mathbf{C}_G(N) \leq \Phi(G)$, and this yields the contradiction $\mathbf{F}(G) \leq \Phi(G)$.)

13) (Huppert [257]) Let G be a solvable group. We define $j_p(G)$ to be

$$j_p(G) = \max\{[U : V] \mid V < U \leq G,\ V \text{ maximal in } U,\ [U : V] \text{ a } p\text{-power}\}.$$

Show that $\log_p(j_p(G)) = r_p(G)$, where $r_p(G)$ is as defined in Definition 5.2.
(Hint: To prove that $r_p(G) \leq \log_p(j_p(G))$, use Exercise 12.)

6 Elementary theory of p-length

Definition 6.1. Let G be p-solvable in the sense of Definition 1.3 b). We define the upper p-series

$$1 = \mathbf{P}_0(G) \trianglelefteq \mathbf{M}_0(G) \lhd \mathbf{P}_1(G) \lhd \mathbf{M}_1(G) \lhd \cdots \lhd \mathbf{P}_l(G) \trianglelefteq \mathbf{M}_l(G) = G$$

of G as follows:

Let $\mathbf{M}_i(G)/\mathbf{P}_i(G)$ be the largest normal subgroup of $G/\mathbf{P}_i(G)$ with order coprime to p, and let $\mathbf{P}_i(G)/\mathbf{M}_{i-1}(G)$ be the largest normal subgroup of $G/\mathbf{M}_{i-1}(G)$ of p-power order. The number l is called the p-length of G, and we write $l = l_p(G)$. We have $l_p(G) = 0$ if and only if $p \nmid |G|$. The groups $\mathbf{P}_i(G)$ and $\mathbf{M}_i(G)$ are clearly characteristic subgroups of G. (Note that $\mathbf{P}_1(G) = \mathbf{F}_p(G)$ is the maximal p-nilpotent normal subgroup of G introduced in Theorem 5.4.)

Lemma 6.2. *Assume G is p-solvable. Suppose we have a normal series*

$$1 = P_0 \trianglelefteq M_0 \trianglelefteq P_1 \trianglelefteq \cdots \trianglelefteq P_m \trianglelefteq M_m = G$$

in which $|M_i/P_i|$ *is coprime to* p $(i = 0, \ldots, m)$ *and* $|P_i/M_{i-1}|$ *is a power of* p $(i = 1, \ldots, m)$. *Then for all* i, *we have*

$$M_i \leq \mathbf{M}_i(G) \quad \textit{and} \quad P_i \leq \mathbf{P}_i(G).$$

Proof. Assume for induction that $P_i \leq \mathbf{P}_i(G)$; this is certainly true when $i = 0$. Then

$$M_i\mathbf{P}_i(G)/\mathbf{P}_i(G) \trianglelefteq G/\mathbf{P}_i(G),$$

and

$$|M_i\mathbf{P}_i(G)/\mathbf{P}_i(G)| = |M_i/(M_i \cap \mathbf{P}_i(G))|$$

is a divisor of $|M_i/P_i|$, so it is therefore coprime to p. Hence, it follows that

$$M_i\mathbf{P}_i(G)/\mathbf{P}_i(G) \leq \mathbf{M}_i(G)/\mathbf{P}_i(G),$$

and so $M_i \leq \mathbf{M}_i(G)$. The next step $P_{i+1} \leq \mathbf{P}_{i+1}(G)$ follows in the same way. □

Lemma 6.3. *Let* $N \trianglelefteq G$, *and suppose that* $N/\Phi(G)$ *is* p*-nilpotent. Then* N *is* p*-nilpotent.*

Proof. Let $K/\Phi(G)$ be the normal p-complement of $N/\Phi(G)$, and let P be the Sylow p-subgroup of $\Phi(G)$. Recall that $\Phi(G)$ is nilpotent by Theorem III.3.6. By the Zassenhaus theorem, there exists a complement V for P in K, and all complements are conjugate in K (Theorem I.18.1 and Theorem I.18.2, pg. 135). For any $g \in G$, V^g is also a complement of P in K. So $V^g = V^x$ for some $x \in K$. It follows that

$$G = \mathbf{N}_G(V)K = \mathbf{N}_G(V)VP = \mathbf{N}_G(V)\Phi(G) = \mathbf{N}_G(V).$$

So V is a normal p-complement of N, and N is p-nilpotent. □

Lemma 6.4. *Assume* G *is* p*-solvable.*

a) *If* N *is a normal subgroup of* G, *then* $l_p(G/N) \leq l_p(G)$.

b) *If* U *is a subgroup of* G, *then* $l_p(U) \leq l_p(G)$.

c) *If* N_1 *and* N_2 *are normal subgroups of* G, *then*

$$l_p(N_1N_2) = \max\{l_p(N_1), l_p(N_2)\}.$$

In particular, we always have

$$l_p(G_1 \times G_2) = \max\{l_p(G_1), l_p(G_2)\}.$$

d) *If* N_1 *and* N_2 *are normal subgroups of* G, *then*

$$l_p(G/(N_1 \cap N_2)) = \max\{l_p(G/N_1), l_p(G/N_2)\}.$$

e) *We have* $l_p(G/\Phi(G)) = l_p(G)$.

Proof. Parts a) and b) are easy consequences of Lemma 6.2.

c) Since $\mathbf{P}_i(N_j)$ and $\mathbf{M}_i(N_j)$ are characteristic subgroups of N_j, the series

$$1 = \mathbf{P}_0(N_1)\mathbf{P}_0(N_2) \trianglelefteq \mathbf{M}_0(N_1)\mathbf{M}_0(N_2) \lhd \mathbf{P}_1(N_1)\mathbf{P}_1(N_2) \lhd \cdots$$

is a normal series of N_1N_2. The index

$$[\mathbf{M}_i(N_1)\mathbf{M}_i(N_2) : \mathbf{P}_i(N_1)\mathbf{P}_i(N_2)] = \frac{[\mathbf{M}_i(N_1) : \mathbf{P}_i(N_1)]\,[\mathbf{M}_i(N_2) : \mathbf{P}_i(N_2)]}{[\mathbf{M}_i(N_1) \cap \mathbf{M}_i(N_2) : \mathbf{P}_i(N_1) \cap \mathbf{P}_i(N_2)]}$$

is coprime to p; likewise, the index

$$[\mathbf{P}_i(N_1)\mathbf{P}_i(N_2) : \mathbf{M}_{i-1}(N_1)\mathbf{M}_{i-1}(N_2)]$$

is a power of p. The assumptions of Lemma 6.2 are therefore satisfied, and so

$$l_p(N_1N_2) \leq \max\{l_p(N_1), l_p(N_2)\}.$$

Equality then follows by b).

d) By Lemma I.9.6 (pg. 50), $G/(N_1 \cap N_2)$ is isomorphic to a subgroup of $G/N_1 \times G/N_2$. It now follows by b) and c) that

$$l_p(G/(N_1 \cap N_2)) \leq l_p(G/N_1 \times G/N_2) = \max\{l_p(G/N_1), l_p(G/N_2)\},$$

and equality then follows by a).

e) If $l_p(G/\Phi(G)) = 0$, then $p \nmid |G/\Phi(G)|$. By Theorem III.3.8 (pg. 286), $p \nmid |G|$ as well, and therefore $l_p(G) = 0$. Now assume that $l_p(G/\Phi(G)) \geq 1$.

By Lemma 6.3, $\mathbf{F}_p(G)/\Phi(G)$ is the maximal p-nilpotent normal subgroup of $G/\Phi(G)$. So $\mathbf{P}_1(G/\Phi(G)) = \mathbf{P}_1(G)/\Phi(G)$, and therefore $\mathbf{P}_i(G/\Phi(G)) = \mathbf{P}_i(G)/\Phi(G)$ and $\mathbf{M}_i(G/\Phi(G)) = \mathbf{M}_i(G)/\Phi(G)$ for all $i \geq 1$, as well. This yields $l_p(G/\Phi(G)) = l_p(G)$. □

We require the following fundamental lemma for the proof of the main theorem in this section:

Lemma 6.5 (Hall–Higman Lemma). *Let G be a p-solvable group, and let U be defined by*

$$U/\mathbf{M}_0(G) = \Phi(\mathbf{P}_1(G)/\mathbf{M}_0(G)).$$

Then $\mathbf{C}_G(\mathbf{P}_1(G)/U) = \mathbf{P}_1(G)$.

Proof. a) We first prove that $\mathbf{C}_G(\mathbf{P}_1(G)/\mathbf{M}_0(G)) \leq \mathbf{P}_1(G)$:

We may assume that $\mathbf{M}_0(G) = 1$, so that G has no nontrivial normal subgroup of order coprime to p. Assume for contradiction that

$$\mathbf{C}_G(\mathbf{P}_1(G))\mathbf{P}_1(G) = K > \mathbf{P}_1(G).$$

Let $M/\mathbf{P}_1(G)$ be a minimal normal subgroup of $G/\mathbf{P}_1(G)$ lying in $K/\mathbf{P}_1(G)$. Then $|M/\mathbf{P}_1(G)|$ is coprime to p. Since $\mathbf{P}_1(G)$ is a p-group by our assumption that $\mathbf{M}_0(G) = 1$, the subgroup $\mathbf{P}_1(G)$ has a complement S in M by the Zassenhaus theorem. So we have

$$M = S\mathbf{P}_1(G) \quad \text{and} \quad S \cap \mathbf{P}_1(G) = 1.$$

Since $|\mathbf{C}_G(\mathbf{P}_1(G))\mathbf{P}_1(G)/\mathbf{C}_G(\mathbf{P}_1(G))|$ is a power of p and $S \leq \mathbf{C}_G(\mathbf{P}_1(G))\mathbf{P}_1(G)$, it follows that $S \leq \mathbf{C}_G(\mathbf{P}_1(G))$. Therefore, $M = S \times \mathbf{P}_1(G)$. So S is a characteristic subgroup of M, and therefore $S \lhd G$. It follows that $S \leq \mathbf{M}_0(G) = 1$ and $M = \mathbf{P}_1(G)$, contradicting our assumption. So $\mathbf{C}_G(\mathbf{P}_1(G)/\mathbf{M}_0(G)) \leq \mathbf{P}_1(G)$.

b) Now we show that $\mathbf{C}_G(\mathbf{P}_1(G)/U) = \mathbf{P}_1(G)$:

We may again assume that $\mathbf{M}_0(G) = 1$. Then $\mathbf{P}_1(G)$ is a p-group, so $U = \Phi(\mathbf{P}_1(G))$ and therefore $\mathbf{P}_1(G)/U$ is elementary abelian. Hence, we certainly have that $\mathbf{C}_G(\mathbf{P}_1(G)/U) \geq \mathbf{P}_1(G)$.

Assume for contradiction that

$$C = \mathbf{C}_G(\mathbf{P}_1(G)/U) > \mathbf{P}_1(G).$$

Since $C \lhd G$, the quotient $C/\mathbf{P}_1(G)$ is not a p-group. So there exists an element $x \in C - \mathbf{P}_1(G)$ of order coprime to p. Since $x \in \mathbf{C}_G(\mathbf{P}_1(G)/\Phi(\mathbf{P}_1(G))$, the element x induces the identity automorphism on $\mathbf{P}_1(G)/\Phi(\mathbf{P}_1(G))$. Hence, x induces the identity automorphism on all of $\mathbf{P}_1(G)$ by Theorem III.3.18 (pg. 290) since the order of x is coprime to p. It follows by a) that

$$x \in \mathbf{C}_G(\mathbf{P}_1(G)) \leq \mathbf{P}_1(G),$$

contradicting our assumption. □

Main Theorem 6.6 (P. Hall, G. Higman [224]). *Assume G is p-solvable.*

a) *If $c(G_p)$ is the nilpotency class of the Sylow p-subgroup G_p of G, then $l_p(G) \leq c(G_p)$.*

b) *If $d(G_p)$ is the minimum number of generators of G_p, then $l_p(G) \leq d(G_p)$.*

c) *If $r_p(G)$ is the p-rank of G, then $l_p(G) \leq r_p(G)$.*

(*The bounds given here are not the best possible. In Volume 2, we will establish much sharper inequalities.*)

Proof. a) The quotient $\mathbf{P}_1(G)/\mathbf{M}_0(G)$ is a normal subgroup of $G/\mathbf{M}_0(G)$ of p-power order, so it is centralized by $\mathbf{Z}(G_p)$. Hence, it follows from Lemma 6.5 that $\mathbf{Z}(G_p) \leq \mathbf{P}_1(G)$. If $c(G_p) = 1$, then $G_p = \mathbf{Z}(G_p) \leq \mathbf{P}_1(G)$, and so $\mathbf{M}_1(G) = G$ and $l_p(G) = 1$.

We prove the general case by induction on $c(G_p)$: Since $\mathbf{Z}(G_p) \leq \mathbf{P}_1(G)$, we have

$$c(G_p\mathbf{P}_1(G)/\mathbf{P}_1(G)) = c(G_p/(G_p \cap \mathbf{P}_1(G))) \leq c(G_p) - 1.$$

So by the induction hypothesis,

$$l_p(G/\mathbf{P}_1(G)) \leq c(G_p) - 1.$$

Since $l_p(G/\mathbf{P}_1(G)) = l_p(G) - 1$, the claim follows for $l_p(G) \geq 1$.

b) We first show that no Sylow p-subgroup of $\mathbf{P}_1(G)$ lies in the Frattini subgroup of a Sylow p-subgroup of G:

Assume for contradiction that P is a Sylow p-subgroup of $\mathbf{P}_1(G)$ and $P \leq \Phi(G_p)$ for some Sylow p-subgroup G_p of G. We may again assume that $\mathbf{M}_0(G) = 1$, as passing from G to the quotient $G/\mathbf{M}_0(G)$ does not affect our assumptions since

$$\Phi(G_p)\mathbf{M}_0(G)/\mathbf{M}_0(G) \leq \Phi(G_p\mathbf{M}_0(G)/\mathbf{M}_0(G)).$$

(In fact, we have equality here by Theorem III.3.14 c), pg. 288). So we now have $\mathbf{P}_1(G) = P \trianglelefteq G$ and $\mathbf{P}_1(G) \leq \Phi(G_p)$. It then follows by Lemma III.3.3 a) (pg. 284) that $\mathbf{P}_1(G) \leq \Phi(G)$. But then using Lemma 6.4 e) we obtain the contradiction

$$l_p(G) = l_p(G/\Phi(G)) \leq l_p(G/\mathbf{P}_1(G)) = l_p(G) - 1.$$

Now we prove part b) by induction on $d(G_p)$. If $d(G_p) = 1$, then G_p is cyclic, so G_p has class 1. Then G has p-length 1 by a).

For the general case, let P be a Sylow p-subgroup of $\mathbf{P}_1(G)$. Then P lies in a Sylow p-subgroup G_p of G. As we have just shown, P does not lie in $\Phi(G_p)$, so

$$G_p \cap \mathbf{P}_1(G) = P > \Phi(G_p) \cap \mathbf{P}_1(G).$$

If we set $d = d(G_p)$, then by Lemma III.3.4 a) (pg. 285), we have

$$\begin{aligned}[G_p\mathbf{P}_1(G)/\mathbf{P}_1(G) : \Phi(G_p\mathbf{P}_1(G)/\mathbf{P}_1(G))] &\leq [G_p\mathbf{P}_1(G) : \Phi(G_p)\mathbf{P}_1(G)] \\ &= \frac{[G_p : \Phi(G_p)]}{[G_p \cap \mathbf{P}_1(G) : \Phi(G_p) \cap \mathbf{P}_1(G)]} \\ &\leq p^{d-1}.\end{aligned}$$

It follows by our induction hypothesis that $l_p(G/\mathbf{P}_1(G)) \leq d - 1$, and therefore $l_p(G) \leq d$.

c) We let $r_p(G) = r$, and we may assume that $r \geq 1$. Let H/K be a p-chief factor of G. Then $|H/K| = p^n$ with $n \leq r$. We consider the automorphism group $\mathbf{Aut}_G(H/K)$ of H/K induced by G. Since H/K is elementary abelian of order p^n, $\mathbf{Aut}_G(H/K)$ is isomorphic to a subgroup of the general linear group $\mathrm{GL}_n(p)$. The Sylow p-subgroups of $\mathrm{GL}_n(p)$ have nilpotency class $n - 1$ by Theorem III.16.3 (pg. 405). So the Sylow p-subgroups of $\mathbf{Aut}_G(H/K)$ have class at most $n - 1$. Hence, $\mathbf{K}_n(G_p) \leq \mathbf{C}_G(H/K)$. Since $n \leq r$, we also have $\mathbf{K}_r(G_p) \leq \mathbf{C}_G(H/K)$, and this holds for all p-chief factors H/K of G. It now follows by Theorem 5.4 that

$$\mathbf{K}_r(G_p) \leq \bigcap_{p||H/K|} \mathbf{C}_G(H/K) = \mathbf{F}_p(G) = \mathbf{P}_1(G).$$

We obtain from a) that

$$l_p(G/\mathbf{P}_1(G)) \leq c(G_p\mathbf{P}_1(G)/\mathbf{P}_1(G)) \leq r-1,$$

and then $l_p(G) \leq r = r_p(G)$. □

The origin of the theory of p-length developed by P. Hall and G. Higman was the restricted Burnside problem for the exponent 6 (see Remark III.6.7, pg. 307). We show:

Theorem 6.7 (P. Hall, G. Higman [224]). *There exists a function f defined on the natural numbers with the following property: If G is a finite group of exponent 6 that can be generated by k elements, then $|G| \leq f(k)$.*

Proof. Since G has exponent 6, we have that $|G| = 2^a3^b$ for some integers a, b. By Theorem V.7.3 (pg. 522), G is solvable. The Sylow 2-subgroups of G have exponent 2, so they are abelian. Hence, $l_2(G) \leq 1$ by Theorem 6.6 a). Let

$$1 \leq \mathbf{M}_0(G) \trianglelefteq \mathbf{P}_1(G) \trianglelefteq \mathbf{M}_1(G) = G$$

be the upper 2-series of G. Then $G/\mathbf{P}_1(G)$ has exponent 3 and can be generated by k elements. Therefore, we have $|G/\mathbf{P}_1(G)| \leq 3^{f_1(k)}$ with $f_1(k) = k + \binom{k}{2} + \binom{k}{3}$ by Theorem III.6.6 (pg. 306). Hence, $\mathbf{P}_1(G)$ can be generated by $2k3^{f_1(k)} = f_2(k)$ elements by Theorem I.19.10 (pg. 151). (A better bound is given by the Schreier Subgroup Theorem I.19.11.) Now, $\mathbf{P}_1(G)/\mathbf{M}_0(G)$ is a group of exponent 2 that can be generated by $f_2(k)$ elements, so $|\mathbf{P}_1(G)/\mathbf{M}_0(G)| \leq 2^{f_2(k)}$. Another application of Theorem III.6.6 yields $|\mathbf{M}_0(G)| \leq f_3(k)$, so $|G| \leq f(k)$ with $f(k) = 3^{f_1(k)}2^{f_2(k)}f_3(k)$. □

Remark 6.8.

a) The exact order of the largest finite group G of exponent 6 generated by k elements can be found by using that both $l_2(G) = 1$ and $l_3(G) = 1$. We will return to the finer aspects of the theory of p-length in Volume 2.

b) M. Hall has shown that the assumption that G is finite in Theorem 6.7 is not necessary; that is, every finitely generated group of exponent 6 is finite (M. Hall [203]). Hence, Burnside's Conjecture has been proved for exponent 6.

The following lemma is sometimes useful.

Lemma 6.9. *Let G be a p-solvable group. Assume that every homomorphic image of G distinct from G has p-length at most k, but $l_p(G) > k$. Then G has the following properties:*

a) $\Phi(G) = 1$.

b) *The maximal p-nilpotent normal subgroup $\mathbf{F}_p(G) = \mathbf{P}_1(G)$ of G is an elementary abelian p-group.*

c) *$\mathbf{F}_p(G)$ is the unique minimal normal subgroup of G, and it has a complement in G.*

d) *We have $\mathbf{C}_G(\mathbf{F}_p(G)) = \mathbf{F}_p(G)$.*

Proof. a) If $\Phi(G) > 1$, then $k < l_p(G) = l_p(G/\Phi(G)) \leq k$ by Lemma 6.4 e), a contradiction. So $\Phi(G) = 1$.

b) If $1 < K \lhd G$ with $p \nmid |K|$, then we obtain the contradiction $k < l_p(G) = l_p(G/K) \leq k$.

So G contains no nontrivial normal subgroup of order coprime to p. In particular, the maximal p-nilpotent normal subgroup $\mathbf{F}_p(G)$ of G is a p-group. Since $\Phi(\mathbf{F}_p(G)) \leq \Phi(G) = 1$, $\mathbf{F}_p(G)$ is elementary abelian.

c) It follows from Lemma 6.4 d), that G contains exactly one minimal normal subgroup N. Since $\Phi(G) = 1$, there exists a maximal subgroup U of G such that $G = NU$ and $N \cap U = 1$. Then $\mathbf{F}_p(G) = N(\mathbf{F}_p(G) \cap U)$ and $\mathbf{F}_p(G) \cap U \trianglelefteq \mathbf{F}_p(G)U = G$. As N is the unique minimal normal subgroup of G, it follows that $\mathbf{F}_p(G) \cap U = 1$ and $N = \mathbf{F}_p(G)$. Therefore, U is a complement of $\mathbf{F}_p(G)$.

d) If $\mathbf{C}_G(\mathbf{F}_p(G)) > \mathbf{F}_p(G)$, then we have $\mathbf{C}_G(\mathbf{F}_p(G)) = \mathbf{F}_p(G) \times R$, where

$$1 < R = \mathbf{C}_G(\mathbf{F}_p(G)) \cap U.$$

But then $R \lhd \mathbf{F}_p(G)U = G$, contradicting the uniqueness of N. So we have $\mathbf{C}_G(\mathbf{F}_p(G)) = \mathbf{F}_p(G)$. □

Lemma 6.10. *Let G be a p-solvable group, let P be a Sylow p-subgroup of G, and let Q be a p-complement of G. If $P'Q = QP'$, then $l_p(G) \leq 1$.*

Proof. Suppose G is a minimal counterexample to the lemma. Since the hypothesis is inherited by homomorphic images, G satisfies the assumptions of Lemma 6.9 for $k = 1$. The subgroup

$$\mathbf{F}_p(G) \cap P'Q = \mathbf{F}_p(G) \cap P'$$

is a normal subgroup of $P'Q$. As $\mathbf{F}_p(G) \cap P'$ is also normal in P, it follows that $\mathbf{F}_p(G) \cap P' \lhd G$. By Lemma 6.9 c), $\mathbf{F}_p(G)$ is a minimal normal subgroup of G. So only two situations are possible:

If $\mathbf{F}_p(G) \cap P' = 1$, then

$$[\mathbf{F}_p(G), P'] \leq \mathbf{F}_p(G) \cap P' = 1.$$

By Lemma 6.9 d), we then have $P' \leq \mathbf{C}_G(\mathbf{F}_p(G)) = \mathbf{F}_p(G)$ and therefore $P' = 1$. By Theorem 6.6 a), we have in this case that $l_p(G) \leq 1$.

If, on the other hand, $\mathbf{F}_p(G) \cap P' = \mathbf{F}_p(G)$, then $\mathbf{F}_p(G) \leq P'$. Then since $\mathbf{F}_p(G) \leq P' \leq \Phi(P)$ and $\mathbf{F}_p(G) \trianglelefteq G$, we conclude with the help of Lemma III.3.3 a) (pg. 284) that $\mathbf{F}_p(G) \leq \Phi(G)$. But this contradicts Lemma 6.9 a). □

Now we can distinguish those groups satisfying $l_p(G) \leq 1$ for all primes p by a permutability condition:

Theorem 6.11 (Huppert [263]). *Let $\{P_1, \ldots, P_r\}$ be a Sylow system of the solvable group G.*

a) *If $P_i'P_j = P_jP_i'$ for all i, j, then $l_p(G) \leq 1$ for all p.*

b) *If $l_p(G) \leq 1$ for all p, then every characteristic subgroup of P_i permutes with every characteristic subgroup of P_j.*

Proof. The claim in a) follows immediately from Lemma 6.10 since P_i' permutes with the p_i-complements $Q_i = \prod_{j \neq i} P_j$ by our assumptions.

If G has p-length at most 1 for all p, then this also holds for the subgroup P_iP_j. Since $l_{p_j}(P_iP_j) \leq 1$, the upper p_j-series of P_iP_j has the form

$$1 \trianglelefteq N \trianglelefteq NP_j \trianglelefteq P_iP_j,$$

where $p_j \nmid |N|$ and $p_j \nmid |P_iP_j/NP_j|$. Let U be a characteristic subgroup of P_j. As NU/N is characteristic in NP_j/N, it follows that $NU \trianglelefteq G$. Hence,

$$P_i(NU) = P_iU$$

is a subgroup of G. By repeating this chain of reasoning, we see that every characteristic subgroup V of P_i permutes with U. □

We now give a further characterization of the groups G with $l_p(G) \leq 1$, which will be important in later sections.

Theorem 6.12. *The group G is p-solvable with $l_p(G) \leq 1$ if and only if for every p-chief factor H/K of G, the automorphism group $\mathbf{Aut}_G(H/K)$ of H/K induced by G has order coprime to p.*

Proof. a) Let $l_p(G) \leq 1$, and let

$$1 \trianglelefteq \mathbf{M}_0(G) \trianglelefteq \mathbf{P}_1(G) \trianglelefteq \mathbf{M}_1(G) = G$$

be the upper p-series of G. By Theorem 5.4 b), we have for every p-chief factor H/K of G that

$$\mathbf{C}_G(H/K) \geq \mathbf{P}_1(G) = \mathbf{F}_p(G).$$

Therefore,

$$|\mathbf{Aut}_G(H/K)| = |G/\mathbf{C}_G(H/K)|$$

is coprime to p, as it is a divisor of $|G/\mathbf{P}_1(G)|$.

b) Conversely, assume that for every chief factor H/K of order divisible by p,

$$|\mathbf{Aut}_G(H/K)| = |G/\mathbf{C}_G(H/K)|$$

is coprime to p. Since $\mathbf{F}_p(G) = \bigcap_{p||H/K|} \mathbf{C}_G(H/K)$, we have that $p \nmid |G/\mathbf{F}_p(G)|$. Therefore, $\mathbf{M}_1(G) = G$ and G has p-length at most 1. □

Definition 6.13. Let $\prec$ be a total order (not necessarily the natural order) on the set of all primes. We say that G has a Sylow tower of type $\prec$ if the following condition is satisfied:

For every prime p and collection of primes π_p of the form

$$\pi_p = \{q \mid q \prec p\},$$

the group G has a normal Hall π_p-subgroup. Hence, if $\{p_1, \dots, p_r\}$ is the set of prime divisors of $|G|$ and $p_1 \prec p_2 \prec \cdots \prec p_r$, then there exist Sylow p_i-subgroups G_{p_i} of G such that

$$G_{p_1} \lhd G_{p_1} G_{p_2} \lhd \cdots \lhd G_{p_1} \cdots G_{p_r} = G$$

is a normal series of G.

It is clear that every group with a Sylow tower has p-length at most 1 for all primes p. The converse is not true, as the direct product $A_4 \times S_3$ of the alternating group A_4 and the symmetric group S_3 shows. However, the following does hold:

Theorem 6.14 (Alperin). *Let G be solvable with $l_p(G) \leq 1$ for all p. Then G is isomorphic to a subgroup of a group H with the following properties:*
(1) *The prime divisors of $|G|$ and $|H|$ are the same.*
(2) *H is a product of normal subgroups K_i of H $(i = 1, \dots, k)$, and every K_i has a Sylow tower; the types of the Sylow towers of the K_i are in general different, however.*

(*It follows trivially from Lemma 6.4 that every subgroup of the group H described above has p-length* 0 *or* 1 *for every p.*)

Proof. Suppose G is a minimal counterexample to the theorem. We show:

(1) G has a unique minimal normal subgroup:

Assume for contradiction that there exist $1 < N_i \lhd G$ $(i = 1, 2)$ with $N_1 \cap N_2 = 1$. Then G is isomorphic to a subgroup of $G/N_1 \times G/N_2$. Since $l_p(G/N_i) \leq 1$ for all p by Lemma 6.4 a), we may embed G/N_i isomorphically in a group H_i that is a product of normal subgroups with Sylow towers. Moreover, $|G/N_i|$ and $|H_i|$ have the same prime divisors. Now $H = H_1 \times H_2$ is a product of normal subgroups with Sylow towers, and G is isomorphic to a subgroup of H. Furthermore, $|G|$ and $|H|$ have the same prime divisors. But this contradicts our choice of G as a minimal counterexample.

(2) G has a normal Sylow subgroup:

Let N be the unique minimal normal subgroup of G, and let $|N| = p^n$. Then G has no nontrivial normal subgroup of order coprime to p. Since $l_p(G) \leq 1$, this means $P = \mathbf{P}_1(G)$ is the normal Sylow p-subgroup of G.

(3) Let Q be a complement of P in G, which exists by the Zassenhaus theorem. Since $|Q| < |G|$, Q can be isomorphically embedded in a group K that is a product of normal subgroups M_j with Sylow towers. Here, $|Q|$ and $|K|$ have the same prime divisors. By Theorem I.15.12 (pg. 107), the group $G = PQ$ can be embedded in the twisted wreath product $H = P \wr_Q K$. By Theorem I.15.11 (pg. 107), H is the semidirect product of a normal subgroup $P^* = P \times \cdots \times P$ (with $[K : Q]$ factors) and a group $K^* = \prod_j M_j^*$ that is isomorphic to K. Therefore, H is the product of the

normal subgroups $P^*M_j^*$, which all have Sylow towers since $p \nmid |M_j^*|$. It is clear that $|H|$ has the same prime divisors as $|G|$. □

7 Formations

Definition 7.1. A class $\mathscr{F}$ of finite groups is called a formation if the following conditions are satisfied:
(1) If $G \in \mathscr{F}$, then every epimorphic image of G also lies in $\mathscr{F}$.
(2) If G/N_1 and G/N_2 lie in $\mathscr{F}$, then $G/(N_1 \cap N_2)$ also lies in $\mathscr{F}$.

In particular, the empty class $\mathscr{F} = \emptyset$ is a formation. If $\mathscr{F} \neq \emptyset$, then $1 \in \mathscr{F}$ by (1). (We do not assume that $\mathscr{F}$ is closed under taking subgroups.)

Definition 7.2. A nonempty formation $\mathscr{F}$ is said to be saturated if whenever $G/\Phi(G) \in \mathscr{F}$, we also have $G \in \mathscr{F}$.

We describe a procedure for constructing saturated formations:

Definition 7.3. Suppose that for every prime p, we have an associated (possibly empty) formation $\mathscr{F}(p)$. We define a class $\mathscr{F}$ of groups by the following procedure: We let $G \in \mathscr{F}$ whenever:
(1) If $\mathscr{F}(p) = \emptyset$, then $p \nmid |G|$.
(2) If $\mathscr{F}(p) \neq \emptyset$ and H/K is a chief factor of G of order divisible by p, then

$$\mathbf{Aut}_G(H/K) \cong G/\mathbf{C}_G(H/K)$$

is a group in $\mathscr{F}(p)$. (If the order of H/K is divisible by multiple primes, which does not happen in solvable groups, then $\mathbf{Aut}_G(H/K)$ must lie simultaneously in multiple formations $\mathscr{F}(p)$.) We say that $\mathscr{F}$ is a local formation described by the formations $\mathscr{F}(p)$.

It is easy to see that $\mathscr{F}$ is actually a formation:

Condition (1) in Definition 7.1 is clearly satisfied. To prove (2), one may assume that $N_1 \cap N_2 = 1$. The chief factors of G are then the chief factors of G in G/N_1 and in N_1. Here, N_1 is isomorphic as a G-group to the normal subgroup $(N_1 \times N_2)/N_2$ of G/N_2.

Lemma 7.4. *Assume $\mathscr{F}$ is a local formation described by the formations $\mathscr{F}(p)$. Then the following are equivalent:*
a) $G \in \mathscr{F}$.
b) *We have $p \nmid |G|$ for $\mathscr{F}(p) = \emptyset$ and $G/\mathbf{F}_p(G) \in \mathscr{F}(p)$ for $\mathscr{F}(p) \neq \emptyset$.*

Proof. By Theorem 5.4 b), $\mathbf{F}_p(G) = \bigcap \mathbf{C}_G(H/K)$, where the intersection is taken over all p-chief factors H/K of G. If $G \in \mathscr{F}$ and $\mathscr{F}(p) \neq \emptyset$, then for every p-chief factor H/K of G we have that

$$G/\mathbf{C}_G(H/K) \in \mathscr{F}(p)$$

by Definition 7.3. By condition (2) in Definition 7.1, it follows that

$$G/\bigcap \mathbf{C}_G(H/K) = G/\mathbf{F}_p(G) \in \mathscr{F}(p).$$

Conversely, if $G/\mathbf{F}_p(G) \in \mathscr{F}(p)$, then for every p-chief factor H/K of G, the quotient $G/\mathbf{C}_G(H/K)$ also lies in $\mathscr{F}(p)$ by Definition 7.1 (1) since it is a homomorphic image of $G/\mathbf{F}_p(G)$. □

Main Theorem 7.5 (Gaschütz [162]). *Every local formation $\mathscr{F}$ is a saturated formation.*

(*The converse of this theorem also holds, as Gaschütz and Lubeseder have shown; see Theorem 7.25.*)

Proof. Let G be a group such that $G/\Phi(G) \in \mathscr{F}$; that is,

(1) $p \nmid |G/\Phi(G)|$ whenever $\mathscr{F}(p) = \emptyset$ and

(2) $G/\Phi(G)/\mathbf{F}_p(G/\Phi(G)) \in \mathscr{F}(p)$ whenever $\mathscr{F}(p) \neq \emptyset$.

It follows from (1) and Theorem III.3.8 (pg. 286) that

(1′) $p \nmid |G|$ whenever $\mathscr{F}(p) = \emptyset$.

By Lemma 6.3, $\mathbf{F}_p(G)/\Phi(G) = \mathbf{F}_p(G/\Phi(G))$ is the maximal p-nilpotent normal subgroup of $G/\Phi(G)$. So it follows from (2) that

(2′) $G/\mathbf{F}_p(G) \in \mathscr{F}(p)$ whenever $\mathscr{F}(p) \neq \emptyset$.

Hence, by (1′), (2′) and Lemma 7.4, the group G lies in the local formation $\mathscr{F}$ described by the $\mathscr{F}(p)$. So $\mathscr{F}$ is a saturated formation. □

Example 7.6.

a) Let $\mathscr{F}$ be the local formation described by $\mathscr{F}(p) = \{1\}$ for all p. Then $\mathscr{F}$ is the class of groups whose chief factors are all central; that is, the class of nilpotent groups. The fact that $\mathscr{F}$ is a saturated formation is exactly the content of Theorem III.3.7 (pg. 286).

b) Let p be a fixed prime. Let $\mathscr{F}$ be the local formation described by $\mathscr{F}(p) = \{1\}$ and $\mathscr{F}(q)$ equal to the class of all groups for primes $q \neq p$. Then $\mathscr{F}$ is the class of those groups whose chief factors of order divisible by p are central. By Theorem IV.4.4 (pg. 455), this is exactly the class of p-nilpotent groups. The fact that G is p-nilpotent whenever $G/\Phi(G)$ is p-nilpotent was proved in Lemma 6.3, which we used in the proof of Theorem 7.5.

c) Let p be a fixed prime. Let $\mathscr{F}$ be the local formation described by the formations $\mathscr{F}(p) = \{G \mid p \nmid |G|\}$ and $\mathscr{F}(q)$ equal to the class of all groups for $q \neq p$. By Theorem 6.12, $\mathscr{F}$ is the class of all p-solvable groups G with $l_p(G) \leq 1$. We already know that $l_p(G) \leq 1$ whenever $l_p(G/\Phi(G)) \leq 1$ from Lemma 6.4 e).

d) Let $\mathscr{F}$ be a local formation described by $\mathscr{F}(p) = \{A \mid A \text{ abelian}\}$ for all p. Since

$$\mathbf{F}(G) = \bigcap \mathbf{C}_G(H/K),$$

where the intersection is taken over all chief factors H/K of G, we have that $G \in \mathscr{F}$ if and only if $G' \leq \mathbf{F}(G)$. So $\mathscr{F}$ is the class of all groups with nilpotent

commutator subgroup, and this collection of groups is a saturated formation. (This fact is also easy to extract from Theorem III.3.5, pg. 285.)

e) Let $\prec$ be a total order on the set of all primes. Let $\mathscr{F}$ be a local formation described by the formations

$$\mathscr{F}(p) = \{A \mid |A| \text{ only divisible by primes } q \text{ with } p \prec q\}.$$

Let $G \in \mathscr{F}$ and write $|G| = \prod_i p_i^{a_i}$ with $p_1 \prec p_2 \prec \cdots \prec p_r$. Assume for induction that G has a normal Hall $\{p_1, \ldots, p_i\}$-subgroup H_i. Let H/K be a p_i-chief factor of G inside H_i. Since

$$G/\mathbf{C}_G(H/K) \in \mathscr{F}(p_i) = \{A \mid |A| \text{ only divisible by primes } q \text{ with } p_i \prec q\}$$

and $|H_i|$ is coprime to $p_{i+1}, \ldots, p_r$, the chief factor H/K is centralized by H_i. So all p_i-chief factors of H_i are central, which means H_i is p_i-nilpotent by Theorem IV.4.4 (pg. 455).

It follows immediately by induction that G has a Sylow tower of type $\prec$. Hence, the collection of groups with a Sylow tower of a fixed type is a saturated formation. If $G/\Phi(G)$ has a Sylow tower of type $\prec$, then so does G.

We will see more examples of saturated formations in the next section.

Lemma 7.7. *Let $\mathscr{F}$ be a saturated formation. Let N be a minimal normal subgroup of the solvable group G, and suppose that $G/N \in \mathscr{F}$ but $G \notin \mathscr{F}$. Then N has a complement in G, and all complements of N in G are conjugate.*

Proof. First assume for contradiction that $N \leq \Phi(G)$. Then since $G/N \in \mathscr{F}$, we also have $G/\Phi(G) \in \mathscr{F}$ by Definition 7.1 (1). But then we obtain the contradiction $G \in \mathscr{F}$ since $\mathscr{F}$ is saturated. So $N \not\leq \Phi(G)$. Let V be a maximal subgroup of G with $N \not\leq V$. Then $G = NV$. Since $N \cap V \lhd NV = G$ and N is minimal normal, we have $N \cap V = 1$. So V is a complement of N in G.

Let V_1 and V_2 be complements of N in G, and let $K = \bigcap_{g \in G} V_1^g$. Certainly $K \lhd G$ and $K \cap N = 1$.

Assume for contradiction that $K \not\leq V_2$. Then it follows immediately from the maximality of V_2 that $G = KV_2$. Since $G/N \cong V_2 \in \mathscr{F}$, we also have

$$G/K \cong V_2/(V_2 \cap K) \in \mathscr{F}$$

by Definition 7.1 (1). But then since $G/K \in \mathscr{F}$ and $G/N \in \mathscr{F}$, we obtain from Definition 7.1 (2) the contradiction

$$G = G/(N \cap K) \in \mathscr{F}.$$

So $K \leq V_2$. The quotient G/K is isomorphic to a primitive permutation group (on the cosets of V_1). The subgroups V_i/K ($i = 1, 2$) are complements of the minimal normal subgroup NK/K of G/K. By Theorem II.3.2 f) (pg. 167), the V_i/K are conjugate in G/K. So V_1 is conjugate to V_2 in G. □

Definition 7.8. Let $\mathscr{F}$ be a nonempty formation, and let G be a finite group.

a) We denote by $G_{\mathscr{F}}$ the intersection of all normal subgroups M of G such that $G/M \in \mathscr{F}$. By Definition 7.1 (2), we have $G/G_{\mathscr{F}} \in \mathscr{F}$.

b) A subgroup F of G is called a covering $\mathscr{F}$-subgroup of G if the following hold:
(1) $F \in \mathscr{F}$.
(2) If $F \leq U \leq G$, then $U = U_{\mathscr{F}}F$.
Clearly, (2) is equivalent to
(2′) If $F \leq U \leq G$ and $U/K \in \mathscr{F}$, then $U = KF$.

Lemma 7.9.

a) *If $U \leq G$ and F is a covering $\mathscr{F}$-subgroup of G with $F \leq U$, then F is a covering $\mathscr{F}$-subgroup of U.*

b) *If $N \trianglelefteq G$ and F is a covering $\mathscr{F}$-subgroup of G, then FN/N is a covering $\mathscr{F}$-subgroup of G/N.*

c) *Let $N \trianglelefteq G$. If $\widetilde{F}/N$ is a covering $\mathscr{F}$-subgroup of G/N and F is a covering $\mathscr{F}$-subgroup of $\widetilde{F}$, then F is a covering $\mathscr{F}$-subgroup of G.*

Proof. a) is trivial.

b) By Definition 7.1 (1), we have

$$FN/N \cong F/(F \cap N) \in \mathscr{F}.$$

Let $FN/N \leq U/N \leq G/N$ with

$$U/N\big/K/N \cong U/K \in \mathscr{F}.$$

Since F is a covering $\mathscr{F}$-subgroup of G, it follows that $U = KF$ and therefore

$$U/N = (K/N)(FN/N).$$

Hence, FN/N is a covering $\mathscr{F}$-subgroup of G/N.

c) Since F is a covering $\mathscr{F}$-subgroup of $\widetilde{F}$, we have $F \in \mathscr{F}$. Since $\widetilde{F}/N \in \mathscr{F}$, it follows immediately that $\widetilde{F} = FN$.

Let $F \leq U \leq G$. Then we have

$$\widetilde{F}/N = FN/N \leq UN/N \leq G/N.$$

By a), $\widetilde{F}/N$ is a covering $\mathscr{F}$-subgroup of UN/N. Since

$$UN/N\big/U_{\mathscr{F}}N/N \cong UN/U_{\mathscr{F}}N \cong U/(U \cap U_{\mathscr{F}}N) \in \mathscr{F},$$

it follows that

$$(\widetilde{F}/N)(U_{\mathscr{F}}N/N) = UN/N.$$

Since $\widetilde{F} = FN$, we now have

$$UN = \widetilde{F}U_{\mathscr{F}}N = FU_{\mathscr{F}}N,$$

and hence

$$U = FU_{\mathscr{F}}(U \cap N).$$

As $F(U \cap N) \leq U \cap \widetilde{F}$, we now have $U = (U \cap \widetilde{F})U_{\mathscr{F}}$. By a), F is a covering $\mathscr{F}$-subgroup of $U \cap \widetilde{F}$. Since

$$(\widetilde{F} \cap U)/(\widetilde{F} \cap U_{\mathscr{F}}) \cong (\widetilde{F} \cap U)U_{\mathscr{F}}/U_{\mathscr{F}} = U/U_{\mathscr{F}} \in \mathscr{F},$$

we have $\widetilde{F} \cap U = F(\widetilde{F} \cap U_{\mathscr{F}})$. Therefore,

$$U = F(\widetilde{F} \cap U_{\mathscr{F}})U_{\mathscr{F}} = FU_{\mathscr{F}}. \qquad \square$$

Theorem 7.10 (Gaschütz [162]). *Let $\mathscr{F}$ be a formation.*

a) *If every group has covering $\mathscr{F}$-subgroups, then $\mathscr{F}$ is saturated.*

b) *If $\mathscr{F}$ is saturated, then every solvable group G has covering $\mathscr{F}$-subgroups, and all covering $\mathscr{F}$-subgroups of G are conjugate in G.*

Proof. a) Let $G/\Phi(G) \in \mathscr{F}$, and let F be a covering $\mathscr{F}$-subgroup of G. Then it follows that $G = \Phi(G)F = F \in \mathscr{F}$. So $\mathscr{F}$ is saturated.

b) We prove the claim by induction on $|G|$. We may assume that $G \notin \mathscr{F}$. Let N be a minimal normal subgroup of G. By our induction hypothesis, the quotient G/N has a covering $\mathscr{F}$-subgroup $\widetilde{F}/N$.

Case 1: Assume $G/N \notin \mathscr{F}$, so $\widetilde{F} < G$. Then $\widetilde{F}$ has a covering $\mathscr{F}$-subgroup F. By Lemma 7.9 c), F is a covering $\mathscr{F}$-subgroup of G.

If F_1 and F_2 are covering $\mathscr{F}$-subgroups of G, then F_1N/N and F_2N/N are covering $\mathscr{F}$-subgroups of G/N by Lemma 7.9 b), so they are conjugate in G/N by the induction hypothesis. Hence, $F_1N = F_2^gN$ for some $g \in G$. Since $G/N \notin \mathscr{F}$, we have $F_1N < G$. Since F_1 and F_2^g are covering $\mathscr{F}$-subgroups of $F_1N < G$, the conjugacy of F_1 and F_2 follows from the induction hypothesis.

Case 2: Assume $G/N \in \mathscr{F}$. By Lemma 7.7, the subgroup N has a complement F in G since $G \notin \mathscr{F}$. We have $F \cong G/N \in \mathscr{F}$. Since F is a maximal subgroup of G, as it is a complement of the minimal normal subgroup N, the only subgroups of G containing F are F and G. It is trivial that $F = FF_{\mathscr{F}}$. Since $G/N \in \mathscr{F}$ and $G \notin \mathscr{F}$, we have $N = G_{\mathscr{F}}$ and so $G = G_{\mathscr{F}}F$. Hence, F is a covering $\mathscr{F}$-subgroup of G.

Every covering $\mathscr{F}$-subgroup of G is a complement of N because $G/N \in \mathscr{F}$ and $G \notin \mathscr{F}$. They are all conjugate in G by Lemma 7.7. $\square$

Lemma 7.11.

a) *Let $\mathscr{F}$ be a formation that contains all cyclic groups of prime order. If F is a covering $\mathscr{F}$-subgroup of the solvable group G, then whenever $F \leq U \leq G$ we always have $\mathbf{N}_G(U) = U$. In particular, $\mathbf{N}_G(F) = F$.*

b) *If $\mathscr{F}$ is a local formation described by formations $\mathscr{F}(p)$ with $\mathscr{F}(p) \neq \emptyset$ for all primes p, then the condition in a) is satisfied.*

Proof. a) Assume for contradiction that $F \leq U \lhd V \leq G$. We may assume that V/U has prime order. Since F is a covering $\mathscr{F}$-subgroup of V by Lemma 7.9 a) and $V/U \in \mathscr{F}$, we obtain the contradiction

$$V = UF = U.$$

Hence, $\mathbf{N}_G(U) = U$.

b) Since $\mathscr{F}(p) \neq \emptyset$, we have $1 \in \mathscr{F}(p)$ for all p by Definition 7.1 (1). Hence, all nilpotent groups are contained in the local formation described by the $\mathscr{F}(p)$. So the condition in a) is satisfied. □

Theorem 7.12 (Carter, Hawkes [94]). *Let $\mathscr{F}$ be a saturated formation, and let G be a solvable group with $G/\mathbf{F}(G) \in \mathscr{F}$. Let H be a subgroup of G with $G = \mathbf{F}(G)H$ and $H \in \mathscr{F}$.*

a) *The subgroup H lies in a covering $\mathscr{F}$-subgroup of G. If $\mathscr{F}$ contains all nilpotent groups, then $\mathbf{N}_G(H)$ lies in a covering $\mathscr{F}$-subgroup of G.*

b) *Suppose $\mathscr{F}$ contains all nilpotent groups. If F_1 and F_2 are covering $\mathscr{F}$-subgroups of G with $H \leq F_1 \cap F_2$, then $F_1 = F_2$.*

Proof. a) We prove the claim by induction on $|G|$. If $G \in \mathscr{F}$, then the theorem is trivial. So we may assume that $G \notin \mathscr{F}$. If N is a minimal normal subgroup of G, then $HN/N \in \mathscr{F}$. Since $\mathbf{F}(G)/N \leq \mathbf{F}(G/N)$, we also have $G/N = \mathbf{F}(G/N)HN/N$. By induction, HN/N (resp. $\mathbf{N}_{G/N}(HN/N) = \mathbf{N}_G(HN)/N$) lies in a covering $\mathscr{F}$-subgroup U/N of G/N. By Lemma 7.9 b), we have $U = FN$ for a covering $\mathscr{F}$-subgroup F of G. So it follows that $H \leq FN$ (resp. $\mathbf{N}_G(H) \leq \mathbf{N}_G(HN) \leq FN$).

First assume that $FN < G$. Since $H \leq FN$, we then have

$$FN = FN \cap \mathbf{F}(G)H = (FN \cap \mathbf{F}(G))H,$$

and clearly $FN \cap \mathbf{F}(G) \leq \mathbf{F}(FN)$. Hence, FN satisfies the assumptions of the theorem. It therefore follows by the induction hypothesis that $H \leq F^x$ (resp. $\mathbf{N}_G(H) = \mathbf{N}_{FN}(H) \leq F^x$) for a suitable $x \in FN$.

So we may assume that $G = FN$. Since N is a minimal normal subgroup of G, it follows that $N \leq \mathbf{Z}(\mathbf{F}(G))$. It follows from $\mathbf{F}(G) \cap F \trianglelefteq F$ and $N \leq \mathbf{C}_G(\mathbf{F}(G) \cap F)$ that $\mathbf{F}(G) \cap F \trianglelefteq FN = G$. If $\mathbf{F}(G) \cap F \neq 1$, then an application of the induction hypothesis to $G/(\mathbf{F}(G) \cap F)$ yields the claim. So assume that $\mathbf{F}(G) \cap F = 1$. It then follows from $N \leq \mathbf{F}(G)$ and $G = FN$ that $N = \mathbf{F}(G)$ and $G = HN$. Since $G \notin \mathscr{F}$ and $H \in \mathscr{F}$, we have $H < G$ and so $N \cap H = 1$. Hence, H is a covering $\mathscr{F}$-subgroup of G.

Continuing the case of the last paragraph, now assume that $\mathscr{F}$ contains all nilpotent groups. Let $\mathbf{N}_G(H) = N_1H$, where $N_1 = N \cap \mathbf{N}_G(H)$. Then $N_1 \trianglelefteq G$. If $N_1 = 1$, then we are done. On the other hand, if $N_1 \neq 1$, then it follows from the minimality of N that $N = N_1$ and $[N, H] \leq N \cap H = 1$. Using the fact that $\mathbf{C}_G(\mathbf{F}(G)) \leq \mathbf{F}(G) = N$, we have $H = 1$. Then $G = N$ is cyclic of prime order, so it lies in $\mathscr{F}$. But this contradicts our assumption that $G \notin \mathscr{F}$.

b) Again let N be a minimal normal subgroup of G. By a), we may assume that $\mathbf{N}_G(H) \leq F_1$. It follows from $HN/N \leq F_1N/N \cap F_2N/N$ by induction that

$F_1N = F_2N$. Since F_1 and F_2 are covering $\mathscr{F}$-subgroups of F_1N, we may clearly assume that $G = F_1N = F_2N$. It then follows from the minimality of N that $N \leq F_1$ or $N \cap F_1 = 1$. If $N \leq F_1$, then we obtain $F_1 = F_2$. So assume that $N \cap F_1 = N \cap F_2 = 1$. As $H \leq F_i$ for $i = 1, 2$, it then follows that

$$H = H(N \cap F_i) = HN \cap F_i.$$

Since F_1 and F_2 are covering $\mathscr{F}$-subgroups of $G = F_1N$, there exists an $x \in N$ such that $F_1^x = F_2$. From this it follows that

$$H^x = HN \cap F_1^x = HN \cap F_2 = H,$$

and so $x \in \mathbf{N}_G(H) \leq F_1$ by a). So $F_2 = F_1^x = F_1$. □

Theorem 7.12 provides a useful characterization of the covering $\mathscr{F}$-subgroups:

Theorem 7.13 (Hawkes). *Assume G is solvable, and let $\mathscr{F}$ be a saturated formation.*

a) *If F is a covering $\mathscr{F}$-subgroup of G and $N \trianglelefteq G$, then $FN/N \in \mathscr{F}$, and no subgroup of G/N properly containing FN/N lies in $\mathscr{F}$.*

b) *Let*

$$1 = N_0 \lhd N_1 \lhd \cdots \lhd N_r = G,$$

where $N_i \trianglelefteq G$ and the factor groups N_{i+1}/N_i are nilpotent. Let $F \leq G$. If $FN_i/N_i \in \mathscr{F}$ $(i = 0, 1, \ldots, r-1)$ and no subgroup of G/N_i properly containing FN_i/N_i lies in $\mathscr{F}$, then F is a covering $\mathscr{F}$-subgroup of G.

Proof. a) By Lemma 7.9 b), $\overline{F} = FN/N$ is a covering $\mathscr{F}$-subgroup of $\overline{G} = G/N$. Suppose $\overline{F} \leq \overline{U} \leq \overline{G}$ and $\overline{U} \in \mathscr{F}$. Then $\overline{U} = \overline{U}_{\mathscr{F}}\overline{F} = \overline{F}$.

b) We prove the claim by induction on r. For $r = 1$, it follows from Theorem 7.12 that F lies in a covering $\mathscr{F}$-subgroup, so F is a covering $\mathscr{F}$-subgroup of G.

Clearly, FN_1/N_1 satisfies the hypotheses in G/N_1. Therefore, FN_1/N_1 is a covering $\mathscr{F}$-subgroup of G/N_1 by induction. Furthermore, F is a covering $\mathscr{F}$-subgroup of FN_1 by Theorem 7.12 a). It follows by Lemma 7.9 c), that F is a covering $\mathscr{F}$-subgroup of G. □

The following lemma has independent interest.

Lemma 7.14. *Suppose A is an abelian normal subgroup of G. Then all complements of A in G are conjugate under automorphisms of G.*

Proof. We let $\overline{G} = G/A$. Let

$$K_i = \{r_i(\overline{g}) \mid \overline{g} \in \overline{G}\} \quad (i = 1, 2),$$

where $\overline{g} = r_i(\overline{g})A$, be complements of A in G. We define a bijective map α from G onto itself by $(ar_1(\overline{g}))^\alpha = ar_2(\overline{g})$ for all $a \in A$ and $\overline{g} \in \overline{G}$. Then we have

$$\begin{aligned}(a_1 r_1(\overline{g_1}) a_2 r_1(\overline{g_2}))^\alpha &= (a_1 a_2^{r_1(\overline{g_1})^{-1}} r_1(\overline{g_1}) r_1(\overline{g_2}))^\alpha \\ &= (a_1 a_2^{r_1(\overline{g_1})^{-1}} r_1(\overline{g_1 g_2}))^\alpha \\ &= a_1 a_2^{r_1(\overline{g_1})^{-1}} r_2(\overline{g_1 g_2}) \\ &= a_1 a_2^{r_1(\overline{g_1})^{-1}} r_2(\overline{g_1}) r_2(\overline{g_2}) \\ &= a_1 r_2(\overline{g_1}) a_2^{r_1(\overline{g_1})^{-1} r_2(\overline{g_1})} r_2(\overline{g_2}).\end{aligned}$$

Since $r_1(\overline{g_1})^{-1} r_2(\overline{g_1}) \in A$ and A is commutative, it follows that

$$\begin{aligned}(a_1 r_1(\overline{g_1}) a_2 r_1(\overline{g_2}))^\alpha &= a_1 r_2(\overline{g_1}) a_2 r_2(\overline{g_2}) \\ &= (a_1 r_1(\overline{g_1}))^\alpha (a_2 r_1(\overline{g_2}))^\alpha.\end{aligned}$$

Hence, α is an automorphism of G that maps K_1 onto K_2. □

The following theorem generalizes results of Schenkmann [365], G. Higman [237], and Carter [88].

Theorem 7.15 (Shult [384], Carter, Hawkes [94]). *Let $\mathfrak{F}$ be a local formation. If G is solvable and $G_{\mathfrak{F}}$ is abelian, then the covering $\mathfrak{F}$-subgroups of G are exactly the complements of $G_{\mathfrak{F}}$ in G.*

Proof. a) Let F be a covering $\mathfrak{F}$-subgroup of G, so $G = G_{\mathfrak{F}} F$. We will show that $G_{\mathfrak{F}} \cap F = 1$, and for the proof we may assume that $G_{\mathfrak{F}}$ is an abelian p-group and $G_{\mathfrak{F}} \cap F$ is a minimal normal subgroup of G. If $\mathfrak{F}(p) = \emptyset$, then $p \nmid |F|$ and therefore $G_{\mathfrak{F}} \cap F = 1$. So assume in what follows that $\mathfrak{F}(p) \neq \emptyset$. We let $\overline{G} = G/G_{\mathfrak{F}}$. Since $\overline{G} \in \mathfrak{F}$, we have $\overline{G}/\mathbf{F}_p(\overline{G}) \in \mathfrak{F}(p)$. So $\overline{G}_{\mathfrak{F}(p)} \leq \mathbf{F}_p(\overline{G})$, and thus $\overline{G}_{\mathfrak{F}(p)}$ is p-nilpotent. Let $\overline{V}$ be the normal p-complement of $\overline{G}_{\mathfrak{F}(p)}$. Since $(|G_{\mathfrak{F}}|, |\overline{V}|) = 1$, we have by Theorem III.13.4 (pg. 370) that

$$G_{\mathfrak{F}} = \mathbf{C}_{G_{\mathfrak{F}}}(\overline{V}) \times [G_{\mathfrak{F}}, \overline{V}].$$

(Here, $\mathbf{C}_{G_{\mathfrak{F}}}(\overline{V})$ and $[G_{\mathfrak{F}}, \overline{V}]$ are constructed as in Definition III.13.2.) If $\overline{K}$ is a chief factor of G inside $\mathbf{C}_{G_{\mathfrak{F}}}(\overline{V})$, then $\overline{K}$ is centralized by $\overline{V}$. Then by Theorem V.5.17 (pg. 515), the normal p-subgroup $\overline{G}_{\mathfrak{F}(p)}/\overline{V}$ of $\overline{G}/\overline{V}$ is trivially represented on $\overline{K}$. Hence, $G/\mathbf{C}_G(\overline{K})$ is a homomorphic image of $\overline{G}/\overline{G}_{\mathfrak{F}(p)}$, and thus it lies in $\mathfrak{F}(p)$. So $\mathbf{C}_{G_{\mathfrak{F}}}(\overline{V})F$ lies in $\mathfrak{F}$, and it follows from Theorem 7.13 a) that $\mathbf{C}_{G_{\mathfrak{F}}}(\overline{V}) \leq F$. Therefore $G = [G_{\mathfrak{F}}, \overline{V}]F$, so $G/[G_{\mathfrak{F}}, \overline{V}] \in \mathfrak{F}$. It then follows that $[G_{\mathfrak{F}}, \overline{V}] = G_{\mathfrak{F}}$ and $\mathbf{C}_{G_{\mathfrak{F}}}(\overline{V}) = 1$. Since $N = G_{\mathfrak{F}} \cap F$ is an abelian normal subgroup of G, it follows from $N \cap \mathbf{C}_{G_{\mathfrak{F}}}(\overline{V}) = 1$ and Theorem III.13.4 that $N = [N, \overline{V}]$. On the other hand, however, N is a minimal normal subgroup of F. Therefore, F and $G = G_{\mathfrak{F}} F$ act as a group of automorphisms of N lying in $\mathfrak{F}(p)$, and we reach the contradiction $[N, \overline{V}] = 1$.

b) If K is a complement of $G_{\mathfrak{F}}$ in G, then there exists by Lemma 7.14 an automorphism α of G such that K^α is a covering $\mathfrak{F}$-subgroup of G. So K is also a covering $\mathfrak{F}$-subgroup of G. □

B. Fischer has shown that the results in this section can be dualized to a large extent. We state without proof a variant of Fischer's results that is due to Gaschütz:

Definition 7.16.

a) A class $\mathscr{K}$ consisting of isomorphism classes of groups is called a Fitting class if the following hold:
(1) If $G \in \mathscr{K}$ and $N \trianglelefteq G$, then $N \in \mathscr{K}$.
(2) If $N_i \trianglelefteq G$ and $N_i \in \mathscr{K}$ $(i = 1, 2)$, then $N_1 N_2 \in \mathscr{K}$.
(Clearly, the collections of nilpotent groups, p-nilpotent groups and groups with $l_p(G) \leq k$ are each Fitting classes.)

b) Let $\mathscr{K}$ be a Fitting class. A subgroup K of G is called a Fischer $\mathscr{K}$-group if the following hold:
(1) $K \in \mathscr{K}$.
(2) If N is a normal subgroup of G, then $N \cap K \in \mathscr{K}$, and no subgroup of N properly containing $N \cap K$ lies in $\mathscr{K}$.
(This is clearly a dualized version of the characterization of covering $\mathscr{F}$-subgroups given in Theorem 7.13.)

The definition of Fischer $\mathscr{K}$-groups given by Fischer himself is somewhat different and is tied directly to Definition 7.8. If $\mathscr{K}$ contains all the subgroups of G whenever G lies in $\mathscr{K}$, then Fischer's definition is equivalent to Definition 7.16. But in general the definitions are different.

Theorem 7.17 (Fischer, Gaschütz, Hartley [138]). *Let $\mathscr{K}$ be a Fitting class, and assume G is solvable. Then:*

a) *G contains exactly one conjugacy class of Fischer $\mathscr{K}$-groups.*
b) *If K is a Fischer $\mathscr{K}$-group of G and $K \leq U \leq G$, then K is a Fischer $\mathscr{K}$-group of U.*
c) *If K is a Fischer $\mathscr{K}$-group of G and $N \trianglelefteq G$, then $N \cap K$ is a Fischer $\mathscr{K}$-group of N.*
d) *If K is a Fischer $\mathscr{K}$-group of G and M/N is a chief factor of G, then either $N(M \cap K) = M$ or $N(M \cap K) = N$.*

For the class $\mathscr{N}$ of nilpotent groups, the Fischer $\mathscr{N}$-groups can be described in the following way:

Theorem 7.18. *Let $\mathscr{N}$ be the class of nilpotent groups, and assume G is solvable. Then the following are equivalent:*

a) *N is a Fischer $\mathscr{N}$-group of G.*
b) *We have $\mathbf{F}(G) \leq N$, and if $N \leq U \leq G$ with U nilpotent, then $U = N$.*
c) *The Sylow p-subgroup of N is a Sylow p-subgroup of $\mathbf{C}_G(\mathbf{F}(G)_{p'})$.*

Conversely, if M_p is a Sylow p-subgroup of $\mathbf{C}_G(\mathbf{F}(G)_{p'})$, then the M_p commute and $\bigtimes_p M_p$ is a Fischer $\mathscr{N}$-group of G.

Our goal for the rest of this section is to prove the converse of Theorem 7.5. For the proof, we will need the following generalization of Theorem V.10.8 (pg. 550):

Theorem 7.19 (Steinberg [397]). *Let G be a finite group, let K be a field and let V be a finite-dimensional K-vector space that is a faithful G-module. Then every irreducible $\mathsf{K}[G]$-module is a composition factor of a tensor product $\mathsf{V} \otimes \underset{r}{\cdots} \otimes \mathsf{V}$ $(r = 1, 2, \dots)$.*

Proof. a) We first show that if $w_1, \dots, w_k$ are pairwise distinct, nonzero vectors lying in a vector space W, then the elements

$$(w_i, w_i \otimes w_i, w_i \otimes w_i \otimes w_i, \dots)$$

in the Cartesian sum of the $\mathsf{W}^{(r)} = \mathsf{W} \otimes \underset{r}{\cdots} \otimes \mathsf{W}$ are linearly independent:

If V_1 and V_2 are vector spaces over K and f is a linear form on V_2, then the map μ defined by $(v_1, v_2)\mu = f(v_2)v_1$ is clearly balanced in the sense of Definition V.9.1 (pg. 539). Therefore, there exists an additive map f^* from $\mathsf{V}_1 \otimes \mathsf{V}_2$ to V_1 defined by $(v_1 \otimes v_2)f^* = f(v_2)v_1$. Clearly, f^* is K-linear.

Suppose that

$$\sum_{i=1}^{n} a_i (w_i \otimes \underset{r}{\cdots} \otimes w_i) = 0 \quad (a_i \in \mathsf{K}; r = 1, 2, \dots) \tag{1}$$

is an expression of linear dependence with n minimal. Then $n \geq 2$, and there exists a linear form f on W that is not constant on $\{w_1, \dots, w_k\}$. If we replace r by $r + s$ in (1) and apply f^* to (1) s-times, then

$$\sum_{i=1}^{n} a_i f(w_i)^s (w_i \otimes \underset{r}{\cdots} \otimes w_i) = 0 \quad (r = 1, 2, \dots; s = 0, 1, \dots, n-1). \tag{2}$$

Let $b_i = f(w_i)$ $(i = 1, \dots, m)$ be the distinct values taken by f on $\{w_1, \dots, w_n\}$. Since

$$\det(b_i^j) \neq 0 \quad (i = 1, \dots, m; j = 0, \dots, m-1),$$

it follows that

$$\sum_{f(w_i) = f(w_1)} a_i (w_i \otimes \underset{r}{\cdots} \otimes w_i) = 0$$

for all $r = 1, 2, \dots$. This contradicts the minimality of n.

b) Let V be a faithful G-module. We let $\mathsf{V}^{(r)} = \mathsf{V} \otimes \underset{r}{\cdots} \otimes \mathsf{V}$ and denote by $\mathbf{T}(\mathsf{V})$ the Cartesian sum of all the $\mathsf{V}^{(r)}$. We define the annihilator of $\mathbf{T}(\mathsf{V})$ to be

$$\mathbf{Ann}(\mathbf{T}(\mathsf{V})) = \{a \in \mathsf{K}[G] \mid \mathbf{T}(\mathsf{V})a = 0\}.$$

We claim that $\mathbf{Ann}(\mathbf{T}(\mathsf{V})) = 0$:

Let $\{v_1, \dots, v_k\}$ be a K-basis of V. We define

$$G_i = \{g \in G \mid v_i g = v_i\}.$$

Since V is a faithful G-module, $\bigcap_{i=1}^{k} G_i = 1$. We set $\mathsf{W} = \mathsf{V} \oplus \underset{k}{\cdots} \oplus \mathsf{V}$ and $w = (v_1, \ldots, v_k)$. Then the elements wg with $g \in G$ are nonzero and pairwise distinct. Let $a = \sum_{g \in G} a(g)\, g$ be an element of the group ring $\mathsf{K}[G]$ such that $\mathsf{W}^{(r)} a = 0$ for $r = 1, 2, \ldots$. For all r, we have

$$0 = (w \otimes \underset{r}{\cdots} \otimes w)a = \sum_{g \in G} a(g)(wg \otimes \underset{r}{\cdots} \otimes wg).$$

It follows from a) that $a = 0$. Since $\mathsf{W}^{(r)}$ is a k^r-fold direct sum of $\mathsf{K}[G]$-modules isomorphic to $\mathsf{V}^{(r)}$, we have that 0 is also the annihilator of $\mathbf{T}(\mathsf{V})$.

c) Let $R = \mathsf{K}[G]$ and $\overline{R} = R/\mathbf{J}(R)$. Let $\overline{R} = \bigoplus_{i=1}^{h} \overline{R}_i$ be the Wedderburn decomposition of $\overline{R}$ into simple algebras $\overline{R}_i$, and let $\mathsf{W}_1, \ldots, \mathsf{W}_h$ be the corresponding irreducible R-modules. The representation of $\overline{R}_i$ on W_i is faithful, and $\mathsf{W}_j \overline{R}_i = 0$ whenever $j \neq i$ (see Theorem V.4.5, pg. 502). By b), we have $\bigcap_r \mathbf{Ann}(\mathsf{V}^{(r)}) = 0$. Since R satisfies the minimal condition, we already have $\bigcap_{r=1}^{s} \mathbf{Ann}(\mathsf{V}^{(r)}) = 0$ for some s. Assume for contradiction that W_i does not appear as a composition factor of $\mathsf{V}^{(r)}$ for any $r = 1, \ldots, s$. If R_i is the preimage of $\overline{R}_i$ in R, then the ideal R_i annihilates every composition factor of $\mathsf{V}^{(r)}$ $(r = 1, \ldots, s)$. For large enough m, it therefore follows that $\mathsf{V}^{(r)} R_i^m = 0$ $(r = 1, \ldots, s)$. Since $\bigcap_{r=1}^{s} \mathbf{Ann}(\mathsf{V}^{(r)}) = 0$, we have $R_i^m = 0$. Then the nilpotent ideal R_i of R lies in the radical $\mathbf{J}(R)$ by Theorem V.2.4 b) (pg. 491). This yields the contradiction $\overline{R}_i = 0$. So W_i is a composition factor of some $\mathsf{V}^{(r)}$. □

Theorem 7.20. *Let G be a group containing no nonidentity normal p-subgroup, and let K be a field of characteristic p. Then there exists a completely reducible, faithful G-module.*

Proof. Let V be the direct sum of all irreducible $\mathsf{K}[G]$-modules. We show that G is faithfully represented on V. Let $g - 1 \in \mathbf{Ann}(\mathsf{V})$. By Definition V.2.1 (pg. 490), the element $g - 1$ then lies in the radical $\mathbf{J}(\mathsf{K}[G])$ of $\mathsf{K}[G]$. By Theorem V.2.4 a), there exists a positive integer k such that $\mathbf{J}(\mathsf{K}[G])^k = 0$. If $p^n \geq k$, then since $\operatorname{char} \mathsf{K} = p$, we have

$$g^{p^n} - 1 = (g - 1)^{p^n} \in \mathbf{J}(\mathsf{K}[G])^{p^n} = 0.$$

Thus, the elements of G that act as the identity map on V constitute a normal p-subgroup of G. By our hypothesis, this is exactly the identity subgroup. □

By Theorem V.5.17 (pg. 515), every normal p-subgroup of G is represented trivially on every irreducible $\mathsf{K}[G]$-module, so the same holds for every completely

reducible $\mathsf{K}[G]$-module. So it follows from Theorem 7.20 that there exists a completely reducible, faithful G-module if and only if G contains no nonidentity normal p-subgroup.

Lemma 7.21. *Let $\mathscr{F}$ be a formation, let G be a group in $\mathscr{F}$, and let A be an abelian normal subgroup of G. If $M \trianglelefteq G$ and $A \leq M \leq \mathbf{C}_G(A)$, then the semidirect product of A with G/M constructed via the automorphism $a^{gM} = a^g$ also lies in $\mathscr{F}$.*

Proof. Let A^* be a G-group isomorphic to A, and let $H = A^*G$ be the semidirect product of A^* with G. If α is a G-isomorphism from A to A^*, then clearly

$$D = \{(a^\alpha, a) \mid a \in A\}$$

is a normal subgroup of H with $A^* \cap D = G \cap D = 1$. It follows from $H = A^*G = DG$ that

$$H/D \cong G/(G \cap D) = G,$$

and so $H/D \in \mathscr{F}$. Since $H/A^* \cong G \in \mathscr{F}$, we obtain $H = H/(A^* \cap D) \in \mathscr{F}$. As A^* is centralized by M, we have $M \lhd H$ and $H/M \in \mathscr{F}$. Clearly, H/M is isomorphic to the semidirect product of A with G/M. □

Lemma 7.22. *Let V_1 and V_2 be $\mathsf{K}[G]$-modules of finite dimension over K, where $\mathsf{K} = \mathrm{GF}(p)$. The set*

$$P = \{(x, y) \mid x \in \mathsf{V}_1 \oplus \mathsf{V}_2, y \in \mathsf{V}_1 \otimes \mathsf{V}_2\}$$

equipped with the multiplication

$$(v_1 + v_2, w)(v_1' + v_2', w') = (v_1 + v_1' + v_2 + v_2', v_1 \otimes v_2' + w + w')$$

is then a p-group, and

$$\{(0, y) \mid y \in \mathsf{V}_1 \otimes \mathsf{V}_2\} = P' \leq \mathbf{Z}(P).$$

If we equip $\mathsf{V}_1 \otimes \mathsf{V}_2$ with the structure of a G-module by defining $(v_1 \otimes v_2)g = v_1g \otimes v_2g$, then P becomes a G-group upon defining

$$(x, y)^g = (xg, yg).$$

Proof. A simple calculation shows that the given multiplication on P is associative, that $(0, 0)$ is the identity element of P, and that $(-v_1 - v_2, v_1 \otimes v_2 - w)$ is the inverse of $(v_1 + v_2, w)$. Since the map $(x, y) \mapsto x$ is an epimorphism from P onto the additive group of $\mathsf{V}_1 \oplus \mathsf{V}_2$, we have

$$P' \leq \{(0, y) \mid y \in \mathsf{V}_1 \otimes \mathsf{V}_2\}.$$

Since

$$(v_1, 0)(v_2, 0)(-v_1, 0)(-v_2, 0) = (0, v_1 \otimes v_2)$$

for all $v_i \in \mathsf{V}_i$ $(i = 1, 2)$, it follows that

$$P' = \{(0, y) \mid y \in \mathsf{V}_1 \otimes \mathsf{V}_2\}.$$

The remaining claims are verified by easy calculations. □

We now arrive at the key lemma:

Lemma 7.23. *Let $\mathscr{F}$ be a saturated formation, and let G be a p-solvable group in $\mathscr{F}$ with p dividing $|G|$. We let $\overline{G} = G/\mathbf{F}_p(G)$. If the elementary abelian p-group V is an irreducible $\overline{G}$-module, then the semidirect product of V with $\overline{G}$ lies in $\mathscr{F}$.*

Proof. We let $\mathbf{F}_p(G) = \mathbf{P}_1(G)$ as in Definition 6.1, and we define U by

$$U/\mathbf{M}_0(G) = \Phi(\mathbf{P}_1(G)/\mathbf{M}_0(G)).$$

By Lemma 6.5, the representation of $\overline{G}$ on $W = \mathbf{F}_p(G)/U$ is faithful.

a) For all $r = 1, 2, \ldots$, the semidirect product of $W^{(r)} = W \otimes \cdots \otimes W$ with $\overline{G}$ lies in $\mathscr{F}$:

We prove the claim by induction on r. We have $G/U \in \mathscr{F}$. Therefore, the semidirect product of W with $\overline{G}$ lies in $\mathscr{F}$ by Lemma 7.21.

Let $r > 1$, and assume for induction that the semidirect product of $W^{(r-1)}$ with $\overline{G}$ lies in $\mathscr{F}$. Following Lemma 7.22, we construct $H = P\overline{G}$, where

$$P = \{(x, y) \mid x \in W^{(r-1)} \oplus W,\ y \in W^{(r-1)} \otimes W = W^{(r)}\}.$$

Then

$$P_1 = \{(x, y) \mid x \in W^{(r-1)},\ y \in W^{(r)}\}$$

and

$$P_2 = \{(x, y) \mid x \in W,\ y \in W^{(r)}\}$$

are normal subgroups of H with $H/P_1 \cong W\overline{G} \in \mathscr{F}$ and $H/P_2 \cong W^{(r-1)}\overline{G} \in \mathscr{F}$. Thus, we have $H/P' \in \mathscr{F}$ as $P_1 \cap P_2 = P'$. Since $\mathscr{F}$ is saturated, it follows from $P' \leq \Phi(P) \leq \Phi(H)$ that $H \in \mathscr{F}$. Applying Lemma 7.21 to H with P' in place of A and P in place of M shows that the semidirect product of P' with $\overline{G}$ lies in $\mathscr{F}$. As P' is a $\overline{G}$-group isomorphic to $W^{(r)}$, we have proved our intermediate claim.

b) By Theorem 7.19, the irreducible $\overline{G}$-module V is a composition factor of some $W^{(r)}$. By a), the semidirect product of $W^{(r)}$ with $\overline{G}$ lies in $\mathscr{F}$. Applying Lemma 7.21 to a suitable homomorphic image of $W^{(r)}\overline{G}$ now shows that the semidirect product of V with $\overline{G}$ lies in $\mathscr{F}$. □

Lemma 7.24. *Let $\mathscr{F}$ be a saturated formation of solvable groups that contains groups of order divisible by p. We denote by $\mathbf{O}_p(G)$ the maximal normal p-subgroup*

of G. Then

$$\mathscr{F}(p) = \{G \mid G/\mathbf{O}_p(G) \cong H/\mathbf{F}_p(H) \text{ for some } H \in \mathscr{F}\}$$

is a formation.

Proof. a) $\mathscr{F}(p)$ is closed under epimorphisms:

Let $G \in \mathscr{F}(p)$, and let α be an epimorphism from G onto $K = G^\alpha$. By the definition of $\mathscr{F}(p)$, there exists some $H \in \mathscr{F}$ such that $G/\mathbf{O}_p(G) \cong H/\mathbf{F}_p(H)$. Since $\mathbf{O}_p(G)^\alpha \leq \mathbf{O}_p(K)$, the group $\overline{K} = K/\mathbf{O}_p(K)$ is isomorphic to a quotient of $H/\mathbf{F}_p(H)$, so it lies in $\mathscr{F}$. If $p \nmid |H|$, then it follows that $H = \mathbf{F}_p(H)$ and $\overline{K} = 1$, so

$$K/\mathbf{O}_p(K) = 1 = H/\mathbf{F}_p(H),$$

and hence K lies in $\mathscr{F}(p)$. So we may now assume that p divides $|H|$. Since $\mathbf{O}_p(\overline{K}) = 1$, there exists a faithful, completely reducible $\overline{K}$-module V over $\mathsf{K} = \mathrm{GF}(p)$ by Theorem 7.20. Let $V = \bigoplus_i V_i$, where the V_i are irreducible $\mathsf{K}[\overline{K}]$-modules. Since p divides $|H|$, the semidirect product of V_i with $H/\mathbf{F}_p(H)$ lies in $\mathscr{F}$ by Lemma 7.23. So $V_i\overline{K}$ also lies in $\mathscr{F}$, as it is a homomorphic image of $V_i(H/\mathbf{F}_p(H))$. Since

$$V\overline{K} \Big/ \bigoplus_{i \neq j} V_j \cong V_i\overline{K} \in \mathscr{F},$$

we have $V\overline{K} \in \mathscr{F}$. As V is a faithful $\overline{K}$-module, we have $\mathbf{M}_0(V\overline{K}) \leq \mathbf{C}_{V\overline{K}}(V) = V$ (see Definition 6.1, pg. 728), so $\mathbf{M}_0(V\overline{K}) = 1$ and therefore $\mathbf{F}_p(V\overline{K}) = V\mathbf{O}_p(\overline{K}) = V$. So $K/\mathbf{O}_p(K) \cong V\overline{K}/\mathbf{F}_p(V\overline{K})$ with $V\overline{K} \in \mathscr{F}$, and hence $K \in \mathscr{F}(p)$.

b) Let G_1 and G_2 be elements of $\mathscr{F}(p)$, and let α_i $(i = 1, 2)$ be epimorphisms from G onto G_i with $\ker\alpha_1 \cap \ker\alpha_2 = 1$. We must prove that $G \in \mathscr{F}(p)$. To this end, we define the following maps:

(1) A monomorphism α from G into $G_1 \times G_2$ defined by $g^\alpha = (g^{\alpha_1}, g^{\alpha_2})$.
(2) The natural epimorphisms β_i $(i = 1, 2)$ from G_i onto $\overline{G_i} = G_i/\mathbf{O}_p(G_i)$, and the epimorphism β from $G_1 \times G_2$ onto $\overline{G_1} \times \overline{G_2}$ defined by $(g_1, g_2)^\beta = (g_1^{\beta_1}, g_2^{\beta_2})$.
(3) By the definition of $\mathscr{F}(p)$, there exist $H_i \in \mathscr{F}$ $(i = 1, 2)$ and epimorphisms γ_i from H_i onto $\overline{G_i}$ with $\ker\gamma_i = \mathbf{F}_p(H_i)$. We define an epimorphism γ from $H_1 \times H_2$ onto $\overline{G_1} \times \overline{G_2}$ by $(h_1, h_2)^\gamma = (h_1^{\gamma_1}, h_2^{\gamma_2})$.
(4) The projections ν_i from $H_1 \times H_2$ onto H_i and the projections μ_i from $\overline{G_1} \times \overline{G_2}$ onto $\overline{G_i}$ $(i = 1, 2)$.

We let $\overline{G} = G^{\alpha\beta}$. Then μ_i induces an epimorphism from $\overline{G}$ onto $\overline{G_i}$ $(i = 1, 2)$ since

$$g^{\alpha\beta\mu_i} = (g^{\alpha_1\beta_1}, g^{\alpha_2\beta_2})^{\mu_i} = g^{\alpha_i\beta_i}$$

and $G^{\alpha_i\beta_i} = \overline{G_i}$.

Let H be the preimage of $\overline{G}$ in $H_1 \times H_2$ with respect to the map γ. The projection ν_1 of $H_1 \times H_2$ onto H_1 induces an epimorphism from H onto H_1:

For any $h_1 \in H_1$, there exists some $(\overline{g_1}, \overline{g_2}) \in \overline{G}$ such that $(\overline{g_1}, \overline{g_2})^{\mu_1} = \overline{g_1} = h_1^{\gamma_1}$ since $\overline{G}^{\mu_i} = \overline{G_i}$. If we choose $h_2 \in H_2$ such that $h_2^{\gamma_2} = \overline{g_2}$, then it follows that $(h_1, h_2) \in H$ and $(h_1, h_2)^{\nu_1} = h_1$. In the same way, ν_2 is an epimorphism from H onto H_2. Since $\ker \nu_1 \cap \ker \nu_2 = 1$ and $H/\ker \nu_i \cong H_i \in \mathscr{F}$, we have $H \in \mathscr{F}$.

Clearly, $\mathbf{F}_p(H_1) \times \mathbf{F}_p(H_2) = \ker \gamma \leq H$. As

$$\mathbf{F}_p(H)^{\nu_i} \leq \mathbf{F}_p(H^{\nu_i}) = \mathbf{F}_p(H_i),$$

we have $\mathbf{F}_p(H) \leq \mathbf{F}_p(H_1) \times \mathbf{F}_p(H_2)$, and so $\ker \gamma = \mathbf{F}_p(H_1) \times \mathbf{F}_p(H_2) = \mathbf{F}_p(H)$. Therefore, $H/\mathbf{F}_p(H) \cong \overline{G}$, and thus $\overline{G}$ has no nontrivial normal p-subgroup. Since

$$\ker \beta \cap G^{\alpha} = (\mathbf{O}_p(G_1) \times \mathbf{O}_p(G_2)) \cap G^{\alpha}$$

is a normal p-subgroup of G^{α}, it now follows that $\mathbf{O}_p(G^{\alpha}) = \ker \beta \cap G^{\alpha}$ and $H/\mathbf{F}_p(H) \cong G/\mathbf{O}_p(G)$. So $G \in \mathscr{F}(p)$. □

We can now prove the converse of Theorem 7.5:

Theorem 7.25 (Lubeseder [321]). *Every saturated formation of solvable groups is a local formation.*

Proof. Let $\mathscr{F}$ be a saturated formation. If $\mathscr{F}$ does not contain any groups of order divisible by p, then we set $\mathscr{F}(p) = \emptyset$; if $\mathscr{F}$ contains groups of order divisible by p, then we define the formation $\mathscr{F}(p)$ as in Lemma 7.24. Let $\mathscr{F}^*$ be the local formation described by the $\mathscr{F}(p)$. By Lemma 7.4, we have $\mathscr{F} \subseteq \mathscr{F}^*$. It remains to show that $\mathscr{F}^* \subseteq \mathscr{F}$.

Assume for contradiction that G is a group of minimum order such that $G \in \mathscr{F}^*$ but $G \notin \mathscr{F}$. If $1 < N \trianglelefteq G$, then $G/N \in \mathscr{F}^*$, which means $G/N \in \mathscr{F}$ as well by the minimality of $|G|$. Hence, G has only one minimal normal subgroup, say N. Since $\mathscr{F}$ is saturated, we must have $\Phi(G) = 1$. Hence, N has a complement K in G by Theorem III.4.4 (pg. 294). As $\mathbf{C}_G(N) \cap K \lhd G$, we have $\mathbf{C}_G(N) = N$. If $|N|$ is a power of the prime p, then $N = \mathbf{O}_p(G)$. As G does not contain any nonidentity normal subgroup of order coprime to p, we have $\mathbf{F}_p(G) = \mathbf{O}_p(G) = N$. Since $G \in \mathscr{F}^*$, we have

$$G/N = G/\mathbf{F}_p(G) \in \mathscr{F}(p).$$

Therefore, there exists an $H \in \mathscr{F}$ such that

$$H/\mathbf{F}_p(H) \cong (G/N)\big/\mathbf{O}_p(G/N) = G/N$$

by the definition of $\mathscr{F}(p)$. By virtue of this isomorphism, N is an irreducible $H/\mathbf{F}_p(H)$-module. So the semidirect product of N with $H/\mathbf{F}_p(H)$, which is isomorphic to G, lies in $\mathscr{F}$ by Lemma 7.23. This is a contradiction. □

8 Rank and the Frattini subgroup

Lemma 8.1. *Let G be a p-solvable group. Suppose that the vector space $\mathsf{V} = \mathsf{V}(n, p)$ of dimension n over $\mathrm{GF}(p)$ is a faithful, irreducible G-module with $(n, |G|) = 1$. Then G is cyclic and $|G|$ divides $p^n - 1$.*

(*It is not possible to remove the assumption that G is p-solvable, as Example V.5.13, pg. 511 shows.*)

Proof. We prove the claim by induction on $|G|$. Let M be a maximal normal subgroup of G. There are two cases:

Case 1: Assume V is a reducible M-module, but not all irreducible constituents of V are isomorphic as M-modules. Then by Theorem V.17.3 (pg. 599), we have $\mathsf{V} = \bigoplus_{i=1}^{m} \mathsf{V}_i$, where the V_i are transitively permuted by G. But then m is a divisor of $(n, |G|) = 1$, contradicting the assumption that $m > 1$.

Case 2: Now assume that V decomposes into $s \geq 1$ irreducible isomorphic M-modules. By the induction hypothesis, M is cyclic. Then it follows by Lemma II.3.11 (pg. 174) that G, when viewed as a permutation group on $\mathsf{V} = \mathsf{V}(n, p)$, is permutation isomorphic to a subgroup of the group of all semilinear maps on the vector space $\mathsf{V}' = \mathsf{V}(s, p^m)$ of dimension s over the field $\mathrm{GF}(p^m)$ with $n = sm$. In addition, $\mathbf{C}_G(M)$ is the group of all maps on V lying in G that are linear over $\mathrm{GF}(p^m)$. So $G/\mathbf{C}_G(M)$ is isomorphic to a subgroup of the automorphism group of $\mathrm{GF}(p^m)$. Since $(m, |G|) = 1$, the group G therefore contains only linear maps. Hence, $G = \mathbf{C}_G(M)$. The quotient G/M is simple. If G/M is cyclic, then it follows that G is commutative. Then by Theorem II.3.10, G is cyclic and $|G|$ divides $p^n - 1$.

If G/M is not cyclic, then G/M has order coprime to p as G is p-solvable. Therefore, the Sylow p-subgroup P of G has a complement Q in G by the Zassenhaus theorem. So $G = P \times Q$. By Clifford's Theorem V.17.3 (pg. 599), V is a direct sum of irreducible Q-modules of the same dimension. Let W be an irreducible Q-submodule of V. Over a splitting field for $\mathrm{GF}(p)[Q]$, the module W decomposes into algebraically conjugate, absolutely irreducible constituents W_i by Theorem V.13.3. Since $p \nmid |Q|$, we have that $\dim \mathsf{W}_i$ divides $|Q|$ by Lemma V.12.6. Altogether, we find that

$$\dim \mathsf{W}_i = (n, |G|) = 1.$$

Thus, Q' is trivially represented on each W_i, which means it is trivially represented on V, as well. But then $Q' = 1$ and G/M is cyclic, a contradiction. □

Definition 8.2. Let

$$1 = G_0 \lhd G_1 \lhd \cdots \lhd G_s = G$$

be a chief series of G.

a) If G $(\neq 1)$ is solvable and $|G_{i+1}/G_i| = p_i^{n_i}$, then we define the arithmetic rank $\overline{r}(G)$ of G to be the least common multiple of the n_i. We set $\overline{r}(1) = 0$.

b) If G ($\neq 1$) is p-solvable and the p-chief factors G_{i+1}/G_i of G in the above chief series have orders $|G_{i+1}/G_i| = p^{n_i}$, then let $\overline{r}_p(G)$ be the least common multiple of these n_i. We again set $\overline{r}_p(1) = 0$.

Lemma 8.3. *Let n be a natural number, let p be a prime and let $\mathscr{F}$ be the formation of groups G with the following properties:*
(1) *G is p-solvable with $\overline{r}_p(G) \mid n$.*
(2) *$(n, |G|) = 1$.*
Then $\mathscr{F}$ is a local formation described by the formations $\mathscr{F}(q)$, where
(i) *$\mathscr{F}(q) = \emptyset$ for $q \mid n$.*
(ii) *$\mathscr{F}(p) = \{A \mid A \text{ abelian}, \exp A \mid (p^n - 1)\}$ if $p \nmid n$.*
(iii) *$\mathscr{F}(q)$ is the class of all groups for all other primes q.*

Proof. a) Let $G \in \mathscr{F}$. Since $(n, |G|) = 1$, condition (i) is satisfied. Let K be a p-chief factor of the p-solvable group G with $|K| = p^k$. Then $k \mid \overline{r}_p(G)$, so $k \mid n$ as well. The quotient

$$H = G/\mathbf{C}_G(K)$$

is isomorphic to a p-solvable group that is represented faithfully and irreducibly on the vector space $\mathsf{V}(k, p)$ of dimension k over $\mathrm{GF}(p)$. Since $(k, |H|)$ divides $(n, |G|)$ and $(n, |G|) = 1$, it follows from Lemma 8.1 that H is a cyclic group of exponent $\exp H$ dividing $p^k - 1$. Since $k \mid n$, we conclude that $H \in \mathscr{F}(p)$. So G lies in the local formation described by the $\mathscr{F}(q)$.

b) Suppose G lies in the local formation described by the $\mathscr{F}(q)$. Let K be a chief factor of G with p dividing $|K|$. Then

$$G/\mathbf{C}_G(K) \in \mathscr{F}(p) = \{A \mid A \text{ abelian}, \exp A \mid (p^n - 1)\}.$$

In particular, $G' \leq \mathbf{C}_G(K)$. Then $K' \leq \mathbf{Z}(K)$, so K is solvable and G is p-solvable.

If $q \mid n$, then $\mathscr{F}(q) = \emptyset$ by (i), so $q \nmid |G|$. This yields $(n, |G|) = 1$.

Let K be a p-chief factor of G with $|K| = p^k$. Then

$$H = G/\mathbf{C}_G(K) \in \mathscr{F}(p),$$

so H is abelian. By Theorem II.3.10 (pg. 174), we have that H is cyclic, $p^k \equiv 1 \bmod |H|$ and k is the smallest natural number such that $p^k \equiv 1 \bmod |H|$. On the other hand, since

$$\mathscr{F}(p) = \{A \mid A \text{ abelian}, \exp A \mid (p^n - 1)\},$$

we have $p^n \equiv 1 \bmod |H|$. Therefore, $k \mid n$. Since this holds for all p-chief factors K of G, we have $\overline{r}_p(G) \mid n$. □

Theorem 8.4 (Huppert [266]). *If G is p-solvable and $(\overline{r}_p(G/\Phi(G)), |G|) = 1$, then $\overline{r}_p(G/\Phi(G)) = \overline{r}_p(G)$.*

Proof. We set $n = \overline{r}_p(G/\Phi(G))$ and define the formation $\mathscr{F}$ to be the class of groups H with the following properties:

(1) H is p-solvable with $\overline{r}_p(H) \mid n$.
(2) $(n, |H|) = 1$.
By Lemma 8.3, $\mathscr{F}$ is a local formation. So $\mathscr{F}$ is a saturated formation by Theorem 7.5. Since $G/\Phi(G) \in \mathscr{F}$, it therefore follows that $G \in \mathscr{F}$. □

Next, we turn to a few special cases of Theorem 8.4.

Definition 8.5.
a) A p-solvable group G is said to be p-supersolvable if $r_p(G) = 1$.
b) If G is solvable and $r(G) = 1$, then G is called supersolvable. Thus, the supersolvable groups are those groups whose chief factors are cyclic.

Theorem 8.6.
a) (*Huppert*) *If $G/\Phi(G)$ is p-supersolvable* (*supersolvable*), *then G is also p-supersolvable* (*supersolvable*).
b) (*Kohler [310]*) *If $r(G/\Phi(G)) = 2$ and $2 \nmid |G|$, then $r(G) = 2$.*

Proof. The claims follow immediately from Theorem 8.4, since if $\overline{r}_p(H) \leq 2$ we have $\overline{r}_p(H) = r_p(H)$. □

Now we give an example to show that the assumption $(\overline{r}_p(G/\Phi(G)), |G|) = 1$ cannot be removed in Theorem 8.4:

Example 8.7. Let p, q, r be prime numbers with $p \mid q-1$ and $pq \mid r-1$. Then there exists a solvable group G of order $p^a q^b r^c$ (for suitable a, b, c) with $\overline{r}(G/\Phi(G)) = p$ and $\overline{r}(G) \equiv 0 \bmod p^{r-1}$.

Proof. Let H be a transitive permutation group of degree q with $|H| = pq$. Let S be a direct product of $k = r-1$ groups H_i, each isomorphic to H. Let F be the free group in the qk free generators f_{ij} $(i = 1, \ldots, k;\ j = 1, \ldots, q)$. We give S the structure of a group of automorphisms of F by defining the following:

If $s \in S$ and $s = h_1 \cdots h_k$ with $h_i \in H_i$, then define

$$f_{ij}^s = f_{ij'}, \quad \text{where} \quad j' = j^{h_i}.$$

Let $\mathbf{K}_r(F)$ be the r-th term of the lower central series of F and $F^r = \langle f^r \mid f \in F \rangle$. We let $R = F/F^r\mathbf{K}_r(F)$. Then R is an r-group of exponent r and nilpotency class at most $r-1$. The group S acts on R in a natural way. Let G be the semidirect product of R with S.

We show:

a) $\overline{r}(G/\Phi(G)) = p$:

Since pq divides $r-1$, the field $\mathrm{GF}(r)$ is a splitting field for H. (It is not necessary to use Brauer's Theorem V.19.11 to prove this, as the representations of the metabelian group H are all monomial by Theorem V.18.5 a).) Since $r \nmid |H_i|$, Maschke's theorem implies that $L_i = \langle f_{ij}F^r\mathbf{K}_2(F) \mid j = 1, \ldots, q \rangle$ is a completely reducible H_i-submodule of $R/\Phi(R)$. The irreducible H_i-submodules of L_i are absolutely irreducible, so they have dimension 1 or $|H_i/H_i'| = p$ by Theorem V.17.10

(pg. 605). Since $H_1 \times \cdots \times H_{i-1} \times H_{i+1} \times \cdots \times H_k$ acts trivially on L_i, the irreducible H_i-submodules of L_i are also irreducible S-modules. Therefore, $R/\Phi(R)$ completely decomposes as an S-module into irreducibles of dimensions 1 or p, and so $\overline{r}(G/\Phi(G)) = r(G/\Phi(G))$ is a divisor of p. Since G is not supersolvable, we have $\overline{r}(G/\Phi(G)) = p$.

b) We let $r_{ij} = f_{ij}F^r\mathbf{K}_r(F)$. Then

$$[r_{12}r_{11}^{-1}, r_{22}r_{21}^{-1}, \ldots, r_{k2}r_{k1}^{-1}] \neq 1 :$$

Let T be a Sylow r-subgroup of the symmetric group of degree r^2. By Theorem III.15.3 (pg. 400), $X = T/\mathbf{K}_r(T)$ is a group of nilpotency class $r-1$ and exponent r. We have $X = \langle a_1, \ldots, a_r, b\rangle$, where

$$a_i^r = b^r = [a_i, a_j] = a_1a_2\cdots a_r = 1 \quad \text{and} \quad a_i^b = a_{i+1} \ (i \bmod r).$$

In particular,

$$[a_1, b, \underset{r-2}{\ldots}, b] = a_1^{-1}a \neq 1 \quad \text{with} \quad a \in \langle a_2, \ldots, a_{r-1}\rangle.$$

Let ϵ be the homomorphism from F to X defined by $f_{12}^{\epsilon} = a_1$, $f_{j2}^{\epsilon} = b$ for $j \geq 2$ and $f_{ji}^{\epsilon} = 1$ otherwise. Since $(F^r\mathbf{K}_r(F))^{\epsilon} = 1$, the map α defined by

$$r_{ij}^{\alpha} = f_{ij}^{\epsilon}$$

is a homomorphism from R to X with

$$[r_{12}r_{11}^{-1}, \ldots, r_{k2}r_{k1}^{-1}]^{\alpha} = [a_1, b, \underset{k-1}{\ldots}, b] \neq 1.$$

c) Let $\mathsf{V}_i = \langle v_{i1}, \ldots, v_{iq}\rangle$ be a representation module over $\mathrm{GF}(r)$ for the permutation representation of H_i of degree q. By Maschke's theorem, the module V_i is completely reducible. We define

$$\mathsf{U}_i = \langle \sum_{j=1}^{q} v_{ij}\rangle \quad \text{and} \quad \mathsf{W}_i = \langle v_{i2} - v_{i1}, \ldots, v_{iq} - v_{i1}\rangle.$$

Then H_i has the trivial representation on the submodule U_i of V_i, and W_i is an H_i-invariant complement of U_i in V_i since $r \neq q$. The representation of H_i' on V_i is the regular representation, which contains the trivial representation of H_i' with multiplicity 1. So U_i is the only one-dimensional H_i-submodule of V_i. As the absolutely irreducible H_i-modules over $\mathrm{GF}(r)$ only have dimensions 1 and p, the submodule W_i decomposes into a direct sum of $s = \frac{q-1}{p}$ absolutely irreducible H_i-modules U_{ij} of dimension p. Then

$$\mathsf{W}_1 \otimes \cdots \otimes \mathsf{W}_k \cong \bigoplus_{j_i} (\mathsf{U}_{1j_1} \otimes \cdots \otimes \mathsf{U}_{kj_k}).$$

By Theorem V.10.3 (pg. 547), every $\mathsf{U}_{1j_1} \otimes \cdots \otimes \mathsf{U}_{kj_k}$ is an absolutely irreducible S-module of dimension p^k.

The map μ from $\mathsf{W}_1 \otimes \cdots \otimes \mathsf{W}_k$ to $\mathbf{K}_k(R)$ defined by

$$\big((v_{1i} - v_{11}) \otimes \cdots \otimes (v_{ki} - v_{k1})\big)\mu = [r_{1i}r_{11}^{-1}, \ldots, r_{ki}r_{k1}^{-1}]$$

is an S-homomorphism since $\mathbf{K}_r(R) = 1$. By b), μ is not the trivial map. So $\mathbf{K}_k(R)$ contains a chief factor of G of order r^{p^k}, and thus $p^k \mid \overline{r}(G)$. □

Exercises

14) If n_1 and n_2 are coprime, then there is an algebra isomorphism

$$\mathrm{GF}(p^{n_1}) \otimes \mathrm{GF}(p^{n_2}) \cong \mathrm{GF}(p^{n_1 n_2}),$$

where the tensor product is constructed over $\mathrm{GF}(p)$.

15) (Huppert [266]) Let n_1 and n_2 be coprime with $n_1 < n_2$. Then there exists a solvable group G with $r(G/\Phi(G)) = n_2$ and $r(G) \geq n_1 n_2$.

(Hint: Let $\mathsf{V}_i = \mathrm{GF}(p^{n_i})^+$ be the vector space of dimension n_i over $\mathrm{GF}(p)$ ($p > 2$), and let $Z = \langle z \rangle$ be a cyclic group of order $p^{n_1 n_2} - 1$. We give V_i the structure of a Z-module in the obvious way. Let $\mathsf{V} = \mathsf{V}_1 \oplus \mathsf{V}_2$, and let W be the homogeneous part of the exterior algebra of V in degree 2. We let $P = \{(v, w) \mid v \in \mathsf{V}, w \in \mathsf{W}\}$, and we make P into a group by defining

$$(v_1, w_1)(v_2, w_2) = (v_1 + v_2, w_1 + w_2 + v_1 \wedge v_2).$$

In a natural way, P is a Z-group. The semidirect product of P with Z yields the desired example. For the proof, use Exercise 14.)

9 Supersolvable groups

In this section, we will investigate supersolvable groups, which were introduced in Definition 8.5. We will show that these groups are closely related to nilpotent groups.

We begin by remarking that every p-nilpotent group is p-supersolvable, and every p-supersolvable group is p-solvable. Furthermore, it follows from Lemma 5.3 that homomorphic images, subgroups and direct products of p-supersolvable groups are again p-supersolvable. By Lemma 8.3, the class of p-supersolvable (supersolvable) groups is a saturated formation.

Theorem 9.1.

a) *If G is p-supersolvable, then the commutator subgroup G' is p-nilpotent.*

b) *If G is supersolvable, then G' is nilpotent.*

c) *If G is supersolvable, then G has a Sylow tower with respect to the reverse of the natural order on the prime numbers. In particular, the Sylow subgroup for the largest prime divisor of $|G|$ is a normal subgroup of G; if q is the smallest prime dividing $|G|$, then G is q-nilpotent.*

d) *If G is solvable with $r(G) \leq 2$ and $p > 3$ is the largest prime divisor of $|G|$, then G has a normal Sylow p-subgroup.*

Proof. a) By our hypothesis, all p-chief factors of G are cyclic of order p. If K is such a chief factor, then $\mathbf{Aut}_G(K)$ is cyclic, as it is a subgroup of the automorphism group of K. Therefore, G' lies in the centralizer $\mathbf{C}_G(K)$ of K. It now follows by Theorem 5.4 b) that

$$G' \leq \bigcap_K \mathbf{C}_G(K) = \mathbf{F}_p(G),$$

where the intersection is taken over all p-chief factors of G. So G' is a subgroup of $\mathbf{F}_p(G)$, and thus G' is p-nilpotent.

b) By a), G' is p-nilpotent for all p, so is nilpotent.

We prove c) and d) together by induction on $|G|$. Let N be a minimal normal subgroup of G. Since $r(G) \leq 2$, we have $|N| = q^n$ for some $n \leq 2$ and prime q. Let p be the largest prime divisor of $|G|$, and let G_p be a Sylow p-subgroup of G. Applying the induction hypothesis to G/N yields $G_pN/N \trianglelefteq G/N$, so $G_pN \trianglelefteq G$. If $q = p$, then we obtain $G_p = G_pN \trianglelefteq G$ and we are finished.

So now assume that $q < p$. Since $|N| = q^n$ with $n \leq 2$, the number of Sylow p-subgroups of G_pN is a divisor of q^2 of the form $1 + kp$. If $k = 0$, then $G_p \trianglelefteq G_pN \trianglelefteq G$, so $G_p \trianglelefteq G$. If $k > 0$, then certainly $1 + kp = q^2$ since $q < p$. So $kp = (q-1)(q+1)$. This necessarily leads to $q = 2$, $p = 3$ and $n = 2$.

The full claim in part c) follows immediately by induction on the number of prime divisors of $|G|$. □

We note that the conclusion of Theorem 9.1 c) is considerably stronger than the conclusion that $l_p(G) \leq 1$ for all p, which follows from Theorem 6.6 c). The conclusion of Theorem 9.1 d) does not hold when $p = 3$, as the alternating group A_4 shows.

Theorem 9.2.

a) *If G is p-supersolvable and U is a maximal subgroup of G, then either $[G : U] = p$ or $[G : U]$ is coprime to p.*

b) *If G is supersolvable, then all maximal subgroups of G have prime index.* (*See also Exercise 13.*)

Proof. a) We prove the claim by induction on $|G|$. Let N be a minimal normal subgroup of G, and let U be a maximal subgroup of G. If $N \leq U$, then U/N is a maximal subgroup of G/N. As G/N is also p-supersolvable, the induction hypothesis implies that the index

$$[G/N : U/N] = [G : U]$$

is either equal to p or coprime to p.

So assume that $N \not\leq U$. If $|N| = p$, then we have $G = NU$ and $N \cap U = 1$. Therefore, $[G : U] = |N| = p$. On the other hand, if $|N|$ is coprime to p, then $G = NU$ and

$$[G : U] = [N : N \cap U].$$

This index is likewise coprime to p.

b) is a trivial consequence of a). □

Theorem 9.3. *Let G be p-solvable. Suppose that for every maximal subgroup U of G, the index $[G : U]$ is either equal to p or coprime to p. Then G is p-supersolvable.*

(*In the simple group of order* 168*, each maximal subgroup has index* 7 *or* 8*, but this simple group is not* 7*-supersolvable. So the condition that G is p-solvable in Theorem 9.3 is necessary.*)

Proof. Suppose G is a counterexample of minimum order. As our hypothesis is inherited by homomorphic images of G, all proper quotients of G are p-supersolvable.

Let N be a minimal normal subgroup of G. If $p \nmid |N|$ or $|N| = p$, then the p-supersolvability of G follows from the p-supersolvability of G/N, contradicting our choice of G as a counterexample. So $|N| = p^k > p$. In particular, N is elementary abelian.

If $N \not\leq \Phi(G)$, then there exists a maximal subgroup U of G with $N \not\leq U$. Then $G = NU$ and $N \cap U = 1$. So $[G : U] = |N| = p^k > p$. However, this contradicts our assumption regarding the indices of the maximal subgroups of G.

Hence, $N \leq \Phi(G)$. Since G/N is p-supersolvable, it follows that $G/\Phi(G)$ is p-supersolvable, as well. Then by Theorem 8.6 a), G is itself p-supersolvable. □

To prove the converse of Theorem 9.2 b), we require the following solvability criterion:

Theorem 9.4 (P. Hall). *If the index of every maximal subgroup of G is prime or the square of a prime, then G is solvable.*

Proof. We prove the claim by induction on $|G|$. The hypothesis is clearly inherited by homomorphic images of G.

If N is a minimal normal subgroup of G, then G/N is solvable by induction. So it suffices to prove that N is solvable. Let p be the largest prime divisor of $|N|$, and let P be a Sylow p-subgroup of N. By Theorem I.7.8 (pg. 36), we have $G = N\mathbf{N}_G(P)$.

If $G = \mathbf{N}_G(P)$, then $P \trianglelefteq G$ and $N = P$ is solvable. Otherwise, $\mathbf{N}_G(P)$ lies in at least one maximal subgroup U of G. By our hypothesis, $[G : U] = q^s$ for some prime q and $s \leq 2$. It now follows from $G = N\mathbf{N}_G(P) = NU$ that

$$[G : U] = [N : N \cap U] = q^s \quad \text{with} \quad s \leq 2.$$

Since $P \leq N \cap U$, we certainly have $p \neq q$. So q is smaller than the largest prime divisor p of $|N|$. Since P is a Sylow p-subgroup of N and of $N \cap U$, it follows by Sylow's theorem that

$$[N : \mathbf{N}_G(P) \cap N] \equiv 1 \mod p \quad \text{and} \quad [U \cap N : \mathbf{N}_G(P) \cap N] \equiv 1 \mod p.$$

Therefore, we have $[N : N \cap U] \equiv 1 \bmod p$ as well. So either $q \equiv 1 \bmod p$ or $q^2 \equiv 1 \bmod p$. The first case cannot occur since $q < p$. In the second case, $q < p$ implies that $p = q + 1$, so $q = 2$ and $p = 3$.

Now we have $[N : N \cap U] = q^2 = 4$. Hence, N has a representation as a transitive permutation group of degree 4. It therefore has a nontrivial solvable factor group. The subgroup N is characteristically simple, as it is a minimal normal subgroup of G, so N is solvable. □

Main Theorem 9.5 (Huppert [257]). *The group G is supersolvable if and only if all maximal subgroups of G have prime index.*

Proof. If G is supersolvable, then all maximal subgroups of G have prime index by Theorem 9.2 b).

If all maximal subgroups of G have prime index, then G is solvable by Theorem 9.4. In particular, G is p-solvable for all p. So G satisfies all the assumptions of Theorem 9.3, and G is therefore p-supersolvable. As this holds for all p, the group G is supersolvable. □

If all maximal subgroups of G have prime or prime-squared index, then G is solvable by Theorem 9.4. However, the rank of G can be greater than two in this case, as Example 8.7 shows for $p = 2$.

Theorem 9.6 (Huppert [257]). *If every maximal subgroup of G is supersolvable, then G is solvable.*

Proof. Let p be the smallest prime divisor of $|G|$. By Theorem 9.1 c), every maximal subgroup of G is p-nilpotent. If G is p-nilpotent, then the normal p-complement of G is supersolvable, so G is solvable. If G is not p-nilpotent, then G is solvable by Theorem IV.5.4 (pg. 462). □

With the help of Theorem 9.6, we can deduce a lattice-theoretic characterization of supersolvable groups:

Theorem 9.7 (Iwasawa [287]). *The group G is supersolvable if and only if every chain of subgroups*

$$1 = U_0 < U_1 < \cdots < U_k = G$$

with U_i maximal in U_{i+1} has the same length k.

(*We remark that, in particular, every group whose lattice of subgroups is modular is supersolvable.*)

Proof. a) First assume that G is supersolvable. By Theorem 9.2 b), the index $[U_{i+1} : U_i]$ in any chain we consider is a prime. So the length of each maximal chain of subgroups is equal to the number of prime divisors of $|G|$ counted with multiplicity. Hence, all chains have the same length.

b) Now assume that all chains of maximal subgroups of G have the same length. We prove the supersolvability of G by induction on $|G|$. Every maximal subgroup of G is supersolvable by induction. Therefore, G is solvable by Theorem 9.6. Then there exists a chain, namely a composition series of G, whose length is equal to the number t of prime divisors of $|G|$. For every maximal subgroup U of G, all of its chain have length $t-1$. Since U is supersolvable, it follows from Theorem 9.2 that $|U|$ has exactly $t-1$ prime divisors, counted with multiplicity. So $[G:U]$ is a prime number. Since this holds for all maximal subgroups U of G, the group G is supersolvable by Theorem 9.5. □

Lemma 9.8 (Baer [25]). *Let N be a normal subgroup of G. Suppose there exists a chain*

$$1 = N_0 \lhd N_1 \lhd \cdots \lhd N_m = N$$

with $N_i \trianglelefteq G$ such that $|N_{i+1}/N_i| = p_i$ is prime. Then $G/\mathbf{C}_G(N)$ is supersolvable.
(*One refers to this situation as a supersolvable embedding of N in G.*)

Proof. We prove the claim by induction on the length m of the given normal series. For $m = 1$, we have $|N_1| = p_1$. Then $G/\mathbf{C}_G(N_1)$ is cyclic, as it is isomorphic to a subgroup of the automorphism group of N_1, so it is certainly supersolvable.

Let $m > 1$. By the induction hypothesis, $G/\mathbf{C}_G(N_{m-1})$ is supersolvable. Applying the induction hypothesis to the group G/N_1 with normal subgroup N/N_1, we conclude that $G/\mathbf{C}_G(N/N_1)$ is supersolvable. We let $C = \mathbf{C}_G(N_{m-1}) \cap \mathbf{C}_G(N/N_1)$. By Lemma 5.3 d), the quotient G/C is supersolvable.

Let $N = \langle N_{m-1}, n_0 \rangle$, and let x be any element of C. Since $x \in \mathbf{C}_G(N_{m-1})$, we have $n^x = n$ if $n \in N_{m-1}$. Furthermore, since $x \in \mathbf{C}_G(N/N_1)$, we have $n^x \equiv n \bmod N_1$ for all $n \in N$. So $n_0^x = n_0 k(x)$ with $k(x) \in N_1$. If y is another arbitrary element of C, then since $y \in \mathbf{C}_G(N_{m-1}) \leq \mathbf{C}_G(N_1)$ we have

$$n_0 k(xy) = n_0^{xy} = (n_0 k(x))^y = n_0^y k(x) = n_0 k(y) k(x).$$

As N_1 is commutative, we see that $k(xy) = k(x)k(y)$. Hence, the map μ defined by $x^\mu = k(x)$ is a homomorphism from C to the cyclic group N_1. The kernel of this homomorphism is clearly $\mathbf{C}_G(N)$. So $C/\mathbf{C}_G(N)$ is cyclic, and therefore $G/\mathbf{C}_G(N)$ is supersolvable. □

Now we can show that a supersolvable embedding of a small part of a chief series of G forces G to be supersolvable:

Theorem 9.9. *Assume G is solvable. Suppose there exists a normal series*

$$\Phi(G) = N_0 \lhd N_1 \lhd \cdots \lhd N_m = \mathbf{F}(G)$$

with $N_i \trianglelefteq G$ such that the factor groups N_{i+1}/N_i have prime order. Then G is supersolvable.

Proof. By Theorem III.4.2 d) (pg. 292), the Fitting subgroup of $\overline{G} = G/\Phi(G)$ is $\mathbf{F}(\overline{G}) = \mathbf{F}(G)/\Phi(G)$. The pair $\overline{G}, \mathbf{F}(\overline{G})$ satisfies the hypothesis of Lemma 9.8. So $\overline{G}/\mathbf{C}_{\overline{G}}(\mathbf{F}(\overline{G}))$ is supersolvable. Since G is solvable, we have that $\mathbf{C}_{\overline{G}}(\mathbf{F}(\overline{G})) \leq \mathbf{F}(\overline{G})$ by Theorem III.4.2 b). So $\overline{G}/\mathbf{F}(\overline{G})$ is supersolvable. Together with the given embedding of $\mathbf{F}(\overline{G})$ in $\overline{G}$, this implies that $\overline{G}$ is supersolvable. Since $\overline{G} = G/\Phi(G)$, it follows from Theorem 8.6 a) that G is supersolvable. □

We conclude this section with a useful observation regarding groups with nilpotent commutator subgroups. This observation holds, in particular, for supersolvable groups by Theorem 9.1 b).

Theorem 9.10.

a) *Assume G' is nilpotent. If H is a Hall subgroup of G, then $H' \trianglelefteq G$.*

b) (*Inagaki [267]*) *Assume G is solvable. If $H' \trianglelefteq G$ for all Hall subgroups H of G, then G' is nilpotent.*

Proof. a) The hypothesis is clearly inherited by homomorphic images. Suppose G is a counterexample of minimum order, and let H be a Hall subgroup of G with $H' \not\trianglelefteq G$.

Assume for contradiction that N_1 and N_2 are nontrivial normal subgroups of G with coprime orders. Then

$$(HN_i/N_i)' = H'N_i/N_i \trianglelefteq G/N_i,$$

and so $H'N_i \trianglelefteq G$ $(i = 1, 2)$. Therefore, $H' = H'N_1 \cap H'N_2 \trianglelefteq G$, contradicting our choice of G and H.

By what we have just shown, the Fitting subgroup $\mathbf{F}(G)$ of G is a p-group for some prime p. Since G' is nilpotent, G' is also a p-group. If $p \nmid |H|$, then $H' \leq H \cap G' = 1$, contradicting $H' \not\trianglelefteq G$. Hence, p divides $|H|$, which means $G' \leq H$ and so $H \trianglelefteq G$. But then $H' \trianglelefteq G$ in this case, as well.

b) The hypothesis in b) is inherited by homomorphic images. If G is a smallest counterexample, then G has exactly one minimal normal subgroup. The Fitting subgroup $\mathbf{F}(G)$ of G is therefore a p-group for some prime p. If $\Phi(G) \neq 1$, then $(G/\Phi(G))' = G'\Phi(G)/\Phi(G)$ is nilpotent. Then G' is also nilpotent by Theorem III.3.5 (pg. 285), contradicting our choice of G. So $\Phi(G) = 1$.

Let P be a Sylow p-subgroup of G. Since $P' \trianglelefteq G$ and $P' \leq \Phi(P)$, it follows from Lemma III.3.3 a) (pg. 284) that $P' \leq \Phi(G) = 1$. So P is abelian. Let Q be a p-complement of G. We have $Q' = 1$ since $Q' \cap \mathbf{F}(G) = 1$. Hence, G is the product of the abelian groups P and Q. By Itô's Theorem 4.4 (pg. 714), G' is abelian. □

Exercises

16) (Doerk [116]) Suppose that every proper subgroup of G is supersolvable, but G itself is not supersolvable. Show that:

a) G contains exactly one normal Sylow subgroup, say G_p.

b) $G_p/\Phi(G_p)$ is a minimal normal subgroup of $G/\Phi(G_p)$ and is not cyclic.
c) There exists a normal series

$$1 = N_0 \lhd N_1 \lhd \cdots \lhd N_k = \Phi(G_p)$$

for G such that each quotient N_{i+1}/N_i is cyclic of prime order.
d) $\Phi(G_p) \leq \mathbf{Z}(G_p)$; in particular, G_p has nilpotency class at most 2.
e) G_p' has exponent at most p.
f) For $p > 2$, G_p has exponent p; for $p = 2$, G_p has exponent at most 4.
g) If Q is a complement of G_p in G, then

$$Q \cap \mathbf{C}_G(G_p/\Phi(G_p)) = \Phi(G) \cap Q.$$

h) Let $\overline{Q} = Q/(Q \cap \mathbf{C}_G(G_p/\Phi(G_p)))$. Then $\overline{Q}$ is a minimal nonabelian group (that is, all proper subgroups of $\overline{Q}$ are abelian, but $\overline{Q}$ itself is not), or $\overline{Q}$ is a cyclic group of prime power order.
i) Q is generated by two elements, and $|Q|$ has at most two distinct prime divisors.

17) Give an example of a group G with the following properties:
a) $|G|$ is divisible by exactly three distinct primes.
b) Every proper subgroup of G is supersolvable, but G itself is not.

18) (Baer) If every subgroup of G generated by two elements is supersolvable, then G is itself supersolvable.

19) (McLain [329]) The group G is supersolvable if and only if for every subgroup U of G and for every divisor d of $|U|$, there exists a subgroup D of U with $|D| = d$. (Compare this exercise with Theorem 1.11.)

20) (Doerk [116]) If G contains four supersolvable subgroups with pairwise coprime indices, then G is supersolvable. (Compare this exercise with Theorem 1.9!)

21) (Baer [19]) Suppose $G = N_1 N_2 \cdots N_k$, where the N_i are supersolvable normal subgroups of G. The group G is supersolvable if and only if G' is nilpotent.

22) Give an example of a group G with the following properties:
a) $G = N_1 N_2$ with $N_i \lhd G$ $(i = 1, 2)$.
b) N_1 and N_2 are supersolvable, but G itself is not.

23) (Gaschütz [162]) Let $\mathscr{F}$ be the saturated formation consisting of all supersolvable groups. The covering $\mathscr{F}$-subgroups of a solvable group G are exactly the supersolvable subgroups F of G with the following property: If $F \leq U < V \leq G$, then $[V : U]$ is not a prime number.

24) Determine the covering $\mathscr{F}$-subgroups of the symmetric group S_4 for the formations $\mathscr{F}$ consisting of nilpotent groups and consisting of supersolvable groups.

25) (Kegel [303]) Suppose $G = AB = BC = CA$. Assume A and B are nilpotent and C is supersolvable. Then G is supersolvable.
(Hint: Show that a minimal counterexample has the following properties:
a) G contains exactly one minimal normal subgroup M, and M is not cyclic.

b) If p is the largest prime divisor of $|G|$ and P is a Sylow p-subgroup of G, then $M = P = A_pB_p = B_pC_p = C_pA_p$, and M is elementary abelian: Use Exercise 10 and Maschke's Theorem.
c) We have $A_p \cap B_p = 1$ and $A_p \neq 1 \neq B_p$.
d) $|A_p| = |B_p| = p$ and $|M| = p^2$:
Let K be an A-invariant complement of A_p in M. Consider a subgroup T such that $T^A = T$ and $[T : M] = p$. If $|B_p| \geq p^2$, then letting $D = T \cap B_p$ we have that D^G is a normal subgroup of G with $M \not\leq D^G$.
e) A and B are cyclic:
$A_{p'}$ acts faithfully on the cyclic group K.)

10 Products of cyclic groups

Theorem 10.1 (J. Douglas [118], N. Itô [277]). *Let $G = Z_1Z_2$ be the product of two cyclic groups Z_1 and Z_2.*
a) *If $|Z_1| \geq |Z_2| > 1$, then there exists a normal subgroup N of G with $1 < N \leq Z_1$.*
b) *G is supersolvable.*

Proof. a) Let $Z_1 = \langle a \rangle$ and $Z_2 = \langle b \rangle$. Assume for contradiction that Z_1 contains no nonidentity normal subgroup of G, so $Z_1 \cap Z_2 = 1$ and $b \notin Z_1$. The permutation encoding the action of a on the cosets of Z_1 has the form

$$\begin{pmatrix} Z_1, & Z_1b, & Z_1b^2, & \cdots \\ Z_1, & Z_1ba, & Z_1b^2a, & \cdots \end{pmatrix}.$$

We consider the cycle $(Z_1b, Z_1ba, \ldots, Z_1ba^{r-1})$ in this permutation. We have $Z_1ba^r = Z_1b$ and therefore $ba^rb^{-1} \in Z_1$. Since

$$[G : Z_1] = [Z_2 : Z_1 \cap Z_2] = |Z_2|$$

and the coset Z_1 is not contained in the cycle we are considering, we have $r \leq |Z_2| - 1 < |Z_1|$. But then $a^r \neq 1$ and $\langle a^r \rangle \lhd \langle a, b \rangle = G$, a contradiction.

b) As every homomorphic image of G is again a product of two cyclic groups, the claim in b) follows immediately by a) and induction on $|G|$. □

Theorem 10.2. *Let $G = G_1 \cdots G_n$ for some pairwise permutable subgroups G_i.*
a) *If every product G_iG_j has a Sylow tower of type $\prec$, then so does G.*
b) *If every product $G_iG_jG_k$ is supersolvable, then so is G.*

Proof. a) Let p be the smallest prime divisor of $|G|$ with respect to the order $\prec$. Then every G_i has exactly one Sylow p-subgroup P_i. By Theorem 4.7, the product P_iP_j is a Sylow p-subgroup of G_iG_j. Therefore, $P = P_1 \cdots P_n$ is a Sylow p-subgroup of G. Since G_iG_j has a Sylow tower of type $\prec$, we have $P_iP_j \trianglelefteq G_iG_j$. Hence, it follows for all $g_j \in G_j$ that

$$P_i^{g_j} \leq P_i P_j \leq P,$$

and so $P \trianglelefteq G$. The full claim regarding the Sylow tower now follows easily by induction.

b) Suppose G is a minimal counterexample. It is clear that every proper homomorphic image of G satisfies the assumptions, so is supersolvable. We show:

(1) G has a Sylow tower with respect to the order $\prec$, which is the reverse of the natural order on the primes:

Every $G_i G_j$ is supersolvable, so has a Sylow tower of type $\prec$ by Theorem 9.1 c). By a), G also has a Sylow tower of type $\prec$.

(2) Let p be the largest prime divisor of $|G|$, and let P be the normal Sylow p-subgroup of G. Then P is a minimal normal subgroup of G and $\mathbf{C}_G(P) = P$:

If $\Phi(G) > 1$, then by Theorem 8.6 a), G is supersolvable since $G/\Phi(G)$ is supersolvable. But this contradicts our choice of G as a counterexample, so $\Phi(G) = 1$.

Let N be a minimal normal subgroup of G with $N \leq P$. Since $\Phi(G) = 1$, there exists a maximal subgroup U of G such that $G = NU$ and $N \cap U = 1$. Then $S = \mathbf{C}_G(N) \cap U \lhd G$. If $S > 1$, then as G/N and G/S are supersolvable, so is $G = G/(N \cap S)$, contradicting our choice of G. So $S = 1$ and $\mathbf{C}_G(N) = N$. Since $1 < \mathbf{Z}(P) \cap N \lhd G$, it then follows that $N = P$.

Up to this point, we have only used the fact that every product $G_i G_j$ is supersolvable.

(3) G/P is abelian:

Since $G_i G_j$ is solvable, there exist p-complements Q_i and Q_j of G_i, resp. G_j, such that $Q_i Q_j$ is a p-complement of $G_i G_j$ by Theorem 4.6. By Theorem 1.8, $Q_i Q_j$ lies in a p-complement Q_{ijk} of the supersolvable group $G_i G_j G_k$. Let P_i be the normal Sylow p-subgroup of G_i. It now follows by Theorem 9.10 that

$$(Q_i Q_j)' \leq Q'_{ijk} \lhd G_i G_j G_k,$$

and since $P_i P_j P_k \trianglelefteq G_i G_j G_k$, we then have

$$(Q_i Q_j)' \leq \mathbf{C}_G(P_i P_j P_k) \leq \mathbf{C}_G(P_k).$$

Using (2), this gives

$$(Q_i Q_j)' \leq \mathbf{C}_G(P) = P,$$

so $(Q_i Q_j)' = 1$. Now G/P is the product of the commuting abelian groups $G_i P/P = Q_i P/P$, so is abelian.

(4) G is supersolvable:

Since $P_i P_j \trianglelefteq G_i G_j$, we have $P_i P_j G_j = P_i G_j \leq G$. So G_j permutes with all the P_i. In particular, $G_j P$ is a subgroup of G.

Suppose that $G = G_j P$ for some j. Then P_j is a normal subgroup of G by (2). Also by (2), $P_j = P$ or $P_j = 1$. If $P_j = P$, then $G = G_j$ is supersolvable. So assume that $P_j = 1$ and p divides $|G_i|$. By the supersolvability of $G_i G_j$ with Sylow

p-subgroup P_i, there exists a subgroup N such that $|N| = p$ and $N \lhd G_iG_j$. Then $N \lhd G_jP = G$, and we are done.

So now assume that $G_jP < G$ for all j. Then by the induction hypothesis,

$$G_jP = P_1 \cdots P_{j-1}G_jP_{j+1} \cdots P_n$$

is supersolvable. So P decomposes completely as a G_j-module into one-dimensional constituents by Maschke's theorem. If we set $\overline{G_j} = G_jP/P$, then since $\mathbf{C}_G(P) = P$, we have that $\exp \overline{G_j}$ divides $p - 1$. Therefore, $\exp \overline{G}$ divides $p - 1$ as well.

Since the abelian group $\overline{G}$ acts faithfully and irreducibly on P, the group $\overline{G}$ is cyclic by Theorem II.3.10 (pg. 174). So it follows that $|\overline{G}| = \exp \overline{G}$ divides $p - 1$. By Theorem II.3.10, we have $|P| = p$ and G is supersolvable. □

Theorem 10.3 (Huppert [256]). *If $G = Z_1 \cdots Z_n$ is the product of pairwise permutable cyclic groups Z_i, then G is supersolvable.*

Proof. By Theorem 10.2, it suffices to prove that every product $Z_1Z_2Z_3$ of three cyclic groups Z_i is supersolvable. Let G be a minimal counterexample. The products Z_iZ_j are supersolvable by Theorem 10.1. Therefore, step (2) in the proof of Theorem 10.2 is applicable. Using the notation of Theorem 10.2, we may assume that $P = P_1P_2P_3$ is a minimal normal subgroup of G, that G/P is supersolvable and $\mathbf{C}_G(P) = P$. Since

$$PZ_j \cap Z_iZ_j = (P \cap Z_iZ_j)Z_j = (P_iP_j)Z_j = P_iZ_j,$$

we have that $P_iZ_j = Z_jP_i$ for all i, j.

Assume first that $PZ_iZ_j = G$ for some pair i, j. As Z_iZ_j is supersolvable, if p divides $|Z_iZ_j|$ then there exists some subgroup $N \trianglelefteq Z_iZ_j$ with $|N| = p$. Since P is commutative, we have $N \lhd G$, and so $N = P$ by the minimality of P. On the other hand, if $p \nmid |Z_iZ_j|$, then it follows that

$$|P| = |P_1P_2P_3| = p.$$

So assume that $PZ_iZ_j < G$ for all i, j. Then

$$PZ_iZ_j = P_kZ_iZ_j,$$

and this product is supersolvable by the induction hypothesis. Let Q_i be the p-complement of the cyclic group Z_i. By Theorem 4.8 b), Q_iQ_j is a p-complement of $P_kZ_iZ_j$. It follows from Theorem 9.10 that

$$(Q_iQ_j)' \leq \mathbf{C}_G(P) = P,$$

so $(Q_iQ_j)' = 1$. Hence, G has an abelian p-complement Q.

Since $PZ_iZ_j < G$ for all i and j, the Q_i are all nonidentity subgroups. Suppose, say, $P_1 \neq 1$. Letting $P_1 = \langle x_1 \rangle$ and $Q_1 = \langle y_1 \rangle$, we then have for all $y \in Q$ that

$$x_1^{yy_1} = x_1^{y_1 y} = x_1^y.$$

By the irreducibility of P as a G/P-module, we have

$$P = \langle x_1^y \mid y \in Q \rangle \leq \mathbf{C}_G(Q_1),$$

and so $y_1 \in \mathbf{C}_G(P) = P$. This leads to $y_1 = 1$, a contradiction. □

For products of two cyclic groups, Theorems 4.4 and 10.3, together with the structure of the Sylow p-subgroups for $p > 2$ given in Theorem III.11.5 (pg. 356), yield an extensive, if not complete, structure theory. In contrast, nothing is known at the time of writing about the products of three or more cyclic groups besides Theorem 10.3. For example, it is an open question whether Theorem 4.4 can be generalized to conclude that the derived length of a product of n pairwise permutable cyclic groups is at most n.

11 System normalizers of solvable groups

Definition 11.1. Let

$$\mathscr{S} = \{G_{p_i} \mid p_i \text{ divides } |G|\} \quad \text{and} \quad \mathscr{K} = \{G_{p_i'} \mid p_i \text{ divides } |G|\}$$

be a Sylow system, resp. complement system, of the solvable group G that correspond according to Theorem 2.2. We define

$$\mathbf{N}(\mathscr{S}) = \bigcap_i \mathbf{N}_G(G_{p_i}) \quad \text{and} \quad \mathbf{N}(\mathscr{K}) = \bigcap_i \mathbf{N}_G(G_{p_i'}),$$

and call $\mathbf{N}(\mathscr{S})$, resp. $\mathbf{N}(\mathscr{K})$, the system normalizer of the Sylow system $\mathscr{S}$, resp. the complement system $\mathscr{K}$.

Clearly, $\mathbf{N}(\mathscr{S}) = \mathbf{N}(\mathscr{K})$.

Theorem 11.2 (P. Hall [213]). *Assume G is solvable.*

a) *Every system normalizer of G is nilpotent. If G_p is the Sylow p-subgroup in $\mathscr{S}$ and $G_{p'}$ is the p-complement of G in the corresponding complement system $\mathscr{K}$, then $\mathbf{N}_G(G_{p'}) \cap G_p$ is the Sylow p-subgroup of $\mathbf{N}(\mathscr{S})$.*

b) *All system normalizers of G are conjugate in G.*

Proof. a) Since $\mathbf{N}_G(G_p)$ has only one Sylow p-subgroup, namely G_p, the subgroup $\mathbf{N}(\mathscr{S})$ of $\mathbf{N}_G(G_p)$ also has only one Sylow p-subgroup, namely $\mathbf{N}(\mathscr{S}) \cap G_p$. So $\mathbf{N}(\mathscr{S})$ is nilpotent.

We have

$$\mathbf{N}(\mathscr{S}) \cap G_p \leq \mathbf{N}_G(G_{p'}) \cap G_p.$$

The group $\mathbf{N}_G(G_{p'}) \cap G_p$ normalizes $G_{p'}$. Furthermore, since for $q \neq p$ we have

$$\mathbf{N}_G(G_{p'}) \cap G_p \leq G_p \leq G_{q'},$$

the group $\mathbf{N}_G(G_{p'}) \cap G_p$ also normalizes the q-complement $G_{q'}$ in $\mathscr{K}$ (see Theorem 2.2). So $\mathbf{N}_G(G_{p'}) \cap G_p$ lies in the Sylow p-subgroup of $\mathbf{N}(\mathscr{S})$.

b) The conjugacy of system normalizers follows immediately from the conjugacy of Sylow systems, which we proved in Theorem 2.4. □

Theorem 11.3 (P. Hall [213]). *Let $\mathbf{N}(\mathscr{K})$ be the system normalizer of the complement system $\mathscr{K} = \{G_{p'} \mid p \text{ divides } |G|\}$ of the solvable group G, and let N be a normal subgroup of G. Then $\mathbf{N}(\mathscr{K})N/N$ is the system normalizer of the complement system $\mathscr{K}^* = \{G_{p'}N/N \mid p \text{ divides } |G/N|\}$ of G/N.*

Proof. We may assume that N is a minimal normal subgroup of G. Then $|N|$ is a prime power, say q^n. It is clear that $\mathbf{N}(\mathscr{K})N/N \leq \mathbf{N}(\mathscr{K}^*)$. Let $xN \in \mathbf{N}(\mathscr{K}^*)$, so $(G_{p'}N)^x = G_{p'}N$ for all Sylow complements $G_{p'}$ in $\mathscr{K}$. For $p \neq q$, we have $G_{p'}^x = G_{p'}$ since $N \leq G_{p'}$.

By the conjugacy of all q-complements of $G_{q'}N$ (see Theorem 1.8), it follows from $G_{q'}^x N = G_{q'}N$ that there exists an $n \in N$ such that $G_{q'}^x = G_{q'}^n$. Since $n \in N \leq G_q \leq G_{p'}$ for $p \neq q$, we also have that

$$G_{p'}^x = G_{p'} = G_{p'}^n.$$

Hence, we have $xn^{-1} \in \bigcap_p \mathbf{N}_G(G_{p'}) = \mathbf{N}(\mathscr{K})$ and then $x \in \mathbf{N}(\mathscr{K})N$. □

The system normalizers introduced in Definition 11.1 are an important link between the arithmetic structure of a solvable group and its normal subgroup structure. The rest of this section is dedicated to exploring this connection. We begin with some preparatory definitions and lemmas:

Definition 11.4. A Sylow set of G is a collection $\mathscr{T} = \{G_p \mid p \text{ divides } |G|\}$ of Sylow subgroups of G that contains exactly one Sylow p-subgroup G_p for every prime divisor p of $|G|$. The intersection $\mathbf{N}(\mathscr{T}) = \bigcap_p \mathbf{N}_G(G_p)$ is the Sylow set normalizer of $\mathscr{T}$.

Definition 11.5. Let $B \leq A \leq G$ and $V \leq G$. Then

$$\frac{|B(A \cap V)|}{|B|} = \frac{|B||A \cap V|}{|B||B \cap V|} = \frac{|A \cap V|}{|B \cap V|} = \frac{|A \cap BV|}{|B|}$$

is the number of cosets of B in A that have the form Bv for some $v \in V$. Since $B \subseteq A \cap BV \subseteq A$, we have immediately that

$$1 \leq \frac{|A \cap V|}{|B \cap V|} \leq \frac{|A|}{|B|}.$$

We say that:

a) V covers A/B if $[A : B] = [A \cap V : B \cap V]$; this is equivalent to $B(A \cap V) = A$, so $A \subseteq BV$.

b) V avoids A/B if $[A \cap V : B \cap V] = 1$; this is equivalent to $B(A \cap V) = B$, so $A \cap V \leq B$.

Lemma 11.6. *Let $G \geq A_1 \geq \cdots \geq A_k$ and $G \geq V$. If V covers (avoids) the A_i/A_{i+1} for $i = 1, 2, \ldots, k-1$, then V also covers (avoids) A_1/A_k.*

Proof. This follows immediately from

$$[A_1 \cap V : A_k \cap V] = \prod_{i=1}^{k-1} [A_i \cap V : A_{i+1} \cap V]. \qquad \square$$

Definition 11.7.

a) The chief factor H/K of G is said to be central if all inner automorphisms of G act as the identity automorphism on H/K; that is, for all $h \in H$ and all $g \in G$, we have $h^g K = hK$. This is clearly equivalent to each of the following two conditions:

$$H/K \leq \mathbf{Z}(G/K) \quad \text{resp.} \quad [H, G] \leq K.$$

b) Every non-central chief factor is called eccentric.

Lemma 11.8. *Let $\mathbf{N}(\mathscr{T})$ be the normalizer of the Sylow set $\mathscr{T}$ of the solvable group G. Then:*

a) *$\mathbf{N}(\mathscr{T})$ avoids every eccentric chief factor of G.*

b) *$\mathbf{N}(\mathscr{T})$ is nilpotent.*

Proof. a) Let $\mathscr{T} = \{G_p \mid p \text{ divides } |G|\}$, and let H/K be a chief factor of G that is not avoided by $\mathbf{N}(\mathscr{T})$. We must show that H/K is central. Without loss of generality, we may assume that $K = 1$ because of the trivial inequality

$$\mathbf{N}(\mathscr{T})K/K \leq \bigcap_p \mathbf{N}_{G/K}(G_p K/K).$$

Let $1 \neq h \in H \cap \mathbf{N}(\mathscr{T})$. We let $L = \langle [h, g] \mid g \in G \rangle$. By Lemma III.1.2 b) (pg. 267), we have for all $x \in G$ that

$$[h, g]^x = [h, x]^{-1}[h, gx] \in L.$$

So $L \trianglelefteq G$. The subgroup H has prime power order, say $|H| = q^n$, as H is a minimal normal subgroup of G. Let $g \in G_p$ with $p \neq q$. Since $h \in \mathbf{N}_G(G_p)$, it then follows that $[h, g] \in H \cap G_p = 1$. Hence, we have

$$\langle G_p \mid p \neq q \rangle \leq \mathbf{C}_G(h),$$

and therefore $[G : \mathbf{C}_G(h)]$ is a power of q. So we have $G = \mathbf{C}_G(h)G_q$, and it follows from the nilpotency of G_q that

$$L = \langle [h, g] \mid g \in G \rangle \leq [H, G_q] < H.$$

As H is a minimal normal subgroup of G and $L \trianglelefteq G$, this forces $L = 1$. Therefore, $1 \neq h \in H \cap \mathbf{Z}(G)$, so $H \leq \mathbf{Z}(G)$ and H is central.

b) is trivial, as $\mathbf{N}(\mathscr{T})$ has only one Sylow p-subgroup, namely $\mathbf{N}(\mathscr{T}) \cap G_p$, since $\mathbf{N}(\mathscr{T}) \leq \mathbf{N}_G(G_p)$. □

Remark 11.9. The conclusion of Lemma 11.8 a) does not necessarily hold for non-solvable groups:

Let G be the alternating group A_5. Consider the Sylow set $\mathscr{T} = \{G_2, G_3, G_5\}$ consisting of the subgroups

$$G_2 = \langle (1,2)(3,4), (1,4)(2,3) \rangle$$
$$G_3 = \langle (1,4,5) \rangle$$
$$G_5 = \langle (1,2,3,4,5) \rangle.$$

Then $(1,4)(2,3) \in \mathbf{N}(\mathscr{T})$, but G clearly does not contain any central chief factors.

Main Theorem 11.10 (P. Hall [213]). *Assume G is solvable. Then every system normalizer of G covers every central chief factor and avoids every eccentric chief factor of G.*

Proof. By Theorem 11.8 a), every eccentric chief factor is avoided by every system normalizer $\mathbf{N}(\mathscr{S})$.

Let H/K be a central chief factor of G. We must prove that $H \leq \mathbf{N}(\mathscr{S})K$. Let

$$\mathscr{S} = \{G_p \mid p \text{ divides } |G|\} \quad \text{and} \quad \mathscr{S}^* = \{G_pK/K \mid p \text{ divides } |G/K|\}.$$

The claim then follows by Theorem 11.3:

$$H/K \leq \mathbf{Z}(G/K) \leq \mathbf{N}(\mathscr{S}^*) = \mathbf{N}(\mathscr{S})K/K.$$ □

Now we extract a few simple consequences from Theorem 11.10:

Theorem 11.11. *Assume G is solvable. Then:*

a) *No proper normal subgroup of G contains a system normalizer of G.*

b) *The group generated by all system normalizers of G is G itself.*
(*In particular, if $G \neq 1$ then the system normalizers of G are not the identity subgroup.*)

c) *If $\mathscr{S}$ is a Sylow system of G, then*

$$\bigcap_{g \in G} \mathbf{N}(\mathscr{S})^g = \bigcap_{g \in G} \mathbf{N}_G(\mathbf{N}(\mathscr{S}))^g = \mathbf{Z}_\infty(G).$$

Proof. a) If M is a maximal normal subgroup of G, then G/M is a central chief factor of G, so it is covered by every system normalizer $\mathbf{N}(\mathscr{S})$ by Theorem 11.10. So $\mathbf{N}(\mathscr{S}) \not\leq M$.

b) Since all system normalizers are conjugate by Theorem 11.2 b), the subgroup they generate is a normal subgroup of G. By a), this subgroup cannot be smaller than G.

c) The hypercenter $\mathbf{Z}_\infty(G)$ of G contains only central chief factors of G, so Theorem 11.10 and Lemma 11.6 imply that $\mathbf{Z}_\infty(G)$ is covered by every system normalizer. This means

$$\mathbf{Z}_\infty(G) \leq \bigcap_{g \in G} \mathbf{N}(\mathscr{S})^g.$$

We let $Z = \mathbf{Z}_\infty(G)$. Let

$$\mathscr{S} = \{G_p \mid p \text{ divides } |G|\} \quad \text{and} \quad \mathscr{S}^* = \{G_p Z/Z \mid p \text{ divides } |G/Z|\}.$$

Then by Theorem 11.3,

$$\mathbf{N}_G(\mathbf{N}(\mathscr{S}))Z/Z \leq \mathbf{N}_{G/Z}(\mathbf{N}(\mathscr{S})Z/Z) = \mathbf{N}_{G/Z}(\mathbf{N}(\mathscr{S}^*)).$$

If we can show that $\bigcap_{g^* \in G/Z} \mathbf{N}_{G/Z}(\mathbf{N}(\mathscr{S}^*))^{g^*} = 1$, then it follows that

$$\bigcap_{g \in G} \mathbf{N}_G(\mathbf{N}(\mathscr{S}))^g \leq Z = \mathbf{Z}_\infty(G),$$

and we are done. So we may assume that $\mathbf{Z}_\infty(G) = 1$.

Let N be a minimal normal subgroup of G that lies in one (and therefore in all) $\mathbf{N}_G(\mathbf{N}(\mathscr{S}))$. Since $\mathbf{Z}_\infty(G) = 1$, the subgroup N is eccentric, and so $\mathbf{N}(\mathscr{S})$ avoids N by Theorem 11.10. Since $N \leq \mathbf{N}_G(\mathbf{N}(\mathscr{S}))$, it follows that

$$[N, \mathbf{N}(\mathscr{S})] \leq N \cap \mathbf{N}(\mathscr{S}) = 1.$$

Then by b), N is also centralized by $\langle \mathbf{N}(\mathscr{S})^g \mid g \in G \rangle = G$. But this contradicts $\mathbf{Z}_\infty(G) = \mathbf{Z}(G) = 1$. □

The covering-avoidance property in Theorem 11.10 does not characterize the system normalizers, as the following example shows:

Example 11.12 (Hawkes [226]). Let $p > 3$. Let $B = \langle b_1, b_2, b_3 \rangle$ be a group with defining relations

(1) $b_i^p = [b_i, b_j] = b_1 b_2 b_3 = 1$ $(i, j = 1, 2, 3)$.

So B is elementary abelian of order p^2. In the same way, we construct $C = \langle c_1, c_2, c_3 \rangle$ with

(2) $c_i^p = [c_i, c_j] = c_1 c_2 c_3 = 1$.

Let $Z = \langle z \rangle$ be the cyclic group of order p. We let $T = B \times C \times Z$. The maps a_i $(i = 1, 2, 3)$ from T to itself defined by

(3) $z^{a_i} = z b_i c_i^{-1}$

(4) $b_i^{a_i} = b_i c_i^{-1}$

(5) $b_j^{a_i} = b_j c_i c_j$ $(i \neq j)$

(6) $c_j^{a_i} = c_j$

clearly define automorphisms of T. A simple calculation shows that

(7) $a_i^p = [a_i, a_j] = a_1 a_2 a_3 = 1$.

The group $A = \langle a_1, a_2, a_3 \rangle$ is elementary abelian of order p^2: Otherwise, A would be cyclic of order p, and we would have $A = \langle a_2 \rangle$ and $a_1 = a_2^k$ for some integer k. But from this would follow the contradiction $b_2 c_1 c_2 = b_2^{a_1} = b_2^{a_2^k} = b_2 c_2^{-k}$.

Let $P = TA$ be the semidirect product of T with A. By the symmetry in the relations (1)–(7), every permutation s in the symmetric group S_3 gives rise to an automorphism α of P defined by

$$a_i^\alpha = a_{i^s}, \quad b_i^\alpha = b_{i^s}, \quad c_i^\alpha = c_{i^s}, \quad z^\alpha = z.$$

Let $G = PS_3$ be the semidirect product of P with S_3. We have $|G| = p^7 \cdot 3 \cdot 2$. The group G has the normal series

$$1 \lhd P \lhd PA_3 \lhd G,$$

where $|P| = p^7$, $|PA_3/P| = 3$ and $|G/PA_3| = 2$; here, A_3 is the alternating group of degree 3. So G has a Sylow tower. The group G has the chief series

$$1 \lhd C \lhd B \times C \lhd T \lhd P \lhd PA_3 \lhd G.$$

The only central chief factors in this chief series are $T/(B \times C)$ and G/PA_3; in particular, S_3 acts on all chief factors of order p^2 according to its two-dimensional irreducible representation. So all system normalizers of G have order $2p$ by Theorem 11.10.

C is the unique minimal normal subgroup of G:

Let N be a minimal normal subgroup of G. If the order of N is not a power of p, then we obtain the contradiction $N \leq S_3 \cap \mathbf{C}_G(P) = 1$. So $N \leq P$. Since $1 < N \cap \mathbf{Z}(P) \lhd G$, we have in fact that $N \leq \mathbf{Z}(P) = C$. So $N = C$.

Let $s = (1,2) \in S_3$. Then the Sylow system $\{P, \langle (1,2,3) \rangle, \langle (1,2) \rangle\}$ of G is normalized by $D = \langle s, z \rangle$. Since $|D| = 2p$, the group D is a system normalizer of G.

Since $[zc_3, s] = 1$, the subgroup $S = \langle zc_3, s \rangle$ of G has order $2p$. The group S avoids the eccentric chief factor C, and since

$$SC/C = DC/C,$$

the subgroup S avoids (covers) all remaining eccentric (central) chief factors of G. But S is not a system normalizer of G:

Assume for contradiction that S is a system normalizer. Then there exists an $x \in G$ such that $D^x = S$. Then $s^x = s$ and $z^x = (zc_3)^k$ for some integer k. The first equality forces

$$x \in \mathbf{C}_G(s) = \langle z, a_3, b_3, c_3, s \rangle.$$

Let $x = z^e b_3^{x_1} c_3^{x_2} a_3^{x_3} s^t$. Then we have

$$z^x = z^{a_3^{x_3} s^t} = z b_3^{x_3} c$$

for some $c \in C$. From the relations

$$z b_3^{x_3} c = z^x = (z c_3)^k$$

it now follows that $x_3 = 0$, and we obtain the contradiction

$$z = z^x = (z c_3)^k.$$

So S has the covering-avoidance property, but S is not a system normalizer of G.

In Theorem 11.3, we showed that for every homomorphism μ of G and for every Sylow system $\mathscr{S}$ of G, we have $\mathbf{N}(\mathscr{S})^\mu = \mathbf{N}(\mathscr{S}^\mu)$. We now show that it is entirely possible that $\mathbf{N}_G(\mathbf{N}(\mathscr{S}))^\mu < \mathbf{N}_{G^\mu}(\mathbf{N}(\mathscr{S}^\mu))$:

Example 11.13 (Alperin [5]). Let $p > 3$, and let $P = \langle x_1, x_2, x_3 \rangle$ be the p-group defined by the relations

$$x_i^p = [x_i, x_j]^p = [[x_i, x_j], x_k] = 1 \quad (i, j, k = 1, 2, 3).$$

Then P is a p-group of exponent p and nilpotency class 2. If we define $x_{ij} = [x_i, x_j]$ for $i \neq j$, then every element $x \in P$ can be written uniquely in the form

$$x = x_1^{n_1} x_2^{n_2} x_3^{n_3} x_{12}^{n_{12}} x_{13}^{n_{13}} x_{23}^{n_{23}}$$

with $0 \leq n_i < p$ and $0 \leq n_{ij} < p$. We have $|P| = p^6$ and

$$P' = \mathbf{Z}(P) = \langle x_{12}, x_{13}, x_{23} \rangle.$$

By the symmetry in the defining relations for P, every permutation $s \in S_3$ gives rise to an automorphism α of P defined by

$$x_i^\alpha = x_{i^s} \quad \text{and} \quad x_{ij}^\alpha = x_{i^s j^s}.$$

Let $G = PS_3$ be the semidirect product of P with S_3. Let $r = (1, 2)$ and $t = (1, 2, 3)$. Then since $p > 3$, the set

$$\mathscr{S} = \{P, \langle t \rangle, \langle r \rangle\}$$

is a Sylow system of G. We let $D = \mathbf{N}(\mathscr{S})$. Clearly, $\langle r \rangle$ is the Sylow 2-subgroup of D. Since PA_3/P is an eccentric chief factor of G, we have $3 \nmid |D|$ by Theorem 11.10. Let

$$x = x_1^{n_1} x_2^{n_2} x_3^{n_3} x_{12}^{n_{12}} x_{13}^{n_{13}} x_{23}^{n_{23}}$$

be an element of the Sylow p-subgroup D_p of D. Since

$$[x, r] \in P \cap \langle r \rangle = 1 \quad \text{and} \quad [x, t] \in P \cap \langle t \rangle = 1,$$

we have $x^r = x^t = x$. This yields

$$\begin{aligned} x = x^r &= x_2^{n_1} x_1^{n_2} x_3^{n_3} x_{21}^{n_{12}} x_{23}^{n_{13}} x_{13}^{n_{23}} \\ &= x_1^{n_2} x_2^{n_1} x_3^{n_3} x_{12}^{-n_{12}-n_1 n_2} x_{13}^{n_{23}} x_{23}^{n_{13}} \end{aligned}$$

and

$$\begin{aligned} x = x^t &= x_2^{n_1} x_3^{n_2} x_1^{n_3} x_{23}^{n_{12}} x_{21}^{n_{13}} x_{31}^{n_{23}} \\ &= x_1^{n_3} x_2^{n_1} x_3^{n_2} x_{12}^{-n_{13}-n_1 n_3} x_{13}^{-n_{23}-n_2 n_3} x_{23}^{n_{12}}. \end{aligned}$$

It follows from this that $n_1 = n_2 = n_3$ and $n_{12} = n_{13} = n_{23} = -n_1 n_2/2$. So $|D_p| = p$, and D_p is generated by

$$d = x_1 x_2 x_3 (x_{12} x_{13} x_{23})^{-e},$$

where $2e \equiv 1 \bmod p$.

Let $g \in \mathbf{N}_G(D)$. Since $D_2 = \langle r \rangle$, we have $r^g = r$. As no Sylow 3-subgroup of G is centralized by r, the order $|\mathbf{N}_G(D)|$ of the normalizer is not divisible by 3. Therefore, $\mathbf{N}_G(D) = D_2 \times L$, where $L = \mathbf{C}_P(r) \cap \mathbf{N}_P(\langle d \rangle)$. It is clear that $x_{13}x_{23} \in L$.

Let $w = w^r \in L$. As above, it follows that

$$w = x_1^a x_2^a x_3^b x_{12}^{-a^2/2} x_{13}^c x_{23}^c$$

for appropriate integers a, b, c. If $w \in \mathbf{N}_P(\langle d \rangle)$, then

$$[w, d] \in \langle d \rangle \cap P' = 1,$$

so in fact $w \in \mathbf{C}_P(d)$. Therefore,

$$\begin{aligned} [w, d] = [x_1^a x_2^a x_3^b, x_1 x_2 x_3] &= x_{12}^a x_{13}^a x_{21}^a x_{23}^a x_{31}^b x_{32}^b \\ &= x_{13}^{a-b} x_{23}^{a-b} = 1. \end{aligned}$$

This forces $a = b$. Hence, $\mathbf{N}_G(D) = \langle r, d, x_{13}x_{23} \rangle$. Since $x_1 x_2 P' \in \mathbf{N}_{G/P'}(DP'/P')$, we have

$$\mathbf{N}_G(D)P'/P' < \mathbf{N}_{G/P'}(DP'/P').$$

Definition 11.14. A subgroup U of G is said to be abnormal in G if $g \in \langle U, U^g \rangle$ for every element $g \in G$.

Theorem 11.15. *The following are equivalent:*

a) *U is abnormal in G.*

b) *Whenever $U \leq H \leq G$ and $U \leq H \cap H^g$, we have $g \in H$.*

(*In particular, if $U \leq H \leq G$, then $\mathbf{N}_G(H) = H$.*)

Proof. a) $\Rightarrow$ b): If $U \leq H \cap H^g$, then it follows from the abnormality of U that

$$g^{-1} \in \langle U, U^{g^{-1}} \rangle \leq H,$$

so $g \in H$.

b) $\Rightarrow$ a): We let $H = \langle U, U^g \rangle$. Then $U \leq H \cap H^{g^{-1}}$, and so $g^{-1} \in H$ by b). □

Lemma 11.16 (P. Hall). *Let $N \trianglelefteq G$, and suppose U is a subgroup of G such that NU is abnormal in G.*

a) *If $\mathbf{N}_{NU}(U) = U$, then $\mathbf{N}_G(U) = U$.*

b) *If U is abnormal in NU, then U is abnormal in G.*

Proof. a) Let $g \in \mathbf{N}_G(U)$. Since NU is abnormal in G, we have

$$g \in \langle NU, (NU)^g \rangle = NU,$$

and so $g \in \mathbf{N}_{NU}(U) = U$, as well.

b) For every $g \in G$, we have

$$g \in \langle NU, (NU)^g \rangle = N\langle U, U^g \rangle.$$

So $g = nh$ for a suitable $n \in N$ and $h \in \langle U, U^g \rangle$. Since U is abnormal in NU, we have

$$n \in \langle U, U^n \rangle = \langle U, U^{gh^{-1}} \rangle \leq \langle U, U^g \rangle.$$

Hence, it also follows that

$$g = nh \in \langle U, U^g \rangle,$$

and so U is abnormal in G. □

Theorem 11.17 (Taunt). *Assume G is solvable, and let $U \leq G$. Suppose that for all subgroups H of G with $U \leq H \leq G$, we have $\mathbf{N}_G(H) = H$. Then U is abnormal in G.*

(*At the time of writing, it is not known if the solvability of G is necessary here.*)

Proof (P. Hall). We prove the claim by induction on $|G|$. Let N be a minimal normal subgroup of G. We may assume that $U < G$. Let

$$UN/N \leq V/N \leq G/N.$$

Then

$$\mathbf{N}_{G/N}(V/N) = \mathbf{N}_G(V)/N = V/N.$$

By the induction hypothesis, we have that UN/N is abnormal in G/N, and hence UN is abnormal in G. If $UN < G$, then it follows by induction that U is abnormal in UN. Then U is abnormal in G by Lemma 11.16 b). On the other hand, if $G = UN$, then U is a maximal, non-normal subgroup of G since G is solvable, and so U is trivially abnormal in G. □

The following generalization of Theorem I.7.6 (pg. 35) is an important tool for constructing abnormal subgroups:

Lemma 11.18. *Let $N \trianglelefteq G$, and let P be a Sylow p-subgroup of N. Then $\mathbf{N}_G(P)$ is abnormal in G.*

Proof. We let $U = \mathbf{N}_G(P)$, and we must prove that $g \in \langle U, U^g \rangle$. Let $H = \langle U, U^g \rangle$. Then P and P^g are Sylow p-subgroups of $N \cap H$, so they are already conjugate in $N \cap H$. Therefore, there exists an $h \in N \cap H$ such that $P^g = P^h$. It follows that

$$gh^{-1} \in \mathbf{N}_G(P) = U \leq H,$$

and so $g \in H = \langle U, U^g \rangle$. Hence, $\mathbf{N}_G(P)$ is abnormal in G. □

We prove a complement to Theorem 11.11 a):

Lemma 11.19. *Assume G is solvable. Let M be a maximal, but non-normal, subgroup of G.*

a) *If $[G : M] = p^m$ and K is a p-complement of M, then $\mathbf{N}_G(K) \leq M$.*

b) *If $\mathscr{K}_G = \{G_{q'} \mid q \text{ divides } |G|\}$ is a complement system of G such that*

$$\mathscr{K}_M = \{M \cap G_{q'} \mid q \text{ divides } |M|\}$$

is a complement system of M, then $\mathbf{N}_G(\mathscr{K}_G) \leq \mathbf{N}_M(\mathscr{K}_M)$.

Proof. a) If M contains a nontrivial normal subgroup N of G, then it follows by induction that

$$\mathbf{N}_G(K)N/N \leq \mathbf{N}_{G/N}(KN/N) \leq M/N,$$

and we are done. So assume that $\bigcap_{g \in G} M^g = 1$. The representation of G as a permutation group on the cosets of M is therefore faithful and primitive. Let N be the unique minimal normal subgroup of G, and let L be a minimal normal subgroup of M. Then $G = NM$ and $N \cap M = 1$, so $|N| = [G : M] = p^m$. By Theorem II.3.2 e) (pg. 167), $|L|$ is a power of a prime q with $q \neq p$. Hence, $L \leq K$. If $g \in \mathbf{N}_G(K)$, then

$$L^g \leq NL \cap K = L,$$

and therefore $g \in \mathbf{N}_G(L) = M$. So we have $\mathbf{N}_G(K) \leq M$.

b) Let $[G : M] = p^m$. Then $G_{p'} \cap M$ is a p-complement of M, and so it is a p-complement of G, as well. It follows that $G_{p'} = G_{p'} \cap M \leq M$. By a), we then have

$$\mathbf{N}_G(\mathscr{K}_G) \leq \mathbf{N}_G(G_{p'}) \leq M$$

and therefore

$$\mathbf{N}_G(\mathscr{K}_G) \leq \mathbf{N}_G(G_{q'}) \cap M \leq \mathbf{N}_M(G_{q'} \cap M)$$

for all q. This yields the claim $\mathbf{N}_G(\mathscr{K}_G) \leq \mathbf{N}_M(\mathscr{K}_M)$. □

Definition 11.20. A subgroup U of G is called subabnormal in G if there exists a chain of subgroups

$$U = U_0 < U_1 < \cdots < U_s = G$$

in which U_i is abnormal in U_{i+1}.

We now prove a peculiar characterization of system normalizers:

Theorem 11.21 (P. Hall [213]). *Assume G is a solvable group, and let $\mathscr{T}$ be the set of subabnormal subgroups of G. Then the system normalizers of G are exactly the minimal elements of $\mathscr{T}$ with respect to the natural partial order on $\mathscr{T}$ given by inclusion.*

Proof. a) We first show that every element of $\mathscr{T}$ contains a system normalizer of G. Let U be subabnormal in G. Then there exists a chain of subgroups

$$U = U_0 < U_1 < \cdots < U_s = G$$

such that U_i is abnormal in U_{i+1}. By refining this chain, we may ensure that U_i is maximal, but not normal, in U_{i+1}. By Lemma 11.19 b), every system normalizer of U_i contains a system normalizer of U_{i+1}. So $U = U_0$ contains a system normalizer of $G = U_s$.

b) It remains to show that every system normalizer of G is subabnormal in G. We follow the proof given by Carter:

If G is nilpotent, then the only subabnormal subgroup of G is G itself, and G is the only system normalizer of G. So assume that G is not nilpotent. Then there exists a chief factor H/K of G such that K is nilpotent, but H is not. Let $|H/K| = p^a$. No Sylow p-subgroup H_p of H is normal in H. Let $T = \mathbf{N}_G(H_p)$. By Lemma 11.18, the subgroup T is abnormal in G. It follows from Theorem I.7.8 (pg. 36) that

$$G = \mathbf{N}_G(H_p)H = \mathbf{N}_G(H_p)H_pK = TK.$$

We show that every system normalizer D of T is a system normalizer of G. By induction, we may assume that D is subabnormal in T. Since T is abnormal in G, the subgroup D is subabnormal in G.

Let

$$\mathscr{S}_K = \{K_q \mid q \text{ divides } |K|\} \quad \text{and} \quad \mathscr{S}_T = \{T_q \mid q \text{ divides } |T|\}$$

be Sylow systems of K, resp. T, and let $D = \mathbf{N}_T(\mathscr{S}_T)$. Since K is nilpotent and $K \lhd G$, we have $K_q \lhd G$ for all q. So the Sylow subgroups K_qT_q of $G = KT$ are pairwise permutable, and

$$\mathscr{S}_G = \{K_qT_q \mid q \text{ divides } |G|\}$$

is a Sylow system of G. Since $K_q \lhd G$, we have $\mathbf{N}_T(\mathscr{S}_T) \leq \mathbf{N}_G(K_q T_q)$, and so $\mathbf{N}_T(\mathscr{S}_T) \leq \mathbf{N}_G(\mathscr{S}_G)$.

Since $\mathbf{N}_T(\mathscr{S}_T) = D$ is subabnormal in G, the subgroup $\mathbf{N}_T(\mathscr{S}_T)$ contains a system normalizer of G by a). So it follows that $\mathbf{N}_T(\mathscr{S}_T) = \mathbf{N}_G(\mathscr{S}_G)$, and we are done. □

Remark 11.22. Theorem 11.21 implies the following: If $\mathbf{N}(\mathscr{S})$ is a system normalizer of the solvable group G, then there exists a chain of subgroups

$$\mathbf{N}(\mathscr{S}) = U_0 < U_1 < \cdots < U_s = G$$

in which U_i is maximal, but not normal, in U_{i+1}. No proper subgroup of $\mathbf{N}(\mathscr{S})$ has this property by Theorem 11.21. However, Theorem 11.21 does not imply that every maximal chain of subgroups

$$U_0 < U_1 < \cdots < U_s = G$$

in which U_i is maximal, but not normal, in U_{i+1} ends in a system normalizer U_0:

Let $G = S_4$ be the symmetric group of degree 4. Then $D = \langle (1,2) \rangle$ is a system normalizer of G. One chain of the form described in Theorem 11.21 is

$$D < \langle (1,2), (1,2,3) \rangle < S_4.$$

However, if P is a Sylow 2-subgroup of S_4, then $P < S_4$ is a maximal chain with each term abnormal in the next, but P is not a system normalizer of S_4. Hence, not all chains with abnormal steps, but rather only suitable ones, descend to the system normalizers.

12 Carter subgroups of solvable groups

Definition 12.1. The nilpotent, self-normalizing subgroups of a solvable group G are called the Carter subgroups of G.

Theorem 12.2 (Carter [87]). *Let G be a solvable group.*

a) *The Carter subgroups of G are exactly the covering $\mathscr{F}$-subgroups of G corresponding to the saturated formation $\mathscr{F}$ of nilpotent groups.*

b) *Carter subgroups of G exist, and all Carter subgroups of G are conjugate in G.*

c) *Every Carter subgroup C of G is abnormal in G.*

Proof. Let $\mathscr{F}$ be the saturated formation consisting of nilpotent groups.

a) If C is a covering $\mathscr{F}$-subgroup of G, then C is nilpotent, and $\mathbf{N}_G(C) = C$ by Lemma 7.11.

Conversely, assume C is a nilpotent subgroup of G with $\mathbf{N}_G(C) = C$. Assume for contradiction that we have a chain of subgroups $C \leq U \leq G$ with U/K nilpotent and $KC < U$. Then there exists a maximal subgroup M of U with $KC \leq M < U$. Since

U/K is nilpotent, $M \lhd U$. By induction on $|G|$, we may assume that C is a covering $\mathscr{F}$-subgroup of M. Then for all $u \in U$, C^u is a covering $\mathscr{F}$-subgroup of M. So C^u is conjugate to C in M by b), which we may assume holds for M by induction. But then we obtain the contradiction

$$U = M\mathbf{N}_G(C) = MC = M.$$

Hence, C is a covering $\mathscr{F}$-subgroup of G.

b) follows immediately from a) and Theorem 7.10 b).

c) For all subgroups U such that $C \leq U \leq G$, we have $\mathbf{N}_G(U) = U$ by Lemma 7.11. Hence, U is abnormal by Theorem 11.17. □

The discovery of Carter subgroups was suggested by the classical theory of Lie algebras. Over an algebraically closed field of characteristic zero, there always exist nilpotent subalgebras, the so-called Cartan subalgebras, that are not contained in any larger subalgebra as an ideal. These Cartan subalgebras are the crucial tool for classifying the simple Lie algebras. However, in finite nonsolvable groups, there need not exist any nilpotent self-normalizing subgroups, as the example of the alternating group A_5 shows. Solvable Lie algebras are supersolvable, so they correspond only to the small class of supersolvable groups. In this case, the Carter subgroups coincide with the system normalizers by Theorem 12.4, and they are therefore well understood. Hence, there are hardly any useful indications in the theory of Lie algebras for how to study the subtle properties of Carter subgroups.

We extract the following lemma from Lemma 7.9 (pg. 740) and Theorem 7.13 (pg. 743).

Lemma 12.3. *Let C be a Carter subgroup of the solvable group G.*

a) *If N is a normal subgroup of G, then CN/N is a Carter subgroup of G/N.*

b) *C is a maximal nilpotent subgroup of G.*

Theorem 12.4 (Carter [86]). *If G contains a nilpotent normal subgroup K with nilpotent quotient G/K, then the system normalizers of G are exactly the Carter subgroups of G.*

Proof. Let D be a system normalizer of G. By Theorem 11.3, DK/K is a system normalizer of the nilpotent group G/K, so $G = KD$. Let

$$\mathscr{S}' = \{K_p \mid p \text{ divides } |K|\} \quad \text{and} \quad \mathscr{S}'' = \{D_p \mid p \text{ divides } |D|\}$$

be Sylow systems of K, resp. D. Since $K_p \trianglelefteq G$,

$$\mathscr{S} = \{K_pD_p \mid p \text{ divides } |G|\}$$

is a Sylow system of G. Clearly, $D \leq \mathbf{N}(\mathscr{S})$, so $D = \mathbf{N}(\mathscr{S})$. For $g \in \mathbf{N}_G(D)$, we have $D_p^g = D_p$, and therefore

$$(K_pD_p)^g = K_p^g D_p^g = K_pD_p.$$

Hence, $g \in \mathbf{N}(\mathscr{S}) = D$. So $\mathbf{N}_G(D) = D$. Since D is nilpotent by Theorem 11.2 a), the subgroup D is a Carter subgroup of G. The claim now follows from the fact that the collection of system normalizers and the collection of Carter subgroups are each a single conjugacy class. □

Our next goal is a characterization of Carter subgroups that mirrors the characterization of system normalizers in Theorem 11.21.

Lemma 12.5. *Let M be a maximal but non-normal subgroup of the solvable group G, and assume that $M/\bigcap_{g\in G} M^g$ is nilpotent. Then M contains a Carter subgroup of G.*

Proof. We let $N = \bigcap_{g\in G} M^g$. If C is a Carter subgroup of G, then CN/N is a Carter subgroup of G/N by Lemma 12.3. As M/N is nilpotent and self-normalizing in G/N, it is a Carter subgroup of G/N. Hence, CN and M are conjugate, and M contains a Carter subgroup of G. □

Lemma 12.6. *Assume G is solvable, but not nilpotent. Then there exists a maximal but non-normal subgroup M of G such that $M/\bigcap_{g\in G} M^g$ is nilpotent.*

Proof. Since G is not nilpotent, we have $\mathbf{K}_\infty(G) \neq 1$. Let $\mathbf{K}_\infty(G)/N$ be a chief factor of G. Since G/N is not nilpotent, there exists a maximal but non-normal subgroup M/N of G/N. So $\mathbf{K}_\infty(G) \not\leq M$, hence $G = M\mathbf{K}_\infty(G)$ and

$$N \leq M \cap \mathbf{K}_\infty(G) \lhd \mathbf{K}_\infty(G).$$

This forces $N = M \cap \mathbf{K}_\infty(G)$, and then

$$M/N \cong M\mathbf{K}_\infty(G)/\mathbf{K}_\infty(G) = G/\mathbf{K}_\infty(G)$$

is nilpotent. Since $M \not\lhd G$ and $N \leq \bigcap_{g\in G} M^g$, the subgroup M satisfies the conditions given in the statement of the lemma. □

Theorem 12.7 (Erber [120]). *Assume G is solvable, and let U be a subgroup of G. Suppose there exists a chain of subgroups*

$$U = U_0 < U_1 < \cdots < U_s = G$$

with the following properties:

(1) *U_i is maximal, but not normal, in U_{i+1} $(i = 0, 1, \ldots, s-1)$.*
(2) *$U_i / \bigcap_{x\in U_{i+1}} U_i^x$ is nilpotent $(i = 0, 1, \ldots, s-1)$.*
(3) *The chain cannot be extended from below while preserving conditions* (1) *and* (2).

Then U is a Carter subgroup of G.

(*Note that here, in contrast to the situation in Theorem 11.21, every maximal chain ends in a Carter subgroup of G.*)

Proof. Assume that we have found a Carter subgroup C of G with $C \leq U_{i+1}$. Then C is a Carter subgroup of U_{i+1}. By Lemma 12.5, some conjugate of C lies in U_i. So U_i also contains a Carter subgroup of G. Hence, U contains a Carter subgroup C^* of G. As the chain cannot be extended, U is nilpotent by Lemma 12.6. Since $\mathbf{N}_U(C^*) = C^*$, it follows that $U = C^*$ by Theorem III.2.3. □

Little is known about the relationship between Carter subgroups and system normalizers. The following theorem gives some information in this direction.

Theorem 12.8 (Carter [87]). *Assume G is solvable. Then every Carter subgroup C of G contains a system normalizer of G. More precisely: If*

$$\mathscr{S} = \{G_q \mid q \text{ divides } |G|\}$$

is a Sylow system of G such that $G_q \cap C = C_q$ for all q (such a Sylow system $\mathscr{S}$ exists by Theorem 2.5), then $\mathbf{N}(\mathscr{S}) \leq C$.

Proof. The first claim follows immediately from Theorem 11.21 since the Carter subgroups are abnormal (see Theorem 12.2).

We prove the sharper statement by induction on $|G|$. Let N be a minimal normal subgroup of G, and let $|N| = p^n$. Then

$$\mathscr{S}^* = \{G_qN/N \mid q \text{ divides } |G/N|\}$$

is a Sylow system of G/N with $C_qN/N \leq G_qN/N$. Here, C_qN/N is the Sylow q-subgroup of the Carter subgroup CN/N of G/N by Lemma 12.3. By the induction hypothesis, we have

$$\mathbf{N}(\mathscr{S})N/N \leq \mathbf{N}(\mathscr{S}^*) \leq CN/N,$$

and so $\mathbf{N}(\mathscr{S}) \leq CN$. Hence, $\mathbf{N}(\mathscr{S})$ normalizes $CN \cap G_q = C_q$ (for $q \neq p$) as well as $CN \cap G_p = NC_p$. Then $\mathbf{N}(\mathscr{S})$ also normalizes the centralizers $\mathbf{C}_G(C_q)$ $(q \neq p)$, so $\mathbf{N}(\mathscr{S})$ likewise normalizes

$$\bigcap_{q \neq p} \mathbf{C}_G(C_q) \cap NC_p.$$

This group has the form TC_p for a suitable $T \leq N$. Therefore,

$$L = \prod_{q \neq p} C_q \times TC_p$$

is a nilpotent subgroup of G containing C. Then by Lemma 12.3, we have $L = C$ and thus $TC_p = C_p$. Therefore, C_p is also normalized by $\mathbf{N}(\mathscr{S})$, and it follows that $\mathbf{N}(\mathscr{S}) \leq \mathbf{N}_G(C) = C$. □

Lemma 12.9.

a) *Let N be a normal subgroup of G with p-power order. Then $N \leq \mathbf{Z}_\infty(G)$ if and only if $|G/\mathbf{C}_G(N)|$ is a power of p.*

b) *If P is a Sylow p-subgroup of G with $P \leq \mathbf{Z}_\infty(G)$, then P is a direct factor of G.*

Proof. a) First assume $N \leq \mathbf{Z}_\infty(G)$, and let Q be a Sylow q-subgroup of G with $q \neq p$. Then NQ is nilpotent, as it admits an upper central series since $N \leq \mathbf{Z}_\infty(G)$. So $Q \leq \mathbf{C}_G(N)$. Since this holds for all primes q distinct from p, we have that $|G/\mathbf{C}_G(N)|$ is a power of p.

Conversely, assume that $|G/\mathbf{C}_G(N)|$ is a power of p, and let P be a Sylow p-subgroup of G. Then $G = \mathbf{C}_G(N)P$. We let $N_i = \mathbf{Z}_i(P) \cap N$. Since

$$\begin{aligned}[N_i, G] &= [N_i, \mathbf{C}_G(N)P] \\ &= [N_i, P] \leq N \cap [\mathbf{Z}_i(P), P] \leq N \cap \mathbf{Z}_{i-1}(P) = N_{i-1},\end{aligned}$$

the N_i constitute a central series of G. So $N_i \leq \mathbf{Z}_i(G)$. For large enough s, it follows that

$$N = N_s \leq \mathbf{Z}_s(G) \leq \mathbf{Z}_\infty(G).$$

b) As $P \leq \mathbf{Z}_\infty(G)$ and $\mathbf{Z}_\infty(G)$ is nilpotent, we have $P \trianglelefteq G$. By a), $|G/\mathbf{C}_G(P)|$ is a power of p. So $G = \mathbf{C}_G(P)P$. By the Zassenhaus theorem, $P \cap \mathbf{C}_G(P) = \mathbf{Z}(P)$ has a complement S in $\mathbf{C}_G(P)$, and it follows that $\mathbf{C}_G(P) = \mathbf{Z}(P) \times S$. Hence, S is characteristic in $\mathbf{C}_G(P)$, and since $\mathbf{C}_G(P) \trianglelefteq G$, it follows that $S \trianglelefteq G$ as well. Therefore, $G = P \times S$. □

Lemma 12.10. *Let N be a normal p-subgroup of the solvable group G, and let C be a Carter subgroup of G. Then $N \leq C$ if and only if the Sylow p-subgroups and the Carter subgroups of $G/\mathbf{C}_G(N)$ coincide; that is, if and only if $G/\mathbf{C}_G(N)$ has self-normalizing Sylow p-subgroups.*

Proof. a) Assume that $N \leq C$. Then the p-complement $C_{p'}$ of C lies in $\mathbf{C}_G(N)$, and so $C\mathbf{C}_G(N)/\mathbf{C}_G(N)$ is a p-group. By Lemma 12.3, the quotient $C\mathbf{C}_G(N)/\mathbf{C}_G(N)$ is a Carter subgroup of $G/\mathbf{C}_G(N)$. Then using Lemma 12.3 again, it follows that $C\mathbf{C}_G(N)/\mathbf{C}_G(N)$ is a Sylow p-subgroup of $G/\mathbf{C}_G(N)$.

b) Now assume that the Carter subgroup $C\mathbf{C}_G(N)/\mathbf{C}_G(N)$ of $G/\mathbf{C}_G(N)$ is a Sylow p-subgroup of $G/\mathbf{C}_G(N)$. Then there exists a Sylow p-subgroup P of G such that

$$P\mathbf{C}_G(N)/\mathbf{C}_G(N) = C\mathbf{C}_G(N)/\mathbf{C}_G(N).$$

As N is a normal p-subgroup of G, we have

$$N \leq P \leq C\mathbf{C}_G(N).$$

We let $U = C\mathbf{C}_G(N)$. Since $U = P\mathbf{C}_G(N)$, $|U/\mathbf{C}_U(N)|$ is a power of p. It therefore follows from Lemma 12.9 that $N \leq \mathbf{Z}_\infty(U)$. Clearly, C is a Carter subgroup of $U = C\mathbf{C}_G(N)$. As $\mathbf{Z}_\infty(U)$ is the intersection of all system normalizers of U by Theorem 11.11 c), the claim $N \leq C$ now follows from Theorem 12.8. □

Now we can determine the intersection $T = \bigcap_{g \in G} C^g$ of all Carter subgroups of G. Clearly, T is a nilpotent, characteristic subgroup of G.

Theorem 12.11 (Huppert). *Assume G is a solvable group, and let C be a Carter subgroup of G. Let*

$$T = \bigcap_{g \in G} C^g = \bigtimes_p T_p$$

be the intersection of the Carter subgroups of G. If P is a Sylow p-subgroup of G, then

$$T_p = \bigcap_{g \in G} \big(\mathbf{Z}_\infty(\mathbf{N}_G(P)) \cap P\big)^g.$$

Proof. Let N be a normal p-subgroup of G. By Lemma 12.10, we have that $N \leq C$ if and only if

$$\mathbf{N}_G(P)\mathbf{C}_G(N)/\mathbf{C}_G(N) = \mathbf{N}_{G/\mathbf{C}_G(N)}(P\mathbf{C}_G(N)/\mathbf{C}_G(N)) = P\mathbf{C}_G(N)/\mathbf{C}_G(N)$$

is a p-group. This is equivalent to the condition that $|\mathbf{N}_G(P)/(\mathbf{N}_G(P) \cap \mathbf{C}_G(N))|$ is a power of p. By Lemma 12.9, this is equivalent to $N \leq \mathbf{Z}_\infty(\mathbf{N}_G(P))$. As this holds for all Sylow p-subgroups P of G, we have $N \leq T_p$ if and only if

$$N \leq \bigcap_{g \in G} \big(\mathbf{Z}_\infty(\mathbf{N}_G(P)) \cap P\big)^g.$$

This means that

$$T_p = \bigcap_{g \in G} \big(\mathbf{Z}_\infty(\mathbf{N}_G(P)) \cap P\big)^g. \qquad \square$$

Example 12.12. Let G be the group of affine semilinear maps $x \mapsto ax^\alpha + b$ over the field $\mathsf{K} = \mathrm{GF}(p^n)$.

a) Let $\pi = \{q \text{ prime} \mid q \text{ divides } p-1\}$, and let Z_π be the Hall π-subgroup of the cyclic group $\mathrm{GF}(p^n)^\times$. Then

$$D = \left\{ \begin{pmatrix} x \\ ax^\alpha \end{pmatrix} \;\middle|\; a \in Z_\pi,\ \alpha \text{ an automorphism of } \mathsf{K} \right\}$$

is a system normalizer of G.

b) If $p > 2$, then D is a Carter subgroup of G. If $p = 2$ and $n = 2^m e$ with $2 \nmid e$, then

$$\left\{ \begin{pmatrix} x \\ x^\alpha + b \end{pmatrix} \;\middle|\; b \in \mathrm{GF}(2^{2^m}),\ \alpha \text{ an automorphism of } \mathsf{K} \right\}$$

is a Carter subgroup of G.

Proof. a) Let μ be the so-called Frobenius automorphism of $\mathsf{K} = \mathrm{GF}(p^n)$ defined by $x^\mu = x^p$. We let $r = \binom{x}{x^\mu}$ and $g = \binom{x}{cx}$, where $c \in \mathsf{K}^\times$. Then

$$[g, r] = \binom{x}{c^{p-1}x}.$$

Therefore, the only chief factors of the cyclic group

$$K = \left\{ \binom{x}{cx} \;\middle|\; c \in \mathsf{K}^\times \right\}$$

that are centralized by $R = \langle r \rangle$ are those of order q with $q \mid (p-1)$. Therefore,

$$D = \left\{ \binom{x}{cx^\alpha} \;\middle|\; c \in Z_\pi,\ \alpha \text{ arbitrary} \right\}$$

is a nilpotent subgroup of G. The group

$$H = \left\{ \binom{x}{cx^\alpha} \;\middle|\; c \in \mathsf{K}^\times,\ \alpha \text{ arbitrary} \right\}$$

satisfies the hypothesis of Theorem 7.12. So there exists a Carter subgroup C of H such that $D \leq C$. By Theorem 12.4, the subgroup C is a system normalizer of H. Furthermore, $|D|$ is the product of the orders of the central chief factors of H in a chief series of H. Then $|C| = |D|$ by Theorem 11.10, so D is a system normalizer of H. Let

$$\mathscr{S} = \{H_q \mid q \text{ divides } |H|\}$$

be a Sylow system of H with $\mathbf{N}(\mathscr{S}) = D$. Since

$$T = \left\{ \binom{x}{x+b} \;\middle|\; b \in \mathsf{K} \right\}$$

is a minimal and eccentric normal subgroup of G and D normalizes the Sylow system

$$\{TH_p, H_q \mid q \text{ divides } |G|,\ q \neq p\}$$

of G, we conclude that D is a system normalizer of G.

b) Let C be a Carter subgroup of G with $D \leq C$. As CT/T is a Carter subgroup and DT/T is a system normalizer of G/T, it follows by Theorem 12.4 that $CT = DT$. So we have $C = D(C \cap T)$.

For all

$$t = \binom{x}{x+b} \in C \cap T \quad \text{and} \quad g = \binom{x}{cx} \in D,$$

we have $(|t|, |g|) = 1$, and so $gt = tg$ since C is nilpotent. If $1 \neq t$, then this forces $g = 1$. If $C \cap T \neq 1$, then $Z_\pi = \{1\}$ and therefore $p = 2$. Hence, D is a Carter subgroup of G for $p > 2$.

Let $p = 2$ and $n = 2^m e$. We define $\sigma = \mu^{2^m}$. Then $s = \left(\begin{smallmatrix} x \\ x^\sigma \end{smallmatrix}\right)$ has odd order e. For $t = \left(\begin{smallmatrix} x \\ x+b \end{smallmatrix}\right) \in C$, it again follows that $ts = st$, and so $b^\sigma = b$. Hence, $b \in \mathrm{GF}(2^{2^m})$. So we obtain

$$C = D(C \cap T) \leq \left\{ \begin{pmatrix} x \\ x^\alpha + b \end{pmatrix} \;\middle|\; b \in \mathrm{GF}(2^{2^m}),\ \alpha \text{ arbitrary} \right\}.$$

As this last group is nilpotent,

$$C = \left\{ \begin{pmatrix} x \\ x^\alpha + b \end{pmatrix} \;\middle|\; b \in \mathrm{GF}(2^{2^m}),\ \alpha \text{ arbitrary} \right\}$$

is a Carter subgroup of G. □

It is easy to construct examples of very complicated solvable groups whose system normalizers have order 2, the smallest possible. By Theorem 11.10, the system normalizers reflect only the central chief factors of G, and a solvable group could in principle contain only one central chief factor, namely G/G'. It has been conjectured, however, that specifying the order of a Carter subgroup of G sharply restricts the structure of G.

Conjecture 12.13. Let G be a solvable group, and let C be a Carter subgroup of G. Write $|C| = p_1 \cdots p_t$, where the primes p_i are not necessarily distinct. Then the nilpotent length $n(G)$ (see Definition III.4.7, pg. 295) of G is bounded above by a function of t.[4]

Remark 12.14. Conjecture 12.13 has been confirmed in some special cases.

a) Suppose the Carter subgroup C of G is a cyclic p-group. Since $\mathbf{N}_G(C) = C$, this means C is a self-normalizing Sylow p-subgroup of G. Therefore, by Burnside's Theorem IV.2.6 (pg. 445), there exists a normal subgroup N of G such that $G = NC$ and $N \cap C = 1$. Let $C = \langle x \rangle$. Since $\mathbf{N}_G(C) = C$, if $n^x = n$ for $n \in N$, then $n = 1$. Hence, x induces a fixed point free automorphism of N. Then if $|C| = p^t$, the nilpotent length of N is bounded by $n(N) \leq t$ (see F. Hoffman [248]).

b) Thompson has shown the following: If the Carter subgroup C is a Hall subgroup of G, then the nilpotent length of G is bounded above by a function depending only on the number of prime divisors of $|C|$ (Thompson [431]).

[4] This conjecture was proved in E. C. Dade, Carter subgroups and Fitting heights of finite solvable groups, Illinois J. Math. **13** (1969), 449–514. *trans.*

Exercises

26) Prove that if G has p-length at most 1 for all primes p, then the intersection of the Carter subgroups of G is the hypercenter of G.
27) Determine the Carter subgroups of the groups constructed in Example 11.12 and Example 11.13.

13 Groups in which the system normalizers are Carter subgroups

Definition 13.1. A group G is called an SC-group whenever G is solvable and the system normalizers of G are self-normalizing. In this case, the system normalizers of G are Carter subgroups of G, so they are abnormal in G by Theorem 12.2. As every subabnormal subgroup of G contains a system normalizer by Theorem 11.21, all subabnormal subgroups of G are abnormal in G.

The class of groups described in Definition 13.1 is closed under homomorphic images:

Theorem 13.2. *If G is an SC-group and $N \trianglelefteq G$, then G/N is also an SC-group.*

Proof. This follows immediately from Theorem 11.3 and Lemma 12.3. □

We give a necessary and sufficient condition for a group G to be an SC-group:

Theorem 13.3 (P. Hall). *Let D be a system normalizer of the solvable group G. Then G is an SC-group if and only if for every eccentric chief factor H/K of G, whenever $h^d K = hK$ for some $h \in H$ and for all $d \in D$, we have $h \in K$. (It suffices to demand that this property is satisfied for all eccentric chief factors in a fixed chief series of G.)*

Proof. a) First assume that G is an SC-group. We consider an eccentric chief factor H/K of G and define

$$T = \{t \in H \mid t^d K = tK \text{ for all } d \in D\}.$$

We will prove that $T = K$.

For all $t \in T$ and $d \in D$, we have

$$t^{-1}t^d = [t, d] \in K.$$

Since

$$d^t = d[d, t] \in DK = KD$$

and $K \trianglelefteq G$, we have $T \leq \mathbf{N}_G(KD)$. It now follows from the abnormality of D that $T \leq \mathbf{N}_G(KD) = KD$. As D avoids the eccentric chief factor H/K, we have $H \cap D \leq K$. Hence, we now obtain

$$T \leq KD \cap H = K(D \cap H) = K,$$

which is what we wanted to show.

b) Assume that $\mathbf{N}_G(D) = L > D$. As D covers all the central chief factors in a given chief series of G, there exists an eccentric chief factor H/K in this chief series that is not avoided by L. So $H \cap L \not\leq K$. We show that $t^d K = tK$ for all $t \in H \cap L$ and all $d \in D$.

Since $H \trianglelefteq G$, we have on the one hand that

$$[H \cap L, D] \leq [H, D] \leq H.$$

On the other hand, since $L = \mathbf{N}_G(D)$ we have

$$[H \cap L, D] \leq [L, D] \leq D.$$

So this implies that

$$[H \cap L, D] \leq H \cap D \leq K,$$

as the eccentric chief factor H/K is avoided by D. Thus, $t^{-1}t^d \in [H \cap L, D] \leq K$ for all $t \in H \cap L$, which means $t^d K = tK$. □

Theorem 13.4.

a) *If G contains a nilpotent normal subgroup F with nilpotent factor group G/F, then G is an SC-group.* (*This was proved already in Theorem 12.4.*)

b) *If the Carter subgroups and the system normalizers of the solvable group G are Hall subgroups of G, then G is an SC-group.*

c) *Assume $l_p(G) \leq 1$ for all primes p. Assume that every chief factor of G is either covered or avoided by the Carter subgroup C.* (*This last is the case, in particular, whenever C is a Hall subgroup of G.*) *Then G is an SC-group.*

Proof. In all three cases, the hypothesis is inherited by homomorphic images of G. In the following proof, H is always a minimal normal subgroup of G.

a) By Theorem 13.3, it suffices to show that if H is eccentric, then no nonidentity element of H is fixed by all elements of the system normalizer D.

By Theorem 11.3, DF/F is a system normalizer of the nilpotent group G/F. So $G = DF$. Since F is nilpotent, $F \leq \mathbf{F}(G)$. Therefore, $F \leq \mathbf{C}_G(H)$ by Theorem III.4.2 e) (pg. 292). If $1 \neq h \in H$ and $h^d = h$ for all $d \in D$, then $h \in \mathbf{Z}(G)$ and H is central. This contradicts our assumption that H is eccentric.

b) We prove the claim by induction on $|G|$. Let D be a system normalizer of G lying in a Carter subgroup C. As DH/H is a system normalizer of G/H by Theorem 11.3 and CH/H is a Carter subgroup of G/H by Lemma 12.3, it follows by induction that $DH/H = CH/H$. As C is a Hall subgroup of G and H has prime power order, we have $H \cap C = 1$ or $H \leq C$. In the first case, it follows that $H \cap D \leq H \cap C = 1$, and hence

$$D = D/(H \cap D) \cong DH/H = CH/H \cong C/(H \cap C) = C,$$

and we are done. So assume that $H \leq C$. If H is central, then we have $H \leq D$ by Theorem 11.10, and we again see that $D = DH = CH = C$. So we may assume that $H \leq C$ and H is eccentric, so $H \cap D = 1$. As D is a Hall subgroup of C, it follows that $(|H|, |D|) = 1$. Then since C is nilpotent, $D \leq \mathbf{C}_G(H) \trianglelefteq G$, and we find that $G = \mathbf{C}_G(H)$ by Theorem 11.11 a). But then H is central, contradicting that $H \cap D = 1$.

c) Letting D be a system normalizer of G contained in a Carter subgroup C, we see that $DH = CH$ by induction, just as in b). If $H \cap C = 1$, then it follows as before that $D = C$. If $H \cap C \neq 1$, then by the assumed covering-avoidance property of C, we certainly have $H \leq C$. Let $|H| = q^m$. Since $l_q(G) \leq 1$, we have $q \nmid |G/\mathbf{C}_G(H)|$ by Theorem 6.12. So H is centralized by all Sylow q-subgroups of G. In particular, H is centralized by C_q. Since C is nilpotent and $H \leq C$, we also have that H is centralized by $C_{q'}$. It follows that $C = C_q \times C_{q'} \leq \mathbf{C}_G(H) \trianglelefteq G$, and so $G = \mathbf{C}_G(H)$ by Theorem 11.11 a). So H is central, and we see that $H \leq D$ by Theorem 11.10. Thus, $D = C$. □

Theorem 13.5 (Carter [86]). *Let N_i $(i = 1, 2)$ be normal subgroups of G, and assume that the factor groups G/N_i are SC-groups. Then $G/(N_1 \cap N_2)$ is also an SC-group.* (*That is, the SC-groups constitute a formation in the sense of Definition 7.1.*)

Proof. Without loss of generality, we may assume that $N_1 \cap N_2 = 1$. As G/N_1 is an SC-group, the chief factors of G above N_1 satisfy the condition given in Theorem 13.3. The map μ defined by $n_1^\mu = n_1 N_2$ is a G-isomorphism from N_1 onto $(N_1 \times N_2)/N_2$ since

$$(n_1^g)^\mu = n_1^g N_2 = (n_1 N_2)^g$$

for all $g \in G$. Under this isomorphism, the chief factors of G inside N_1 are mapped one-to-one onto the chief factors of G/N_2 in $(N_1 \times N_2)/N_2$, and the fixed points of D pass to fixed points of DN_2/N_2. As G/N_2 is an SC-group, the system normalizer DN_2/N_2 of G/N_2 has only the trivial fixed point on an eccentric chief factor of G/N_2. So D has only the trivial fixed point on an eccentric chief factor inside N_1. Therefore, G is an SC-group by Theorem 13.3. □

The usual method of proving that $G/(N_1 \cap N_2)$ has group property $\mathscr{E}$ whenever the quotients G/N_1 and G/N_2 have this property uses the embedding of $G/(N_1 \cap N_2)$ in $G/N_1 \times G/N_2$ (see Theorem I.9.7, pg. 50). This method requires that the property $\mathscr{E}$ is inherited by subgroups. However, we will see in Theorem 13.7 that subgroups of SC-groups are by no means always SC-groups. Hence, the proof of Theorem 13.5 cannot be carried out with the help of Theorem I.9.7.

Example 13.6. The collection of SC-groups is not a saturated formation:

Let H be the imprimitive permutation group $H = \langle n_1, n_2, n_3, t \rangle$ of degree 6 with

$$n_1 = (1, 2),\ n_2 = (3, 4),\ n_3 = (5, 6),\ t = (1, 3, 5)(2, 4, 6).$$

Let P be the p-group $(p > 3)$ of class 2 and exponent p defined by the generators x_i $(1 \leq i \leq 6)$ and the defining relations

$$x_i^p = [x_1, x_2] = [x_3, x_4] = [x_5, x_6] = [x_i, x_j]^p = [x_i, x_j, x_k] = 1.$$

We define $x_{ij} = [x_i, x_j]$. Then P' is generated by the elements

$$x_{13},\ x_{23},\ x_{14},\ x_{24},\ x_{15},\ x_{25},\ x_{16},\ x_{26},\ x_{35},\ x_{36},\ x_{45},\ x_{46}.$$

So $|P/P'| = p^6$ and $|P| = p^{18}$. By defining

$$x_i^h = x_{i^h} \quad \text{and} \quad x_{ij}^h = x_{i^h j^h},$$

we give H the structure of a group of automorphisms of P. Let G be the semidirect product of P with H. We show:

(1) $\overline{G} = G/P'$ is an SC-group:

We use the bar notation $\overline{G} = G/P'$ to denote passing to the quotient of G by P', and we let

$$\overline{P} = P/P',\ \overline{N} = \langle \overline{n_1}, \overline{n_2}, \overline{n_3} \rangle,\ \overline{T} = \langle \overline{t} \rangle.$$

Then $\overline{\mathscr{S}} = \{\overline{P}, \overline{N}, \overline{T}\}$ is a Sylow system of $\overline{G}$. It is clear that

$$\mathbf{N}(\overline{\mathscr{S}}) \geq \langle \overline{x_1}\,\overline{x_2} \cdots \overline{x_6}, \overline{n_1}\,\overline{n_2}\,\overline{n_3}, \overline{t} \rangle.$$

As $G/P \cong H$ has exactly one central 2-chief factor, namely $\langle n_1 n_2 n_3 \rangle$, the subgroup $\langle \overline{n_1}\,\overline{n_2}\,\overline{n_3} \rangle \times \langle \overline{t} \rangle$ is the p-complement of $\mathbf{N}(\overline{\mathscr{S}})$ by the covering-avoidance property of Theorem 11.10.

Let $\overline{x} \in \overline{P} \cap \mathbf{N}(\mathbf{N}(\overline{\mathscr{S}}))$. Then $[\overline{x}, \overline{t}] \in \overline{P} \cap \overline{T} = 1$ and $[\overline{x}, \overline{n_1}\,\overline{n_2}\,\overline{n_3}] \in \overline{P} \cap \overline{N} = 1$. By the transitivity of $\langle n_1 n_2 n_3, t \rangle$ on $\{1, \ldots, 6\}$, this forces, for a suitable integer a,

$$\overline{x} = (\overline{x_1}\,\overline{x_2} \cdots \overline{x_6})^a \in \mathbf{N}(\overline{\mathscr{S}}).$$

As $\overline{G}/\overline{P}$ is an SC-group by Theorem 13.4 a), we have

$$\overline{P}\mathbf{N}(\mathbf{N}(\overline{\mathscr{S}})) = \overline{P}\mathbf{N}(\overline{\mathscr{S}}),$$

and so

$$\mathbf{N}(\mathbf{N}(\overline{\mathscr{S}})) = \big(\overline{P} \cap \mathbf{N}(\mathbf{N}(\overline{\mathscr{S}}))\big)\mathbf{N}(\overline{\mathscr{S}}) = \mathbf{N}(\overline{\mathscr{S}}).$$

Hence, $\overline{G}$ is an SC-group.

(2) G is not an SC-group:

Let D be a system normalizer of G. As DP/P is a system normalizer of G/P, we may assume that $\langle n_1 n_2 n_3, t \rangle$ is the p-complement of D. Let

$$f = x_{13}\, x_{35}\, x_{51}\, x_{24}\, x_{46}\, x_{62}.$$

Then

$$f^t = f^{n_1 n_2 n_3} = f.$$

Since $f \in P' \leq \mathbf{Z}(P)$, we have $f \in \mathbf{C}_G(D)$.

By Maschke's theorem, P' is a completely reducible G/P-module. If

$$1 \neq x = \prod_{i<j} x_{ij}^{m_{ij}} \in \mathbf{Z}(G) \cap P',$$

then it follows from $x^{n_i} = x = x^t$ $(i = 1, 2, 3)$ that, for a suitable integer c,

$$x = (x_{13}\, x_{23}\, x_{14}\, x_{24}\, x_{15}^{-1}\, x_{25}^{-1}\, x_{16}^{-1}\, x_{26}^{-1}\, x_{35}\, x_{45}\, x_{36}\, x_{46})^c.$$

Therefore, $\langle x \rangle$ is the unique central chief factor of G inside P'. As f does not lie in $\langle x \rangle$, the system normalizer D has a nontrivial fixed point on an eccentric chief factor of G. Thus, G is not an SC-group by Theorem 13.3.

Since G/P' is an SC-group and $P' \leq \Phi(P) \leq \Phi(G)$, the factor group $G/\Phi(G)$ is an SC-group, but G is not.

Subgroups of SC-groups are generally not SC-groups. In fact, we have the following theorem:

Theorem 13.7 (Alperin, Thompson). *Let G be an arbitrary solvable group. Then G can be embedded in an SC-group.*

Proof. Let

$$1 = G_0 \lhd G_1 \lhd \cdots \lhd G_m = G$$

be a chief series of G. We define $K_i = G_i/G_{i-1}$ $(i = 1, \ldots, m)$. Let $|K_i| = p_i^{k_i}$. Let p be a prime that is distinct from all p_i, and choose integers n_i $(i = 1, \ldots, m)$ such that $k_i \leq n_i$ and $p_i^{n_i} \equiv 1 \bmod p$. We let $N_i = \mathrm{GF}(p_i^{n_i})^+$, and we can embed K_i isomorphically in N_i.

a) We first show that G is isomorphic to a subgroup of the iterated regular wreath product $(\cdots (N_1 \wr_r N_2) \wr_r \cdots) \wr_r N_m$:

We prove the claim by induction on m. Assume for induction that

$$G_{m-1} \cong T \leq (\cdots (N_1 \wr_r N_2) \wr_r \cdots) \wr_r N_{m-1}.$$

By Theorem I.15.9 and Lemma I.15.8 (pg. 105), it follows that (up to isomorphism)

$$G \leq G_{m-1} \wr_r G/G_{m-1} \leq (\cdots (N_1 \wr_r N_2) \wr_r \cdots \wr_r N_{m-1}) \wr_r N_m.$$

b) Every $N_i = \mathrm{GF}(p_i^{n_i})^+$ admits a fixed point free automorphism α_i of order p given by multiplication with a p-th root of unity in $\mathrm{GF}(p_i^{n_i})$. We prove the following claim by induction on k:

The automorphism group of the group $H_k = (\cdots (N_1 \wr_r N_2) \wr_r \cdots \wr_r N_k)$ contains an elementary abelian group A_k of order $|A_k| = p^k$ satisfying the following property: There does not exist a nonidentity $h \in H_k$ such that $h^\alpha = h$ for all $\alpha \in A_k$.

For $k = 1$, we set $A_1 = \langle \alpha_1 \rangle$. As α_1 fixes only the identity element of $H_1 = N_1$, the claim holds for $k = 1$.

Assume for induction that we have found an A_k with the desired properties. We have $H_{k+1} = H_k \wr_r N_{k+1}$. By Lemma I.15.7 b) (pg. 104), the automorphism $\mu = \alpha_{k+1}$ of N_{k+1} gives rise to an automorphism $\overline{\mu}$ of order p of H_{k+1} of the form

$$(f, n)^{\overline{\mu}} = (f', n^{\mu}) \quad \text{with} \quad f'(n) = f(n^{\mu^{-1}}).$$

(Here, we are using the notation of Chapter I, §15.) For $(f, n) \in H_k \wr_r N_{k+1}$ with $1 \neq n \in N_{k+1}$, we have (for a suitable g)

$$(f, n)^{\overline{\mu}}(f, n)^{-1} = (g, n^{\mu} n^{-1}) \notin H_k \wr 1.$$

So $\overline{\mu}$ acts as a fixed point free automorphism on $H_{k+1}/(H_k \wr 1)$.

For $\alpha \in A_k$, we define the automorphism $\overline{\alpha} \in \mathbf{Aut}(H_{k+1})$ by

$$(f, n)^{\overline{\alpha}} = (f', n) \quad \text{with} \quad f'(x) = f(x)^{\alpha}$$

according to Lemma I.15.7 a) (pg. 104). Then $\overline{A}_k = \{\overline{\alpha} \mid \alpha \in A_k\}$ is a group of automorphisms of H_{k+1} that is isomorphic to A_k. If $(f, 1)^{\overline{\alpha}} = (f, 1)$ for all $\overline{\alpha} \in \overline{A}_k$, then $f(n)^{\alpha} = f(n)$ for all $n \in N_{k+1}$. Applying the induction hypothesis to the action of A_k on H_k forces $f(n) = 1$ for all $n \in N_{k+1}$.

We define $A_{k+1} = \langle \overline{A}_k, \overline{\mu} \rangle$. By induction, $\overline{A}_k$ is elementary abelian of order p^k. By Lemma I.15.7 c) (pg. 104), the $\overline{\alpha} \in \overline{A}_k$ commute with $\overline{\mu}$. As $\overline{A}_k$ acts on $H_{k+1}/(H_k \wr 1)$ as the trivial automorphism, but $\overline{\mu}$ does not, it follows immediately that $|A_{k+1}| = |\overline{A}_k||\langle \overline{\mu} \rangle| = p^{k+1}$.

Let $(f, n) \in H_{k+1}$ with $(f, n)^{\tau} = (f, n)$ for all $\tau \in A_{k+1}$. For $\tau = \overline{\mu}$, it follows that $n = 1$. For $\tau = \overline{\alpha} \in \overline{A}_k$, we obtain $f(n)^{\alpha} = f(n)$, so $f(n) = 1$ for all $n \in N_{k+1}$. Hence, no nonidentity element of H_{k+1} is fixed by all automorphisms in A_{k+1}.

c) Let $H = H_m A_m$ be the semidirect product of H_m with A_m. By our choice of p, we have $(|H_m|, |A_m|) = 1$. Let $x \in H_m \cap \mathbf{N}_H(A_m)$. Since $H_m \lhd H$, we then have for all $a \in A_m$ that

$$[x, a] \in H_m \cap A_m = 1.$$

So $x \in \mathbf{C}_H(A_m) \cap H_m = 1$, and therefore $\mathbf{N}_H(A_m) = A_m$. Hence, A_m is a nilpotent (even abelian) self-normalizing subgroup of H, so it is a Carter subgroup of H. As $H/H_m \cong A_m$ is abelian, the system normalizers of H have order at least $|A_m| = p^m$ by the covering-avoidance property of Theorem 11.10. So A_m is a system normalizer of H, and H is an SC-group. By a), G can be isomorphically embedded in H. □

In SC-groups, the system normalizers are characterized by the covering-avoidance property. Namely, we have:

Theorem 13.8 (Huppert). *Let G be an SC-group. Then the system normalizers of G are the only subgroups of G satisfying the covering-avoidance property of Theorem 11.10; that is, system normalizers are the only subgroups that cover all*

central chief factors of G and avoid all eccentric chief factors of G. (The proof will show that it suffices to satisfy the covering-avoidance property for all chief factors in a given chief series of G.)

Proof. We prove the theorem by induction on $|G|$. Let S be a subgroup of G that satisfies the covering-avoidance property with respect to a chief series of G beginning with the minimal normal subgroup N. Then SN/N satisfies the covering-avoidance property for a chief series of G/N. By Theorem 13.2, G/N is an SC-group. By Theorem 11.3, every system normalizer of G/N has the form DN/N for a suitable system normalizer D of G. Hence, G/N satisfies the assumptions of the theorem. Therefore, we have by induction that

$$SN/N = DN/N,$$

so $SN = DN$ for a suitable system normalizer D of G. If N is a central chief factor of G, then N is covered by S and D as the covering-avoidance property holds for these subgroups. So then

$$S = SN = DN = D,$$

and we are done. On the other hand, if N is eccentric, then $S \cap N = D \cap N = 1$. We conclude from

$$S \cong SN/N = DN/N \cong D$$

that S is nilpotent. We have $\mathbf{N}_{SN}(S) = S \times L$ for an appropriate subgroup L of N. Here,

$$L \leq \mathbf{Z}(SN) = \mathbf{Z}(DN) \leq \mathbf{N}_G(D) = D,$$

and so $L \leq N \cap D = 1$. Hence, S is self-normalizing inside $SN = DN$. This means S and D are Carter subgroups of SN, so they are conjugate in SN. So S is a system normalizer of G. □

It is also possible to say something about the relationship between Carter subgroups and system normalizers for a somewhat larger class of groups:

Theorem 13.9 (Carter [89]). *Suppose the solvable group G contains a nilpotent normal subgroup F such that G/F is an SC-group. Then every system normalizer of G lies in exactly one Carter subgroup of G.*

Proof. Let D be a system normalizer of G, and let C be a Carter subgroup of G with $D \leq C$. As G/F is an SC-group, we have $DF/F = CF/F$, so $C = D(F \cap C)$. Since C is nilpotent, we have

$$F_p \cap C \leq F_p \cap \mathbf{C}_G(D_{p'}).$$

We define

$$J = \bigtimes_p (F_p \cap \mathbf{C}_G(D_{p'})).$$

Then $D \leq \mathbf{N}_G(J)$, and therefore

$$C = D(F \cap C) = D \bigtimes_p (F_p \cap C) \leq DJ \leq G.$$

The Sylow p-subgroup D_pJ_p of DJ is normal in DJ since $D \leq \mathbf{N}_G(D_p) \cap \mathbf{N}_G(J_p)$ and $J_{p'} \leq \mathbf{C}_G(D_p)$. So DJ is nilpotent. By Lemma 12.3, $C = DJ$ and so C is uniquely determined by D. □

Theorem 13.10 (Carter [89]). *Suppose the solvable group G has a nilpotent normal subgroup F such that G/F is an SC-group. Let D be a system normalizer of G, and let C be the unique (by Theorem 13.9) Carter subgroup of G containing D. Then C is generated by all subgroups H of G that can be connected to D by a chain*

$$D = D_0 \trianglelefteq D_1 \trianglelefteq \cdots \trianglelefteq D_r = H.$$

(*Since C is nilpotent, there certainly exists such a chain from D to C.*)

Proof. We show that if

$$D = D_0 \triangleleft D_1 \triangleleft \cdots \triangleleft D_r = H,$$

then $H \leq C$, where C is the unique Carter subgroup of G containing D.

Assume for induction that $D_i \leq C$. If $g \in D_{i+1}$ is an arbitrary element, then we have $D_i^g = D_i \leq C$, and so $D_i \leq gCg^{-1}$. Hence, we have

$$D \leq C \cap gCg^{-1}.$$

As D is contained in only one Carter subgroup of G by Theorem 13.9, it follows that $C = gCg^{-1}$, so $g \in \mathbf{N}_G(C) = C$. Hence, D_{i+1} lies in C, and in the end we arrive at $H \leq C$, as claimed. □

The hypotheses of Theorems 13.9 and 13.10 hold, in particular, when G has nilpotent length at most 3 by Theorem 12.4. For groups of nilpotent length 4, the conclusion of Theorem 13.10 need not hold, as the following example shows:

Example 13.11. Let Z be the cyclic group of order 5, and let $G = Z \wr S_4$ be the wreath product of Z with the symmetric group S_4, where S_4 is understood in the natural way as a permutation group of degree 4. Then $N = Z \wr 1$ is an elementary abelian normal subgroup of G of order 5^4. This subgroup has a basis $\{n_1, n_2, n_3, n_4\}$ for which the n_i are permuted by the inner automorphisms of G.

Let C be a Carter subgroup of G. Then CN/N is a Carter subgroup of G/N, which is isomorphic to S_4. So CN/N has order 8. If P is a Sylow 2-subgroup of G lying in C, then $C = P(C \cap N)$. As C is nilpotent, the subgroup $C \cap N$ is centralized by P. This means

$$C \cap N \leq \mathbf{C}_G(P) \cap N = \langle z \rangle,$$

where $z = n_1 n_2 n_3 n_4$. Since $z \in \mathbf{Z}(G)$, we certainly have $z \in C$. So $|C| = 40$. If t is the element corresponding to the permutation $(1, 2)$, then $D = \langle z, t \rangle$ is a system normalizer of G. We have $D < C$ and

$$D \lhd \langle n_1 n_2, n_3, n_4, t \rangle \not\leq C.$$

14 Solvable groups with abelian Sylow subgroups

Definition 14.1. A group G is called an A-group if G is solvable and all of its Sylow subgroups are abelian.

The following lemma is trivial.

Lemma 14.2. *Every subgroup and every factor group of an A-group is again an A-group. Every nilpotent subgroup of an A-group is abelian. Direct products of A-groups are again A-groups.*

The following theorem is fundamental to the theory of A-groups.

Theorem 14.3 (Taunt [426]).
a) *If G has an abelian Sylow p-subgroup P, then*

$$G' \cap \mathbf{Z}(G) \cap P = 1.$$

So p is not a divisor of $|G' \cap \mathbf{Z}(G)|$.
b) *If all Sylow subgroups of G are abelian, then $G' \cap \mathbf{Z}(G) = 1$.*

Proof. We already proved a) in Theorem IV.2.2 (pg. 443), and b) is a trivial consequence of a). □

We specialize our study of system normalizers in §11 to A-groups.

Theorem 14.4. *Let G be an A-group, and let D be a system normalizer of G. Then D is a complement of G' in G.*

Proof. As $G'D/G'$ is a system normalizer of the nilpotent group G/G' by Theorem 11.3, we have $G = G'D$. We show that every chief factor of G inside G' is eccentric:

Let $K \lhd H \trianglelefteq G'$, and let H/K be a central chief factor of G. Then we have:

$$H/K \leq (G/K)' \cap \mathbf{Z}(G/K),$$

contradicting Theorem 14.3. So all chief factors of G inside G' are eccentric, and therefore Theorem 11.10 implies they are avoided by D. By Lemma 11.6, D avoids all of G'. Therefore, $G' \cap D = 1$, and D is a complement of G' in G. □

Theorem 14.5. *If G is a solvable group with abelian system normalizers, then $\mathbf{Z}(G) = \mathbf{Z}_\infty(G)$. (In particular, this holds for A-groups.)*

Proof. By Theorem 11.11 c), the intersection of all system normalizers of G is the hypercenter $\mathbf{Z}_\infty(G)$. As the system normalizers are abelian by assumption, $\mathbf{Z}_\infty(G)$ is centralized by every system normalizer. As the system normalizers generate all of G by Theorem 11.11 b), it follows that $\mathbf{Z}_\infty(G) \leq \mathbf{Z}(G)$, so $\mathbf{Z}_\infty(G) = \mathbf{Z}(G)$. □

Theorem 14.6 (Taunt [426]).

a) *Assume G is solvable. Let D be a system normalizer of G, and let N be a normal subgroup of G. Then*

$$N = (N \cap G')(N \cap D).$$

b) *If G is an A-group and N is an abelian normal subgroup of G, then*

$$N = (N \cap G') \times (N \cap \mathbf{Z}(G)).$$

Proof. a) Since $[N, G] \leq N \cap G'$, all chief factors of G between N and $N \cap G'$ are central, so they are covered by D. Then $N \leq (N \cap G')D$ by Lemma 11.6. The claim follows using the Dedekind identity

$$N = (N \cap G')(N \cap D).$$

b) We show that if N is an abelian normal subgroup, then $N \cap D = N \cap \mathbf{Z}(G)$. The inclusion $N \cap \mathbf{Z}(G) \leq N \cap D$ is trivial since $\mathbf{Z}(G) \leq D$. So it remains to prove that $N \cap D \leq \mathbf{Z}(G)$. It suffices to prove that this holds for all Sylow subgroups of the abelian group N. We may therefore assume that N is a p-group.

Let G_p be a Sylow p-subgroup, and let $G_{p'}$ be a p-complement in a complement system $\mathscr{K}$ of G with $D = \mathbf{N}(\mathscr{K})$. As N is a subgroup of the abelian group G_p, we have

$$[N \cap D, G_p] = 1.$$

Furthermore,

$$[N \cap D, G_{p'}] \leq N \cap G_{p'} = 1.$$

So $N \cap D$ is centralized by $G = G_p G_{p'}$, and it follows that $N \cap D \leq \mathbf{Z}(G)$. □

Theorem 14.6 gives a useful decomposition of the normal abelian subgroups (or normal nilpotent subgroups, which coincide in A-groups) of G:

Theorem 14.7 (Taunt [426]).

a) *If G is an A-group with $G^{(k-1)} > G^{(k)} = 1$ and A is an abelian normal subgroup of G, then*

$$A = (A \cap \mathbf{Z}(G)) \times (A \cap \mathbf{Z}(G')) \times \cdots \times (A \cap \mathbf{Z}(G^{(k-1)})).$$

b) *If G is an A-group with $G^{(k)} = 1$, then the Fitting subgroup $\mathbf{F}(G)$ has the form*

$$\mathbf{F}(G) = \mathbf{Z}(G) \times \mathbf{Z}(G') \times \cdots \times \mathbf{Z}(G^{(k-1)}).$$

Proof. a) By Theorem 14.6 b), we have

$$A = (A \cap \mathbf{Z}(G)) \times (A \cap G').$$

The commutator subgroup G' is an A-group with $|G'| < |G|$. By induction on $|G|$, we may therefore assume that

$$A \cap G' = (A \cap \mathbf{Z}(G')) \times \cdots \times (A \cap \mathbf{Z}(G^{(k-1)})).$$

This yields the claim.

b) is a special case of a). □

The next theorem follows immediately from Theorem 6.6 a):

Theorem 14.8. *If G is an A-group, then $l_p(G) \leq 1$ for all p.*

In the next theorems, we investigate the relationships between system normalizers and Carter subgroups of A-groups.

Theorem 14.9 (Carter [89]). *Let G be an A-group.*

a) *Distinct system normalizers of G have distinct normalizers.*

b) *Every Carter subgroup of G contains exactly one system normalizer of G.*

c) *If D is a system normalizer of G, then*

$$\mathbf{N}_G(D) = D \times (G' \cap \mathbf{C}_G(D));$$

in particular, $\mathbf{N}_G(D) = \mathbf{C}_G(D)$.

Proof. a) Let D be a system normalizer of G. As all system normalizers of G are conjugate, the index $[G : \mathbf{N}_G(D)]$ is the number of system normalizers of G.

Let C be a Carter subgroup of G, and let D be a system normalizer of G lying in C; there exists such a D by Theorem 12.8. As C is abelian, we have $D \trianglelefteq C$, so $C \leq \mathbf{N}_G(D)$. By Theorem 12.2, C is abnormal in G. So $\mathbf{N}_G(D)$ is self-normalizing in G. Therefore, $[G : \mathbf{N}_G(D)]$ is also the number of distinct normalizers of system normalizers.

b) Suppose that D and D^g are two system normalizers of G lying in the same Carter subgroup C of G. As C is abelian, it follows that $C \leq \mathbf{N}_G(D) \cap \mathbf{N}_G(D)^g$. Since C is abnormal, we conclude by Theorem 11.15 that

$$\mathbf{N}_G(D) = \mathbf{N}_G(D)^g = \mathbf{N}_G(D^g).$$

It then follows by a) that $D = D^g$.

c) By Theorem 14.4, $G = G'D$ and $G' \cap D = 1$. It follows that

$$\mathbf{N}_G(D) = D(G' \cap \mathbf{N}_G(D)).$$

Since

$$[G' \cap \mathbf{N}_G(D), D] \leq G' \cap D = 1,$$

we have $G' \cap \mathbf{N}_G(D) = G' \cap \mathbf{C}_G(D)$. □

Lemma 14.10. *Let D be a system normalizer of the A-group G. Then the Sylow p-subgroup D_p of D normalizes every p-complement $G_{p'}$ of G.*

Proof. Let G_p be an element of a Sylow system of G, and let $G_{p'}$ be an element of the corresponding complement system $\mathscr{K}$ of G with $D = \mathbf{N}(\mathscr{K})$. We have by Theorem 11.2 that

$$D_p = G_p \cap \mathbf{N}_G(G_{p'}),$$

and $G = G_p G_{p'}$. If $G^*_{p'}$ is another p-complement of G, then we have $G^*_{p'} = G^x_{p'}$ for some $x \in G_p$. Then for all $g \in D_p$, it follows that

$$(G^*_{p'})^g = (G^x_{p'})^g = (G^g_{p'})^x = (G_{p'})^x = G^*_{p'}.$$ □

Lemma 14.11. *Let $\mathscr{S} = \{G_p \mid p \text{ divides } |G|\}$ be a Sylow system of the A-group G, and let D be a system normalizer of G with $D \cap G_p = D_p$ for all p. Then $\mathbf{N}(\mathscr{S}) = D$.*

Proof. Let $G_{p'} = \prod_{q \neq p} G_q$. Then

$$D \cap G_{p'} \geq D \cap G_q = D_q$$

for all $q \neq p$. It therefore follows that

$$D_{p'} = \prod_{q \neq p} D_q \leq G_{p'} \leq \mathbf{N}_G(G_{p'}).$$

By Lemma 14.10, $D_p \leq \mathbf{N}_G(G_{p'})$. So we have for all p that

$$D = D_p \times D_{p'} \leq \mathbf{N}_G(G_{p'}),$$

and hence $D \leq \mathbf{N}(\mathscr{S})$. A comparison of group orders shows that $D = \mathbf{N}(\mathscr{S})$. □

Lemma 14.12. *If D is a system normalizer of the A-group G and $D \leq H \leq G$, then D lies in a system normalizer of H.*

Proof. By Theorem 2.5 a), there exist Sylow systems

$$\begin{aligned}
\mathscr{S}_G &= \{G_p \mid p \text{ divides } |G|\} \text{ of } G,\\
\mathscr{S}_H &= \{H_p \mid p \text{ divides } |H|\} \text{ of } H \text{ and}\\
\mathscr{S}_D &= \{D_p \mid p \text{ divides } |D|\} \text{ of } D
\end{aligned}$$

such that $G_p \cap H = H_p$ and $H_p \cap D = D_p$ for all p. Then since $G_p \cap D = D_p$, we have $D = \mathbf{N}(\mathscr{S}_G)$ by Lemma 14.11. As $D \leq H$, the subgroup $H \cap G_p = H_p$ is also normalized by D, and so $D \leq \mathbf{N}(\mathscr{S}_H)$. □

Lemma 14.13. *Let D be a system normalizer of the A-group G, and suppose we have a chain of subgroups*

$$D = D_0 \lhd D_1 \lhd \cdots \lhd D_s = H \leq G.$$

Then $D \trianglelefteq H$.

Proof. By refining the chain if necessary, we may assume that D_{i+1}/D_i is abelian.

We have $D = D_0 \leq \mathbf{N}_G(D)$. Assume for induction that $D_{i-1} \leq \mathbf{N}_G(D)$. By Lemma 14.12, the subgroup D lies in a system normalizer D^* of D_i. As D^* is abelian, we have $D \trianglelefteq D^*$, and so $D^* \leq \mathbf{N}_G(D)$. By Theorem 11.10 and Lemma 11.6, the central factor D_i/D_{i-1} of D_i is covered by D^*. Hence, we have

$$D_i = D_{i-1}D^* \leq \mathbf{N}_G(D),$$

and then finally $H = D_s \leq \mathbf{N}_G(D)$. □

For A-groups, we can give a peculiar characterization of Carter subgroups:

Theorem 14.14 (Carter [89]). *Let G be an A-group with $G^{(k)} = 1$, and let D be a system normalizer of G. We set $D_0 = 1$, $N_0 = G$ and define the subgroups D_i and N_i for $i \geq 1$ recursively in the following way:*

Let D_i be a system normalizer of N_{i-1}, and let $N_i = \mathbf{N}_G(D_i)$. Then

$$D_0 \leq D_1 \leq \cdots \leq D_k = N_k = N_{k-1} \leq \cdots \leq N_1 \leq N_0,$$

and D_k is a Carter subgroup of G.

(It is proved in Carter [89] that if $G^{(2k+1)} = 1$, then N_k is already a Carter subgroup of G. In addition, it is shown that if $G^{(2k)} = 1$, then D_k is already a Carter subgroup of G. These statements cannot be improved, as the examples given in Carter [89] show.)

Proof. It is clear that

$$1 = D_0 \leq D_1 \leq N_1 = \mathbf{N}_G(D_1) \leq N_0 = G.$$

Furthermore, we have by Theorem 14.9 c) that

$$N_1 = \mathbf{N}_G(D_1) = D_1 \times (G' \cap \mathbf{N}_G(D_1)),$$

and therefore $N_1' \leq G''$ since $D_1' = 1$.

Assume for induction that

$$D_{i-1} \leq D_i \leq N_i \leq N_{i-1}$$

and $N_i' \leq G^{(i+1)}$. Then we have $N_i = \mathbf{N}_G(D_i) \leq N_{i-1}$, so $N_i = \mathbf{N}_{N_{i-1}}(D_i)$. It then follows by Theorem 14.9 c) that $D_i \leq \mathbf{Z}(N_i)$. Therefore, D_i certainly lies in the system normalizer D_{i+1} of N_i. Furthermore, we have

$$N_{i+1} = \mathbf{N}_G(D_{i+1}) \geq D_{i+1}.$$

We show that $N_{i+1} \leq N_i$:

We have $N_{i+1} \leq N_0 = G$. Assume for induction that there exists some j with $j < i$ such that $N_{i+1} \leq N_j$. As the system normalizer D_{i+1} of N_i is abelian, it follows that

$$D_{j+1} \trianglelefteq D_{i+1} \trianglelefteq \mathbf{N}_G(D_{i+1}) = N_{i+1} \leq N_j.$$

Applying Lemma 14.13 to N_j with system normalizer D_{j+1} yields $D_{j+1} \trianglelefteq N_{i+1}$. Therefore,

$$N_{i+1} \leq \mathbf{N}_G(D_{j+1}) = N_{j+1}.$$

By induction, we conclude that $N_{i+1} \leq N_i$.

By Theorem 14.9 c), we have

$$N_{i+1} = \mathbf{N}_{N_i}(D_{i+1}) = D_{i+1} \times (N_{i+1} \cap N_i').$$

Since $N_i' \leq G^{(i+1)}$, it then follows that

$$N_{i+1}' \leq N_i'' \leq G^{(i+2)}.$$

Eventually we arrive at $N_{k-1}' \leq G^{(k)} = 1$, and N_{k-1} is abelian. Therefore, $D_k = N_{k-1}$. Since

$$\mathbf{N}_G(D_k) = N_k \leq N_{k-1} = D_k,$$

the subgroup D_k is a Carter subgroup of G. □

A-groups are not always SC-groups; groups of semilinear maps over finite fields provide easy examples. So Theorem 13.8 cannot be applied to A-groups. However, there is a similar theorem that holds for A-groups:

Theorem 14.15 (Huppert). *Let G be an A-group, and let T be a subgroup of G that avoids every eccentric chief factor of G in every chief series of G. Then T lies in a system normalizer of G.*

(*Note the difference between this theorem and Theorem 13.8: In that theorem, only one chief series was considered, but it was required that the central chief factors of this chief series were covered.*)

Proof. We prove the theorem by induction on $|G|$. Let N be a minimal normal subgroup of G. Then N has prime power order, say $|N| = p^n$. Let T be a subgroup of

G that avoids all eccentric chief factors of G in every chief series of G. Then TN/N has the same property with respect to G/N:

Namely, if $H/N \big/ K/N$ is an eccentric chief factor of G/N, then H/K is an eccentric chief factor of G. Therefore, we have $H \cap T \leq K$, and so then

$$H/N \cap TN/N = (H \cap TN)/N = N(H \cap T)/N \leq K/N.$$

By induction, TN/N therefore lies in a system normalizer of G/N, which has the form DN/N for a suitable system normalizer D of G by Theorem 11.3. So we have $T \leq DN$. If N is central, then $N \leq D$, and we obtain $T \leq D$, as desired. Thus, we may assume that N is an eccentric normal subgroup of G. Then N is avoided by T and D. It follows from

$$T \cong TN/N \leq DN/N \cong D$$

that T is nilpotent.

The p-complement $T_{p'}$ of T lies in a p-complement of ND, so it lies in a conjugate of $D_{p'}$. By passing to a conjugate of D if necessary, we may assume that $T_{p'} \leq D_{p'}$.

We show that the Sylow p-subgroup T_p of T normalizes every p-complement $G_{p'}$ of G:

By Theorem 14.8, G has p-length at most 1. Therefore, the upper p-series of G has the form

$$1 \trianglelefteq M \trianglelefteq MG_p \trianglelefteq G,$$

where G_p is a Sylow p-subgroup of G and $M = \mathbf{M}_0(G)$ is the maximal normal subgroup of G with order coprime to p. Since $p \nmid |M|$, we have $M \leq G_{p'} \leq \mathbf{N}_G(G_{p'})$ for every p-complement $G_{p'}$ of G.

The group $\overline{T} = TM/M$ avoids all eccentric chief factors of $\overline{G} = G/M$. So T_pM/M avoids the eccentric p-chief factors of G/M. The quotient $\overline{G}_p = G_pM/M$ is an abelian normal subgroup of $\overline{G}$. Therefore, we have by Theorem 14.6 b) that

$$\overline{G}_p = (\overline{G}_p \cap \overline{G}') \times (\overline{G}_p \cap \mathbf{Z}(\overline{G})).$$

The $\overline{G}$-group $\overline{G}_p/(\overline{G}_p \cap \mathbf{Z}(\overline{G}))$, which is isomorphic to $\overline{G}_p \cap \overline{G}'$, contains only eccentric chief factors of $\overline{G}$, so it is avoided by $\overline{T}_p$. Hence, we obtain

$$\overline{T}_p \leq \overline{G}_p \cap \mathbf{Z}(\overline{G}) \leq \mathbf{N}_{\overline{G}}(\overline{G}_{p'}).$$

And from this follows our intermediate claim

$$T_p \leq T_pM \leq \mathbf{N}_G(G_{p'}M) = \mathbf{N}_G(G_{p'}).$$

Let G_p and $G_{p'}$ be elements of a complement system $\mathscr{K}$ of G with $D = \mathbf{N}(\mathscr{K})$. Then we have by Theorem 11.2 that

$$D_p = G_p \cap \mathbf{N}_G(G_{p'}).$$

As N is a normal p-subgroup of G, we have $ND_p \leq G_p$. Now, T_p is a p-subgroup of ND since $T \leq ND$, so T_p lies in the unique Sylow p-subgroup ND_p of ND. So $T_p \leq ND_p \leq G_p$. Together with the observation that $T_p \leq \mathbf{N}_G(G_{p'})$, which we have already proved, it follows that

$$T_p \leq G_p \cap \mathbf{N}_G(G_{p'}) = D_p.$$

With $T_{p'} \leq D_{p'}$, this yields $T \leq D$. □

Up to this point, we have applied the theory developed in §11 and §12 to A-groups. The following theorems are connected to §5.

Theorem 14.16 (Hall, Higman [224]).

a) *Let $G = \prod_{i=1}^{r} A_i$ be the product of pairwise permutable nilpotent groups A_i of pairwise coprime orders. If c_i is the nilpotency class of A_i, then we have the following bound for the derived length $k(G)$ of G:*

$$k(G) \leq \sum_{i=1}^{r} c_i.$$

(*At the time of writing, it is unknown if one can remove the assumption that the orders of the A_i are coprime.*)

b) *If G is an A-group whose order is divisible by exactly r distinct primes, then G has derived length at most r.*

Proof. a) We prove the claim by induction on $|G|$. By Theorem 4.3, G is solvable. We may assume that $G \neq 1$, so that we have $\mathbf{F}(G) \neq 1$ for the Fitting subgroup of G, as well.

If $\mathbf{F}(G)$ does not have prime power order, then there exist primes p and q with $p \neq q$ such that $\mathbf{F}(G)_p \neq 1 \neq \mathbf{F}(G)_q$. Then the factor groups $G/\mathbf{F}(G)_p$ and $G/\mathbf{F}(G)_q$ satisfy all our assumptions. If we let $k = \sum_{i=1}^{r} c_i$, then we have by induction that $G^{(k)} \leq \mathbf{F}(G)_p \cap \mathbf{F}(G)_q = 1$, and therefore $k(G) \leq \sum_{i=1}^{r} c_i$.

It remains to handle the case that the order of $\mathbf{F}(G)$ is a prime power p^f. Say p divides $|A_1|$. As the $|A_i|$ are coprime, we have $\mathbf{F}(G) \leq A_1$. We therefore have that $\mathbf{Z}(A_1) \leq \mathbf{C}_G(\mathbf{F}(G))$ and $\mathbf{C}_G(\mathbf{F}(G)) \leq \mathbf{F}(G)$ by Theorem III.4.2 b). It follows that $\mathbf{Z}(A_1) \leq \mathbf{Z}(\mathbf{F}(G))$. Therefore, $A_1\mathbf{Z}(\mathbf{F}(G))/\mathbf{Z}(\mathbf{F}(G))$ has nilpotency class at most $c_1 - 1$. Applying the induction hypothesis to $G/\mathbf{Z}(\mathbf{F}(G))$, we obtain

$$k(G/\mathbf{Z}(\mathbf{F}(G))) \leq \left(\sum_{i=1}^{r} c_i\right) - 1$$

and then $k(G) \leq \sum_{i=1}^{r} c_i$.

b) The A-group G has a Sylow system consisting of pairwise permutable Sylow subgroups G_{p_i} with $c(G_{p_i}) = 1$. Applying a) immediately yields $k(G) \leq r$. □

In the last theorem, we bounded the derived length $k(G)$ of an A-group G by the number of distinct prime divisors of $|G|$. Our next goal is to bound the derived length of an A-group by the number of generators of the Sylow subgroups. In preparation, we prove a lemma regarding A-groups of linear maps.

Lemma 14.17. *Let G be an A-group consisting of linear maps of an n-dimensional vector space over a field K. Assume the characteristic of K does not divide $|G|$. Then $G^{(n)} = 1$.*

Proof. We prove the claim by induction on n. For $n = 1$, G is abelian and the theorem holds. We may assume that K is algebraically closed, as passing to a larger field does not change the structure of G as an abstract group, but it does allow more freedom in establishing normal forms for the linear maps. By Theorem V.18.5 c) (pg. 616), we may assume that G consists of monomial matrices.

Let D be the normal subgroup of G consisting of all diagonal matrices of the monomial group G. Then D is abelian, and G/D is isomorphic to a permutation group H of degree n. As a permutation group, H contains the identity representation at least once by Theorem V.20.2 (pg. 633). The characteristic of K does not divide the order of H, so H is completely reducible by Maschke's Theorem I.17.7 (pg. 132). Therefore, the permutation group H decomposes completely into the trivial representation and a faithful representation of H of degree $n - 1$. Thus, $H^{(n-1)} = 1$ by induction. It follows that $G^{(n-1)} \leq D$, and so $G^{(n)} = 1$. □

Theorem 14.18 (Huppert [257]). *Let G be an A-group. Then:*

a) *$k(G) \leq r(G) + 1$, where $r(G)$ is the rank of G.*

b) *If every Sylow subgroup of G can be generated by at most d elements, then $k(G) \leq d + 1$.*

Proof. a) We consider a chief factor H/K of G with order p^s and the automorphism group

$$\mathbf{Aut}_G(H/K) \cong G/\mathbf{C}_G(H/K)$$

of H/K induced by G. As the Sylow p-subgroups of G are abelian, H/K is centralized by all Sylow p-subgroups. So $p \nmid |\mathbf{Aut}_G(H/K)|$. If we consider $\mathbf{Aut}_G(H/K)$ as a subgroup of $\mathrm{GL}_s(p)$, then the assumptions of Lemma 14.17 are satisfied. Thus, $\mathbf{Aut}_G(H/K)^{(s)} = 1$. Since $s \leq r(G) = r$, it also follows that $\mathbf{Aut}_G(H/K)^{(r)} = 1$, and so $G^{(r)} \leq \mathbf{C}_G(H/K)$. As this holds for all chief factors H/K of G, we obtain by Theorem 5.4 that

$$G^{(r)} \leq \bigcap_{H/K} \mathbf{C}_G(H/K) = \mathbf{F}(G).$$

The Fitting subgroup $\mathbf{F}(G)$ is abelian since it is a nilpotent normal subgroup of the A-group G. So $G^{(r+1)} = 1$.

b) Every p-chief factor of G is a homomorphic image of a subgroup of a Sylow p-subgroup of G. If all (abelian) Sylow subgroups of G have at most d generators, then this also holds for the chief factors. So $r(G) \leq d$, and the claim follows by a).□

Remark 14.19. Let G be a group whose Sylow subgroups are all cyclic. By Theorem IV.2.9 (pg. 447), G is solvable. It therefore follows by Theorem 14.18 that $G'' = 1$, which we already knew by Theorem IV.2.11. Incidentally, the theory of supersolvable groups leads to the same conclusion: Clearly, G is supersolvable. As G' is a nilpotent group with cyclic Sylow subgroups by Theorem 9.1 b), G' is cyclic.

Exercises

28) Let P be an abelian p-group, and let $G \leq \mathbf{Aut}(P)$ with $p \nmid |G|$. Let p^m be the smallest invariant of P. (See Theorem I.13.12, pg. 85 for the definition of the invariants of a finitely generated abelian group.) Then there exists a direct decomposition $P = P_1 \times P_2$ of P into G-groups P_i, where all invariants of P_1 are larger than p^m and all invariants of P_2 are exactly p^m.
29) Let P be an abelian p-group, and let $G \leq \mathbf{Aut}(P)$ with $p \nmid |G|$. Then:
   a) There exists a decomposition $P = P_1 \times \cdots \times P_n$, where every P_i is a G-group in which all the invariants are equal.
   b) $\Omega_1(P)$ and $P/\mho_1(P)$ are isomorphic G-groups.
30) (Huppert [257]) Let G be an A-group.
   a) Let U_i $(i = 1, \ldots, s)$ be the maximal subgroups of G, and let $[G : U_i] = p_i^{k_i}$. Then $r(G) = \max k_i$.
   b) We have $r(G/\Phi(G)) = r(G)$.
31) (Huppert [257]) Let G be an A-group. Assume that every minimal normal subgroup of G can be generated by k elements. Then $r(G) \leq k$.
32) (Carter [89]) Let G be an A-group, and let D be a system normalizer of G. Then $\mathbf{N}_G(D)$ covers every D-central D-composition factor and avoids every D-eccentric D-composition factor of G.
33) (Carter [89]) Let D be a system normalizer and N a normal subgroup of the A-group G. Then $\mathbf{N}_{G/N}(DN/N) = \mathbf{N}_G(D)N/N$ (see also Example 11.13).

15 Sylow systems and Carter subgroups

Definition 15.1. Let $\mathscr{K}_G = \{G_{p'} \mid p \text{ divides } |G|\}$ be a complement system of the solvable group G, and let H be a subgroup of G. We say that $\mathscr{K}_G$ is reducible in H if

$$\mathscr{K}_H = \{H \cap G_{p'} \mid p \text{ divides } |H|\}$$

is a complement system of H. In this case, we say that $\mathscr{K}_H$ is the reduction of $\mathscr{K}_G$ in H.

Let $\mathscr{S}_G = \{G_p \mid p \text{ divides } |G|\}$ be the Sylow system of G constructed from $\mathscr{K}_G$ according to Theorem 2.2, so $G_p = \bigcap_{p \neq q} G_{q'}$. Then the subgroups

$$G_p \cap H = \bigcap_{p \neq q} (G_{q'} \cap H)$$

are together the Sylow system $\mathscr{S}_H$ of H corresponding to $\mathscr{K}_H$. We call $\mathscr{S}_H$ the reduction of $\mathscr{S}_G$ in H.

Lemma 15.2. *Assume G is solvable, and let $\mathscr{S}$ be a Sylow system of G. Then:*
a) *$\mathscr{S}$ is reducible in every normal subgroup of G.*
b) *$\mathscr{S}$ is reducible in its system normalizer $D = \mathbf{N}(\mathscr{S})$.*
c) *If $\mathscr{S}$ is reducible in A and B and $A \leq \mathbf{N}_G(B)$, then $\mathscr{S}$ is reducible in AB.*

Proof. Part a) was proved in Theorem 2.5 b).

b) Let $G_{p'} = \prod_{q \neq p} G_q$ with $G_q \in \mathscr{S}$ be the Sylow complements in the complement system of G corresponding to $\mathscr{S}$ according to Theorem 2.2. By Theorem 11.2 a), the Sylow p-subgroup of $D = \mathbf{N}(\mathscr{S})$ is $G_p \cap \mathbf{N}_G(G_{p'})$. So it follows that $D \cap G_p = D_p$, and hence $\mathscr{S}$ is reducible in D.

c) Let $G_{p'}$ be the p-complement in the complement system corresponding to $\mathscr{S}$ constructed according to Theorem 2.2. Since $B \trianglelefteq AB$, $B(AB \cap G_{p'})$ is a subgroup of AB. We have

$$\begin{aligned}[AB : AB \cap G_{p'}] &= [AB : B(AB \cap G_{p'})][B(AB \cap G_{p'}) : AB \cap G_{p'}] \\ &= [A : A \cap B(AB \cap G_{p'})][B : B \cap G_{p'}].\end{aligned}$$

As $\mathscr{S}$ is reducible in B, the second factor is a power of p. Since

$$A \cap B(AB \cap G_{p'}) \geq A \cap G_{p'},$$

the first factor certainly divides $[A : A \cap G_{p'}]$, so it is also a power of p as $\mathscr{S}$ is reducible in A. Hence, $[AB : AB \cap G_{p'}]$ is a power of p, and $AB \cap G_{p'}$ is therefore a p-complement of AB. So $\mathscr{S}$ is reducible in AB. □

Lemma 15.3. *Assume G is solvable, and suppose H is a maximal, but non-normal, subgroup of G. Let $[G : H] = p^n$.*
a) *If $G_{p'}$ is a p-complement of G lying in H, then $\mathbf{N}_G(G_{p'}) \leq H$.*
b) *The complement system $\mathscr{K} = \{G_{q'} \mid q \text{ divides } |G|\}$ of G is reducible in H if and only if $G_{p'} \leq H$. Every complement system of H is the reduction of the same number of complement systems of G.*
c) *Let D be a system normalizer of G. The number of Sylow systems of G that are reducible in H is $[G : D]/[G : H]$.*

Proof. Part a) was proved already in Lemma 11.19 a).

b) Let $q \neq p$. Since the indices of $G_{q'}$ and H are coprime, we have $G = G_{q'}H$, and therefore

$$[H : H \cap G_{q'}] = [G : G_{q'}]$$

is a power of q. So $H \cap G_{q'}$ for $q \neq p$ is a q-complement of H. The complement system $\mathscr{K} = \{G_{q'} \mid q \text{ divides } |G|\}$ of G is therefore reducible in H if and only if $H \cap G_{p'}$ is a p-complement of H. Since $[G : H] = p^n$, this forces $G_{p'} \leq H$.

As all complement systems of H are conjugate in H by Theorem 2.4, every complement system of H is the reduction of the same number of complement systems of G.

c) The number of complement systems (Sylow systems) of G is

$$[G : D] = \prod_q [G : \mathbf{N}_G(G_{q'})]$$

by Theorem 2.4. By b), the number of complement systems of G that are reducible in H is equal to the number of complement systems $\{G_{q'} \mid q \text{ divides } |G|\}$ with $G_{p'} \leq H$. So by a),

$$[H : \mathbf{N}_G(G_{p'})] \prod_{q \neq p} [G : \mathbf{N}_G(G_{q'})] = \frac{[G : D]}{[G : H]}. \qquad \square$$

Theorem 15.4. *Assume G is solvable and H is abnormal in G. Let D be a system normalizer of G with $D \leq H$.*

a) *The number of Sylow systems of G that are reducible in H is $[H : D]$.*

b) *Every two Sylow systems of G that are reducible in H are conjugate by an element of H.*

Proof. a) Let

$$1 = N_0 \lhd N_1 \lhd \cdots \lhd N_s = G$$

be a composition series of G as an H-group. (In particular, $H \leq \mathbf{N}_G(N_i)$, and the characteristically simple, hence elementary abelian, factor groups N_{i+1}/N_i are irreducible H-modules.) We consider the chain of subgroups

$$H = H_0 \leq H_1 \leq \cdots \leq H_s = G$$

with $H_i = HN_i$.

First we show that if $H_i < H_{i+1}$, then H_i is a maximal subgroup of H_{i+1}. Let

$$H_i = HN_i \leq K < HN_{i+1} = H_{i+1}.$$

Then $K = H(K \cap N_{i+1})$, and it follows that $N_i \leq K \cap N_{i+1} < N_{i+1}$. Since $H \leq K$, $K \cap N_{i+1}$ is invariant under H. As N_{i+1}/N_i is an H-composition factor of G, it follows that $K = H_i$. So H_i is a maximal subgroup of H_{i+1}.

Let D_i be a system normalizer of H_i. Assume for induction that

$$n_i = \frac{[G : D]}{[G : H_i]}$$

Sylow systems of G are reducible in H_i and that every Sylow system of H_i is the reduction of exactly

$$m_i = \frac{[G : D]}{[G : D_i]}$$

Sylow systems of G. We show that the corresponding statements for H_{i-1} also hold. If $H_{i-1} = H_i$, then there is nothing to prove. So assume that $H_{i-1} < H_i$. By the abnormality of H, the subgroup H_{i-1} is maximal, but not normal, in H_i.

Let $\mathscr{S}_{i-1}$ be a Sylow system of H_{i-1} that is a reduction of the Sylow system $\mathscr{S} = \{G_p \mid p \text{ divides } |G|\}$ of G. By applying Lemma 15.2 a) repeatedly, we see that $\mathscr{S}$ is reducible in all N_j. So $\mathscr{S}$ is then also reducible in

$$H_{i-1}N_i = (HN_{i-1})N_i = HN_i = H_i$$

by Lemma 15.2 c). Let

$$\mathscr{S}_i = \{G_p \cap H_i \mid p \text{ divides } |H_i|\}.$$

Then by Lemma 15.3 c), $\mathscr{S}_i$ is one of the

$$[H_i : D_i]/[H_i : H_{i-1}]$$

Sylow systems of H_i that are reducible in H_{i-1}. As $\mathscr{S}_i$ is the reduction of

$$m_i = [G : D]/[G : D_i]$$

Sylow systems of G, there are exactly

$$\frac{[G : D][H_i : D_i]}{[G : D_i][H_i : H_{i-1}]} = \frac{[G : D]}{[G : H_{i-1}]} = n_{i-1}$$

Sylow systems of G that are reducible in H_{i-1}.

Each of the $[H_{i-1} : D_{i-1}]$ Sylow systems of H_{i-1} is the reduction of the same number, say k_{i-1}, of Sylow systems of H_i by Lemma 15.3 b). So we have by Lemma 15.3 c) that

$$k_{i-1}[H_{i-1} : D_{i-1}] = \frac{[H_i : D_i]}{[H_i : H_{i-1}]}.$$

This yields $k_{i-1} = [D_{i-1} : D_i]$. Therefore, every Sylow system of H_{i-1} is the reduction of exactly

$$k_{i-1}m_i = \frac{[G : D]}{[G : D_{i-1}]} = m_{i-1}$$

Sylow systems of G.

b) Changing notation if necessary, let

$$H = H_0 < H_1 < \cdots < H_k = G$$

be the chain of groups we considered in a) but with the repetitions removed. So H_i is maximal, but not normal, in H_{i+1}. We recall from Lemma 15.2 that every Sylow system of G that is reducible in H is also reducible in all H_i. Let

$$\mathscr{S} = \{G_p \mid p \text{ divides } |G|\} \quad \text{and} \quad \mathscr{S}_i = \{G_p \cap H_i \mid p \text{ divides } |H_i|\}.$$

By Lemma 11.19 b), we have $\mathbf{N}_{H_i}(\mathscr{S}_i) \leq \mathbf{N}_{H_{i-1}}(\mathscr{S}_{i-1})$. So $\mathbf{N}(\mathscr{S}) \leq H_0 = H$.

Let $H = \bigcup_{j=1}^{s} \mathbf{N}(\mathscr{S})h_j$ with $s = [H : \mathbf{N}(\mathscr{S})]$. Then the s Sylow systems

$$\mathscr{S}^{h_j} = \{G_p^{h_j} \mid p \text{ divides } |G|\}$$

are pairwise distinct and reducible in H. By a), we have that

$$\frac{[G : D]}{[G : H]} = [H : \mathbf{N}(\mathscr{S})]$$

is the number of Sylow systems of G that are reducible in H. So all such Sylow systems are conjugate by an element of H. □

Lemma 15.5 (Alperin [6]). *Assume G is solvable, and let P be a p-subgroup of $\mathbf{N}_G(G_{p'})$.*

a) *Then $G_{p'} \cap \mathbf{C}_G(P)$ is a p-complement of $\mathbf{C}_G(P)$.*

b) *If $H \leq G$ with $P \leq \mathbf{C}_G(H)$ and $p \nmid |H|$, then there exists a p-complement $G_{p'}^*$ of G with $H \leq G_{p'}^*$ and $P \leq \mathbf{N}_G(G_{p'}^*)$.*

Proof. a) We prove the claim by induction on $|G|$. Let N be a minimal normal subgroup of G and $|N| = q^k$.

Case 1: Assume $p \neq q$. We have by induction that

$$G_{p'}/N \cap \mathbf{C}_{G/N}(PN/N) = T/N$$

is a p-complement of $\mathbf{C}_{G/N}(PN/N) = \mathbf{C}_G(PN/N)/N$. So T is a p-complement of $\mathbf{C}_G(PN/N)$.

We first show that $T = \mathbf{C}_T(P)N$:

Let $t \in T$. Then P^t is a Sylow p-subgroup of PN. So $P^t = P^n$ for some $n \in N$. Therefore, it follows that

$$x = tn^{-1} \in \mathbf{N}_G(P) \cap T.$$

Since $x \in \mathbf{C}_G(PN/N) \cap \mathbf{N}_G(P)$ and $(|N|, |P|) = 1$, we have for all $y \in P$ that

$$[x, y] \in P \cap N = 1.$$

So it follows that $x \in \mathbf{C}_T(P)$ and $t \in \mathbf{C}_T(P)N$.

Now let K be a p-complement of $\mathbf{C}_G(P)$. By Theorem 1.7, there exists a $y \in \mathbf{C}_G(P)$ such that $\mathbf{C}_T(P)^y \leq K$. So

$$T^y = \mathbf{C}_T(P)^y N \leq KN.$$

As T^y is a p-complement of $\mathbf{C}_G(PN/N)$ and KN is a subgroup of $\mathbf{C}_G(PN/N)$ with order coprime to p, we have $T^y = KN$. Since $y \in \mathbf{C}_G(P)$, this yields $K^{y^{-1}} \leq T$, where $K^{y^{-1}}$ is a p-complement of $\mathbf{C}_G(P)^{y^{-1}} = \mathbf{C}_G(P)$. So we have

$$K^{y^{-1}} \leq T \leq G_{p'},$$

and $K^{y^{-1}} = G_{p'} \cap \mathbf{C}_G(P)$ is a p-complement of $\mathbf{C}_G(P)$.

Case 2: Assume $|N| = p^k$. By the induction hypothesis,

$$G_{p'}N/N \cap \mathbf{C}_{G/N}(PN/N) = TN/N,$$

and $T = G_{p'} \cap \mathbf{C}_G(PN/N)$ is a p-complement of $\mathbf{C}_{G/N}(PN/N) = \mathbf{C}_G(PN/N)/N$. Since $|N| = p^k$, the subgroup T is a p-complement of $\mathbf{C}_G(PN/N)$. Since $P \leq \mathbf{N}_G(G_{p'})$, the subgroup $T = G_{p'} \cap \mathbf{C}_G(PN/N)$ is normalized by P. It follows that

$$[P, T] \leq N \cap T = 1.$$

So $T \leq \mathbf{C}_G(P) \cap G_{p'}$. As T is a p-complement of $\mathbf{C}_G(PN/N)$, T is also a p-complement of $\mathbf{C}_G(P)$, so $T = \mathbf{C}_G(P) \cap G_{p'}$.

b) We may clearly assume that H is a p-complement of $\mathbf{C}_G(P)$. By a), $\mathbf{C}_G(P) \cap G_{p'}$ is a p-complement of $\mathbf{C}_G(P)$. By Theorem 1.7, there exists an $x \in \mathbf{C}_G(P)$ such that

$$H = (\mathbf{C}_G(P) \cap G_{p'})^x = \mathbf{C}_G(P) \cap G_{p'}^x.$$

Then the claim holds upon letting $G_{p'}^* = G_{p'}^x$, since $P = P^x \leq \mathbf{N}_G(G_{p'}^x)$. □

Theorem 15.6 (Alperin [6]). *Assume G is solvable.*

a) *Let D be a system normalizer of G, and let C be a Carter subgroup of G. Then $D \leq C$ if and only if there exists a Sylow system $\mathscr{S}$ of G that is reducible in C and whose system normalizer $\mathbf{N}(\mathscr{S})$ is equal to D.*

b) *If D_1 and D_2 are system normalizers of G that both lie in the Carter subgroup C of G, then D_1 and D_2 are conjugate in C.*

Proof. a) If the Sylow system $\mathscr{S}$ of G is reducible in C, then $\mathbf{N}(\mathscr{S}) \leq C$ by Theorem 12.8.

Now let D be a system normalizer of G with $D \leq C$. Since $D_p \leq \mathbf{C}_G(C_{p'})$, there exists by Lemma 15.5 b) a p-complement $G_{p'}$ of G with $C_{p'} \leq G_{p'}$ and $D_p \leq \mathbf{N}_G(G_{p'})$. It follows trivially from $D_{p'} \leq C_{p'} \leq G_{p'}$ that $D_{p'} \leq \mathbf{N}_G(G_{p'})$. Therefore,

$$\mathscr{K} = \{G_{p'} \mid p \text{ divides } |G|\}$$

is a complement system of G that is reducible in C and satisfies $D \leq \mathbf{N}(\mathscr{K})$, so we have $D = \mathbf{N}(\mathscr{K})$.

b) Let D_1 and D_2 be system normalizers with $D_i \leq C$ $(i = 1, 2)$. By a), there exist Sylow systems $\mathscr{S}_1$ and $\mathscr{S}_2$ of G that are reducible in C with $D_i = \mathbf{N}(\mathscr{S}_i)$. As C is abnormal in G by Theorem 12.2, the Sylow systems $\mathscr{S}_1$ and $\mathscr{S}_2$ are conjugate by an element $c \in C$ by Theorem 15.4 b). It follows that

$$D_1 = \mathbf{N}(\mathscr{S}_1) = \mathbf{N}(\mathscr{S}_2^c) = \mathbf{N}(\mathscr{S}_2)^c = D_2^c. \qquad \square$$

Theorem 15.7. *Assume G is solvable, and let $\mathscr{S}$ be a Sylow system of G with $D = \mathbf{N}(\mathscr{S})$. Then:*
a) *(P. Hall) $D \leq \mathbf{Z}_\infty(\mathbf{N}_G(D))$.*
b) *(Alperin) $\mathscr{S}$ is reducible in $\mathbf{N}_G(D)$.*

Proof. a) Let

$$1 = G_0 \lhd G_1 \lhd \cdots \lhd G_s = G$$

be a chief series of G. Let $D_i = D \cap G_i$.

If G_{i+1}/G_i is central, then

$$[D_{i+1}, \mathbf{N}_G(D)] \leq D \cap [G_{i+1}, G] \leq D \cap G_i = D_i.$$

On the other hand, if G_{i+1}/G_i is eccentric, then D avoids G_{i+1}/G_i by Theorem 11.10, and we have $D_{i+1} = D_i$.

So all the factor groups D_{i+1}/D_i are centralized by $\mathbf{N}_G(D)$, and it follows that

$$D \leq \mathbf{Z}_\infty(\mathbf{N}_G(D)).$$

b) Let $\mathscr{K} = \{G_{p'} \mid p \text{ divides } |G|\}$ be a complement system of G, let $D = \mathbf{N}(\mathscr{K})$, and let $N = \mathbf{N}_G(D)$. Let $N_{p'}$ be a p-complement of N. Since $D_p \trianglelefteq \mathbf{N}_G(D) = N$ and

$$D_p \leq D_p N_{p'} \cap \mathbf{Z}_\infty(N) \leq \mathbf{Z}_\infty(D_p N_{p'}),$$

we have by Lemma 12.9 b) that $D_p N_{p'} = D_p \times N_{p'}$. By Lemma 15.5 b), there exists a p-complement $G_{p'}^*$ of G with $D_p \leq \mathbf{N}_G(G_{p'}^*)$ and $N_{p'} \leq G_{p'}^*$. Since $D_{p'} \trianglelefteq N$, we have $D_{p'} \leq N_{p'} \leq G_{p'}^*$, and it follows that $D \leq \mathbf{N}_G(G_{p'}^*)$. Let

$$\mathscr{K}^* = \{G_{p'}^* \mid p \text{ divides } |G|\}.$$

Then $D = \mathbf{N}(\mathscr{K}) = \mathbf{N}(\mathscr{K}^*)$. By Theorem 2.4, there exists an $x \in G$ such that $G_{p'}^x = G_{p'}^*$ for all p. Hence,

$$D = \mathbf{N}(\mathscr{K}^*) = \mathbf{N}(\mathscr{K}^x) = \mathbf{N}(\mathscr{K})^x = D^x,$$

and so $x \in \mathbf{N}_G(D)$. It follows that

$$G_{p'} \cap \mathbf{N}_G(D) = (G_{p'}^*)^{x^{-1}} \cap \mathbf{N}_G(D) = (G_{p'}^* \cap \mathbf{N}_G(D))^{x^{-1}} = N_{p'}^{x^{-1}},$$

and this is a p-complement of $\mathbf{N}_G(D)$. Hence, the complement system $\mathscr{K}$ is reducible in $\mathbf{N}_G(D)$. Then the Sylow system corresponding to $\mathscr{K}$ is also reducible in $\mathbf{N}_G(D)$.□

Theorem 15.8 (Alperin [6]). *Assume G is solvable. Let D be a system normalizer of G, and let C be a Carter subgroup of G with $D \leq C$. Then:*

a) $[C : \mathbf{N}_G(D) \cap C]$ *is the number of system normalizers of G lying in C.*

b) $[\mathbf{N}_G(D) : \mathbf{N}_G(D) \cap C]$ *is the number of Carter subgroups of G that contain a given system normalizer.*

c) $[\mathbf{N}_C(D) : D]$ *is the number of Sylow systems of G that are reducible in C and normalized by D.*

Proof. a) This follows immediately from Theorem 15.6 b), as all system normalizers of G that lie in C are conjugate in C.

b) Let $D \leq C \cap C^x$ with $x \in G$. Then we have $D^{x^{-1}} \leq C$. By Theorem 15.6 b), there exists a $c \in C$ such that $D^{x^{-1}} = D^c$. Therefore, $cx \in \mathbf{N}_G(D)$, so $x \in C\mathbf{N}_G(D)$. Conversely, it is clear that if $x \in C\mathbf{N}_G(D)$ and $D \leq C$, then $D \leq C^x$. So

$$[C\mathbf{N}_G(D) : C] = [\mathbf{N}_G(D) : C \cap \mathbf{N}_G(D)]$$

is the number of C^x with $D \leq C^x$.

c) Let $\mathscr{S}$ be a Sylow system of G that is reducible in C and whose system normalizer $\mathbf{N}(\mathscr{S})$ is equal to D. Assume that $\mathscr{S}^x$ is reducible in C and $\mathbf{N}(\mathscr{S}^x) = D^x = D$. Since C is abnormal in G, there exists by Theorem 15.4 b) a $c \in C$ such that $\mathscr{S}^x = \mathscr{S}^c$. It follows that

$$D = \mathbf{N}(\mathscr{S})^x = \mathbf{N}(\mathscr{S})^c = D^c.$$

So $c \in \mathbf{N}_C(D)$, and the claim is proved. □

Theorem 15.9 (Alperin [6]). *Assume G is solvable. For every Sylow system $\mathscr{S}$ of G, there exists exactly one Carter subgroup of G in which $\mathscr{S}$ is reducible.*

Proof. As all Sylow systems of G are conjugate, every $\mathscr{S}$ is reducible in at least one Carter subgroup. If $\mathscr{S}$ is reducible in C, then $\mathbf{N}(\mathscr{S}) \leq C$ by Theorem 12.8.

Let n be the number of pairs $(\mathscr{S}, C)$ consisting of a Sylow system $\mathscr{S}$ and a Carter subgroup C in which $\mathscr{S}$ is reducible. Let k be the number of C in which a fixed $\mathscr{S}$ is reducible. As $[G : D]$ is the number of Sylow systems of G, we have $n = [G : D]k$. On the other hand, $n = n_1 n_2 n_3$, where

n_1 = number of system normalizers = $[G : \mathbf{N}_G(D)]$,

n_2 = number of Carter subgroups C with $D \leq C$

$= [\mathbf{N}_G(D) : \mathbf{N}_G(D) \cap C]$ (by Theorem 15.8 b)),

n_3 = number of Sylow systems $\mathscr{S}$ such that $\mathscr{S}$ is reducible in C and $D = \mathbf{N}(\mathscr{S})$

$= [\mathbf{N}_G(D) \cap C : D]$ (by Theorem 15.8 c)).

Altogether, it follows that

$$[G : D]k = n = n_1 n_2 n_3 = [G : D],$$

and therefore $k = 1$. Hence, every Sylow system of G is reducible in exactly one Carter subgroup of G. □

Notes for Chapter VI

§1: Exercise 1 follows the original proof of Theorem 1.8, which can be found in P. Hall [206].

§2: Our proof of Theorem 2.4 stems from a lecture of H. Wielandt. It is a considerable simplification of the proof in P. Hall [212].

§11: The theory of system normalizers was developed in P. Hall [213]. In parts, our proofs follow a lecture of H. Wielandt and Baer [27]. The proof of Theorem 11.21 b) was shared with me by R. Carter. It is a considerable simplification of the proof in P. Hall [213].

§12: Lemma 12.5 through Theorem 12.7 follow an unpublished dissertation by F. Erber (Tübingen 1965).

§13: This section reproduces the most important results in Carter [86]. Theorem 13.7 is due to a conversation with Alperin.

§14: In this section, the discussion of A-groups up to Theorem 14.7 follows Taunt [426], with a few simplifications.

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B. Huppert, *Finite Groups I*, Grundlehren der mathematischen Wissenschaften 364,
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$p^n \top a$: p is a prime with $p^n \mid a$ and $p^{n+1} \nmid a$.

(a, b) is the greatest common divisor of the integers a and b.

$\mathsf{V}(n, \mathsf{K})$ is the vector space of dimension n over the field K.

V^* is the vector space dual to V, and a^T is the transpose of the linear map a.

$\langle v_1, \ldots, v_n \rangle$ is the subspace spanned by the vectors v_i.

$\mathrm{GL}_n(\mathsf{K}), \mathrm{PGL}_n(\mathsf{K}), \mathrm{SL}_n(\mathsf{K}), \mathrm{PSL}_n(\mathsf{K});$
$\mathrm{GL}_n(p^f), \mathrm{PGL}_n(p^f), \mathrm{SL}_n(p^f), \mathrm{PSL}_n(p^f)$ (II.6.1; pg. 185)

$U \leq G, U < G$	(I.2.1; pg. 4)	$\Phi(G)$	(III.3.1; pg. 283)
$U \trianglelefteq G, U \lhd G$	(I.3.1; pg. 12)	$\Omega_i(G), \mathfrak{V}_i(G)$	(III.10.4; pg. 342)
$U \lhd \lhd G$	(pg. 325)	$\mathbf{F}(G)$	(III.4.2; pg. 292)
1	(I.2.1; pg. 4)	$n(G)$	(III.4.7; pg. 295)
$\lvert M \rvert$	(I.1.7; pg. 4)	$\mathbf{F}_p(G)$	(VI.5.4; pg. 726)
$[G : U]$	(I.2.5; pg. 5)	$\mathbf{Aut}(G)$	(I.4.1; pg. 19)
$\langle M \rangle$	(I.2.4; pg. 5)	$\mathbf{Inn}(G)$	(I.4.2; pg. 19)
m^g	(I.2.14; pg. 8)	$\bigtimes_i G_i$	(I.9.1; pg. 46)
M^g	(I.2.15; pg. 9)	$A \curlyvee B$	(I.9.10; pg. 51)
M^G	(I.3.14; pg. 18)	$A \curlywedge B$	(I.9.11; pg. 52)
$\mathbf{N}_G(M), \mathbf{C}_G(M)$	(I.2.17; pg. 10)	$A \wr B$	(I.15.1; pg. 101)
$\mathbf{C}_G(N/M)$	(I.4.3; pg. 19)	$A \wr_r B$	(I.15.5; pg. 103)
$\lvert g \rvert, \exp G$	(I.2.8; pg. 6)	$A \wr_U B$	(I.15.10; pg. 106)

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https://doi.org/10.1007/978-3-031-87529-8

$[a,b]$, $[A,B]$, G', G'', $G^{(i)}$	(I.8.1; pg. 38)	G_a, G_Δ, G^Δ, Δ^G	(I.5.9; pg. 27)
$k(G)$	(I.8.7; pg. 40)	Δ^g	(I.5.1; pg. 25)
$\mathbf{K}_i(G)$, $\mathbf{K}_\infty(G)$	(III.2.2; pg. 274)	$\mathrm{sgn}(G)$	(I.5.4; pg. 26)
$c(G)$	(III.2.7; pg. 277)	$S[G]$	(I.16.6; pg. 112)
$\mathbf{Z}(G)$	(I.2.17; pg. 10)	χ^G, V^G, $\mathbf{D}^G$	(V.16.1; pg. 586)
$\mathbf{Z}_i(G)$, $\mathbf{Z}_\infty(G)$	(III.2.2; pg. 274)	χ_U, V_U, $\mathbf{D}_U$	(V.16.4; pg. 588)
$l_p(G)$	(VI.6.1; pg. 728)		

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